



RCR STD-27

ENGLISH TRANSLATION

PERSONAL DIGITAL CELLULAR TELECOMMUNICATION SYSTEM ARIB STANDARD

RCR STD-27 L

Fascicle 2

ISSUED	A P R I L	30, 1991
REVISION A	J A N U A R Y	30, 1992
REVISION B	D E C E M B E R	10, 1992
REVISION C	N O V E M B E R	10, 1994
REVISION D	J U N E	27, 1995
REVISION E	S E P T E M B E R	18, 1996
REVISION F	F E B R U A R Y	18, 1997
REVISION G	M A Y	29, 1998
REVISION H	F E B R U A R Y	2, 1999
REVISION I	J U L Y	25, 2000
REVISION I-1	M A R C H	27, 2001
REVISION J	M A Y	30, 2002
REVISION K	J U L Y	29, 2003
REVISION L	N O V E M B E R	30, 2005

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Fascicle 3

Annex 1 Authentication, Ciphering and Subscriber Data Registration Standard for the Personal Digital Cellular Telecommunication System

Annex 2 Data Transmission Standard for the Personal Digital Cellular Telecommunication System (G3 facsimile and modem (V.42 ANNEX))

Annex 3 High Speed Data Transmission Standard for the Personal Digital Cellular Telecommunication System

Annex 4 Standard for Interface between Mobile Station and Subscriber Information Module

Annex 5 Half-duplex packet communications standard for the Personal Digital Cellular Telecommunication System

Annex 6 Standard for the Radio Interface of the Wireless Local Loop System

Attachment Amendment History

Appendix A SDL Diagrams

A.1 Outline

The purpose of these reference materials and flow charts is to provide an example of a SDL representation of the point-to-point procedures for each layer to assist in the understanding of the specifications.

The representation is a peer-to-peer model of the point-to-point procedures for each layer and is applicable to the entities at both the user and network sides. (See Fig. A.1)

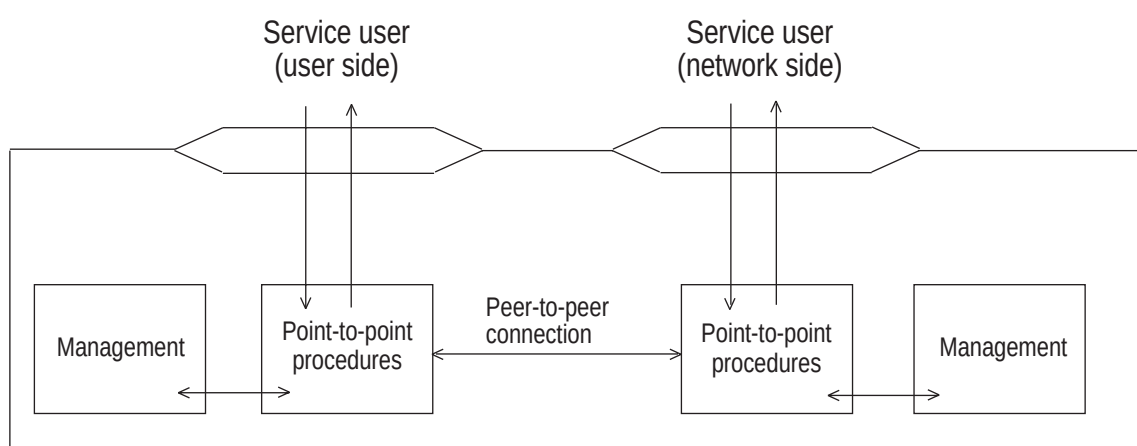
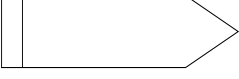
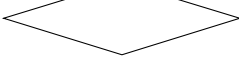
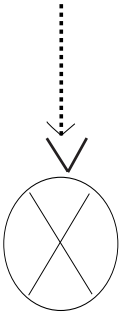


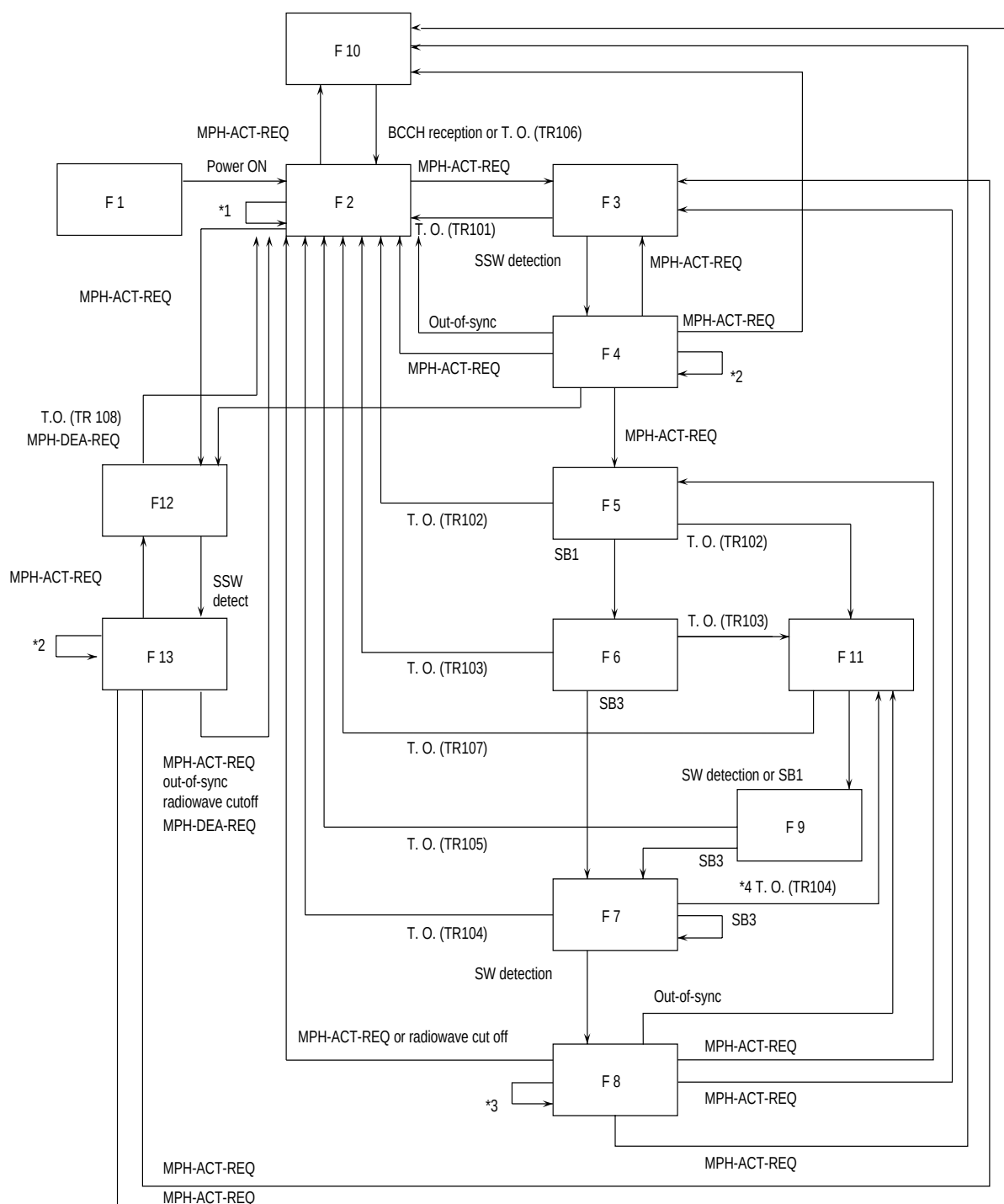
Fig. A.1 Peer-to-peer model of the point-to-point procedures

A. 2 Definition of symbols

The following symbols and abbreviations are used within this description. A full description of the symbols and their meaning and application can be found in the ITU-T Series Z Recommendations (Fascicles 0.1 and 0. 2).

a)		State
b)		Signal reception
c)		Signal generation
d)		Primitive reception
e)		Primitive transmission
f)		Save a signal (until completion of a transition to a new state)
g)		Process description
h)		Test
i)		Procedure call
j)		Procedure definition
		
k)	***	To mark an internal event or signal requiring a layer entity as a result of the representation approach adopted
l)	RC	Retransmission counter

A.3 Layer 1 SDL Diagrams



*1 : MPH-INF-REQ

*2 : MPH-INF-REQ, PH-DATA-REQ Layer 2 signal reception, State report

*3 : MPH-INF-REQ, PH-DATA-REQ, Layer 2 signal reception, State report, Power control, Time alignment designation, Downlink transmission power change report

*4 : Optional (Refer to Fig. A.3-10.)

Fig. A.3-1 State transition diagram (user side)

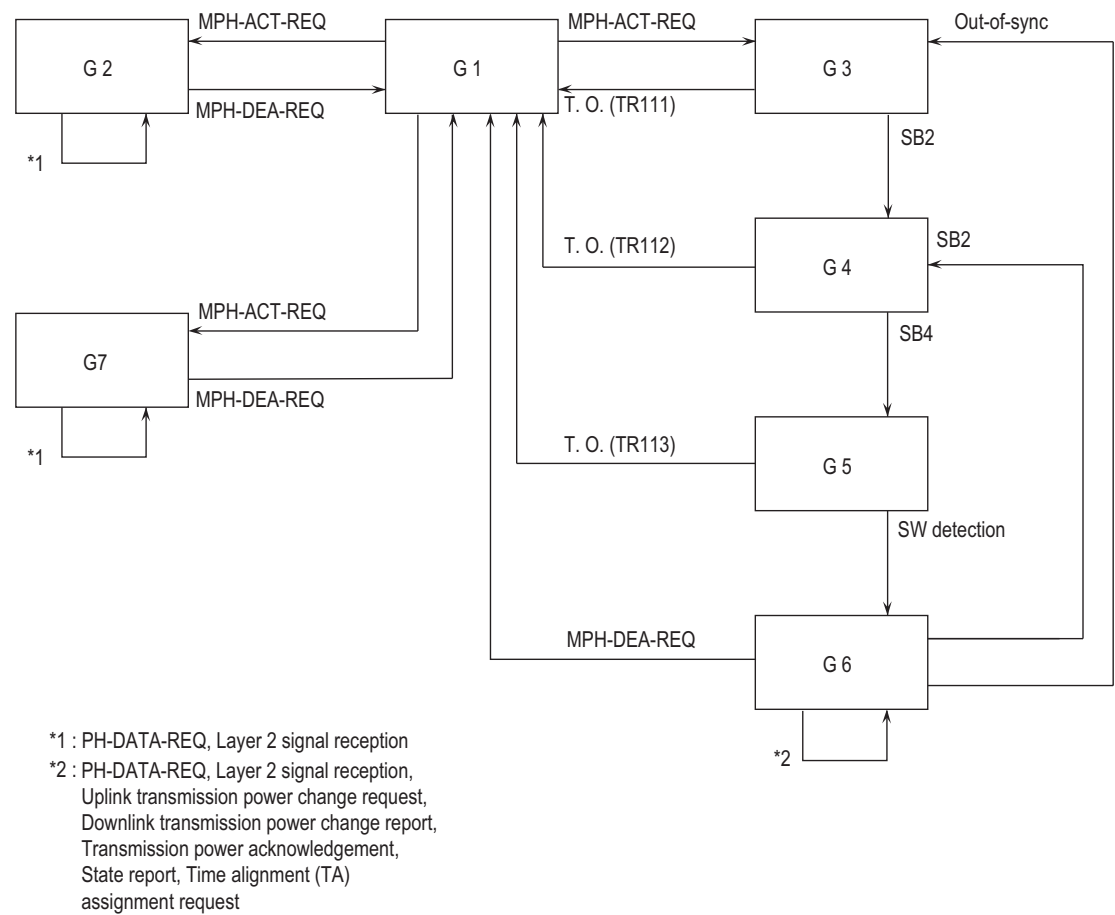


Fig. A. 3-2 State transition diagram (NW side)

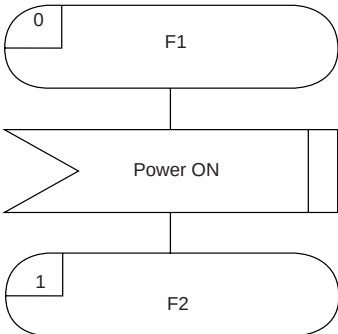


Fig. A. 3-3 Layer 1 SDL diagram (user side) (1/14)

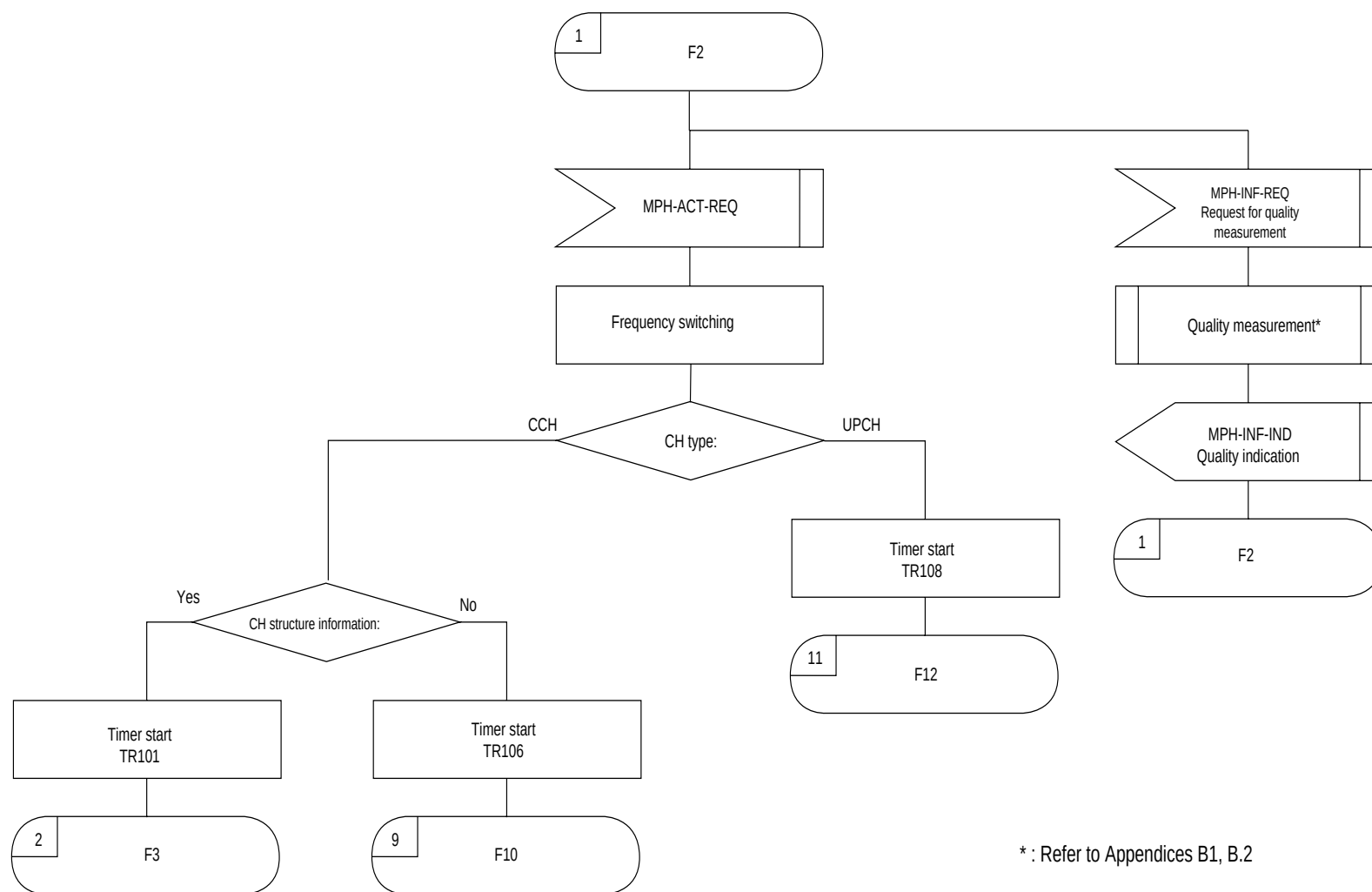


Fig. A. 3-4 Layer 1 SDL diagram (user side) (2/14)

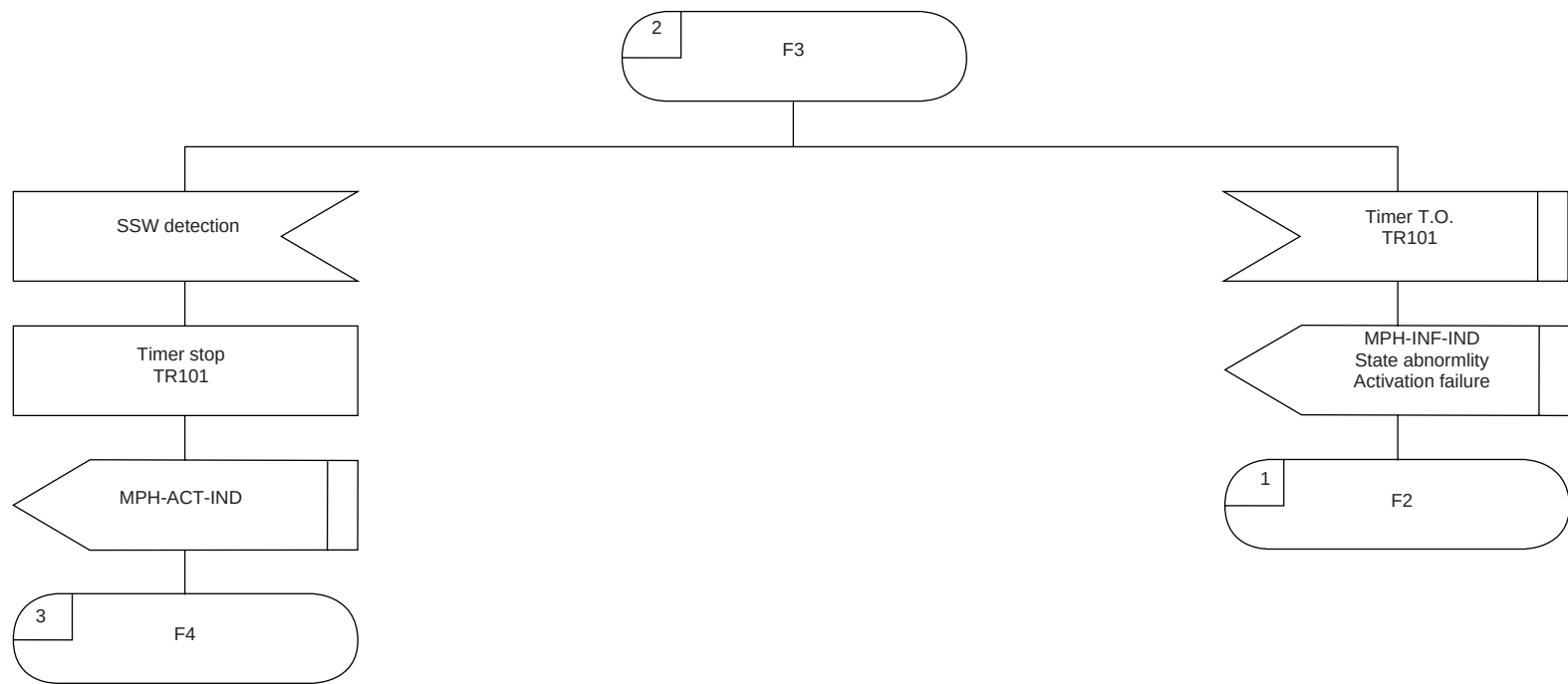


Fig. A. 3-5 Layer 1 SDL diagram (user side) (3/14)

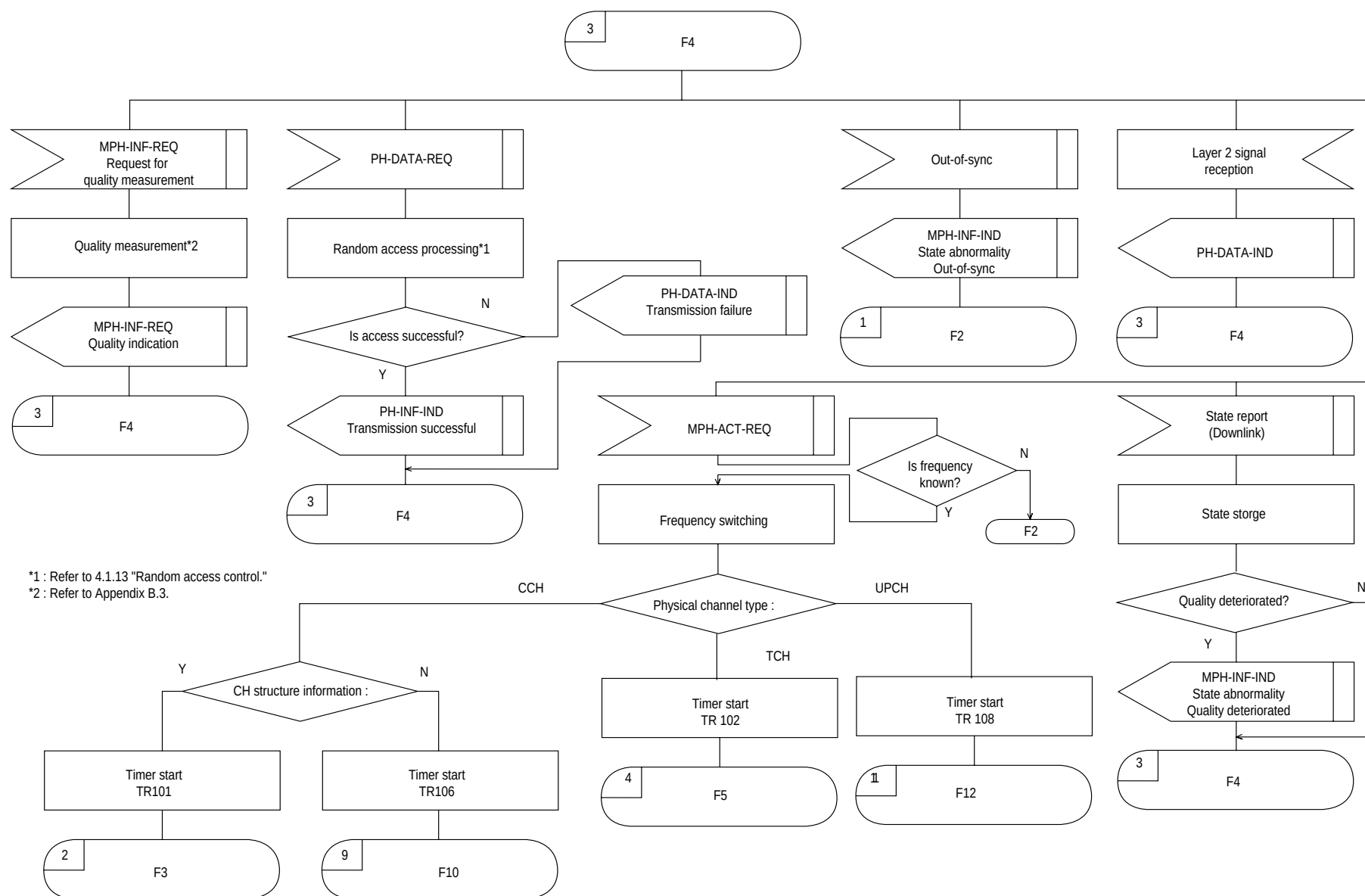


Fig. A.3-6 Layer 1 SDL diagram (User side) (4/14)

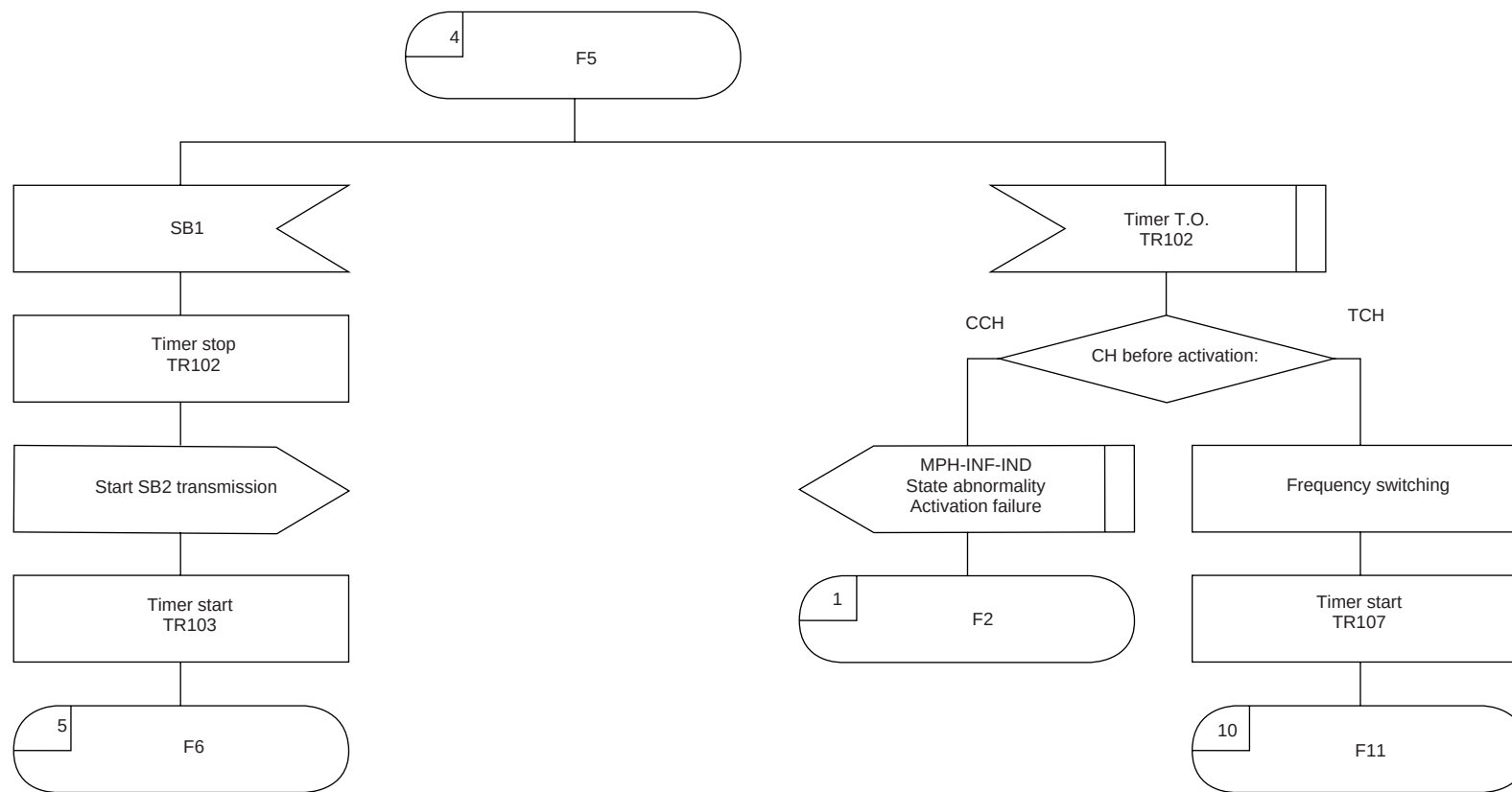


Fig. A. 3-7 Layer 1 SDL diagram (user side) (5/14)

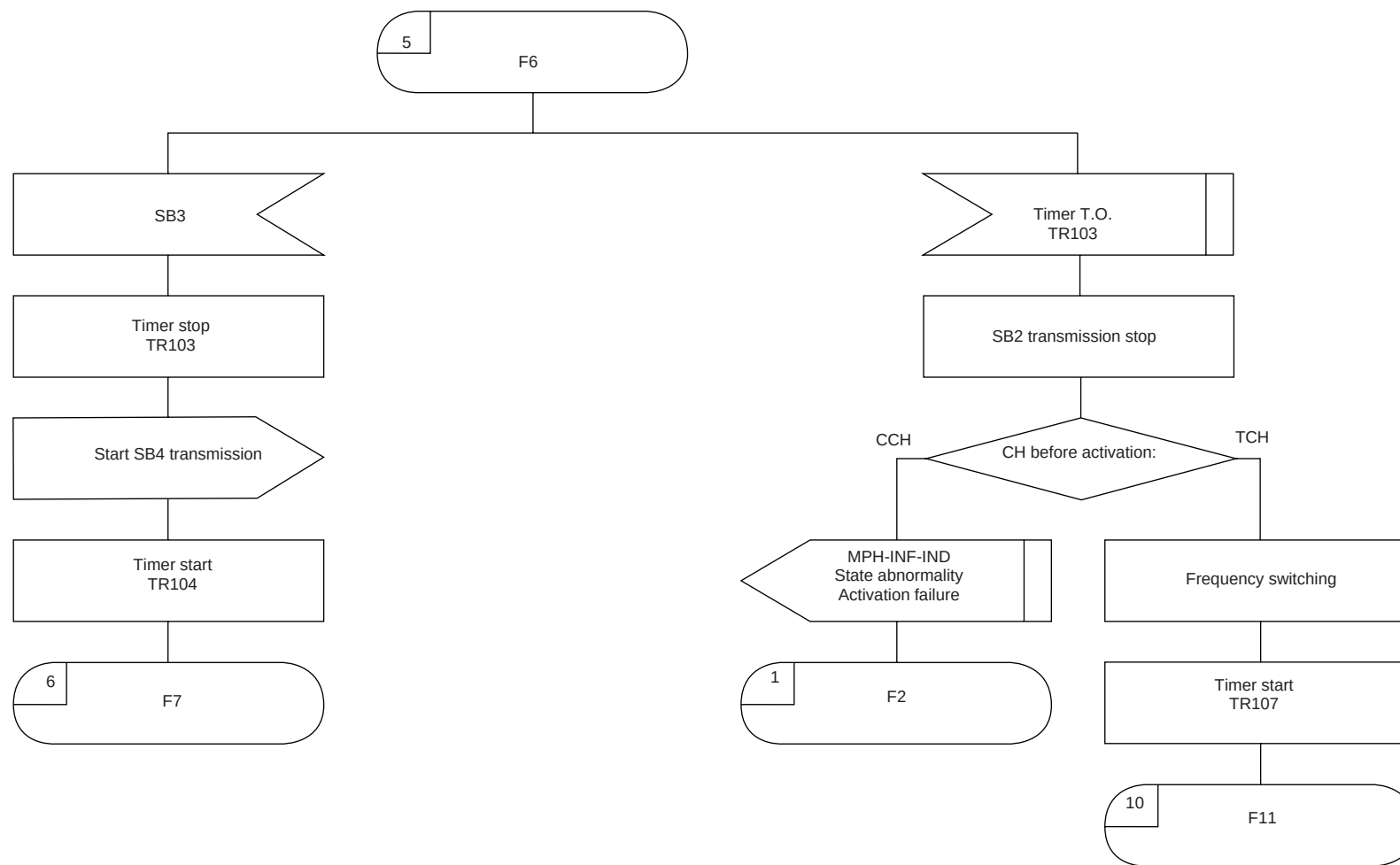


Fig. A. 3-8 Layer 1 SDL diagram (user side) (6/14)

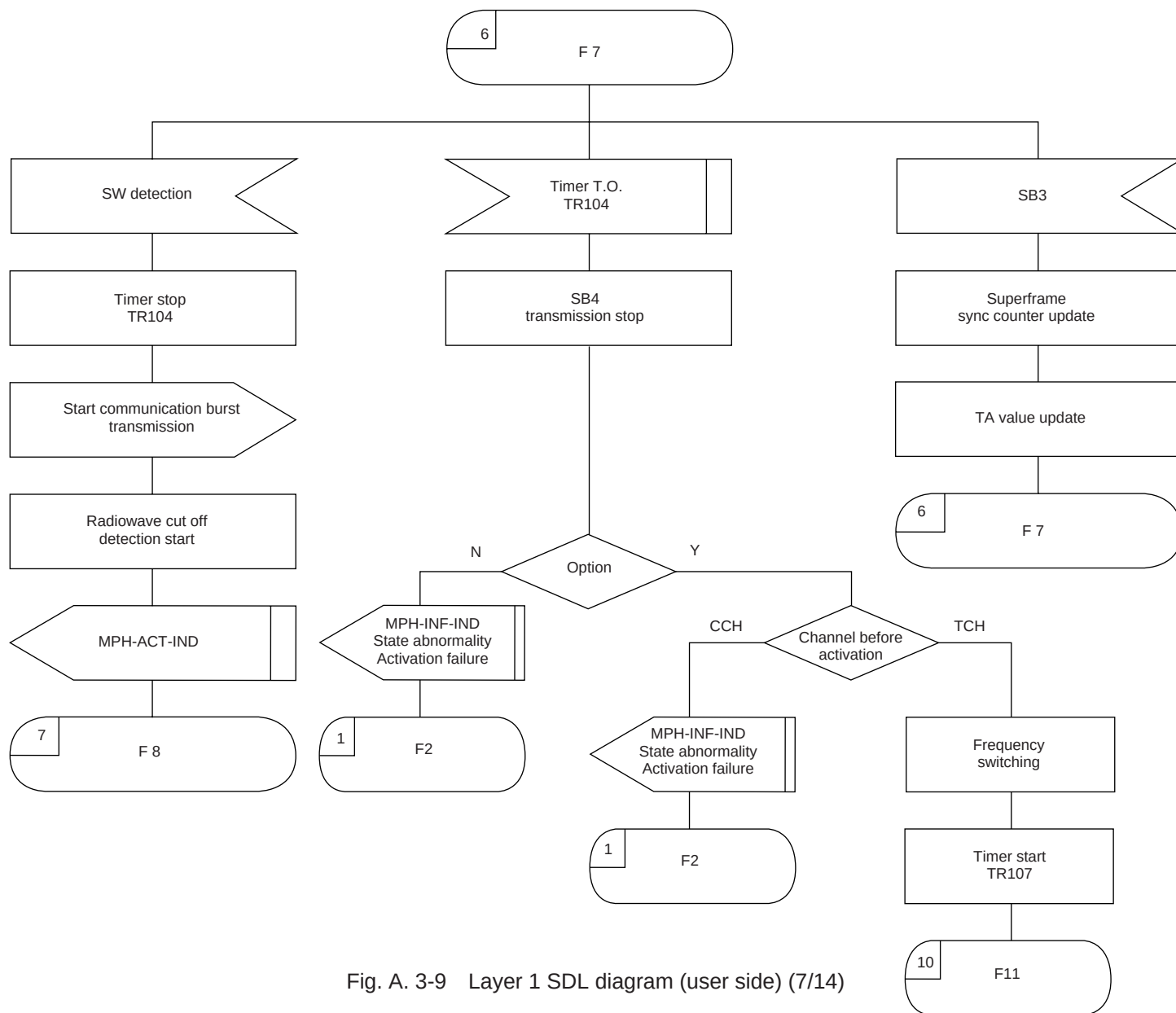


Fig. A. 3-9 Layer 1 SDL diagram (user side) (7/14)

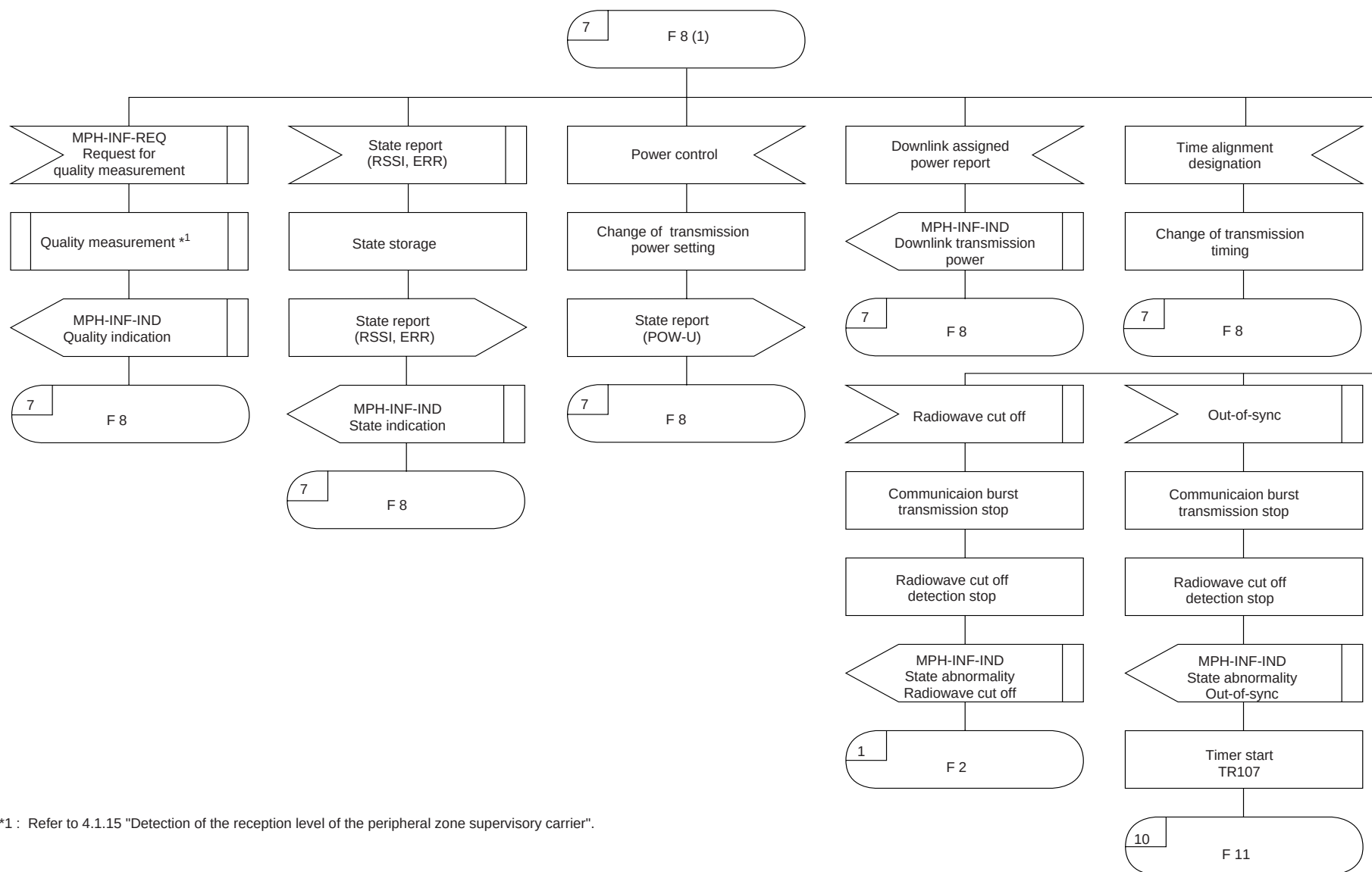


Fig. A. 3-10 Layer 1 SDL diagram (user side) (8/14)

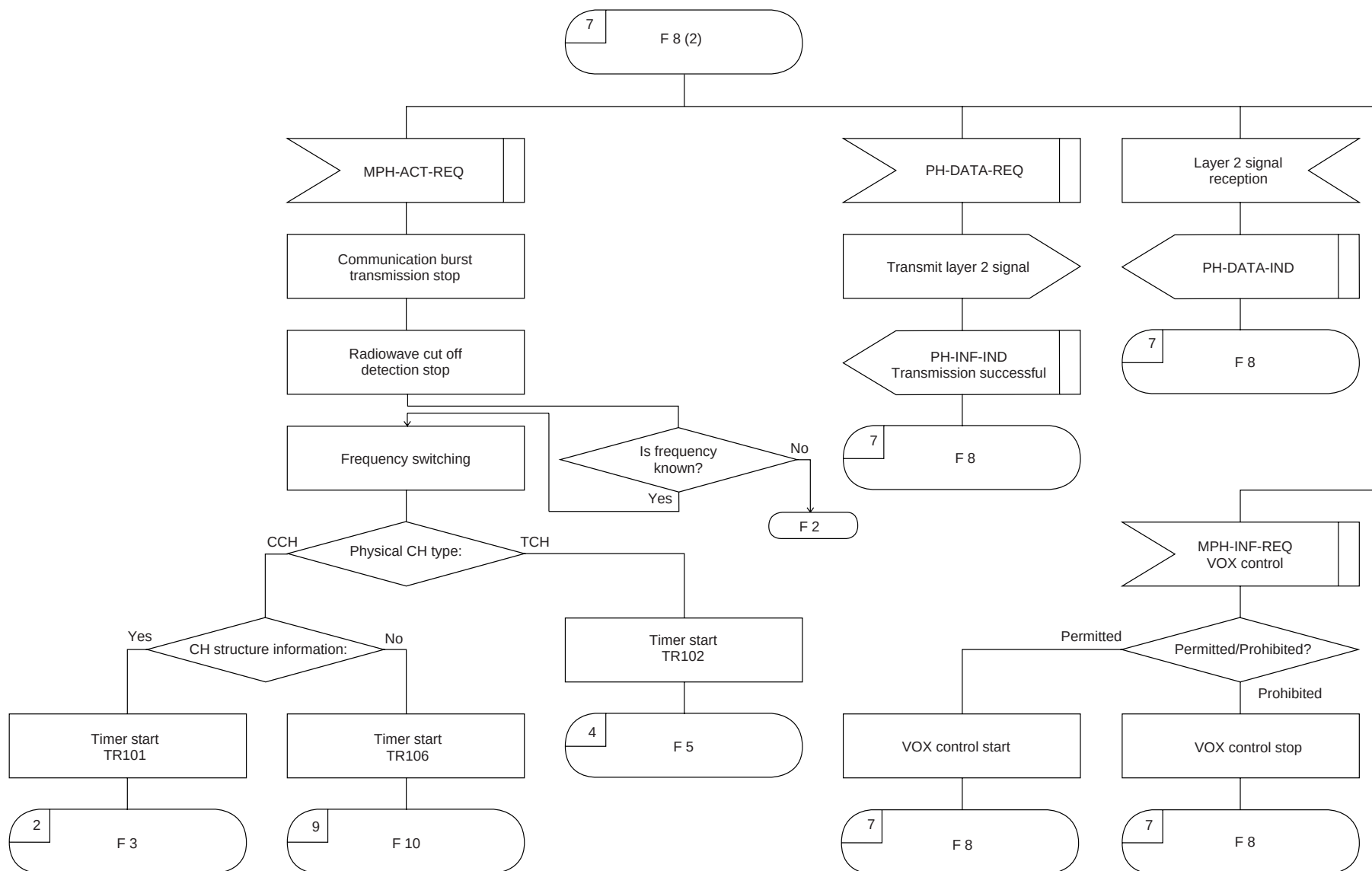


Fig. A. 3-11 Layer 1 SDL diagram (user side) (9/14)

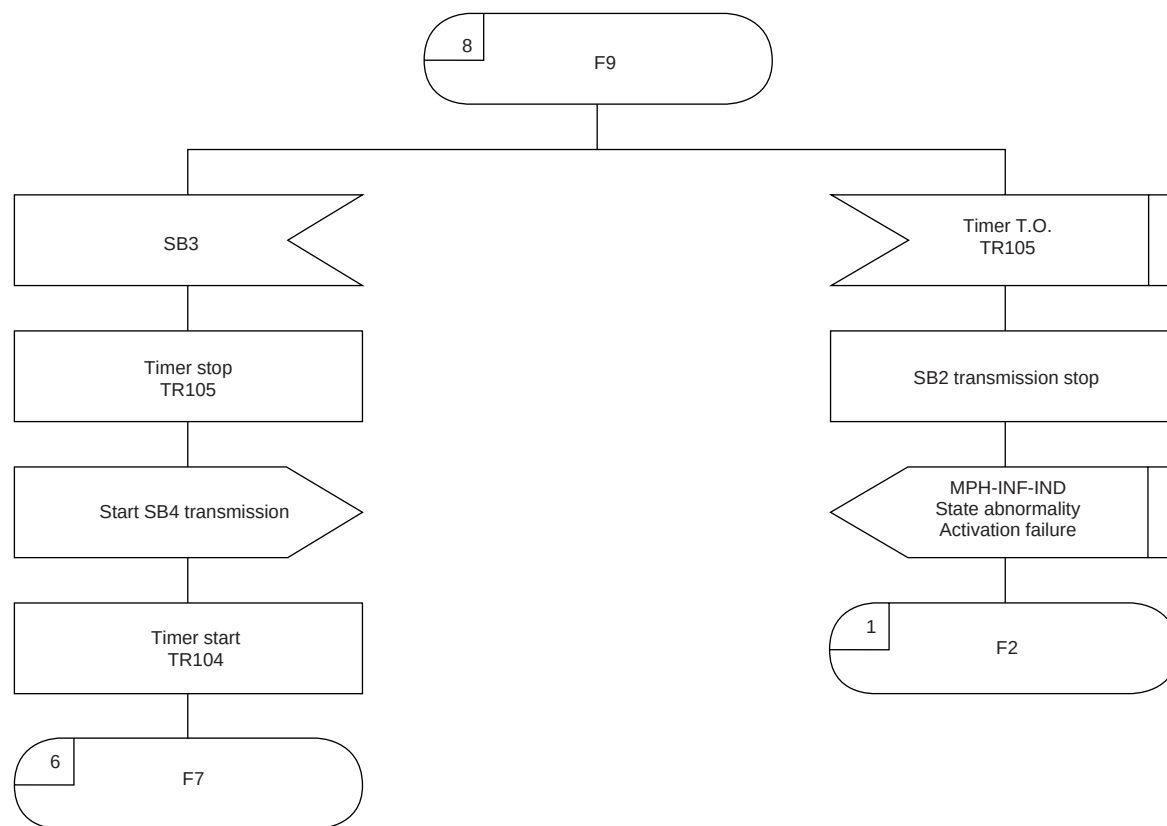


Fig. A. 3-12 Layer 1 SDL diagram (user side) (10/14)

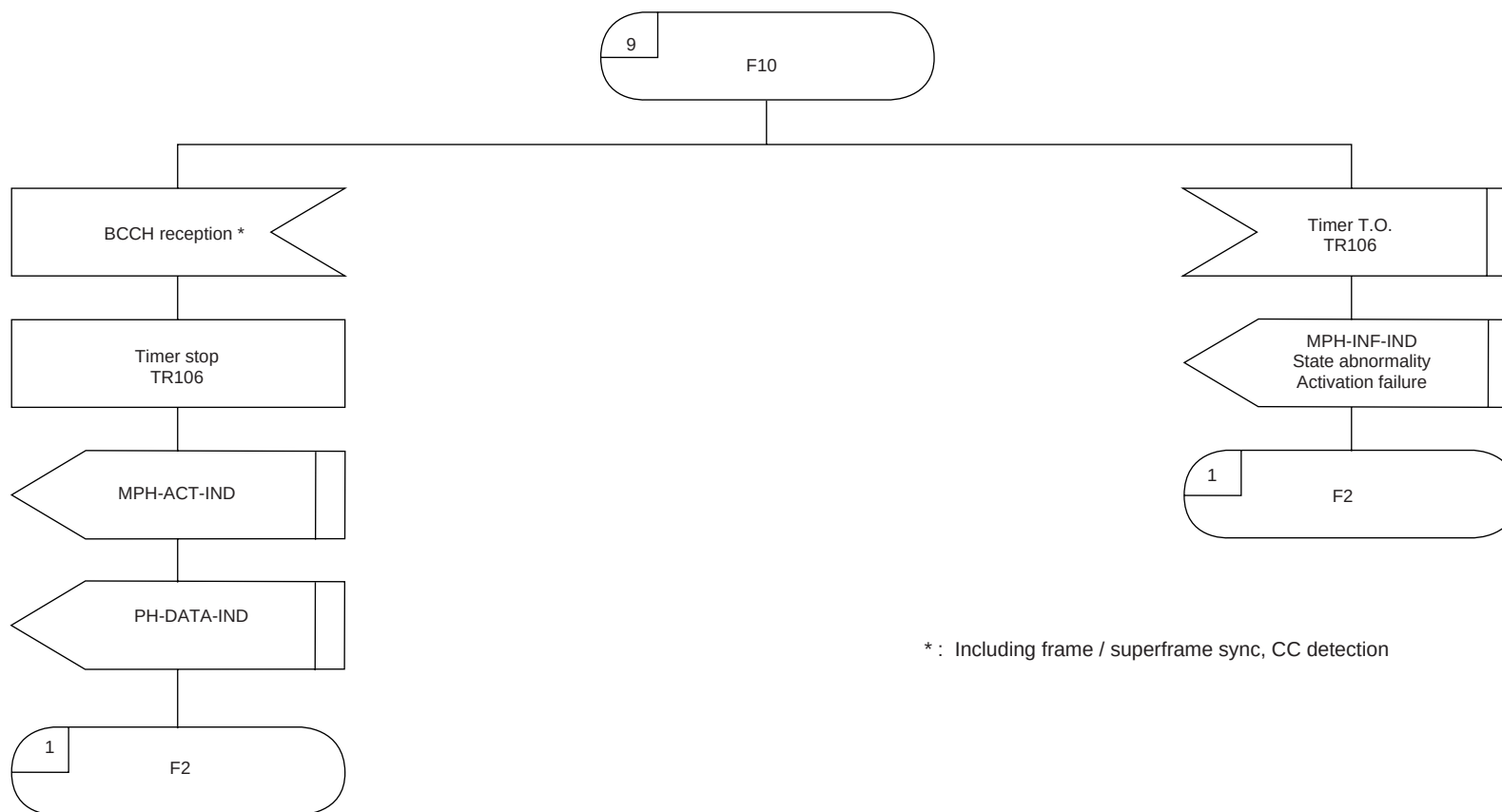


Fig. A. 3-13 Layer 1 SDL diagram (user side) (11/14)

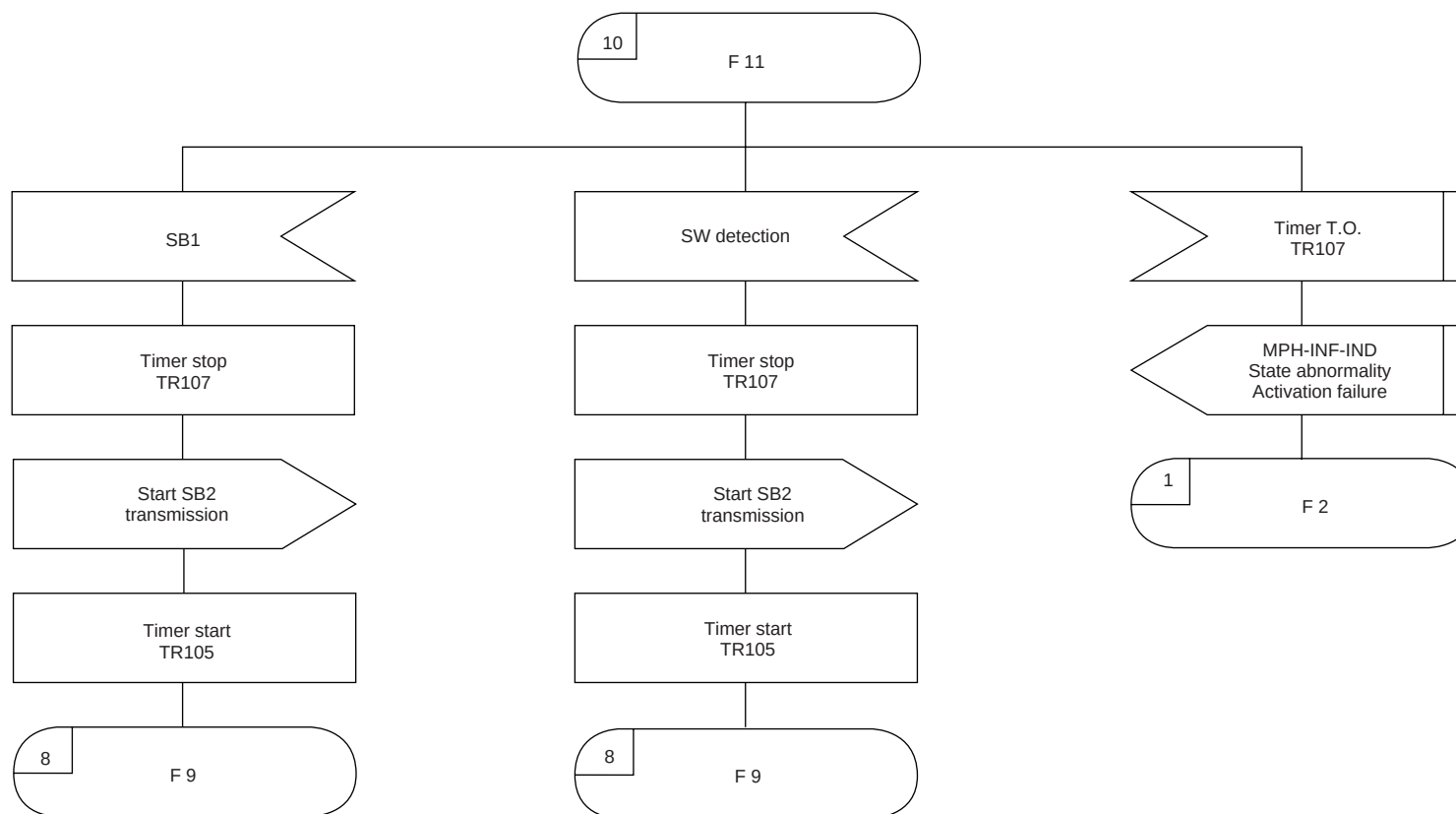


Fig. A. 3-14 Layer 1 SDL diagram (user side) (12/14)

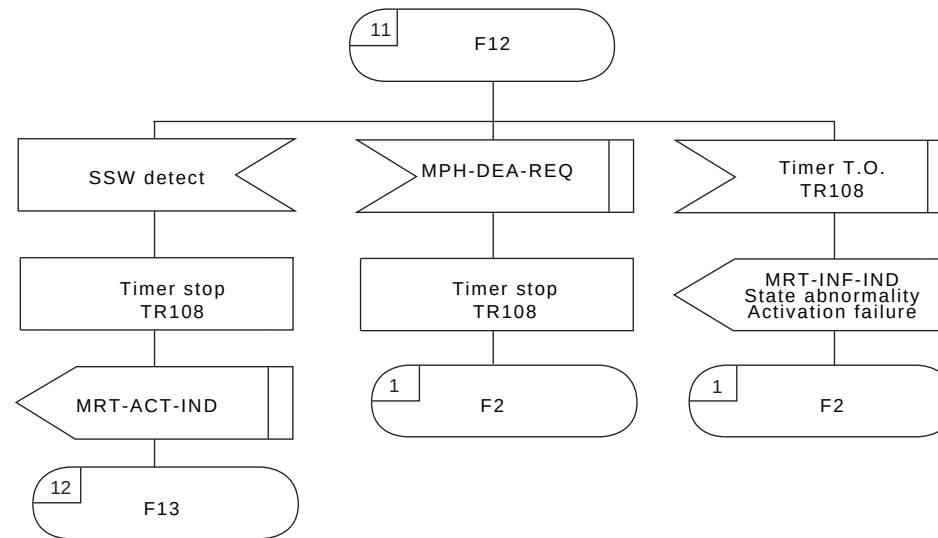


Fig. A. 3-15 Layer 1 SDL diagram (user side) (13/14)

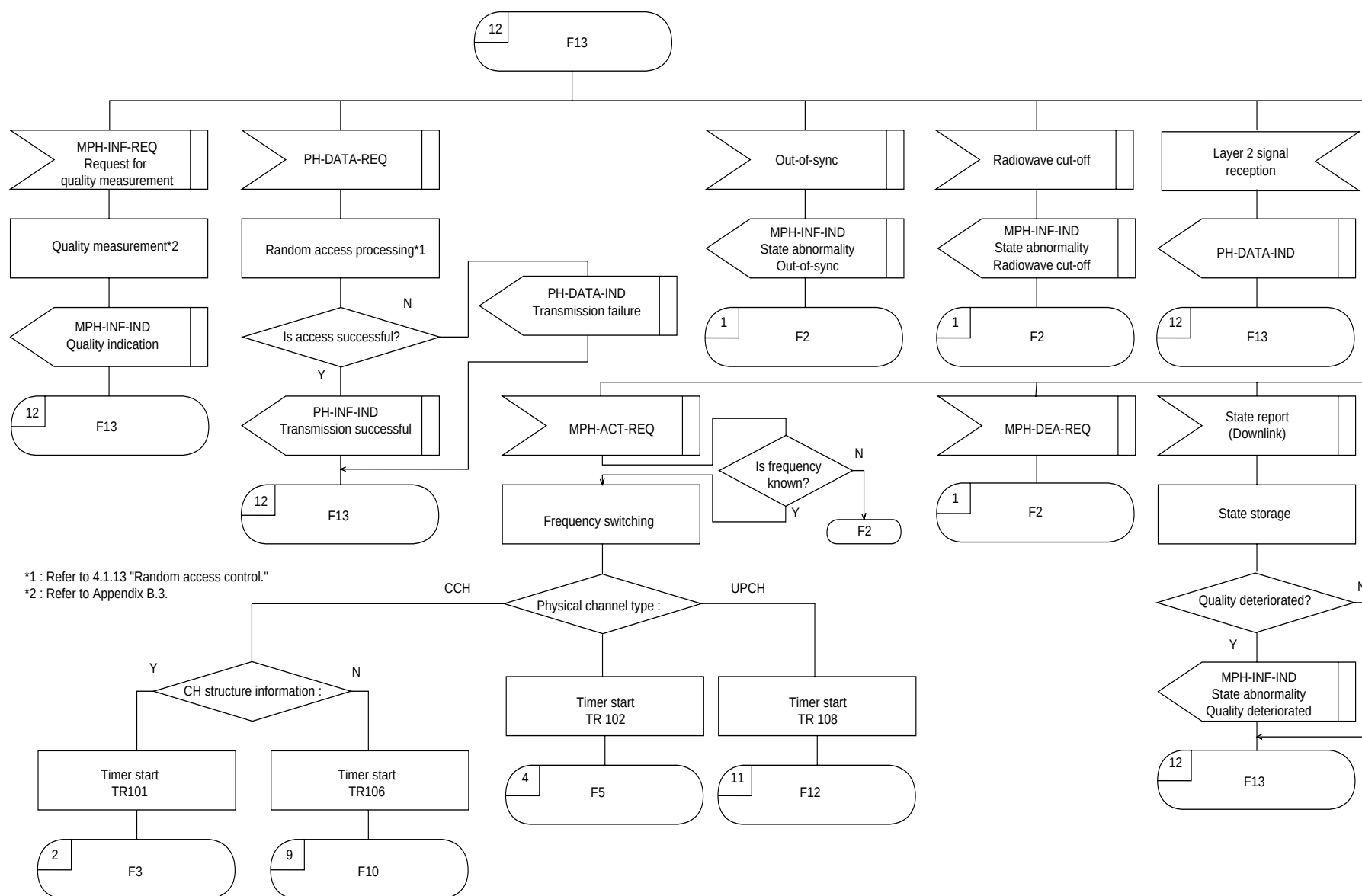


Fig. A. 3-16 Layer 1 SDL diagram (user side) (14/14)

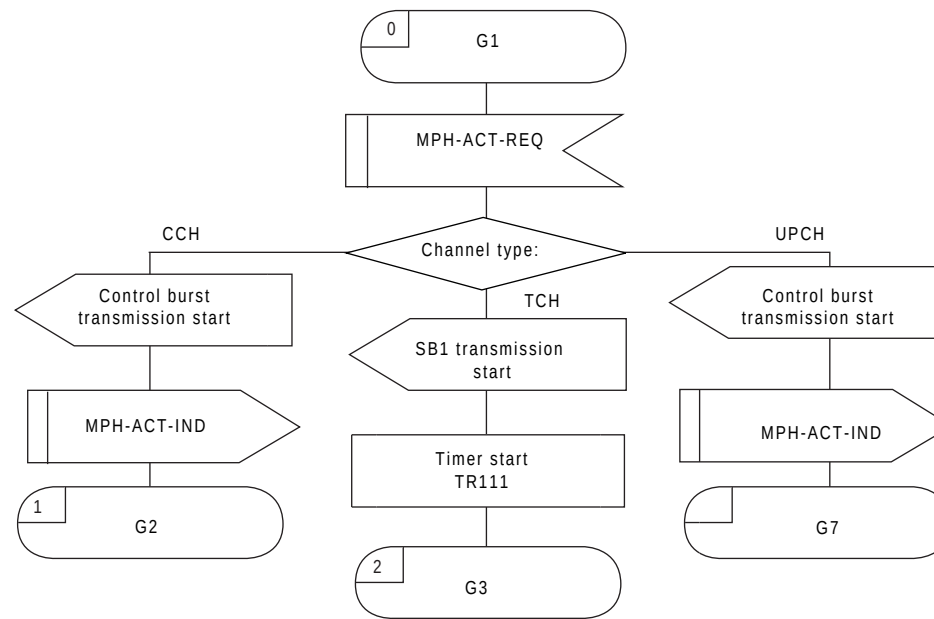
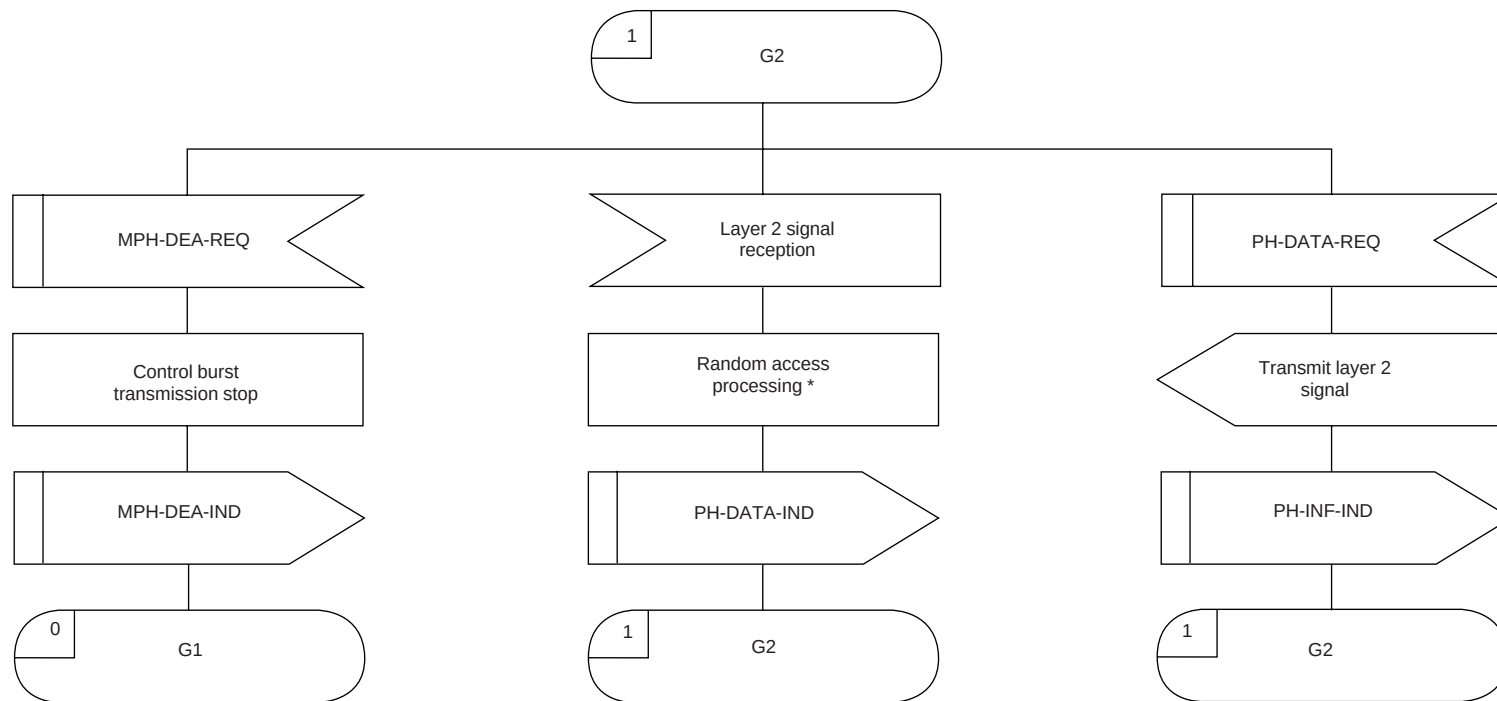


Fig. A. 3-17 Layer 1 SDL diagram (NW side) (1/8)



* Refer to 4.1.13 "Random access control"

Fig. A. 3-18 Layer 1 SDL diagram (NW side) (2/8)

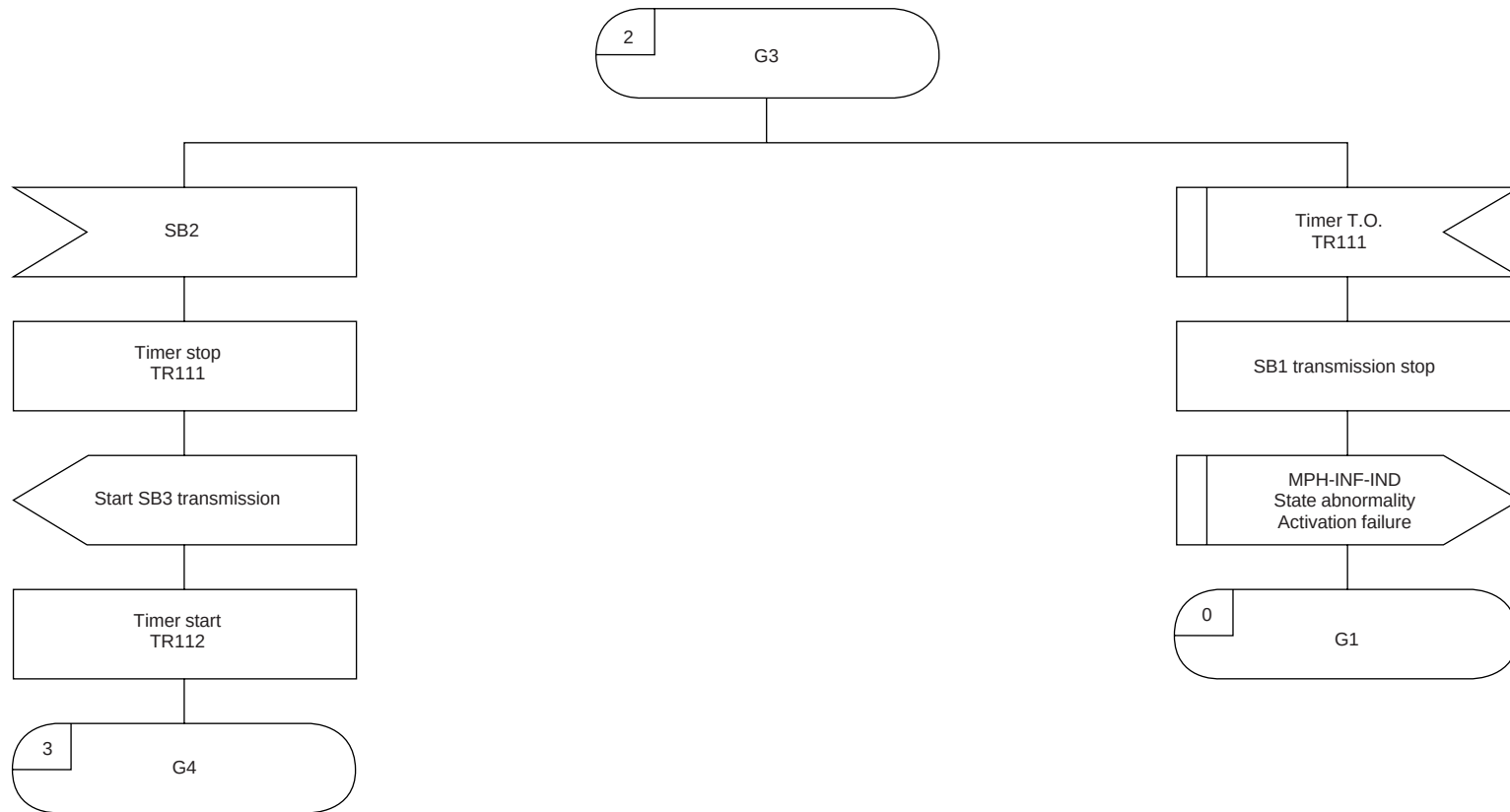


Fig. A. 3-19 Layer 1 SDL diagram (NW side) (3/8)

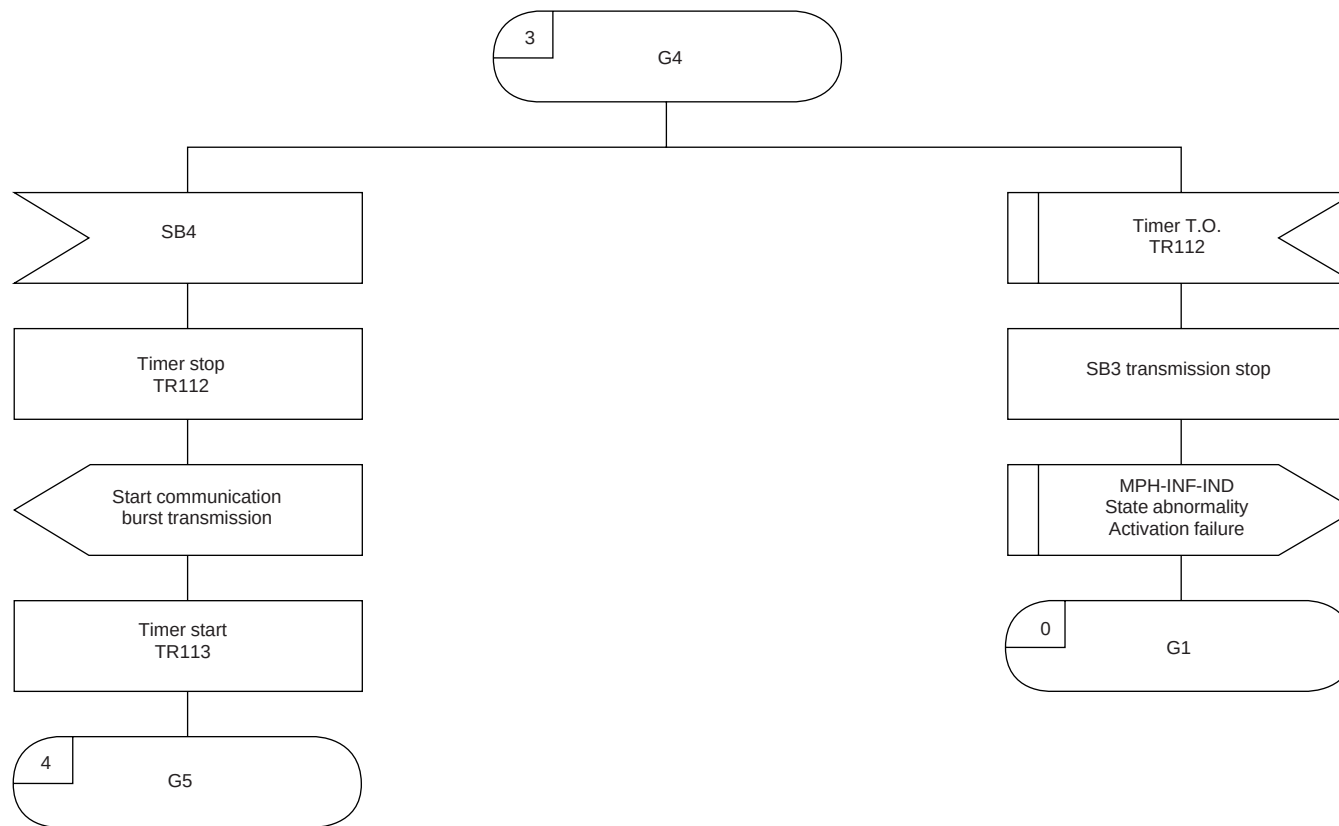
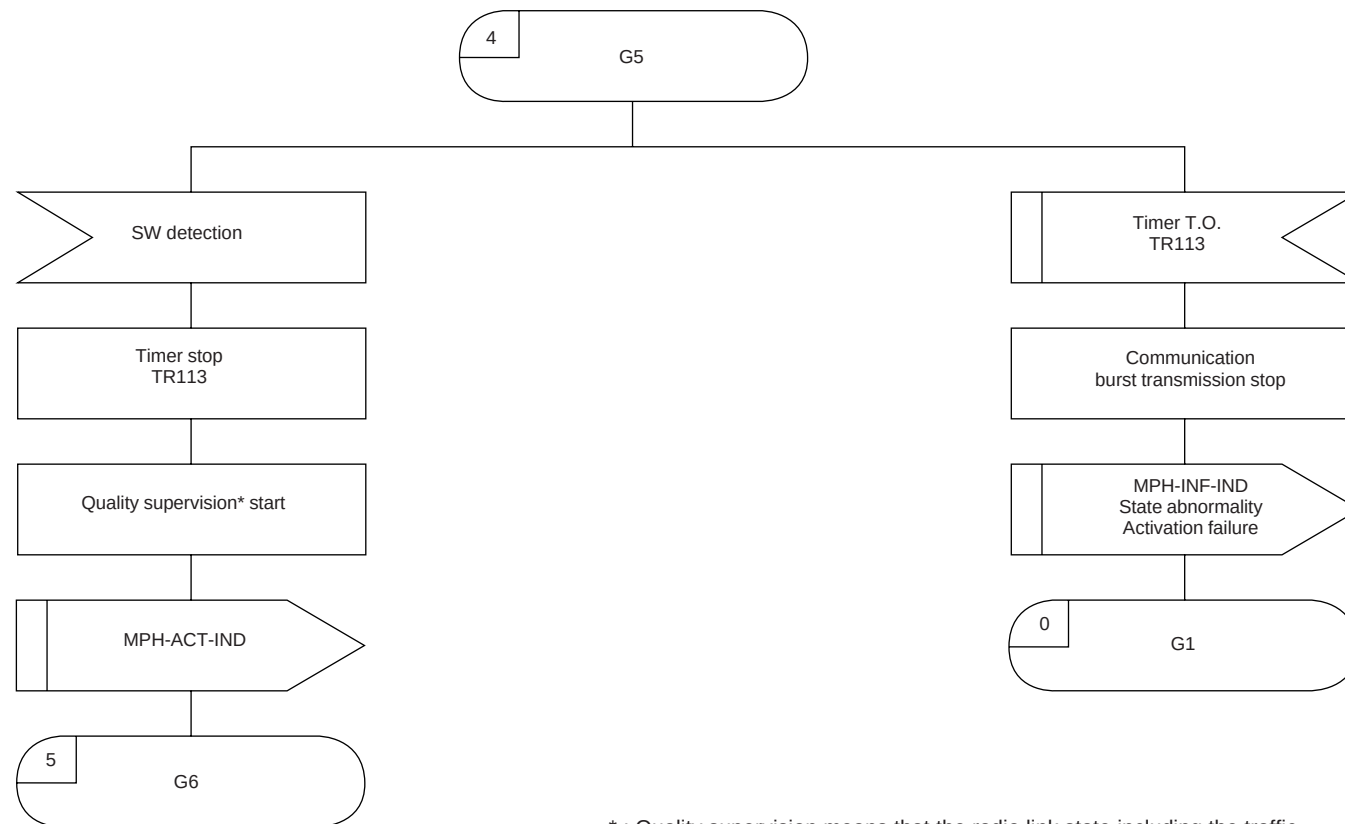


Fig. A. 3-20 Layer 1 SDL diagram (NW side) (4/8)



* : Quality supervision means that the radio link state including the traffic channel reception level, BER, CC error, etc. is monitored.

Fig. A. 3-21 Layer 1 SDL diagram (NW side) (5/8)

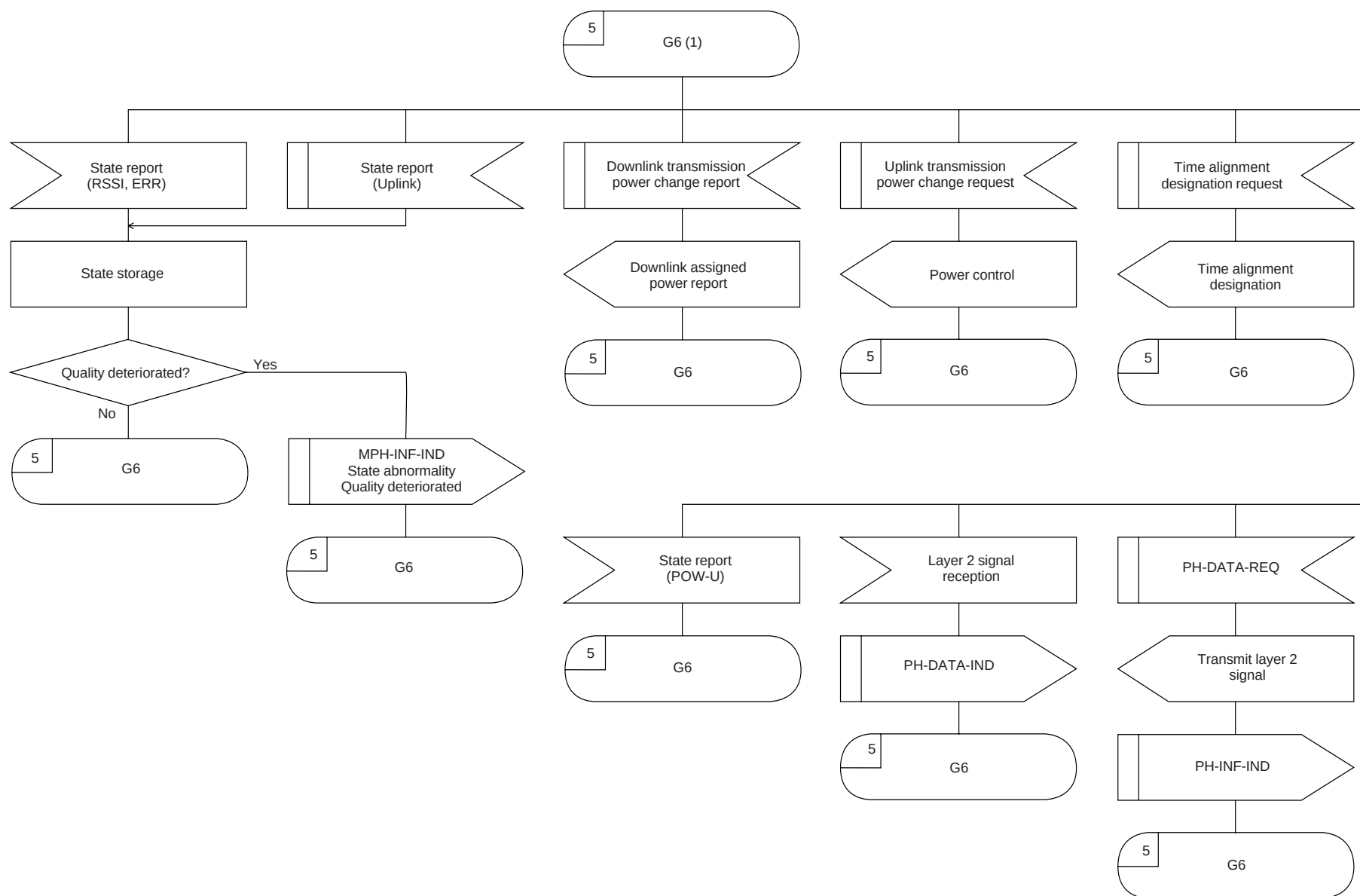


Fig. A. 3-22 Layer 1 SDL diagram (NW side) (6/8)

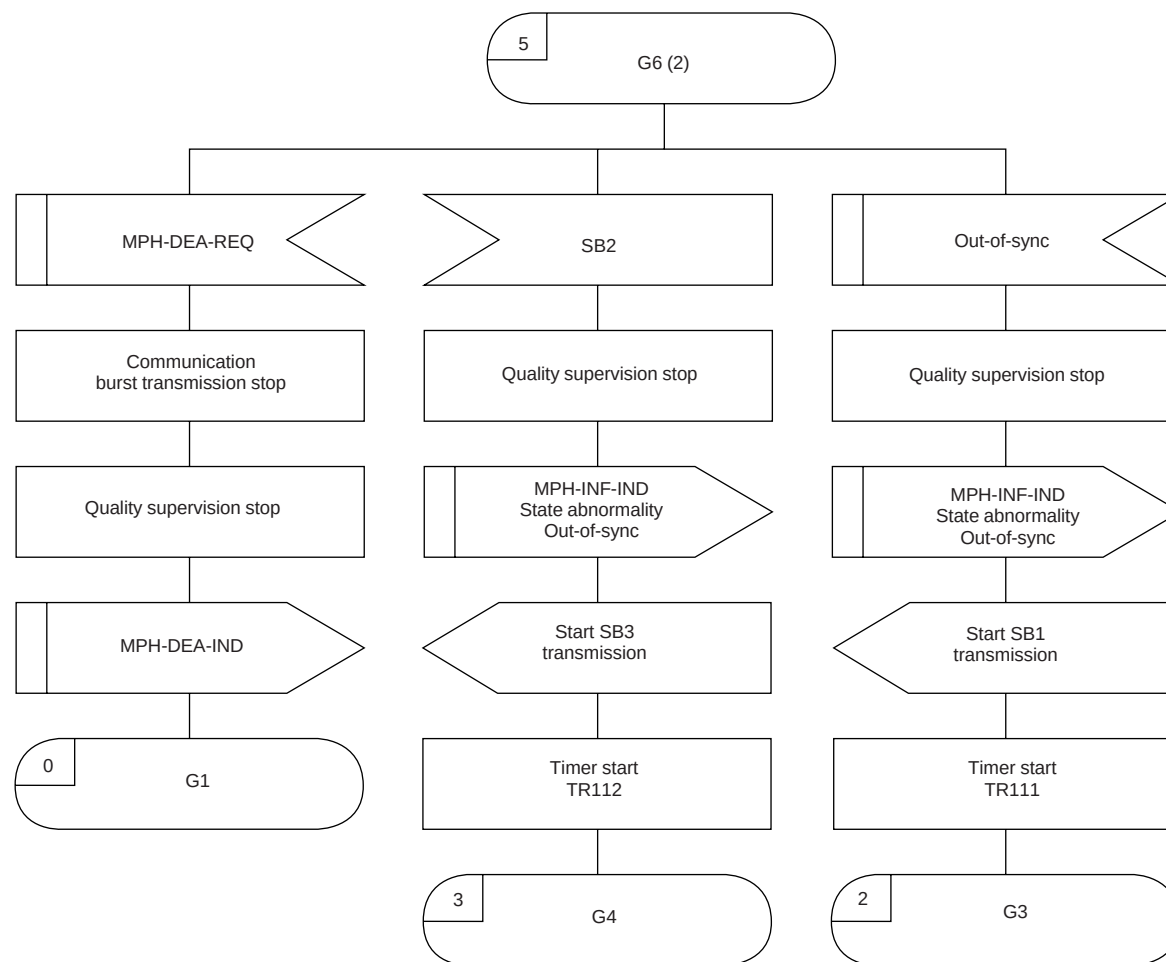
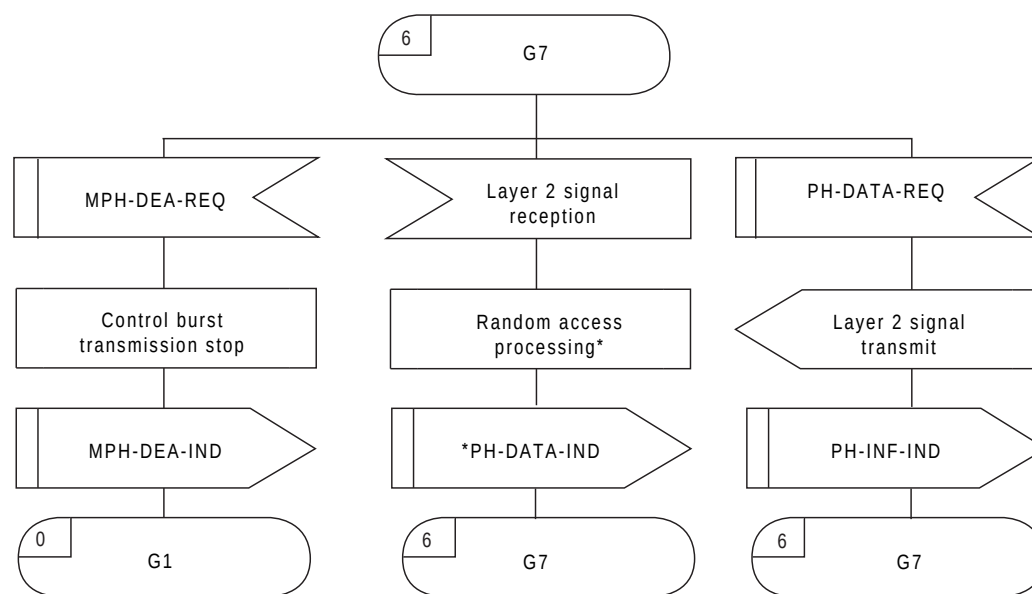


Fig. A. 3-23 Layer 1 SDL diagram (NW side) (7/8)



* : Refer to 4.1.13 "Random access control."

Fig. A. 3-24 Layer 1 SDL diagram (NW side) (8/8)

Table A. 3-1 shows specified timer values.

Table A. 3-1 Timer values

Timer name	Value
TR 101	3 s
TR 102	500 ms
TR 103	500 ms
TR 104	500 ms
TR 105	500 ms
TR 106	3 s
TR 107	5 s
TR 108	3 s

A.4 Layer 2 SDL diagrams

A.4.1 Operations on the control physical channel and the traffic physical channel (SAPI=0, 7)

A.4.1.1 Outline of the states for point-to-point data link layer entity

Six basic states are used in conjunction with the SDL expressions for determining point-to-point procedures. All six states are expanded forms derived from the two basic states described in Section 4.2.1.3.5.2

State 1 : Multiple frame unestablished

State 2 : Standby for link establishment

State 3 : Standby for link release

State 4 : Multiple frame established

State 5 : Timer recovery

State 6 : Link stop

The interrelationship between these states is outlined in Fig. A. 4-1. This diagram is used basically to introduce the SDL expressions and does show the full interrelationship between states.

All data link layer entities start with multiple frame unestablished state (State 1). For all of the states 1 to 5 the data link layer is capable of providing services directly in response to the unit data request. When the data link layer entity receives the link establish request in the multiple frame unestablished state (State 1), the link establish procedure is initiated and the data link layer entity switches to the standby state for link establishment state (State 2). The data link layer entity switches to the multiple frame established state (State 4) on completion of the LAP establishment procedure. The data link layer entity on the other side switches from the multiple frame unestablished state (State 1) directly to the multiple frame established state (State 4).

During the multiple frame established state (State 4), the data link layer entity can provide the required service in response to an acknowledged type data transfer request under the conditions applying to the procedure in progress. When timer T200, which is used for both flow control and the data transfer in the data link layer entity procedure expires, switching to the timer recovery state (State 5) begins. When the timer recovery procedure is completed, the data link layer entity returns to the multiple frame established state (State 4).

During States 4 and 5 for the SDL expressions, the following conditions, all of which are defined by the Standard, can be generated.

- a) Peer receiver busy
- b) Reject exception
- c) Own receiver busy

In order to avoid the need to define additional states, several other conditions are used. Combining the five states for use in SDL expressions with the conditions for the above categories serves as the basis for state transition expressions for the data link layer entity. When the data link layer entity on the other side begins LAP release, the data link layer entity switches directly to the multiple frame unestablished state (State 1). When a link release is requested, the data link layer entity switches to the standby for link release state (State 3) and then to State 1.

When a link stop request is received in any state from 1 through 5, the link stop procedure is performed and the state shifts to the link stop state (state 6).

In the link stop state (state 6) all information transfer services are suspended. When a link restart request is received, the link restart process starts, and the state shifts to the multiple frame unestablished state. When a link reconnect request is received, the link reconnect procedure starts, and the state returns to the state in effect before the link stop state.

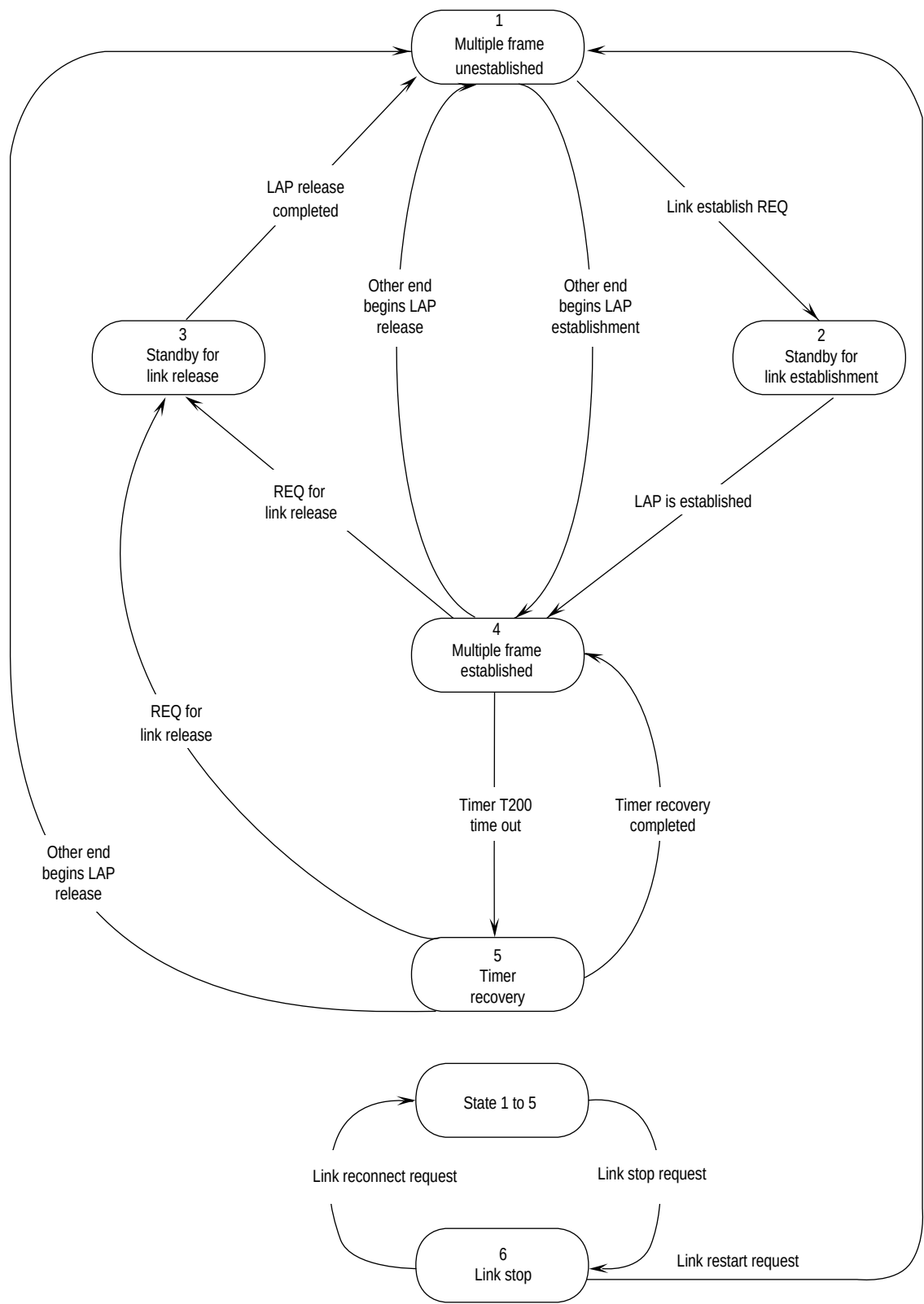


Fig. A. 4.1-1 Outline of point-to-point procedure states

A.4.1.2 Using queues

To provide an understandable description of the data link layer entity, the conceptional queues for transferring UI, I and I' frames are clearly described. Usage of these conceptional queues is limited, but their depth is not limited; also, they do not limit implementation of point-to-point procedures. Three additional signals (i.e., Connect UI frame to queue and Connect I' frame to queue) are provided for activating the queues.

These services are provided by the queue manager. The queue manager must control to assign priority according to Section 4.2.1.4.2.2.

A.4.1.3 Grouping of unit numbers

To simplify description of partial retransmission control, the variables used for partial retransmission control in the SDL diagrams are expressed for groups as shown below.

Send divided unit sequence number

$$n(S) = \{n1(S), n2(S), \dots, nj(S), ni(S) < nj(S), i < j\}$$

Receive divided unit sequence number

$$n(R) = \{n1(R), n2(R), \dots, nj(R), ni(R) < nj(R), i < j\}$$

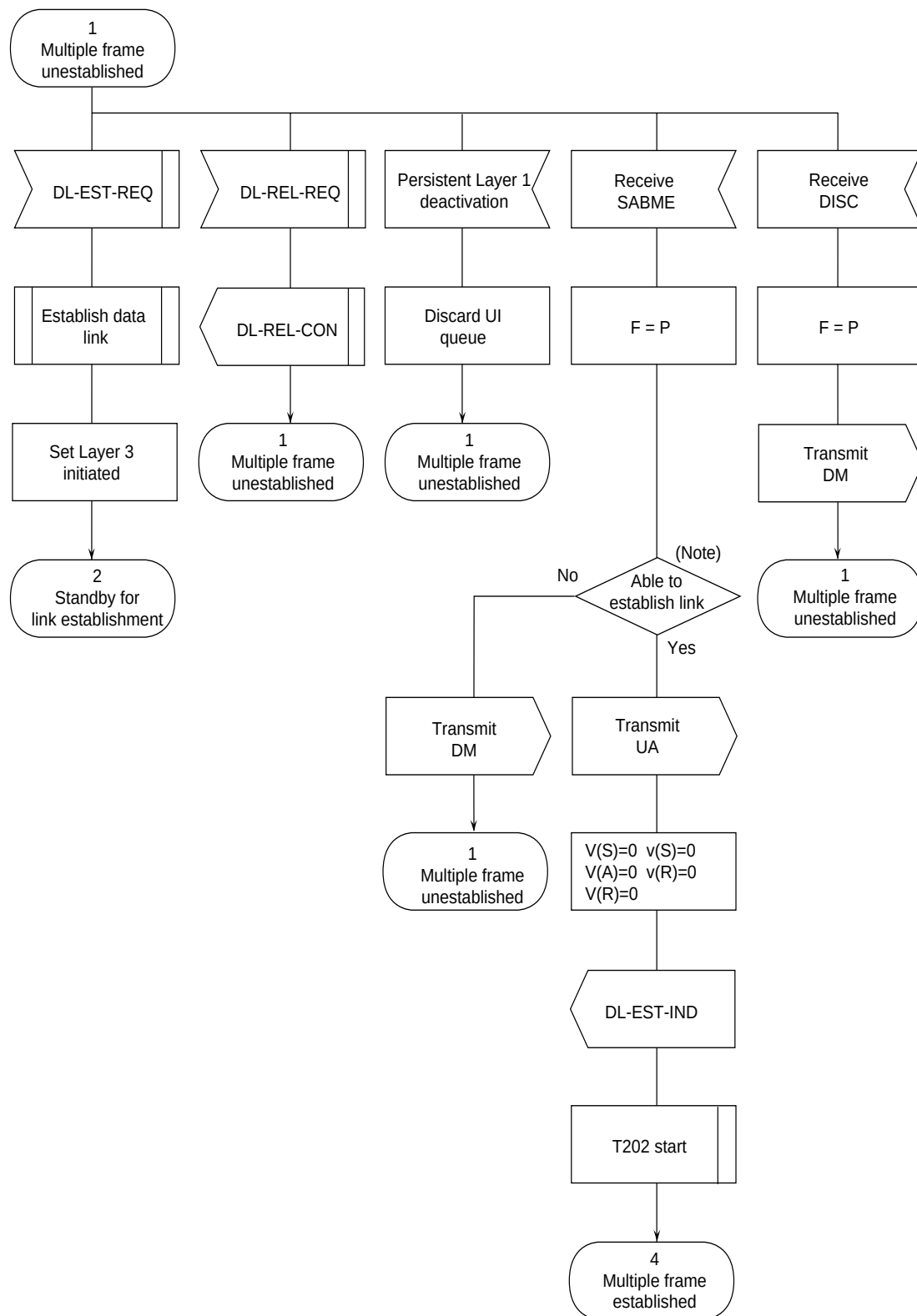
Divided send state variable

$$v(S) = \{v1(S), v2(S), \dots, vi(S)\}$$

Divided receive state variable

$$v(R) = \{v1(R), v2(R), \dots, vi(R)\}$$

The send divided unit sequence number $n(S)$ matching the divided receive state variable $v(R)$ is a correct $n(S)$. "Aborting illegal $n(S)$ " refers to aborting the $n(S)$ and the information portions which are not matching $v(R)$. Also, "updating $v(R)$ " refers to removing $n(S)$ which have been successfully received from the $v(R)$.



Note: Availability of a data link is subject to whether the data link entity can communicate messages and primitives with L3 and L1 and the management entity.

Fig. A. 4.1-2 Layer 2 SDL diagram (1/41)

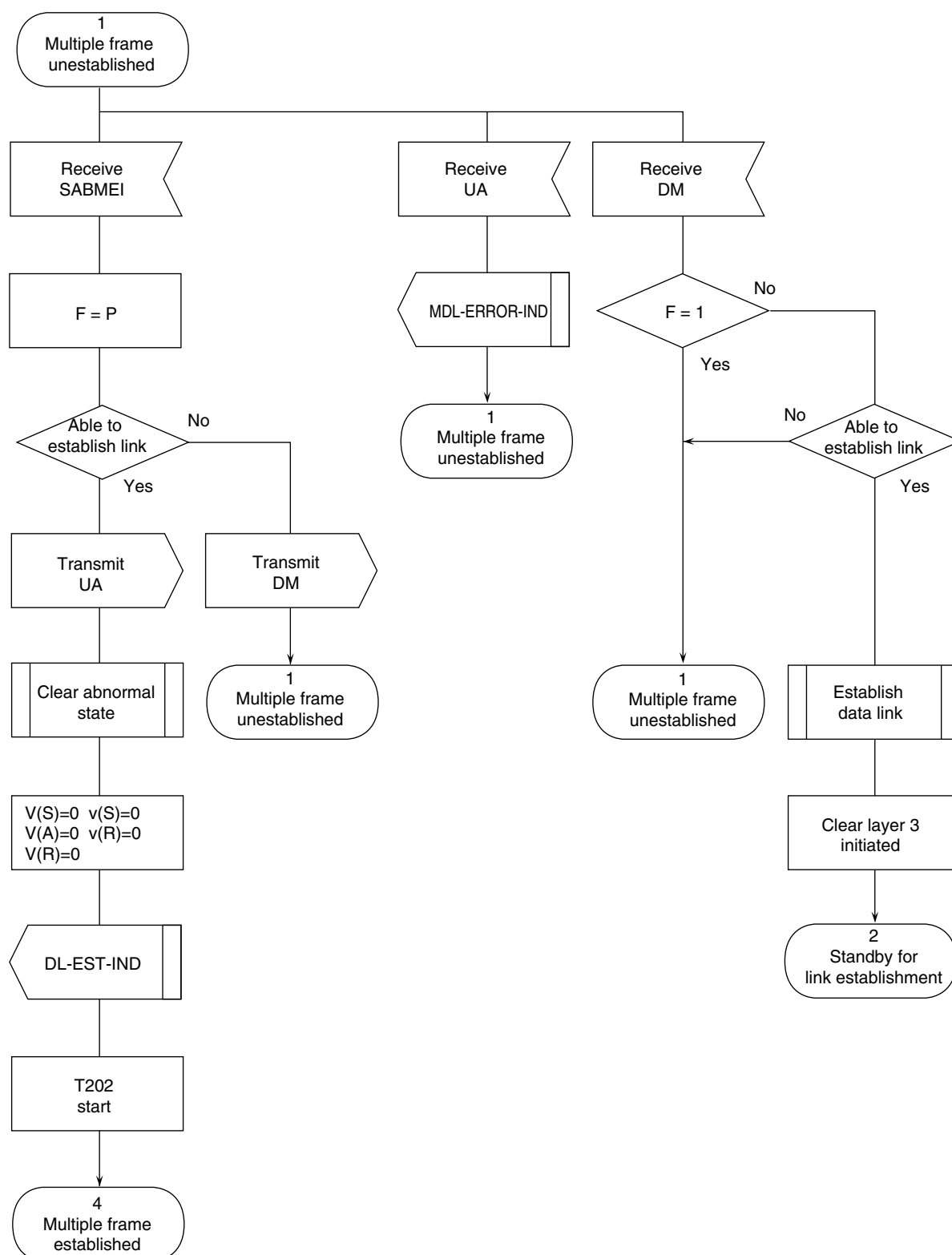
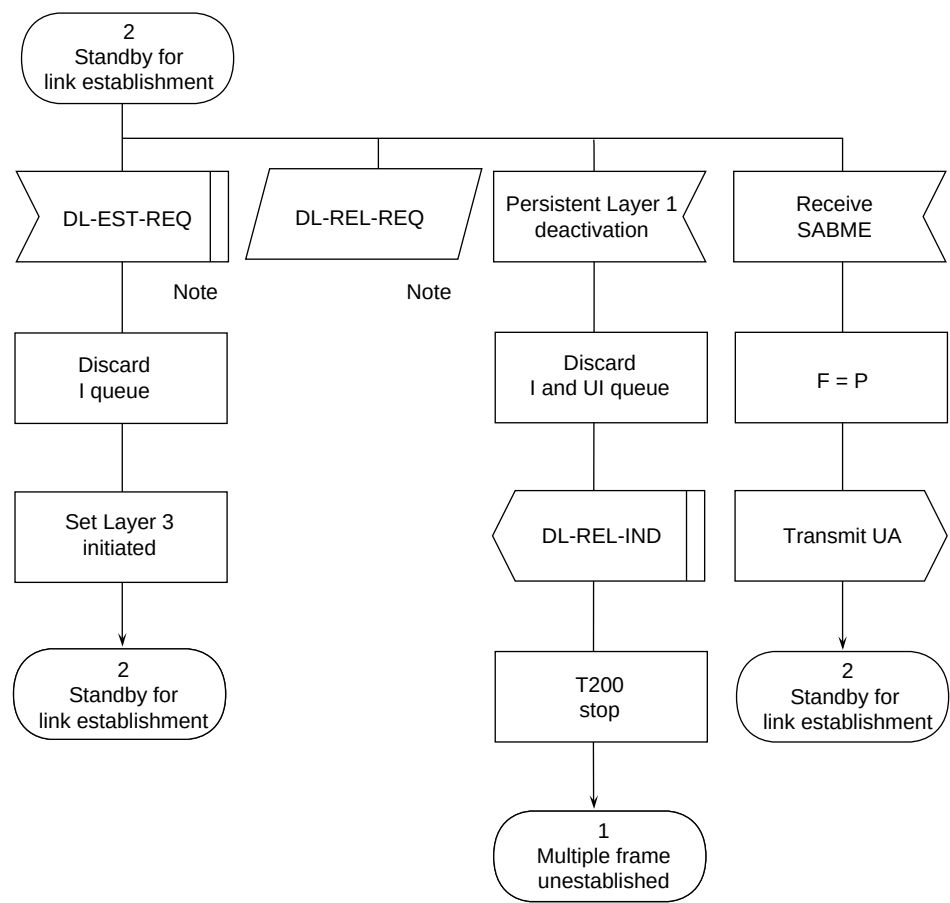


Fig. A. 4.1-3 Layer 2 SDL diagram (2/41)



Note: Can be generated only when the re-establishment procedure is initiated by layer 3.

Fig. A. 4.1-4 Layer 2 SDL diagram (3/41)

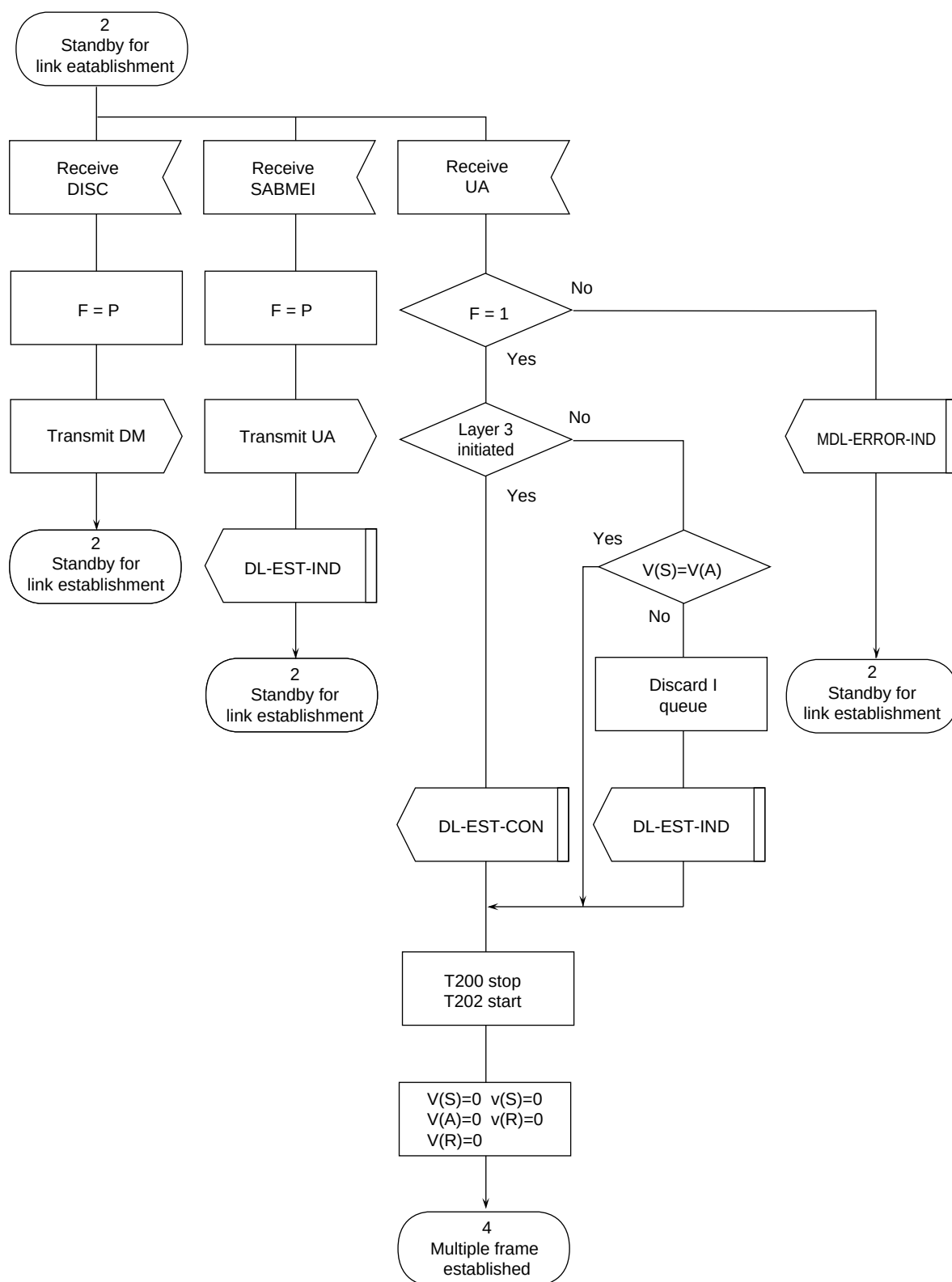
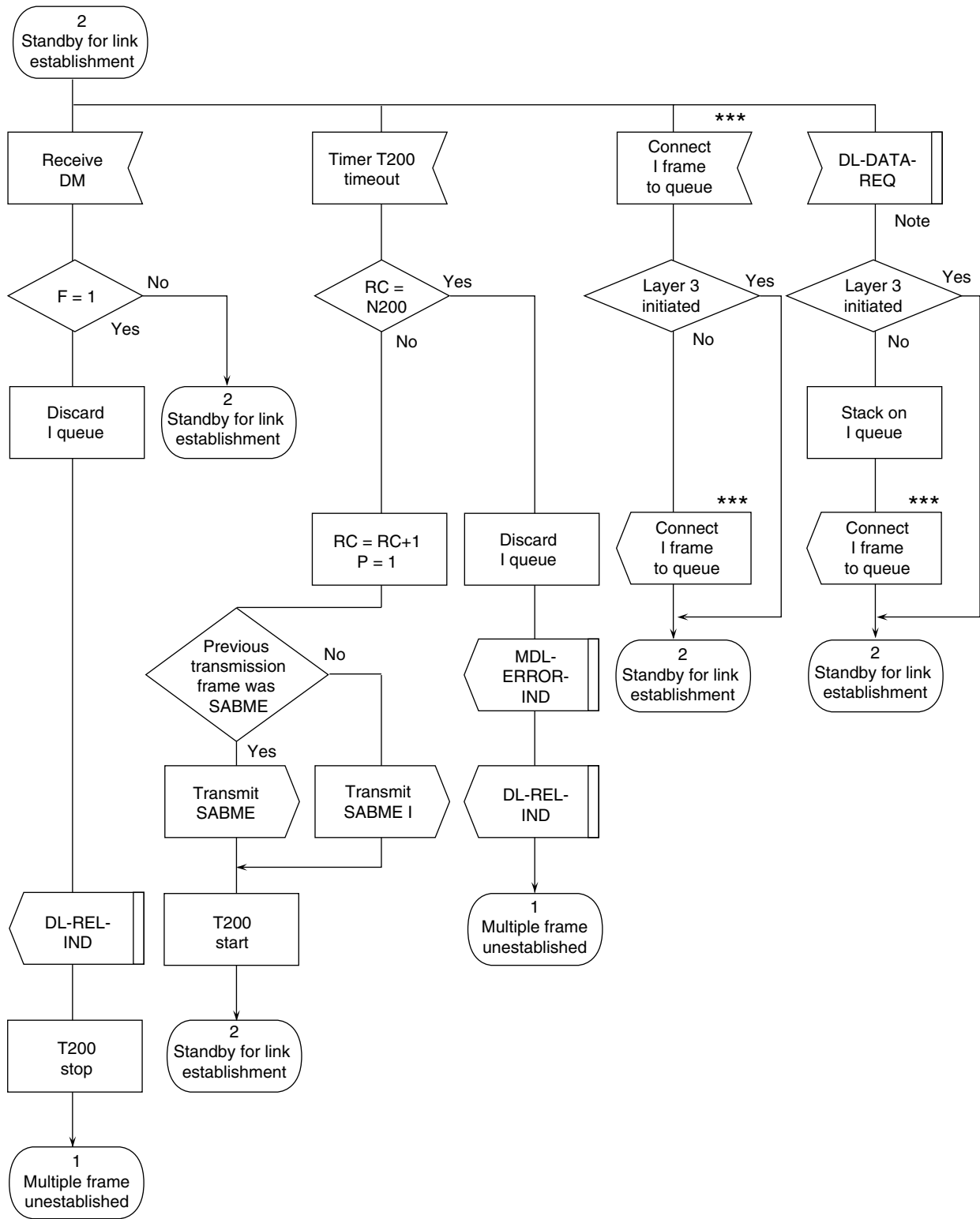


Fig. A. 4.1-5 Layer 2 SDL diagram (4/41)



Note: Can be generated only when the reestablishing procedure is activated by layer 2.

Fig. A. 4.1-6 Layer 2 SDL diagram (5/41)

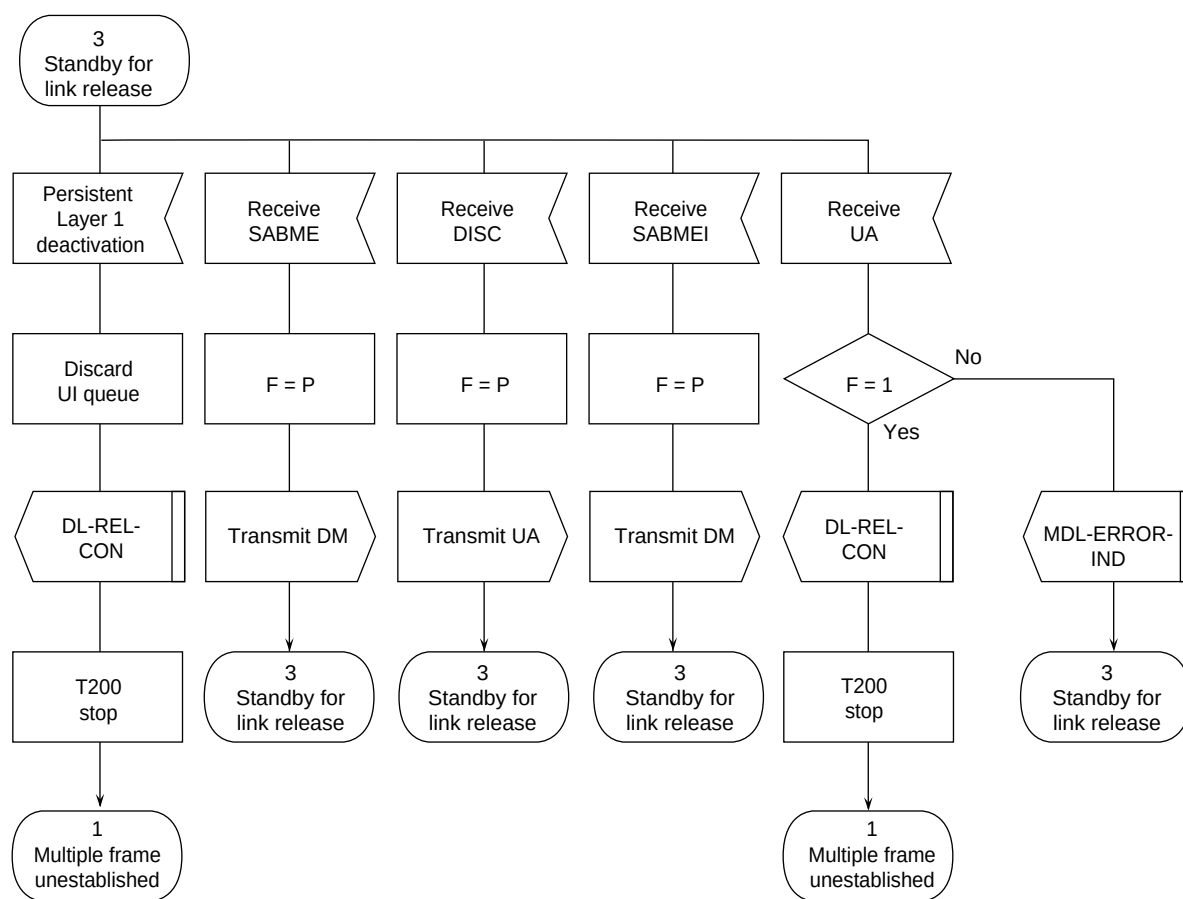


Fig. A. 4.1-7 Layer 2 SDL diagram (6/41)

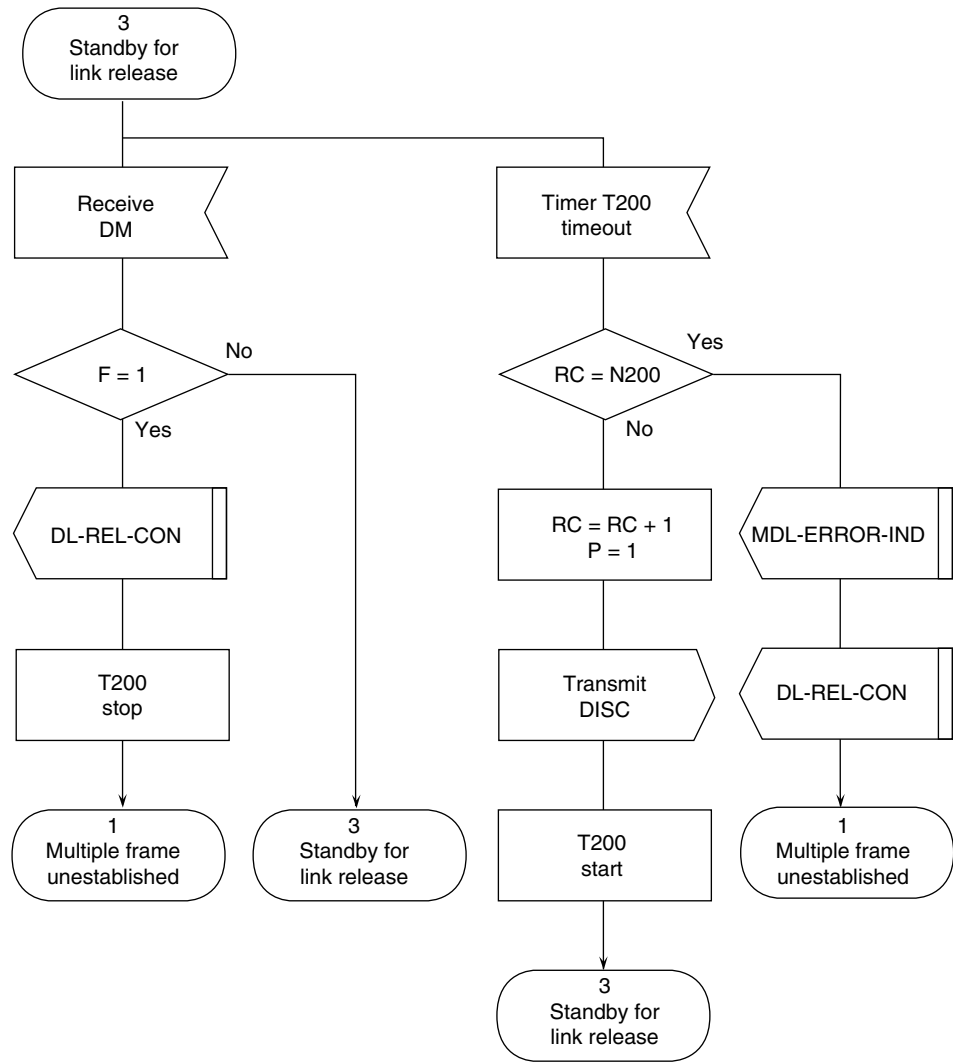


Fig. A. 4.1-8 Layer 2 SDL diagram (7/41)

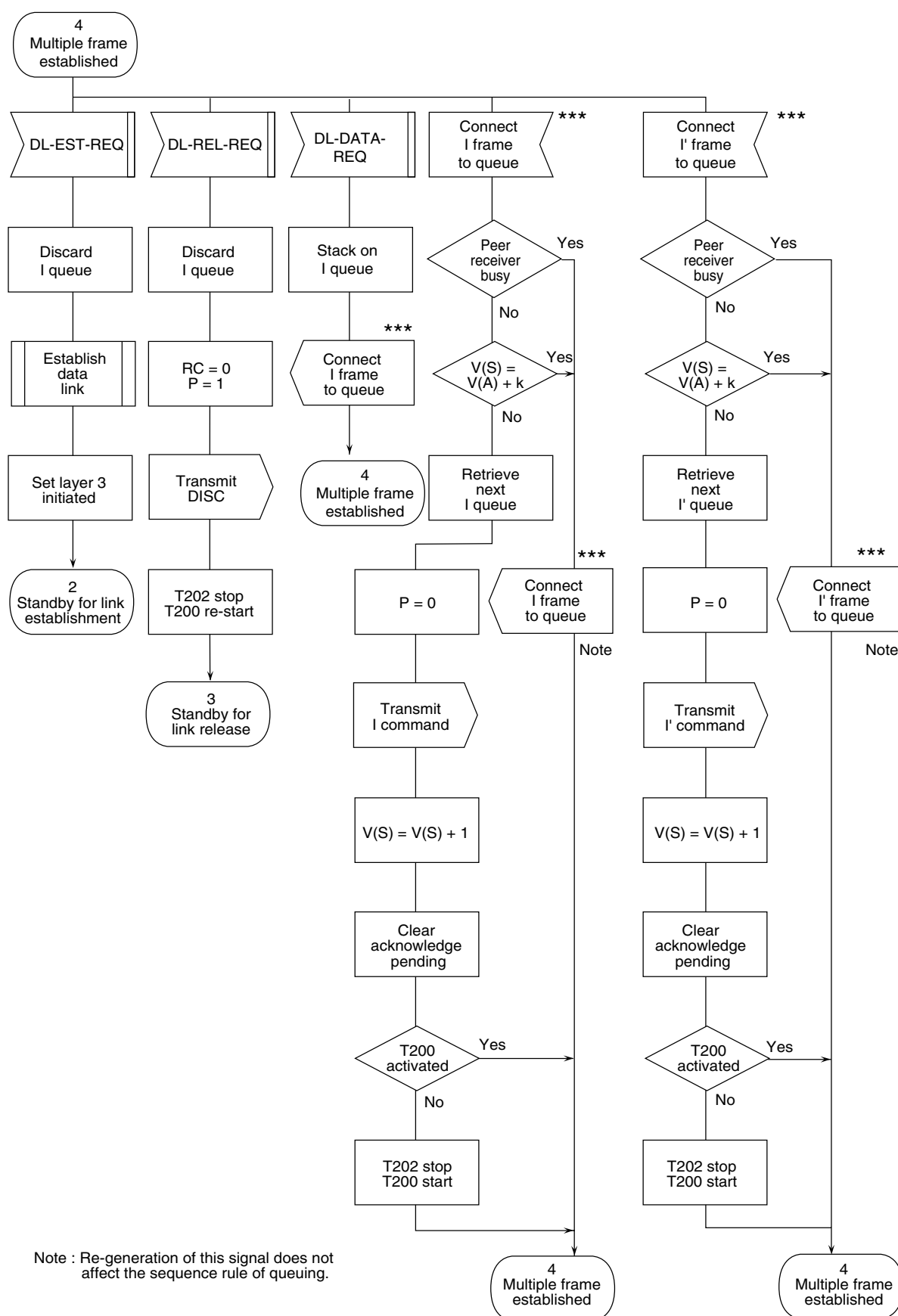


Fig. A. 4.1-9 Layer 2 SDL diagram (8/41)

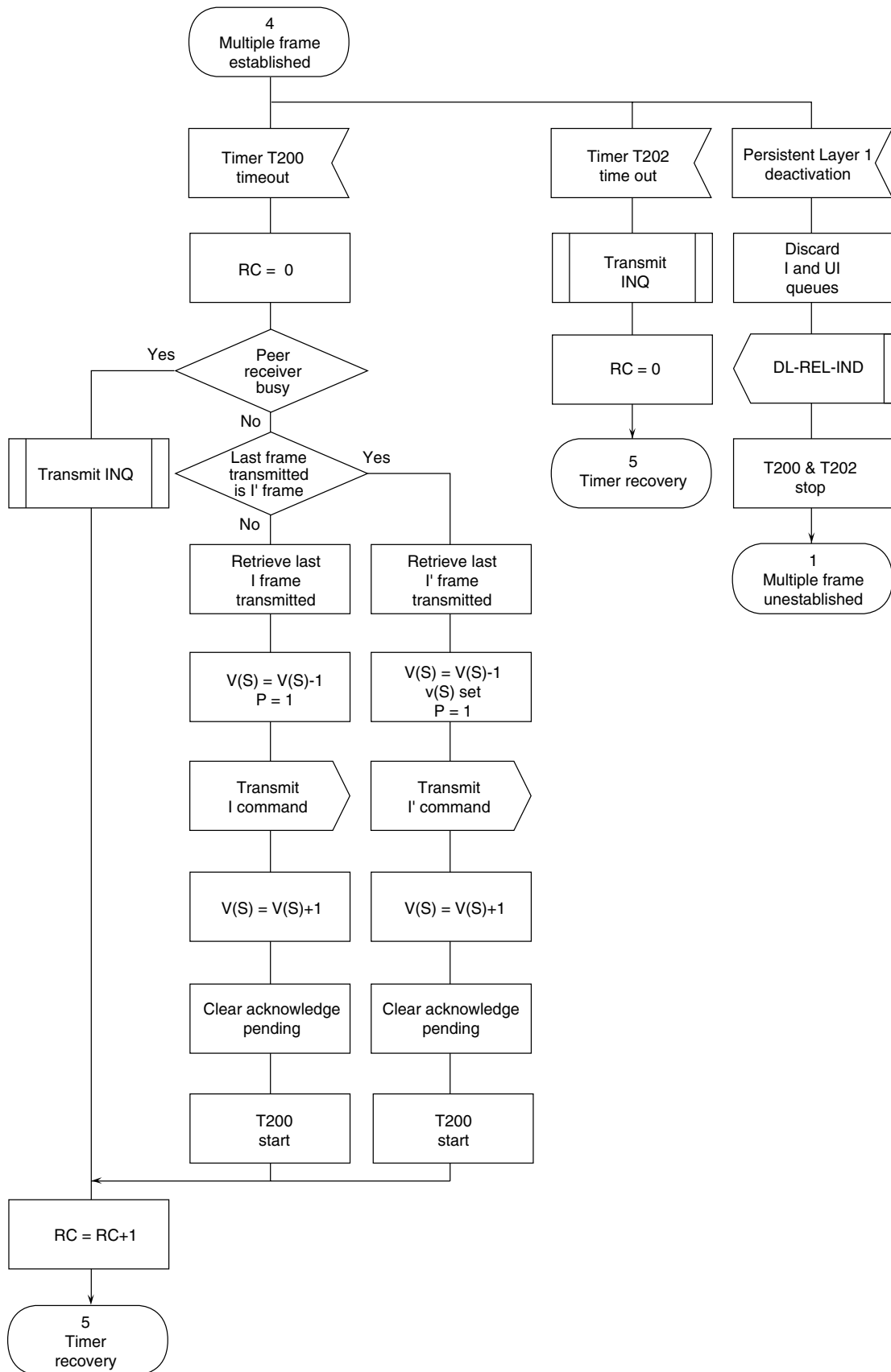


Fig. A. 4.1-10 Layer 2 SDL diagram (9/41)

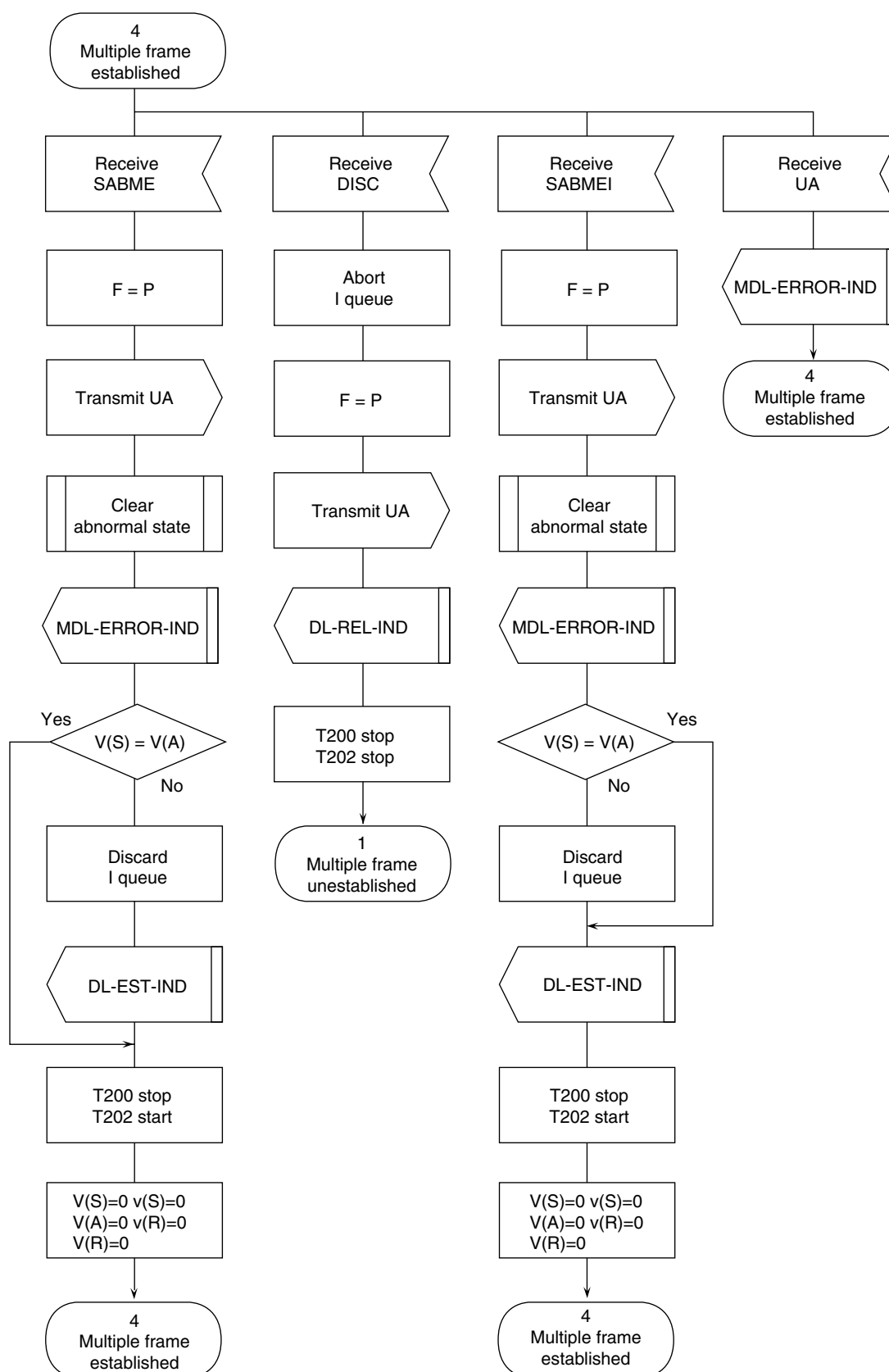
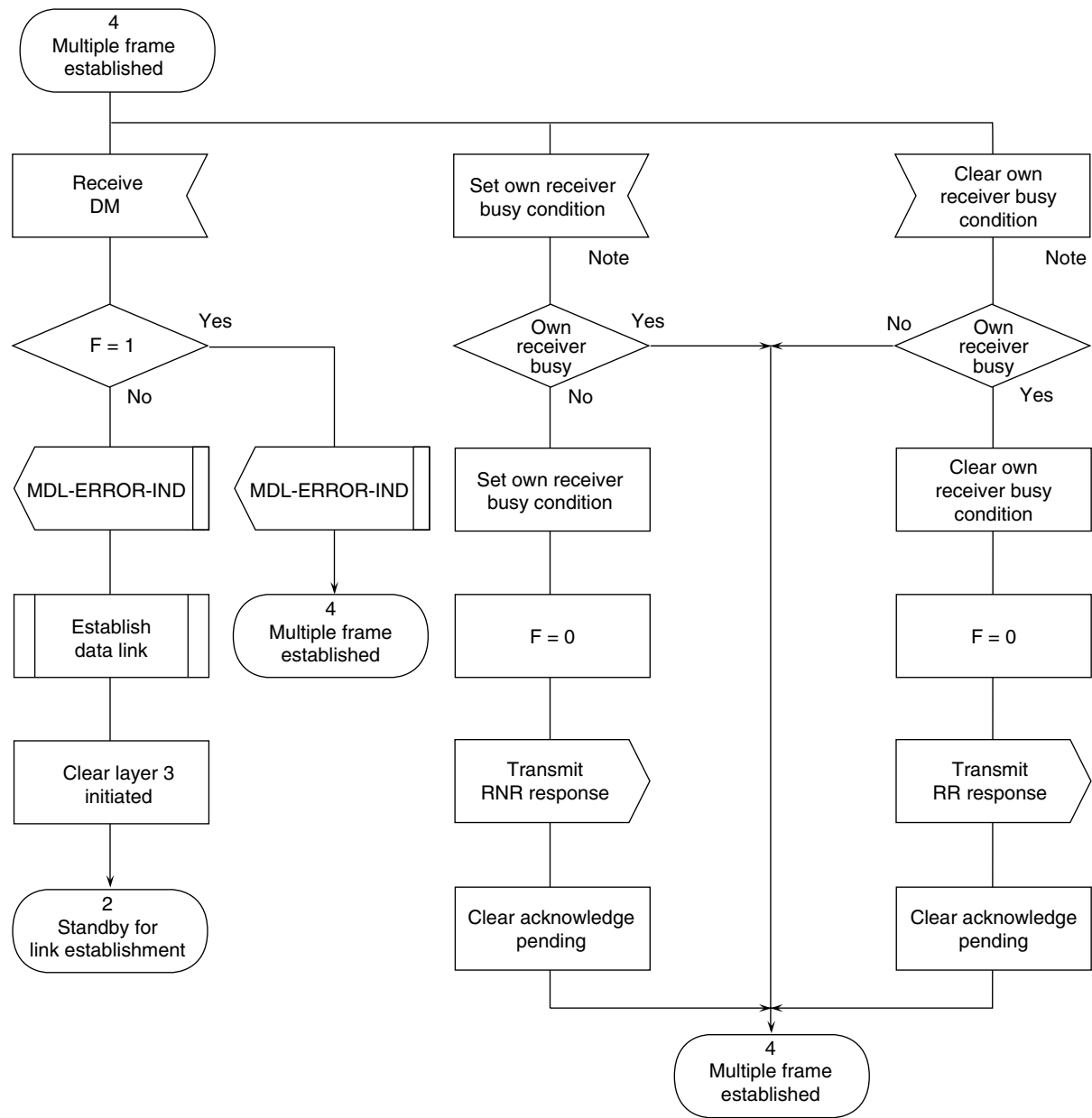


Fig. A. 4.1-11 Layer 2 SDL diagram (10/41)



Note: These signals are generated outside the SDL expression category and can also be generated in the connection management entity.

Fig. A. 4.1-12 Layer 2 SDL diagram (11/41)

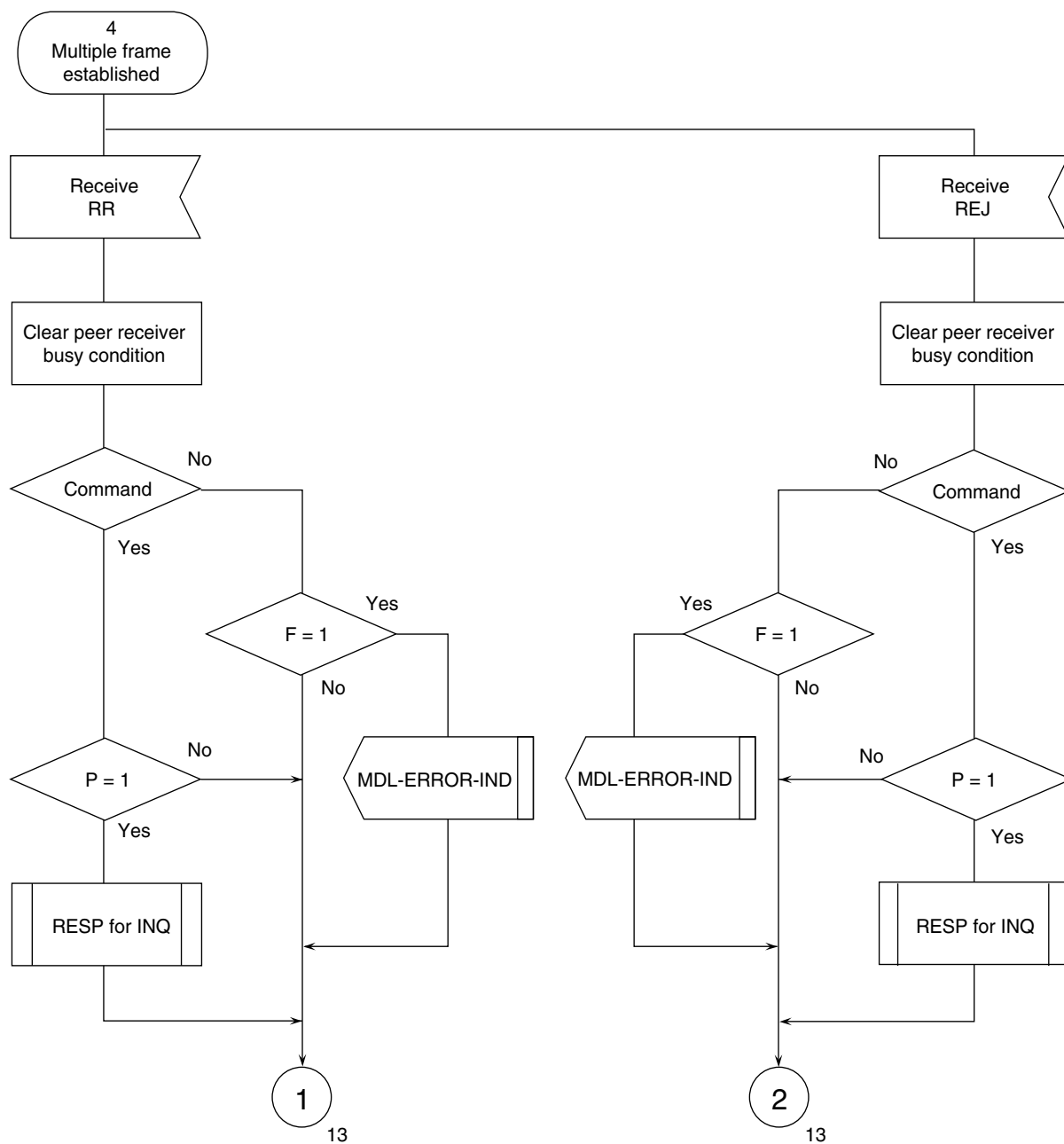


Fig. A. 4.1-13 Layer 2 SDL diagram (12/41)

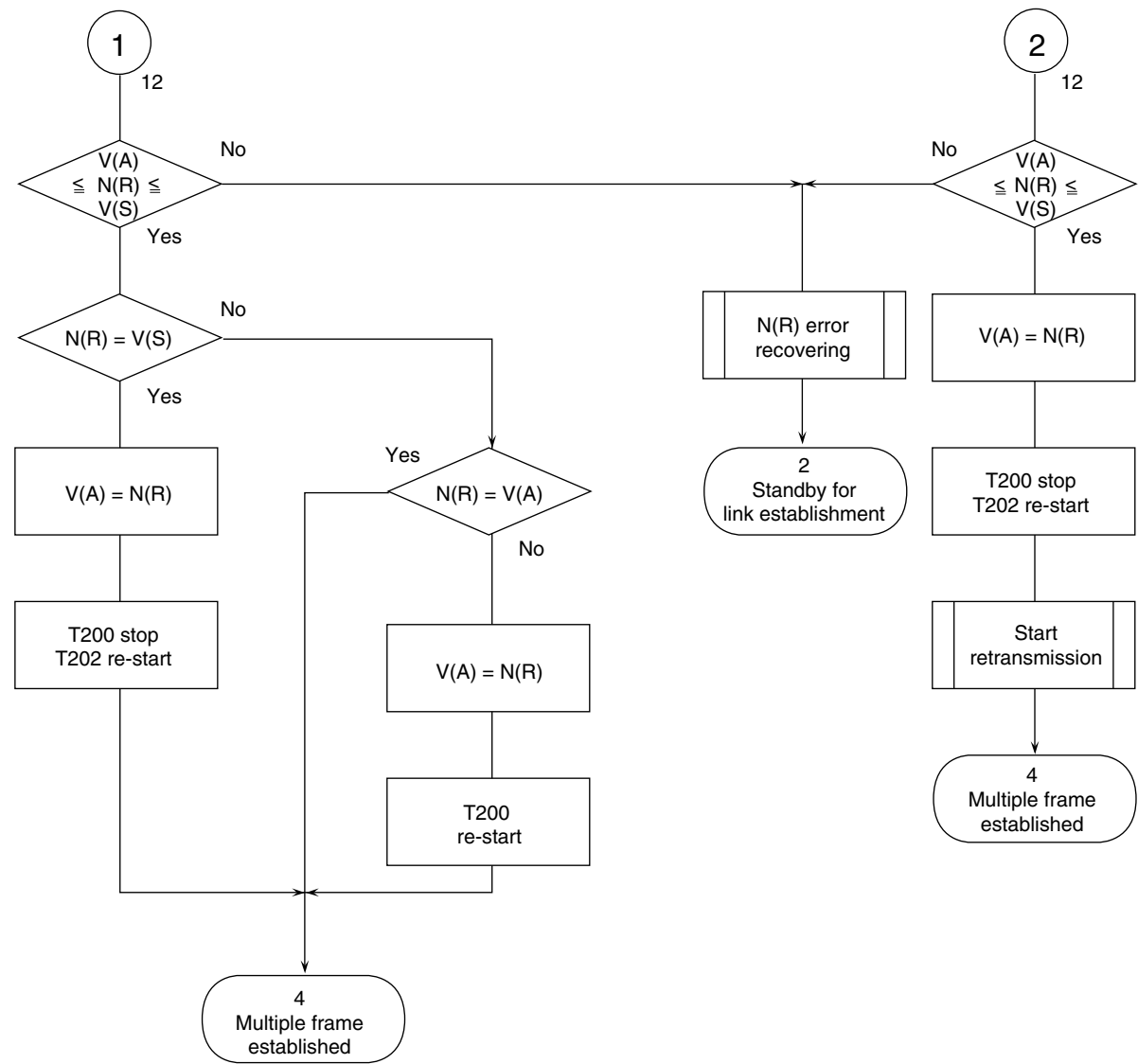


Fig. A. 4.1-14 Layer 2 SDL diagram (13/41)

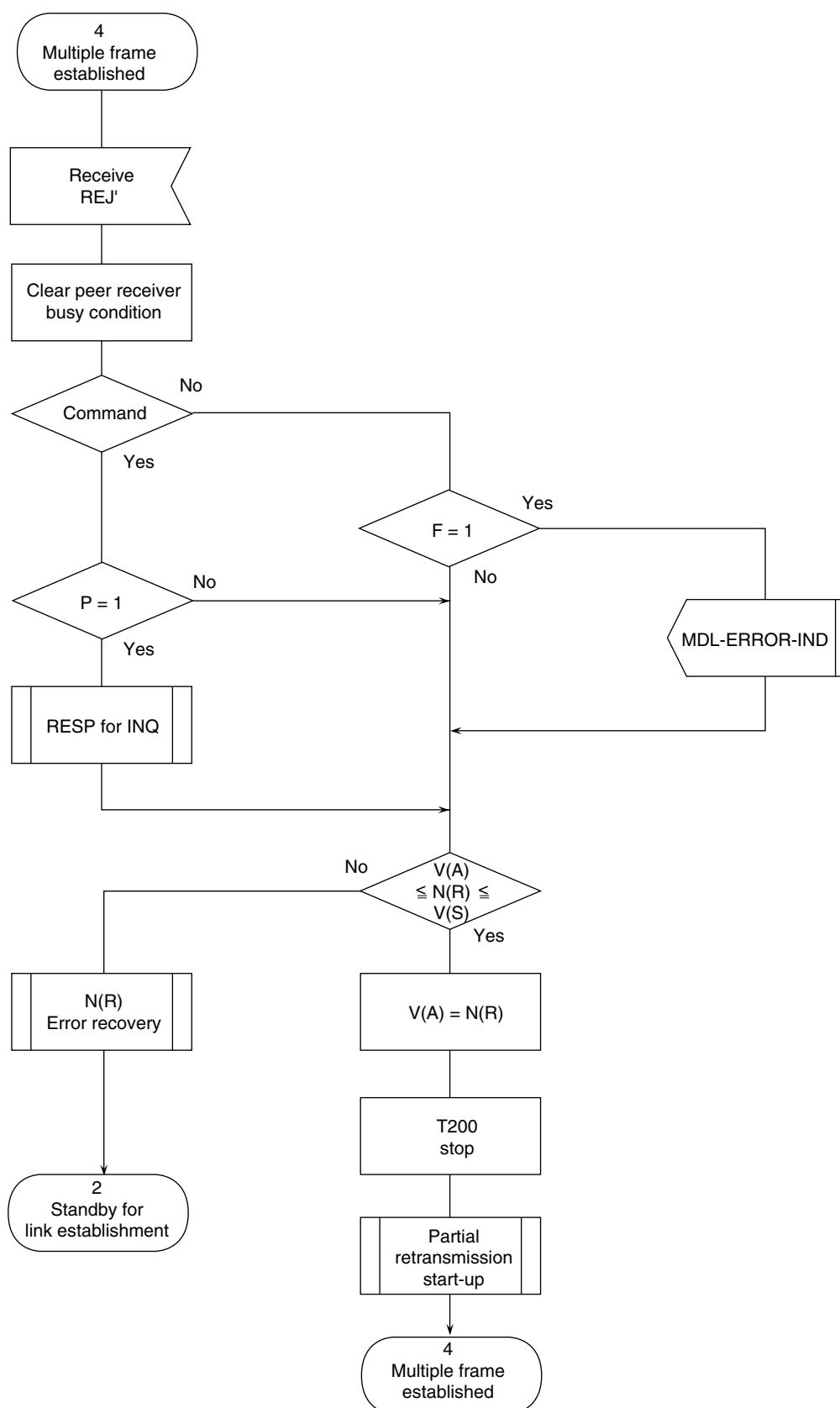


Fig. A. 4.1-15 Layer 2 SDL diagram (14/41)

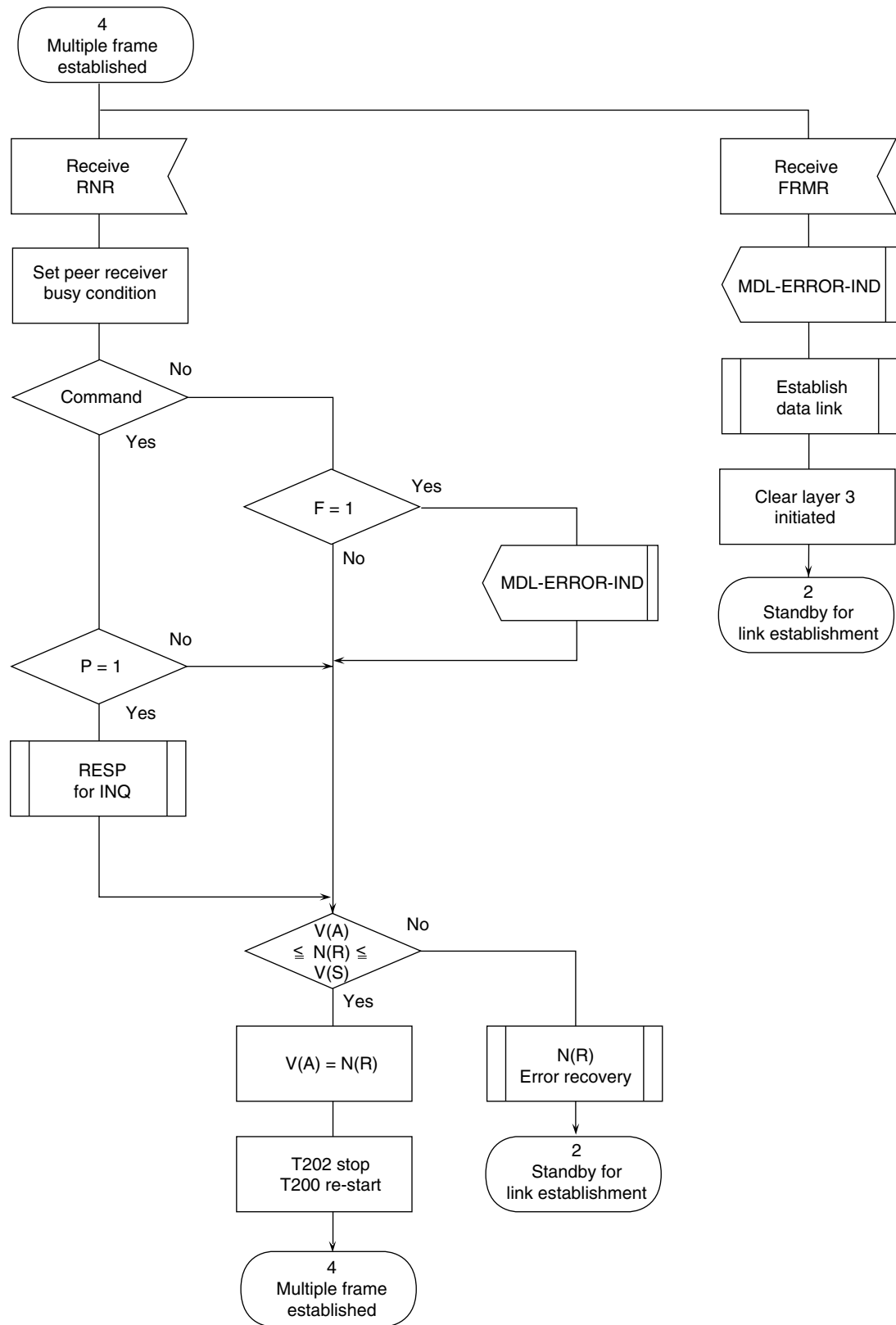


Fig. A. 4.1-16 Layer 2 SDL diagram (15/41)

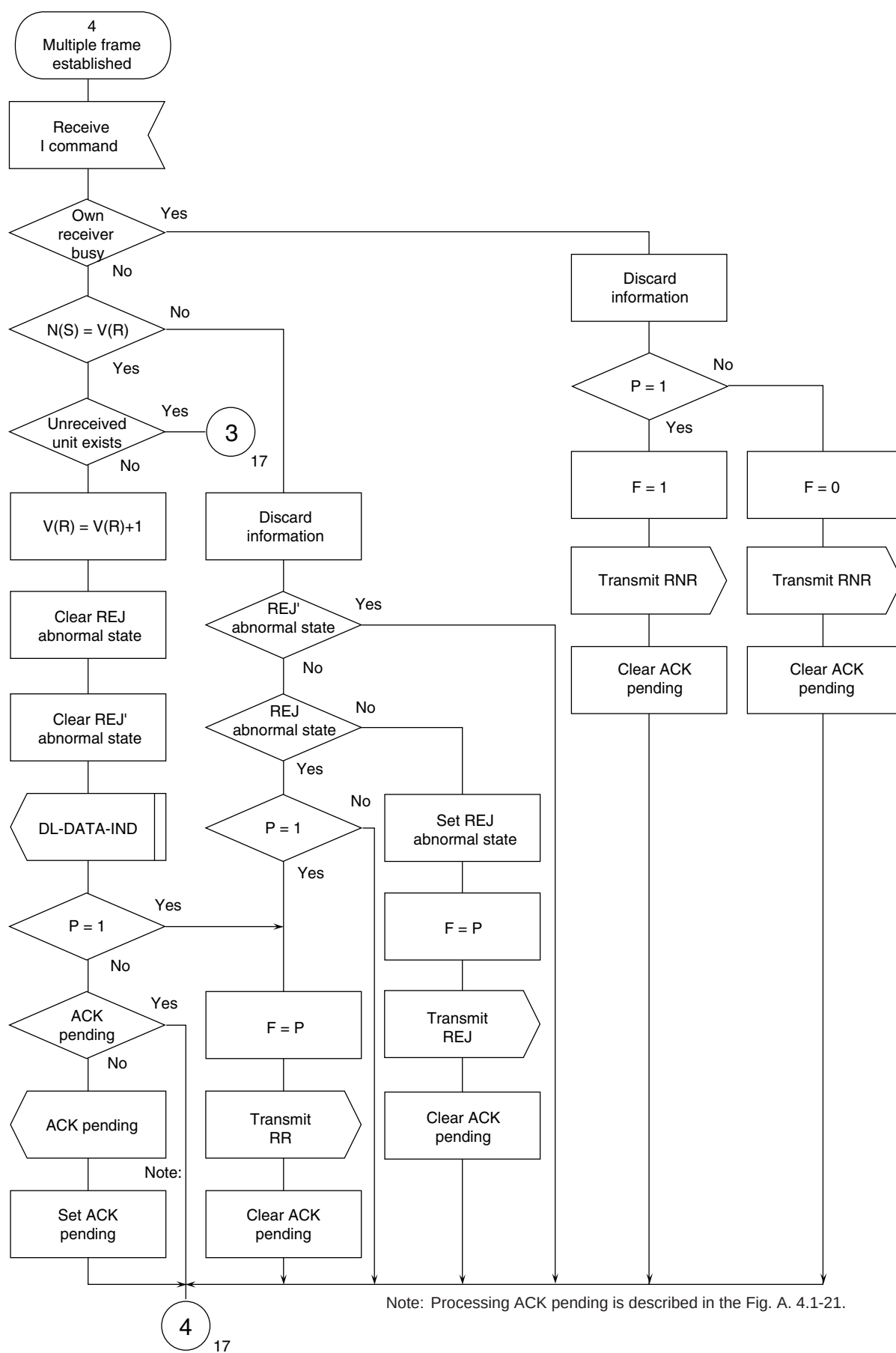


Fig. A. 4.1-17 Layer 2 SDL diagram (16/41)

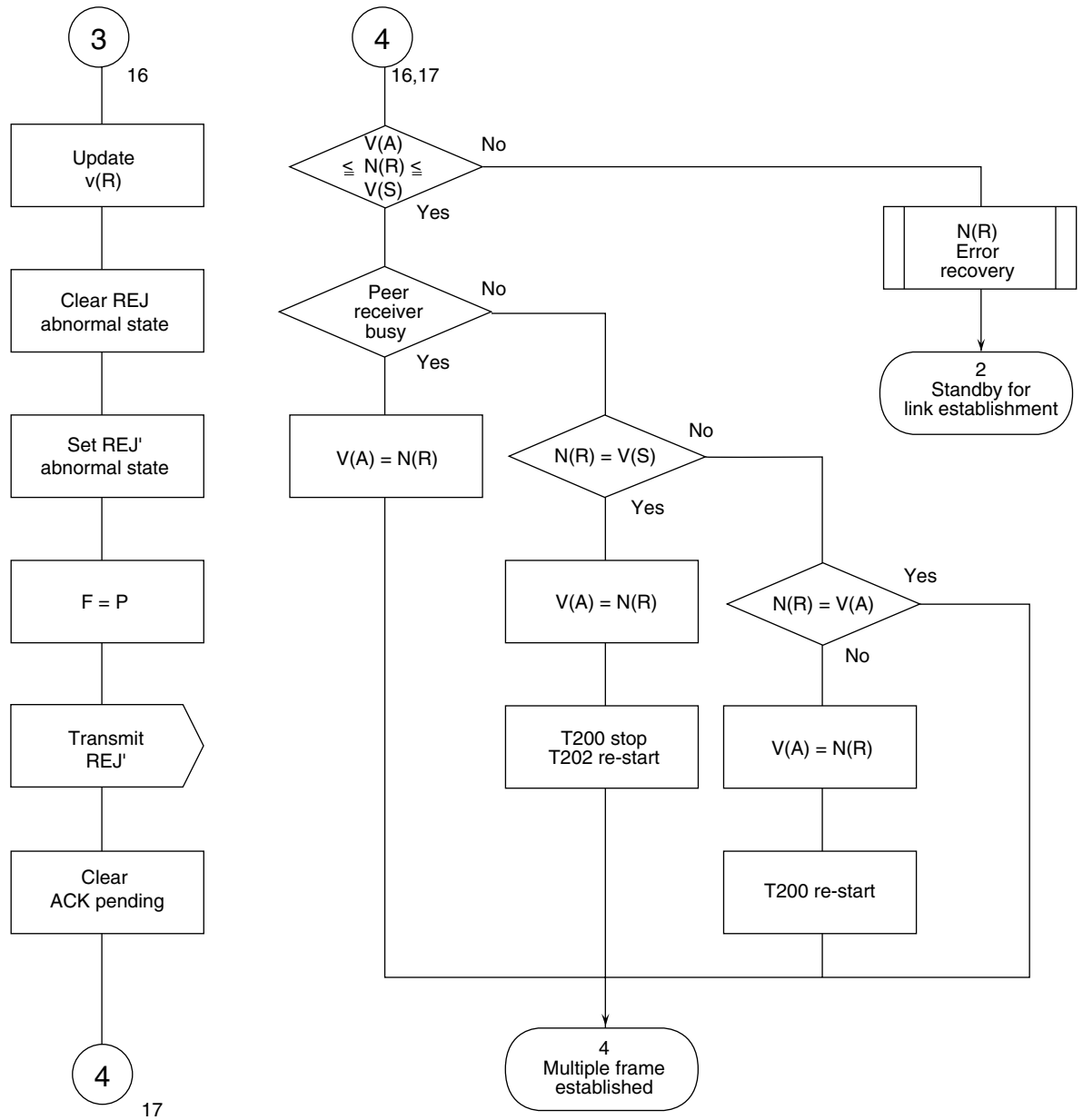


Fig. A. 4.1-18 Layer 2 SDL diagram (17/41)

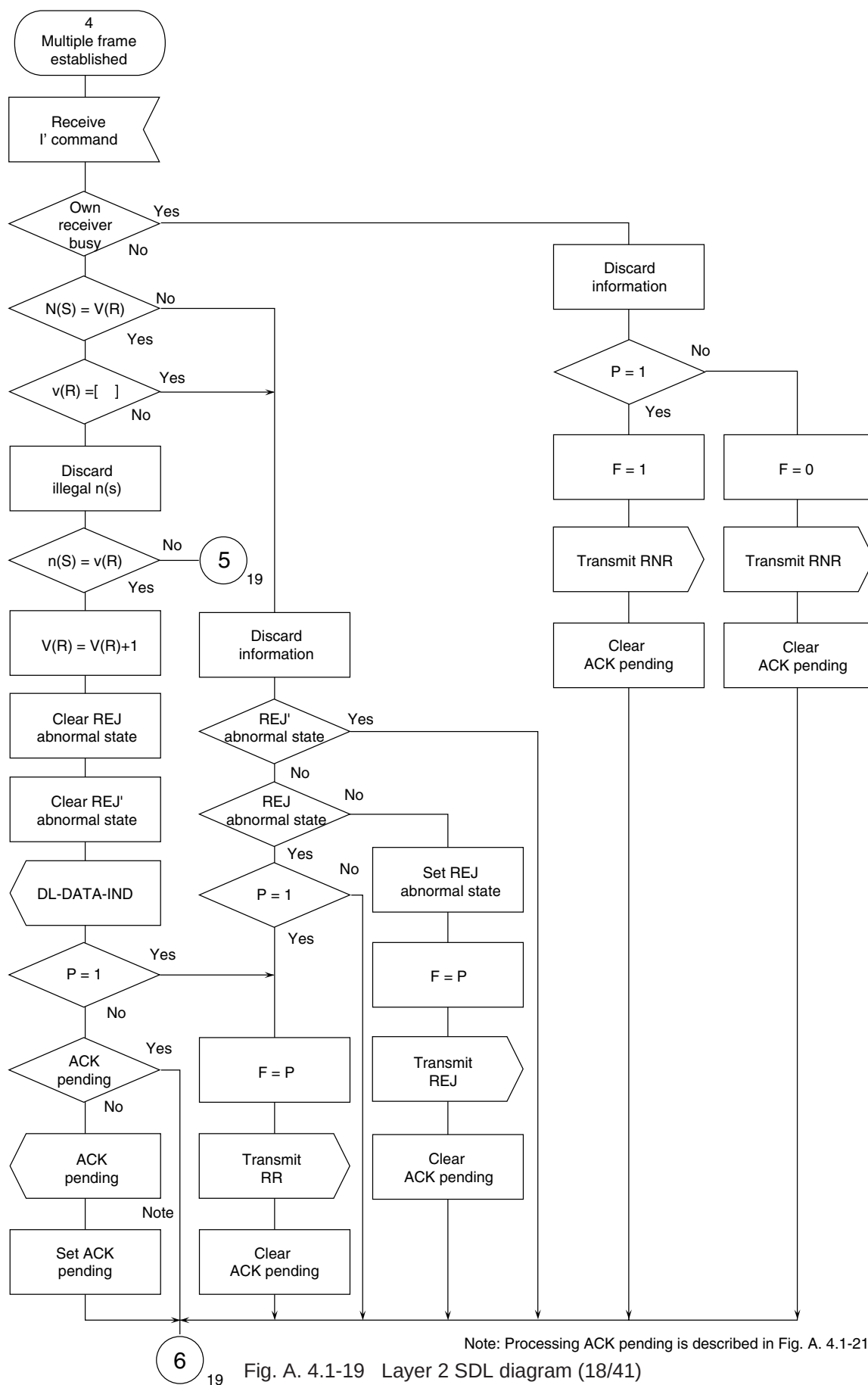


Fig. A. 4.1-19 Layer 2 SDL diagram (18/41)

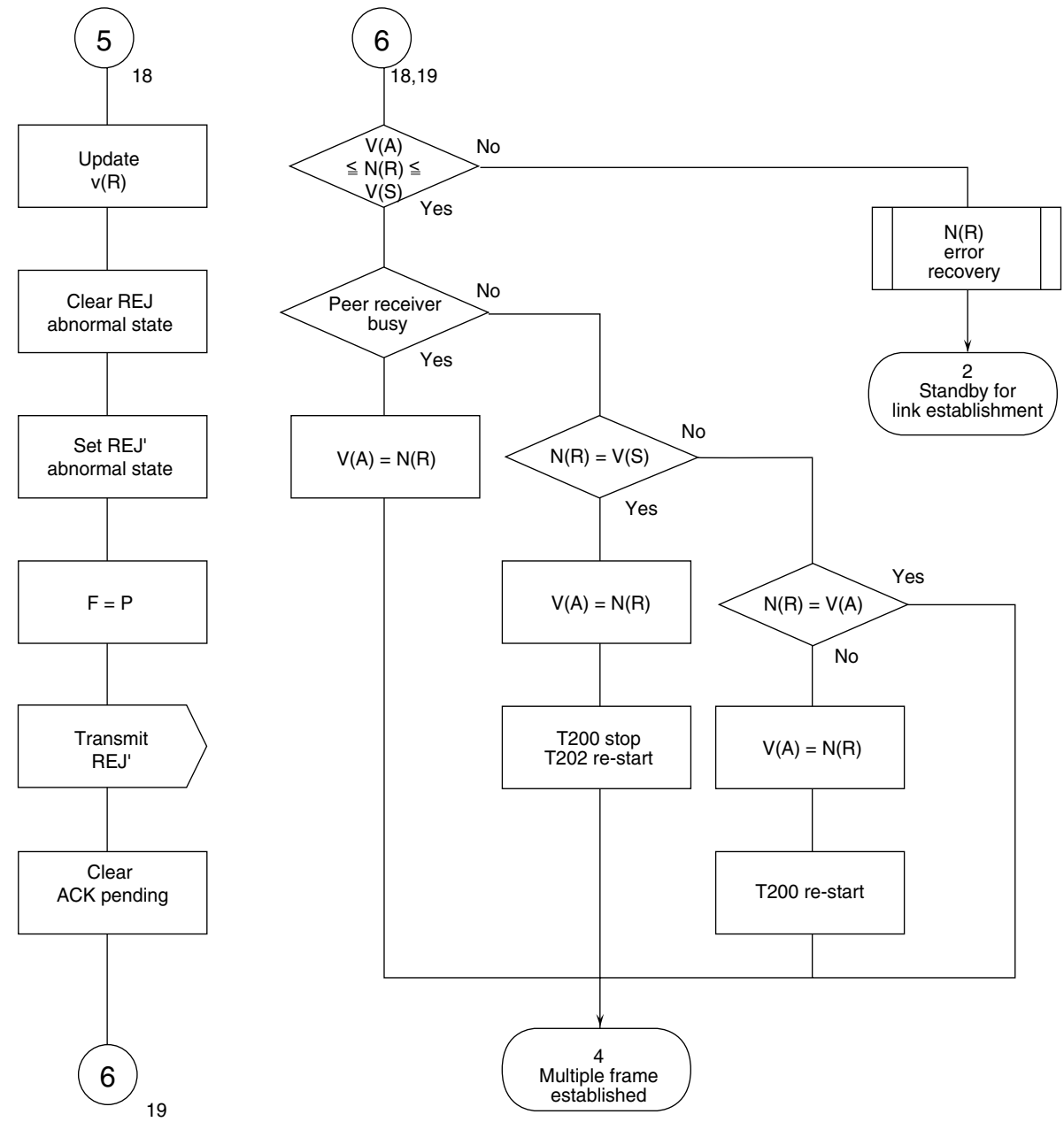


Fig. A. 4.1-20 Layer 2 SDL diagram (19/41)

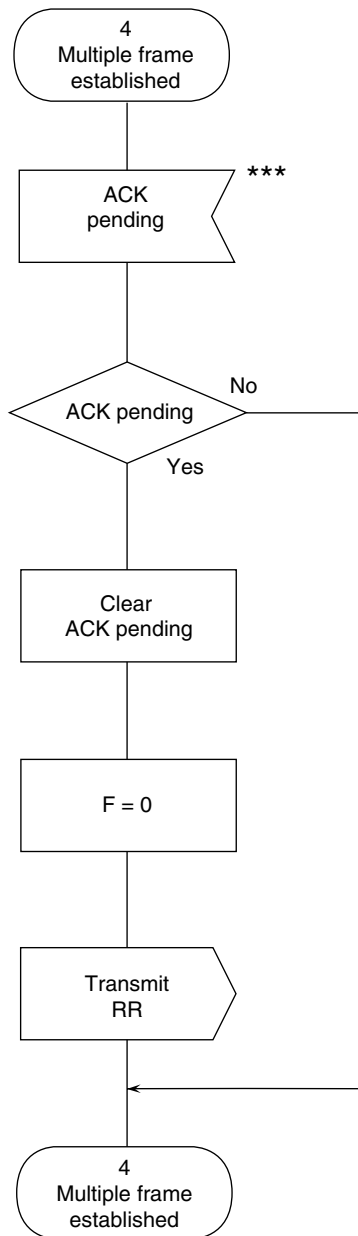


Fig. A. 4.1-21 Layer 2 SDL diagram (20/41)

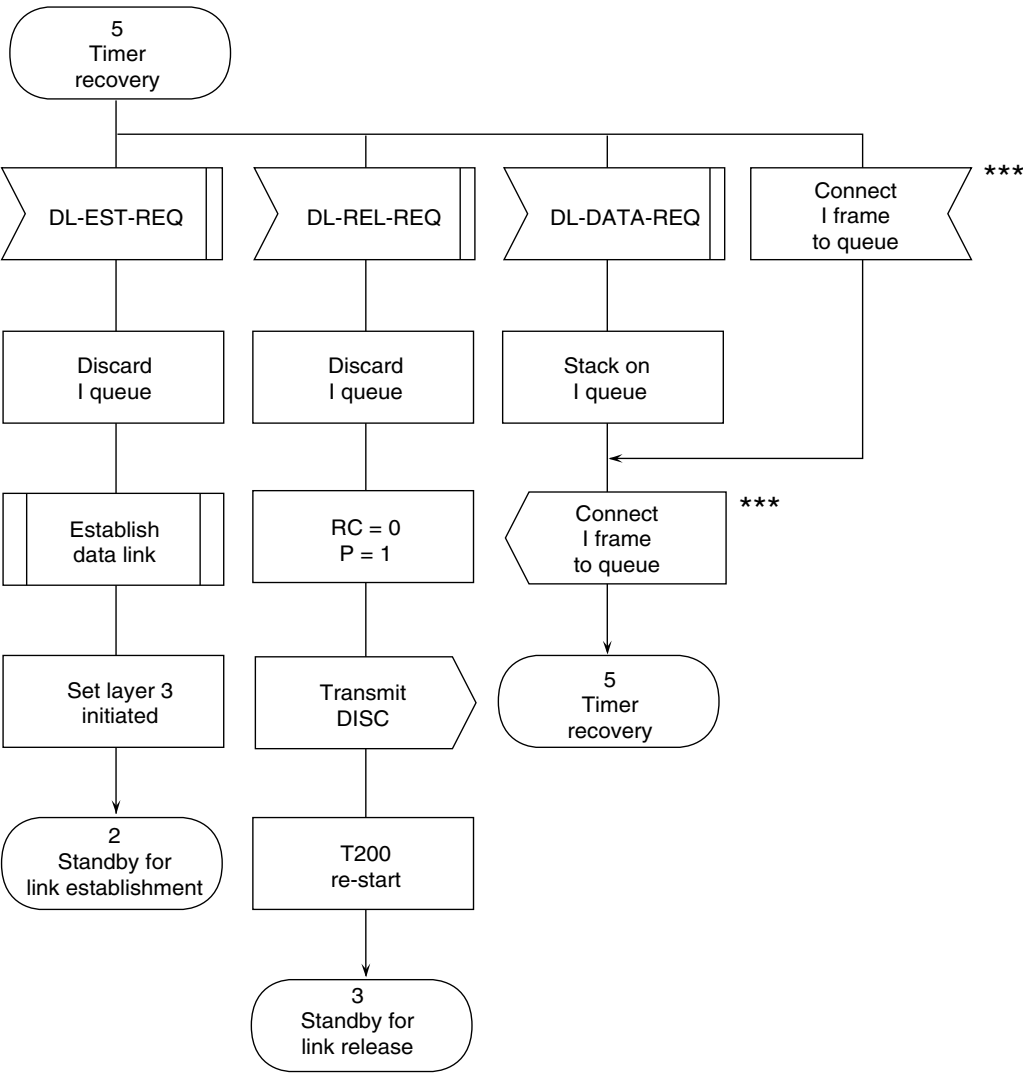


Fig. A. 4.1-22 Layer 2 SDL diagram (21/41)

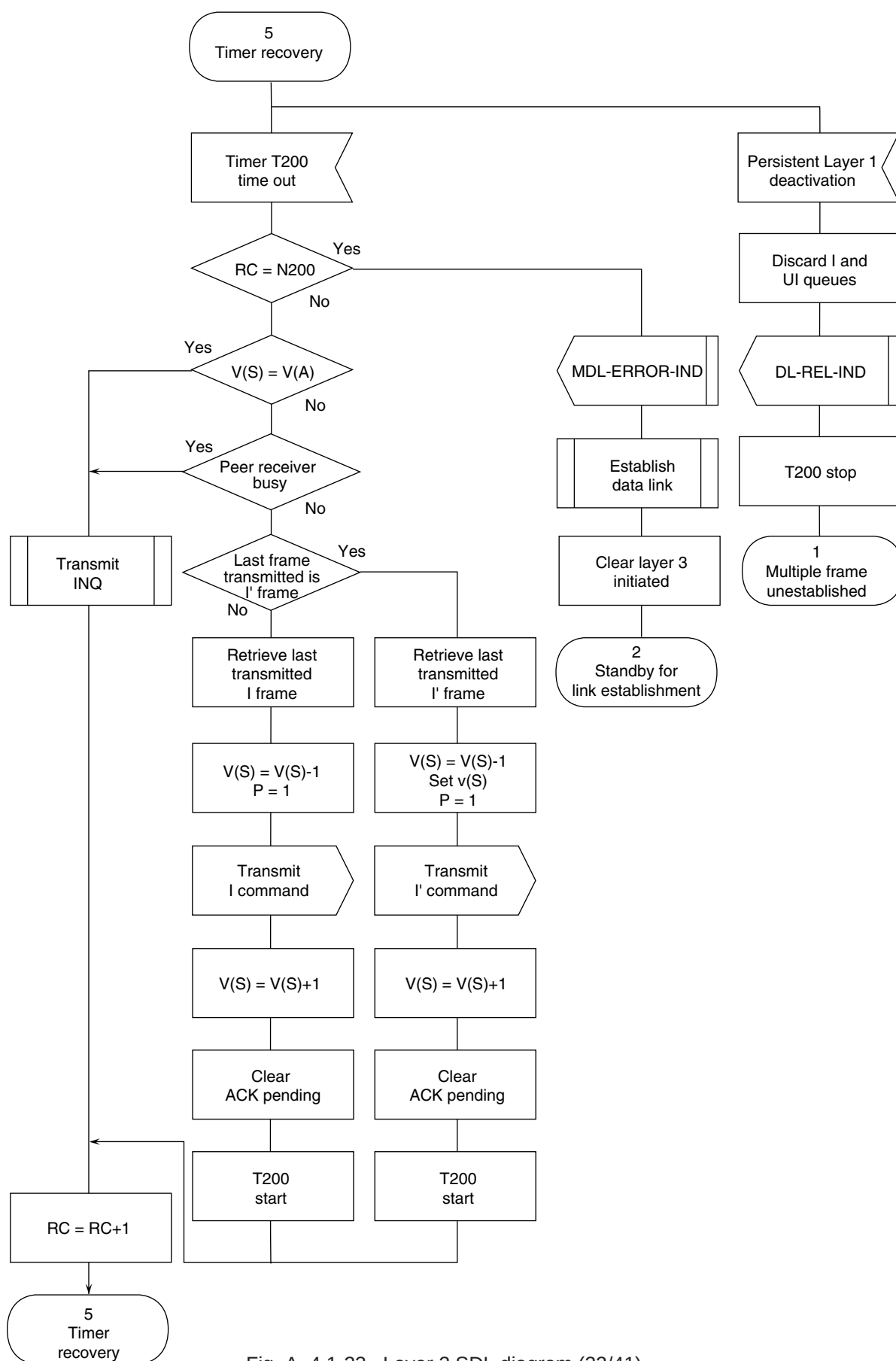


Fig. A. 4.1-23 Layer 2 SDL diagram (22/41)

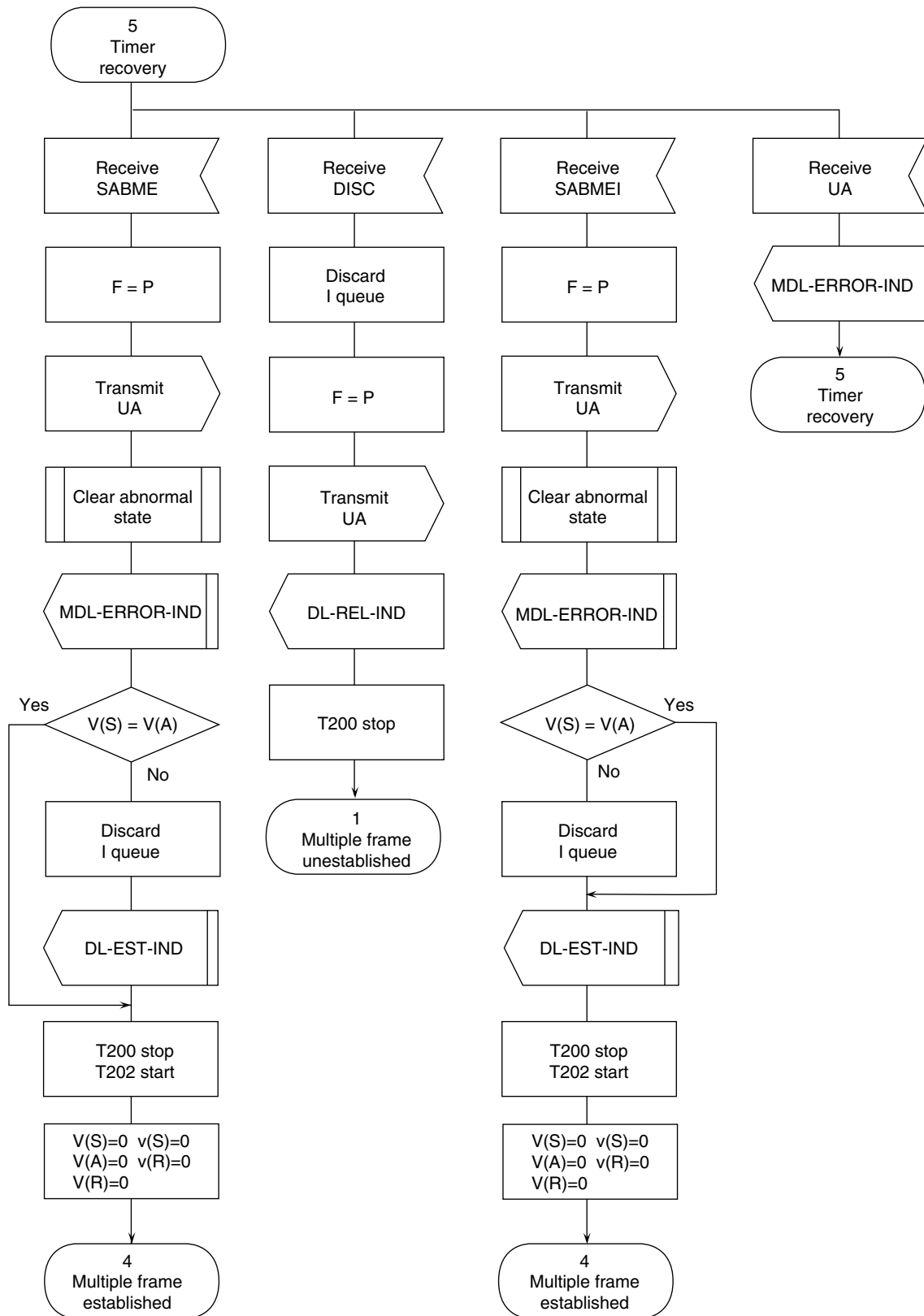
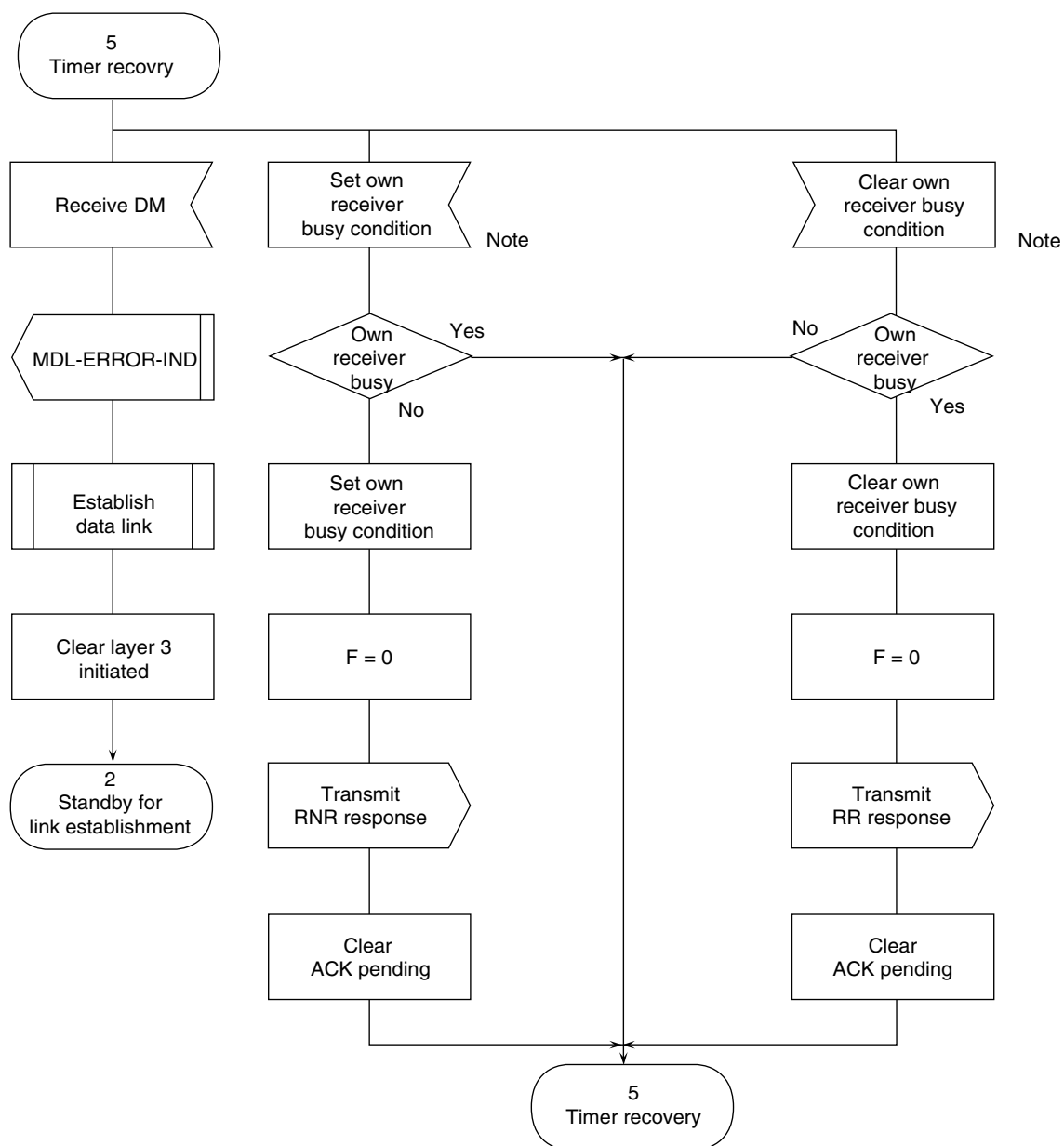


Fig. A. 4.1-24 Layer 2 SDL diagram (23/41)



Note: These signals are generated outside of the SDL expression category and can also be generated in the connection management entity.

Fig. A. 4.1-25 Layer 2 SDL diagram (24/41)

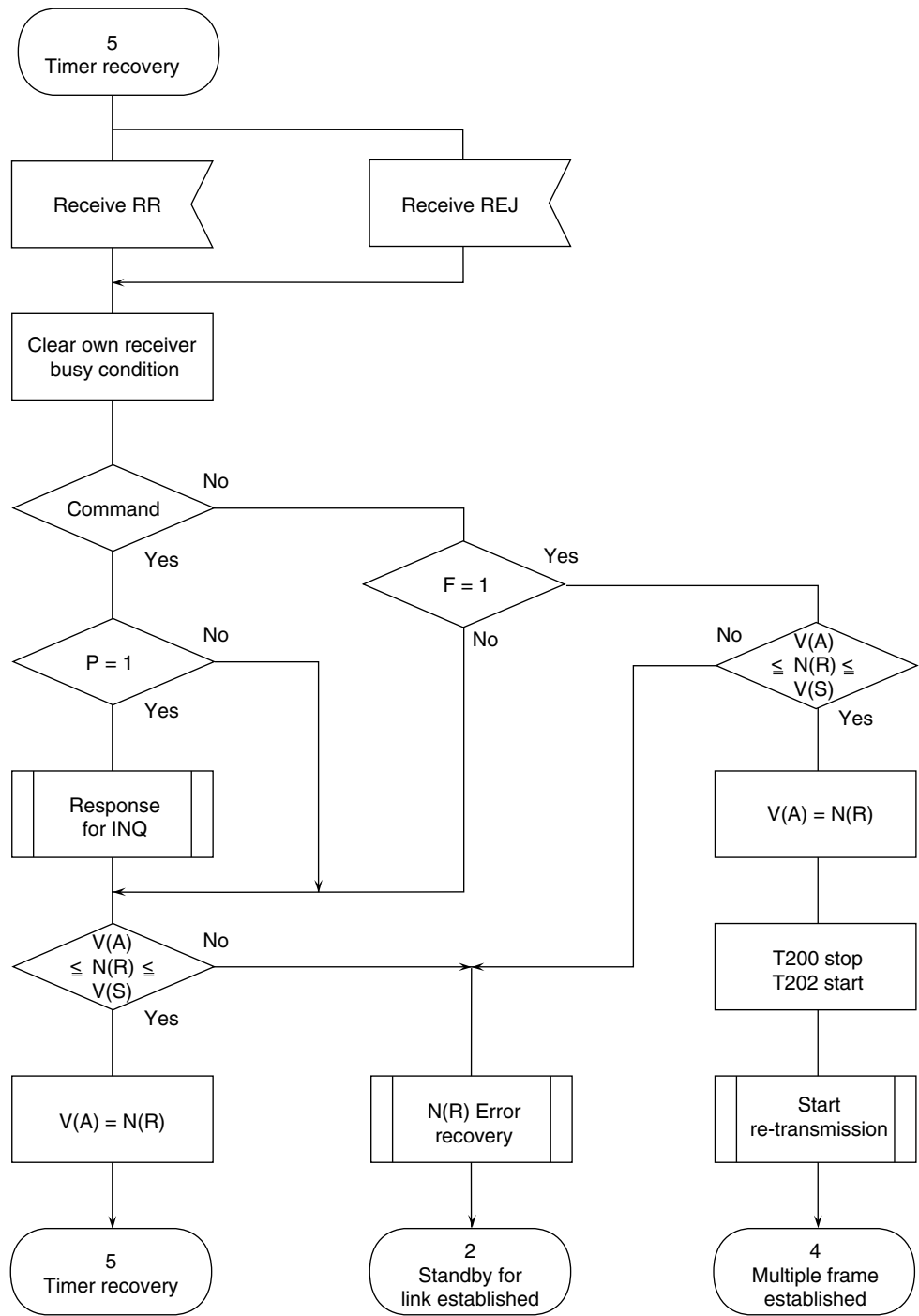


Fig. A. 4.1-26 Layer 2 SDL diagram (25/41)

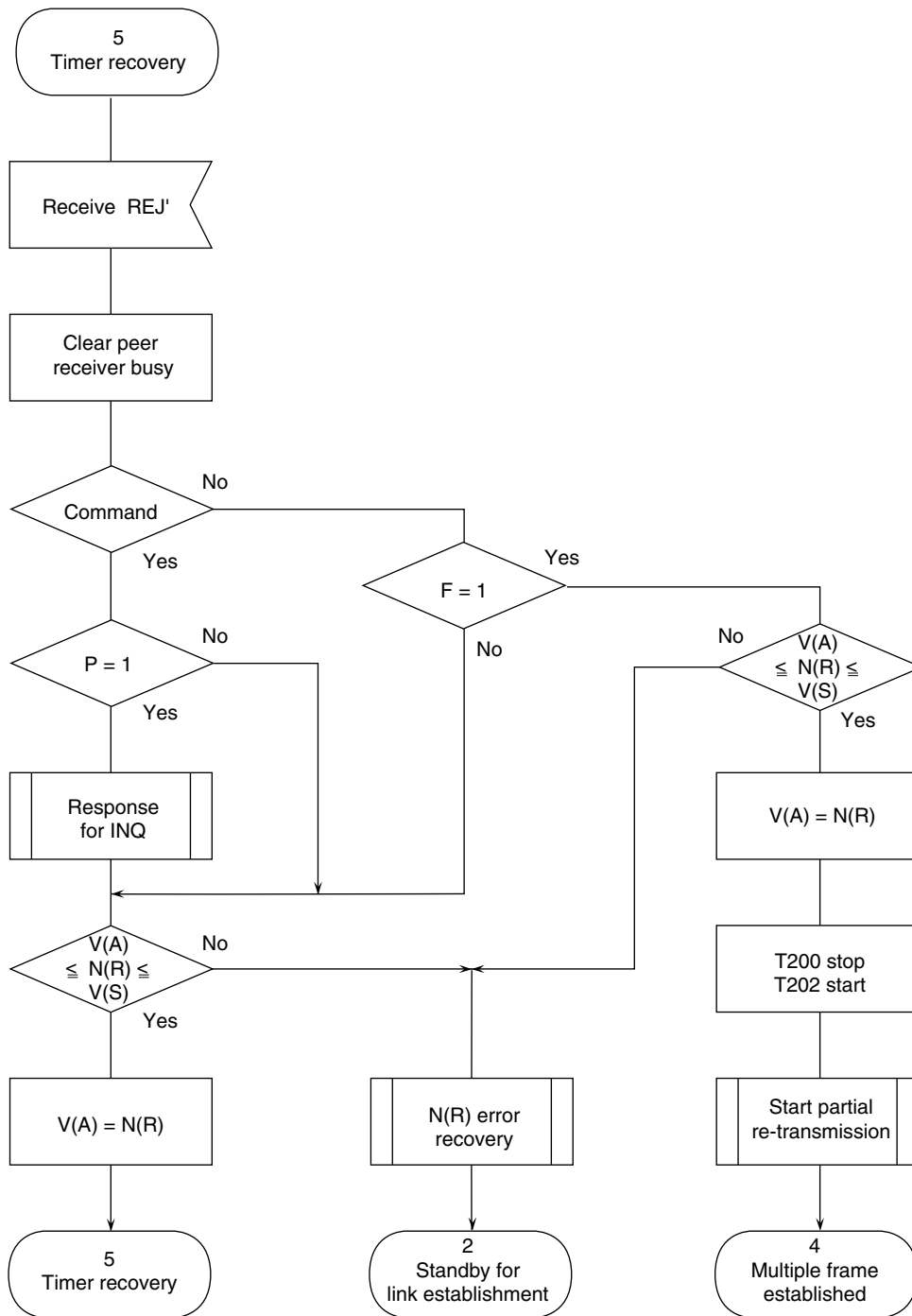


Fig. A. 4.1-27 Layer 2 SDL diagram (26/41)

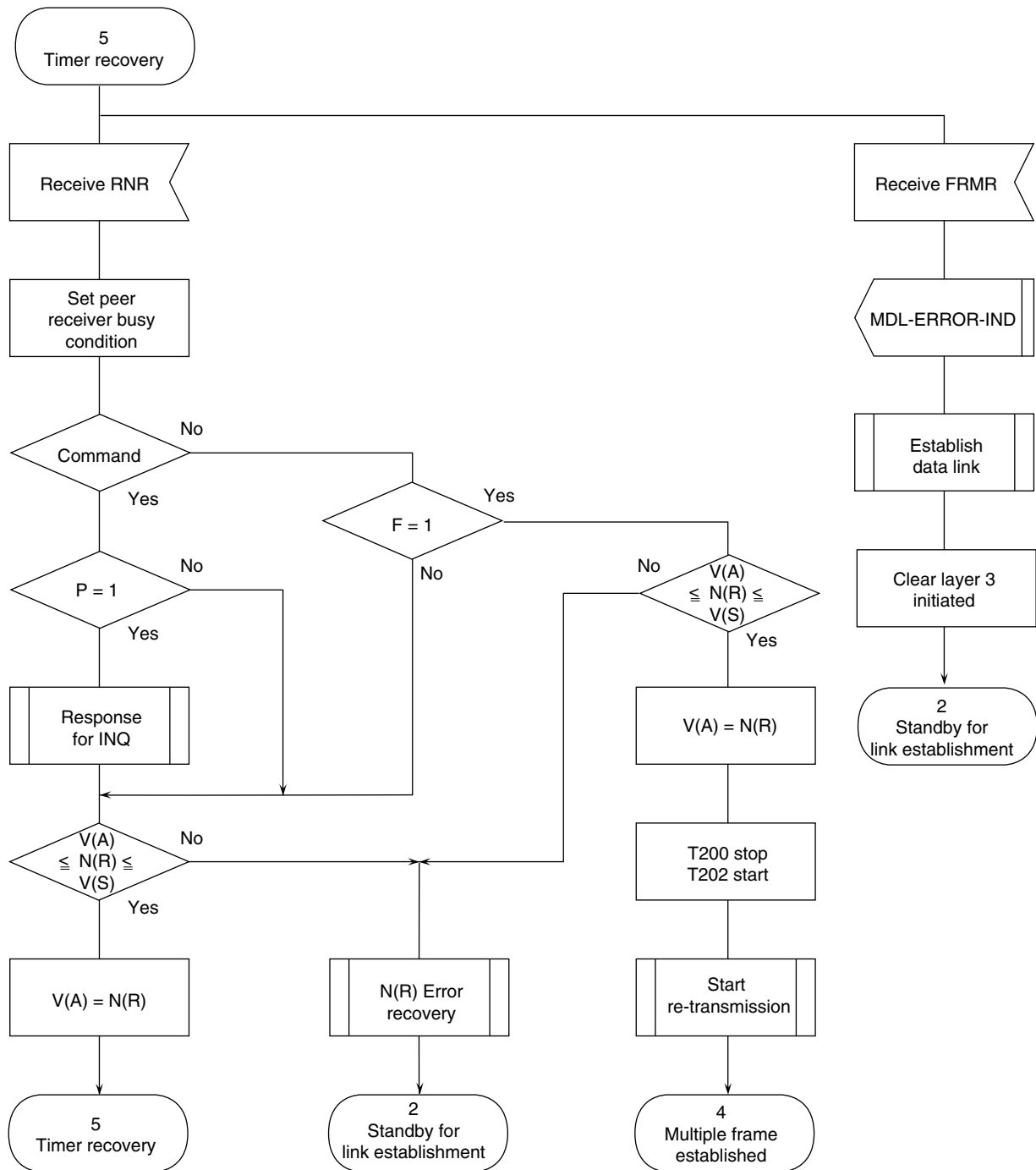


Fig. A. 4.1-28 Layer 2 SDL diagram (27/41)

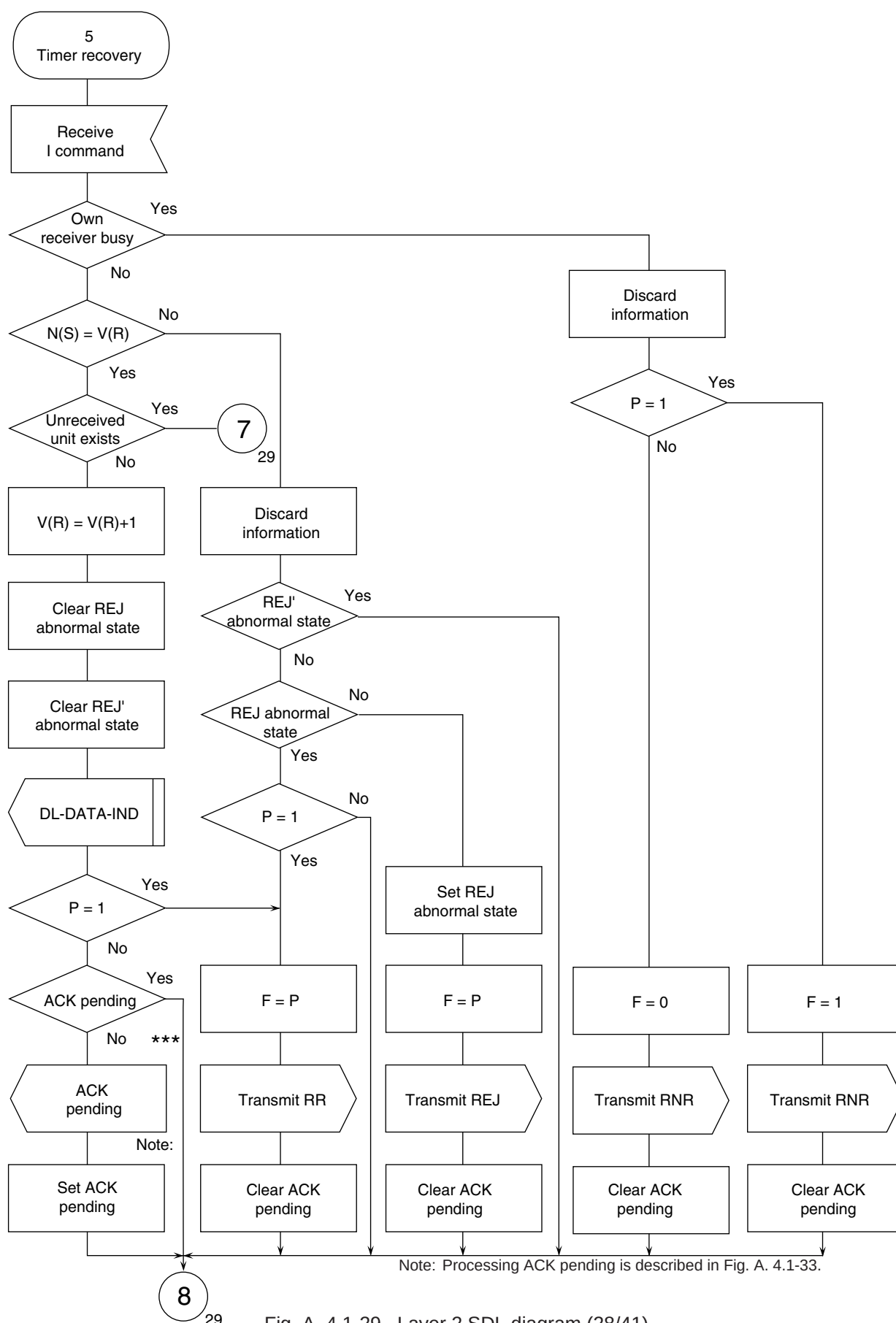


Fig. A. 4.1-29 Layer 2 SDL diagram (28/41)

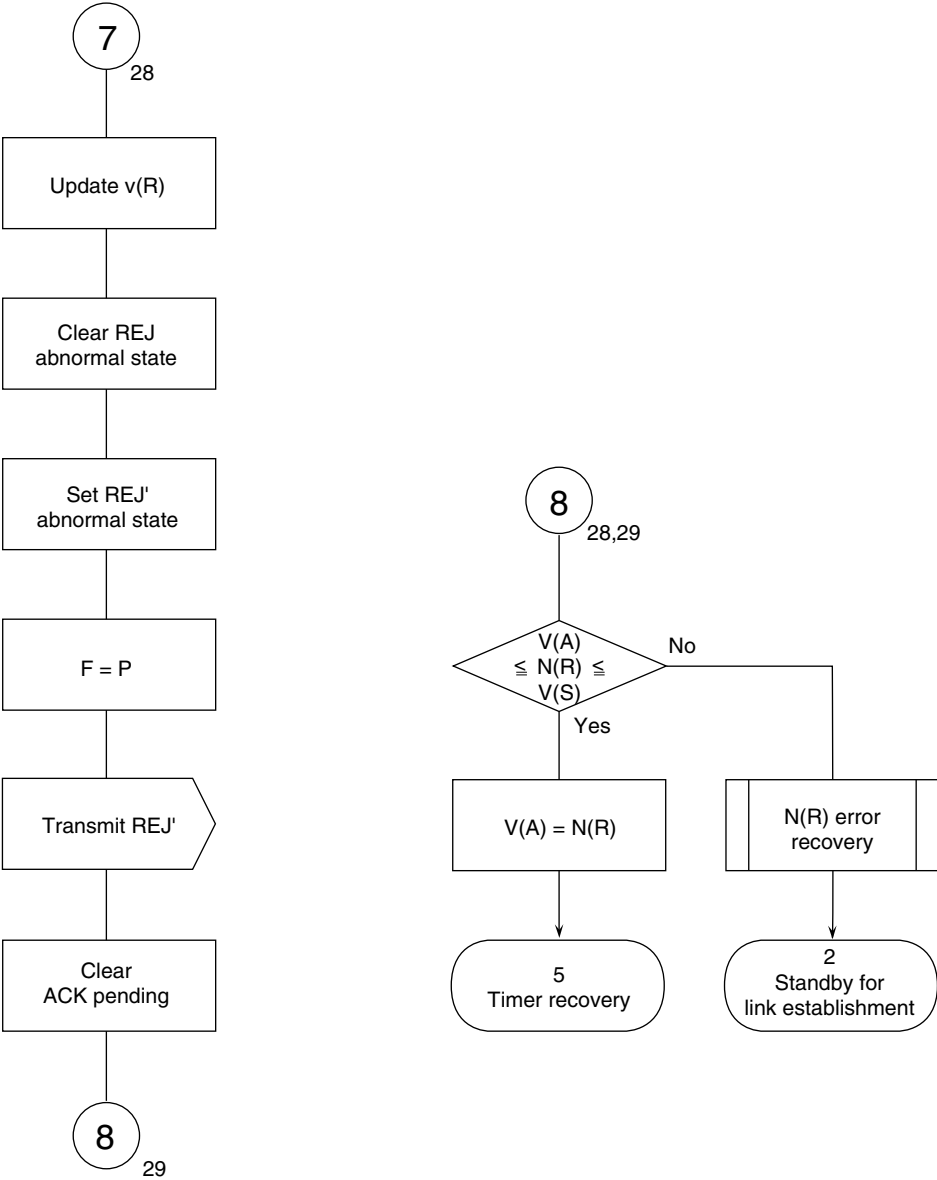


Fig. A. 4.1-30 Layer 2 SDL diagram (29/41)

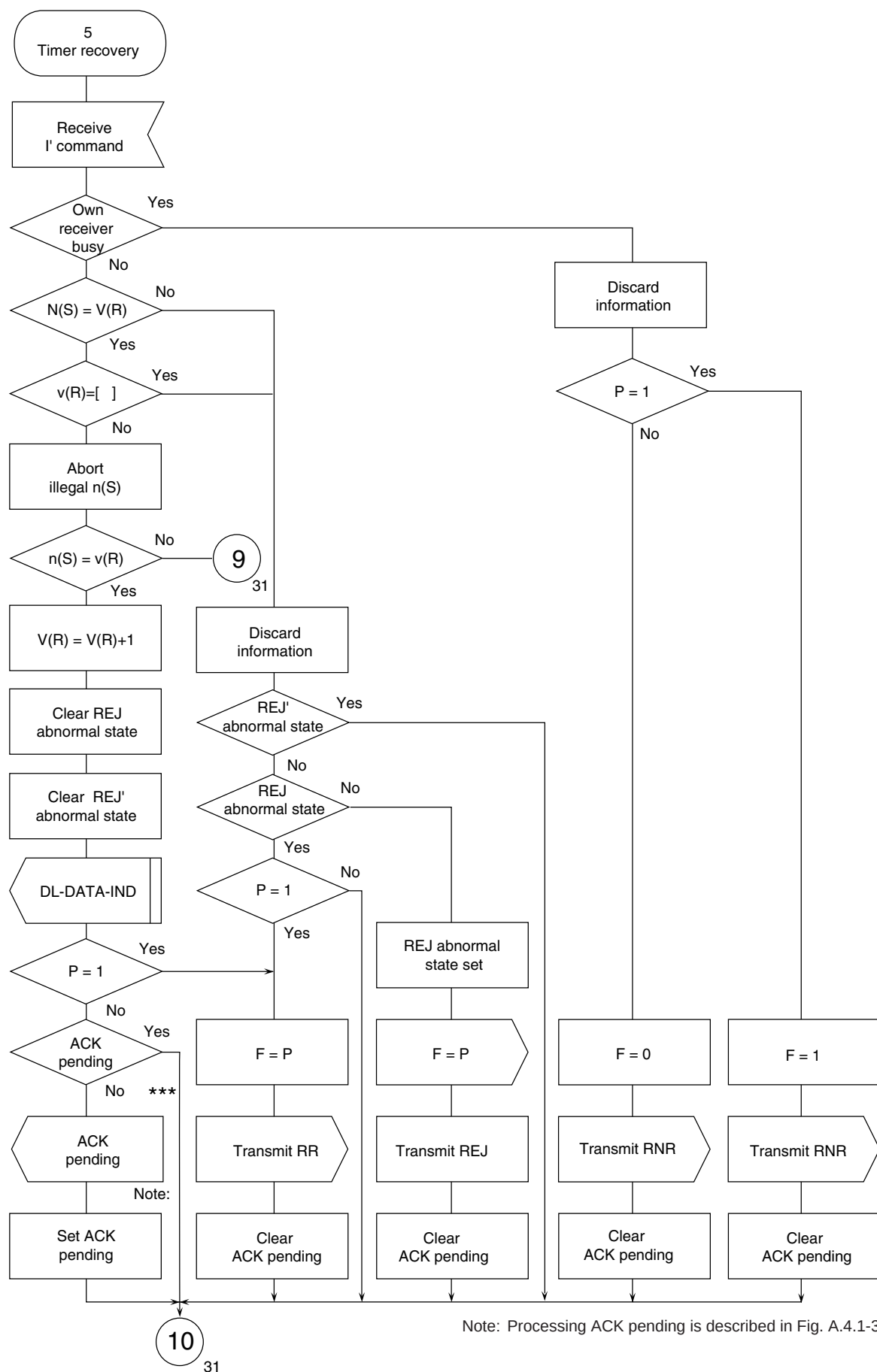


Fig. A. 4.1-31 Layer 2 SDL diagram (30/41)

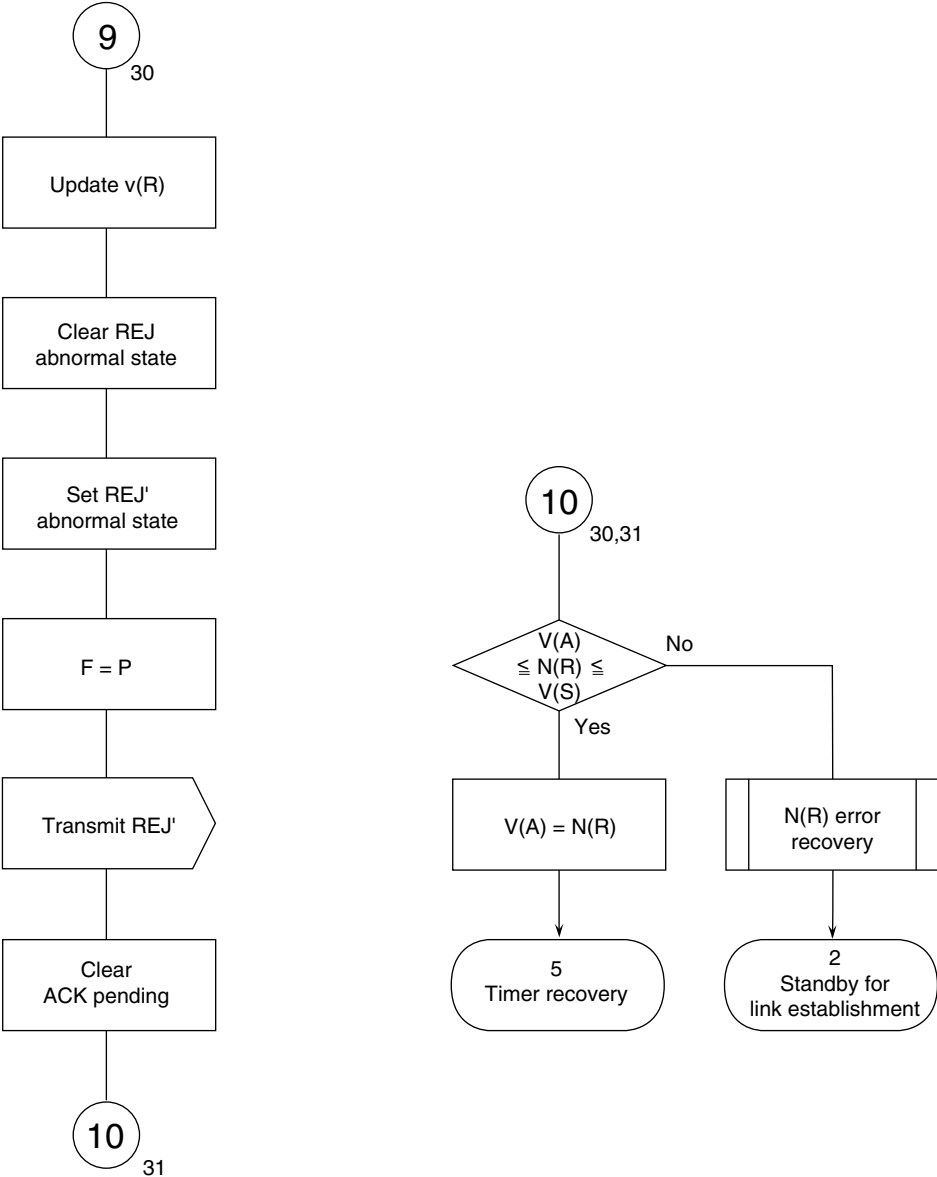


Fig. A. 4.1-32 Layer 2 SDL diagram (31/41)

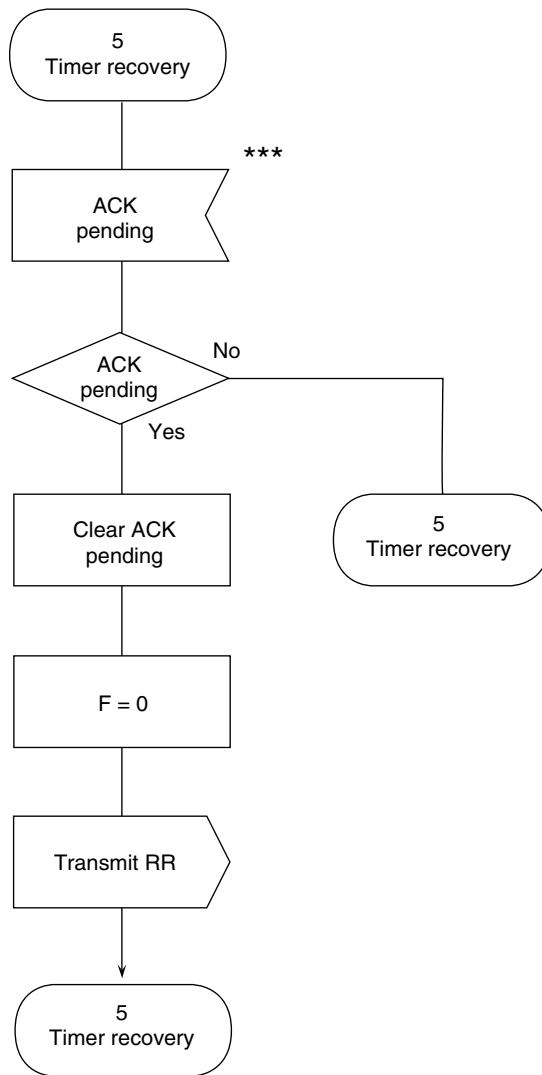
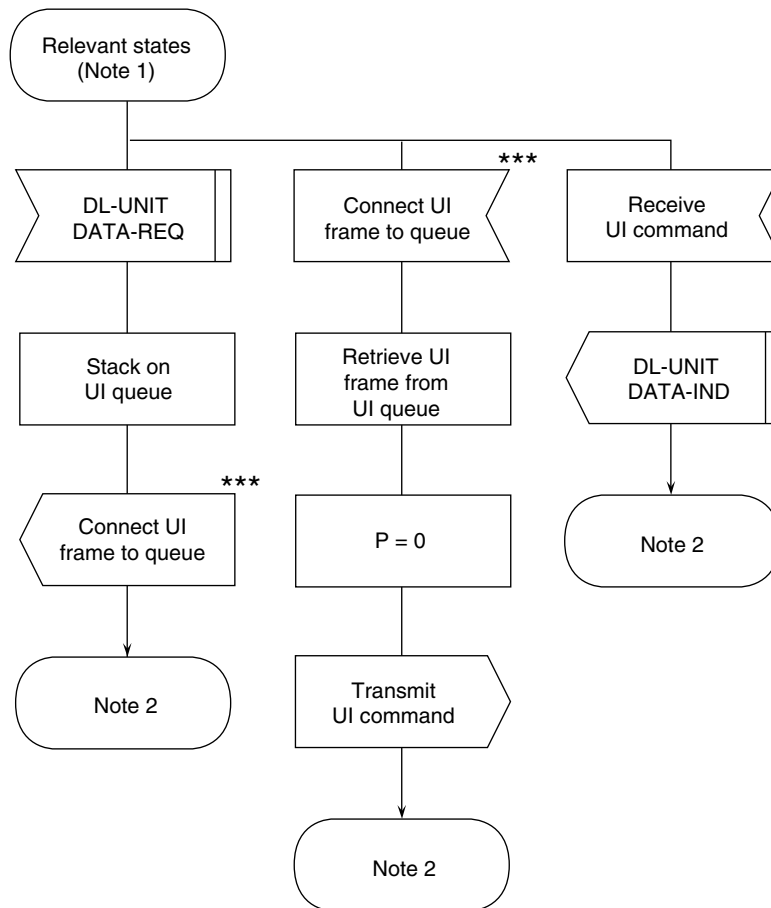


Fig. A. 4.1-33 Layer 2 SDL diagram (32/41)

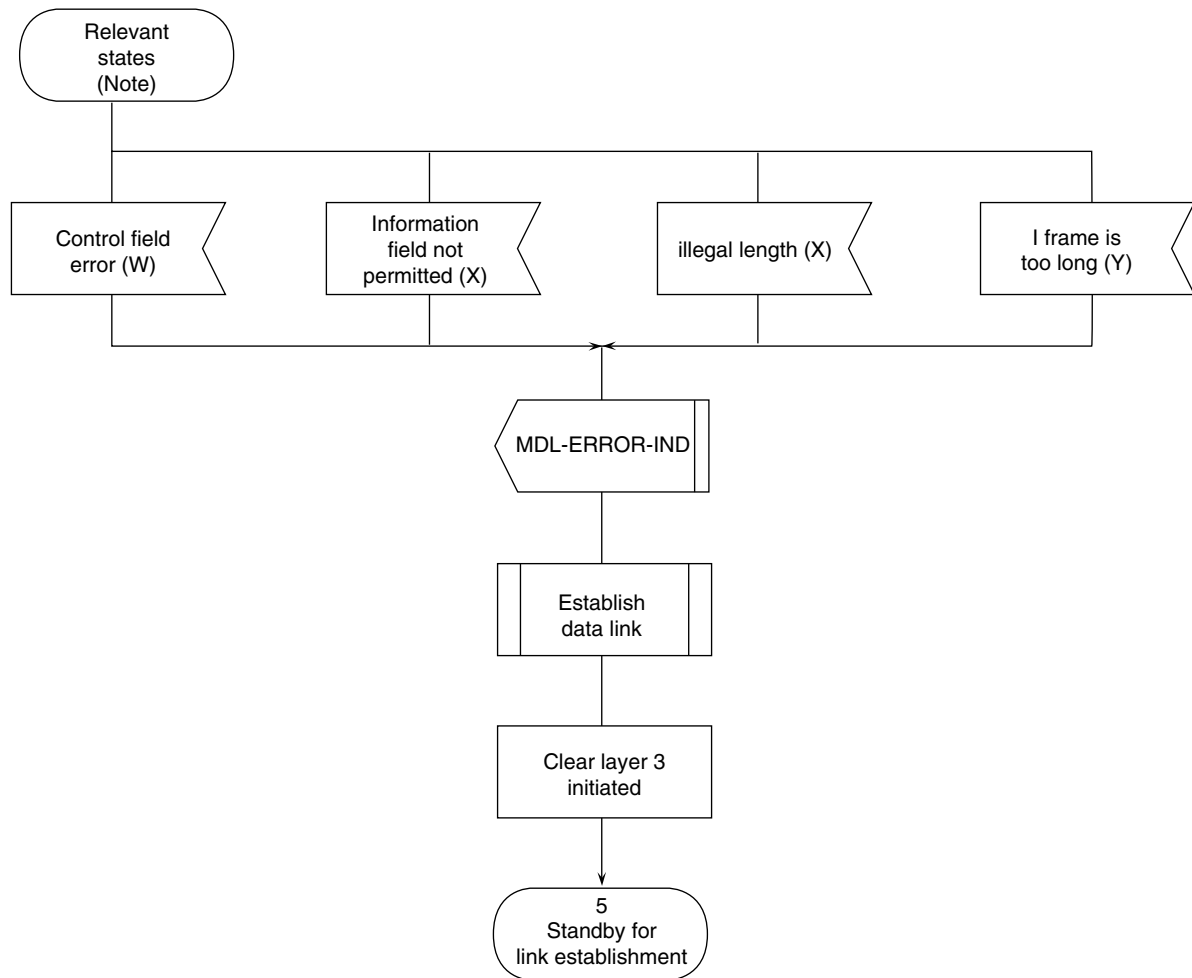


Note 1: "Relevant states" are listed below:

1. Multiple frame unestablished
2. Standby for link establishment
3. Standby for link release
4. Multiple frame established
5. Timer recovery

Note 2: Data link layer resets to the previously set state of the above states.

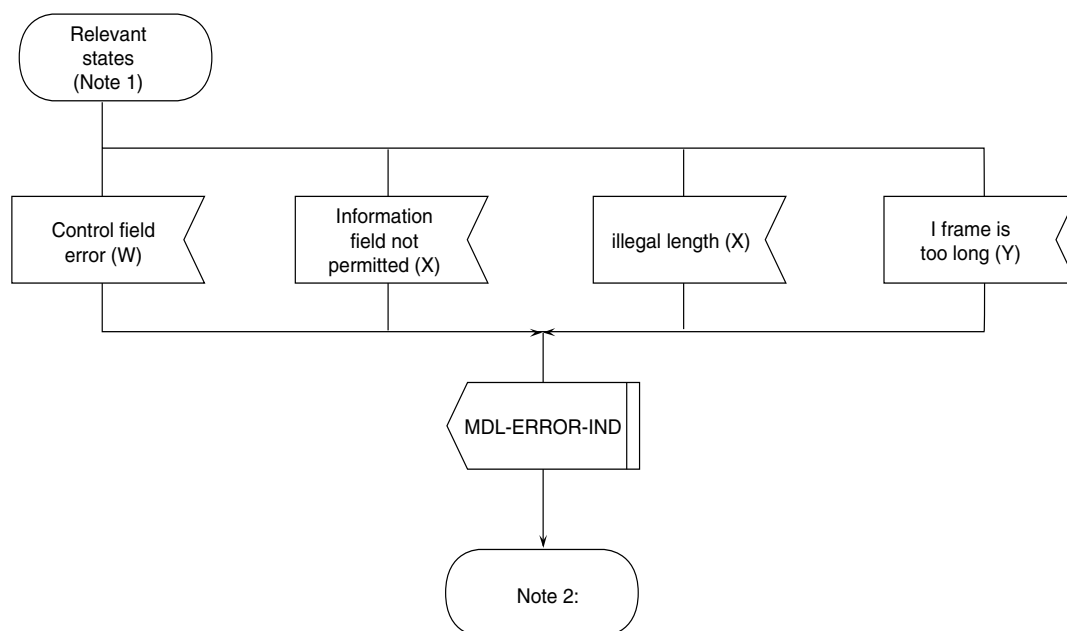
Fig. A. 4.1-34 Layer 2 SDL diagram (33/41)



Note: "Relevant states" are as follows;

- 4. Multiple frame established
- 5. Timer recovery

Fig. A. 4.1-35 Layer 2 SDL diagram (34/41)



Note 1: "Relevant states" are as follows;

1. Multiple frame unestablished
2. Standby for link establishment
3. Stand by for link release

Note 2: The data link layer resets to the state previously set among the above state.

Fig. A. 4.1-36 Layer 2 SDL diagram (35/41)

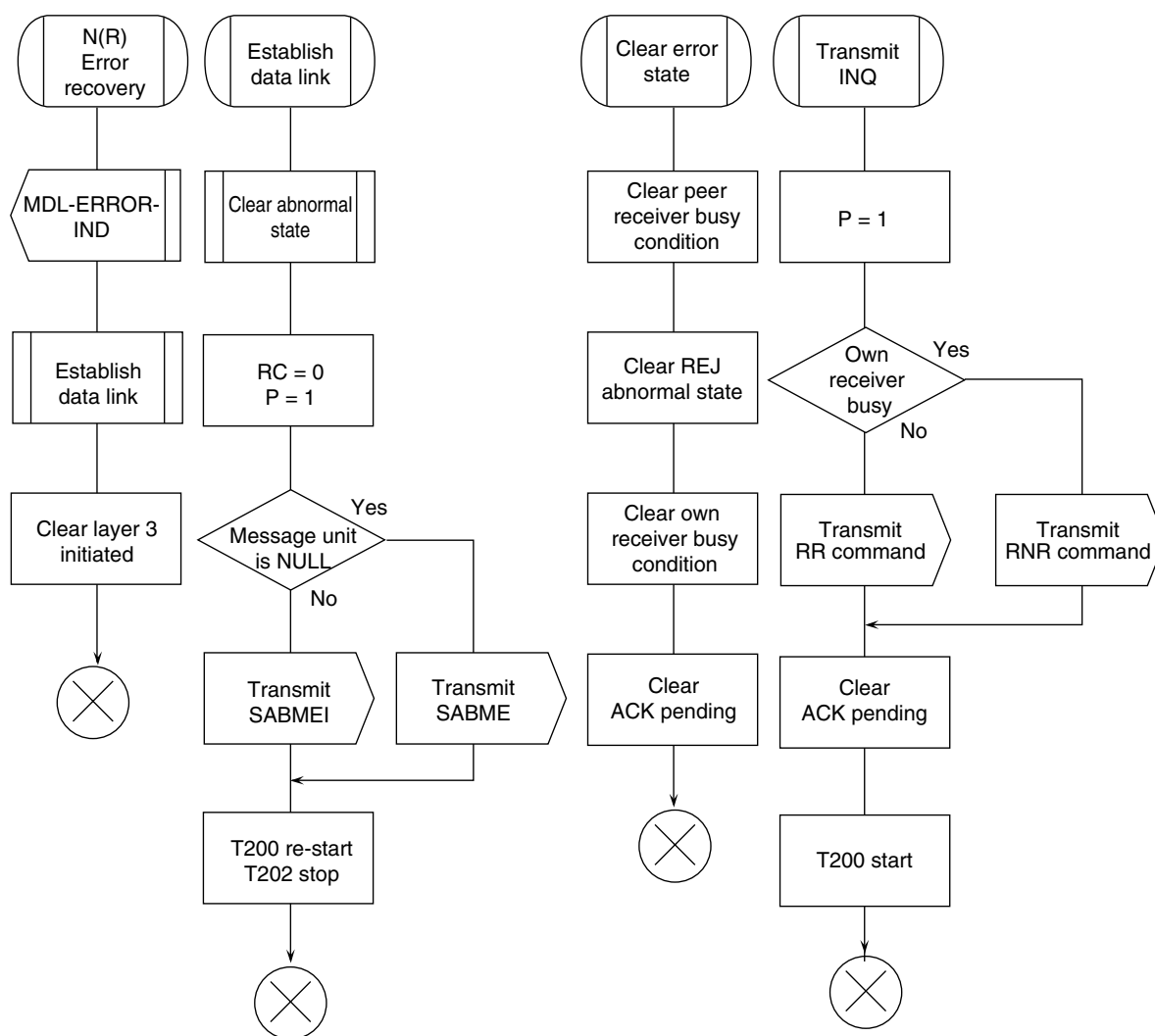


Fig. A. 4.1-37 Layer 2 SDL diagram (36/41)

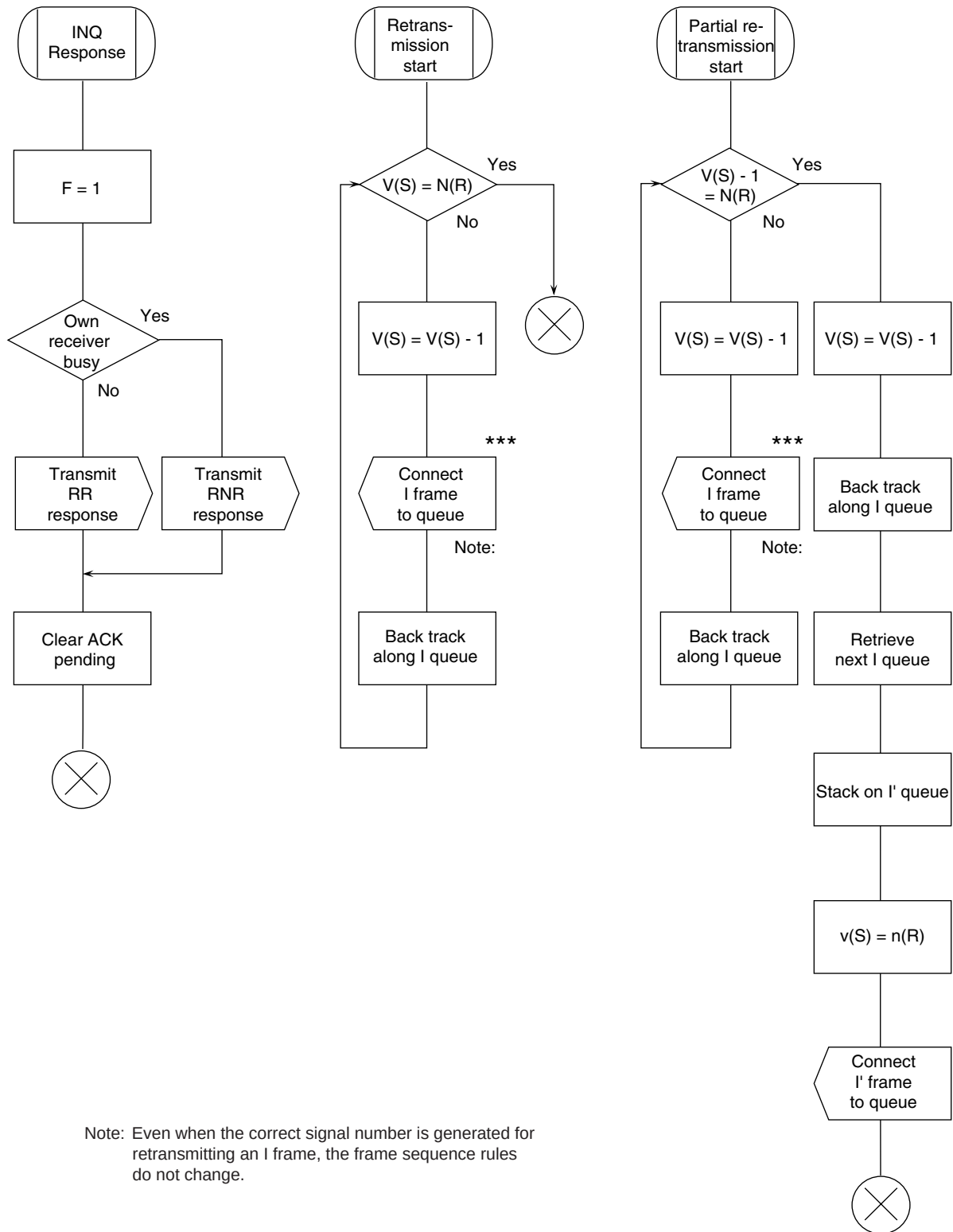


Fig. A. 4.1-38 Layer 2 SDL diagram (37/41)

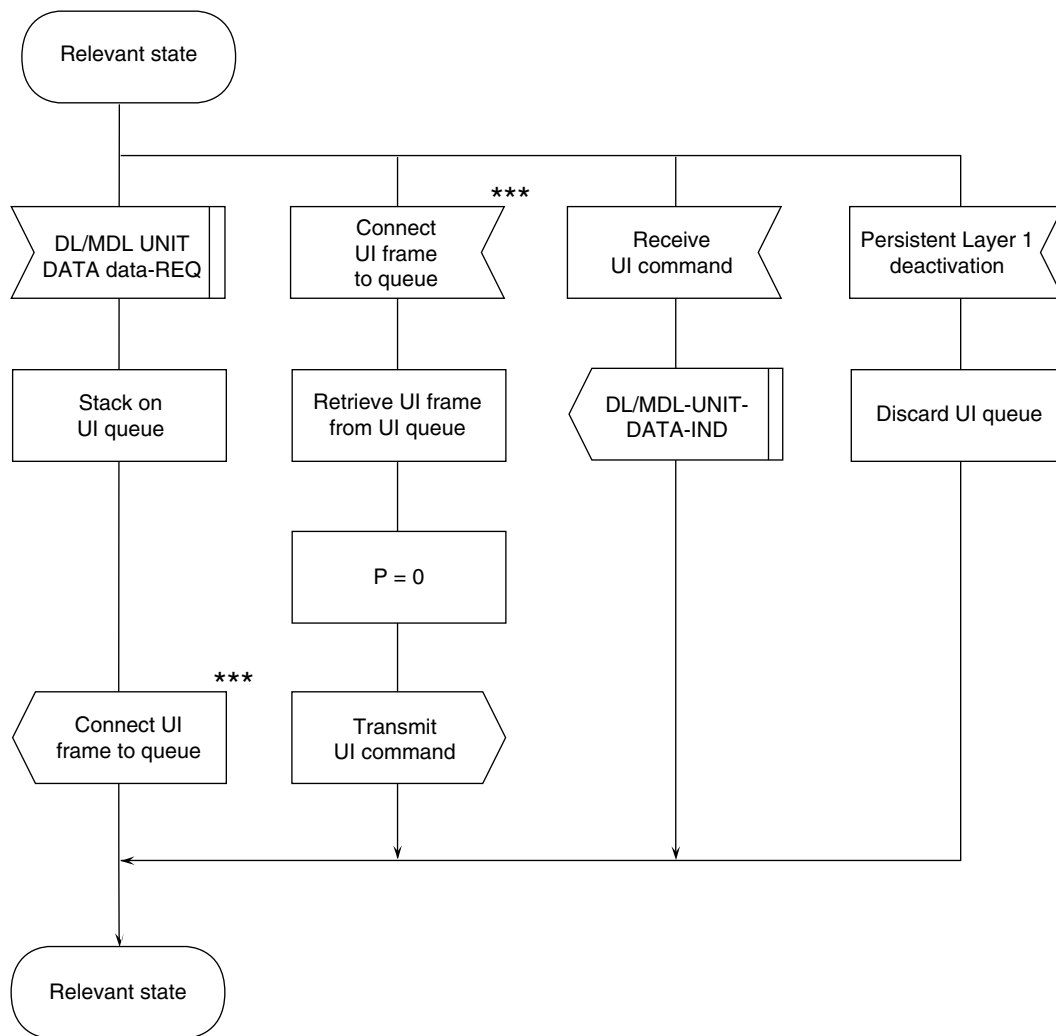
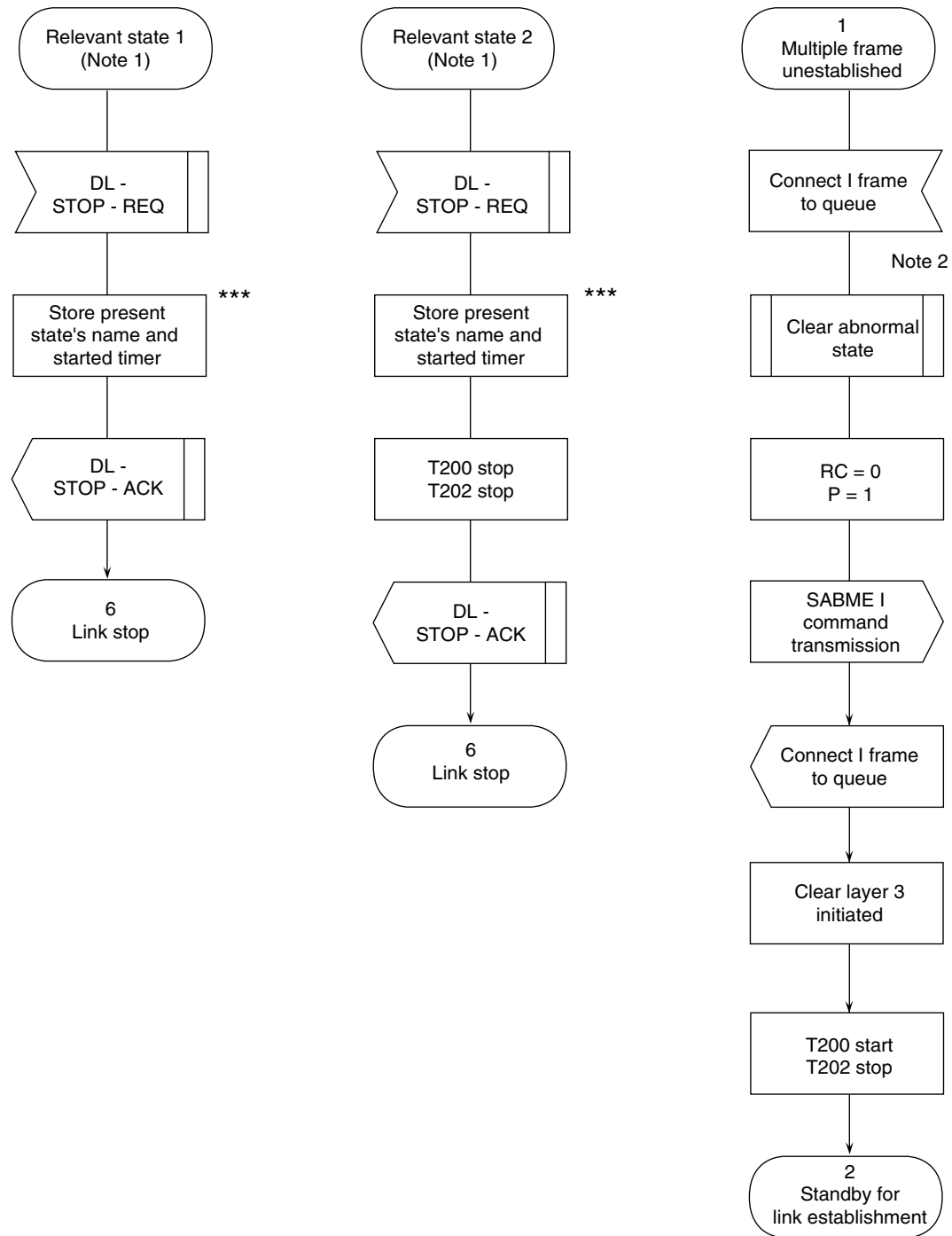


Fig. A. 4.1-39 Layer 2 SDL diagram (38/41)



Note 1: Relevant states are listed below:

- | | |
|-----------------------------------|-------------------------------|
| Relevant states 1 | Relevant states 2 |
| 1. Multiple frame unestablished | 4. Multiple frame established |
| 2. Standby for link establishment | 5. Timer recovery |
| 3. Standby for link release | |

Note 2: Can be generated only when the reestablishment procedure is activated by layer 2.

Fig. A. 4.1-40 Layer 2 SDL diagram (39/41)

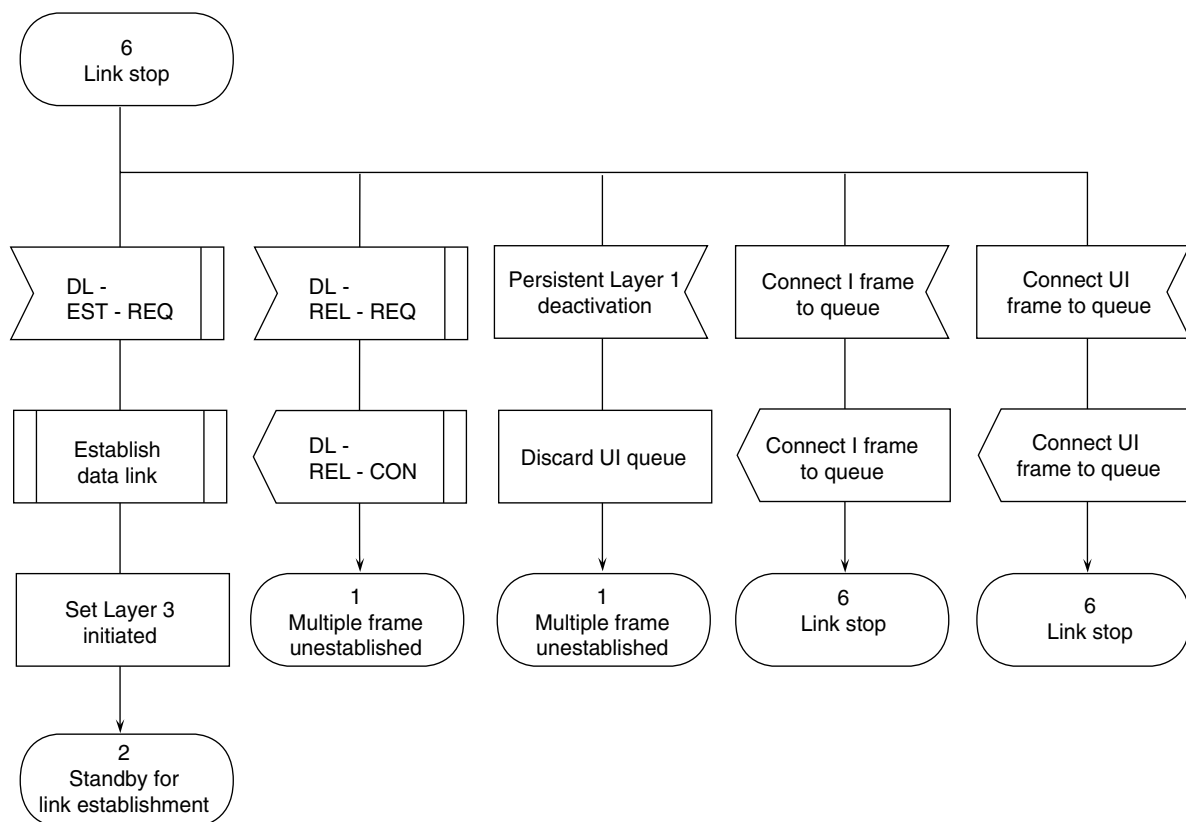
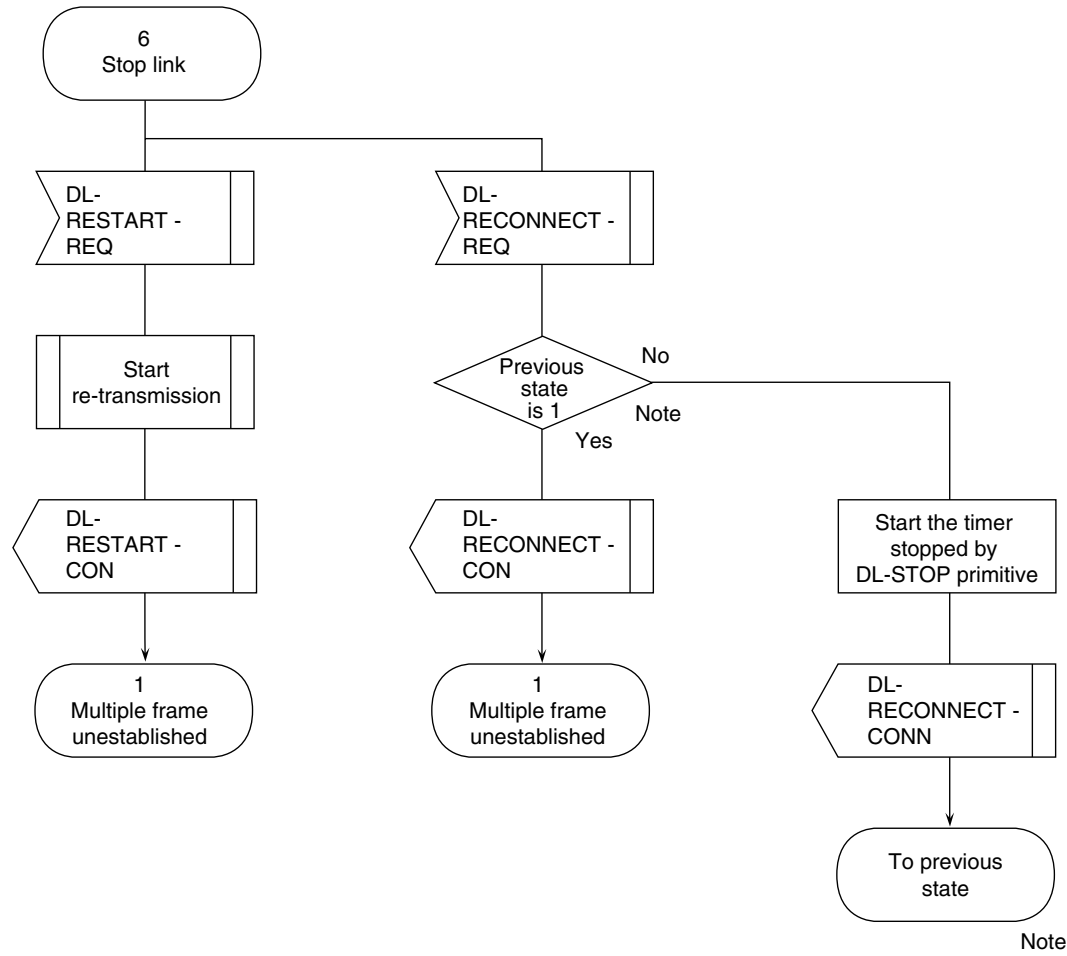


Fig. A. 4.1-41 Layer 2 SDL diagram (40/41)



Note: This "Previous state" means the state just before "Stop link" (State 6).

Fig. A. 4.1-42 Layer 2 SDL diagram (41/41)

A.4.2 Operations on the packet physical channel (SAPI= 1, 6)

A.4.2.1 Outline of the states for the point to point data link layer entity

Six basic states are used in conjunction with the SDL expressions for determining point-to-point procedures. All six states are expanded forms derived from the two basic states described in section 4.2.1.3.5.2

State 1 : Multiple frame unestablished

State 2 : Standby for link establishment

State 3 : Standby for link release

State 4 : Multiple frame established

State 5 : Timer recovery

State 6 : Link stop

The interrelationship between these states is outlined in Fig. A.4.2-43. This diagram is used basically to introduce the SDL expressions and does not show the full interrelationship between states.

All data link layer entities start with the multiple frame unestablished state (State 1). For all of states 1 through 5, the data link layer is capable of providing services directly in response to the unit data request. When the data link layer entity receives the link establish request in the multiple frame unestablished state (State 1), the link establishing procedure is initiated and the data link layer entity switches to the standby state for the link establishment state (State 2). The data link layer entity switches to the multiple frame established state (State 4) on completion of the LAP establishment procedure. The data link layer entity on the other side switches from the multiple frame unestablished state (State 1) directly to the multiple frame established state (State 4).

During the multiple frame established state (State 4), the data link layer entity can provide the required service in response to an acknowledged type data transfer request under the conditions applying to the procedure in progress. When timer T200, which is used for both flow control and the data transfer in the data link layer entity procedure expires, switching to the timer recovery state (State 5) begins. When the timer recovery procedure is completed, the data link layer entity returns to the multiple frame established state (State 4).

During States 4 and 5 for the SDL expressions, the following conditions, all of which are defined by the Standard, can be generated.

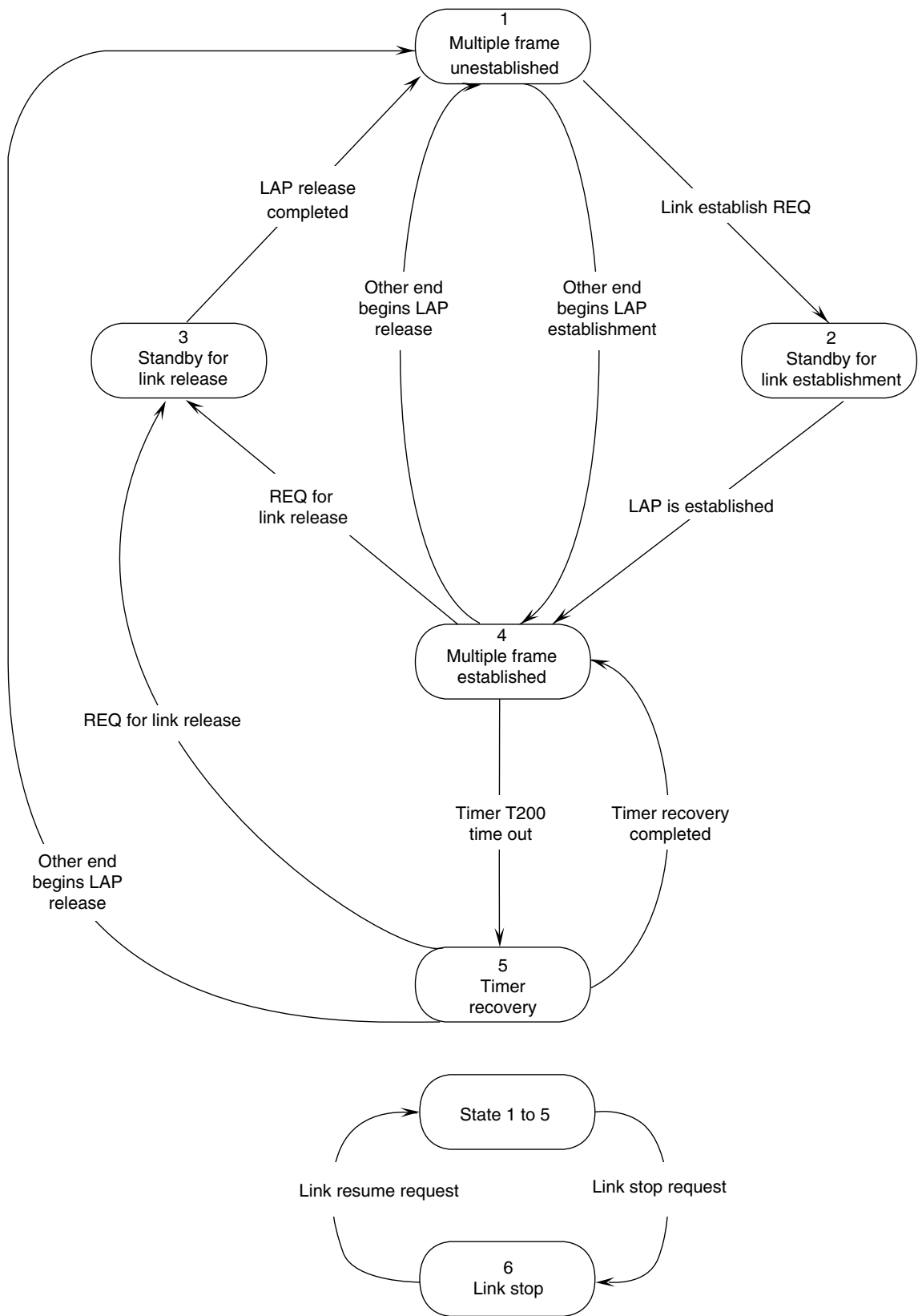


Fig. A.4.2-1 Outline of point-to-point procedure states (SAPI = 1, 6)

- a) Peer receiver busy
- b) SREJ exception and SREJ' exception
- c) Own receiver busy

In order to avoid the need to define additional states, several other conditions are used. Combining the five states for use in SDL expressions with the conditions for the above categories serves as the basis for state transition expressions for the data link layer entities. When the data link layer entity on the other side begins LAP release, the data link layer entity switches directly to the multiple frame unestablished state (State 1). When a link release is requested, the data link layer entity switches to the standby for link release state (State 3) and then to State 1.

When a link stop request is received in any state from 1 through 5, the link stop procedure is performed and the state shifts to the link stop state (State 6).

In the link stop state (State 6) the multiple frame operation is halted. When a link restart request is received, the link restart process starts, and the state shifts to the multiple frame unestablished state. When a link reconnect request is received, the link reconnect procedure starts, and the state returns to the state before the link stop state. When link resume request is received, the link resume procedure starts and the user returns to the state before Link stop.

A.4.2.2 Using queues

A.4.2.2.1 Transmission queue

To provide an understandable description of the data link layer entity, the conceptual queues for transferring UI, I and I' frames are clearly described. Usage of these conceptual queues is limited, but their depth is not limited; also, they do not limit implementation of point-to-point procedures. Three additional signals (i.e., Connect UI frame to queue, Connect I frame to queue and Connect I' frame to queue) are provided for activating the queues.

These services are provided by the queue manager. The queue manager must control to assign priority according to Section 4.2.1.4.2.2.

A.4.2.2.2 Reception queue

A conceptual reception queue is provided in order to manage I and I' frames which have already been received when the SREJ and SREJ' control is performed.

Usage of this conceptual reception queue is limited, but the depth is not limited; also it does not limit implementation of point-to-point procedures.

Two additional signals are defined as below:

"Connect to the receive queue" means that if the frames to be connected are I frames, they are connected so that they are arranged in the order of $V(R)$ to $V(R)+k$. If the frames to be connected are I' frames, it means to fill those units which have not been received in the already connected I frames with units contained in the I' frames.

"Retrieve from receive queue" refers to retrieve an I frame from the top of the receive queue.

A.4.2.2.3 SREJ queue

A conceptual SREJ queue is provided to manage SREJ error state and SREJ' error state of the frames which have already been transmitted when the SREJ and SREJ' control is performed.

Usage of this conceptual SREJ queue is limited, but the depth is not limited; also it does not limit implementation of point-to-point procedures. This queue manages the following information items as a set:

- Type of error state which indicates whether the state is the SREJ error state or the SREJ' error state.
- The sequence number requested for retransmission contained in the SREJ or SREJ' frame.
- Divided receive state variable $v(R)$ for the SREJ' error state.

Two additional signals are defined as below:

"Connect to SREJ queue" refers to connecting the SREJ error state or SREJ' error state and the receive sequence number $N(R)$ which was activated by the retransmission request using the SREJ or SREJ' frame to the end of the SREJ queue.

"Retrieve from SREJ queue" refers to retrieving the error state type and receive sequence number $N(R)$ from the top of the SREJ queue.

A.4.2.3 Grouping of unit numbers

To simplify description of partial retransmission control, the variables used for partial retransmission control in the SDL diagrams are expressed for groups as shown below.

Send divided unit sequence number

$$n(S) = \{n1(S), n2(S), \dots, nj(S), ni(S) < nj(S), i < j\}$$

Receive divided unit sequence number

$$n(R) = \{n1(R), n2(R), \dots, nj(R), ni(R) < nj(R), i < j\}$$

Divided send state variable

$$v(S) = \{v1(S), v2(S), \dots, vi(S)\}$$

Divided receive state variable

$$v(R) = \{v1(R), v2(R), \dots, vi(R)\}$$

These groups are managed for each outstanding sending frame.

The send divided unit sequence number $n(S)$ matching the divided receive state variable $v(R)$ among others is a correct $n(S)$. "Discard illegal $n(S)$ " refers to discarding the $n(S)$ and the information portions

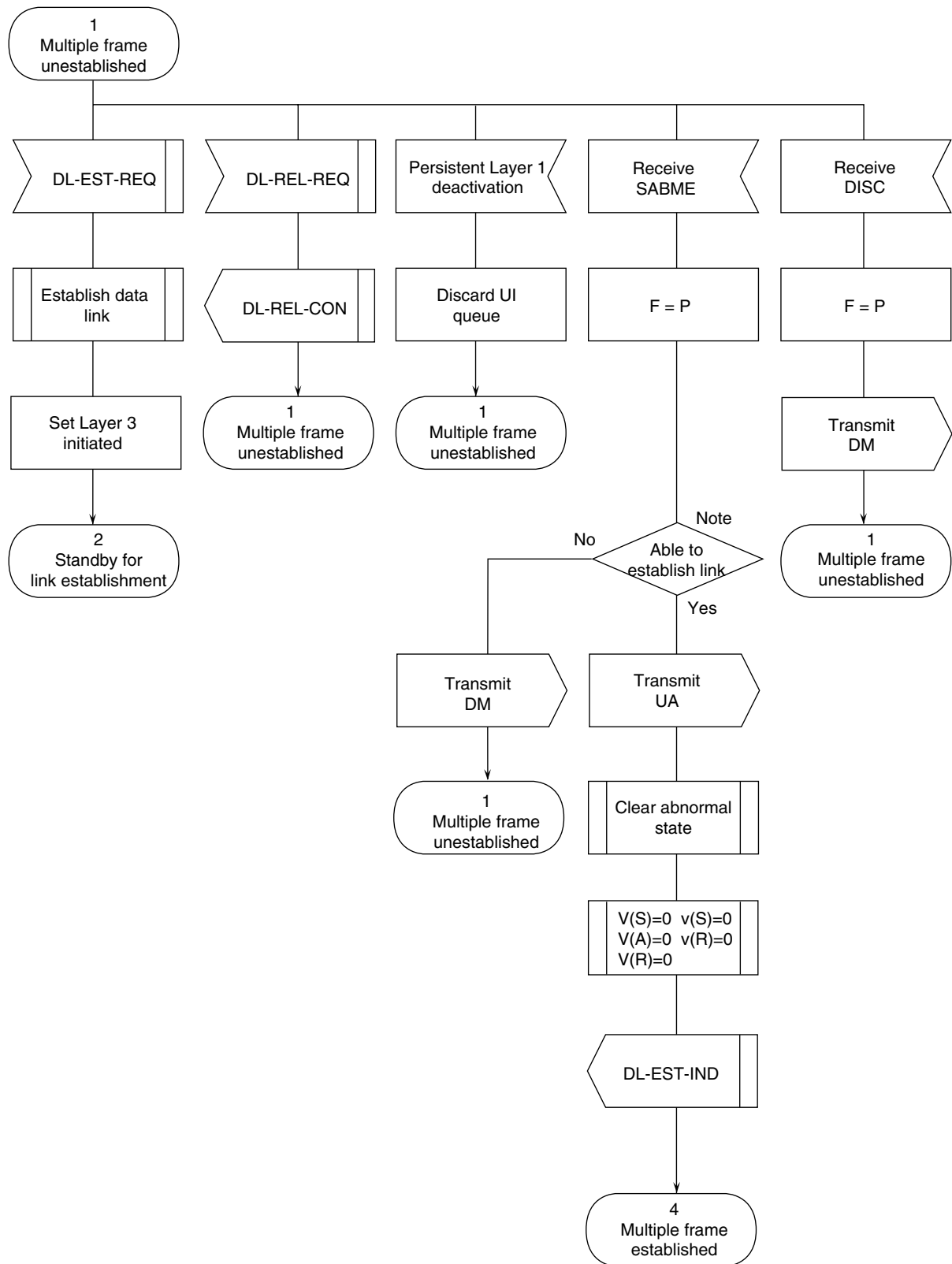
which are not matching $v(R)$. Also, "Update $v(R)$ " refers to removing $n(S)$ which has been successfully received from the $v(R)$.

A.4.2.4 Largest received sequence number

The largest received sequence number is provided to manage received I and I' frames when the SREJ and SREJ' control is performed.

The largest received sequence number indicates the largest sequence number which has already been received from among $V(R)$ to $V(R) + k$ when the reception frames exist within the receive queue.

This number is not defined when no reception frame exists in the receive queue.



Note : Whether or not data link can be set is decided depending on whether or not the data link layer entity can receive/transmit messages and primitives normally with the L3, L1 and management entity.

Fig. A. 4.2-2 Layer 2 SDL diagram (1/58)

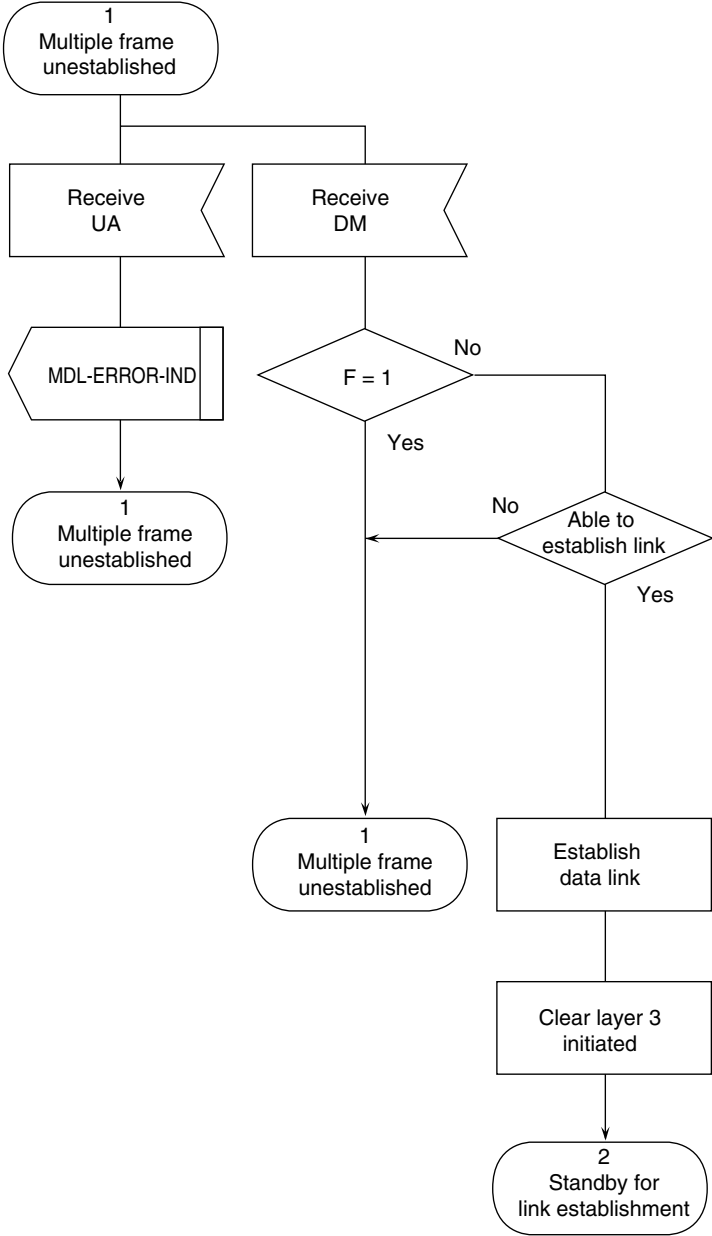
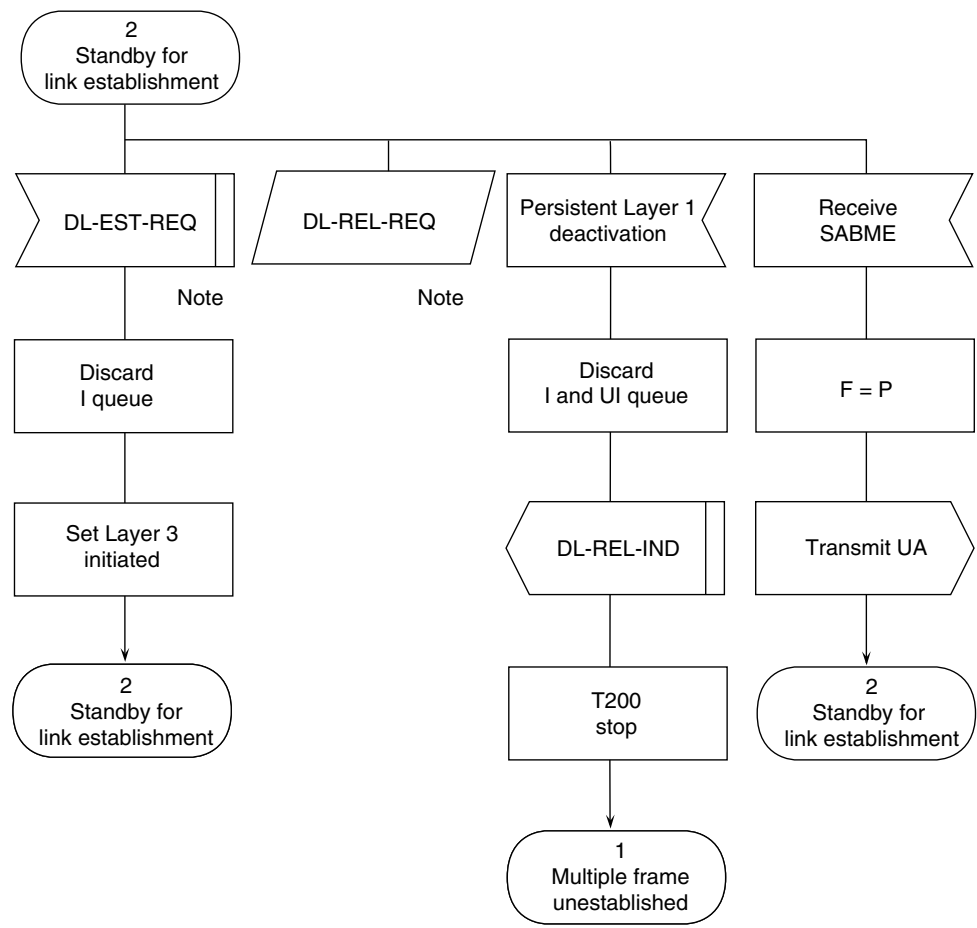


Fig. A. 4.2-3 Layer 2 SDL diagram (2/58)



Note : Can be generated only when the re-establishment procedure is initiated by layer 3

Fig. A. 4.2-4 Layer 2 SDL diagram (3/58)

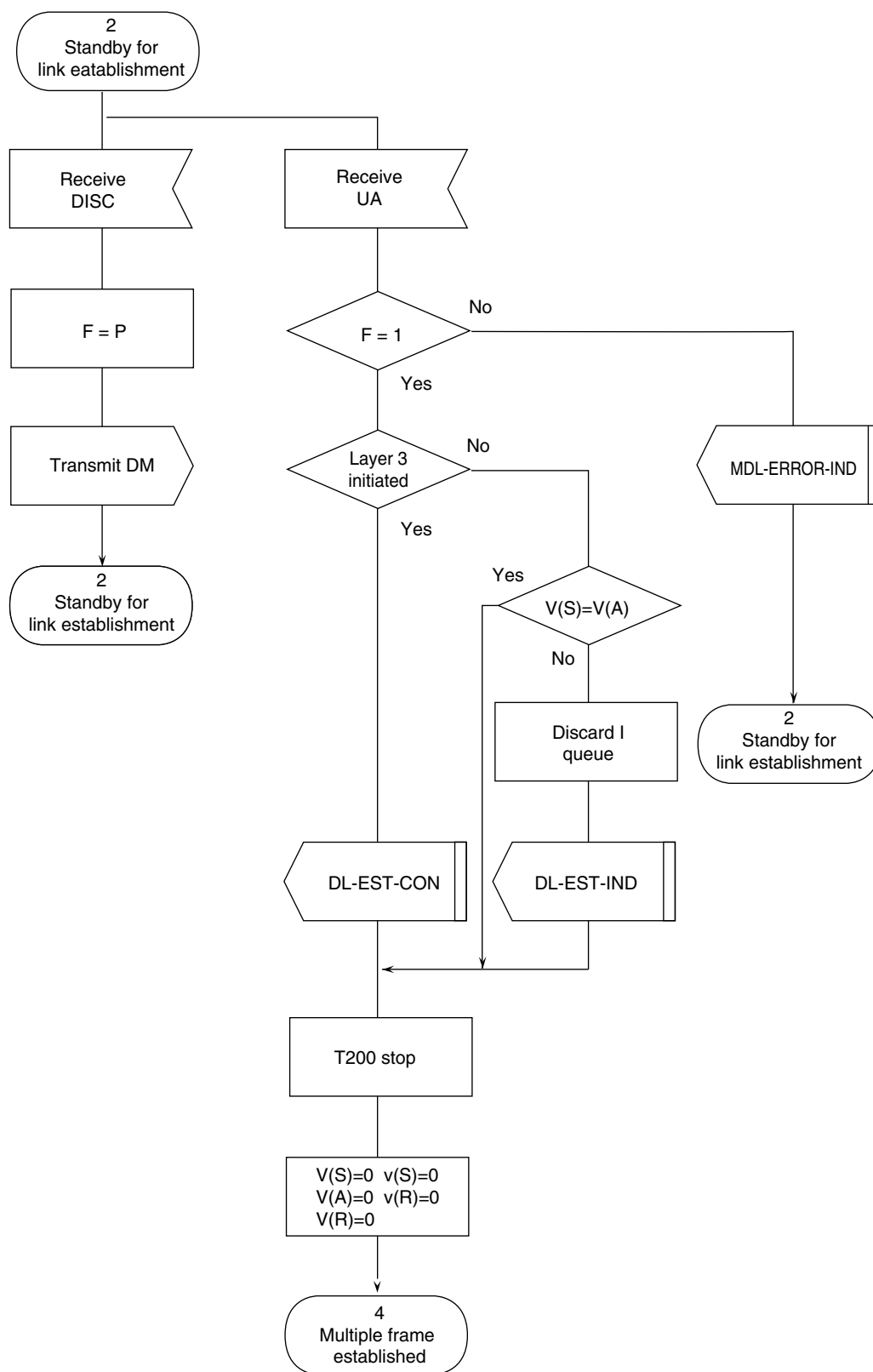
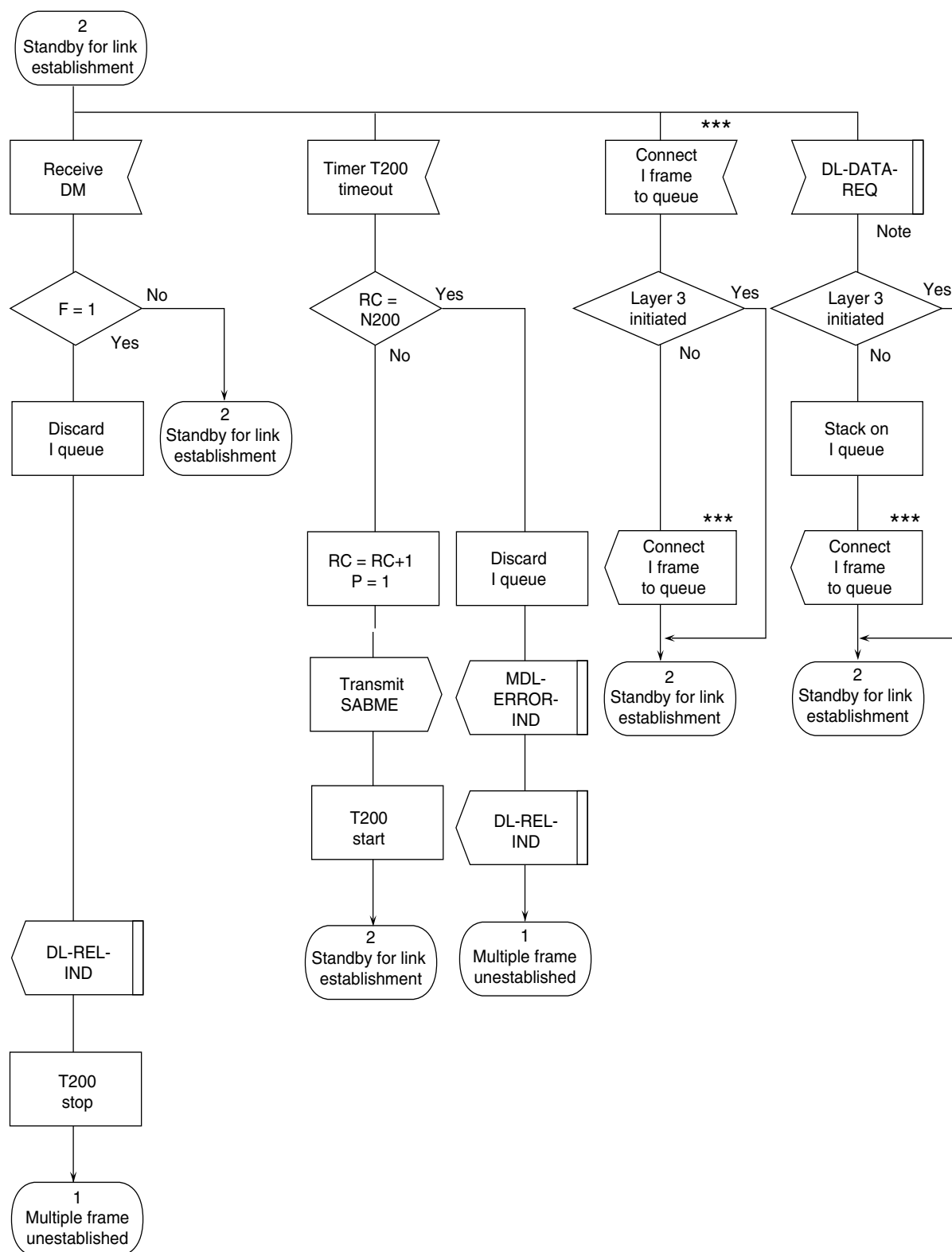


Fig. A. 4.2-5 Layer 2 SDL diagram (4/58)



Note : Can be generated only when the re-establishment procedure is initiated by layer 3

Fig. A. 4.2-6 Layer 2 SDL diagram (5/58)

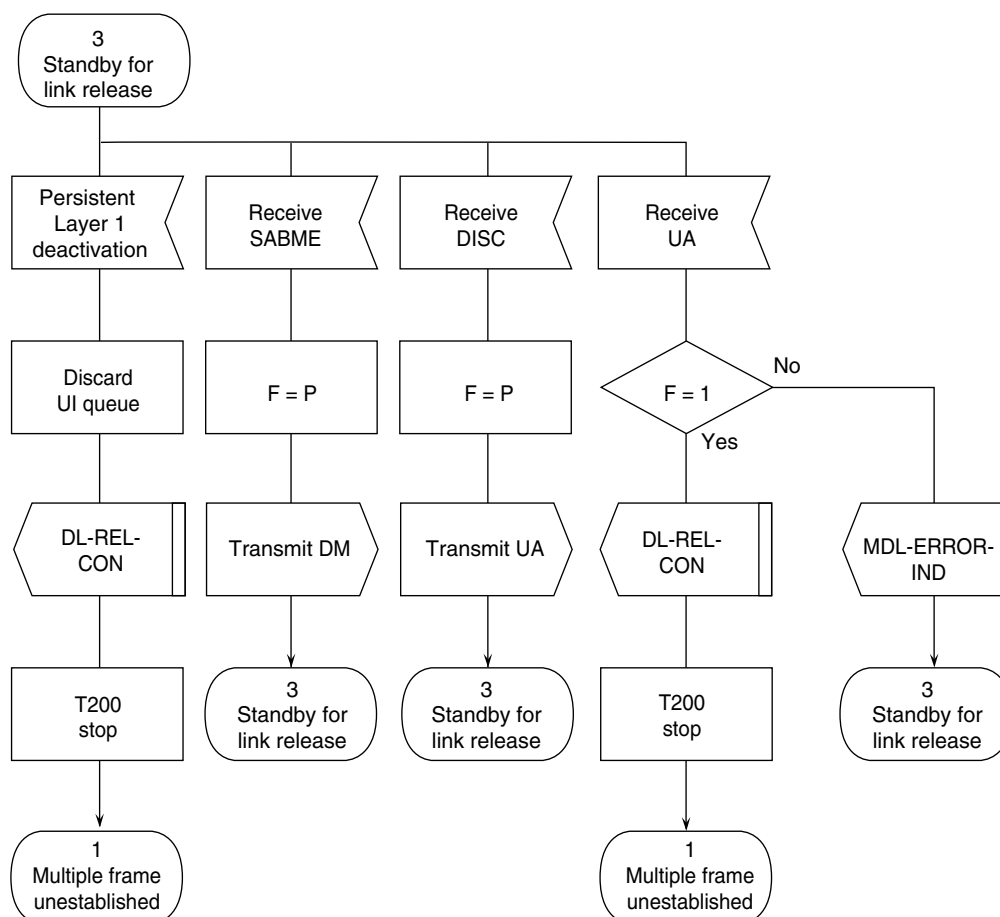


Fig. A. 4.2-7 Layer 2 SDL diagram (6/58)

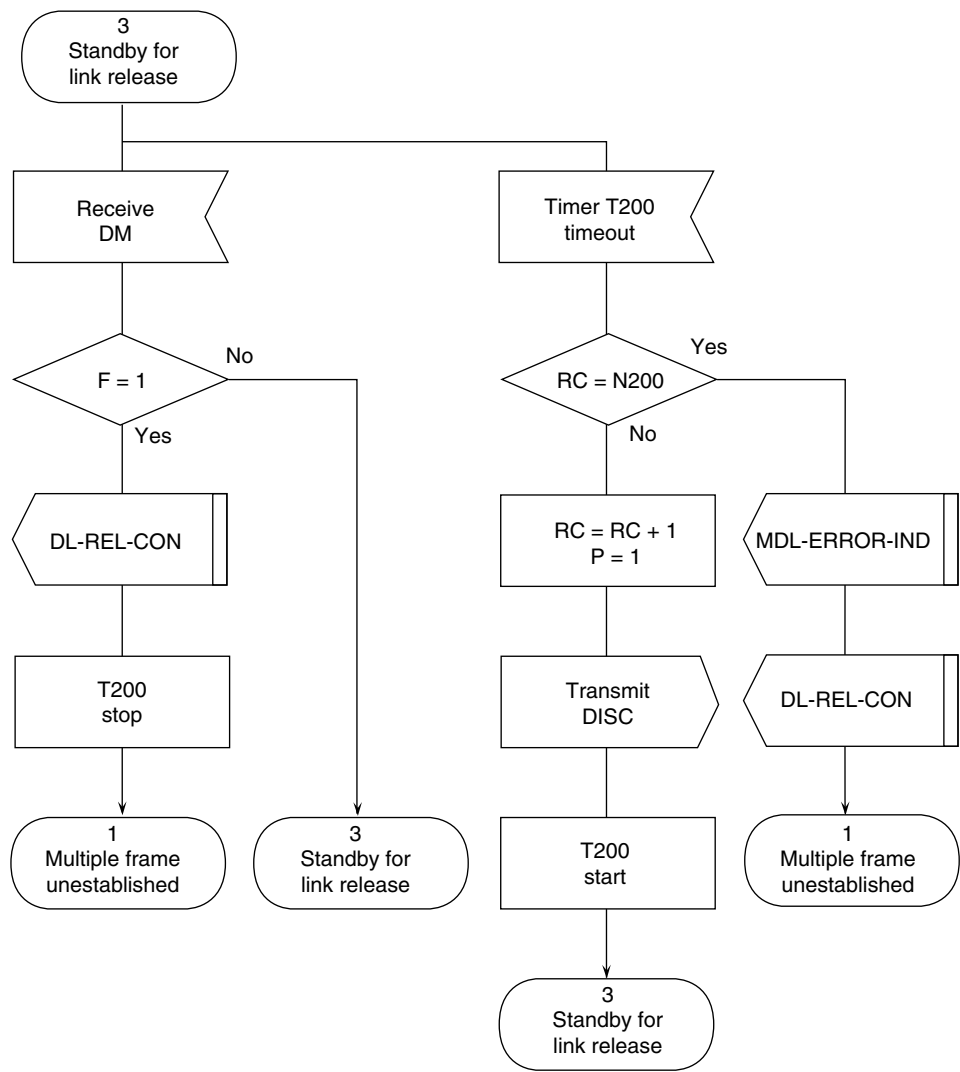


Fig. A. 4.2-8 Layer 2 SDL diagram (7/58)

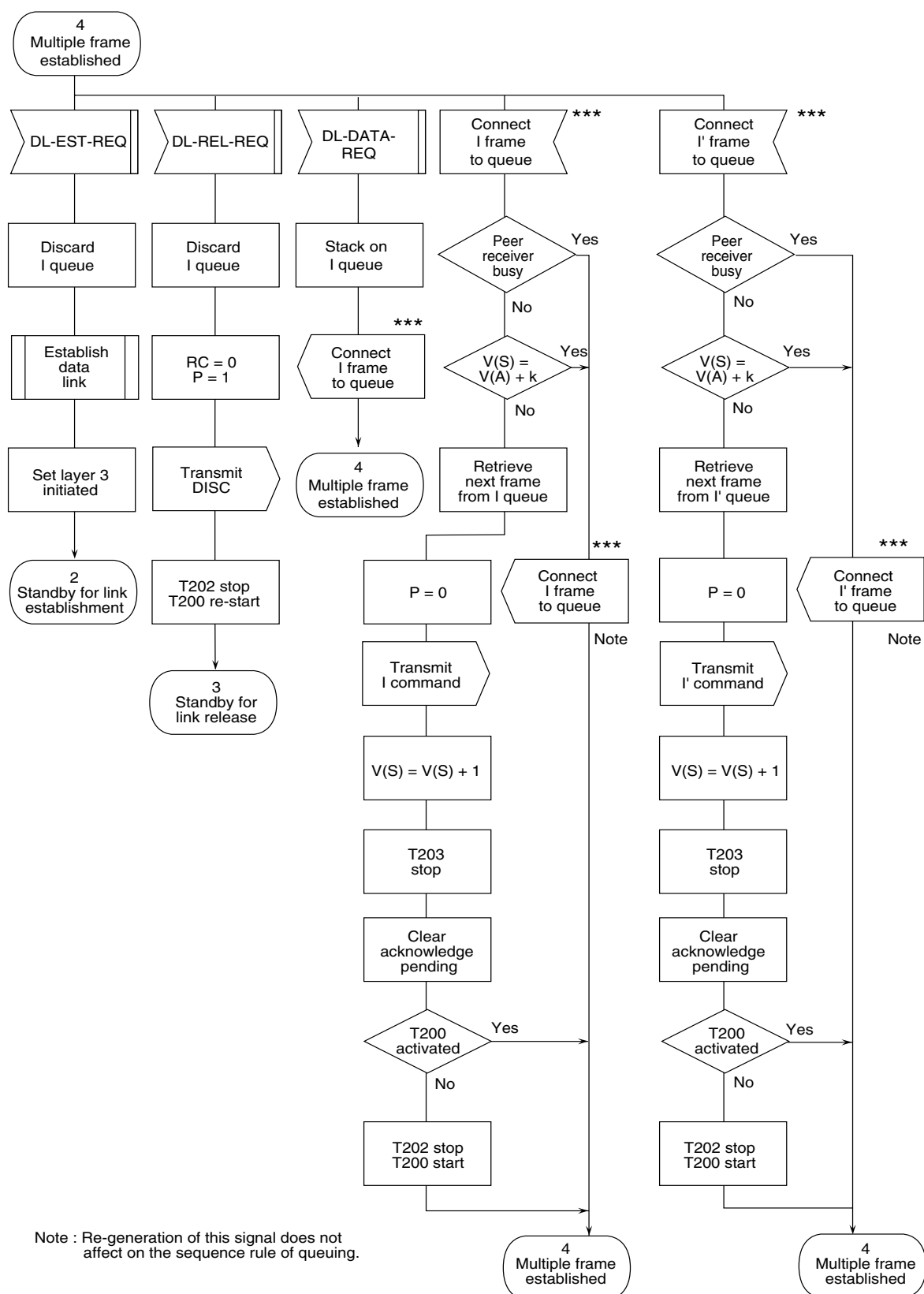


Fig. A. 4.2-9 Layer 2 SDL diagram (8/58)

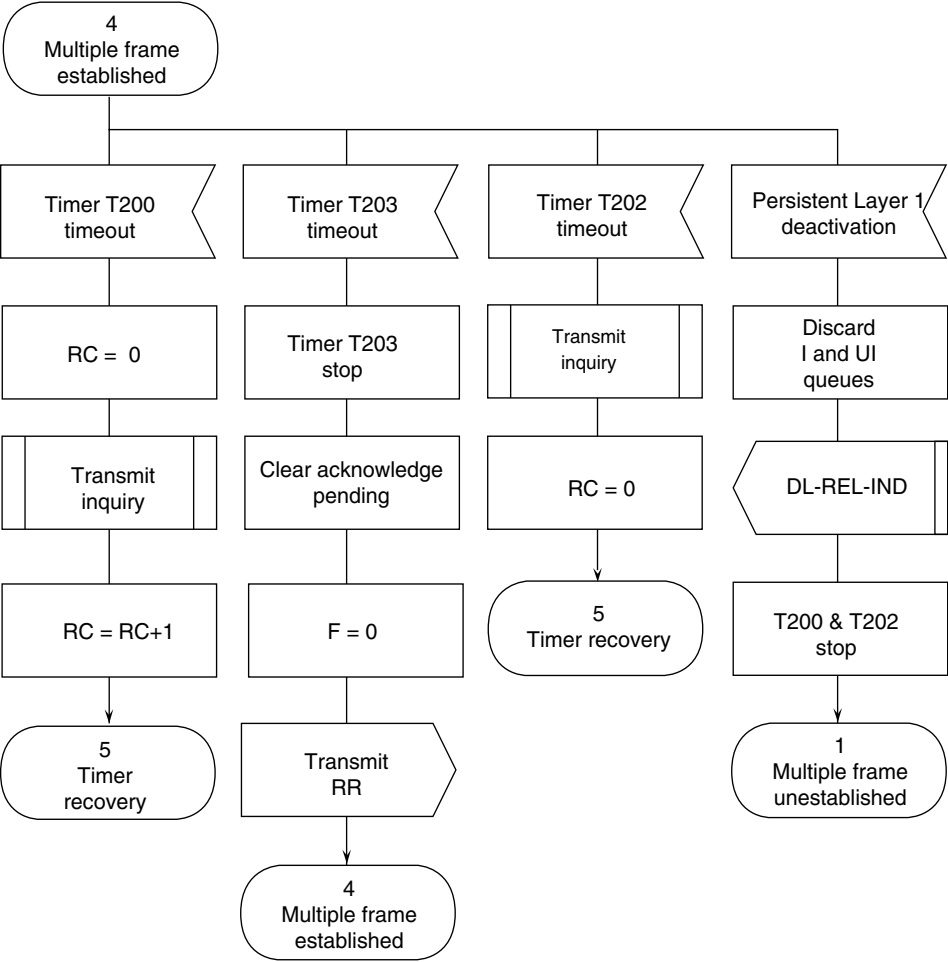


Fig. A. 4.2-10 Layer 2 SDL diagram (9/58)

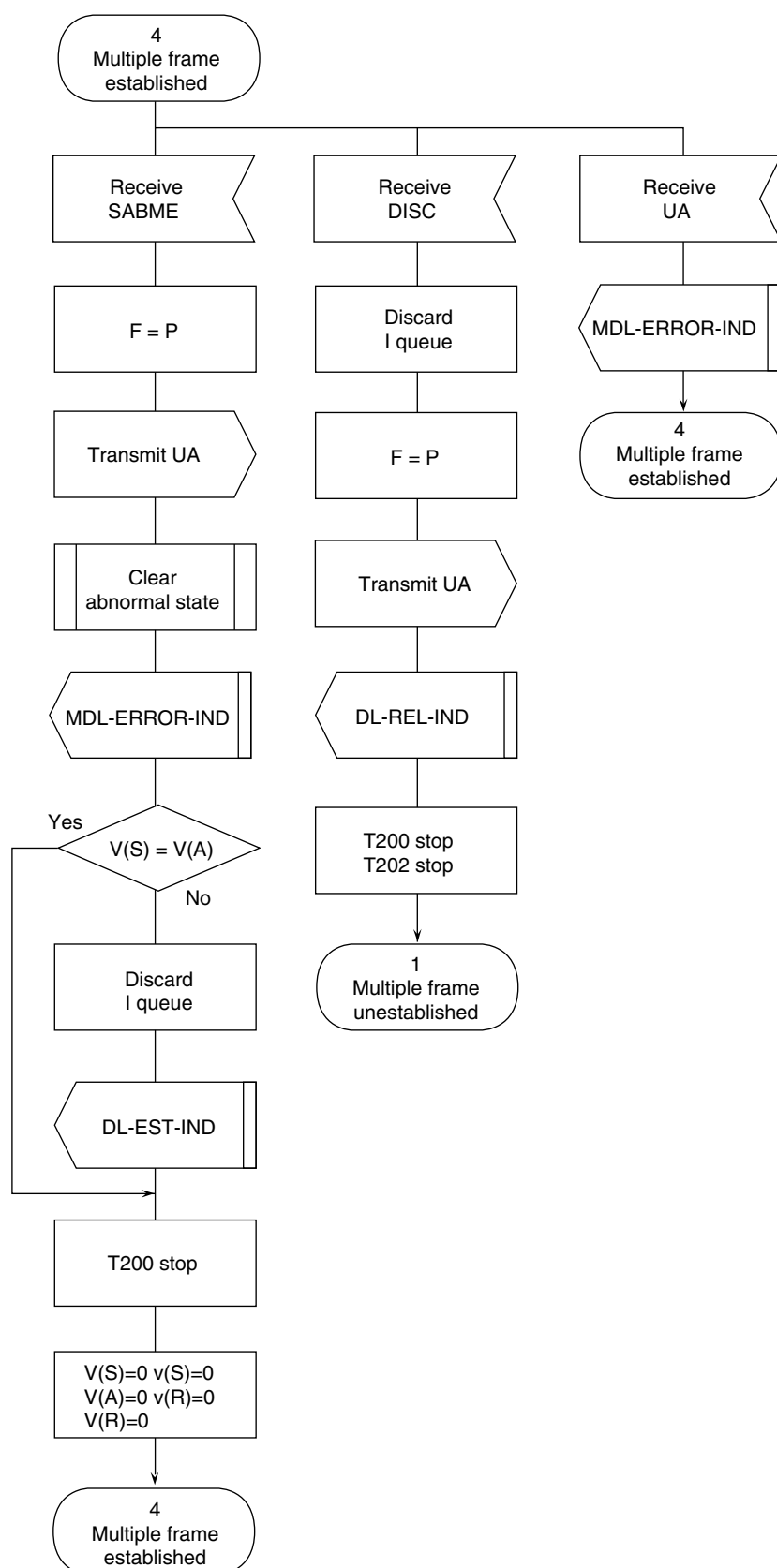


Fig. A. 4.2-11 Layer 2 SDL diagram (10/58)

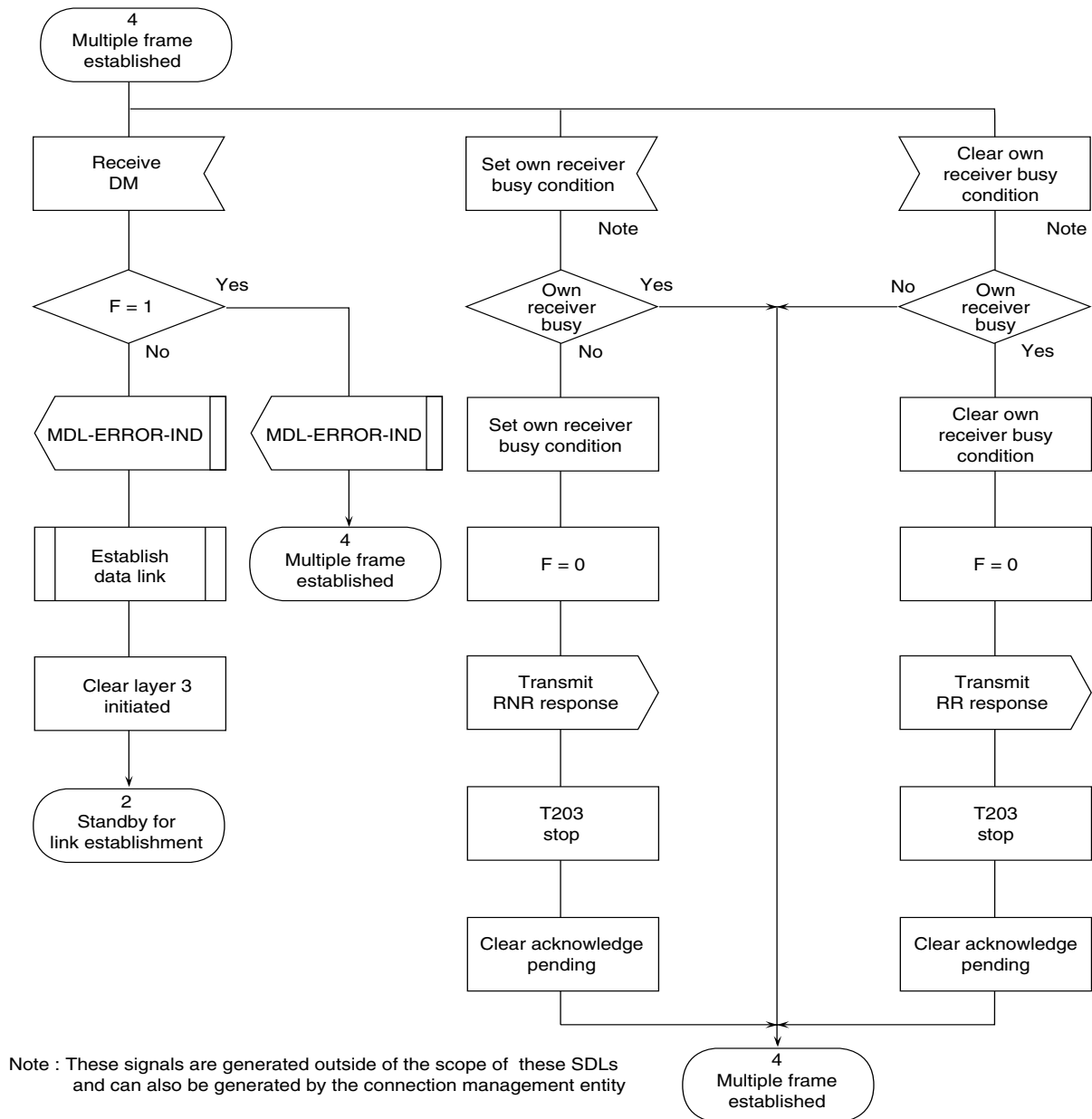


Fig. A. 4.2-12 Layer 2 SDL diagram (11/58)

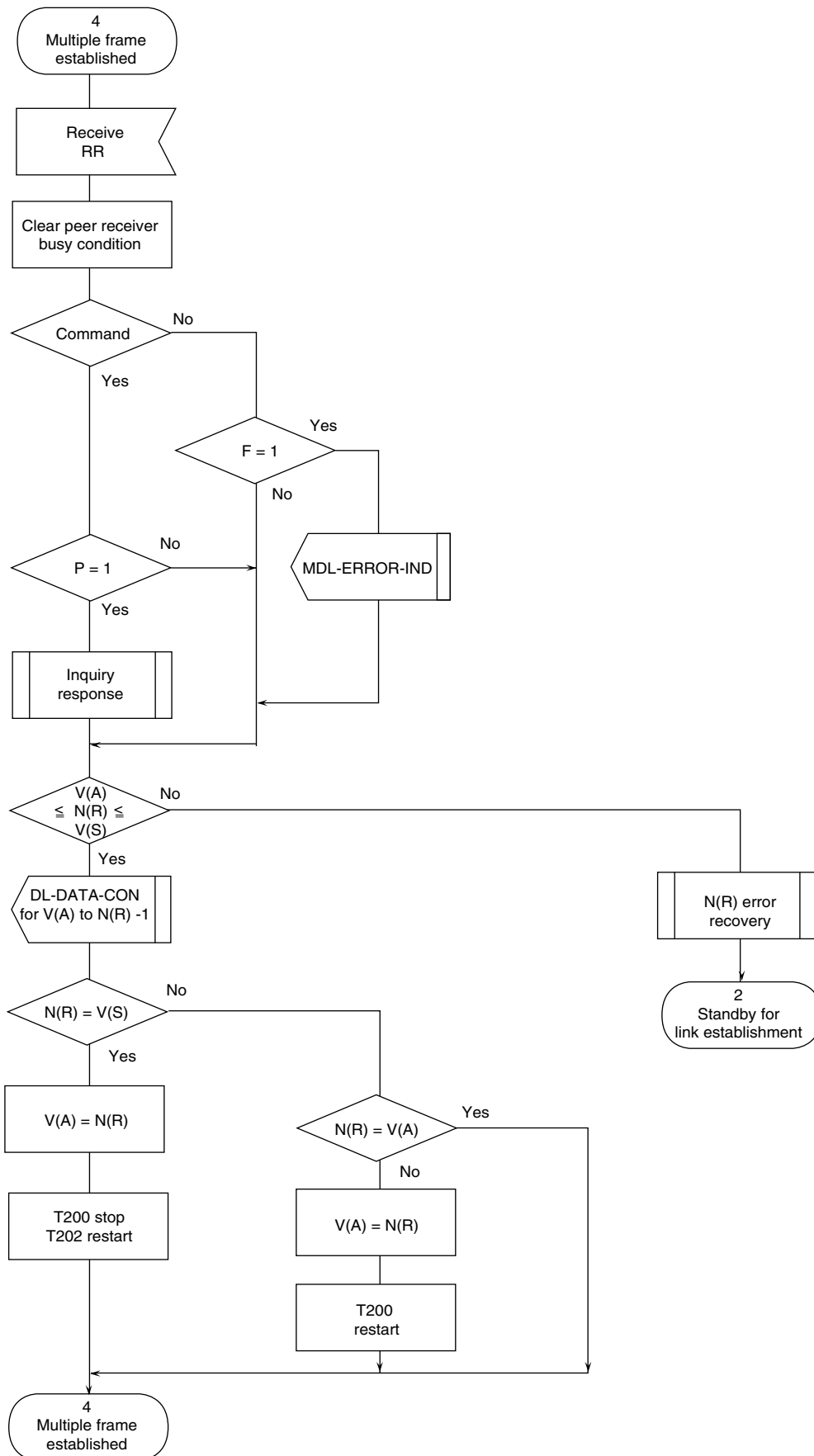
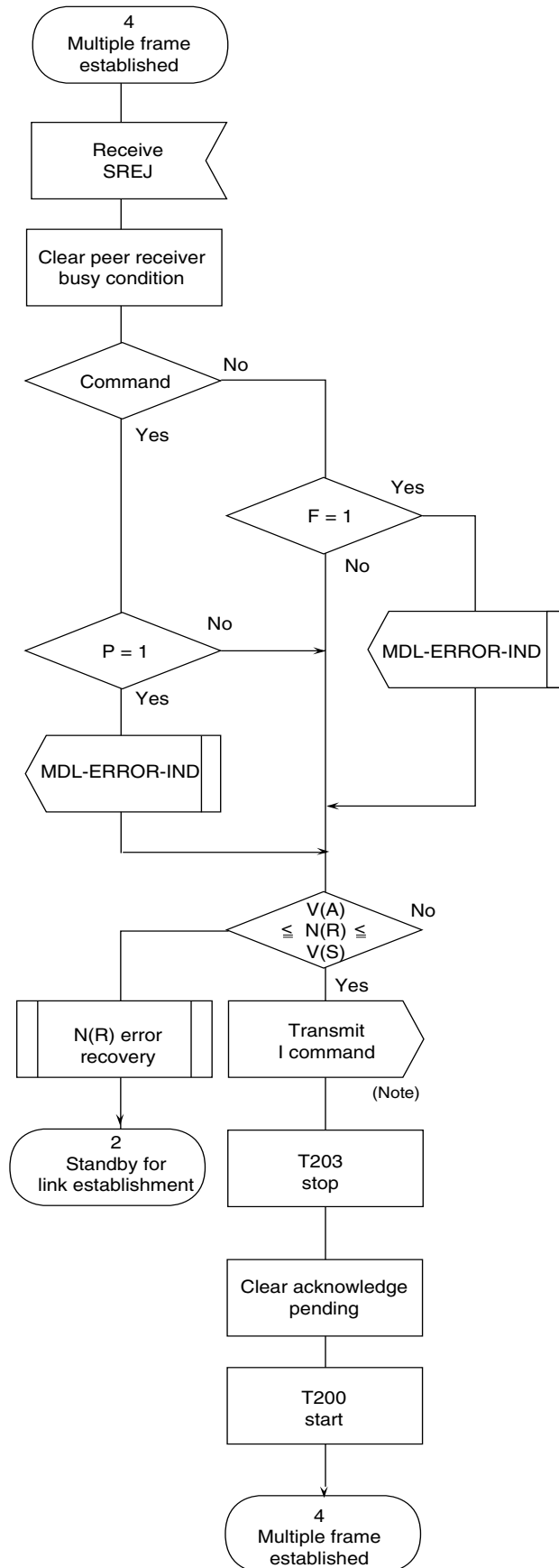


Fig. A. 4.2-13 Layer 2 SDL diagram (12/58)



Note : I frame designated by N(R) of SREJ is transmitted.

Fig. A. 4.2-14 Layer 2 SDL diagram (13/58)

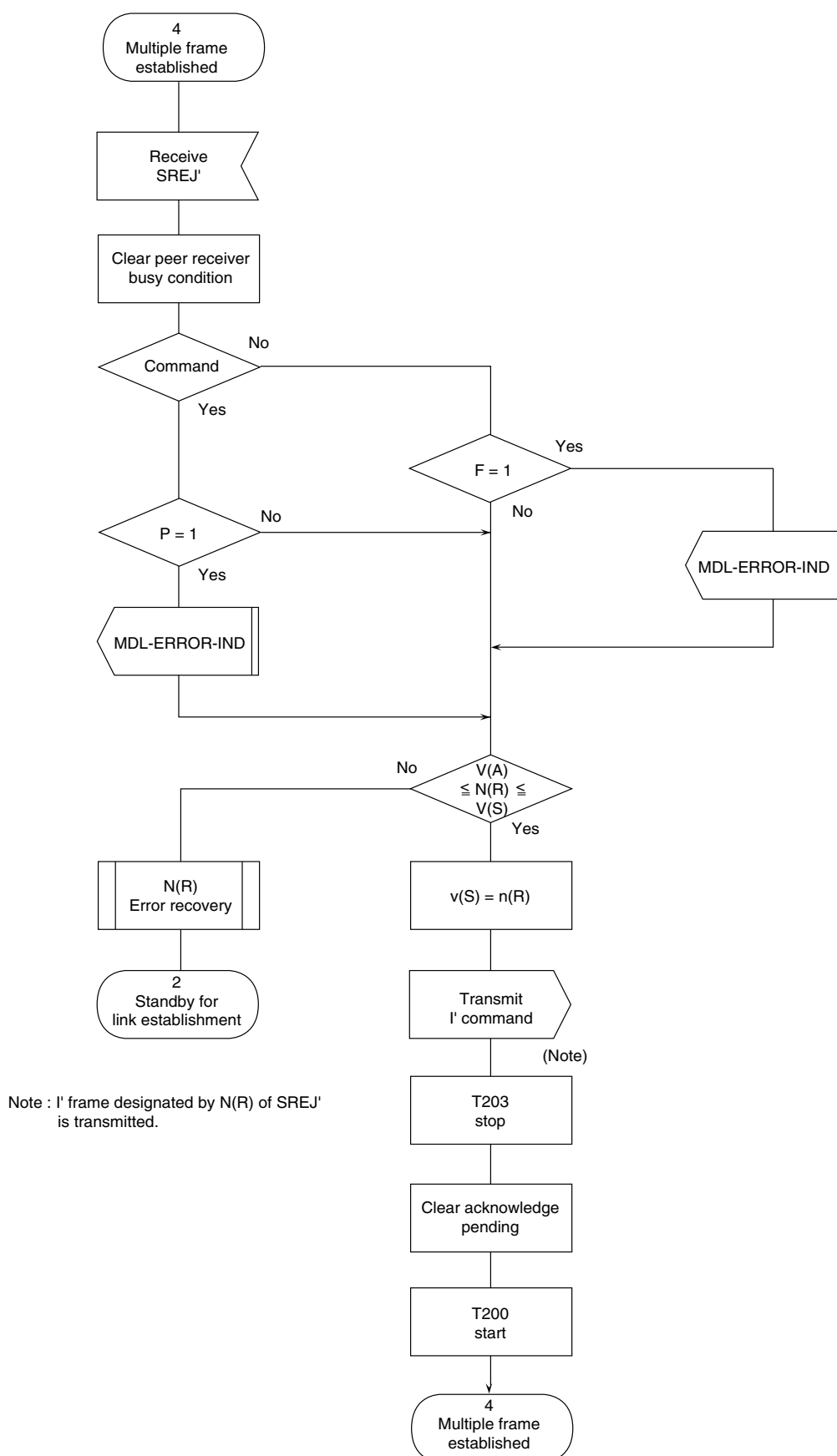


Fig. A. 4.2-15 Layer 2 SDL diagram (14/58)

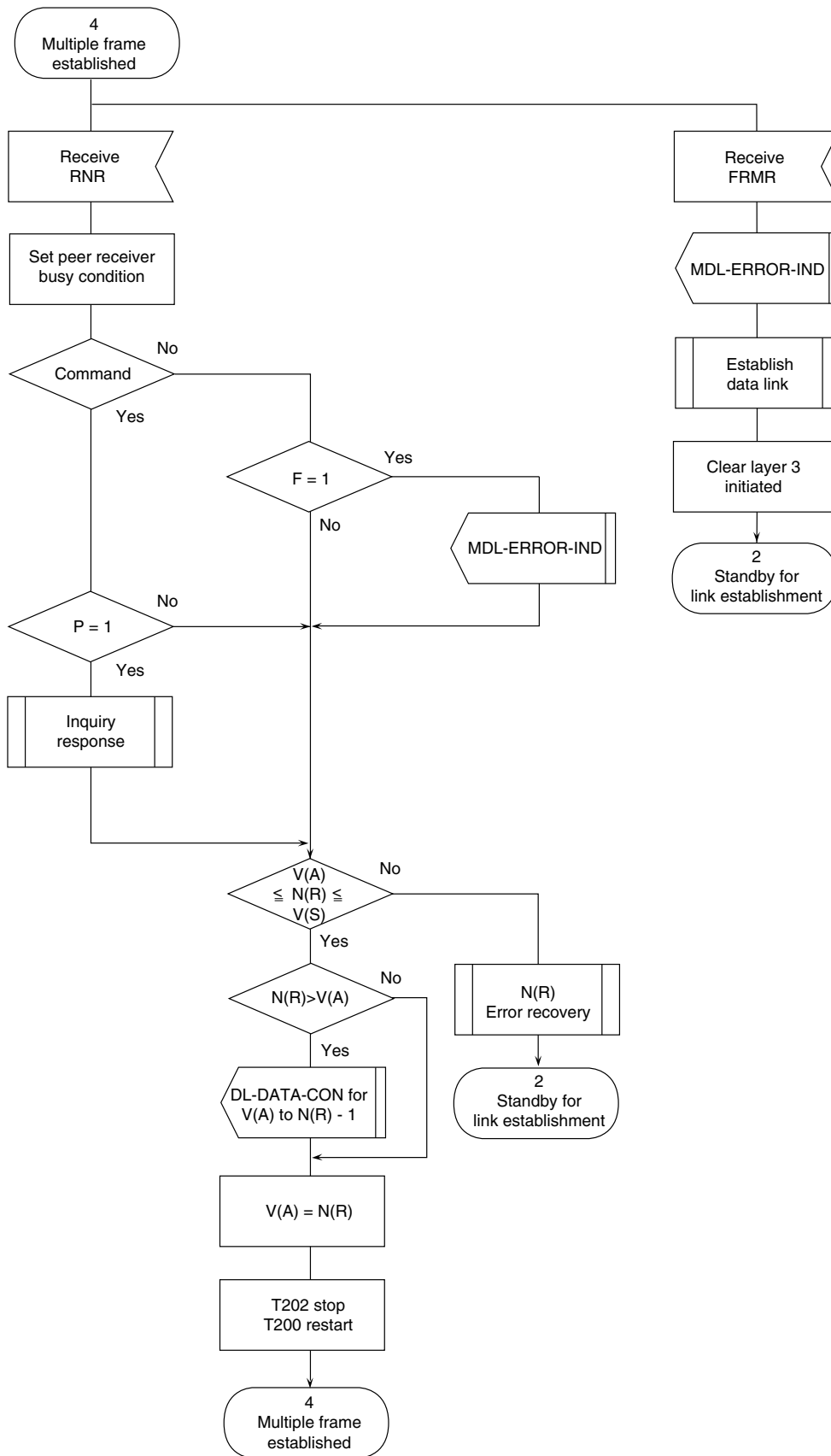


Fig. A. 4.2-16 Layer 2 SDL diagram (15/58)

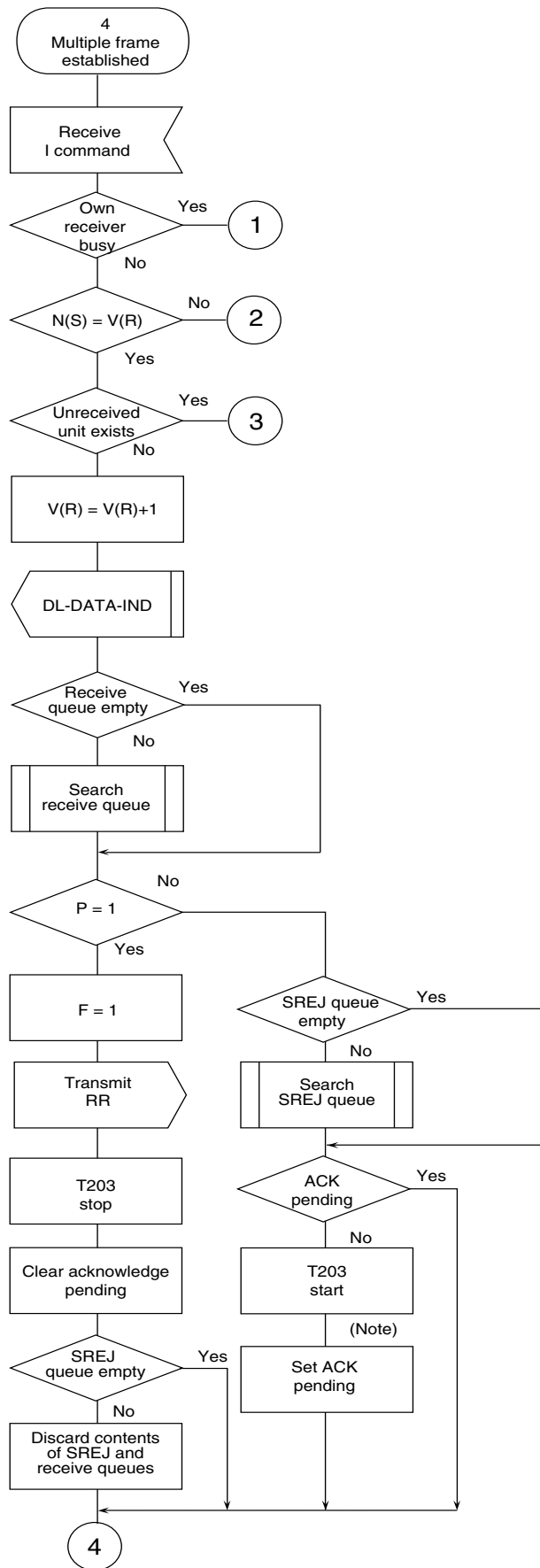


Fig. A. 4.2-17 Layer 2 SDL diagram (16/58)

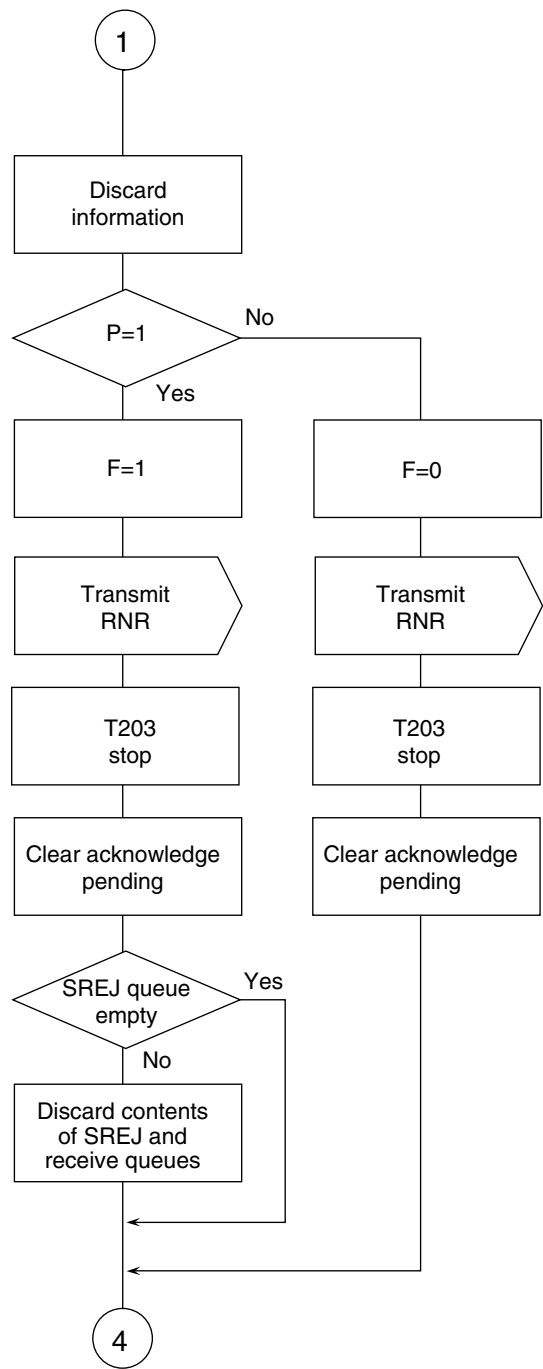
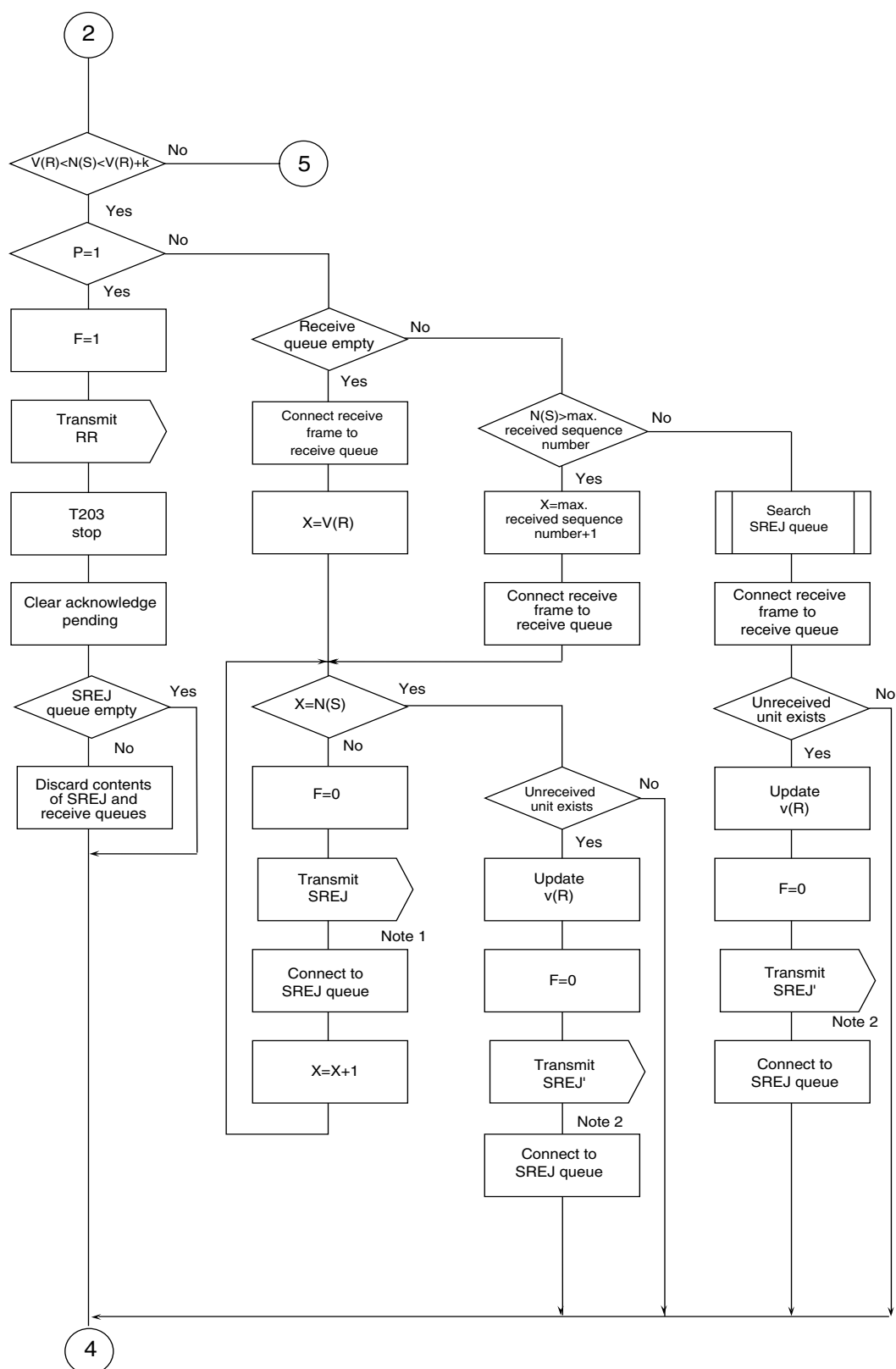


Fig. A. 4.2-18 Layer 2 SDL diagram (17/58)



Note 1 : Set $N(R)$ to X .
 Note 2 : Set $N(R)$ to $N(S)$.

Fig. A. 4.2-19 Layer 2 SDL diagram (18/58)

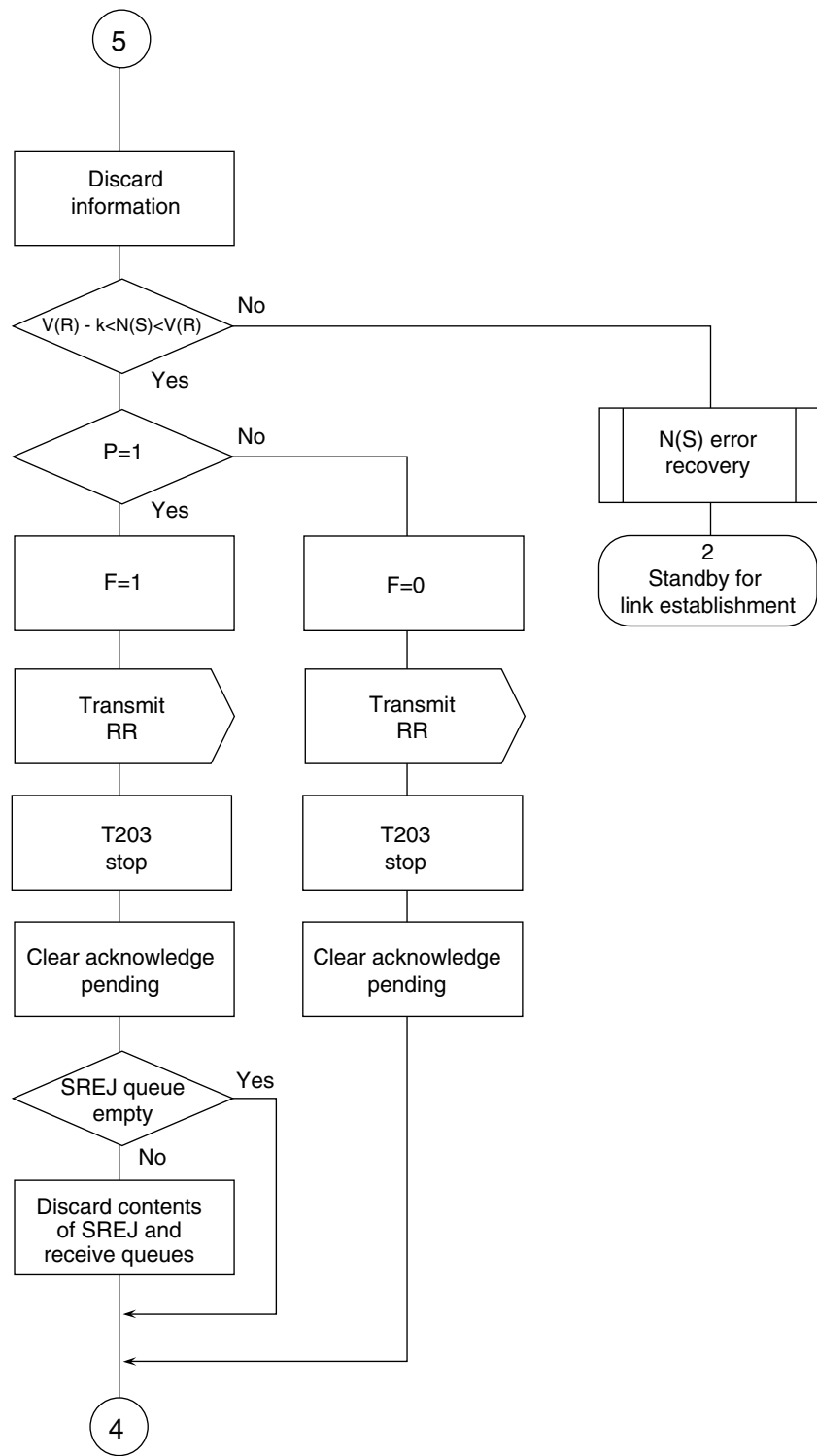
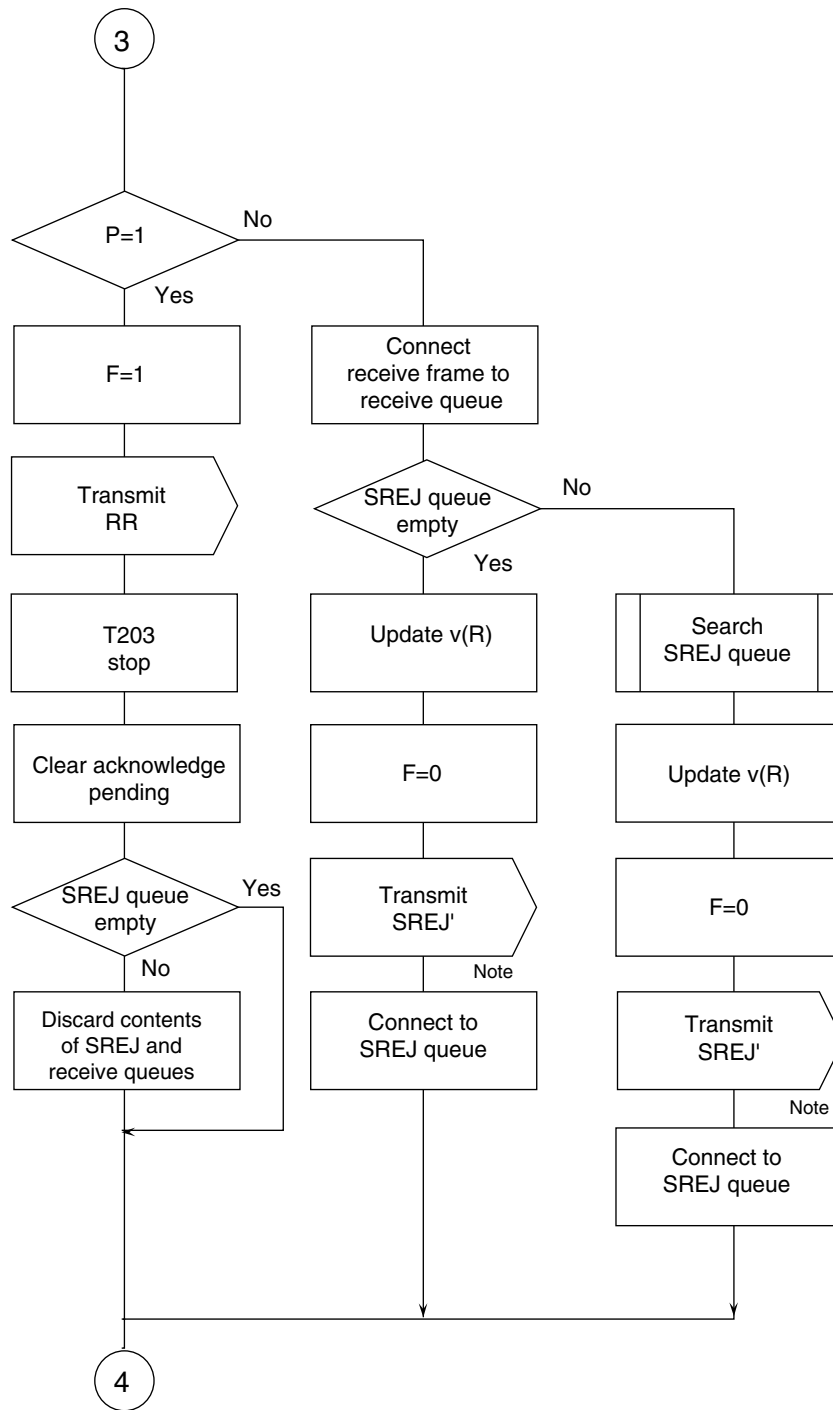


Fig. A. 4.2-20 Layer 2 SDL diagram (19/58)



Note : Set $N(R)$ to $N(S)$

Fig. A. 4.2-21 Layer 2 SDL diagram (20/58)

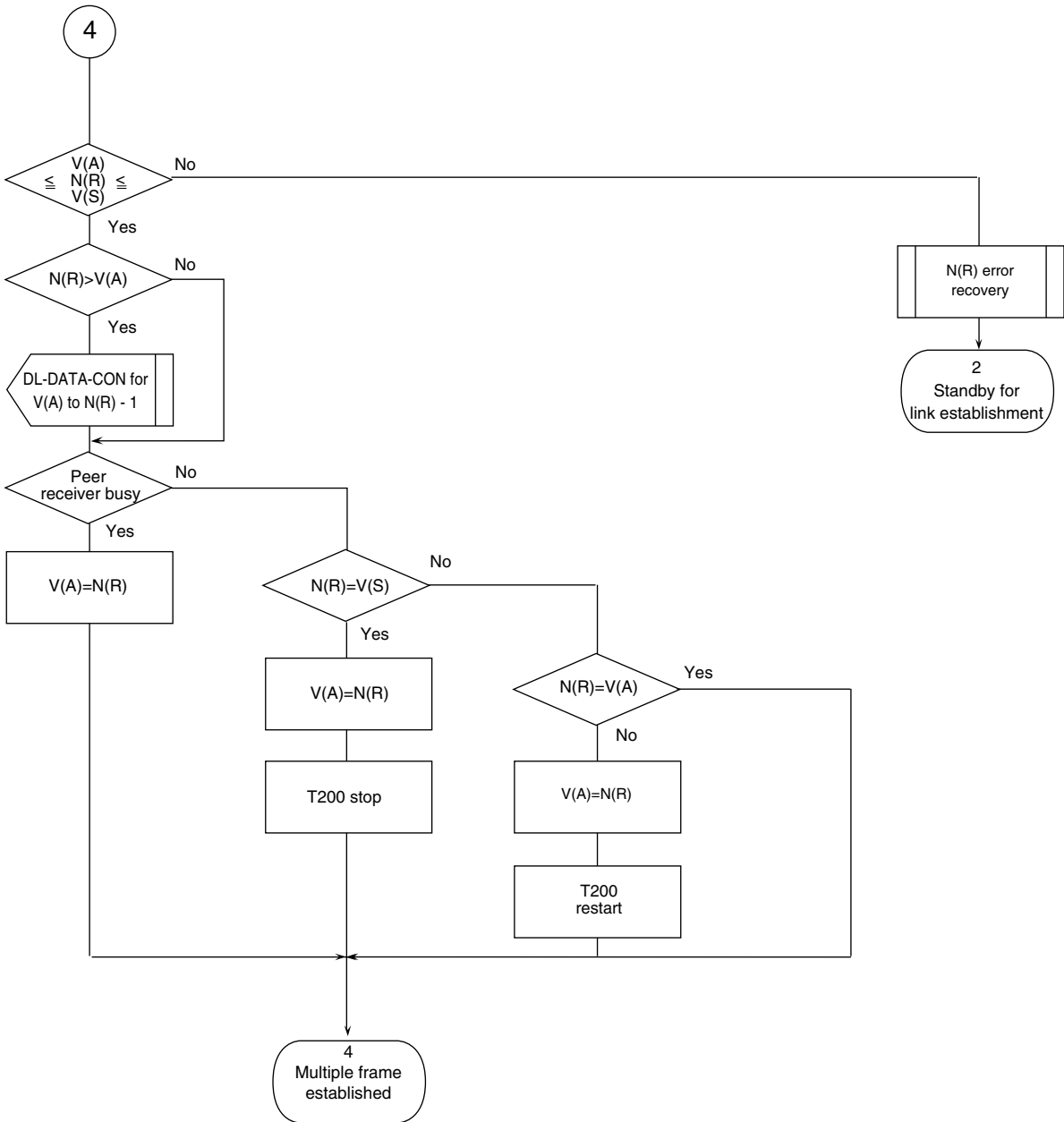


Fig. A. 4.2-22 Layer 2 SDL diagram (21/58)

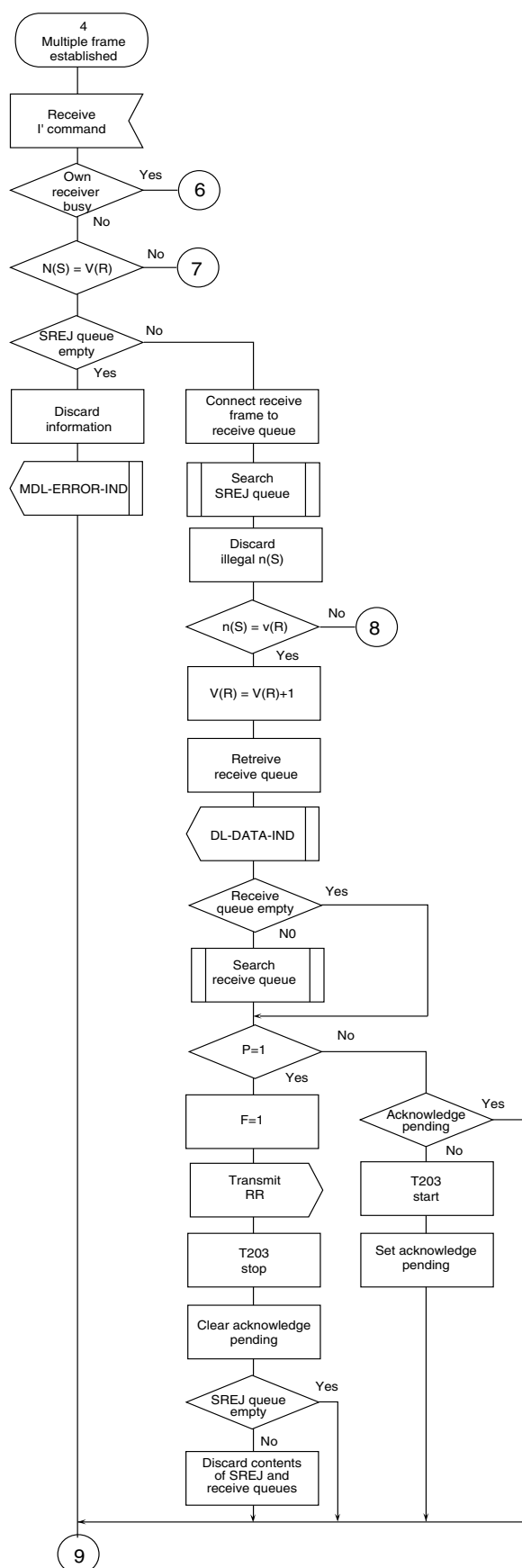


Fig. A. 4.2-23 Layer 2 SDL diagram (22/58)

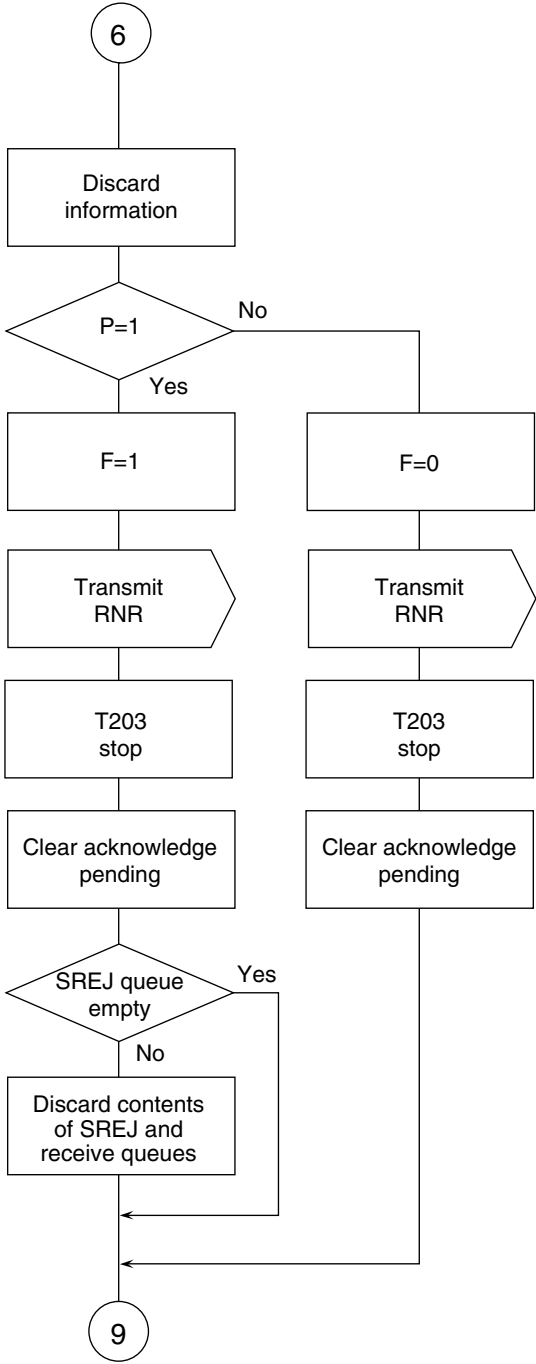
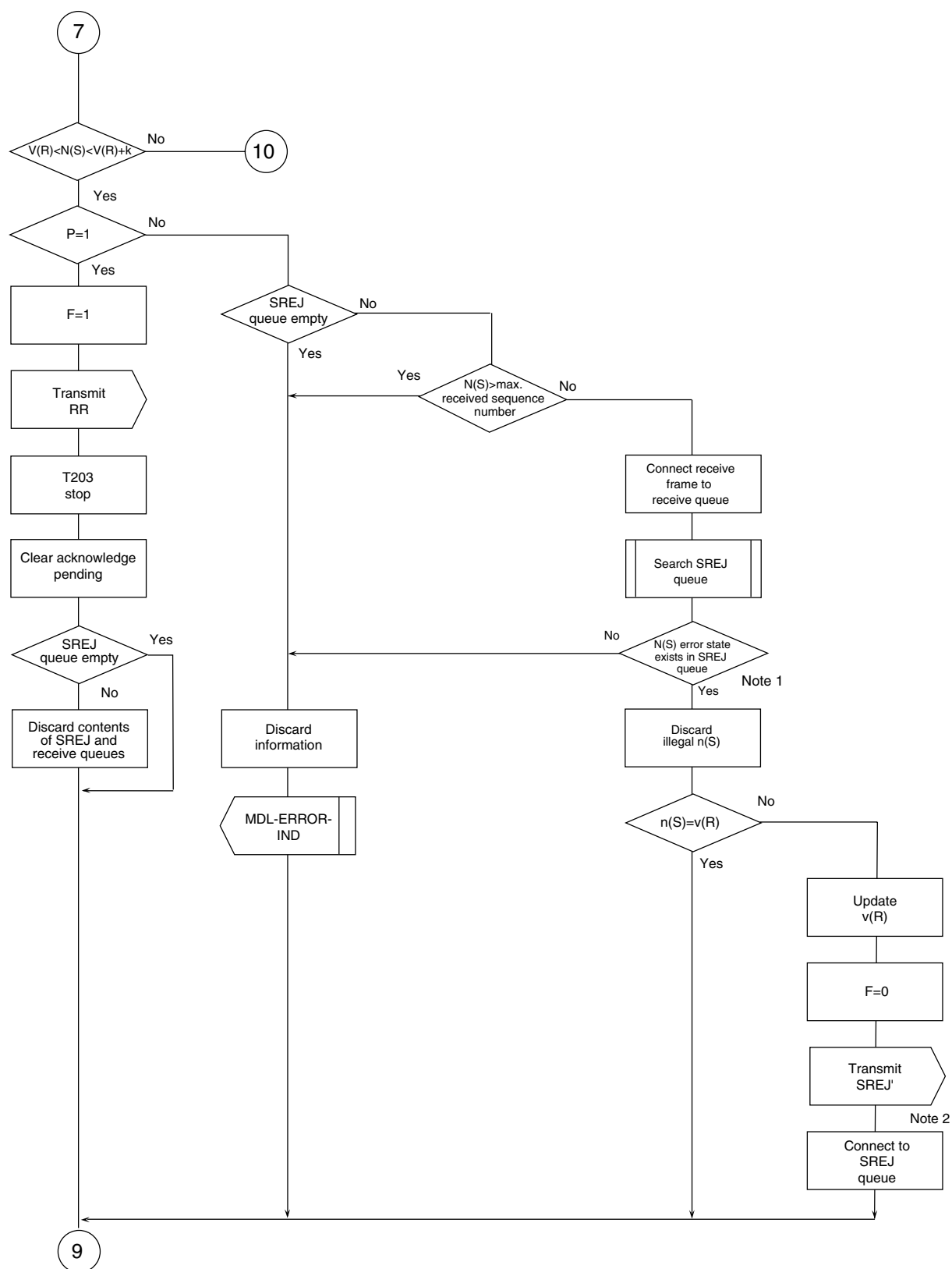


Fig. A. 4.2-24 Layer 2 SDL diagram (23/58)



Note 1 : A method for reporting the result of SREJ queue search is outside the scope of the SDLs

Note 2 : Set $N(R)$ to $N(S)$

Fig. A. 4.2-25 Layer 2 SDL diagram (24/58)

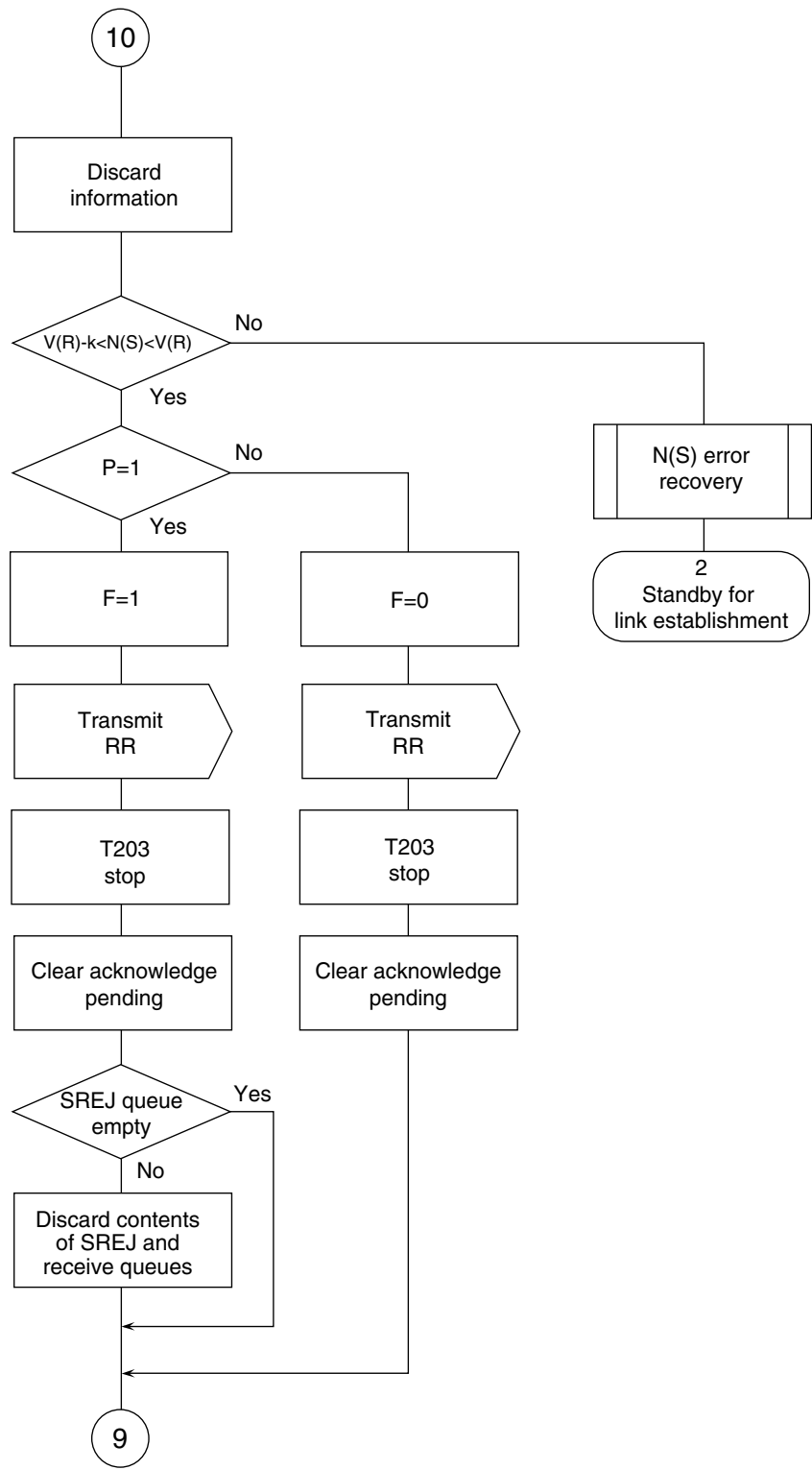
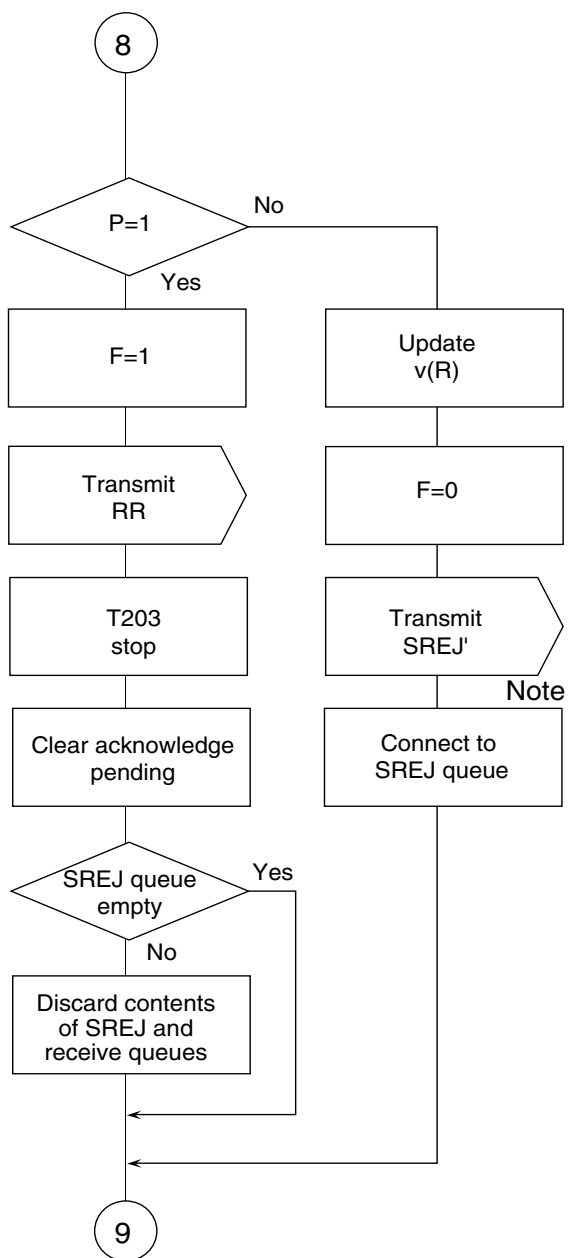


Fig. A. 4.2-26 Layer 2 SDL diagram (25/58)



Note : Set $N(R)$ to $N(S)$

Fig. A. 4.2-27 Layer 2 SDL diagram (26/58)

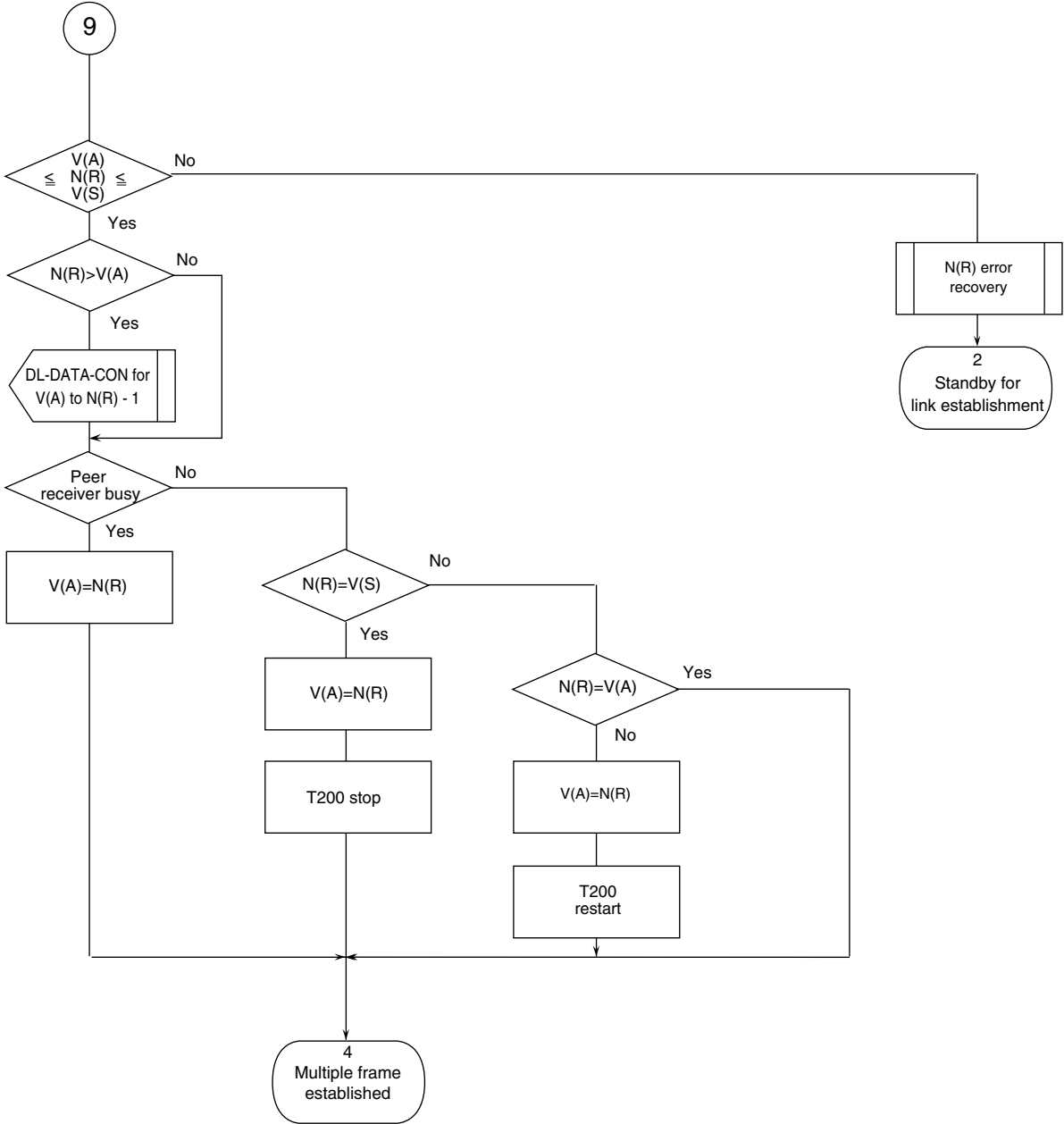


Fig. A. 4.2-28 Layer 2 SDL diagram (27/58)

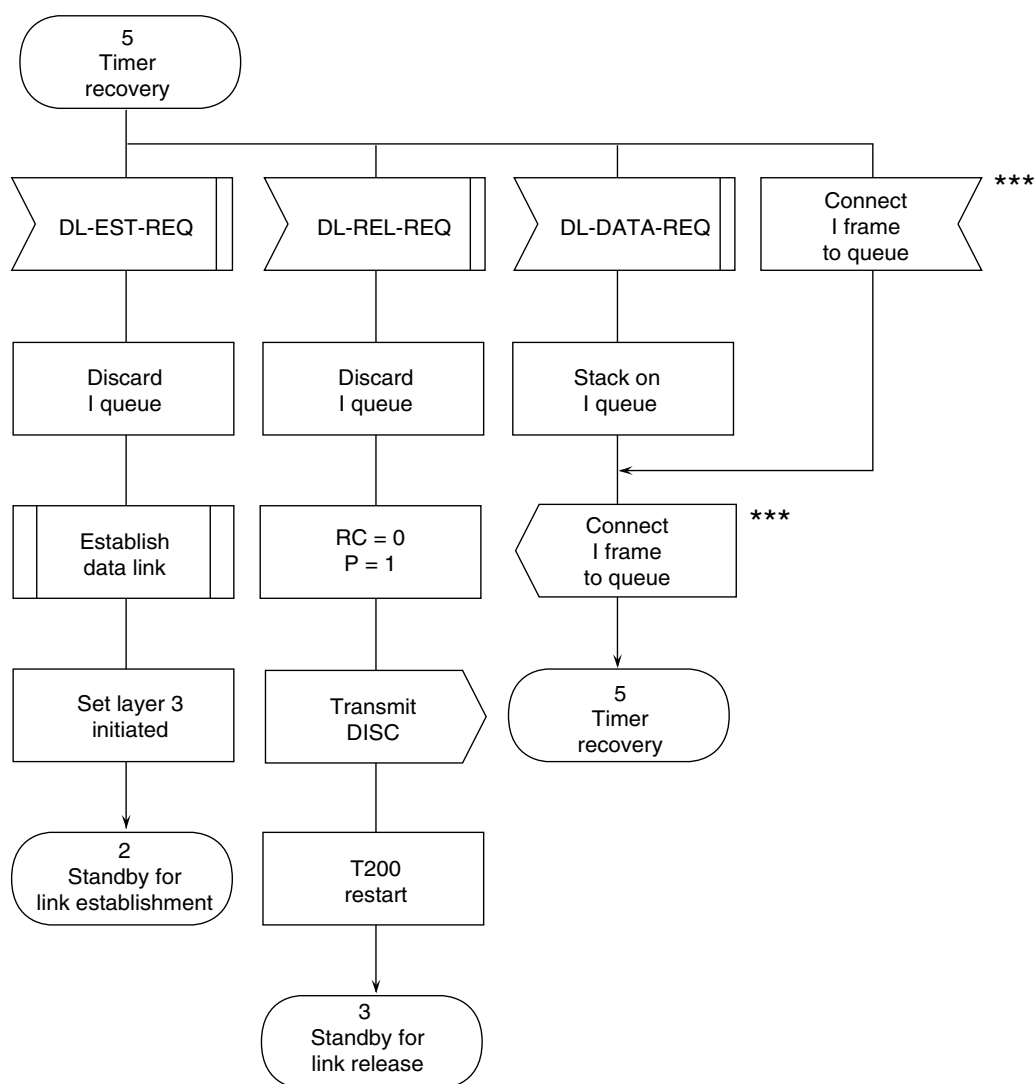


Fig. A. 4.2-29 Layer 2 SDL diagram (28/58)

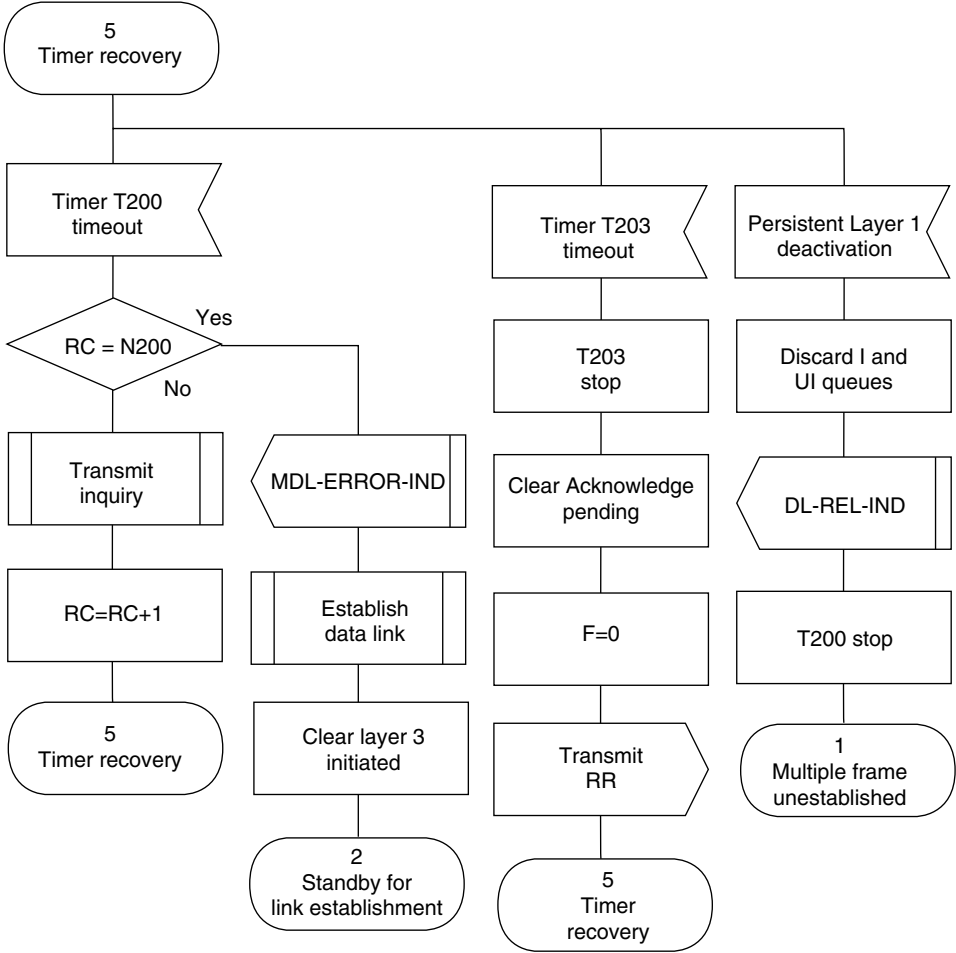


Fig. A. 4.2-30 Layer 2 SDL diagram (29/58)

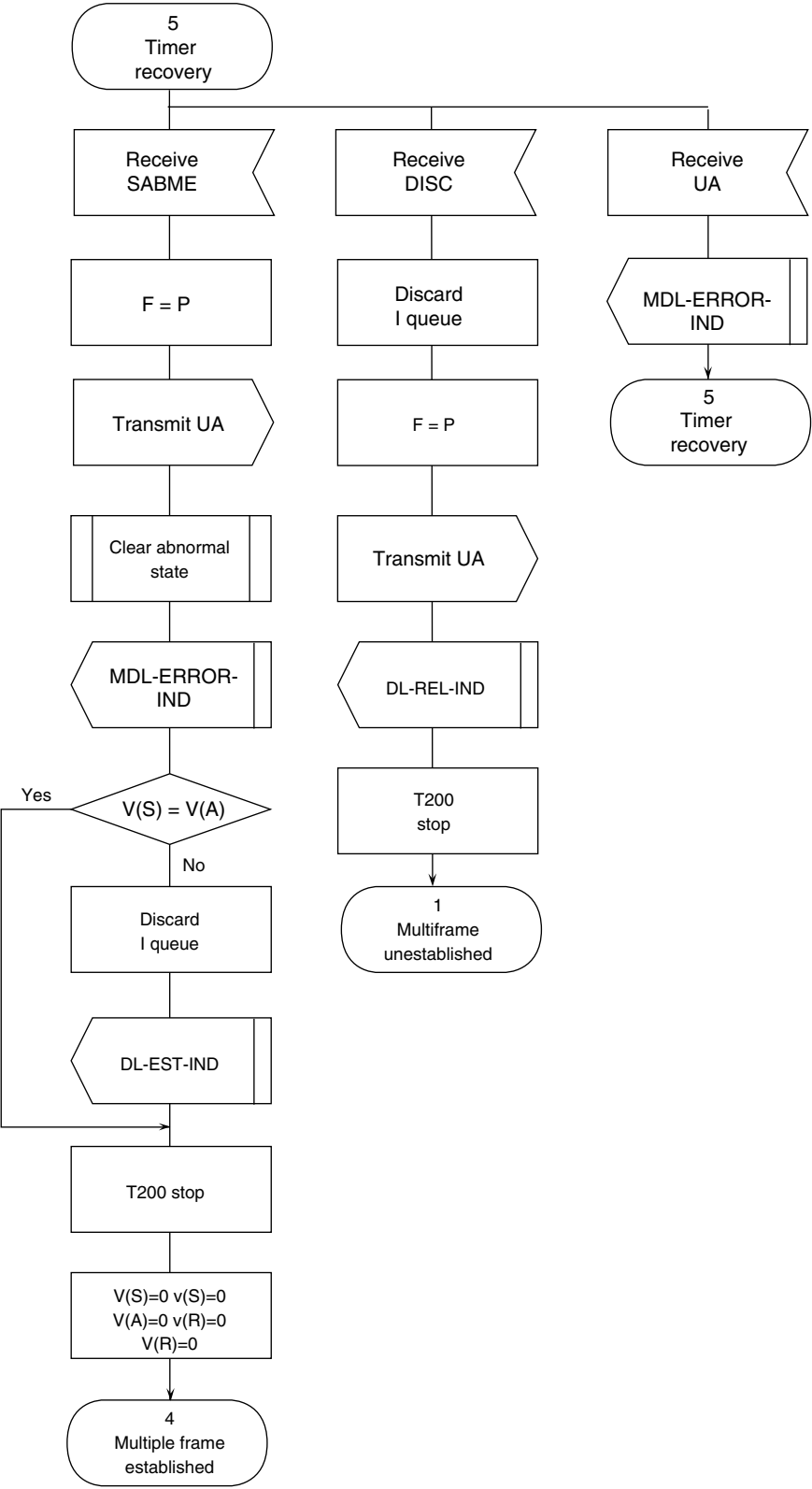
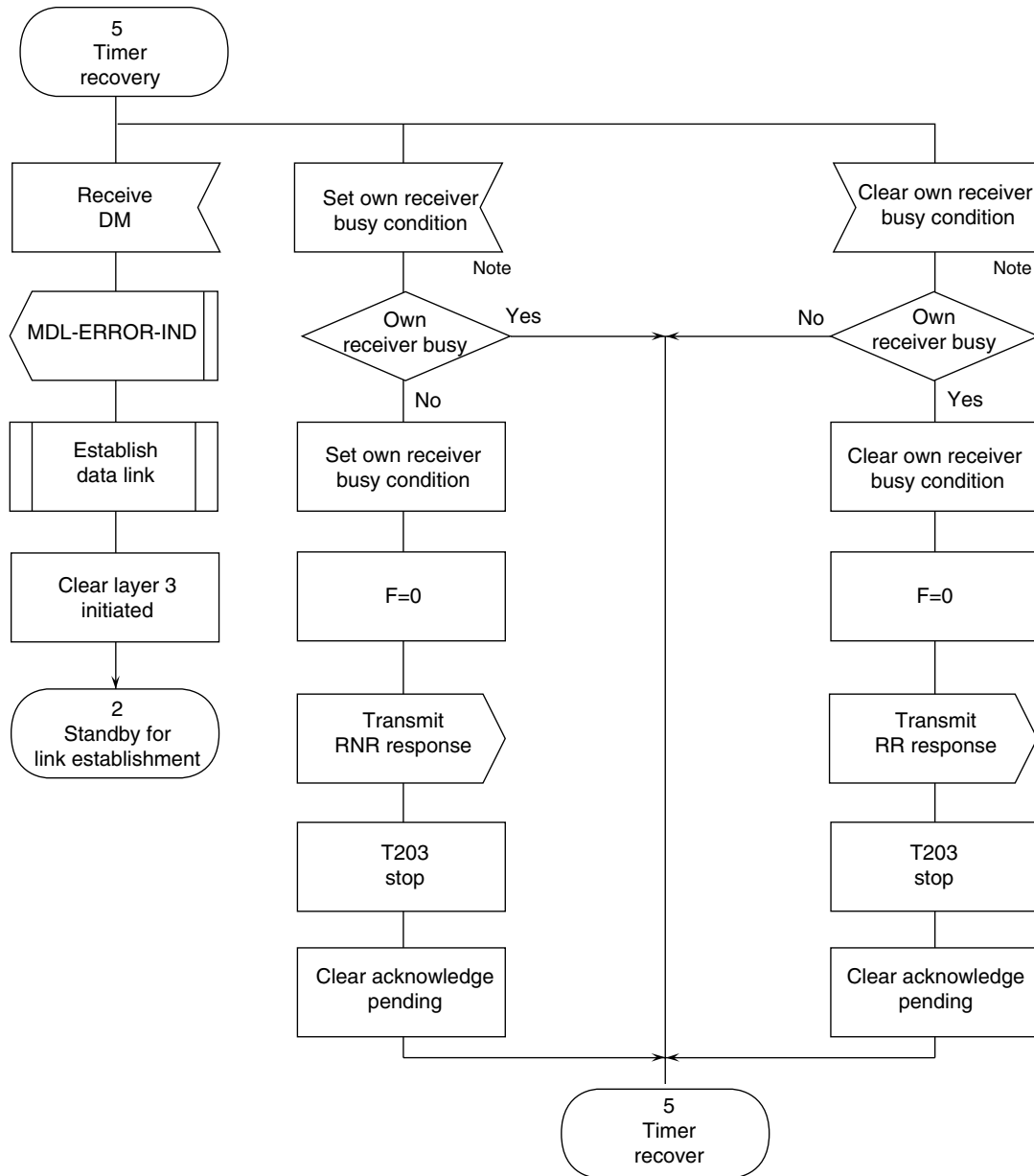


Fig. A. 4.2-31 Layer 2 SDL diagram (30/58)



Note : These signals are generated outside of the scope of these SDLs and can also be generated by the connection management entity

Fig. A. 4.2-32 Layer 2 SDL diagram (31/58)

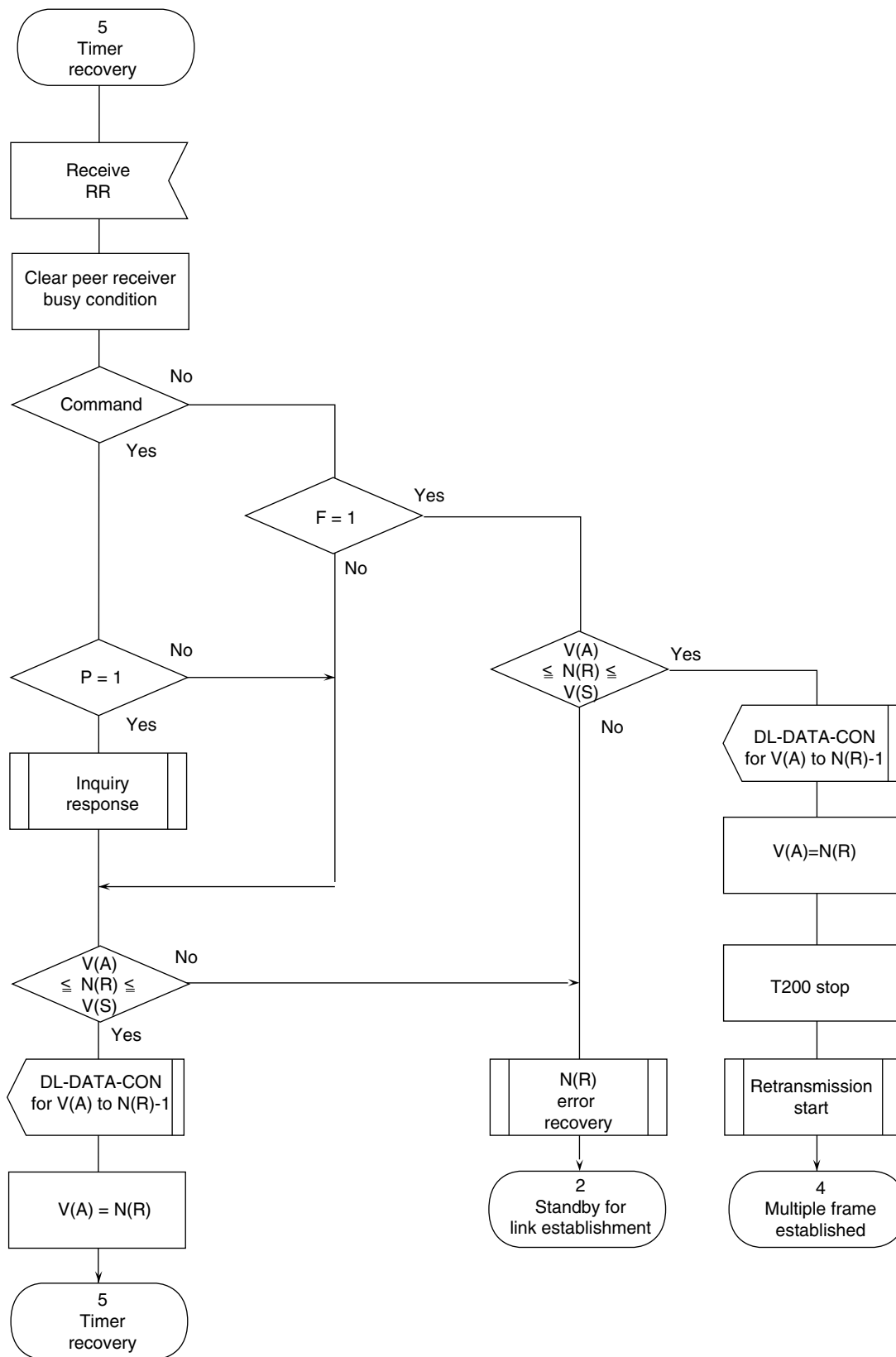


Fig. A. 4.2-33 Layer 2 SDL diagram (32/58)

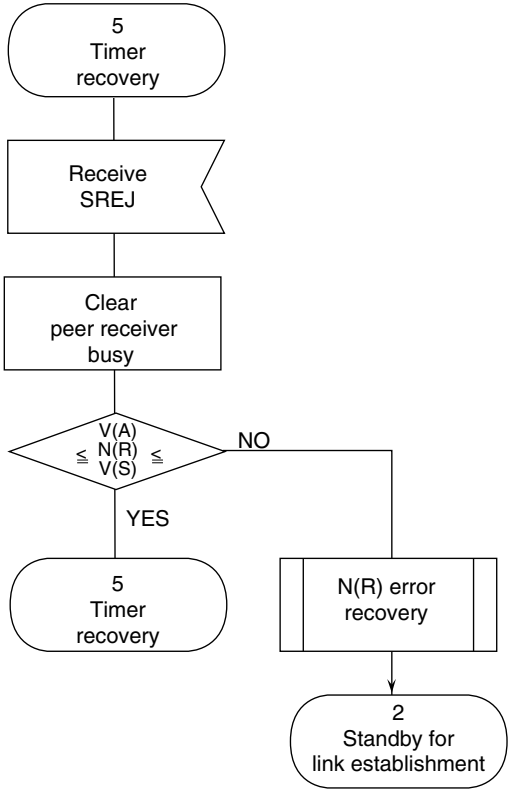


Fig. A. 4.2-34 Layer 2 SDL diagram (33/58)

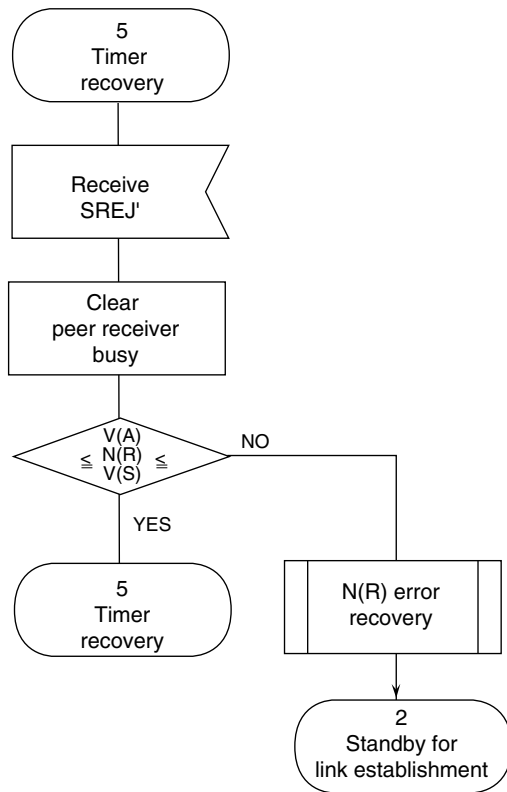


Fig. A. 4.2-35 Layer 2 SDL diagram (34/58)

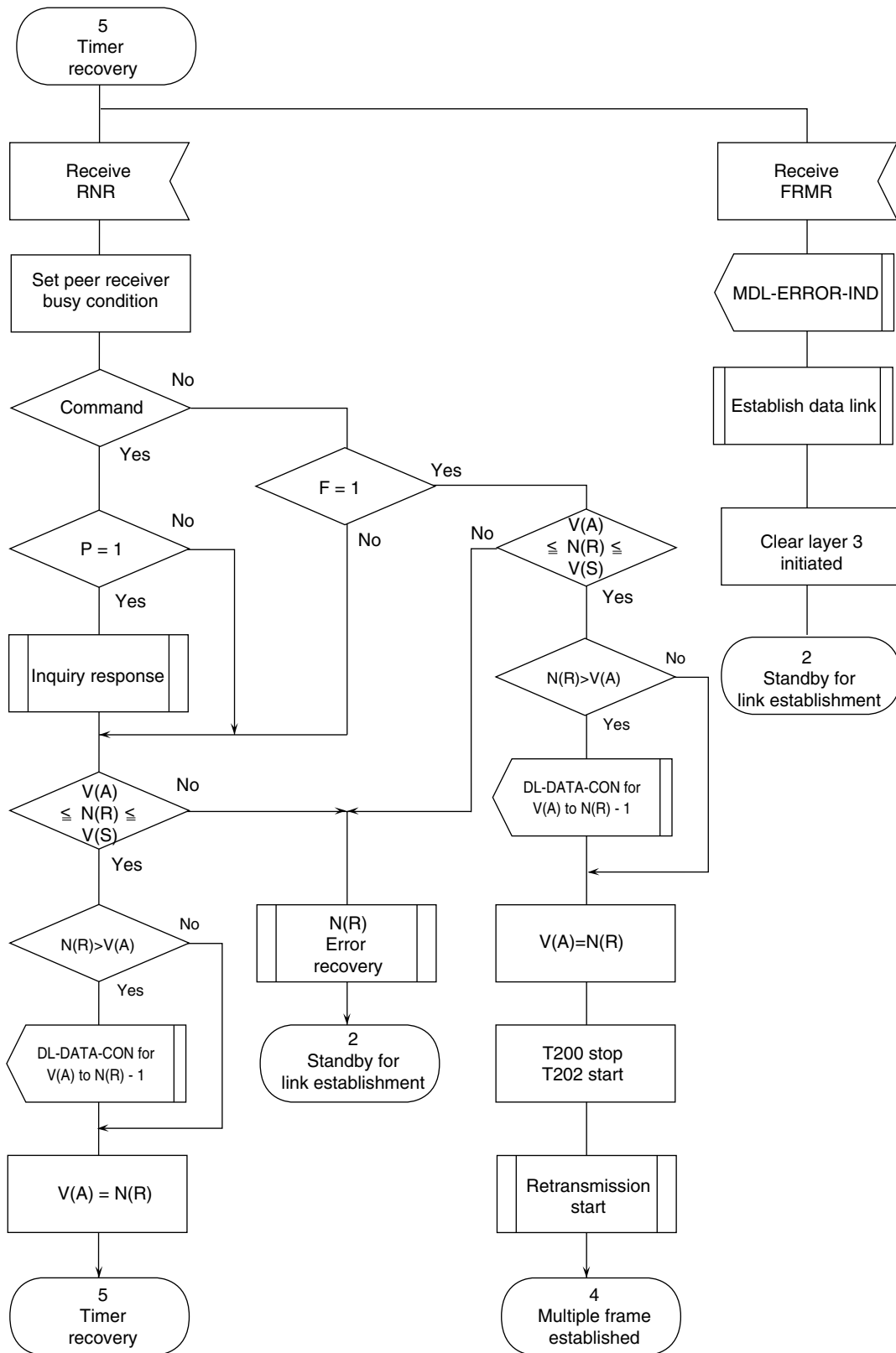


Fig. A. 4.2-36 Layer 2 SDL diagram (35/58)

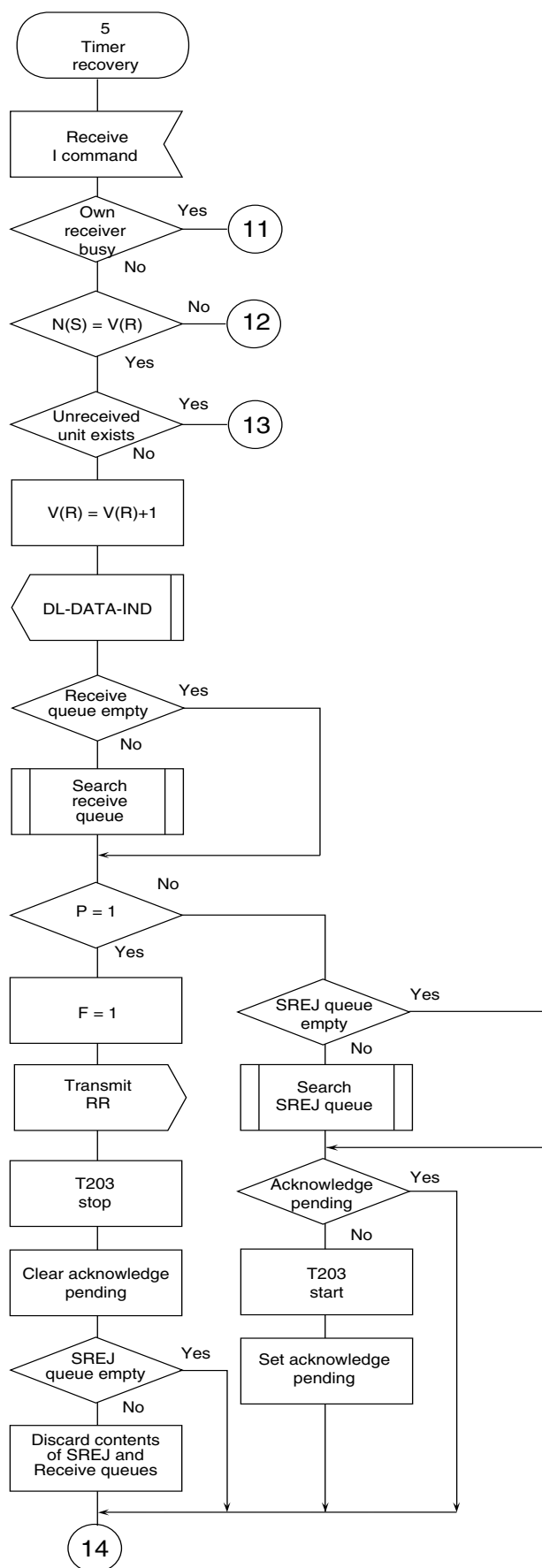


Fig. A. 4.2-37 Layer 2 SDL diagram (36/58)

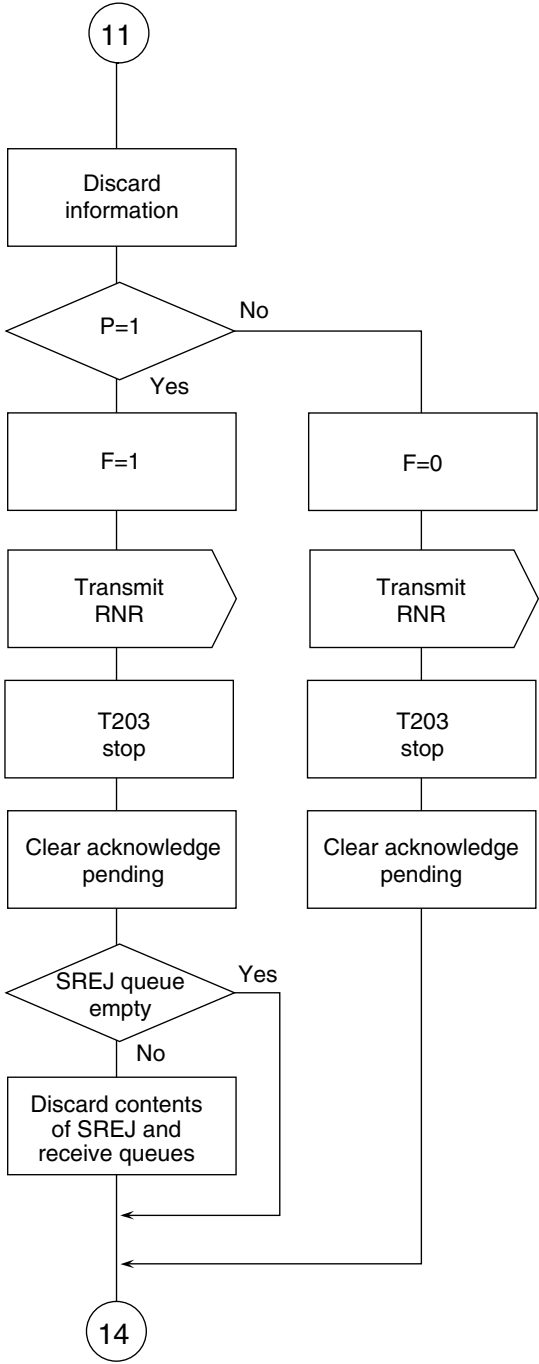


Fig. A. 4.2-38 Layer 2 SDL diagram (37/58)

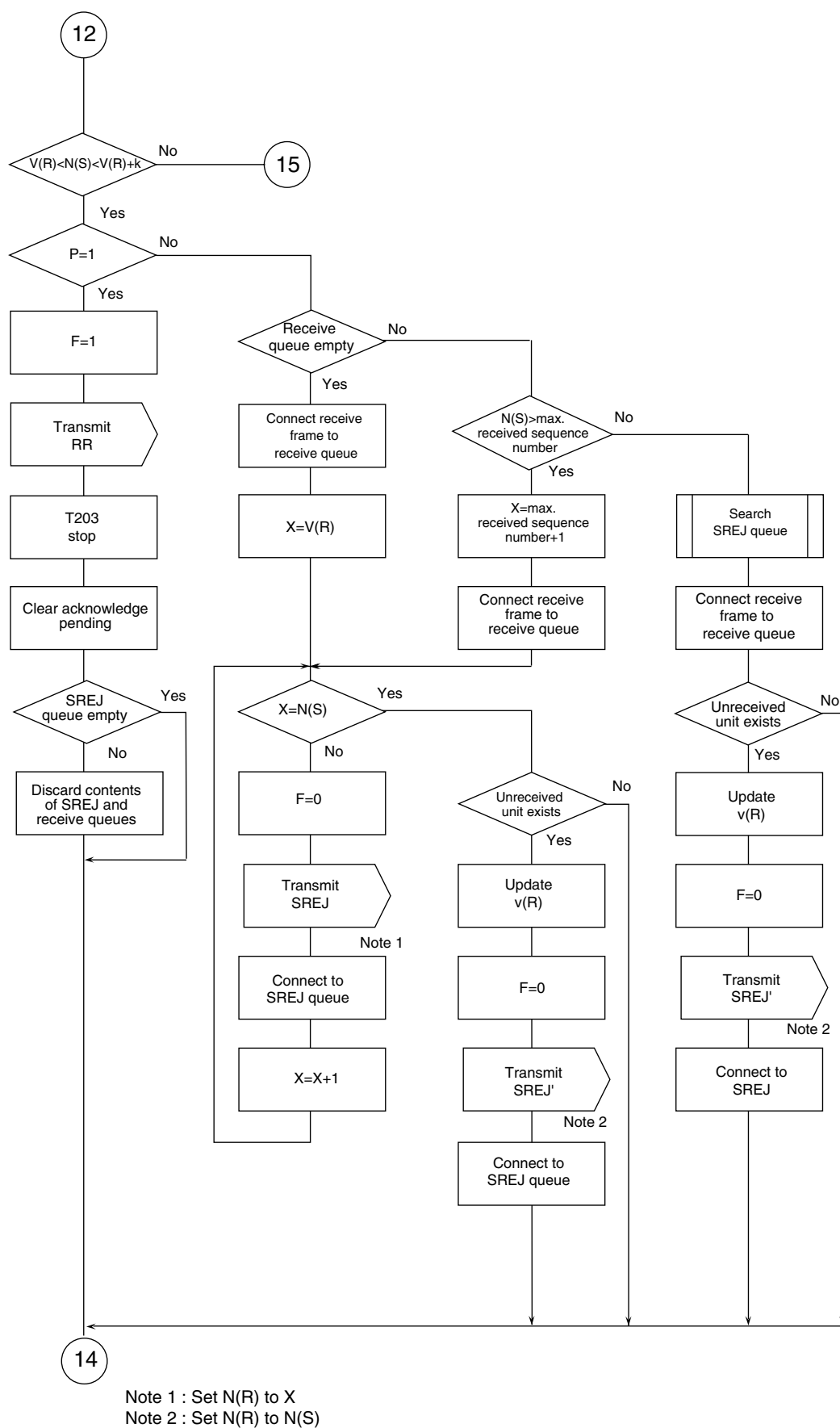


Fig. A. 4.2-39 Layer 2 SDL diagram (38/58)

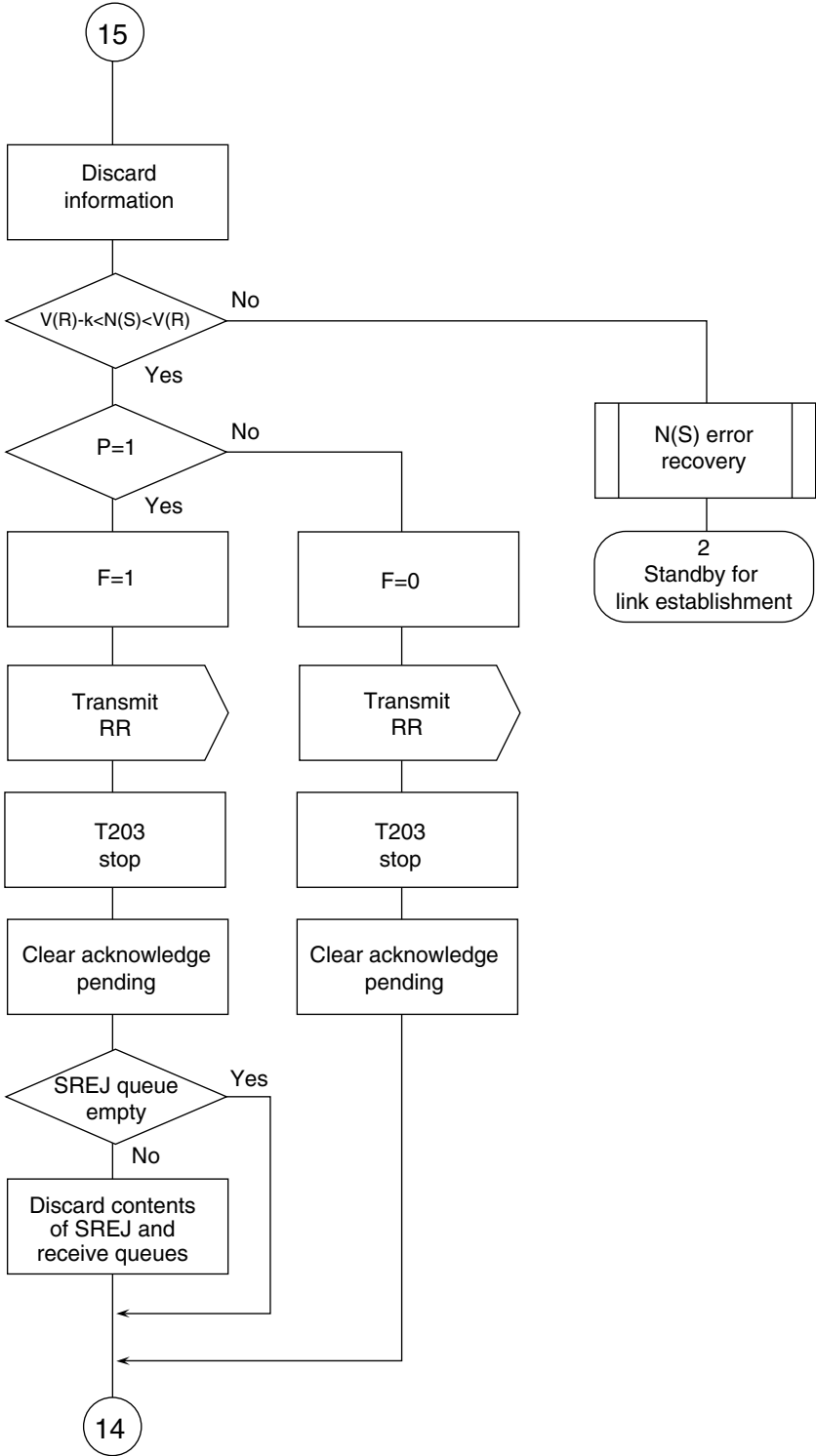
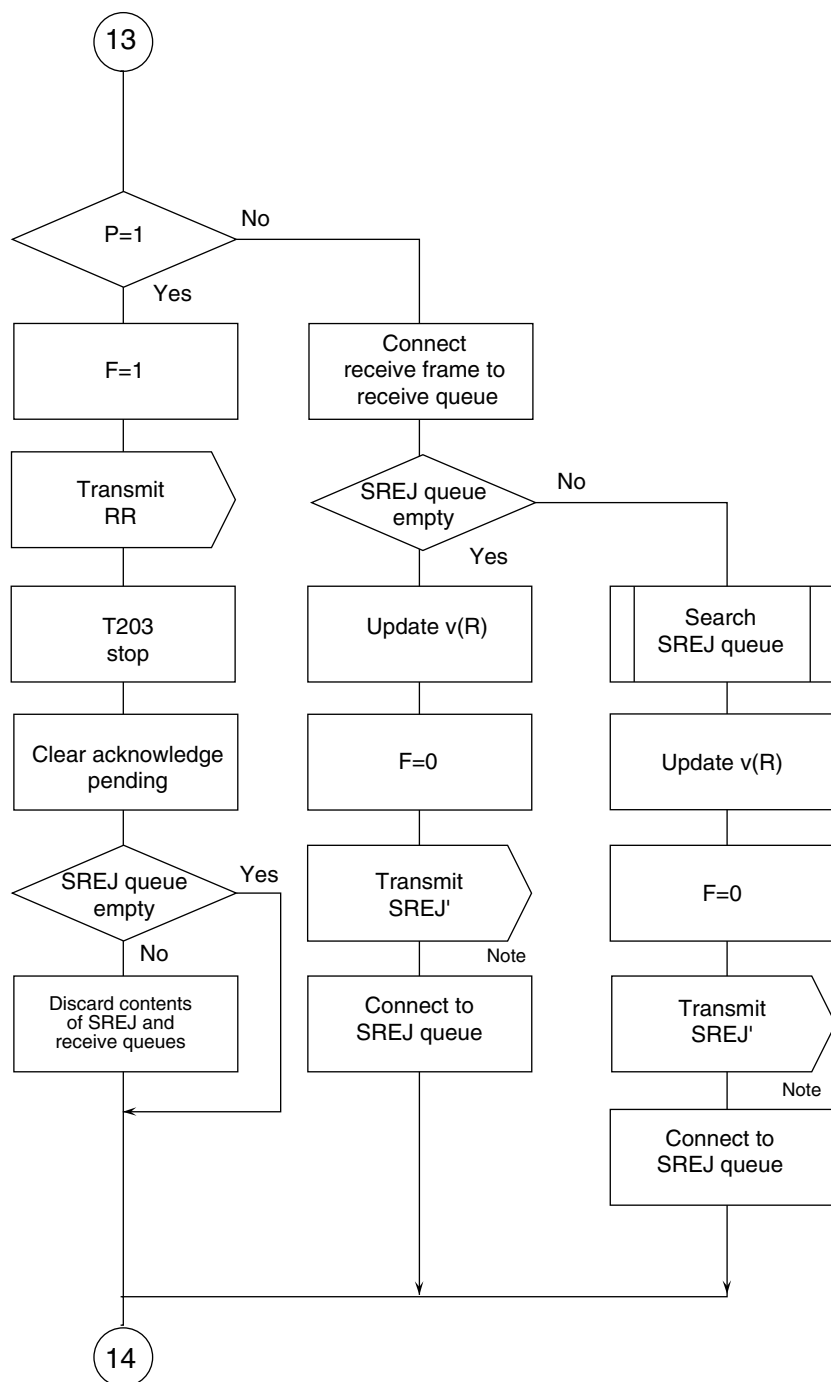


Fig. A. 4.2-40 Layer 2 SDL diagram (39/58)



Note : Set N(R) to N(S)

Fig. A. 4.2-41 Layer 2 SDL diagram (40/58)

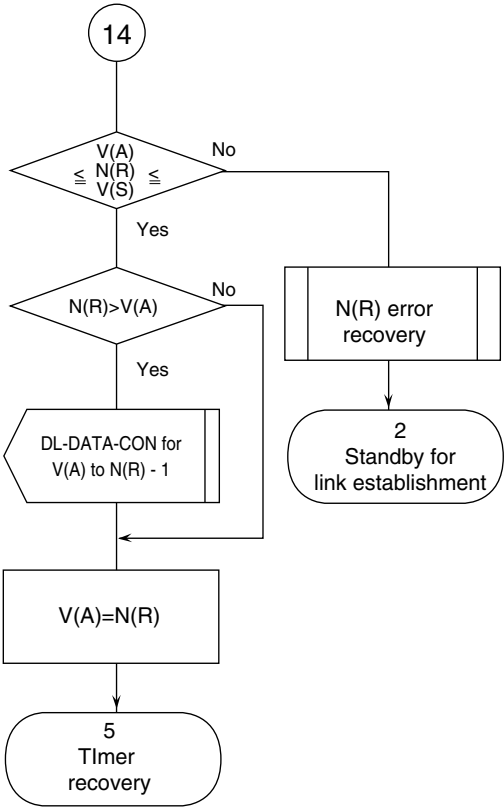


Fig. A. 4.2-42 Layer 2 SDL diagram (41/58)

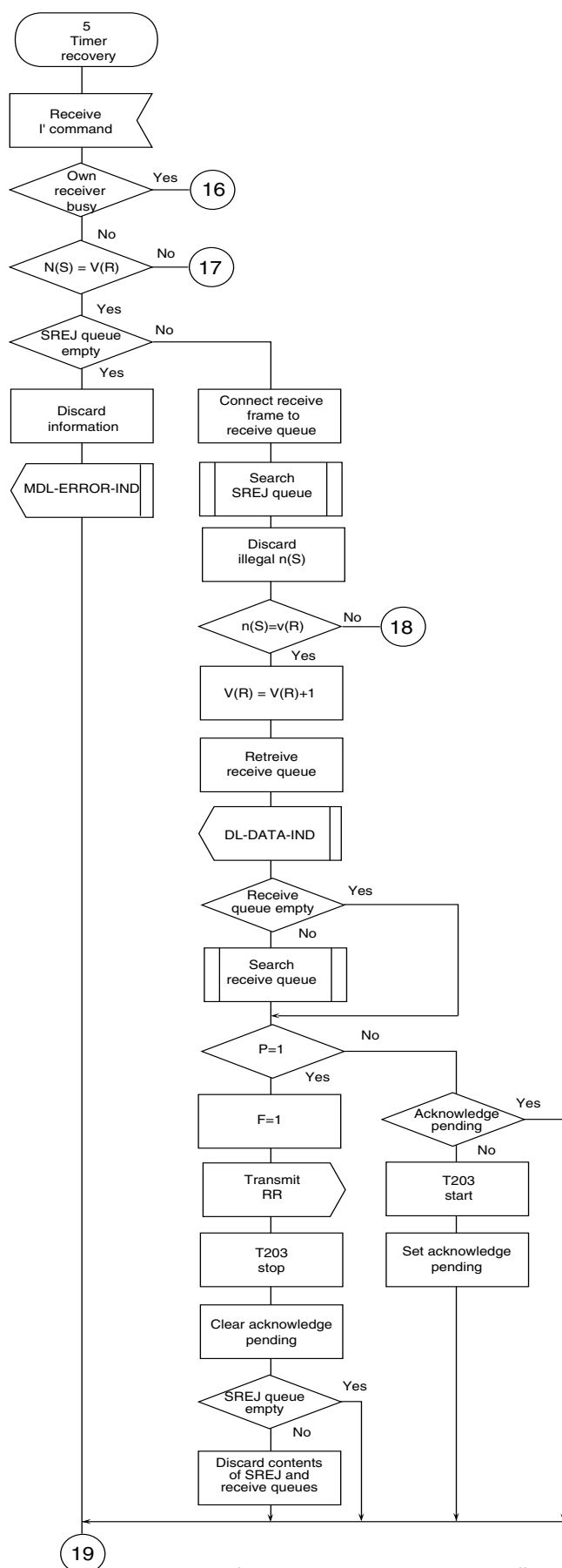


Fig. A. 4.2-43 Layer 2 SDL diagram (42/58)

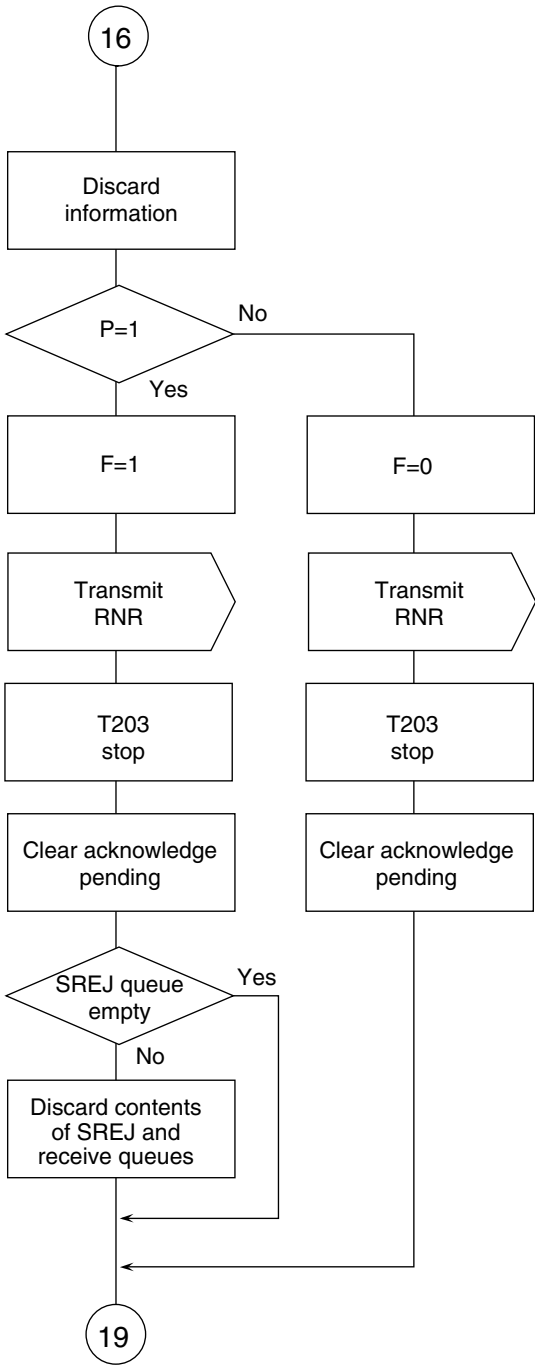
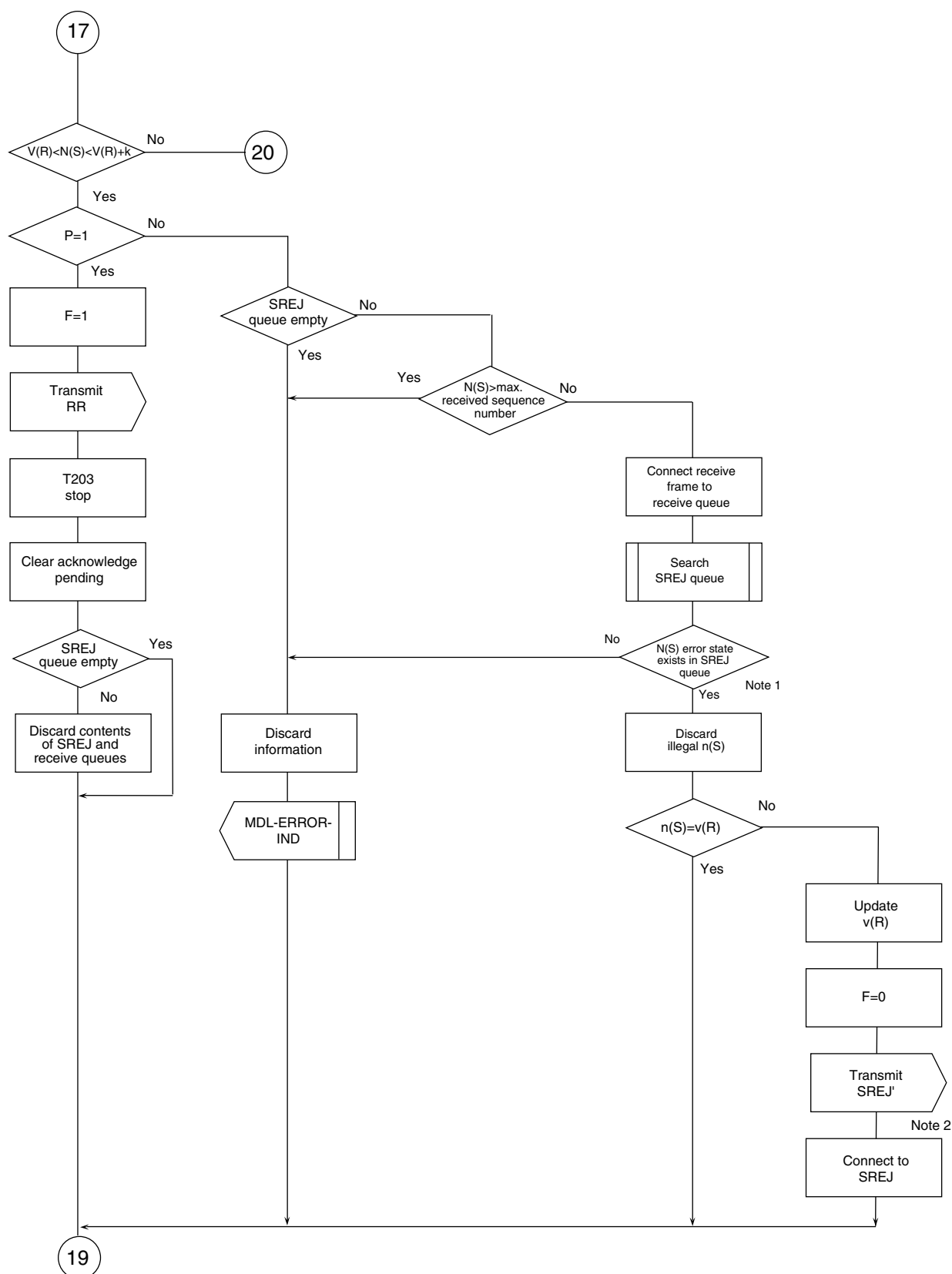


Fig. A. 4.2-44 Layer 2 SDL diagram (43/58)



Note 1 : A method for reporting the result of SREJ queue search is outside the scope of the SDLs

Note 2 : Set N(R) to N(S)

Fig. A. 4.2-45 Layer 2 SDL diagram (44/58)

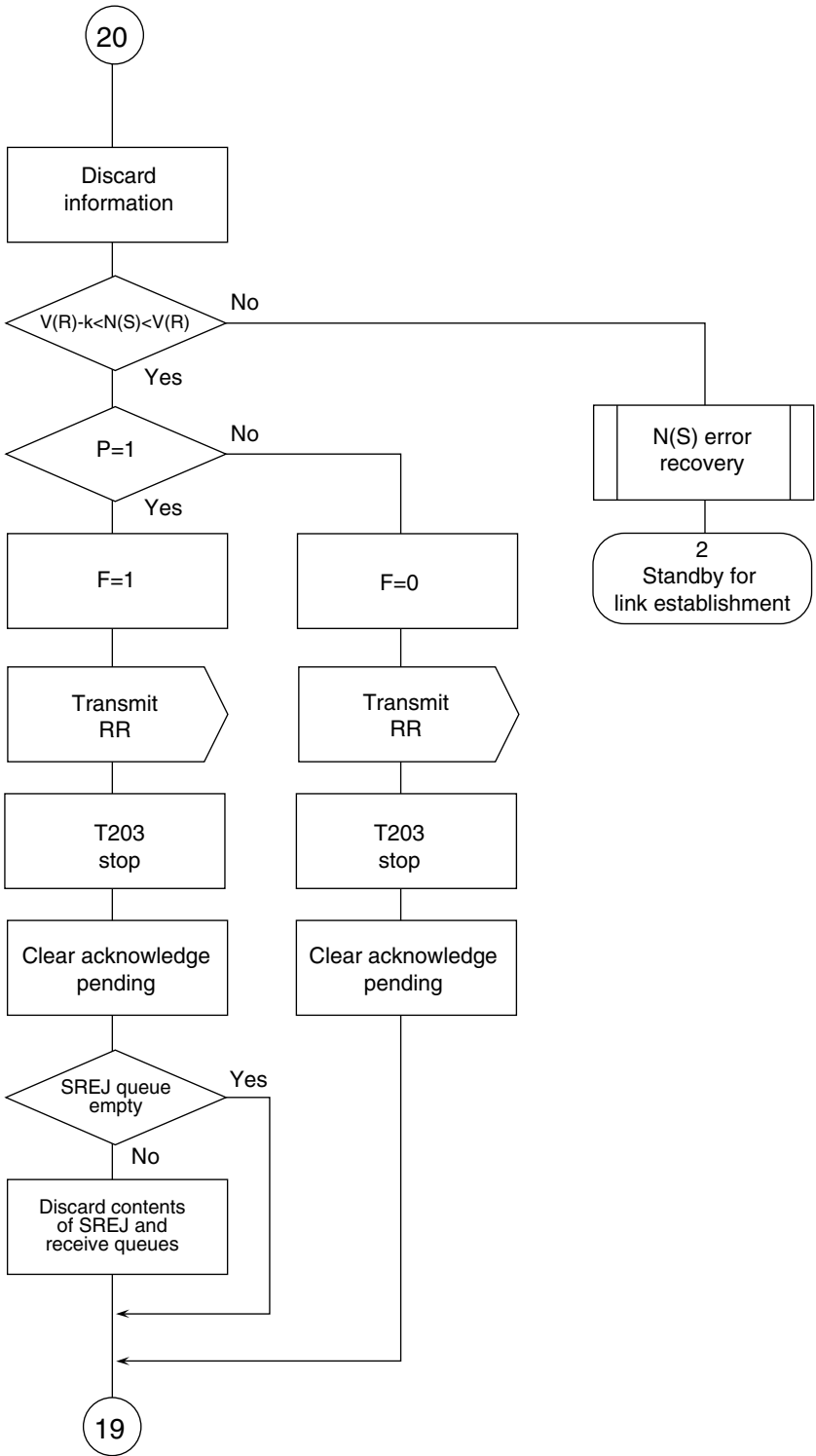
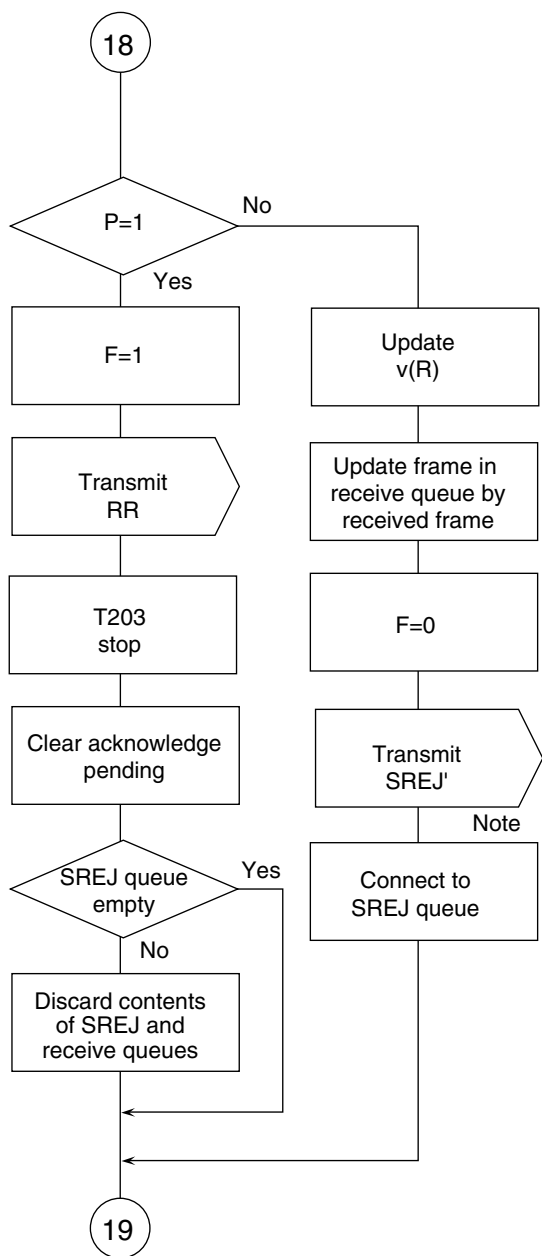


Fig. A. 4.2-46 Layer 2 SDL diagram (45/58)



Note : Set N(R) to N(S)

Fig. A. 4.2-47 Layer 2 SDL diagram (46/58)

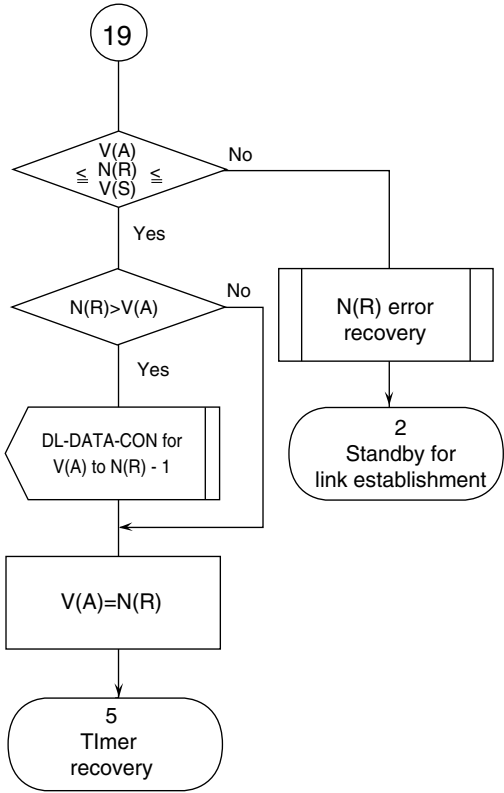
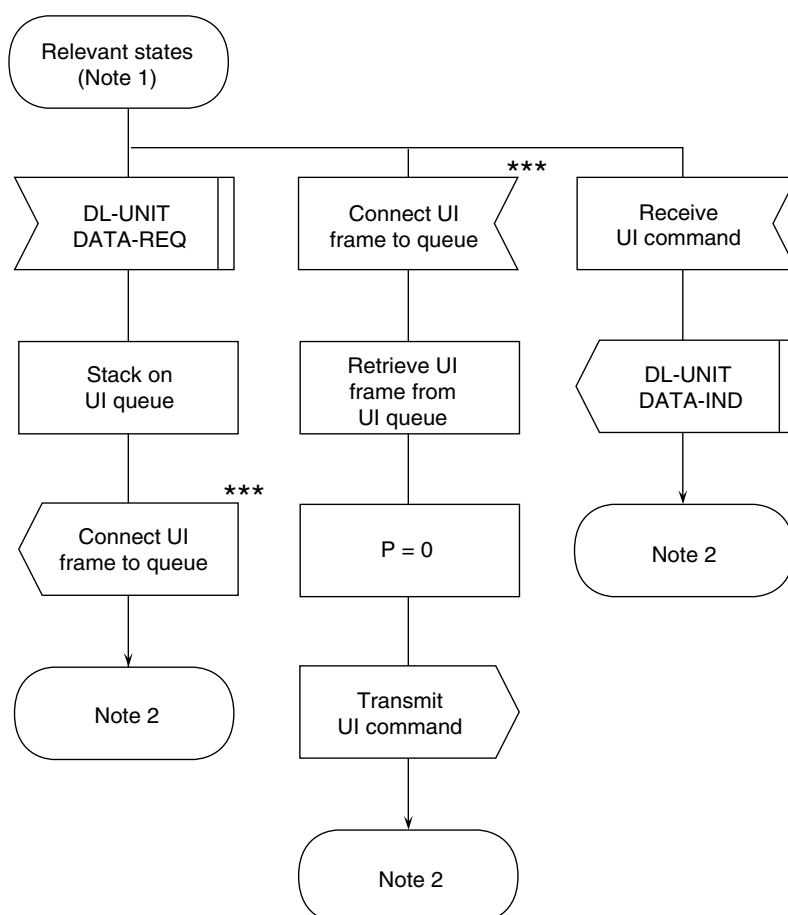


Fig. A. 4.2-48 Layer 2 SDL diagram (47/58)

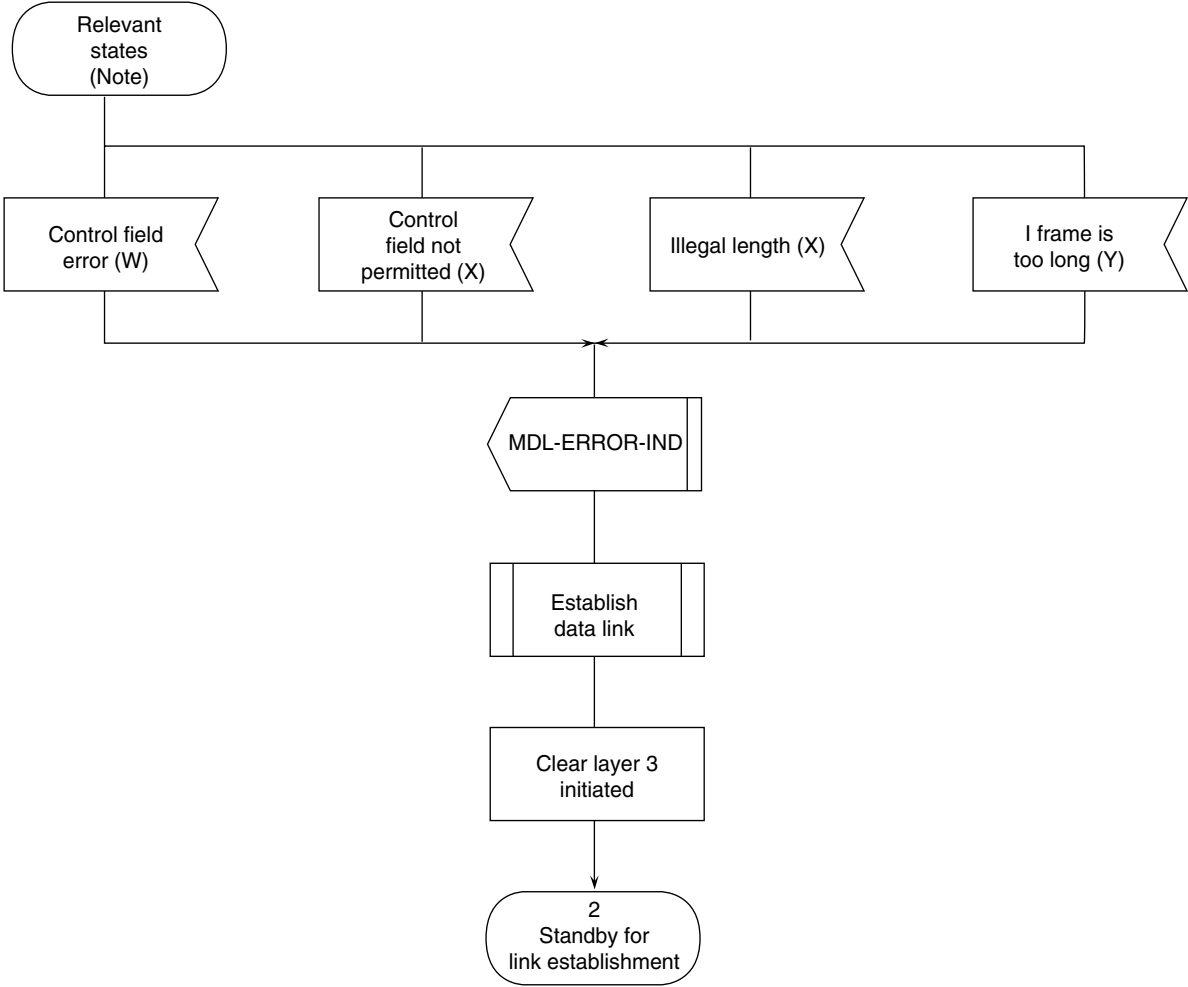


Note 1 : "Relevant states" are listed below.

1. Multiple frame unestablished
2. Standby for link establishment
3. Standby for link release
4. Multiple frame established
5. Timer recovery

Note 2 : Data link layer resets to the previously set state of the above states.

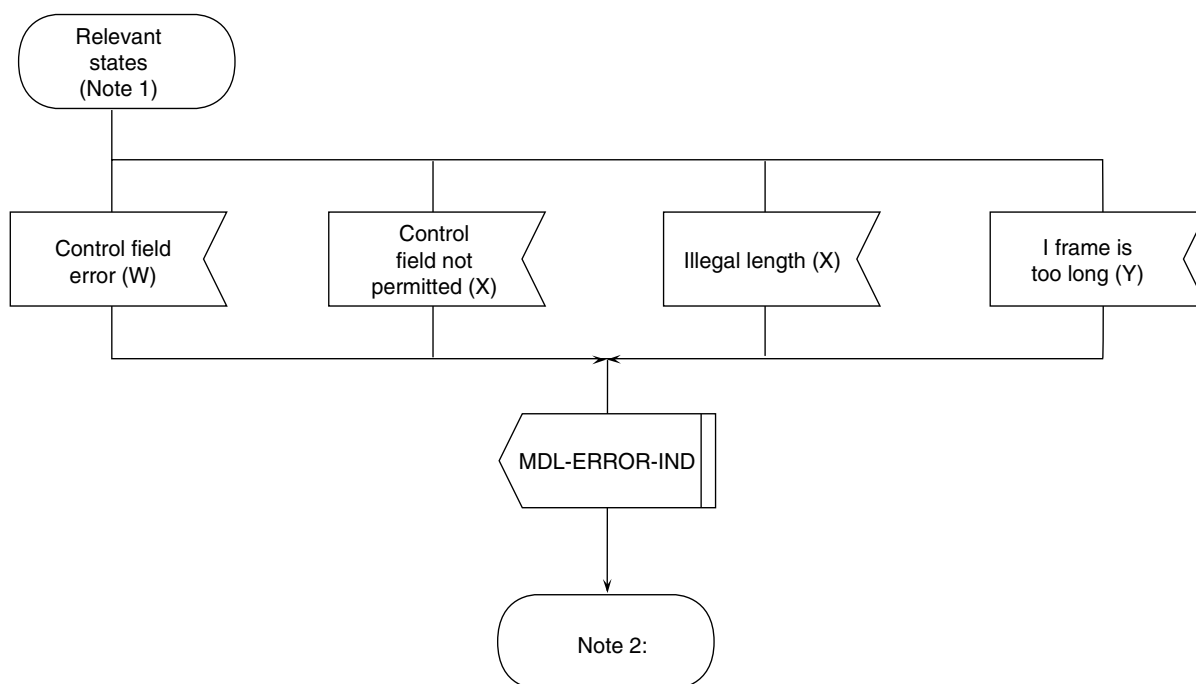
Fig. A. 4.2-49 Layer 2 SDL diagram (48/58)



Note : "Relevant states" are as follows :

- 4. Multiple frame established
- 5. Timer recovery

Fig. A. 4.2-50 Layer 2 SDL diagram (49/58)



Note 1 : "Relevant states" are as follows:

1. Multiple frame unestablished
2. Standby for link establishment
3. Standby for link release

Note 2 : The data link layer resets to the state previously set among the above states.

Fig. A. 4.2-51 Layer 2 SDL diagram (50/58)

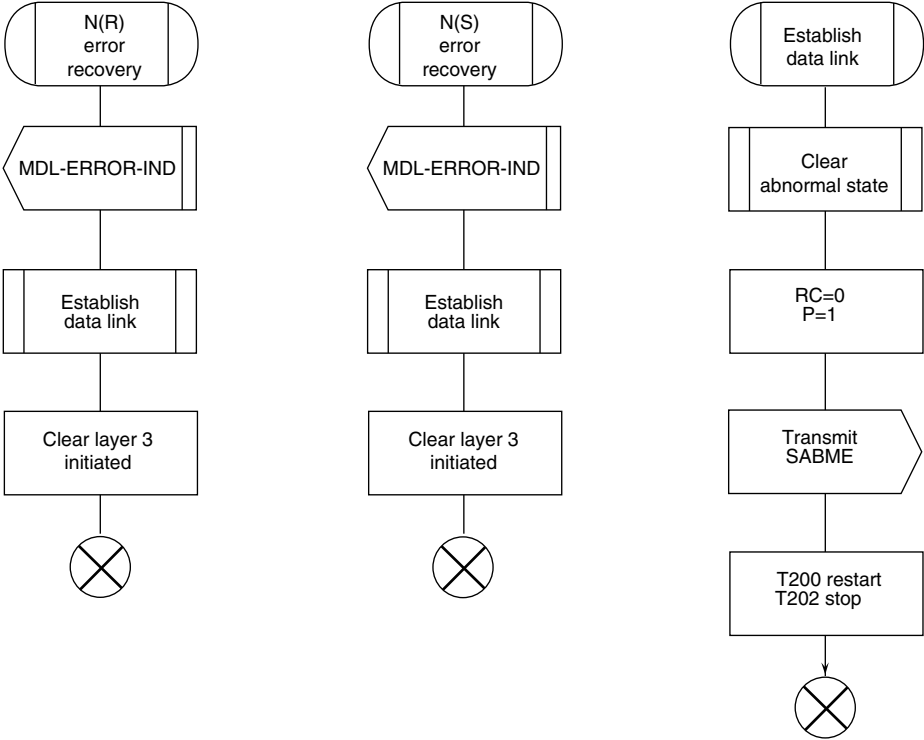


Fig. A. 4.2-52 Layer 2 SDL diagram (51/58)

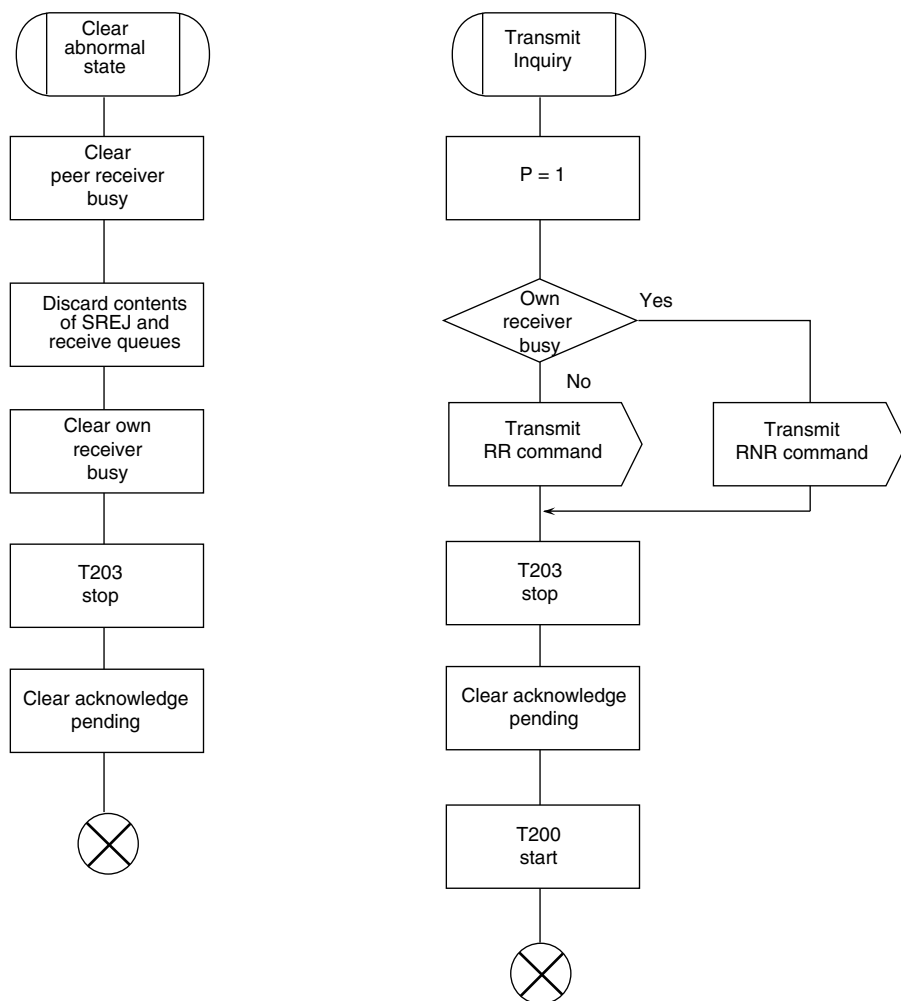


Fig. A. 4.2-53 Layer 2 SDL diagram (52/58)

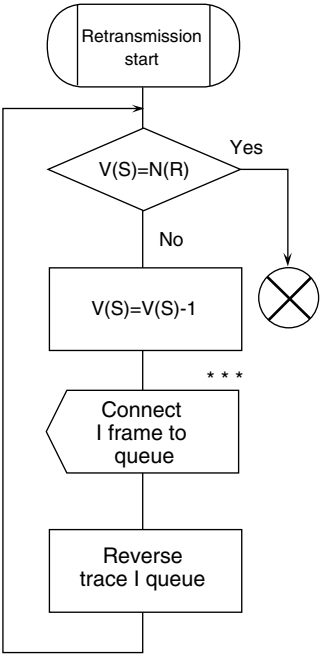
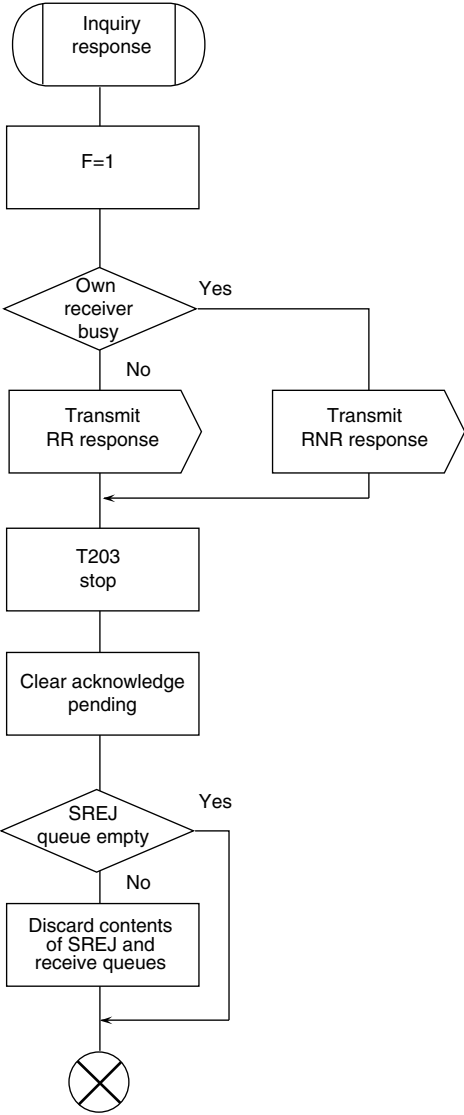
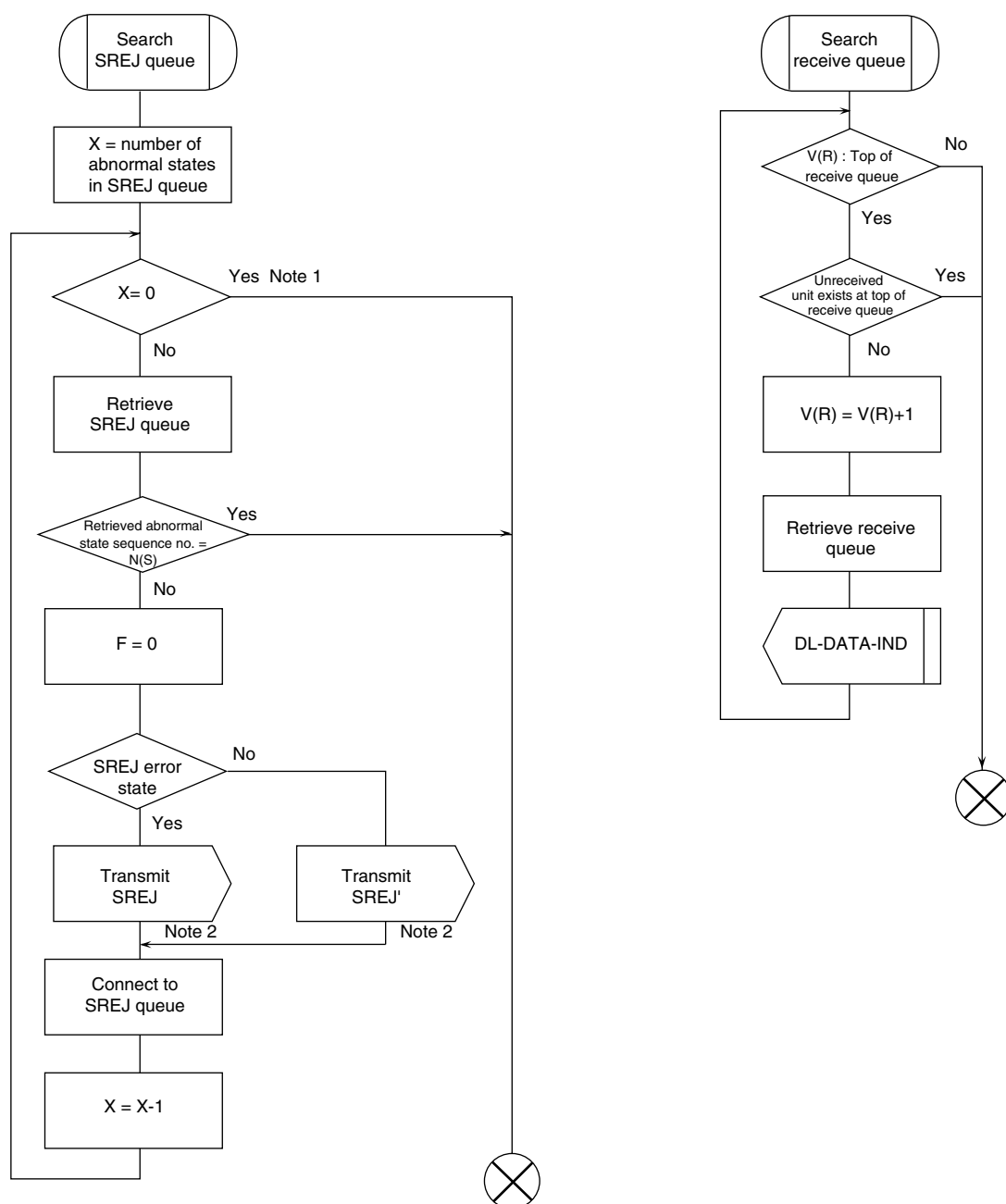


Fig. A. 4.2-54 Layer 2 SDL diagram (53/58)



Note 1 : This occurs when an I or I' frame containing the N(S) which is not connected to the SREJ queue is received
 Note 2 : Set N(R) to the retrieved abnormal state sequence number.

Fig. A. 4.2-55 Layer 2 SDL diagram (54/58)

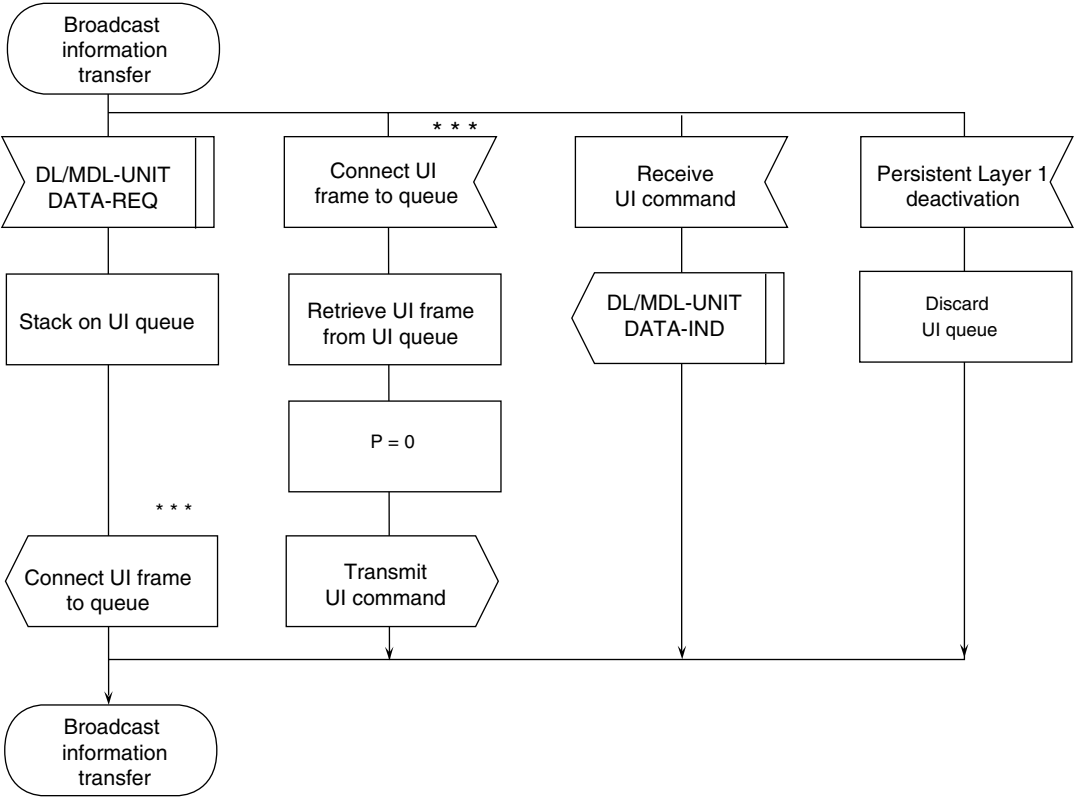
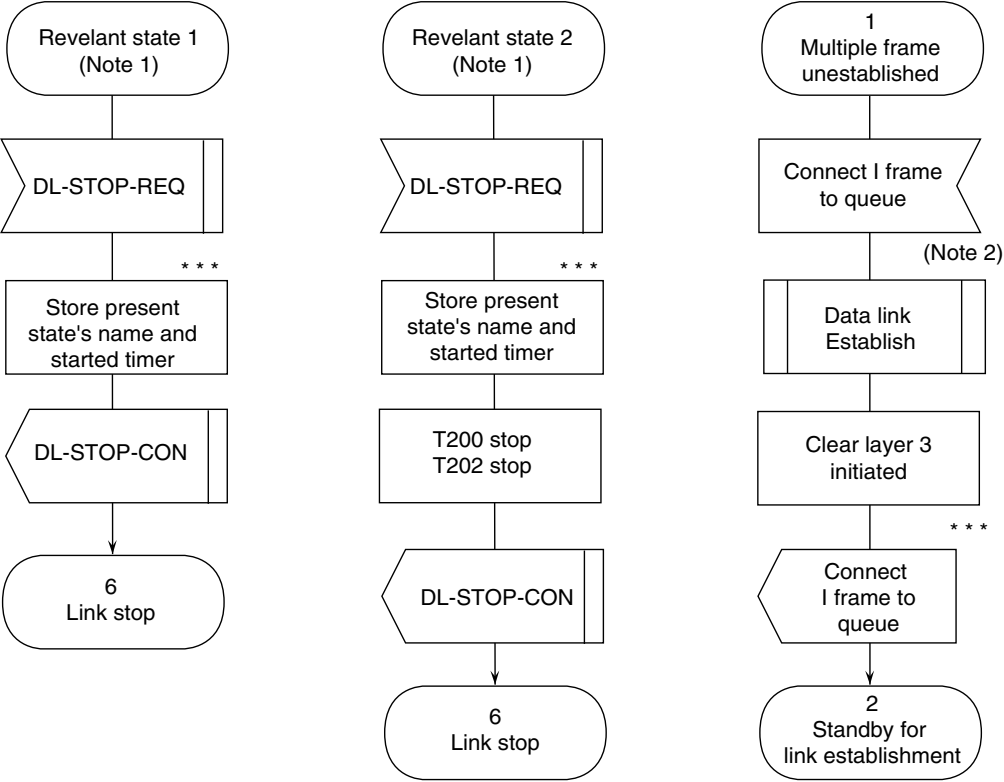


Fig. A. 4.2-56 Layer 2 SDL diagram (55/58)



Note 1 : "Relevant states" are as follows:

Relevant states 1

1. Multiple frame unestablished

Relevant states 2

- 2. Standby for link establishment
- 3. Standby for link release
- 4. Mutiple frame established
- 5. Timer recovery

Note 2 :This can be generated only when the reestablishment procedure is activated by layer 2.

Fig. A. 4.2-57 Layer 2 SDL diagram (56/58)

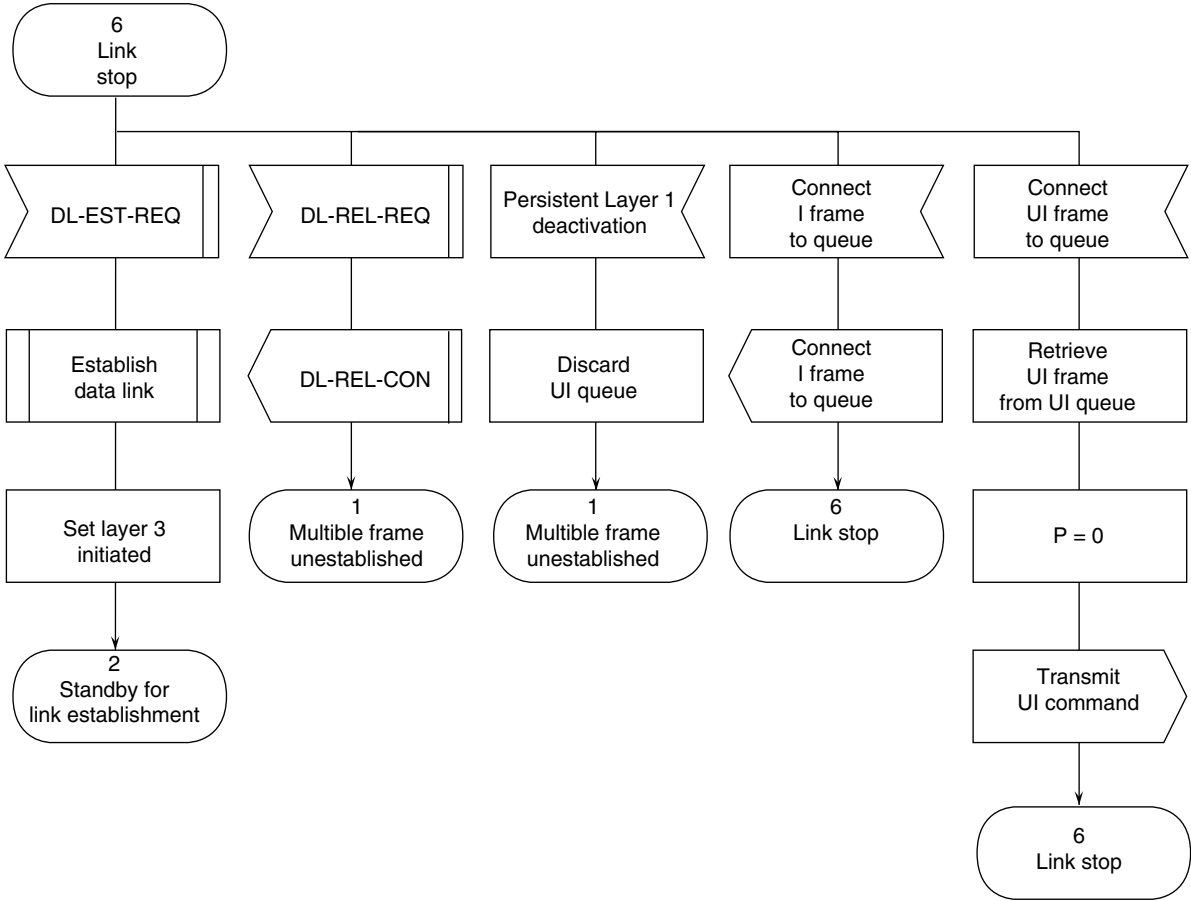
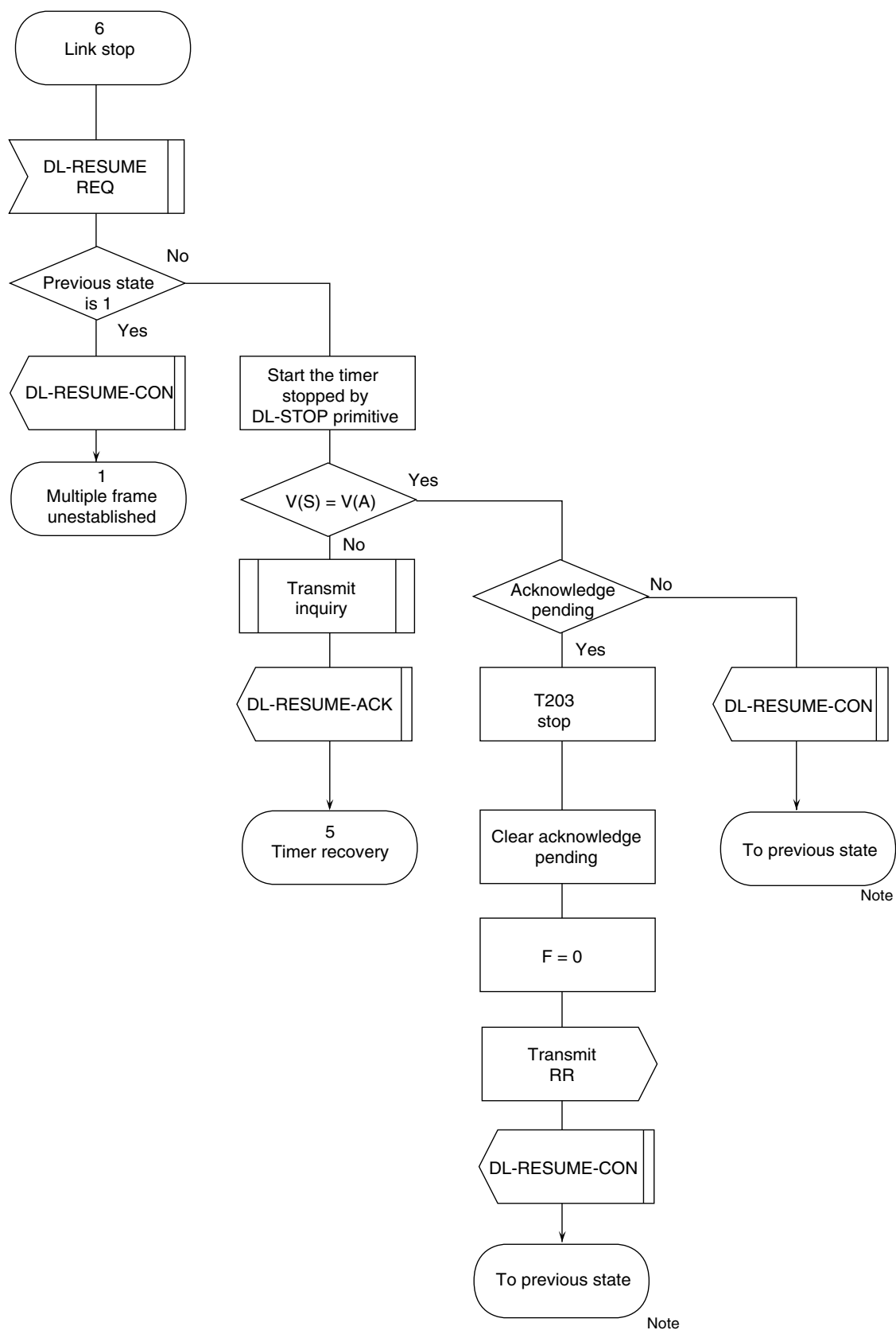


Fig. A. 4.2-58 Layer 2 SDL diagram (57/58)



Note : "Previous state" refers to the state before switching to the link stop state (6).

Fig. A. 4.2-59 Layer 2 SDL diagram (58/58)

A.5 Layer 3 SDL diagrams

A.5.1 Layer 3 Control procedures

This section specifies the procedures of the layer 3 functions RT (radio frequency transmission management), MM (mobility management) and CC (call control) and the relationship among them. The CC procedures must accord with I.451. (This section does not go into detail on the CC procedures.)

(1) Broadcast information (RT)

The network periodically transmits the Broadcast Information message (RT) using DL-UNIT DATA-REQ primitive to all users over the BCCH of control channel in the home zone.

The user (mobile station) receives the Broadcast Information message (RT) in the BCCH on the perch channel of the zone where the mobile station is located when the power is turned on or while the MS is moving between zones during the standby state (RT). The user then switches to the standby state (RT) based on the information used for specifying user operation.

(2) Location registration

After the location identity in the Broadcast Information message (RT) has determined to meet the location registration requirements, the user begins location registration. While location registration is performed, Layer 3 messages are transmitted using DL-UNIT DATA-REQ Primitive over the SCCH.

a. Location registration request (MM)

The user transmits the Location Registration Request message (MM) to the network.

When the location registration is regulated by the restriction information contained in the Broadcast Information message (RT), the user performs the operation according to the restriction information.

When the network receives the Location Registration Request message (MM), the network performs authentication for the user.

b. Authentication (MM)

The network generates the authentication random pattern, and transmits the Authentication Request message (MM) to the user to report the pattern. When the user receives the Authentication Request message (MM), the user converts the random pattern into ciphering pattern using its own ciphering key and reports the authentication ciphering pattern to the network in the Authentication Response message (MM). When the network receives the Authentication Response message (MM), it verifies that the authentication ciphering pattern generated by using the authentication random pattern and by the user's ciphering key in the home memory matches up to the authentication ciphering pattern reported from the user.

c. Location registration acknowledgment (MM)

When the authentication ciphering pattern during the location registration is correct, the network overwrites the location identity in the user's home memory with that of the zone to which the location registration is requested and transmits the Location Registration Acknowledgement message (MM) to the user.

(3) Call setup at the originating end interface

The RT and MM Layer 3 messages in these procedures are sent using DL-UNIT DATA-REQ primitive. The CC Layer 3 messages are sent using DL-ESTABLISH-REQ primitive (the SETUP message (CC), and ALERT message (CC)), or using DL-DATA-REQ primitive (CC messages other than above). The RT message uses the primitive of CC when transmitted along with the CC message by the common platform. The messages for these procedures are transmitted over SCCH before the radio channel setting is completed, but over SACCH or FACCH after it is completed.

a. Call initiation request (CC, RT)

When the user transmits to the network the SETUP message (CC) together with the Originating Condition Report message (RT), which is requested to transmit together with the SETUP message (CC), using the common platform, call setup is initiated.

When origination is regulated by the restriction information contained in the Broadcast Information message (RT), the user performs the operation as indicated in the restriction information.

When the SETUP message (CC) is retransmitted, the Originating Condition Report message (RT) is transmitted together with the SETUP message (CC) by the common platform function.

b. Outgoing call proceeding (CC)

When the network receives the SETUP message (CC) and verifies the call, the network transmits the CALL PROC message (CC) to the user to indicate that the call is in progress.

When the network receives the SETUP message (CC), the network performs authentication of the user.

c. Authentication (MM)

The network generates the authentication random pattern and transmits the Authentication Request message (MM) to the user to report the pattern. The Authentication Request message (MM) may be transmitted together with other messages (RT/CC) by the common platform. When the user receives the Authentication Request message (MM), it converts the received authentication random pattern into the ciphering pattern using its own ciphering keys, and then transmits the authentication ciphering pattern to the network in the Authentication Response message (MM). When the network receives the Authentication Response message (MM), it verifies that the authentication ciphering pattern generated by using the authentication random pattern and by the user's ciphering key in the home memory matches up to the authentication ciphering pattern reported from the user. When the result of the authentication is unacceptable, the call release procedure is initiated according to call clearing specification. When the result of the authentication is acceptable, the call connection is proceeded.

d. Radio traffic channel selection (RT)

When the network receives an Originating Condition Report message, the network selects the radio traffic channel.

To select the radio traffic channel, the network may or may not request the user to measure the reception level.

e. Reception level measurement (RT)

The network may transmit the Reception-level Measurement Request message (RT) to the user to request the level measurement (RT) of the radio traffic channel. When the user receives the Reception-level Measurement Request message (RT), the user performs the level measurement of the specified channel for the specified period of time. Then the user transmits the result of the measurement to the network using the Reception-level Measurement Response message (RT). Then the network rates the result of the measurement received in the Reception-level Measurement Response message (RT). When the result of the measurement is unacceptable, the network reselects the radio traffic channel or assumes that no available channel exists and initiates the call release procedure according to the regulations of call clearing specification.

f. Radio channel set (RT)

After the network selects the radio traffic channel, the network transmits the Radio-channel Set message (RT) to the user. If the Authentication Response message (MM) has been transmitted completely by the user who receives the Radio-channel Set message (RT), the user switches to the designated channel. If the Authentication Response message (MM) transmission has not yet been completed, the user does not perform any operation for the Radio-channel Set message (RT).

g. Call delivered (CC)

When the network receives the report that alerting for the terminating user has started, the network transmits the ALERT message (CC) to the user. This message is not transmitted until the Radio-channel set (RT) is completed.

h. Call received indication (CC)

When the network receives the indication that the call is accepted by the terminating user, the network transmits the CONN message (CC) to the user. This message is not transmitted until the Radio-channel set (RT) is completed.

i. Call reject (RT, MM, CC)

When the terminating user or the network indicates that the incoming call cannot be connected, the network starts the call release procedure according to the call clearing regulations.

(4) Call setup at the terminating end interface

For these procedures, RT and the MM Layer 3 messages are sent using DL-UNIT DATA-REQ primitive. The CC Layer 3 message is transmitted using DL-ESTABLISH-REQ primitive (SETUP message (CC)) or DL-DATA-REQ primitive (ALERT message and the successive messages (CC)).

The layer 3 messages, excluding the Paging message (RT), for these procedures are transmitted over the SCCH before the radio channel set is completed, but over SACCH or FACCH after the radio channel set is completed. The Paging message (RT) is transmitted over the PCH.

a. Paging (RT)

When a mobile terminated call is requested from the network, the network transmits the Paging message (RT) to the user. In the Stand by state (RT) the user monitors the specific PCH slot assigned for the group including the user. If the network transmits a Paging message with the Mobile station number information element in that PCH slot and it is matched up, then the user will respond with a Terminating Condition Report message (RT) to the network.

If the network does not receive any response to this message within the specified period of time, the Paging message (RT) is re-transmitted. When the network receives a Terminating Condition Report message (RT), it starts to perform an authentication for the user.

b. Authentication (MM)

The network generates the authentication random pattern and transmits the Authentication Request message (MM) to the user to report the pattern. When the user receives the Authentication Request message (MM), the user converts the received authentication random pattern into the ciphering pattern using the ciphering keys that the user owns. Then the user transmits the authentication ciphering pattern to the network over the Authentication Response message (MM). When the network receives the Authentication Response message (MM), it verifies that the authentication ciphering pattern generated by using the authentication random pattern and by the user's ciphering key in the home memory matches up to the authentication ciphering pattern reported from the user. When the authentication ciphering pattern is unacceptable, the network initiates the call release procedure according to the call clearing specification. When the authentication ciphering pattern is acceptable, call connection is proceeded.

c. Radio traffic channel select (RT)

When the network receives the Terminating Condition Report message (RT), the network selects radio traffic channel.

To select the radio traffic channel, the network may or may not request the reception-level measurement to the user.

d. Reception-level measurement (RT)

The network may transmit the Reception-level Measurement Request message (RT) to the user to request the Reception-level Measurement (RT) of the selected radio traffic channel. When the user receives the Reception-level Measurement Request message (RT), the user measures the reception-level of the specified channel for the specified period of time. Then the user transmits the result of the measurement to the network over the Reception-level Measurement Response message (RT). Then the network rates the result of the measurement received in the Reception-level Measurement Response message (RT). When the result of the measurement is unacceptable, the network reselects the radio traffic channel or assumes that no available channel exists and initiates the call release procedure according to the call clearing specification.

e. Radio-channel set (RT)

After the radio traffic channel has been selected (RT), the network transmits the Radio-channel Set message (RT) to the user. If the Authentication Response message (MM) has been transmitted completely by the user who received the Radio-channel set message, the user switches the channel to the specified channel. If it has not yet been completed, the user does not perform any operation for the Radio-channel Set message (RT) that the user has received.

f. Call present (CC)

The network transmits the SETUP message (CC) to inform the user that a call is received. This message is not transmitted until the Radio-channel set (RT) is completed.

g. Alerting acknowledgment (CC)

When the user receives the SETUP message (CC) which contains the sufficient call establishment information and meets the requirements for compatibility, the user transmits the ALERT message.

h. Connect received (CC)

The user transmits the CONN message (CC) to the network to report that the call is accepted.

i. Active (CC)

The network transmits the CONN ACK message (CC) to the user on reception of the CONN message (CC).

(5) Call clearing

When the call clearing procedure is initiated, the radio channel set (RT) procedure is immediately halted. Thus, the consecutive layer 3 signals used for this procedure do not span over SCCH and SACCH or SCCH and FACCH. The RT Layer 3 message in this procedure is sent using DL-UNIT DATA-REQ primitive. The CC Layer 3 message is sent using DL-DATA-REQ primitive, or using DL-ESTABLISH-REQ primitive if data link connection is not set up. If CC and RT messages are transmitted together by the common platform they are sent using the CC primitive. The layer 3 messages for this procedure are transmitted over SCCH before radio channel set (RT) is completed, but over SACCH or FACCH after it is completed.

When call clearing is caused by CC, the release procedure for CC is performed. However when the call clearing is caused by RT or MM, CC or MM is released by disconnecting layer 1 by the RT release procedure. The RT release procedure is to release layer 1 by transmitting the Mobile station Release message (RT) from the network to the user before the Radio channel set (RT) is completed or the Radio channel Disconnect message (RT) after it is completed. The RT is autonomously released by the user without reporting the layer 3 message to the network.

At the end of a call, DL-RELEASE-REQ primitive is not required for layer 2, but the release of layer 2 by disconnecting layer 1 is expected.

a. Call clearing before the Radio-channel set (RT) is completed:

i) Call clearing (CC, RT, MM) caused by CC

When the call clearing procedure is initiated by the user, the user transmits the DISC message (CC) to the network. When the network receives the DISC message (CC) from the user, the network, then, transmits the REL message (CC) to the user. When the user receives REL message (CC), the user returns the REL COMP message (CC) to the network, then switches to the Null state (CC). When the network receives the REL COMP message, the network also switches to the Null state (CC) and transmits the Mobile Station Release message (RT) to the user, then switches to the Standby state (RT) and the Null state (MM). When the user receives the Mobile Station Release message (RT), the user switches to the Standby state (RT) and the Null state (MM).

When the call clearing procedure is initiated by the network, the network transmits the DISC message (CC) to the user. When the user receives this message, the user returns the REL message (CC) to the network. When the network receives the REL message (CC) from the user, the network returns the REL COMP message (CC) along with the Mobile Station Release message (RT) to the user by the common platform function, then the network switches to the Null state (CC), Standby state (RT) and the Null state (MM). When the user receives the common platform message consisting of the REL COMP message (CC) and the Mobile Station Release message (RT), the user switches to the Null state (CC), Standby state (RT), and the Null state (MM).

ii) Call clearing (CC, RT, MM) caused by RT or MM

When call clearing procedure is initiated by the user, the user switches to the Null state (CC), Standby state (RT) and the Null state (MM) autonomously. The user does not transmit any layer 3 message to the network.

When call clearing procedure is initiated by the network, the network transmits the Mobile Station Release message (RT) to the user, then switches to the Null state (CC), Standby state (RT) and the Null state (MM). When the user receives the Mobile Station Release (RT) message, the user switches to the Null state (CC), Standby state (RT), and the Null state (MM).

b. Call clearing after the radio-channel set (RT) is completed

i) Call clearing (CC, RT, MM) caused by CC

When call clearing procedure is initiated by the user, the user transmits the DISC message (CC) to the network. When the network receives this message, the network transmits the REL message (CC) to the user. When the user receives this message, the user returns the REL COMP message (CC) to the network, then the CC entity at the user side switches to the Null state (CC). When the network receives the REL COMP message (CC) from the user, the CC entity at the network side switches to the Null state (CC) and transmits either a SETUP message (CC) or a Radio channel Disconnect message (RT). If the user receives a SETUP message (CC), the user proceeds as outlines in item (4) g. Otherwise if the user receives the Radio channel Disconnect message (RT), the user returns the Radio-channel Disconnect Acknowledgment message (RT) to the network and releases the radio traffic channel and switches to the Standby state (RT) and the Null state (MM). When the network receives the Radio channel Disconnect Acknowledge message (RT) from the user, the network releases the radio traffic channel and switches to the Standby state (RT) and the Null state (MM).

When call clearing procedure is initiated by the network, the network transmits the DISC message (CC) to the user. When the user receives this message, the user transmits the REL message (CC) to the network. When the network receives the REL message sent from the user, the network returns the REL COMP message (CC) together with the Radio channel Disconnect message (RT) to the user and switches to the Null state (CC). When the user receives the REL COMP message (CC), the user switches to the Null state (CC). When the user receives the Radio-channel Disconnect message (RT), the user returns the Radio channel Disconnect Acknowledgment message (RT) to the network and releases the radio traffic channel and switches to the Standby state (RT) and the Null state (MM). When the network receives the Radio-channel Disconnect Acknowledgment message (RT), the network releases the radio traffic channel and switches to the Standby state (RT) and the Null state (MM).

ii) Call clearing (CC, RT, MM) caused by RT and MM

When the call clearing procedure is initiated by the user, the user autonomously releases the radio traffic channel and switches to the Null state (CC), Standby state (RT) and the Null state (MM). The user does not report any layer 3 messages to the network.

When the call disconnection recovery procedure (including squelch) is initiated by the network, the network transmits the Radio-channel Disconnect message (RT) to the user. Then the network operator can decide the timing at which the network releases the radio traffic channel and switches to the Null state (CC), Standby state (RT) and the Null state (MM). When the user receives the Radio channel Disconnect message (RT), the user returns the Radio-channel Disconnect Acknowledgment message (RT) to the network and releases the radio traffic channel and switches to the Null state (CC), Standby state (RT) and the Null state (MM).

(6) Frequency management procedures during call in progress

The RT layer 3 messages for these procedures are transmitted over SACCH or FACCH using DL-UNIT DATA-REQ primitive.

The user must perform the following procedures when requested by the network.

a. Condition report information set (RT)

The network sends the Condition Report Information message (RT) to the user to report the information required for the mobile assisted handover (MAHO) performed by the mobile station. When the user receives the Condition Report Information message (RT), the user returns the Condition Report Information Acknowledgment message (RT) to acknowledge the reception of the message to the network. At the same time the user starts mobile assisted handover.

This procedure is necessary for channel handover.

b. Periodical condition report (RT)

After the Condition report information is set (RT), the user transmits the Condition Report 1 message (RT) to the network periodically at the specified time interval to report the reception level information of the home and peripheral zones.

When the network receives Condition Report 1 message (RT), the network determines the channel handover based on the Reception level information contained in the message. When the information meets the condition for the channel handover, the network initiates the channel handover (RT). When it does not meet the condition, the network stores the reception level information and continues communication.

c. Strong electric field detection report (RT)

When the user meets the strong electric field detection condition after the Condition report information is set (RT), the user transmits the Condition Report 2 message (RT) to the network. When the network receives the Condition Report 2 message (RT), the network transmits the Condition Report Acknowledgment message (RT) as the reception acknowledgement. At the same time, the network initiates the channel handover (RT).

d. Channel handover (RT)

When the periodical condition report (RT), strong electric field detection report (RT) or the result of quality supervision for layer 1 meets the channel handover condition, the network initiates this procedure.

i) Radio traffic channel selection (RT)

After initiating the channel handover procedure, the network selects the destined handover channel. To select the radio traffic channel (TCH), the network may or may not request the user to measure the reception level.

ii) Reception level measurement (RT)

The network can transmit the Reception-level Measurement Request message (RT) to the user to request the user for reception-level measurement (RT) of the radio traffic channel that the network selected. When the user receives this message, the user measures the reception level of the specified channel for the specified period of time. Then the user transmits the Reception-level Measurement Response message (RT) to the network to report the result of the measurement. During the measurement, the user halts the MAHO. When the network receives the Reception-level Measurement Response message (RT), the network rates the measured reception level. When it is unacceptable, the network re-selects the radio traffic channel or decides that no available channel exists and stops the channel handover procedure.

iii) Radio channel set (RT)

After selecting the radio traffic channel, the network transmits the Handover Radio-channel Set message (RT) to the user. When the user receives this message, the user switches to the channel specified by the message. Before transmitting the Handover Radio-channel Set message (RT), DL-RELEASE-REQ primitive is not requested to layer 2. The message in the first "I frame", after handover is transmitted using DL-DATA-REQ primitive.

e. System information set (RT)

During the call clearing, the network transmits the System Information message (RT) to the user to report the information required for switching to the Standby state (RT) without receiving the Broadcast Information message (RT). When the user receives the System Information message (RT), the user returns the System Information Acknowledgment message (RT) to the network as the reception acknowledgment. The information to be reported through this procedure differs by each zone, and is reported after the radio channel set (RT) until the call clearing.

f. Condition report state set (RT)

To change the MAHO report condition after the radio condition report information (RT) is set, the network transmits the Condition Report State message (RT) to the user. When the user receives this message, the user returns the Condition Report State Acknowledgment message (RT). At the same time, the user changes the succeeding report conditions as specified.

g. Condition inquiry (RT)

To inquire the reception level information after the condition report information set (RT), the network transmits the Condition Inquiry message (RT) to the user. When the user receives this message, the user transmits the Condition Report 1 message (RT) to the network to report the latest detected information that the user has.

h. VOX control (RT)

To enable the VOX control at the user, the network transmits the VOX Control message (RT) to the user. When the user receives the VOX Control message (RT), the user returns the VOX Control Acknowledgment message (RT) to the network as the reception acknowledgment. At the same time, the user performs the VOX control according to the specified information.

(7) Authentication during call in progress

The MM layer 3 messages for these procedures are transmitted over SACCH or FACCH using DL-UNIT DATA-REQ primitive. The network can initiate this procedure in any timing of TCH active (RT), Null state (MM) or Active state (CC). The user must initiate this procedure when it is requested by the network.

The network generates the authentication random pattern and transmits the Authentication Request message (MM) to the user to report the authentication random pattern. When the user receives the Authentication Request message (MM), the user converts the received authentication random pattern into the ciphering pattern using the ciphering keys that the user owns.

Then the user transmits the authentication ciphering pattern to the network over the Authentication Response message (MM). When the network receives the Authentication Response message (MM), it verifies that the authentication ciphering pattern generated by using the authentication random pattern and by the user's ciphering key in the home memory matches up to the authentication ciphering pattern reported from the user.

(8) Packet origination procedure

When packet communications are started by a request from the user, the user starts packet communication registration. The layer 3 messages during packet communications registration are transmitted to the UPCH using the DL-UNITDATA-REQ primitive.

a. Packet communication registration request (MM)

When the user receives the indication which notifies that the packet physical channel has been activated, the user sends the Packet Communication Registration Request message (MM) to the network.

b. Packet Authentication (MM)

The network generates the authentication random pattern and sends the Packet Authentication Request message (MM) to the user to report this authentication random pattern. In return the user reports the authentication ciphering pattern to the network using the Packet Authentication Response message (MM).

c. Packet Communication Registration Acknowledgment (MM)

When the result of authentication is OK, the network sends the Packet Communication Registration Acknowledgment message (MM) to the user. On reception of this message, the user starts packet transfer.

d. Packet Communications Registration Reject (MM)

When the network rejects a request for packet communication registration, the network transmits the Packet Communication Registration Reject message (MM) to the user.

(9) Packet termination procedure

When packet communications are started by a request from the network, the network starts packet paging. The Paging message is transmitted to the PCH* by using the DL-UNITDATA-REQ primitive and the layer 3 messages during packet communications registration are transmitted to the UPCH by using the DL-UNITDATA-REQ primitive.

a. Paging (RT)

The network transmits the Paging message (RT) to the user.

b. Packet Communication Registration Request (MM)

When the user receives the indication for notifying that the packet physical channel has been activated, the user sends the Packet Communication Registration Request message (MM) to the network.

c. Packet Authentication (MM)

The network generates the authentication random pattern and sends the Packet Authentication Request message (MM) to the user to report this authentication random pattern. In return the user reports the authentication ciphering pattern to the network using the Packet Authentication Response message (MM).

d. Packet Communication Registration Acknowledgment (MM)

When the result of authentication is OK, the network sends the Packet Communication Registration Acknowledgment message (MM) to the user. On reception of this message, the user starts packet transfer.

e. Packet Communication Registration Reject (MM)

When the network rejects a request for packet communication registration, the network transmits the Packet Communication Registration Reject message (MM) to the user.

(10) Packet channel registration procedure (active)

When the user requests for switching the state from the packet standby to the active state or when the packet cycle registration timer sets for time out, the user starts packet channel registration. The layer 3 messages during packet channel registration are transmitted to the UPCH by using the DL-UNITDATA-REQ primitive.

a. Packet Channel Registration Request (MM)

After packet transfer by the CC entity stops, the user sends the Packet Channel Registration Request message (MM).

As authentication is network optional, there are cases when authentication is performed and when not performed.

b. Packet Authentication (MM)

When authentication is performed, the network generates the authentication random pattern and sends the Packet Authentication Request message (MM) to the user to notify this authentication random pattern. In return the user reports the authentication ciphering pattern to the network by using the Packet Authentication Response message (MM).

c. Packet Channel Registration Acknowledgment (MM)

The network sends the Packet Channel Registration Acknowledgment message (MM) to the user. On reception of this message, the user starts packet transfer.

d. Packet Channel Registration Reject (MM)

When the network rejects packet channel registration, the network sends the Packet Channel Registration Reject message (MM) to the user.

(11) Packet channel registration procedure (packet standby)

When the user requests for switching the state from the active to the packet standby states (the packet standby timer sets for time out) or when the packet cycle registration timer sets for time out, the user starts packet channel registration. The layer 3 messages during packet channel registration are transmitted to the UPGH by using the DL-UNITDATA-REQ primitive.

a. Packet Channel Registration Request (MM)

After packet transfer by the CC entity stops, the user sends the Packet Channel Registration Request message (MM). As authentication is network optional, there are cases when authentication is performed and when not performed.

b. Packet Authentication (MM)

When authentication is performed, the network generates the authentication random pattern and sends the Packet Authentication Request message (MM) to the user to notify this authentication random pattern. In return the user reports the authentication ciphering pattern by using the Packet Authentication Response message (MM).

c. Packet Channel Registration Response (MM)

The network sends the Packet Channel Registration Response message (MM) to the user. On reception of this message, the user starts packet transfer.

d. Packet Channel Registration Reject (MM)

When the network rejects packet channel registration, the network sends the Packet Channel Registration Reject message (MM) to the user.

(12) Packet System Information Report procedure

The network starts transmitting the Packet System Information Report to inform the user of the information regarding packet communications. The layer 3 messages during transmission of the Packet System Information Report are transmitted to the UPCH by the DL-UNITDATA-REQ primitive.

a. Packet System Information Report (RT)

The network transmits the Packet System Information Report message (RT) to the user. When packet channel registration is required, the user starts packet channel registration.

(13) Packet communication deregistration procedure (activated by the user)

When the user requests for ending packet communications, the network starts packet communications deregistration. The layer 3 messages during packet communications deregistration are transmitted to the UPCH by using the DL-UNITDATA-REQ primitive.

a. Packet Communication Deregistration Request (MM)

After the CC entity stops packet transfer, the user sends the Packet Communication Deregistration Request message (MM).

As authentication is network optional, there are cases when authentication is performed and when not performed.

b. Packet authentication (MM)

When authentication is performed, the network generates the authentication random pattern and sends the Packet Authentication Request message (MM) to the user to notify this authentication random pattern. In return the user reports the authentication ciphering pattern to the network by using the Packet Authentication Response message (MM).

c. Packet Communication Deregistration Response (MM)

The network sends the Packet Communication Deregistration Response message (MM) to the user. On reception of this message, the user stops CC packet transfer.

d. Packet Communications Deregistration Reject (MM)

When the network rejects a request for packet communication deregistration, the network transmits the Packet Communications Deregistration Reject message (MM) to the user.

(14) Packet communication registration state inquiring procedure

The network starts inquiring the packet communication state to detect the state of the user. The layer 3 messages during packet communications registration are sent to the UPCH by using the DL-UNITDATA-REQ primitive.

a. Packet Communication Inquiry (MM)

The network sends a Packet Communication State Inquiry message (MM). If the network receives any layer 3 message from the user, it ends the packet communication inquiring procedure.

If no sequence procedure described in Section 4.3.8 is activated on reception of the Packet Communication Inquiry message (MM), the user starts the packet communications registration procedure after packet transfer by the CC entity is terminated.

b. Packet communication registration request (MM)

The user sends the Packet Communication Registration Request message (MM) to the network when it receives the indication which notifies that the packet physical channel has been activated.

c. Packet authentication (MM)

The network generates the authentication random pattern and sends the Packet Authentication Request message (MM) to the user to notify the authentication random pattern. In return the user reports the authentication ciphering pattern to the network by using the Packet Authentication Response message (MM).

d. Packet Communications Registration Acknowledgment (MM)

When the result of authentication is OK, the network sends the Packet Communication Registration Response message (MM) to the user. On reception of this message, the user starts packet transfer.

e. Packet Communications Registration Reject (MM)

When the network rejects a request for packet communication registration, the network transmits the Packet Communications Registration Reject message (MM) to the user.

(15) Packet paging procedure

When the network requests the user to switch the state from the packet standby state to the active state, the network activates packet paging. The Packet Paging message is sent to the UPCH using the DL-UNITDATA-REQ primitive.

a. Packet Paging (RT)

The network sends the Packet Paging message (RT) to the user.

When the user receives this Packet Paging message (RT), the user stops packet transfer by the CC entity and then activates the packet channel registration procedure.

b. Packet Channel Registration Request (MM)

After the user stops packet transfer by the CC entity, the user transmits the Packet Channel Registration Request message (MM).

As authentication is network optional, there are cases when authentication is performed and when not performed.

c. Packet authentication (MM)

The network generates the authentication random pattern and sends the Packet Authentication Request message (MM) to the user to notify the authentication random pattern. In return, the user reports the authentication ciphering pattern to the network by using the Packet Authentication Response message (MM).

d. Packet Channel Registration Response (MM)

When the result of authentication is OK, the network sends the Packet Channel Registration Response message (MM) to the user. On reception of this message, the user starts CC packet transfer.

e. Packet Channel Registration Reject (MM)

When the network rejects a request for packet communication registration, the network transmits the Packet Channel Registration Reject message (MM) to the user.

(16) Zone information report procedure (RT)

When the user obtains information regarding packet communications in the active state, the user activates the zone information report procedure. The layer 3 messages during zone information reporting are sent to the UPCH by using the DL-UNITDATA-REQ primitive.

a. Zone Information Request

The user sends the Zone Information Request message (RT) to the network.

b. Zone Information Notification

The network in return sends the Zone Information Notification message (RT) to the user.

(17) Packet communications disconnecting procedure (activated by the network)

When the network requests to end packet communications, packet communications disconnection procedure is activated. Layer 3 messages during packet communication disconnecting procedure are sent to the UPCH by using the DL-UNITDATA-REQ primitive.

a. Packet Communications Disconnection Request (MM)

The network sends the Packet Communications Disconnection Request message (MM) to the user.

b. Packet Communications Disconnection Acknowledgment (MM)

The user sends a Packet Communications Disconnection Acknowledgment message (MM) to the network to end packet transfer by the CC entity.

(18) Packet transfer

Packet Transfer message (CC) is transmitted on the UPCH by using DL-DATA-REQ primitive.

(19) Procedure for Switched Virtual Circuit connection(user activated)

When the user requests Switched Virtual Circuit connection during packet communications, this procedure is performed. In cases when the mobile station switches channels between terminal registration areas during virtual circuit connection and the Communication state indication of the "Communication state" information element indicates the "Communication initial state" in the Packet Communication Registration Response message from the network, the communication state on the network side is not always maintained. In such cases, the mobile station can also perform this Switched Virtual Circuit connection procedure by assigning the connection destination code of the previous communications. However, if virtual circuit connection is not permitted in the Zone service information in the Broadcast Information message (RT), Zone Information Report message (RT) or Packet System Information Report message (RT), it is not possible to perform this Switched Virtual Circuit connection procedure.

Layer 3 messages during Switched Virtual Circuit connection are transmitted on the UPCH by using the DL-DATA-REQ primitive.

a. Virtual Circuit Connection Request (CC)

Only if the use of the protocol for Switched Virtual Circuit connection is explicitly permitted by the network by means of the Packet Communications Registration Response message, the user can transmit a Virtual Circuit Connection Request (CC).

b. Virtual Circuit Connection Response (CC)

Upon receipt of a Virtual Circuit Connection Request message (CC), the network sends a Virtual Circuit Connection Response message (CC) only when it permits the connection.

c. Virtual Circuit Connection Reject (CC)

Upon receipt of a Virtual Circuit Connection Request message (CC), the network sends a Virtual Circuit Connection Reject message (CC) when it does not permit the connection.

When the user terminates packet communications upon receipt of the Virtual Connection Reject message (CC) or on time out as the Virtual Connection Response message (CC) cannot be received, the user must perform a packet communications deregistration procedure.

(20) Procedure for selective virtual circuit connection release, (network activated)

This procedure is performed when the user requests Switched Virtual Circuit connection during packet communications. Layer 3 messages during Switched Virtual Circuit connection are transmitted on the UPCH by using DL-DATA-REQ primitive.

a. Virtual Circuit Connection Request (CC)

The network transmits a Virtual Circuit Connection Request message (CC).

b. Virtual Circuit Connection Response (CC)

Upon receipt of a Virtual Circuit Connection Request message (CC), the user sends a Virtual Circuit Connection Response message (CC) when it permits the connection.

c. Virtual Circuit Connection Reject (CC)

Upon receipt of a Virtual Circuit Connection Request message (CC), the user sends a Virtual Circuit Connection Reject message (CC) when it does not permit the connection.

When the user terminates packet communications upon receipt of the Virtual Circuit Connection Reject message (CC), the user must perform a packet communication registration release procedure.

(21) Procedure for switched virtual circuit connection (user activated)

This procedure is performed when the user requests Switched Virtual Circuit connection release while Switched Virtual Circuit connection is in progress. However, to terminate packet communications by releasing the connection when there is only one active connection on the user side, the packet communications deregistration processing is performed instead of this procedure.

Layer 3 messages during selective virtual circuit connection release are transmitted on the UPCH by using DL-DATA-REQ primitive.

a. Virtual Circuit Connection Release Request (CC)

The user transmits a Virtual Circuit Connection Release Request message (CC).

b. Virtual Circuit Connection Release Response (CC)

Upon receipt of a Virtual Circuit Connection Release Request message (CC), the network sends a Virtual Circuit Connection Release Response message (CC).

(22) Procedure for Switched Virtual Circuit connection release (network activated)

This procedure is performed when the network requests the Switched Virtual Circuit connection release while Switched Virtual Circuit connection is in progress. Layer 3 messages during Switched Virtual Circuit connection release are sent on the UPCH by using DL-DATA-REQ primitive.

a. Virtual Circuit Connection Release Request (CC).

The network transmits a Virtual Circuit Connection Release Request (CC).

b. Virtual Circuit Connection Release Response (CC).

Upon receipt of a Virtual Circuit Connection Release Request message (CC), the user sends a Virtual Circuit Connection Release RESP message (CC).

(23) Switched Virtual Circuit connection user information transfer procedure

After Switched Virtual Circuit connection, "User Information" messages are transmitted over the UPCH using DL-DATA-REQ primitive.

(24) Voice Terminating Method 2 procedures during packet communications

This procedure is activated when a voice call is received for the user in packet communications under the conditions wherein both the mobile station and the network support the Voice Terminating Method 2. The layer 3 messages are transmitted over the UPCH by means of a DL-UNITDATA-REQ primitive during voice paging 2 during packet communications.

Instead of starting a voice paging procedure using UPCH Voice Paging 2 message described in sub-clause a, the network has the option to send a UPCH Radio Channel Assignment at the Radio Channel Assignment state described in sub-clause c(3), immediately to a mobile station that is in the packet active state. Such MS can then either start to establish TCH and accept the call, or send a UPCH Voice Paging 2 Response when it decides not to continue the voice call setup signalling.

a. UPCH Voice Paging 2 (RT)

The network sends a UPCH Voice Paging 2 message (RT) or a UPCH Voice Paging 2 Inquiry message (RT) to the user.

Upon receipt of either the UPCH Voice Paging 2 message (RT) or the UPCH Voice Paging 2 Inquiry message (RT), the user sends a UPCH Voice Paging 2 Acknowledgement message (RT). If the UPCH Voice Paging 2 Response message (RT) or a UPCH Terminating Condition Report message (RT) is sent within the timer value defined in Section 4.3.5.4, transmission of the UPCH Voice Paging 2 Acknowledgement message (RT) is omitted.

Note that the communication mode during packet communications registration or packet channel registration must remain in the "active" state after the user sends the UPCH Voice Paging 2 Acknowledgement message (RT) either until the UPCH Voice Paging 2 Response procedure, or the UPCH Voice Paging Connection 2 procedure is completed normally or erroneously due to expiration or other reason, or until the user receives the UPCH Voice Paging Stop message (RT) from the network.

Also, upon receipt of the UPCH Voice Paging 2 Paging message (RT), the mobile station must perform low-speed continuous reception using the slot in which the UPCH Voice Paging 2 Acknowledge message (RT), UPCH Voice Paging 2 Response message (RT) or the UPCH Terminating Condition Report message (RT) are transmitted in until packet communications registration or packet channel registration is completed normally. When sending the CC message until the packet communication registration procedure or the packet channel registration procedure is completed, the CC message must be sent after the state switches to the active state by the packet channel registration.

b. UPCH Voice Paging 2 Response (RT)

When the user decides not to continue the voice call setup signalling following the reception of the UPCH Voice Paging 2 message (RT), the user sends a UPCH Voice Paging 2 Response message (RT) to the network.

Upon receipt of the UPCH Voice Paging 2 Response message (RT), if offering service identified by the termination response information from the user, sets the same content and sends the message to the user. If the service is not offered, the network sets the information for prompting re-response or for notifying that the remote end party has been disconnected in the UPCH Voice Paging 2 Response Acknowledgement message (RT), then sends this message to the user.

Upon receipt of the UPCH Voice Paging 2 Response Acknowledgement message (RT), if the information for prompting re-response is included, the user maintains the UPCH reception 2 response hold state. Also, the user may set different terminating response information in the UPCH Voice Paging 2 Response message (RT), and send it again to the network until the user receives the UPCH Voice Paging Stop message from the network.

c. UPCH Voice Paging 2 Connection (RT)

When the user decides to continue the voice call setup signalling following the reception of the UPCH Voice Paging 2 message (RT), the user sends a UPCH Terminating Condition Report message (RT) to the network.

i) Radio Channel selection (RT)

The network which has started paging connection selects a radio channel for connection. When selecting the radio channel, the network requests reception level measurement or not case by case.

ii) Reception Level Measurement (RT)

To request the reception level measurement (RT) of the selected radio channel, the network can send the Reception Level Measurement Request message (RT) to the user. When the user receives this message, the user measures the reception level of the specified channel for the specified period of time and sends the result of the measurement to the network by the Reception Level Measurement Response message (RT). Upon receipt of this response message, the network rates the result of the measurement. If the result is NG, the network reselects a radio channel or decides that there are no available channels and stops terminating connection.

iii) Radio Channel Assignment (RT)

After selecting the radio channel, the network sends the UPCH Radio Channel Set message (RT) to the user. When the user receives this message, the user switches to the channel specified by this message.

iv) Call Setup (CC)

The call setup procedures after switching to the traffic channel must comply with the procedures described in (f) to (l) in section (4).

To start packet communications again after user has switched to the traffic channel by UPCH terminating connection and ended communications with the traffic channel, the user must perform the packet originating procedure. In this instance, the user can execute the packet originating procedure either in the communications initial state or communications active state, however, if the communication initial state is notified by the response from the network, the user must execute the processing corresponding to the communications initial state.

(25) Packet channel handover procedure

When requesting assignment of a packet physical channel to the user after the packet communication has been initiated upon request from the user or the network, the network initiates the packet channel handover procedure. Layer 3 messages during the packet channel handover procedure are transmitted on the UPCH using the DL-UNITDATA-REQ primitive.

a. Packet Channel Handover Request (RT)

The network sends a Packet Channel Handover Request message (RT) to the user.

b. Packet Communication Registration Request (MM)/ Packet Channel Registration Request (MM)

Upon receipt of an indication that the activation of the packet physical channel notified by the Packet Channel Handover Request is completed, the user sends a Packet Communication Registration Request message (MM) or Packet Channel Registration Request message (MM).

c. Packet Authentication (MM)

If authentication is performed, the network generates an authentication random pattern and sends a Packet Authentication Request message (MM) to the user to notify the authentication random pattern. The user reports the authentication ciphering pattern to the network by means of the Packet Authentication Response message (MM).

d. Packet Communication Registration Response (MM)/ Packet Channel Registration Response (MM)

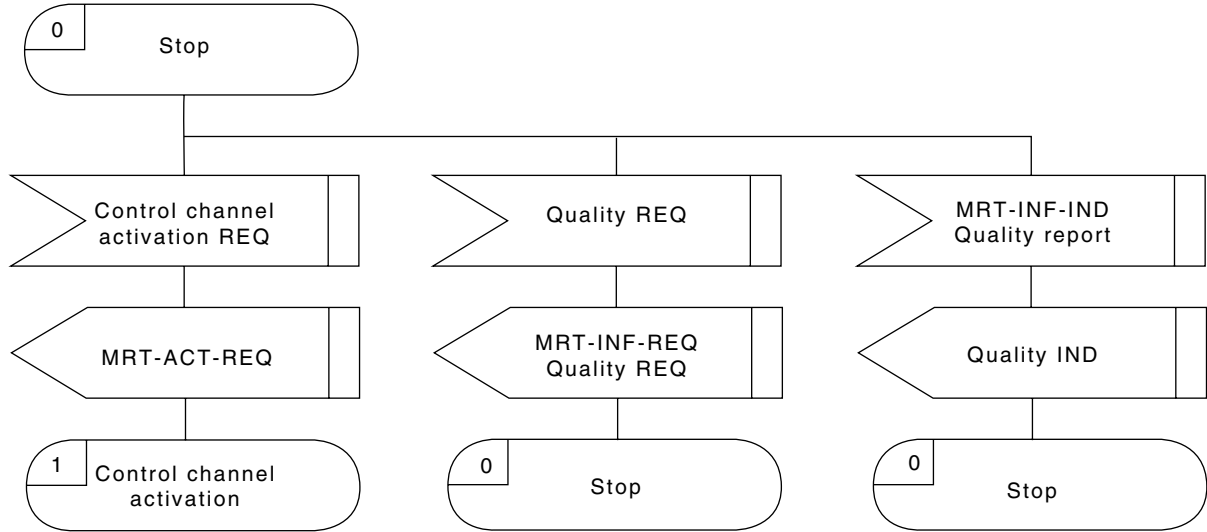
The network sends a Packet Communication Registration Response message (MM) or Packet Channel Registration Response message (MM) to the user. When the user receives this message, the user starts packet transmission in CC.

e. Packet Communication Registration Reject (MM)/ Packet Channel Registration Reject (MM)

When rejecting packet channel registration, the network sends a Packet Communication Registration Reject message (MM) or Packet Channel Registration Reject message (MM) to the user.

A.5.2 Layer 3 SDL diagram

Figures from Fig.A.5-1 to Fig.A.5-4 are blank



- Note 1. : The SDL diagrams shown on this and the following pages describe Layer-3 protocol controls.
- Note 2. : "Unexpected messages" indicated on this and the following pages mean messages that are not supposed to be received under each applicable condition.
- Note 3. : Messages received will be discarded if they are not defined in this standard.

Fig. A. 5-5 Layer 3 RT SDL diagram (User side) (1/43)

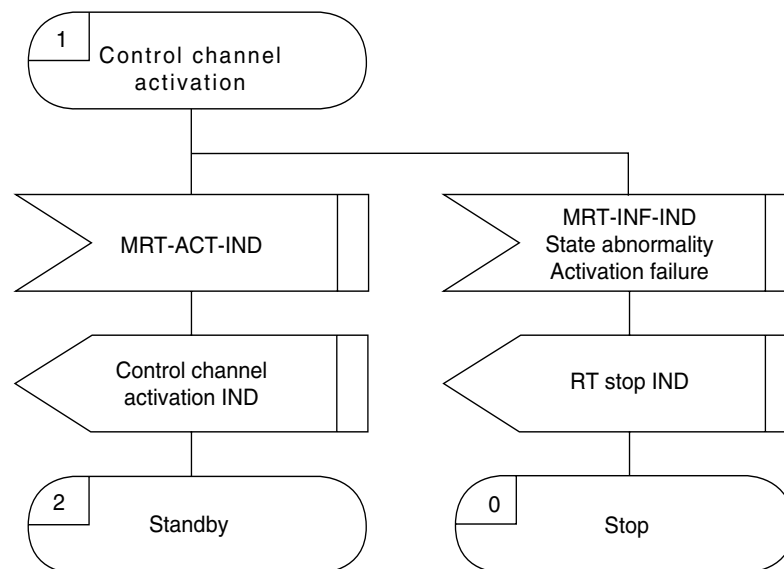


Fig. A. 5-6 Layer 3 RT SDL diagram (User side) (2/43)



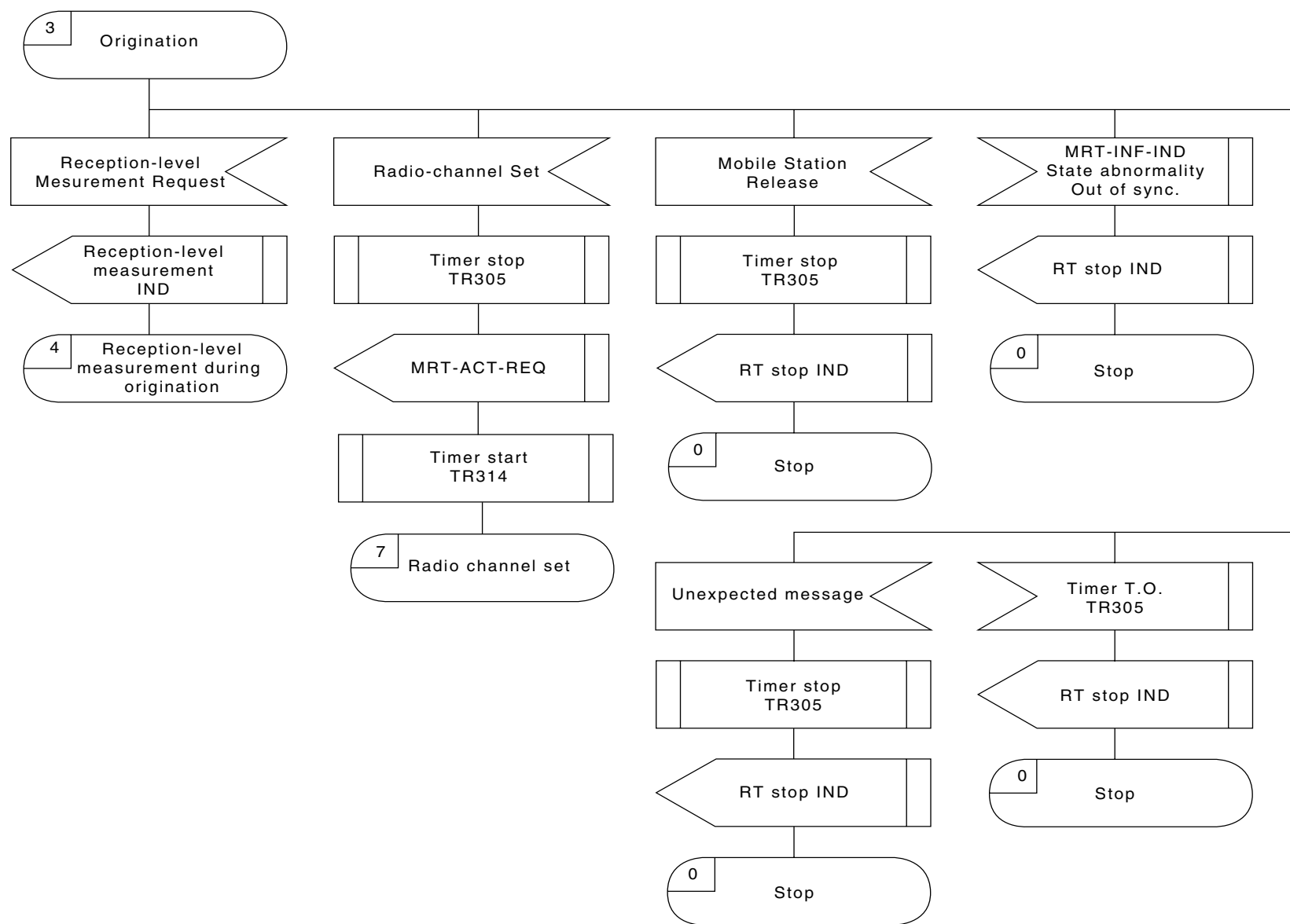


Fig. A. 5-8 Layer 3 RT SDL diagram (User side) (4/43)

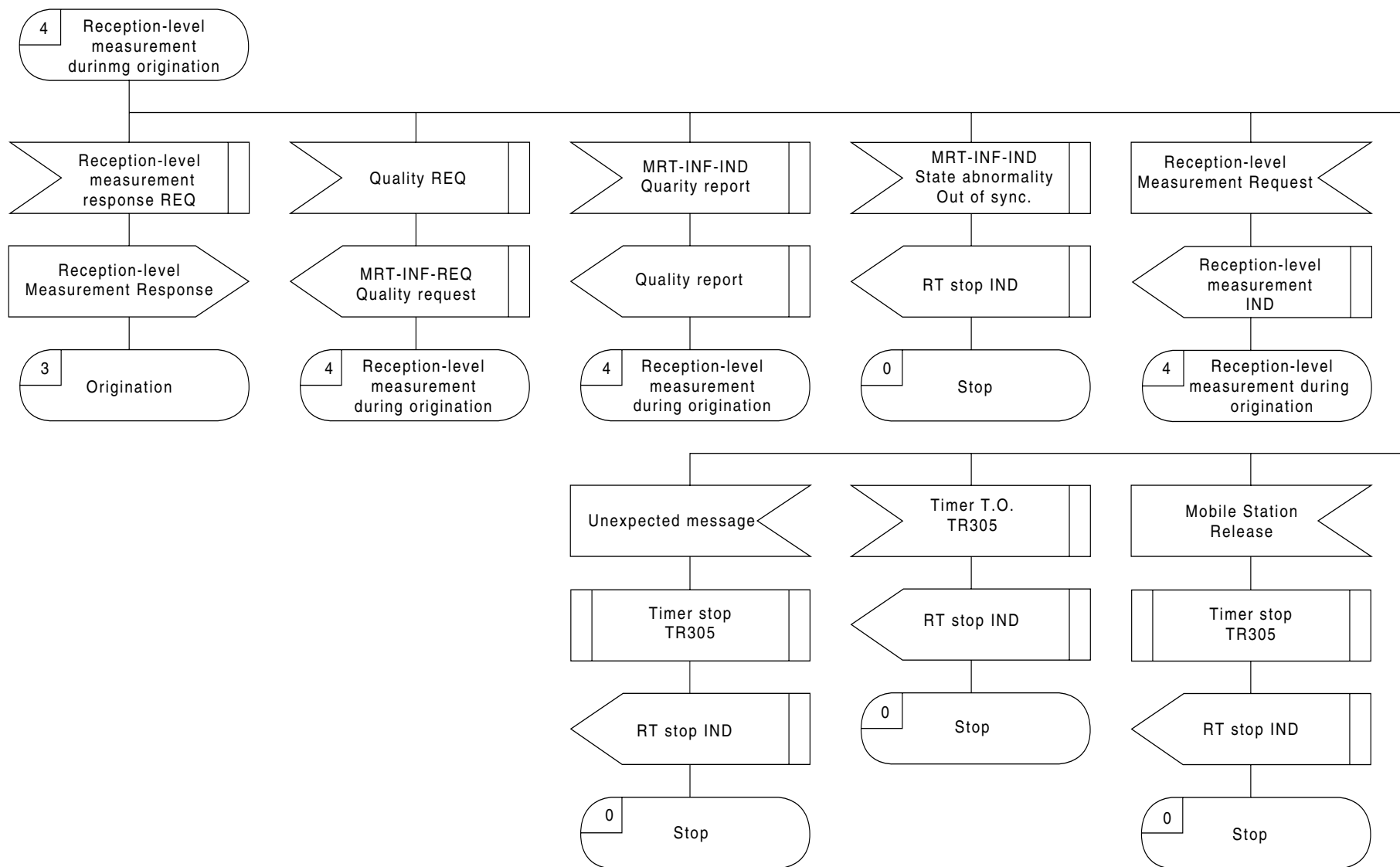


Fig. A. 5-9 Layer 3 RT SDL diagram (User side) (5/43)

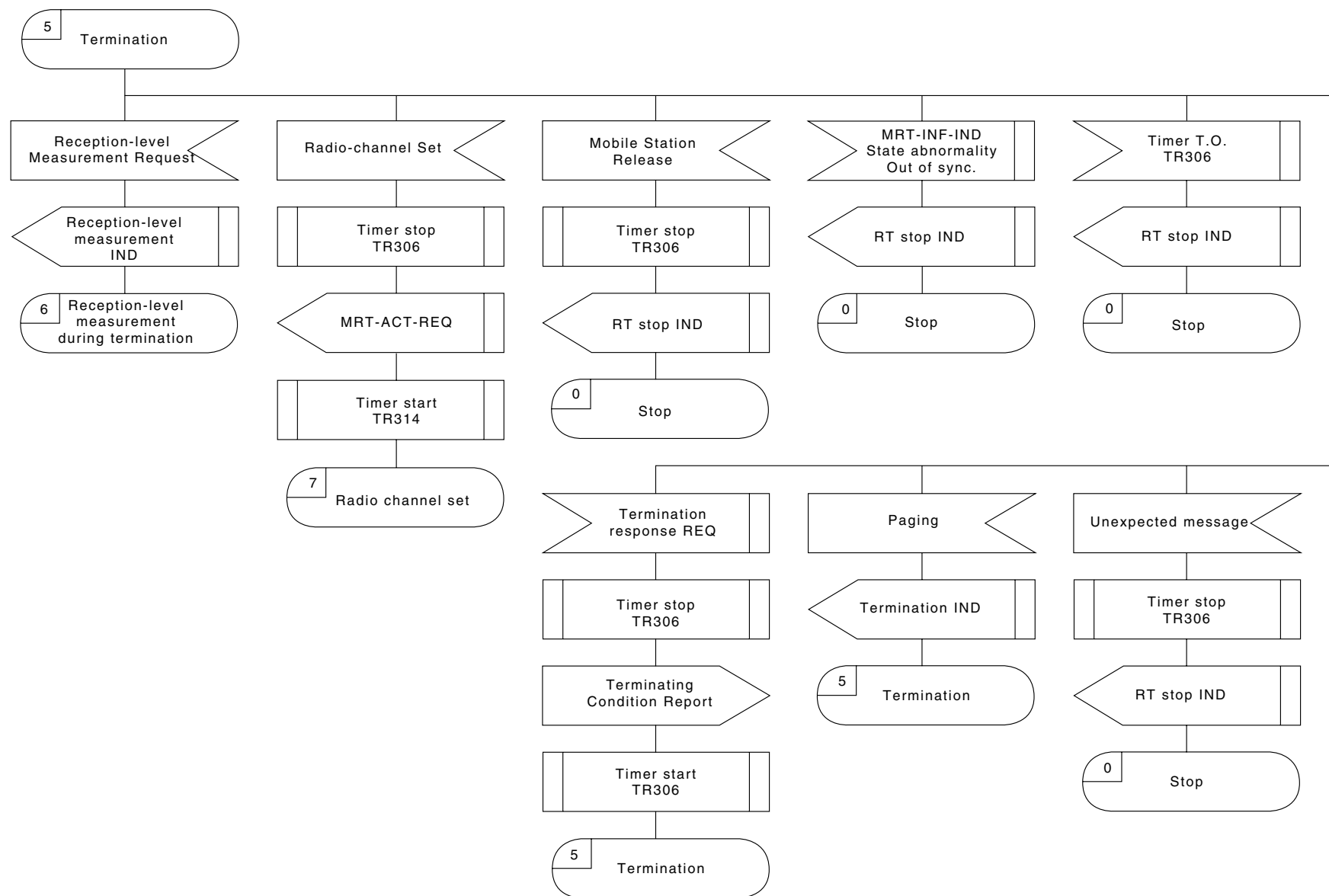


Fig. A. 5-10 Layer 3 RT SDL diagram (User side) (6/43)

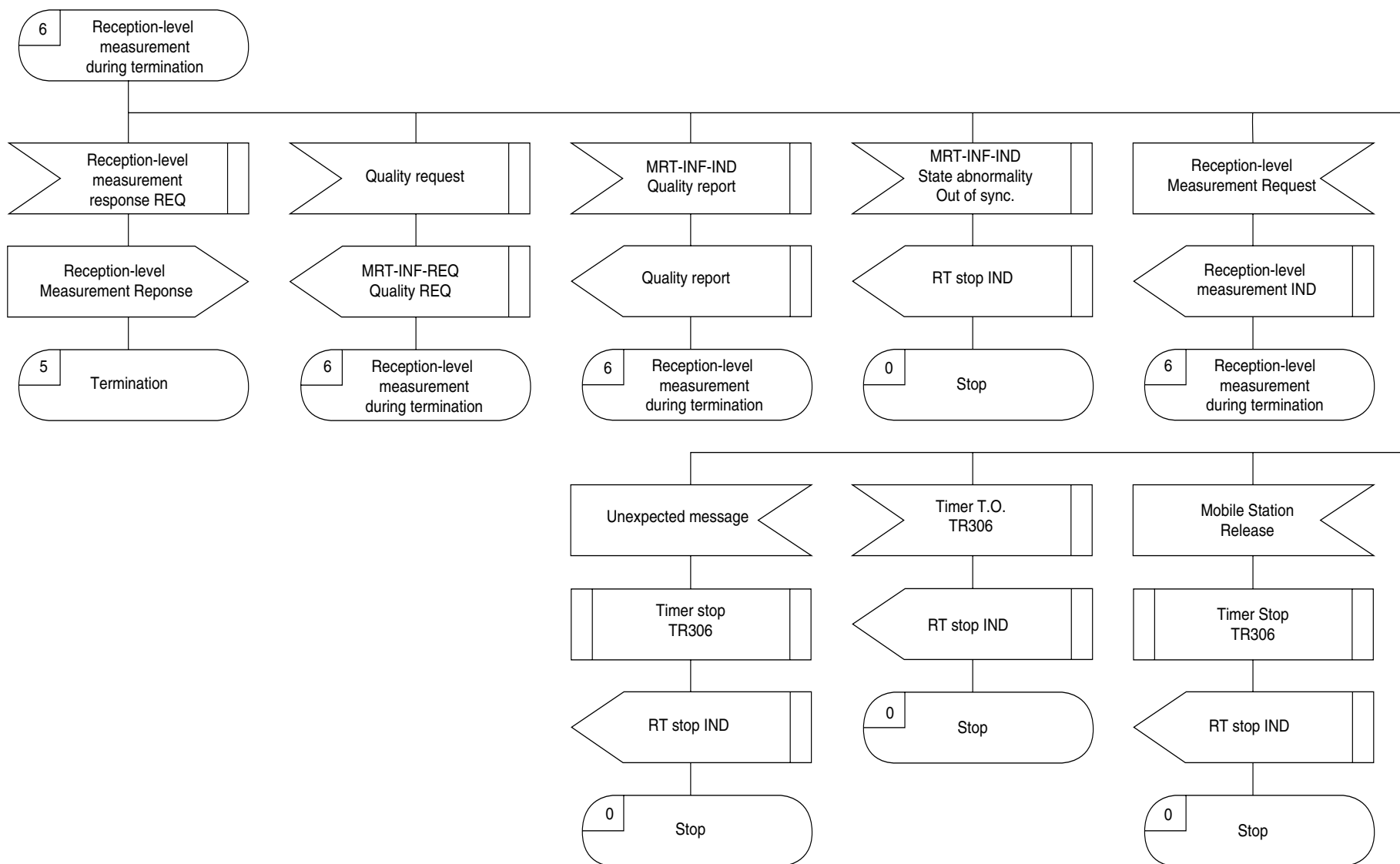


Fig. A. 5-11 Layer 3 RT SDL diagram (User side) (7/43)

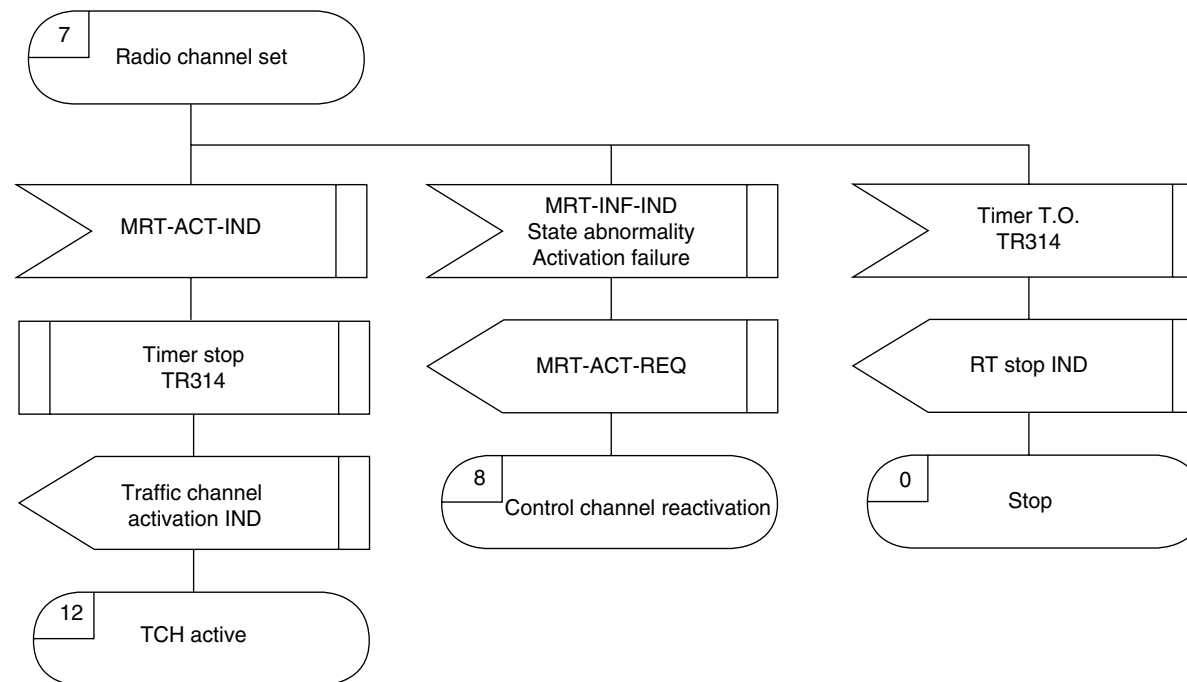


Fig. A. 5-12 Layer 3 RT SDL diagram (User side) (8/43)

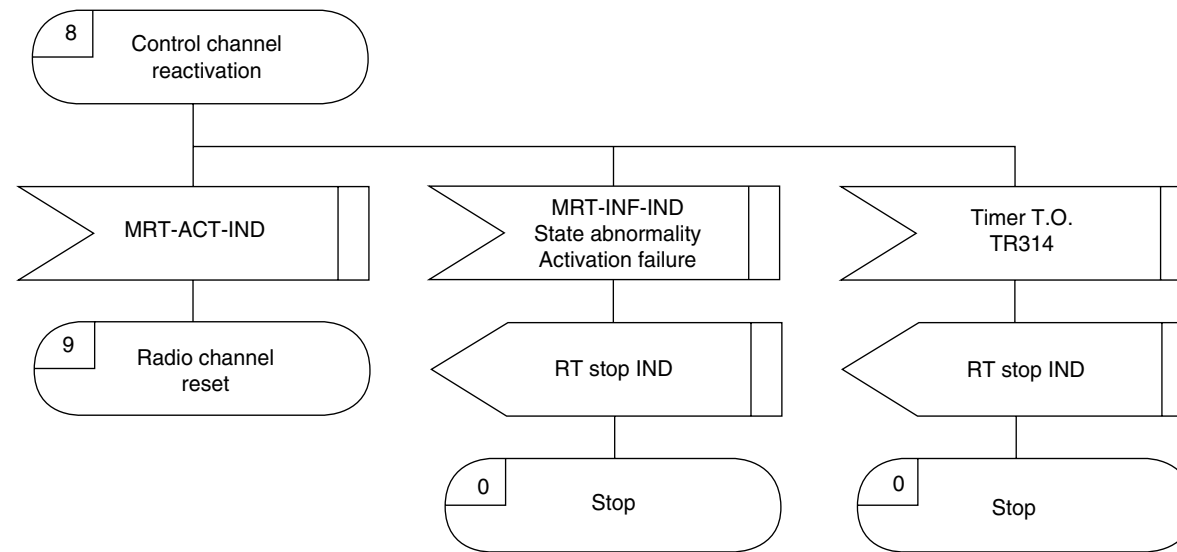


Fig. A. 5-13 Layer 3 RT SDL diagram (User side) (9/43)

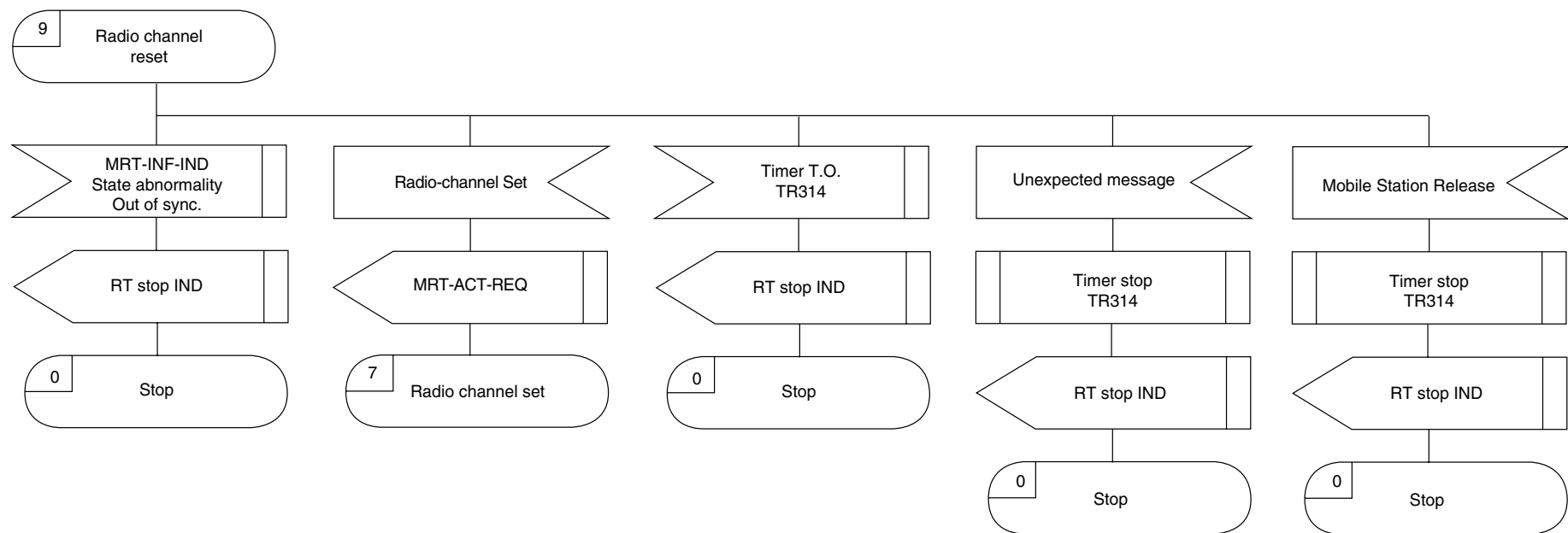


Fig. A. 5-14 Layer 3 RT SDL diagram (User side) (10/43)

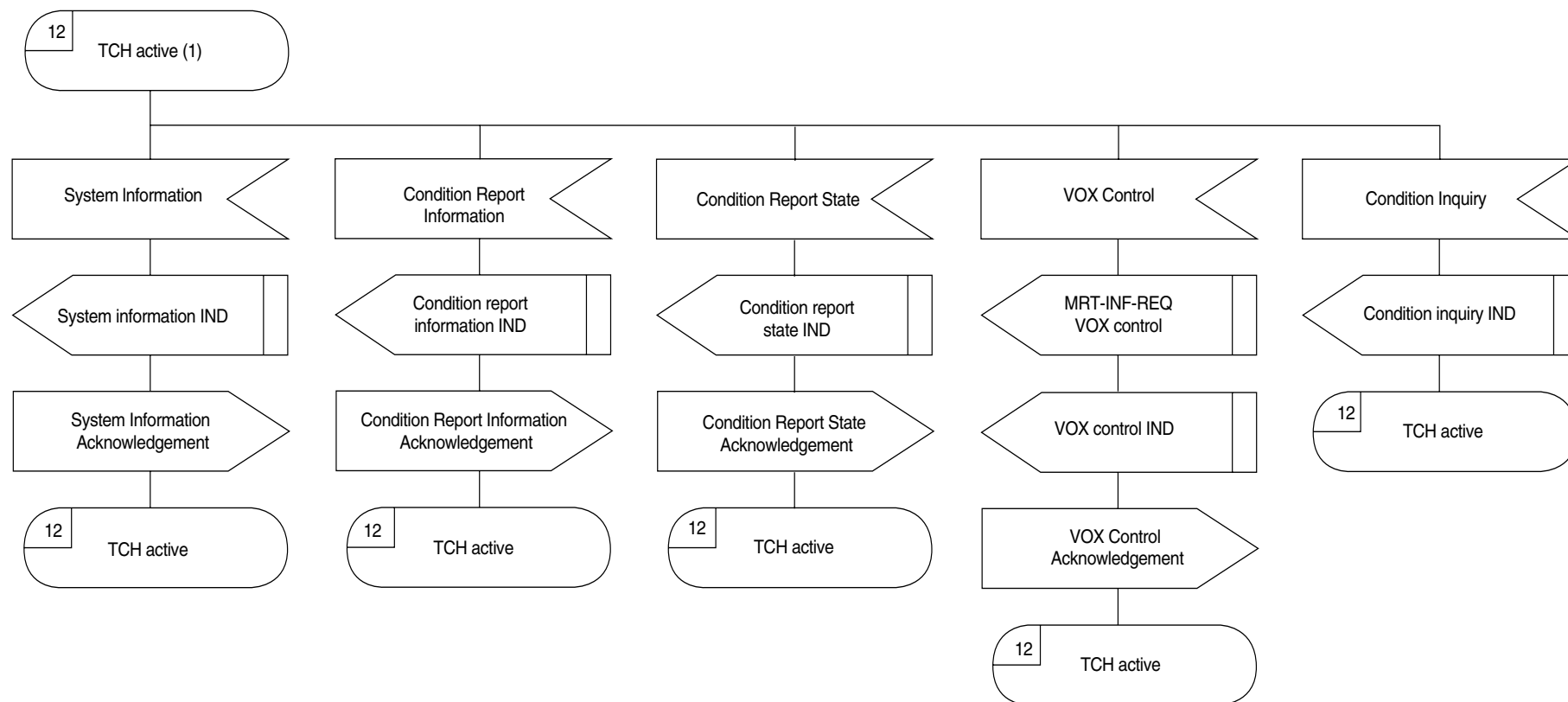


Fig. A. 5-15 Layer 3 RT SDL diagram (User side) (11/43)

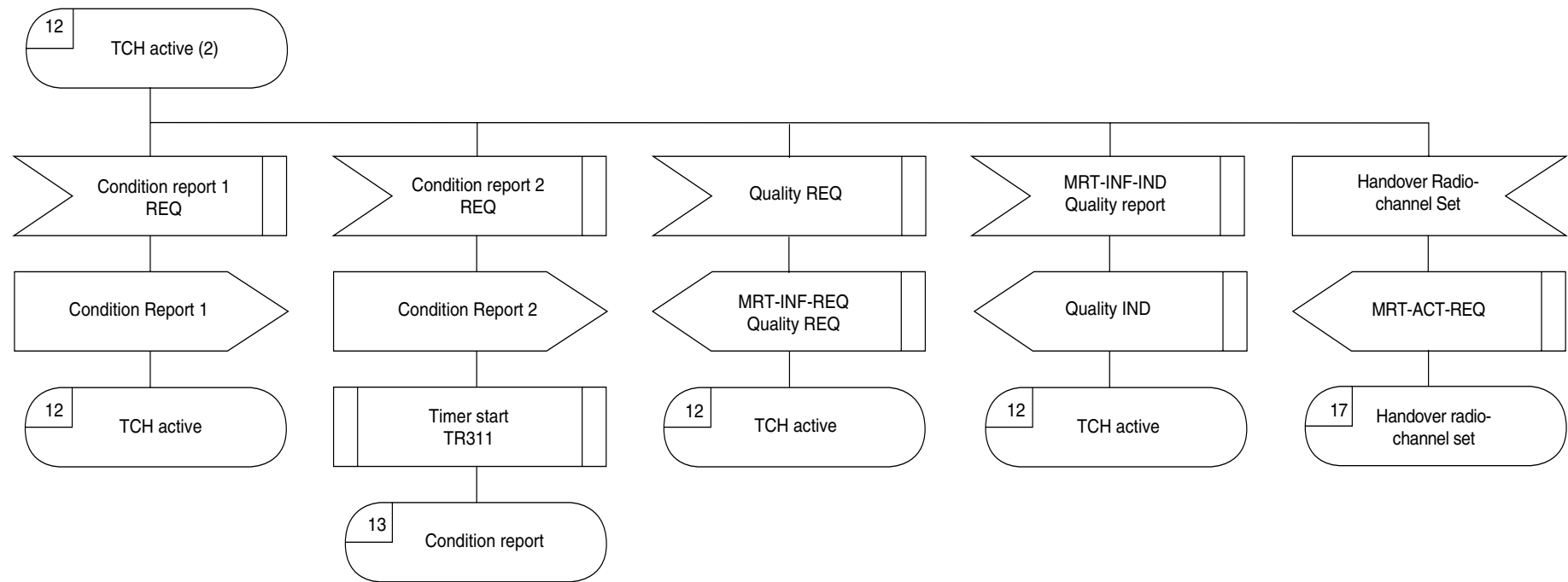


Fig. A. 5-16 Layer 3 RT SDL diagram (User side) (12/43)

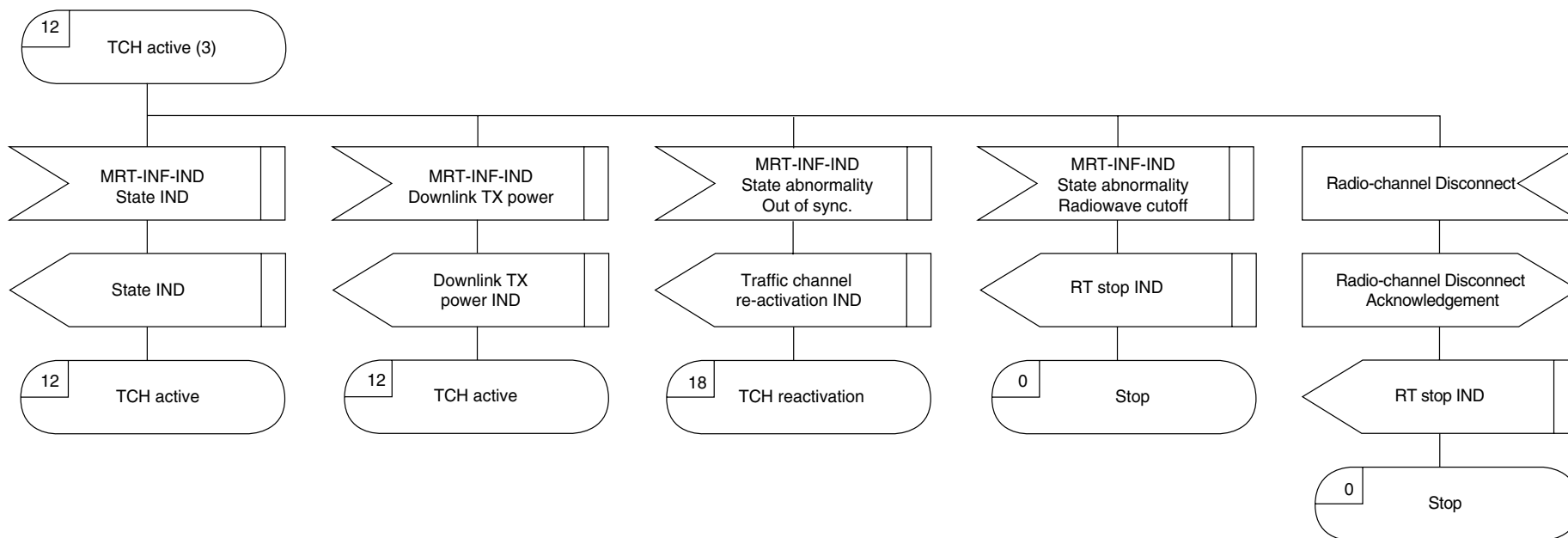


Fig. A. 5-17 Layer 3 RT SDL diagram (User side) (13/43)

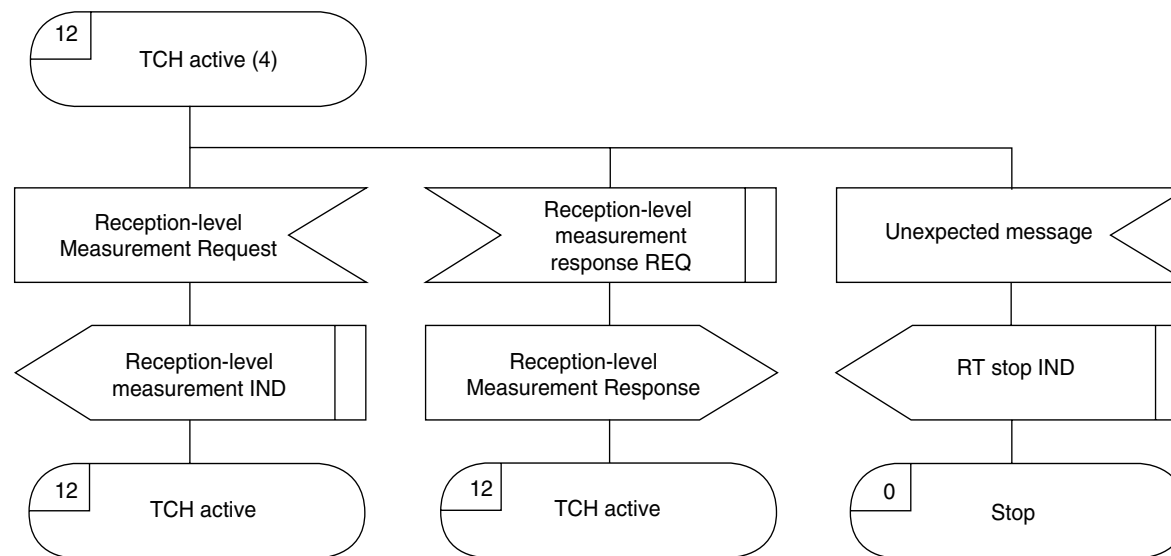


Fig. A. 5-18 Layer 3 RT SDL diagram (User side) (14/43)

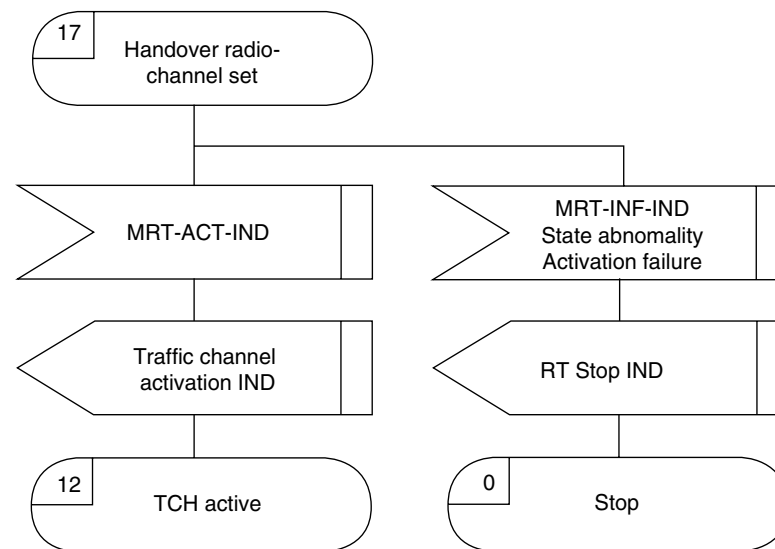


Fig. A. 5-19 Layer 3 RT SDL diagram (User side) (15/43)

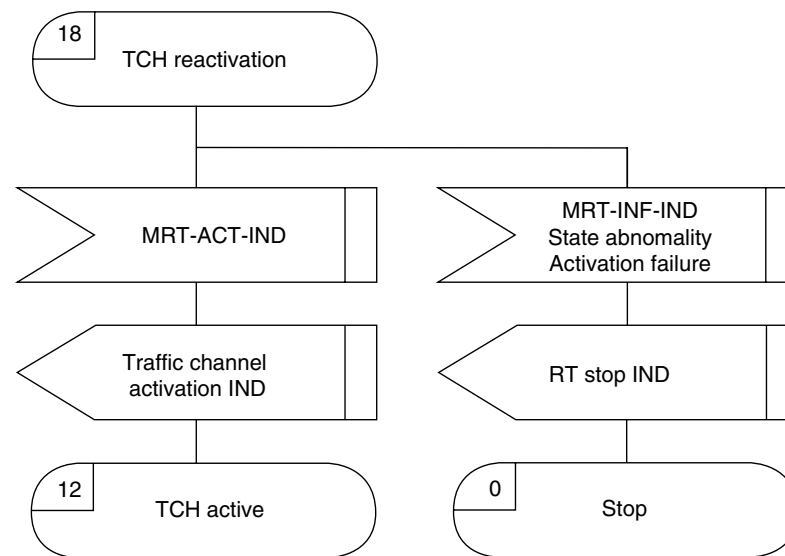


Fig. A. 5-20 Layer 3 RT SDL diagram (User side) (16/43)

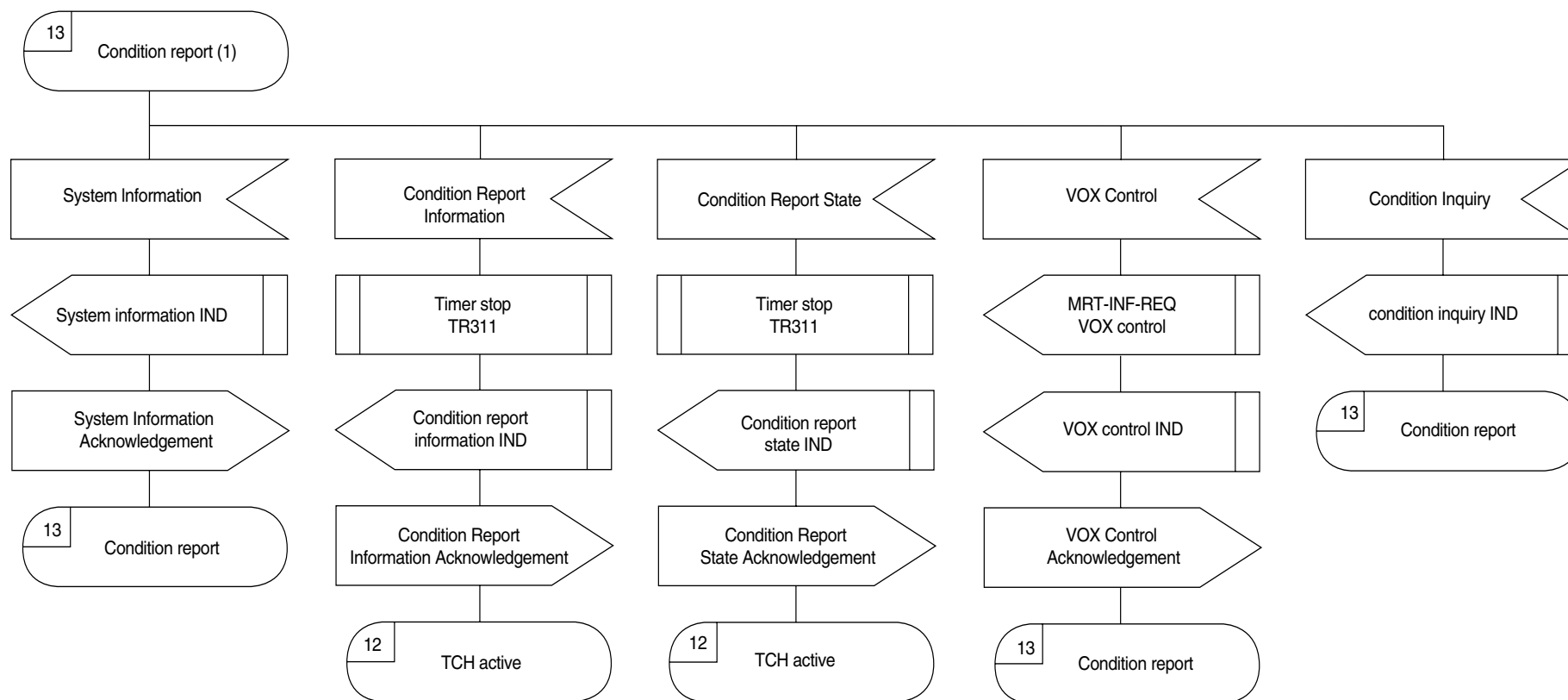


Fig. A. 5-21 Layer 3 RT SDL diagram (User side) (17/43)

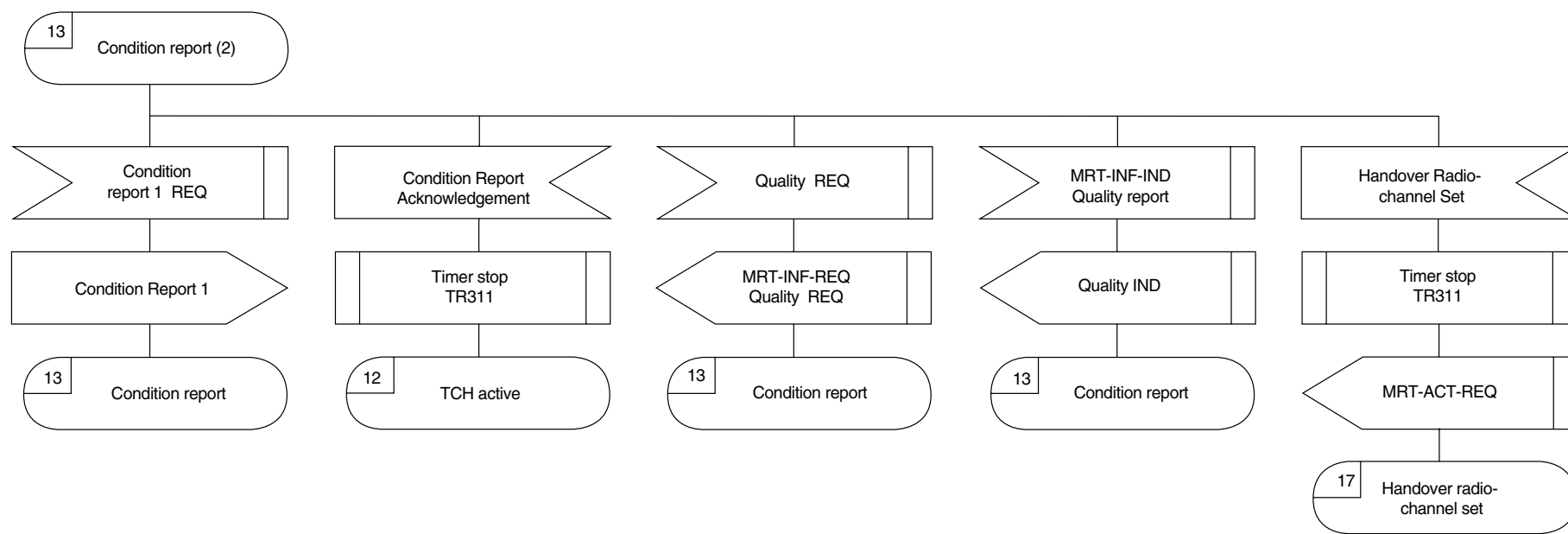


Fig. A. 5-22 Layer 3 RT SDL diagram (User side) (18/43)

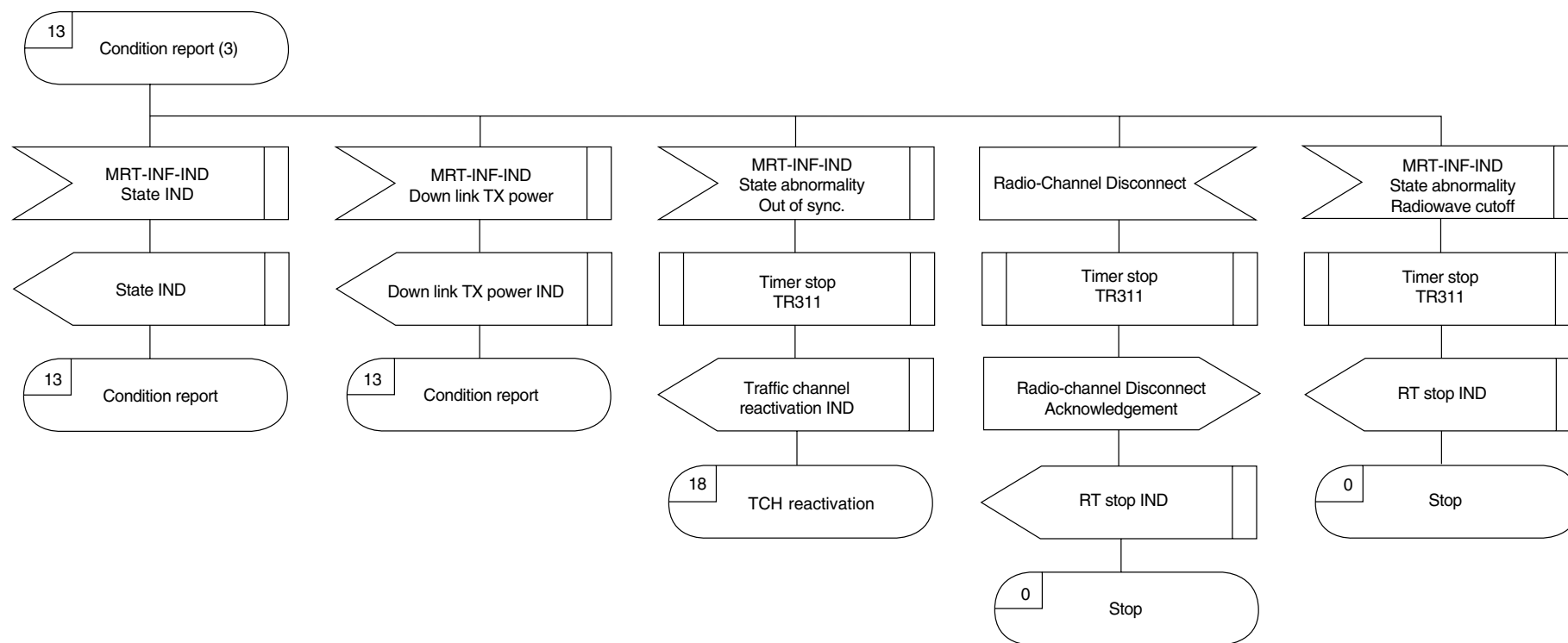


Fig. A. 5-23 Layer 3 RT SDL diagram (User side) (19/43)

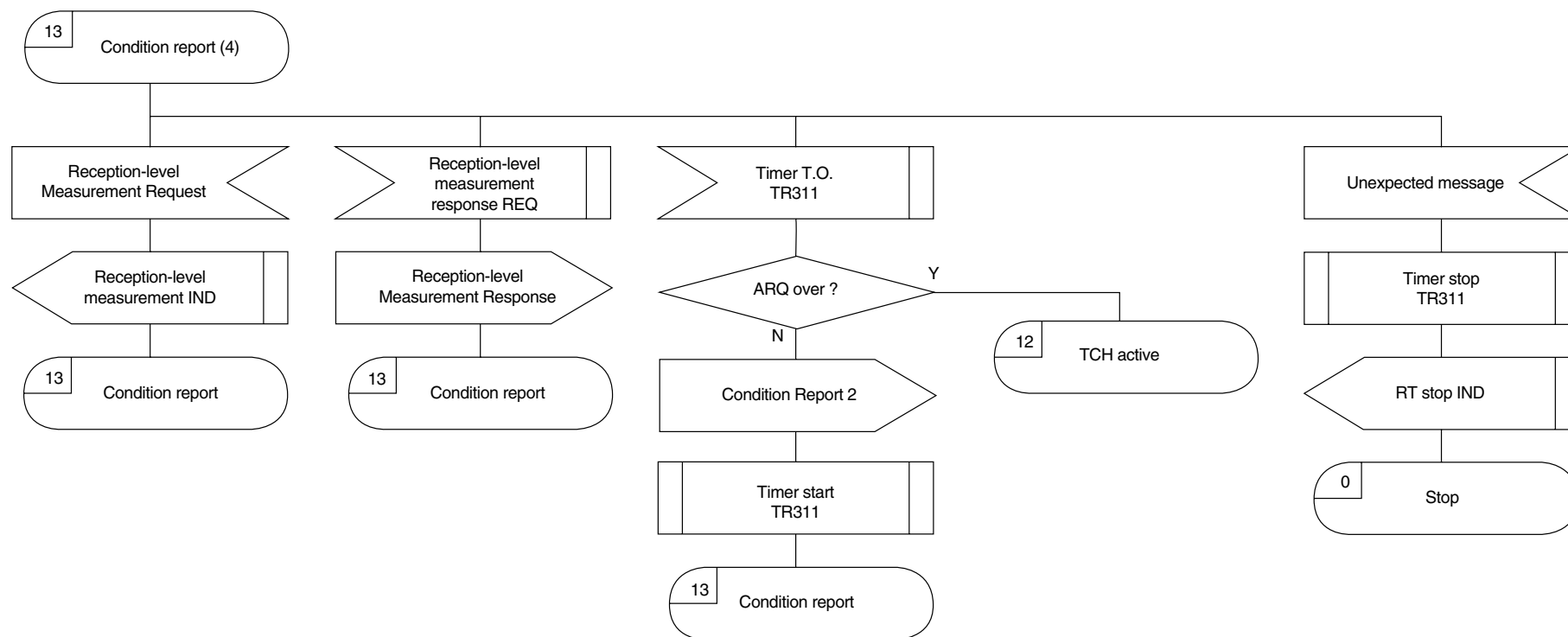


Fig. A. 5-24 Layer 3 RT SDL diagram (User side) (20/43)

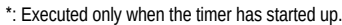
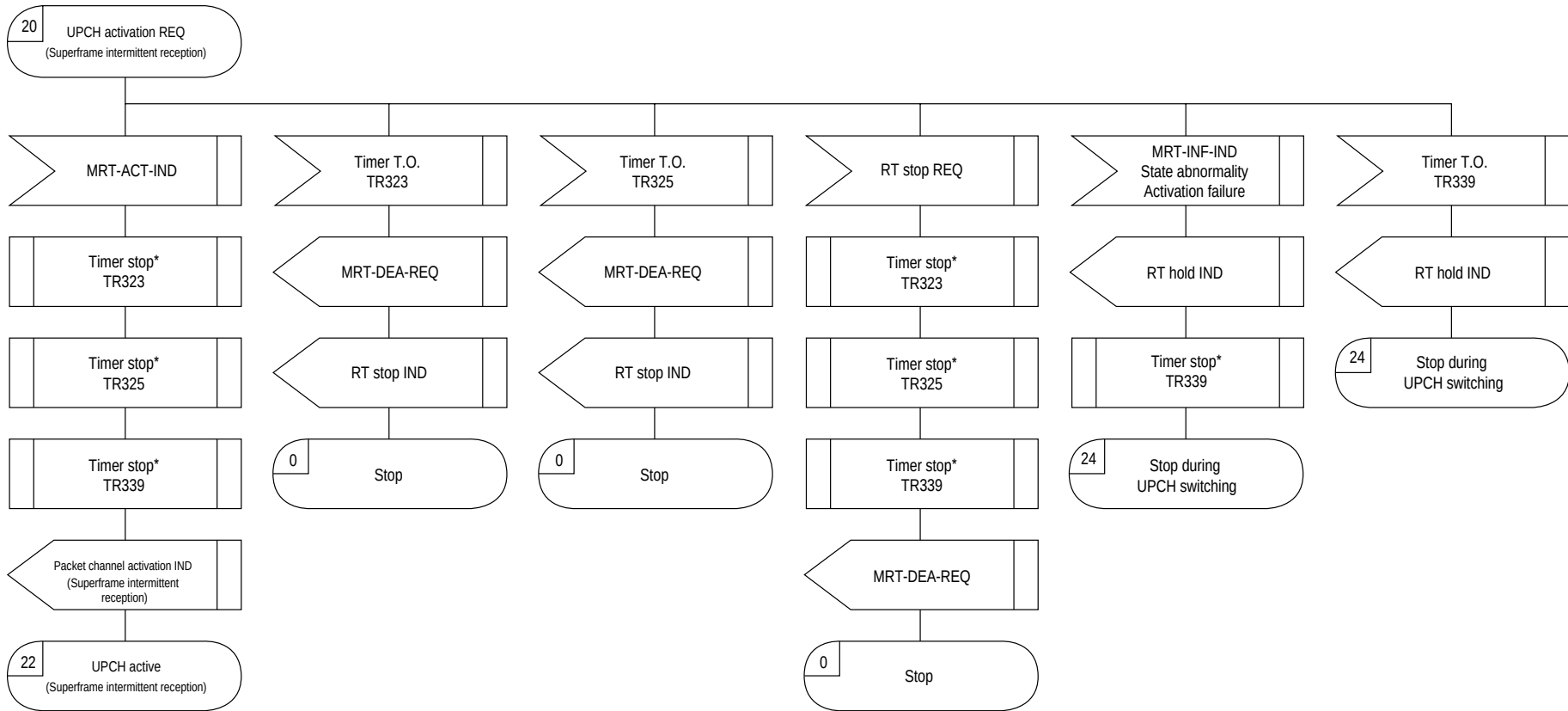


Fig. A. 5-25 Layer 3 RT SDL diagram (User side) (21/43)



*: Executed only when the timer has started up.

Fig. A. 5-26 Layer 3 RT SDL diagram (User side) (22/43)

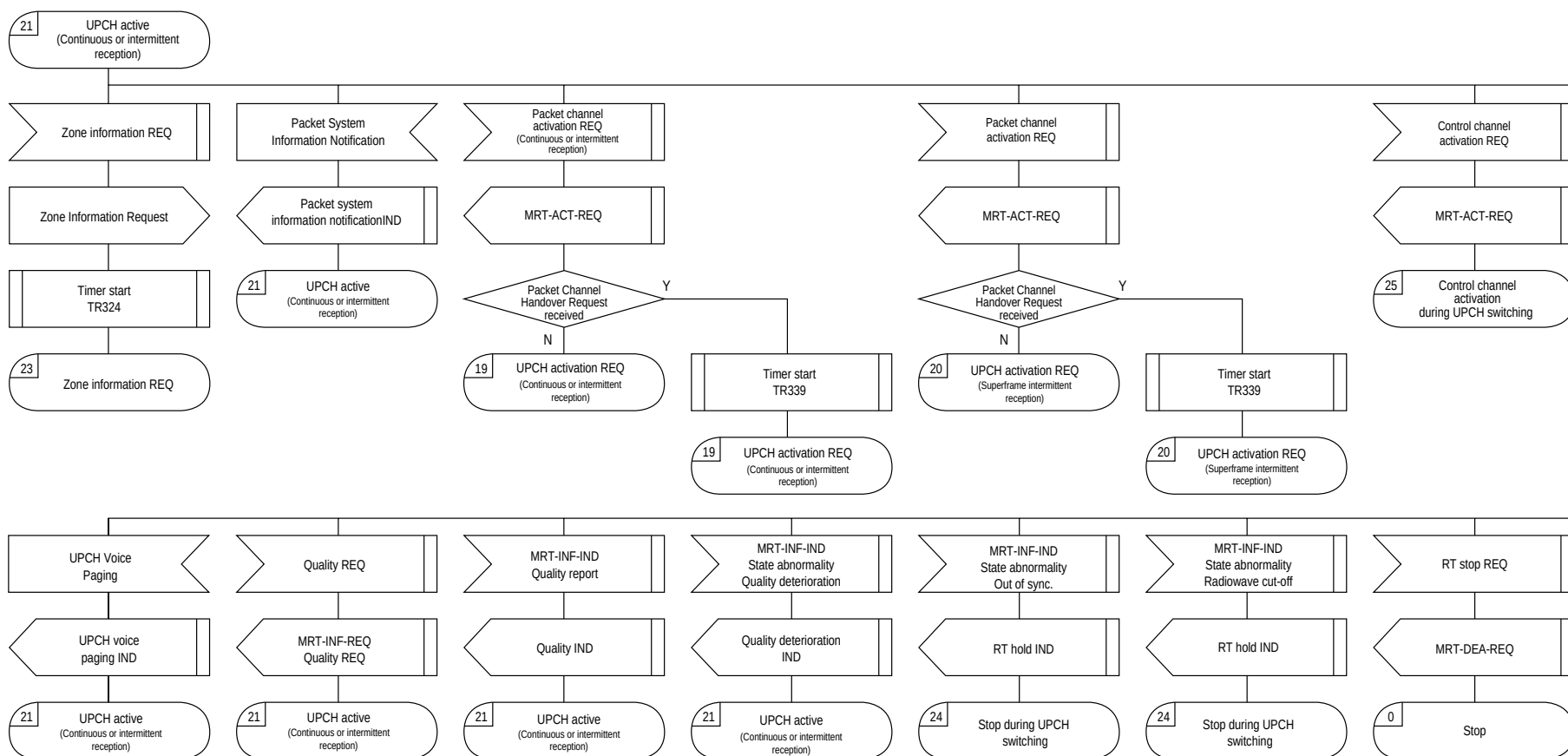


Fig. A. 5-27 Layer 3 RT SDL diagram (User side) (23/43)

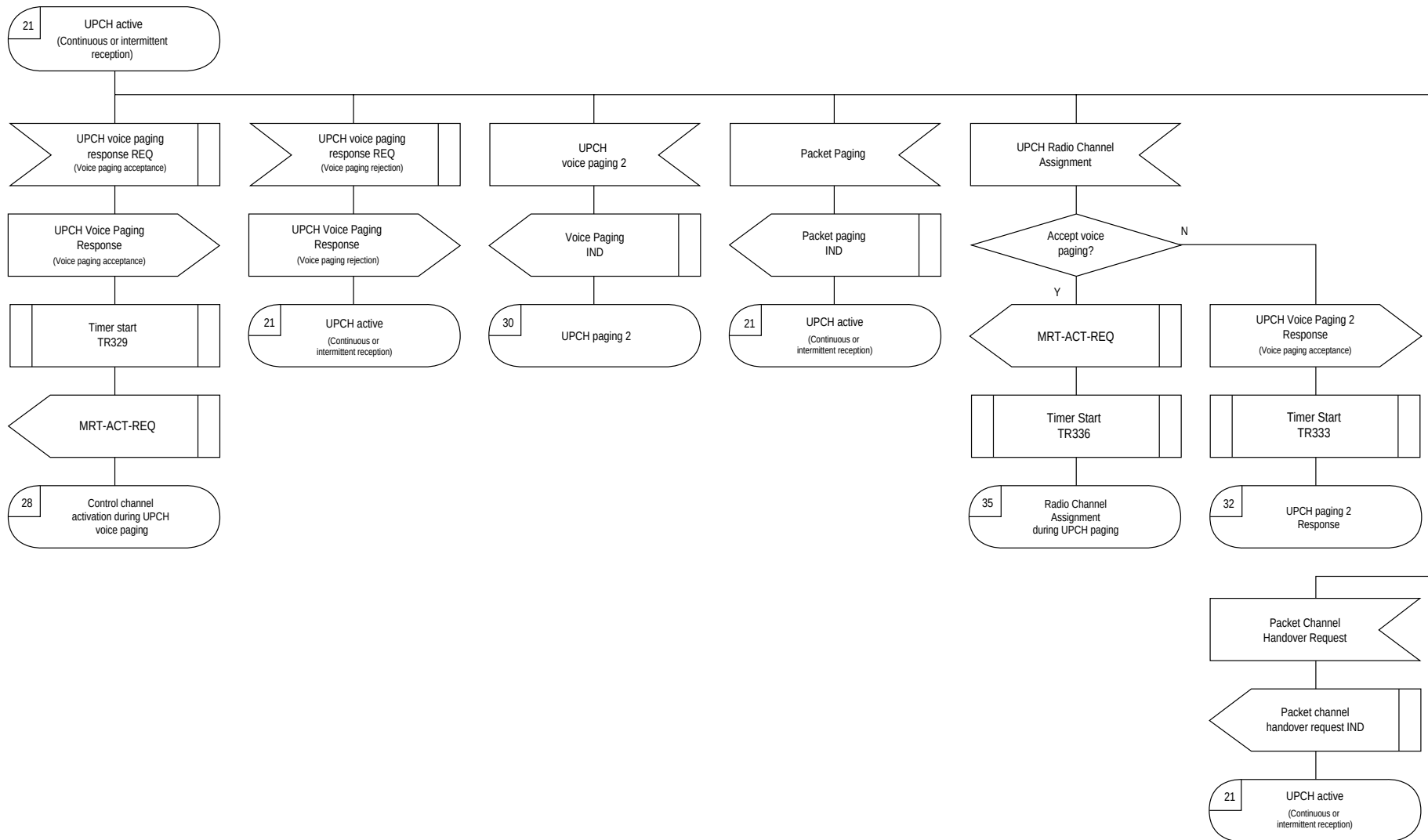


Fig. A. 5-28 Layer 3 RT SDL diagram (User side) (24/43)

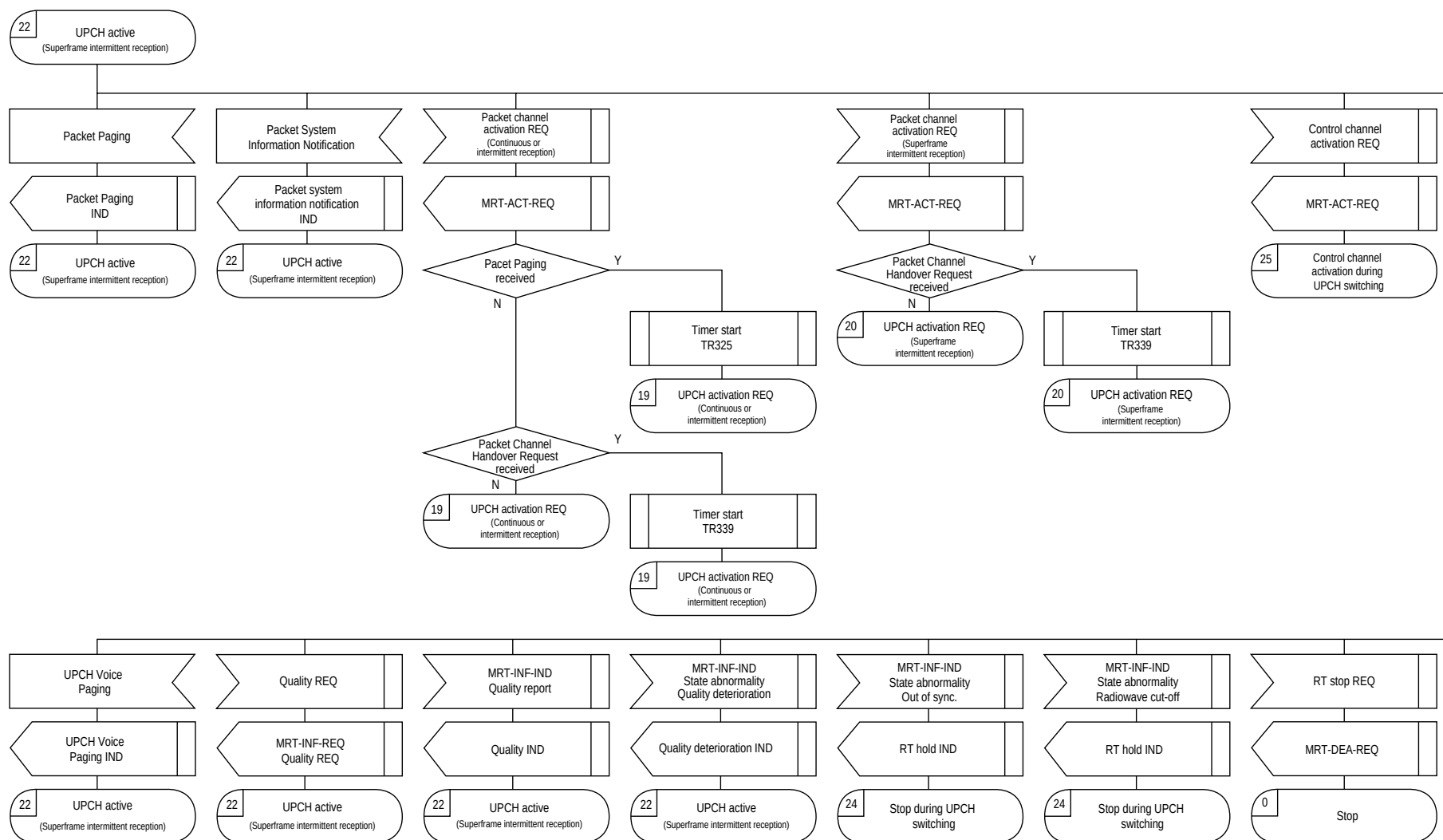


Fig. A. 5-29 Layer 3 RT SDL diagram (User side) (25/43)

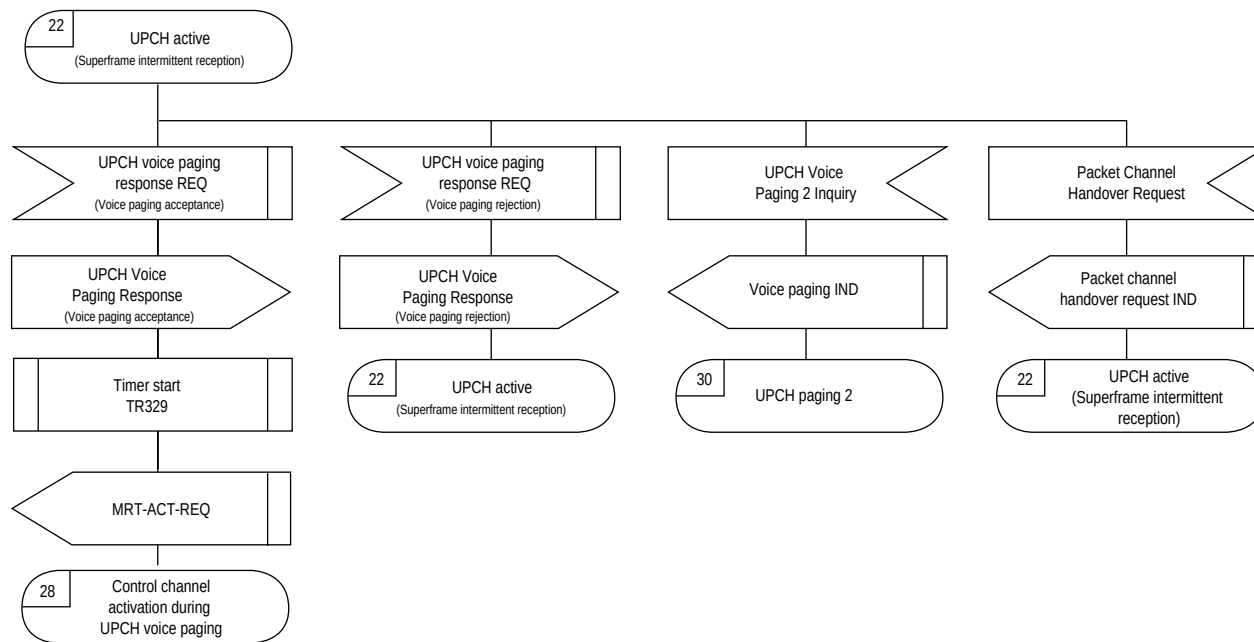


Fig. A. 5-30 Layer 3 RT SDL diagram (User side) (26/43)



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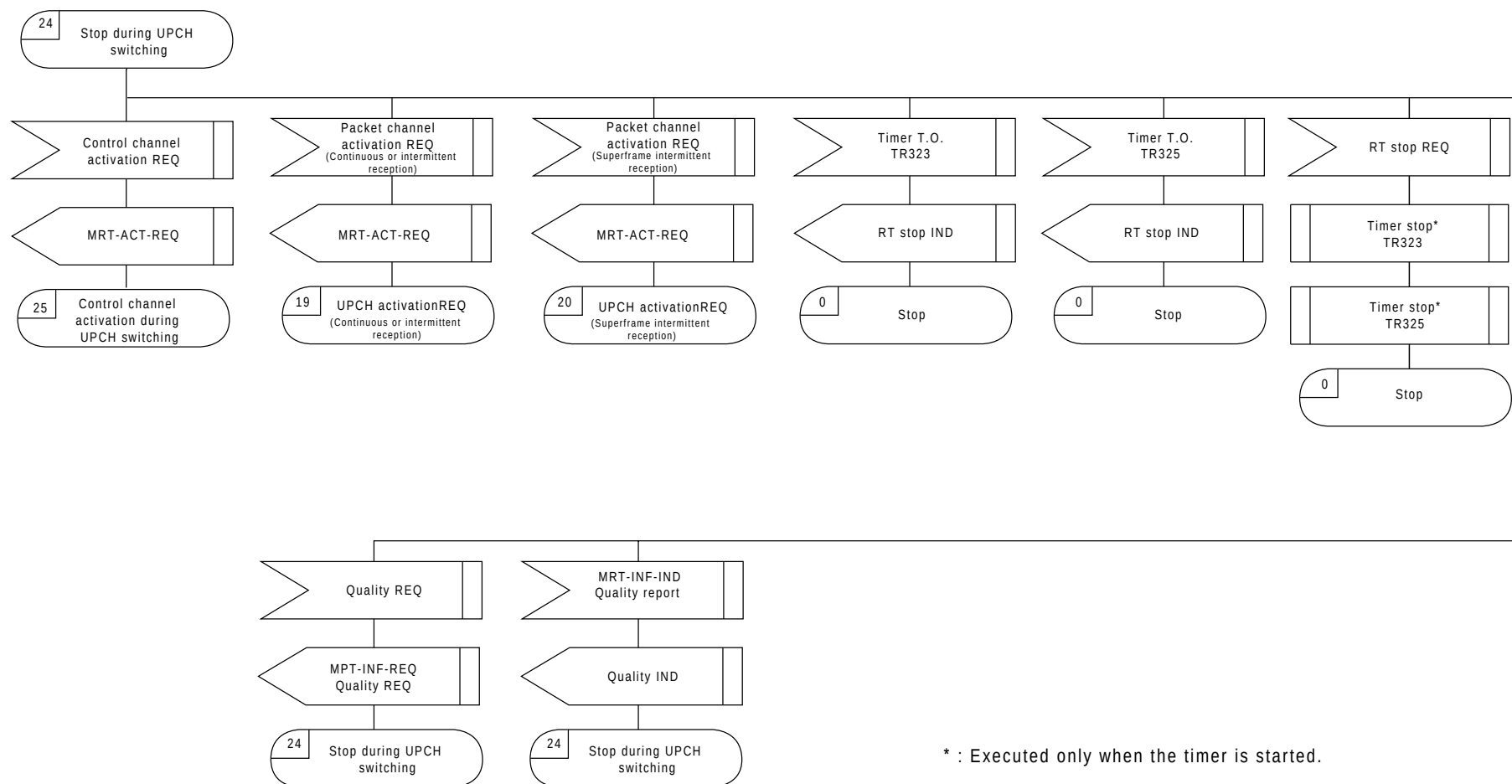
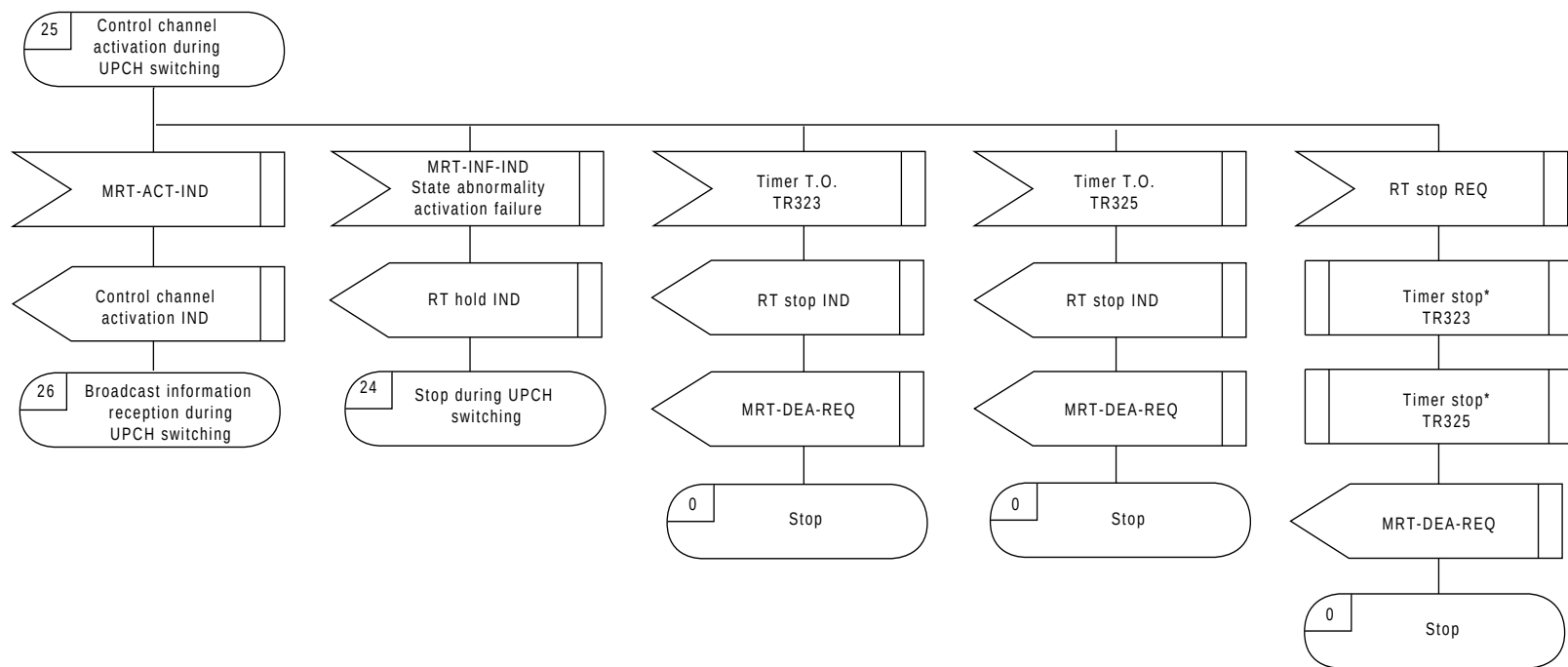
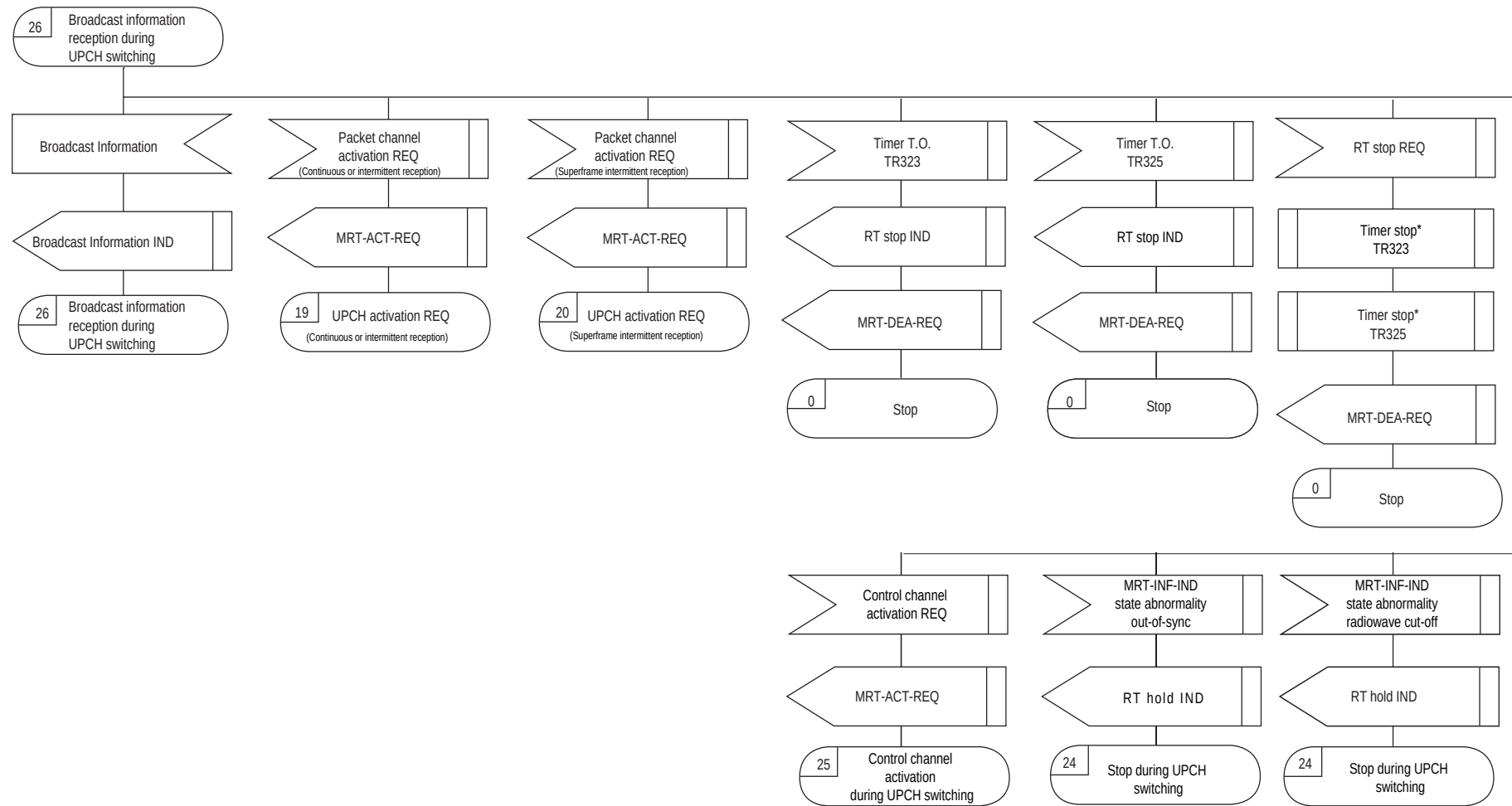


Fig. A. 5-32 Layer 3 RT SDL diagram (User side) (28/43)



* Executed only when the timer has started

Fig. A. 5-33 Layer 3 RT SDL diagram (User side) (29/43)



* Executed only when the timer has started.

Fig. A. 5-34 Layer 3 RT SDL diagram (User side) (30/43)

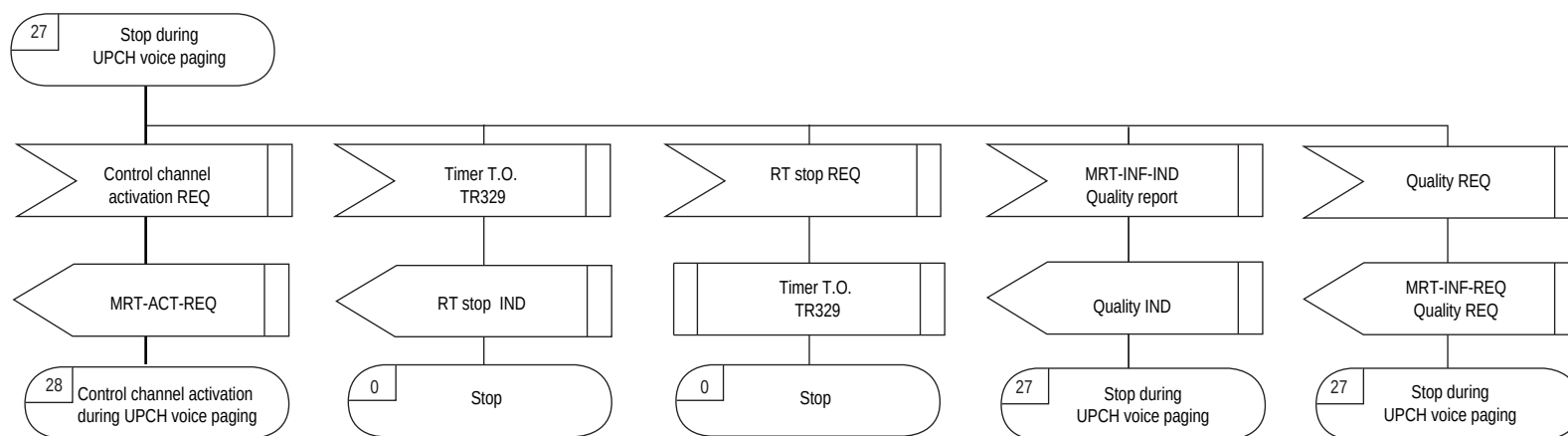


Fig. A. 5-35 Layer 3 RT SDL diagram (User side) (31/43)

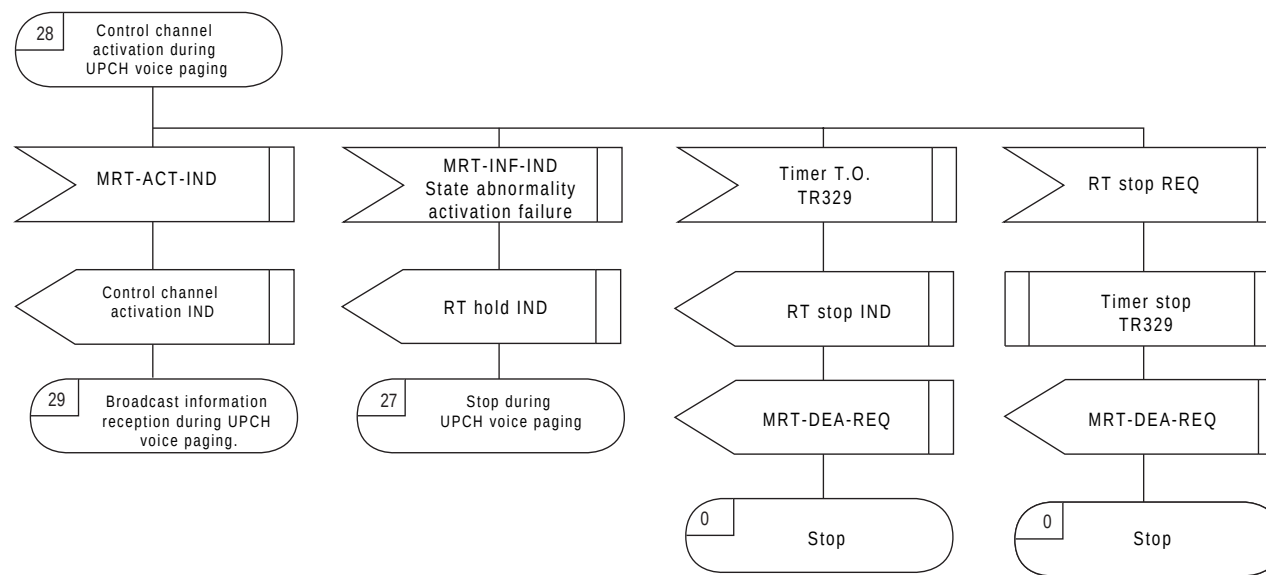


Fig. A. 5-36 Layer 3 RT SDL diagram (User side) (32/43)

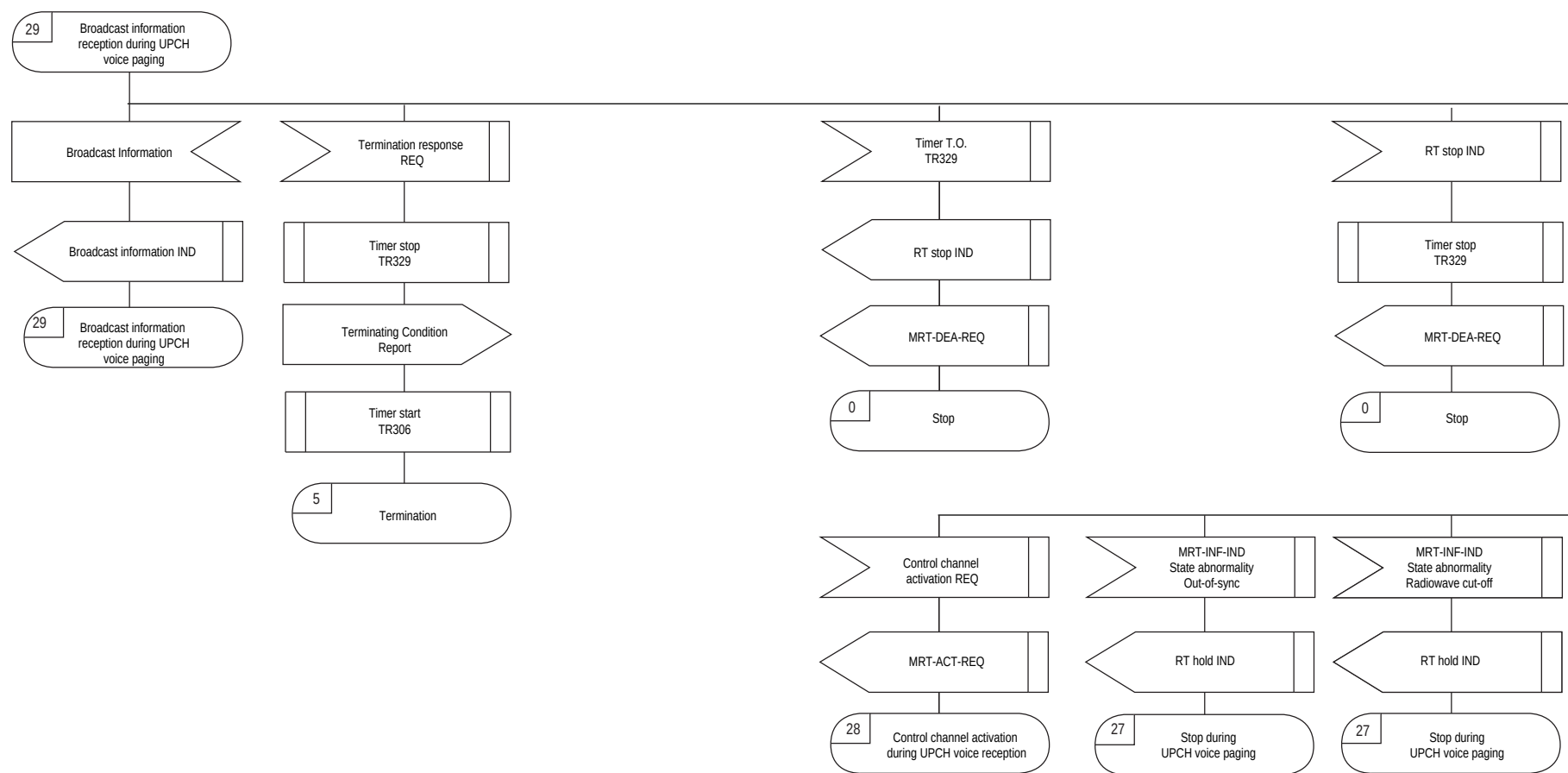


Fig. A. 5-37 Layer 3 RT SDL diagram (User side) (33/43)

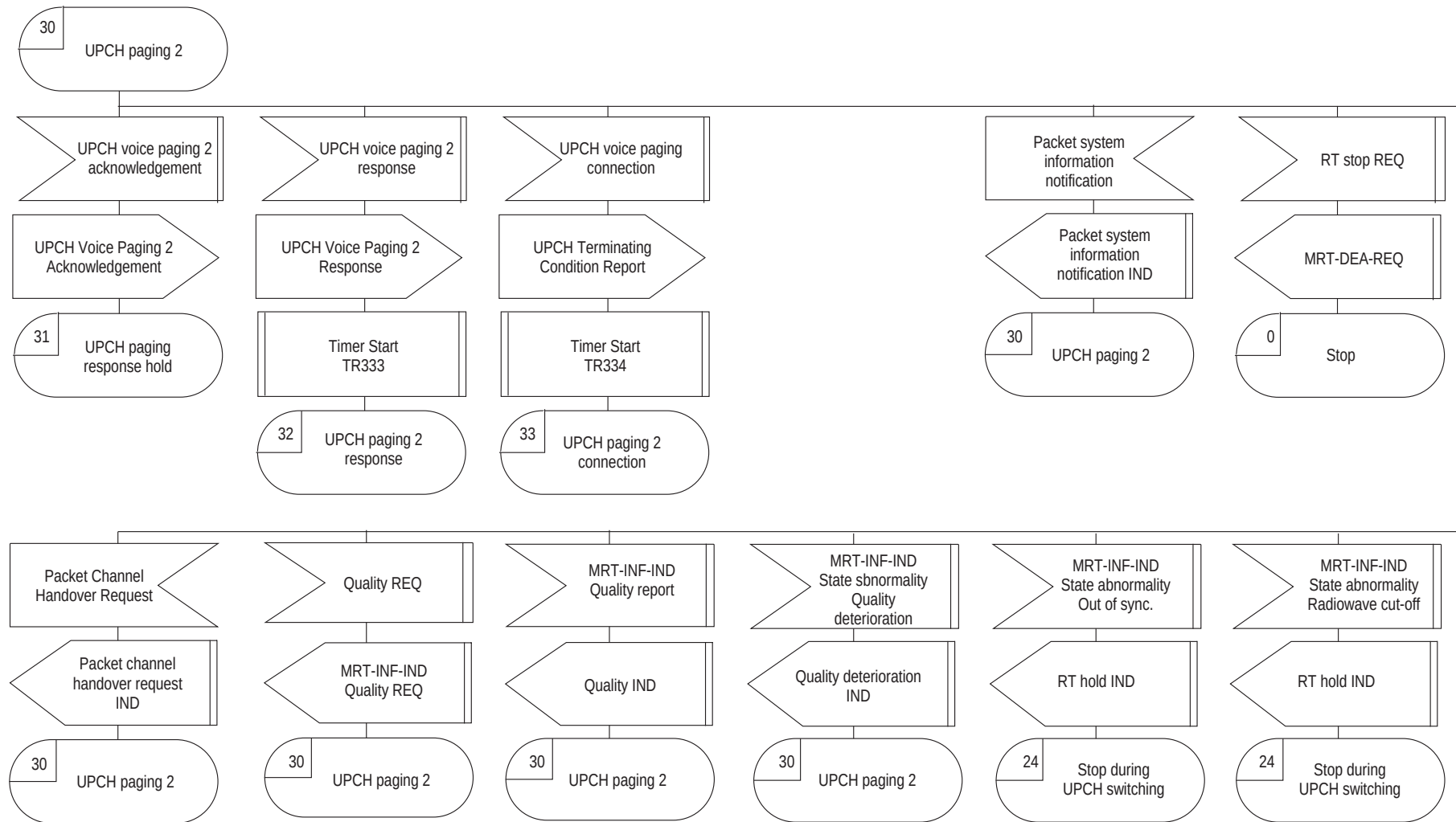


Fig. A. 5-38 Layer 3 RT SDL diagram (User side) (34/43)

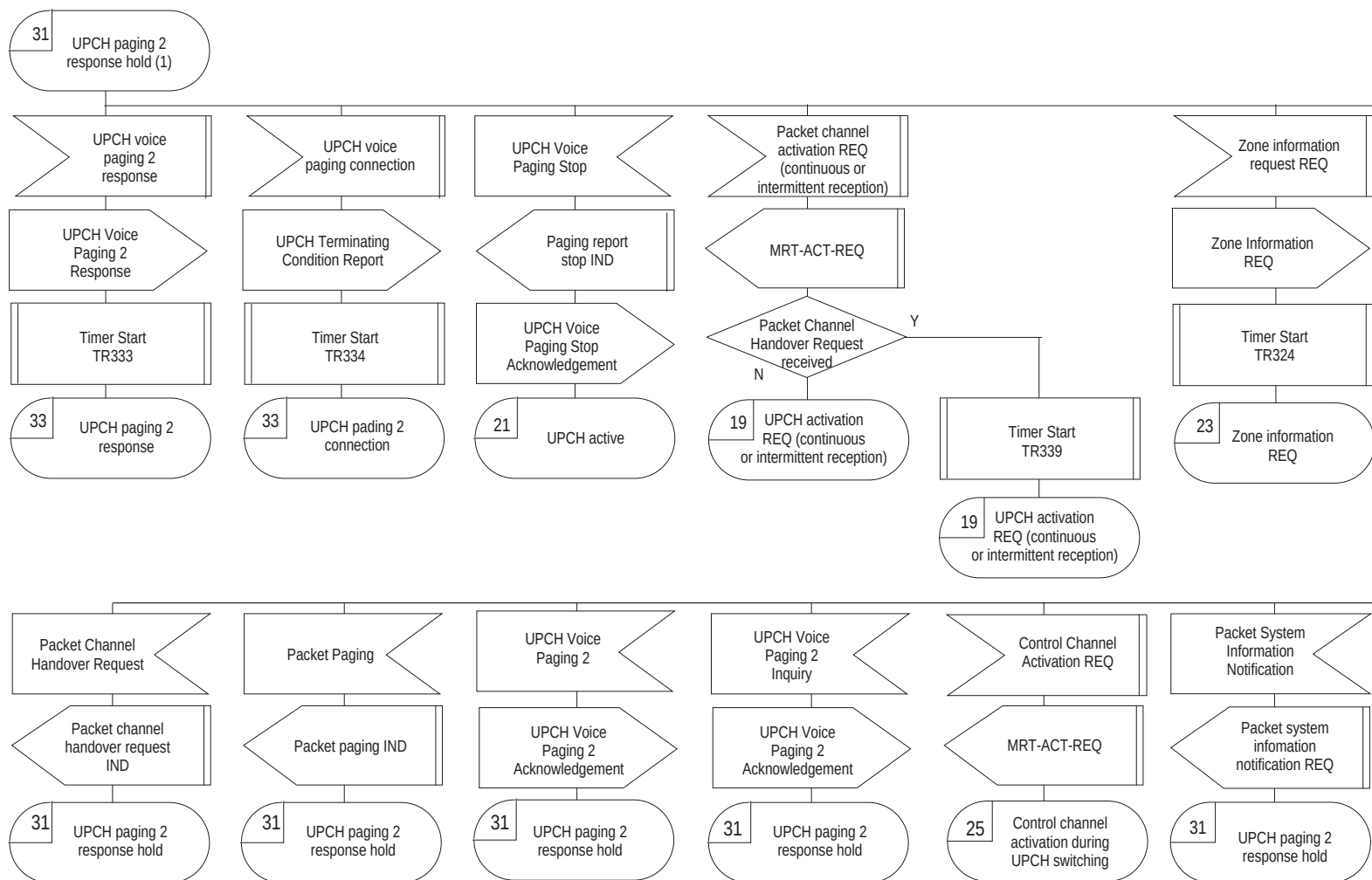


Fig. A. 5-39 Layer 3 RT SDL diagram (User side) (35/43)

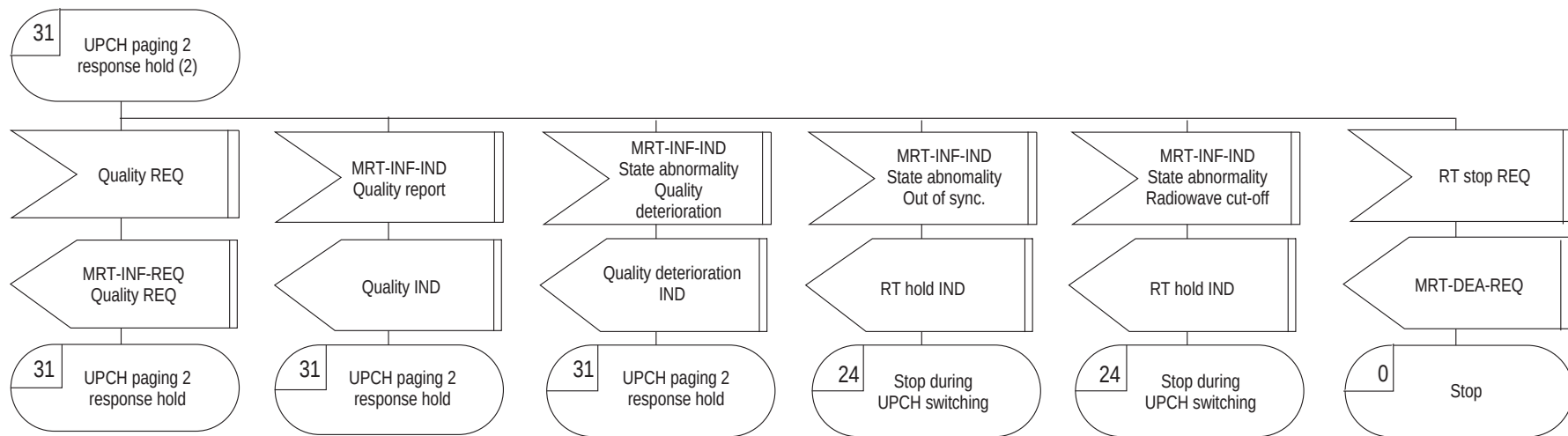


Fig. A. 5-40 Layer 3 RT SDL diagram (User side) (36/43)

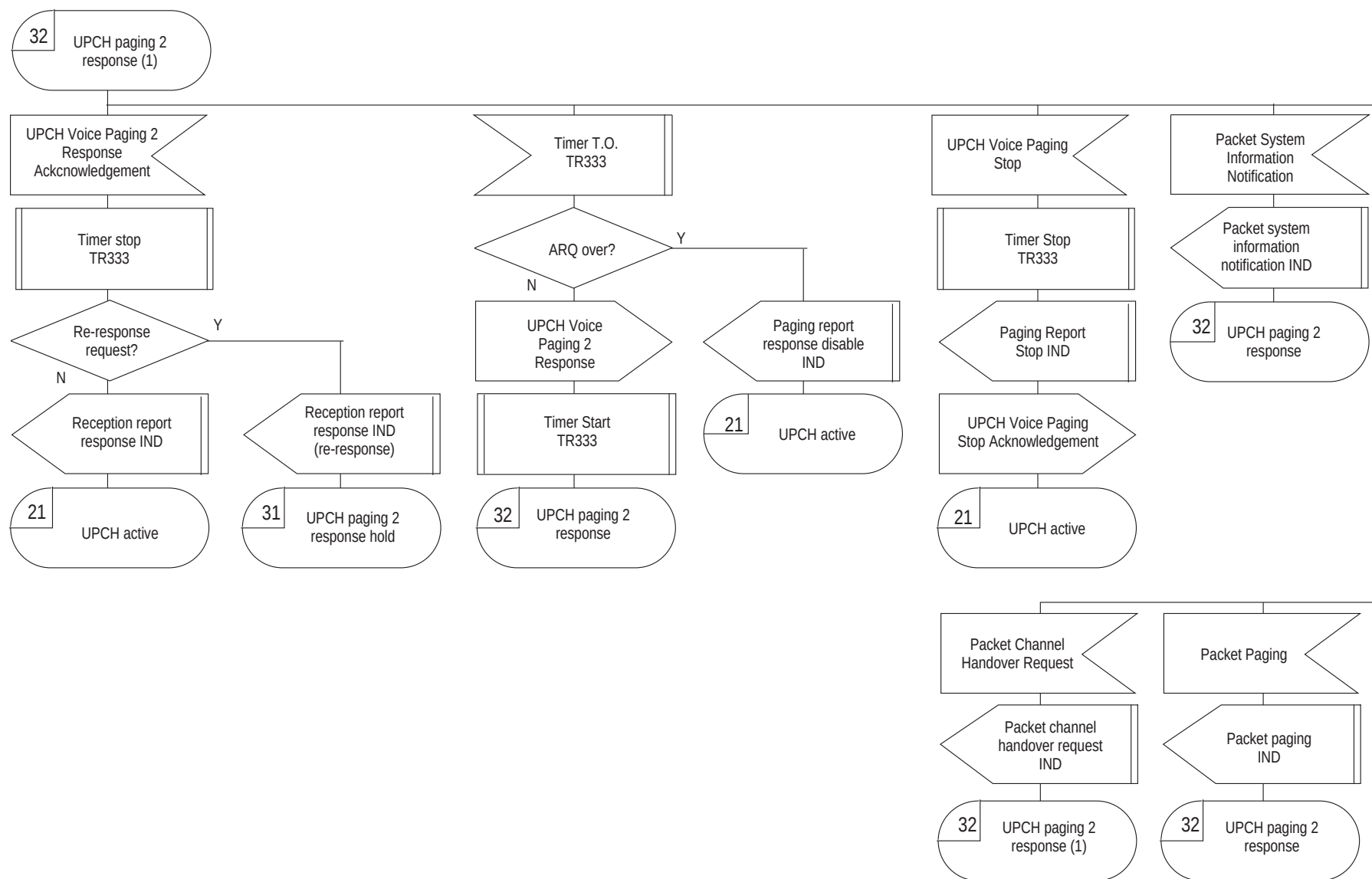


Fig. A. 5-41 Layer 3 RT SDL diagram (User side) (37/43)

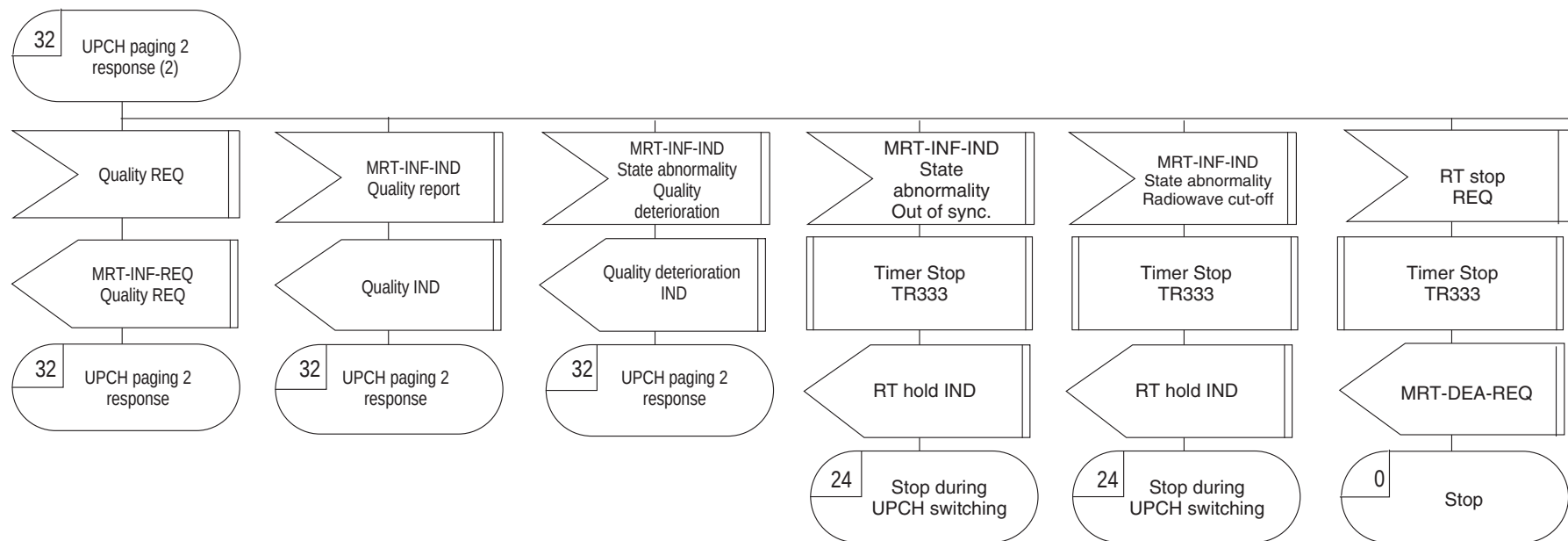


Fig. A. 5-42 Layer 3 RT SDL diagram (User side) (38/43)

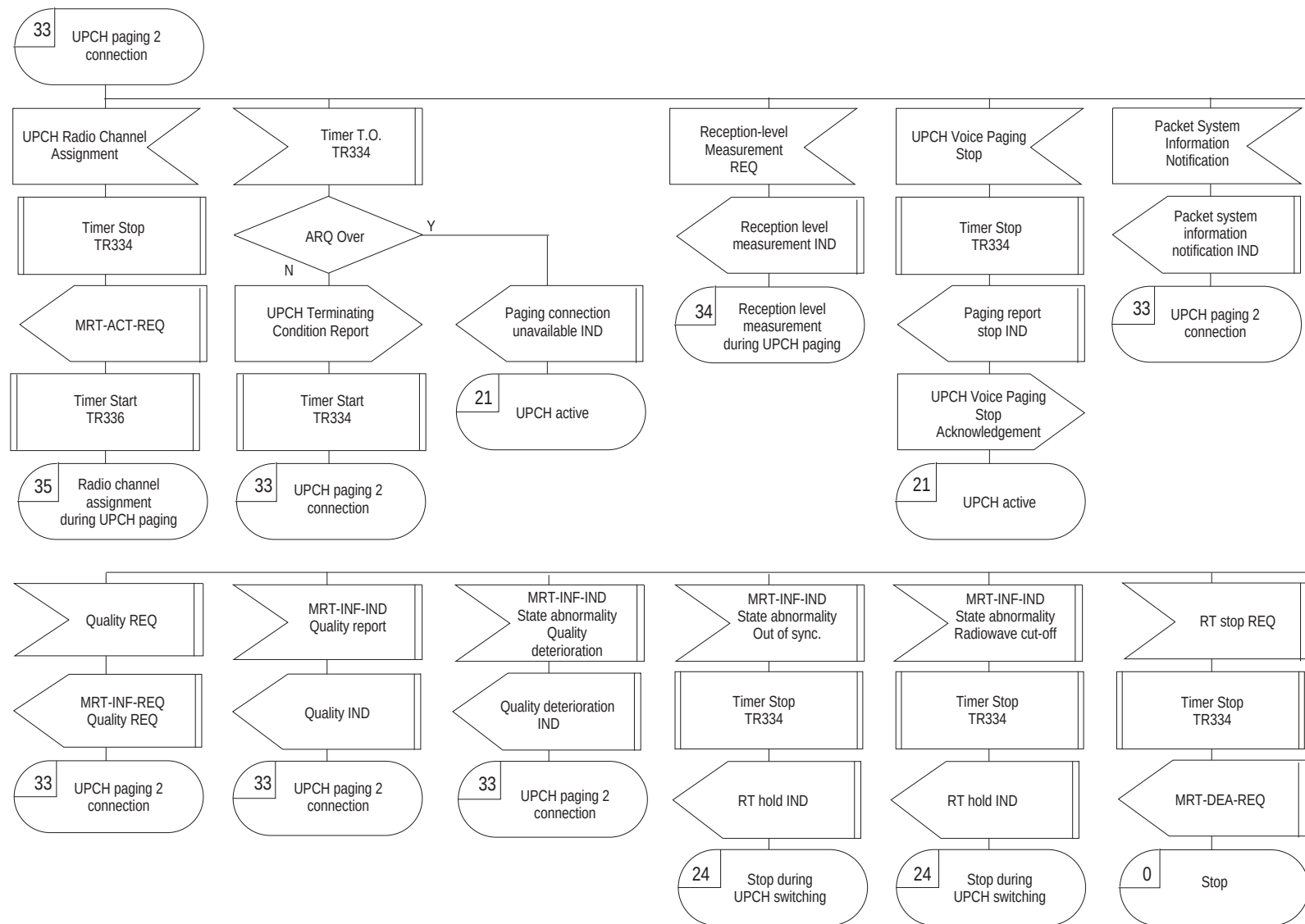


Fig. A. 5-43 Layer 3 RT SDL diagram (User side) (39/43)

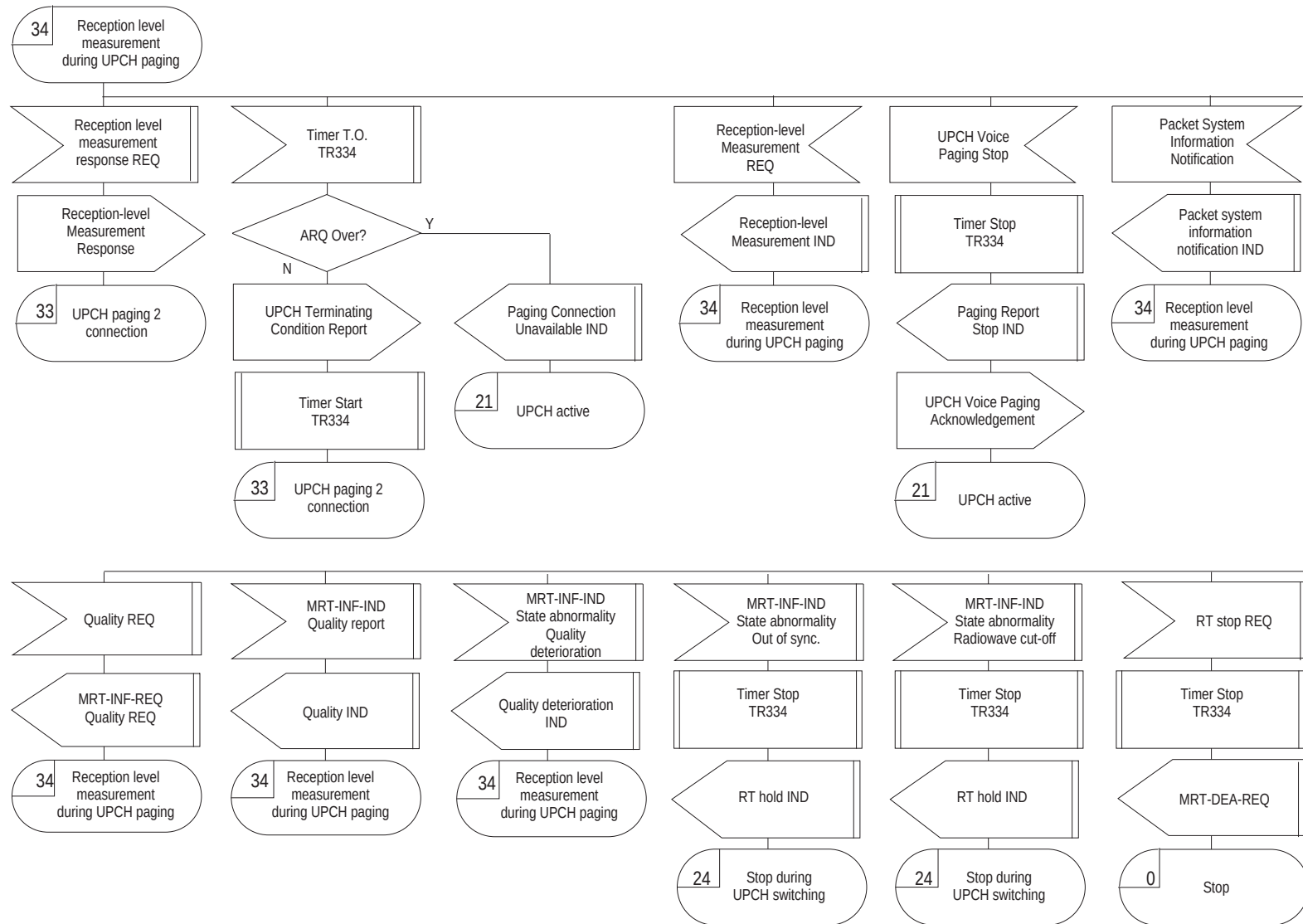


Fig. A. 5-44 Layer 3 RT SDL diagram (User side) (40/43)

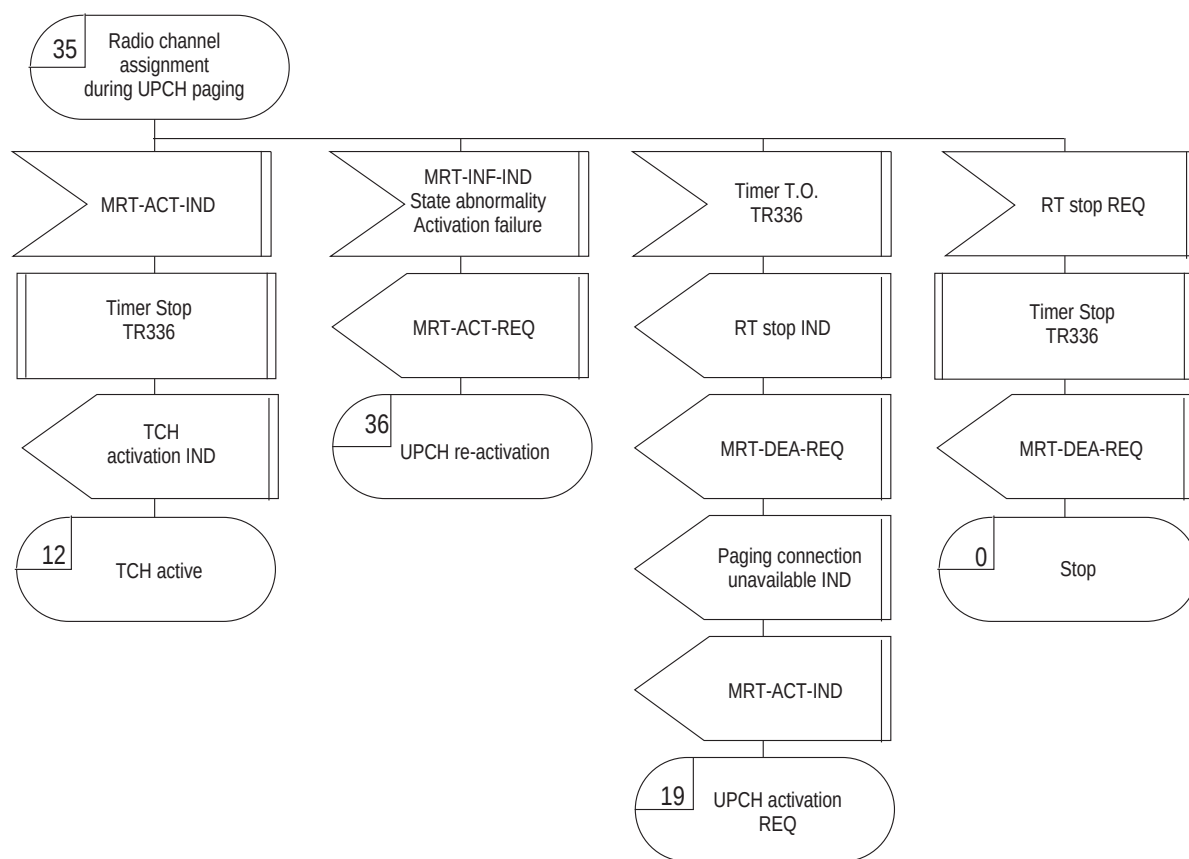


Fig. A. 5-45 Layer 3 RT SDL diagram (User side) (41/43)

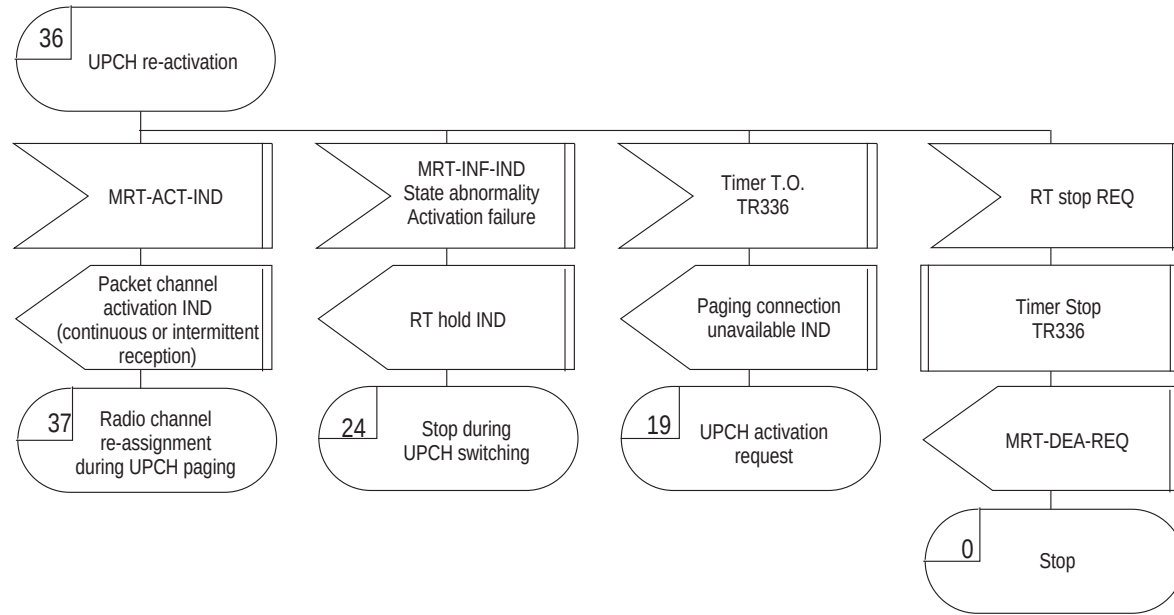


Fig. A. 5-46 Layer 3 RT SDL diagram (User side) (42/43)

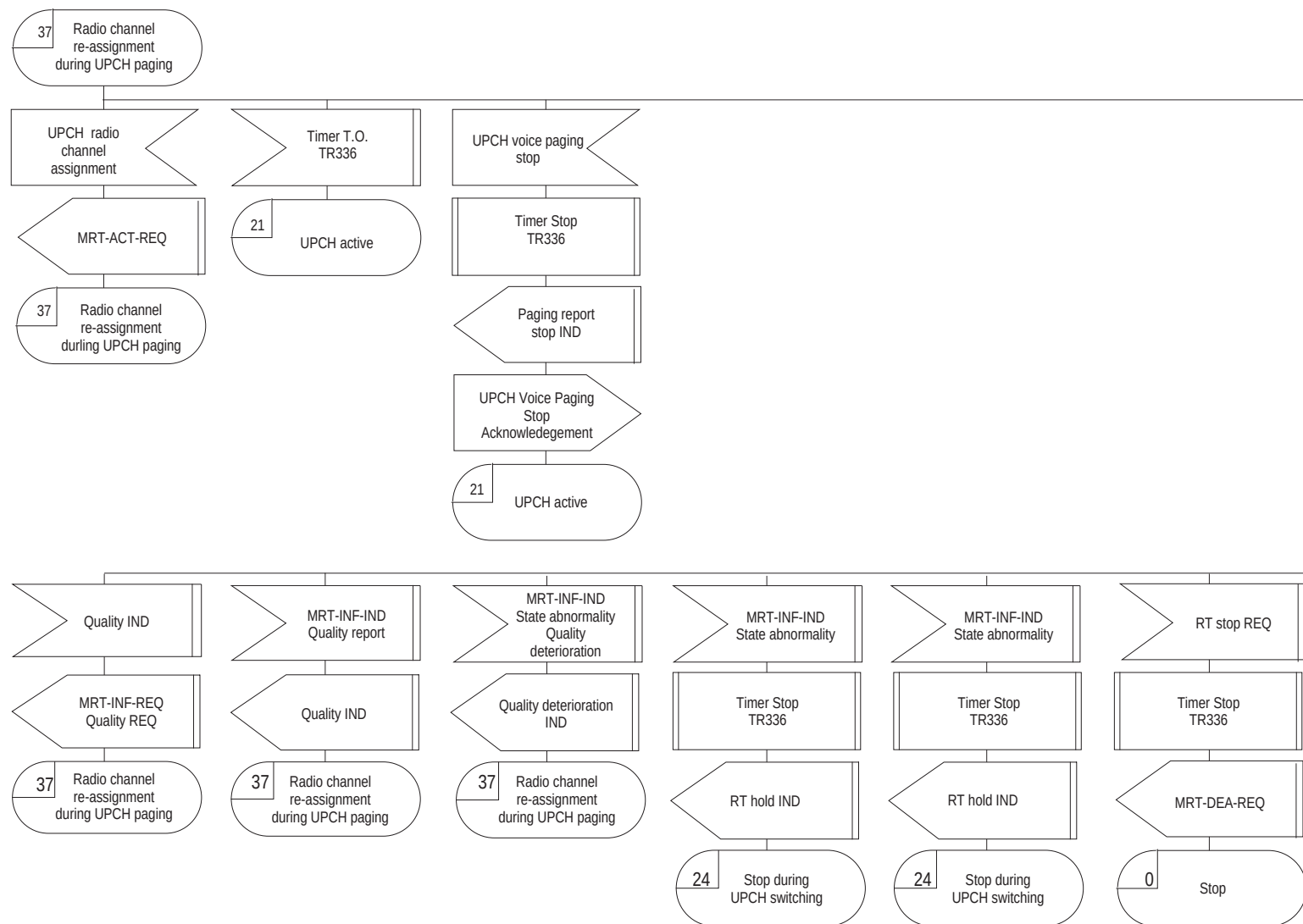


Fig. A. 5-47 Layer 3 RT SDL diagram (User side) (43/43)

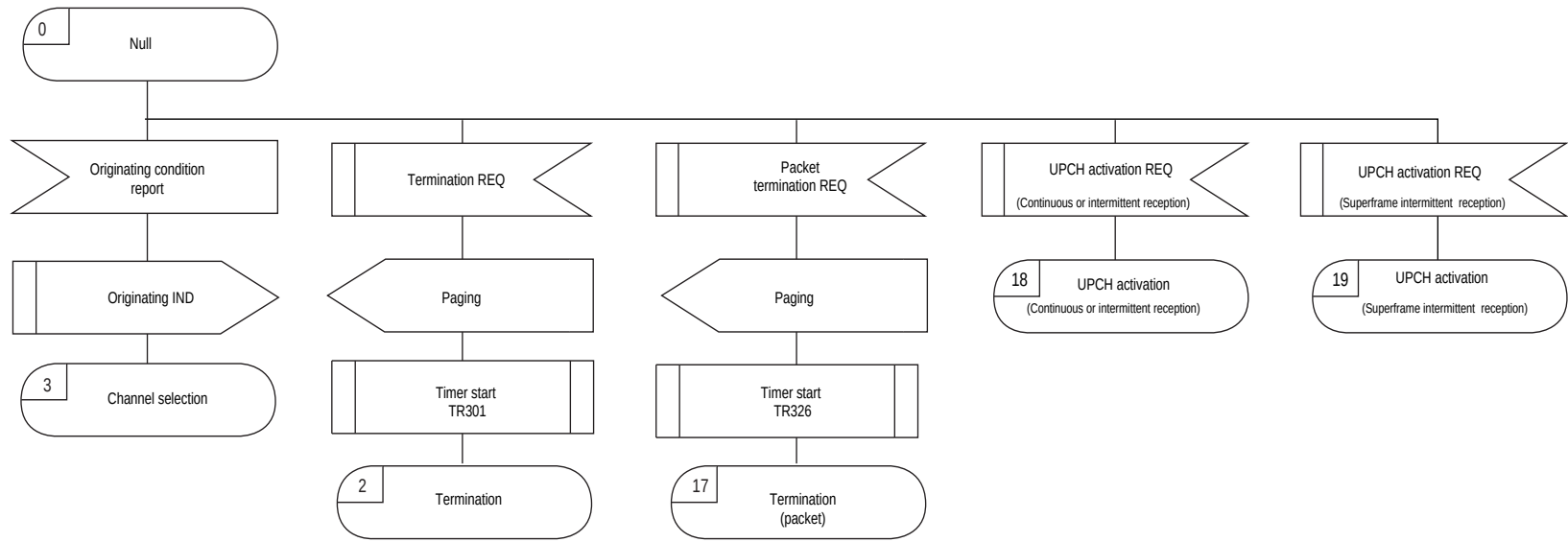


Fig. A. 5-48 Layer 3 RT SDL diagram (Network side) (1/34)

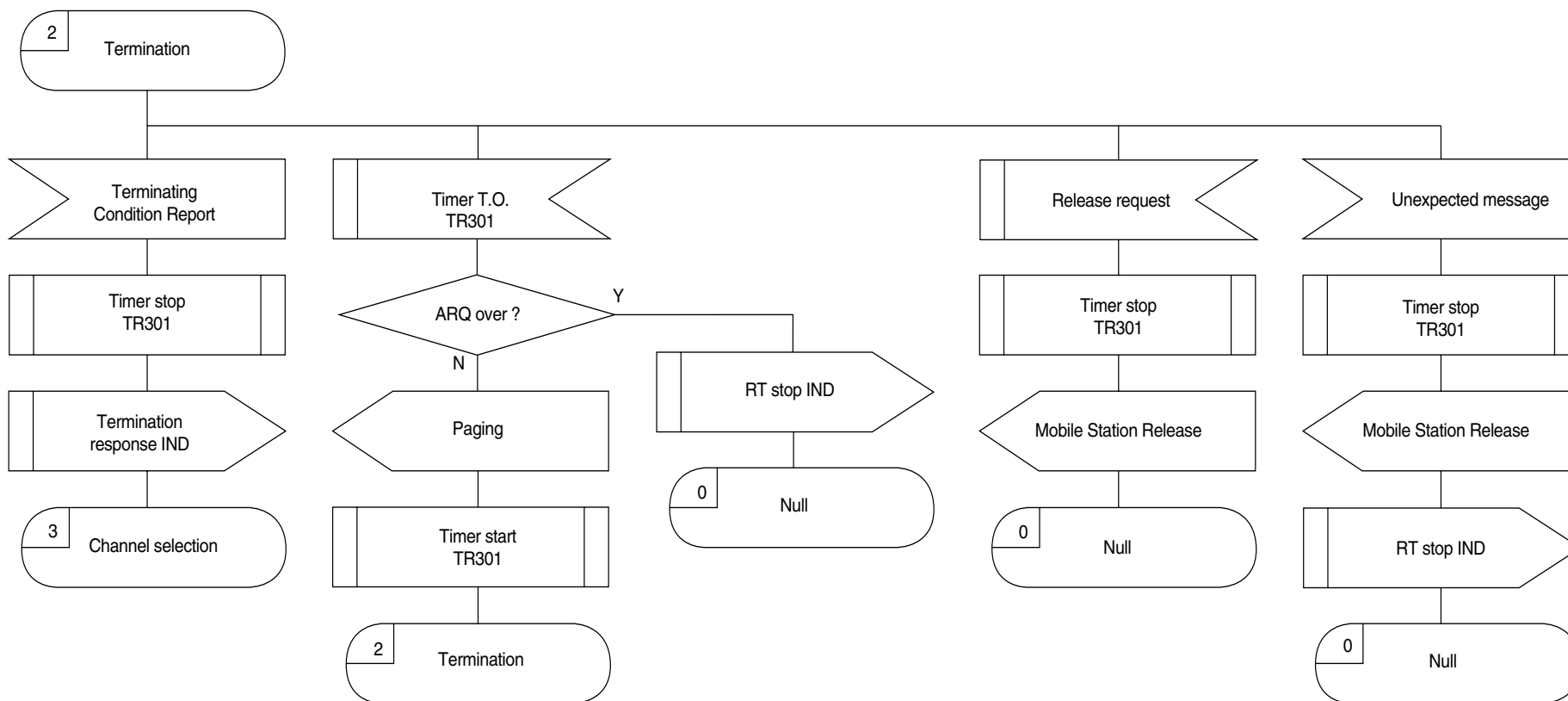


Fig. A. 5-49 Layer 3 RT SDL diagram (Network side) (2/34)

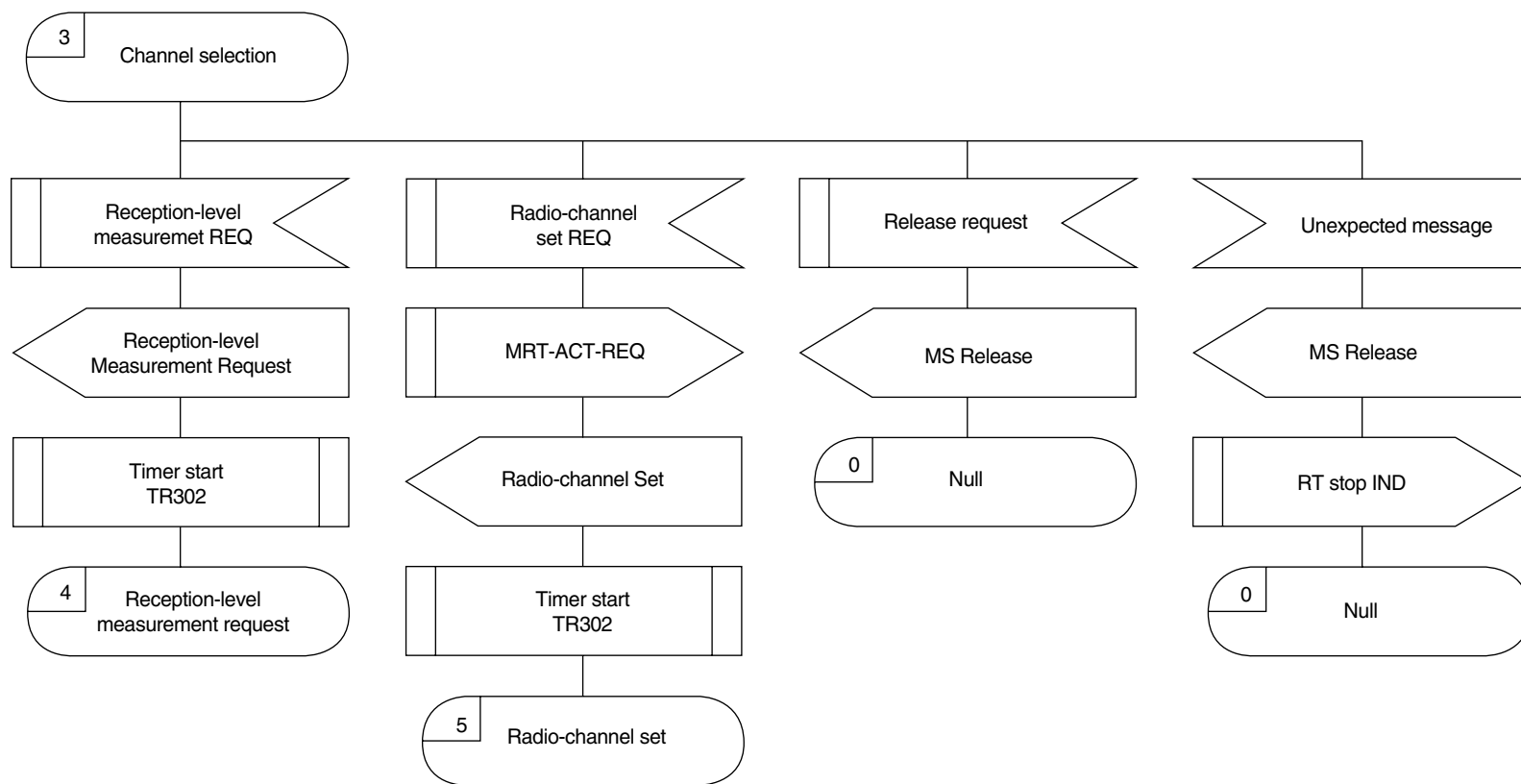


Fig. A. 5-50 Layer 3 RT SDL diagram (Network side) (3/34)

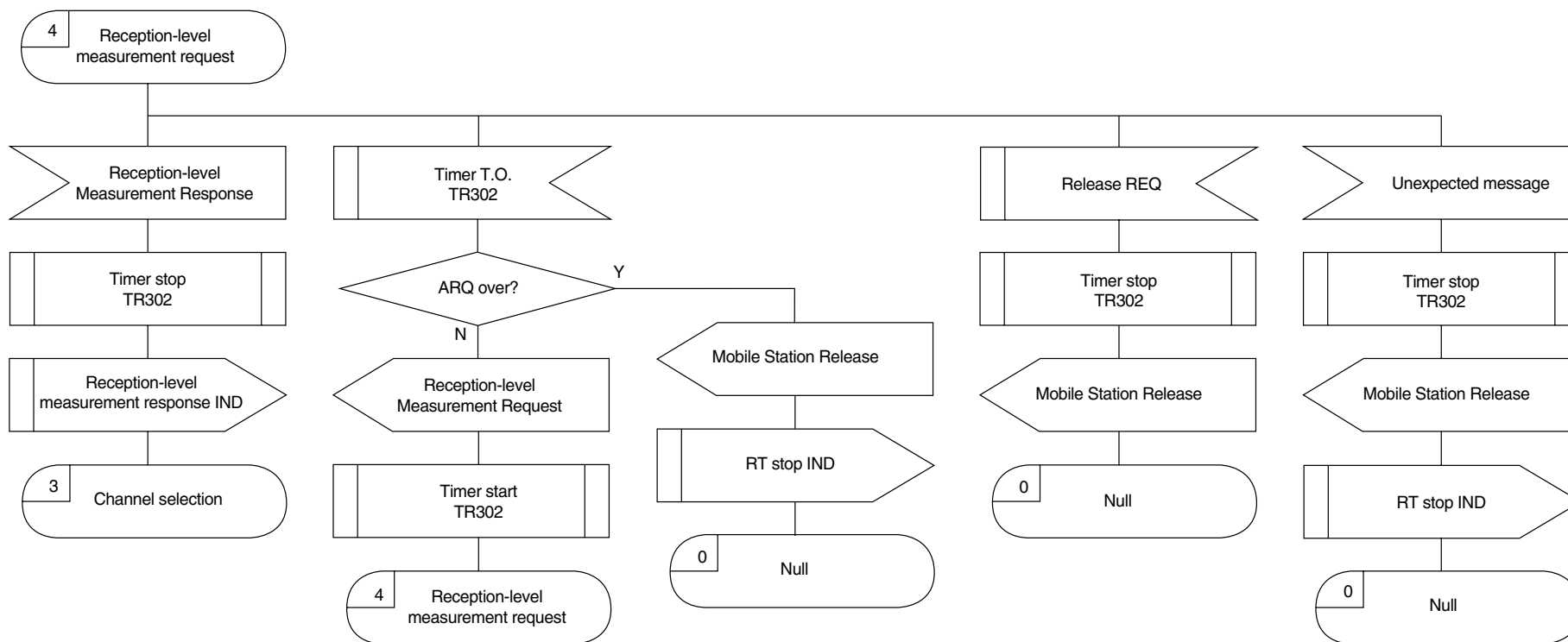


Fig. A. 5-51 Layer 3 RT SDL diagram (Network side) (4/34)

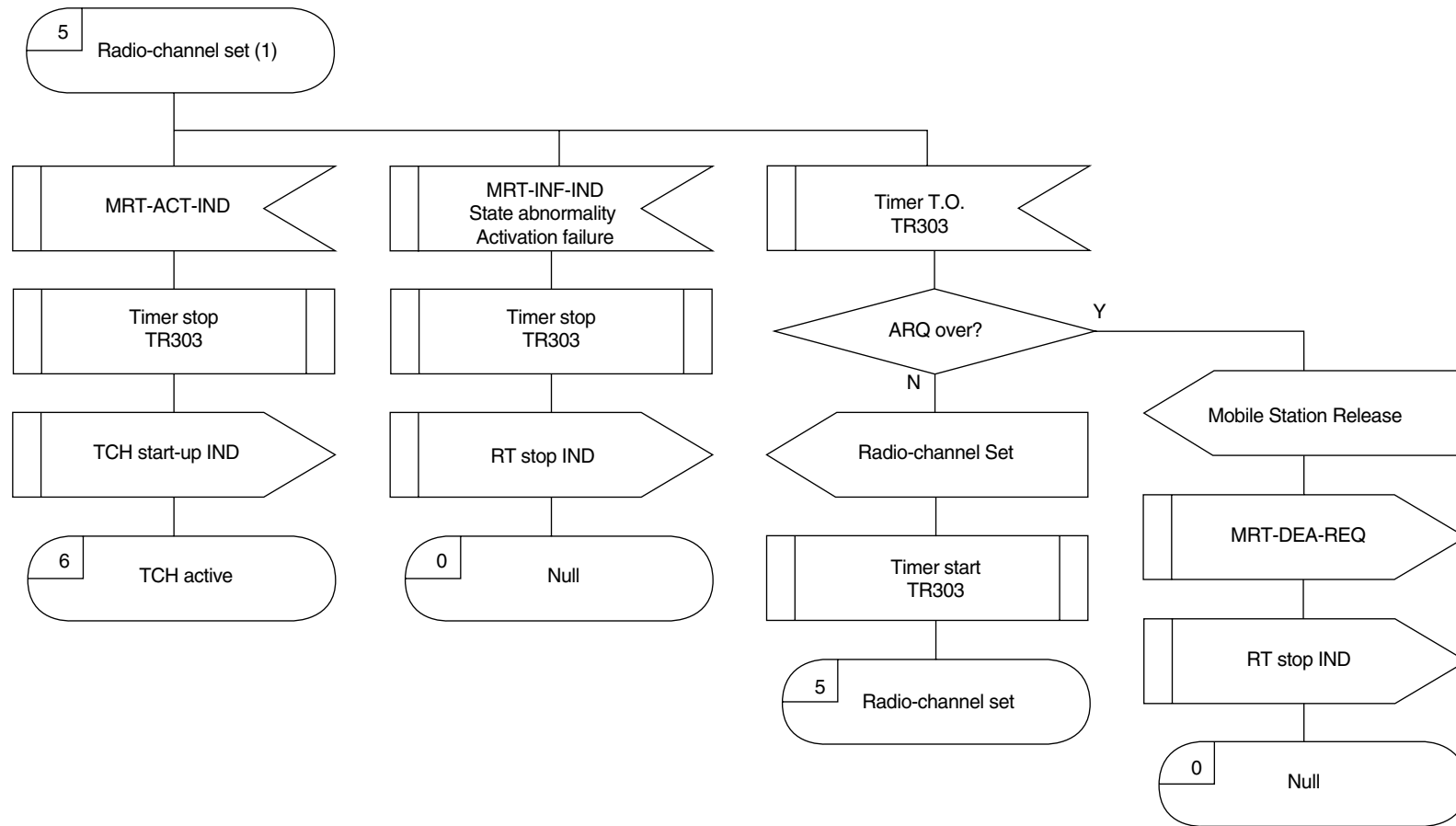


Fig. A. 5-52 Layer 3 RT SDL diagram (Network side) (5/34)

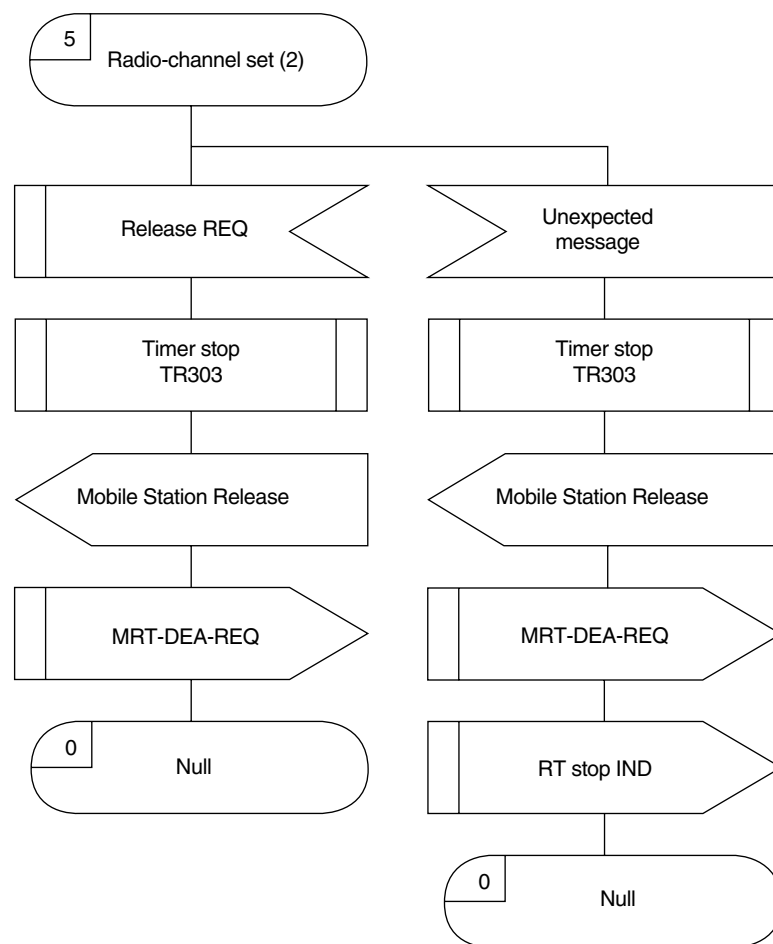


Fig. A. 5-53 Layer 3 RT SDL diagram (Network side) (6/34)

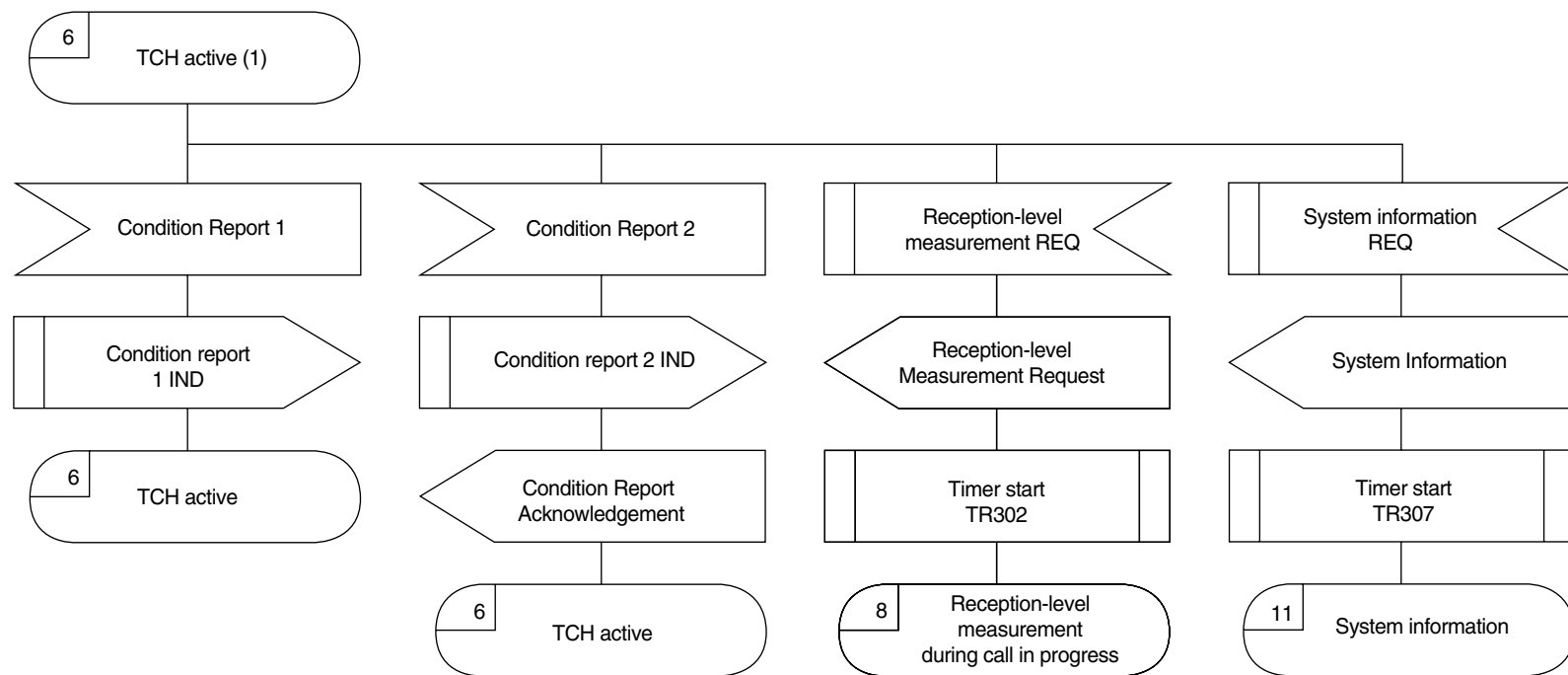


Fig. A. 5-54 Layer 3 RT SDL diagram (Network side) (7/34)

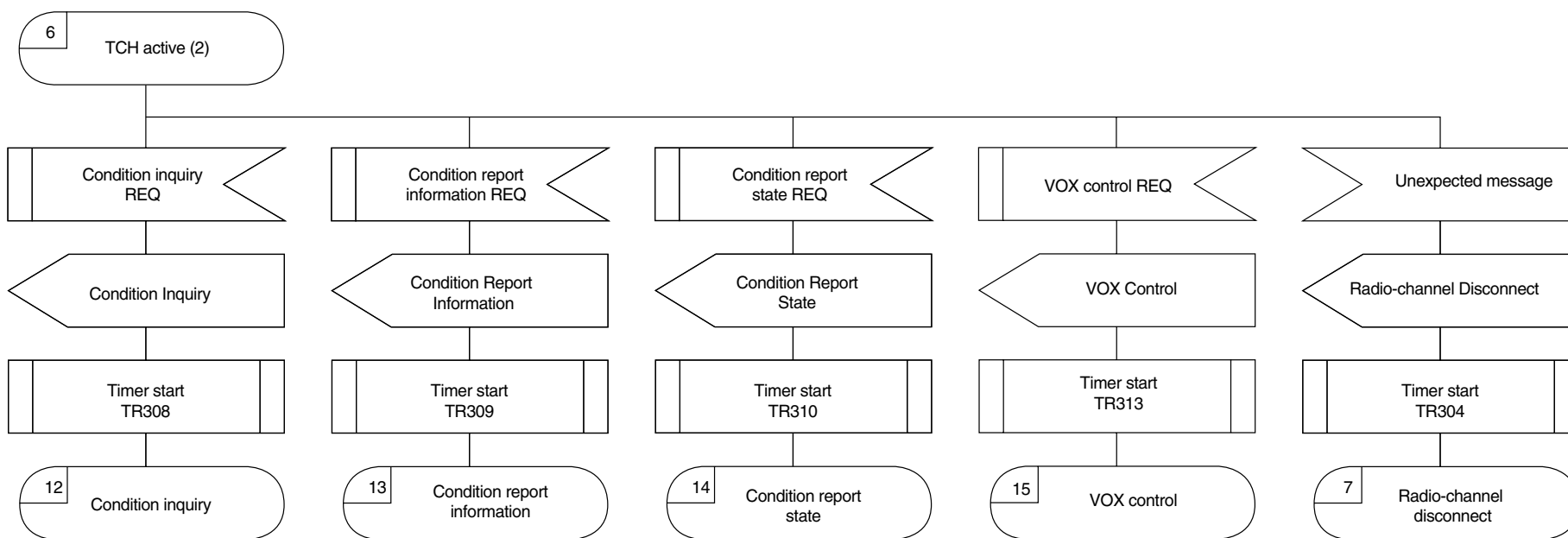


Fig. A. 5-55 Layer 3 RT SDL diagram (Network side) (8/34)

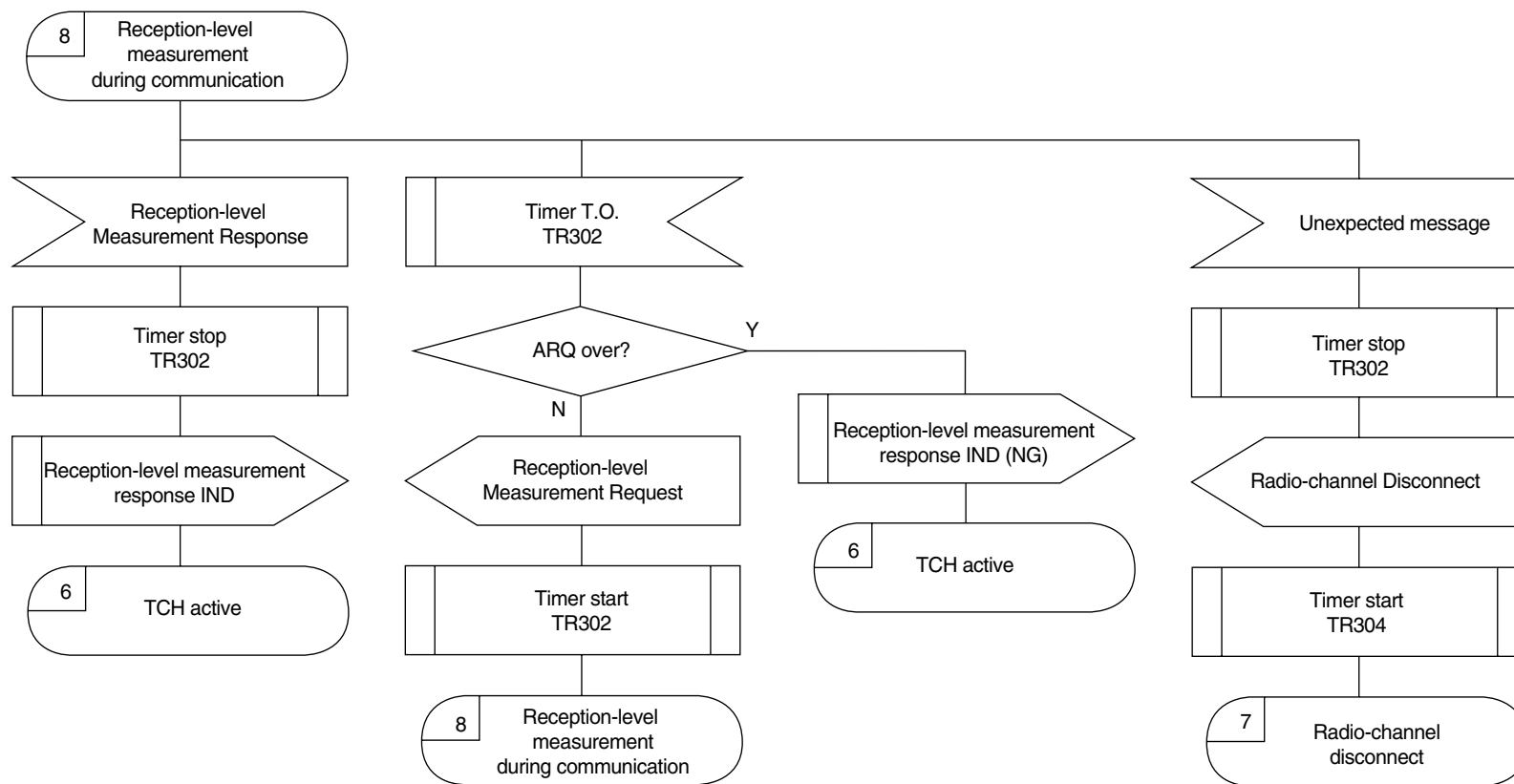


Fig. A. 5-56 Layer 3 RT SDL diagram (Network side) (9/34)

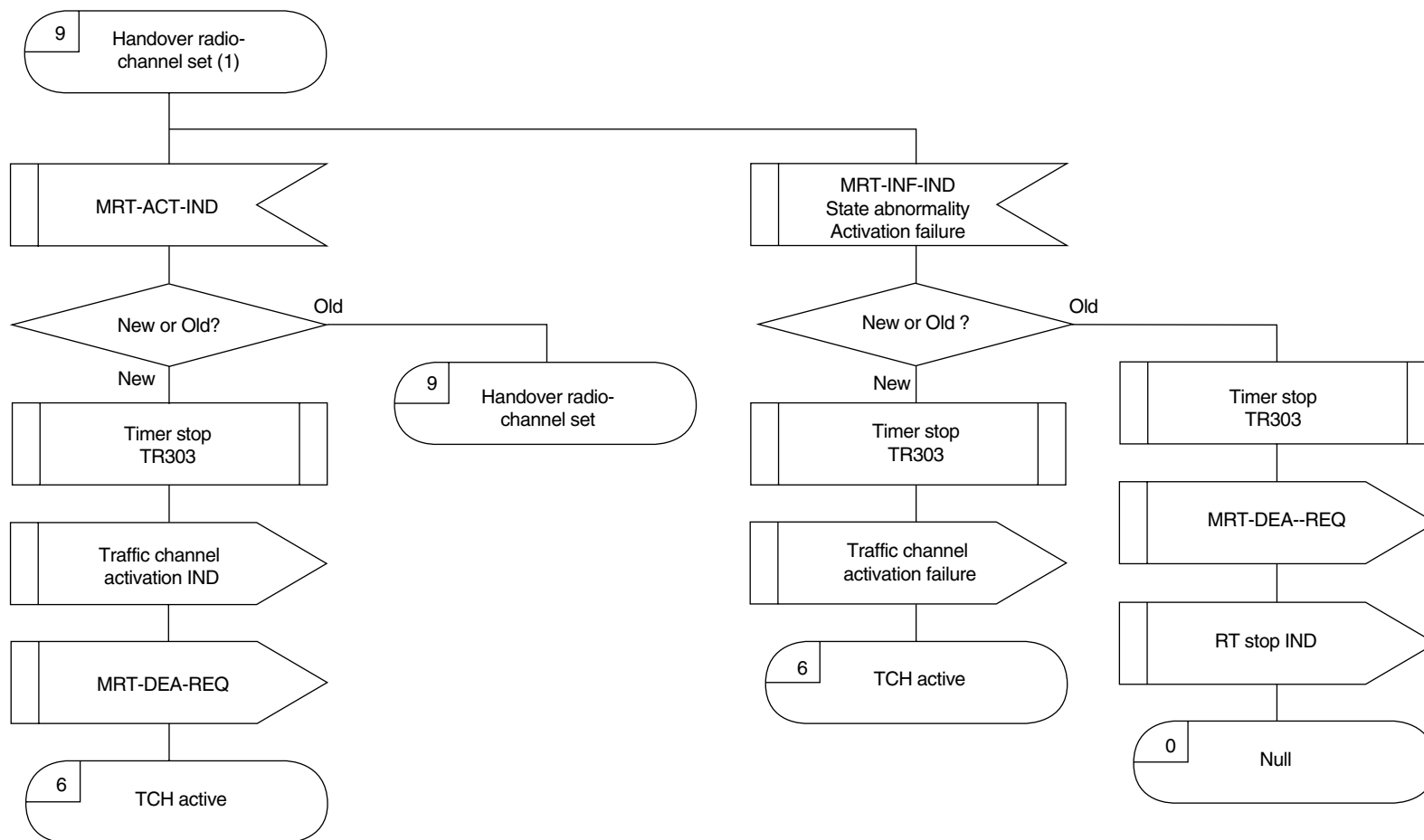


Fig. A. 5-57 Layer 3 RT SDL diagram (Network side) (10/34)

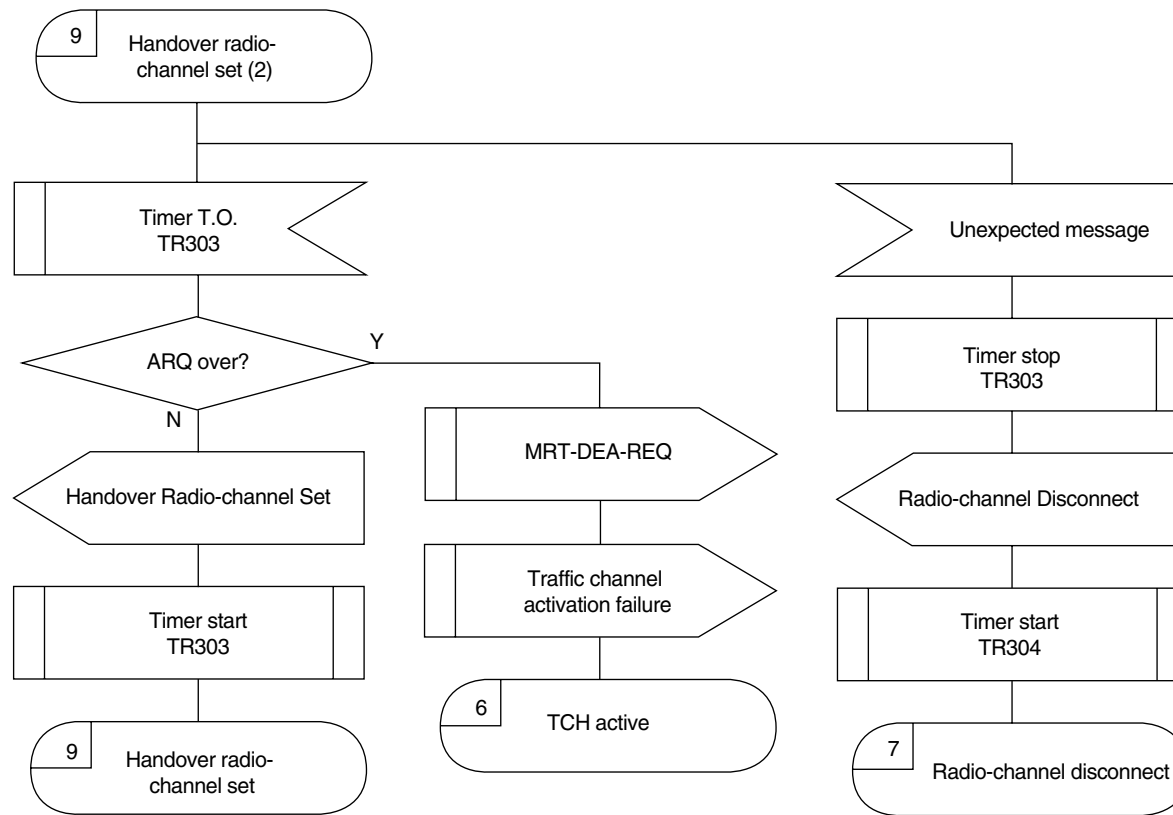


Fig. A. 5-58 Layer 3 RT SDL diagram (Network side) (11/34)

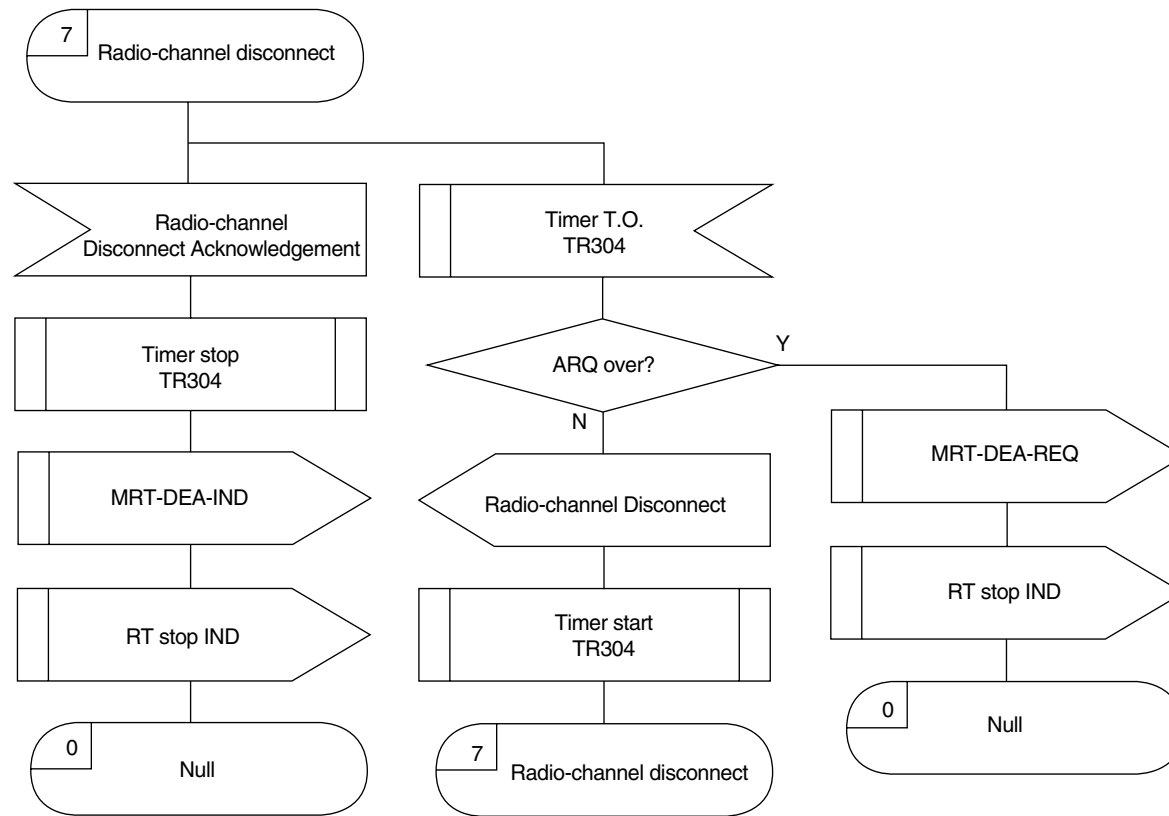


Fig. A. 5-59 Layer 3 RT SDL diagram (Network side) (12/34)

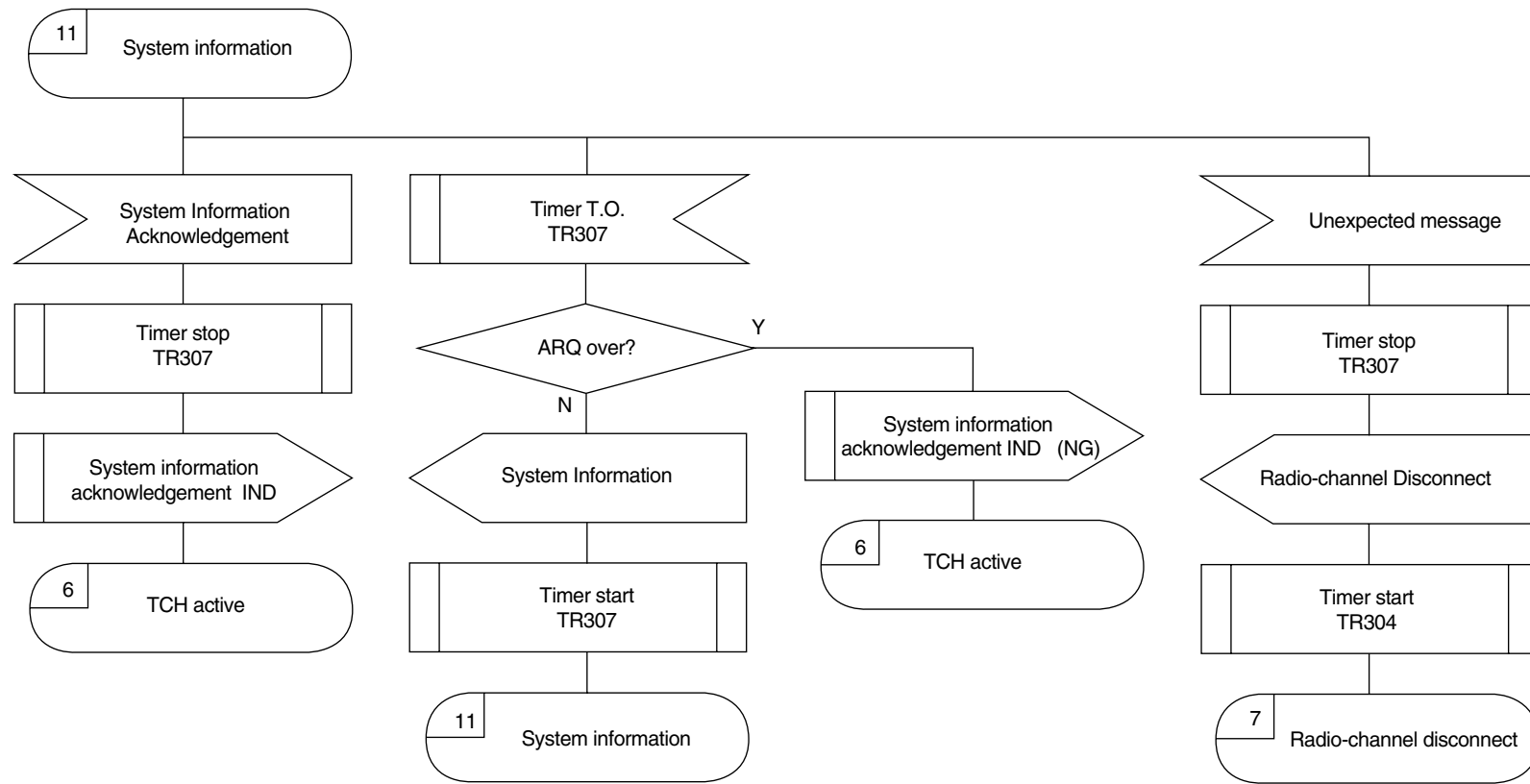


Fig. A. 5-60 Layer 3 RT SDL diagram (Network side) (13/34)

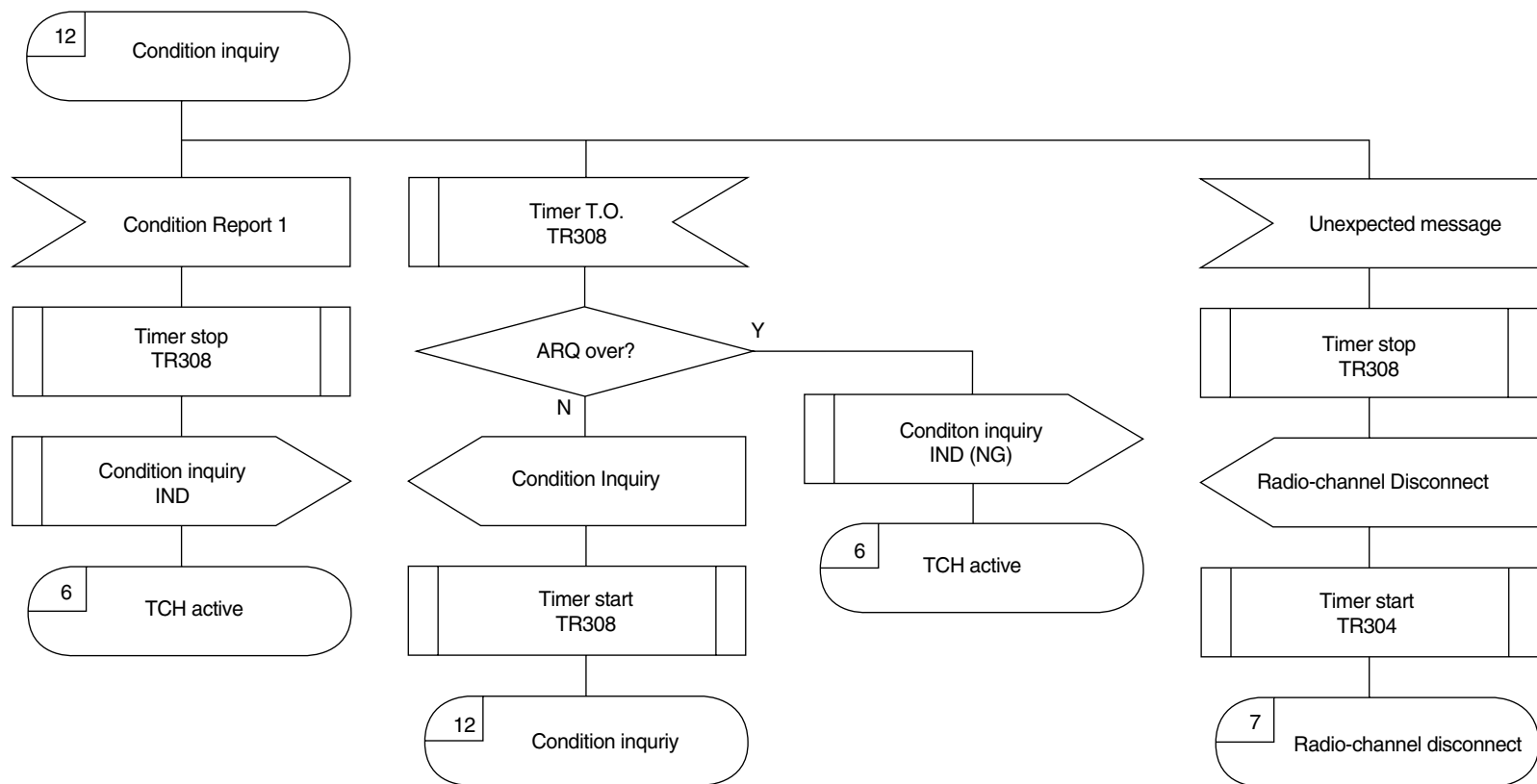


Fig. A. 5-61 Layer 3 RT SDL diagram (Network side) (14/34)

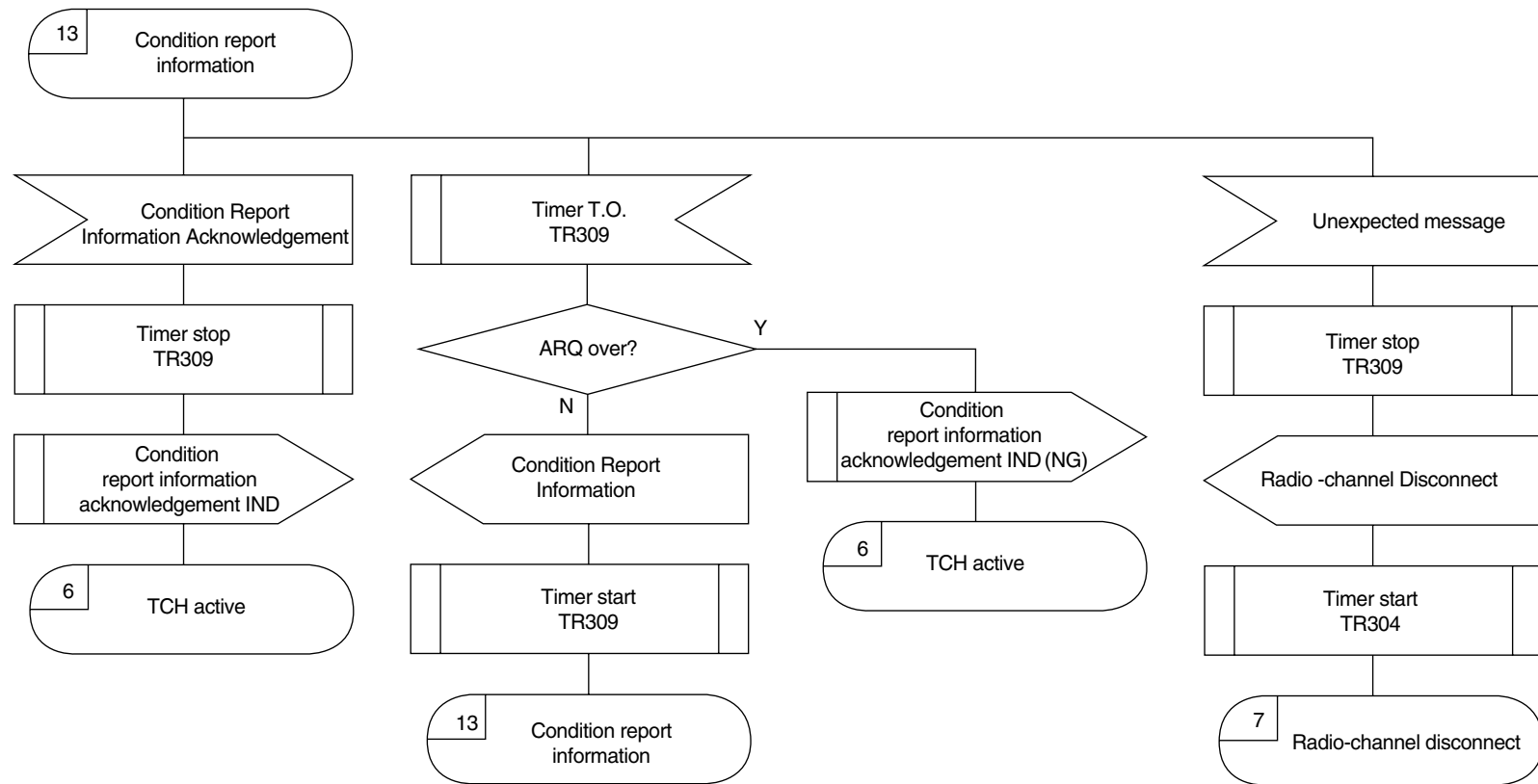


Fig. A. 5-62 Layer 3 RT SDL diagram (Network side) (15/34)

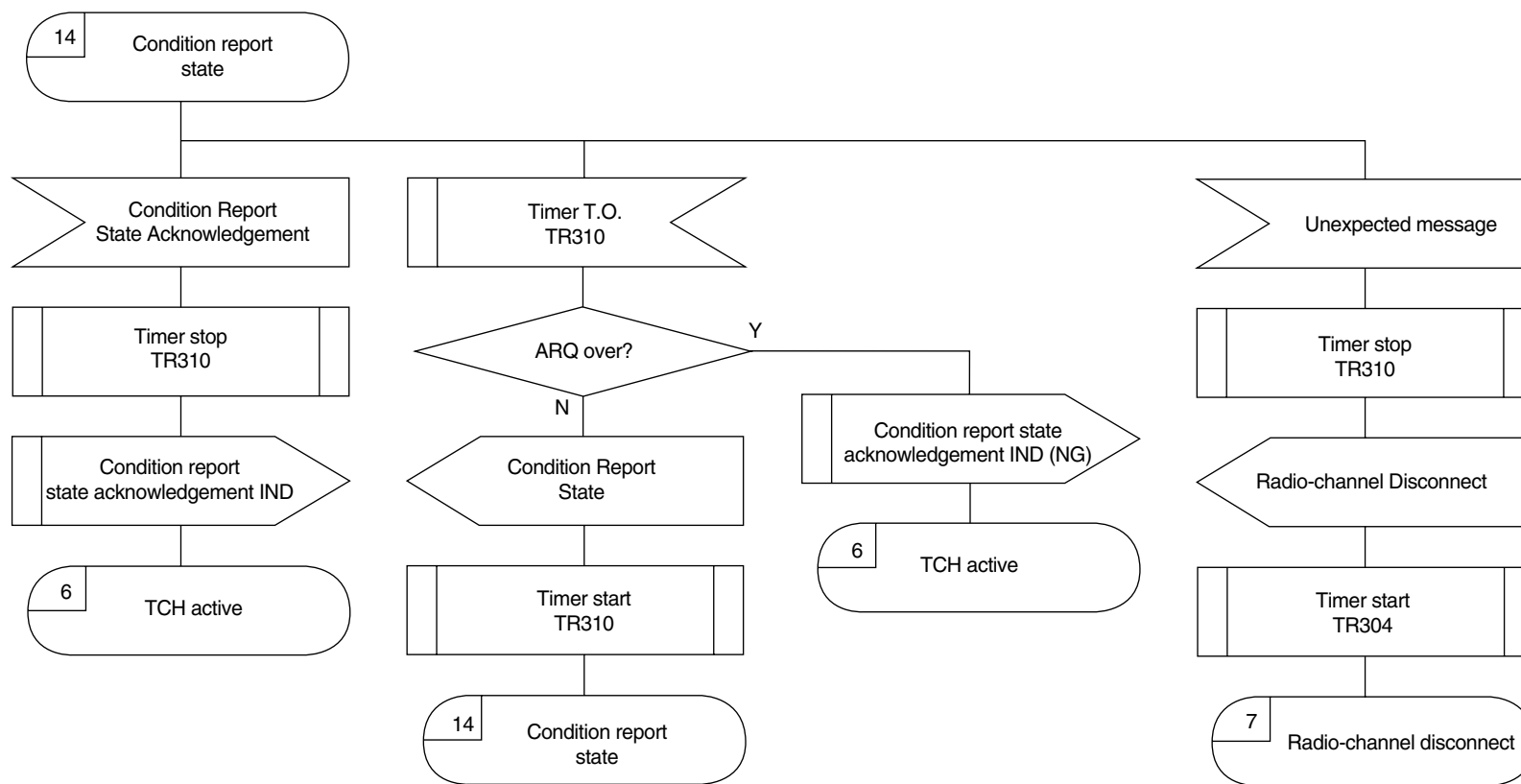


Fig. A. 5-63 Layer 3 RT SDL diagram (Network side) (16/34)

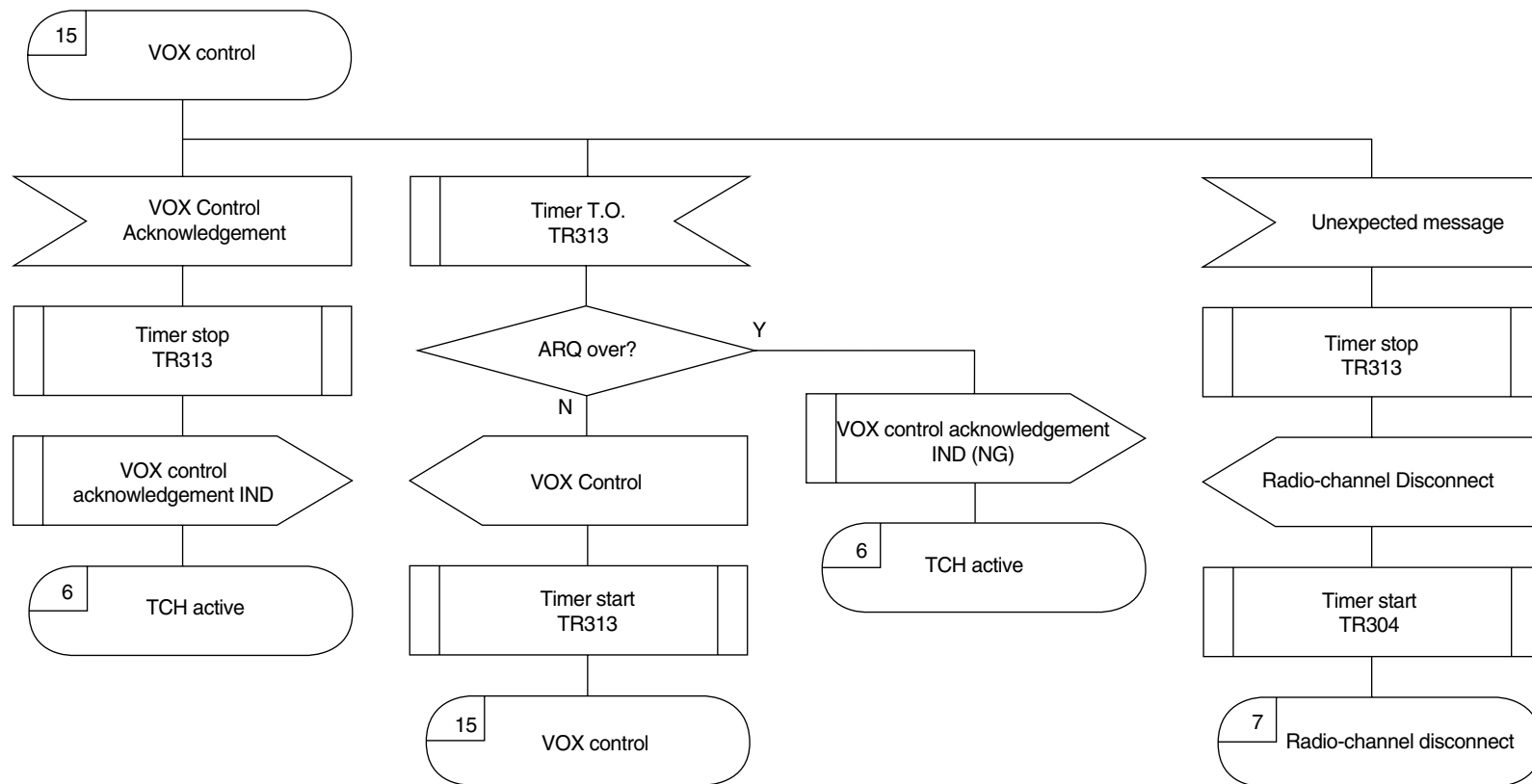


Fig. A. 5-64 Layer 3 RT SDL diagram (Network side) (17/34)

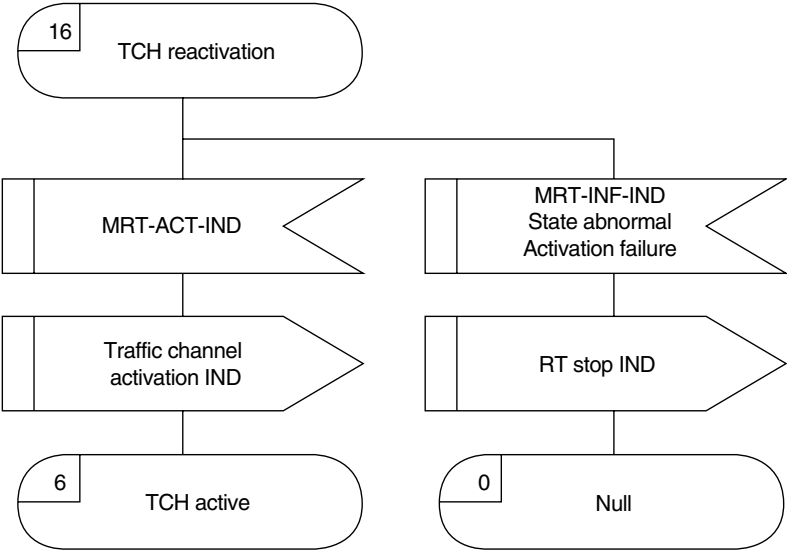


Fig. A. 5-65 Layer 3 RT SDL diagram (Network-side) (18/34)

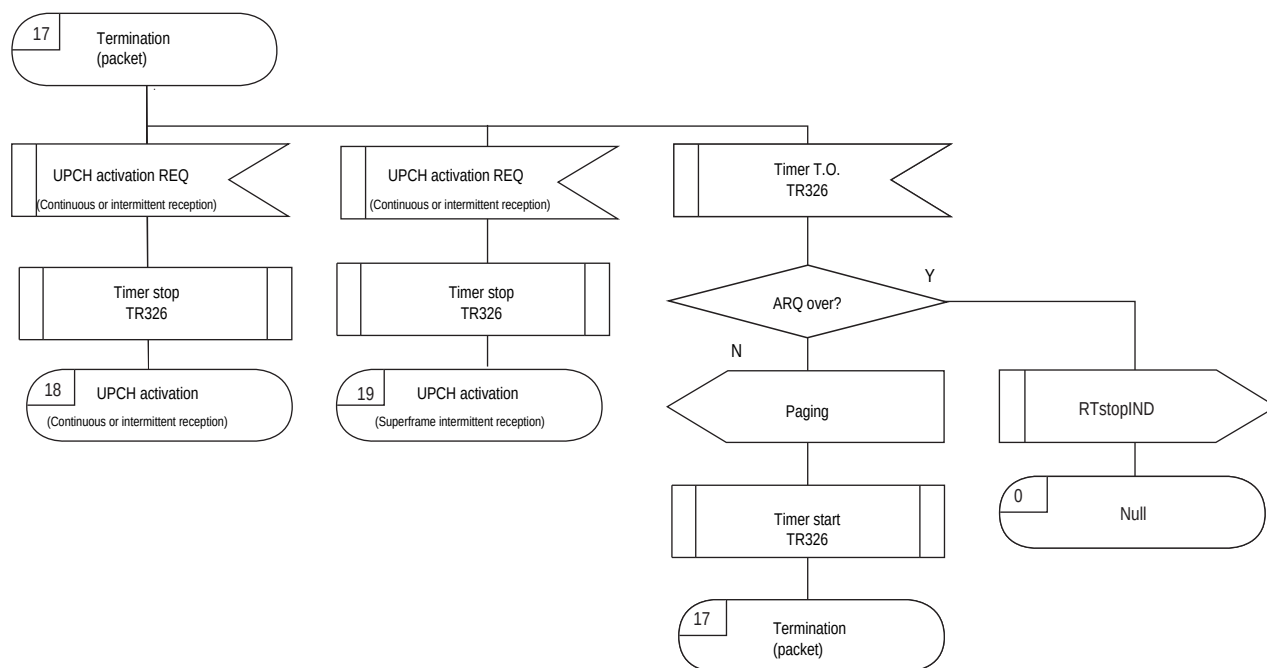


Fig. A. 5-66 Layer 3 RT SDL diagram (Network side) (19/34)

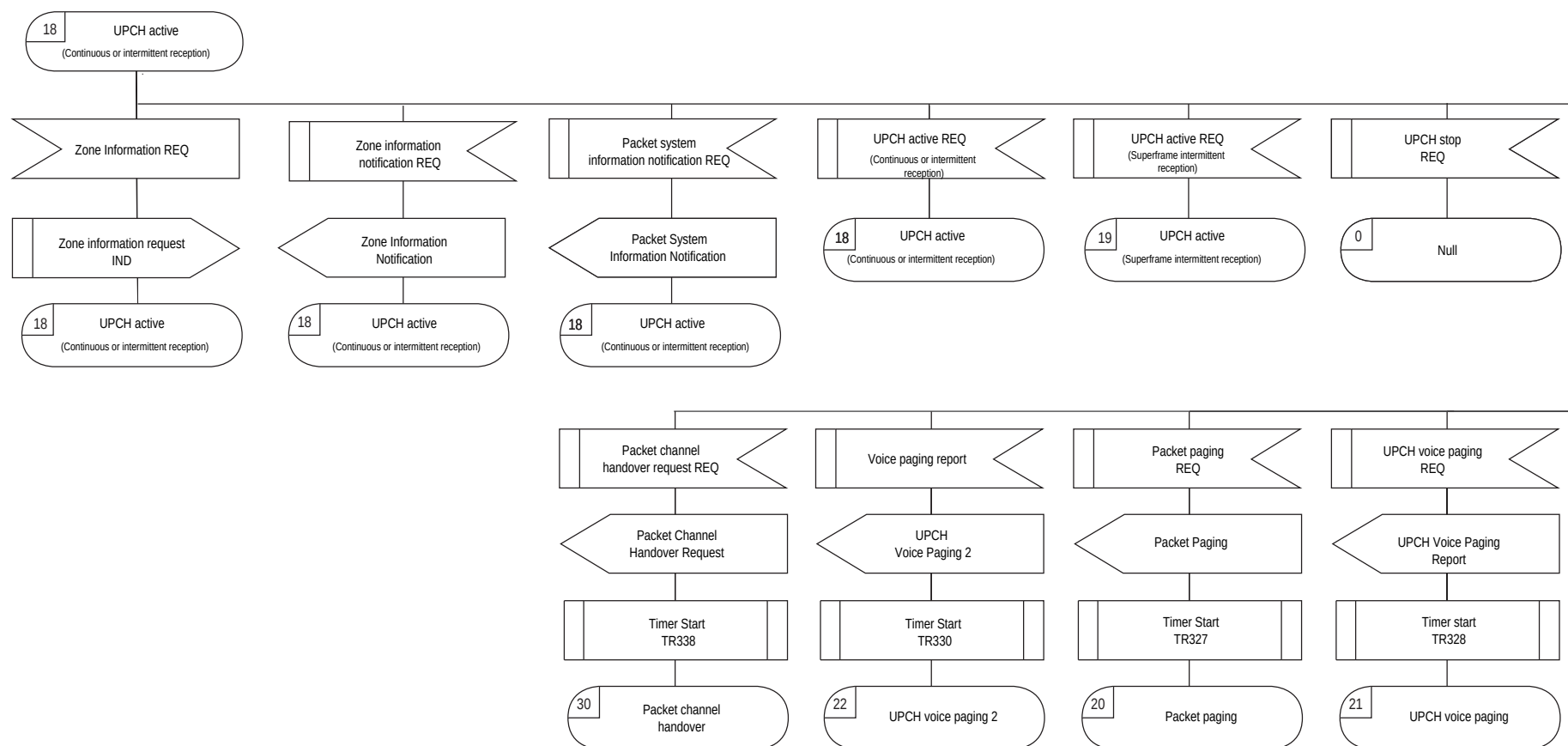


Fig. A. 5-67 Layer 3 RT SDL diagram (Network side) (20/34)

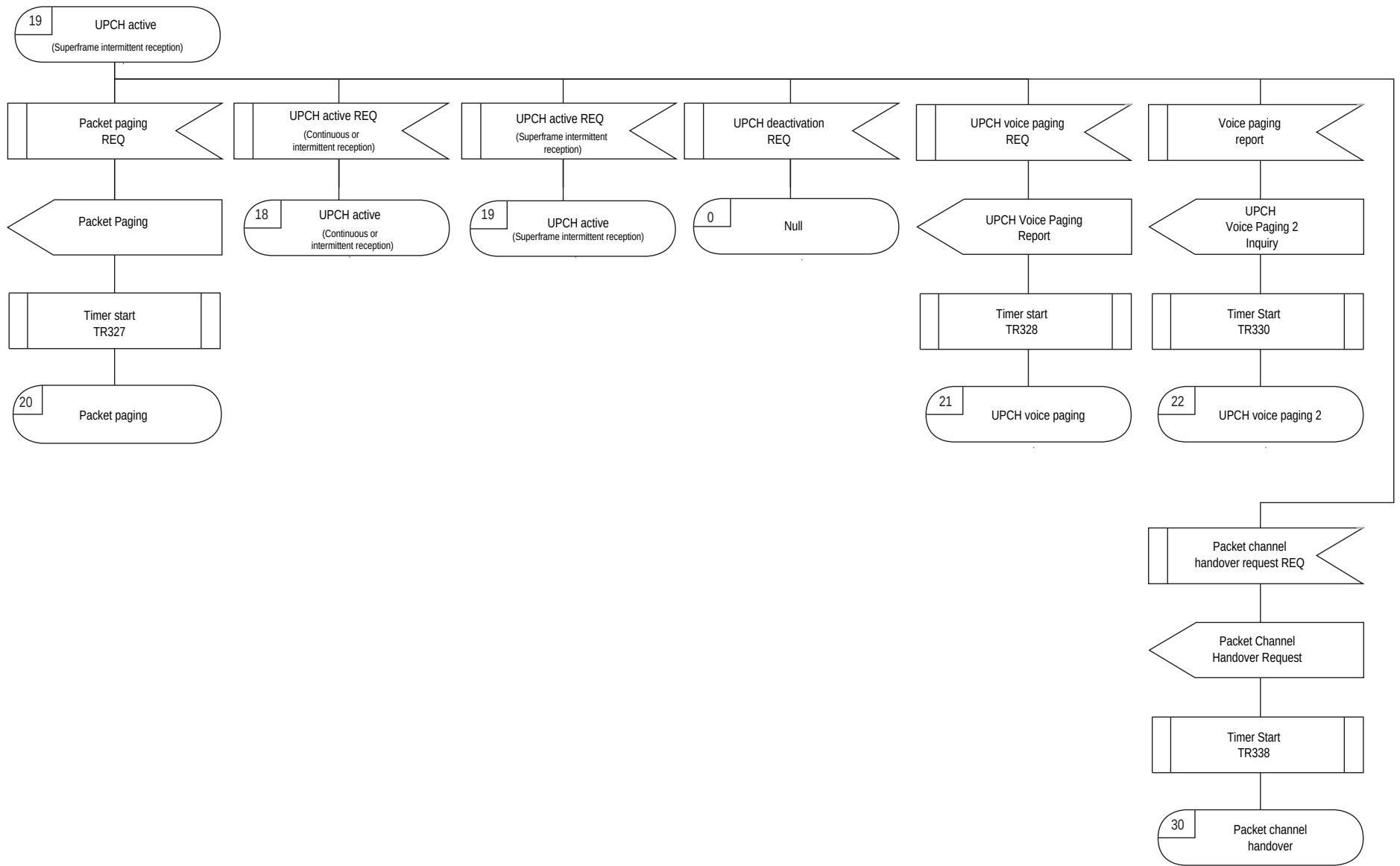
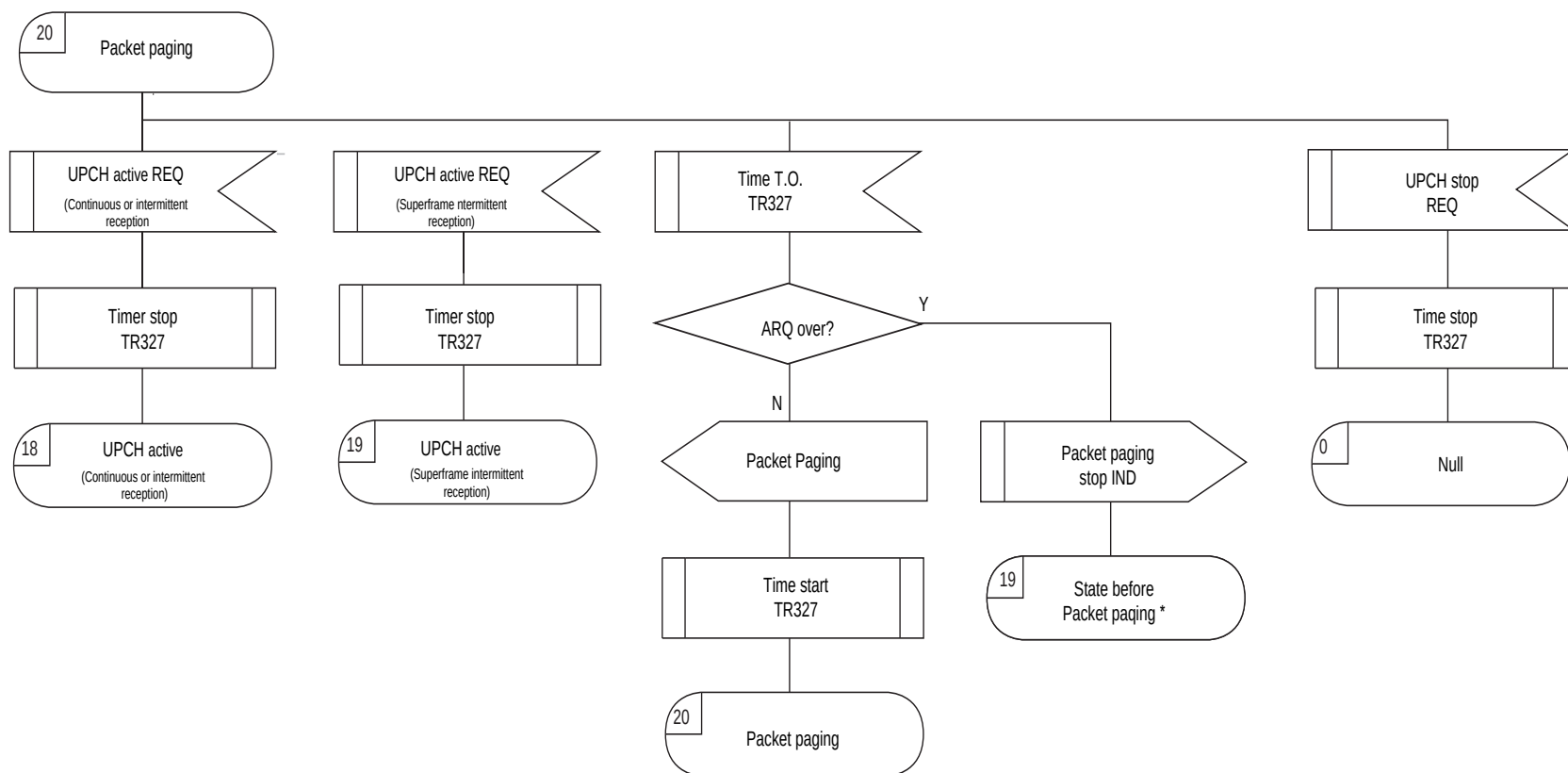


Fig. A. 5-68 Layer 3 RT SDL diagram (Network side) (21/34)

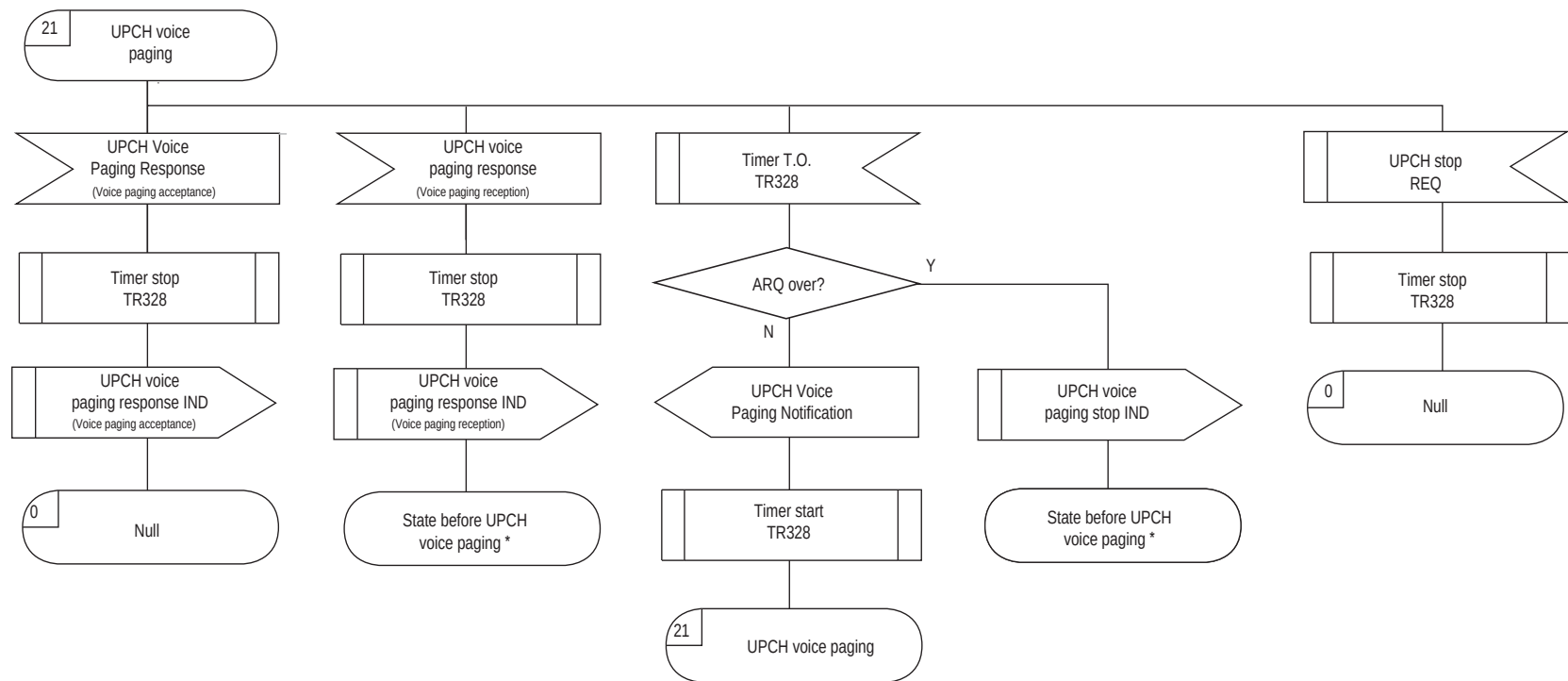


* Possible states are :

18: UPCH active (Continuous or intermittent reception)

19: UPCH active (Superframe intermittent reception)

Fig. A. 5-69 Layer 3 RT SDL diagram (Network side) (22/34)



* States allowed at this point are
 18: UPCH active (Continuous or intermittent reception)
 19: UPCH active (Superframe intermittent reception)

Fig. A. 5-70 Layer 3 RT SDL diagram (Network side) (23/34)

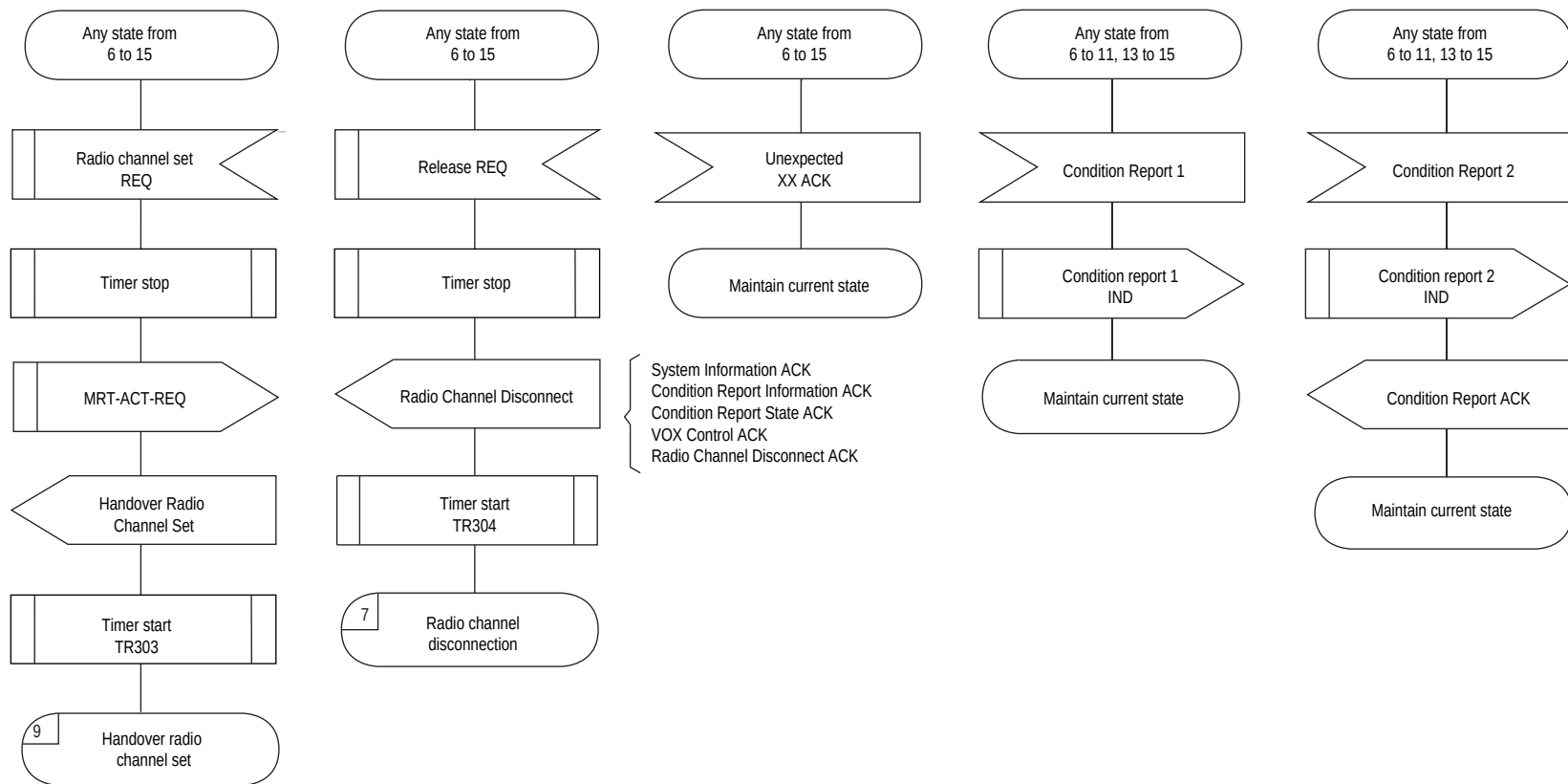


Fig. A. 5-71 Layer 3 RT SDL diagram (Network side) (24/34)

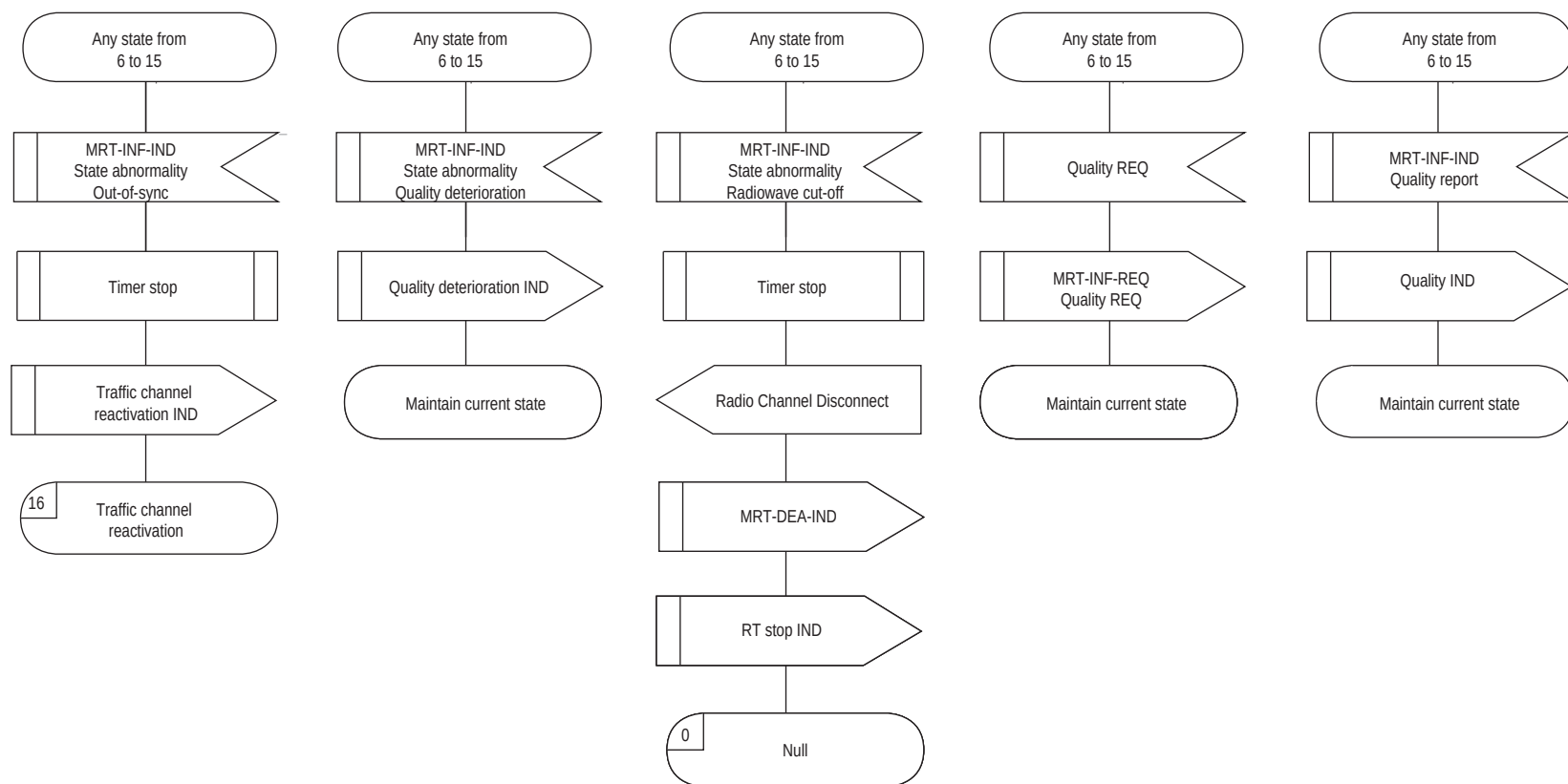


Fig. A. 5-72 Layer 3 RT SDL diagram (Network side) (25/34)

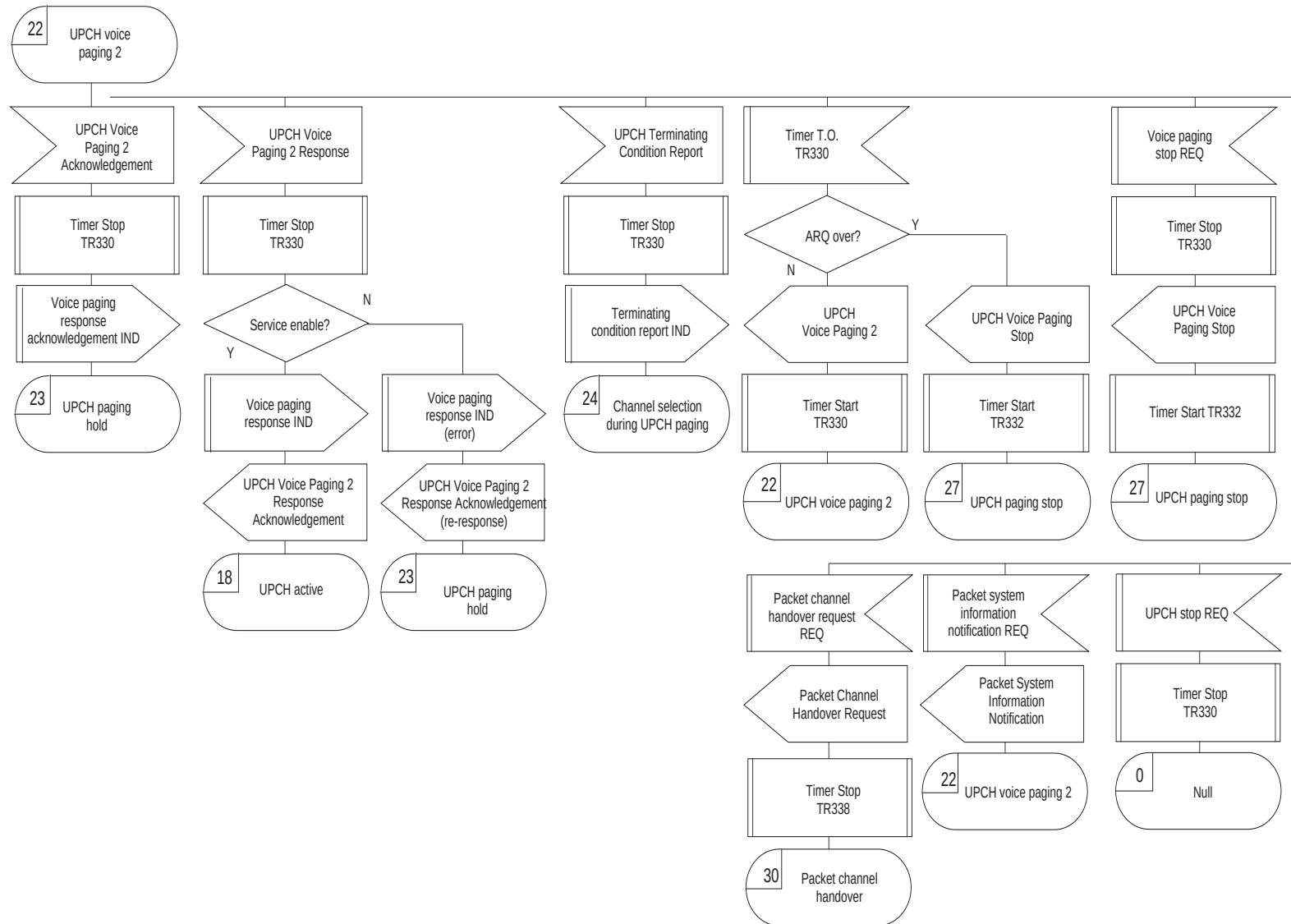


Fig. A. 5-73 Layer 3 RT SDL diagram (Network side) (26/34)

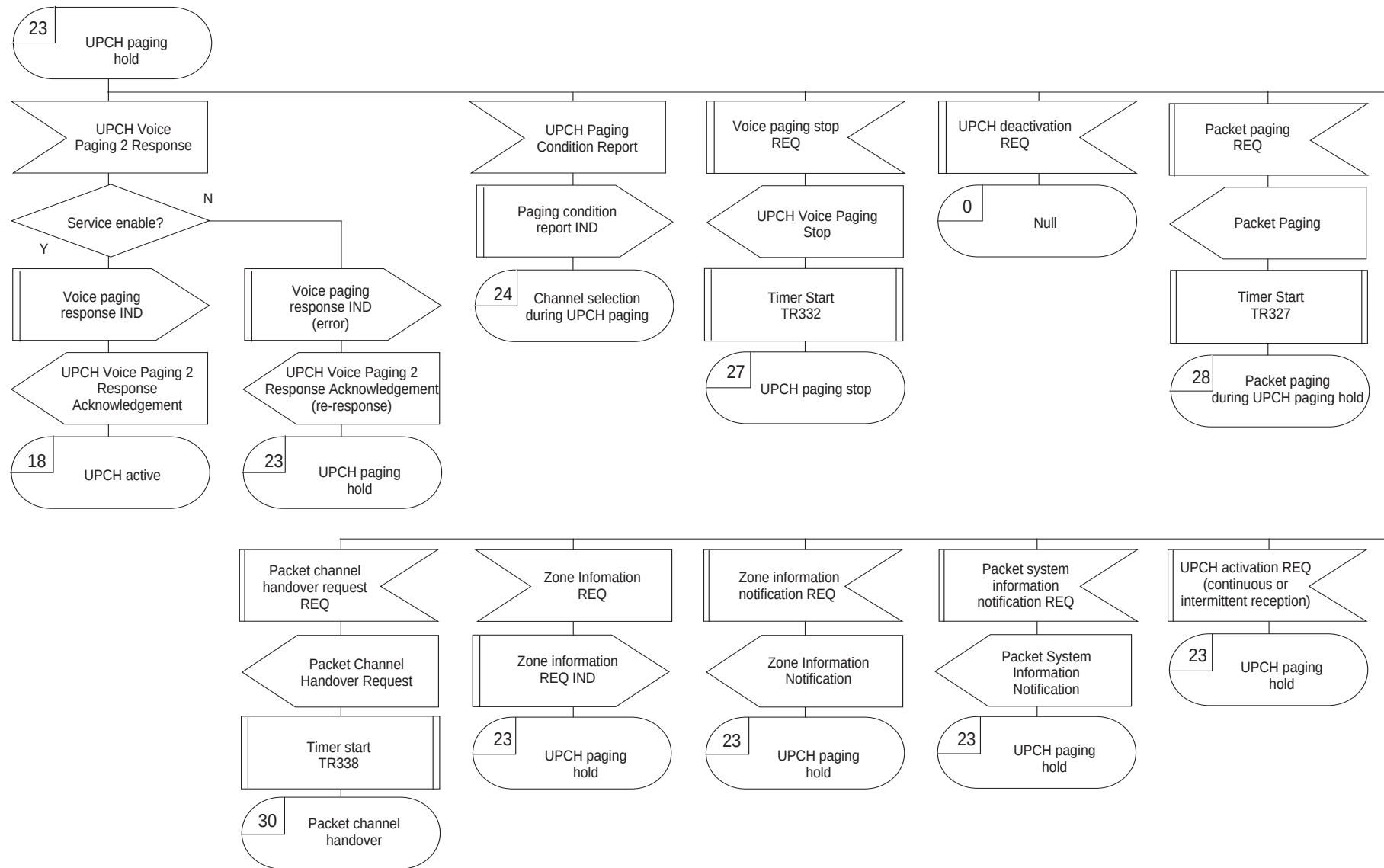


Fig. A. 5-74 Layer 3 RT SDL diagram (Network side) (27/34)

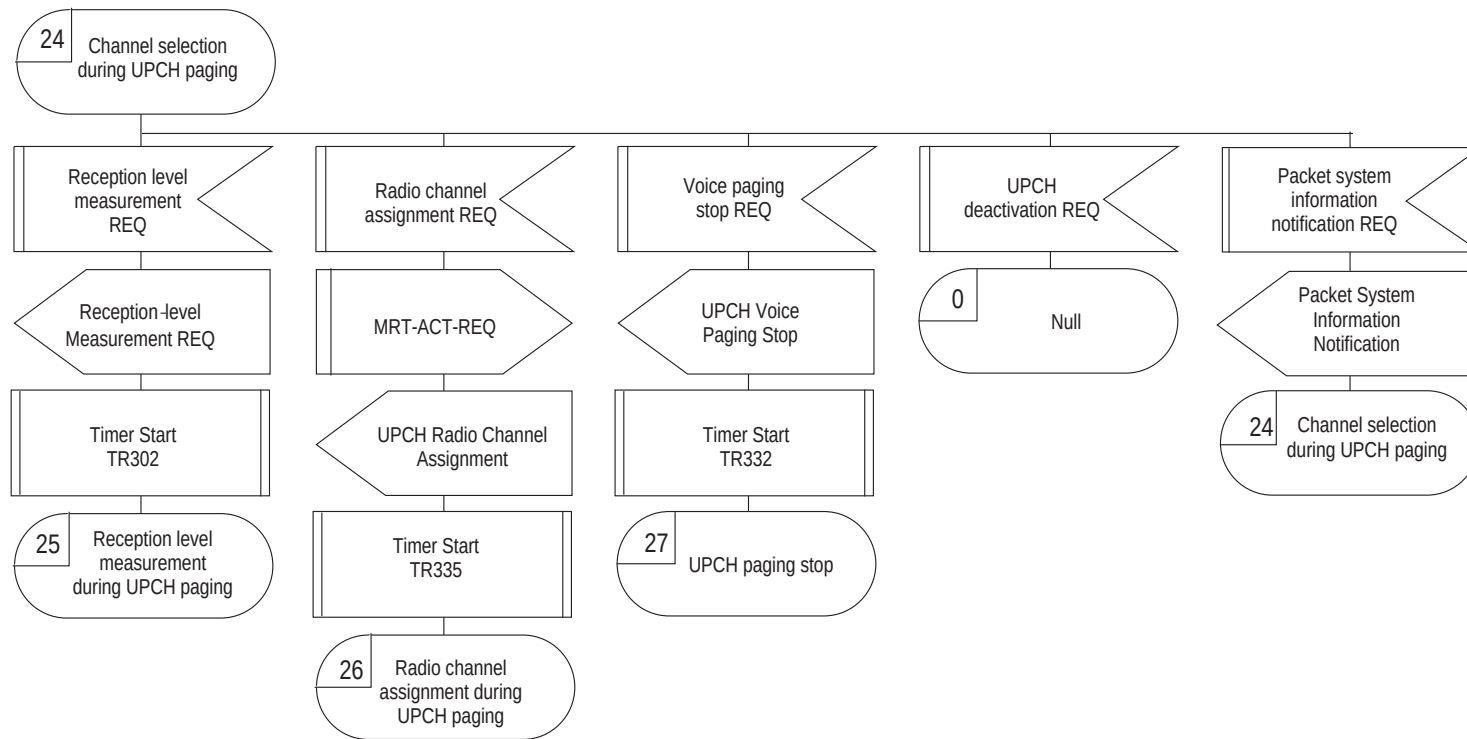


Fig. A. 5-75 Layer 3 RT SDL diagram (Network side) (28/34)

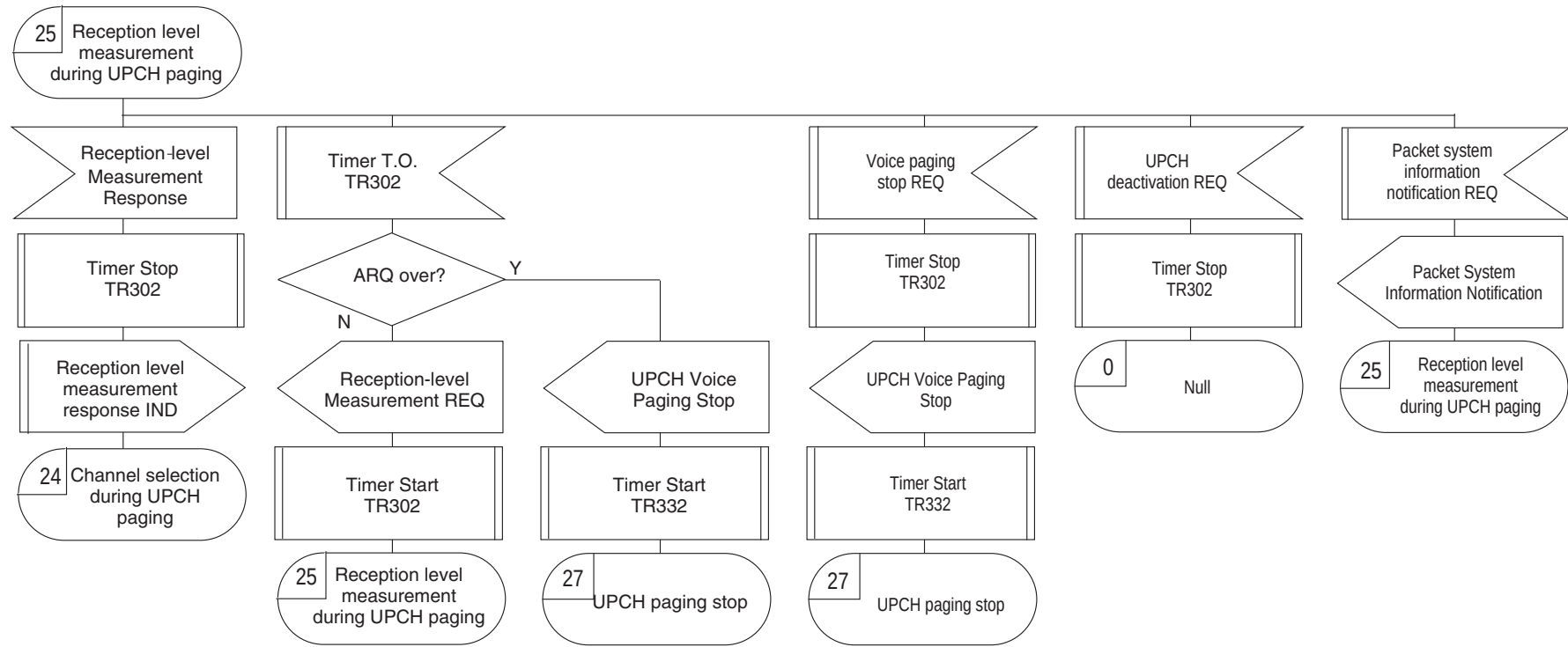


Fig. A. 5-76 Layer 3 RT SDL diagram (Network side) (29/34)

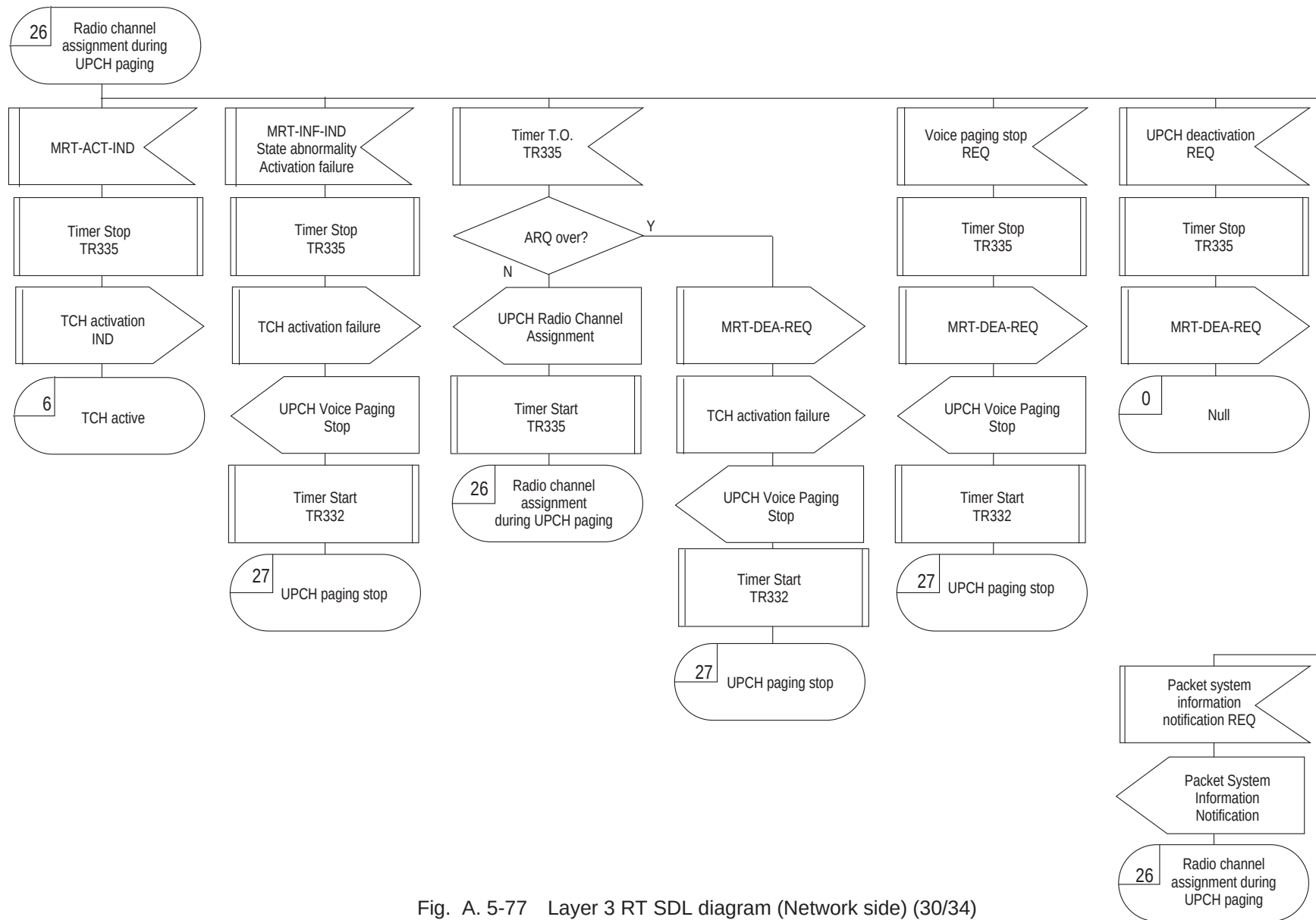


Fig. A. 5-77 Layer 3 RT SDL diagram (Network side) (30/34)

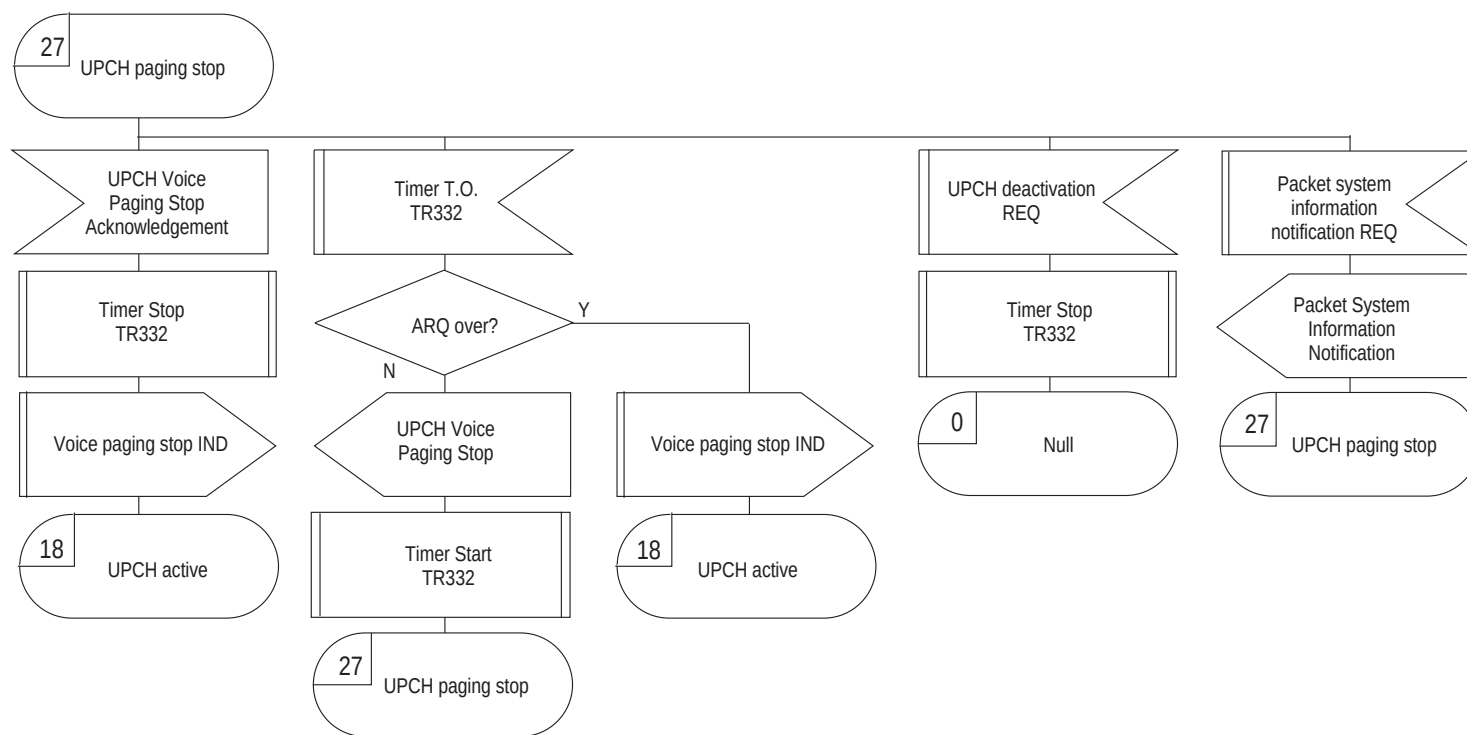


Fig. A. 5-78 Layer 3 RT SDL diagram (Network side) (31/34)



1

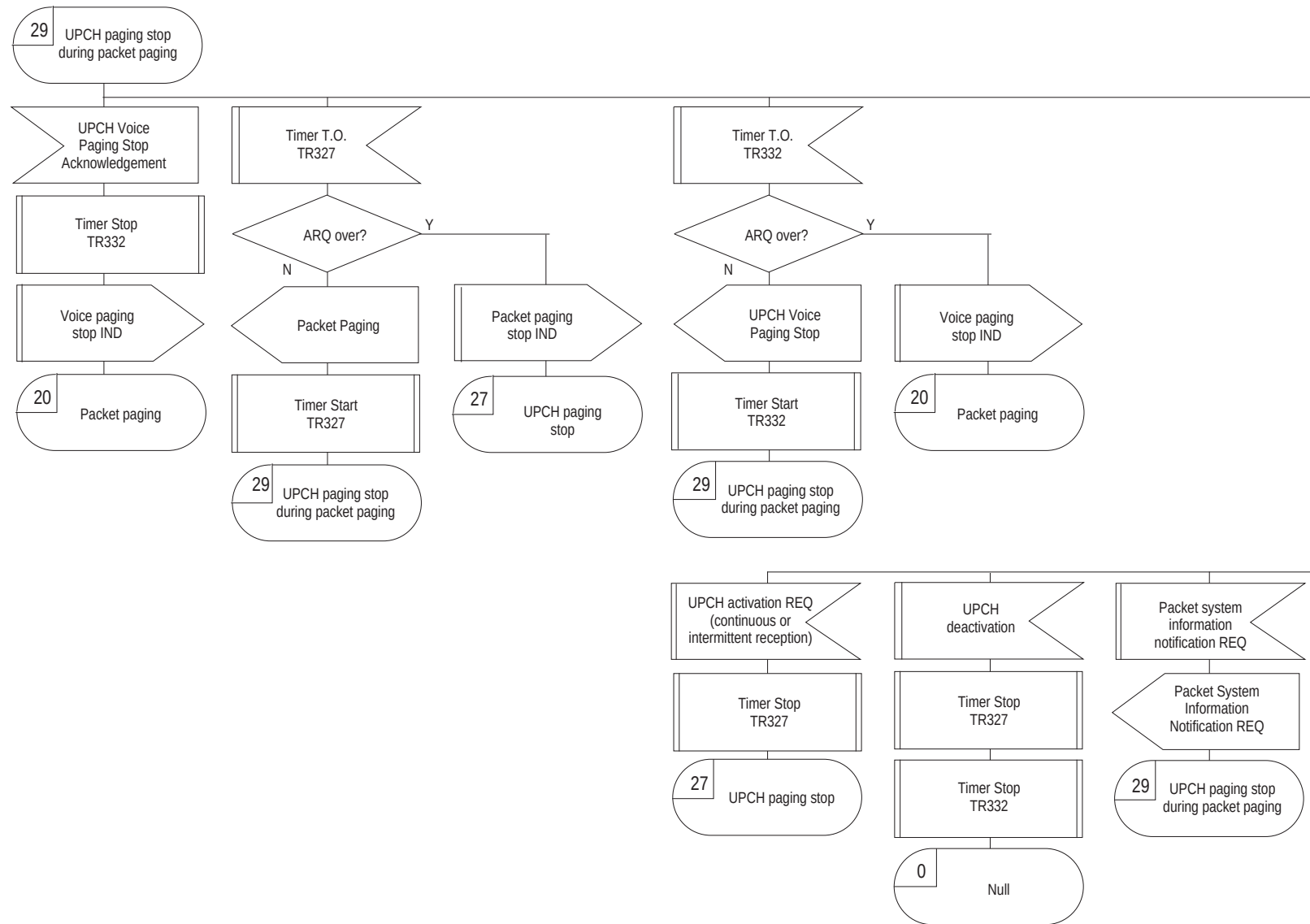
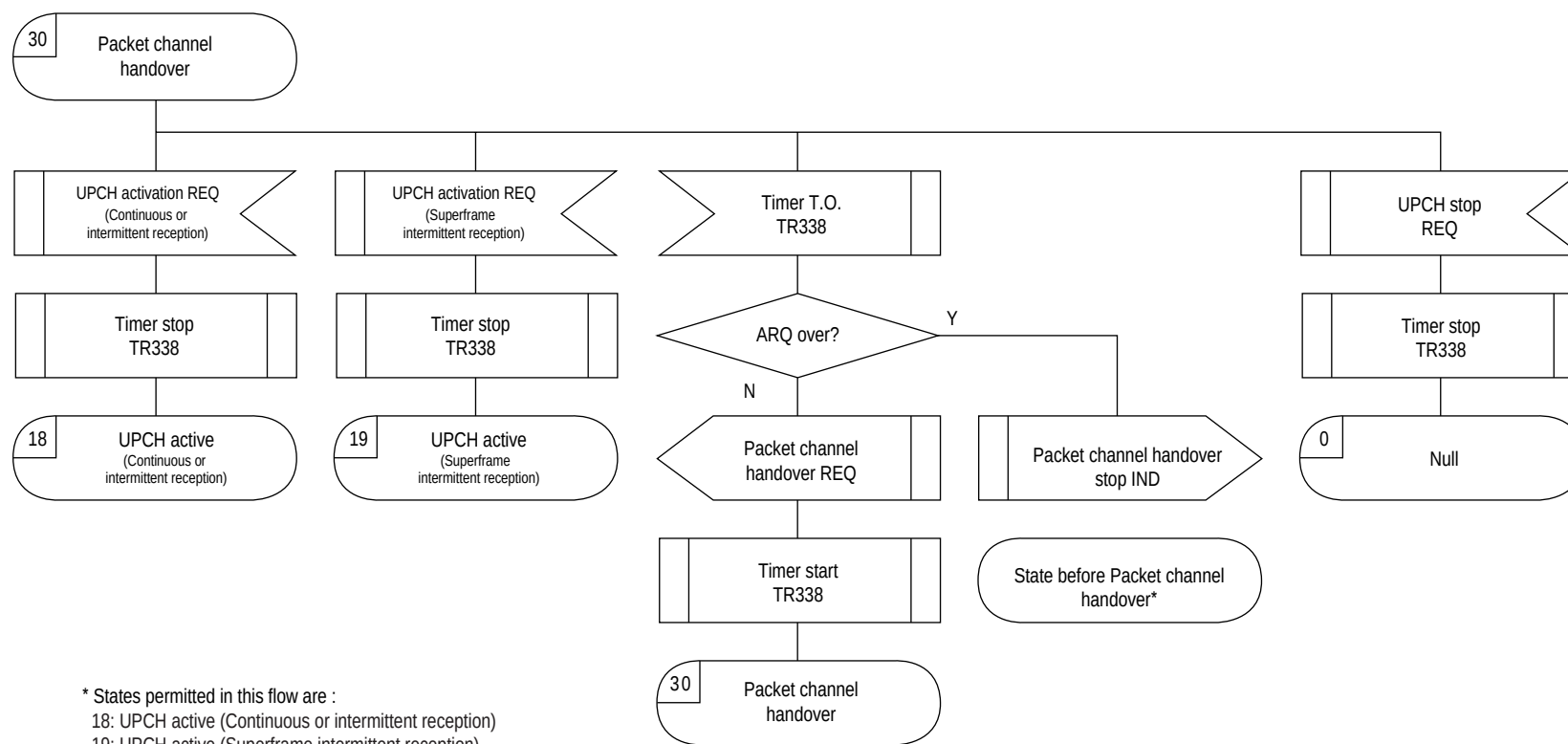


Fig. A. 5-80 Layer 3 RT SDL diagram (Network side) (33/34)



* States permitted in this flow are :
 18: UPCH active (Continuous or intermittent reception)
 19: UPCH active (Superframe intermittent reception)
 22: UPCH voice paging 2
 23: UPCH paging hold

Fig. A. 5-81 Layer 3 RT SDL diagram (Network side) (34/34)

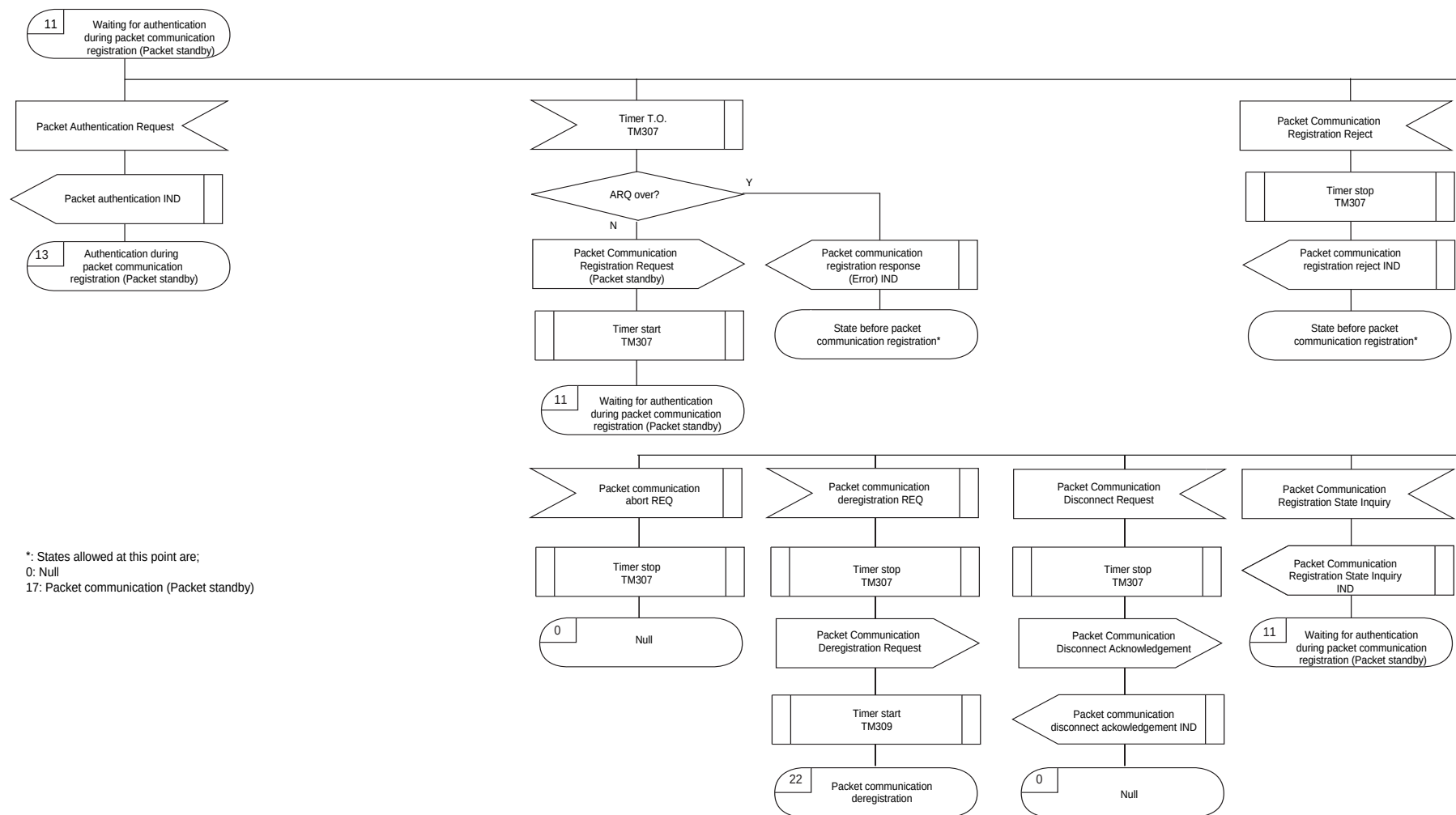


Fig. A. 5-82 Layer 3 MM SDL diagram (User side) (1/24)

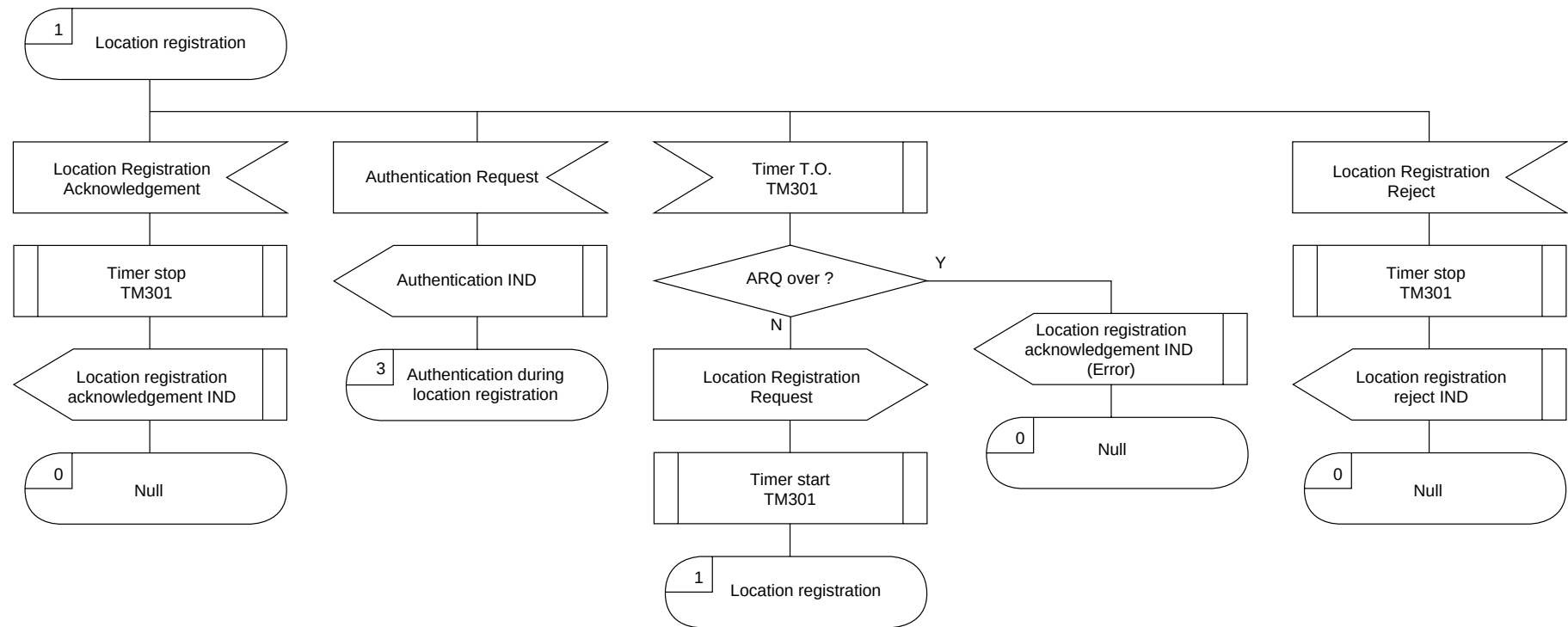


Fig. A. 5-83 Layer 3 MM SDL diagram (User side) (2/24)

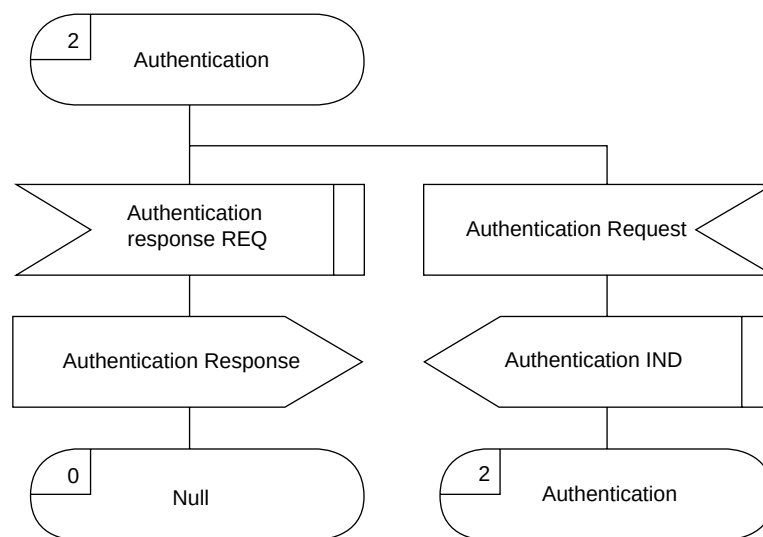


Fig. A. 5-84 Layer 3 MM SDL diagram (User side) (3/24)

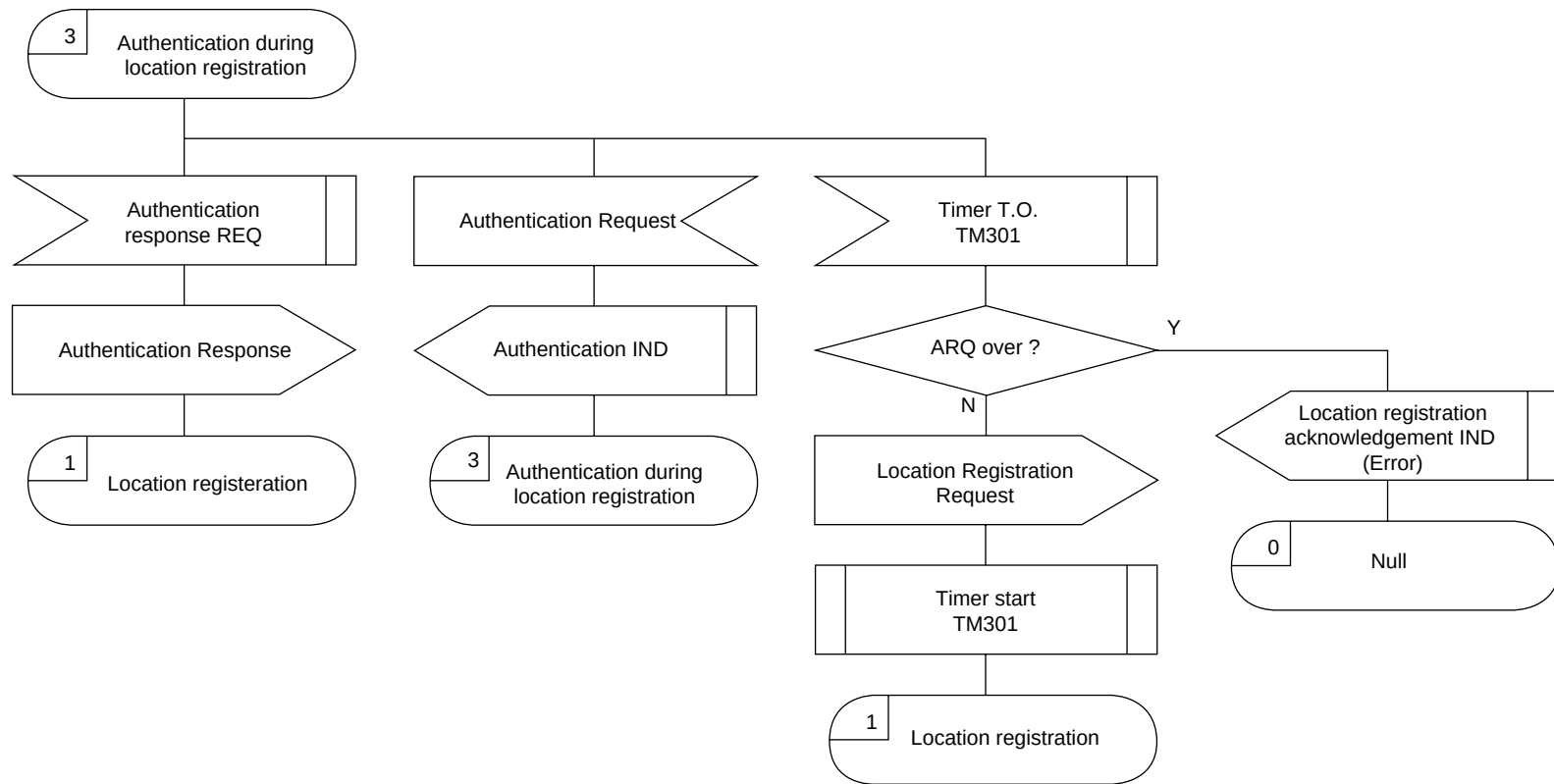


Fig. A. 5-85 Layer 3 MM SDL diagram (User side) (4/24)

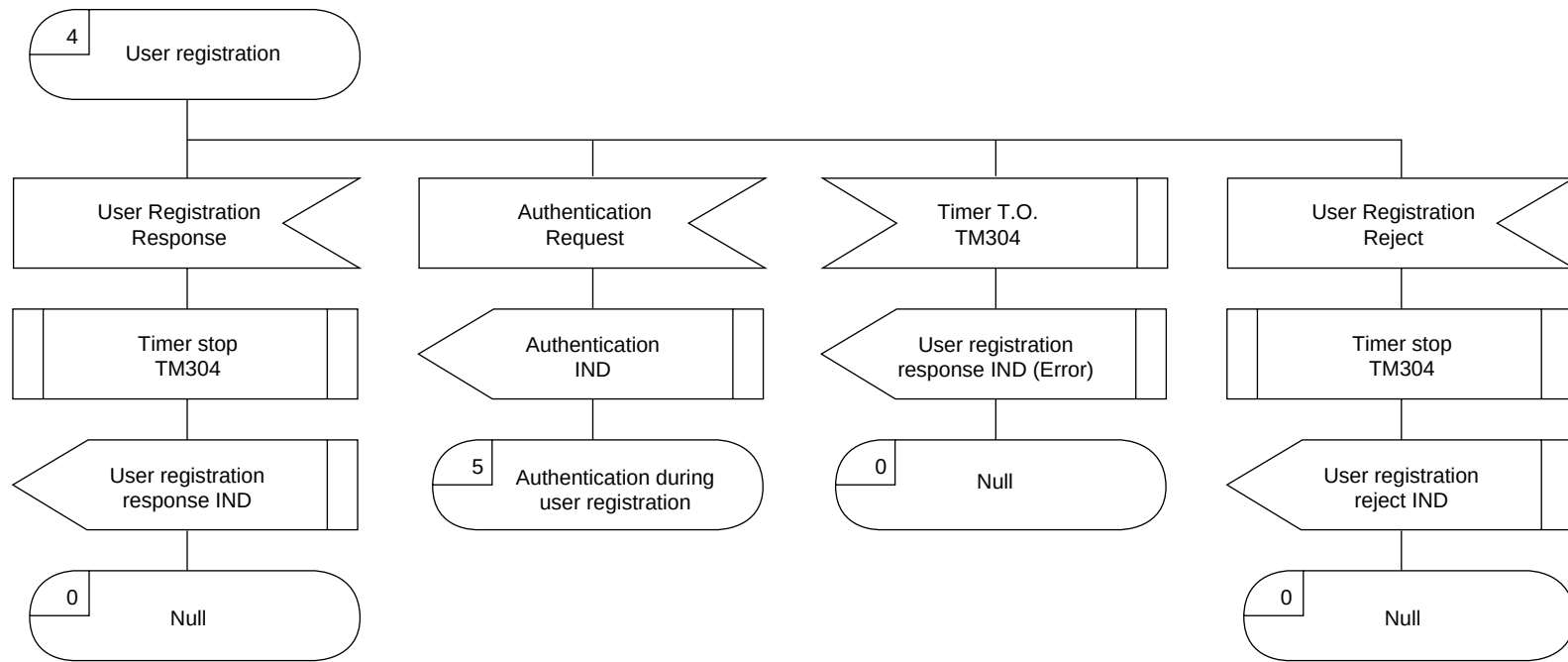


Fig. A. 5-86 Layer 3 MM SDL diagram (User side) (5/24)

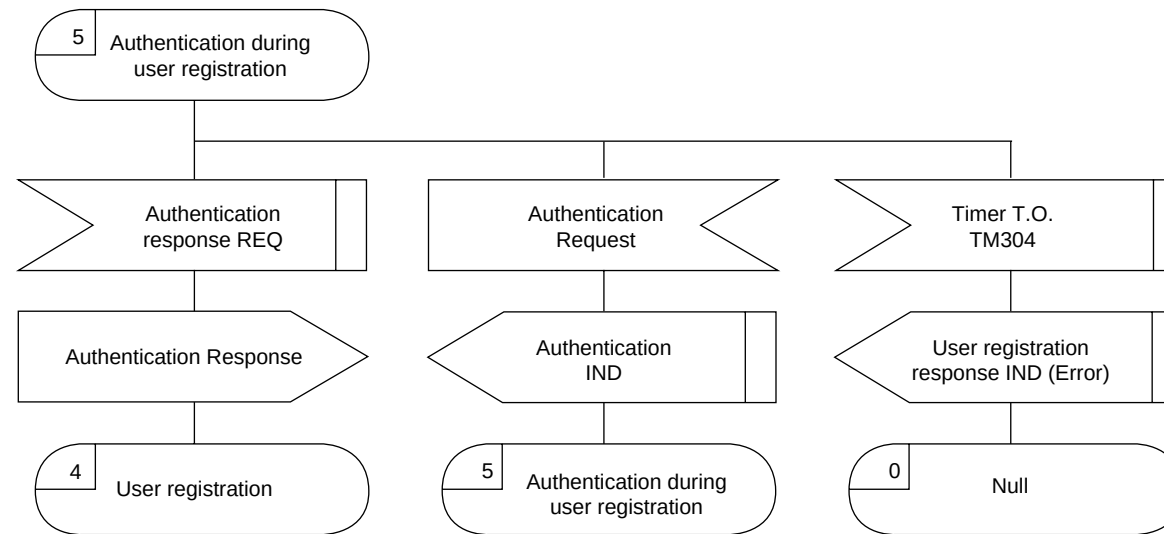


Fig. A. 5-87 Layer 3 MM SDL diagram (User side) (6/24)

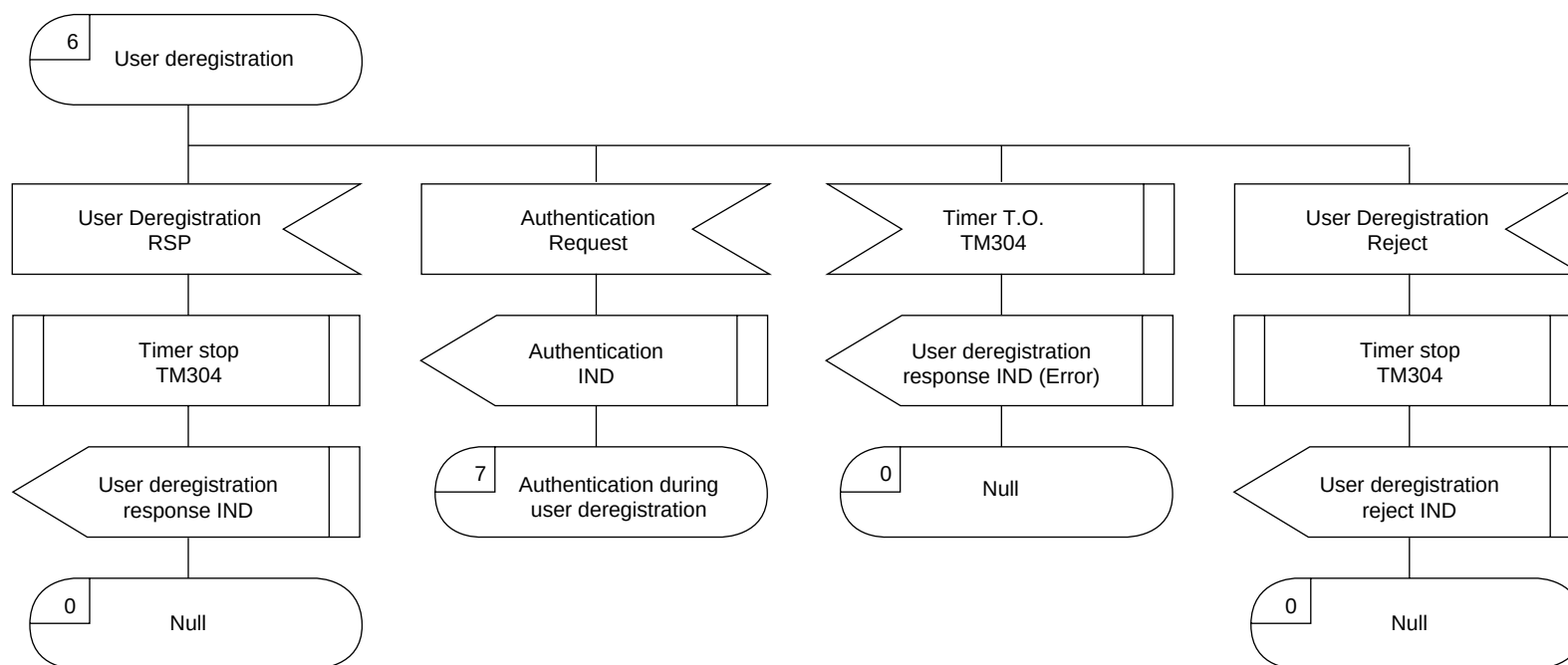


Fig. A. 5-88 Layer 3 MM SDL diagram (User side) (7/24)

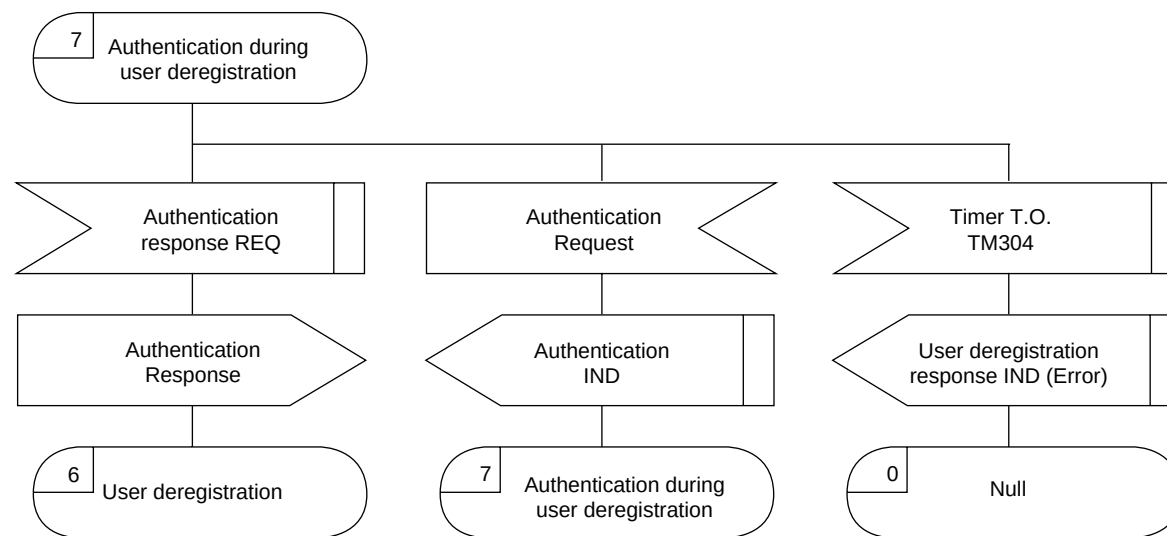


Fig. A. 5-89 Layer 3 MM SDL diagram (User side) (8/24)

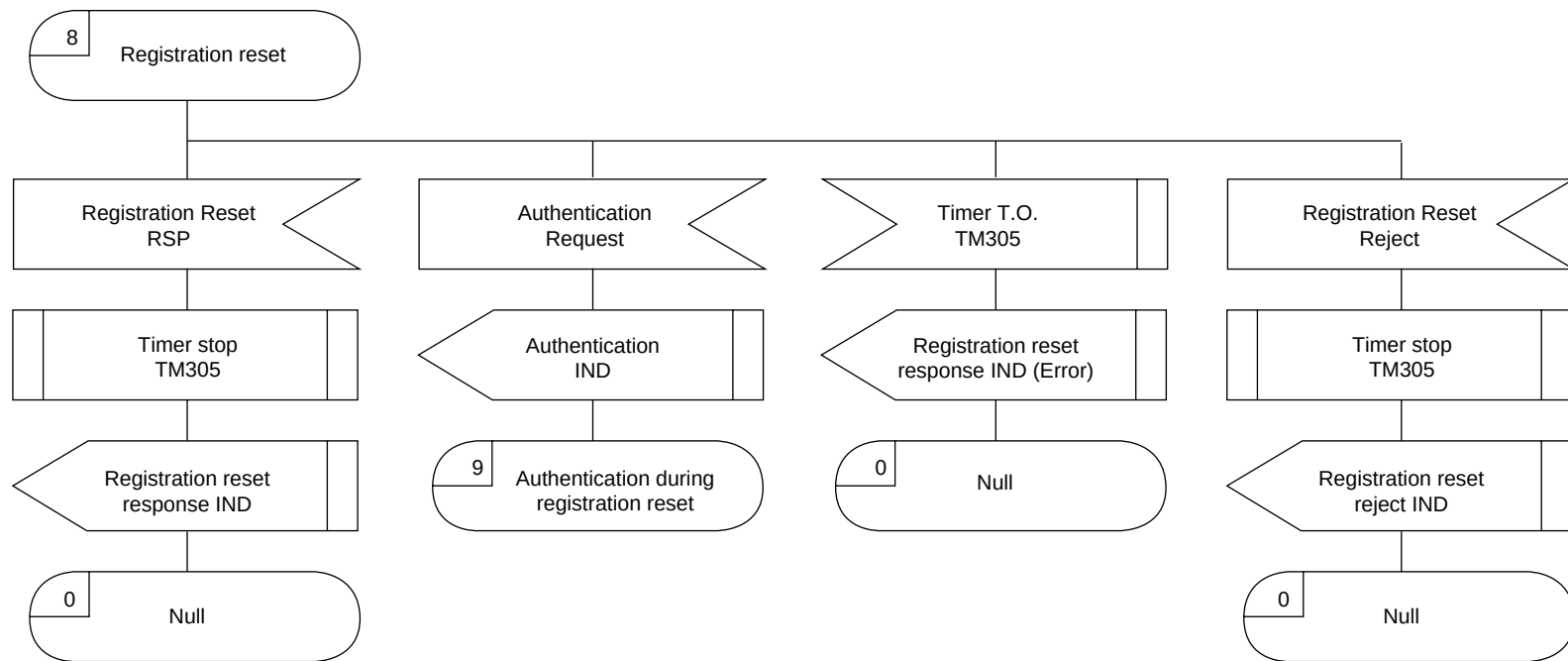


Fig. A. 5-90 Layer 3 MM SDL diagram (User side) (9/24)

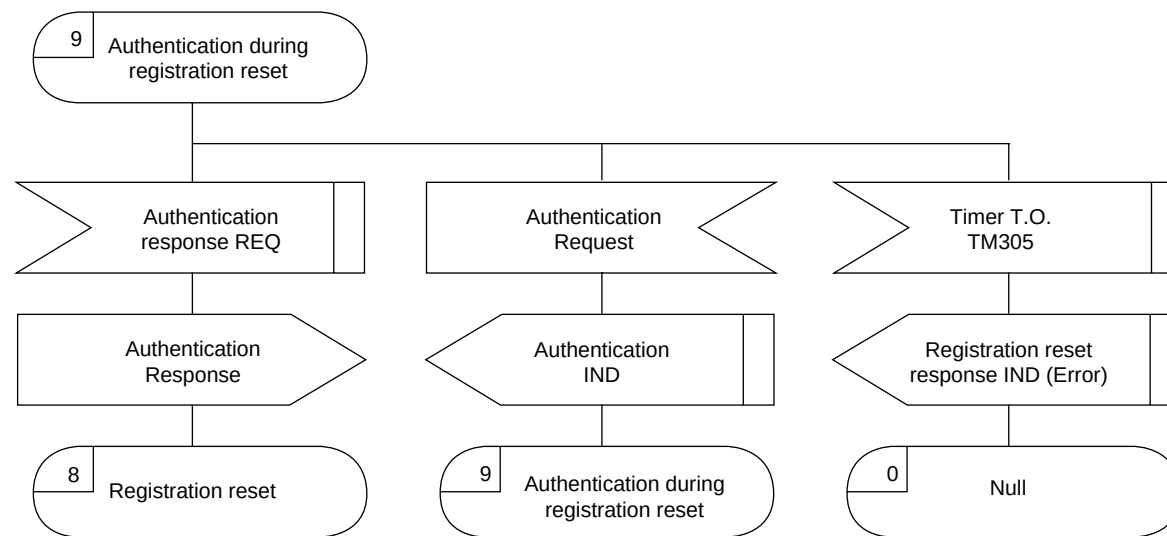


Fig. A. 5-91 Layer 3 MM SDL diagram (User side) (10/24)

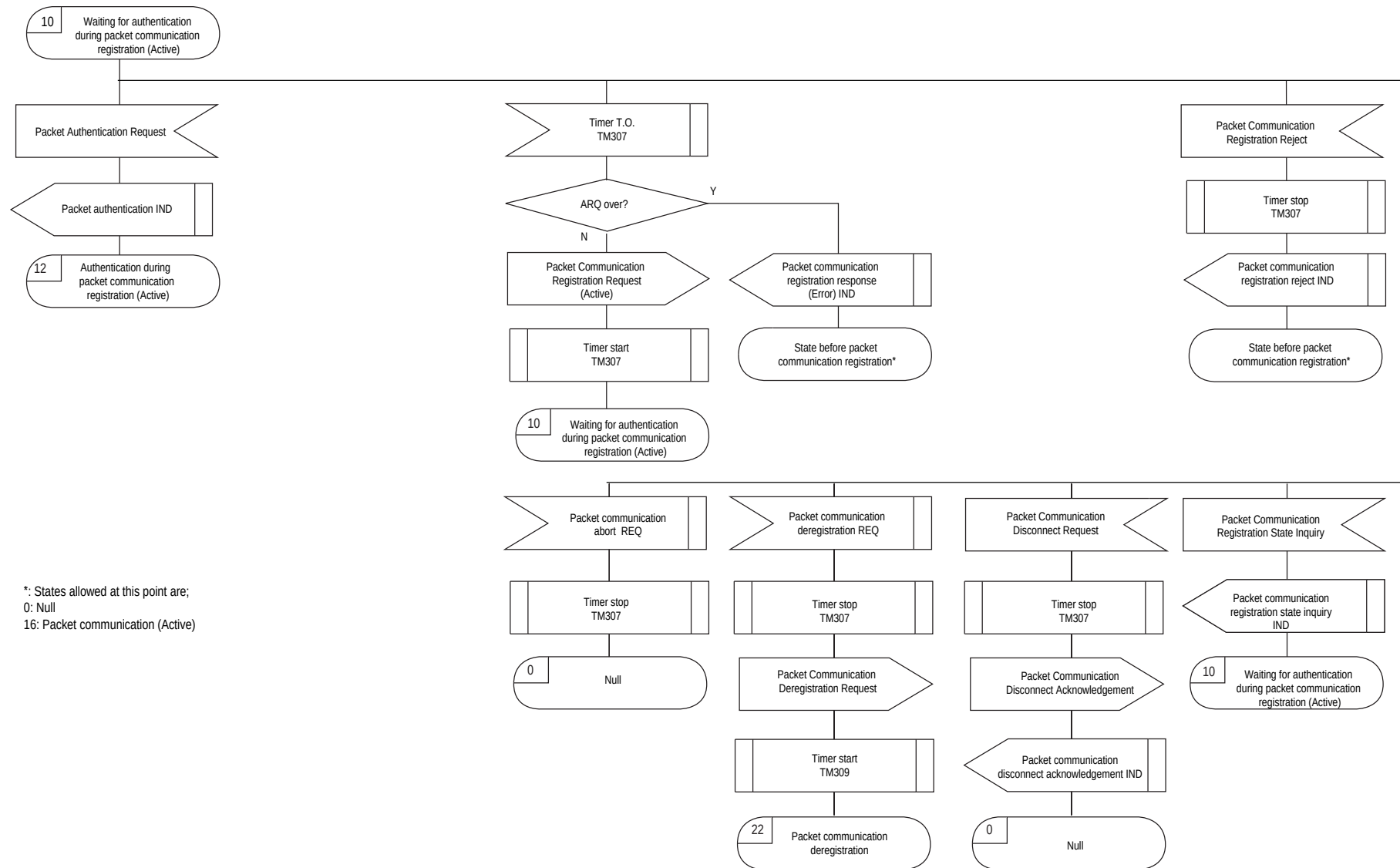


Fig. A. 5-92 Layer 3 MM SDL diagram (User side) (11/24)

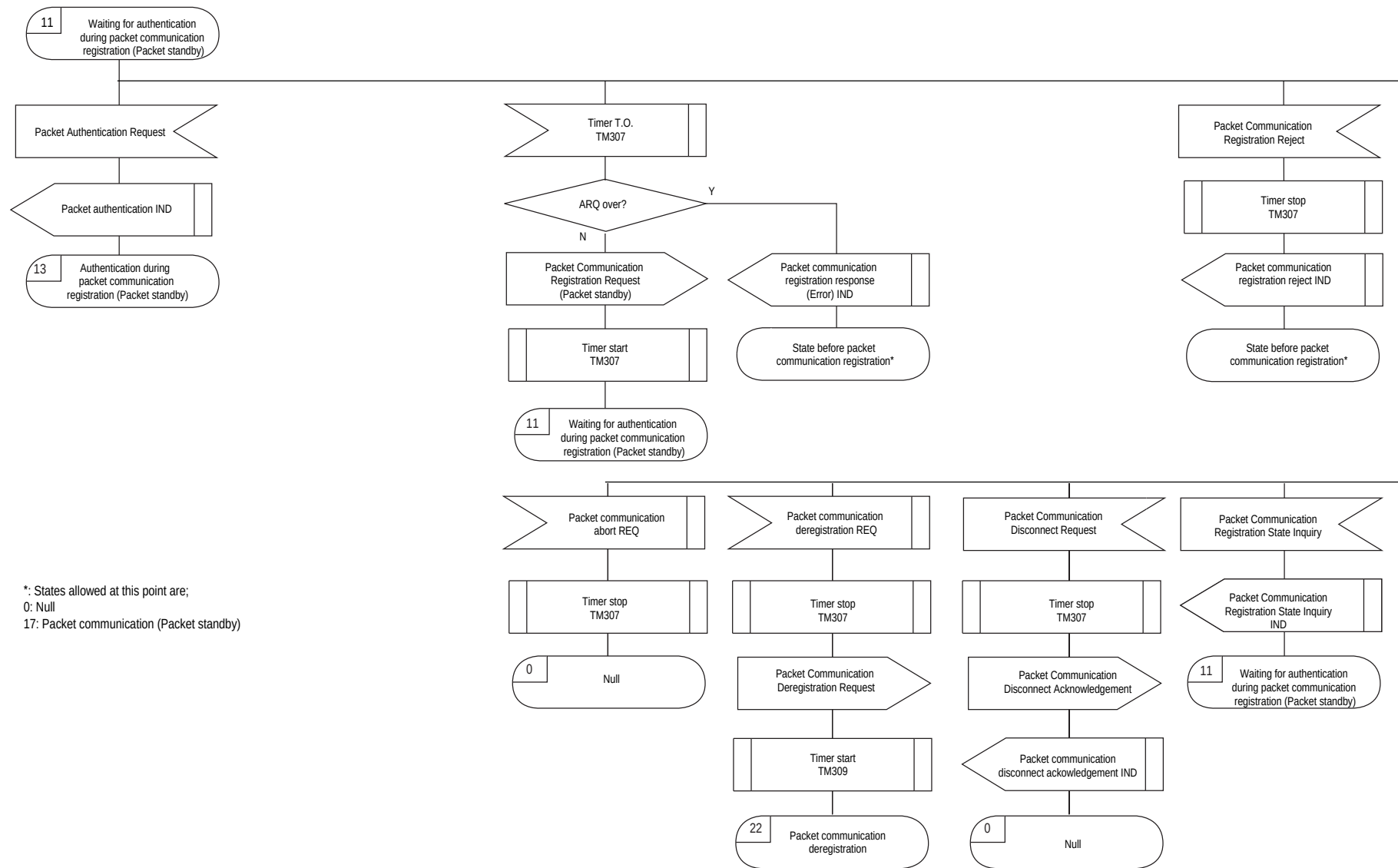


Fig. A. 5-93 Layer 3 MM SDL diagram (User side) (12/24)

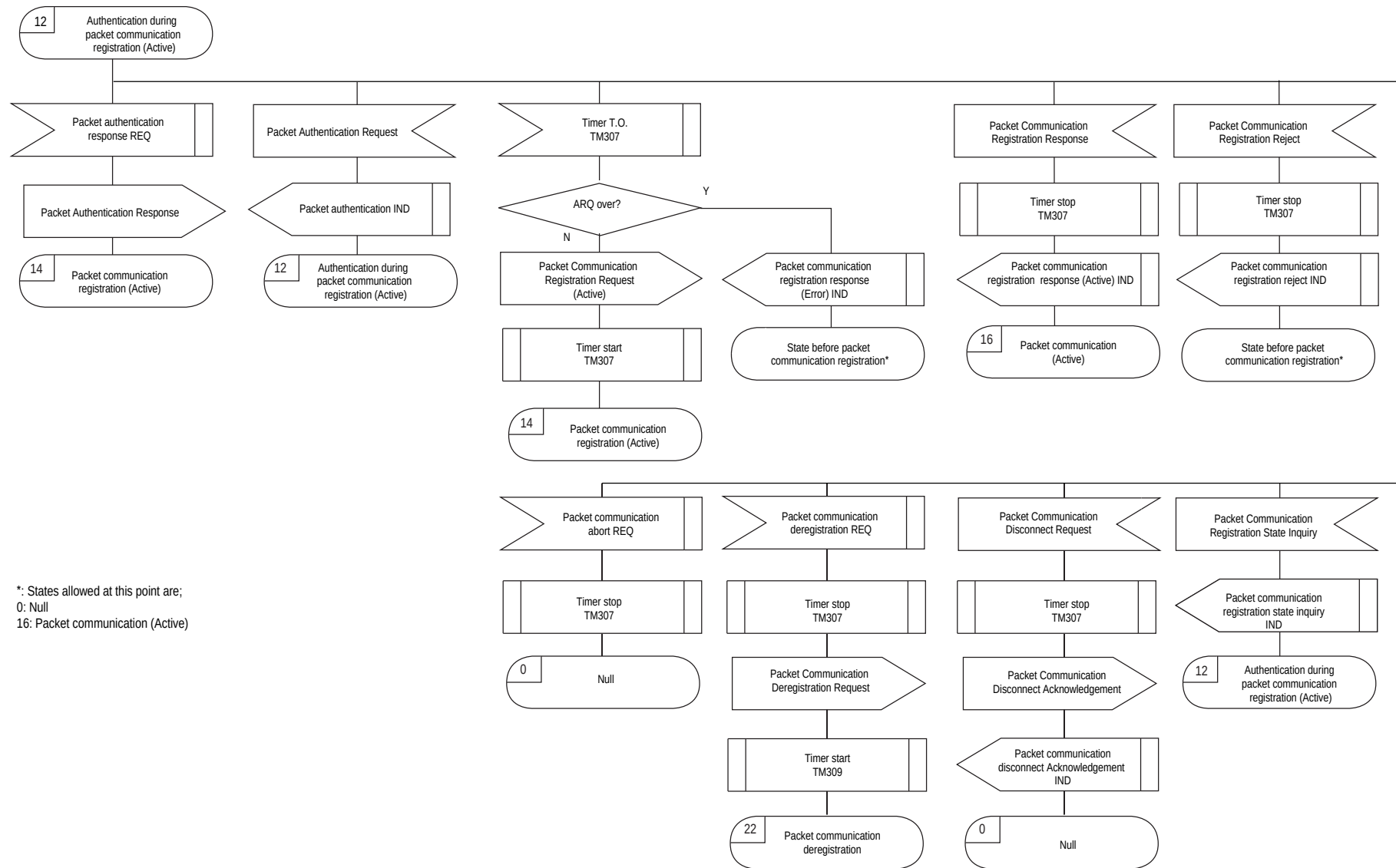


Fig. A. 5-94 Layer 3 MM SDL diagram (User side) (13/24)

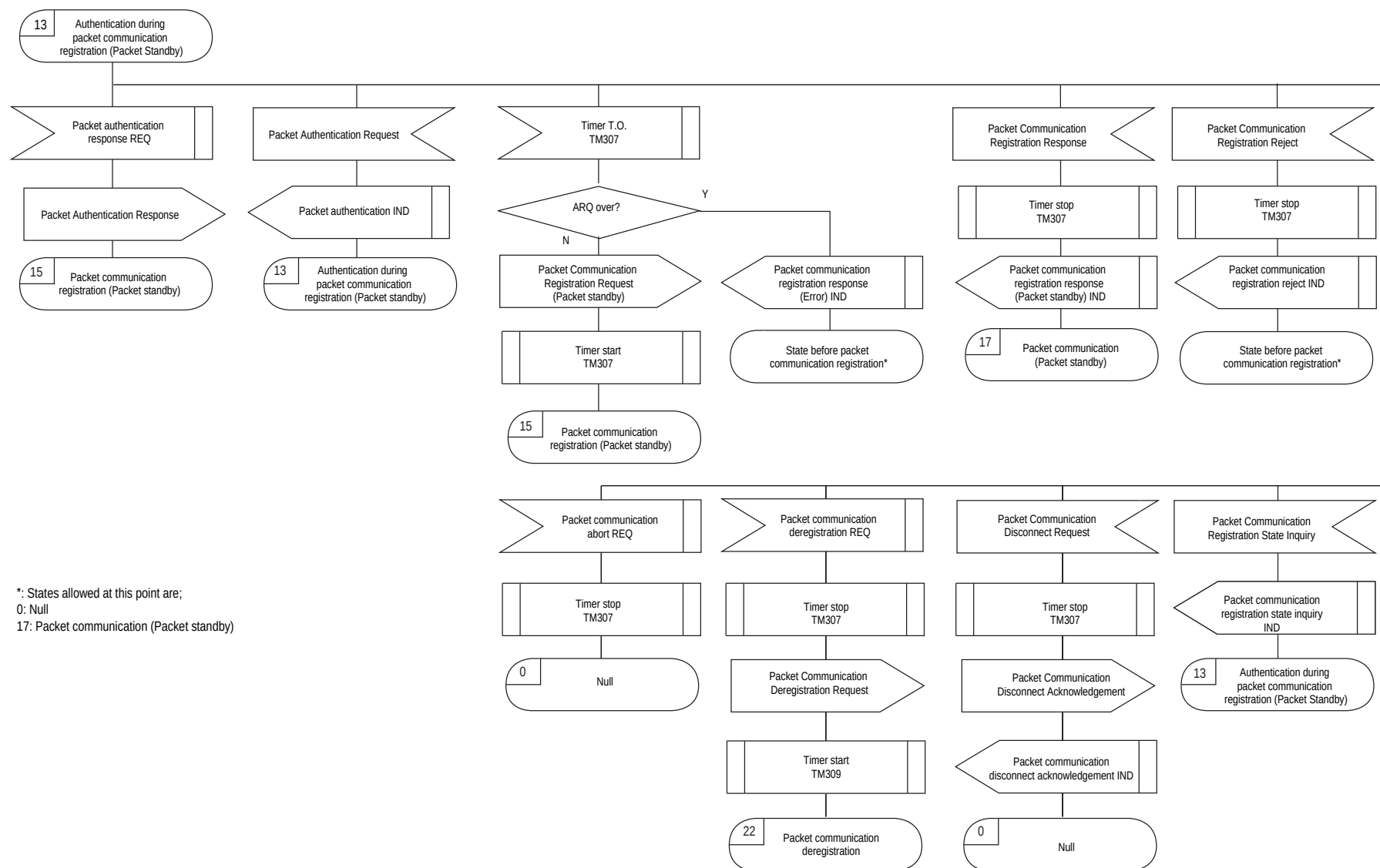


Fig. A. 5-95 Layer 3 MM SDL diagram (User side) (14/24)

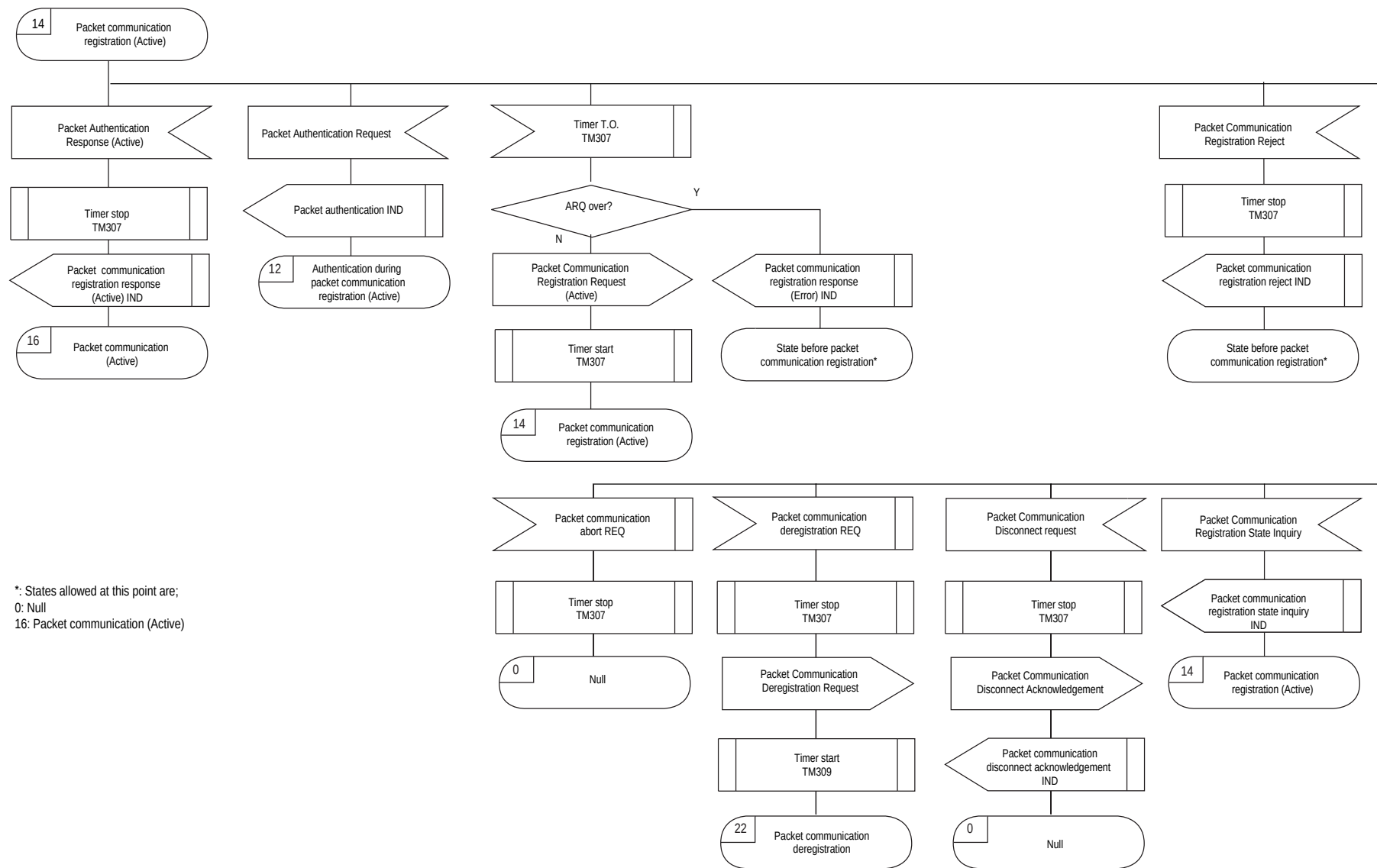


Fig. A. 5-96 Layer 3 MM SDL diagram (User side) (15/24)

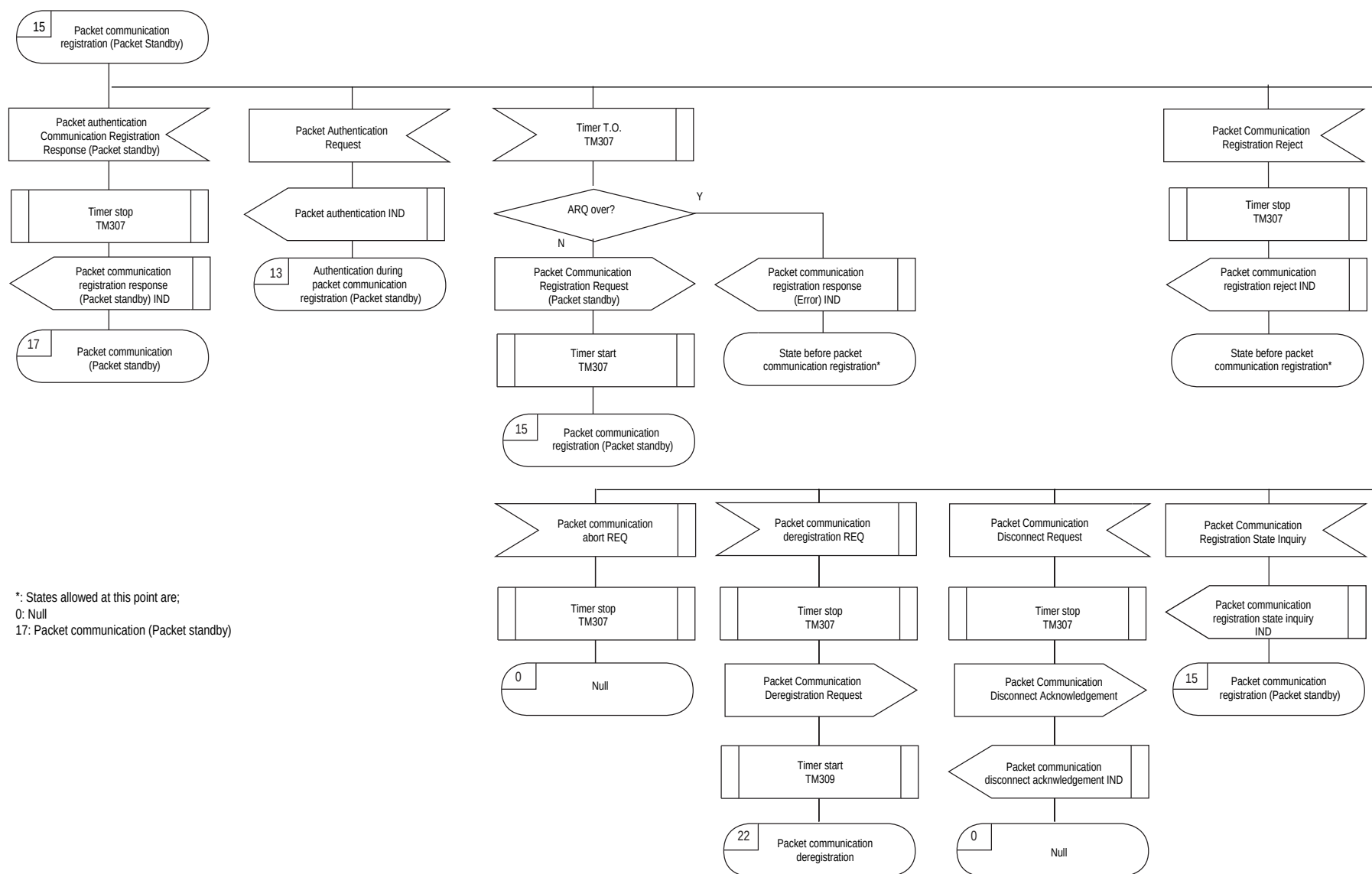


Fig. A. 5-97 Layer 3 MM SDL diagram (User side) (16/24)

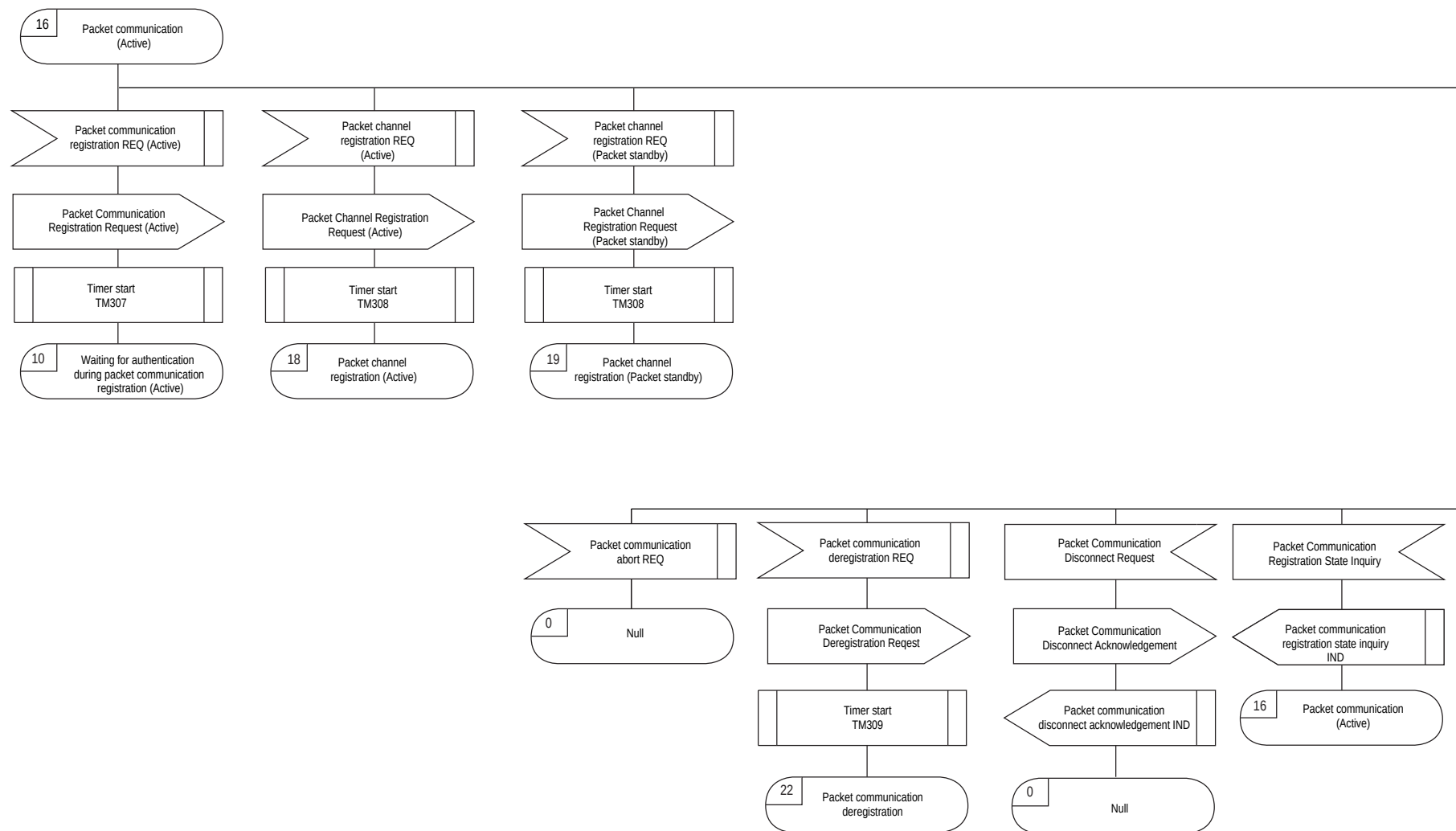


Fig. A. 5-98 Layer 3 MM SDL diagram (User side) (17/24)

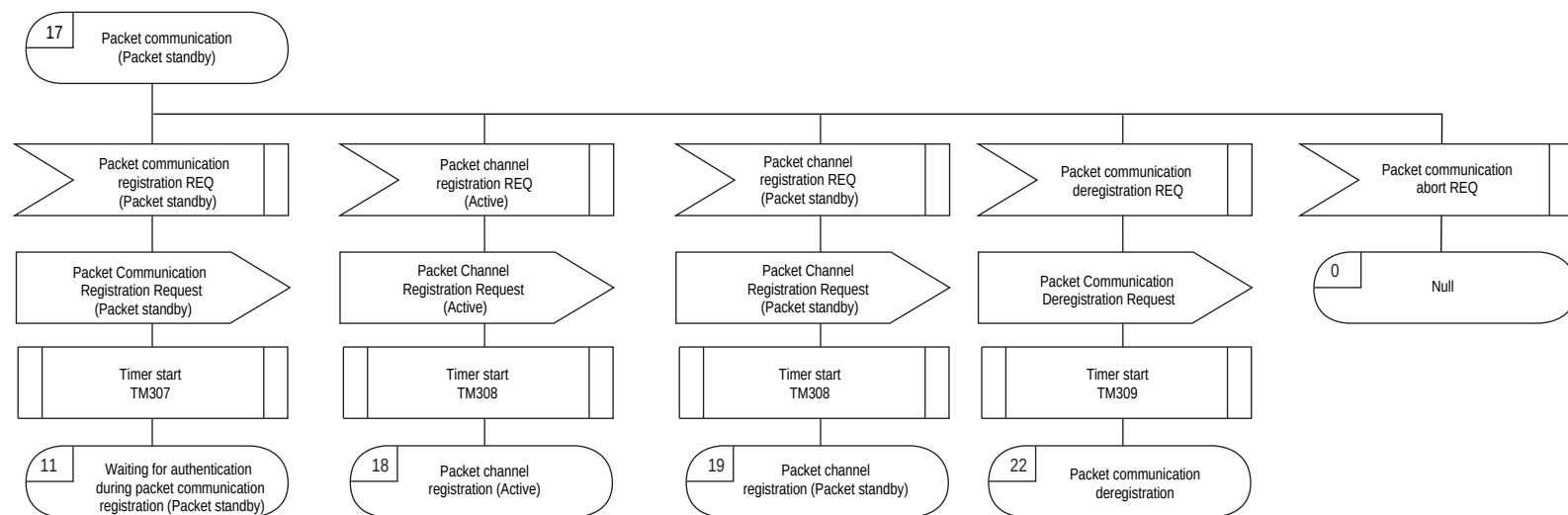


Fig. A. 5-99 Layer 3 MM SDL diagram (User side) (18/24)

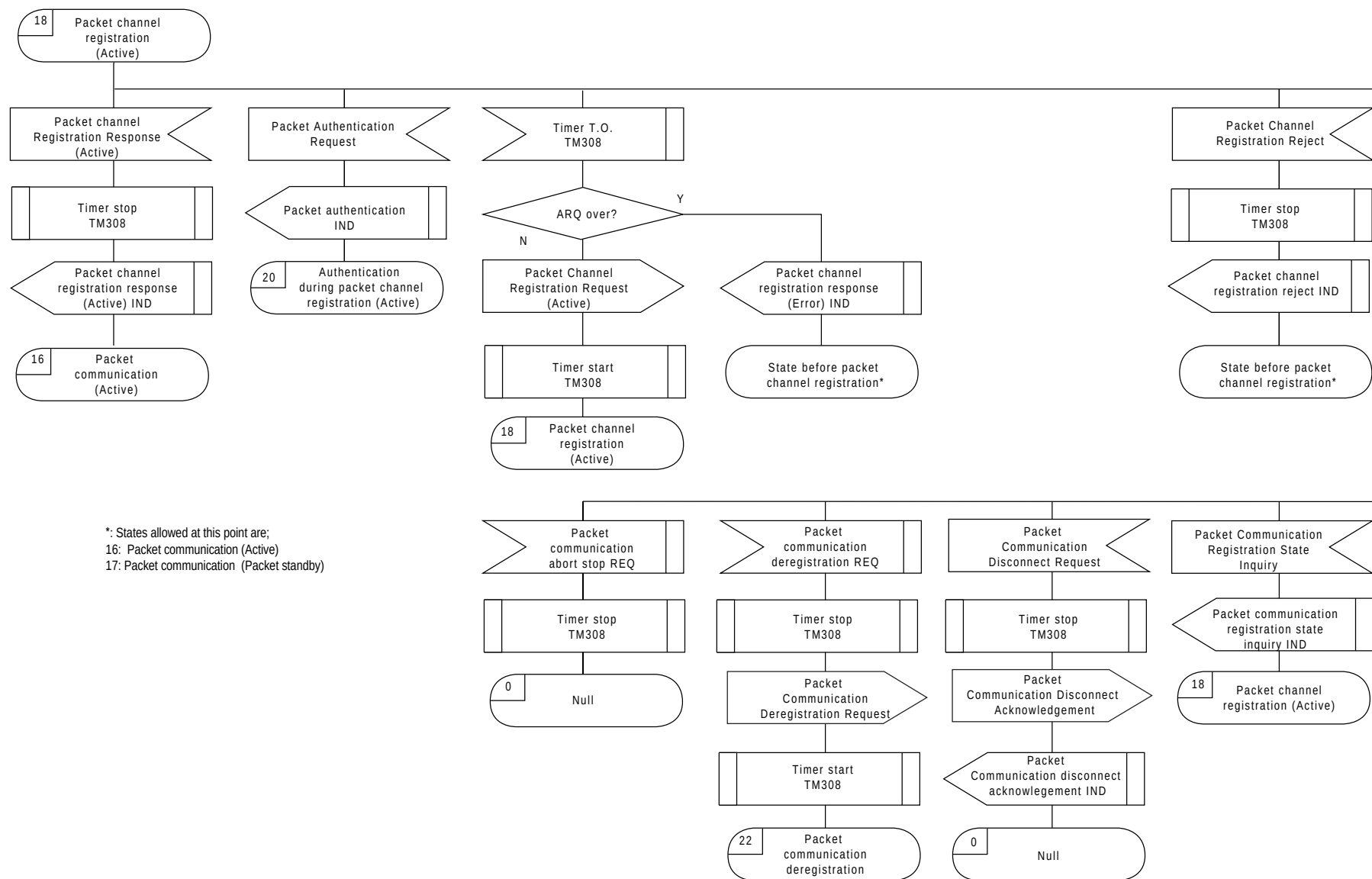


Fig. A. 5-100 Layer 3 MM SDL diagram (User side) (19/24)

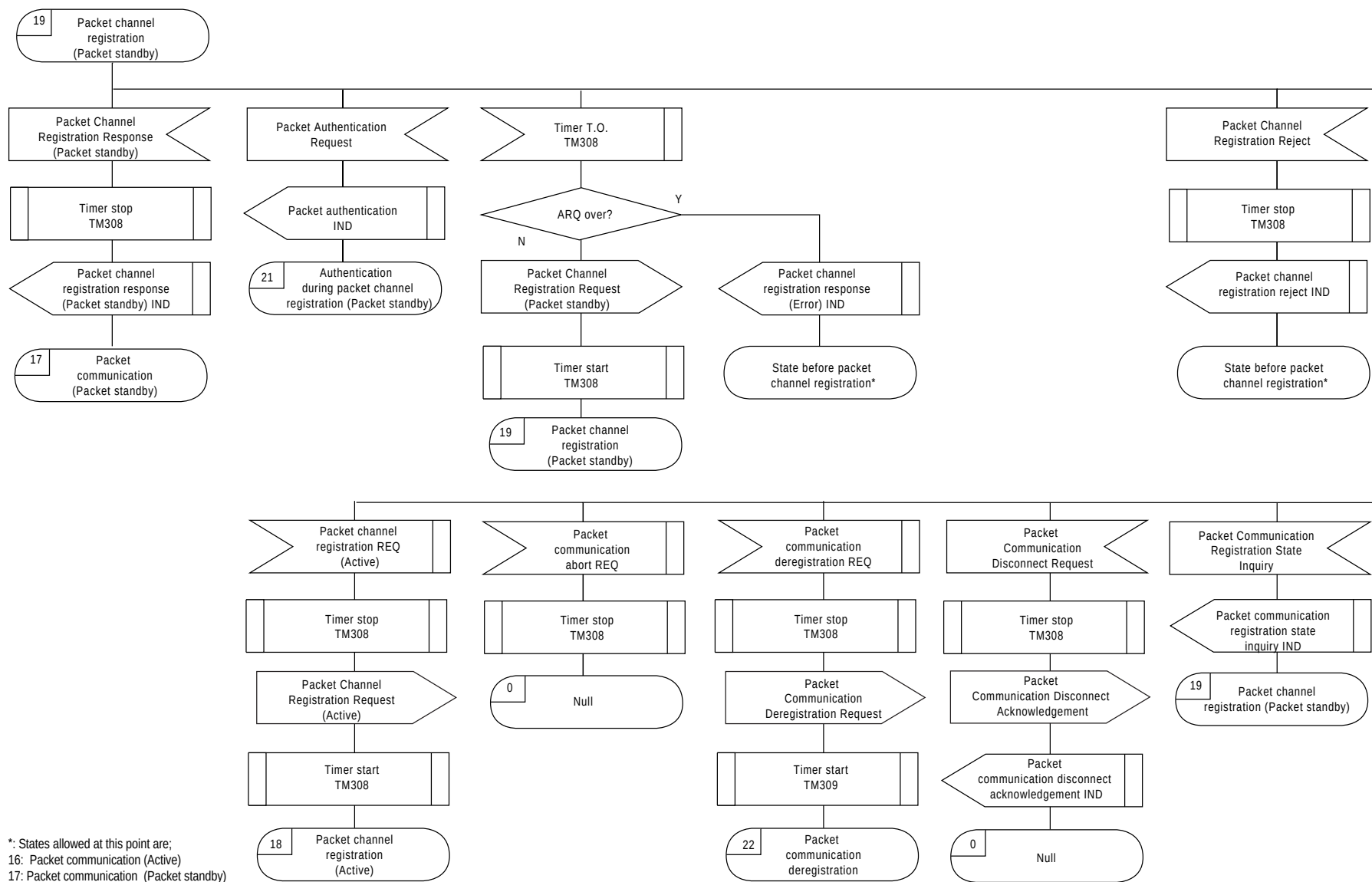


Fig. A. 5-101 Layer 3 MM SDL diagram (User side) (20/24)

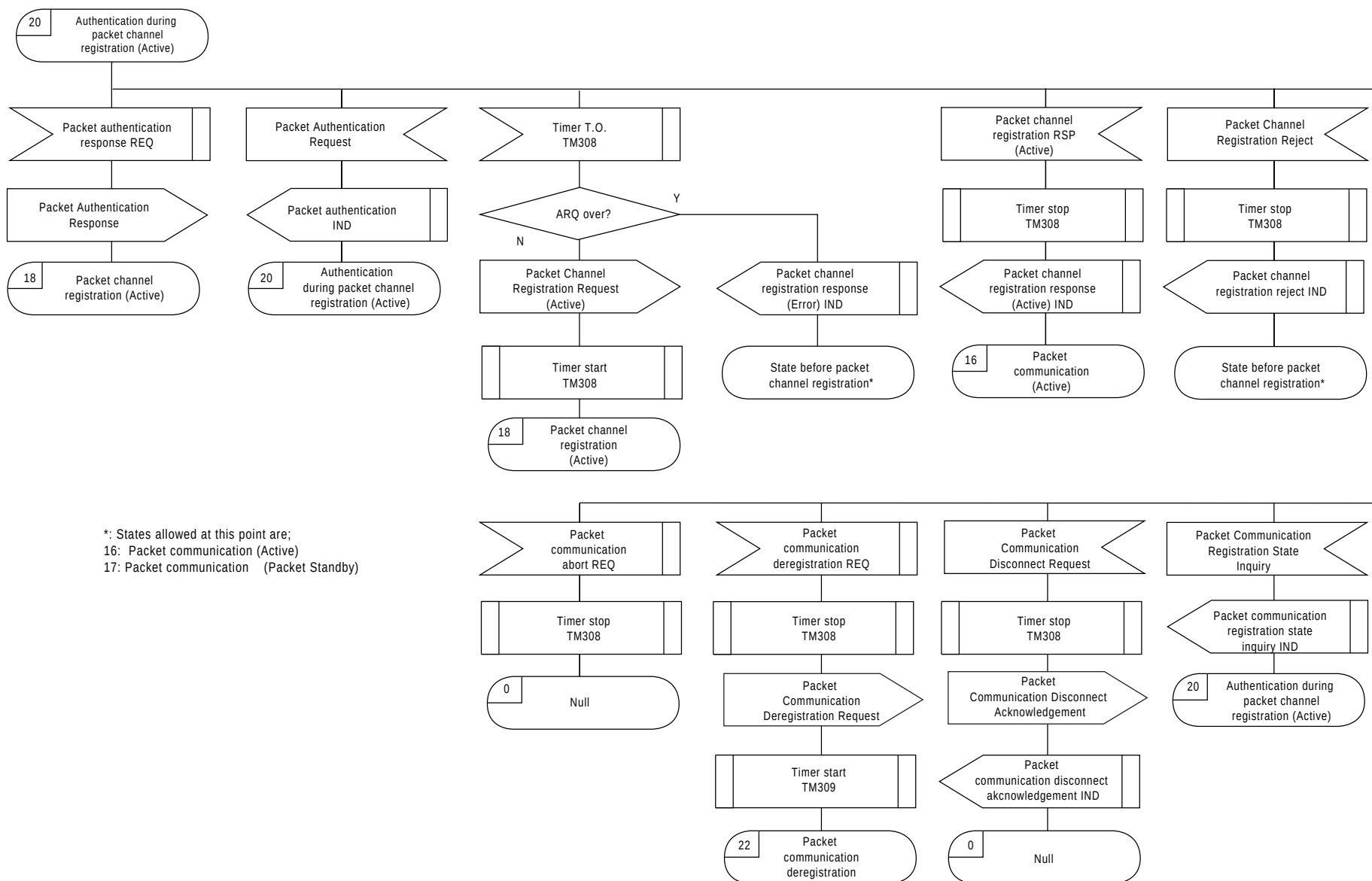


Fig. A. 5-102 Layer 3 MM SDL diagram (User side) (21/24)

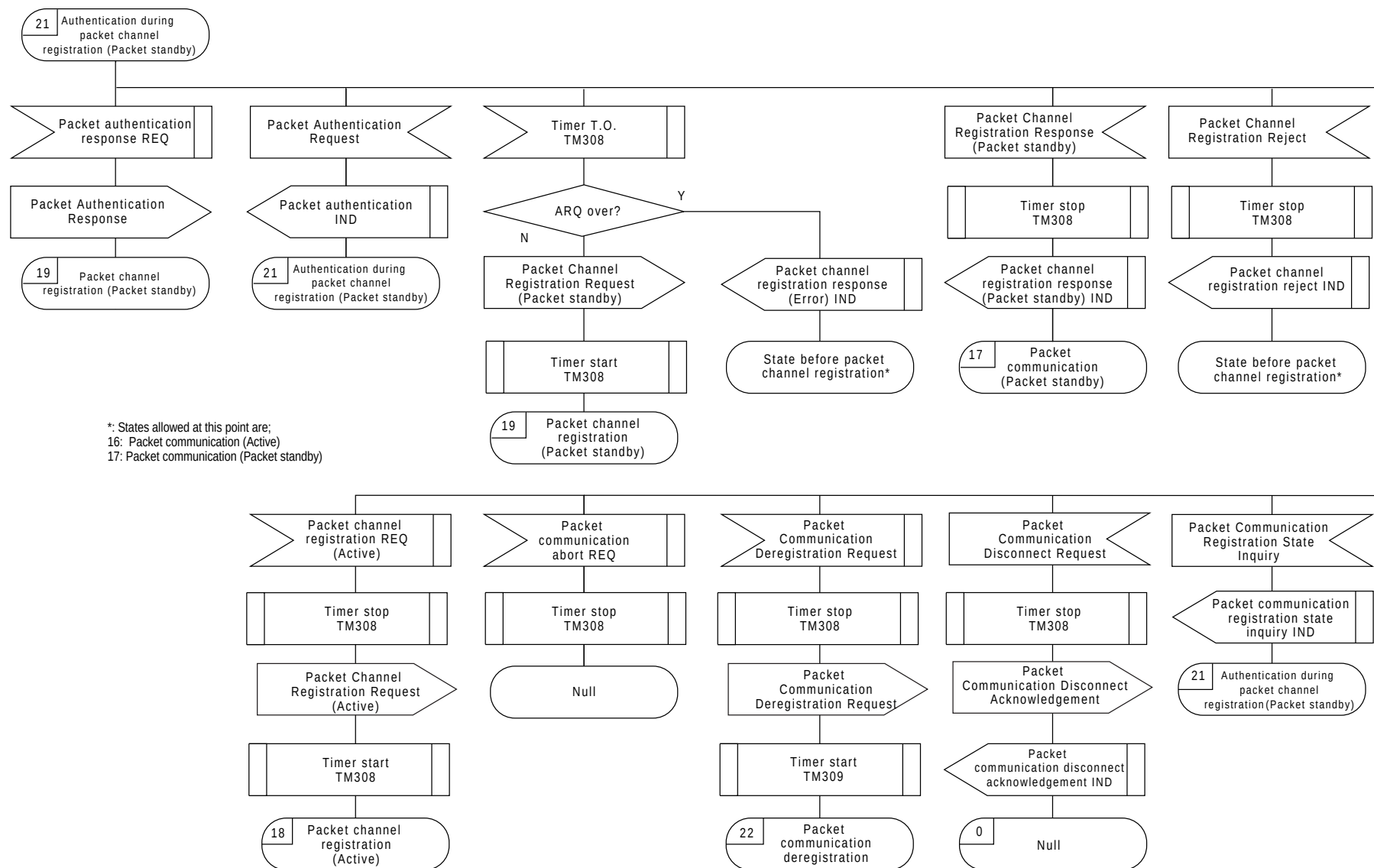


Fig. A. 5-103 Layer 3 MM SDL diagram (User side) (22/24)

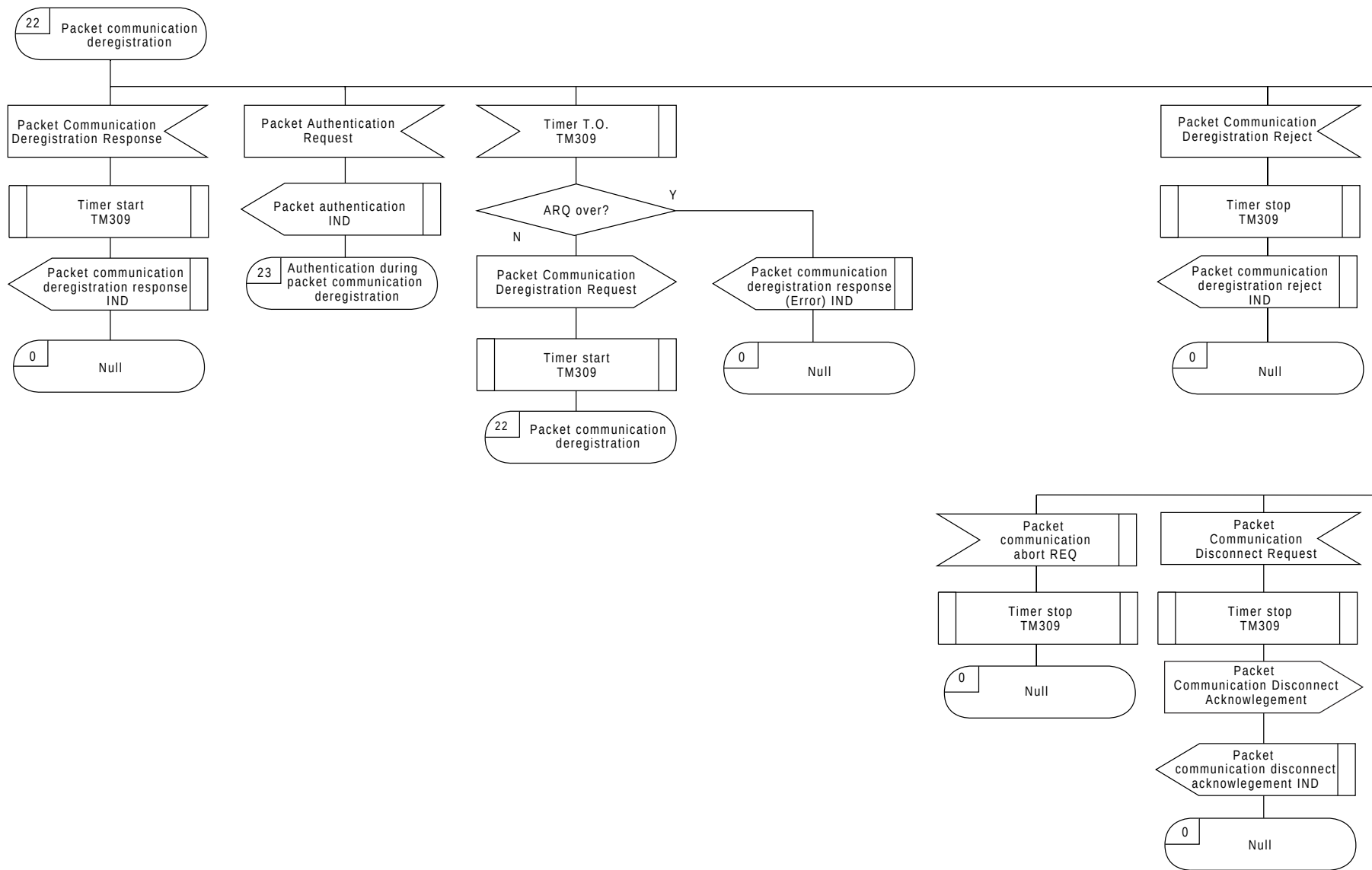


Fig. A. 5-104 Layer 3 MM SDL diagram (User side) (23/24)

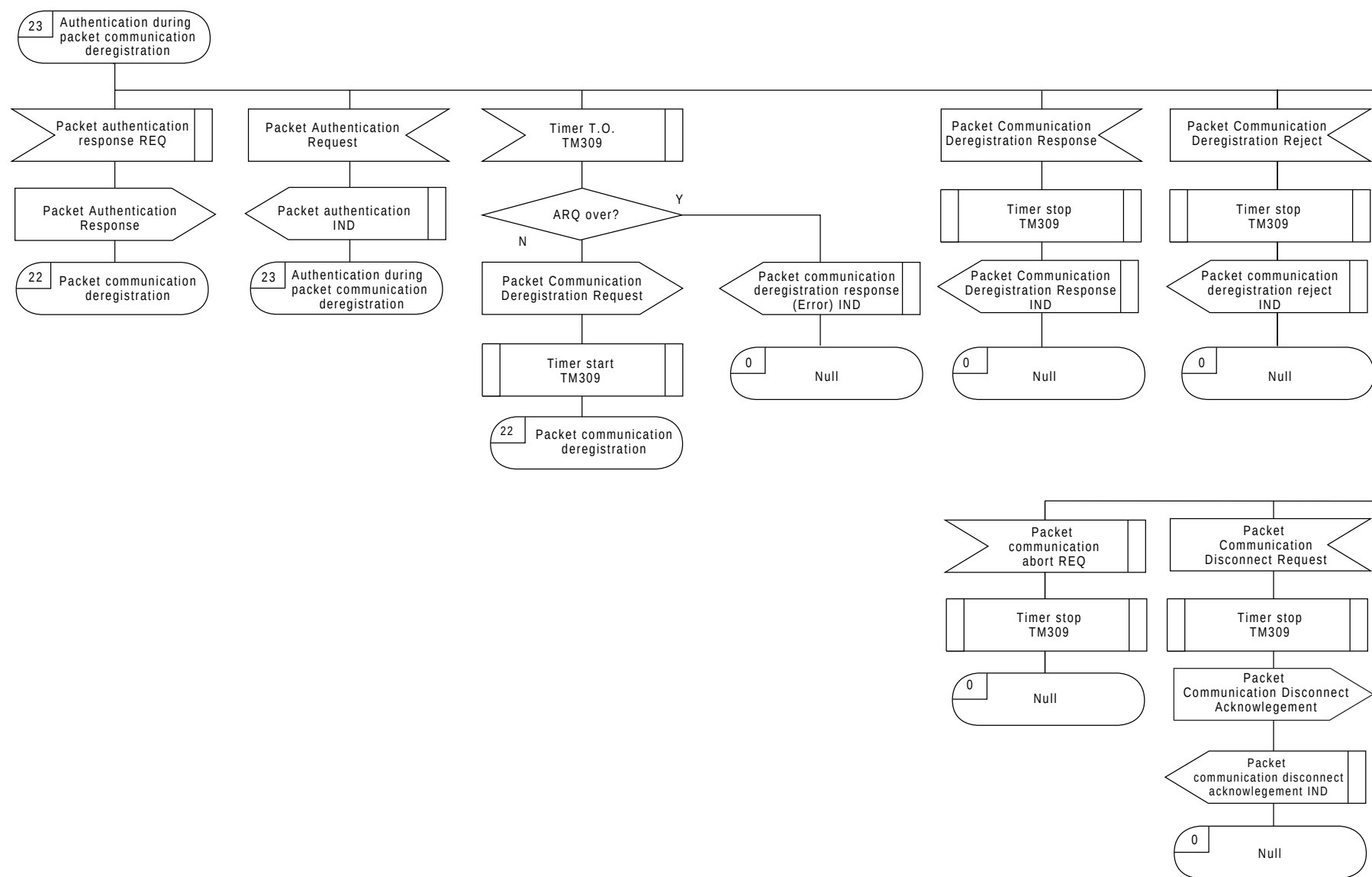


Fig. A. 5-105 Layer 3 MM SDL diagram (User side) (24/24)

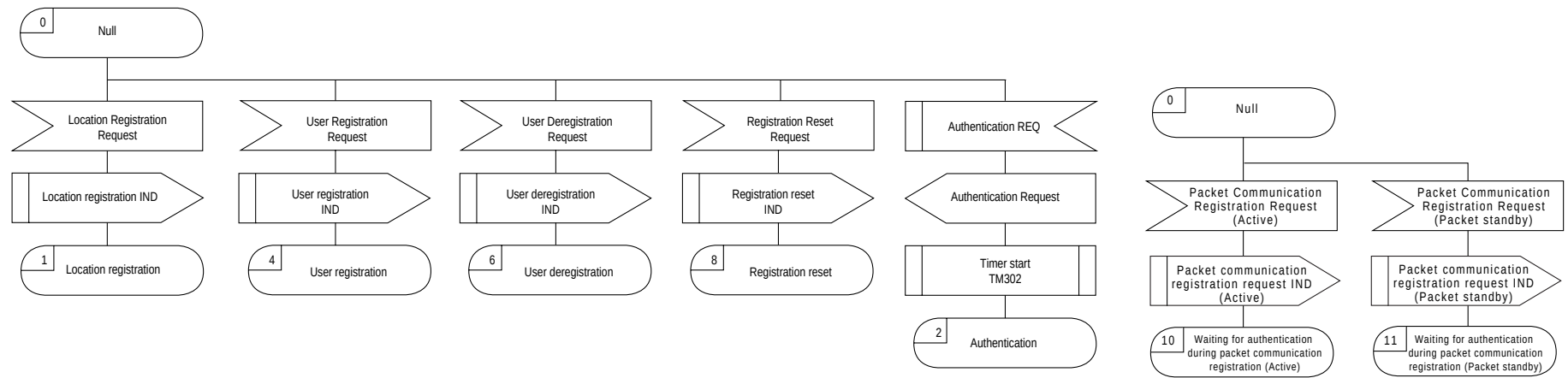


Fig. A. 5-106 Layer 3 MM SDL diagram (Network side) (1/26)

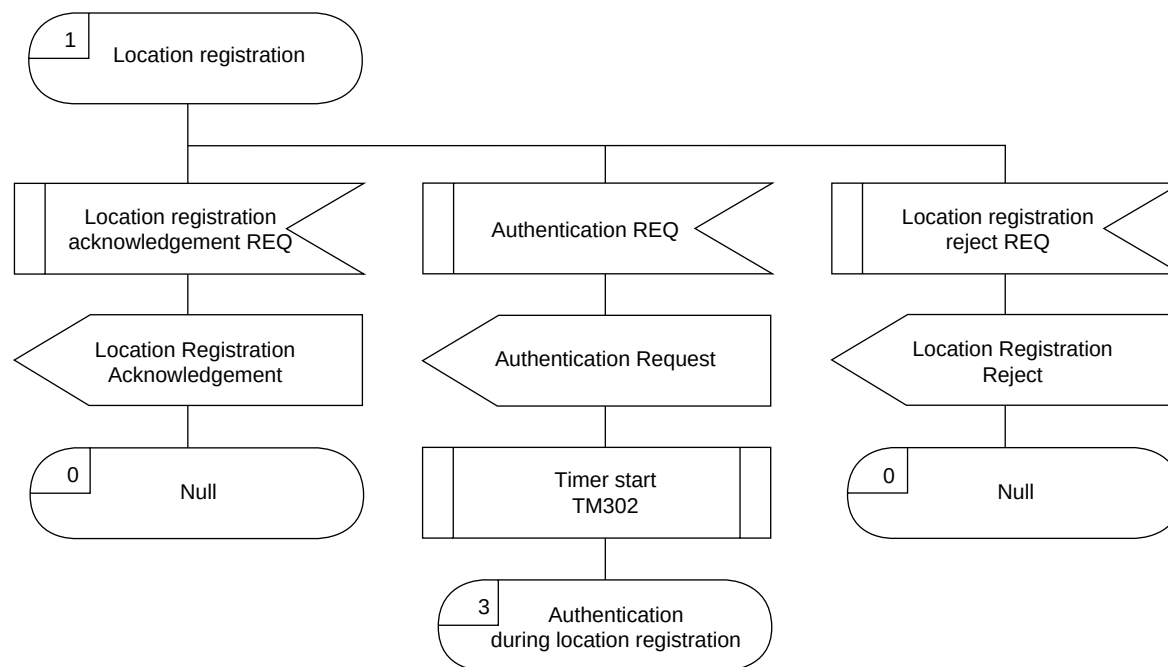


Fig. A. 5-107 Layer 3 MM SDL diagram (Network side) (2/26)

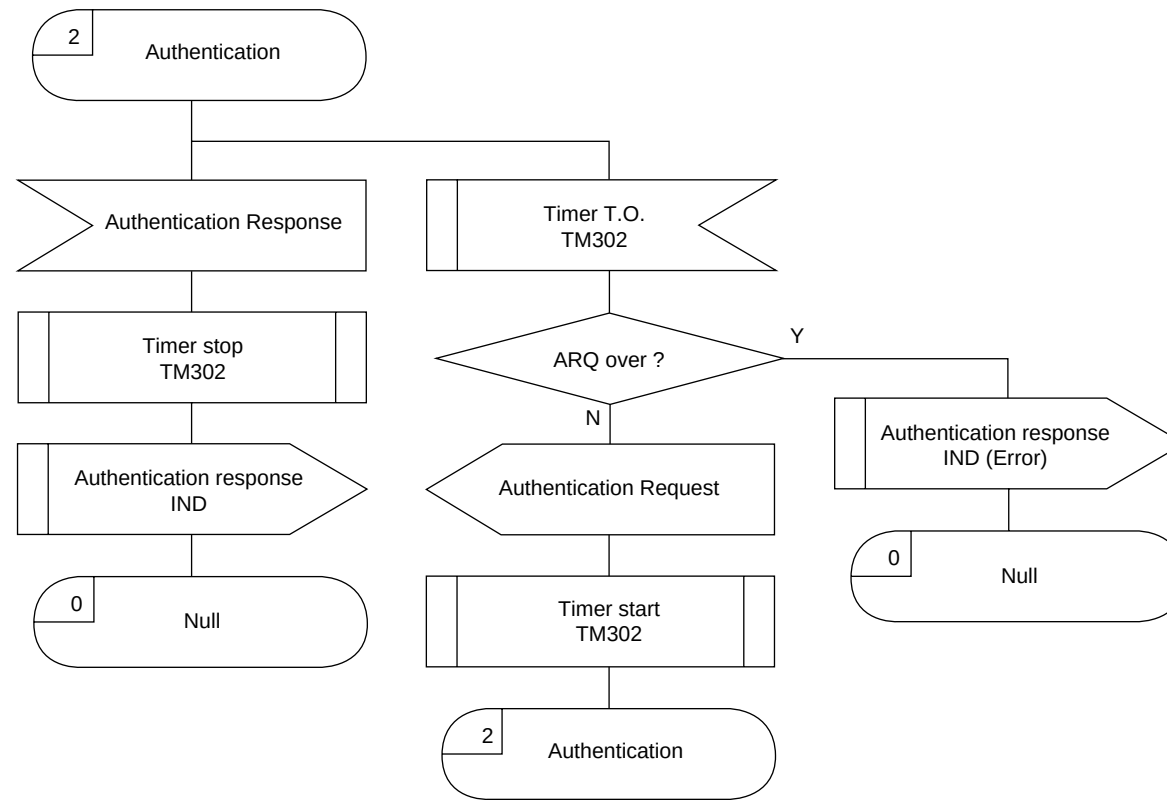


Fig. A. 5-108 Layer 3 MM SDL diagram (Network side) (3/26)

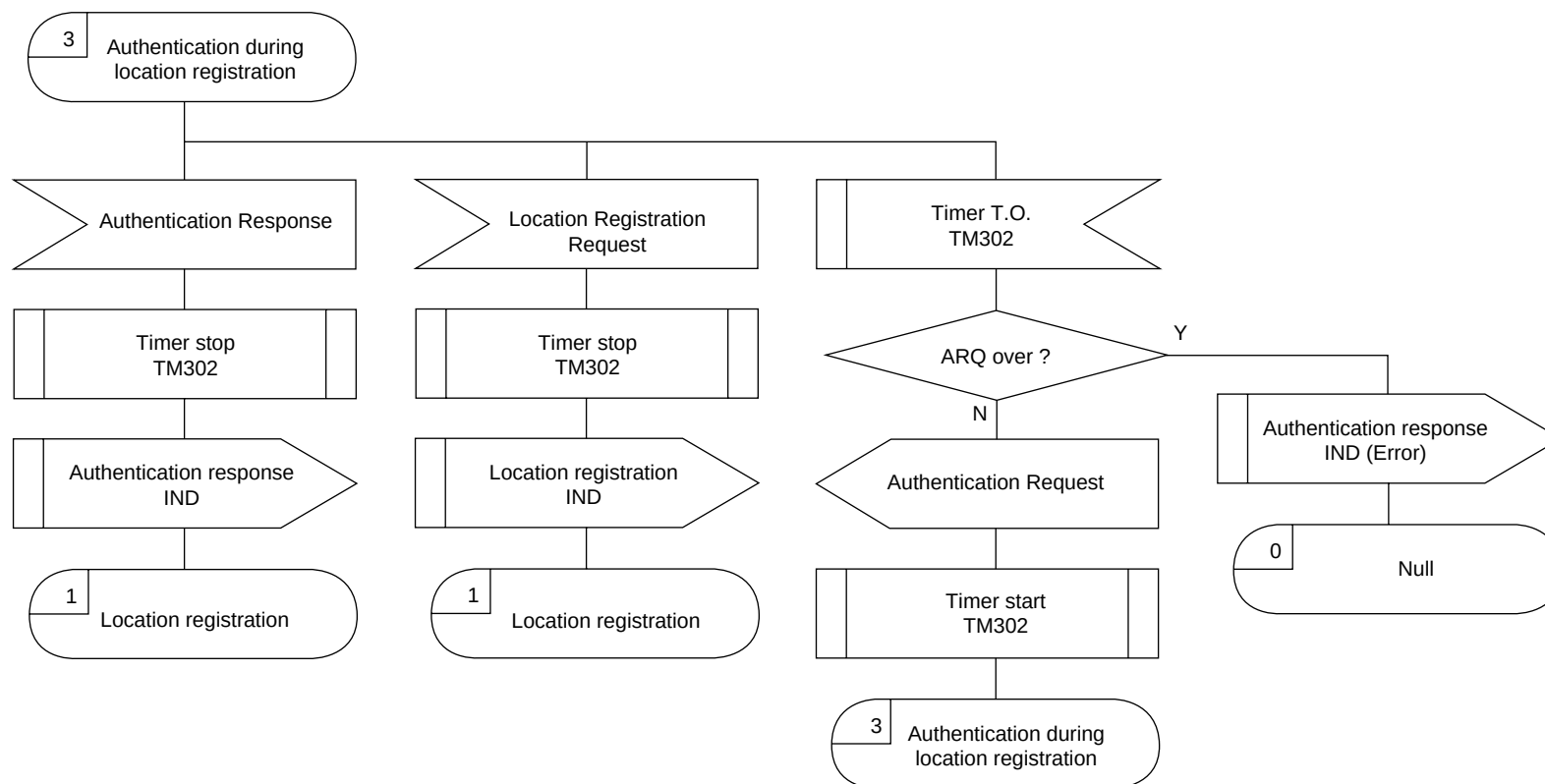


Fig. A. 5-109 Layer 3 MM SDL diagram (Network side) (4/26)

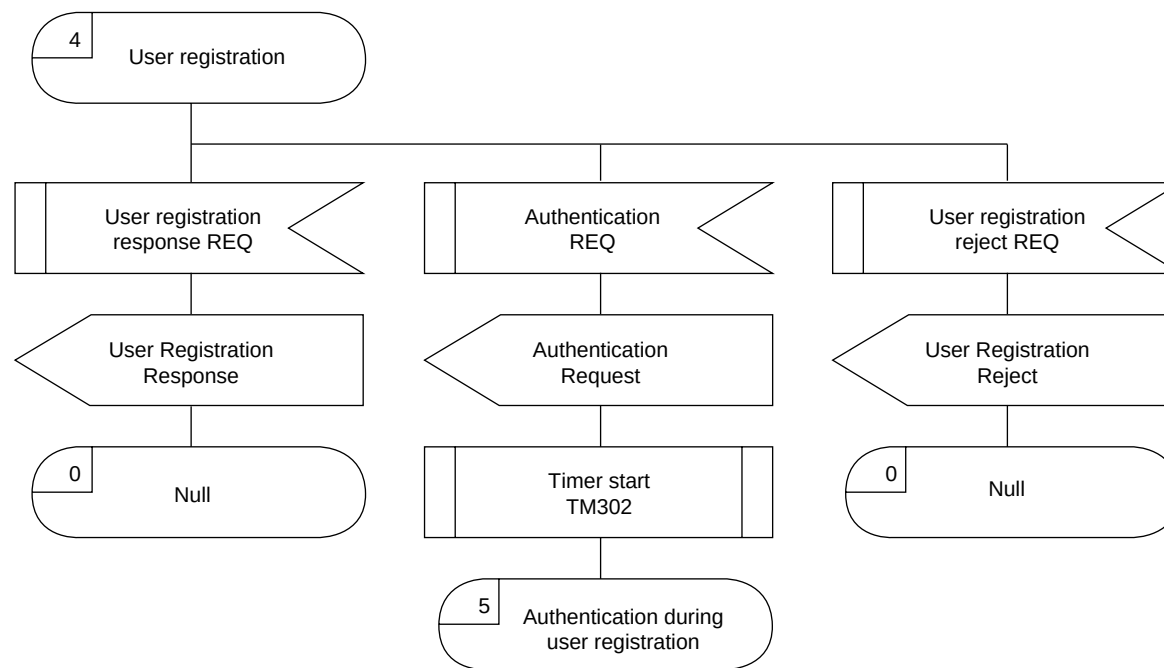


Fig. A. 5-110 Layer 3 MM SDL diagram (Network side) (5/26)

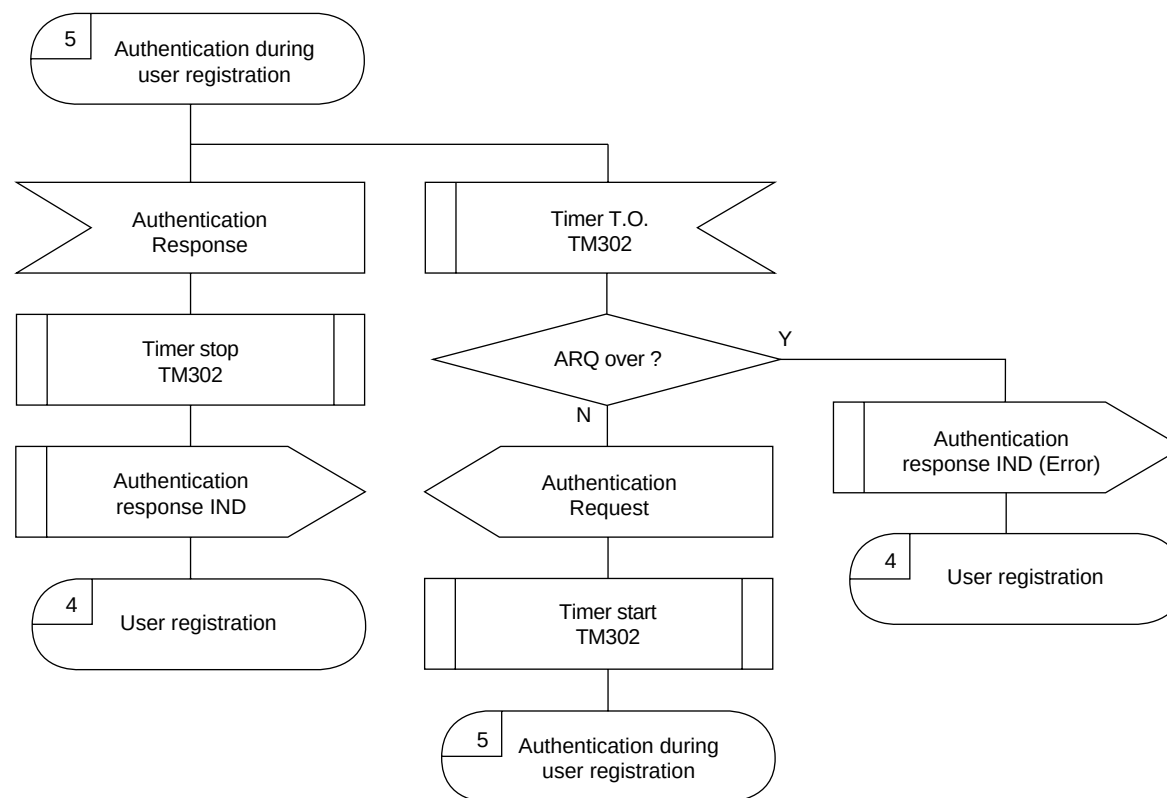


Fig. A. 5-111 Layer 3 MM SDL diagram (Network side) (6/26)

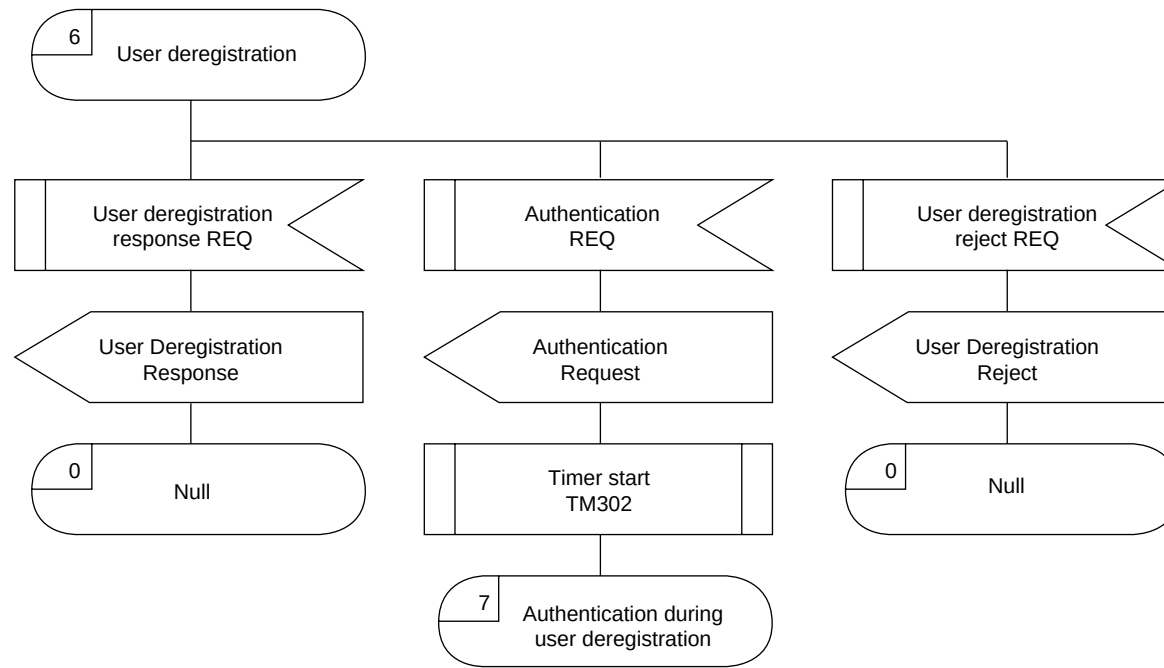


Fig. A. 5-112 Layer 3 MM SDL diagram (Network side) (7/26)

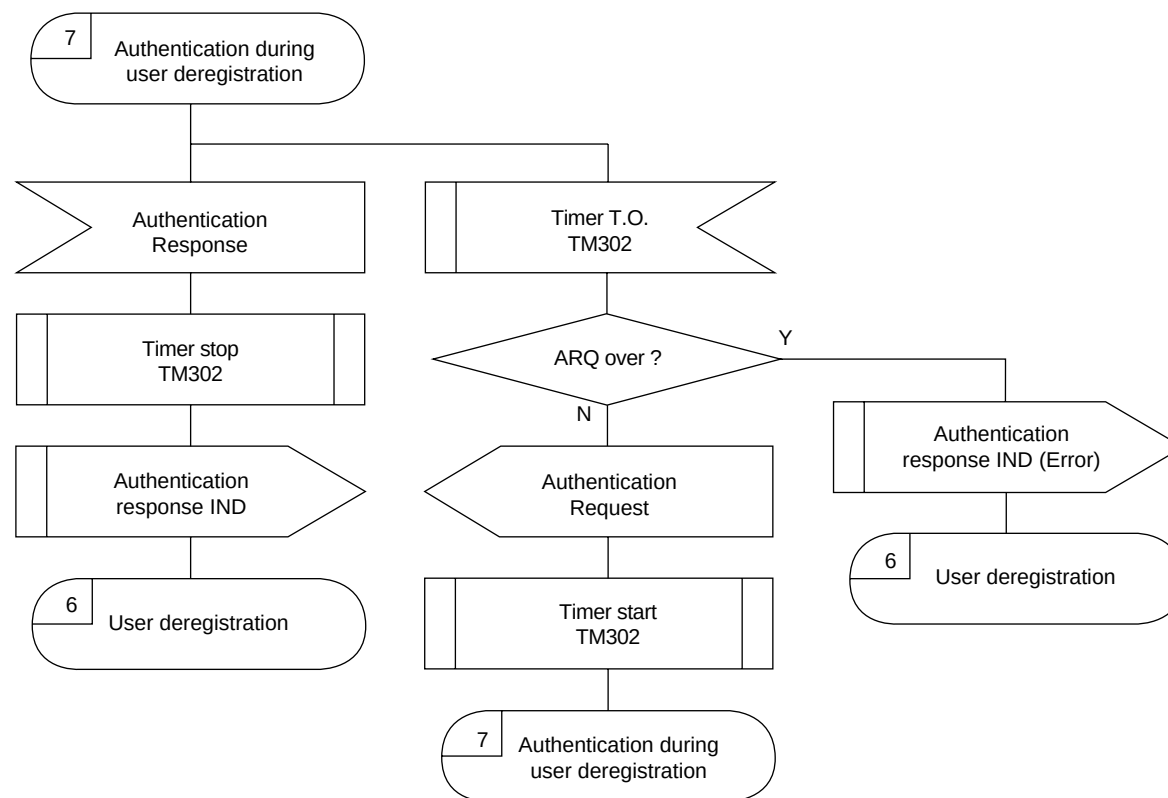


Fig. A. 5-113 Layer 3 MM SDL diagram (Network side) (8/26)

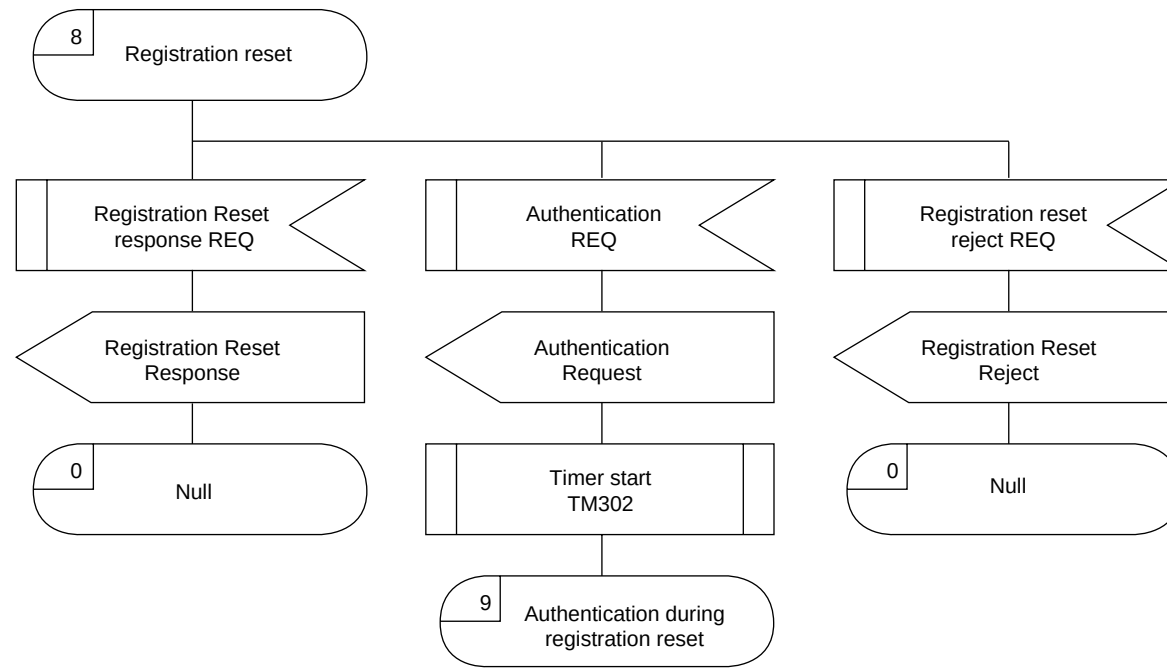


Fig. A. 5-114 Layer 3 MM SDL diagram (Network side) (9/26)

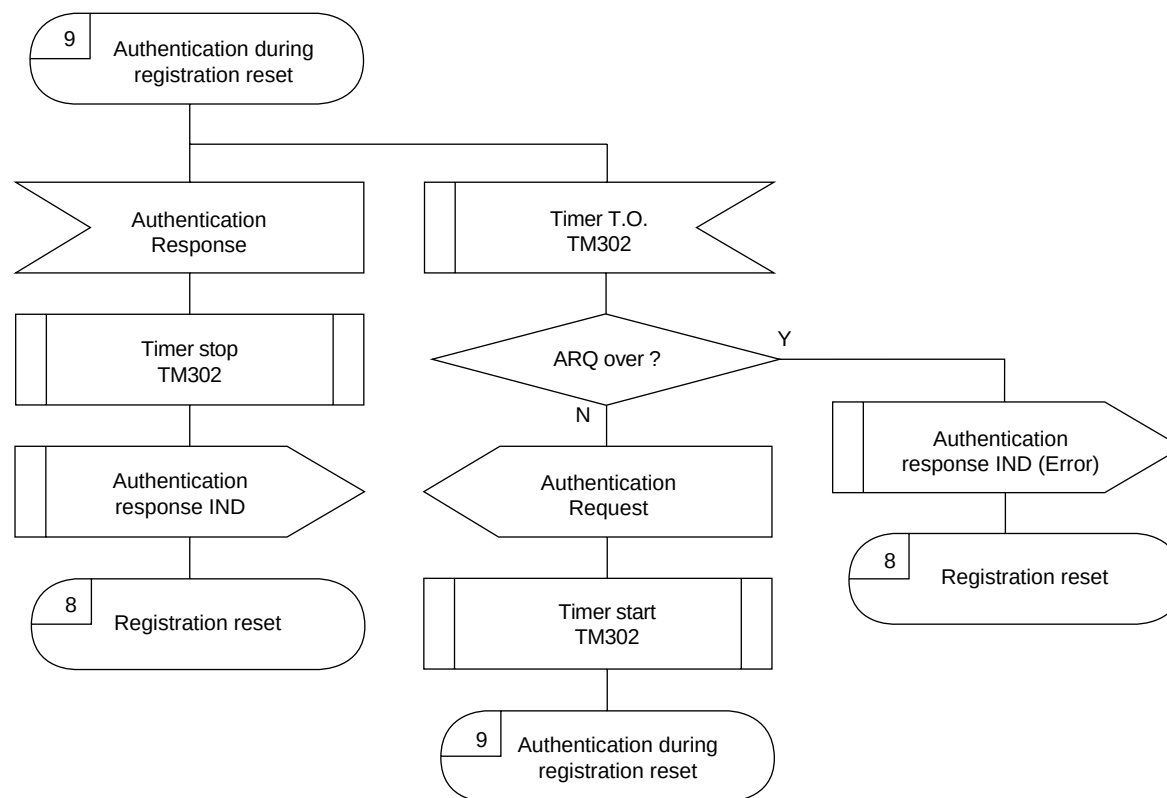
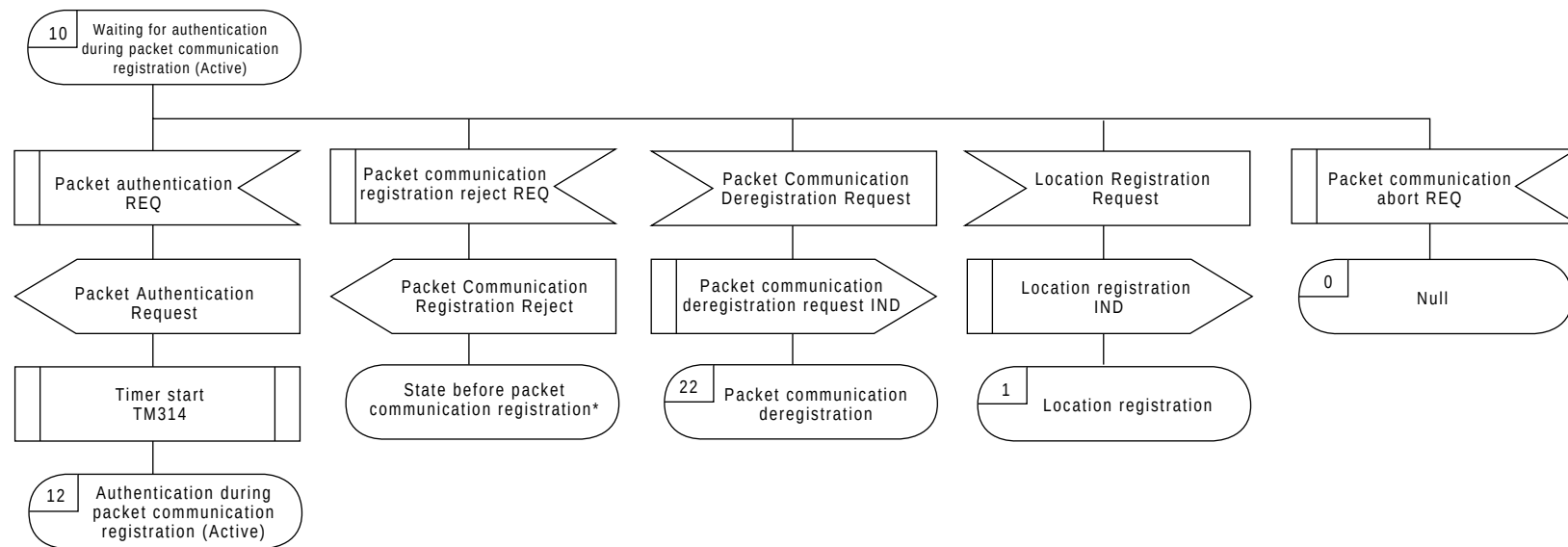
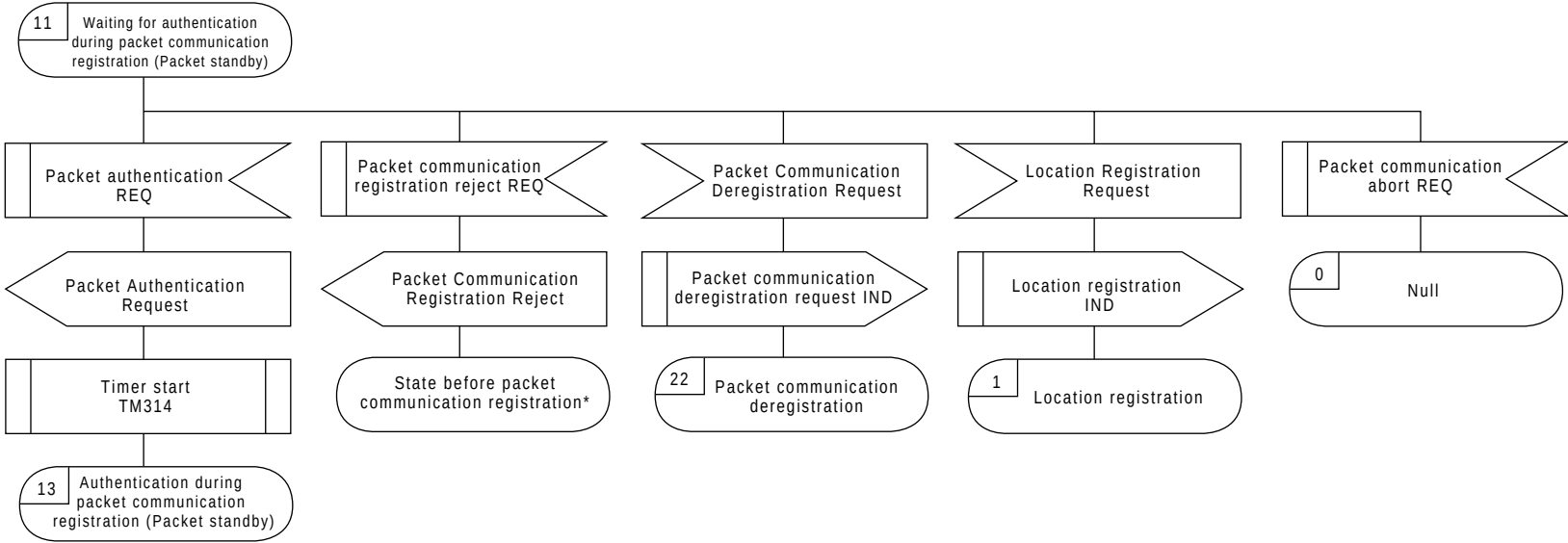


Fig. A. 5-115 Layer 3 MM SDL diagram (Network side) (10/26)



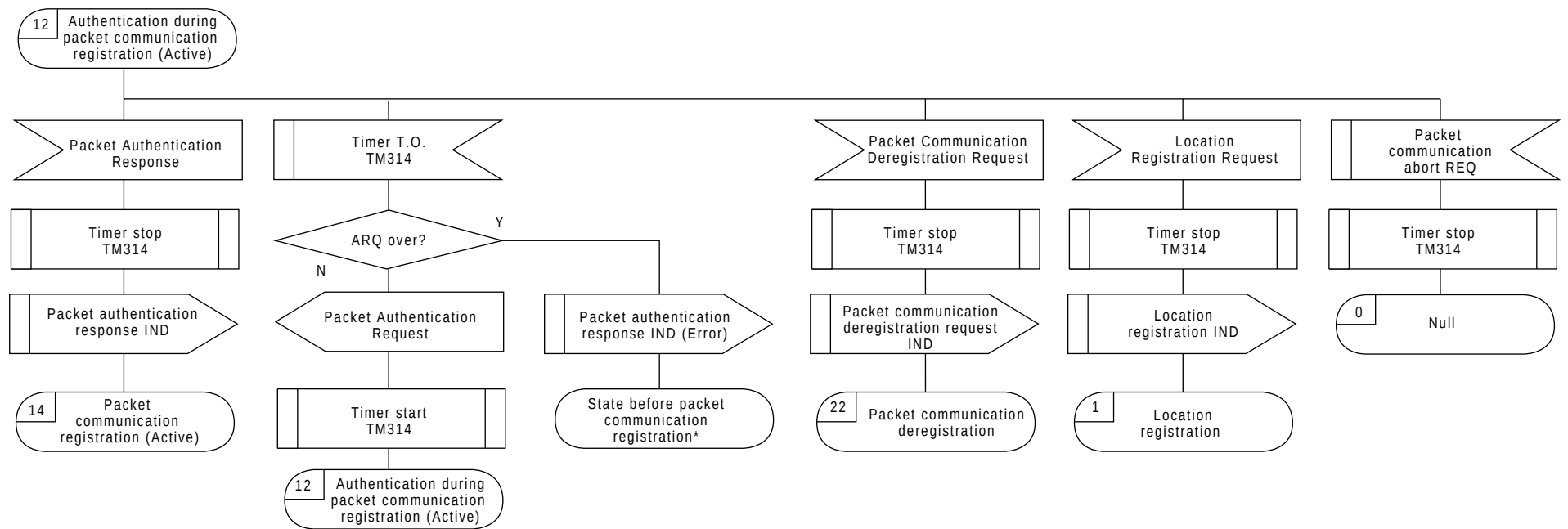
* States allowed at this point are;
 0 : Null
 16 : Packet communication (Active)
 17 : Packet communication (Packet standby)

Fig. A. 5-116 Layer 3 MM SDL diagram (Network side) (11/26)



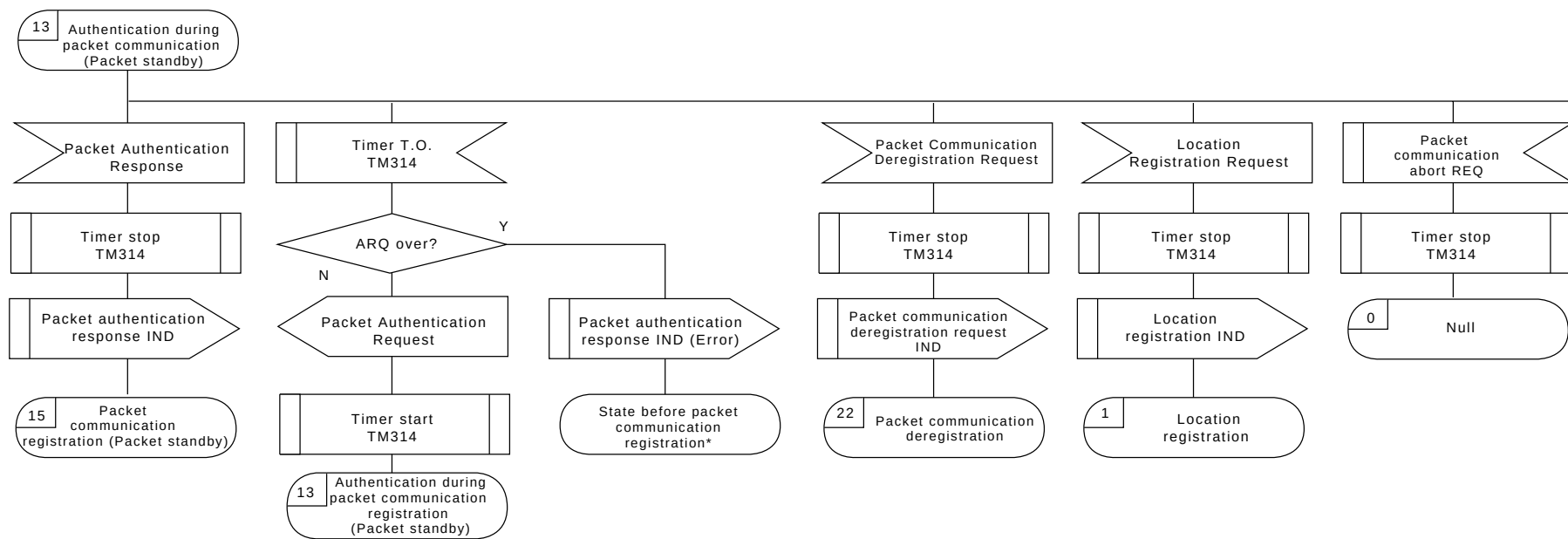
* States allowed at this point are;
0 : Null
16 : Packet communication (Active)
17 : Packet communication (Packet standby)

Fig. A. 5-117 Layer 3 MM SDL diagram (Network side) (12/26)



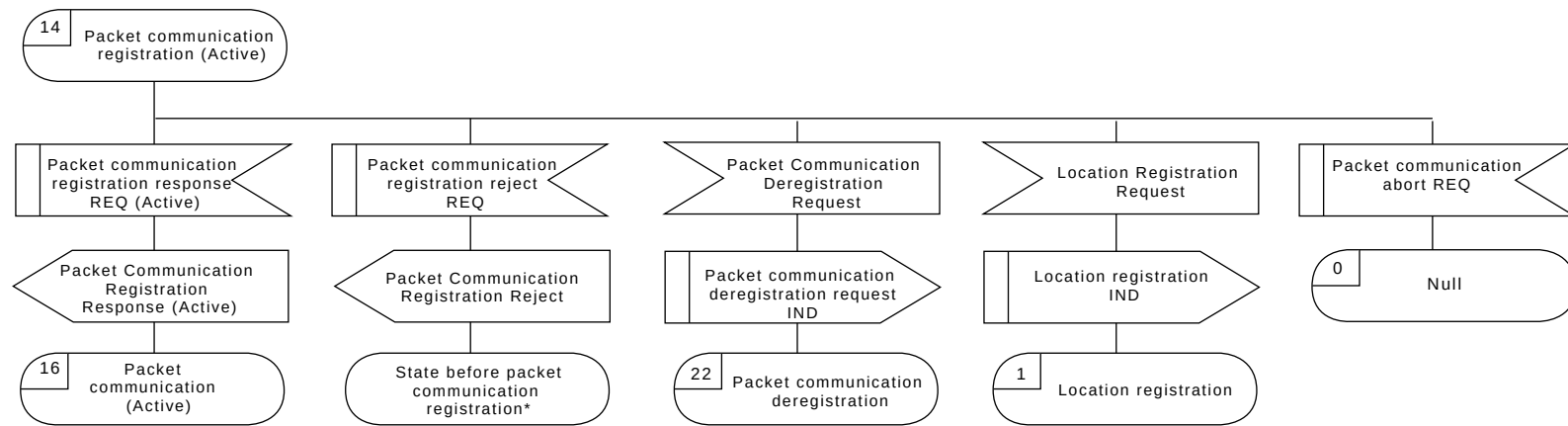
*: States allowed at this point are;
 0: Null
 16: Packet communication (Active)
 17: Packet communication (Packet standby)

Fig. A. 5-118 Layer 3 MM SDL diagram (Network side) (13/26)



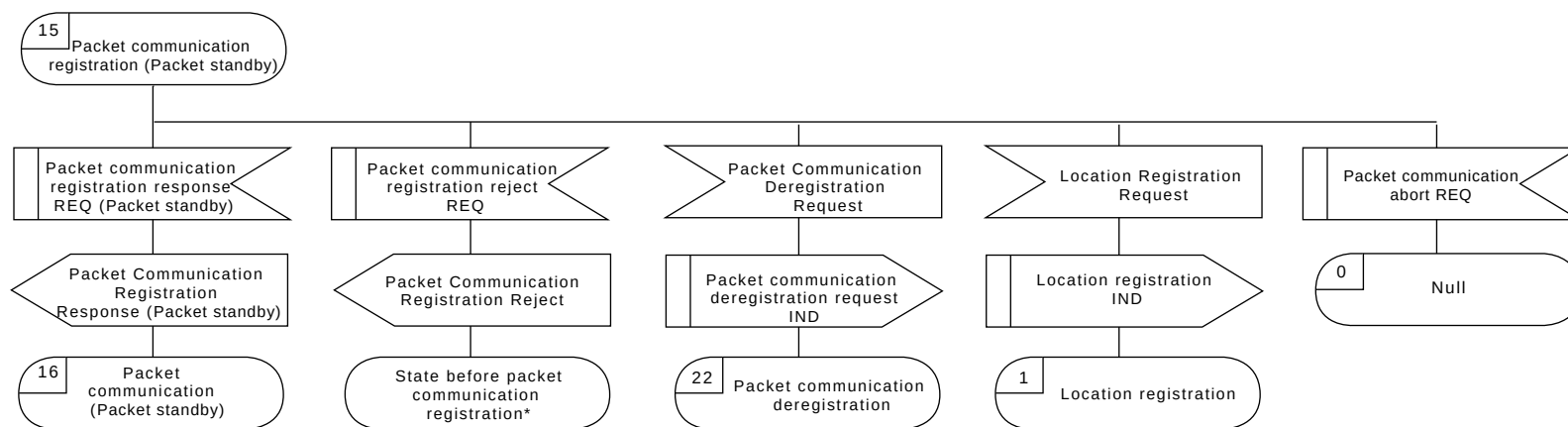
*: States allowed at this point are;
 0: Null
 16: Packet communication (Active)
 17: Packet communication (Packet standby)

Fig. A. 5-119 Layer 3 MM SDL diagram (Network side) (14/26)



*: States allowed at this point are;
 0: Null
 16: Packet communication (Active)
 17: Packet communication (Packet standby)

Fig. A. 5-120 Layer 3 MM SDL diagram (Network side) (15/26)



*: States allowed at this point are;
 0: Null
 16: Packet communication (Active)
 17: Packet communication (Packet standby)

Fig. A. 5-121 Layer 3 MM SDL diagram (Network side) (16/26)

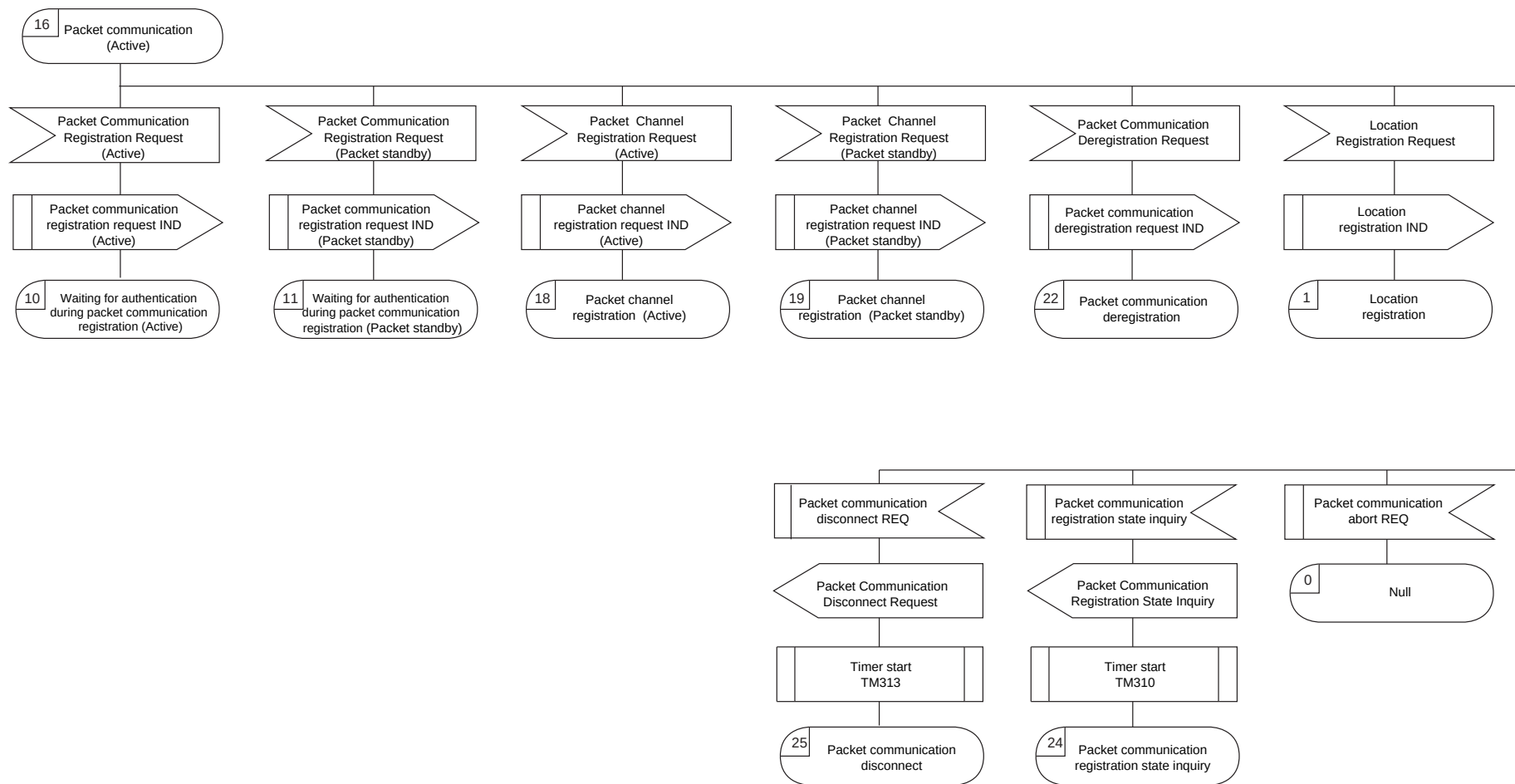


Fig. A. 5-122 Layer 3 MM SDL diagram (Network side) (17/26)

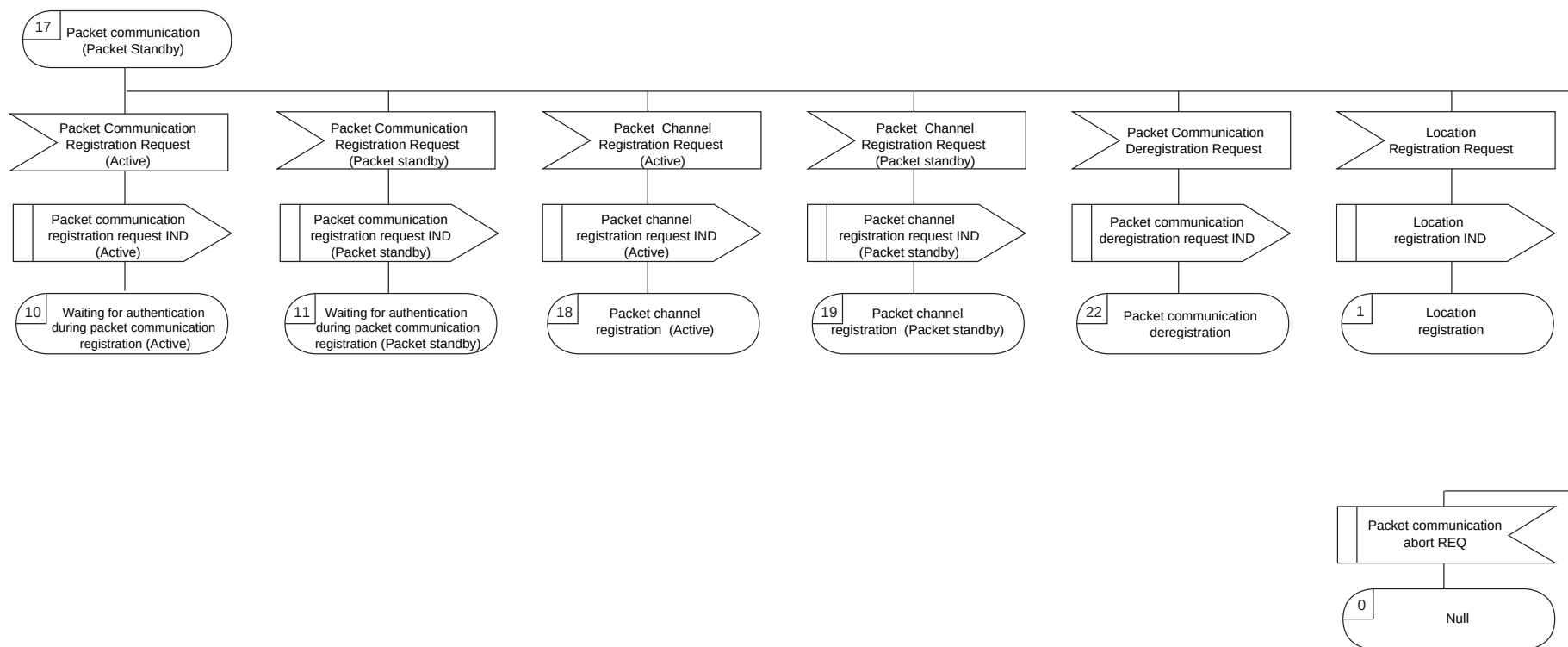


Fig. A. 5-123 Layer 3 MM SDL diagram (Network side) (18/26)

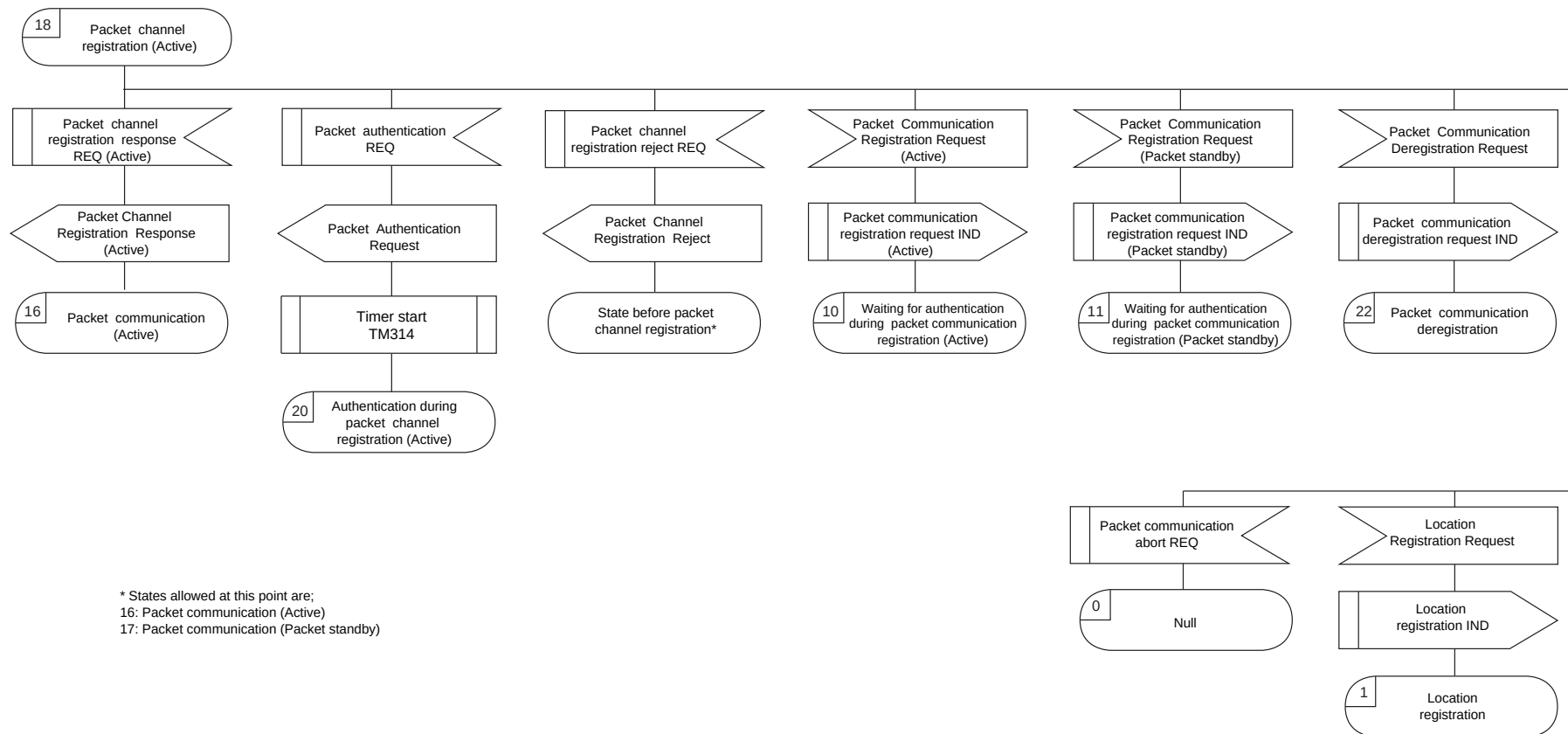


Fig. A. 5-124 Layer 3 MM SDL diagram (Network side) (19/26)

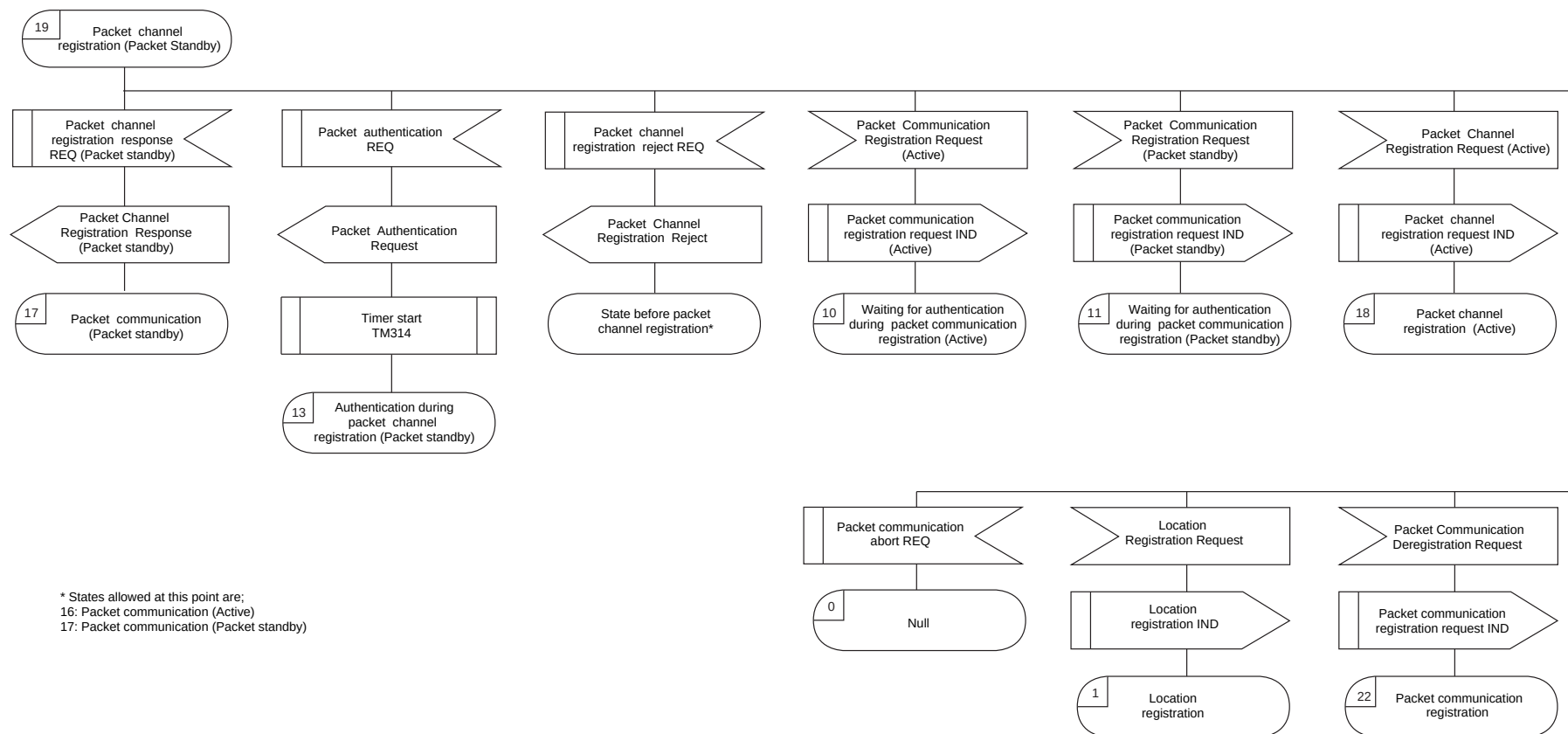


Fig. A. 5-125 Layer 3 MM SDL diagram (Network side) (20/26)

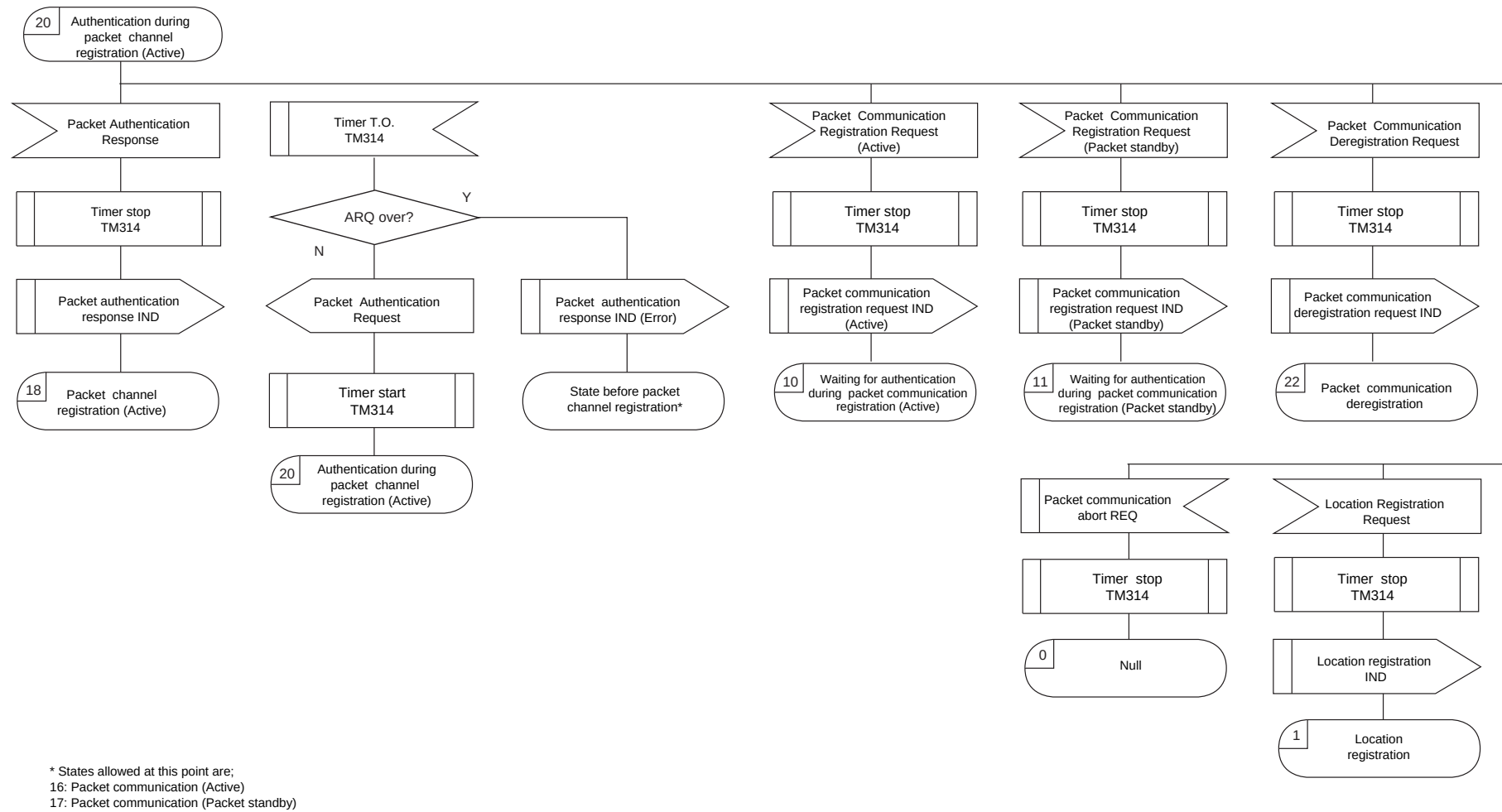
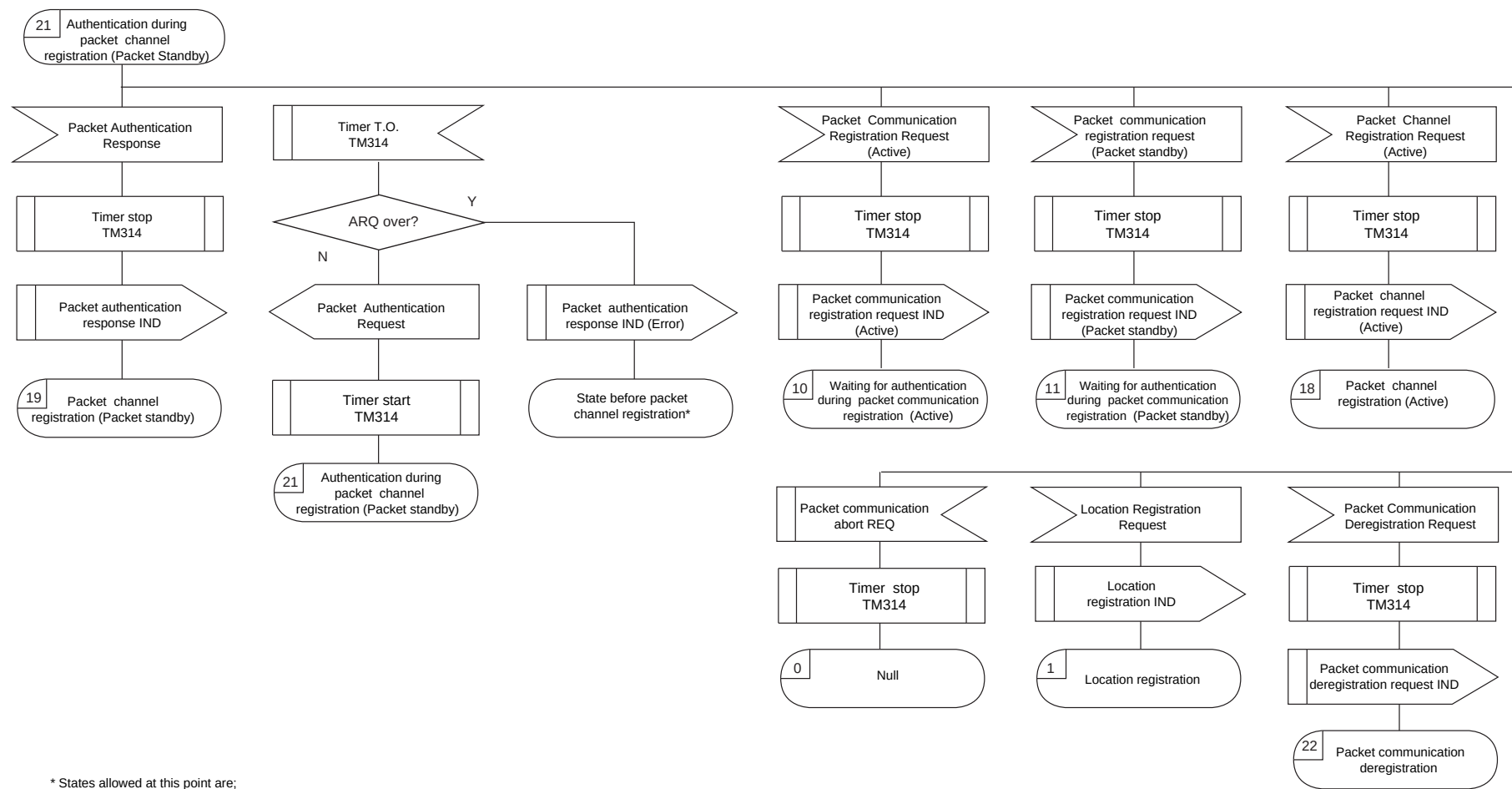
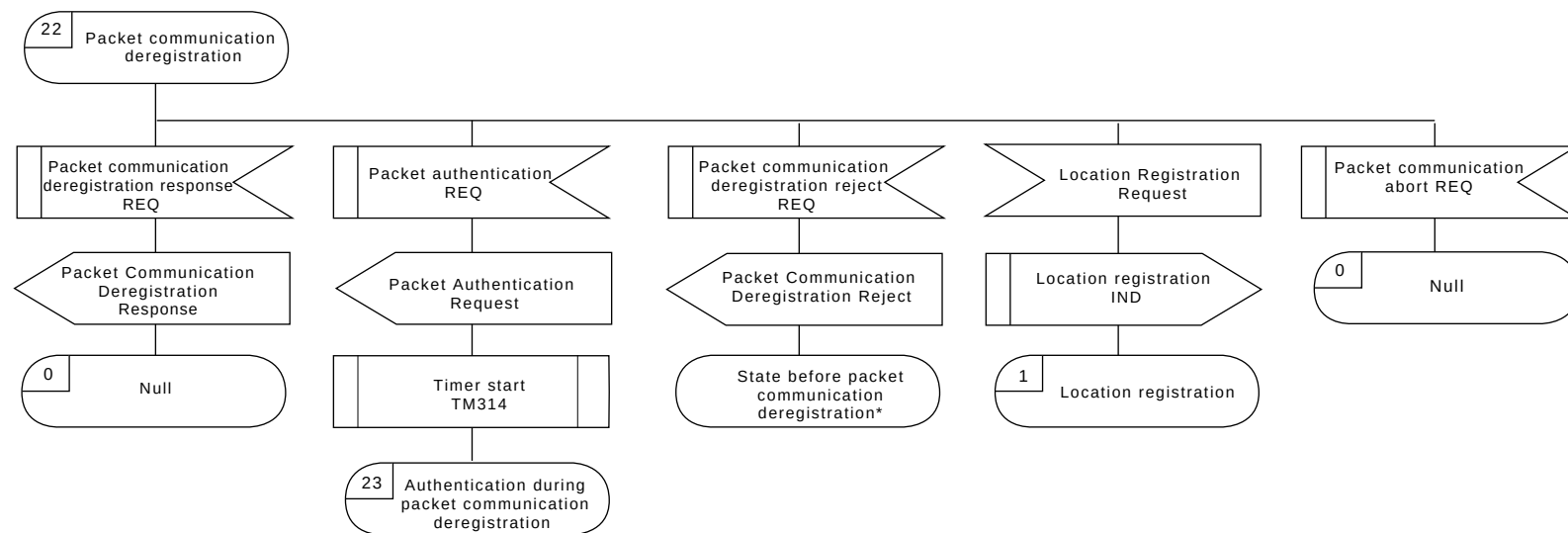


Fig. A. 5-126 Layer 3 MM SDL diagram (Network side) (21/26)



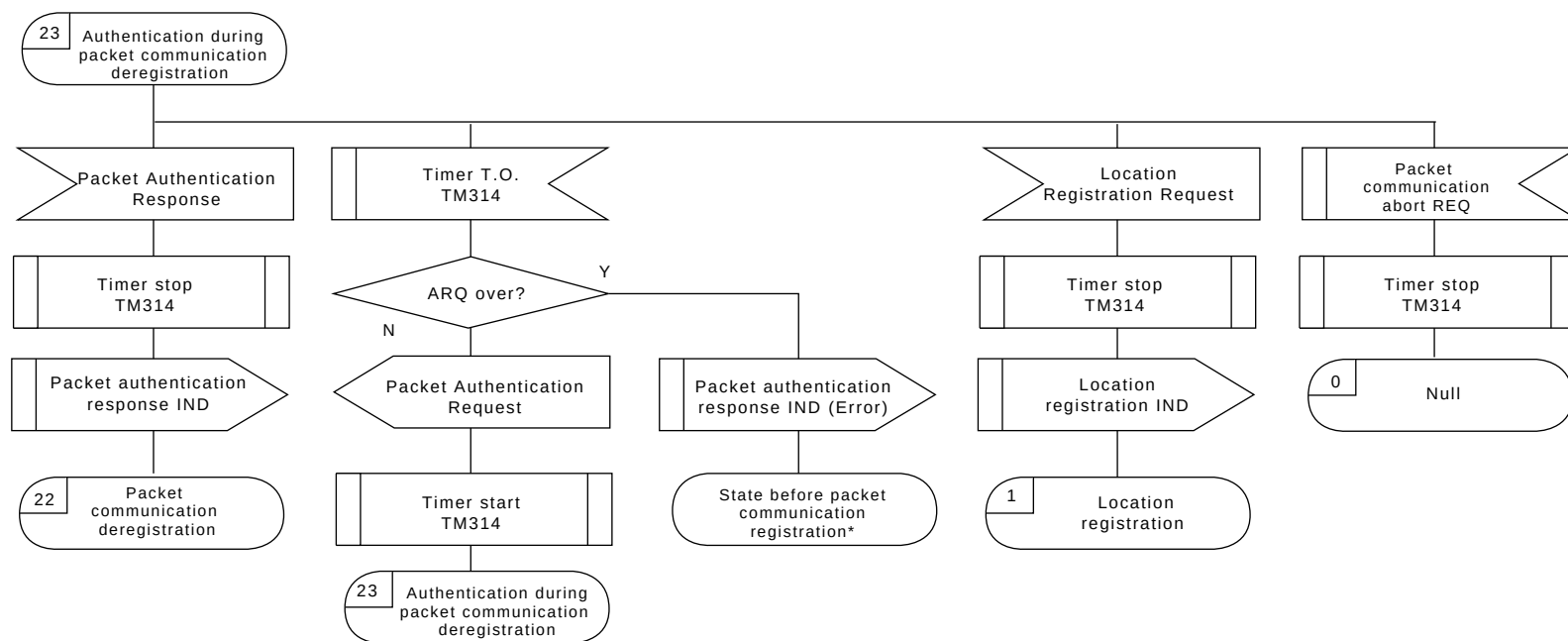
* States allowed at this point are;
 16: Packet communication (Active)
 17: Packet communication (Packet standby)

Fig. A. 5-127 Layer 3 MM SDL diagram (Network side) (22/26)



*: States allowed at this point are;
 16: Packet communication (Active)
 17: Packet communication (Packet standby)

Fig. A. 5-128 Layer 3 MM SDL diagram (Network side) (23/26)



*: States allowed at this point are;
 16: Packet communication (Active)
 17: Packet communication (Packet standby)

Fig. A. 5-129 Layer 3 MM SDL diagram (Network side) (24/26)

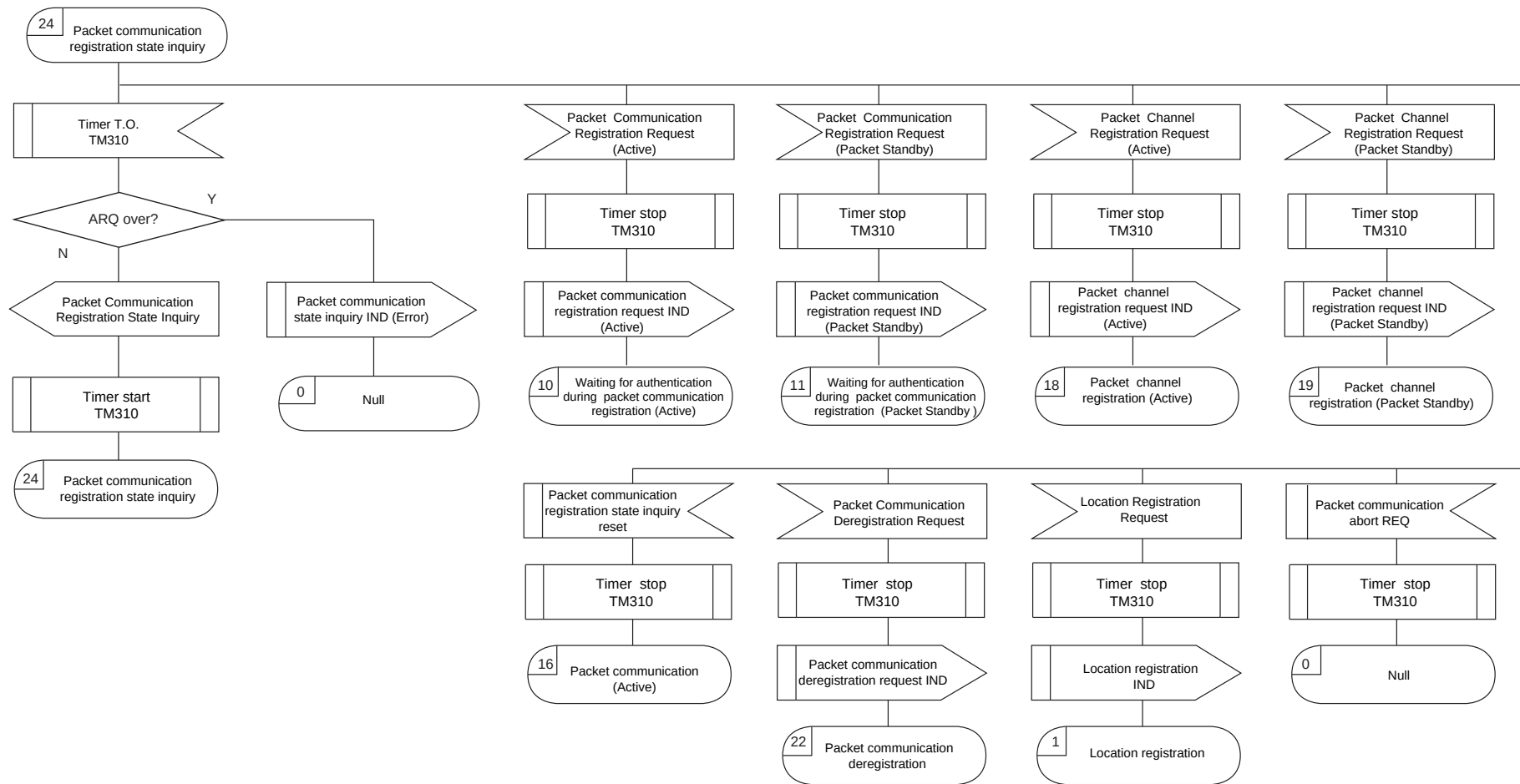


Fig. A. 5-130 Layer 3 MM SDL diagram (Network side) (25/26)

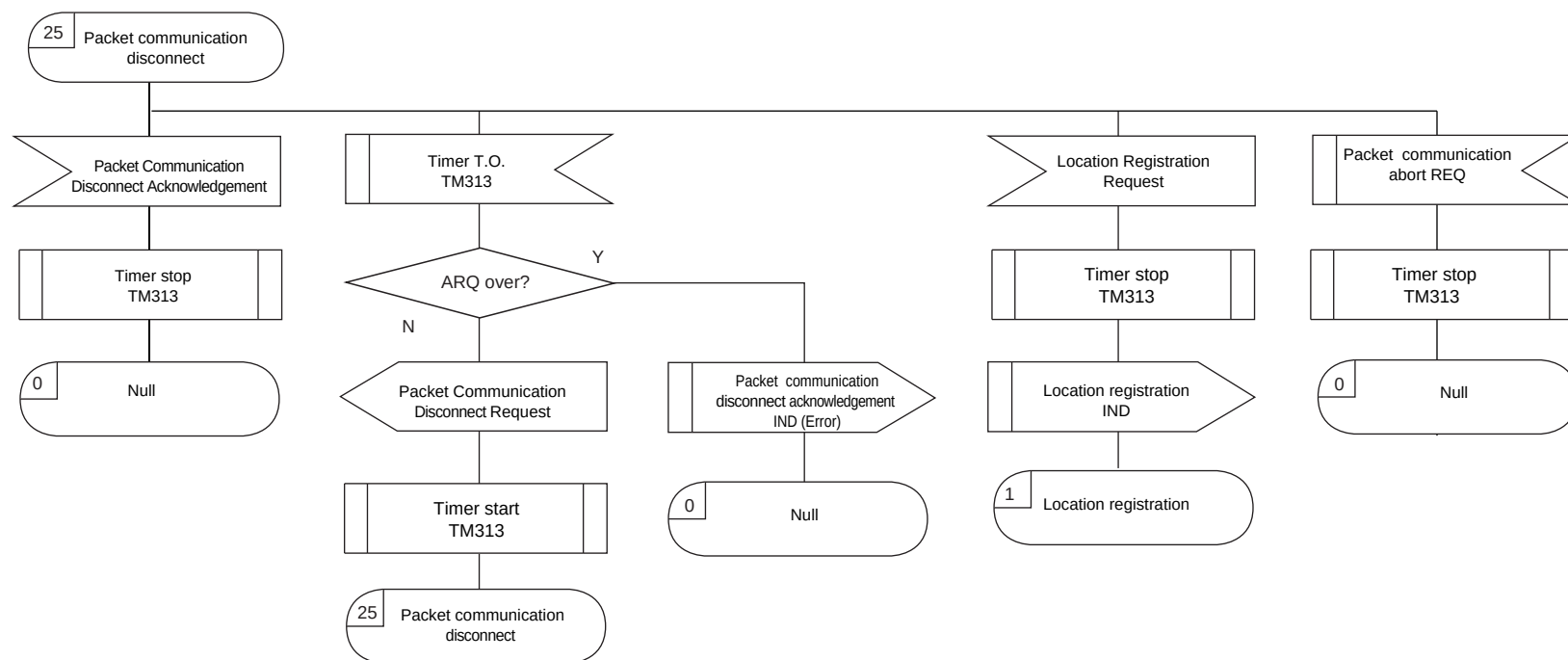


Fig. A. 5-131 Layer 3 MM SDL diagram (Network side) (26/26)

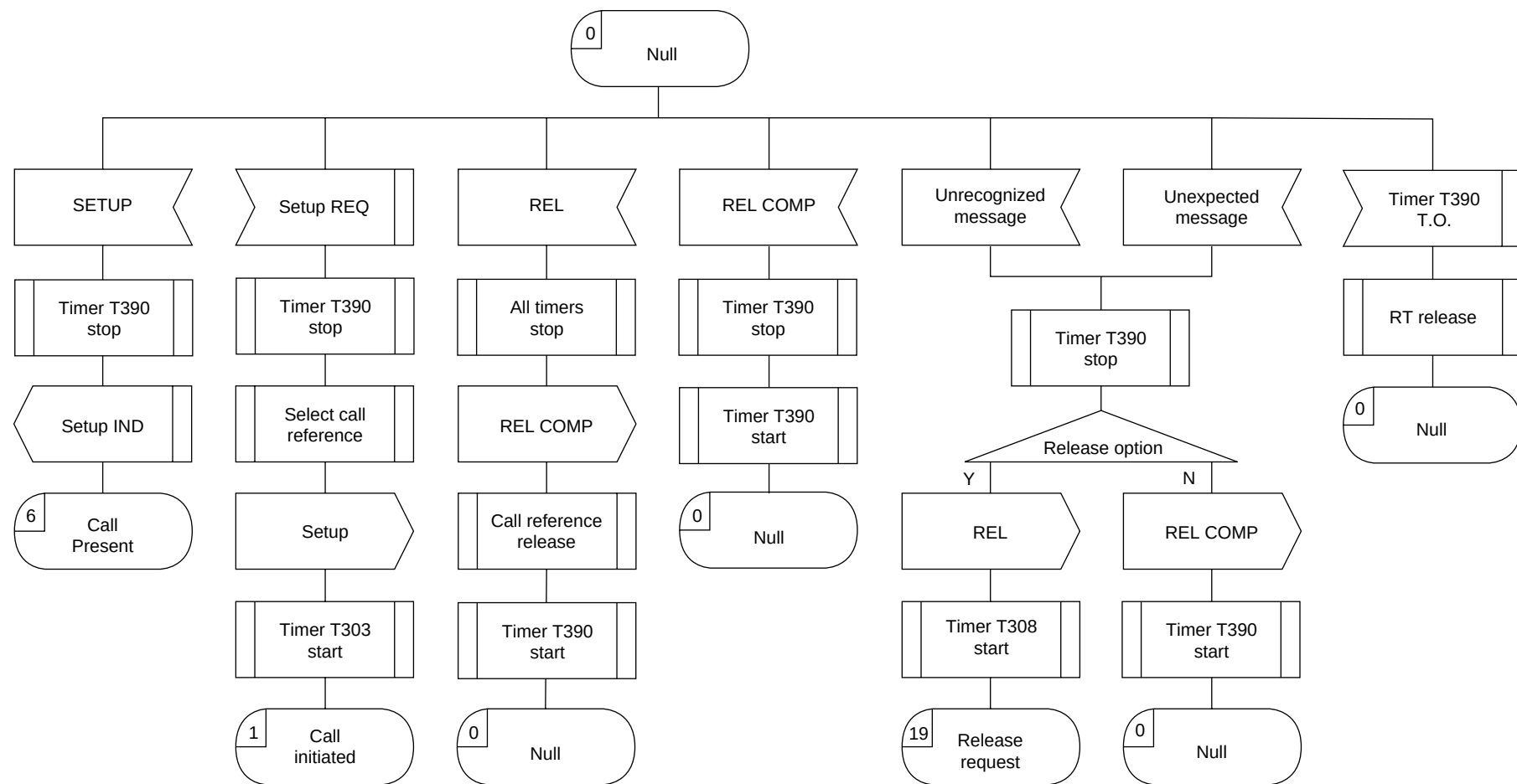


Fig. A. 5-132 Layer 3 CC SDL diagram (User side) (1/9)

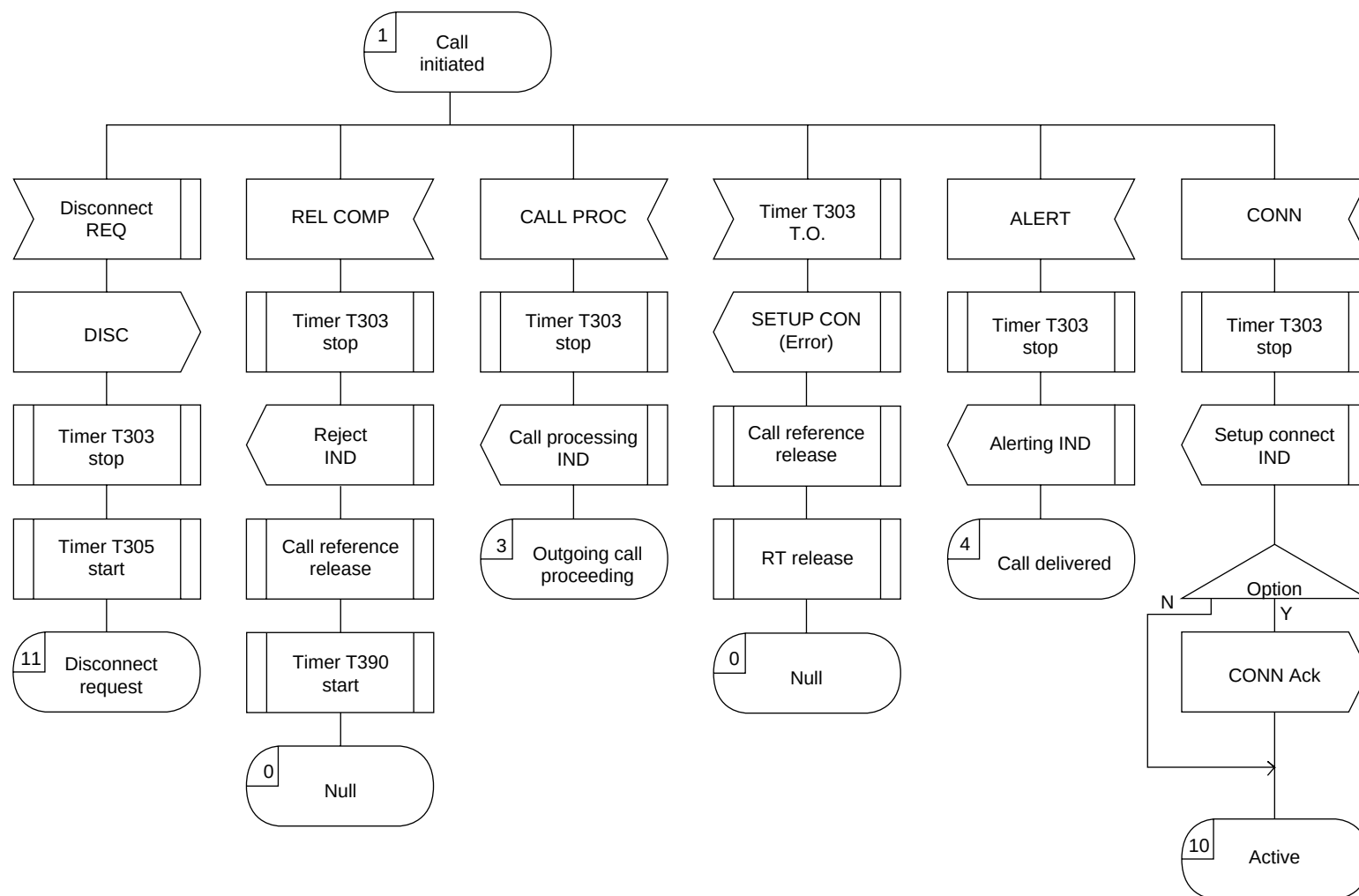


Fig. A. 5-133 Layer 3 CC SDL diagram (User side) (2/9)

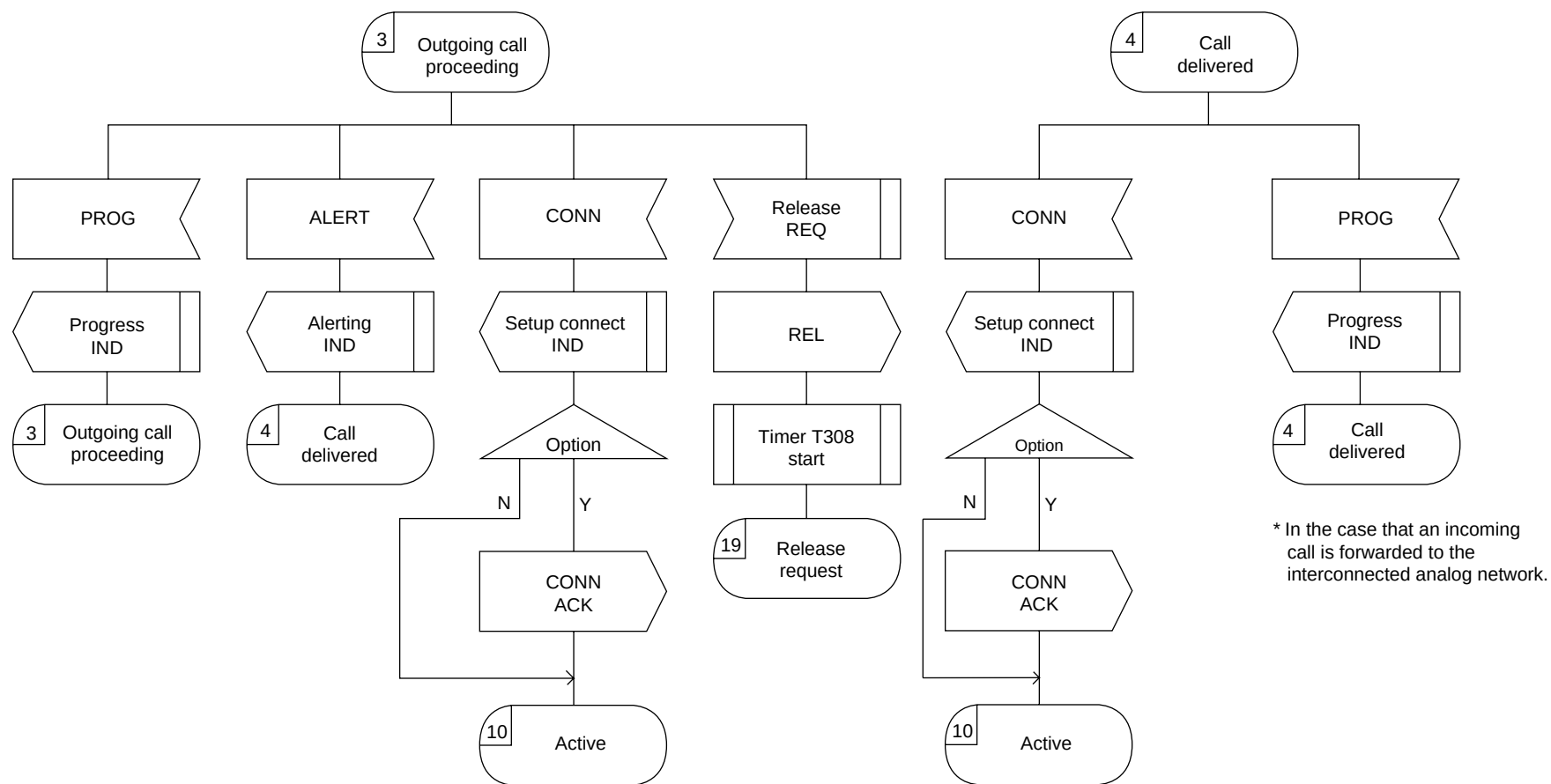


Fig. A. 5-134 Layer 3 CC SDL diagram (User side) (3/9)

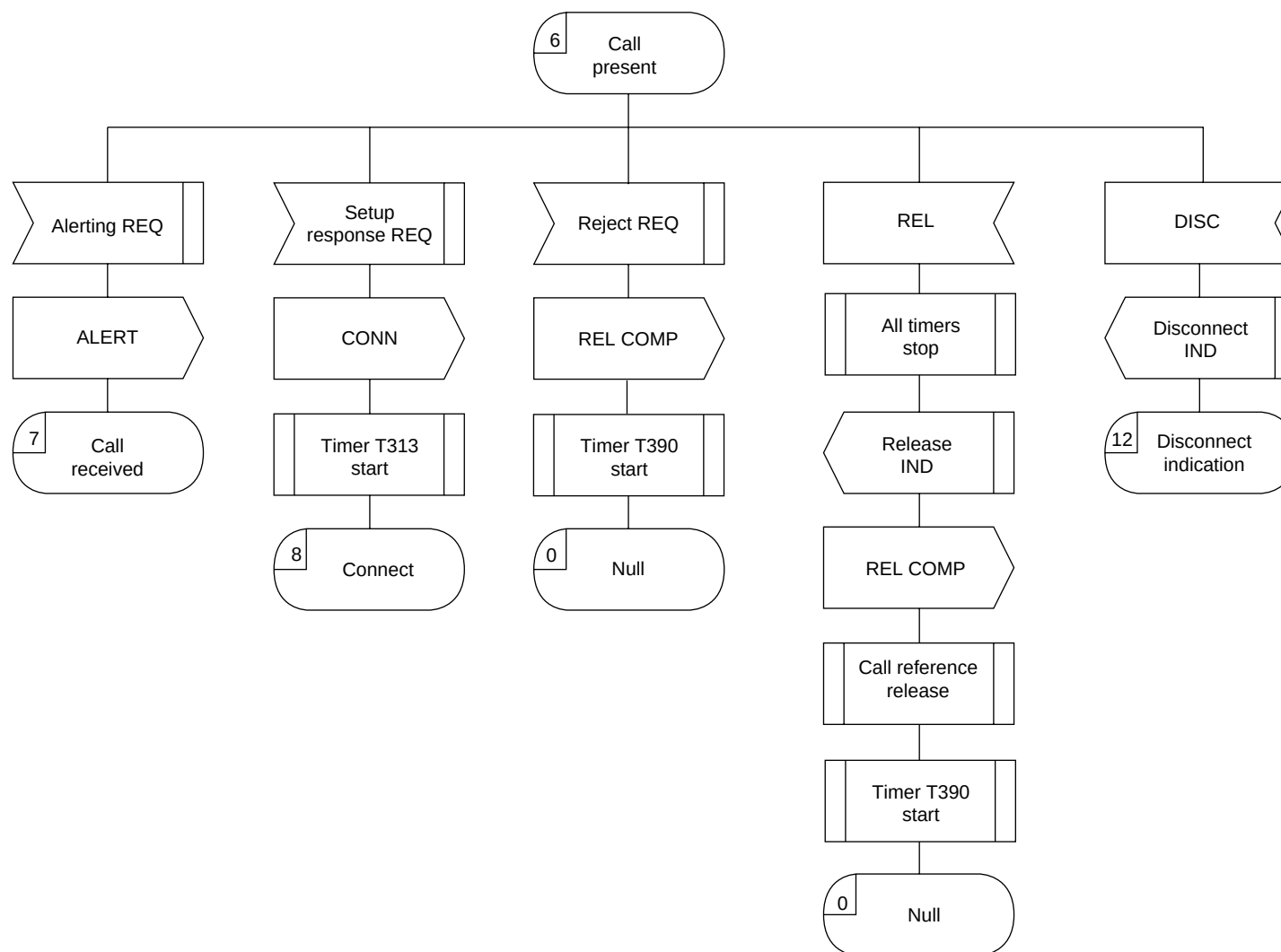


Fig. A. 5-135 Layer 3 CC SDL diagram (User side) (4/9)

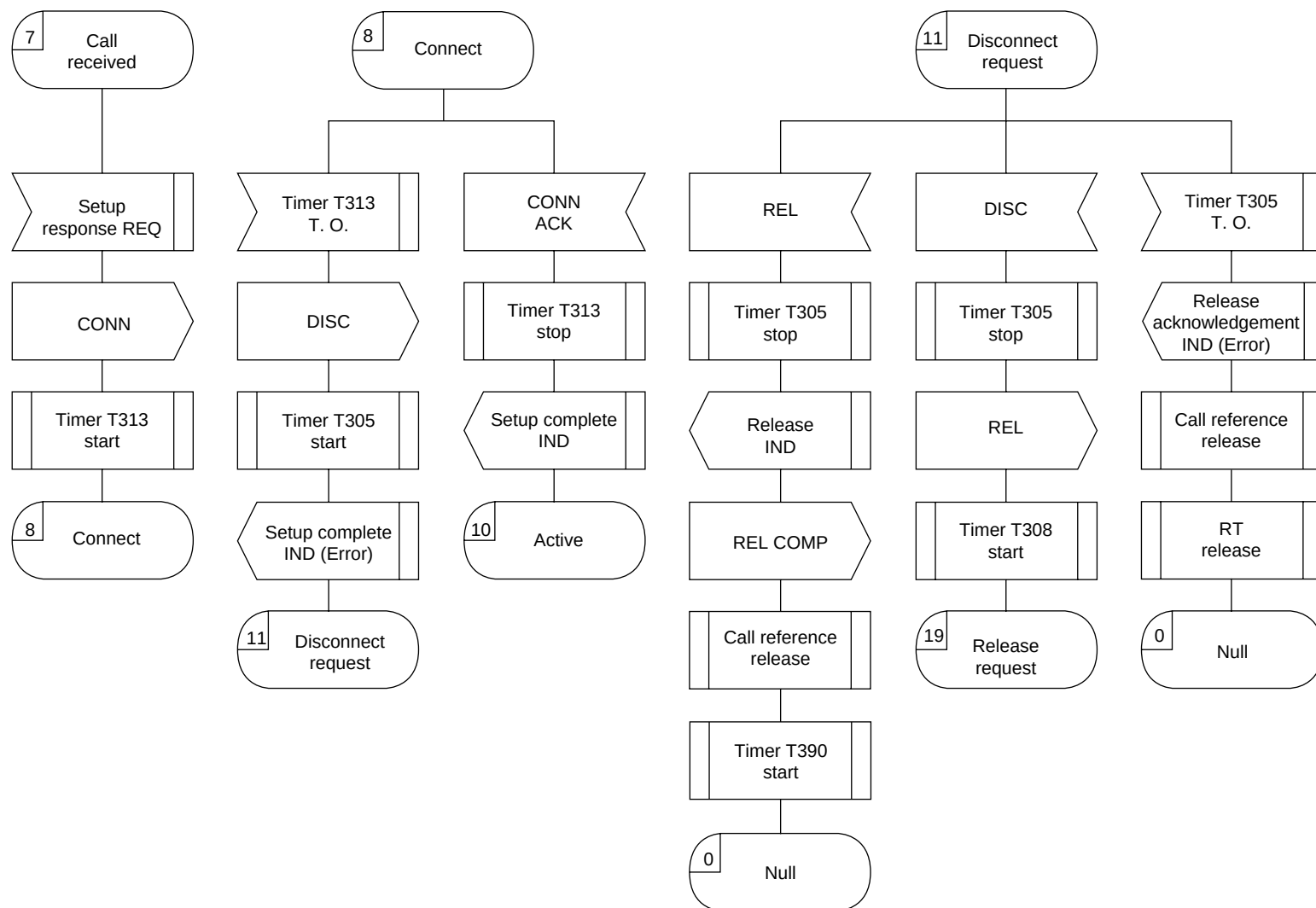


Fig. A. 5-136 Layer 3 CC SDL diagram (User side) (5/9)

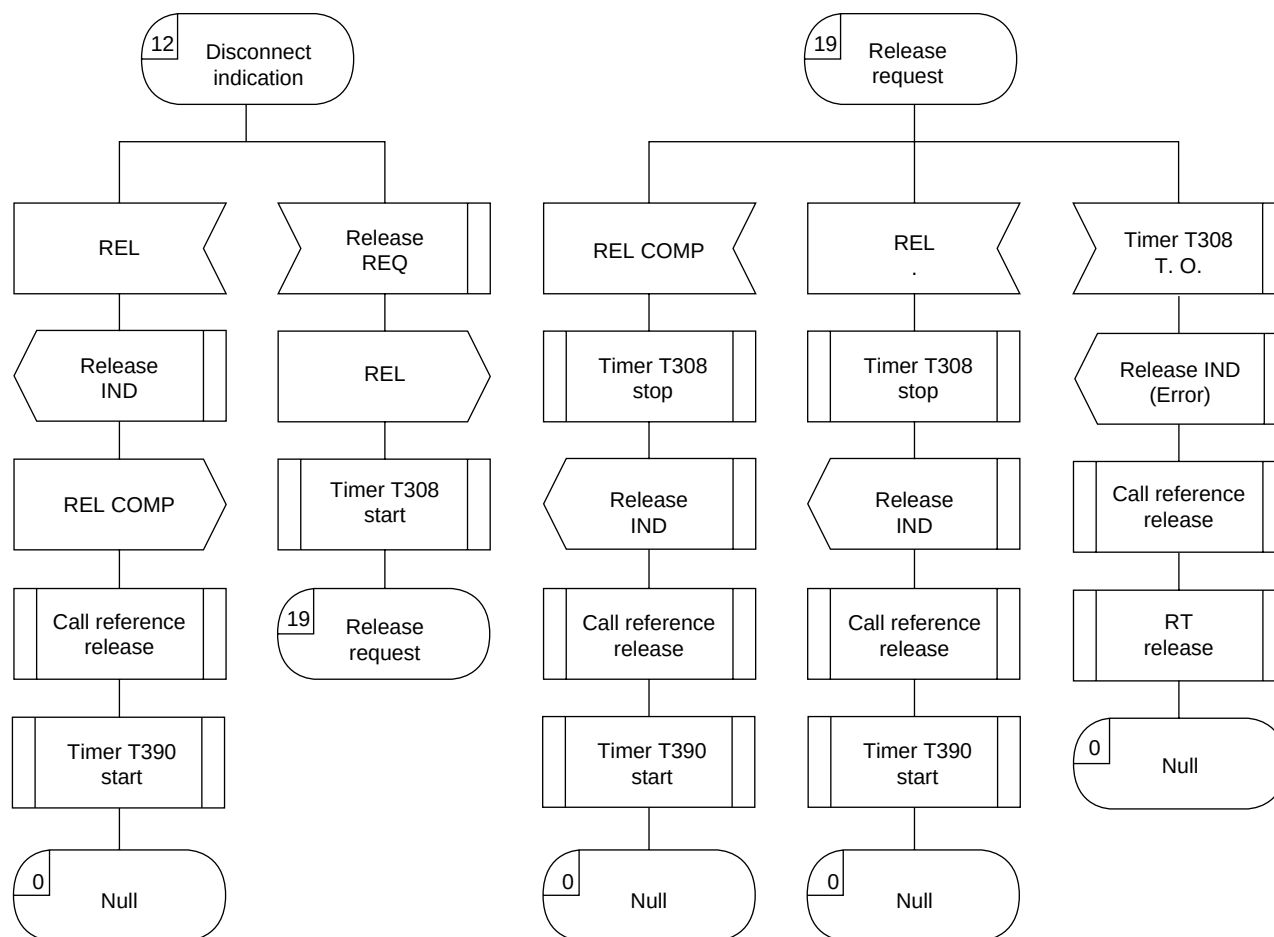


Fig. A. 5-137 Layer 3 CC SDL diagram (User side) (6/9)

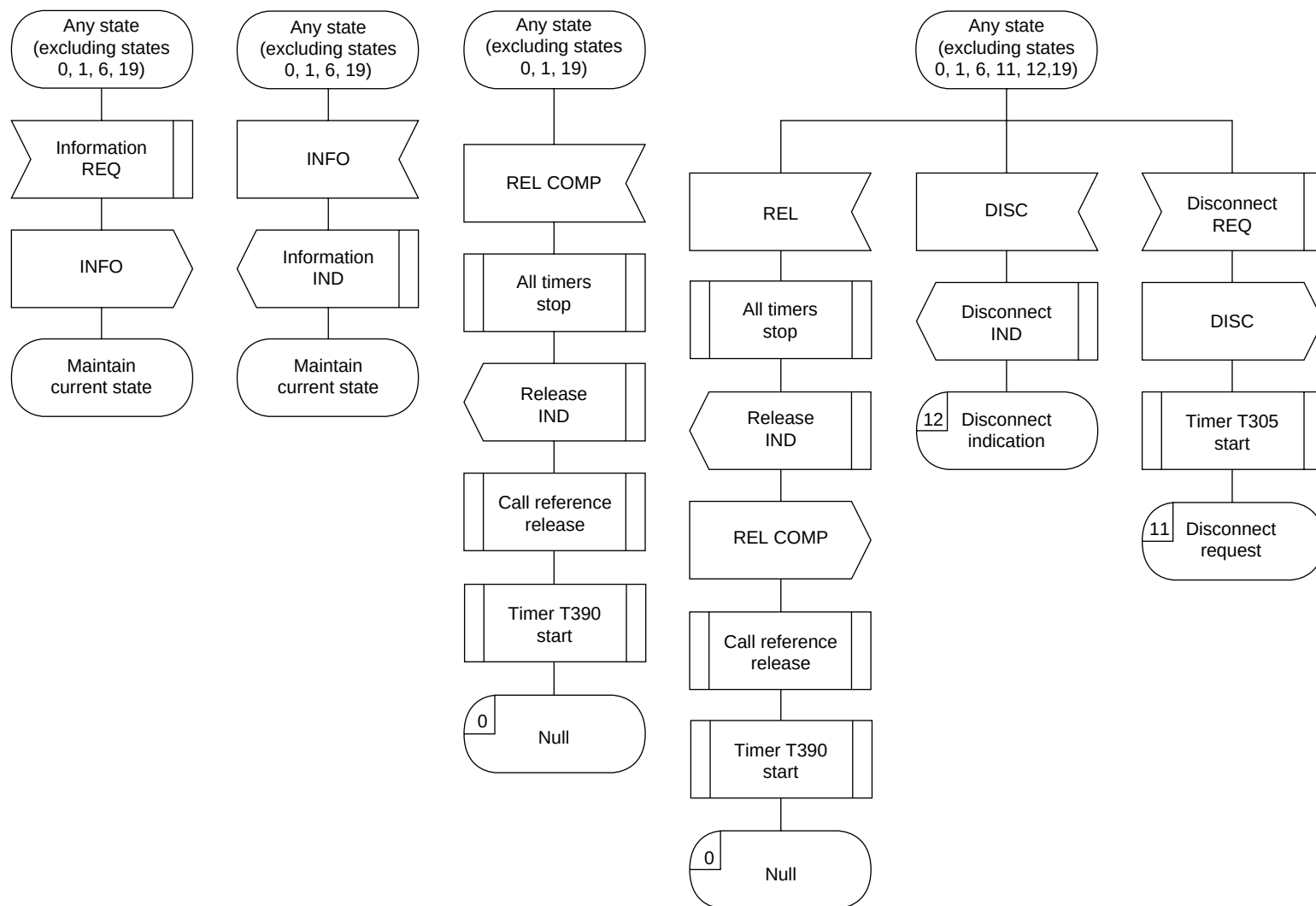
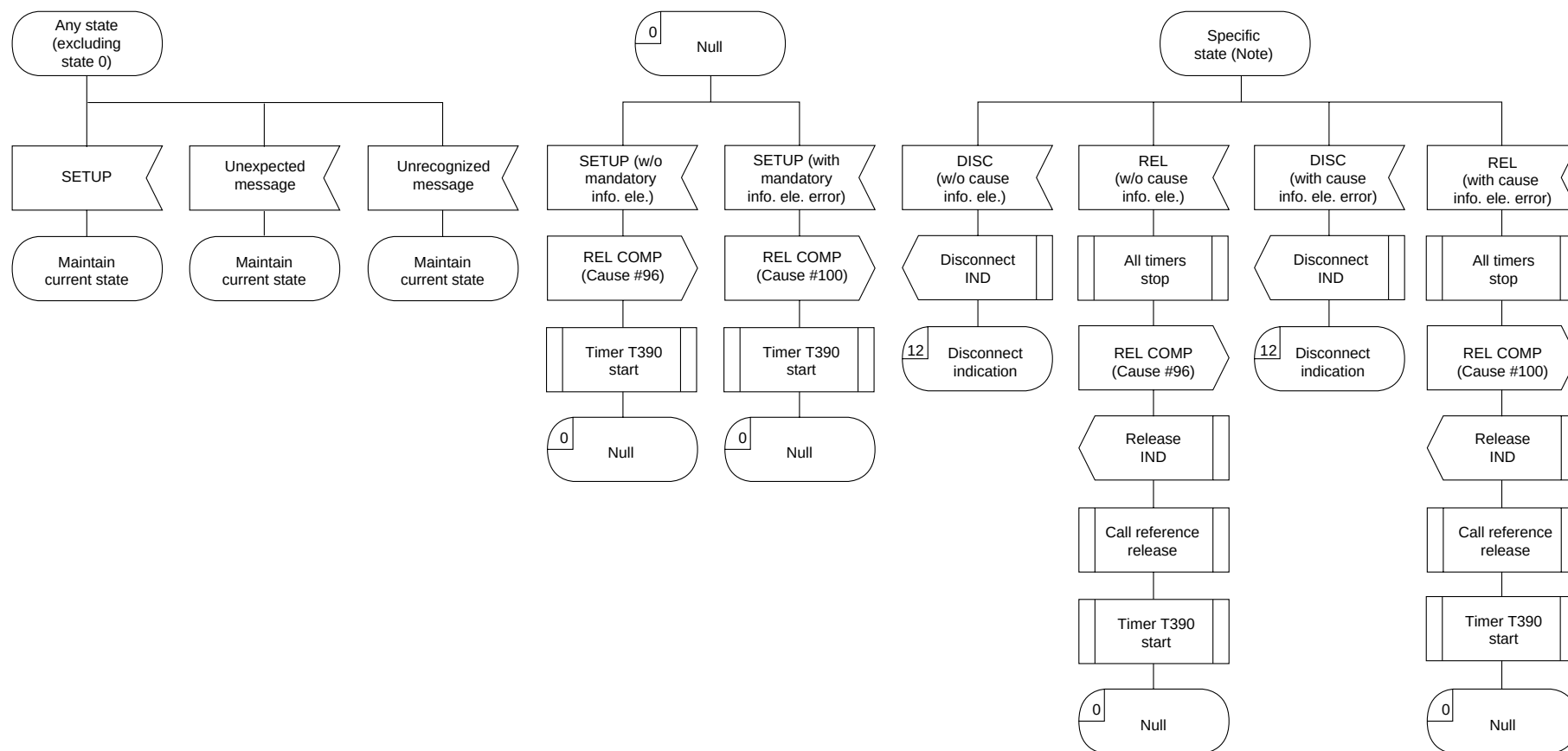


Fig. A. 5-138 Layer 3 CC SDL diagram (User side) (7/9)



Note : Specific state means UCC 0, 6, 11, 12 or 19.

Fig. A. 5-139 Layer 3 CC SDL diagram (User side) (8/9)

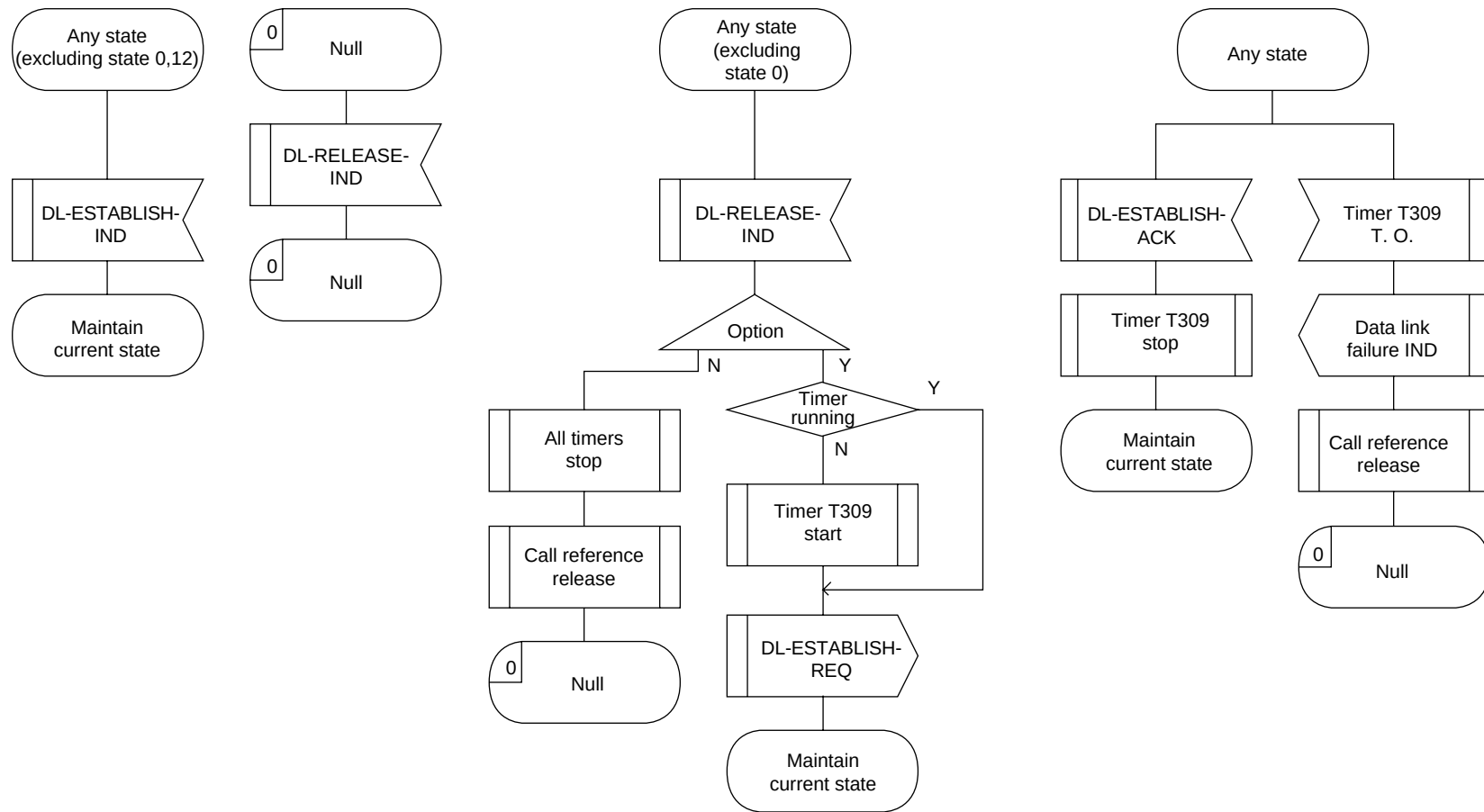


Fig. A. 5-140 Layer 3 CC SDL diagram (User side) (9/9)

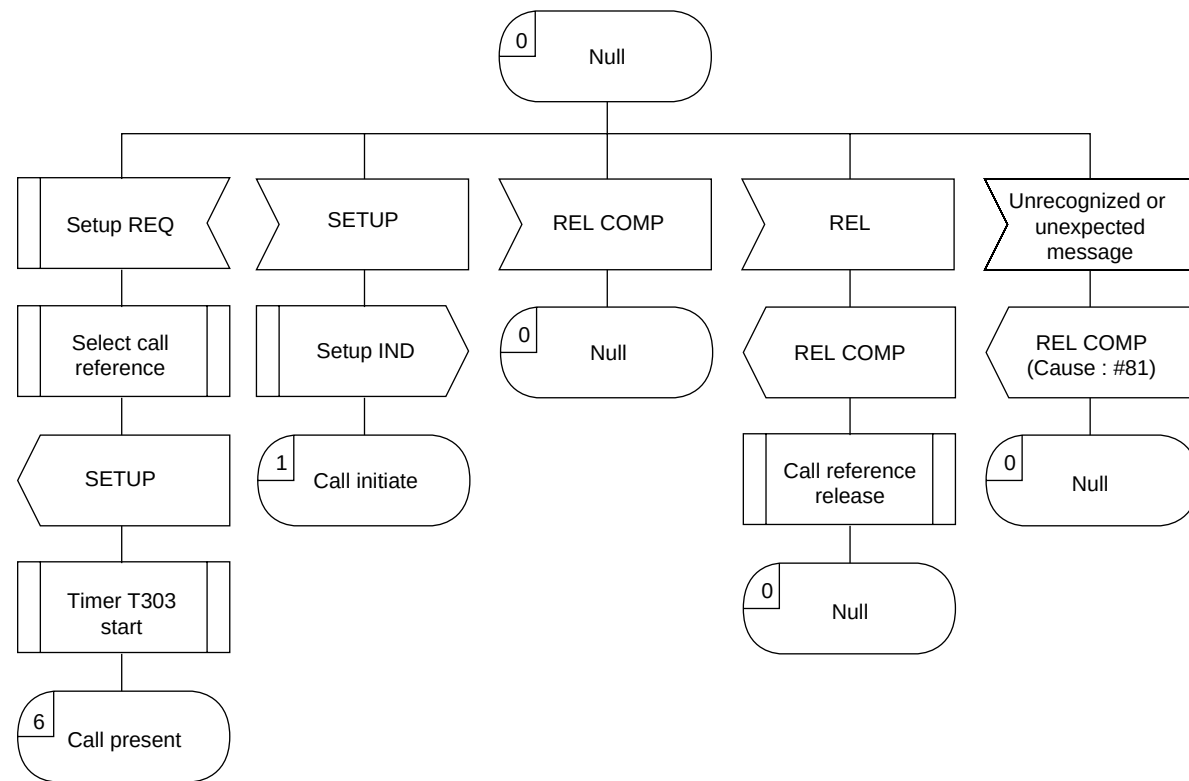


Fig. A. 5-141 Layer 3 CC SDL diagram (Network side) (1/10)

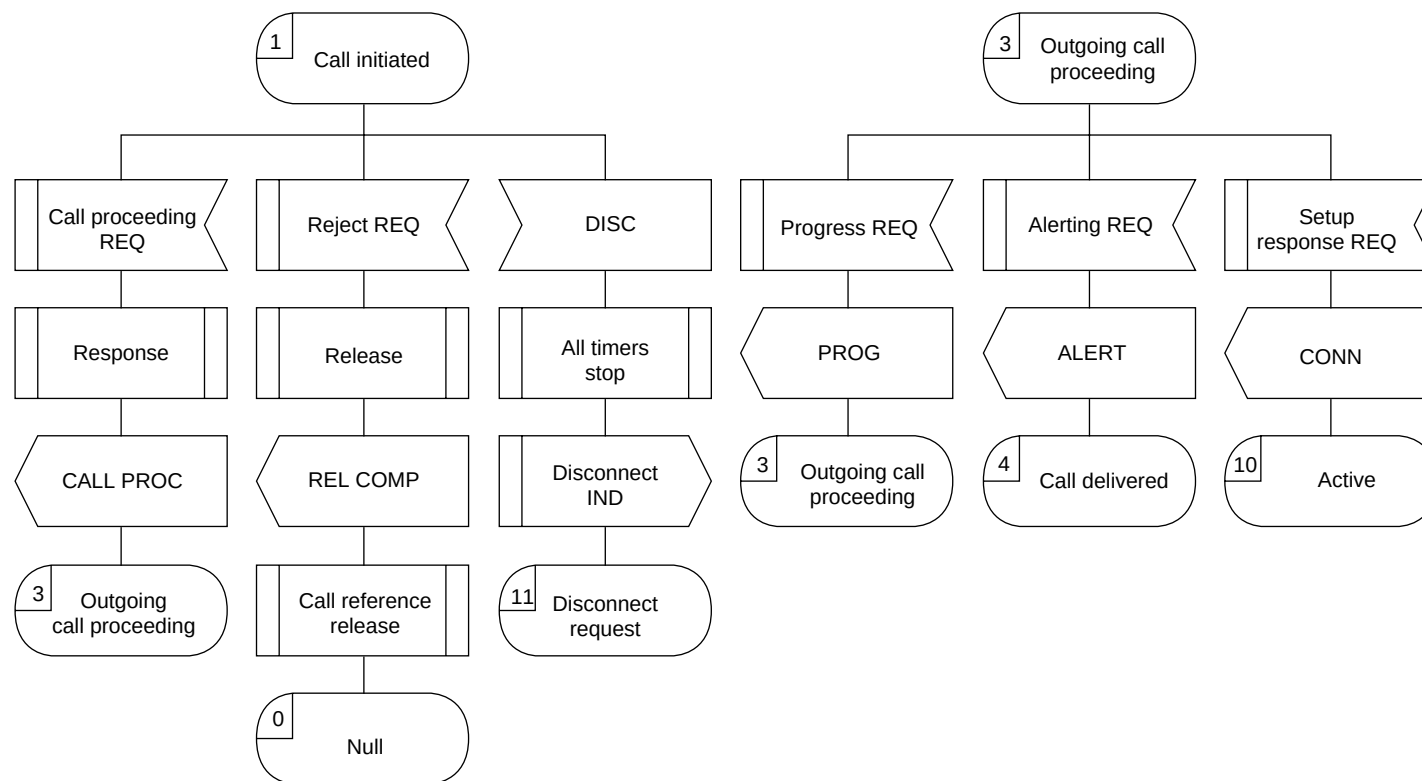


Fig. A. 5-142 Layer 3 CC SDL diagram (Network side) (2/10)

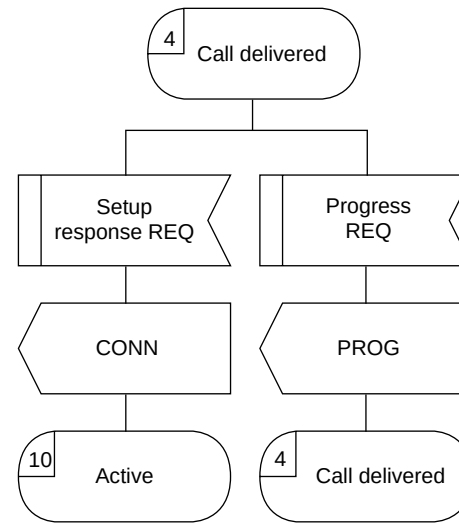


Fig. A. 5-143 Layer 3 CC SDL diagram (Network side) (3/10)

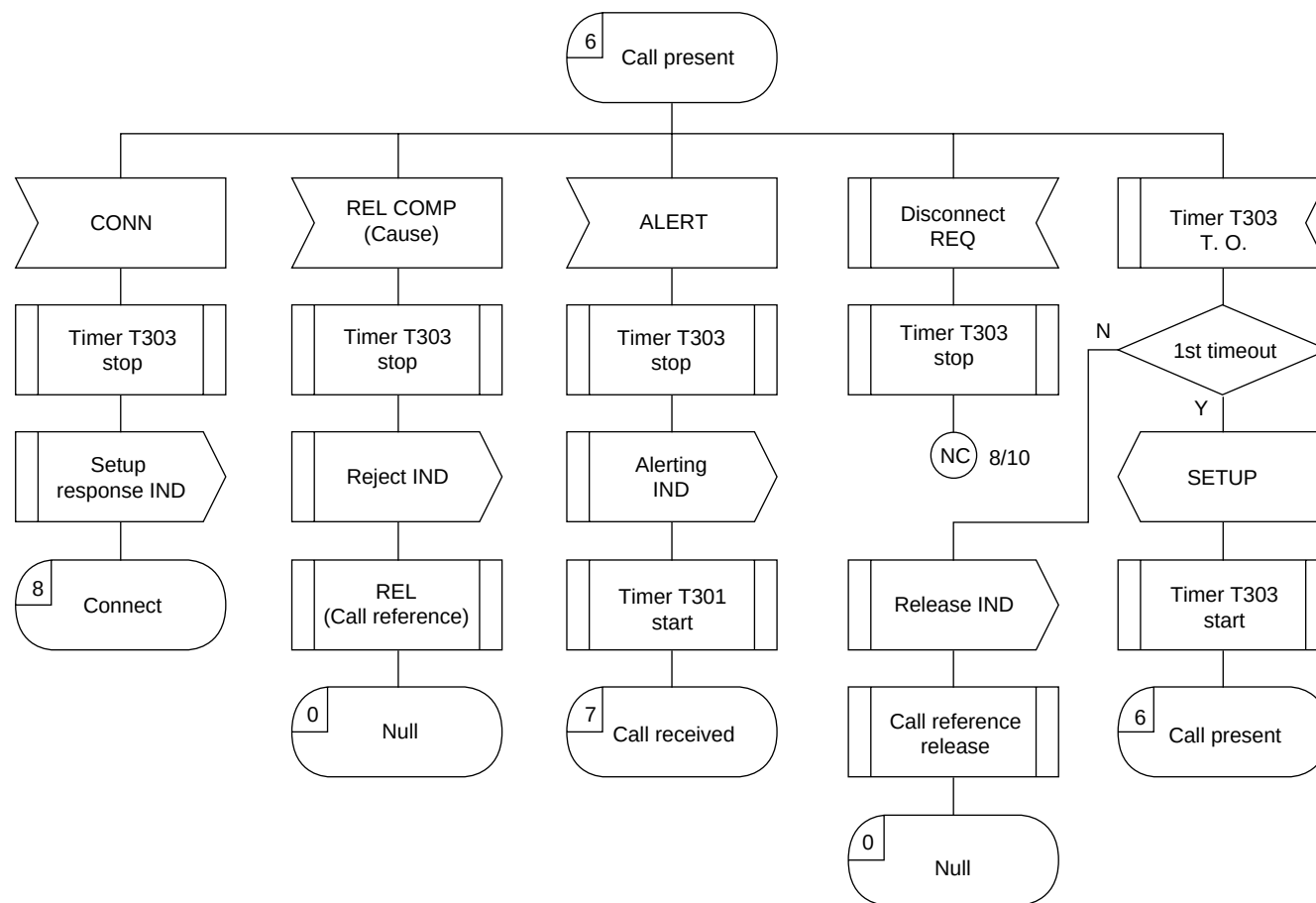


Fig. A. 5-144 Layer 3 CC SDL diagram (Network side) (4/10)

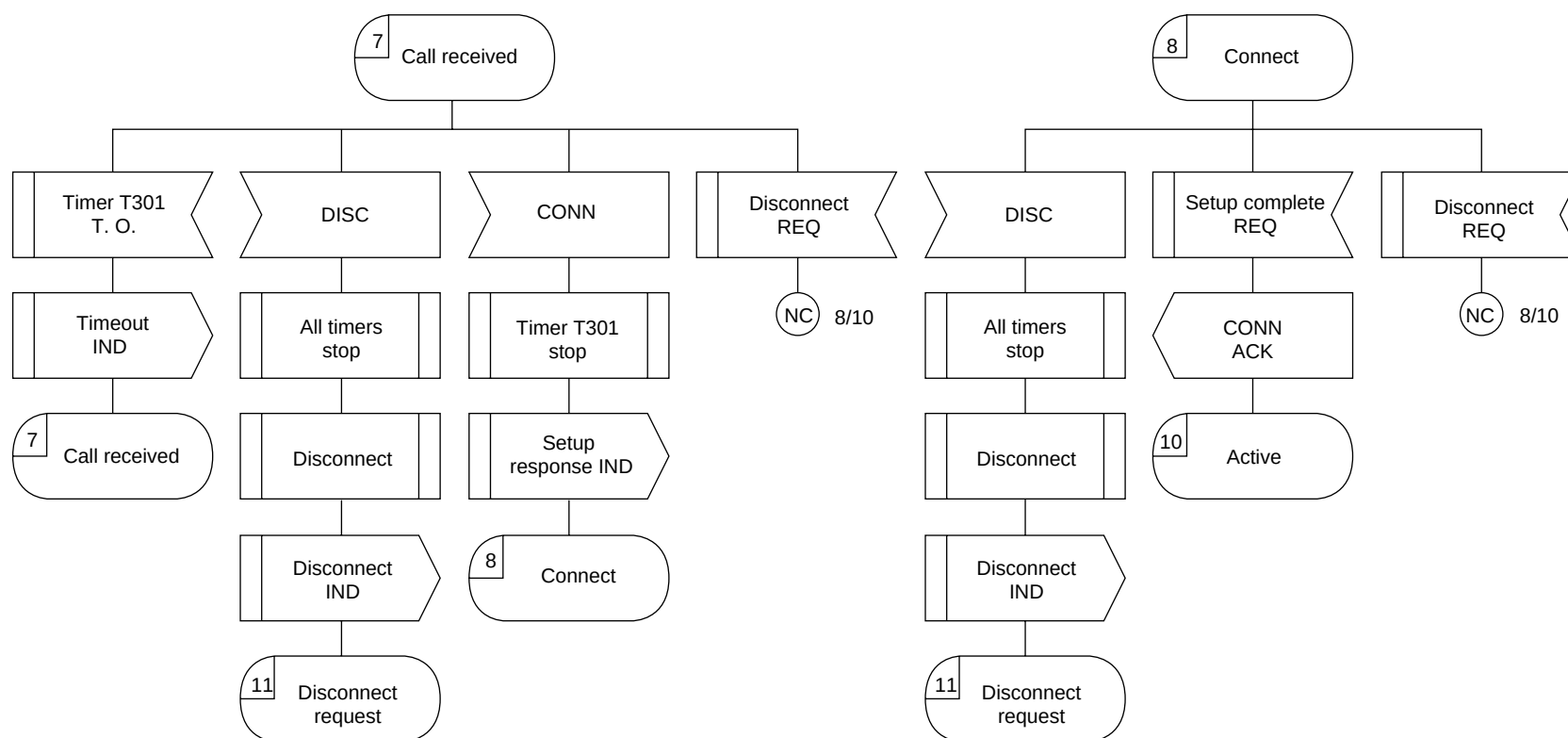


Fig. A. 5-145 Layer 3 CC SDL diagram (Network side) (5/10)

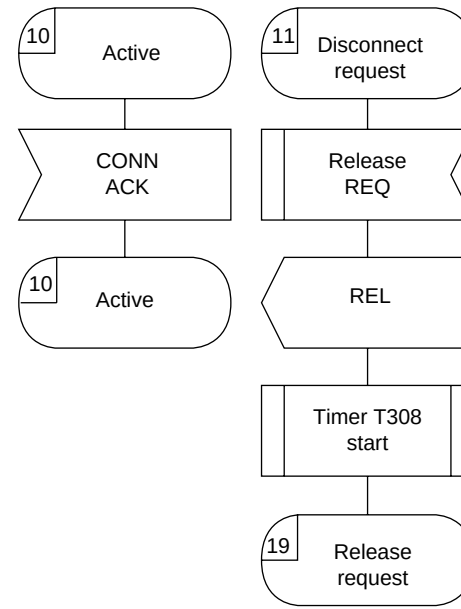


Fig. A. 5-146 Layer 3 CC SDL diagram (Network side) (6/10)

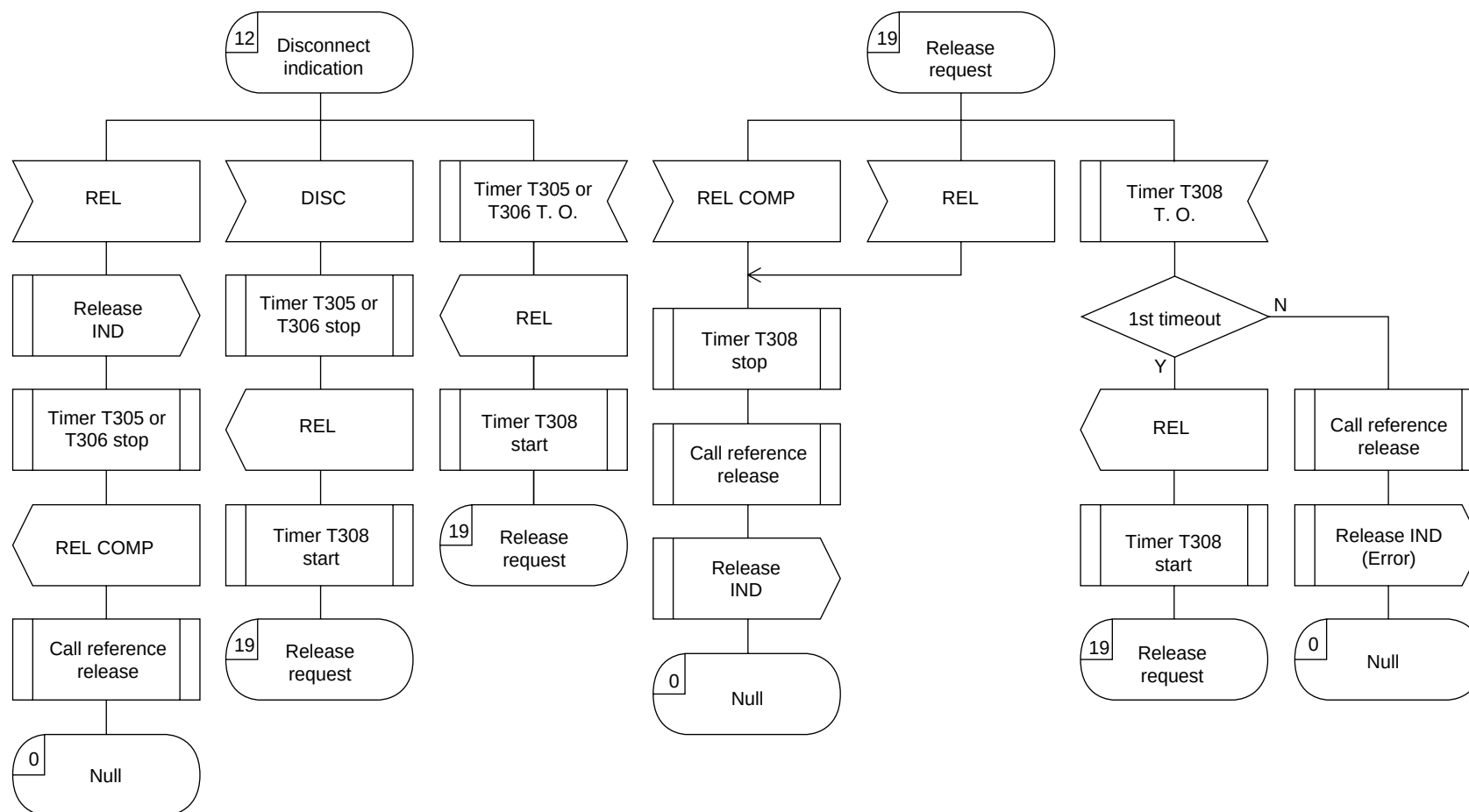


Fig. A. 5-147 Layer 3 CC SDL diagram (Network side) (7/10)

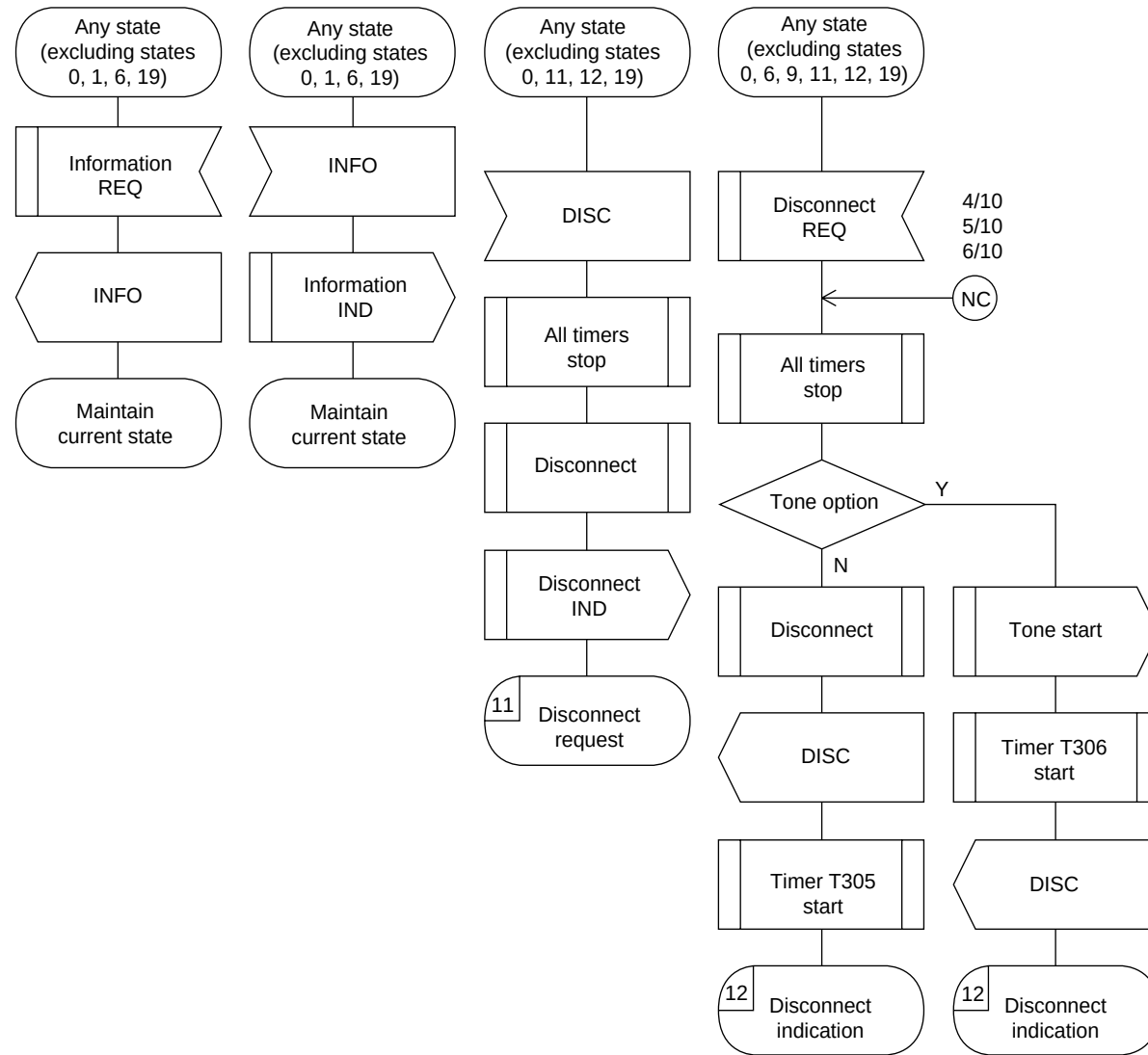


Fig. A. 5-148 Layer 3 CC SDL diagram (Network side) (8/10)

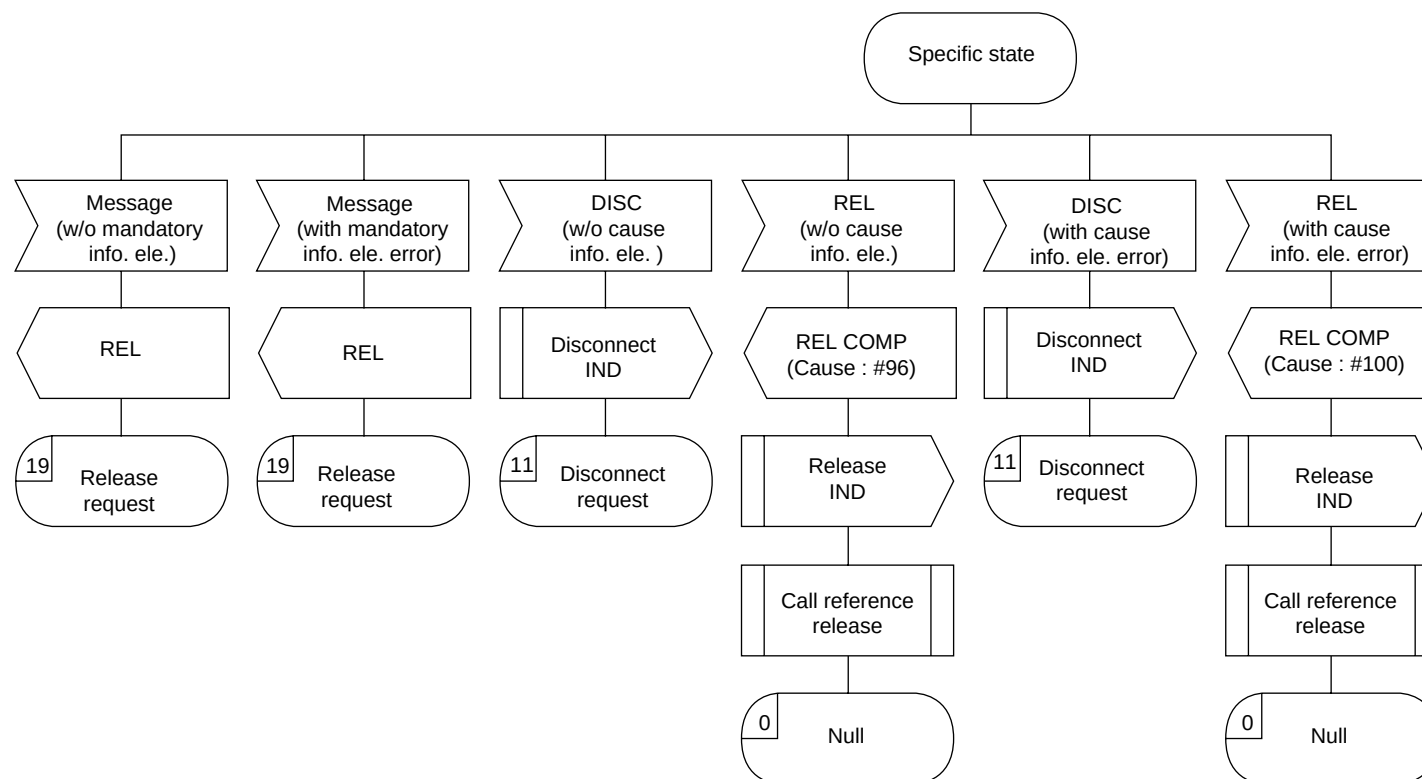
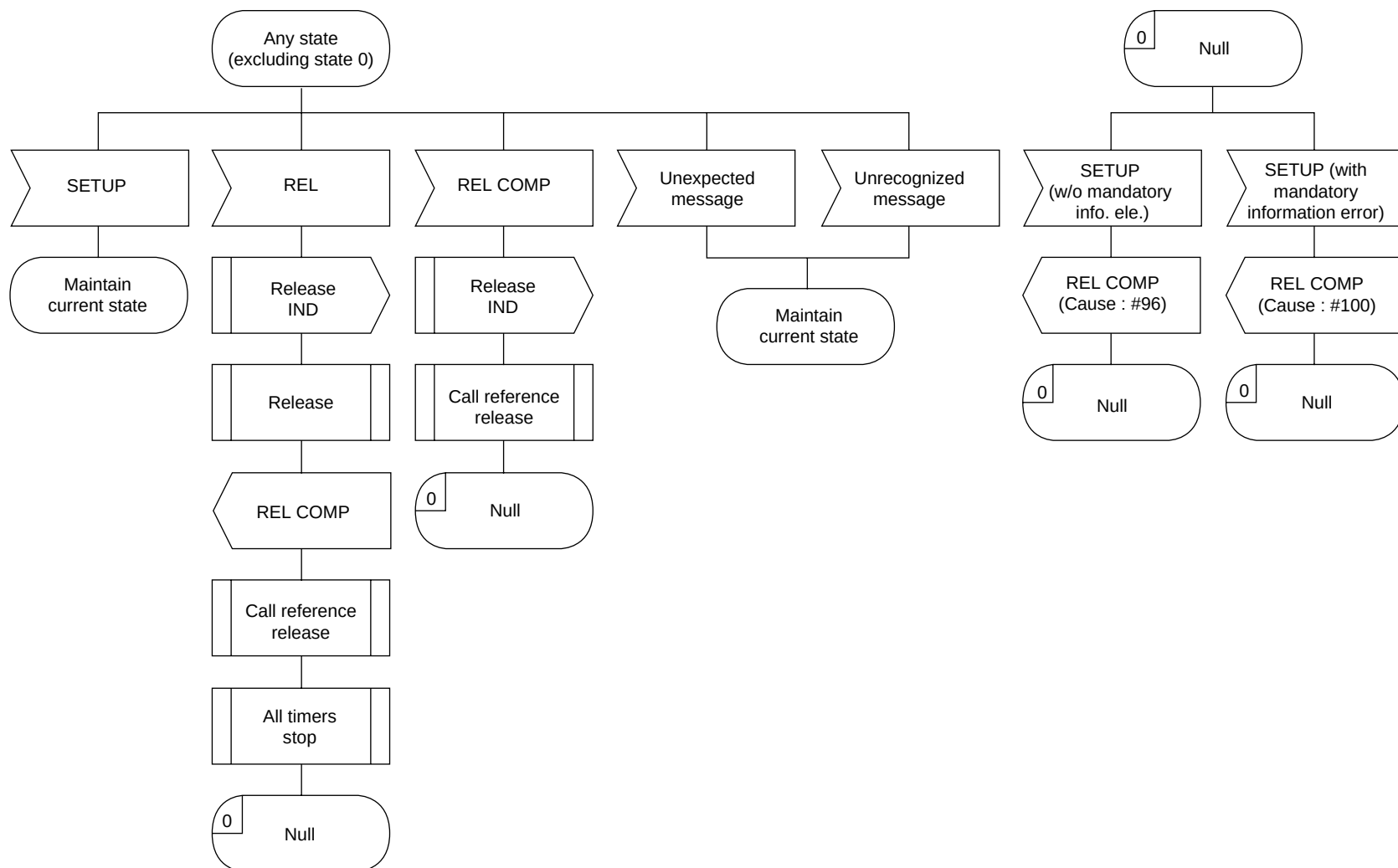


Fig. A. 5-149 Layer 3 CC SDL diagram (Network side) (9/10)





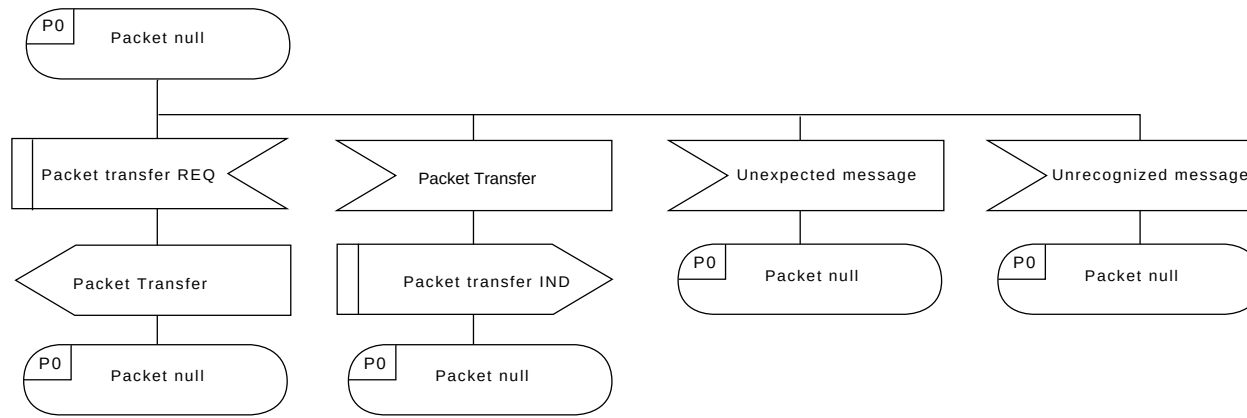


Fig. A. 5-152 Layer 3 CC (Packet communication function) SDL diagram (Network side) (1/1)

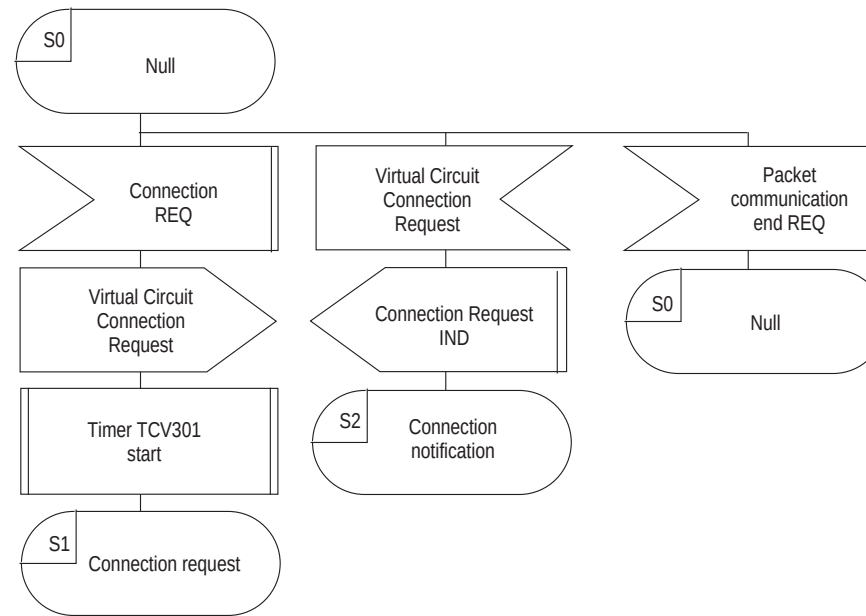


Fig. A. 5-153 Layer 3 CC (SVC function) SDL diagram (User side) (1/5)

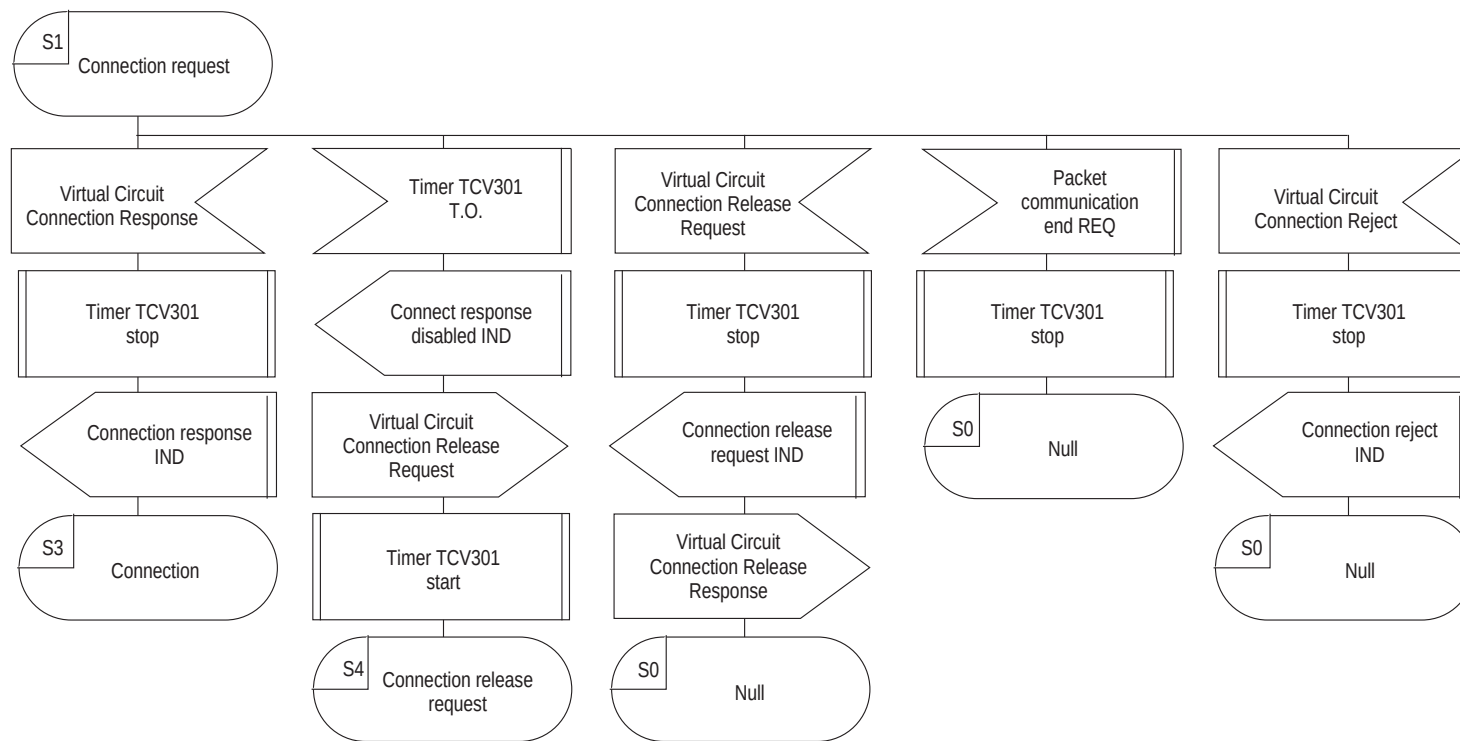


Fig. A. 5-154 Layer 3 CC (SVC function) SDL diagram (User side) (2/5)

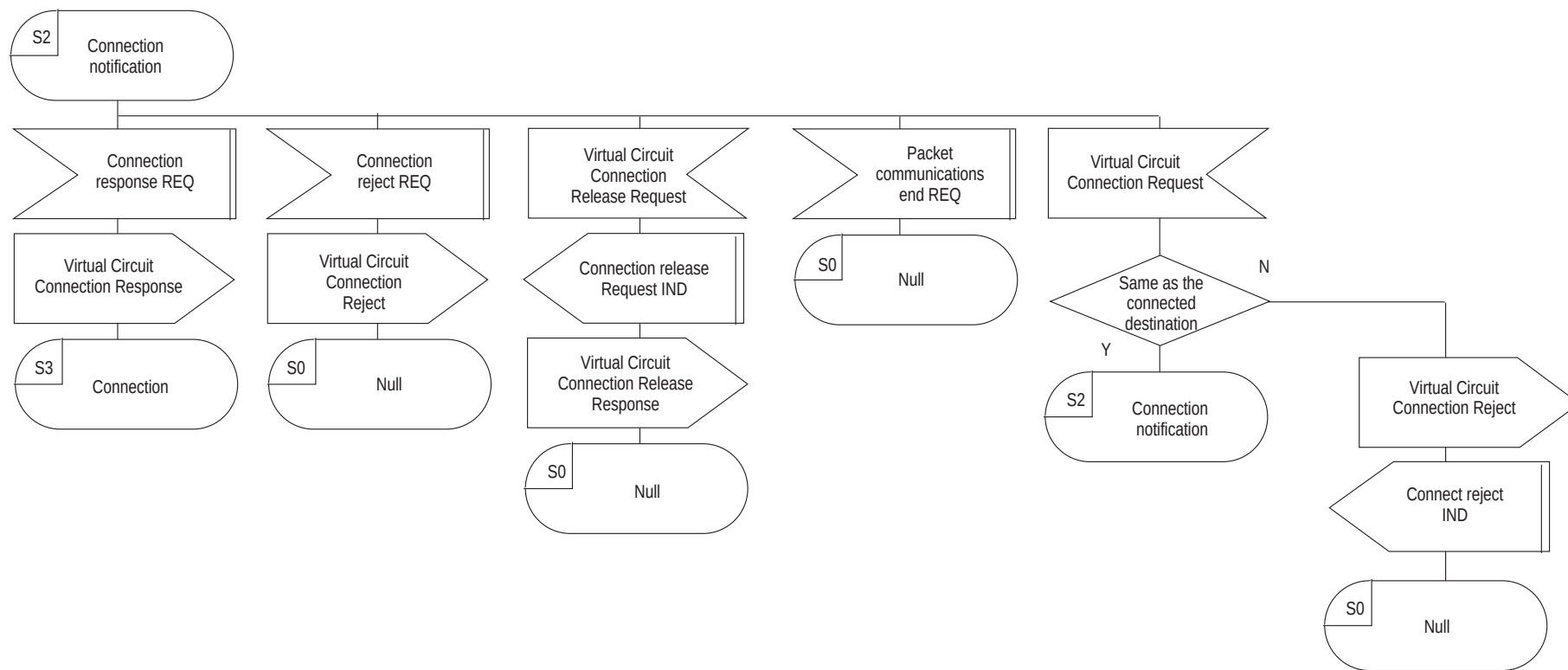


Fig. A. 5-155 Layer 3 CC (SVC function) SDL diagram (User side) (3/5)

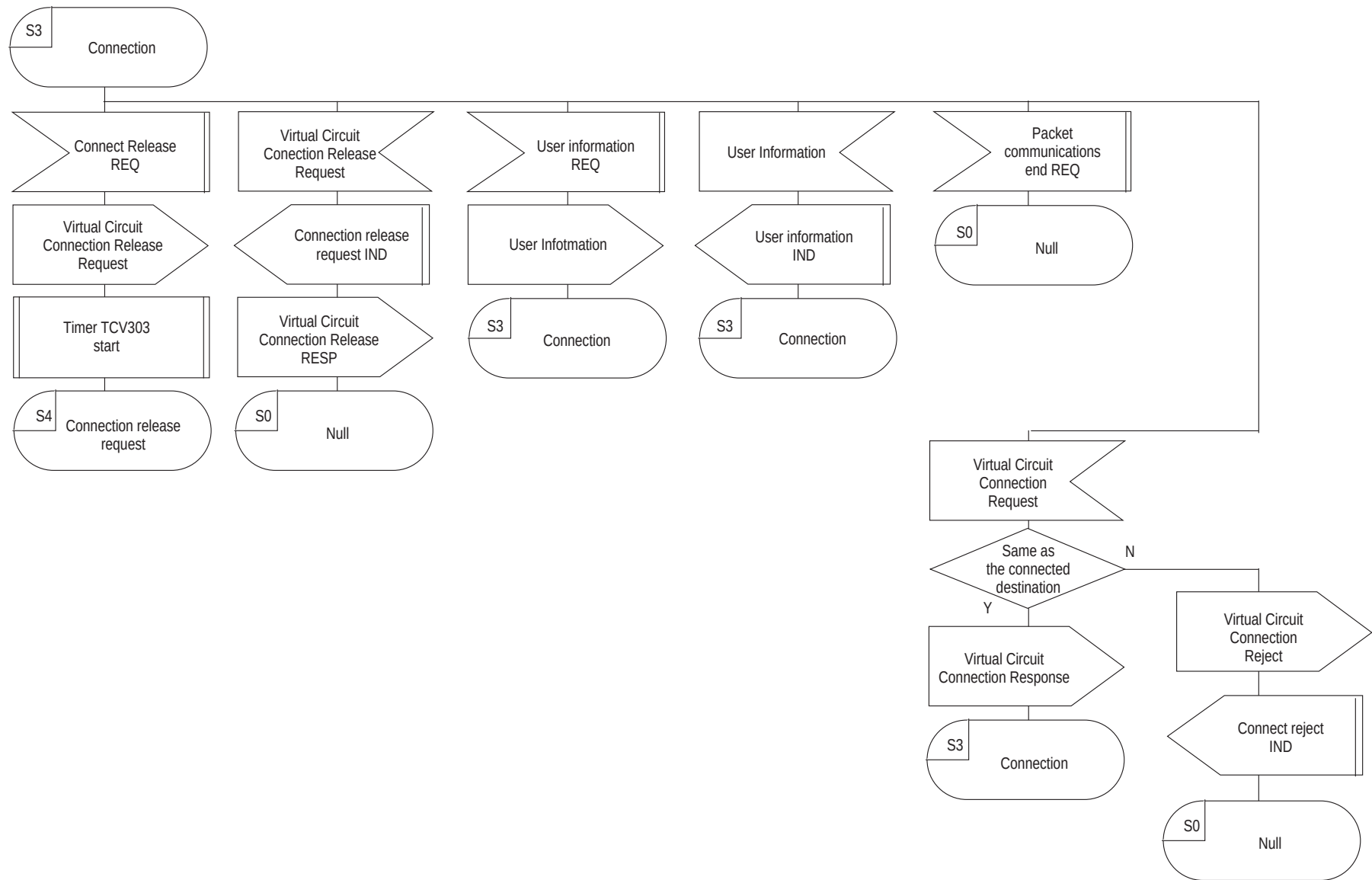


Fig. A. 5-156 Layer 3 CC (SVC function) SDL diagram (User side) (4/5)

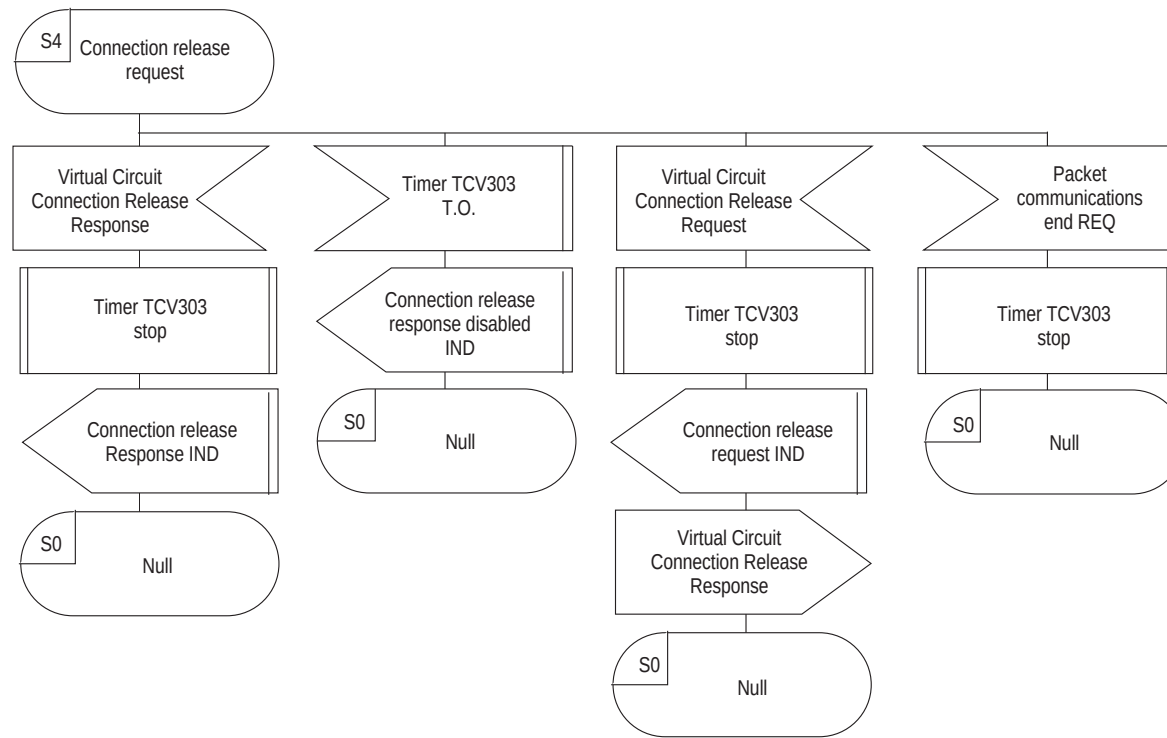


Fig. A. 5-157 Layer 3 CC (SVC function) SDL diagram (User side) (5/5)

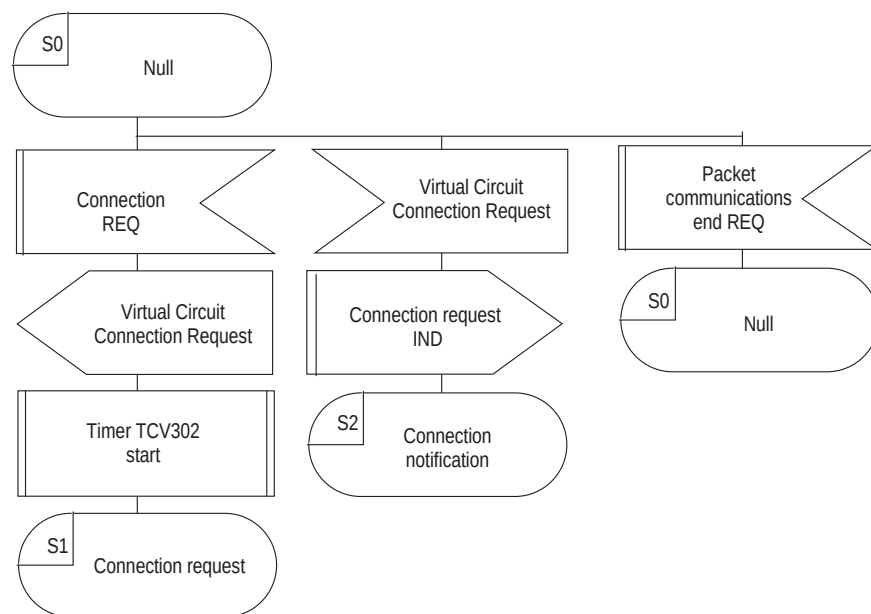


Fig. A. 5-158 Layer 3 CC (SVC function) SDL diagram (Network side) (1/5)

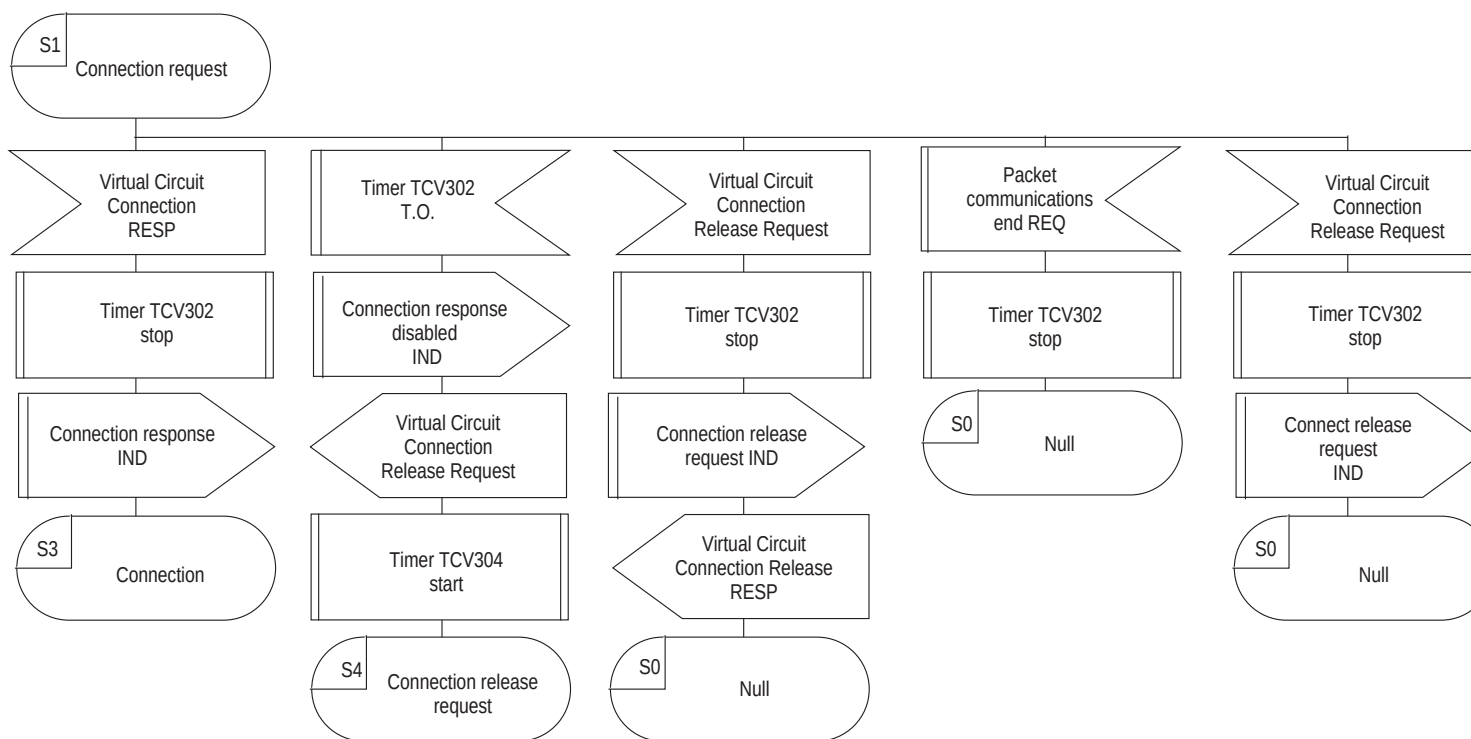


Fig. A. 5-159 Layer 3 CC (SVC function) SDL diagram (Network side) (2/5)

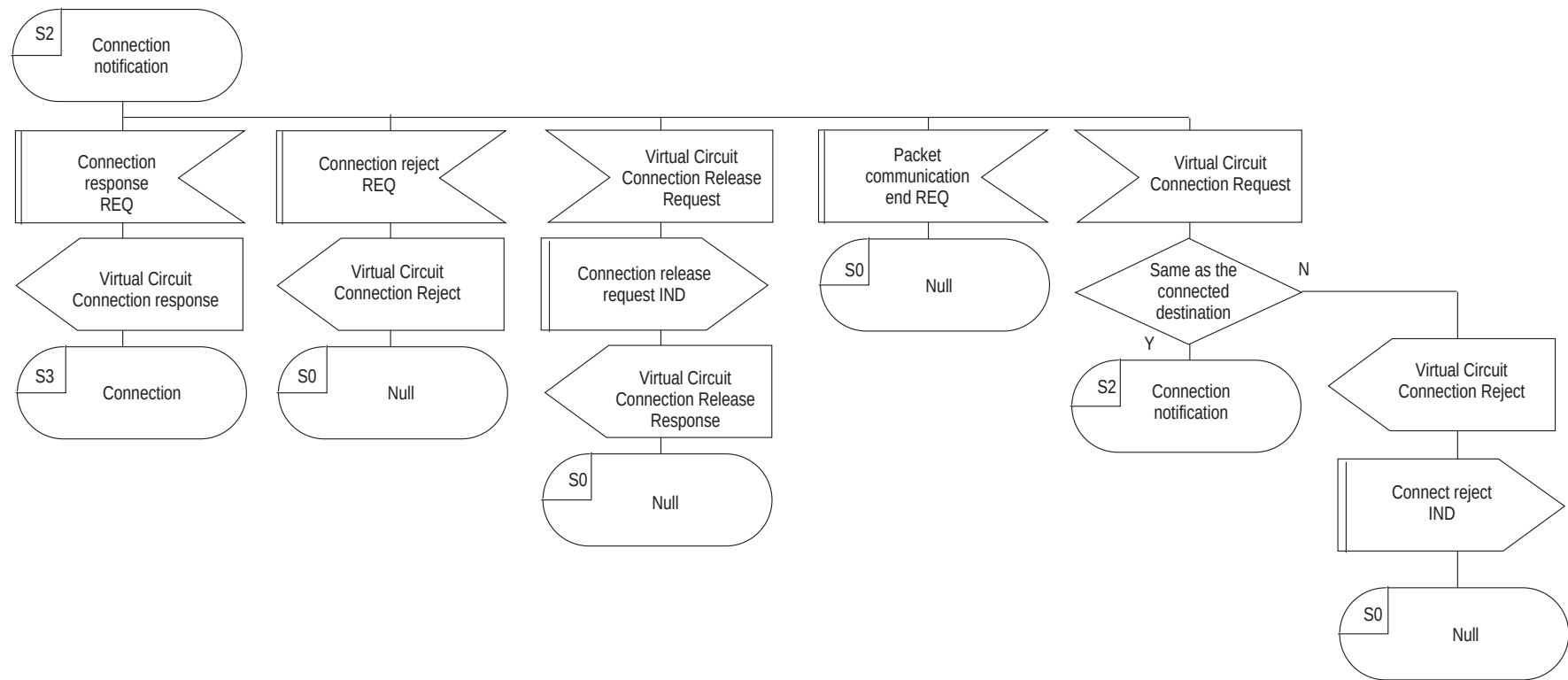


Fig. A. 5-160 Layer 3 CC (SVC function) SDL diagram (Network side) (3/5)



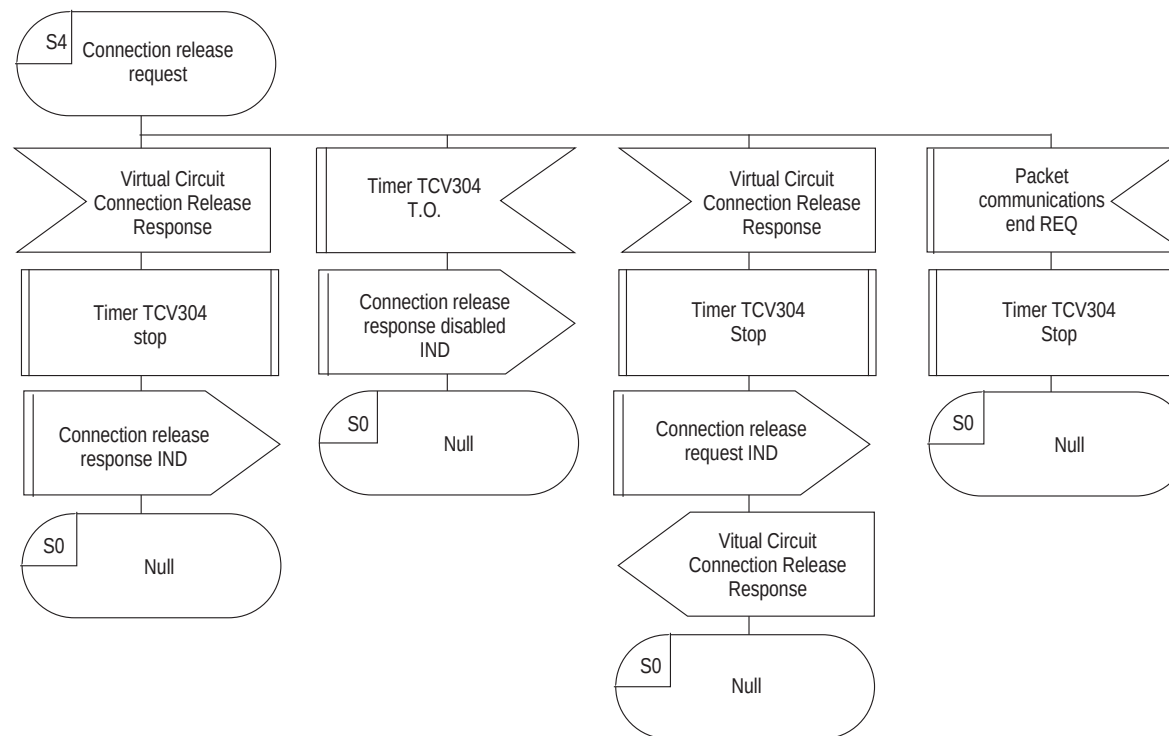


Fig. A. 5-162 Layer 3 CC (SVC function) SDL diagram (Network side) (5/5)

A.6 State transition table (network side)

A.6.1 Description method of supplementary service state transition table for network side

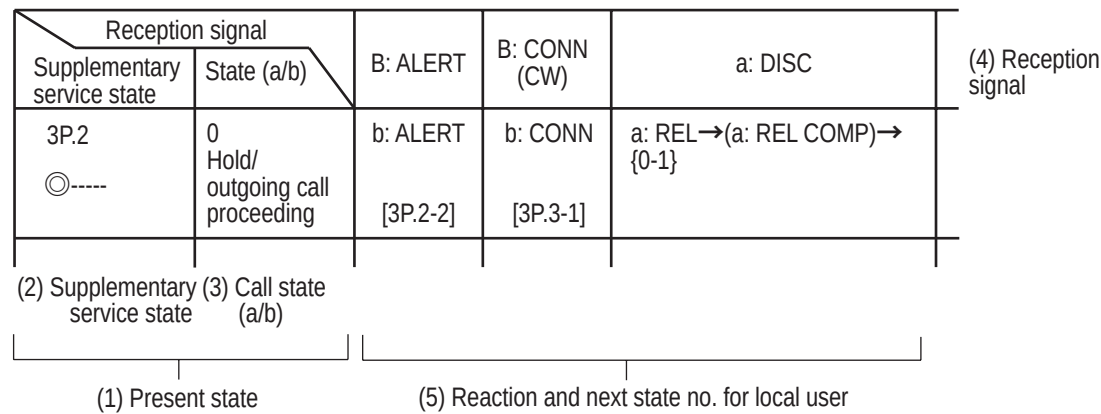


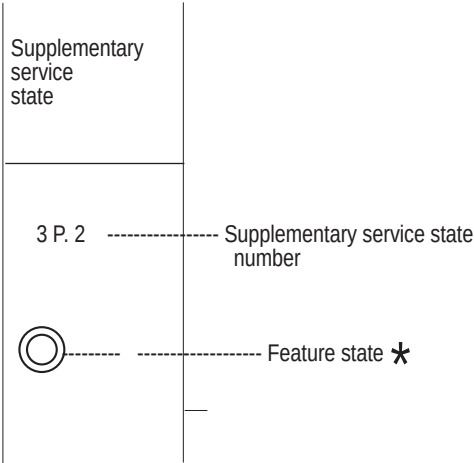
Fig. A.6-1: Format of supplementary service state transition table (network side)

(1) Present state

The present state is a combination of supplementary state and call condition.

(2) Supplementary service state

The supplementary service states are classified by the feature states. The upper half of the following figure is the supplementary service number while the lower half is the corresponding feature state.



* Feature state

The feature state indication is described based on the supposition that the terminal has the lamp corresponding to the feature identification number and transition of the state of the corresponding lamp is reported by the feature indication information element. (Refer to Table A.6-1.)

Table A.6-1 : Feature state indication

Symbol	Feature state
●	Non-active state (lamp off)
○	Active state (lamp on)
*	Prompt state (lamp flashes rapidly)
◎	Execution state (lamp flashes slowly)

Serial numbers are assigned to each function as supplementary service state number. Abbreviations of function names are listed in Table A.6-2.

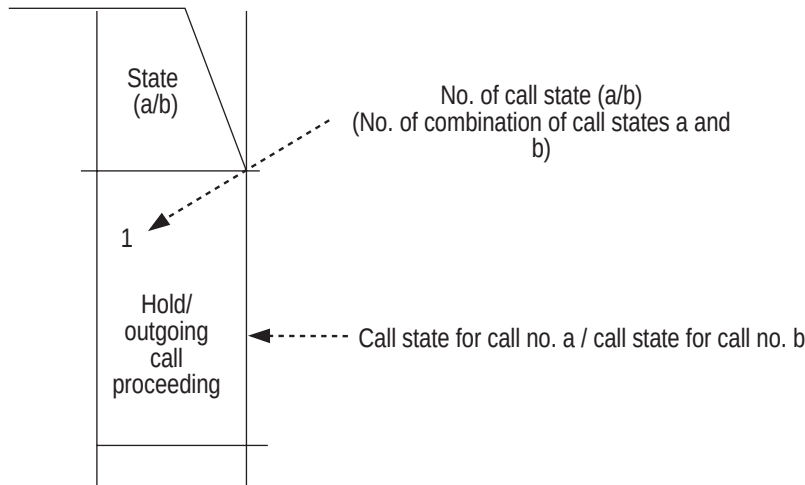
Table A.6-2 : Abbreviation and function name

Abbreviation	Function name
No abbreviations used for service numbers (numbers only)	Basic call control
CW	Call waiting
3P	3-party call (switching mode)

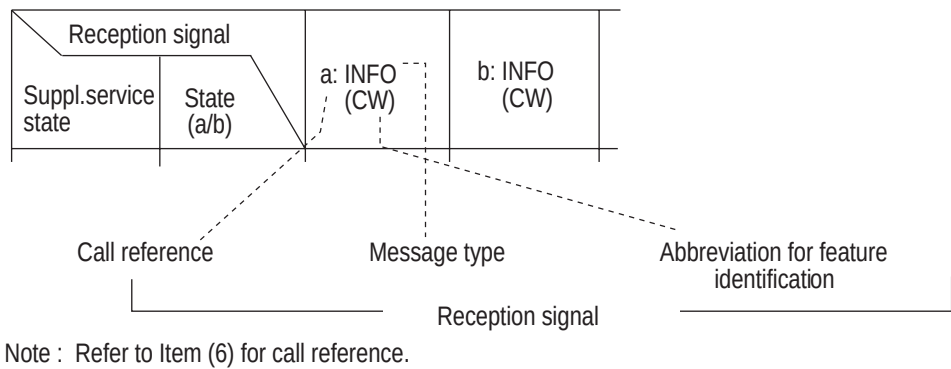
Feature states are as listed below from the left to right.

- 3-party call (switching mode)
- Call waiting

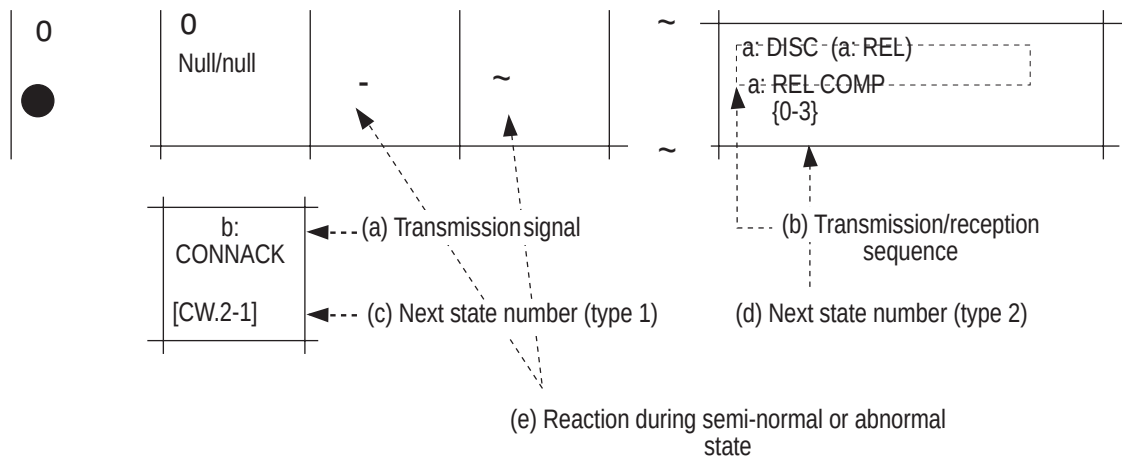
(3) Call state (a/b)



(4) Reception signal



(5) Reactions and next state number for the local user



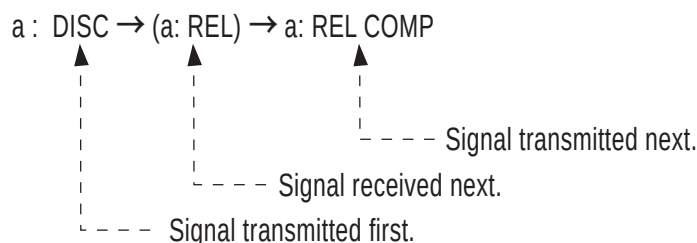
In reference to the above, the reaction (transmission signal) to the local user corresponding to the reception signal in each state and succeeding state number is indicated as follows :

a. Transmission signal

$\begin{array}{ccc} \underline{b} & : & \underline{\text{CONN ACK}} \\ \text{Call reference} & & \text{Message type} \end{array}$

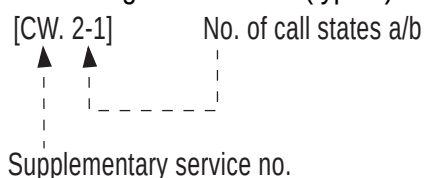
Refer to Item (6) for call reference.

b. Transmission/reception sequence



→ indicates the succeeding state; the item in "()" is a reception signal.

c. Succeeding state number (type 1)



The state number in "{ }" in the diagrams indicates that call states (a/b) should be reversed.

Example: When call state (a/b) "3" is "call present/null", it should be read as "null/call present".

d. Reaction during semi-normal and abnormal states

- ~ : Accord with the error procedure or semi-normal procedure in the RCR standard.
- : As not permitted by the supplementary service procedure, reception signal is discarded.

(6) Call reference

- a : Call reference for local user
- b : Call reference for local user
- A : Call reference for other party of call a
- B : Call reference for other party of call b

A.6.2 State transition tables for 3-party call

Table A.6-3 : State transition table for 3-Party call function (network-side) (1/4)

Reception signal		a: INFO (3P)	b: SETUP	b: ALERT	b: CONN	a: DISC	A: DISC
Supplementary service state	State (a/b)						
0 	0 Null/null	~	b: CALL PROC {0-1}	{0-3}	b: CONN ACK {0-4}	~	~
	1 Outgoing call proceeding/ Null	~	~	[CW.0-2]	~	a: REL→ (a: REL COMP) → [0-0]	a: DISC→(a: REL) →a: REL COMP [0-0]
	2 Call delivered/ Null	~	~	[CW.0-3]	~	a: REL→ (a: REL COMP) → [0-0]	a: DISC→(a: REL) →a: REL COMP [0-0]
	3 Call received/ Null	~	~	[CW.0-4]	b: CONN ACK {CW:0-1}	a: REL→ (a: REL COMP) → [0-0]	a: DISC→(a: REL) →a: REL COMP [0-0]
	4 Active/ Null	a: INFO (DT) [3P.1-1]	~	[CW.0-1]	~	a: REL→ (a: REL COMP) → [0-0]	a: DISC→(a: REL) →a: REL COMP [0-0]
	5 Disconnect IND/ Disconnect/ IND	~	~	~	~	~	~

Table A.6-4 : State transition table for 3-Party call function (network-side) (2/4) Part I

Reception signal		a: INFO (3P)	b: SETUP (3P)	b: SETUP	b: ALERT	b: CONN	B: ALERT	B: CONN
Supplementary service state	State (a/b)							
3P.1 -----*●-----	1 Hold/Null	a: INFO [0-4]	~	b: CALL PROC [3P.2-1]	~	~	~	~

3P

AO state

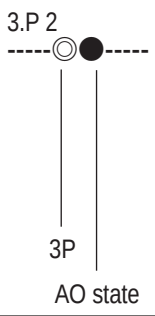
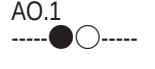
Table A.6-4 : State transition table for 3-Party call function (network-side) (2/4) Part 2

Reception signal		a: DISC	A: DISC	b: DISC	B: DISC
Supplementary service state	State (a/b)				
3P.1 -----*●-----	0 Hold/Null	a: REL→(a: REL COMP) → [0-0]	a: DISC→ (a: REL)→ a: REL COMP [0-0]	~	~

Table A.6-5 : State transition table for 3-party call function (network-side) (3/4)

Reception signal		a: INFO (3P)	a: INFO (AO)	b: INFO (3P)	b: INFO (AO)	b: SETUP	b: ALERT	b: CONN
Supplementary service state	State (a/b)							
3P.2 -----○●----- 3P AO state	1 Hold/Outgoing call proceeding	—	~	~	~	~	~	~
	2 Hold/Call delivered	—	~	~	~	~	~	~
3P.3 -----○●-----	1 Hold/Active	{3P.3-1}	a: INFO [AO. 1-1]	—	—	~	~	~
AO.1 -----●○-----	1 Hold/Active	a: INFO [3P.3-1]	—	—	—	~	~	~

Table A.6-6 : State transition table for 3-party call function (network-side) (4/4)

Reception signal		B: ALERT	B: CONN	a: DISC	A: DISC	b: DISC	B: DISC
Supplementary service state	State (a/b)						
3.P 2 	1 Hold/outgoing call proceeding	b: ALERT [3P.2-2]	b: CONN [3P.3-1]	a: REL→ (a: REL COMP) →{0-1}	a: DISC→ (a: REL)→ a: REL COMP {0-1}	b: REL→(b: REL COMP)→ a: INFO(CWT) [3P.1-1]	b: DISC→(b: REL) →b: REL COMP a: INFO (CWT) [3P.1-1]
	2 Hold/outgoing call proceeding	~	b: CONN [3P.3-1]	a: REL→ (a: REL COMP) →{0-2}	a: DISC→ (a: REL)→ a: REL COMP {0-2}	b: REL→(b: REL COMP)→ a: INFO(CWT) [3P.1-1]	b: DISC→(b: REL) →b: REL COMP a: INFO (CWT) [3P.1-1]
3P.3 	1 Hold/Active	~	~	a: REL→ (a: REL COMP) →{0-4}	a: DISC→ (a: REL)→ a: REL COMP {0-4}	b: REL→(b: REL COMP)→ a: INFO(CWT) [3P.1-1]	b: DISC→(b: REL) →b: REL COMP a: INFO (CWT) [3P.1-1]
AO.1 	1 Hold/Active	~	~	a: REL→ (a: REL COMP) →{0-4}	a: DISC→ (a: REL)→ a: REL COMP {0-4}	b: REL→(b: REL COMP)→ [0-4]	b: DISC→(b: REL) →b: REL COMP [0-4]

A.6.3 State transition tables for call waiting

Table A.6-7 : State transition table for CW function (network-side) (1/4)

Reception signal		a: INFO (CW)	a: INFO (CW)	b: SETUP	b: ALERT	b: CONN	a: DISC	A: DISC
Supplementary service state	State (a/b)							
0 	0 Null/null	~	~	b: CALL PROC {0-1}	{0-3}	b: CONN ACK {0-4}	~	~
	1 Outgoing call proceeding/ Null	~	~	~	[CW.0-2]	~	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]
	2 Call delivered/Null	—	~	~	[CW.0-3]	~	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]
	3 Call received/Null	—	~	~	[CW.0-4]	b: CONN ACK {w:0-1}	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]
	4 Active/Null	—	~	~	[CW.0-1]	~	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]
	5 Disconnect IND/ Disconnec IND	—	~	~	~	~	~	~

Table A.6-8 : State transition table for CW function (network-side) (2/4)

Reception signal		a: INFO (CW)	b: INFO (CW)	b: SETUP	b: ALERT	b: CONN
Supplementary service state	State (a/b)					
CW. 0 ●-----	1 Active/Call received	—	—	~	~	CONN ACK b: INFO [CW.2-1]
	2 Outgoing call proceeding/ Call received	—	—	~	~	~
	3 Call delivered/ Call received	—	—	~	~	~
	4 Call received/ Call received	—	—	~	~	~

Table A.6-9 : State transition table for CW function (network-side) (3/4)

Reception signal		B: ALERT	B: CONN	a: DISC	A: DISC	b: DISC	B: DISC
Supplementary service state	State (a/b)						
CW. 0 ●-----	1 Active/ Call received	—	—	a: REL→ (a: REL COMP)→ b: INFO(RGT) {0-3}	a: DISC→ (a: REL)→ a: REL COMP b: INFO(RGT) {0-3}	b: REL→ (b: REL COMP)→ → [0-4]	b: DISC→ (b: REL)→ b: REL COMP [0-4]
	2 Outgoing call proceeding/ Call received	—	—	a: REL→ (a: REL COMP)→ b: INFO(RGT) {0-3}	a: DISC→ (a: REL)→ a: REL COMP b: INFO(RGT) {0-3}	b: REL→ (b: REL COMP)→ → [0-1]	b: DISC→ (b: REL)→ b: REL COMP [0-1]
	3 Call delivered/ Call received	—	—	a: REL→ (a: REL COMP)→ b: INFO(RGT) {0-3}	a: DISC→ (a: REL)→ a: REL COMP b: INFO(RGT) {0-3}	b: REL→ (b: REL COMP)→ → [0-2]	b: DISC→ (b: REL)→ b: REL COMP [0-2]
	4 Call received/ Call received	—	—	a: REL→ (a: REL COMP)→ b: INFO(RGT) {0-3}	a: DISC→ (a: REL)→ a: REL COMP b: INFO(RGT) {0-3}	b: REL→ (b: REL COMP)→ → [0-3]	b: DISC→ (b: REL)→ b: REL COMP [0-3]

Fig. A.6-10 : State transition table for call waiting (network side) (4/4) Part 1

Reception signal		a: INFO (CW)	b: INFO (CW)	b: SETUP	b: ALERT	b: CONN	B: ALERT	B: CONN
Supplementary service state	State (a/b)							
CW. 1 *-----	1 Hold/Null	a: INFO [0-4]	~	~	—	—	~	~
CW. 2 ○-----	2 Hold/Active	{CW.2-1}	{CW.2-1}	~	~	~	~	~

Fig. A.6-10 : State transition table for call waiting (network side) (4/4) Part 2

Reception signal		a: DISC	A: DISC	b: DISC	B: DISC
Supplementary service state	State (a/b)				
CW. 1 *-----	1 Hold/Null	a: REL→ (a: REL COMP) → [0-0]	a: DISC→ (a: REL)→ a: REL COMP [0-0]	~	~
CW. 2 ○-----	2 Hold/Active	a: REL→ (a: REL COMP) → {0-4}	a: DISC→ (a: REL)→ a: REL COMP {0-4}	b: REL→(b: REL COMP)→ a: INFO(CWT) [CW.1-1]	b: DISC→(b: REL) →b: REL COMP a: INFO(CWT) [CW.1-1]

A.6.4 State transition tables for call transfer function

Table A.6-11 : State transition table of CW, 3P, CT functions (network-side) (1/3) Part 1

Reception signal		a: INFO (3P)	a: INFO (CW)	a: INFO (AO)	a: INFO (CT)	b: INFO (3P)	b: INFO (CW)	b: INFO (AO)	b: INFO (CT)	b: SETUP	b: ALERT	b: CONN
Supplementary service state	State (a/b)											
0 CT AO 3P CW state	0 Null/null	~	~	~	~	~	~	~	~	b: CALL PROC {0-1}	{0-3}	b: CONN ACK {0-4}
	1 Outgoing call proceeding/ Null	—	—	—	—	~	~	~	~	~	{CW.0-2}	~
	2 Call delivered/ Null	—	—	—	—	~	~	~	~	~	[CW.0-3]	~
	3 Call received/ Null	—	—	—	—	~	~	~	~	~	[CW.0-4]	b: CONN ACK {CW.0-1}
	4 Active/ Null	a: INFO (DT) [3P.1-1]	—	—	a: INFO (DT) [CT.1-1]	~	~	~	~	~	[CW.0-1]	~
	5 Disconnect IND/	—	—	—	—	—	—	—	—	~	~	~

Table A.6-12 : State transition table of CW, 3-party and CT functions (network side) (1/3) Part 2

Reception signal		B: ALERT	B: CONN	a: DISC	A: DISC	b: DISC	B: DISC
Supplementary service state	State (a/b)						
0 ●●●●●	0 Null/Null	~	~	~	~	~	~
	1 Outgoing proceeding/ Null	~	~	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]	~	~
	2 Call delivered/ Null	~	~	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]	~	~
	3 Call received/ Null	~	~	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]	~	~
	4 Active/Null	~	~	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]	~	~
	5 Disconnect IND/Discon- nect IND	~	~	~	~	~	~

Table A.6-13 : State transition table for call transfer function (network side) (2/3)

Reception signal		a: INFO (CT)	a: INFO (CT)	b: INFO	b: INFO	b: SETUP	B: ALERT	B: CONN
Supplementary service state	State (a/b)							
CT. 1 -----*	1 Hold/Null	a: INFO [0-4]	~	b: CALL PROC [CT.2-1]	~	~	~	~
	2 Hold/ Call delivered	a: DISC b: DISC [0-5]	—	~	~	~	~	b: CONN [CT.3-1]
CT. 2 -----◎	1 Hold/Out- going call proceed-ing	—	~	~	~	~	b: ALERT [CT.1-2]	b: CONN [CT.3-1]
CT. 3 -----*	1 Hold/ Active	a: DISC b: DISC [0-5]	a: DISC b: DISC [0-5]	~	~	~	~	~

Table A.6-14 : State transition table of call transfer function (network side) (3/3)

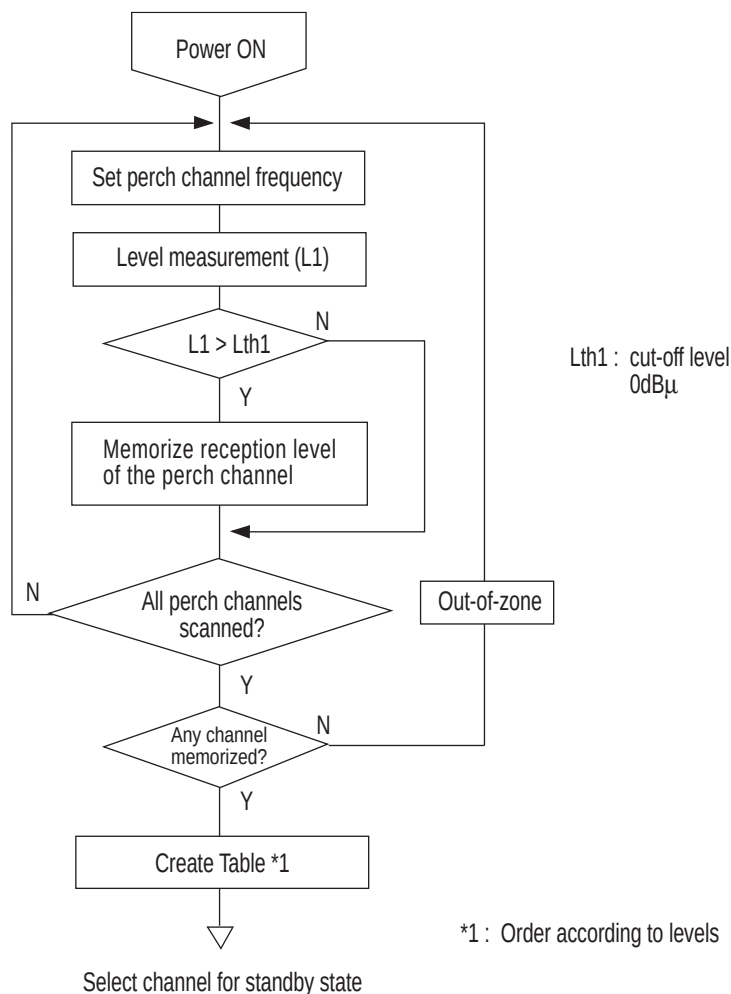
Reception signal		a: DISC	A: DISC	b: DISC	B: DISC
Supplementary service state	State (a/b)				
CT. 1 ----- *	1 Hold/Null	a: REL → (a: REL COMP) → [0-0]	a: DISC → (a: REL) → a: REL COMP [0-0]	~	~
	2 Hold/Call delivered	a: REL → (a: REL COMP) → [0-2]	a: DISC → (a: REL) → a: REL COMP [0-2]	b: REL → (b: REL COMP) → a: INFO(CWT) [CT.1-1]	b: DISC → (b: REL) → b: REL COMP a: INFO(CWT) [CT.1-1]
CT. 2 ----- ◎	1 Hold/Outgoing call proceeding	a: REL → (a: REL COMP) → [0-1]	a: DISC → (a: REL) → a: REL COMP [0-1]	b: REL → (b: REL COMP) → a: INFO(CWT) [CT.1-1]	b: DISC → (b: REL) → b: REL COMP a: INFO(CWT) [CT.1-1]
CT. 3 ----- *	1 Hold/Active	a: REL → (a: REL COMP) → {0-4}	a: DISC → (a: REL) → a: REL COMP {0-4}	b: REL → (b: REL COMP) → a: INFO(CWT) [CT.1-1]	b: DISC → (b: REL) → b: REL COMP a: INFO(CWT) [CT.1-1]

Appendix B Operation of the Mobile Station When Switching to the Standby State

This appendix outlines operational procedures for the mobile station from the time when the power is turned on until it switches to the standby state.

B.1 Perch channel scanning

After the power for the mobile station is turned on, the mobile station begins scanning the perch channels written in the ID-ROM and creates a table that lists up the channels above the cut-off level in the order of their levels.



Note: Level detection specifications for perch channel scanning

- Detection algorithm is not specified.
- Detection accuracy must be such that the standard deviation of the level detection of a signal with 5Hz Rayleigh fading is less than or equal to 2.5 dB.

Fig. B.1 : MS perch channel scanning flow

B.2 Selection of the channel for the standby state

B.2.1 Outline of operation

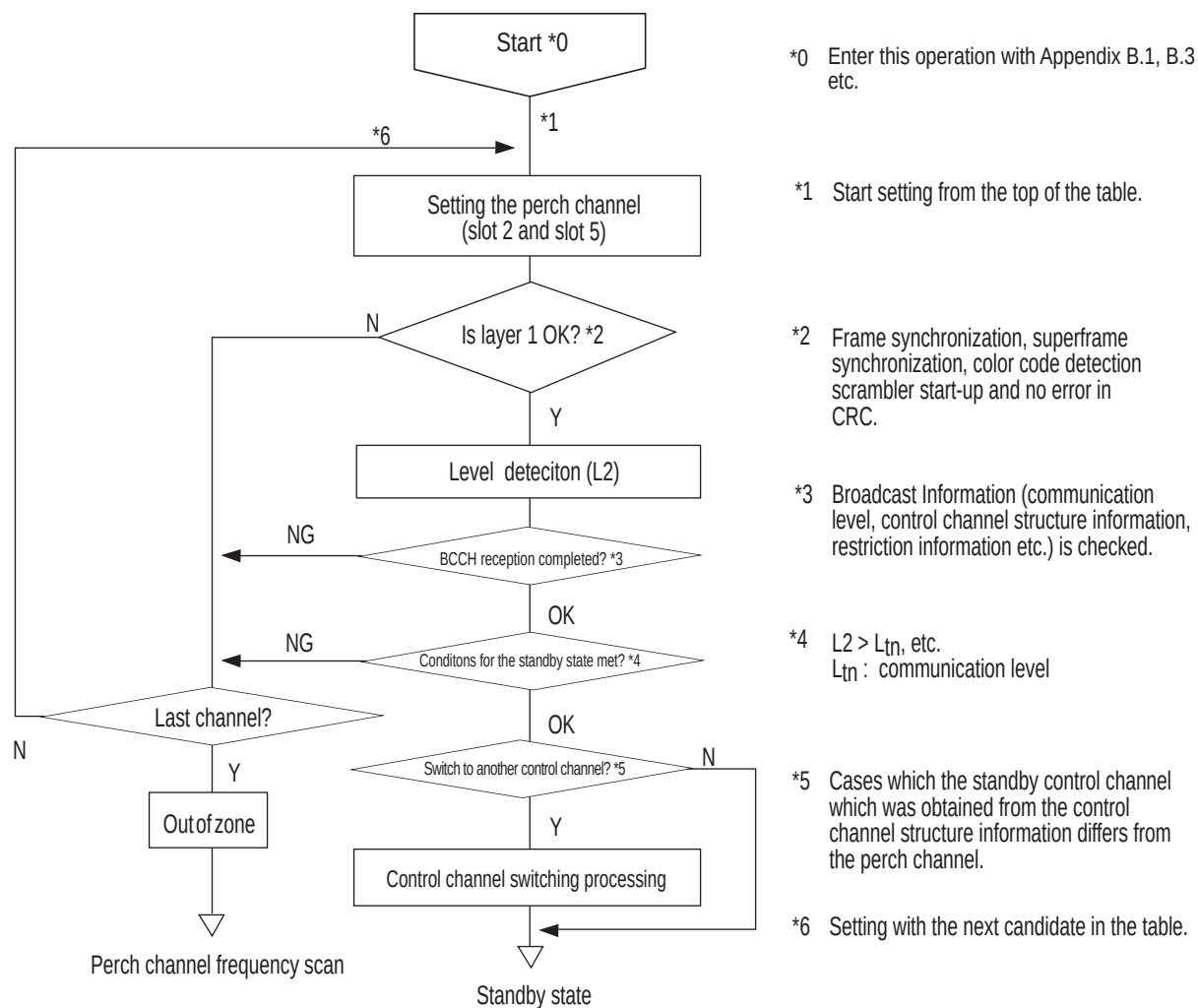


Fig. B.2 : Flow chart for selecting the channel to be used for the standby state.

- (1) Slot 2 and slot 5 of a perch frequency is set as a perch channel. MS should switch to the frequency of the perch channel at the top of the table listed in the order of signal level and establish synchronization (frame and superframe synchronization)
- (2) After synchronization has been established, this channel is used to detect the color code, and the scramble code to start up the scrambler is obtained from the color code.
- (3) The mobile station checks the condition for switching to the standby state. If all the conditions are met, the mobile station is switched to the standby state on this channel. However, in cases which more than one control channel exist within the same zone, the mobile station needs to be switched from the perch channel to another control channel corresponding to the mobile station group.

- (4) When the above conditions are not met, the next frequency in the table is checked by the same procedure.

If none of the channels in the table could be used, the mobile station is rated as in the "out-of-range" zone. In this case, the mobile station starts scanning the perch channels.

B.2.2 Conditions for the standby state

(1) Reception level detection

The mobile station detects the reception level of the home slot. When the reception level is below the communication level broadcast by the BCCH, the channel is rejected.

Detection accuracy must be such that the standard deviation of the level detection of a signal with 5Hz Rayleigh fading is less than or equal to 2.5 dB.

(2) BCCH reception

The mobile station receives Broadcast Information in the BCCH and analyzes the following information to determine whether it is available channel.

i) Location registration information

The number of multiplexed areas and location registration number are broadcast as information for location registration.

ii) Information to specify the channel structure

The control channel structure information is broadcast, enabling the mobile station to recognize the position of the BCCH, PCH and the SCCH. The intermittent reception timing, which is derived from the position of the home group PCH for the station, the number of control channel within the zone, the channel number and so forth are known.

The common access channel (CAC) normally meets the relationship indicated by the following equation:

$$A_B + A_{S1} + (A_P + A_{S2}) * N_P \leq 36$$

The rest in the channel are SCCH. If the relationship indicated by the following equation occurs, the pertinent common access channel is discarded.

$$A_B + A_{S1} + (A_P + A_{S2}) * N_P > 36$$

iii) Information for peripheral zone supervision

The perch channels for peripheral zone supervision in the standby state are recognized.

iv) Information on the home zone

The maximum transmission power, communication level, communication-out level and so forth for the home zone are recognized.

v) Information on various restriction

Recognizes restriction information and displays it to the user, or initiates the control to reject the channel etc. Refer to Appendix E.

vi) Operation of an MS supporting the packet communication function

The MS gets information in the zone by receiving Packet channel structure information, Terminal registration area code, Packet paging area code and zone service information. In case that the zone does not provide packet service, the MS can select a zone with a channel, the reception level of which is larger than (the top channel reception level of the order table of the zone - packet standby permitted level difference).

B.2.3 Switching to home group control channel

For a zone where traffic is concentrated, multiple control channels are used. The control channel to be used is determined by the respective mobile station group. If necessary, the perch channel is switched to the specified control channel according to the control channel structure information contained within the Broadcast Information.

The following information is broadcast as the control channel structure information:

$F_1(S) \sim F_{N_c}(S)$: Frequency number (slot number) for the control channels in the pertinent zone.
 N_c : No. of control channels in the pertinent zone.
 N_p : No. of PCH grouping per control channel.

Where the lower two octets for Mobile Station Number in decimal expression is represented by "n", the control channel frequency number is represented by " I_F " and the number for PCH group is represented by " I_P ". The control channel where the mobile station should standby is obtained using the following formula.

$$\begin{aligned} n' &= (n-1) \bmod (N_c \times N_p) + 1 \\ I_F &= (n'-1) \bmod N_c + 1 \\ I_P &= [(n'-1)/N_c] + 1 \end{aligned}$$

Note: 1) $n-1=2^{16}-1$ shall be set if $n=0$

Note: 2) $[X]$ represents the maximum integer which does not exceed X.

In the following example, $N_c = 3$ and $N_p = 3$. Mobile stations are classified into 9 groups, represented as " $N_c \times N_p = 9$."

Assuming that Mobile Station Identifier $n=9$, the following result is obtained from the above equation:

$$\begin{aligned} n' &= 9 \\ I_F &= 3 \\ I_P &= 3 \end{aligned}$$

The control channel to be used is $F_3(S)$ and the home PCH groups for this control channel is "3".

The common access channel structure is shown in Fig. B.3.

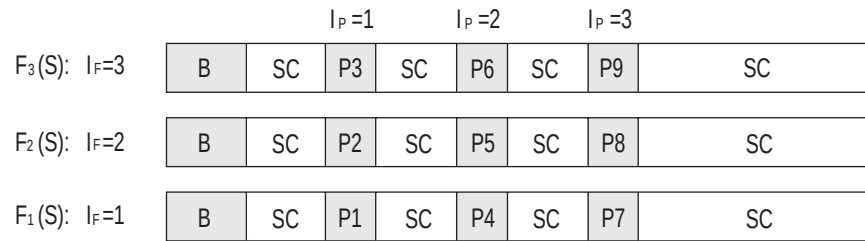


Fig. B.3 : Common access channel structure

B.3 Standby state

B.3.1 Outline

The outline of standby operation is shown in Fig. B.4.

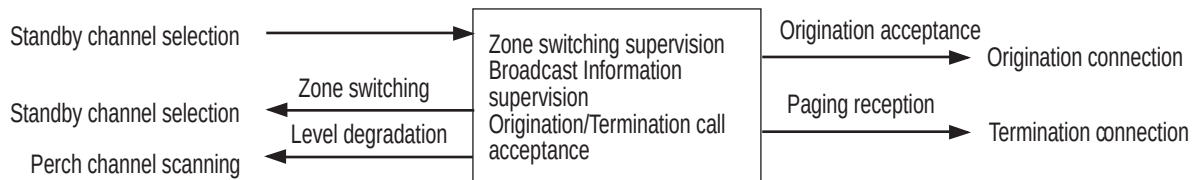


Fig. B.4 : Standby operation

- (1) The mobile station detects the reception levels of the home zone and the peripheral zones and determines the occurrence of the condition for zone switching and level degradation.
- (2) When the mobile station receives information from the BCCH and the Broadcast Information has been changed, it performs the appropriate processing.
- (3) When paging reception on the PCH or origination acceptance occurs, the mobile station switches to origination/termination connection control.
- (4) Detection precession for (1) through (3) above must be such that the standard deviation of the level detection of a signal with 5Hz Rayleigh fading is less than or equal to 2.5 dB.

B.3.2 Intermittent reception

An example of intermittent reception control for receiving PCH and BCCH in the home group is described below.

If the number of slots for respective PCH and BCCH in the home group are represented as A_p and A_b , the ratio of power ON time when the power for the mobile station is turned on is given as :

$$(A_p + A_b) \times 20/720$$

To detect the reception level using empty slots, the power must be turned ON not only for the home station slot but for one subframe. The ratio of time that the power is ON shown in the following example ($A_p = 1$, $A_b = 2$) is 0.083.

The value of A_p and A_b is determined in relation to the respective region. The ratio of time that the power is ON for the mobile station above can be obtained from the structure of the control channel. In reality, the average time ratio that the power is ON needs to be obtained in relation to the hardware start-up time and the frequency of zone switching.

The BCCH may be turned on intermittently, rather than continuously to further improve the power on time ratio. (Refer to Appendix G.)

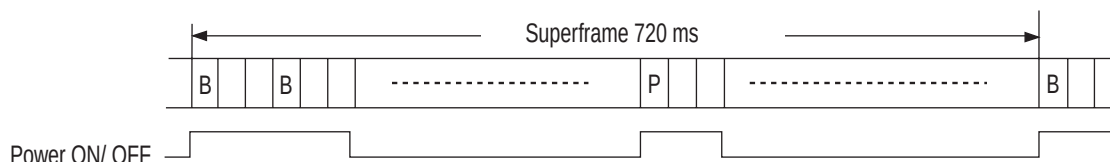


Fig. B.5 : Power ON/OFF timing during intermittent reception

B.3.3 Home station slot and peripheral zone level detection

The reception level for the home station slot and the peripheral zone is detected during intermittent reception. Home station slot level detection is performed using the slots for BCCH and the home station's PCH. The reception level for the peripheral zone is detected with the empty slots associated to the BCCH and the home station's PCH.

Peripheral zone level detection is performed for the perch channel frequencies lying in the peripheral zone, which are broadcast by the BCCH, using empty slots in the order of $F_1, F_2, \dots, F_n$ (n is maximum 20). Level detection for F_1 to F_n is performed several times to obtain the average.

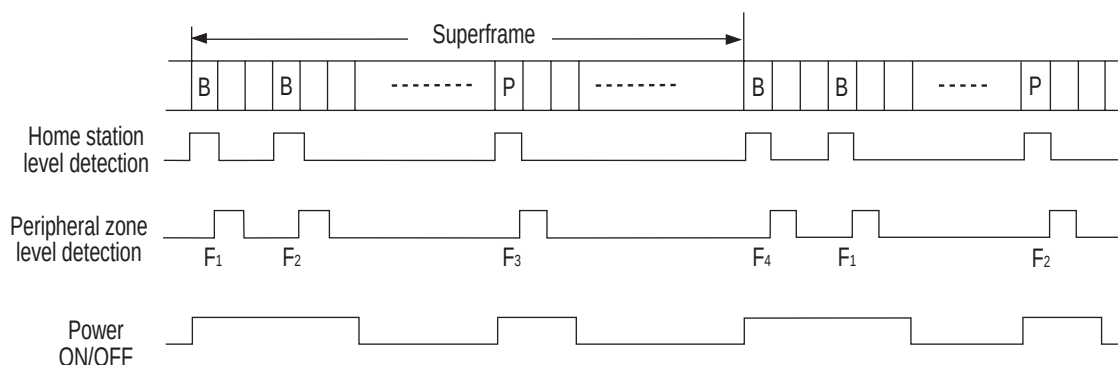


Fig . B.6 : Level detection during intermittent reception

Conditions for zone switching are as follows:

Home station slot reception level: L_0

Peripheral zone reception level: $L_1, \dots, L_n$, and the maximum is $\max(L_i)$

Communication level: L_{tn}

Communication deteriorated level: L_{th}

Zone switching level difference: ΔL shall be $(L_{tn} - L_{th})$ or greater

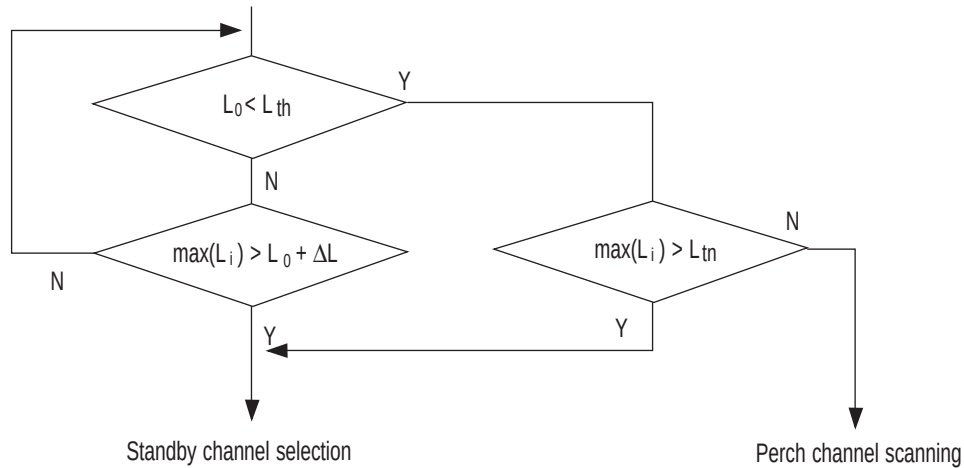


Fig. B.7 : Zone switching flow chart

B.3.4 Reception of broadcast information

The mobile station first receives the information from the BCCH. If the information received differs from the broadcast information memorized through the standby channel selecting operation, the mobile station performs proper controls. Refer to Appendix G for details on the reception timing.

B.4 Roaming control method

Regarding the roaming control method by the MS, the following 2 points are regulated:

- (1) The requirement for which the MS sends a "Location Registration Request" to the network after comparing the network identity in the Broadcast Information notified by the network with the roaming information memorized in the non-volatile memory in the MS.
- (2) The operation by the MS when the MS receives "Location Registration Acknowledgment" or "Location Registration Reject" (roaming not allowed) in response to the "Location Registration Request" sent by the MS.

Concerning the 4 modes (roaming OFF mode, home operator group network selection mode, roaming operator group network selection mode and auto mode), the above operations (1) and (2) are described in the 2 cases: the respective operation of the MS on turning the power ON and on activating these modes manually by the user.

B.4.1 Roaming information

The following roaming information are memorized in a non-volatile memory of the MS each time a subscription contract is signed.

(1) Home operator group network information

- Information which identifies the country code of the home operator group
- Information which identifies the operator group code of the home operator group
- Information which identifies the network identity in a group for the home local network
- Information which allows or does not allow location registration on the network (for each local network) in each group
[Roaming allowed/not allowed information]
- Information which controls whether or not the MS updates roaming information on reception of the "roaming not allowed information"
[Roaming information update/not update information]
- Information which identifies whether or not the MS operates based on the "roaming allowed/not allowed information" on turning the power ON
[Roaming information dependency ON/OFF information]

(2) Roaming operator group network information (multiple information can be registered)

- Information which identifies the country code of the roaming operator group
- Information which identifies the operator group code of the roaming operator group
- Information which allows or does not allow location registration operation on the network (for each local network) in each group
[Roaming allowed/not allowed information]
- Information which controls whether or not it updates roaming information on reception of the "roaming not allowed information"
[Roaming information update/not update information]
- Information which identifies whether or not the MS is to operate based on the "roaming allowed/not allowed information" on turning the power ON
[Roaming information dependency ON/OFF information]

B.4.2 Network selection operation by the MS

The MS is to operate in any of the following 4 modes. The roaming OFF mode is a mandatory function; other modes are optional functions.

(a) Roaming OFF mode

(1) Operation of the MS on turning the power ON

- The MS scans the perch channel on the home operator group network only and sets standby on the network based on the "roaming allowed/not allowed information" and the "information which identifies the network identity in the group for the home local network," regardless of setting of the "roaming information dependency ON/OFF information" on the home operator group network.
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

(2) Operation of the MS when this mode is selected by the user

- The MS scans the perch channel on the home operator group network only and sets standby on the network based on the "roaming allowed/not allowed information" and the "information which identifies the network identity in the group for the home local network," regardless of setting of the "roaming information dependency set/not set information" on the home operator group network.
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

(b) Home operator group network selection mode

(1) Operation of the MS on turning the power ON

[When the "roaming information dependency ON/OFF information" is "ON"]

- The MS scans the perch channel on the home operator group network only and sets standby on the network based on the "roaming allowed/not allowed information."
- If the network identity in the group is specified, the MS scans the perch channel on the home operator group network only and sets standby only with the network identity in the specified group based on the "roaming allowed/not allowed information" on the home operator group network.
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

[When the "roaming information dependency ON/OFF information" is "OFF"]

- The MS scans the perch channel on the home operator group network only and sets standby, regardless of setting of the "roaming allowed/not allowed information" on the home operator group network.
- If the network identity in the group is specified, the MS scans the perch channel on the home operator group network only and sets standby only with the network identity in the specified group regardless of setting of the "roaming allowed/not allowed information" on the home operator group network.
- When the "Location Registration Acknowledge" is received by the network, for which the "roaming not allowed information" is set, in response to the "Location Registration Request", the MS updates the "not allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "allowed" only when the "roaming information update/not update information" is set for "update."
- When the "Location Registration Reject" (roaming not allowed) is received by the network for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

(2) Operation of the MS when this mode is selected by the user

- The MS scans the perch channel on the home operator group network only and sets standby, regardless of setting of the "roaming allowed/not allowed information" on the home operator group network.

- If the network identity in the group is specified, the MS scans the perch channel on the home operator group network only and sets standby only with the network identity in the specified group, regardless of setting of the "roaming allowed/not allowed information" on the home operator group network.
- When the "Location Registration Acknowledge" is received by the network, for which the "roaming not allowed information" is set, in response to the "Location Registration Request", the MS updates the "not allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "allowed" only when the "roaming information update/not update information" is set for "update."
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

(c) Roaming operator group network selection mode

(1) Operator of the MS on turning the power ON

[When the "roaming information dependency ON/OFF information" is "ON"]

- The MS scans the perch channel on the specified roaming operator group network only and sets standby on the specified roaming operator group network based on the "roaming allowed/not allowed information."
- If the network identity in the group is specified, the MS scans the perch channel on the specified roaming operator group network only and sets standby only with the network identity in the specified group based on the "roaming allowed/not allowed information" on the specified roaming operator group network.
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

[When the "roaming information dependency ON/OFF information" is "OFF"]

- The MS scans the perch channel on the specified roaming operator group network only and sets standby, regardless of setting of the "roaming allowed/not allowed information" on the specified roaming operator group network.
- If the network identity in the group is specified, the MS scans the perch channel on the specified roaming operator group network only and sets standby only with the network identity in the specified group regardless of setting of the "roaming allowed/not allowed information" on the specified roaming operator group network.
- When the "Location Registration Acknowledge" is received by the network, for which the "roaming not allowed information" is set, in response to the "Location Registration Request", the MS updates the "not allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "allowed" only when the "roaming information update/not update information" is set for "update."
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

(2) Operation of the MS when this mode is selected by the user

- The MS scans the perch channel on the specified roaming operator group network only and sets

standby, regardless of setting of the "roaming allowed/not allowed information" on the specified roaming operator group network.

- If the network identity in the group is specified, the MS scans the perch channel on the specified roaming operator group network only and sets standby only with the network identity in the specified group, regardless of setting of the "roaming allowed/not allowed information" on the specified roaming operator group network.
- When the "Location Registration Acknowledge" is received by the network, for which the "roaming not allowed information" is set, in response to the "Location Registration Request", the MS updates the "not allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "allowed" only when the "roaming information update/not update information" is set for "update."
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

(d) Auto mode

(1) Operation of the MS on turning the power ON

- The MS scans the perch channel on the home operator group network and sets standby on the network based on the "roaming allowed/not allowed information," regardless of setting of the "roaming information dependency ON/OFF information" on the home operator group network.
- When the MS is out of the home operator group network zone, it automatically switches to scanning of the perch channel on the roaming operator group network and sets standby on the network based on the "roaming allowed/not allowed information," regardless of setting of the "roaming information dependency ON/OFF information" on the roaming operator group network.
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update".

(2) Operation of the Ms when this mode is selected by the user

When this mode is selected by the user, the MS performs the same operation on turning the power ON or the following operation:

- The MS scans the perch channel on the home operator group network and sets standby on the network, regardless of setting of the "roaming allowed/not allowed information."
- When the MS is out of the home operator group network zone, it automatically switches to scanning of the perch channel on the roaming operator group network and sets standby, regardless of setting of the "roaming allowed/not allowed information."
- When the "Location Registration Acknowledge" received by the network, for which the "roaming not allowed information" is set, in response to the "Location Registration Request," the MS updates the "not allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "allowed" only when the "roaming information update/not update information" is set for "update."
- When the "Location Registration Reject" (roaming not allowed) is received by the network, for which the "roaming allowed information" is set, in response to the "Location Registration Request", the MS updates the "allowed" setting for the pertinent network identity in the "roaming allowed/not allowed information" to "not allowed" only when the "roaming information update/not update information" is set for "update."

In any of the above four modes, from (a) to (d), the mode can be changed, and the same mode, which includes cases when the specified operator, the network identity in the specified group, etc., are identical, can be reactivated by the user operation.

B.4.3 Control conditions for the auto mode

If an "out-of -zone" condition occurs due to communication out level, etc., while the MS is set standby on the home operator group network, 5 seconds later, the MS scans the perch channel on the home operator group network again. If the "out-of-zone" is registered again, the MS switches to scanning the perch channel for the roaming operator group network.

Also, the MS has a function for scanning the perch channel periodically on the home operator group network.

Appendix C Mobile Assisted Handover

This appendix describes operation of the mobile station for Mobile Assisted Hand Over (MAHO) during communication.

C.1 Outline

- (1) To detect the zone switching, the mobile station detects the reception level of the peripheral zone using empty slots. The information on the perch channel used for peripheral zone supervision is provided in the form of a Condition Report Information. When the mobile station receives the Condition Report Information, it transmits the Condition Report Information Acknowledgment and it simultaneously begins detecting the reception level.
- (2) The following information is contained in the Condition Report Information.
 - Frequency code (F1, F2,...,Fm: max. 20) of the perch channel for peripheral zone supervision. The perch channel number is reported. The mobile station has a table of perch channel numbers and corresponding frequency code.
 - Periodical report time interval (T2)
 - Time interval criteria (T1)
 - Report threshold level criteria (A)
 - Threshold for the amount of state change (B)
 - Level measurement time (Ts)
 - Maximum number of reporting channels. (N)
- (3) After the first reception level detection for the peripheral zone has been completed, the mobile station must always transmit Condition Report 1 to the network. Following this, the mobile station transmits Condition Report 1 at preset time interval T2. The Condition Report 1 message reports channels with the reception level of 0 dBμ or higher. The number of zone selection shall be the maximum number of reporting channel N specified by the Condition Report Information. However, if the number of channels with a reception level 0 dBμ or higher does not reach N, the zone selection number will be a value smaller than N. The network does not transmit an acknowledgment signal in response to Condition Report 1.
- (4) In addition to the periodical report, the mobile station compares the reception level for the home station (the value revised by the downlink set power report POW-D reported by the downlink RCH) and the maximum reception level for the peripheral zone. If the reception level for the peripheral zone is higher than that of the home station by A, the "event occurrence" state is assumed and the mobile station then transmits Condition Report 2. The Condition Report 2 reports channels with a reception level of 0 dBμ or higher. The number of zone selection shall be the maximum number of reporting channel N specified by the Condition Report Information. However, if the number of channels with a reception of 0 dBμ or higher does not reach N, the zone selection number will be a value smaller than N.
- (5) Condition Report 1 is a signal which the mobile station periodically transmits the condition reporting. When the quality of the uplink channel begins to deteriorate, the network may transmit a Condition Inquiry to the mobile station to request transmission of the condition report. If the mobile station receives the Condition Inquiry, it transmits Condition Report 1. The number of zone selection in Condition Report 1 shall be the same as in (3).
- (6) The report condition is informed by the Condition Report Information after communication starts.

However, this information may be changed during communication. In such instances, a Condition Report State is transmitted from the network and when the mobile station receives the Condition Report State, it transmits a Condition Report State Acknowledgment.

At the same time it also begins detecting the reception level for the peripheral zone under the new conditions. The condition report sequence is shown in Figs. C.1 and C.2.

Note: The network can include the perch channel of the home zone in F1, F2 ... Fm. Even in such instances, the mobile station shall perform the processing mentioned above without discriminating the perch channel of the home zone from other perch channels.

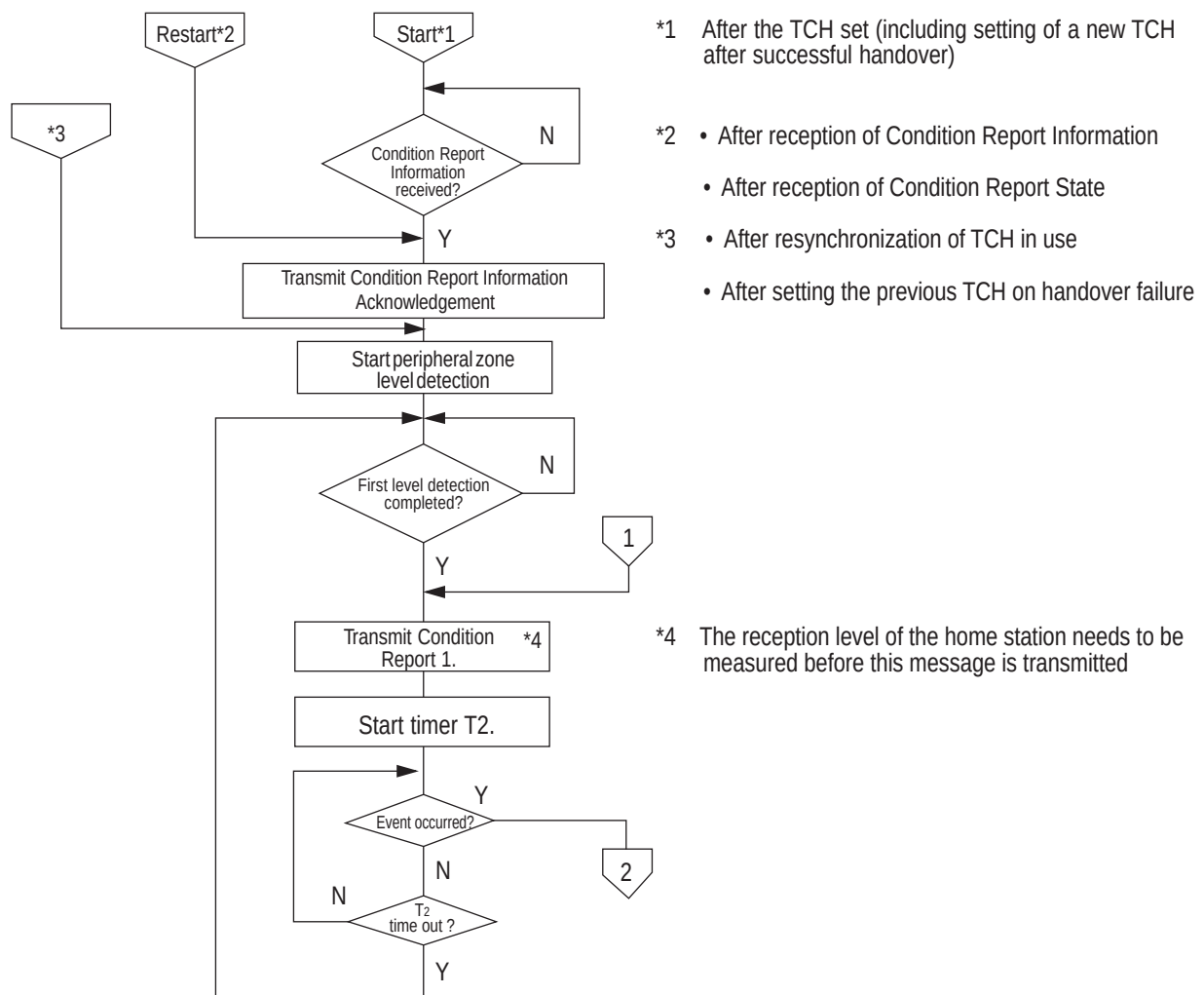


Fig. C.1 : Condition report sequence (1/2)

Note 1 : When the MS receives Condition Inquiry, the MS transmits Condition Report 1 with the information at this point. This control is performed separately from the above sequence flow. Accordingly, timers T1 and T2 remain unaffected on reception of Condition Inquiry.

Note 2 : When the MS receives Condition Report Information during the MAHO, the MS halts the MAHO and restarts the MAHO under new condition. When the number of perch channels contained in the Condition Report Information is "0", the MAHO is stopped.

Note 3 : When the MS receives Reception-level Measurement Request during the MAHO, the MAHO is halted from the start of the measurement until it is completed.

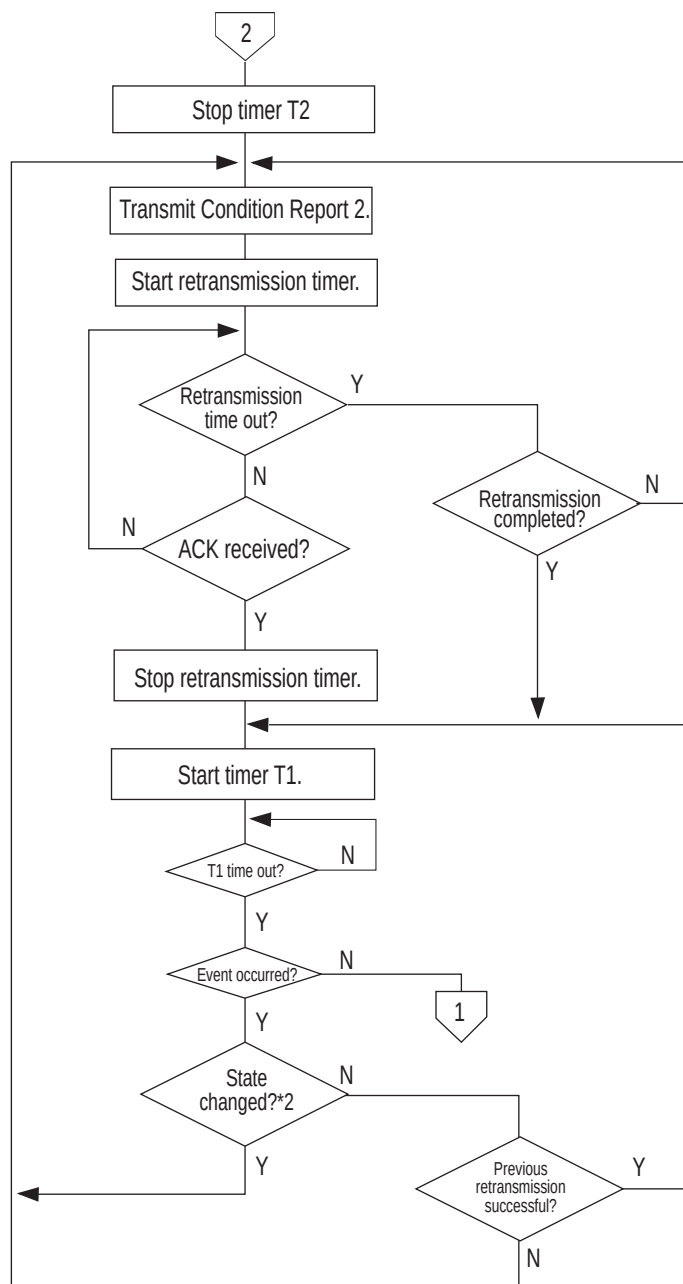


Fig C.1 : Condition report sequence (2/2)

*2 When the difference between the present reception level and the reception level which was reported on the previous detection is more than B; "no state change" is stated only when all of the following three conditions are met.

- (1) The number of selected zones reported previously = the number of selected zones reporting this time.
- (2) The order of perch channels reported previously does not differ from that reporting this time.
- (3) With respect to the home zone reception level and the reception level of reported perch channel, the level difference between the previous and the present receptions is below the state change threshold (B).

Appendix D Communication Connections with Peripheral Zone Selection

This appendix describes how selective communication connections are processed for other available zones. Such processing assigns an available channel in the peripheral zone when no TCH is available in the zone where the mobile station is presently located either when originating or terminating a call.

D.1 Outline of processing

- (1) The mobile station monitors the perch channels of the peripheral zones indicated in the Broadcast Information and measures the reception levels of those perch channels for zone switching recognition during the standby state. When the mobile station sends an origination request to the network (when transmitting an Originating Condition Report message), or responds to paging from the network (when transmitting the Terminating Condition Report message), the mobile station reports the reception levels of those peripheral zones obtained through peripheral zone level measurement.
- (2) When the network receives an Originating (Terminating) Condition Report and no vacant TCH is available in the zone where the mobile station is located (the zone whose control channel has establishment with the mobile), the network inquires the availability of TCH in the zones to which the connection can be made based on the above reception level, and reports it. Then the network specifies the channel to be assigned to the mobile station and establishes communication.

D.2 Report conditions for selecting peripheral candidate zones

The network indicates the perch channel numbers for peripheral zone level detection in the Broadcast Information. The candidate zones for peripheral zone selection are selected also from these perch channel numbers. At the same time, in the Broadcast Information, the network specifies the maximum number of candidate zones as the maximum number of reporting channels. The mobile station sets the total number of candidate zones detected within the maximum number of reporting channels as the selected zone number, and reports it to the network along with the reception levels and perch channel numbers. Fig D. 1 shows the sequence of reporting peripheral candidate zones.

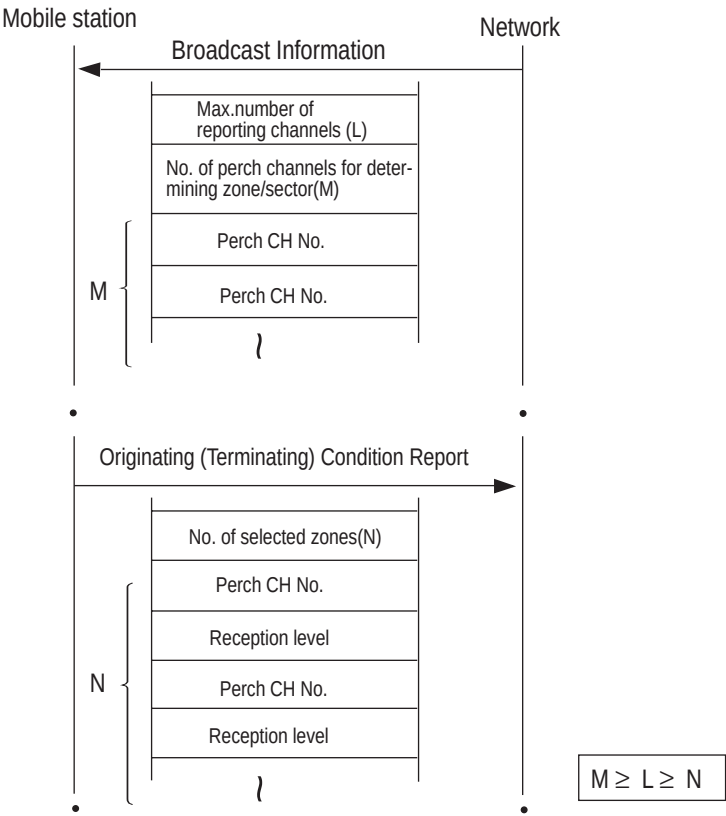


Fig. D.1 : Peripheral zone selection report sequence

Appendix E Restriction Control

This appendix describes the contents of the restriction control mentioned in this standard and the operation of a mobile station controlled by restriction information contained in the Broadcast Information.

The "origination" referred below means voice communication origination. If the 'Packet restriction condition' bit in [Packet channel structure information] in [[Broadcast Information]] is set to '0 : Refer to the restriction condition', "origination" also includes the packet communication origination, and packet communication registration and packet channel registration procedures which are performed in the selected zone after receiving Broadcast Information in the Active state or in the Packet standby state. If the 'Packet restriction condition' bit in [Packet channel structure information] in [[Broadcast Information]] is set to '1 : Not refer to the restriction condition', the packet operation procedures above mentioned are not restricted.

E.1 Access group restriction

(1) Outline of the restricting processes

General mobile stations are divided into eight groups. The traffic for mobile stations accessing a network is controlled by restricting (prohibiting) the origination/location registration of the mobile stations. Eight different restricting levels can be set from 0% to 100%, with 12.5% intervals by means of adjusting the number of groups accessing the network simultaneously. In addition, the groups under restriction must also be changed periodically so as not to bias the restriction to certain groups.

(2) Restriction method

The method of restricting the operation of mobile stations is controlled based on an overload level of the network zone. The network broadcasts home zone access restriction information (MS operation information), General MS location registration/ origination restriction information (restriction information) and restriction enforcement group information (restriction group set). When a general mobile station recognizes that home zone access restriction is in effect, the MS judges whether it is contained in the restriction enforcement group. If the MS is contained in the restriction enforcement group (*1), the MS prohibits operations specified in the restriction information (call origination/location registration).

In the case of originating a circuit switched call from a general mobile station to the police (110), fire and rescue (119) or maritime safety (118), the equivalent access control to the circuit switched call origination from a priority mobile station is applied. In this case however, mobile station type shall remain to be set general mobile station and the access cycle restriction shall be valid.

In order not to bias the restriction over certain groups, the restriction enforcement group are changed periodically while the access group restriction is performed. The method of changing the groups to be restricted is shown in Fig. E-1.

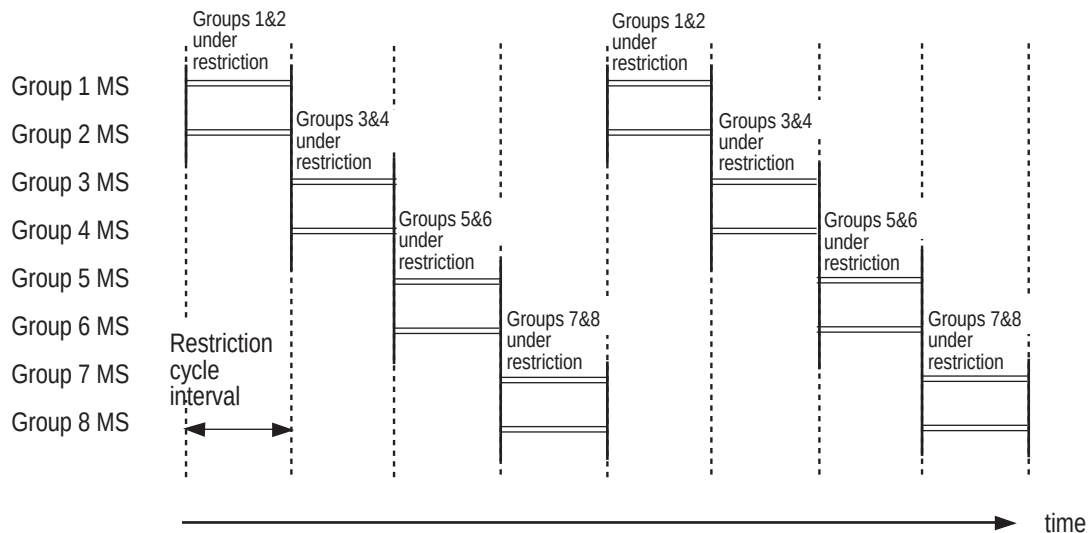


Fig. E.1 : An example changing the restriction enforcement group

During the period that access group restriction is being enforced, the mobile stations, which ignore the restriction and perform call origination or location registration, are detected, and further processes for those mobile stations will be suspended.

(*1) : This occurs if the adding 1 to the lower 3 bits in the octet obtained by taking ExOR of the lower 2 octets of the mobile station number in binary expression equals to the restriction enforcing group.

E.2 Access cycle restriction

(1) Outline of the restriction processing

When the restriction is canceled, all the mobile stations in the group for which the restriction is canceled will initiate call origination or location registration requests simultaneously, which could result in an overload on the SCCH. In order to disperse call origination and location registration requests from mobile stations, access cycle restriction is performed. Figs. E2 and E3 give examples of dispersing traffic by the access cycle restriction after the restriction has been canceled.

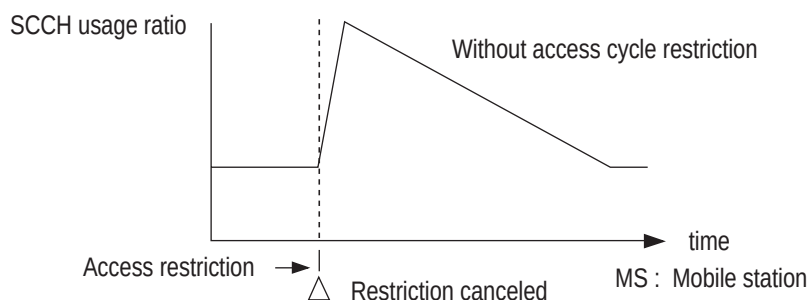


Fig. E.2 : SCCH usage ratio when regulation is canceled without the access cycle restriction .

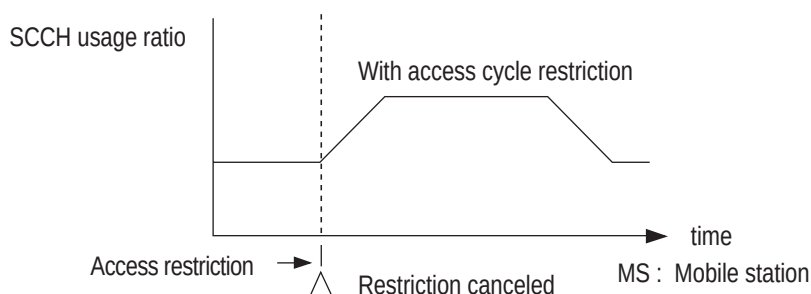


Fig. E.3 : SCCH usage ratio when restriction is canceled with the access cycle restriction .

(2) Restriction method

The network broadcasts the existing home zone access restriction information (MS operation information) and the access cycle interval information (access cycle interval) by means of restriction information. When the mobile station recognizes that home zone access restriction is undertaken, the mobile station detects whether the access cycle interval is "0" or not. If the access cycle interval is other than "0", the mobile station prohibits itself from accessing the network during the specified cycle interval after a call origination/location registration request is initiated. As a result of such control, SCCH overload can be prevented by varying the SCCH access timing for each MS.

Fig. E-4 is an example of how the mobile stations operate in this restriction. While restriction on call origination and location registration for mobile stations in groups 1 and 2 is being carried out by the network, access cycle restriction is also performed. The mobile stations in groups 1 and 2 recognize the access group restriction and access cycle interval through the Broadcast Information. In cases when call origination or location registration requests are initiated at the mobile stations in groups 1 or 2, call origination or location registration is prohibited by means of the access group restriction. At the same time, mobile stations are prohibited from accessing the network during the access cycle interval after the requests are made, regardless of the access group restriction.

This prevents all the mobile stations from accessing the network simultaneously at the time of canceling access group restriction thus preventing the sudden rise in SCCH usage ratio. Note that the access cycle restriction is independent of other restrictions. In the above example, the mobile stations in other groups which are not under access group restriction are also prohibited from accessing the network

during the access cycle interval once they have accessed the network.

Access cycle restriction is valid only for general mobile stations, but is invalid for high priority mobile stations.

The access cycle restriction timer starts after originating a call or location registration. When the mobile station switches to the standby state (Refer to Appendix B.3) following origination or location registration, the following operation occurs:

- (1) When the mobile station performs "selection of the channel for the standby state" (Refer to Appendix B.2), the access cycle restriction timer starts based on the Broadcast Information.
- (2) When the mobile station directly switches to the standby state, the access restriction cycle timer starts based on the previously memorized access cycle interval.

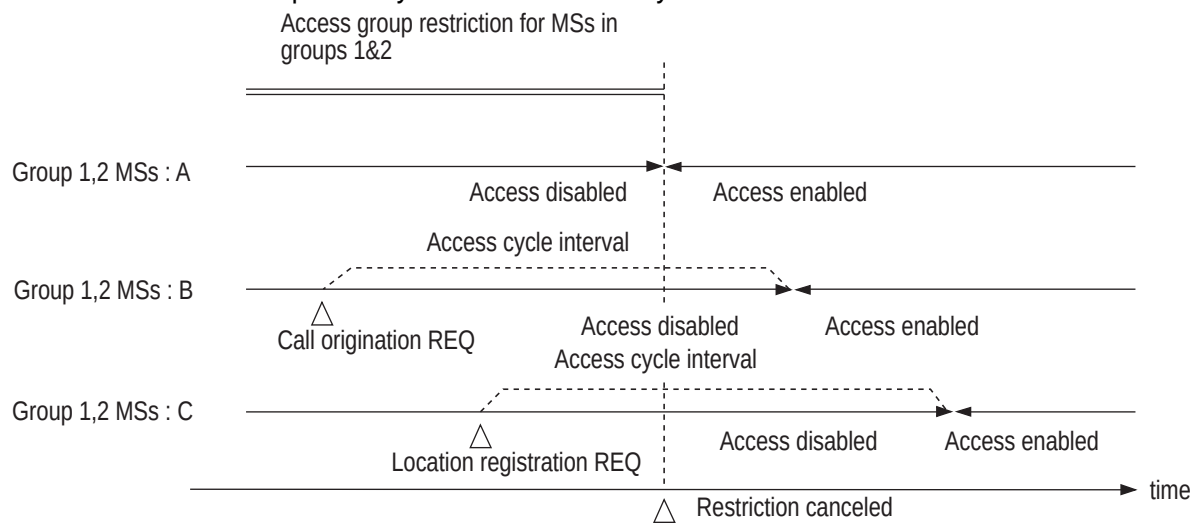


Fig. E.4 : MS operation during access cycle restriction

E.3 Peripheral zone access control

(1) Outline of the restriction processing

Peripheral zone access control forces mobile stations locating at the outer edge of a zone to call the adjacent zones, which are not malfunctioning or congested, while prohibiting access to the home zone. The purpose of this control is to maintain call connections as much as possible in such instances when the home zone is malfunctioning to the extent that it does not cause interference with other zones. Also, this control disperses traffic in the congested zone to the adjacent zones and reduces the number of call connection requests to the congested zone.

(2) Restriction Method

The network broadcasts the zone selection correction state (mobile station operation information) and the zone selection correction level (zone selection correction level) in the Broadcast Information.

When a MS recognizes that a zone selection correction state is in effect, the MS refers to the zone selection correction level information.

When the reception level difference between the home zone and the peripheral zone is below the zone selection correction level, the MS (shown in area B in Fig. E.5 below) normally remains in the home zone. At the call origination request, however, the MS prohibits itself from accessing in the home zone and follows the Broadcast Information for the adjacent zone. If the peripheral zone is permitted to access with the home zone access enabled, no peripheral zone access restriction and the home zone selection enabled and furthermore if the pertinent mobile station is neither under access group restriction nor high priority MS access restriction, the mobile station will be permitted to originate calls in the adjacent zone. Note that location registration is not controlled by the peripheral zone access restriction. When the level difference is greater than the zone selection correction level, the MS (shown in area A in Fig. E.5 below) refers to the home zone access restriction information (MS operation information) contained in the Broadcast Information. If access is disabled, call origination and location registration are prohibited in the home zone. If access is enabled, origination and location registration are enabled in the home zone.

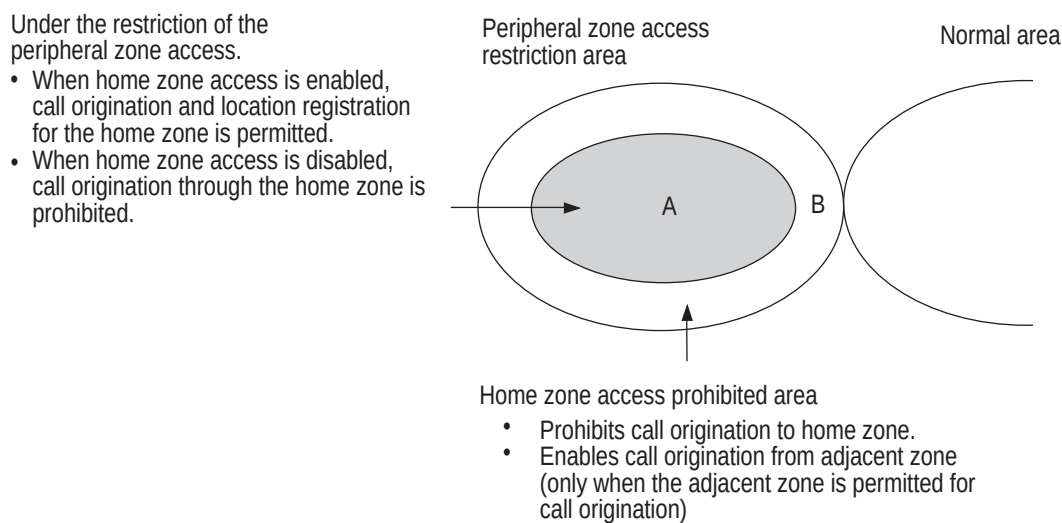


Fig.E.5 : Zone selection correction area diagram

E.4 High priority MS access restriction

(1) Outline of the restriction processing

In order to reinforce access restriction control, this method restricts high priority mobile stations after 100% restriction of general mobile stations has been carried out.

(2) Restriction method

High priority mobile stations are restricted for call origination and location registration respectively under this restriction. The network broadcasts the state of the home zone access restriction (MS operation information) and the high priority MS restriction information regarding call origination and location registration (restriction information) in the Broadcast Information. When a high priority mobile station recognizes that the home zone access restriction exists, it follows the high priority MS restriction information (on call origination/location registration) and prohibits its operation accordingly.

E.5 Maintenance restriction

(1) Outline of the restriction processing

This restriction is executed to enable the mobile station for maintenance to access the network during construction or other periods when the network is not under normal operation. If normal operation zone is located adjacent to the maintenance zone, mobile stations other than those used for maintenance are moved to the normal operation zone.

(2) Restriction method

The network broadcasts home zone selection state (MS operation information) using the restriction information in the Broadcast Information. When a mobile station other than maintenance mobile station recognizes that the home zone selection is disabled, it will scan the control channels in order to switch to standby state in other zone.

E.6 Access restriction

(1) Outline of the restriction processing

When Broadcast Information can be transmitted but the access from the mobile station with the origination/ location registration request is unable (due to malfunction of the network etc.), this restriction is used to prohibit access from mobile stations.

(2) Items to be restricted

The network broadcasts home zone access state (MS operation information) in the restriction information contained in the Broadcast Information. When the mobile station recognizes that the home zone access is disabled, it stops accessing the home zone and switches to the standby state.

E.7 Mobile station operation procedure by restriction information

The "standby state" in the following figure refers to the state described in Appendix B.3.

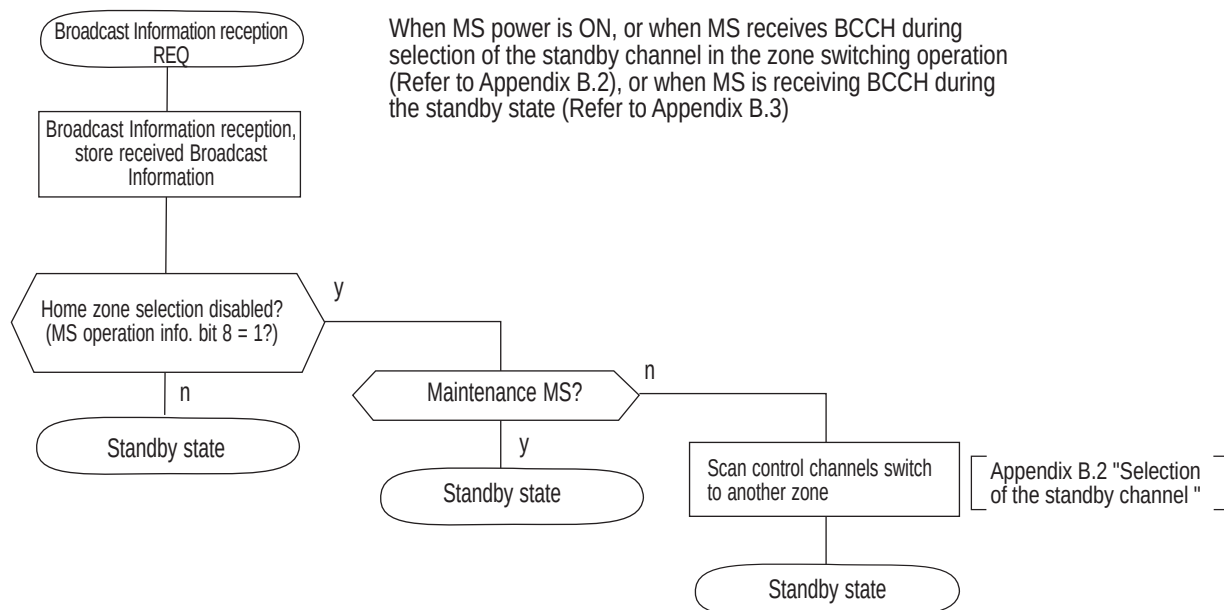


Fig. E.6 : MS operation procedure (on reception of Broadcast Information)

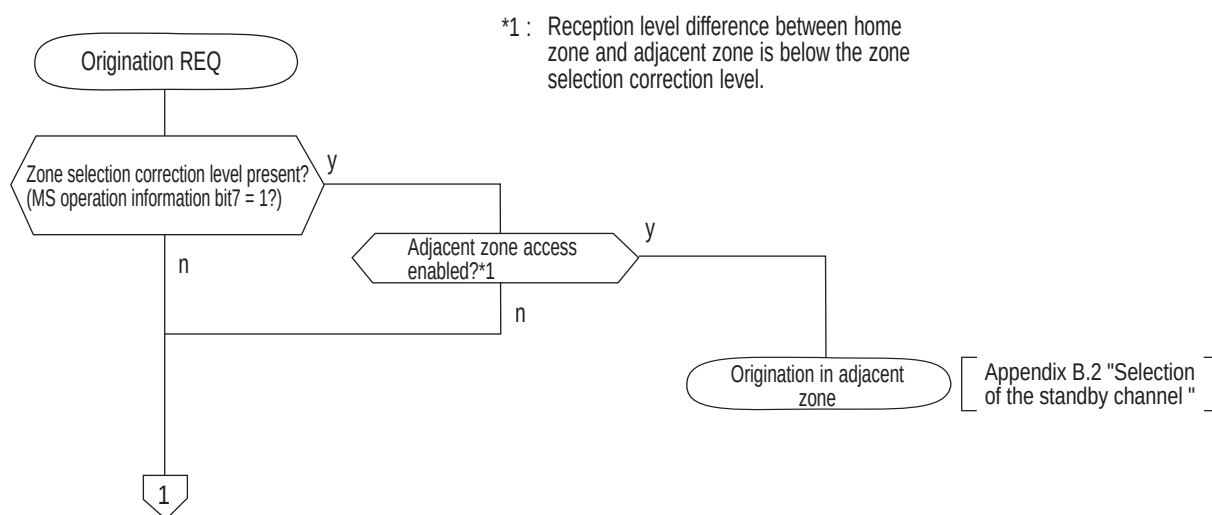


Fig. E.7 : MS operation procedure (on reception of call origination request)

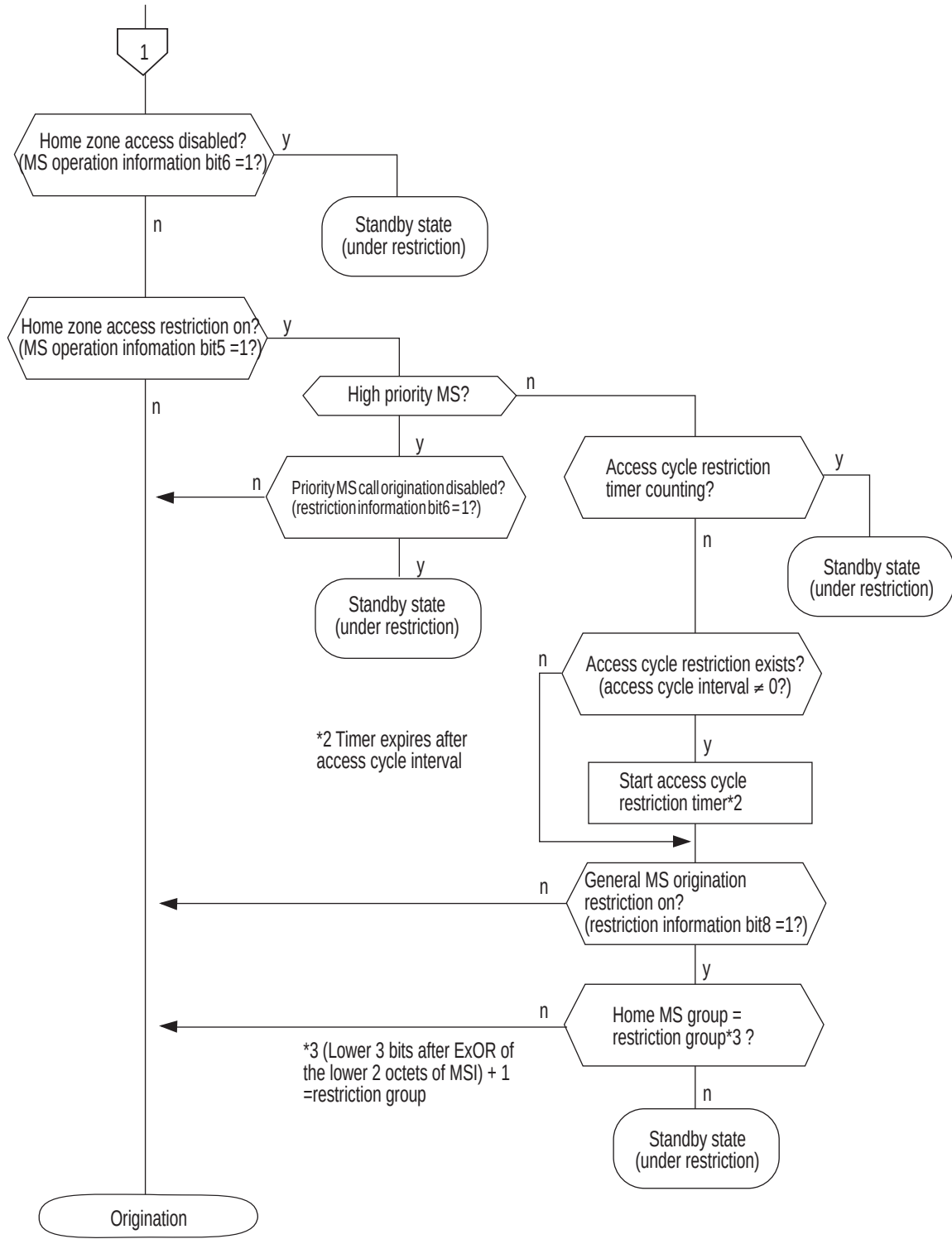


Fig. E.8 : MS operation procedure (on reception of call origination request.)

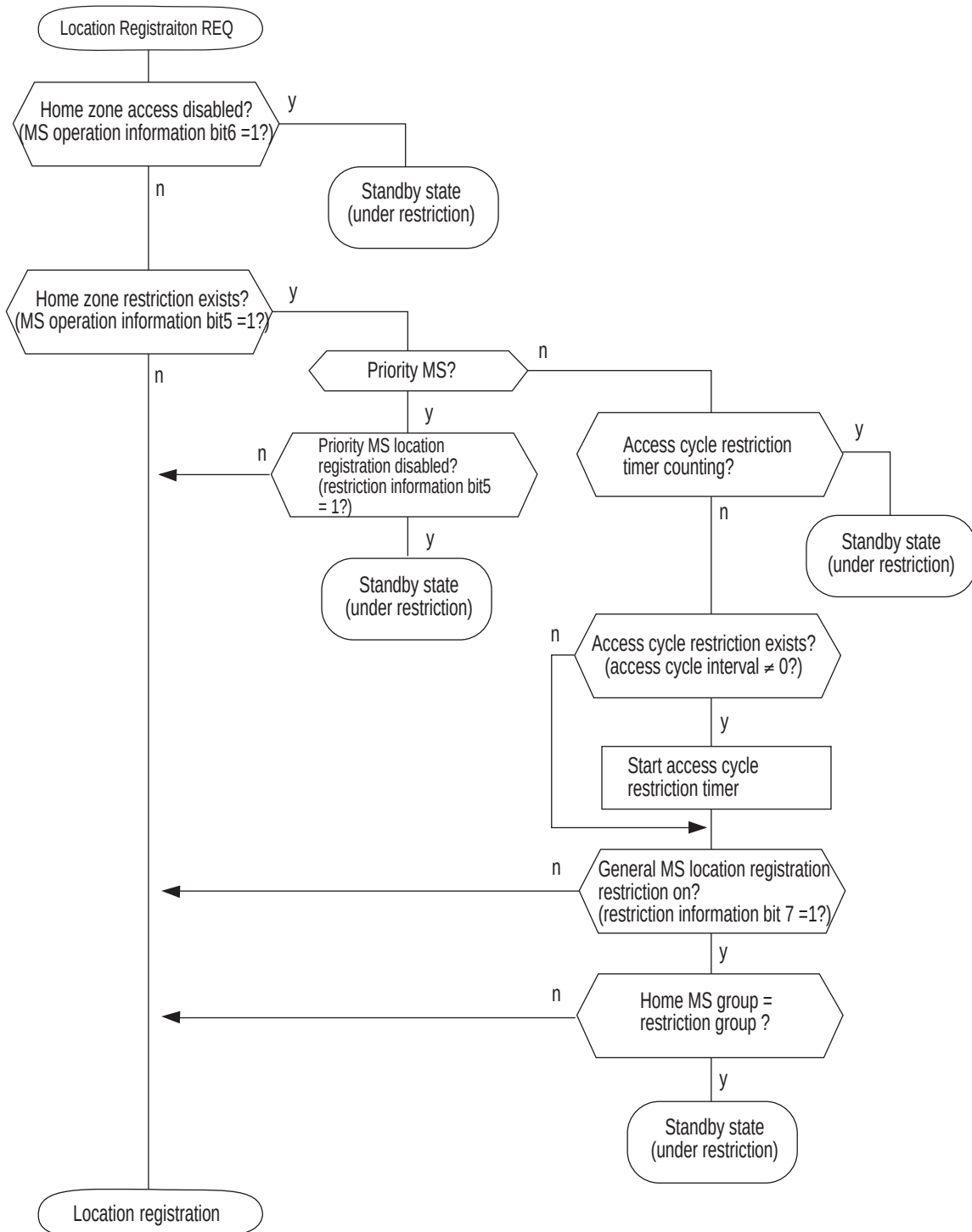


Fig. E.9 : MS operation procedure (on reception of Location Registration Request)

Appendix F Location Registration Control

This appendix describes the methods used for location registration control.

F.1 Outline

What determines the limit of a control channel during extreme minimization of zone is the signal traffic for paging and location registration. In order to cope with the increase in location registration signal traffic, the following method is used in the digital mobile communication system.

- Define a sector (the smallest area that one control channel covers) as the smallest unit of area for location registration.
- Multiplexing of the location registration area. (Separate mobile stations into groups, and by shifting the location registration area for each group, the concentration of registration traffic in a specific zone of the area boundary can be avoided. Since the registration areas are also overlapped and the mobile station always chooses to register the location around the center of which the mobile locates, the probability of crossing a registration area decreases. In addition it brings the location registration system the property of hysteresis, so that there is no frequent back-and-forth registration between registration areas.)

The above method is possible because all the control channels have zone specific frequencies. This method is effective for preventing frequent location registrations which are accompanied by the fluctuation in determining the home zone caused by the simultaneous judgments of home zone switching and location registration area switching.

F.2 Method of location registration control

F.2.1 Configuration of Broadcast Information

Fig. F.1 shows configuration of Broadcast Information.

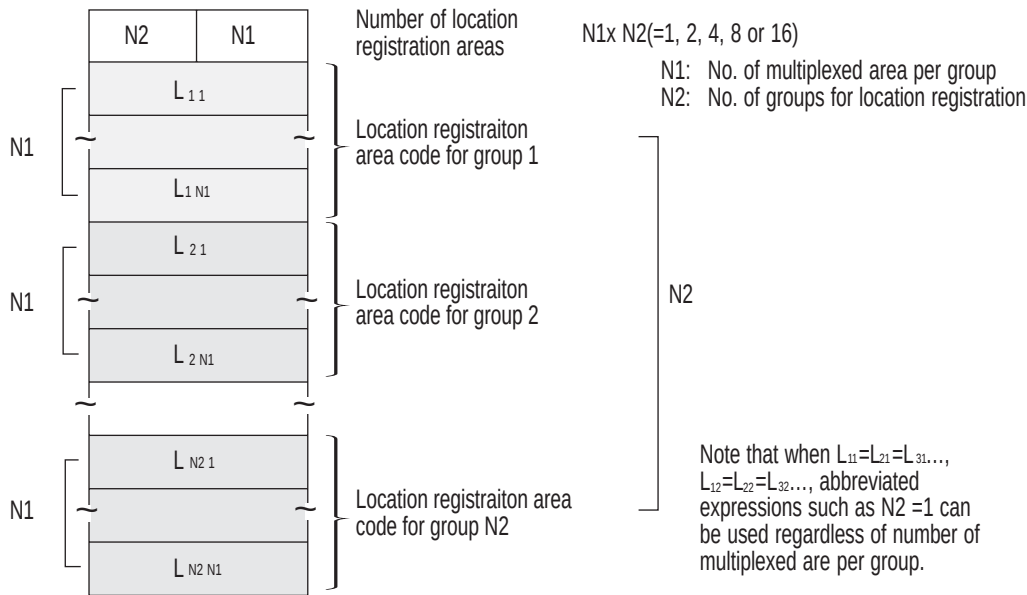


Fig.F.1 : Configuration of Broadcast Information

L=255 shall be defined as dummy code. This code has no meaning as location identity, thus, its use as a dummy code for the top location identity, L_{i1} , is prohibited. Furthermore when $L_{ij}=255$, $L_{ij+1}\dots L_{iN1}$ shall all be set to 255. An example of this use is shown in the figure below. Although this figure exemplifies the case for $N1=8$, it can be used as the equivalent to $N1=5$ by setting the last 3 octets to L=255.

N1 = 8	L ₁₁
	L ₁₂
	L ₁₃
	L ₁₄
	L ₁₅
	L ₁₆ = 255
	L ₁₇ = 255
	L ₁₈ = 255

F.2.2 Operation of mobile station

[Location registration]

- (1) Calculate group number g from a portion of the Mobile station number.

" g " is obtained as follows:

(After ExOR of the lower 2 octets of the Mobile station number expressed in binary, the modulo N_2 of the upper 4 bits of the resulting value) + 1 is defined as a group number.

- (2) Only location registration codes L_{g1} to L_{gN1} for the mobile's own group are checked for location registration.
- (3) When memorized L_0 is not included in the codes above, or when the memorized network identity does not match up with the one in the received Broadcast Information, the mobile station performs location registration.
- (4) The mobile station performs one of the following three operations according to the result of the location registration initiation.

- (i) Receives a Location Registration Acknowledge message:

This message is received when location registration was successful. The mobile station memorizes the location identity (L_{g1} as L_0) and the network identity in its non-volatile memory. Additionally, the mobile station complying with RCR STD-27F or a subsequent version shall also memorize the terminating service registration type set in Location registration request in its non-volatile memory.

- (ii) Receives a Location Registration Reject message:

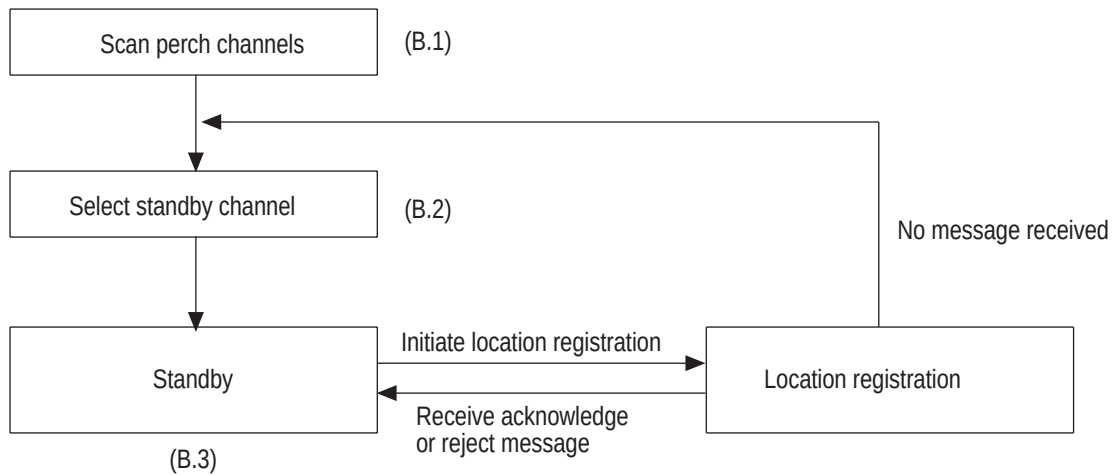
This message is received when location registration fails. The mobile station does not update the information in its non-volatile memory. In this instance, the operation of the mobile station depends on its cause value included in the Location Registration Reject message as described in 4.3.6.3.3.5.

- (iii) Receives no message:

This is the case that no response (neither a Location Registration Acknowledgement or Location Registration Reject) is received in reply to a Location Registration Request message, including retransmission of the message. In this instance, the mobile station clears the data in the non-volatile memory and retries location registration, regardless of the location identity codes in the BCCH.

If the receiving conditions on the SCCH channel degrades to a degree that the radio connection is lost according to the criteria in chapter 4.1, then the mobile station shall enter the state "Select standby channel (B.2)" immediately.

- (5) If the mobile station receives an Mobile Station Release message which includes the cause "MS roaming unregistered" on origination, it initiates location registration after having switched to the standby state.
- (6) The relationship between the standby state of the mobile station and the operation (4) above is illustrated in the following figure.



(B.*) refer to the section in the appendix.

- (7) During the location registration procedure (i.e. including the possible up to 3 retransmissions), the mobile shall not make any cell reselections or any mobile origination or termination calls. Before the location registration procedure, the mobile must read the actual information of the BCCH. In addition, it shall not update this information before it evaluates the result as defined in item (4) above.

[Updating processing for L_o on origination]

On origination of the MS, after the MS receives a Radio-channel Set message, the MS updates the network identity and location identity (L_{g1} as L_o). However, when the MS switches to the standby state without reception of the Radio-channel Set message, then it activates location registration again. In this instance, however, if the MS satisfies the following condition, the MS switches to the standby state without initiating location registration.

Condition : The network identity and L_{g1} in the Broadcast Information received before origination are identical to those memorized by the MS.

F.3 Example of configuration

o When $N1 = 2, N2 = 2$:

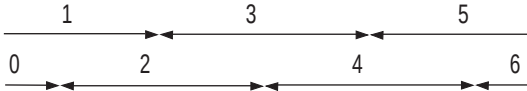
Radio zone allocation :



Location registration area for group 1 :



Location registration area for group 2 :



Configuration of Broadcast Information :

Zone number	1	2	3	4	5	6	7	8	9	10
L11	1	2	2	3	3	4	4	5	5	6
L12	2	1	3	2	4	3	5	4	6	5
L21	1	1	2	2	3	3	4	4	5	5
L22	0	2	1	3	2	4	3	5	4	6

* On location registration, mobile stations in group 1 select L11 while mobile stations in group 2 select L21.

Example of the operation of the MS in group 1:

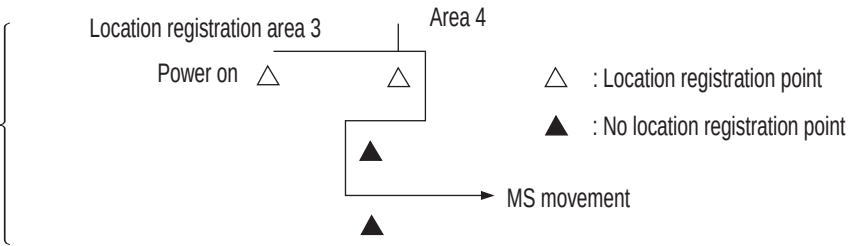


Fig. F.2 : Example of configuration

F.4 Terminating service registration during standby

F.4.1 Overview

If the MS performs periodical location registration during the Standby state, the network can identify whether the power for the MS is turned ON or OFF (refer to Appendix P). In addition, if the MS can detect whether packet communications are allowed or not, the network can identify the ON/OFF state for each service by registering the services which satisfy both the services requested by the MS and the services provided by the zone where the MS is currently located. This enables the network to reduce the incoming traffic on a service-by service basis.

F.4.2 Operation of the MS

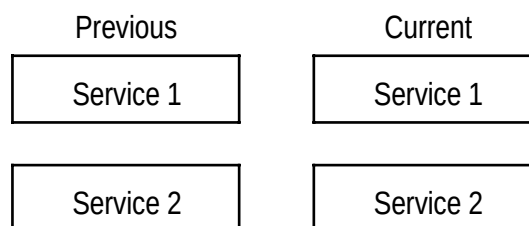
When any of the services requested by the MS have been changed, or when the state of the current zone has been changed, or also when the MS moves into another zone, the MS determines whether the terminating service is provided or not.

The MS determines the services which satisfy both the services requested by the MS and the services provided in the current zone and sets the pertinent service (s) in the [Terminating service registration type] information element in the [Location Registration Request] message.

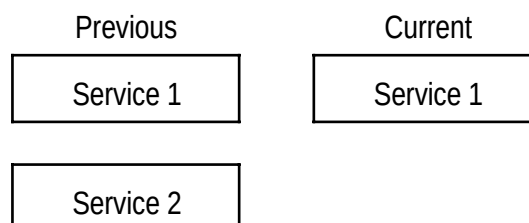
However if all of the services determined by the MS are included in the previous [Terminating service registration type] information element, or if they are perfectly match up with those included in the previous [Terminating service registration type] information element, location registration traffic can be suppressed without terminating loss by setting the result of determination in the [Terminating service registration type] information element at the periodical location registration timing. Note that in the network which does not require periodical location registration, location registration according to the determination will not be performed.

(Examples of when location registration is not performed)

Example 1 When the previous [Terminating service registration type] information element matches up with the results of determination of the services currently recognized by the MS:

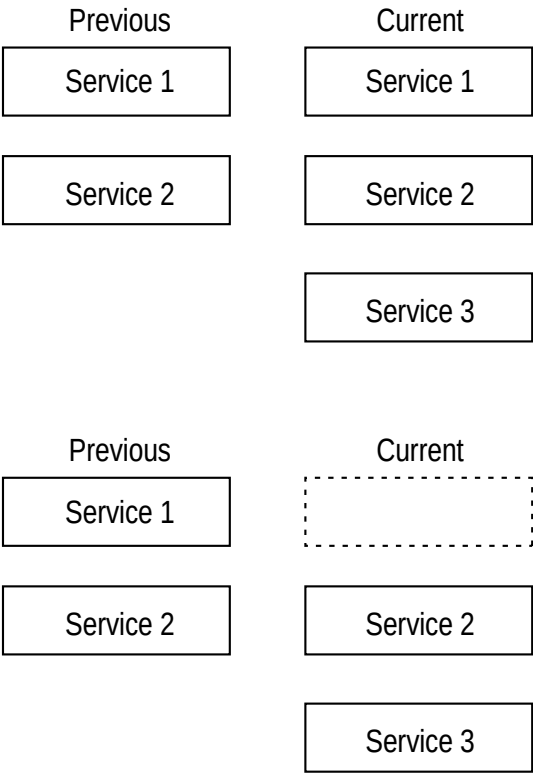


Example 2 When any of the currently recognized services are included in the services set in the previous [Terminating service registration type] information element:



(Examples of when location registration is performed)

Example 3 When any of the currently recognized services is not included in the previous [Terminating service registration type] information element:



F.5 Location registration for switching from the system other than PDC to PDC

In case of switching from the Standby state of a system other than PDC to that of the PDC system, the MS performs a location registration procedure for requesting the network for the system switching. When the MS performs the location registration procedure in order to switch between Standby states, the MS sets a "System switching registration" for the Network registration identifier in the Network registration type information element included in the Location Registration Request message regardless of the Network identity and Location identity stored in the MS.

Appendix G Mobile Station BCCH Reception

This appendix describes the method to command mobile stations to receive the BCCH using paging message in the PCH so as to ensure reception of information by the mobile station when the BCCH information (Broadcast Information) is changed, without affecting the intermittent reception ratio.

G.1 Concrete method for BCCH reception

(1) Operation of mobile station

- The mobile station receives Broadcast Information in the BCCH when the power is turned ON.
- Following this, reception of the BCCH by the mobile station during the standby state is not obligated. (Both periodical reception function and its reception cycle can be set optionally.)
- When CRC errors occur consecutively in the standby state for receiving the PCH, the mobile station receives the BCCH. The time limit for CRC errors is not specified.
- When there is a zone switching during the standby state, reception of BCCH for the new zone is essential.
- When the mobile station receives the paging message through the PCH, the mobile station receives the BCCH consecutively during the segment of time specified by the BCCH reception period information element.
- When a BCCH is not received within the specified BCCH reception period, the mobile station switches to standby channel selection process.

(2) Operation of the network

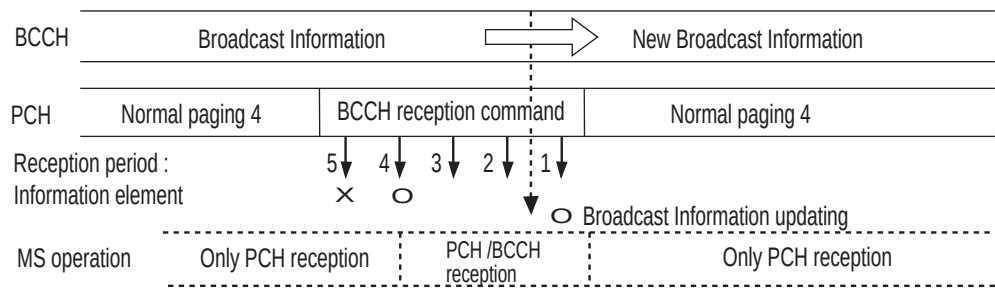
- When the BCCH Broadcast Information needs to be changed, the BCCH reception command is executed by the respective PCH which are divided into intermittent reception groups.
- After the network transmits a BCCH reception command, the network changes the Broadcast Information in the BCCH at an appropriate timing, which accords with the value of the BCCH reception period information element.

(3) Feature

As BCCH reception for a multiple superframe period can be specified to mobile stations, this scheme is suited for the situation where a BCCH reception command message is not received. Also, in the layer 3 processing by the network, strict timing control between BCCH reception command transfer and the Broadcast Information modification is not necessary.

(4) Example of operation

BCCH reception of mobile station is shown in Fig G.1



The BCCH reception period for the PCH is normally set to "0" and the MS does not receive BCCH. When the BCCH reception period is not set to "0," the MS receives BCCH for the specified period of time. The above figure shows that the MS does not receive the PCH when the BCCH reception period is "5" but receives the next PCH correctly (i.e., when the BCCH reception period is "4"). BCCH reception begins at this point. The network updates BCCH just before BCCH reception period is set to "1". Therefore, the MS can receive the updated BCCH during the BCCH reception period.

Note : Detailed network operations are left up to the individual operators. Above is just an example.

When the length of the BCCH reception segment is fixed, for example, the mobile station will receive the BCCH more times than necessary, but this will not interfere with the operation of the system. Only a minute degree of deterioration of the intermittent reception ratio for the mobile station occurs.

Fig . G.1 : Mobile station BCCH reception

Appendix H VOX Control

VOX (Voice Operated Transmitter) is a method for saving power of the mobile station. The mobile station with VOX control function detects the voice input and turns the transmitter on only while voice is present. When no voice is present, the transmitter is turned off. This appendix describes the control mechanism for starting/ending VOX control.

H.1 Assumptions

VOX is an option for both the MS and the BS. Within the same network, there is no need for all traffic channels to have VOX capability. It is also possible to use a VOX-capable channel without VOX operation (throughout a whole telephone conversation or for a specific period of the conversation.)

H.2 Control structure

(1) Mobile station type information element

Through the VOX capable/incapable indication bit, the mobile can tell the network whether it has VOX function or not. (The mobile that has VOX inoperable indication in the VOX capability bit may be connected to a non VOX-capable channel.)

(2) Channel information information element

In order to realize the VOX function setting (at start of communication) in individual TCH as mentioned in the assumption above, (normally uplink) VOX control disabled/enabled for each TCH is indicated by the network to the MS through the channel information information element in the Radio-channel Set message or the Handover Radio-channel Set message. In accordance with the above assumption, the indication can be changed for the same channel at arbitrary timing.

(3) VOX enable/disable during call in progress

As in (2) above, in order to realize the VOX function setting for individual TCH (during call in progress), the following are provided:

(i) Control on the network side

VOX enabling or disabling is controlled by the VOX control signal.

When the VOX control signal (enable) is received by a MS, VOX operation can be started by the decision of the MS only in the case that both the MS and the TCH are able to perform VOX operation. A MS that has received the VOX control signal (disable) goes into the VOX disabled state, and will terminate VOX operation if it is in VOX operation.

(ii) Control on the MS side

If a TCH is a VOX-capable channel, the MS is able to start or stop VOX operation arbitrarily.

Appendix J Paging Identification Number

This appendix describes the purpose and procedure for paging identification numbers which are used to identify paging incoming calls from when paging starts until the call is connected to the destination.

J.1 Purpose

Since the zone where the destination mobile station is located cannot be specified in paging, paging is performed over several zones for an incoming call. Thus, the number of pagings performed by the network paging control is greater than the number of calls to be connected and many of the pagings do not receive the responses. Normally, management of the links between the network and mobile stations is performed using the layer 2 addresses (MSI, SMSI). However it is not efficient to page with the layer 2 addresses because of the above reason. Thus, the "Paging" signal is transmitted in the broadcast type data link connection, and the paging identification number is used in layer 3 as a number to identify the incoming call from when the paging starts until the response returns. The number of paging identification numbers which need to be managed by each zone is not large because the period of their lifetime is short, which makes it easier for the network to manage numbers.

J.2 Usage

J.2.1 Paging and response (setting the paging identification number)

On generating the call request to be received by the mobile station, the network selects an unused number and reports it to the mobile station as layer 3 information in the "Paging" message. The mobile station responds to the paging by returning the paging identification number as layer 3 information in the "Terminating Condition Report" message to the network.

J.2.2 Reception of paging response (search and release by paging identification number)

When the network receives the "Terminating Condition Report" message, it searches for the corresponding paging call using the paging identification number, after which it releases the paging identification number and puts the call through.

Appendix K Handling of Spare Bits

The bit patterns indicated as "spare bit" in this Standard are used as listed in the following table according to the version of the digital cellular telecommunication system.

Item	Spare Bits	Processing on reception
Systems based on this version and preceding versions of this Standard (MS and network)	All set to "0 (zero)"	Both of "0" and "1" are Ignored
Systems based on the next and following versions of this Standard (MS and network)	Not fixed	If all spare bits are set to "0", it is recognized as a system based on this version and preceding versions of this standard, and ignored. Other cases than all "0" indicate specific usages to be defined in the future.

Table K.1: Handling of spare bits

Note : Equipment which accord with this version and those of the future versions may coexist by the handling of spare bits. In addition, equipment of the Future versions may use the spare bits for new functions.

Appendix L Operation of the Data Link Layer During Channel Handover Failure

When channel handover fails, the mobile station and base station transmit or receive messages over the data link layer of the previous channel. The data link which has been established before channel handover continues to be used.

Fig. L.1 shows the operation of the data link layer in the case of channel handover failure. When channel handover begins, the base station starts transmitting SB1, t_3 after the synchronization has been lost on the previous channel. The mobile station starts synchronization establishment operation on the new channel, but if synchronization cannot be established within t_1 sec., it restarts sync establishment operation on the previous channel. If synchronization cannot be established on the previous channel and an out-of-sync condition continues for no less than t_2 , continuous stop of layer 1 is detected and layer 2 data link connection is released. If synchronization is established within t_2 on the previous channel, layer 2 having not invoked any action, messages can be transmitted and received over the previous data link without resetting the multiple frame operation.

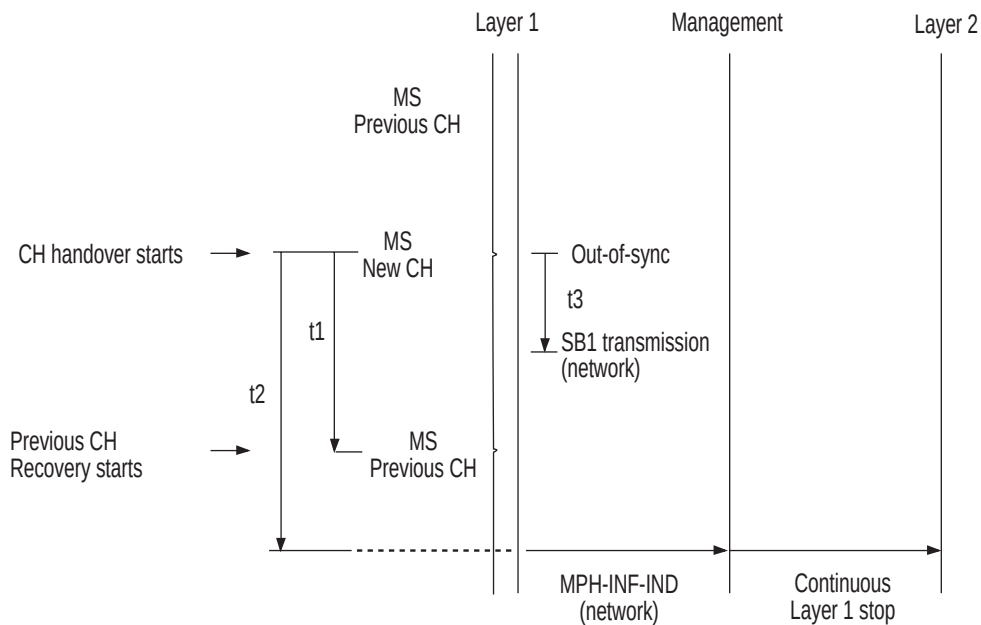


Fig. L.1 : Operation during channel handover failure

Appendix M Division of Messages During Partial Retransmission Control

The purpose of this appendix is to give an example of partial retransmission control procedure so as to help in understanding these standards.

Fig. M.1 shows the error recovery procedure using partial retransmission control when an I frame is transmitted from the mobile station to the network. Fig. M.2 shows the configuration of I and I' frames as transmitted in Fig. M.1. Procedures using partial retransmission control are explained below according to the sequence shown in Fig. M.1

The mobile station and network are assumed in the multiple frame set state and neither of them are in the receiver busy condition. The I frame shown in Fig. M.2 " is transmitted from the mobile station. This I frame has a transmission sequence number of $N(S)=i$ and reception sequence number of $N(R)=j$ and consists of 5 units. Unit sequence numbers $n(S)=0, 1, 2, 3, 4$ are assigned to the units which make up the I frame, while only $N(S)$ and $N(R)$ exist in the control field. In the radio link, the 3rd and 5th units are received as errors, while the 1st, 2nd and 4th units are received correctly. The network transmits the REJ' frame, which contains $N(R)=i$, $n(R)=2,4$ to request the 3rd and 5th units (the units with $n(S)=2, 4$).

When the mobile station receives the REJ' frame, it transmits the I' frame shown in Fig. M.2 # . This I' frame consists of 3 units and its control field contains $N(S)=i$, $N(R)=j$, $n(S)=2,4$. The 2nd unit [$n(S)=2$] of # contains exactly the same information as 3rd unit [$n(S)=2$] of " , While 3rd unit [$n(S)=4$]# contains exactly the same information as contained in the 5th unit [$n(S)=4$] of " . Here, it should be noted that the information field for 1st unit of # is set as "Don't care".

When the network receives the 3rd unit of the I' frame transmitted over the radio link as an error, but receives the 1st and 2nd units correctly, the network transmits REJ' frame which contains $N(R)=i$ and $n(R)=4$ in order to request retransmission of the 3rd frame (the unit with $n(S)=4$).

When the mobile station receives this REJ' frame, the mobile station transmits the I' frame shown in Fig. M.2 \$. This I' frame consists of 2 units and its control field contains $N(S)=i$, $N(R)=j$, $n(S)=4$. Exactly the same information is contained in the 2nd unit [$n(S)=4$] of \$, 3rd unit [$n(S)=4$] of # , the 5th unit [$n(S)=4$] of " . When the network receives the 1st and 2nd unit \$ its correctly, the network transmits an RR frame which contains $N(R)=i+1$ to inform the mobile station that the network has received the messages up to $N(R)=i$ correctly. If the network possesses an I frame to be transmitted at this time, the network should then transmit the I frame containing the $N(R)=i+1$.

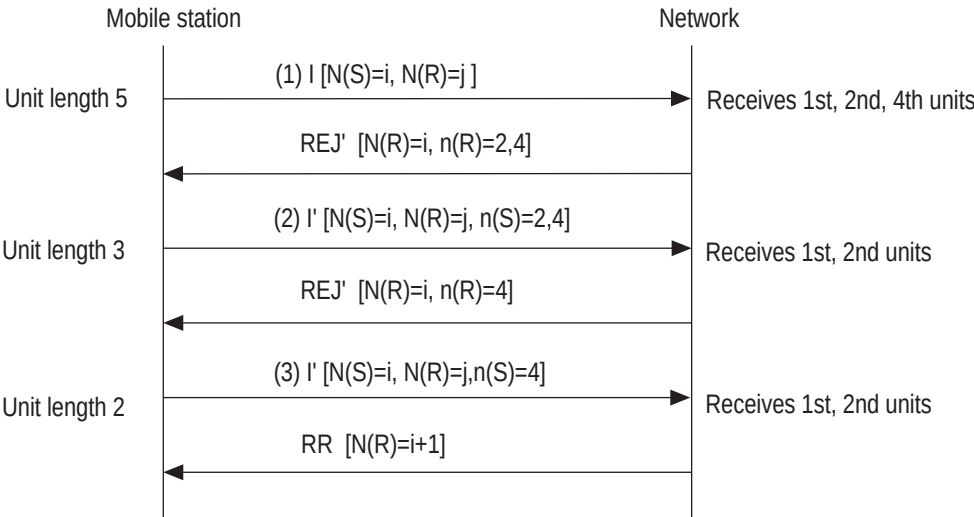


Fig.M.1 : Example of partial retransmission control procedure

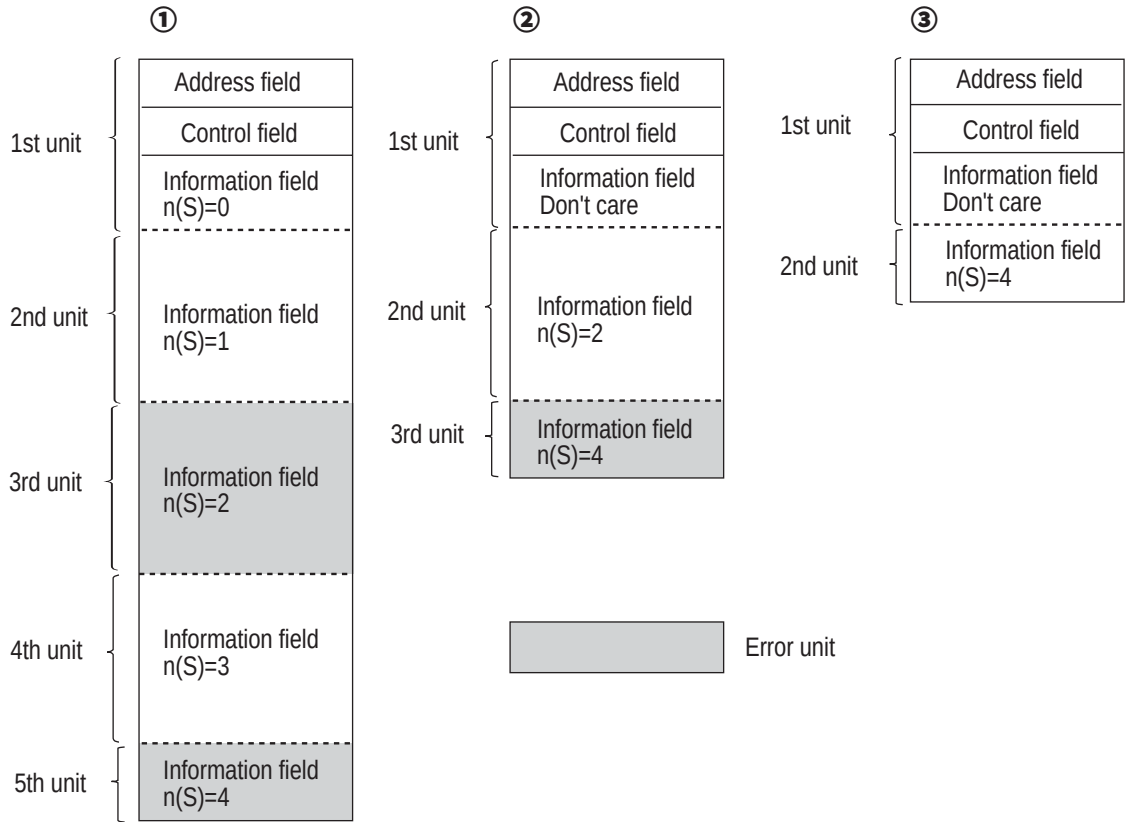


Fig. M.2 : Configuration of frames

Appendix N Definition of Cause Indications

The purpose of this Appendix is to define the various cause indications (RT) in Chapter 4.3.5.3.3.10.

N.1 Normal class

N.1.1 Cause indication #16 "Normal disconnection"

This indication shows the pertient user disconnects a call.

N.1.2 Cause indication #24 "MS roaming unregistered"

This indicates that the MS is not registered and thus it requires registration.

N.1.3 Cause indication #27 "MS abnormal state"

This indication is used to indicate when the MS is not approved as legal (including authentication failure by MM).

N.1.4 Cause indication #31 "Other normal events"

This indication is used to report normal events when the event cannot be indicated by other normal class cause indications, etc.

N.2 Resource unusable class

N.2.1 Cause indication #34 "No available channel"

This is used to indicate that there is no available channel which the subscriber can use (including radio channel busy.)

N.2.2 Cause indication #38 "Network failure"

This cause indication signifies that the network is not functioning normally.

N.2.3 Cause indication #41 "Temporary failure"

This signifies that interface in the network connection is temporarily experiencing failure. (This includes the radio channel squelch disconnected state, etc.)

N.2.4 Cause indication #42 "Device congestion"

This indicates that the network devices are experiencing high traffic congestion.

N.2.5 Cause indication #44 "Requested channel unusable"

This indicates that the selected channel between the mobile switching center and mobile base station cannot be accessed at either party.

N.2.6 Cause indication #47 "Other resource unusable class"

This is to report a cause as a resource unusable event when other reasons of the resource unusable class cannot be applied.

N.3 Procedure error class

N.3.1 Cause indication #97 "Message type undefined or not provided"

Indicates that a message, whose type is not defined nor provided, has been received.

N.3.2 Cause indication #100 "Information element content invalid"

Indicates that a message which contains an invalid information element has been received.

N.3.3 Cause indication #101 "Mismatch of call condition and message."

Indicates that a message which does not match with the call condition has been received.

N.3.4 Cause indication #102 "Recovery by time up"

Indicates that the error recovery procedure has been initiated by time up of the timer.

N.3.5 Cause indication #111 "Other procedure error class"

This cause indication is used to report the procedure error when other cause indications of procedure error class cannot be applied.

Appendix O The Correspondence Between ITU-T I-Interface Recommendations and Pertinent Sections of the Layer 2 and Layer 3 Interfaces in the RCR Standard

RCR Standard	I-Interface
4.2.1. Layer 2 Standards Outline	
1.1 Overview	Q.920 1. (Slightly different)
1.2. Principle of operation and terminology	Q.920 1. (Identical)
1.3. LAPDM function and overview	
1.3.1. Outline	Q.920 3.1 (Slightly different)
1.3.2 Unacknowledgemented operating mode	Q.920 3.2 (Identical)
1.3.3 Acknowledgemented information transfer	Q.920 3.2 (Identical)
1.3.3.1 Basic re-transfer control	Q.920 (Not covered)
1.3.3.2 Partial re-transfer control	Q.920 (Not covered)
1.3.4 Information transfer by function channels	Q.920 (Not covered)
1.3.5 Data link connection identification	
1.3.5.1 Data link connection structure	Q.920 3.4.1 (Slightly different)
1.3.5.2 Data link state	Q.920 3.4.2 (Slightly different)
1.3.5.3 SMSI management	Q.920 (Not covered)
1.4. Service characteristics	
1.4.1 Outline	Q.920 4.1 (Slightly different)
1.4.2 Services provided to Layer 3	
1.4.2.1 Outline	Q.920 4.2 (Slightly different)
1.4.2.2 Priority orders	Q.920 (Not covered)
1.4.2.3 Unacknowledgemented information transfer service	Q.920 4.2.1 (Identical)
1.4.2.4 Acknowledgemented information transfer service	Q.920 4.2.2. (Slightly different)
1.4.3 Services provided for the layer management	Q.920 4.3 (Identical)
1.4.4 Services provided for requests to the physical layer	Q.920 4.6 (Slightly different)
1.4.5 Management functions	Q.920 (Not covered)
1.4.5.1 Outline	Q.920 (Not covered)
1.4.5.2 Definition of primitives for management functions	Q.920 (Not covered)
1.5 Outline of data link layers and management entities	
1.5.1 Functional configuration	Q.920 5. (Slightly different)
1.5.2 Identification of data link connection end points	Q.920 (Not covered)
1.5.3 Data link entity	Q.920 (Not covered)
1.5.4 Data link assignment entity	Q.920 (Not covered)
1.5.5 Management structure	Q.920 5. (Slightly different)
1.6 Special requirements	Q.920 (Not covered)
1.6.1 Operating modes and applicable SAPI	Q.920 (Not covered)
1.6.2 Acknowledgemented operating mode	Q.920 (Not covered)
1.6.2.1 Outstanding number "K"	Q.920 (Not covered)
1.6.2.2 Processing capacity	Q.920 (Not covered)
Chapter 4.2.2 Layer 2 Specifications	
2.1. Overview	Q.921 1. (Slightly different)
2.2 The frame structure used for communications between layers of the same rank.	
2.2.1 Outline	Q.921 2.1 (Slightly different)
2.2.2 Address field	Q.921 2.3 (Slightly different)
2.2.3 Control fields	Q.921 2.4 (Slightly different)

2.2.4 Information fields	Q.921 2.5 (Slightly different)
2.2.5 Transparency	Q.921 2.6 (Slightly different)
2.2.6 Frame indications	Q.921 (Not covered)
2.2.7 Format rules	
2.2.7.1 Numbering rules	Q.921 2.8.7 (Identical)
2.2.7.2 Bits send out order	
2.2.7.3 Field mapping rules	Q.921 2.8.3 (Identical)
2.2.2.8 Invalid frames	Q.921 2.9 (Slightly different)
2.3 Procedure elements and field formats for communications between layers of the same rank	
2.3.1 Outline	Q.921 3.1 (Identical)
2.3.2 Address field formats	Q.921 (Not covered)
2.3.3 Address field variables	
2.3.3.1 Command/response field bit (C/R)	Q.921 3.3.2 (Slightly different)
2.3.3.2 SAPI (Service access point identifiers)	Q.921 3.3.3 (Slightly different)
2.3.3.3 ID control field (AC)	Q.921 (Not covered)
2.3.3.4 ID indication field (AI)	Q.921 (Not covered)
2.3.3.5 Address field extension bits (EA)	Q.921 3.3.1 (Slightly different)
2.3.3.6 MSI (Mobile station identifiers) and SMSI (shortened mobile station identifiers)	Q.921 (Not covered)
2.3.3.6.1 Broadcast type data link connections	Q.921 (Not covered)
2.3.3.6.2 Point-to-point data link connections	Q.921 (Not covered)
2.3.3.7 Relationship with channel types	Q.921 (Not covered)
2.3.4 Control field formats	Q.921 3.4 (Slightly different)
2.3.4.1 (I) format for information transfer	Q.921 3.4.1 (Identical)
2.3.4.2 (S) format for supervisory functions	Q.921 3.4.2 (Identical)
2.3.4.3 The (U) format for unnumbered transmissions and control	Q.921 3.4.3 (Identical)
2.3.5 "Status variables" for indicating state for control field parameters	Q.921 3.5 (Identical)
2.3.5.1 P/F bits	Q.921 3.5.1 (Identical)
2.3.5.2 EC Control field extension bit (EC)	Q.921 (Not covered)
2.3.5.3 Multiframe operation variables and sequence numbers	
2.3.5.3.1 Modulo n	Q.921 3.5.2.1 (Slightly different)
2.3.5.3.2 V(S) Transmission associated state variable	Q.921 3.5.2.2 (Slightly different)
2.3.5.3.3 V(A) Acknowledgement associated state variable	Q.921 3.5.2.3 (Slightly different)
2.3.5.3.4 N(S) Transmission sequence number	Q.921 3.5.2.4 (Slightly different)
2.3.5.3.5 V(R) Reception associated state variables	Q.921 3.5.2.5 (Slightly different)
2.3.5.3.6 N(R) Reception sequence number	Q.921 3.5.2.6 (Slightly different)
2.3.5.4 Unacknowledgement operation variables and parameters	Q.921 3.5.3 (Slightly different)
2.3.6 Partial re-transmission control field formats	Q.921 (Not covered)
2.3.6.1 Information transfer (I) format	Q.921 (Not covered)
2.3.6.2 Supervisory (S) format	Q.921 (Not covered)
2.3.7 Partial re-transmission control field parameters and corresponding associated state variables	Q.921 (Not covered)

2.3.7.1 Multiframe operation variables and sequence numbers	Q.921	(Not covered)
2.3.7.1.1 v(S) Divided transmission associated state variable	Q.921	(Not covered)
2.3.7.1.2 n(S) Transmission divided unit sequence number	Q.921	(Not covered)
2.3.7.1.3 v(R) Divided transmission associated state variable	Q.921	(Not covered)
2.3.7.1.4 n(R) Reception divided unit sequence number	Q.921	(Not covered)
2.3.8 Commands and responses	Q.921 3.6.1	(Slightly different)
2.3.8.1 "I" command (Information)	Q.921 3.6.2	(Slightly different)
2.3.8.2 SABME command (Set asynchronous balanced mode extended)	Q.921 3.6.2	(Identical)
2.3.8.3 SABMEI command (Set asynchronous balanced mode extended with information)	Q.921	(Not covered)
2.3.8.4 DISC command (Disconnect)	Q.921 3.6.4	(Identical)
2.3.8.5 UI command (Unnumbered information)	Q.921 3.6.5	(Identical)
2.3.8.6 Reception ready (RR) command and response	Q.921 3.6.6	(Identical)
2.3.8.7 REJ (reject) command and response	Q.921 3.6.7	(Identical)
2.3.8.8 Reception not ready (RNR) command/response	Q.921 3.6.8	(Identical)
2.3.8.9 UA response (Unnumbered acknowledge)	Q.921 3.6.9	(Identical)
2.3.8.10 DM response (Disconnect mode)	Q.921 3.6.10	(Identical)
2.3.8.11 FRMR response (Frame reject)	Q.921 3.6.11	(Identical)
2.3.8.12 XID command and response (Exchange identification data)	Q.921 3.6.12	(Slightly different)
2.3.8.13 Partial re-transmission control information (I') format	Q.921	(Not covered)
2.3.8.14 Partial re-transmission control reject (REJ') command/response	Q.921	(Not covered)
2.4 Elements for interlayer communications		
2.4.1 Outline	Q.921 4.1	(Identical)
2.4.1.1 General name	Q.921 4.1.1	(Identical)
2.4.1.1.1 DL-SET primitive	Q.921 4.1.1.1	(Identical)
2.4.1.1.2 DL-REL (release) primitive	Q.921 4.1.1.2	(Identical)
2.4.1.1.3 DL-DATA primitive	Q.921 4.1.1.3	(Identical)
2.4.1.1.4 DL-UNIT DATA primitive	Q.921 4.1.1.4	(Identical)
2.4.1.1.5 DL-STOP primitive	Q.921	(Not covered)
2.4.1.1.6 DL-RESTART primitive	Q.921	(Not covered)
2.4.1.1.7 DL-RECONNECT primitive	Q.921	(Not covered)
2.4.1.1.8 DL-ASSIGN primitive	Q.921 4.1.1.5	(Slightly different)
2.4.1.1.9 DL-CANCEL primitive	Q.921 4.1.1.6	(Identical)
2.4.1.1.10 DL-ERROR primitive	Q.921 4.1.1.7	(Slightly different)
2.4.1.1.11 MDL-UNIT DATA primitive	Q.921 4.1.1.8	(Identical)
2.4.1.1.12 MDL-XID primitive	Q.921 4.1.1.9	(Identical)
2.4.1.1.13 MDL-INFO primitive	Q.921	(Not covered)
2.4.1.1.14 PH-DATA primitive	Q.921 4.1.1.10	(Identical)
2.4.1.1.15 PH-INFO (information) primitive	Q.921	(Not covered)
2.4.1.1.16 MPH-START primitive	Q.921 4.1.1.13	(Identical)
2.4.1.1.17 MPH-STOP primitive	Q.921 4.1.1.14	(Identical)

2.4.1.1.18 MPH-INFO primitive	Q.921 4.1.1.15	(Identical)
2.4.1.2 Primitive types	Q.921 4.1.2	(Identical)
2.4.1.2.1 (REQ) Request primitives	Q.921 4.1.2.1	(Identical)
2.4.1.2.2 (IND) Indication primitives	Q.921 4.1.2.2	(Identical)
2.4.1.2.3 (RSP) Response primitives	Q.921 4.1.2.3	(Identical)
2.4.1.2.4 (ACK) Acknowledgement primitives	Q.921 4.1.2.4	(Identical)
2.4.1.3 Definition of parameters		
2.4.1.3.1 Priority order identifier	Q.921 4.1.3.1	(Slightly different)
2.4.1.3.2 Channel type	Q.921	(Not covered)
2.4.1.3.3 ID control	Q.921	(Not covered)
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Appendix P Control Method for Periodical Location Registration

This appendix describes the controlling method for periodical location registration. With this function, mobile stations which are turned on, repeat location registration periodically and it allows the network to recognize the power on/off of mobile stations, enabling a reduction of terminating traffic.

P.1 Pre-conditions

- (1) This function is optional for network, and mandatory for mobile stations. Mobile stations based on RCR STD-27B may not have this function. They shall nevertheless accept BCCH message in the normal way independent of the value in the location registration timer information element. When the network which supports this function requests mobile stations for periodical location registration, and if one mobile station without this function based on RCR STD-27B does not perform periodical location registration, the network must be designed so that it does not recognize that this mobile station is turned off.
- (2) Periodical location registration is used for informing the network of the mobile stations which are turned on, and the location registration procedure described in Chapter 4.3.8.5 is used as air interface procedure for periodical location registration.
- (3) The network does not permit coexistence of zones with periodical location registration and zones without periodical location registration in a group of zones where the same location identity is sent in the [Broadcast Information] message. (i.e., in a paging area). Thus whether or not to perform periodical location registration can be changed for each paging area.

P.2 Operation of the mobile station

P.2.1 Operation conditions

The mobile station must have memorized the location registration timer information element received in the Broadcast Information message, (initial timer, repeating cycle timer) for the mobile with voice communication capability, or the packet user registration timer information element (initial timer, repeating cycle timer) for the mobile with packet communication capability.

P.2.2 Mobile station monitoring timer

For the mobile station which has voice communication capability only, set the location registration timer as a monitoring timer. For the mobile station which has packet communication capability only, set the packet registration timer as a monitoring timer.

For the mobile station which has both voice and packet communication capability, the mobile station makes the determination according to the operating service types of the current zone and the service types of the mobile station, and changes the monitoring timer by the service in Terminating Service Registration Type Information element of Location Registration Request message. Refer to Appendix F for the service type determination.

- When the terminating service registration type is [Voice], set the location registration timer as a monitoring timer
- When the terminating service registration type is [Packet], set the packet user registration timer as monitoring timer

- When the terminating service registration type is [Voice], Packet], compare the location registration timer and packet user registration timer, and decide the timer for monitoring.
- Application of the initial timer
 - (1) If only the initial timer value of the location registration timer is 0, apply the initial timer value of the packet user registration timer as the initial timer.
 - (2) If only the initial timer value of the packet user registration timer is 0, apply the initial timer value of the location registration timer as the initial timer.
 - (3) If both initial timer value are 0, apply 0 as the initial timer.
 - (4) If the initial timer value of either the location registration timer or the packet user registration timer or both is 7, apply 7 as the initial timer.
 - (5) If initial timer value are deferent from the values in (1)-(4), compare both initial timer values and apply the one with the shorter initial timer value.
- Application of the repeating cycle timer
 - (1) If only the repeating cycle timer value of the location registration timer is 0, apply the repeating cycle timer value of the packet user registration timer as the repeating cycle timer.
 - (2) If only the repeating cycle timer value of the packet user registration timer is 0, apply the repeating cycle timer value of the location registration timer as the repeating cycle timer.
 - (3) If both repeating cycle timer values are 0, apply 0 as the repeating cycle timer.
 - (4) If both repeating cycle timer values are deferent from the value in (1)-(3), compare the both repeating cycle timer values and apply the one with shorter repeating cycle timer value.

P.2.3 Operation of the mobile station

- (1) When the mobile station receives a BCCH after the power for the mobile station is turned on, the mobile station starts the initial timer. When the initial and repeating cycle timers are set to "0", the mobile station does not perform periodical location registration.

When the initial timer is set to 7 ("111"), the mobile station immediately initiates the location registration procedure without starting the initial timer, then starts the repeating cycle timer after the operation is completed.

- (2) When the initial timer expires, the mobile station initiates the location registration procedure. After the location registration procedure is completed, the mobile station starts repeating the cycle timer. If the repeating cycle timer is set to "0," the mobile station does not start the repeating cycle timer.

Note : If the location area, or network identity memorized in the last successful location registration, is not included in the received BCCH information, then the mobiles station shall perform a normal location registration immediately, without waiting for the expiry of the initial timer.

- (3) Following the above operation, each time the repeating cycle timer expires, the mobile station initiates the location registration procedure. When the location registration procedure is completed, the mobile station restarts the repeating cycle timer.
- (4) If the normal location registration, origination or termination described in Appendix F is initiated while the initial timer or the repeating cycle timer is on active, the mobile station once stops the initial timer or the repeating cycle timer, and then starts the repeating cycle timer after the relevant operation is completed.
- (5) If the mobile station sends or receives packets while the initial timer or repeating cycle timer is active, the mobile station stops the pertinent timer and performs each control operation. After the communications are completed, the mobile station performs the procedure according to the available service.
- (6) When the mobile station receives the Broadcast Information message while the initial timer or repeating cycle timer is active, the mobile station maintains the pertinent timer in the active state, even if the mobile station detects that the Location Registration Timer information element and Packet User Registration Timer information element have changed while the network identity or location identity remains unchanged. Following this the mobile station activates another repeating cycle timer when location registration is completed on expiration of the timer.
- (7) When the mobile station detects that the state has changed from the state with no periodical location registration to the state with periodical location registration (when the repeating cycle timer value is changed from "0" to another value) by broadcast information message, the mobile station immediately starts the repeating cycle timer, then switches to the normal periodical location registration operation after this timer expires.
- (8) If the services which may be activated while the initial timer or repeating cycle timer is active are changed, the mobile station stops the pertinent timer and switches to the each processing according to the changed service.
- (9) After the mobile station receives a Location Registration Acknowledgment message from the network, the mobile station memorizes L_{gi} and L_o . (Refer to Appendix F.)
- (10) The initial timer and the repeating cycle timer stop when the power for the mobile station is turned off.

P.3 Operation of the network

- (1) When the network does not request the mobile station for periodical location registration, both the initial and repeating cycle timers need to be set to "0".
When the network does not request the mobile station for activating the initial timer, the initial tier needs to be set to "0." When the network does not request the mobile station for activating repeating cycle timer, the repeating cycle timer needs to be set to "0."
- (2) When the mobile station does not perform any access to the network (periodical location registration, normal location registration, origination or termination of calls, sending or receiving of packets,) for a certain period, the network assumes that the mobile station is turned off. The network does not need to activate termination operation for mobile stations which are turned off. The network does not need to activate termination operation for mobile stations which are turned off, thus reducing the termination traffic.

Appendix Q Layer 2 Frame Unit Structure When the Address and Control Field Spanning Over Multiple Units

The purpose of this appendix is to aid in understanding these standards by giving an example of the unit structure for layer 2 frames when the address and control field span over multiple units.

Fig. Q shows an I' frame to be retransmitted according to the I frame and REJ' frame request. Size of 1 unit is 8 octets and an address field is 2 octets. The I frame consists of 20 units; the 1st unit consists of a 2-octet address field, 2-octet control field and 4-octet information field. Unit sequence numbers $n(S)=0, 1, 2, 3, \dots, 18, 19$ are assigned to them.

When four units $n(S) = 1, 3, 5, 7$ become errors on the RF channel, partial retransmission of the units is requested by a REJ' frame and the I' frame ② is retransferred.

The control field is extended to 6 octets to indicate $n(S) = 1, 3, 5, 7$. Thus, the 1st unit consists of a 2-octet address field and 6-octet control field. The information field for $n(S)=1, 3, 5, 7$ is positioned in the 2nd and the successive units.

When six units $n(S) = 1, 3, 5, 7, 9, 11$ become errors on the RF channel, partial retransmission of the units is requested by a REJ' frame in the same manner as above, and the I' frame ③ is retransferred. The control field is extended to 8 octets to indicate $n(S) = 1, 3, 5, 7, 9, 11$ and the total size will be 10 octets including the address field. Note that the control field spans over 1st and 2nd units. The 1st unit consists of a 2-octet address field and 6-octet control field while the 2nd unit consists of the rest of the 2-octet control field and a 6-octet information field (Don't care). The information field for $n(S) = 1, 3, 5, 7, 9, 11$ is positioned in the 3rd and the successive units.

As described above, it is important to note that the beginning of the information field of the I' frame differs depending on the number of units to be retransmitted.

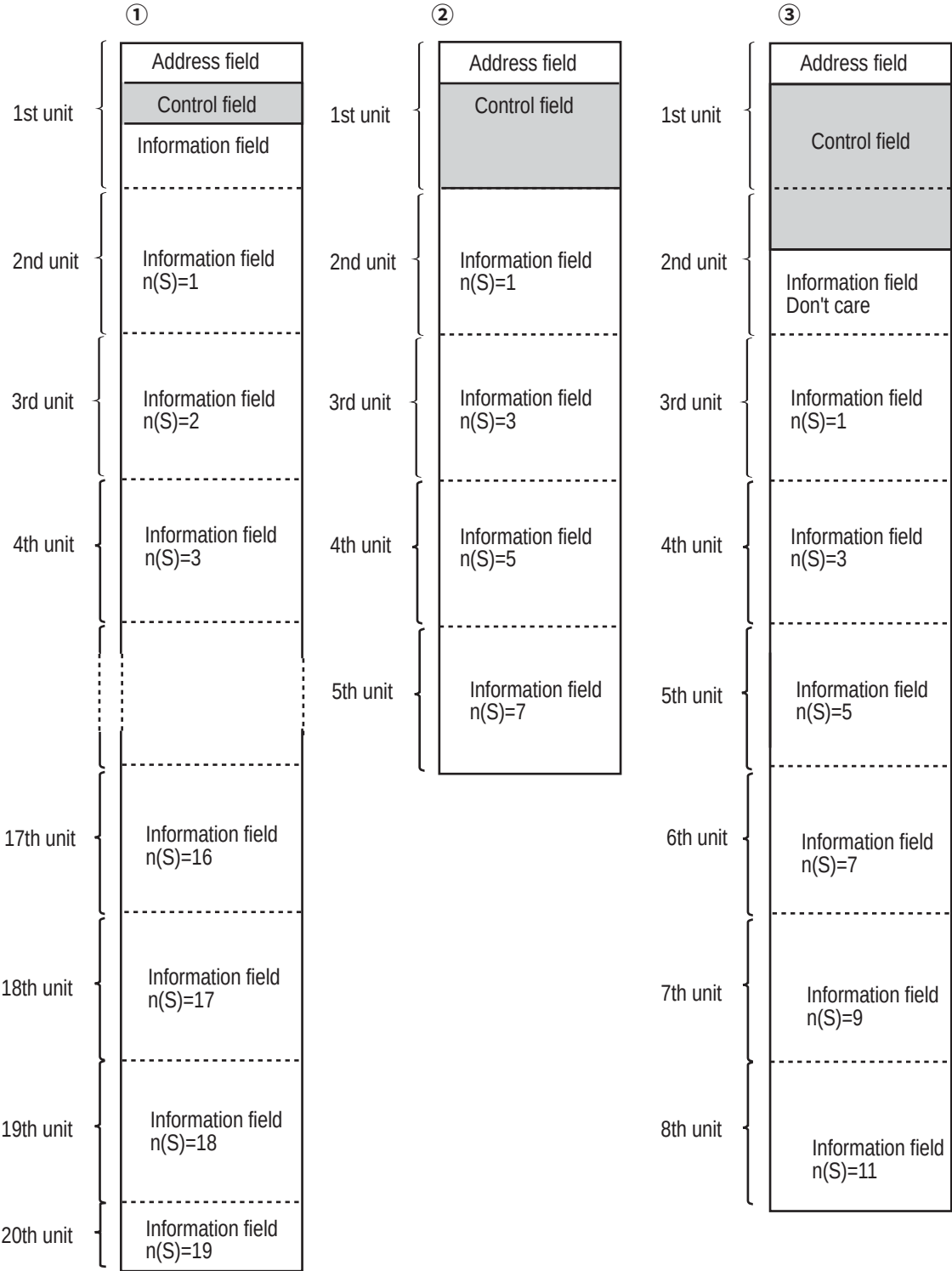


Fig. Q : Configuration of I and I' frames

Appendix R Reception Conditions for RT and MM Messages

In this appendix, reception conditions for RT and MM messages are specified.

R.1 Reception conditions for RT message

R.1.1 Undefined message

When message with undefined message type is received, this messages is ignored.

R.1.2 Unexpected messages

When unexpected message is received, the disposal conforms to the SDL diagrams in Appendix A.5.

R.1.3 Processing of information element

When the received message is longer than the expected message length, excessive information elements in the message are ignored and the pertinent message is processed within the received information element.

R.1.4 Missing information element

- When mandatory information element is missing, the pertinent message is ignored. However, the MS Release and the Radio-channel Disconnect messages are not ignored even if mandatory information element is missing.
- When optional information elements is missing, the pertinent message is ignored. However if this message can be processed or used regardless of the missing optional information, the reception side continues processing the message.

R.1.5 Information element data error

- When mandatory information element has invalid data, the pertinent message is ignored. However, the MS Release and the Radio-channel Disconnect messages are not ignored even if mandatory information element has invalid data.
- When optional information element has invalid data, the pertinent message is ignored. However if this message can be processed or used regardless of the invalid data in the optional information element, the reception side continues processing the message.
- When a reserved value is received, the pertinent information element has invalid data.

R.2 Reception conditions for MM messages

R.2.1 Undefined message

When message of undefined message type is received, this messages is ignored.

R.2.2 Unexpected message

When unexpected message is received, this messages is ignored.

R.2.3 Processing of information element

When the received message is longer than the expected message length, excessive information elements in the message are ignored and the pertinent message is processed within the received information element.

R.2.4 Missing information element

- When mandatory information element is missing, the pertinent message is ignored, or this message is rejected with Cause value (# 96: mandatory information element missing).
- When optional information element is missing, the pertinent message is ignored. However if this message can be processed or used regardless of the missing optional information, the reception side continues processing the message.

R.2.5 Information element data error

- When mandatory information element has invalid data, the pertinent message is ignored, or this message is rejected with Cause value (# 100: invalid information element data).
- When optional information elements have invalid data, the pertinent message is ignored. However if this message can be processed or used regardless of the invalid data in the optional information element, the reception side continues processing the message.
- When a reserved value is received, the pertinent information element has invalid data.

R.3 Additional information element

Information is added to the message due to additional function.

Appendix S User-User Information Transfer (UUS) Supplementary Services

S.1 Definition

The user-user information transfer service allows the user to transmit and receive a limited amount of information over the signaling channel related to the call which is in progress with the other user.

S.2 General

S.2.1 Overview

The UUS supplementary service, provides a means of communication between two users by using as a basis the layer 3 call control protocol. The UUS information transfer procedure is used to exchange information between users. User-user signal exchange is limited by the flow control procedure provided by the network or the user. The exchange of user-user information is not a network acknowledgment service. Any acknowledgment procedure shall be controlled at a higher layer between users. The network can provide the following three types of UUS services associated with the circuit-switched calls to users.

- (1) Service 1: User-user information exchanged by the transfer of the user-user information element included in the call control messages in the call establishment and disconnection recovery phases.
- (2) Service 2: User-user information included in the USER INFO message exchanged between the ALERT and CONN messages during call establishment.
- (3) Service 3: User-user information included in the USER INFO message exchanged during a call in progress.

All three services may be used separately or in any combination in association with a signal call. As an option, at the call establishment, users may be able to specify that the requested user-to-user signaling service(s) is (are) required for the call, i.e., the call should not be completed if user-to-user information cannot be passed.

S.2.2 Special terms

"User" is an entity on the user side over the RF interface.

"Network" is an entity on the network side over the RF interface.

"Called user" is the user who received a call terminated over the interface on the termination side.

"Calling user" is the user who originated a call over the interface on the origination side.

"Service user" is the user who requested UUS supplementary services.

S.3 Requirements for operation

S.3.1 Subscribing and canceling services

Services 1 and 2 of UUS supplementary services shall be provided to a call at least when the calling user subscribes to the UUS services. The UUS service 3 shall be provided to users who subscribe to the service. Whether these services are provided to the user as an individual supplementary service or in a free combination is optional for the party who provide the services.

S.3.2 Requirements for the originating network

The basic call control procedure in A.5.1 is applicable.

S.3.3 Requirements for the terminating network

The basic call control procedure in A.5.1 is applicable.

S.3.4 Virtual terminal

Terminals which are capable of using the UUS service 1 must feature a function for generating and receiving the user-user information element described in Appendix S.5.2.1. When the service is activated evidently, the procedure in Appendix S.5.2.1 needs to be provided. Terminals which are capable of using the UUS service 2 must have a function for generating and/or receiving the USER INFO message including the user-user information element in Appendix S.5.2.2.2. Also, these terminals must provide the procedure outlined in Appendix S.5.2.2.

Terminals which are capable of using the UUS supplementary service 3 must have a function for generating and/or receiving the USER INFO message including the user-user information element in Appendix S.5.2.3.3.

Also, these terminals must provide the procedure outlined in Appendix S.5.2.3. When the congestion control is used, terminals must be capable of providing the CONG CON message and the congestion control level information element and must accord with the procedure outlined in Appendix S.5.2.3.5.

S.4 Requirements for coding

S.4.1 Messages

The following messages shall be applied to the activating procedure for services 1, 2 and 3.

- | | |
|---------|-------------|
| • SETUP | • CALL PROC |
| • PROG | • ALERT |
| • CONN | • DISC |
| • REL | • REL COMP |

The FAC message shall be applied for the services 3 activating procedure.

The following messages are applied for the service 1 activating procedure.

- | | |
|---------|------------|
| • SETUP | • ALERT |
| • CONN | • DISC |
| • REL | • REL COMP |

The following message is applied for information transfer for services 2 and 3.

- USER INFO

When congestion control is used, the following message shall be applied to congestion control for service 3.

- CONG CON

S.4.2 Code group

All the information elements are included in code group 0.

S.4.3 Information elements

The facility information element is applicable to the functional service start-up procedure for services 1, 2 and 3.

The user-user information element is applicable to services 1, 2 and 3.

The more-data information element is applicable to service 2 and 3.

The congestion control level information element is applicable to service 3 control.

S.4.4 Code point

S.4.4.1 Syntax for the service activating procedure using the facility information element

Definitions of the operations required for UUS supplementary service are outlined in the table on the following page using ASN. 1 described in ITU-T Rec. X.208 and the OPERATION macro defined in Fig. 4 in Rec. X.219 in the attached table of Appendix S.4.1.

The coding for the invoke component operation original field is shown in the attached figure of Appendix S.4-1 and the attached table of Appendix S.4-2.

Table S.4-1 Definition of operation and errors

User-User Singnalling	{ITU-T recommendation q 957 user-to-user-singnalling (1) operations-and errors (1)}
DEFINITIONS	:: =
BEGIN	
EXPORTS	UserUserService;
IMPORTS	OPERATION,ERROR
	FROM Remote-Operation-Notation
	{joint-iso-ITU-T remote-operations (4) notation (0)}
	rejectedByNetwork,rejectedByUser
	FROM General-Error-List
	{ITU-T recommendation q 950 general-error-list(1)};
UserUserService	OPERATION
ARGUMENT SEQUENCE	{[1] IMPLICIT Service,
	[2] IMPLICIT Preferred}
RESULT	
ERRORS	{rejectedByNetwork,rejectedByUser}
Service	:: = INTEGER {Service1(1), Service2(2), Service3(3)} (1...3)
Preferred	:: = BOOLEAN{
	preferred (TRUE),
	required (FALSE)}
UserUserService	UserUserService :: = 1
END	

S.4.4.2 Coding for user-user information transfer (UUS) supplementary service operation component

8	7	6	5	4	3	2	1
0	0	1	1	0	0	0	0
Class		Type	Sequence tag				
0 Length format	Sequence length						
1	0	0	0	0	0	0	1
Class		Type	Service tag				
0 Length format	Service length						
Service							
1	0	0	0	0	0	1	0
Class		Type	Preference degree tag				
0 Length format	0	0	0	0	0	0	1
Preference degree length							
Preference degree							

Fig. S.4-1: Operation original field for the invoke component and UUS supplementary service included in the facility information element

Table. S.4-2: Operation original field for the invoke component and UUS supplementary service included in the facility information element

(1) Class (Octets 6.6, 6.8, 6.8.3)

Bit	8	7	
	0	0	Universal
	1	0	Context original
	Other		Reserved

(2) Type (Octets 6.6, 6.8, 6.8.3)

Bit	6	
	0	Primitive
	1	Constructor

(3) Length format (Octets 6.7, 6.8.1, 6.8.4)

Bit	8	
	0	Length is 1 octet
	Other	Reserved

(4) Sequence length (Octet 6.7)

This field indicates the overall length for the sequence for the fields which follow this field (i.e., Octet 6.8 and its subsequent fields). The number of octets for the service is coded in binary and the least significant bit (LSB) is Bit 1.

(5) Service length (Octet 6.8.1)

This field indicates the overall length of the data in the service field (i.e., Octet 6.8.2). The number of octets for the service is coded in binary and the least significant bit (LSB) is Bit 1.

(6) Service (Octet 6.8.2)

Bit	8	7	6	5	4	3	2	1	
	0	0	0	0	0	0	0	1	Service 1
	0	0	0	0	0	0	1	0	Service 2
	0	0	0	0	0	0	1	1	Service 3
									Reserved
									Other

(7) Preference length (Octet 6.8.4)

This field indicates overall length of the data in the preference field (i.e., Octet 6.8.5). the number of octets in this field is coded in binary and the least significant bit (LSB) is bit 1.

(8) Preference (Octet 6.8.5)

Bit	8	7	6	5	4	3	2	1	
	0	0	0	0	0	0	0	0	False (services are required)
									True (services are preferred) (see note.)
									Other

Note: Usually "0 0 0 0 0 0 0 1" is used.

S.5 Requirements for signals

S.5.1 Activation/deactivation/registration of services

Services 1,2 and 3 shall, in principle, be provided for each call in response to explicit requests by the user. The facility information element is used for the explicit standard activating procedure.

Service 1 can also be activated implicitly as described in Appendix S.5.2.1.1.1. In such instances, it is impossible to distinguish between the service activating and information transfer procedures.

The SETUP message transmitted by the originating user on call request should include a request for respective services 1,2 or 3.

The SETUP message which is transmitted from the network to the terminated user must also include the respective service request. When the terminated user can support the display service requested by the evident mechanism, acceptance of a specific service should be reported by the appropriate message as defined in Appendix S.5.2.2 and S.5.2.3.

The deactivating procedure is not necessary for providing these services.

S.5.2 Invocation and operation

S.5.2.1 User-User information transfer (USS) supplementary service 1

S.5.2.1.1 Call establishment phase

S.5.2.1.1.1 Implicit operation

Service 1 may be implicitly requested by including a User-user information element with variable length in the SETUP message transferred across the RF interface of the calling side. This information element is transported by the network and delivered unchanged in the User-user information element included in the SETUP message transferred across the RF interface of the called side. For invocation purpose, this information element must be at least three octets long.

The user-user information element may be included in the ALERT message and/or the CONN message which is transferred across the interface at the called side. This information element may also be included in the DISC and REL COMP message. The content of this information element is transported by the network and delivered in the corresponding message which is transferred across the RF interface at the calling side.

S.5.2.1.1.2 Explicit operation (preferred or required)

The call establishment procedures is identical to Appendix S.5.2.1.1.1, except for the following points. For call originating request, explicit service 1 request is included in the SETUP message sent by the calling user.

The explicit service 1 request is also included in the SETUP message sent from the network to the called side.

The user-user information element may be included in the SETUP message sent from the calling user. The user-user information element is included in the SETUP message sent from the network to the called side when this information element is included in the SETUP message sent from the calling user.

When the user-user information element in the call establishment or clearing message can be transferred by the called user, the service acceptance confirmation for service 1 is included in the ALERT or CONN message before the communication state is set or in the first clearing message.

The acceptance confirmation for the explicit service 1 included in the ALERT, CONN or DISC message is transported by the network and transferred to the calling user.

The user-user information element may be included in the ALERT and/or CONN message which is transferred across the RF interface at the calling side.

S.5.2.1.2 Call clearing phase

The user-user information element is included in the first message used for initiating the normal call clearing.

The information included in such information elements is transferred to the peer user by the first clearing message. This information transfer can be performed only when the information is received before the disconnection recovery message is transferred by the peer user's switching unit to the peer user; otherwise the information will be discarded without notice.

The user-user information element may be included in the first clearing message sent by the called user during a call establishment across the interface at the called side.

If the called user rejects a call by the disconnection recovery message which includes the user-user information element, the network sends the user-user information element in the DISC message to the calling user. However, if the network determines that the disconnection recovery procedure is not activated while inband information is being provided to the calling user, the network may send the user-user information element in the PROG message to the calling user.

S.5.2.1.3 Exceptional procedures

S.5.2.1.3.1 Reject to implicit service requests

When the network receives the user-user information element in the SETUP message from the calling user but the calling user has not subscribed to service 1, the network will discard the user-user information element. Even after discarding this information element, the network continues the call establishment processing. The network may notify the calling user that the UUS service request is not allowed by sending the "cause #50: request facility not subscribed" or "cause #43: access information discarded" information in the STAT message.

The called user may not interpret the content of the user-user information element. In such instances, the user will discard this information without interrupting the normal call processing. In relation to this, the network does not provide the original signaling procedure.

S.5.2.1.3.2 Reject to explicit service requests

When the service 1 requested is indicated as "preferred" and the calling user does not subscribe to the service 1, or the network does not support the service 1, the service 1 rejection which indicates the "rejected by network" will be included in the CALL PROC, PROG, ALERT, or CONN message in accordance with the call state when the network determines that the service cannot be supported.

When the service 1 requested is indicated as "required" and the network cannot support the service (i.e. when the service is not supported or temporarily cannot be provided), the service 1 rejection which indicates the "rejected by network" and "cause #47: other resource unavailable class" or "cause #69: unsupported facility request" will be included in the DISC, or REL COMP message which is sent by the network.

When the service 1 requested is indicated as "required" and the calling user does not subscribe to the service, the service 1 rejection which indicates "rejected by network" and "cause #50: requested facility not subscribed" will be included in the DISC, or REL COMP message which is sent by the network.

When the called user does not wish to accept the service 1 indication but service 1 is requested as "preferred", the service 1 rejection which indicates "rejected by user" will be included in the ALERT or CONN message, or in the message of the first disconnection recovery before the communication state is set. Such explicit service 1 rejection is sent by the network to the calling user in the ALERT, CONN or DISC message.

When the called user does not wish to accept the service 1 indication but service 1 is requested as "required", the service 1 rejection which indicates "rejected by user" and "cause #69: unsupported facility request" will be included in the DISC or REL COMP message. Such explicit service 1 rejection is sent by the network to the calling user in the DISC message.

If the service request is set for "preferred" and "accepted" or "rejected" for service 1 is not included in the ALERT, CONN, DISC or REL COMP message sent by the called user, the service 1 explicit rejection which indicates "rejected by user" will be included in the CONN or DISC message which is sent by the network to the calling user.

If the service request is set for "required" and "accepted" or "rejected" for service 1 is not included in the ALERT or CONN message sent by the called user, the service 1 rejection which indicates "rejected by user" and "cause #69: unsupported facility request" will be included in the ALERT or CONN message which is sent by the network to the calling user. In addition, "cause #31: other normal class" will be included in the DISC message which is sent by the network to the called user.

If the service request is set for "required" and "accepted" or "rejected" for service 1 is not included in the DISC or REL COMP message sent by the called user, the service 1 rejection which indicates "rejected by user" will be included in the DISC message which is sent by the network to the calling user.

S.5.2.1.3.3 Unexpected user-user information elements included in call control messages

When the user-user information element is contained in the ALERT or CONN message sent by called user (in spite of the fact that the UUS service request is not included (explicitly or implicitly) in the SETUP message reported to the called user) the network will discard this information element. When

the user-user information element is discarded, the network performs the operation according to other information elements included in the message. Then, the network transmits a STAT message including "cause#43: access information discarded: to the called user.

When the user-user information element is included in the DISC, REL or REL COMP message sent from either of the user (in spite of the fact that the UUS service request is not included (explicitly or implicitly) in the SETUP message reported to the terminating user). The network will discard this information element. When the user-user information element is discarded, the network performs the operation according to other information elements included in the received message. At this point if the user who activated disconnection recovery has sent a DISC (or REL) message, the network transmits a REL (or REL COMP) message including "cause #43: access information discarded" to the user who activated disconnection recovery. If the user who activated disconnection recovery has sent a REL COMP message, the network assumes that disconnection recovery for the call is completed and does not perform any other operations.

The network will discard the following user-user information element under any of the following conditions:

- When the user-user information element is longer than 131 octets (or 35) and service 1 is activated evidently or implicitly.
- When the network received the message including the user-user information element, however, this message is not allowed to include the user-user information element defined in this STD.

When the network discarded the user-user information element, the network performs operation according to other information element included in the message sent from the user, then transmits a STAT message including "cause #43: access information discarded." However, if the network discard the user-user information element received by a disconnection recovery message., "cause #43: access information discarded" is included in the disconnection recovery message that the network will transmit. If the network discards the user-user information element included in the REL COMP message, the network assumes that call disconnection recovery is completed by the user who transmitted the REL COMP message and performs no other operation.

S.5.2.2 User-user information (UUS) supplementary service 2

S.5.2.2.1 Call establishment phase

At call initiating request, an explicit service 2 request is included in the SETUP message sent from the originating user. The explicit service 2 request is also included in the SETUP message sent from the network on the called side.

When the called user can process the USER INFO message during the call establishment state, the acceptance for service 2 is included in the ALERT message sent from the network. This explicit acceptance notice is transferred in the ALERT message sent from the network to the calling user.

S.5.2.2.2 USER INFO message transfer

On reception of the ALERT message, information transfer is enabled by transferring the USER INFO message between the two users over the RF interface. The network transfers these messages from the calling side to/from the called side. The USER INFO message includes a call reference, protocol discriminator, user-user information element. The "more data" information element can also be included in this message to report the peer user that other USER INFO message including information

which belongs to the same block will continue to be transmitted. Use of the "more data" information element cannot be monitored by the network.

When service 2 is provided, a maximum of two USER INFO messages are transferred in either direction during the period after the ALERT message is transmitted until the CONN message is transmitted.

The USER INFO message may be transferred during the call in progress as a network option.

The call state will never be changed by transmission or reception of the USER INFO message.

S.5.2.2.3 Exceptional procedure

S.5.2.2.3.1 Service rejected/not accepted

When the service 2 requested is indicated as "preferred" and the calling user does not subscribe to the service 2, or the network does not support the service 2, the service 2 rejection which indicates "rejected by network" will be included in the SETUP, CALL PROC, PROG or ALERT message in accordance with the call state when the network determines that the service cannot be supported.

When the service 2 requested is indicated as "required" and the network cannot support the service (i.e. when the service is not supported or temporarily cannot be provided), the service 2 rejection which indicates "rejected by network" and "cause#47: other resource unavailable class" or "cause#69: unsupported facility request" will be included in the DISC, or REL COMP message which is sent by the network.

When the service 2 requested is indicated as "required" and the calling user does not subscribe to the service, the service 2 rejection which indicates "rejected by network" and "cause#50: requested facility not subscribed" will be included in the DISC, or REL COMP message which is sent by the network.

When the called user does not wish to accept the service 2 indication but service 2 is requested as "preferred", the service 2 rejection which indicates "rejected by user" will be included in the ALERT message. Such explicit service 2 rejection is sent by the network to the calling user in the ALERT message.

When the called user does not wish to accept the service 2 indication but service 2 is requested as "required", the service 2 rejection which indicates "rejected by user" and "cause#69: unsupported facility request" will be included in the DISC or REL COMP message. Such explicit service 2 rejection is sent by the network to the calling user in the DISC message.

If the service request is set for "preferred" and "accepted" or "rejected" for service 1 is not included in the ALERT message sent by the called user, the service 2 explicit rejection which indicates "rejected by user" will be included in the ALERT or DISC message which is sent by the network to the calling user.

If the service requests is set for "required" and "accepted" or "rejected" for service 1 is not included in the ALERT message sent by the called user, the service 2 rejection which indicates "rejected by user" and "cause#69: unsupported facility request" will be included in the DISC message which is sent by the network to the calling user. In addition, "cause#31: other normal class" will be included in the DISC message which is sent by the network to the called user.

If the service request is set for "required" and "rejected" for service 2 is not included in the DISC or REL COMP message sent by the called user, the service 2 rejection which indicated "rejected by user" and

"cause#31: other normal class" will be included in the DISC message which is sent by the network to the calling user.

S.5.2.3.2 Unsupported USER INFO message

The network discards the USER INFO message under any of the following conditions and transmits a STAT message including "cause #43: access information discarded" :

- When the network receives the USER INFO message but service 2 has not been activated.
- When the network receives three or more USER INFO messages.
- When the USER INFO message is received by the network in a call state other than alerting.

S.5.2.3 User-user information transfer (UUS) supplementary service 3

This service is requested during the call establishment state or during the call in progress.

S.5.2.3.1 Call establishment phase

At call initiating request, an explicit service 3 request is included in the SETUP message sent from the calling user. The explicit service 3 request is also included in the SETUP message sent from the network at the called side.

When the called user can process the USER INFO message during the call-in-progress state, the acceptance for service 3 is included in the CONN message.

S.5.2.3.2 Communication phase

When the call is in progress, the user is allowed to request "service 3" only. The user who requested the service sends a FAC message to the network, which indicates a request for service 3. Then the network notifies the user who has not requested service 3 of the service 3 request by sending a FAC message and starts timer T1 for the RF interface for the user.

When the user who has not requested service 3 is in the call-in-progress state and can process the USER INFO message, acceptance of service 3 is returned in a FAC message. This explicit acceptance of the service is transferred to the user who requested the service in the FAC message, then the network stops timer T1.

S.5.2.3.3 Transfer of the USER INFO message

When the call (for which service 3 is activated) is in progress, either the calling or called user may transfer information in a USER INFO message over the RF interface to the peer user and the network will transfer these messages.

The USER INFO message includes the call reference, protocol discriminator and the user-user information element. The "more data" information element can also be included in this message to report to the peer user that another USER INFO message including information which belongs to the same block will continue to be transmitted. Use of the "more data" information element cannot be monitored by the network. Perfection of blocks which are divided by this procedure cannot be ensured.

S.5.2.3.4 Congestion control for the USER INFO message

The network flow control mechanism controls flow of the USER INFO messages for the respective directions after the call is in progress.

The burst capability which transmits USER INFO messages corresponding to the value of N in the respective directions is first assigned to each user. In this instance, the initial value for N equals the value of parameter X. The value of N decrements by 1 each time a USER INFO message is transmitted and N is supplemented by Y at given interval T within the range that N does not exceed X.

Burst parameter X is $X=16$ while supplement parameter $Y=8$.

Note: The value of X and Y can be greater depending on network. However, X and Y shall be as specified above for overseas calls. Unless a special agreement is signed between the operators, networks which utilize greater value should take the appropriate measures for compatibility with other networks.

If the network receives USER INFO messages in excess of the value of N within an interval T, these messages will be discarded. the network returns the congestion control level information element "receiver not receivable" included in the CONG CON message in response to the message which is discarded first. "Cause#43: access information discarded" is also included in the CONG CON message. Following this, the USER INFO message will be discarded (by the network) without any indication to users.

When flow control restriction is canceled (at timer T time out) and the USER INFO message for flow control restriction is discarded, the CONG CON message which indicates "receiver receivable" is transmitted to the user, but nothing is transmitted unless the USER INFO message is not discarded.

If the network receives the CONG CON message from the user, the network transmits STAT message including "cause#111: other procedure error class" to the user and does not perform any other operations.

S.5.2.3.5 Exceptional procedure

S.5.2.3.5.1 Rejection of the service requested during call establishment

When the service 3 requested is indicated as "preferred" and the calling user does not subscribe to the service 3, or the network does not support the service 3, the service 3 rejection which indicates "rejected by network" will be included in the CALL PROC, PROG, ALERT or CONN message in accordance with the call state when the network determines that the service cannot be supported.

When the service 3 requested is indicated as "required" and the network cannot support the service (i.e. when the service is not supported or temporarily cannot be provided), the service 3 rejection which indicates "rejected by network" and "cause #47: other resource unavailable class" or "cause #69: unsupported facility request" will be included in the DISC, or REL COMP message which is sent by the network.

When the service 3 requested is indicated as "required" and the calling user does not subscribe to the service, the service 3 rejection which indicates "rejected by network" and "cause #50: requested facility not subscribed" will be included in the DISC, or REL COMP message which is sent by the network.

When the called user does not wish to accept the service 3 indication but service 3 is requested as "preferred", the service 3 rejection which indicates "rejected by user" will be included in the CONN message. Such explicit service 3 rejection is sent by the network to the calling user in the CONN message.

When the called user does not wish to accept the service 3 indication but service 3 is requested as "required", the service 3 rejection which indicates "rejected by user" and "cause #69: unsupported facility request" will be included in the DISC or REL COMP message. Such explicit service 3 rejection is sent by the network to the calling user in the DISC message.

If the service request is set for "preferred" and "accepted" or "rejected" for service 3 is not included in the CONN message sent by the called user or "rejected" for service 3 is not included in the DISC or REL COMP message sent by the called user, the service 2 explicit rejection which indicates "rejected by user" will be included in the CONN or DISC message which is sent by the network to the calling user.

If the service request is set for "required" and "accepted" or "rejected" for service 3 is not included in the CONN message sent by the called user, the service 3 rejection which indicates "rejected by user" and "cause #69: unsupported facility request" will be included in the DISC message which is sent by the network to the calling user. Then the network performs a disconnect recovery procedure. In addition, "cause #31": other normal class" will be included in the DISC message which is sent by the network to the called user.

If the service request is set for "required" and the service 3 rejection which indicates "rejected by user" is not included in the DISC or REL COMP message sent by the called user, the service 3 rejection will be included in the DISC message which is sent by the network to the calling user.

S.5.2.3.5.2 Rejection of the service requested after call establishment

When the network cannot support service 3 or calling user has not subscribed to the service, yet the user requests service 3, the service 3 rejection which indicates "rejected by network" is included in the FAC message.

when the user which has not requested service 3 does not wish to accept the service 3 indication, the user returns the FAC message including the service 3 rejection which indicates "rejected by user" and the network stops timer T1.

If the user who has requested to provide service 3 does not respond to the service 3 request before timer T1 out, the network returns the service 3 rejection which indicates "rejected by user" to the user requesting the service.

If the next request is sent from the peer user when the service 3 requesting processing has not yet been completed, the service 3 requests will collide. In such instances, the entity (the user or the network) which has detected the collision will return the FAC message including the service 3 rejection which indicates "rejected by user" or "rejected by network" to reject the 2nd request. However, as a result, both of the requests will be rejected.

S.5.2.3.5.3 Reception of unexpected USER INFO message

On reception of the USER INFO message one or both of the following conditions will occur:

- a) The service is not allowed in the present call state (i.e., when only service 3 is activated in the any states other than the call-in-progress state).

and/or:

- b) The appropriate user-user information transfer (UUS) supplementary service has not been activated, the message will be discarded by the network. At the same time, the network will send a STAT message including "cause#43: access information discarded".

S.6 Parameter values

Timers T and T1 for the network must be set for 10 seconds.

Appendix T Examples of operation of the MS for packet communications

T.1 Overview

T.1.1 Purpose

This appendix is intended to provide an example of the basic operation of mobile stations which can perform packet communication (hereinafter referred to as [packet communication MS]).

T.1.2 Example configuration of packet communication MS

The configuration of the packet communication MS can be defined as shown in Fig. T.1 below, similarly to Section 2.2. This appendix explains the operation of the packet communication MS by using Fig. T.1. Note that the operation of the packet communication MS for voice communications is not explained in this appendix.

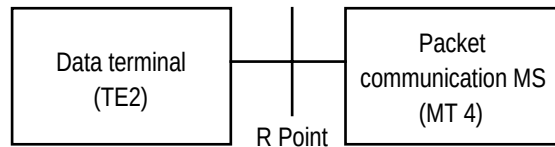


Fig. T.1 : Configuration example of packet communication MS

T.1.3 Overview of the operation of the packet communication MS

Fig. T.2 is an example of the operation of packet communication MS. The basic common processing used for each operation is described in T.2.1; also, the operations marked with the section numbers in this figure are described in Sections T.2.2. and T.2.3.

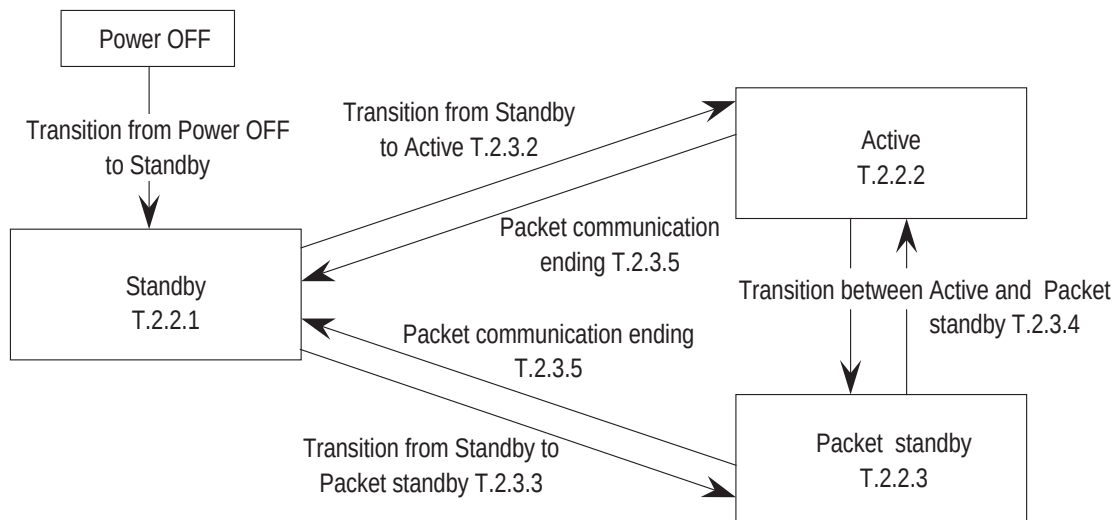


Fig. T.2 State transition diagram (Standby/Active/Packet standby)

T.1.3.1 Standby state (control physical channel)

The state wherein the mobile station waits for a request for starting packet communications from the user or the network.

T.1.3.2 Active state (packet physical channel)

A state wherein packets are transmitted/received

T.1.3.3 Packet standby state (packet physical channel)

A state wherein no packets are transmitted or received for a long period of time and the mobile station performs intermitted reception.

T.2 Description of operation

The operation of the packet communication mobile stations is described. The following symbols are used in the text.

[[]]	:	Message
[]	:	Information element
' '	:	The content of the information element
< >	:	Basic processing (refer to T.2.1)
{ }	:	State in Fig. T.2
" "	:	State transition in Fig. T.2

T.2.1 Definition of basic processing

This section defines the processing commonly required for explaining the operation, included in each state and state transition (T.2.2 and T.2.3).

T.2.1.1 Perch channel scanning processing

The reception levels for perch channels written in the MS are measured sequentially to create a table listing all levels sequentially.

- (1) The mobile station performs the operation based on Appendix B.1.

T.2.1.2 Standby channel selection processing

After switching to the top frequency in the table listing the levels in order, the MS establishes synchronization then measures the reception level and decodes the [[Broadcast Information]] message. The mobile station selects the standby channel based on the [Control channel structure information] information element in the [[Broadcast Information]] message and then switches the channel if necessary.

- (1) The mobile station performs the operation based on Appendix B.2.

T.2.1.3 Packet physical channel selection processing

The mobile station selects a packet physical channel from the information included in the [[Broadcast Information]] message, [[Zone Information Notification]] message, [[Packet System Information Notification]] message or [[Packet Channel Handover Request]] message, then establishes synchronization on this channel and measures the reception level.

- (1) The mobile station performs the operation based on Section 4.1.10.3 and Appendix U.2 for selecting the packet physical channel.

- (2) The mobile station performs the operation under the conditions described in Section 4.1.17 for establishing synchronization.
- (3) The reception level is measured according to Section 4.1.14.2 to determine if the reception level is higher than the packet communication level.

T.2.1.4 Packet communication registration processing

The mobile station registers the packet physical channel to be used, and either that it has entered the Active state or that it has newly switched zones, via the selected packet physical channel.

- (1) The mobile station performs the operation based on the sequence described in Section 4.3.8.13.

T.2.1.5 Packet channel registration processing

The mobile station registers the packet physical channel to be used, or whether the mobile station is in the Active state or in the Standby state, via the selected packet physical channel.

- (1) The mobile station performs the operation based on the sequence described in Section 4.3.8.15.

T.2.1.6 Zone information notification processing

The mobile station obtains the peripheral zone information, via the packet physical channel currently being used.

- (1) The mobile station performs the operation based on the sequence described in Section 4.3.8.20.

T.2.1.7 Packet communication deregistration processing

The mobile station requests to exit the Active or Packet standby state, via the packet physical channel currently being used.

- (1) The mobile station performs the operation based on the sequence described in Section 4.3.8.17.

T.2.1.8 Perch channel reception processing

The mobile station switches to the top frequency in the table listing the peripheral zone levels in order then establishes synchronization on the pertinent channel, measures the reception level and receives the [[Broadcast Information]] message.

- (1) The mobile station establishes synchronization under the conditions described in Section 4.1.17.
- (2) The reception level is measured based on the operation described in Section 4.1.14.1 to determine whether the reception level is higher than the communication level.

T.2.2 Operation in each state

T.2.2.1 Operation in the Standby state

The mobile station performs the following operation in the {Standby} state in addition to the operation described in Appendix B.3.

(1) Paging monitoring

- i) When the [Number of packet paging mobile stations] information element in the [[Paging]] message is set to '1' or higher and the own mobile station number is included in the [Mobile station number [Packet]] information element, the mobile station performs "Transition from Standby to Active"

(2) Packet communication start monitoring

- i) When the mobile station starts packet communications, the mobile station performs "Transition from Standby to Active" or "Transition from Standby to Packet standby".

T.2.2.2 Operation during the Active state

The mobile station performs the following operations during the Active state.

(1) Reception level monitoring

- i) The mobile station detects the reception level at a fixed cycle based on the operations described in Section 4.1.14.1.2.1 and Section 4.1.15.
- ii) Information required to detect whether the zone has changed or the communication level has deteriorated is obtained from the [Communication quality information] and [Connection zone decision information] information elements included in the [[Packet Communication Registration Response]] and the [[Packet Channel Registration Response]] messages. During packet communication procedure for packet origination or packet termination, before Packet Communication Registration Response is received, the Communication level and the Communication deteriorated level in the Broadcast Information is used for Packet communication level (L_z) and Packet communication deteriorated level (L_{th}), respectively. During packet communication registration procedure or Packet channel registration procedure resulting from zone switching, etc., before the corresponding response message is received, the communication quality information in the last received response message is used.

The information items included in the [Communication quality information] information element are listed below:

- Packet communication level (L_z)
- Packet communication deteriorated level (L_{th})
- No. of times packet communication deteriorated level detected (N_r)

The information items included in the [Connection zone decision information] information element are listed below:

- Connection permitted level difference (ΔL_s)
- Strong electric field decision level difference (ΔL_k)

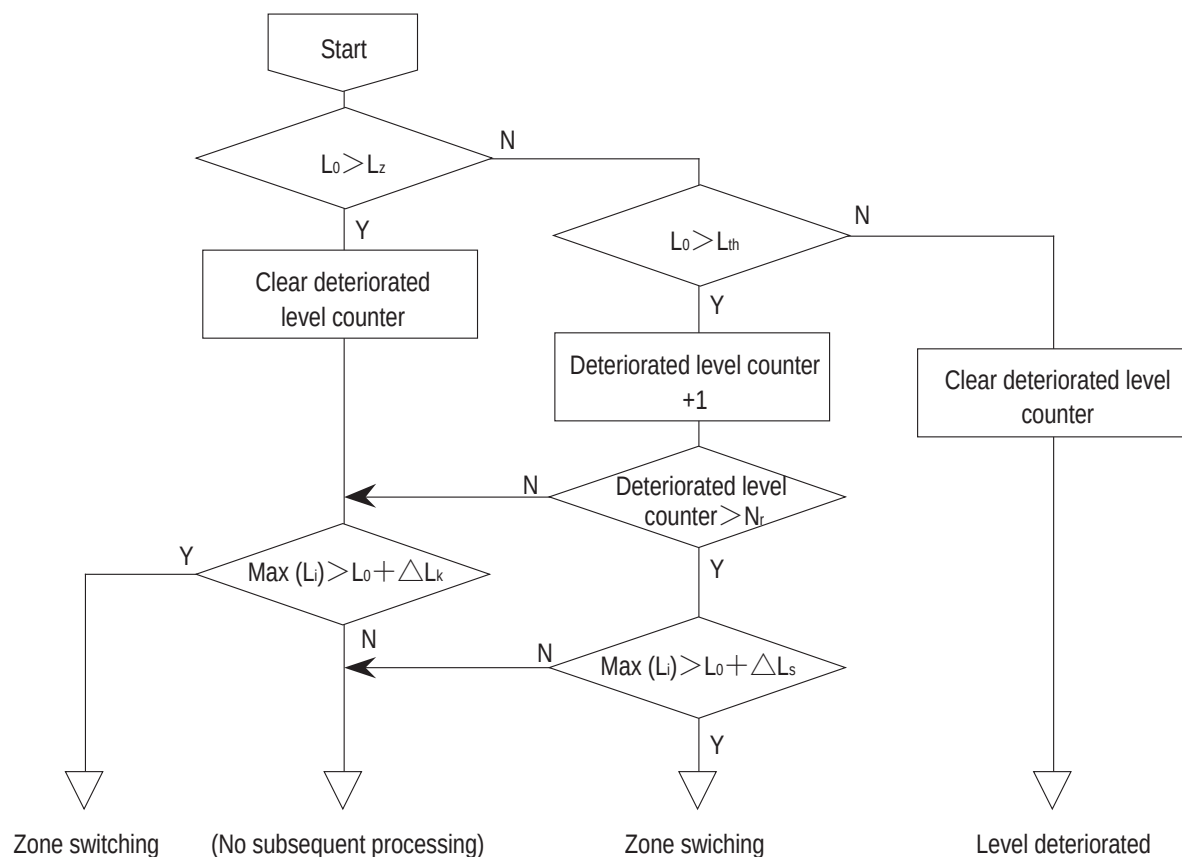


Fig. T.3 : Zone switching decision processing flow

iii) When the result of decision is "zone switching", the mobile station performs the necessary operation to switch to a packet physical channel in a peripheral zone. Either of the following zone switching methods can be used. If zone switching cannot be completed within the time set by the channel switching timer, the mobile station performs the "packet communication ending"

(Method 1)

- Perform the <zone information notification processing>.
- Perform the <packet physical channel selection processing>.
- Perform the <packet communication registration processing> or the <packet channel registration processing> based on the decision described in T.3, Table 1.

(Method 2)

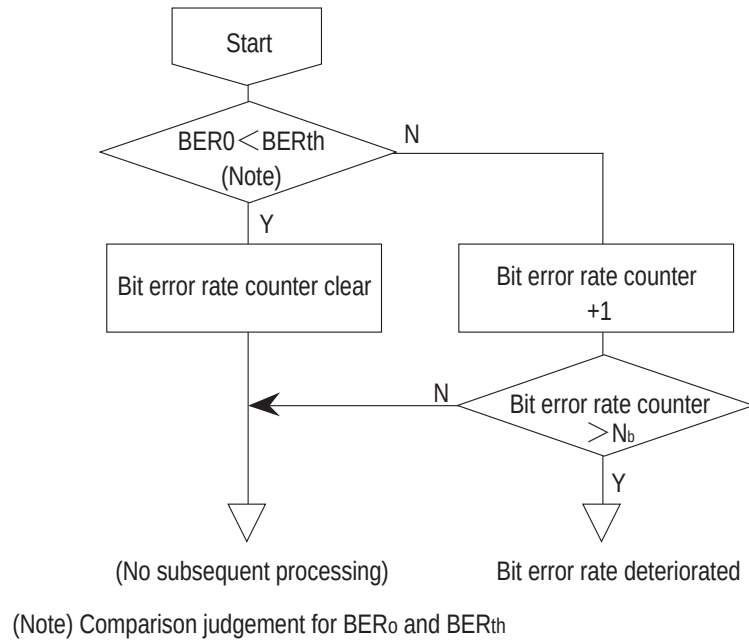
- Perform the <perch channel reception processing>.
 - Perform the <packet physical channel selection processing>.
 - Perform the <packet communication registration processing> or the <packet channel registration processing> based on the decision described in T.3, Table 1.
- iv) When the result of the decision is "Level deteriorated", the mobile station performs the following operation. If the operation cannot be completed within the time set by the channel switching timer, the mobile station performs the "packet communications ending".
- Perform the <perch channel scanning processing>.
 - Perform the <perch channel reception processing>.
 - Perform the <packet physical channel selection processing>.
 - Perform the <packet communication registration processing> or the <packet channel registration processing> based on the decision described in T.3, Table 1.

(2) Bit error rate (BER) monitoring

- i) Detection of the BER for the own mobile station is performed with a fixed cycle and is based on Section 4.1.14.1.2.2.
- ii) The information required for detection of deterioration of BER is obtained from the [Communication quality information] information element included in the [[Packet Communication Registration Response]] or the [[Packet Channel Registration Response]] messages.

The information items included in the [Communication quality information] information element are listed below:

- Deterioration bit error rate (BER_{th})
- No. of times bit error rate deteriorated detected (N_b)



		Value of BER_{th}			
		less than 0.3%	0.3% or more but less than 1%	1% or more but less than 3%	3% or more
Value of BER_0	less than 0.3%	=	<	<	<
	0.3% or more but less than 1%	>	=	<	<
	1% or more but less than 3%	>	>	=	<
	3% or more	>	>	>	=

Fig. T.4 : BER deterioration decision processing flow

iii) When the result of the detection is "BER deteriorated", the mobile station selects a channel from packet physical channels other than the one where the BER deterioration is detected.

- When the processing cannot be completed within the time set by the channel switching timer, the mobile station performs "packet communication ending".

(3) Layer 3 message transmission/reception monitoring

- When packets are not transmitted or received during the time set by the packet cycle registration timer, the mobile station performs the <packet channel registration processing>.
- When packets are not transmitted or received during the time set by the packet standby switching timer, the mobile station performs the "Transition between Active and Packet standby".
- On reception of the [[Packet Communication Disconnect Request]] message, the mobile station transmits the [[Packet Communication Disconnect Acknowledgment]] message, then performs the "packet communication ending".

- iv) On reception of the [[Packet Communication Registration State Inquiry]] message, the mobile station performs the <packet communication registration processing>. Note that the [Communication state] information element in the [[Packet Communication Registration Request]] message is set to [Communication initial state].
- v) On reception of the [[Packet System Information Notification]] message, the mobile station performs the operation described in Appendix U.3. As a result if the mobile station has to select another packet physical channel, the mobile station performs the following processing to select another channel. Note that if this processing cannot be completed within the time set by the channel switching timer, the mobile station performs the "packet communication ending".

When the number of packet physical channels in a cell of the [Packet channel structure] information element is set to a value other than '0', the mobile station performs the following operations:

- <Packet physical channel selection processing>
- <Packet channel registration processing>

When the number of packet communication physical channels in a cell of the [Packet channel structure] information element is set to '0', the mobile station performs the following operations:

- <Perch channel reception processing>
- <Packet communication physical channel selection processing>
- The mobile station performs the <packet communication registration processing> or <packet channel registration processing> based on the decision described in T.3, Table 1.

(4) Power monitoring

- i) When the power is turned off, the mobile station performs the "packet communication ending".

(5) Packet communication ending monitoring

- i) To end packet communication, the mobile station performs the "packet communication ending".

(6) Voice paging monitoring

- i) The mobile station monitors [[UPCH Voice Paging]] and [[UPCH Voice Paging 2]] messages sent from the network.

T.2.2.3 Operation during the Packet standby state

The mobile station performs the following operation during the Packet standby state.

(1) Reception level monitoring

- i) The same operation in the {Active state} (T.2.2.2 (1)) is performed for reception level detection and decision in the packet standby state.

ii) When the result of the decision is "zone switching", the mobile station switches to the packet communication physical channel in a peripheral zone. If this operation cannot be completed within the time set by the channel switching timer, the mobile station performs the "packet communication ending".

- <Perch channel reception processing>
- <Packet communication physical channel selection processing>
- The mobile station performs the <packet communication registration processing> or the <packet channel registration processing> based on the decision described in T.3, Table 1.

iii) When the result of decision is "communication out level", the mobile station performs the following operations. If these operations cannot be performed within the time set by the channel switching timer, the mobile station performs the "packet communication ending".

- <Perch channel scanning processing>
- <Perch channel reception processing>
- <Packet communication physical channel selection processing>
- The mobile station performs the <packet communication registration processing> or the <packet channel registration processing> based on the decision described in T.3, Table 1.

(2) Packet paging monitoring

- i) The mobile station performs the [[Packet Paging]] message reception based on the operation described in Section 4.1.10.2.2.2.
- ii) When the [[Number of packet paging MSs]] information element in the [Packet Paging] message is set to '1' or higher and the [Mobile station number (MSI)] information elements contain the own Mobile station number (MSI) the mobile station performs the "transition between Active and Packet standby states".
- iii) When the [[Common information reception segment length]] information element in the [[Packet Paging]] message is set to '1' or higher, the mobile station receives the common information (Refer to Section 4.1.10.2.2.2 for reception timing.).
- iv) When the mobile station receives the [[Packet System Information Notification]] message, the mobile station performs the same operation as in the {Active} state (Refer to T.2.2.2 (3) v)).

(3) Power monitoring

- i) When the power is turned off, the mobile station performs "packet communication ending".

(4) Monitoring the state transition to the Active state

- i) To switch to the active state, the mobile station performs the "switching between Packet standby and Active states".

(5) Packet communication ending monitoring

i) To end packet communication, the mobile station performs the "packet communication ending".

(6) Voice paging monitoring

i) The mobile station monitors [[UPCH Voice Paging]] and [[UPCH Voice Paging 2]] messages sent from the network.

T.2.2.4 Channel switching timer start and stop conditions

The channel switching timer starts and stops under the following conditions:

(Start condition)

- When the frame synchronization is released for channel handover (including state transition between the Packet standby and Active states) on the UPCH currently used.

(Stop condition)

(1) In Packet standby state, when the MS switches to another channel in the same packet paging area.
 - When frame synchronization is established in the destination UPCH:

(2) When packet communication registration processing or packet channel registration processing is performed in cases other than the above.
 - When a Packet Communication Registration Response or a Packet Channel Registration Response message is received in the destination UPCH

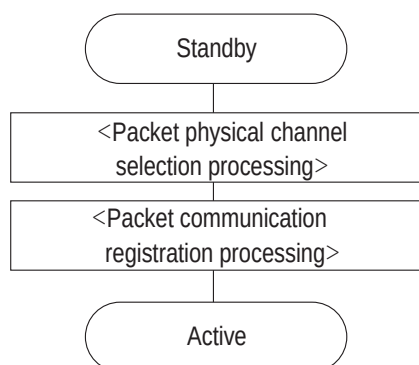
T.2.3 Operations for transition between states

T.2.3.1 Transition from Power OFF to Standby

To Enter the Standby state after the power for the MS is turned on, the MS performs the processing based on the operation described in Appendix B.1 and B.2.

T.2.3.2 Transition from Standby to Active

To switch from the Standby state to the Active state, the MS performs the processing as shown in the figure below.



Note 1 : The [Communication state] information element in the [[Packet Communications Registration Request]] message is set to the 'Communication initial state'. [Communication mode] information element is set to the Active • 'Continuous reception' or 'Intermittent reception'.

Fig. T.5 : Transition from Standby to Active

T.2.3.3 Transition from Standby to Packet standby

To switch from the Standby state to the Packet standby state, the MS performs the processing shown in the figure below.

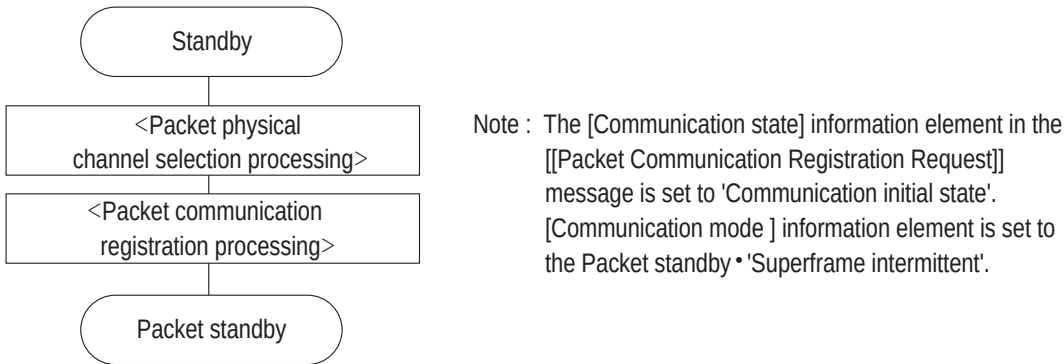


Fig. T.6 : Transition from Standby to Packet standby

T.2.3.4 Transition between Active and Packet standby

To switch from the Active state to the Packet standby state or from the Packet standby state to the Active state, the mobile station performs the processing as shown in the figure below.

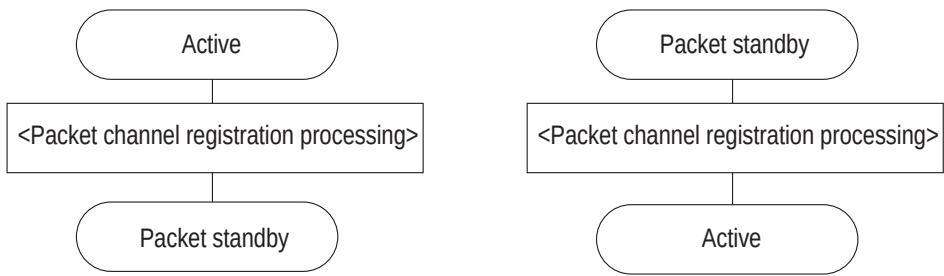


Fig. T.7 : Transition between Active and Packet standby

T.2.3.5 Packet communication ending

To switch from the Activate state or the Packet standby state to the Standby state, the mobile station performs the processing as shown in the figure below.

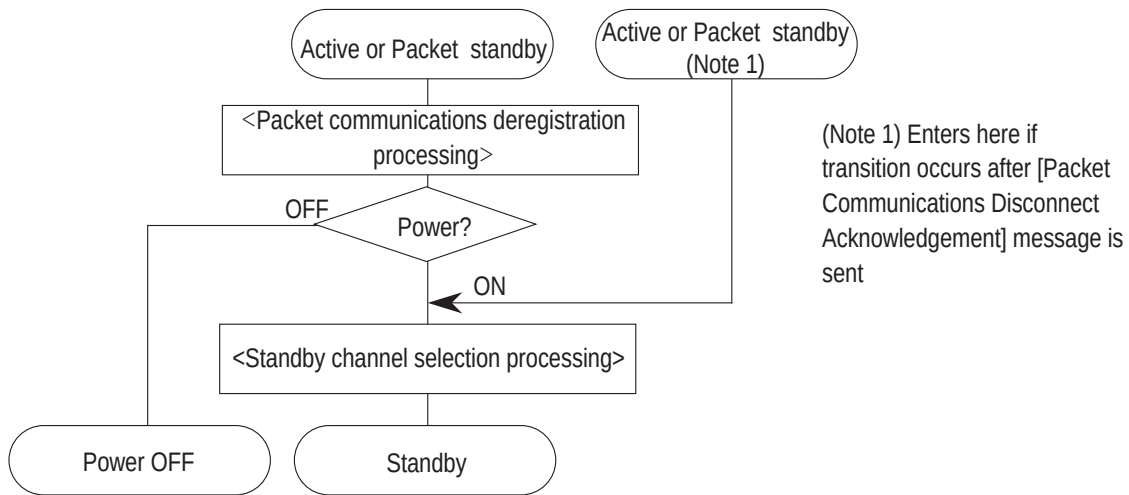


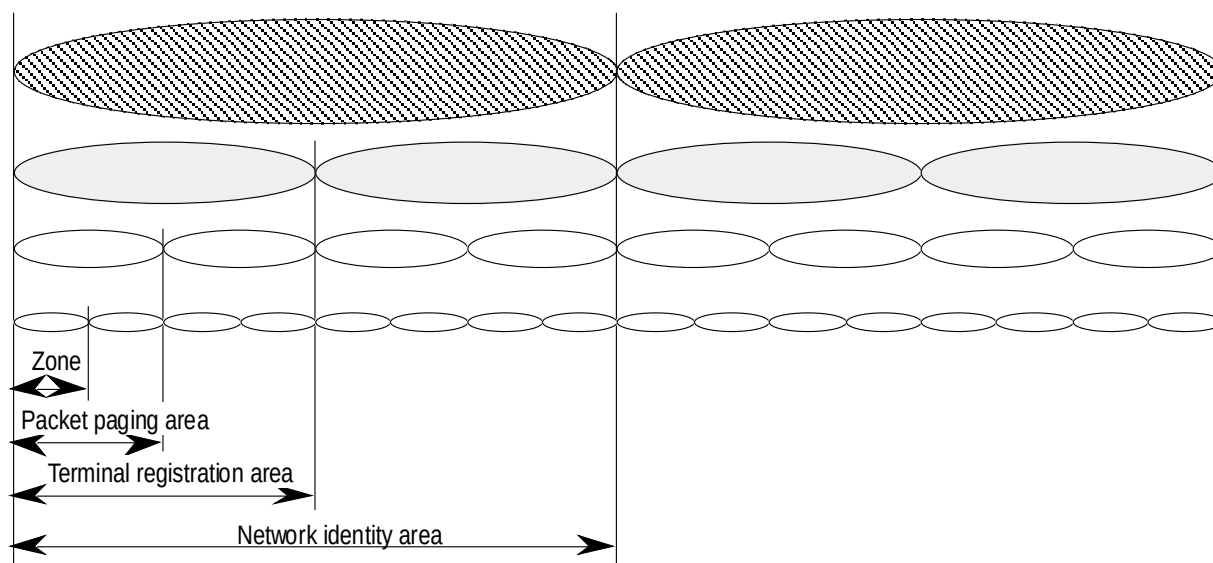
Fig. T.8 : Packet communication

T.3 Zone structure and zone switching operation during packet communications

During packet communications, the mobile station operates based on the following logical control areas in addition to the zones.

- Network identity area (areas identified by the Network identity information element)
- Terminal registration area (areas identified by the Terminal registration area code information element)
- Packet paging area (areas identified by the Packet paging area code information element)

The relationship between each control area is shown below.



- Network identity areas

Areas set by each operator who provides services

- Terminal registration areas

Areas where data link and packet communication registration state are maintained and one or more terminal registration areas can be set by zone units within one operator area.

- Packet paging areas

Areas where paging is performed in the Packet standby state and one or more packet paging areas can be set by zone units in one terminal registration area.

Fig. T.9 : Structure of control areas

The network notifies the information (codes, etc.) which identify each control area to the mobile station by the [[Broadcast Information]] message, the [[Zone Information Notification]] message or [[Packet System Information Notification]] message.

The MS compares the codes contained in the message which is received with each code value memorized in the MS at channel switching and then performs the following processing depending on the changes of the code. If multiple codes change, the mobile station places higher priority on upper codes and performs processing accordingly.

The configuration concept and the group calculation method of “Packet paging area code multiplex number” and “Packet paging area code” information elements which are notified in Broadcast Information, Zone Information Notification or Packet System Information Notification messages, shall follow the description for the location registration area described in F.2 and F.3 of Appendix F. At referring these sections, “Packet paging area code multiplex number” and “Packet paging area code” information elements shall replace “Number of location registration areas” information element and Location registration area code, respectively. In addition to this, the operation of the MS during packet communication shall be as described below, and the packet paging area code memorized in the MS shall be invalidated after the packet communication ends.

- (1) The location registration procedure described in F.2.2(1) and (2) of Appendix F shall be replaced with the packet channel registration procedure, then performed accordingly.
- (2) The operation after the MS initiates packet channel registration.
 - i) When the MS receives a Packet Channel Registration Response message:
Packet channel registration was completed successfully and the packet channel area code is memorized in the memory of the MS.
 - ii) When the MS receives a Packet Channel Registration Reject message:
This means that packet channel registration has failed and the memory of the MS will not be updated.
 - iii) When no message is received:
This is the case when the MS receives no response (neither Packet Channel Registration Response, Packet Channel Registration Reject, nor retransmitted messages) in reply to the Packet Channel Registration Request. In such cases, the MS clears the memory and executes the corresponding processing according to the network identity, terminal registration area code, packet paging area code and UPCH functional channel for the packet physical channel included in the message that the MS receives next.
If the reception condition of the UPCH deteriorates and the radio connection is disconnected as stipulated in Section 4.1, the MS immediately shall activate the packet physical channel starting with “Selection of the channel for the standby state” (Appendix B.2)
- (3) During the packet channel registration procedure (including a maximum of 3 retransmissions), the MS must not reselect a cell. Also, before the result defined in (2) above is evaluated, the MS must not update the pertinent information.

Table 1 Relationship between control areas and codes and the processing performed by the mobile station

State of mobile station Changed code (switched control area etc.)	Active	Packet standby
Network identity (network identity area)	<Packet communication registration processing>	<Packet communication registration processing>
Terminal registration area code (terminal registration area)	<Packet communication registration processing>	<Packet communication registration processing>
Packet paging area code (packet paging area)	<Packet communication registration processing>	<Packet channel registration processing>
UPCH functional channel	<Packet channel registration processing>	No processing (Note)

Note : If the mobile station in Packet standby state cannot use the channel Calculated from the mobile station identifier (MSI) based on the selection algorithm specified in Section 4.1.10.3 and re-selects another channel in the zone, a packet channel registration shall be performed.

T.4 An example of information flow for notification at reference point R of the channel switching between terminal registration areas (for the case of Packet communication environment with DTE notified.)

The standard specifies that the network notifies the user whether or not the packet communication mobile station needs to notify the data terminal DTE that the channel has been switched between the terminal registration areas by using the Packet communication environment information element in the Packet Communication Registration Response message.

When the packet communication mobile station needs to report the channel switching to the data terminal DTE, the Packet communication environment information element included in the packet Communication Registration Response message sent from the network to the user is set to "Method 2 (DTE notified)" and notification of channel switching between terminal registration areas from the packet communication mobile station to the data terminal DTE is mandatory. In this instance, the packet communication mobile station performs the Packet Communication Registration Request at the timing of start-up by the data terminal and reestablishes the data link.

An example of information flow of notification the channel switching between terminal registration areas when the Packet communication environment information element is set to Method 2 (DTE notified) is given in the following sections. Also, the information flows at point R when the data terminal DTE starts packet communication and cancels packet communication are explained.

T.4.1 Starting packet communication

The mobile station sends the Packet Communication Request to the network according to the Packet communication start request from the data terminal. In return the network performs the authentication processing for the mobile station and sends the Packet Communication Registration Response. When the mobile station receives this Response, it notifies the data terminal that packet communication is enabled.

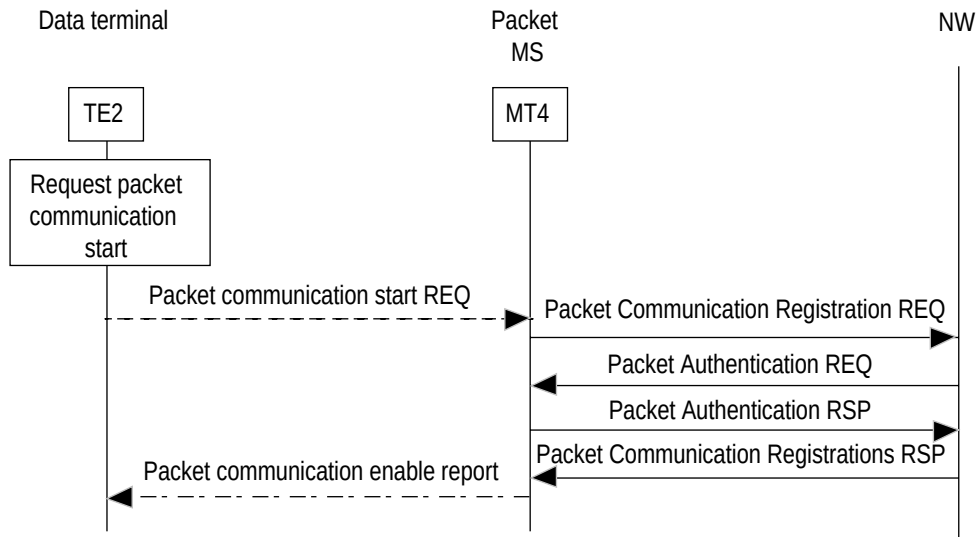


Fig. T.10 : Activation of packet communication

T.4.2 Canceling packet communication

The mobile station sends the Packet Communication Deregistration Request to the network according to the Packet communication abort request from the data terminal. In return the network performs the authentication processing for the mobile station and sends the Packet Communication Deregistration Response. When the mobile station receives this Response, it notifies the data terminal that packet communication is disabled.

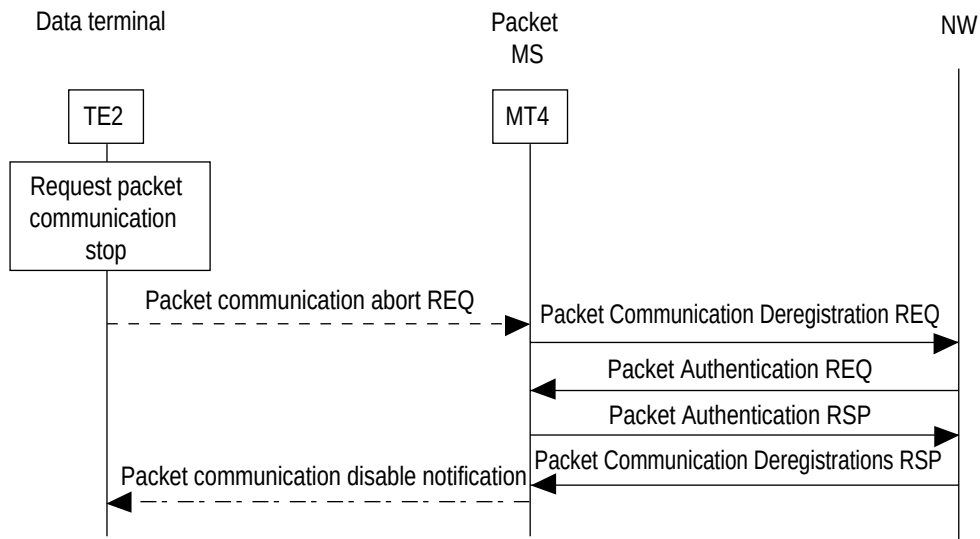


Fig. T.11 : Deactivation of packet communication

T.4.3 Channel switching between terminal registration areas

When the mobile station recognizes that the UPCH to which the MS will switch is in a different registration area, it notifies the data terminal that packet communication is disabled (halted). If the data terminal continues to transmit the Packet Communication Start Request even after the data terminal is notified that packet communication is disabled (halted), the mobile station sends the Packet Communication Registration Request to the network to re-establish the data path. When the mobile station receives the Packet Communication Registration response from the network, it notifies the data terminal that packet communication is enabled.

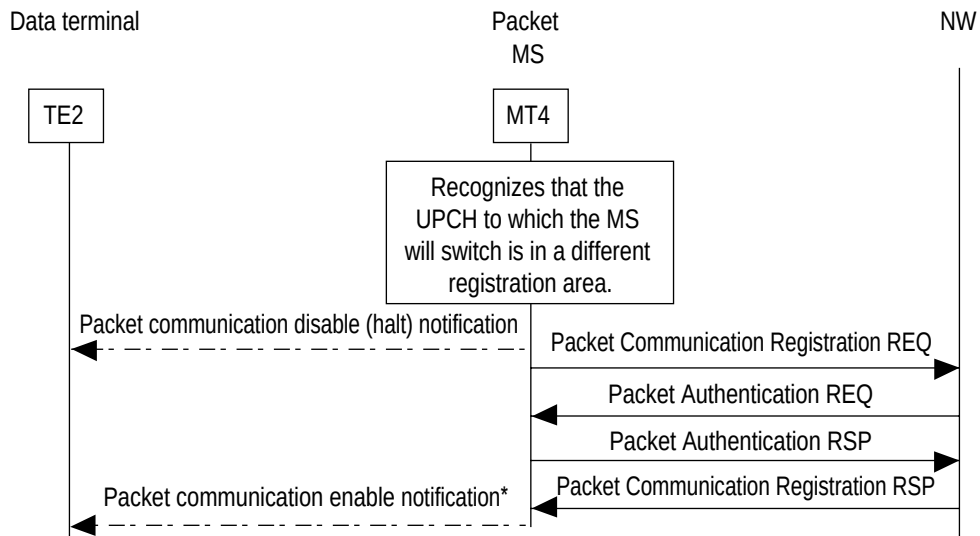


Fig. T.12 : Channel switching between terminal registration areas

Appendix U Restriction Control for Packet Communication Physical channel

U.1 Overview

This appendix describes the restriction control for the packet communication physical channel and the operation of the mobile station. The [Communication Disable Channel] used in this appendix refers to packet communication physical channels which cannot be selected by the mobile station due to restriction setting or other causes (detection of radiowave cut-off, etc.); the [Communication Enable Channel] refers to packet communication physical channels that can be selected by the mobile station.

U.2 Selection restriction

(1) Overview

This restriction is used by the network to restrict selection of packet communication physical channels by the mobile station.

(2) Operation

The network assigns the [Channel Restriction Information] information element according to the state of each packet communication physical channel and reports this information element in the [[Broadcasting Information]], [[Zone Information Notification]] or [[Packet System Information Notification]] message.

On reception of the [[Broadcasting Information]], [[Zone Information Notification]] or [[Packet system Information Notification]] message ^(note), the mobile station determines the restriction condition for each packet communication physical channel by the [Channel Restriction Information] information element.

- When the [channel restriction information] information element is set for '100%', it is determined as a communication disable channel.
- When the [channel restriction information] information element is set for 'less than 100%', the mobile station calculates uniform random number X within a range of 1 to 100 and compares the X with the [Channel Restriction Information] information element to determine whether the channel is selectable or not.
- When the channel restriction information is less than X, the channel is determined as a communication enable channel.
- When the channel is equal to or more than X, the channel is determined as a communication disable channel.

Note : This operates when another packet communication physical channel is selected due to [U.3 Usage Restriction].

U.3 Usage restriction

(1) Overview

This restriction is used by the network to restrict use of a packet communication physical channels for the mobile station and force the mobile station to switch to another packet communication physical channel.

(2) Operation

The network transmits the [[Packet System Information Notification]] message containing the [Operation Probability] information element.

The mobile station receives the [[Packet System Information Notification]] message and calculates the uniform random number X within a range of 1 to 100 and then compares the X with the [Operation Probability] information element.

- When the operation probability is equal to or more than X , the mobile station selects another packet communication physical channel.
- When the operation probability is less than X , the mobile station remains on the pertinent packet communication physical channel and performs packet communications.

Appendix V Independent notification procedure for packet physical channel

V.1 Overview

This appendix describes the method of notifying the mobile station of a request individually for packet physical channel handover to each mobile station during packet communication when necessary, and the operation of the mobile station upon receipt of this request.

Note that the 'basic channel' used in this appendix refers to the packet physical channels which are commonly notified to all mobile stations in the zone by means of [[Broadcast Information]] message and also by [[Zone Information Notification]] message or [[Packet System Information Notification]] message. The 'extended channel' refers to the packet physical channels which are notified individually by means of the [[Packet Channel Handover Request]] message or the [[Zone Information Notification]] message when necessary, and these channels are those indicated by the 'Number of extended channels' information element in the messages.

V.2 Notification method

V.2.1 Overview

The network is capable of notifying the mobile station during packet communication of a request to handover to the packet physical channel to be used for communication when necessary.

V.2.2 Packet physical channel handover notification method

When requesting the mobile station during packet communication individually to handover the packet physical channel, the network can notify this request to the mobile station by the following methods:

(1) During packet channel handover request processing

The network notifies the mobile station of the channels in the current zone by the [[Packet Channel Handover Request]] message via the active packet physical channel .

Note that this message can be notified with the [[Packet Communication Registration Response]] message or [[Packet Channel Registration Response]] message by means of the common platform function.

(2) During zone information notification processing

During the zone information notification processing by which the mobile station obtains the information on the peripheral zones via the active packet physical channel, the channels of the peripheral zones are notified by the [[Zone Information Notification]] message.

V.3 Operation of mobile station

V.3.1 Overview

The operation of the mobile station upon receipt of a handover request of the packet physical channel from the network is stipulated as below:

V.3.2 Packet physical channel selection

If a handover request of the packet physical channel to be used is notified by means of the [[Packet Channel Handover Request]] message, the mobile station performs the operation defined in Table V.3.2.1, if the request is notified by the [[Zone Information Notification]] message, the mobile station performs the operation according to Table V.3.2.2, depending on the methods described in V.2.2. If selecting a packet physical channel in the zone as defined in Section 4.1.10.4.2, the basic channel and extended channel groups are maintained independently so that extended channels are not included during the basic channel selection, and basic channels are not included during the extended channel selection.

If information of multiple channels of other zones was notified by the message described above, the zone is selected from among the zones having the reception levels higher than the [Communication level] in the order of the higher reception level to the lower reception level.

[Channel handover information]		[Number of reported zones] in other zones	Number of basic channels in other zones	[Number of extended channels] in other zones	Mobile station selection
'Perch channel re-selection'	'Number of extended channels' in the current zone				
Performed	-	-	-	-	Perch channel scanning
Not Performed	0	0	0	-	Basic channels in the current zone (1-Npc)
Not Performed	1-n	0	0	-	Extended channels in the current zone (1-n) --> Previous basic channel in the current zone
Not Performed	0	1-N	1-Npc	0	Basic channels in zone 1 (1-Npc) --> Basic channels in zone 2 (1-Npc) -->... Basic channels in zone N (1-Npc) --> Previous basic channel in the current zone
Not Performed	0	1-N	1-Npc	1-n	Extended channels in zone 1 (1-n) --> Basic channels in zone 1 (1-Npc) --> Extended channels in zone 2 (1-n) --> Basic channels in zone 2 (1-Npc) -->... Extended channels in zone N (1-n) --> Basic channels in zone N (1-Npc) --> Previous basic channel in the current zone
Not Performed	1-n	1-N	1-Npc	0	Extended channels in the current zone (1-n) --> Basic channels in zone 1 (1-Npc) --> Basic channels in zone 2 (1-Npc) -->... Basic channels in zone N (1-Npc) --> Previous basic channel in the current zone
Not Performed	1-n	1-N	1-Npc	1-n	Extended channels in the current zone (1-n) --> Extended channels in zone 1 (1-n) --> Basic channels in zone 1 (1-Npc) --> Extended channels in zone 2 (1-n) --> Basic channels in zone 2 (1-Npc) -->... Extended channels in zone N (1-n) --> Basic channels in zone N (1-Npc) --> Previous basic channel in the current zone

Table V.3.2.1 Operation on reception of Packet Channel Handover Request

[Number of reported zones]	Number of basic channels in other zones	[Number of extended channels] in other zones	Mobile station selection
0	-	-	Perch channel scanning
1-N	1-Npc	-	Basic channels in zone 1 (1-Npc) --> Basic channels in zone 2 (1-Npc) -->... Basic channels in zone N (1-Npc) --> Perch channel scanning
1-N	1-Npc	0	Basic channels in zone 1 (1-Npc) --> Basic channels in zone 2 (1-Npc) -->... Basic channels in zone N (1-Npc) --> Perch channel scanning
1-N	1-Npc	1-n	Extended channels in zone 1 (1-n) --> Basic channels in zone 1 (1-Npc) --> Extended channels in zone 2 (1-n) --> Basic channels in zone 2 (1-Npc) -->... Extended channels in zone N (1-n) --> Basic channels in zone N (1-Npc) --> Perch channel scanning

Table V.3.2.2 Operation on reception of Zone Information Notification

V.3.3 Registration

V.3.3.1 Channel handover is successful

When the handover channel was activated successfully as a result of packet physical channel selection according to V.3.2, the mobile station certainly performs the <packet communication registration processing> or <packet channel registration processing> and starts transmission/reception of packets on the pertinent channel thereafter.

V.3.3.2 Channel handover fails

When registration processing on the handover channel has failed (due to TR339 time out, etc.) as a result of packet physical channel selection according to V.3.2, the mobile station switches the candidate channels sequentially according to V.3.2 and performs the <packet physical channel selection processing>. After registration processing is completed due to handover to the assigned channel, switching back to the previous channel, or re-selection following perch channel scanning, etc., the extended channel information which was notified by the [[Packet Channel Handover Request]] message or [[Zone Information Notification]] message before registration processing is invalidated, and the mobile station is allowed for selection only with the basic channel until the extended channel information is newly notified. In case where the mobile station failed to handover to the extended channel in the pertinent zone and activated the previous basic channel in the zone again, the mobile station shall necessarily perform the <packet communication registration processing> or <packet channel registration processing>. The [Registration identifier] included in the [[Packet Communication Registration Request]] message or [[Packet Channel Registration Request]] message at this point must be set to 'Re-registration'.

V.3.3.3 Frequency switching count

The 'Frequency switching count' which is set in the [Registration identifier] included in the [[Packet Communication Registration Request]] message or [[Packet Channel Registration Request]] message, indicates the number of times that the packet physical channels (basic channels and extended channels) and the control physical channels are tried to be activated until registration processing on a packet physical channel is completed. This number of times of activation of the packet physical channel and control physical channel is not dependent on the frequency value. Also, the timing to reset the 'Frequency switching count' is as follows:

- When packet communication ends.
- When switching from packet communication to voice communication mode.
- Upon receipt of the [[Packet Communication Registration Response]] message.
- Upon receipt of the [[Packet Channel Registration Response]] message.

V.3.4 Switching to Packet standby

When the mobile station does not transmit/receive packets for a long period of time on the extended channel and switches to the superframe intermittent reception mode upon time out of the Packet standby switching timer, switching to the Packet standby state on the extended channel is prohibited if the [Standby condition] in the latest [[Packet Channel Handover Request]] message or [[Zone Information Notification]] message is set to 'Standby not allowed', and the mobile station shall switch to a basic channel in the pertinent zone and performs <packet channel registration processing> to switch to the Packet standby state. If the [Standby condition] is set to 'Standby allowed', the mobile station performs <packet channel registration processing> to allow for switching to the Packet standby state.

V.3.5 Reception of [[Packet System Information Notification]] message

Upon receipt of the [[Packet System Information Notification]] message during packet communication on a basic channel, the mobile station, if the [Operation probability] information element included in this message does not pertain to the uniform random pattern X as a result of comparison with the uniform pattern X, continues communication on the basic channel as is. However, if this basic channel is not included in the [[Packet System Information Notification]] message, the mobile station selects another packet physical channel from among the notified channels.

If the [Operation probability] information element pertains to the uniform random pattern X as a result of the comparison, the mobile station selects the packet physical channel from among the notified channels.

Upon receipt of the [[Packet System Information Notification]] message during packet communication on the extended channel, the mobile station, if the [Operation probability] information element included in this message does not pertain to the uniform random pattern X as a result of comparison with the uniform pattern X, continues communication on the extended channel as is. If this extended channel is not included in the [[Packet System Information Notification]] message, the mobile station still continues communication on the same extended channel as is.

If the [Operation probability] information element pertains to the uniform random pattern X as a result of the comparison, the mobile station selects the notified packet physical channel.

Chapter 5 Speech Coding Method (CODEC)

5.1 Full-rate speech CODEC

5.1.1 Speech coding

The speech coding algorithm is a member of a class of speech coders known as Code Excited Linear Predictive Coding (CELP). These techniques use codebooks to vector quantize the excitation (residual) signal. The speech coding algorithm is a variation on CELP called Vector-Sum Excited Linear Predictive Coding (VSELP). VSELP uses a codebook which has a predefined structure such that the computations required for the codebook search process can be significantly reduced. VSELP calculates the reflection coefficients γ_i for determination of short term filter coefficients α_i from input speech signal, and frame energy $R(0)$ is determined, too. Then long term lag L and codeword I is computed, followed by computation of long term prediction coefficients β and gain γ . Moreover, processing stages for these computed parameters are necessary respectively before sending them to channel coding process described in later section.

5.1.1.1 Definitions and basic coder parameters

Figure 5.1.1.1-1 shows a block diagram of the speech decoder. These symbols in this figure 5.1.1.1-1 indicates the various parameters which must be determined and encoded by the speech coder.

These parameters are defined as follows:

α_i : short term filter coefficients

β_q : quantized long term predictor coefficients β

γ_q : quantized gain γ

L : long term lag

I : codeword selected from VSELP codebook

The remaining symbols are described as follows:

$b_L(n)$: output of long term filter state "codebook" for Lag L

$e_x(n)$: combined excitation signal

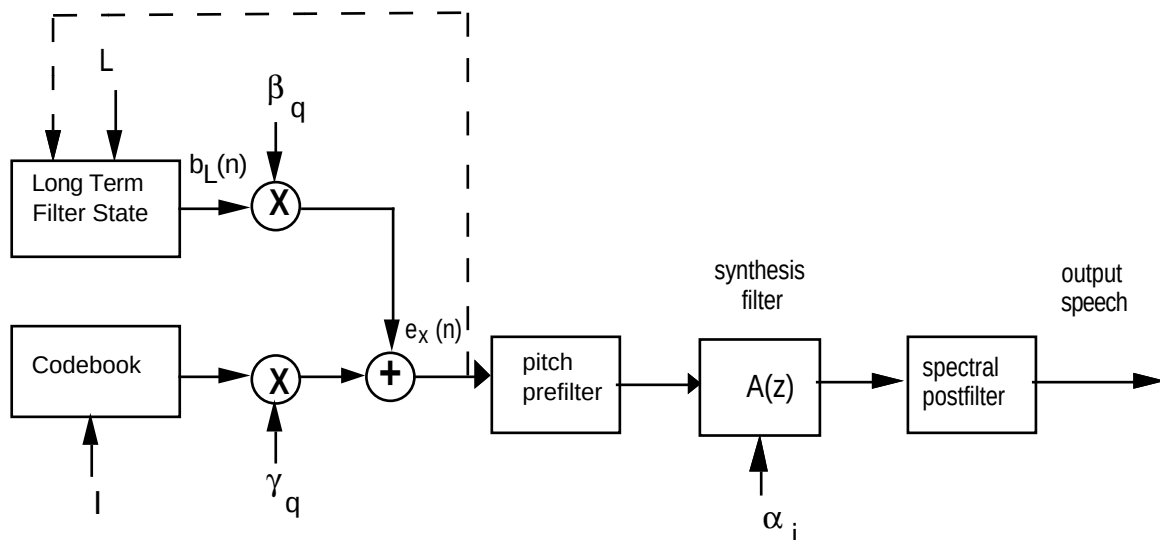


Fig. 5.1.1.1-1 Structure of the speech decoder

Table 5.1.1.1-1 shows the basic parameters for the 6700 bps speech coder and decoder where 20 ms for one speech frame consisting of 4 subframes (5 ms).

Table 5.1.1.1-1 Basic Parameters

Symbol	Description	data
	sampling rate	8 kHz
N_F	frame length	160 samples (20 ms)
N	subframe length	40 samples (5 ms)
N_P	short term predictor order	10
	long term predictor order	1
M	# of bits in codeword (# of basis vectors)	9

The basic data rate of the speech coder is 6700 bps. There are 134 bits per speech frame (20 ms) for the speech coder. These 134 bits are allocated as shown in the table 5.1.1.1-2:

Table 5.1.1.1-2 Bit allocation of speech frame

Parameter	bits/subframe	bits/frame
short term filter coefficients, α_i		36bits/frame
soft-interpolation bit		1 bit/frame
frame energy, $R(0)$		5 bits/frame
lag, L	7 bits/subframe	28 bits/frame
codewords, l	9 bits/subframe	36 bits/frame
gains β, γ	7 bits/subframe	28 bits/frame

The table 5.1.1.1-3 shows the list of all the parameter codes transmitted per speech frame per 20 ms. for channel coding described later. The 10 short term filter coefficients are transformed to 10 reflection coefficients which are coded for transmission. The gains β, γ are transformed to the parameters GS and $P0$ which are vector quantized together as one 7 bit code. The following code descriptions correspond to the parameters in the preceding table 5.1.1.1-2.

Table 5.1.1.1-3 List of parameters sent per speech frame

Symbol	Bit	Description
R0	5 bits	frame energy
SOFINT	1 bit	soft interpolation bit
LPC1	5 bits	1st reflection coefficient
LPC2	5 bits	2nd reflection coefficient
LPC3	4 bits	3rd reflection coefficient
LPC4	4 bits	4th reflection coefficient
LPC5	4 bits	5th reflection coefficient
LPC6	3 bits	6th reflection coefficient
LPC7	3 bits	7th reflection coefficient
LPC8	3 bits	8th reflection coefficient
LPC9	3 bits	9th reflection coefficient
LPC10	2 bits	10th reflection coefficient
LAG_1	7 bits	lag for first subframe
LAG_2	7 bits	lag for second subframe
LAG_3	7 bits	lag for third subframe
LAG_4	7 bits	lag for fourth subframe
CODE_1	9 bits	codebook code, I, for first subframe
CODE_2	9 bits	codebook code, I, for second subframe
CODE_3	9 bits	codebook code, I, for third subframe
CODE_4	9 bits	codebook code, I, for fourth subframe
GSP0_1	7 bits	{GS, P0} code for first subframe
GSP0_2	7 bits	{GS, P0} code for second subframe
GSP0_3	7 bits	{GS, P0} code for third subframe
GSP0_4	7 bits	{GS, P0} code for fourth subframe

5.1.1.2 Audio interface

The following speech processing condition shall be satisfied prior to speech coding.

- (1) Transmit Level Adjustment
- (2) Bandpass Filter Insertion
- (3) Analog to Digital Conversion

5.1.1.2.1 Transmit level adjustment

The sensitivity shall be such that when a 97dBspl+/- 8 dB C-message white noise acoustic signal from an artificial mouth is applied to the audio-to-electric transducer of the subscriber unit, a value of 24 shall be produced in the R0 bit field of the transmit speech coder. R0 equal to 24 represents that the frame energy is 18dB down from full scale. For details see reference[1].

However, when a -20dBs+/-4dB C-message white noise (or ITU-T recommendation Artificial Voice P.50) is used as a input which produces a value of 24 in R0 bit for speech coder, the acoustic signal mentioned above may not be required.

The relationship between the transmit transducer output signal and the value in the transmit coder R0 field shall be such that a linear relationship within +/-2dB is maintained between the output from the transducer and the value in the R0 field for a +14,-28 dB range around the level which produced a value of 24 in R0.

5.1.1.2.2 Bandpass filter insertion

The filter is inserted for avoiding aliasing distortion of the input signal. The attenuation of the filter shall comply with ITU-T Red Book G.714 sending filter.

5.1.1.2.3 Analog to digital converter

The sampling rate of A/D conversion is 8 kHz. The A/D function shall be performed according to either of the following:

- * by direct conversion analog signal to a uniform PCM format with a minimum resolution of 13 bits.
- * or by converting analog signal to an 8-bit/ μ law format followed by a μ law/uniform code conversion.

The A/D conversion is based on the standard 8-bit/ μ law codec specified in ITU-T Red Book G.711.

The μ law/uniform code conversion is performed according to definition in ITU-T Red Book G.721 section 4.2.1 sub-block EXPAND. The parameter LAW shall be set to LAW=0.

5.1.1.3 Pre-processing

It may be desirable in some instances to provide additional high-pass filtering after analog to digital conversion. A fourth order Chebyshev type II highpass with a filter response which is 3 dB down at 120 Hz and 40 dB down at 60 Hz may be used. The transfer function of the high-pass filter is given by:

$$H_{hp}(z) = \frac{\sum_{i=0}^4 a_i z^{-i}}{1 - \sum_{i=1}^4 b_i z^{-i}} \quad (5.1.1.3-1)$$

where:

$$\begin{array}{ll}
 a_0 = 0.898025036 & b_1 = 3.78284979 \\
 a_1 = -3.59010601 & b_2 = -5.37379122 \\
 a_2 = 5.38416243 & b_3 = 3.39733505 \\
 a_3 = -3.59010601 & b_4 = -0.806448996 \\
 a_4 = 0.898024917 &
 \end{array}$$

5.1.1.4 Short-term predictor coefficients

The short-term filter is equivalent to the traditional LPC synthesis filter. The transfer function for the short-term filter is given by.

$$A(z) = \frac{1}{1 - \sum_{i=1}^{N_p} \alpha_i z^{-i}} \quad (5.1.1.4-1)$$

The short term predictor parameters are the α_i 's of the short term or synthesis filter. These are standard LPC direct form filter coefficients. The short term predictor parameters are computed from the input speech. No pre-emphasis is used. This analysis interval should be centered with respect to the center of the fourth subframe of each frame. The order of the predictor is 10 ($N_p = 10$).

5.1.1.4.1 Solution for reflection coefficients

An fixed point covariance lattice algorithm, FLAT, may be used for determination of the short -term filter coefficients. Let the samples of the input speech which fall in the analysis interval be represented by $s(n)$; $0 \leq n \leq N_A-1$. The analysis length used for computation of the parameters is 170 samples ($N_A = 170$).

Defining $b_j(n)$ to be the backward residual out of stage j of the inverse lattice filter and $f_j(n)$ to be the forward residual out of stage j of the inverse lattice filter we can define:

the autocorrelation of $f_j(n)$;

$$F_j(i, k) = \sum_{n=N_p}^{N_A-1} f_j(n-i) f_j(n-k) \quad (5.1.1.4.1-1)$$

the autocorrelation of $b_j(n-1)$ and:

$$B_j(i, k) = \sum_{n=N_p}^{N_A-1} b_j(n-i-1) b_j(n-k-1) \quad (5.1.1.4.1-2)$$

the cross correlation between $f_j(n)$ and $b_j(n-1)$.

$$C_j(i, k) = \sum_{n=N_p}^{N_A-1} f_j(n-i) b_j(n-k-1) \quad (5.1.1.4.1-3)$$

Let r_j represent the reflection coefficient for stage j of the inverse lattice filter. Then:

$$F_j(i, k) = F_{j-1}(i, k) + r_j (C_{j-1}(i, k) + C_{j-1}(k, i)) + r_j^2 B_{j-1}(i, k) \quad (5.1.1.4.1-4)$$

and

$$B_j(i, k) = B_{j-1}(i+1, k+1) + r_j (C_{j-1}(i+1, k+1) + C_{j-1}(k+1, i+1)) + r_j^2 F_{j-1}(i+1, k+1) \quad (5.1.1.4.1-5)$$

and

$$C_j(i, k) = C_{j-1}(i, k+1) + r_j (B_{j-1}(i, k+1) + F_{j-1}(i, k+1)) + r_j^2 C_{j-1}(k+1, i) \quad (5.1.1.4.1-6)$$

r_j can be expressed as:

$$r_j = -2 \frac{C_{j-1}(0, 0) + C_{j-1}(N_P - j, N_P - j)}{F_{j-1}(0, 0) + B_{j-1}(0, 0) + F_{j-1}(N_P - j, N_P - j) + B_{j-1}(N_P - j, N_P - j)} \quad (5.1.1.4.1-7)$$

The FLAT algorithm can now be stated as follows.

step 1. First compute the covariance (autocorrelation) matrix from the input speech:

$$\phi(i, k) = \sum_{n=N_P}^{N_A-1} s(n-i) s(n-k) \quad (5.1.1.4.1-8)$$

for $0 \leq i, k \leq N_P$.

(Note: see also 5.1.4.2).

$$\text{step 2. } F_0(i, k) = \phi(i, k) \quad 0 \leq i, k \leq N_P - 1 \quad (5.1.1.4.1-9)$$

$$B_0(i, k) = \phi(i+1, k+1) \quad 0 \leq i, k \leq N_P - 1 \quad (5.1.1.4.1-10)$$

$$C_0(i, k) = \phi(i, k+1) \quad 0 \leq i, k \leq N_P - 1 \quad (5.1.1.4.1-11)$$

step 3. set $j = 1$

step 4. Compute r_j using (5.1.4.1-7)

step 5. Quantize r_j (see section 5.1.4.3)

step 6. If $j = N_P$ then done.

$$\text{step 7. Compute } F_j(i, k) \quad 0 \leq i, k \leq N_P - j - 1 \quad \text{using (5.1.1.4.1-4)}$$

$$\text{Compute } B_j(i, k) \quad 0 \leq i, k \leq N_P - j - 1 \quad \text{using (5.1.1.4.1-5)}$$

$$\text{Compute } C_j(i, k) \quad 0 \leq i, k \leq N_P - j - 1 \quad \text{using (5.1.1.4.1-6)}$$

step 8. $j = j+1$; go to step 4.

This algorithm can be simplified by noting that the ϕ , F and B matrices are symmetric such that only the upper triangular part of the matrices need to be computed or updated. In addition $\phi(i, j)$ can be efficiently computed from $\phi(i-1, j-1)$. Also, if step 7 is done so that $F_j(i, k)$, $B_j(i-1, k-1)$, $C_j(i, k-1)$, and $C_j(k, i-1)$ are updated together, then common terms can be computed once and the recursion can be done in place.

5.1.1.4.2 Bandwidth expansion

The speech coder should provide for a small amount of bandwidth expansion of the short-term filter coefficients prior to quantization. Windowing of the autocorrelations prior to the solution of the reflection coefficients is one technique which may be used.

Prior to solving for the reflection coefficients, the ϕ array is modified by windowing the autocorrelation functions:

$$\phi'(i, k) = \phi(i, k)w(|i-k|) \quad (5.1.1.4.2-1)$$

The window used is a binomial window with an effective bandwidth of 80 Hz. The values of $w(i)$ are:

$w(0)$	1.000000
$w(1)$	0.999644
$w(2)$	0.998577
$w(3)$	0.996802
$w(4)$	0.994321
$w(5)$	0.991141
$w(6)$	0.987268
$w(7)$	0.982710
$w(8)$	0.977478
$w(9)$	0.971581
$w(10)$	0.965032

(5.1.1.4.2-2)

5.1.1.4.3 Quantization and encoding of coefficients

As stated in 5.1.1.4.1, the quantization of the reflection coefficients may be done within the FLAT recursion.

The reflection coefficients are quantized using codebooks designed for each reflection coefficient. The bit allocations for the reflection coefficient quantizers are shown in the table 5.1.1.4.3-1

Table 5.1.1.4.3-1 Bit allocation for the reflection coefficient

Refection	Bits
r_1	5 bits
r_2	5 bits
r_3	4 bits
r_4	4 bits
r_5	4 bits
r_6	3 bits
r_7	3 bits
r_8	3 bits
r_9	3 bits
r_{10}	2 bits

The 10 reflection coefficients ($r_1 - r_{10}$) which represent the short term predictor parameters are each independently quantized. Table 5.1.1.4.3-2 provides the 10 codebooks for quantization of the 10 reflection coefficients to provide codes LPC1 through LPC10. The quantization may be performed by finding the codebook value which is closest (minimum absolute error) from the unquantized reflection coefficient. The code for that codebook entry (which appears in the left column) is the code for that reflection coefficient. Note that no transformation (such as log area ratio or arc sine) of the reflection coefficients is required during the quantization process.

Table 5.1.1.4.3-2 Quantization codebook of reflection coefficient

32 QUANTIZATION LEVELS FOR r_1 FOLLOW:

0	-0.9824126363E+00
1	-0.9735090137E+00
2	-0.9624753594E+00
3	-0.9502775669E+00
4	-0.9373488426E+00
5	-0.9229213595E+00
6	-0.9075114727E+00
7	-0.8909863830E+00
8	-0.8725319505E+00
9	-0.8524736762E+00
10	-0.8291744590E+00
11	-0.8042194247E+00
12	-0.7773172855E+00
13	-0.7462986708E+00
14	-0.7086332440E+00
15	-0.6654584408E+00
16	-0.6129908562E+00
17	-0.5543900728E+00
18	-0.4910933077E+00
19	-0.4234648049E+00
20	-0.3498347402E+00
21	-0.2670036852E+00
22	-0.1731550395E+00
23	-0.5800925940E-01
24	-0.7369286567E-01
25	0.2092804909E+00
26	0.3275107443E+00
27	0.4384389818E+00
28	0.5466716886E+00
29	0.6594731212E+00
30	0.7793787718E+00
31	0.9039739370E+00

32 QUANTIZATION LEVELS FOR r_2 FOLLOW:

0	-0.7547348738E+00
1	-0.5826729536E+00
2	-0.4569368660E+00
3	-0.3481135964E+00
4	-0.2492762953E+00
5	-0.1585778296E+00
6	-0.7726432383E-01
7	-0.5096863955E-02
8	0.6527176499E-01
9	0.1329884380E+00
10	0.1978287548E+00
11	0.2600678802E+00
12	0.3186267912E+00
13	0.3747462034E+00
14	0.4288900495E+00
15	0.4810178876E+00
16	0.5302551389E+00

17	0.5760165453E+00
18	0.6193220615E+00
19	0.6593915224E+00
20	0.6967787147E+00
21	0.7315257788E+00
22	0.7650170326E+00
23	0.7966732979E+00
24	0.8267812133E+00
25	0.8543012142E+00
26	0.8798117638E+00
27	0.9037305117E+00
28	0.9251338840E+00
29	0.9448361397E+00
30	0.9636774063E+00
31	0.9816107750E+00

16 QUANTIZATION LEVELS FOR r_3 FOLLOW:

0	-0.8205619454E+00
1	-0.7210606337E+00
2	-0.6306993961E+00
3	-0.5443551540E+00
4	-0.4633429348E+00
5	-0.3844017684E+00
6	-0.3091120422E+00
7	-0.2334560007E+00
8	-0.1566459835E+00
9	-0.7615681738E-01
10	0.7552316878E-02
11	0.9473155439E-01
12	0.1914243251E+00
13	0.3015856445E+00
14	0.4329566360E+00
15	0.6044923663E+00

16 QUANTIZATION LEVELS FOR r_4 FOLLOW:

0	-0.4505536556E+00
1	-0.2582354248E+00
2	-0.1276191175E+00
3	-0.1891620830E-01
4	0.7355944812E-01
5	0.1581759751E+00
6	0.2381238639E+00
7	0.3138437271E+00
8	0.3864109516E+00
9	0.4573097229E+00
10	0.5282919407E+00
11	0.5986876488E+00
12	0.6679506302E+00
13	0.7368153930E+00
14	0.8054329753E+00
15	0.8715001345E+00

16 QUANTIZATION LEVELS FOR r_5 FOLLOW:

0	-0.6570090652E+00
1	-0.5339469314E+00
2	-0.4366474450E+00
3	-0.3503953516E+00
4	-0.2729992270E+00
5	-0.2044571489E+00
6	-0.1395732164E+00
7	-0.7738068700E-01
8	-0.1441930979E-01
9	0.4928108677E-01
10	0.1149446666E+00
11	0.1853068322E+00
12	0.2629969418E+00
13	0.3558021188E+00
14	0.4658669233E+00
15	0.6091661453E+00

8 QUANTIZATION LEVELS FOR r_6 FOLLOW:

0	-0.3351013660E+00
1	-0.1157702208E+00
2	0.3265872598E-01
3	0.1545225680E+00
4	0.2685563564E+00
5	0.3836385608E+00
6	0.5067840815E+00
7	0.6525010467E+00

8 QUANTIZATION LEVELS FOR r_7 FOLLOW:

0	-0.5834863186E+00
1	-0.4363777936E+00
2	-0.3007811308E+00
3	-0.1750739664E+00
4	-0.5848890170E-01
5	0.5718512833E-01
6	0.1880214661E+00
7	0.3601035774E+00

8 QUANTIZATION LEVELS FOR r_8 FOLLOW:

0	-0.3832900822E+00
1	-0.2111644298E+00
2	-0.7242984325E-01
3	0.4901395738E-01
4	0.1637183726E+00
5	0.2812094688E+00
6	0.4174390733E+00
7	0.5824319720E+00

8 QUANTIZATION LEVELS FOR r_9 FOLLOW:

0	-0.5373110771E+00
1	-0.3927696943E+00

2	-0.2594878078E+00
3	-0.1391013712E+00
4	-0.2935140580E-01
5	0.8255130798E-01
6	0.2107033432E+00
7	0.3958446980E+00

4 QUANTIZATION LEVELS FOR r_{10} FOLLOW:

0	-0.1508204788E+00
1	0.4652437568E-01
2	0.1958054304E+00
3	0.3621688187E+00

5.1.1.4.4 Conversion to direct form coefficients

The quantized reflection coefficients must be converted to direct form filter coefficients, α_i 's.

5.1.1.4.5 Soft interpolation of coefficients

Two sets of LPC coefficient vectors are generated. The first corresponds to an interpolated coefficients set, the second to an uninterpolated coefficient set. The frame's speech samples are inverse filtered by each of the two coefficient sets, and the residual frame energy corresponding to each set is computed. The coefficient set yielding the lower frame residual energy is then selected to be used. If the residual energies are equal, the uninterpolated coefficient set is used. SOFINT, the soft interpolation bit, is set to 1 when interpolation is selected or to 0 otherwise.

5.1.1.4.5.1 Interpolated coefficients

To generate the interpolated coefficient set, the speech coder linearly interpolates the a_i 's for the first, second and third subframes of each frame. The fourth subframe uses the uninterpolated α_i 's for that frame. For all α_i 's:

$$\begin{aligned}
 \alpha_i &= .75\alpha_i(\text{previous}) + .25\alpha_i(\text{current}) && \text{for subframe 1} \\
 \alpha_i &= .5 \alpha_i(\text{previous}) + .5 \alpha_i(\text{current}) && \text{for subframe 2} \\
 \alpha_i &= .25\alpha_i(\text{previous}) + .75\alpha_i(\text{current}) && \text{for subframe 3} \\
 \alpha_i &= \alpha_i(\text{current}) && \text{for subframe 4}
 \end{aligned}
 \tag{5.1.1.4.6.1-1}$$

where $\alpha_i(\text{previous})$ is the i^{th} direct form coefficient from the previous frame and $\alpha_i(\text{current})$ is the i^{th} direct form coefficient from the current frame.

For interpolated subframes, the α_i 's are converted to reflection coefficients to check for filter stability. If the resulting filter is unstable (any reflection coefficient having a magnitude equal to or greater than 1.0,) then uninterpolated coefficients are used for that subframe.

5.1.1.4.5.2 Uninterpolated coefficients

The uninterpolated coefficients used for subframe 1 are the previous frame's coefficients. The uninterpolated coefficients used for subframe 3 are the current frame's coefficients. Subframe 2 uses either the previous frame's or the current frame's coefficients, choosing those coefficients which correspond to the frame with the higher value for R_0 (see section 5.1.1.5). If both frames have the same R_0 then subframe 2 uses the previous frame's coefficients.

5.1.1.5 Frame energy

An energy value is computed and encoded once per frame. This energy value, $R(0)$, reflects the average signal power in the input speech over a 20 msec. interval which is centered with respect to the middle of the fourth subframe.

5.1.1.5.1 Computation of frame energy

$R(0)$, may be computed during the computation of the short term predictor parameters.

$$R(0) = \frac{\phi(0,0) + \phi(N_P, N_P)}{2(N_A - N_P)} \quad (5.1.1.5.1-1)$$

where $\phi(i,k)$ is defined by (5.1.4.1-8). Equation 5.1.5.1-1 can be rewritten as:

$$R(0) = \frac{\phi(0,0) + \phi(10,10)}{320} \quad (5.1.1.5.1-2)$$

5.1.1.5.2 Quantization and encoding of frame energy

$R(0)$ is converted into dB relative to full scale (full scale, $R_{\max}$, is defined as the square of the maximum sample amplitude).

$$R_{dB} = 10 \log_{10} (R(0) / R_{\max}) \quad (5.1.1.5.2-1)$$

R_{dB} is then quantized to 32 levels. The 32 quantized values for R_{dB} range from a minimum of -66 (corresponding to a code of 0 for R_0) to a maximum of -4 (corresponding to a code of 31 for R_0). The step size of the quantizer is 2 (2 dB steps). R_0 is obtained by minimizing the following equation (5.1.1.5.2-2)

$$\text{abs}(R_0 - (R_{dB} + 66)/2) \quad (5.1.1.5.2-2)$$

where R_0 can take on the integer values from 0 to 31 corresponding to the 32 codes for R_0 .

The quantized value of R_0 , $R_q(0)$ is given by:

$$R_q(0) = R_{\max}(10^{(2R_0-66)/10}) \quad (5.1.1.5.2-3)$$

5.1.1.5.3 Interpolation of frame energy

Define $R'_q(0)$ to be the quantized value of $R(0)$ to be used for the subframe and $R_q(0)$ to be the quantized value of $R(0)$. Then:

$$R'_q(0) = R_q(0)_{\text{previous frame}} \quad \text{for subframe 1} \quad (5.1.1.5.3-1a)$$

$$R'_q(0) = R_q(0)_{\text{current frame}} \quad \text{for subframes 3 and 4} \quad (5.1.1.5.3-1b)$$

$$R'_q(0) = \max [R_q(0)_{\text{previous frame}}, R_q(0)_{\text{current frame}}] \quad \text{for subframe 2} \quad (5.1.1.5.3-1c)$$

5.1.1.6 Subframe processing

The 20 msec. speech frame is subdivided into four 5 msec. subframes. For each subframe the speech coder must determine and code the long-term predictor lag, L , the codeword, I , and the gains, β and γ

5.1.1.6.1 Weighting of input speech

The speech coder utilizes a perceptual noise weighting filter of the form:

$$W(z) = \frac{1 - \sum_{i=1}^{N_p} \alpha_i z^{-i}}{1 - \sum_{i=1}^{N_p} \alpha_i \lambda^i z^{-i}} \quad (5.1.1.6.1-1)$$

where α_i 's are the filter coefficients for the subframe (see 5.1.1.4.5) and λ is the noise weighting parameter. A value of $\lambda = .8$ may be used. The input speech for the subframe must be filtered by the perceptual weighting filter. The filter is implemented as a cascade of a 10th order inverse (all pole) direct form filter corresponding to the numerator of equation 5.1.1.6.1-1, followed by a 10th order (all pole) direct form filter corresponding to the denominator. The filter coefficients will change for each subframe. The states of the filters should be preserved from one subframe to the next. (Note: other weighting filter implementations may adversely affect performance. Consequently, care must be taken to achieve or exceed the performance level that is realized by the above filter configuration.)

5.1.1.6.2 Subtraction of zero input response

The speech encoder is the coding system which is utilizing a synthesis is coding system (A-b-s.). Therefore a version of the speech decoder is used in the speech encoder. The form of the synthesis filter used in the speech encoder is given by:

$$H(z) = \frac{1}{1 - \sum_{i=1}^{N_p} \alpha_i \lambda^i z^{-i}} \quad (5.1.1.6.2-1)$$

Note that the synthesis filter used in the speech encoder is different than that used in the speech decoder. The synthesis for the speech encoder includes the same noise weighting parameter, λ , as the weighting filter in section 5.1.1.6.1. The synthesis filter used in the speech coder is therefore called the weighted synthesis filter. A weighted synthesis filter is used to match the weighting applied to the input speech. The weighted synthesis filter will have a filter state associated with it at the start of each subframe. In order to remove the effects of the weighted synthesis filter's initial state from the subframe parameter determinations, the zero input response of the weighted synthesis filter shall be computed and subtracted from the weighted input speech for the subframe. The weighted synthesis filter should be implemented with a direct form filter.

5.1.1.6.3 Long-term predictor lag

The speech coder uses the closed loop approach to choosing the long term predictor lag. In the closed loop case the lag is determined from only past output of the long term filter and the current input speech. The long term filter response can be expressed as:

$$B_n(z) = \frac{1}{1 - \beta z^{-\lfloor \frac{n+L}{L} \rfloor L}} \quad (5.1.1.6.3-1)$$

where $\lfloor x \rfloor$ is the floor function of x which evaluates to the largest integer $\leq x$ and n is the sample number in the subframe; $0 \leq n \leq N-1$. The lag L is used as the first L samples of the subframe. If $L < N$ then a lag etc of $2L$ is used for samples $n = L$ to $2L-1$. In this way the lag is always greater than n , and used for only long term filter states existing at the start of the subframe are used.

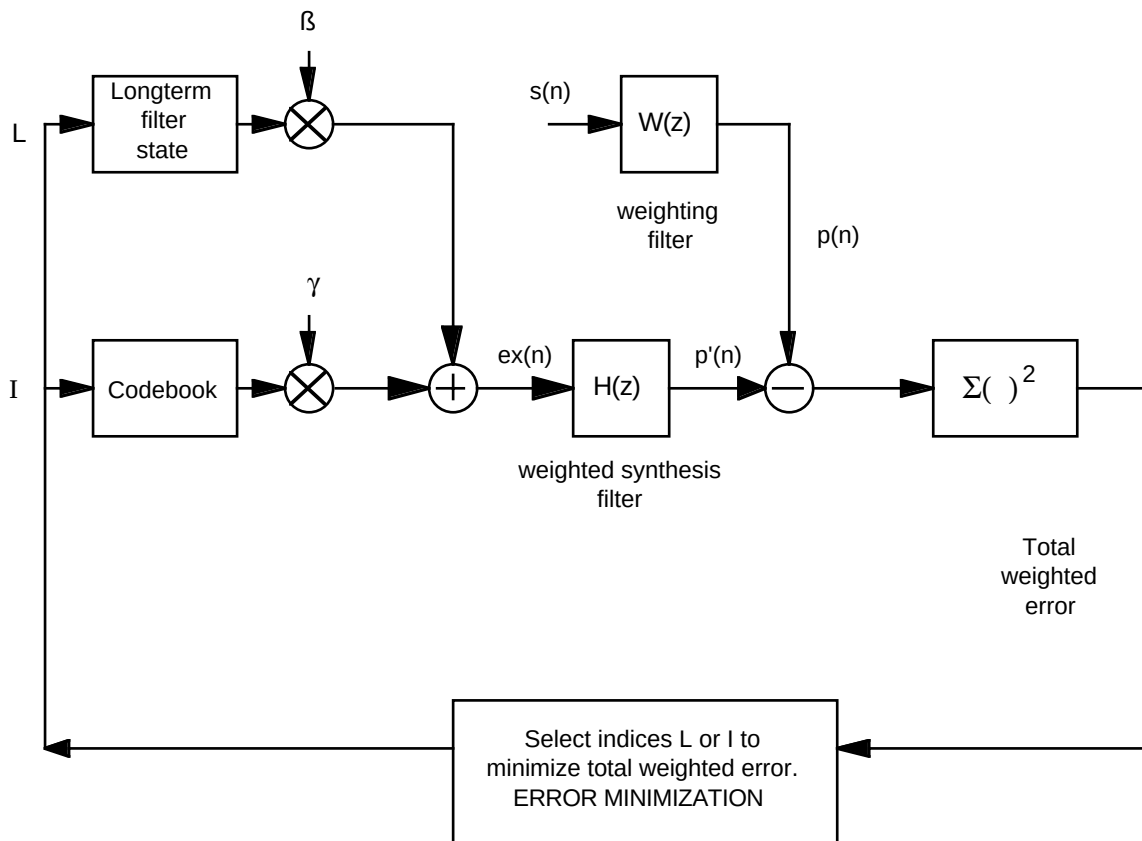


Fig. 5.1.1.6.3-1 Analysis by Synthesis Procedure for Long Term Predictor Lag and Code Search

Fig. 5.1.1.6.3-1 illustrates how the long term predictor lag search and code search can be formulated utilizing the long term filter just described. The input $p(n)$ in this case is the value which is obtained by subtracting the weighted input speech for the subframe from the zero input response of the weighted synthesis filter, $H(z)$. The long term lag optimization looks just like a codebook search where this long term filter state codebook is defined by the long term filter state and the specific vector in the long term filter state codebook is specified the long term predictor lag, L . These two "codebook" (the long term filter state and the excitation codebook) are searched sequentially. The long term predictor lag is determined first assuming no input from the excitation codebook.

5.1.1.6.3.1 Computation of lag

Necessary parameters for computation of lag is defined as follows:

- $L_{\min}$: minimum possible value for long term lag L ($L_{\min} = 20$)
- $r(n)$: long term filter state; $n < 0$ (past outputs of long term filter)
- $b_L(n)$: output of long term filter state “codebook” for lag L
- $h(n)$: impulse response of $H(z)$
- $b'_L(n)$: $b_L(n)$ filtered by $H(z)$ (convolved with $h(n)$)

$$G_L = \sum_{n=0}^{N-1} (b'_L(n))^2 \quad (5.1.1.6.3.1-1)$$

$$C_L = \sum_{n=0}^{N-1} b'_L(n) p(n) \quad (5.1.1.6.3.1-2)$$

Then the lag, L , which will minimize the total weighted error (with optimal β) should be chosen as that which maximizes the following equation.

$$(C_L)^2 / G_L \quad (5.1.1.6.3.1-3)$$

Since we are restricting β to be positive, only lags with positive C_L are considered. L is coded with 7 bits and can take on a value from 20 to 146. One of the 128 coded lag values is reserved to indicate that the long term predictor is not used. This allows the long term predictor to be disabled when a positive correlation cannot be found. $b_L(n)$ may be calculated as follows:

$z_L(n)$ is defined as following equation,

$$z_L(n) = \sum_{i=0}^{\min(n, L-1)} r(i-L) h(n-i) \quad (5.1.1.6.3.1-4)$$

then, $b'_L(n)$ becomes as follows

$$b'_L(n) = \sum_{i=0}^{\left\lfloor \frac{n}{L} \right\rfloor} z_L(n-iL) \quad (5.1.1.6.3.1-5)$$

where $\lfloor x \rfloor$ is the floor function of x which evaluates to the largest integer $\leq x$

$z_L(n)$ can be computed from $z_{L-1}(n)$ using (5.1.1.6.3.1-6) and (5.1.1.6.3.1-7).

$$z_L(n) = z_{L-1}(n-1) + r(-L) h(n) \quad (5.1.1.6.3.1-6)$$

for $1 \leq n \leq N-1$ and

$$z_L(0) = r(-L) h(0) \quad (5.1.1.6.3.1-7)$$

A further computation can be reduced if a truncated impulse response is used in the computations associated with the long term lag determination process. The truncated impulse response is shown as follows.

$$h'(n) = \begin{cases} h(n) & : 0 \leq n \leq N_T - 1 \\ 0 & : N_T \leq n \end{cases} \quad (5.1.1.6.3.1-8)$$

A value of 21 for N_T , corresponding to a impulse response is truncated by 21 samples long, may be used.

Replacing $h(n)$ with $h'(n)$ in equations 5.1.1.6.3.1-4 and 5.1.1.6.3.1-6. These equations can be replaced by the following set of equations

$$z_{L_{\min}}(n) = \sum_{i=\max(0, n - N_T + 1)}^{\min(n, L_{\min} - 1)} r(i - L_{\min}) h(n-i) \quad (5.1.1.6.3.1-9)$$

$$z_L(n) = \begin{cases} z_{L-1}(n-1) + r(-L)h(n) & : 1 \leq n \leq N_T - 1 \\ z_{L-1}(n-1) & : N_T \leq n \leq N-1 \end{cases} \quad (5.1.1.6.3.1-10)$$

Equation 5.1.1.6.3.1-9 is computed only for the minimum lag value and the array $z_L(n)$ for all other values of L is computed using 5.1.1.6.3.1-10.

Further computational reduction is possible when $L > N-1$ since then equation 5.1.1.6.3.1-5 reduces to

$$b'_L(n) = z_L(n) \quad (5.1.1.6.3.1-11)$$

and computation of the energy term, G_L , in equation (5.1.1.6.3.1-1) is simplified to

$$G_L = E_L + \sum_{n=0}^{N_T-1} z_L^2(n) \quad (5.1.1.6.3.1-12)$$

where E_L and E_N is shown as following equation.

$$E_L = E_{L-1} + z_{L-1}^2(N_T - 1) - z_{L-1}^2(N-1) \quad , L > n \quad (5.1.1.6.3.1-13)$$

and

$$E_N = \sum_{n=N_T}^{N-1} (z_N(n))^2 \quad , L > n \quad (5.1.1.6.3.1-14)$$

These equations are used to determine G_L and C_L and the determination of the long term predictor lag, L , is based on maximizing equation 5.1.1.6.3.1-3. Once the lag is determined, the untruncated impulse response, $h(n)$, is used to compute the weighted long term lag vector, $b'_L(n)$. This is done by computing $b'_L(n)$ as the zero state response of $H(z)$ to $b_L(n)$ where $b_L(n)$ is:

$$b_L(n) = r(n - \lfloor (n+L)/L \rfloor L) \quad 0 \leq n \leq 39 \quad (5.1.1.6.3.1-15)$$

5.1.1.6.3.2 Encoding of lag

The lag, L , for each of the four subframes takes on the value of 20 to 146. This corresponds to 127 possible codes. Seven bits are used to encode each lag. The 128th code is used to indicate that the pitch predictor for that subframe is deactivated. The lag value is converted to the lag code as follows:

$$\begin{aligned} \text{LAG}_x &= L - 19 && \text{if predictor activated} \\ \text{LAG}_x &= 0 && \text{if predictor deactivated} \end{aligned} \quad (5.1.1.6.3.2-1)$$

where x is the numerals 1 through 4 for the 4 subframes.

5.1.1.6.4 Codebook excitation

The VSELP code uses an excitation codebook of 2^M code vectors which is constructed from a sets of M basis vectors, where $M = 9$. Defining $v_m(n)$ as the m-th basis vector of the codebook and $u_i(n)$ as the i-th code vector in the codebook, then:

$$u_i(n) = \sum_{m=1}^M \theta_{im} v_m(n) \quad (5.1.1.6.4-1)$$

where $0 \leq i \leq 2^M-1$; $0 \leq n \leq N-1$.

In other words, each code vector in the codebook is constructed as a linear combination of the M basis vectors. The linear combinations are defined by the θ_{im} parameters. θ_{im} is defined as:

$$\begin{aligned} \theta_{im} &= +1 \text{ if bit } m \text{ of codeword } i = 1 \\ \theta_{im} &= -1 \text{ if bit } m \text{ of codeword } i = 0 \end{aligned}$$

The codebook construction for the VSELP code can restated as follows. Code vector i is constructed as the sum of the M basis vectors where the polarity (plus or minus) of each basis vector is determined by the state of the corresponding bit in codeword i. Note that if we complement all the bits in codeword i, the corresponding code vector is the negative of code vector i. Therefore, for every code vector, its negative is also a code vector in the codebook. These pairs are called complementary code vectors since the corresponding codewords are complements of each other. The basis vectors ($v_m(n)$) which specify the VSELP codebook are given in Table 5.1.1.6.4-1.

Table 5.1.1.6.4-1

FORMAT IN WHICH THE BASIS VECTOR SAMPLES ARE GIVEN:

1	2	3	4
5	6	7	8
.	.	.	.
.	.	.	.
.	.	.	.
33	34	35	36
37	38	39	40

– BASIS VECTOR # 1

0.6268516779E+00	−0.4542078972E+00	−0.1760918051E+00	−0.5861995816E+00
0.1378517002E+00	−0.1823240072E+00	−0.2775301933E+00	−0.2964571118E+00
−0.2311124057E+00	−0.3477011919E+00	−0.7365584606E−03	−0.3974846005E+00
0.7419414818E−01	−0.1460401062E−01	0.1091039032E+00	0.2842747867E+00
0.1790350974E+00	−0.1180389971E+00	0.2658933997E+00	−0.2497244254E−01
0.2913824916E+00	−0.7639420684E−02	−0.1302794069E+00	−0.9610164165E−01
−0.7092607766E−01	0.1249964982E+00	−0.3015049100E+00	−0.2726641893E+00
−0.2149710953E+00	−0.1119261011E+00	0.2582789958E+00	0.3643897176E−01
0.3608868122E+00	0.4519217014E+00	0.2646971047E+00	0.6711819023E−01
−0.3257541060E+00	0.2701312006E+00	−0.1926795989E+00	0.1213812009E+00

– BASIS VECTOR # 2

−0.6768730879E+00	0.1942162961E+00	−0.2322226390E−01	0.3822003007E+00
0.1149367355E−01	0.1851516962E+00	−0.9766262025E−02	0.2865583897E+00
−0.2562691867E+00	−0.6563412398E−01	−0.2626664042E+00	−0.1768510602E−01
0.1316096038E+00	0.3285849094E+00	−0.2817670107E+00	0.1763696969E+00
−0.2744578123E+00	−0.4173304141E+00	−0.5507235602E−01	−0.2101140022E+00
−0.5379775912E−01	−0.1204859987E+00	−0.2512671947E+00	−0.4590868056E+00
−0.2901743054E+00	−0.5972300470E−01	−0.1642322987E+00	0.1331884181E−02
0.7630597800E−01	−0.1746048033E+00	0.2357622981E+00	−0.3006218970E+00
0.2828867137E+00	−0.1119790003E+00	0.1880459934E+00	0.3702029884E+00
0.5028694272E+00	0.2539961040E+00	0.4348948002E+00	0.1730542928E+00

– BASIS VECTOR # 3

−0.3045909107E−01	−0.9361042082E−01	0.2468007952E+00	0.5261582136E−01
−0.5794353783E−01	−0.5306274816E−01	−0.3414047062E+00	0.2482274026E+00
−0.2673354149E+00	0.3823634088E+00	0.2244946957E+00	0.3500150144E+00
0.4617719948E+00	0.5547925830E+00	0.1304437965E+00	−0.2909902036E+00
0.2724617124E+00	−0.2662397027E+00	0.4453388974E−01	−0.3728887141E+00
0.1674896926E+00	−0.1169330999E+00	0.2741321921E+00	−0.1578143984E+00
0.3770607114E+00	0.8141511679E−01	0.1834725775E−01	−0.4516902566E−01
−0.4783500358E−01	−0.1556749940E+00	0.7163218409E−01	0.5144109726E+00
0.7860491984E−02	0.2075532973E+00	0.2995041013E+00	0.1335048974E+00
0.1867219955E+00	−0.1592454016E+00	−0.2922371030E+00	−0.2735118195E−01

– BASIS VECTOR # 4

–0.6781588793E+00	0.5470318198E+00	0.2255750000E+00	0.4176881909E+00
0.1862290949E+00	0.5442212820E+00	–0.2990111336E–01	0.1008564979E+00
0.1529103965E+00	–0.1012031455E–01	0.2418514043E+00	–0.2152020931E+00
0.1231288984E+00	–0.2608709931E+00	–0.2222152948E+00	–0.1940249056E+00
0.1628644019E+00	0.2152141035E+00	0.2346480042E+00	0.1390030980E+00
0.3159368932E+00	0.1414815038E+00	–0.1107136980E+00	0.3299163878E+00
–0.1430862993E+00	–0.4246554896E–01	–0.4862520099E+00	–0.2819257081E+00
–0.1770402938E+00	–0.2981126010E+00	–0.2827757895E+00	0.9456578642E–01
–0.1576932073E+00	0.4485046864E+00	–0.8093725890E–01	0.1420958042E+00
0.3595492244E–01	–0.1189899966E+00	0.1931491941E+00	–0.7398038357E–01

– BASIS VECTOR # 5

–0.9664934874E+00	–0.1308428049E+00	–0.3007073998E+00	0.1161881983E+00
–0.1512849033E+00	0.2704344094E+00	–0.8542487025E–01	0.1896096021E+00
0.2657969892E+00	0.3013361096E+00	0.2235424966E+00	0.4019590914E+00
0.2418593019E+00	–0.1880683005E+00	0.7497540861E–01	0.6694605201E–01
0.2233978957E+00	–0.6115953624E–01	0.9584994614E–01	0.2826313078E+00
0.3384723514E–01	–0.4036342353E–01	–0.2207729965E+00	–0.3311130106E+00
–0.1693277061E+00	–0.2735322118E+00	0.2574662864E+00	–0.3160803914E+00
0.2936640084E+00	–0.1631772965E+00	0.1612997055E+00	0.1175545976E+00
0.9666328877E–01	–0.3569331393E–01	–0.2369606048E+00	–0.9164055437E–01
–0.5905457139E+00	0.8020491898E–01	0.5376718938E–02	0.1375676226E–01

– BASIS VECTOR # 6

0.7454934716E+00	0.1441866010E+00	0.3293612972E–01	–0.3181879595E–01
–0.3218258023E+00	0.1645569950E+00	0.4332777113E–01	–0.1343419999E+00
0.1119066030E+00	–0.1878135055E+00	0.7401214540E–01	0.2256347984E+00
0.5185933709E+00	–0.2301809378E–02	0.3925162852E+00	0.1791860014E+00
0.4263260067E+00	0.2082957029E+00	0.6929016858E–01	0.1308410019E+00
0.2770608068E+00	0.1205063015E+00	0.8444339037E–01	–0.6009020805E+00
–0.6238225847E–01	0.1929809526E–01	–0.3643352091E+00	0.1030322015E+00
–0.4868617281E–01	–0.2442118973E+00	–0.1846102253E–01	–0.3636698127E+00
–0.1745346934E+00	–0.3676941097E+00	–0.1865134984E+00	0.5001406372E–01
0.2847849131E+00	0.5577248335E–01	–0.1189483032E+00	–0.6568223238E–01

– BASIS VECTOR # 7

–0.1480803967E+01	–0.7920601219E–01	–0.2616612911E+00	0.4103611410E–01
0.1389960945E+00	–0.3839070722E–01	0.8559114486E–01	0.1734882966E–01
–0.3655524924E–01	–0.3239679039E+00	0.2493936010E–01	–0.1115060002E+00
0.1592901945E+00	0.1122876033E+00	0.1940158010E+00	0.3329296112E+00
–0.1035078336E–01	0.2379682958E+00	0.1591711044E+00	–0.2001806051E+00
0.1303105056E+00	0.1022555977E+00	0.3698922098E+00	0.1627361029E+00
0.4245679080E+00	–0.4154393449E–01	0.6320028752E–01	–0.2823280916E–01
–0.2555764914E+00	0.8544488996E–01	–0.1576953977E+00	–0.2879562974E+00
–0.1565037966E+00	–0.8424925804E–01	–0.1198138967E+00	–0.8562137932E–01
0.9463231266E–01	–0.6978236139E–01	–0.1084283646E–01	–0.6704759598E–01

– BASIS VECTOR # 8

–0.7696284056E+00	0.5744875073E+00	0.1776295342E–01	0.5335813761E+00
0.2096530050E+00	0.8879880607E–01	0.2314195037E+00	–0.1802335978E+00
–0.3412540853E+00	–0.7997681946E–01	–0.3156605959E+00	–0.1392517984E+00
–0.1293537207E–01	0.1169421002E+00	0.3880879879E+00	0.3939177990E+00
0.8509635180E–01	–0.9309069067E–01	0.3394401073E+00	–0.3983358666E–01
0.1640962958E+00	–0.1400365978E+00	–0.5516054854E–01	–0.2546899915E+00
–0.1267561018E+00	0.1686787009E+00	–0.1605554074E+00	0.3370665014E+00
0.1954406947E+00	0.3195438087E+00	–0.3434188664E–01	0.3679802120E+00
0.1715907007E+00	–0.9389920160E–03	–0.1815388054E+00	0.1689489037E+00
–0.4949538112E+00	–0.1757726818E–01	–0.3213627934E+00	–0.3482311964E+00

– BASIS VECTOR # 9

–0.6175625920E+00	0.9708964825E–01	–0.1585091054E+00	–0.2045332938E+00
0.2081142925E–01	–0.2946406901E+00	–0.1383146048E+00	–0.8538375050E–01
–0.3891209066E+00	0.7004356384E–01	–0.2962988615E–01	0.1176202968E+00
0.1210985035E+00	–0.4984315932E+00	–0.2204985023E+00	–0.2623530030E+00
0.2123277932E+00	0.2498406023E+00	0.5423782393E–01	0.5100889802E+00
0.2310080975E+00	0.2161350995E+00	0.9698209167E–01	0.1712574959E+00
–0.2572568953E+00	0.2769351006E+00	0.1303689927E+00	0.4648959935E+00
0.3099783063E+00	–0.1619630307E–01	0.4056490064E+00	–0.1091113896E–02
0.1733437926E+00	–0.3201962113E+00	0.3038722873E+00	0.1945876032E+00
0.4797148108E+00	0.1241720989E+00	–0.4200162739E–01	–0.1429350972E+00

The excitation codebook search procedure takes place after the long term predictor lag, L , has been determined. Define:

I = codeword selected from VSELP codebook

5.1.1.6.4.1 Filtering of basis vectors

To perform the codebook search, the zero state response of each basis vector to $H(z)$ must be computed. Define $q_m(n)$ to be the zero state response of $H(z)$ whose input is the basis vector $v_m(n)$; $0 \leq n \leq N-1$. From the definition of the VSELP codebook (5.1.1.6.4-1), the zero state response of each code vector $f_i(n)$, can be expressed as:

$$f_i(n) = \sum_{m=1}^M \theta_m q_m(n) \quad (5.1.1.6.4.1-1)$$

5.1.1.6.4.2 Orthogonalization of filtered basis vectors

The selection of the code vector is processed after the selection of the long term predictor lag, L . Decouple the selection process for the codebook excitation vectors from previously determined excitation components, utilizing the orthogonalization procedures.

Prior to the codebook search, each filtered basis vector for the codebook, $q_m(n)$, is made orthogonal to $b'_L(n)$ which is the zero state response of $H(z)$ to the long term prediction vector $b_L(n)$.

Defining the Γ and Ψ_m as following equations for $1 \leq m \leq M$.

$$\Gamma = \sum_{n=0}^{N-1} (b'_L(n))^2 \quad (5.1.1.6.4.2-1)$$

and

$$\Psi_m = \sum_{n=0}^{N-1} b'_L(n) q_m(n) \quad (5.1.1.6.4.2-2)$$

for $1 \leq m \leq M$ and $0 \leq n \leq N-1$; then $q'_m(n)$, the orthogonalized filtered basis vectors, can be computed by the following equation.

$$q'_m(n) = q_m(n) - \left(\frac{\Psi_m}{\Gamma}\right) b'_L(n) \quad (5.1.1.6.4.2-3)$$

The orthogonalized filtered code vectors can now be expressed as:

$$f'_i(n) = \sum_{m=1}^M \theta_m q'_m(n) \quad (5.1.1.6.4.2-4)$$

for $0 \leq i \leq 2^M - 1$ and $0 \leq n \leq N-1$.

Using the orthogonalized filtered codebook vectors, $f'_i(n)$, the equation to be minimized is:

$$E'_i = \sum_{n=0}^{N-1} (p(n) - \gamma' f'_i(n))^2 \quad (5.1.1.6.4.2-5)$$

where γ' is optimized for each code vector i .

5.1.1.6.4.3 VSELP codebook search

The codebook search procedure should find the codeword i which minimizes the following equation.

$$E_i = \sum_{n=0}^{N-1} (p(n) - \gamma' f'_i(n))^2 \quad (5.1.1.6.4.3-1)$$

where γ' is optimal for each code vector i .

C_i and G_i is defined as following equation.

$$C_i = \sum_{n=0}^{N-1} f'_i(n) p(n) \quad (5.1.1.6.4.3-2)$$

and

$$G_i = \sum_{n=0}^{N-1} (f'_i(n))^2 \quad (5.1.1.6.4.3-3)$$

then the code vector shall be chosen as the one which maximizes the following equation.

$$\frac{(C_i)^2}{G_i} \quad (5.1.1.6.4.3-4)$$

The search process should evaluate the equation (5.1.1.6.4.3-4) for each code vector. The code vector which maximizes (5.1.1.6.4.3-4) is then chosen. Using properties of the VSELP codebook construction, the computations required for computing C_i and G_i can be greatly simplified.

defining:

$$R_m = 2 \sum_{n=0}^{N-1} q'_m(n) p(n) \quad (5.1.1.6.4.3-5)$$

for $1 \leq m \leq M$ and

$$D_{mj} = 4 \sum_{n=0}^{N-1} q'_m(n) q'_j(n) \quad (5.1.1.6.4.3-6)$$

for $1 \leq m \leq j \leq M$

C_i can be expressed as following equation.

$$C_i = \frac{1}{2} \sum_{m=1}^M \theta_{im} R_m \quad (5.1.1.6.4.3-7)$$

and G_i can be expressed as:

$$G_i = \frac{1}{2} \sum_{j=2}^M \sum_{m=1}^{j-1} \theta_{im} \theta_{ij} D_{mj} + \frac{1}{4} \sum_{j=1}^M D_{jj} \quad (5.1.1.6.4.3-8)$$

Assuming that codeword u differs from codeword i in only one bit position, $\theta_{um} = \theta_{im}$ for a positive V position when $\theta_{uv} = -\theta_{iv}$ and $m \neq v$. Then C_u and G_u are formed respectively as follows.

$$C_u = C_i + \theta_{uv} R_v \quad (5.1.1.6.4.3-9)$$

and

$$G_u = G_i + \sum_{j=1}^{v-1} \theta_{uj} \theta_{uv} D_{jv} + \sum_{j=v+1}^M \theta_{uj} \theta_{uv} D_{vj} \quad (5.1.1.6.4.3-10)$$

If the codebook search is structured such that each successive codeword evaluated differs from the previous codeword in only one bit position, then (5.1.1.6.4.3-9) and (5.1.1.6.4.3-10) can be used to update C_i and G_i in a very efficient manner. Sequencing of the codewords is accomplished in this manner using a binary Gray code.

Note that complementary codewords (see 5.1.1.6.4) will have equivalent values for (5.1.1.6.4.3-4). Therefore only half of the code vectors need to be evaluated. Once the code vector which maximizes (5.1.1.6.4.3-4) is found, the sign of C_i for that code vector will determine whether that code vector or its complement will yield a positive gain. If C_i is positive then i is the selected codeword, if C_i is negative the ones complement of i is selected as the codeword.

5.1.1.6.4.4 Encoding of excitation codewords

The code value for the codebook, $CODE_x$, is the codeword I as derived by the codebook search procedure in section 4. The least significant bit corresponds to the first basis vector for the codebook.

5.1.1.6.5 Quantization of gains

The weighted error per sample in a subframe is given by

$$e(n) = p(n) - \beta c'_0(n) - \gamma c'_1(n) \quad : \quad 0 \leq n \leq N-1 \quad (5.1.1.6.5-1)$$

where $p(n)$: The value obtained after subtracting zero input response of $H(z)$ from the weighted input to be matched.

$c'_0(n)$: The weighted long term prediction vector - $b'_L(n)$

$c'_1(n)$: The weighted code vector selected from the codebook - $f_l(n)$

β : The long term predictor coefficient

γ : The gain scaling the code vector

Consequently the total weighted error squared for a subframe is given by the following equation.

$$E = \sum_{n=0}^{N-1} e^2(n) = \sum_{n=0}^{N-1} (p(n) - \beta c'_0(n) - \gamma c'_1(n))^2 \quad (5.1.1.6.5-2)$$

To simplify the equation (5.1.1.6.5-2), E may be expressed in terms of correlations among vectors $p(n)$, $c'_0(n)$, and $c'_1(n)$,

Let

$$R_{pp} = \sum_{n=0}^{N-1} p(n)p(n) \quad (5.1.1.6.5-3)$$

$$R_{pc}(k) = \sum_{n=0}^{N-1} p(n)c'_k(n) \quad k = 0, 1 \quad (5.1.1.6.5-4)$$

$$R_{cc}(k,j) = \sum_{n=0}^{N-1} c'_k(n) c'_j(n) \quad k = 0, 1, \quad j = k, 1 \quad (5.1.1.6.5-5)$$

$$R_{cc}(k,j) = R_{cc}(j,k) \quad (5.1.1.6.5-6)$$

Incorporating the correlations into the equation (5.1.1.6.5-2) yields

$$E = R_{pp} - 2\beta R_{pc}(0) - 2\gamma R_{pc}(1) + 2\beta\gamma R_{cc}(0,1) + \beta^2 R_{cc}(0,0) + \gamma^2 R_{cc}(1,1) \quad (5.1.1.6.5-7)$$

Minimizing the weighted error correspond to optimizing the long term predictor coefficient β and the term gain γ , and also correspond to minimize the equation 5.1.1.6.5-7.

5.1.1.6.5.1. Transformation of gains to GS and P0

Define $ex(n)$ to be the excitation function at a given subframe. $ex(n)$ is a linear combination of the long term predictor vector scaled by the long term predictor coefficient β , and the code vector scaled by its gain γ .

$$ex(n) = \beta c_0(n) + \gamma c_1(n) \quad 0 \leq n \leq N-1 \quad (5.1.1.6.5.1-1)$$

where $c_0(n)$: the unweighted long term prediction vector, $b_L(n)$
 $c_1(n)$: the unweighted code vector selected from the codebook, $u_l(n)$

The energy in each excitation vector is given by

$$R_x(k) = \sum_{n=0}^{N-1} c_k^2(n) \quad k = 0, 1 \quad (5.1.1.6.5.1-2)$$

Let RS be the approximate residual energy at a given subframe. RS is a denominator of N, $R'q(0)$, and of the normalized prediction gain of the LPC filter.

$$RS = N \cdot R'q(0) \prod_{i=1}^{N^p} (1 - r_i^2) \quad (5.1.1.6.5.1-3)$$

where r_i is the i-th reflection coefficient for the subframe corresponding to the set of direct form filter coefficients (α_i 's) for the subframe.

GS, the energy offset parameter, is a coded parameter which adjusts the estimated value of RS. Define:

$$R = GS \cdot RS \quad (5.1.1.6.5.1-4)$$

Define P_0 , the energy contribution of the long term prediction vector as a fraction of the total excitation energy at a subframe, as

$$P_0 = \frac{\beta^2 R_x(0)}{R} \quad \text{where } 0 \leq P_0 \leq 1 \quad (5.1.1.6.5.1-5)$$

Thus β and γ are replaced by two new parameters: P_0 and GS . The transformations relating β and γ to GS and P_0 are given by

$$\beta = \sqrt{\frac{RS \, GS \, P_0}{R_x(0)}} \quad (5.1.1.6.5.1-6)$$

$$\gamma = \sqrt{\frac{RS \, GS \, (1-P_0)}{R_x(1)}} \quad (5.1.1.6.5.1-7)$$

5.1.1.6.5.2 Vector quantization and encoding of GS and P_0

Replacing the β and γ in (5.1.1.6.5-7) by the equivalent equations in terms of GS , P_0 , and $R_x(k)$, the equation (5.1.1.6.5-7) becomes as follows.

$$E = R_{pp} - a\sqrt{GS \, P_0} - b\sqrt{GS(1-P_0)} + c \, GS\sqrt{P_0(1-P_0)} + d \, GS \, P_0 + e \, GS \, (1-P_0) \quad (5.1.1.6.5.2-1)$$

where

$$a = 2R_{pc}(0)\sqrt{\frac{RS}{R_x(0)}} \quad (5.1.1.6.5.2-2)$$

$$b = 2R_{pc}(1)\sqrt{\frac{RS}{R_x(1)}} \quad (5.1.1.6.5.2-3)$$

$$c = \frac{2R_{cc}(0,1) \, RS}{\sqrt{R_x(0)R_x(1)}} \quad (5.1.1.6.5.2-4)$$

$$d = \frac{RS \, R_{cc}(0,0)}{R_x(0)} \quad (5.1.1.6.5.2-5)$$

$$e = \frac{RS \, R_{cc}(1,1)}{R_x(1)} \quad (5.1.1.6.5.2-6)$$

GS and P0 are vector quantized. The first step in quantizing vector {GS, P0} consists of calculating the parameters required by the equation 5.1.1.6.5.2-1:

$$\begin{aligned} R_{cc}(k,j) \quad & k = 0, 1; \quad j = k, 1 \\ R_x(0), R_x(1) \\ RS \\ R_{pc}(k) \quad & k = 0, 1 \\ a, b, c, d, e \end{aligned}$$

Next equation (5.1.1.6.5.2-1) is evaluated for each vector in the {GS, P0}codebook, and the vector which minimizes the weighted error is selected. The {GS, P0} codebook is given in Table 5.1.1.6.5.2-1.

Table 5.1.1.6.5.2-1 {GS, P0} codebook

INDEX #	GS	P0
0	0.6869540666E-03	0.4476251900E+00
1	0.1436048187E-02	0.2861252129E+00
2	0.2056100406E-02	0.6649292707E+00
3	0.3014685120E-02	0.1721598059E+00
4	0.3906219034E-02	0.6965826750E+00
5	0.4988078028E-02	0.4867439866E+00
6	0.7128443569E-02	0.6385692954E+00
7	0.1012304146E-01	0.7445229292E+00
8	0.9183008224E-02	0.3202570081E+00
9	0.1473146770E-01	0.4356988966E+00
10	0.1383242756E-01	0.6932892799E+00
11	0.1885624230E-01	0.6018816829E+00
12	0.1596315578E-01	0.1067145020E+00
13	0.2770061977E-01	0.1127054989E+00
14	0.2540918998E-01	0.3960103989E+00
15	0.3587478027E-01	0.4326250851E+00
16	0.2285091206E-01	0.7630308270E+00
17	0.3081010655E-01	0.6933609247E+00
18	0.3373205289E-01	0.8621196151E+00

19	0.4454405606E-01	0.8086944818E+00
20	0.4042430222E-01	0.6082698703E+00
21	0.5205642805E-01	0.5698894262E+00
22	0.5724171177E-01	0.7135738730E+00
23	0.7463405281E-01	0.6977375746E+00
24	0.4404250905E-01	0.1780095994E+00
25	0.5672956631E-01	0.3668105900E+00
26	0.6864905357E-01	0.9826055169E-01
27	0.8920857310E-01	0.1655579954E+00
28	0.6916176528E-01	0.4874900877E+00
29	0.8691659570E-01	0.4830487967E+00
30	0.9913752228E-01	0.5978199840E+00
31	0.1195112988E+00	0.5629572868E+00
32	0.5645768717E-01	0.8710960150E+00
33	0.7304289937E-01	0.9161574244E+00
34	0.7630827278E-01	0.8216353059E+00
35	0.9198626876E-01	0.8889883757E+00
36	0.9578324854E-01	0.7449157834E+00
37	0.1139025986E+00	0.8013312817E+00
38	0.1123045012E+00	0.9158360958E+00
39	0.1344650984E+00	0.9465798736E+00
40	0.1089958027E+00	0.2585754991E+00
41	0.1417811066E+00	0.2649452090E+00
42	0.1519473046E+00	0.4224564135E+00
43	0.1964756995E+00	0.4381014109E+00
44	0.1369020045E+00	0.6498267055E+00
45	0.1654966027E+00	0.6849172115E+00
46	0.1920738965E+00	0.6129153967E+00

47	0.2217566073E+00	0.6894881129E+00
48	0.1348659992E+00	0.7870811224E+00
49	0.1545130014E+00	0.8275581002E+00
50	0.1521859020E+00	0.8977674246E+00
51	0.1761115938E+00	0.9405922890E+00
52	0.1777600944E+00	0.8131462932E+00
53	0.2011377960E+00	0.8337951899E+00
54	0.2059223950E+00	0.9291666746E+00
55	0.2213090956E+00	0.9766067863E+00
56	0.2265692949E+00	0.8644598722E+00
57	0.2620548010E+00	0.8804330826E+00
58	0.2408834994E+00	0.9317544103E+00
59	0.2622093856E+00	0.9621856809E+00
60	0.2892234027E+00	0.9264010787E+00
61	0.3182702959E+00	0.9531841278E+00
62	0.2943232059E+00	0.9758533239E+00
63	0.3376874924E+00	0.9837425947E+00
64	0.2073526978E+00	0.1345687956E+00
65	0.3402079046E+00	0.1264445037E+00
66	0.2715196908E+00	0.3424046040E+00
67	0.3454430997E+00	0.4101729095E+00
68	0.2527894974E+00	0.5479403734E+00
69	0.2949407995E+00	0.6495116949E+00
70	0.3585202992E+00	0.6219326854E+00
71	0.4089600146E+00	0.6932708025E+00
72	0.4873662889E+00	0.2380438000E+00
73	0.4617147148E+00	0.4731487036E+00
74	0.7496141791E+00	0.2872014940E+00

75	0.7256929278E+00	0.5270823836E+00
76	0.5085545778E+00	0.6109396815E+00
77	0.5838373899E+00	0.7006831765E+00
78	0.6889815927E+00	0.7469186187E+00
79	0.8430874944E+00	0.7804297209E+00
80	0.2370706946E+00	0.7655134797E+00
81	0.2718037069E+00	0.7923678160E+00
82	0.3113563061E+00	0.7989513874E+00
83	0.3554876149E+00	0.8334587812E+00
84	0.3161123991E+00	0.8889619112E+00
85	0.3565140963E+00	0.9034569860E+00
86	0.4013800025E+00	0.8952671289E+00
87	0.4174228013E+00	0.9366862774E+00
88	0.3558124900E+00	0.9439828992E+00
89	0.3790760934E+00	0.9644626975E+00
90	0.4237785041E+00	0.9721646905E+00
91	0.4056538939E+00	0.9868369102E+00
92	0.4654931128E+00	0.9469987750E+00
93	0.4958581924E+00	0.9618533850E+00
94	0.4900814891E+00	0.9784963727E+00
95	0.4963087142E+00	0.9908313751E+00
96	0.4094735086E+00	0.7861018181E+00
97	0.4881666899E+00	0.7994922996E+00
98	0.4567241967E+00	0.8743832111E+00
99	0.4880503118E+00	0.9119582772E+00
100	0.5745701790E+00	0.8464691043E+00
101	0.6282662153E+00	0.8929378986E+00
102	0.5513637066E+00	0.9188777208E+00

103	0.5593187213E+00	0.9491345286E+00
104	0.6391062140E+00	0.9401535988E+00
105	0.6293665767E+00	0.9670869708E+00
106	0.5638924241E+00	0.9725283980E+00
107	0.5873979926E+00	0.9881426096E+00
108	0.7092357874E+00	0.9560046792E+00
109	0.7587450147E+00	0.9752473235E+00
110	0.6742975712E+00	0.9823337793E+00
111	0.7517855167E+00	0.9916964769E+00
112	0.1050716996E+01	0.5055761933E+00
113	0.1494652033E+01	0.6326776743E+00
114	0.1176993966E+01	0.7874556184E+00
115	0.1125803947E+01	0.8668910861E+00
116	0.7642365098E+00	0.8648762703E+00
117	0.7707328796E+00	0.9073584080E+00
118	0.7981277704E+00	0.9335256815E+00
119	0.8304246068E+00	0.9594538808E+00
120	0.9789227843E+00	0.9259408712E+00
121	0.9895892143E+00	0.9652320743E+00
122	0.8809021115E+00	0.9833607078E+00
123	0.1041756034E+01	0.9887707233E+00
124	0.1339319944E+01	0.9154701233E+00
125	0.1190603971E+01	0.9647362232E+00
126	0.1761451006E+01	0.9547418952E+00
127	0.1403365970E+01	0.9860584140E+00

Note that in conducting the code search Rpp correlation may be ignored in the equation (5.1.1.6.5.2-1), since it is a constant, thus eliminating the need to compute it. The quantized long term predictor coefficient β_q , and the quantized gain γ_q for the code vector selected from the VSELP codebook, are obtained from the following equations.

$$\beta_q = \sqrt{\frac{RS \text{ GS}_{vq} \text{ P0}_{vq}}{R_x(0)}} \quad (5.1.1.6.5.2-7)$$

$$\gamma_q = \sqrt{\frac{RS \text{ GS}_{vq} (1 - \text{P0}_{vq})}{R_x(1)}} \quad (5.1.1.6.5.2-8)$$

where GS_{vq} and P0_{vq} , are the elements of the vector chosen from the $\{\text{GS}, \text{P0}\}$ codebook. The index of the corresponding codebook entry selected is then assigned to GSP0_x , where x is the current subframe number (1 to 4).

A special case occurs when the long term predictor is disabled for a certain subframe. This occurs when no positive correlation is found during the lag search, or when the state of the long term predictor is populated entirely by zeroes (e.g., at the first subframe).

When the long term predictor is deactivated, a modified form of (5.1.1.6.5.2-1) is used:

$$E = R_{pp} - b \sqrt{\text{GS}(1 - \text{P0})} + e \text{GS}(1 - \text{P0}) \quad (5.1.1.6.5.2-9)$$

For this case the quantized code vector gains are:

$$\beta_q = 0 \quad (5.1.1.6.5.2-10)$$

$$\gamma_q = \sqrt{\frac{RS \text{ GS}_{vq} (1 - \text{P0}_{vq})}{R_x(1)}} \quad (5.1.1.6.5.2-11)$$

5.1.1.6.6 Update filter states

After all subframe parameters have been determined and quantized, the long term filter state and the weighted synthesis filter state must be updated in preparation for processing the next subframe. Figure 5.1.1.6.6-1 shows the weighted synthesizer employed in the speech encoder.

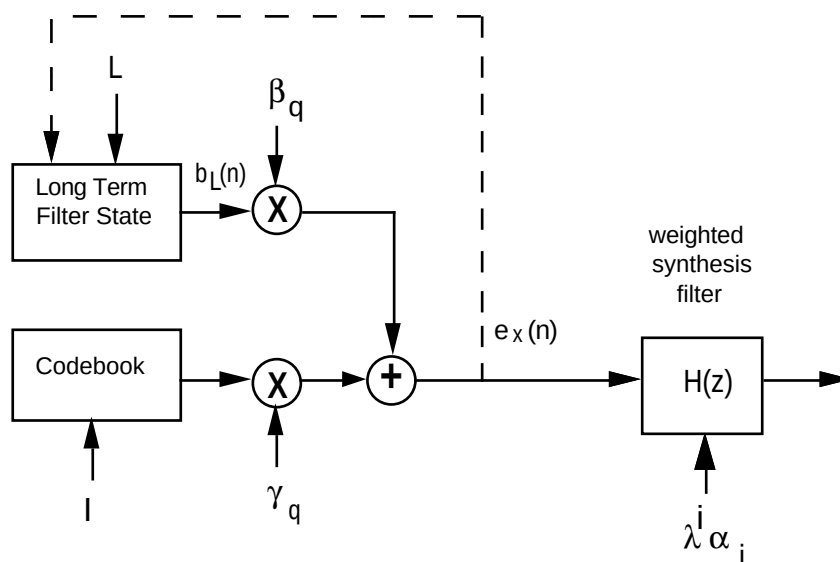


Fig. 5.1.1.6.6-1

The combined excitation, $ex(n)$, shall be computed as:

$$ex(n) = \beta_q b_L(n) + \gamma_q u_l(n) \quad (5.1.1.6.6-1)$$

for $0 \leq n \leq 39$

The long term predictor state, $r(n)$, is updated by:

$$r(n) = r(n+40) \quad \text{for } -146 \leq n \leq -41 \quad (5.1.1.6.6-1a)$$

$$r(n) = ex(n+40) \quad \text{for } -40 \leq n \leq -1 \quad (5.1.1.6.6-1b)$$

The weighted synthesis filter is updated by inputting the 40 samples of $ex(n)$ into the weighted synthesis filter. The state of the weighted synthesis filter should reflect the filter's state at the end of the previous subframe processing prior to filtering the excitation signal $ex(n)$ for the current subframe. A direct form filter should be used for the weighted synthesis filter.

5.1.2 Channel coding

The channel error control for the speech coder data employs three mechanisms for the mitigation of channel errors. The first is to use a rate 9/17 convolutional code to protect the more vulnerable bits of the speech coder data stream. The second technique interleaves the transmitted data for each speech coder frame over two time slots to mitigate the effects of Rayleigh fading. The third technique employs the use of a cyclic redundancy check over some the most perceptually significant bits of the speech coder. After the error correction is applied at the receiver, these cyclic redundancy bits are checked to see if the most perceptually significant bits were received properly.

5.1.2.1 Definition of terms, nomenclature, and assumptions

$a(X)$	- the forty third order input polynomial to the CRC.
$a'(X)$	- the forty third order CRC input polynomial at the receiver which is effected by channel conditions.
$b(X)$	- the sixth order CRC parity polynomial.
$b'(n)$	- the sixth order CRC parity polynomial received which is effected by channel conditions.
bit position	- in speech coder parameters. The least significant bit (lsb) is bit 0, the most significant bit (msb) is bit n-1 where there are n bits in the parameter; for example, the speech parameter R0 has 5 bits, the msb is bit 4, the lsb is bit 0.
bit position in channel	- the order the bits are transmitted on the channel. The first bit transmitted is designated as bit 0 and the last transmitted bit is designated bit 223.
bit position in class1	- order that the bits in CL1[i] are encoded. CL1[0] is the first bit to be encoded and CL1[86] is the last.
$cc0[i]$	- the output of $g_0(D)$ to input bit CL1[i].
$cc1[i]$	- the output of $g_1(D)$ to input bit CL1[i].
CL1[i]	- Input bit array to the convolutional encoder where i ranges from 0 to 86.
CL2[i]	- class 2 bits (unencoded bits) where i ranges from 0 to 58.
class 1	- speech coder bits which are convolutionally encoded.

class 2	- speech coder bits which are not convolutionally encoded.
crc[a'(X)]	- the sixth order CRC parity polynomial generated from the received input bits a'(X).
CRC	- Cyclic Redundancy Checking code.
CRC generator	- the CRC generator polynomial.
CRC's	- the CRC parity bits.
frame x	- the first of the two speech coder frames.
frame y	- the speech coder frame occurring after frame x.
g ₀ (D)	- The first of the two convolutional code generator polynomials where $g_0(D) = 1 + D + D^3 + D^5$ (65 octal).
g ₁ (D)	- The second of the two convolutional code generator polynomials where $g_1(D) = 1 + D^2 + D^3 + D^4 + D^5$ (57 octal).
g _{crc} (X)	- The CRC generator polynomial. $g_{crc}(X) = 1 + X^4 + X^5 + X^6 + X^7$
interleaving	- Ordering of the bits on the channel.
memory order, m	- memory order of the convolutional code, where 2^m = the number of convolutional states. For this system, $m = 5$.
parameter names	- the subframe parameters are named using the convention xxx_n where 'n' implies the nth subframe so code_3 is subframe 3's code vector.
q(X)	- the CRC quotient.
subframe	- the subdivision of a speech frame. There are 4 subframes in a frame - each subframe is 5 milliseconds in duration. For each subframe, the speech coder generates 3 parameters: a codevector, a long term predictor lag, and a GSP0.

5.1.2.2 Speech data classes

The first step in the error correction process is the separation of the 134 bits of speech coder frame's information into class 1 and class 2 bits. The contents of 134 bits as speech coder frame is 75 bits for class 1 and 54 bits for class 2. The class 1 bits represent that portion of the speech data stream to which the convolutional coding is applied. A 7-bit CRC is used for error detection purposes and is computed over the 44 most perceptually significant bits of the class 1 bits for each frame. Class 2 bits are transmitted without any error protection. The process is depicted in figure 5.1.2.2-1.

The 7 bit CRC is computed over the 44 most perceptually significant bits in the speech coder output data stream. The purpose of the CRC bits is to detect when the error correction was unable to correct all the channel errors occurring in the transmission of the 44 most perceptually significant bits. Since an error in any of these 44 bits can cause a very serious deterioration of speech quality, it is very desirable not to use these 44 bits if they are known to be incorrect. If an error is detected, it is preferable to replace the received data for this frame with data from a previous frame. This effectively masks the occurrence of isolated frames of data in error so that there is almost no perceived reduction in speech quality.

The forward error correction (FEC) is a rate 9/17 convolutional code with memory length 5 ($R=9/17$, $m=5$). The convolutional code is applied to 75 bits of the 134 bits speech coder output bits, the 7 bits CRC, and 5 overhead bits required by the decoder. These 87 bits are encoded to 165 bits by the convolutional processing. This added bits as redundancy raises the bit rate from 6.7 kbps (134 bits/20 ms) to 11.2 kbps (224 bits/20 ms).

The interleaving time multiplexes the data of one 20 ms frame into two adjacent time slots. Half of the 224 bits are sent in the first time slot, the rest of the frame's bits are sent in the subsequent time slot.

At the receiver, the frame of data is first deinterleaved. The convolutionally encoded data stream is then decoded using Viterbi decoding. Using the decoded received bits a new set of CRC bits is calculated. These new CRC bits are compared to the received CRC bits. If the two CRC bits are same, the frame of data is passed directly to the speech decoder. If the two sets of CRC bits differ, then the repeat method is applied.

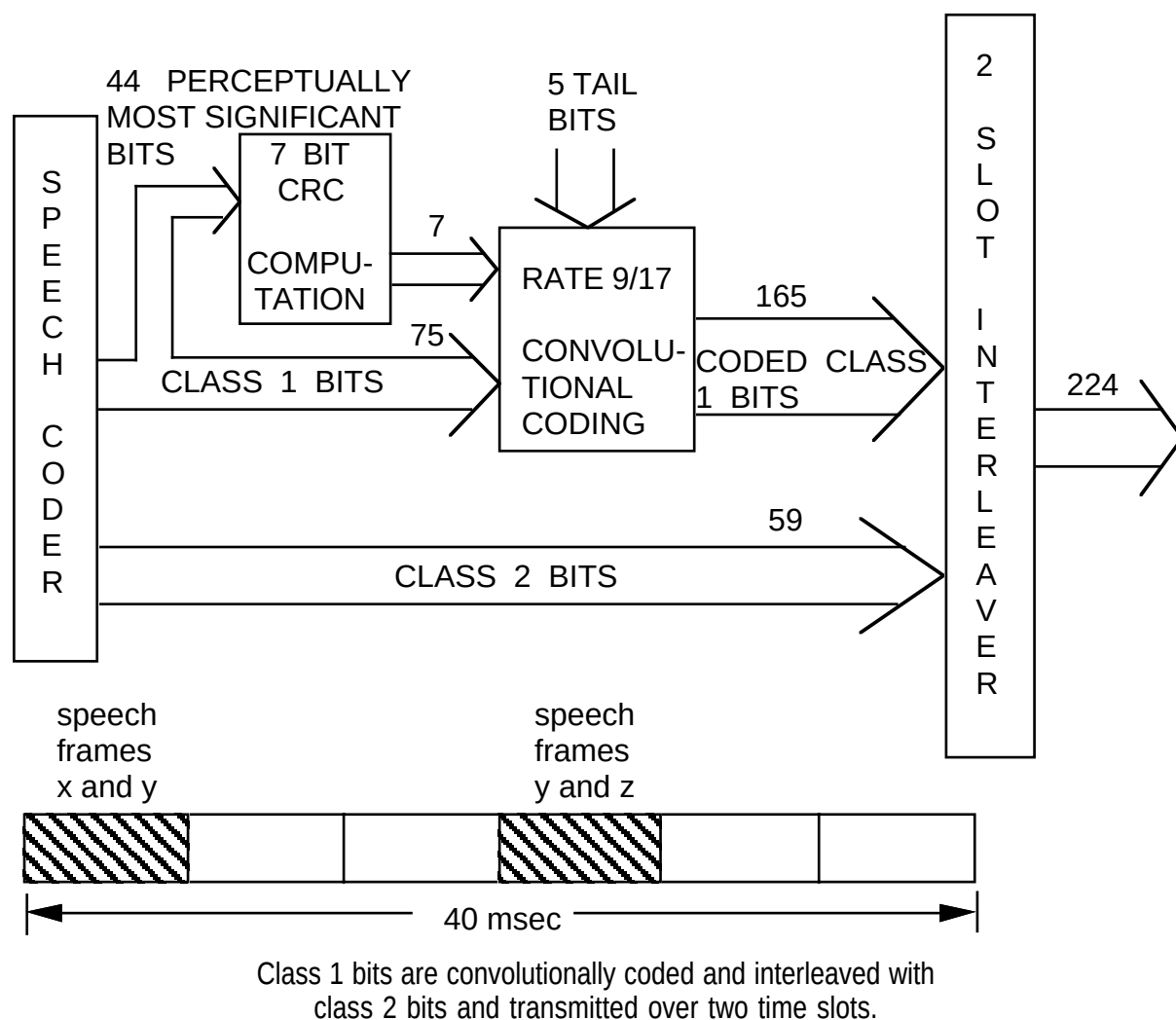


Fig. 5.1.2.2-1 : Error Correction For Speech Coder

Table 5.1.2.2-1 describes the bit allocation among the classes for the parameter bits of the speech coder. The numbers in the third and fifth columns correspond to the number of most significant bits while the fourth column corresponds to the number of least significant bits for the parameter code. The last column specifies the 44 perceptually most significant bits.

Table 5.1.2.2-1: Bit Allocations within Classes 1 and 2

Parameter Name	Total Number of Bits for Parameter	Number of Bits in Class 1	Number of Bits in Class 2	Number of Bits Covered by CRC
LPC1	5	4	1	3
LPC2	5	3	2	2
LPC3	4	3	1	2
LPC4	4	2	2	1
LPC5	4	1	3	1
LPC6	3	1	2	0
LPC7	3	0	3	0
LPC8	3	0	3	0
LPC9	3	0	3	0
LPC10	2	0	2	0
R0	5	4	1	3
SOFINT	1	1	0	0
LAG_1	7	7	0	7
GSP0_1	7	7	0	1
CODE_1	9	0	9	0
LAG_2	7	7	0	7
GSP0_2	7	7	0	1
CODE_2	9	0	9	0
LAG_3	7	7	0	7
GSP0_3	7	7	0	1
CODE_3	9	0	9	0
LAG_4	7	7	0	7
GSP0_4	7	7	0	1
CODE_4	9	0	9	0

5.1.2.3 Cyclic redundancy check (CRC)

A seven bit CRC is computed for the 44 most perceptually significant bits in the frame.

The generator polynomial for the CRC is

$$g_{crc}(X) = 1 + X^4 + X^5 + X^6 + X^7 \quad (5.1.2.3-1)$$

The 44 most perceptually significant bits of the frame are identified in Table 5.1.2.2-1 as the bits covered by the CRC. These bits form the input polynomial, $a(X)$, which is defined as

$$a(X) = CL1[78]X^{43} + CL1[4]X^{42} + CL1[77]X^{41} + CL1[5]X^{40} + CL1[76]X^{39} + CL1[6]X^{38} + \\ CL1[75]X^{37} + \dots + CL1[58]X^3 + CL1[24]X^2 + CL1[57]X^1 + CL1[25]X^0 \quad (5.1.2.3-2)$$

The parity polynomial $b(X)$ is the remainder of the division of the input polynomial and the parity polynomial:

$$\frac{a(X) * X^7}{g_{crc}(X)} = q(X) + \frac{b(X)}{g_{crc}(X)} \quad (5.1.2.3-3)$$

Where $q(X)$ is the quotient of the division, $b(X)$ the remainder. The quotient here is discarded and only the parity bits are sent. The parity polynomial is mapped into the class 1 array. This is done using the following equation.

$$b(X) = CL1[0]X^6 + CL1[81]X^5 + CL1[1]X^4 + CL1[80]X^3 + CL1[2]X^2 + CL1[79]X^1 + CL1[3]X^0 \quad (5.1.2.3-4)$$

5.1.2.4 Convolutional encoding

The rate 9/17 convolutional code is generated by puncturing a rate 1/2 convolutional code. The rate 1/2 convolutional mother code has memory order 5 memory ($R = 1/2$, $m=5$). There are 32 states in this code. Table 5.1.2.4-1 shows all the states and their outputs to a given input.

The polynomials are defined as $g_0 = 65$ octal and $g_1 = 57$ octal or

$$g_0(D) = 1 + D + D^3 + D^5 \quad (5.1.2.4-1)$$

$$g_1(D) = 1 + D^2 + D^3 + D^4 + D^5 \quad (5.1.2.4-2)$$

Table 5.1.2.4-1 Convolutional Encoder States

INPUT = 0			INPUT = 1		INPUT = 0			INPUT = 1	
state	g ₀	g ₁	g ₀	g ₁	state	g ₀	g ₁	g ₀	g ₁
0	0	0	1	1	16	1	1	0	0
1	1	0	0	1	17	0	1	1	0
2	0	1	1	0	18	1	0	0	1
3	1	1	0	0	19	0	0	1	1
4	1	1	0	0	20	0	0	1	1
5	0	1	1	0	21	1	0	0	1
6	1	0	0	1	22	0	1	1	0
7	0	0	1	1	23	1	1	0	0
8	0	1	1	0	24	1	0	0	1
9	1	1	0	0	25	0	0	1	1
10	0	0	1	1	26	1	1	0	0
11	1	0	0	1	27	0	1	1	0
12	1	0	0	1	28	0	1	1	0
13	0	0	1	1	29	1	1	0	0
14	1	1	0	0	30	0	0	1	1
15	0	1	1	0	31	1	0	0	1

The output from the rate 1/2 convolutional coder alternates between these two polynomials starting with $g_0(D)$. Initially the encoders memory elements are cleared; that is, the encoder starts at state 0. The bits in the class 1 buffer are input to the convolutional encoder starting with CL1[0] and concluding with bit CL1[88]. Sequentially the outputs from $g_0(D)$ and $g_1(D)$ are referred to as $cc_0[i]$ and $cc_1[i]$, respectively. For each input bit, CL1[i], the two output bits, $cc_0[i]$ and $cc_1[i]$, are produced.

87 bits are input bits to the convolutional coder. Of these 87 bits, 75 are class 1 bits from the speech coder which are placed into CL1[4] through CL1[78]. Bits CL1[0] through CL1[3] and CL1[79] through CL1[81] are filled by the CRC bits for the frame. CL1[82] through CL1[86] are filled with zeros. These 5 tail bits return the encoder to its initial state at the end of encoding.

The makeup of CL1[i] is shown in Table 5.1.2.4-2. The first column indicates the index (i) into the CL1 array where a particular bit for a parameter is to be placed. The second column indicates the parameter from which the bit is taken and the last column indicates the bit position in the parameter. Where a bit index of 0 represents the least significant bit.

Table 5.1.2.4-2: Bit Ordering Into Convolutional Coder

CL1[i] i	Parameter Name	bit index	CL1[i] i	Parameter Name	bit index	CL1[i] i	Parameter Name	bit index
0	CRC	6	28	SOFINT	0	55	GSP0_3	5
1	CRC	4	29	GSP0_1	4	56	GSP0_1	5
2	CRC	2	30	GSP0_3	4	57	LAG_3	0
3	CRC	0	31	LPC1	1	58	LAG_1	0
4	R0	3	32	LPC3	1	59	LAG_3	1
5	R0	2	33	GSP0_2	3	60	LAG_1	1
6	LPC3	3	34	GSP0_4	3	61	LAG_3	2
7	LPC4	3	35	GSP0_2	2	62	LAG_1	2
8	LPC1	2	36	GSP0_4	2	63	GSP0_3	6
9	LPC5	3	37	GSP0_2	1	64	GSP0_1	6
10	LAG_2	6	38	GSP0_4	1	65	LAG_3	3
11	LAG_4	6	39	GSP0_2	0	66	LAG_1	3
12	LAG_2	5	40	GSP0_4	0	67	LAG_3	4
13	LAG_4	5	41	LPC6	2	68	LAG_1	4
14	LAG_2	4	42	GSP0_3	0	69	LAG_3	5
15	LAG_4	4	43	GSP0_1	0	70	LAG_1	5
16	LAG_2	3	44	GSP0_3	1	71	LAG_3	6
17	LAG_4	3	45	GSP0_1	1	72	LAG_1	6
18	GSP0_2	6	46	GSP0_3	2	73	LPC3	2
19	GSP0_4	6	47	GSP0_1	2	74	LPC2	3
20	LAG_2	2	48	GSP0_3	3	75	LPC1	3
21	LAG_4	2	49	GSP0_1	3	76	LPC2	4
22	LAG_2	1	50	LPC2	2	77	LPC1	4
23	LAG_4	1	51	GSP0_4	4	78	R0	4
24	LAG_2	0	52	GSP0_2	4	79	CRC	1
25	LAG_4	0	53	LPC4	2	80	CRC	3
26	GSP0_2	5	54	R0	1	81	CRC	5
27	GSP0_4	5						

To achieve a rate of 9/17, the output of the convolutional coder is punctured prior to transmission. Every 18th output bit of the convolutional coder is not transmitted. This leads to 9 punctures over the 174 bits output by the R=1/2 mother code. Only the outputs from $g_1(D)$ are punctured. The punctures of the output array $cc1(i)$ occur at $i=8,17,26,35,44,53,62,71,80$.

5.1.2.5 Interleaving

Before transmission, the encoded speech data is interleaved over two time slots with the speech data from an adjacent speech frame. Stated another way, each time slot contains information from two speech coder frames. Table 5.1.2.5-1 shows the placement of the bits within the class 2 array, $CL2[i]$. The transmission order of the coded Class 1 bits from the convolutional coder, $cc0(i)$ and $cc1(i)$, and the Class 2 bits, $CL2[i]$ is shown in Table 5.1.2.5-2. The two speech frames are referred to as x and y where x is the previous speech frame and y is the present or most recent speech frame. The order of transmission over the channel is indicated by the Transmit Order columns where 0 indicates the first bit for transmission and 223 indicates the last transmitted bit in a time slot for the speech data. Puncture indicates that the bit is not transmitted.

Table 5.1.2.5-1: Bit Ordering for Class 2 Bits

CL2[i]	i	Parameter Name	Bit Index	CL2[i]	i	Parameter Name	Bit Index	CL2[i]	i	Parameter Name	Bit Index
0		LPC1	0	20		LPC10	1	40		CODE_2	8
1		LPC2	1	21		LPC10	0	41		CODE_3	7
2		LPC2	0	22		R0	0	42		CODE_3	6
3		LPC3	0	23		CODE_1	7	43		CODE_3	5
4		LPC4	1	24		CODE_1	6	44		CODE_3	4
5		LPC4	0	25		CODE_1	5	45		CODE_3	3
6		LPC5	2	26		CODE_1	4	46		CODE_3	2
7		LPC5	1	27		CODE_1	3	47		CODE_3	1
8		LPC5	0	28		CODE_1	2	48		CODE_3	0
9		LPC6	1	29		CODE_1	1	49		CODE_3	8
10		LPC6	0	30		CODE_1	0	50		CODE_4	7
11		LPC7	2	31		CODE_1	8	51		CODE_4	6
12		LPC7	1	32		CODE_2	7	52		CODE_4	5
13		LPC7	0	33		CODE_2	6	53		CODE_4	4
14		LPC8	2	34		CODE_2	5	54		CODE_4	3
15		LPC8	1	35		CODE_2	4	55		CODE_4	2
16		LPC8	0	36		CODE_2	3	56		CODE_4	1
17		LPC9	2	37		CODE_2	2	57		CODE_4	0
18		LPC9	1	38		CODE_2	1	58		CODE_4	8
19		LPC9	0	39		CODE_2	0				

Table 5.1.2.5-2: Transmission Order

Bit Name	Frame	Transmit Order	Bit Name	Frame	Transmit Order	Bit Name	Frame	Transmit Order
cc0[0]	x	112	cc0[39]	x	184	cc0[78]	x	33
cc1[0]	y	113	cc1[39]	y	185	cc1[78]	y	34
cc0[1]	x	126	cc0[40]	x	198	cc0[79]	x	47
cc1[1]	y	127	cc1[40]	y	199	cc1[79]	y	48
cc0[2]	x	140	cc0[41]	x	212	cc0[80]	x	61
cc1[2]	y	141	cc1[41]	y	213	cc1[80]	puncture	
cc0[3]	x	154	cc0[42]	x	2	cc0[81]	y	62
cc1[3]	y	155	cc1[42]	y	3	cc1[81]	x	75
cc0[4]	x	168	cc0[43]	x	16	cc0[82]	y	76
cc1[4]	y	169	cc1[43]	y	17	cc1[82]	x	89
cc0[5]	x	182	cc0[44]	x	30	cc0[83]	y	90
cc1[5]	y	183	cc1[44]	puncture		cc1[83]	x	103
cc0[6]	x	196	cc0[45]	y	31	cc0[84]	y	104
cc1[6]	y	197	cc1[45]	x	44	cc1[84]	x	10
cc0[7]	x	210	cc0[46]	y	45	cc0[85]	y	11
cc1[7]	y	211	cc1[46]	x	58	cc1[85]	x	24
cc0[8]	x	0	cc0[47]	y	59	cc0[86]	y	25
cc1[8]	puncture		cc1[47]	x	72	cc1[86]	x	38
cc0[9]	y	1	cc0[48]	y	73	CL2[0]	y	137
cc1[9]	x	14	cc1[48]	x	86	CL2[1]	x	95
cc0[10]	y	15	cc0[49]	y	87	CL2[2]	y	102
cc1[10]	x	28	cc1[49]	x	100	CL2[3]	x	130
cc0[11]	y	29	cc0[50]	y	101	CL2[4]	y	200
cc1[11]	x	42	cc1[50]	x	12	CL2[5]	x	177
cc0[12]	y	43	cc0[51]	y	13	CL2[6]	y	107
cc1[12]	x	56	cc1[51]	x	26	CL2[7]	x	186
cc0[13]	y	57	cc0[52]	y	27	CL2[8]	y	109
cc1[13]	x	70	cc1[52]	x	40	CL2[9]	x	123
cc0[14]	y	71	cc0[53]	y	41	CL2[10]	y	94
cc1[14]	x	84	cc1[53]	puncture		CL2[11]	x	165
cc0[15]	y	85	cc0[54]	x	54	CL2[12]	y	39
cc1[15]	x	98	cc1[54]	y	55	CL2[13]	x	214

cc0[16]	y	99	cc0[55]	x	68	CL2[14]	y	121
cc1[16]	x	7	cc1[55]	y	69	CL2[15]	x	144
cc0[17]	y	8	cc0[56]	x	82	CL2[16]	y	79
cc1[17]	puncture		cc1[56]	y	83	CL2[17]	x	67
cc0[18]	x	21	cc0[57]	x	96	CL2[18]	y	9
cc1[18]	y	22	cc1[57]	y	97	CL2[19]	x	74
cc0[19]	x	35	cc0[58]	x	110	CL2[20]	y	158
cc1[19]	y	36	cc1[58]	y	111	CL2[21]	x	207
cc0[20]	x	49	cc0[59]	x	124	CL2[22]	y	178
cc1[20]	y	50	cc1[59]	y	125	CL2[23]	x	205
cc0[21]	x	63	cc0[60]	x	138	CL2[24]	y	122
cc1[21]	y	64	cc1[60]	y	139	CL2[25]	x	151
cc0[22]	x	77	cc0[61]	x	152	CL2[26]	y	150
cc1[22]	y	78	cc1[61]	y	153	CL2[27]	x	116
cc0[23]	x	91	cc0[62]	x	166	CL2[28]	y	206
cc1[23]	y	92	cc1[62]	puncture		CL2[29]	x	136
cc0[24]	x	105	cc0[63]	y	167	CL2[30]	y	4
cc1[24]	y	106	cc1[63]	x	180	CL2[31]	x	172
cc0[25]	x	119	cc0[64]	y	181	CL2[32]	y	193
cc1[25]	y	120	cc1[64]	x	194	CL2[33]	x	149
cc0[26]	x	133	cc0[65]	y	195	CL2[34]	y	32
cc1[26]	puncture		cc1[65]	x	208	CL2[35]	x	108
cc0[27]	y	134	cc0[66]	y	209	CL2[36]	y	60
cc1[27]	x	147	cc1[66]	x	222	CL2[37]	x	93
cc0[28]	y	148	cc0[67]	y	223	CL2[38]	y	46
cc1[28]	x	161	cc1[67]	x	117	CL2[39]	x	53
cc0[29]	y	162	cc0[68]	y	118	CL2[40]	y	65
cc1[29]	x	175	cc1[68]	x	131	CL2[41]	x	80
cc0[30]	y	176	cc0[69]	y	132	CL2[42]	y	135
cc1[30]	x	189	cc1[69]	x	145	CL2[43]	x	219
cc0[31]	y	190	cc0[70]	y	146	CL2[44]	y	191
cc1[31]	x	203	cc1[70]	x	159	CL2[45]	x	37
cc0[32]	y	204	cc0[71]	y	160	CL2[46]	y	23
cc1[32]	x	217	cc1[71]	puncture		CL2[47]	x	163
cc0[33]	y	218	cc0[72]	x	173	CL2[48]	y	164

cc1[33]	x	114	cc1[72]	y	174	CL2[49]	x	51
cc0[34]	y	115	cc0[73]	x	187	CL2[50]	y	192
cc1[34]	x	128	cc1[73]	y	188	CL2[51]	x	88
cc0[35]	y	129	cc0[74]	x	201	CL2[52]	y	221
cc1[35]	puncture		cc1[74]	y	202	CL2[53]	x	81
cc0[36]	x	142	cc0[75]	x	215	CL2[54]	y	179
cc1[36]	y	143	cc1[75]	y	216	CL2[55]	x	66
cc0[37]	x	156	cc0[76]	x	5	CL2[56]	y	52
cc1[37]	y	157	cc1[76]	y	6	CL2[57]	x	220
cc0[38]	x	170	cc0[77]	x	19	CL2[58]	y	18
cc1[38]	y	171	cc1[77]	y	20			

5.1.3 Channel decoding

The channel decoding function is performed at the receiver and follows in the reverse processing order at the transmitter. The data stream is first deinterleaved, then all the coded bits are convolutionally decoded and the CRC is checked to detect errors in the 44 perceptually most significant bits. A bad frame masking process is also employed. The output of the channel decoder is a set of parameter codes which are then passed to the speech decoder.

5.1.3.1 De-interleaving

De-interleaving of the received data stream is the first step in the error correction process. Each time slot contains the interleaved information from parts of two speech coder frames. See section 5.1.2.5 for details. The nomenclature at the decoder is slightly different from the encoder, where frame x is the present speech coder frame and frame y is the next speech coder frame. Each time slot contains data for both speech frames. The ordering of the data within the time slot as well as the speech frame to which it belongs is shown in Table 5.1.3.1-1. Table 5.1.3.1-1 contains the same information as Table 5.1.2.5-2 but it is arranged in the order that the bits were transmitted.

Table 5.1.3.1-1: Ordering of Transmitted Bits

Transmit Order	Bit Name	Frame	Transmit Order	Bit Name	Frame	Transmit Order	Bit Name	Frame
0	cc0[8]	x	75	cc1[81]	x	150	CL2[26]	y
1	cc0[9]	y	76	cc0[82]	y	151	CL2[25]	x
2	cc0[42]	x	77	cc0[22]	x	152	cc0[61]	x
3	cc1[42]	y	78	cc1[22]	y	153	cc1[61]	y
4	CL2[30]	y	79	CL2[16]	y	154	cc0[3]	x
5	cc0[76]	x	80	CL2[41]	x	155	cc1[3]	y

6	cc1[76]	y	81	CL2[53]	x	156	cc0[37]	x
7	cc1[16]	x	82	cc0[56]	x	157	cc1[37]	y
8	cc0[17]	y	83	cc1[56]	y	158	CL2[20]	y
9	CL2[18]	y	84	cc1[14]	x	159	cc1[70]	x
10	cc1[84]	x	85	cc0[15]	y	160	cc0[71]	y
11	cc0[85]	y	86	cc1[48]	x	161	cc1[28]	x
12	cc1[50]	x	87	cc0[49]	y	162	cc0[29]	y
13	cc0[51]	y	88	CL2[51]	x	163	CL2[47]	x
14	cc1[9]	x	89	cc1[82]	x	164	CL2[48]	y
15	cc0[10]	y	90	cc0[83]	y	165	CL2[11]	x
16	cc0[43]	x	91	cc0[23]	x	166	cc0[62]	x
17	cc1[43]	y	92	cc1[23]	y	167	cc0[63]	y
18	CL2[58]	y	93	CL2[37]	x	168	cc0[4]	x
19	cc0[77]	x	94	CL2[10]	y	169	cc1[4]	y
20	cc1[77]	y	95	CL2[1]	x	170	cc0[38]	x
21	cc0[18]	x	96	cc0[57]	x	171	cc1[38]	y
22	cc1[18]	y	97	cc1[57]	y	172	CL2[31]	x
23	CL2[46]	y	98	cc1[15]	x	173	cc0[72]	x
24	cc1[85]	x	99	cc0[16]	y	174	cc1[72]	y
25	cc0[86]	y	100	cc1[49]	x	175	cc1[29]	x
26	cc1[51]	x	101	cc0[50]	y	176	cc0[30]	y
27	cc0[52]	y	102	CL2[2]	y	177	CL2[5]	x
28	cc1[10]	x	103	cc1[83]	x	178	CL2[22]	y
29	cc0[11]	y	104	cc0[84]	y	179	CL2[54]	y
30	cc0[44]	x	105	cc0[24]	x	180	cc1[63]	x
31	cc0[45]	y	106	cc1[24]	y	181	cc0[64]	y
32	CL2[34]	y	107	CL2[6]	y	182	cc0[5]	x
33	cc0[78]	x	108	CL2[35]	x	183	cc1[5]	y
34	cc1[78]	y	109	CL2[8]	y	184	cc0[39]	x
35	cc0[19]	x	110	cc0[58]	x	185	cc1[39]	y
36	cc1[19]	y	111	cc1[58]	y	186	CL2[7]	x
37	CL2[45]	x	112	cc0[0]	x	187	cc0[73]	x
38	cc1[86]	x	113	cc1[0]	y	188	cc1[73]	y
39	CL2[12]	y	114	cc1[33]	x	189	cc1[30]	x
40	cc1[52]	x	115	cc0[34]	y	190	cc0[31]	y

41	cc0[53]	y	116	CL2[27]	x	191	CL2[44]	y
42	cc1[11]	x	117	cc1[67]	x	192	CL2[50]	y
43	cc0[12]	y	118	cc0[68]	y	193	CL2[32]	y
44	cc1[45]	x	119	cc0[25]	x	194	cc1[64]	x
45	cc0[46]	y	120	cc1[25]	y	195	cc0[65]	y
46	CL2[38]	y	121	CL2[14]	y	196	cc0[6]	x
47	cc0[79]	x	122	CL2[24]	y	197	cc1[6]	y
48	cc1[79]	y	123	CL2[9]	x	198	cc0[40]	x
49	cc0[20]	x	124	cc0[59]	x	199	cc1[40]	y
50	cc1[20]	y	125	cc1[59]	y	200	CL2[4]	y
51	CL2[49]	x	126	cc0[1]	x	201	cc0[74]	x
52	CL2[56]	y	127	cc1[1]	y	202	cc1[74]	y
53	CL2[39]	x	128	cc1[34]	x	203	cc1[31]	x
54	cc0[54]	x	129	cc0[35]	y	204	cc0[32]	y
55	cc1[54]	y	130	CL2[3]	x	205	CL2[23]	x
56	cc1[12]	x	131	cc1[68]	x	206	CL2[28]	y
57	cc0[13]	y	132	cc0[69]	y	207	CL2[21]	x
58	cc1[46]	x	133	cc0[26]	x	208	cc1[65]	x
59	cc0[47]	y	134	cc0[27]	y	209	cc0[66]	y
60	CL2[36]	y	135	CL2[42]	y	210	cc0[7]	x
61	cc0[80]	x	136	CL2[29]	x	211	cc1[7]	y
62	cc0[81]	y	137	CL2[0]	y	212	cc0[41]	x
63	cc0[21]	x	138	cc0[60]	x	213	cc1[41]	y
64	cc1[21]	y	139	cc1[60]	y	214	CL2[13]	x
65	CL2[40]	y	140	cc0[2]	x	215	cc0[75]	x
66	CL2[55]	x	141	cc1[2]	y	216	cc1[75]	y
67	CL2[17]	x	142	cc0[36]	x	217	cc1[32]	x
68	cc0[55]	x	143	cc1[36]	y	218	cc0[33]	y
69	cc1[55]	y	144	CL2[15]	x	219	CL2[43]	x
70	cc1[13]	x	145	cc1[69]	x	220	CL2[57]	x
71	cc0[14]	y	146	cc0[70]	y	221	CL2[52]	y
72	cc1[47]	x	147	cc1[27]	x	222	cc1[66]	x
73	cc0[48]	y	148	cc0[28]	y	223	cc0[67]	y
74	CL2[19]	x	149	CL2[33]	x			

5.1.3.2 Convolutional decoding

After de-interleaving, data for one entire speech coder frame becomes available. The de-interleaved data that was convolutionally encoded must now be decoded. The choices of decoding algorithm impact the coding gain and the computational complexity. The choices made are implementation method dependent. To achieve acceptable decoding capability a Viterbi algorithm may be used. The Viterbi algorithm has a path memory of 29 or more, and use a majority vote to reach a bit decision. The distance metric is derived from the mean square distance. The punctured bits are not used in the calculation of the distance metric. Soft decision decoding may be used if additional coding gain is required.

5.1.3.3 Cyclic Redundancy Check (CRC)

After decoding the class 1 bits, the received CRC bits are checked to determine if an error has been detected in the 44 most perceptually significant bits received in each frame. This is done by generating a CRC based on the received 44 most perceptually significant bits. These 44 bits form a new input polynomial $a'(X)$ defined as

$$a'(X) = CL1[78]X^{43} + CL1[4]X^{42} + CL1[77]X^{41} + CL1[5]X^{40} + CL1[76]X^{39} + CL1[6]X^{38} \\ + CL1[75]X^{37} + \dots + CL1[58]X^3 + CL1[24]X^2 + CL1[57]X^1 + CL1[25]X^0 \quad (5.1.3.3-1)$$

The parity polynomial $crc[a'(X)]$ is generated based on the decoded input polynomial $a'(X)$ which may contain channel errors. The received CRC for the frame, $b'(X)$, is compared with $crc[a'(X)]$. If the two differ then an error has been detected in the 44 most perceptually significant bits for that speech frame.

5.1.3.4 Bad frame masking

Based on the CRC comparison, an error in the 44 most perceptually significant bits of the speech frame may be detected. This CRC comparison failure can occur because the data was corrupted so severely by channel errors that correction was not possible. Use of corrupted data for the generation of the speech signal can cause severe degradation to the speech quality. To prevent this problem, a bad frame masking strategy is employed.

The bad frame masking system is based on a 7 state machine as shown in Figure 5.1.3.4-1. The lines with arrows denote possible transition paths. A 1 on the path indicates that a CRC error was detected on the current frame, a 0 indicates no CRC error was detected. On every decode of a speech frame, the state machine can change state. State 0 occurs most often and implies that the CRC comparison was successful. State 6 occurs when there were at least 6 consecutive frames which failed the CRC check. The action taken at each of these states varies as well. At state 0 no action is taken. States 1 and 2 are simple frame repeats. States 3, 4 and 5 repeat and attenuate the speech. States 6 and 7 completely mute the speech.

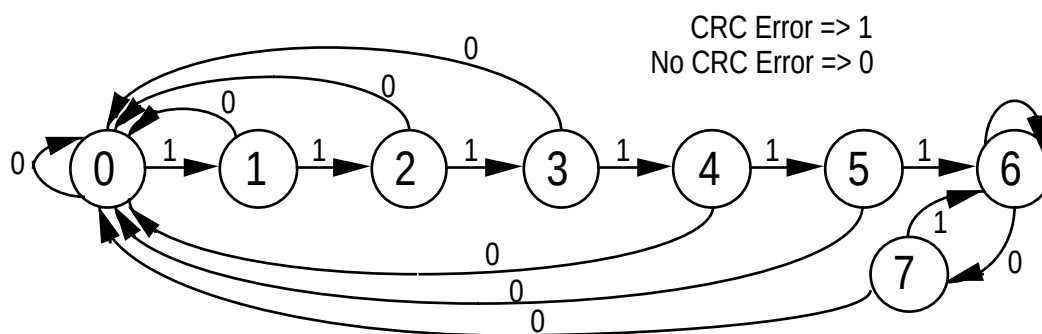


Fig. 5.1.3.4-1 Bad Frame Masking Strategy

The state count, with two exceptions, indicates how many consecutive frames had CRC comparison failures. For example, state 5 indicates 5 consecutive frames (including the current frame) have failed the comparison. The only exceptions are states 6 and 7 which may be preceded by indefinitely many corrupted frames. In any state (except state 6) agreement between the received and regenerated CRC's returns the state machine to the starting state, state 0. From state 6 two consecutive frames with no errors detected are required to return to state 0. This is used to provide additional protection during prolonged intervals of very poor channel conditions which might cause the CRC to occasionally falsely indicate valid speech data.

State 0, the error free state, is the initial state of the system. The state machine stays at this state unless a CRC error is detected where CRC error is defined as a disagreement of the regenerated CRC, $\text{crc}[a'(X)]$, and the received CRC, $b'(X)$. On each successive speech frame detected in error, the state machine moves to the next higher numbered state. As soon as the CRC detects no error for a speech frame, the machine returns to state 0 unless it was in state 6. If the machine is in state 6, a single frame without detected errors moves the machine to state 7 or two successive frames with no detected errors will cause the state machine to return to state 0. In each state the following actions are followed:

- State 0 - No CRC error is detected. The received decoded speech data is used.
- State 1 - A CRC error has been detected in the frame. The parameter values for $R(0)$, the LPC coefficients, $GSP0$, lags and soft interpolation are replaced with the corresponding values from the last frame that was in state 0. The remaining decoded bits for the frame are passed to the speech decoder without modification.
- State 2 - same action is taken as in state 1.
- State 3 - As in state 1 and 2, a frame repeat is done, except that the value of $R0$ is decremented by 2.
- State 4 - same as state 3. $R0$ is again decremented by 2, so now the level for $Rq(0)$ is 8 dB lower than the original value of $Rq(0)$.
- State 5 - $R0$ is decremented by an additional 2.
- State 6 - Again the frame is repeated, but this time $R0$ is set to zero. Alternatively, comfort noise could be inserted in place of the speech signal.
- State 7 - The speech signal remains muted - $R0$ remains at zero.

In states 3 through 5, after the value of $R0$ is decremented by 2 it is checked to ensure that its value is ≥ 0 . If the decremented value of $R0$ is < 0 then $R0$ is set equal to 0.

5.1.4 Speech decoding

The speech decoder takes the 6700 bps data from the channel decoder and generates the received speech signal. The speech decoder must operate correctly in conjunction with the speech coder described in 5.1.1.

5.1.4.1 Definitions and basic decoder parameters

See 5.1.1.1

5.1.4.2 Short-term predictor coefficients

The short-term filter is equivalent to the traditional LPC synthesis filter. The transfer function for the short-term filter is given by.

$$A(z) = \frac{1}{1 - \sum_{i=1}^{N_p} \alpha_i z^{-i}} \quad (5.1.4.2-1)$$

The short term predictor parameters are the α_i 's of the short term or synthesis filter. These are standard LPC direct form filter coefficients.

5.1.4.2.1 Decoding of coefficients

The short term predictor coefficients are coded as quantized reflection coefficients. These codes (LPC1 - LPC10) can be decoded into the 10 reflection coefficients (r_1 - r_{10}) using the Table 5.1.1.4.3-2.

5.1.4.2.2 Conversion to direct form coefficients

See 5.1.1.4.4

5.1.4.2.3 Interpolation of coefficients

See 5.1.1.4.5

5.1.4.3 Frame energy

An energy value is computed and encoded once per frame. This energy value, $R(0)$, reflects the average signal energy in the input speech over a 20 msec. interval which is centered with respect to the middle of the fourth subframe.

DECODING OF FRAME ENERGY

The quantized value of $R(0)$, $R_q(0)$, is determined from $R0$ (the transmitted code for $R_q(0)$) using equations 5.1.1.5.2-3.

INTERPOLATION OF FRAME ENERGY

See 5.1.1.5.3

5.1.4.4 Subframe processing

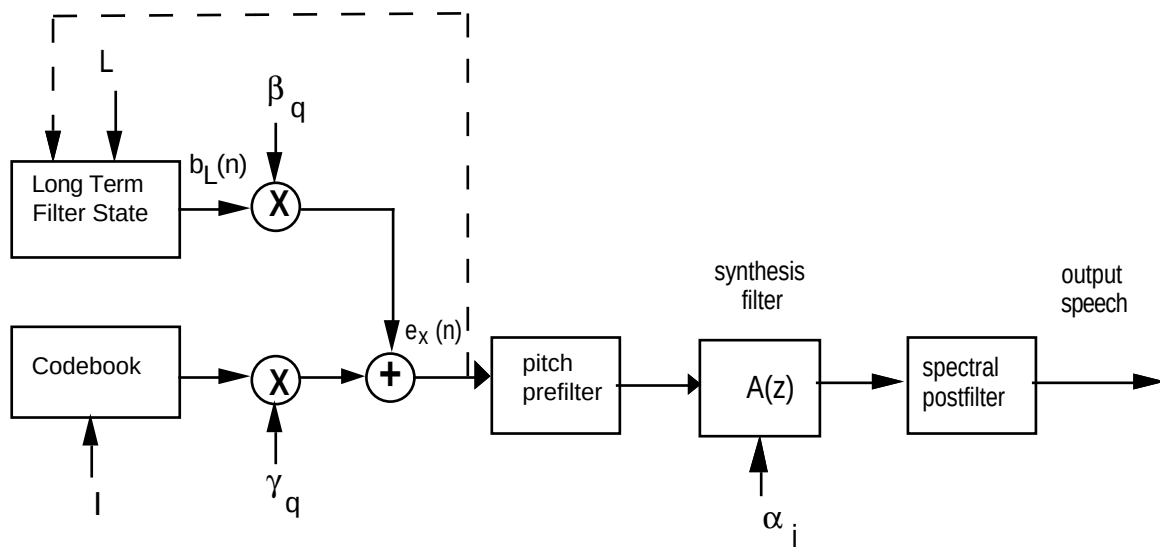


Fig. 5.1.4.4-1 is a block diagram of the speech decoder.

5.1.4.4.1 Decoding of lag

The lag, L , for subframe x can be determined from the code LAG_x as follows:

$$L = LAG_x + 19 \quad \text{if } LAG_x \neq 0 \quad (5.1.4.4.1-1)$$

long term predictor deactivated if $LAG_x = 0$

5.1.4.4.2 Decoding of excitation codewords

See 5.1.1.6.4.4

5.1.4.4.3 Decoding of GS and P0

The $GSP0_x$ code for subframe x is decoded into GS and P0 using Table 5.1.1.6.5.2-1.

5.1.4.4.4 Transformation of GS and P0 to gains

GS and P0 are transformed into β_q and γ_q using equations 5.1.1.6.5.2-7, and 5.1.1.6.5.2-8 if the long term predictor is activated. If the long term predictor is deactivated equations 5.1.1.6.5.2-10 and 5.1.1.6.5.2-11 are used.

5.1.4.4.5 Generation of combined excitation

The combined excitation, $ex(n)$, shall be computed as:

$$ex(n) = \beta_q b_L(n) + \gamma_q u_l(n) \quad (5.1.4.4.5-1)$$

for $0 \leq n \leq 39$

where $b_L(n)$ is defined by 5.1.1.6.3.1-15 and the codebook excitation vector, $u_l(n)$, is defined by 5.1.1.6.4-1 where l is the decoded codeword for the codebook.

5.1.4.4.6 Updating of long-term filter state

The long term predictor state, $r(n)$, is updated by:

$$r(n) = r(n+40) \quad \text{for } -146 \leq n \leq -41 \quad (5.1.1.6.6-1a)$$

$$r(n) = ex(n+40) \quad \text{for } -40 \leq n \leq -1 \quad (5.1.1.6.6-1b)$$

5.1.4.4.7 Synthesis filter

The combined excitation, $ex(n)$, is filtered by the synthesis filter to generate the speech signal. The synthesis filter is a tenth order all pole filter. The filter coefficients for the subframe are the α_i 's defined in 5.1.4.2.3. The filter coefficients will change from subframe to subframe. The filter state must be preserved from subframe to subframe. A direct form filter should be used for the synthesis filter.

5.1.4.4.8 Adaptive pitch prefilter

A pitch prefilter may be used to enhance the quality of the encoded speech. $H_p(z)$, the pitch prefilter transfer function, is given by

$$H_p(z) = \frac{1}{1 - \xi z^{-L}} \quad (5.1.4.4.8-1)$$

where

$$\xi = \varepsilon \text{ Min} \left[\beta, \sqrt{P_0} \right] \quad (5.1.4.4.8-2)$$

$$\varepsilon = 0.4$$

L is the long term predictor lag, and β is the long term predictor coefficient. Thus given $ex(n)$ as the input, $exp(n)$, the pitch prefiltered output, is defined by

$$exp(n) = ex(n) + \xi exp(n-L) \quad 0 \leq n \leq N-1 \quad (5.1.4.4.8-3)$$

To ensure unity power gain between the input and the output of the pitch prefilter, a gain scale factor is computed and is used to scale the pitch prefiltered excitation, prior to applying it the LPC synthesis filter. P_{scale} , the gain scale factor, is

$$P_{scale} = \sqrt{\frac{\sum_{n=0}^{N-1} ex^2(n)}{\sum_{n=0}^{N-1} exp^2(n)}} \quad (5.1.4.4.8-4)$$

The scale factor, P_{scale} , is the square root of the ratio of the input signal energy to output signal energy over the subframe. Thus $ex_{ps}(n)$, the gain corrected pitch prefiltered excitation which drives the LPC synthesis filter, is given by

$$ex_{ps}(n) = P_{scale} exp(n) \quad 0 \leq n \leq N-1 \quad (5.1.4.4.8-5)$$

5.1.4.4.9 Adaptive spectral postfilter

The perceptual quality of the synthetic speech may be enhanced by using an adaptive spectral postfilter as the final processing step. The form of the postfilter is

$$\hat{H}(z) = \frac{1 - \sum_{i=1}^{10} \eta_i z^{-i}}{1 - \sum_{i=1}^{10} v^i \alpha_i z^{-i}}, \quad 0 \leq v < 1 \quad (5.1.4.4.9-1)$$

where the α_i 's are the coefficients of the synthesis filter.

The numerator polynomial in 5.1.4.4.9-1 is a spectrally smoothed version of the denominator polynomial.

To derive the numerator coefficients, the autocorrelation of the impulse response of the all pole filter corresponding to the denominator of 5.1.4.4.9-1 is calculated for lags 0 through 10. The autocorrelation sequence is computed directly from the direct form coefficients via a recursion related to the Levinson recursion. The autocorrelation sequence is then windowed by a binomial window and the numerator coefficients (η_i for $i = 1$ to 10) are calculated from the windowed autocorrelation sequence via the Levinson recursion.

To have more control over postfiltered speech "brightness", a first order filter is used of the form:

$$\tilde{H}(z) = 1 - u z^{-1} \quad (5.1.4.4.9-2)$$

This filter is cascaded with filter 5.1.4.4.9-1 and is considered part of the adaptive spectral postfilter.

The following postfilter parameters may be used. Note that B_{eq} is the bandwidth expansion factor which specifies the degree of smoothing which is performed on the denominator, to generate the numerator.

$$\begin{aligned} v &= 0.8 \\ B_{eq} &= 1200\text{Hz} \\ u &= 0.4 \end{aligned}$$

The spectral smoothing coefficients (autocorrelation window), $wp(i)$, for 1200 Hz are:

$wp(0)$	1.000000
$wp(1)$	0.923077
$wp(2)$	0.725275
$wp(3)$	0.483516
$wp(4)$	0.271978
$wp(5)$	0.127990
$wp(6)$	0.049774
$wp(7)$	0.015718
$wp(8)$	0.003930
$wp(9)$	0.000748
$wp(10)$	0.000102

(5.1.4.4.9-3)

In order to reduce the computations needed to compute the spectrally smoothed numerator coefficients, one may perform the spectral smoothing operation once per frame on the denominator coefficients corresponding to the uninterpolated coefficients. This will yield the coefficients for the numerator of the spectral postfilter for subframe four. The numerator coefficients for subframes one, two, and three are interpolated using the same interpolation scheme that is used for the LPC synthesis coefficients as described in 5.1.1.4.6.

To ensure unity power gain between the input, $\hat{S}(n)$, and the output, $\hat{S}_p(n)$, of the spectral postfilter, a gain scale factor is computed and is used to scale the spectral postfiltered signal. S_{scale} , the postfilter scale factor, is

$$S_{scale} = \sqrt{\frac{\sum_{n=0}^{N-1} \hat{S}^2(n)}{\sum_{n=0}^{N-1} \hat{S}_p^2(n)}} \quad 0 \leq n \leq N-1 \quad (5.1.4.4.9-4)$$

The scale factor, S_{scale} , is the square root of the ratio of the input signal energy to output signal energy over the subframe. This scale factor is not used directly. The scale factor is passed through a first order low pass filter. This filtering is given by :

$$S'_{scale}(n) = (.9875 * S'_{scale}(n-1)) + (.0125 * S_{scale}) \quad (5.1.4.4.9-5)$$

The output of the spectral postfilter, $\hat{S}_p(n)$, is then multiplied by $S'_{scale}(n)$ as the last step in reconstructing the speech signal in the speech decoder.

5.1.4.5 Audio interface

The function of the audio interface at the mobile receiver is to convert the signal from the speech decoder to an analog speech signal.

The speech coder shall be succeeded by the following voice processing stages.

- * Digital to Analog Converter
- * Reconstruction filter
- * Receive Level Adjustment

The characteristics of these stages are described in the following sections.

5.1.4.5.1 Digital-to-analog converter

The D/A converter shall be performed at an 8 kHz rate according to either of the following.

- * by direct conversion from PCM to analog
- * or by making an uniform/ μ law code conversion succeeded by a standard codec D/A converter.

The uniform/ μ law code conversion is performed according to definition in ITU-T Red Book G.721, section 5.1.4.2.7 sub-block COMPRESS. The parameter LAW shall be set to LAW=0. The D/A conversion is based on the standard specified in ITU-T Red Book G.711.

5.1.4.5.2 Reconstruction filter

The function of the filter is to reconstruct the analog band-limited speech signal from the D/A converter. The attenuation of the filter shall comply with ITU-T Red Book G.714 receiving filter.

Note : The filter specification in G.714 is concerned with PCM equipment. In some cases more attenuation will be needed in a terminal equipment.

5.1.4.5.3 Reception level adjustment

The sensitivity shall be such that when an encoded C-message weighted white noise acoustic signal with a value of 24 in the R0 bit field is received by the receive speech coder an acoustic level of at least 97 dBspl shall be produced at the electric-to-audio transducer when measured by an artificial ear. R0 equal to 24 represents the average signal energy during a frame which is 18 dB down from full scale. The pressure developed in the artificial ear is to be measured in accordance with "IEEE Standard Method for Measuring Transmission Performance of Telephone Sets," IEEE std 269-1983. Higher levels may be required for certain applications.

However, when an encoded C-message white noise (or ITU-T recommendation Artificial P.50) with a value of 24 in R0 bits fields is received by receive speech coder and an electric signal output level of -25.5 dBs $\pm$ 4dB is produced, the acoustic signal mentioned above may not required.

The relationship between the value in the receive decoder R0 field and the receive transducer input signal shall be such that a linear relationship within ± 2 dB is maintained between the value in the R0 field and the input to the transducer for a +14, -28 dB range around the level which produced a value of 24 in R0.

[1] "IEEE Standard Method for Measuring Transmission Performance of Telephone Sets", IEEE std 269-1983.

5.2 Half-rate speech CODEC

Speech coding algorithm specified in this chapter is the PSI-CELP (Pitch Synchronous Innovation-Code Excited Linear Prediction). This is a speech coding method compressing redundant information in speech by using codebook parameters to generate the speech in the same manner as in the VSELP. However, as its feature, the PSI-CELP further improves the quantization efficiency by processing the codebook synchronizing with the fundamental period of the speech.

The speech codec is described as a series of algorithmic steps .

Some of these algorithmic steps are optional and may be implemented by means other than that described in this document. If the implementation employs algorithms which differ from those algorithms described as optional here, care must be exercised to ensure that the resulting speech quality is no worse than the examples described in this document. This requires a speech codec validation test for the assurance. In that test, encoder unit, decoder unit and the overall codec performances are tested on various conditions.

5.2.1 Speech coding

5.2.1.0 Definitions

$s_{in}(i)$	input speech
$s_{nc}(i)$	noise canceller output
$s(i)$	pre-processed input speech
$N_S=4$	the number of subframes
$N_{SUB}=80$	subframe length
$N_A=160$	LPC analysis length
$N_P=10$	LPC order
$S_{ana}(n,i)$	LPC analyzer input for the n-th subframe
$s_W(n,i)$	hamming windowed $S_{ana}(n,i)$
$W_{H160}(i)$	160 sample length hamming window
$\phi(n,j)$	j-th order autocorrelation of $s_W(n,i)$
$\phi_W(n,j)$	lag windowed $\phi(n,j)$
$W_L(i)$	lag window
$\alpha(n,i)$	short term predictor coefficients for the n-th subframe
$\omega(n,i)$	LSP coefficients for the n-th subframe
$l_{spn}(i)$	normalized LSP coefficients for the n-th subframe
$l_{spqn}(i)$	quantized $l_{spn}(i)$
$C_{LSPL}()$	low frequency LSP coefficients codebook
$C_{LSPH}()$	high frequency LSP coefficients codebook
$C_{LSPM1}()$	1st stage middle frequency LSP coefficients codebook
$C_{LSPM2}()$	2nd stage middle frequency LSP coefficients codebook
$\alpha_q(n,i)$	quantized short term predictor coefficients for the n-th subframe
$k_q(n,i)$	quantized PARCOR coefficients for n-th subframe
$C_{pow}()$	power codebook
P_n	quantized power for the n-th subframe
$N_{LTP}=256$	analysis length of the long term predictor lag
$F(n,z)$	LPC inverse filter for the n-th subframe
$res(n,i)$	LPC residual signal for the long term predictor lag analysis for the n-th subframe
$res_W(n,i)$	hamming windowed $res(n,i)$
$W_{H256}(i)$	256-sample length hamming window
$\phi_{res_{int}}(n,i)$	j-th order autocorrelation of $res_W(n,i)$

$\phi_{res_{zip}}(n,i)$	zero interpolated $\phi_{res_{int}}(n,i)$
$\phi_{res_{ups}}(n,i)$	four times oversampled $\phi_{res_{int}}(n,i)$
$W_{LIP}(i)$	interpolation window for $\phi_{res_{ups}}(n,i)$ generation
$PSEL(n,i)$	long term predictor lag pre-selection candidates for the n--th subframe
$cov(n,i)$	3-order pitch predictor coefficients for the n-th subframe
$l(z)$	spectral weighting filter
$\alpha_{fir}(i)$	spectral weighting filter coefficients
$Q(z)$	pitch weighting filter
$W(z)$	perceptual weighting filter
$G(z)$	poll-zero spectral weighting filter
$H(z)$	perceptually weighted synthesis filter
$spw(i)$	perceptually weighted input signal
$r(i)$	reference speech after zero input response subtraction
ACB	adaptive codebook
FCB	fixed codebook
$dq(i)$	adaptive codebook state
$lag_i(j)$	long term predictor lag candidates (integral part)
$lag_f(j)$	long term predictor lag candidates (fractional part)
$NCAN_{LTP}$	number of long term predictor lag candidates
$P_{ACB}(i,j)$	adaptive codebook output candidates
$W_n(i)$	interpolation window for ACB output
$rh(i)$	time reverse synthesis of $r(i)$
$NCAN_{ACB}$	number of ACB pre-selection candidates
$P_{FCB}(i,j)$	fixed codebook output candidates
$P(i)$	ACB/FCB output
$SYN_p(i)$	synthesized ACB/FCB output with perceptual weighting
SCB0	stochastic codebook #0
SCB1	stochastic codebook #1
lag_i	ACB lag (integral part)
lag_f	ACB lag (fractional part)
$PSIm()$	PSI processed SCBm
$C_{SCB0}()$	SCB0 table
$C_{SCB1}()$	SCB1 table
$ph(i)$	time reverse synthesis of $SYN_p(i)$
$SYN_{PSIm}()$	synthesized $PSIm()$ with perceptual weighting
$SYNO_{PSIm}()$	orthogonalized $SYN_{PSIm}()$
$S0, S1$	signs for SCB0 and SCB1 output
$SC(i)$	SCB output
$SYN_{SC}(i)$	synthesized SCB output with perceptual weighting
RS	estimated residual power
$Cg0(i), Cg1(i)$	gain codebook
$g0$	gain for ACB/FCB output
$g1$	gain for SCB output
$ex(i)$	excitation signal
$Ndq=98$	adaptive codebook length

5.2.1.1 Encoder block diagram

Figure 5.2.1-1 shows the block diagram of the encoder.

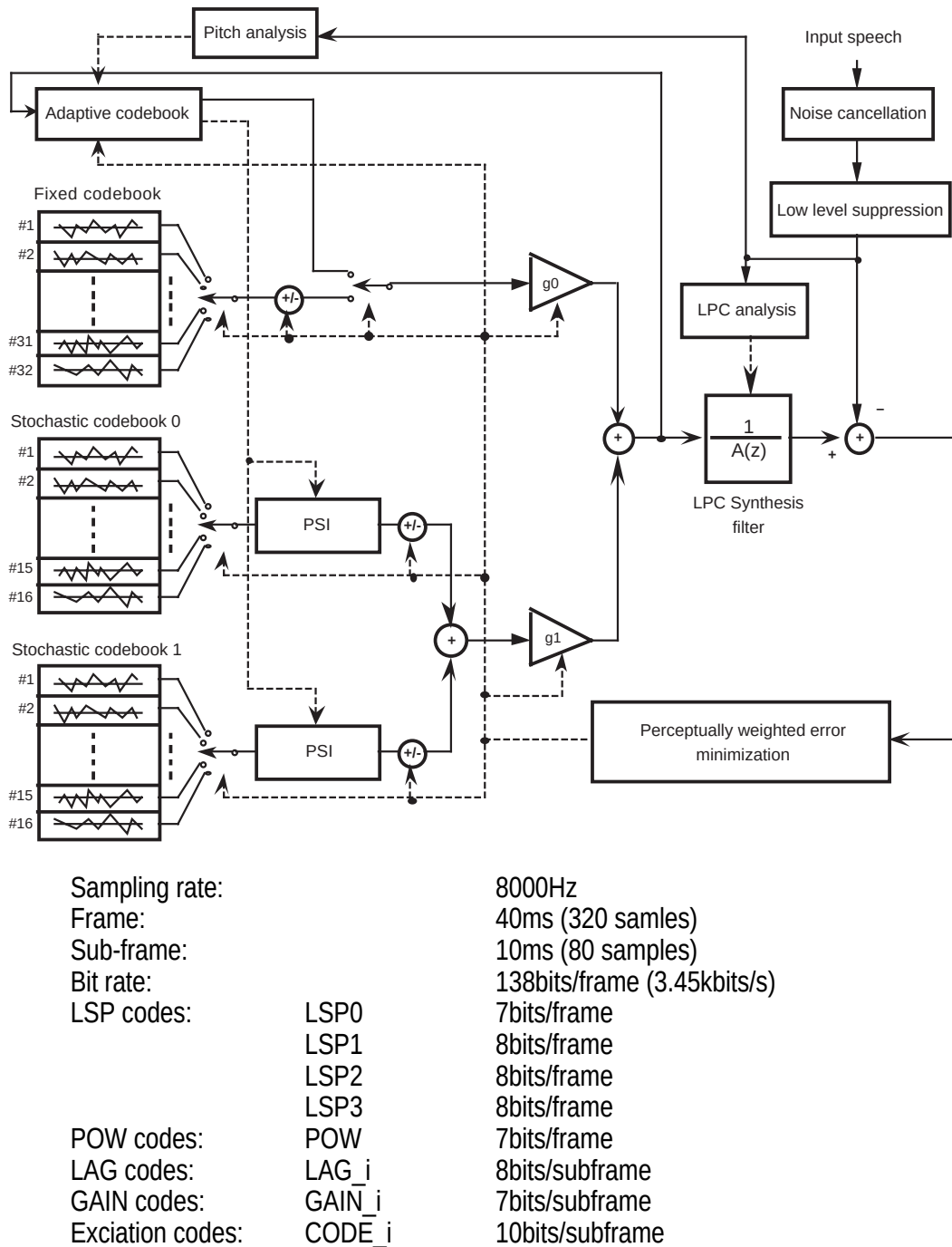


Fig.5.2.1.1-1 : Block Diagram of Encoder

5.2.1.2 Noise canceller

Noise canceller may be used to remove the audio background noise from the input speech signal. The inclusion of the noise canceller in the codec is optional. If a noise canceller is included in the codec, the noise canceller described in this section and the low level suppression in Section 5.2.1.3 are optional.

The noise canceller input $S_{in}()$ is represented in the-16 bit 2's complemental linear PCM (amplitude=-32768~+32767) form.

State transition occurs in every frame as shown in Figure 5.2.1.2-1. Noise cancellation with the Kalman filter is carried out only in states 20, 21 or 22. In other states, the output signal is exactly the same as the input.

The state transition rules are defined in Table 5.2.1.2-1. Table 5.2.1.2-2 explains updating method of the variables in terms of the state transition shown in Table 5.2.1.2-1.

Figure 5.2.1.2-2 depicts the block diagram of the noise canceller. The noise canceller consists of Kalman filter and the input parameter (i.e. a_1 , a_2 , b_1 , b_2 , V_w and V_e) calculation blocks for it.

Kalman filtering algorithm used in the noise canceller is realized in accordance with Section 3. C of a reference [J.D.Gibson, B. Koo and S. D. Gray, "Filtering of Colored Noise for Speech Enhancement and Coding", IEEE Trans. SP, vol.39, No.8]. However, $\sigma^2 w$, $\sigma^2 \eta$ in the reference are redefined as V_w , V_e , respectively, while a_1 , a_2 , b_1 and b_2 are the same definition of the reference. Kalman filter coefficients are updated by using the first 2 samples in every frame. Kalman filter coefficients calculated by the first 2 samples are used constantly for the remaining 318 samples. The input-parameter calculations are shown in Table 5.2.1.2-3.

The processing-order symbols (A~H) in the Table 5.2.1.2-1 to 5.2.1.2-3 indicate the processing order. First, the process of the symbol A are executed, then B, C, •••, and H are done in turn. Kalman filtering is the following process. The processes with the same processing-order symbol are executed in the order as described in the tables.

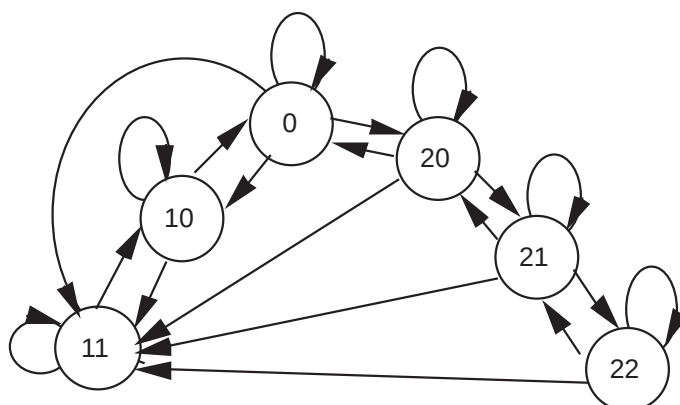


Fig. 5.2.1.2-1 : State Transition of the Noise Canceller

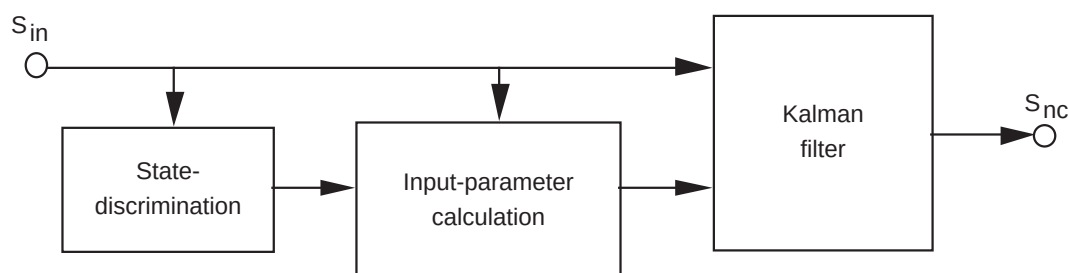


Fig. 5.2.1.2-2 : Block Diagram of the Noise Canceller Algorithm

Table 5.2.1.2-1 State transition rule

st Before trans.	st After trans.	Transi- tion	Transition definition	Processing- order symbol
0	0	a1 or a2	a1 : $P \leq P_{sl}$ a2 : $(st_cnt = 0) \text{ and } (P > P_{sl}) \text{ and } (P \leq spg \times P_{sp}) \text{ and } (Pow10_cnt > 20)$	C
0	10	b1 or b2	b1 : $(st_cnt > 0) \text{ and } (P_{sl} < P \leq P_n)$ b2 : $(st_cnt = 0) \text{ and } (P > P_{sl}) \text{ and } (P \leq spg \times P_{sp}) \text{ and } (Pow10_cnt \leq 20)$	
0	11	c1	c1 : $(st_cnt = 0) \text{ and } (P > spg \times P_{sp})$	
0	20	d1	d1 : $(st_cnt > 0) \text{ and } (P > P_n)$	
10	0	f1 or f2	f1 : $P \leq P_{sl}$ f2 : $(st_cnt = 0) \text{ and } (P > P_{sl}) \text{ and } (P \leq spg \times P_{sp}) \text{ and } (Pow10_cnt > 20)$	
10	10	g1 or g2	g1 : $(st_cnt > 0) \text{ and } (P_{sl} < P \leq P_{sp})$ g2 : $(st_cnt = 0) \text{ and } (P > P_{sl}) \text{ and } (P \leq spg \times P_{sp}) \text{ and } (Pow10_cnt \leq 20)$	
10	11	h1 or h2	h1 : $(st_cnt > 0) \text{ and } (P > P_{sp})$ h2 : $(st_cnt = 0) \text{ and } (P > spg \times P_{sp})$	
11	10	i1 or i2	i1 : $(st_cnt > 0) \text{ and } (P \leq P_{sp})$ i2 : $(st_cnt = 0) \text{ and } (P \leq spg \times P_{sp})$	
11	11	j1 or j2	j1 : $(st_cnt > 0) \text{ and } (P > P_{sp})$ j2 : $(st_cnt = 0) \text{ and } (P > spg \times P_{sp})$	
20	0	k1	k1 : $(st_cnt > 0) \text{ and } ((P = 0) \text{ or } (Pow20 < 0.5 \times P_n))$	
20	11	l1 or l2	l1 : $(st_cnt = 0)$ l2 : $(flg_kp > 3)$	F
20	20	m1	m1 : $(st_cnt > 0) \text{ and } (P_r < 2.2)$	
20	21	n1	n1 : $(st_cnt > 0) \text{ and } (P_r \geq 2.2)$	
21	11	o1 or o2	o1 : $(st_cnt = 0)$ o2 : $(flg_kp > 3)$	
21	20	p1 or p2	p1 : $(st_cnt > 0) \text{ and } (P_r < 2.4) \text{ and } (sstt \geq 8)$ p2 : $(st_cnt > 0) \text{ and } (P < 0.125 \times P_n)$	
21	21	q1 or q2	q1 : $(st_cnt > 0) \text{ and } (P \geq 0.125 \times P_n) \text{ and } (2.4 \leq P_r \leq 10.0)$ q2 : $(st_cnt > 0) \text{ and } (P \geq 0.125 \times P_n) \text{ and } (P_r \leq 10.0) \text{ and } (sstt < 8)$	
21	22	r1	r1 : $(st_cnt > 0) \text{ and } (P \geq 0.125 \times P_n) \text{ and } (P_r > 10.0)$	
22	11	s1 or s2	s1 : $(st_cnt = 0)$ s2 : $(flg_kp > 3)$	
22	21	t1	t1 : $(st_cnt > 0) \text{ and } (P_r < 2.2) \text{ and } (sstt \geq 2)$	
22	22	u1 or u2	u1 : $(st_cnt > 0) \text{ and } (P_r \geq 2.2)$ u2 : $(st_cnt > 0) \text{ and } (sstt = 0)$	

Table 5.2.1.2-2 Explanation of Variables

Variable/ Constant	Explanation	Processing-order symbol
Nf	Number of samples per frame, Nf = 320.	----
st	Variable to express noise canceller state. Initial value is 0. Either one of 0,10,11,20,21 or 22.	C or F (Table-5.2.1.2-1)
P	Power of one input signal sample averaged over the frame.	A
P_ratio	$P_ratio = P/(P' + \varepsilon)$: (for $P' = 0$), $P_ratio = P/P'$: (for $P' \neq 0$). where, "P'" is the value of "P" in the previous frame, "ε" is a sufficiently small positive constant, $\varepsilon = 0.001$. P' : Initial value is 0.	A
st_cnt	Transition control auxiliary variable. Initial value is 24. If st = 0,10, or 11 before transition and st_cnt > 0, subtract ONE after transition. If state transition is caused by "a2", "f2" condition, it is initialized again.	G
Psl, Pn, Psp	Constant. Thresholds for the mean power of input speech. $Psl = 3.0$ $Pn = 640.0$ $Psp = 2500.0$	----
P_min	Initial value is 0.0. "Condition-A" is defined as " $P > Psp$ and $0.3 < P_ratio < 1.0/0.3$ ". When st = 11 and st_cnt = 0, , if Condition-A is satisfied in both the previous and present frames, $P_min = \text{MIN}(P_min, P)$, where MIN(A,B) represents the smaller one of A or B. When st = 11 and st_cnt = 0, , if Condition-A is not satisfied in the previous frame but satisfied in the present frame, then $P_min = P$.	B
spg	A dimensionless variable.. Initial value is 1.0. When st = 11 and st_cnt = 0, if Condition-A is not satisfied, then $spg = 1.0$. if Condition-A is satisfied, then "spg" shall be maintained. if "Condition-A and st = 11 and st_cnt = 0" is satisfied for 11 consecutive frames including the present frame, then $spg = 3.0 \times P_min/Psp$.	B
	When st = 0, 10, , if Condition " $st_cnt = 0$ and $k0 < 0.1$ ", or " $st_cnt = 0$ and $Psl < P \leq spg \times Psp$ and $Pow10 \leq 2.0 \times Pn$ " is satisfied, then $spg = 1.0$. Besides, if the transition is caused by "a2", "f2" condition, it resets as $spg = 1.0$.	D

Variable/ Constant	Explanation	Processing-order symbol
Pow10, Pow10_cnt	Initial values are 0.0 and 0, respectively. When st = 0,10, and if Condition "st_cnt = 0 and $P_{s1} < P \leq spg \times P_{sp}$ " is satisfied, Pow10 = $0.9 \times Pow10 + 0.1 \times P$ and then Pow10_cnt = Pow10_cnt + 1 (if $Pow10 > 2.0 \times P_n$) Pow10_cnt = 0 (if $Pow10 \leq 2.0 \times P_n$). In the other cases, the values shall be maintained.	B
	If transition is caused by "a2", "f2" condition, it resets as Pow10_cnt = 0.	D
P_r	$P_r = 1.0$ (if st = 0 and Condition "d1" is satisfied)	B
	$P_r = P/Pow20$ (in other cases than above)	E
Pow20	$Pow20 = P$ (if st = 0 and Condition "d1" is satisfied)	B
	$Pow20 = 0.9 \times Pow20 + 0.1 \times P$ (if st = 20 and Condition "m1" is satisfied)	F
	$Pow20 = \text{Min}(P)$ (,if st=22 is repeated 51 times or more in succession and $sst \geq 8$ in the present frame.) where, Min(P) represents the minimum P value on those successive frames.	G
K0	If st_cnt = 0, then $K0 = (1.0 - ref01 \times ref01) \times (1.0 - ref02 \times ref02)$ is calculated. Where, the "ref01", "ref02" are the reflection coefficients in the 2nd order analysis, obtained by windowing the input Sin as $Sin(i) \times W_{H320}(i)$ ($0 \leq i \leq N_f - 1$) and applying autocorrelation method to that output.	B
W _{H320}	Hamming window applied to input signal. $W_{H320}(i) = 0.54 - 0.46 \times \cos(2\pi i / (N_f - 1))$ ($0 \leq i \leq N_f - 1$)	A
K, alf1, alf2	If st = 20, 21, 22, then $K = (1.0 - ref1 \times ref1) \times (1.0 - ref2 \times ref2)$ is calculated. Where, the "ref1", "ref2", are the reflection coefficients in the 2nd order analysis, obtained by windowing the input Sin as $Sin(i) \times W_{H320}(i)$ ($0 \leq i \leq N_f - 1$) and applying autocorrelation method to that output. The "alf1", "alf2" are linear predictor coefficients corresponding to the "ref1", "ref2".	E
flg_kp	Initial value is 0. If st = 20, 21, 22, the following updating shall be executed. flg_kp = flg_kp + 1, (,if (st = 20 and $K < 0.2$) or ($P_r < 0.01$)) flg_kp = 0, (,in the other cases.)	E
Var, Var'	Var is residual signal power per sample averaged over the frame. This residual is obtained by applying the input "Sin" to a 2nd order inverse filter with "alf1", "alf2" coefficients. Var' is the value of Var one frame before. Initial value is 0.	E

Variable/ Constant	Explanation	Processing-order symbol
Var20	If transition is caused by the condition "m1", then Var20 = Var is assigned. In other cases, the value shall be maintained. Initial value is 0.	F
Ratio	It is defined by $\text{Ratio} = \text{Var}' / (\text{Var} + \varepsilon \times \varepsilon)$: (for $\text{Var} = 0.0$), or $\text{Ratio} = \text{Var}' / \text{Var}$: (for $\text{Var} \neq 0.0$). The " ε " is the positive constant mentioned beforehand.	E
ssst	Either one of 0, 2, 4, 6, 8, 10. Initial value is 0. If st is 20, 21, 22, the value is updated by the following calculations. If ssst = 0 or ssst = 2 : $\begin{array}{ll} \text{ssst} = \text{ssst} + 2 & (0.6 < \text{Ratio} < 1.4) \\ \text{ssst} = 2 & (0.5 < \text{Ratio} \leq 0.6 \text{ or } 1.4 \leq \text{Ratio} < 1.5) \\ \text{ssst} = 0 & (\text{Ratio} \leq 0.5 \text{ or } 1.5 \leq \text{Ratio}) \end{array}$ If ssst = 4 : $\begin{array}{ll} \text{ssst} = 6 & (0.7 < \text{Ratio} < 1.3) \\ \text{ssst} = 4 & (0.6 < \text{Ratio} \leq 0.7 \text{ or } 1.3 \leq \text{Ratio} < 1.4) \\ \text{ssst} = 2 & (0.5 < \text{Ratio} \leq 0.6 \text{ or } 1.4 \leq \text{Ratio} < 1.5) \\ \text{ssst} = 0 & (\text{Ratio} \leq 0.5 \text{ or } 1.5 \leq \text{Ratio}) \end{array}$ If ssst = 6 : $\begin{array}{ll} \text{ssst} = 8 & (0.8 < \text{Ratio} < 1.2) \\ \text{ssst} = 6 & (0.7 < \text{Ratio} \leq 0.8 \text{ or } 1.2 \leq \text{Ratio} < 1.3) \\ \text{ssst} = 4 & (0.6 < \text{Ratio} \leq 0.7 \text{ or } 1.3 \leq \text{Ratio} < 1.4) \\ \text{ssst} = 2 & (0.5 < \text{Ratio} \leq 0.6 \text{ or } 1.4 \leq \text{Ratio} < 1.5) \\ \text{ssst} = 0 & (\text{Ratio} \leq 0.5 \text{ or } 1.5 \leq \text{Ratio}) \end{array}$ If ssst = 8 or ssst = 10 : $\begin{array}{ll} \text{ssst} = 10 & (0.9 < \text{Ratio} < 1.1) \\ \text{ssst} = 8 & (0.8 < \text{Ratio} \leq 0.9 \text{ or } 1.1 \leq \text{Ratio} < 1.2) \\ \text{ssst} = 6 & (0.7 < \text{Ratio} \leq 0.8 \text{ or } 1.2 \leq \text{Ratio} < 1.3) \\ \text{ssst} = 4 & (0.6 < \text{Ratio} \leq 0.7 \text{ or } 1.3 \leq \text{Ratio} < 1.4) \\ \text{ssst} = 2 & (0.5 < \text{Ratio} \leq 0.6 \text{ or } 1.4 \leq \text{Ratio} < 1.5) \\ \text{ssst} = 0 & (\text{Ratio} \leq 0.5 \text{ or } 1.5 \leq \text{Ratio}) \end{array}$	E

Table 5.2.1.2-3 Input-parameter calculation

st (After trains.)	20	21	22	Processing- order symbol
a1, a2	a1 = $-0.9 * \text{alf1}$ a2 = $-0.81 * \text{alf2}$	a1 = $- \text{alf1}$ a2 = $- \text{alf2}$	a1 = $- \text{alf1}$ a2 = $- \text{alf2}$	H
b1, b2	b1 = $- \text{alf1}$ b2 = $- \text{alf2}$	Same as the previous frame (No change)	Same as the previous frame (No change)	H
Coeffi- cients calcu- lation	w20 = $0.85 * w20 + 96 / \text{pow20}$ (pow20 < 12800) w20 = $0.85 * w20 + 0.0075$ (pow20 $\geq$ 12800) w21 = 1.0	w21 = $0.8 * w21 + 0.14$ (P/P' < 1.0) w21 = 1.0 (P/P' $\geq$ 1.0) w20 = 0.9	w21 = 1.0 w20 : No change	H
Vw, Ve	Vw = $w20 * \text{Var20}$ Ve = Var20	Vw = $w21 * \text{Var}$ Ve = $0.1 * \text{Var}$ (pow20 < $0.01 * P$) Ve = Var20 (pow20 $\geq$ $0.01 * P$)	Vw = $w21 * \text{Var}$ Ve = Var20	H

If transition is caused by the condition l2, o2 or s2, then w20 = 0.9 is set after the transition.

Initial value :
w20 = 0.0

5.2.1.3 Low level signal suppression

Noise canceller output $S_{nc}(n)$ may be processed by a filter below, or other one which possesses speech quality being no worse than that.

$$s(n) = \begin{cases} S_{nc}(n) & (S_{nc}(n) < -Th1) \\ (S_{nc}(n) + Th0) \cdot \frac{Th1}{Th1 - Th0} & (-Th1 \leq S_{nc}(n) < -Th0) \\ 0 & (-Th0 \leq S_{nc}(n) \leq Th0) , \\ (S_{nc}(n) - Th0) \cdot \frac{Th1}{Th1 - Th0} & (Th0 < S_{nc}(n) \leq Th1) \\ S_{nc}(n) & (Th1 < S_{nc}(n)) \end{cases} \quad (5.2.1.3-1)$$

Where $Th1=200$ and $Th0=5$. In the following sections, this $s(n)$ is referred as 'input signal'.

5.2.1.4 LPC analysis

From the $s(n)$, LPC parameters are derived for every subframe in the manner described in Fig. 5.2.1.4-1 using the autocorrelation method. The number of subframes N_s is 4, the subframe length N_{SUB} is 80, the analysis length N_A is 160, the analysis period is centered on the middle of each subframe and the order of the predictor N_p is 10. The samples for the n -th subframe analysis are represented by $S_{ana}(n,i)$ ($0 \leq n \leq N_s-1$, $0 \leq i \leq N_A-1$). In order to accomplish the LPC analysis of subframe #3, the procedures described in Section 5.2.1.2 and 5.2.1.3 are performed 40 samples ahead of these frame procedures.

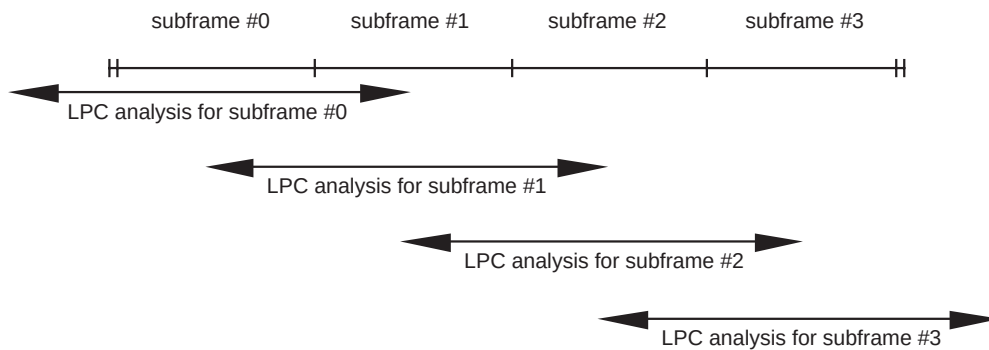


Fig. 5.2.1.4-1 : LPC analysis

5.2.1.4.1 Hamming window

$$s_W(n,i) = S_{ana}(n,i) \cdot W_{H160}(i) \quad (0 \leq n \leq N_s-1, 0 \leq i \leq N_A-1). \quad (5.2.1.4.1-1)$$

$$W_{H160}(i) = 0.54 - 0.46 \cdot \cos\left(\frac{2\pi(i + 0.5)}{N_A}\right) \quad (0 \leq i \leq N_A-1). \quad (5.2.1.4.1-2)$$

5.2.1.4.2 Auto-correlation

$$\phi(n,i) = \sum_{k=0}^{N_A-1-i} s_W(n,k) \cdot s_W(n,k+i) \quad (0 \leq n \leq N_s-1, 0 \leq i \leq N_p). \quad (5.2.1.4.2-1)$$

5.2.1.4.3 Lag window

$$\phi_w(n,i) = \phi(n,i) \cdot W_L(i) \quad (0 \leq n \leq N_s - 1, 0 \leq i \leq N_p). \quad (5.2.1.4.3-1)$$

$$\begin{aligned} W_L(0) &= 1.0000000 \\ W_L(1) &= 0.9994438 \\ W_L(2) &= 0.9977772 \\ W_L(3) &= 0.9950056 \\ W_L(4) &= 0.9911382 \\ W_L(5) &= 0.9861880 \\ W_L(6) &= 0.9801714 \\ W_L(7) &= 0.9731081 \\ W_L(8) &= 0.9650213 \\ W_L(9) &= 0.9559375 \\ W_L(10) &= 0.9458861 \end{aligned} \quad (5.2.1.4.3-2)$$

5.2.1.4.4 Short-term predictor coefficients

The windowed autocorrelation function $\phi_w(n,i)$ is converted into short term predictor coefficients $\alpha(n,i)$

5.2.1.5 Quantization of short-term predictor coefficients

The short term predictor coefficients of the first and the third subframe $\alpha(1,i)$, $\alpha(3,i)$ are quantized in LSP domain, and are used as LPC synthesis filter coefficients. The short term predictor coefficients for the zero and the second subframe are derived by linear interpolation in LSP domain.

5.2.1.5.1 Quantization of LSP parameters

The short term predictor coefficients $\alpha(1,i)$, $\alpha(3,i)$ are converted into the Linear Spectrum Pair parameters $\omega(1,i)$, $\omega(3,i)$ ($1 \leq i \leq N_p$). These parameters are normalized into the range from 0 to 1.

$$lspn(i) = \frac{\omega(n,i)}{\pi} \quad (n=1,3, 1 \leq i \leq N_p). \quad (5.2.1.5.1-1)$$

5.2.1.5.1.1 Quantization of low-frequency LSP parameters

The low frequency LSP code LSP0 is the index "j" which minimizes the Euclidian distance between a set of low frequency LSP parameters LSPL(i) and $C_{LSPL}(j,i)$ shown in Table 5.2.1.5-1.

$$LSPL(i) = \begin{cases} lsp1(i) & (i=1,2) \\ lsp3(i-2) & (i=3,4). \end{cases} \quad (5.2.1.5.1.1-1)$$

The value $C_{LSPL}(j,i)$ corresponding to the selected index is defined as quantized low frequency LSP parameter (LSPQL(i)), and the normalized LSP parameters $lspq1(i)$ and $lspq3(i)$ are computed as follows:

$$lspq1(i) = LSPQL(i) \quad (i=1,2), \quad (5.2.1.5.1.1-2)$$

$$lspq3(i) = LSPQL(i+2) \quad (i=1,2). \quad (5.2.1.5.1.1-3)$$

5.2.1.5.1.2 Quantization of high-frequency LSP parameters

The high frequency LSP code LSP3 is the index "j" which minimizes the Euclidian distance between a set of high frequency LSP parameters LSPH(i) and $C_{LSPH}(j,i)$ shown in Table 5.2.1.5-2.

$$LSPH(i) = \begin{cases} lsp1(i+6) & (1 \leq i \leq 4), \\ lsp3(i+2) & (5 \leq i \leq 8). \end{cases} \quad (5.2.1.5.1.2-1)$$

The value $C_{LSPH}(j,i)$ corresponding to the selected index is defined as quantized high frequency LSP parameter (LSPQH(i)) and the normalized LSP parameters $lspq1(i)$, $lspq3(i)$ are computed as follows:

$$lspq1(i) = LSPQH(i-6) \quad (7 \leq i \leq 10), \quad (5.2.1.5.1.2-2)$$

$$lspq3(i) = LSPQH(i-2) \quad (7 \leq i \leq 10). \quad (5.2.1.5.1.2-3)$$

5.2.1.5.1.3 Quantization of mid-frequency LSP parameters

MM=4 candidates with the lowest quantization error at the first quantization stage are selected. At the second stage, the quantization errors of the selected candidates are quantized. Finally the selected candidate with the minimal total error is used.

5.2.1.5.1.3.1 First-stage quantization of mid-frequency LSP parameters

A set of middle frequency LSP parameters LSPM(i) is computed as follows:

$$LSPM(i) = \begin{cases} lsp1(3) - lspq1(2) & (i=1), \\ lsp1(i+2) & (2 \leq i \leq 4), \\ lsp3(3) - lspq3(2) & (i=5), \\ lsp3(i-2) & (6 \leq i \leq 8). \end{cases} \quad (5.2.1.5.1.3.1-1)$$

MSEL(k) ($0 \leq k \leq MM-1$) are MM number of index "j" which minimize the Euclidian distance between LSPM(i) and $C_{LSPM1}(j,i)$ of Table 5.2.1.5-3. Then the values of $C_{LSPM1}(j,i)$ corresponding to the selected indices are denoted as LSPQM(k,i).

5.2.1.5.1.3.2 Second-stage quantization of mid-frequency LSP parameters

The error vector LSPE(k,i) for the k-th candidate selected at the first quantization stage is computed as follows:

$$LSPE(k,i) = LSPM(i) - LSPQM(k,i). \quad (5.2.1.5.1.3.2-1)$$

Next, for each "k" value, a set of $\theta(k,j)$ minimizing the Euclidian distance between $C_{LSPE}(k,i)$ shown in Eq. 5.2.1.5.1.3.2-2 and LSPE(k,i) is searched.

$$C_{LSPE}(k,i) = \sum_{j=0}^7 \theta(k,j) C_{LSPM2}(j,i), \quad (5.2.1.5.1.3.2-2)$$

where $\theta(k,j)$ is -1 or 1. $C_{LSPM2}(j,i)$ are shown in Table 5.2.1.5-4. Then MM candidates of quantized LSP parameters LSPC1(k,i) and LSPC3(k,i) are computed as follows:

$$LSPC1(k,i) = \begin{cases} lspq1(i) & (1 \leq i \leq 2), \\ LSPQM(k,i-2) + C_{LSPE}(k,i-2) + lspq1(2) & (i=3), \end{cases}$$

$$\begin{aligned}
& \text{LSPQM}(k, i-2) + C_{\text{LSPE}}(k, i-2) & (4 \leq i \leq 6), \\
& \text{lspq1}(i) & (7 \leq i \leq 10), \quad (5.2.1.5.1.3.2-3) \\
& \text{LSPC3}(k, i) = \text{lspq3}(i) & (1 \leq i \leq 2), \\
& \text{LSPQM}(k, i+2) + C_{\text{LSPE}}(k, i+2) + \text{lspq3}(2) & (i=3), \\
& \text{LSPQM}(k, i+2) + C_{\text{LSPE}}(k, i+2) & (4 \leq i \leq 6), \\
& \text{lspq3}(i) & (7 \leq i \leq 10). \quad (5.2.1.5.1.3.2-4)
\end{aligned}$$

5.2.1.5.1.3.3 LSP parameter stabilization

Parameters $\text{LSPCn}(k, i)$ ($n=1,3$) are stabilized in accordance with the procedure below.

- (1) If $(\text{LSPCn}(k, 3) - \text{LSPCn}(k, 2) < 0.02)$ is satisfied, then $\text{LSPCn}(k, 3)$ is replaced by $(\text{LSPCn}(k, 2) + 0.02)$,
- (2) The following are processed with $i=4,5,6$ in this order.

If $(\text{LSPCn}(k, i) - \text{LSPCn}(k, i-1) < 0.02)$ is satisfied, then $\text{LSPCn}(k, i)$ and $\text{LSPCn}(k, i-1)$ are replaced by $(\frac{\text{LSPCn}(k, i-1) + \text{LSPCn}(k, i)}{2} + 0.01)$ and $(\frac{\text{LSPCn}(k, i-1) + \text{LSPCn}(k, i)}{2} - 0.01)$ respectively.

- (3) If $(\text{LSPCn}(k, 7) - \text{LSPCn}(k, 6) < 0.02)$ is satisfied, then $\text{LSPCn}(k, 6)$ is replaced by $(\text{LSPCn}(k, 7) - 0.02)$.
- (4) If $(\text{LSPCn}(k, 6) < \text{LSPCn}(k, 5))$ is satisfied, then $\text{LSPCn}(k, 6)$ and $\text{LSPCn}(k, 5)$ are swapped.

5.2.1.5.1.3.4 LSP code determination

With regard to the stabilized $\text{LSPC1}(k, i)$ and $\text{LSPC3}(k, i)$, select the index "k" which minimizes the squared sum of the Euclidian distances to $\text{lsp1}(i)$ and $\text{lsp3}(i)$ respectively. The selected $\text{LSPC1}(k, i)$, $\text{LSPC3}(k, i)$ are denoted as $\text{lspq1}(i)$, $\text{lspq3}(i)$.

$\text{MSEL}(k)$ corresponding to the selected index "k" is represented as middle frequency LSP code #1 (LSP1). The middle frequency LSP code #2 (LSP2) is computed as follows, from the $\theta(k, j)$ corresponding to this "k".

$$\text{LSP2} = \sum_{j=0}^7 \frac{\theta(k, j) + 1}{2} \cdot 2^j. \quad (5.2.1.5.1.3.4-1)$$

5.2.1.5.2 LSP parameter interpolation

The $\text{lspq0}(i)$ and $\text{lspq2}(i)$ are computed as follows:

$$\text{lspq0}(i) = \frac{\text{lspq1}(i) + \text{lspq3}(i)}{2}, \quad (5.2.1.5.2-1)$$

$$\text{lspq2}(i) = \frac{\text{lspq3}(i) + \text{lspq1}(i)}{2}, \quad (5.2.1.5.2-2)$$

where $\text{lspq3}(i)$ is the $\text{lspq3}(i)$ of the previous frame.

Then, these $\text{lspqn}(i)$ ($0 \leq n \leq N_s - 1$) are converted to short-term predictor coefficients $\alpha_q(n, i)$ and PARCOR coefficients $k_q(n, i)$.

Table 5.2.1.5-1 Low Frequency LSP (1/3)

i	$C_{LSP}(i, 1)$	$C_{LSP}(i, 2)$	$C_{LSP}(i, 3)$	$C_{LSP}(i, 4)$
0	0.0616622	0.0816617	0.0617540	0.0817532
1	0.1038835	0.1604962	0.0761430	0.0961425
2	0.0939042	0.1139035	0.1044250	0.1244245
3	0.0896392	0.1581922	0.0879990	0.15 29584
4	0.0808655	0.1165662	0.0687500	0.0887494
5	0.0741210	0.1077010	0.0739762	0.1072280
6	0.0716719	0.0916715	0.0912949	0.1375089
7	0.1130005	0.1834074	0.1181122	0.1801189
8	0.0662247	0.0862317	0.0754179	0.1046567
9	0.1201875	0.1880995	0.0805145	0.1005140
10	0.1102005	0.1302075	0.1095519	0.1295515
11	0.1293960	0.1646772	0.1256500	0.1605572
12	0.0858475	0.1058469	0.0842072	0.1042067
13	0.1149690	0.1476102	0.1124969	0.1452374
14	0.0757537	0.0974239	0.1017167	0.1747174
15	0.1148697	0.1732144	0.1398790	0.2011610
16	0.0747084	0.0947157	0.0663240	0.0863310
17	0.0969330	0.1693844	0.0894257	0.1217084
18	0.1115890	0.1350447	0.0936295	0.1136290
19	0.1123902	0.1868940	0.0988862	0.1395382
20	0.0926147	0.1379209	0.0720154	0.0920225
21	0.0891357	0.1316420	0.0956285	0.1427732
22	0.0848252	0.1111417	0.0959944	0.1456572
23	0.1305709	0.1926237	0.1276259	0.1908154
24	0.0809190	0.1009185	0.0790802	0.0990797
25	0.1339660	0.1984067	0.1179442	0.1528744
26	0.0913177	0.1314434	0.1059967	0.1670500
27	0.1472029	0.1736495	0.1452957	0.1726652
28	0.0896470	0.1096465	0.0969177	0.1169247
29	0.1096740	0.1608090	0.1067367	0.1567884
30	0.0964447	0.1530882	0.1256424	0.1665692
31	0.1510407	0.2061812	0.1537489	0.2092787
32	0.0672927	0.0872922	0.0695204	0.0895200
33	0.1175857	0.1657377	0.0935302	0.1153149
34	0.1072860	0.1272932	0.1024489	0.1224562
35	0.1152282	0.1661572	0.1184707	0.1702925
36	0.1008317	0.1214719	0.0732667	0.0932660
37	0.0855727	0.1242720	0.0882355	0.1275067
38	0.0807282	0.1007277	0.1050812	0.1268277
39	0.1247344	0.1688657	0.1333479	0.1821105
40	0.0689484	0.0889477	0.0964522	0.1168409
41	0.1410004	0.1943097	0.0971314	0.1333737
42	0.1089874	0.1289869	0.1180434	0.1380965

Table 5.2.1.5-1 Low Frequency LSP (2/3)

43	0.1373155	0.1627927	0.1419539	0.1669355
44	0.0890365	0.1090360	0.0893417	0.1093412
45	0.1295867	0.1583829	0.1174560	0.1448637
46	0.0879990	0.1180617	0.1236969	0.1944395
47	0.1236512	0.1837124	0.1540620	0.2300152
48	0.0759065	0.0959057	0.0753265	0.0953260
49	0.1311049	0.1676832	0.1039825	0.1330227
50	0.1232697	0.1498837	0.0967117	0.1191984
51	0.1354842	0.1867109	0.1045395	0.1715969
52	0.0910584	0.1372037	0.0831085	0.1158490
53	0.1048752	0.1390042	0.1049820	0.1393247
54	0.0914017	0.1342055	0.1160295	0.1432767
55	0.1320435	0.2026409	0.1359267	0.2096602
56	0.0886092	0.1086087	0.0765320	0.0965315
57	0.1800400	0.2539792	0.1084915	0.1558194
58	0.1108414	0.1370665	0.1238647	0.1534314
59	0.1463180	0.1925475	0.1441665	0.1913799
60	0.1030747	0.1230742	0.0954072	0.1154064
61	0.1237427	0.1542249	0.1342177	0.1668287
62	0.0993667	0.1397672	0.1399552	0.1949507
63	0.1584639	0.2252392	0.1643462	0.2291914
64	0.0666825	0.0866819	0.0648057	0.0848050
65	0.1271607	0.1634184	0.0790267	0.0993009
66	0.0976655	0.1176650	0.1118712	0.1375317
67	0.1008622	0.1747174	0.0985504	0.1709257
68	0.0850769	0.1245389	0.0766692	0.1024290
69	0.0818954	0.1192517	0.0801560	0.1165435
70	0.0738160	0.0938154	0.1122147	0.1472745
71	0.1117797	0.1940505	0.1149002	0.2027402
72	0.0727707	0.0927702	0.0815980	0.1191220
73	0.1468827	0.2044797	0.0824600	0.1040235
74	0.1160064	0.1360135	0.1135802	0.1335874
75	0.1445480	0.1720395	0.1330277	0.1603894
76	0.0815597	0.1015592	0.0927519	0.1127515
77	0.1245514	0.1465727	0.1258559	0.1483274
78	0.0773329	0.0978512	0.1297012	0.1747785
79	0.1362777	0.1832930	0.1572205	0.2037167
80	0.0812012	0.1012007	0.0713287	0.0913282
81	0.1116730	0.1581770	0.0998017	0.1385695
82	0.1132447	0.1391492	0.1028000	0.1247067
83	0.1208817	0.1752057	0.1130614	0.1531492
84	0.1098267	0.1419874	0.0806977	0.1007505
85	0.0972457	0.1446195	0.0939042	0.1373945
86	0.0862289	0.1108517	0.1176392	0.1558347

Table 5.2.1.5-1 Low Frequency LSP (3/3)

87	0.1518417	0.2142377	0.1311127	0.1929287
88	0.0804382	0.1004377	0.0849242	0.1049237
89	0.1334929	0.2207532	0.1193787	0.1698422
90	0.0949340	0.1260342	0.1320817	0.1720777
91	0.1541762	0.1796079	0.1549697	0.1806607
92	0.0968872	0.1168942	0.0970627	0.1170622
93	0.1172424	0.1540569	0.1203017	0.1583295
94	0.1062637	0.1531797	0.1181579	0.1867642
95	0.1713042	0.2081495	0.1733490	0.2098889
96	0.0716872	0.0916867	0.0724122	0.0924115
97	0.1353242	0.1782805	0.0903487	0.1161314
98	0.1039062	0.1239057	0.1078660	0.1278654
99	0.1306702	0.1774107	0.1262604	0.1703230
100	0.1032882	0.1236462	0.0865722	0.1065717
101	0.0960480	0.1318402	0.0931107	0.1287657
102	0.0829634	0.1172757	0.1108947	0.1319089
103	0.1375520	0.1826214	0.1363692	0.1818510
104	0.0781570	0.1087919	0.0864349	0.1251875
105	0.1572814	0.2207837	0.0970244	0.1338467
106	0.1196992	0.1397137	0.1202942	0.1403012
107	0.1338897	0.1680722	0.1522537	0.1855054
108	0.0948960	0.1149030	0.0915085	0.1115079
109	0.1337220	0.1553845	0.1328674	0.1543090
110	0.0860840	0.1125990	0.1533677	0.2121092
111	0.1326690	0.1960264	0.1859300	0.2358747
112	0.0736022	0.0936017	0.0816285	0.1016279
113	0.1421525	0.1753584	0.1158615	0.1474499
114	0.1235442	0.1494185	0.1103607	0.1314970
115	0.1556717	0.1973310	0.1270232	0.1699492
116	0.1057907	0.1481214	0.0882507	0.1199079
117	0.1016327	0.1468167	0.1056762	0.1527752
118	0.1023347	0.1223339	0.1232239	0.1519894
119	0.1415345	0.2239730	0.1408860	0.2240034
120	0.0939347	0.1139417	0.0836732	0.1036724
121	0.1784225	0.2381560	0.1379180	0.2017255
122	0.1157394	0.1392027	0.1359572	0.1638684
123	0.1614625	0.1929669	0.1609055	0.1931347
124	0.1008392	0.1208387	0.1018919	0.1218914
125	0.1181044	0.1516157	0.1462342	0.1814159
126	0.1054854	0.1530499	0.1562514	0.2178464
127	0.1822602	0.2360119	0.1971144	0.2431302

Table 5.2.1.5-2 High Frequency LSP (1/12)

i	$C_{LSPH}(i, 1)$	$C_{LSPH}(i, 2)$	$C_{LSPH}(i, 3)$	$C_{LSPH}(i, 4)$
0	0.4720375	0.5610854	0.6633090	0.7274634
1	0.5002815	0.5698667	0.6936205	0.7901924
2	0.4562065	0.5650069	0.6863877	0.8008659
3	0.5793143	0.6505247	0.7496430	0.8002021
4	0.5336525	0.5972182	0.6470812	0.7219397
5	0.5380165	0.6324049	0.7409685	0.7989432
6	0.5218498	0.6531339	0.7285630	0.7741172
7	0.6218405	0.6753126	0.7461870	0.7877966
8	0.5195762	0.5860944	0.6530472	0.7743614
9	0.5433343	0.5942044	0.7269305	0.7756812
10	0.4682915	0.6018644	0.7606599	0.7977759
11	0.5330955	0.6096082	0.7201937	0.7809301
12	0.5308983	0.5945174	0.6643007	0.7510381
13	0.5921623	0.6331679	0.7660845	0.7992712
14	0.4887763	0.5871169	0.7107712	0.7708594
15	0.6520987	0.6889312	0.7525957	0.7919319
16	0.4727470	0.6041762	0.7074297	0.7559896
17	0.5431205	0.5868802	0.7438295	0.8053366
18	0.4457007	0.5437819	0.7470872	0.8053976
19	0.5888357	0.6485486	0.7397554	0.7749487
20	0.5908805	0.6292234	0.6719072	0.7259756
21	0.5932685	0.6480679	0.7562045	0.8048484
22	0.5466072	0.6392409	0.7277162	0.7668386
23	0.6118690	0.6781889	0.7760790	0.8070684
24	0.5472175	0.6045499	0.7367189	0.8075644
25	0.5764685	0.6112714	0.7868517	0.8258216
26	0.5060187	0.6477324	0.7753312	0.8092199
27	0.5657188	0.6440322	0.7132890	0.7639014
28	0.5801382	0.6228071	0.7170960	0.7680672
29	0.6110298	0.6446042	0.7783982	0.8163229
30	0.5564795	0.6527601	0.7674960	0.8006979
31	0.6701880	0.7062116	0.7654739	0.8000114
32	0.5016778	0.6108366	0.6812532	0.7323004
33	0.5101157	0.6096234	0.7580277	0.7937934
34	0.4951010	0.6192059	0.6949097	0.8011786
35	0.6111517	0.6681334	0.7458512	0.7926794
36	0.5719750	0.6240049	0.6869217	0.7839287
37	0.5908195	0.6709257	0.7788712	0.8211829
38	0.5199272	0.6527754	0.7573260	0.7899557
39	0.6545403	0.7023589	0.7648942	0.8024374
40	0.5682975	0.6147504	0.6927429	0.7874304
41	0.5634300	0.6095166	0.7623690	0.7983482
42	0.4505378	0.6374174	0.7824647	0.8200004

Table 5.2.1.5-2 High Frequency LSP (2/12)

43	0.5569677	0.6162001	0.7511002	0.8010719
44	0.5600730	0.6037031	0.6815737	0.7937247
45	0.6244498	0.6619994	0.7685945	0.7986686
46	0.5149223	0.6100737	0.7678087	0.8163687
47	0.6517325	0.6887252	0.7767502	0.8111502
48	0.5480110	0.6379134	0.7006625	0.7618797
49	0.5376808	0.6232572	0.7850740	0.8171394
50	0.4916833	0.5678679	0.7744157	0.8157964
51	0.6260902	0.6735197	0.7381380	0.7880637
52	0.5819922	0.6559874	0.7105120	0.7596671
53	0.6407998	0.6930281	0.7857682	0.8206489
54	0.5788870	0.6764417	0.7357422	0.7817236
55	0.6599875	0.7094006	0.8112274	0.8336341
56	0.5587990	0.6082502	0.7843874	0.8288199
57	0.5604010	0.6374479	0.7992112	0.8267142
58	0.5002205	0.6874739	0.7819460	0.8134924
59	0.5972357	0.6596342	0.7733017	0.8196799
60	0.5593252	0.6378982	0.7496202	0.8517309
61	0.6408303	0.6927304	0.7930162	0.8230901
62	0.5824042	0.6958739	0.7749955	0.8044364
63	0.6825400	0.7181591	0.7838227	0.8126836
64	0.4983513	0.5787474	0.7012727	0.7548681
65	0.5200797	0.5713239	0.7518404	0.8100517
66	0.4880440	0.5625881	0.7281205	0.7862402
67	0.6127158	0.6878401	0.7755982	0.8166891
68	0.5611565	0.6091809	0.6596924	0.7577139
69	0.6080467	0.6776471	0.7758500	0.8200004
70	0.5496665	0.6678511	0.7513902	0.7917334
71	0.6139518	0.6916851	0.7802444	0.8158956
72	0.5398553	0.5863539	0.7081620	0.8031012
73	0.5671760	0.6016662	0.7304322	0.8083274
74	0.4908820	0.6629987	0.7550370	0.7914359
75	0.5275642	0.6566739	0.7478577	0.7882162
76	0.5667258	0.6201596	0.6748827	0.7469337
77	0.5881110	0.6308636	0.7601487	0.8073279
78	0.4632177	0.5882764	0.7629487	0.8105552
79	0.6283028	0.6829876	0.7507875	0.7917717
80	0.5061485	0.6130339	0.7307297	0.7713629
81	0.5409155	0.5794569	0.7850360	0.8237311
82	0.4726402	0.5915951	0.7893539	0.8236471
83	0.6115485	0.6490447	0.7454927	0.7859122
84	0.6205130	0.6569409	0.6982362	0.7437444
85	0.6034690	0.6652724	0.7585772	0.8058019
86	0.5584860	0.6505934	0.7548997	0.7942282

Table 5.2.1.5-2 High Frequency LSP (3/12)

87	0.6427070	0.7226759	0.7821594	0.8102577
88	0.5638648	0.6198544	0.7462100	0.8098684
89	0.5959083	0.6332899	0.7953127	0.8261724
90	0.5360632	0.6511427	0.7893465	0.8144689
91	0.5836250	0.6379896	0.7417314	0.8068319
92	0.5994483	0.6367611	0.7395569	0.8066872
93	0.6200248	0.6562849	0.7938782	0.8224341
94	0.5780480	0.6670272	0.7910477	0.8187186
95	0.6987755	0.7286801	0.7849442	0.8117912
96	0.5399468	0.6337629	0.6981369	0.7523351
97	0.5591040	0.6391034	0.7348039	0.7957237
98	0.4967568	0.6343199	0.7269990	0.8221367
99	0.6155768	0.6637846	0.7681597	0.8170859
100	0.5940315	0.6340834	0.6847549	0.7810521
101	0.5934745	0.6761672	0.7946260	0.8256841
102	0.5260917	0.6754577	0.7666262	0.7980354
103	0.6871862	0.7278792	0.7912232	0.8186117
104	0.5878288	0.6353422	0.6998460	0.8317724
105	0.5813210	0.6224637	0.7266940	0.7985617
106	0.4582970	0.6316954	0.7904984	0.8231284
107	0.5796120	0.6506696	0.7835557	0.8164832
108	0.5737525	0.6503569	0.7285249	0.7819219
109	0.6262505	0.6693387	0.7581575	0.8037116
110	0.5001212	0.6334197	0.7920624	0.8183294
111	0.6662512	0.7011914	0.7952057	0.8222586
112	0.5434332	0.6442686	0.7047899	0.8032691
113	0.5750420	0.6090132	0.7942827	0.8263327
114	0.5038445	0.5879636	0.7820909	0.8173607
115	0.6021185	0.6791196	0.7404650	0.7818456
116	0.6044302	0.6453519	0.7111147	0.7729346
117	0.6218558	0.7028241	0.8133942	0.8518684
118	0.5728295	0.6662259	0.7506120	0.7865606
119	0.6670677	0.7332806	0.7965792	0.8208854
120	0.5772010	0.6255462	0.7844864	0.8241889
121	0.6192237	0.6787839	0.8068102	0.8328254
122	0.5448447	0.6870009	0.7941607	0.8195044
123	0.6392510	0.6977964	0.7894380	0.8201299
124	0.5760717	0.6753202	0.7643297	0.8650442
125	0.6243735	0.6880919	0.7972657	0.8253561
126	0.5701363	0.6800582	0.7938555	0.8195121
127	0.7078620	0.7421384	0.7966782	0.8187034
128	0.5093985	0.6034057	0.6783922	0.7798086
129	0.5135873	0.6093869	0.7283570	0.8005834
130	0.4584265	0.5996136	0.7162492	0.8321844

Table 5.2.1.5-2 High Frequency LSP (4/12)

131	0.5994177	0.6617551	0.7324922	0.7826621
132	0.5544425	0.6273542	0.6806429	0.7310339
133	0.5618430	0.6459699	0.7610262	0.8078087
134	0.5209647	0.6654632	0.7344072	0.7755287
135	0.6426918	0.6895566	0.7619569	0.8000036
136	0.5633843	0.6060606	0.6660172	0.8005071
137	0.5593788	0.6281707	0.7400910	0.7986532
138	0.4936440	0.6236922	0.7720887	0.8058327
139	0.5573187	0.6534924	0.7312487	0.8183371
140	0.5328360	0.6357846	0.7015855	0.7667854
141	0.5988838	0.6643949	0.7678697	0.7985769
142	0.4869148	0.6357619	0.7256562	0.7810369
143	0.6709588	0.7092557	0.7797409	0.8107839
144	0.4632713	0.6349759	0.7274567	0.7812887
145	0.5581352	0.5993544	0.7492692	0.8193824
146	0.4293510	0.5815474	0.7867142	0.8233114
147	0.6091300	0.6810574	0.7514055	0.7888646
148	0.5961142	0.6398589	0.6946352	0.7442024
149	0.6065055	0.6710629	0.7799090	0.8203284
150	0.5454475	0.6618544	0.7288455	0.7660377
151	0.6070243	0.7068754	0.7878817	0.8178946
152	0.5743475	0.6129041	0.7242295	0.8105934
153	0.5921013	0.6341672	0.7942217	0.8256767
154	0.5299523	0.6742369	0.8034914	0.8303381
155	0.5928642	0.6459852	0.7002732	0.7742927
156	0.5764837	0.6218839	0.7425097	0.7812811
157	0.6109077	0.6579481	0.8066804	0.8355491
158	0.5642082	0.6552779	0.7778107	0.8085792
159	0.6777565	0.7081036	0.7922379	0.8214424
160	0.5087272	0.6373259	0.7073535	0.7525259
161	0.5332252	0.6264692	0.7581880	0.7970054
162	0.5329962	0.6537977	0.7270752	0.8366094
163	0.6285010	0.6850249	0.7870577	0.8239981
164	0.5984183	0.6395612	0.7165544	0.8056114
165	0.5914680	0.6905026	0.7953737	0.8271491
166	0.5525353	0.6825071	0.7558762	0.7904594
167	0.6715690	0.7188381	0.7976015	0.8246694
168	0.5854103	0.6245314	0.7050112	0.7962119
169	0.5568152	0.6165129	0.7774979	0.8146139
170	0.4594718	0.6690031	0.8017899	0.8273474
171	0.5456762	0.6194884	0.7862642	0.8239904
172	0.5799857	0.6229214	0.6953142	0.8033071
173	0.6290045	0.6640439	0.7790392	0.8213506
174	0.5121605	0.6423612	0.7910019	0.8251044

Table 5.2.1.5-2 High Frequency LSP (5/12)

175	0.6530830	0.6982542	0.8069092	0.8338324
176	0.5739510	0.6586194	0.7184999	0.7711874
177	0.5758125	0.6519362	0.7908190	0.8240361
178	0.5132285	0.6118359	0.7978227	0.8251044
179	0.6401055	0.6823469	0.7450349	0.7830436
180	0.5921852	0.6738097	0.7257097	0.7709966
181	0.6675482	0.7178999	0.7961977	0.8242652
182	0.5989982	0.6933789	0.7517259	0.7895514
183	0.6689980	0.7362944	0.8167207	0.8367239
184	0.5572425	0.6468396	0.7901474	0.8323219
185	0.5750115	0.6706129	0.8116242	0.8338707
186	0.5144570	0.6971479	0.8134247	0.8334282
187	0.6063452	0.6733519	0.7870882	0.8228536
188	0.5619270	0.6679654	0.7740189	0.8622977
189	0.6452627	0.6790586	0.8010042	0.8272329
190	0.5980445	0.7185254	0.7897432	0.8137136
191	0.6768333	0.7362639	0.8132417	0.8332449
192	0.5382300	0.6020704	0.7004717	0.7626959
193	0.5576775	0.5978741	0.7764910	0.8243949
194	0.4967568	0.5918699	0.7689377	0.8239829
195	0.6224053	0.6891981	0.7687700	0.8050086
196	0.5784752	0.6266447	0.6733567	0.7336356
197	0.6221762	0.6960646	0.7974717	0.8418052
198	0.5592795	0.6809432	0.7706544	0.8046804
199	0.6444543	0.7151379	0.7908342	0.8203894
200	0.5489035	0.6090819	0.7018982	0.8357779
201	0.5867835	0.6189467	0.7542665	0.8210837
202	0.5072853	0.6542096	0.7812440	0.8075339
203	0.5305167	0.6547894	0.7620562	0.8037802
204	0.6070547	0.6481749	0.7028902	0.7640159
205	0.6011650	0.6502499	0.7679459	0.8157507
206	0.5119392	0.6018034	0.7444017	0.8056724
207	0.6587670	0.6960646	0.7668244	0.8052757
208	0.5236960	0.6242414	0.7439514	0.7882774
209	0.5542975	0.5994687	0.8050477	0.8312156
210	0.4876700	0.6318404	0.8058642	0.8341604
211	0.6302178	0.6716734	0.7446077	0.8090369
212	0.6262505	0.6669432	0.7204987	0.7605217
213	0.6245032	0.6778684	0.7751787	0.8148581
214	0.5825415	0.6653409	0.7662294	0.8007284
215	0.6299278	0.7212949	0.8111817	0.8311849
216	0.5940543	0.6344572	0.7677399	0.8214729
217	0.6052010	0.6455884	0.8120439	0.8349007
218	0.5396872	0.6680647	0.8147142	0.8379829

Table 5.2.1.5-2 High Frequency LSP (6/12)

219	0.5834570	0.6704296	0.7464312	0.7966391
220	0.6097558	0.6472746	0.7441195	0.8139426
221	0.6272575	0.6649747	0.8138519	0.8358237
222	0.6030952	0.6793332	0.7897585	0.8187339
223	0.7013390	0.7354247	0.8094499	0.8322379
224	0.5538627	0.6413924	0.7064607	0.7785499
225	0.5746528	0.6457944	0.7718599	0.8155142
226	0.5128090	0.6579177	0.7593705	0.8440711
227	0.6324150	0.6840254	0.7968614	0.8300329
228	0.5953895	0.6501584	0.7184617	0.7977454
229	0.6213218	0.7057461	0.8036362	0.8331991
230	0.5449515	0.7019544	0.7781999	0.8055731
231	0.6971198	0.7390714	0.8093277	0.8303304
232	0.5991127	0.6571469	0.7295930	0.8465812
233	0.5917350	0.6262252	0.7578752	0.8170171
234	0.4981988	0.6585967	0.8068255	0.8313147
235	0.5912850	0.6496932	0.7978227	0.8276221
236	0.6031028	0.6638152	0.7368792	0.7897422
237	0.6368322	0.6740689	0.7864777	0.8178032
238	0.5106422	0.6741911	0.7956789	0.8203666
239	0.6763680	0.7122006	0.8123415	0.8339164
240	0.5692818	0.6689039	0.7290819	0.8145986
241	0.5826332	0.6282087	0.8168505	0.8368537
242	0.5245810	0.6251111	0.8035372	0.8314902
243	0.6304542	0.7043349	0.7641007	0.7997901
244	0.6143485	0.6571546	0.7172030	0.7974706
245	0.6457358	0.7410551	0.8381365	0.8581321
246	0.5816107	0.6930739	0.7664125	0.7974022
247	0.6937553	0.7572142	0.8205659	0.8405617
248	0.6088860	0.6459012	0.7871567	0.8304144
249	0.6126472	0.6877867	0.8143022	0.8362129
250	0.5584785	0.7079891	0.8131730	0.8331764
251	0.6330710	0.7062726	0.8058182	0.8318259
252	0.6029350	0.6787306	0.7643067	0.8192221
253	0.6470788	0.6906094	0.8151110	0.8359762
254	0.5968620	0.7252852	0.8153627	0.8353659
255	0.7208855	0.7614484	0.8201157	0.8401192

Table 5.2.1.5-2 High Frequency LSP (7/12)

i	$C_{LSPH}(i, 5)$	$C_{LSPH}(i, 6)$	$C_{LSPH}(i, 7)$	$C_{LSPH}(i, 8)$
0	0.4702828	0.5623212	0.6614245	0.7357567
1	0.5112145	0.5823027	0.7489259	0.8198629
2	0.4556037	0.5577969	0.6730057	0.7816854
3	0.5183477	0.6106076	0.6986330	0.7610479
4	0.5350715	0.5981412	0.6486909	0.7213674
5	0.4740667	0.5818144	0.7155777	0.7816474
6	0.4958792	0.6406369	0.7248857	0.7678534
7	0.5771170	0.6506161	0.7407777	0.7902457
8	0.5261680	0.5912749	0.6605852	0.7864387
9	0.5474770	0.5979351	0.7358567	0.7838294
10	0.4712898	0.5995451	0.7594622	0.7966087
11	0.5829612	0.6488997	0.7562960	0.8015296
12	0.5592720	0.6111722	0.7152955	0.7993552
13	0.5911627	0.6305051	0.7653292	0.7991797
14	0.5349875	0.6393172	0.7384964	0.7948692
15	0.6554100	0.6920514	0.7543047	0.7942816
16	0.4764547	0.6098142	0.7114962	0.7627417
17	0.5408545	0.5834624	0.7621477	0.8147437
18	0.4482720	0.5533414	0.7482165	0.8121954
19	0.5909035	0.6499371	0.7380997	0.7731406
20	0.5925665	0.6300474	0.6718614	0.7212149
21	0.5576697	0.6097379	0.7437912	0.8020484
22	0.5465233	0.6374937	0.7368562	0.7762992
23	0.6227255	0.6871839	0.7803362	0.8098989
24	0.5342628	0.5967526	0.6611802	0.7411811
25	0.5718222	0.6052137	0.7692507	0.8215109
26	0.4821540	0.6000104	0.7626282	0.7993704
27	0.6224433	0.6770979	0.7460345	0.7876594
28	0.5776360	0.6197097	0.7153872	0.7653967
29	0.6087180	0.6418424	0.7748887	0.8145454
30	0.5663293	0.6437652	0.7718827	0.8047722
31	0.6761008	0.7114454	0.7688692	0.8023841
32	0.5172340	0.6208617	0.6859377	0.7433784
33	0.5133735	0.6135679	0.7570664	0.7948462
34	0.4935828	0.6170164	0.6883637	0.7917487
35	0.5747748	0.6406447	0.7036837	0.7596214
36	0.5674583	0.6219449	0.6693972	0.7425924
37	0.4799567	0.5907864	0.7859667	0.8193214
38	0.5478125	0.6641584	0.7601870	0.7929769
39	0.6014472	0.6643874	0.7287537	0.7760396
40	0.5689385	0.6145597	0.7013719	0.7975164
41	0.5639412	0.6092496	0.7642077	0.8008506
42	0.4446555	0.6274229	0.7788332	0.8197104

Table 5.2.1.5-2 High Frequency LSP (8/12)

43	0.6053840	0.6734814	0.7662752	0.8041389
44	0.5629492	0.6062284	0.7602707	0.8222052
45	0.6279288	0.6656844	0.7694337	0.7985921
46	0.5650855	0.6430936	0.7354447	0.7959219
47	0.6480780	0.6862074	0.7761400	0.8114476
48	0.5130912	0.6233486	0.6906984	0.7497489
49	0.5334160	0.6113707	0.7756365	0.8115547
50	0.4951162	0.5663954	0.7732637	0.8158194
51	0.6071157	0.6606184	0.7135560	0.7649999
52	0.5766822	0.6546904	0.7088870	0.7601172
53	0.5694573	0.6258132	0.7558687	0.8081137
54	0.5600042	0.6714216	0.7345522	0.7773214
55	0.6247855	0.6735807	0.8035982	0.8286826
56	0.5534202	0.5991941	0.6806582	0.7932286
57	0.5647728	0.6475722	0.8064667	0.8308037
58	0.4928962	0.6862301	0.7821594	0.8135839
59	0.6537695	0.6983687	0.7715012	0.8073202
60	0.5579367	0.6365019	0.7450807	0.8426139
61	0.6140357	0.6637846	0.7806642	0.8145301
62	0.5555335	0.6852232	0.7769562	0.8064657
63	0.6986992	0.7314192	0.8010500	0.8260734
64	0.4960168	0.5705839	0.7012497	0.7605369
65	0.5275338	0.5801816	0.7320497	0.7990346
66	0.4872353	0.5572782	0.7268465	0.7874151
67	0.5342322	0.6517454	0.7264040	0.7726751
68	0.5647193	0.6123624	0.6618592	0.7629096
69	0.5079567	0.5924497	0.7284105	0.7887274
70	0.5056755	0.6499067	0.7460724	0.7830129
71	0.5855780	0.6775022	0.7586230	0.7961662
72	0.5371620	0.5859266	0.6941240	0.8019949
73	0.5718148	0.6064266	0.7312639	0.8111274
74	0.4897225	0.6633879	0.7567995	0.7964026
75	0.5955955	0.6816602	0.7689377	0.8081899
76	0.5926963	0.6374937	0.7513979	0.8072592
77	0.6129218	0.6548734	0.7785357	0.8171012
78	0.5410530	0.6580702	0.7664429	0.8217857
79	0.6618567	0.7086684	0.7783907	0.8101279
80	0.5048362	0.6035734	0.7277237	0.7697759
81	0.5403435	0.5789762	0.7864929	0.8252341
82	0.4738685	0.5962109	0.7942980	0.8263251
83	0.6180108	0.6542784	0.7503144	0.7887656
84	0.6210777	0.6572234	0.6981065	0.7432867
85	0.5596610	0.6283001	0.7794055	0.8192297
86	0.5578987	0.6472134	0.7387482	0.7770392

Table 5.2.1.5-2 High Frequency LSP (9/12)

87	0.6397545	0.7213484	0.7842425	0.8125311
88	0.5785362	0.6309706	0.6987017	0.7617116
89	0.6007605	0.6393094	0.7997757	0.8285604
90	0.5356970	0.6511274	0.7896592	0.8147819
91	0.6393652	0.6941192	0.7885607	0.8199317
92	0.5990667	0.6370589	0.7239167	0.7991111
93	0.6228400	0.6572844	0.7962662	0.8244482
94	0.5870277	0.6544004	0.7777955	0.8101202
95	0.6906653	0.7230802	0.7753617	0.8039099
96	0.5610648	0.6437804	0.7074067	0.7625204
97	0.5284950	0.6388974	0.7806717	0.8262717
98	0.5202857	0.6421399	0.7275789	0.8240669
99	0.6140205	0.6600539	0.7259997	0.7888189
100	0.5945502	0.6342894	0.6853655	0.7893759
101	0.5511085	0.6479002	0.7797945	0.8142249
102	0.5310128	0.6891981	0.7704712	0.8040244
103	0.6268837	0.6893964	0.7754532	0.8110816
104	0.5913460	0.6377989	0.7010819	0.8333516
105	0.5822365	0.6224104	0.7701127	0.8196952
106	0.4973670	0.6577726	0.7957857	0.8335807
107	0.6155768	0.6769759	0.7783754	0.8115547
108	0.5777807	0.6567884	0.7639940	0.8107382
109	0.6376105	0.6818892	0.7981354	0.8282706
110	0.5419302	0.6805691	0.8012634	0.8282706
111	0.6605370	0.6954694	0.7991959	0.8261724
112	0.5410530	0.6433379	0.7021654	0.7863547
113	0.5827018	0.6198851	0.8004397	0.8294454
114	0.5282280	0.6258742	0.7878052	0.8227317
115	0.6329185	0.6949049	0.7594392	0.7984549
116	0.6082680	0.6486097	0.7161730	0.7765661
117	0.5827780	0.6604504	0.7649552	0.8058019
118	0.6015160	0.6820951	0.7563492	0.7927481
119	0.6672585	0.7329604	0.7952897	0.8194282
120	0.5927803	0.6424374	0.7354372	0.7958456
121	0.5847465	0.6345106	0.8026902	0.8299262
122	0.5363303	0.6768384	0.7911852	0.8166739
123	0.6819373	0.7244306	0.7918032	0.8193214
124	0.5947945	0.6740462	0.7625215	0.8157431
125	0.6311790	0.6949277	0.8021562	0.8265539
126	0.6017448	0.7131161	0.8056352	0.8270882
127	0.7141182	0.7466474	0.7998522	0.8221747
128	0.4912482	0.5883984	0.6600664	0.7326436
129	0.5095055	0.6035504	0.7843414	0.8269812
130	0.4552300	0.5981184	0.7267167	0.8360069

Table 5.2.1.5-2 High Frequency LSP (10/12)

131	0.5332175	0.6359371	0.7382142	0.7919242
132	0.5550377	0.6288344	0.6830310	0.7318274
133	0.5166922	0.5924801	0.7586762	0.8054434
134	0.5267173	0.6703764	0.7369935	0.7807089
135	0.6061240	0.6622359	0.7603777	0.8061911
136	0.5622245	0.6055569	0.6678255	0.8035132
137	0.5507805	0.6135527	0.7281205	0.7965704
138	0.4868460	0.6393477	0.7795807	0.8109747
139	0.5981437	0.6634946	0.7324160	0.7991264
140	0.5498725	0.6495481	0.7311494	0.8355949
141	0.5981133	0.6590619	0.7676942	0.7978141
142	0.5699682	0.6617247	0.7351245	0.7923744
143	0.6428367	0.6865811	0.7522982	0.7891089
144	0.4735253	0.6420101	0.7374512	0.7932594
145	0.5635445	0.6053511	0.7893235	0.8305672
146	0.4329367	0.5849806	0.7842119	0.8238379
147	0.6091682	0.6783949	0.7451342	0.7829061
148	0.5977318	0.6409117	0.6964510	0.7477424
149	0.5657722	0.6263241	0.7402437	0.7985769
150	0.5650932	0.6709944	0.7333620	0.7743766
151	0.6286308	0.7090422	0.7845779	0.8134239
152	0.5648948	0.6075254	0.6751955	0.7876594
153	0.5859747	0.6243407	0.7778412	0.8191612
154	0.4924538	0.6293759	0.7919632	0.8198249
155	0.6425010	0.6941192	0.7857454	0.8195044
156	0.5836555	0.6279341	0.7464082	0.7854087
157	0.6075278	0.6507232	0.7742022	0.8182837
158	0.5676108	0.6824079	0.7863174	0.8142249
159	0.6771537	0.7079587	0.7819460	0.8134924
160	0.5210183	0.6442077	0.7113892	0.7619482
161	0.5372460	0.6388059	0.7665727	0.8015754
162	0.4907752	0.6411252	0.7174242	0.8195427
163	0.5819160	0.6366469	0.6989457	0.7746894
164	0.5966560	0.6383864	0.6834657	0.7453087
165	0.4969855	0.6580244	0.7972887	0.8246314
166	0.5566245	0.6861006	0.7563034	0.7905814
167	0.6002112	0.6622586	0.7382220	0.7910926
168	0.5905677	0.6276289	0.7275559	0.8073202
169	0.5737220	0.6403014	0.7790697	0.8153312
170	0.4548408	0.6632046	0.7990510	0.8272407
171	0.6122810	0.6811797	0.7993792	0.8277672
172	0.6007073	0.6478696	0.7854632	0.8300939
173	0.6345740	0.6679426	0.7731950	0.8181464
174	0.5847083	0.6581921	0.7731417	0.8167884

Table 5.2.1.5-2 High Frequency LSP (11/12)

175	0.6499625	0.6906402	0.7722947	0.8091894
176	0.5407707	0.6453672	0.7092912	0.7587516
177	0.5461113	0.6052671	0.7930237	0.8241049
178	0.5011740	0.5847822	0.7872102	0.8212896
179	0.6340095	0.6811491	0.7400147	0.7794349
180	0.5960380	0.6771132	0.7291352	0.7748877
181	0.5772162	0.6446042	0.7862564	0.8208701
182	0.5767737	0.6901137	0.7514589	0.7904746
183	0.6394035	0.7106902	0.8100832	0.8312841
184	0.5404425	0.6337324	0.7402969	0.7993782
185	0.5824653	0.6699261	0.8098315	0.8329704
186	0.5081855	0.6921124	0.8115784	0.8315742
187	0.6674415	0.7260787	0.8092667	0.8311622
188	0.5625525	0.6657149	0.7702807	0.8573539
189	0.6397620	0.6716657	0.7970675	0.8260582
190	0.5798560	0.7098814	0.7891710	0.8118674
191	0.7015373	0.7516447	0.8174302	0.8374334
192	0.5436317	0.6036879	0.6970842	0.7596824
193	0.5539238	0.5944561	0.7369554	0.8153847
194	0.4949790	0.5901534	0.7056597	0.7955099
195	0.5297462	0.6663404	0.7644060	0.8006522
196	0.5825340	0.6300549	0.6948487	0.7854621
197	0.5175240	0.6119199	0.7602862	0.8073432
198	0.5236123	0.6530729	0.7647187	0.7982567
199	0.5975333	0.6957442	0.7855089	0.8168722
200	0.5474770	0.6106076	0.7072542	0.8364721
201	0.5878745	0.6201444	0.7445924	0.8178412
202	0.5152045	0.6497999	0.7753617	0.8045204
203	0.6439887	0.7086301	0.7869507	0.8170019
204	0.6095192	0.6531417	0.7546937	0.8143164
205	0.6132805	0.6643034	0.8034532	0.8323751
206	0.5796273	0.6649594	0.7870120	0.8276906
207	0.6733695	0.7088209	0.8019350	0.8304527
208	0.5271828	0.6322294	0.7313249	0.7755897
209	0.5553048	0.6001401	0.8024157	0.8308494
210	0.4816505	0.6232876	0.8013247	0.8304754
211	0.6316977	0.6744047	0.7488725	0.8120656
212	0.6327200	0.6723676	0.7270372	0.7674186
213	0.5908195	0.6464659	0.7853792	0.8210532
214	0.5717383	0.6616789	0.7511920	0.7874304
215	0.6406165	0.7308394	0.8116167	0.8316199
216	0.5803595	0.6218077	0.6943452	0.7921454
217	0.6064827	0.6481292	0.8137604	0.8359382
218	0.5357658	0.6633344	0.8112274	0.8373571

Table 5.2.1.5-2 High Frequency LSP (12/12)

219	0.6814720	0.7277571	0.7961137	0.8220984
220	0.6070700	0.6456572	0.7548922	0.8172919
221	0.6279748	0.6661649	0.8136689	0.8352819
222	0.5991660	0.6740994	0.7918947	0.8187566
223	0.6915885	0.7248731	0.7966477	0.8226706
224	0.5855475	0.6529356	0.7163102	0.7869422
225	0.5128622	0.6503949	0.7911622	0.8228157
226	0.5241995	0.6572614	0.7643602	0.8543249
227	0.6237938	0.6680647	0.7460954	0.7987449
228	0.5730812	0.6428036	0.7005022	0.7927099
229	0.5455620	0.6684309	0.8009890	0.8275381
230	0.5642767	0.7068754	0.7793442	0.8105704
231	0.6552193	0.7005811	0.7823122	0.8116689
232	0.5982200	0.6604659	0.7424332	0.8570029
233	0.5910102	0.6280256	0.7889497	0.8296286
234	0.5016852	0.6636549	0.8079012	0.8327872
235	0.6368095	0.6927609	0.8052995	0.8293844
236	0.6044228	0.6668974	0.7848147	0.8228309
237	0.6581490	0.6937756	0.7975252	0.8246161
238	0.5746910	0.6953626	0.8004242	0.8282554
239	0.6788245	0.7135739	0.8128832	0.8341681
240	0.5567237	0.6670499	0.7298830	0.8298194
241	0.5783760	0.6185726	0.8129289	0.8352666
242	0.5266640	0.6319851	0.8064134	0.8328939
243	0.6193152	0.6994824	0.7611712	0.7973869
244	0.6196890	0.6607634	0.7230775	0.8048789
245	0.6050635	0.6712079	0.7862794	0.8191764
246	0.6032783	0.7038924	0.7739044	0.8049551
247	0.6654273	0.7375379	0.8139285	0.8339239
248	0.5983268	0.6345182	0.7325532	0.8180471
249	0.6054527	0.6835676	0.8162552	0.8363881
250	0.5497580	0.7031827	0.8147677	0.8347709
251	0.6931983	0.7480284	0.8159272	0.8359229
252	0.5791848	0.6759076	0.7663362	0.8642814
253	0.6567833	0.6973539	0.8156144	0.8358619
254	0.5921928	0.7225614	0.8148819	0.8348851
255	0.7155830	0.7585264	0.8168505	0.8368537

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (1/12)

i	$C_{LSPM1}(i, 1)$	$C_{LSPM1}(i, 2)$	$C_{LSPM1}(i, 3)$	$C_{LSPM1}(i, 4)$
0	0.0451913	0.2149502	0.2535239	0.2995555
1	0.0444361	0.1940609	0.3016350	0.4404322
2	0.0447641	0.2079464	0.2445822	0.3487269
3	0.0455576	0.2591319	0.4065620	0.4593379
4	0.0382868	0.1979594	0.3111107	0.3866754
5	0.0360209	0.1813501	0.3635092	0.4997202
6	0.0408351	0.2159266	0.3651497	0.4131419
7	0.0472284	0.2829356	0.4539177	0.4964320
8	0.0470453	0.2367626	0.2987205	0.3389002
9	0.0369898	0.1983486	0.3912879	0.4576977
10	0.0580929	0.2108454	0.2654105	0.4824779
11	0.0566888	0.2685771	0.3582680	0.4017894
12	0.0402551	0.2798077	0.3387900	0.3780544
13	0.0391109	0.2287669	0.4600745	0.5089672
14	0.0584209	0.2256084	0.3066244	0.4662577
15	0.1029001	0.3378367	0.4381020	0.5126979
16	0.0350213	0.2173992	0.2836142	0.3347575
17	0.0575511	0.2157589	0.3401327	0.5224407
18	0.0483806	0.2695384	0.3170309	0.4166284
19	0.0749766	0.2948756	0.4272300	0.5007350
20	0.0440318	0.2458109	0.3278112	0.3664347
21	0.0361506	0.1952357	0.4150687	0.5219372
22	0.0421168	0.2562557	0.3830557	0.4216870
23	0.0753808	0.3523249	0.4547874	0.4925104
24	0.0520961	0.2233806	0.2563924	0.3383587
25	0.0382258	0.1996914	0.3610297	0.4676769
26	0.0481136	0.2559277	0.3745794	0.4529979
27	0.0662028	0.3287044	0.4266272	0.4692789
28	0.1097513	0.3190227	0.3711462	0.4275004
29	0.0415141	0.2080301	0.4692449	0.5452067
30	0.0916926	0.3200831	0.4036932	0.4472147
31	0.0712611	0.3759989	0.4887534	0.5227077
32	0.0454356	0.2478329	0.2899084	0.3339182
33	0.0508448	0.2176662	0.2754049	0.4111202
34	0.0497309	0.2100444	0.2512732	0.3901239
35	0.0665383	0.2859951	0.4069434	0.5004070
36	0.0441081	0.2355799	0.3747322	0.4269587
37	0.0401331	0.2008814	0.3787832	0.5560709
38	0.0475031	0.2382731	0.3648749	0.4208094
39	0.0594508	0.3392176	0.4792929	0.5106304
40	0.0386988	0.2490687	0.3425359	0.3802592
41	0.0373636	0.2379909	0.4008095	0.5022532
42	0.0501811	0.2061687	0.3185797	0.5255304

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (2/12)

43	0.0474726	0.2490154	0.4029990	0.4467417
44	0.0409496	0.2865596	0.3715429	0.4045895
45	0.0452066	0.2772974	0.4682914	0.5048244
46	0.0907619	0.3062664	0.4046240	0.4930292
47	0.1012904	0.3283916	0.4439765	0.5228222
48	0.0553918	0.2230067	0.2677985	0.4079005
49	0.0426431	0.2252192	0.4351569	0.5255687
50	0.0644481	0.2569804	0.3053047	0.4638544
51	0.1011379	0.3360439	0.4904699	0.5715739
52	0.0904338	0.2486339	0.3538122	0.4081217
53	0.0390041	0.2422329	0.4513922	0.5598935
54	0.0419261	0.2797619	0.3911734	0.4243420
55	0.0994288	0.3973689	0.4665747	0.5176647
56	0.0756251	0.2898479	0.3410712	0.3898647
57	0.0413081	0.2179941	0.4168234	0.5314127
58	0.0881374	0.2909694	0.3447179	0.4795177
59	0.0884576	0.3227229	0.4403450	0.5134990
60	0.0963846	0.3237912	0.3768912	0.4295910
61	0.0401331	0.2498926	0.4961462	0.5490289
62	0.1026026	0.3394389	0.4344245	0.5121487
63	0.1221644	0.3856959	0.5045920	0.5852000
64	0.0437188	0.2354884	0.2789069	0.3156382
65	0.0420633	0.2307049	0.3459082	0.4677607
66	0.0483958	0.2449031	0.2900687	0.3785122
67	0.0451686	0.2690884	0.4291450	0.4702097
68	0.0536218	0.2266612	0.3192817	0.4181545
69	0.0529428	0.2161557	0.3845207	0.5497462
70	0.0391794	0.2403254	0.4067527	0.4449490
71	0.0511806	0.3065181	0.4432974	0.4760919
72	0.0474726	0.2753214	0.3210440	0.3781307
73	0.0431086	0.2198634	0.4218969	0.4730097
74	0.0570094	0.2297739	0.3084555	0.4727885
75	0.0622813	0.2470012	0.3217612	0.3916727
76	0.0493724	0.2994227	0.3477852	0.4136685
77	0.0461908	0.2531734	0.4595174	0.5123699
78	0.0840251	0.2819514	0.3489752	0.4856060
79	0.0974909	0.3495022	0.4758750	0.5564524
80	0.0416056	0.2645412	0.3151159	0.3535335
81	0.0735346	0.2494274	0.3971015	0.4786022
82	0.0526836	0.2824549	0.3357002	0.4359767
83	0.0999019	0.3161999	0.4168617	0.5111262
84	0.0392634	0.2543711	0.3635397	0.4011637
85	0.0373788	0.1994471	0.4473867	0.5865047
86	0.0420864	0.2829051	0.4176780	0.4537837

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (3/12)

87	0.0991009	0.3183894	0.4276649	0.4970652
88	0.0640971	0.2514414	0.3028862	0.3918939
89	0.0468166	0.2160717	0.3795387	0.5140482
90	0.0747781	0.2407832	0.3915855	0.5084177
91	0.0821481	0.3660502	0.4300069	0.4851862
92	0.0584971	0.3073269	0.3599464	0.4550424
93	0.0459849	0.2514874	0.4848015	0.6033884
94	0.1004816	0.3272014	0.4438392	0.5029627
95	0.1030908	0.4147107	0.4871817	0.5285289
96	0.0846506	0.2450251	0.2927924	0.3514507
97	0.0420633	0.2083506	0.3277580	0.4515102
98	0.0590388	0.2437664	0.2858420	0.4186122
99	0.1065013	0.2763057	0.4287025	0.5177562
100	0.0483501	0.2692409	0.4128484	0.4560114
101	0.0518596	0.2410501	0.4184409	0.5685450
102	0.0759379	0.2442622	0.4066077	0.4630382
103	0.0615566	0.3274989	0.5158072	0.5521189
104	0.0575891	0.2817759	0.3596487	0.4082440
105	0.0468851	0.2658381	0.4350119	0.5227230
106	0.0702769	0.2447124	0.3446647	0.5112330
107	0.0778911	0.3056637	0.3923560	0.4463985
108	0.0512491	0.3123547	0.3813545	0.4206492
109	0.0483196	0.2907099	0.4993202	0.5389887
110	0.0909144	0.3390957	0.4350882	0.5055872
111	0.0986353	0.3403391	0.4918280	0.5695674
112	0.0629603	0.2619624	0.3018942	0.3831584
113	0.0564448	0.2464442	0.4860297	0.5587260
114	0.0648448	0.2819287	0.3308402	0.4894739
115	0.1027094	0.3775172	0.4971305	0.5449169
116	0.0742976	0.2800289	0.3908607	0.4292782
117	0.0706814	0.2512507	0.4658042	0.5765712
118	0.0453746	0.2965542	0.4196617	0.4542492
119	0.1312968	0.4000087	0.4745857	0.5300777
120	0.0793254	0.2936854	0.3488455	0.4131725
121	0.0458628	0.2399669	0.4674750	0.5569102
122	0.1249338	0.3152841	0.3879767	0.5082425
123	0.1418406	0.3665004	0.4305792	0.5027035
124	0.0821788	0.3161769	0.3814077	0.4285532
125	0.0537286	0.2838589	0.4905845	0.5804087
126	0.0949046	0.3610071	0.4586630	0.5266292
127	0.1197383	0.3996501	0.5147087	0.5867717
128	0.0632961	0.2264934	0.2607719	0.3226192
129	0.0359524	0.1914212	0.3399954	0.4554470
130	0.0584819	0.2315746	0.2681420	0.3570049

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (4/12)

131	0.0434749	0.2538372	0.4393149	0.4843469
132	0.0514934	0.2194591	0.3401099	0.3882242
133	0.0433069	0.2153544	0.3534232	0.4997127
134	0.0659054	0.2343594	0.3828879	0.4312085
135	0.0576121	0.3039316	0.4846260	0.5311382
136	0.0617624	0.2596354	0.3126290	0.3522594
137	0.0525691	0.2255854	0.3909827	0.4821269
138	0.0658519	0.2195889	0.2704459	0.5154597
139	0.0692011	0.2911067	0.3505164	0.4099222
140	0.0523858	0.2768701	0.3511649	0.3863552
141	0.0614343	0.2999417	0.4769812	0.5217540
142	0.0828501	0.2645259	0.3361960	0.4479472
143	0.1400323	0.3426662	0.4453422	0.5227764
144	0.0450771	0.2350079	0.3200674	0.3582407
145	0.0723291	0.2473751	0.3684302	0.5475794
146	0.0689036	0.2826839	0.3270407	0.4335124
147	0.1051814	0.3399502	0.4693825	0.5448252
148	0.0509134	0.2508007	0.3432837	0.3803964
149	0.0340221	0.2241969	0.4286642	0.5383174
150	0.0600689	0.2680432	0.3977577	0.4372967
151	0.0713983	0.3643641	0.4611885	0.4989039
152	0.0556741	0.2374874	0.2799827	0.3415782
153	0.0509363	0.2138514	0.3624334	0.4744822
154	0.0722453	0.2670664	0.3749077	0.4775112
155	0.0649286	0.3371424	0.4171287	0.4578959
156	0.0749766	0.3255001	0.3976507	0.4574687
157	0.0369363	0.2094189	0.4905540	0.5813777
158	0.1244531	0.3563609	0.4209509	0.4650217
159	0.1032894	0.3728481	0.4960090	0.5399042
160	0.0667826	0.2595974	0.3080207	0.3501002
161	0.0617243	0.2156749	0.2951042	0.4538905
162	0.0561931	0.2123867	0.2561790	0.4289194
163	0.1038691	0.3101117	0.4317160	0.4839657
164	0.0446879	0.2273554	0.3878547	0.4616650
165	0.0588709	0.2273707	0.3633337	0.5873667
166	0.0644481	0.2646479	0.3844289	0.4392270
167	0.0666376	0.3357997	0.4912025	0.5223872
168	0.0639978	0.2549969	0.3493872	0.3864847
169	0.0501123	0.2466349	0.3946602	0.5267970
170	0.0695748	0.2279429	0.3338232	0.5723674
171	0.0613046	0.2624964	0.4268942	0.4860485
172	0.0522104	0.2983851	0.3946602	0.4309412
173	0.0609156	0.3003001	0.4646217	0.5065562
174	0.1243768	0.3381877	0.4321585	0.5127894

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (5/12)

175	0.1136271	0.3499674	0.4610587	0.5372722
176	0.0701166	0.2563702	0.3166572	0.4101742
177	0.0540111	0.2465586	0.4570610	0.5410717
178	0.0948206	0.2636639	0.3109885	0.4657010
179	0.1301371	0.3837276	0.5230017	0.5883204
180	0.0884424	0.2560267	0.3748617	0.4460322
181	0.0506616	0.2743069	0.4587012	0.5527979
182	0.0537593	0.3058619	0.4042197	0.4388760
183	0.1420771	0.4097056	0.4815284	0.5404079
184	0.1131768	0.2915264	0.3621895	0.4116160
185	0.0543314	0.2344662	0.4135962	0.5263239
186	0.0987269	0.3194041	0.3790352	0.4953867
187	0.1048991	0.3517147	0.4605780	0.5298792
188	0.1110941	0.3447794	0.4023277	0.4532802
189	0.0539806	0.2628932	0.5278844	0.5675152
190	0.1420466	0.3660121	0.4420692	0.5138882
191	0.1423061	0.4260097	0.5355064	0.6027934
192	0.0677134	0.2411571	0.2886572	0.3295542
193	0.0790963	0.2335506	0.3437949	0.4709957
194	0.0768076	0.2395396	0.2822487	0.3976162
195	0.0560023	0.2789989	0.4320137	0.4810282
196	0.0540948	0.2466959	0.3526602	0.4309032
197	0.0579173	0.2299799	0.4242165	0.5674922
198	0.0549876	0.2525477	0.4222632	0.4803492
199	0.0721996	0.3076244	0.4560309	0.4969127
200	0.0695826	0.2750774	0.3319390	0.3804424
201	0.0572076	0.2362666	0.4140465	0.4972332
202	0.0838344	0.2507319	0.3049842	0.5097299
203	0.0722148	0.2744746	0.3596412	0.4171169
204	0.0671336	0.3008724	0.3634024	0.4069850
205	0.0779901	0.2620464	0.4817725	0.5377682
206	0.1156183	0.3019709	0.3748542	0.4947307
207	0.1321209	0.3586269	0.4942542	0.5660732
208	0.0604809	0.2714916	0.3265450	0.3746135
209	0.0889613	0.2776257	0.4353399	0.5320232
210	0.0843226	0.2976069	0.3471672	0.4500760
211	0.1189523	0.3380274	0.4546730	0.5431087
212	0.0642726	0.2688364	0.3703147	0.4120890
213	0.0518824	0.2255016	0.4090034	0.5907619
214	0.0691783	0.2807154	0.4235374	0.4640910
215	0.1040674	0.3624112	0.4452430	0.5029474
216	0.0732676	0.2757946	0.3288259	0.4301937
217	0.0792719	0.2675472	0.4033957	0.5099742
218	0.0964838	0.2646174	0.3881139	0.5339304

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (6/12)

219	0.0928751	0.3680416	0.4323645	0.4805324
220	0.0767388	0.3299174	0.3898687	0.4806927
221	0.0535226	0.2606884	0.5145180	0.6097210
222	0.1158853	0.3584437	0.4687797	0.5241955
223	0.0952556	0.4154582	0.5170280	0.5577724
224	0.0977426	0.2690044	0.3189002	0.3735225
225	0.0721461	0.2476116	0.3376762	0.4980952
226	0.0746028	0.2365871	0.2807610	0.4542872
227	0.0865733	0.2935252	0.4499427	0.5462520
228	0.0653941	0.2954632	0.4334249	0.4890850
229	0.0897393	0.2622141	0.4279852	0.5880687
230	0.0783488	0.2575984	0.4400627	0.4963787
231	0.0716501	0.3561396	0.5253669	0.5573375
232	0.0567423	0.2781061	0.3830710	0.4238690
233	0.0712686	0.2499919	0.4377815	0.5422999
234	0.1021144	0.2735667	0.3359672	0.5311152
235	0.0795619	0.3015589	0.4030372	0.4674557
236	0.0685908	0.3274301	0.3903494	0.4405010
237	0.0908151	0.3084179	0.5015632	0.5535152
238	0.1142603	0.3644099	0.4692070	0.5299710
239	0.1103463	0.3745951	0.5313712	0.5954692
240	0.0875574	0.2641521	0.3147879	0.4196727
241	0.0593821	0.2648921	0.4942390	0.5784557
242	0.1073098	0.2823939	0.3523627	0.4942729
243	0.1432749	0.4196927	0.5219869	0.5877407
244	0.1031139	0.2856746	0.4044867	0.4553402
245	0.0985208	0.2863842	0.4762182	0.5888469
246	0.0735728	0.3224559	0.4410925	0.4826687
247	0.1665599	0.4394756	0.5055762	0.5678127
248	0.1291681	0.3061901	0.3638527	0.4402567
249	0.0590693	0.2868954	0.4809332	0.5399807
250	0.1251704	0.3167644	0.3940039	0.5335184
251	0.1227671	0.3749156	0.4440070	0.5117367
252	0.1163813	0.3373867	0.3957664	0.4587962
253	0.0724131	0.3132777	0.5286857	0.5954997
254	0.1374536	0.3973384	0.4977332	0.5619457
255	0.1613413	0.4476466	0.5428000	0.6197995

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (7/12)

i	$C_{LSPM1}(i, 5)$	$C_{LSPM1}(i, 6)$	$C_{LSPM1}(i, 7)$	$C_{LSPM1}(i, 8)$
0	0.0449548	0.2150721	0.2533712	0.2987392
1	0.0437418	0.1923366	0.3024664	0.4403864
2	0.0442226	0.2072521	0.2435370	0.3497035
3	0.0465113	0.2594372	0.4174872	0.4714305
4	0.0385461	0.1970744	0.3103247	0.3886135
5	0.0337549	0.1797481	0.3600150	0.4860790
6	0.0401026	0.2147137	0.3694297	0.4176127
7	0.0494943	0.2868954	0.4198982	0.4595592
8	0.0467554	0.2411951	0.3290397	0.3722180
9	0.0368221	0.1981044	0.3883582	0.4540127
10	0.0560098	0.2111124	0.2607259	0.4640224
11	0.0611599	0.2847744	0.4075232	0.4540659
12	0.0397289	0.2812649	0.3402015	0.3773907
13	0.0383936	0.2263636	0.4511482	0.5005672
14	0.0536601	0.2236781	0.2715597	0.3773907
15	0.0906931	0.3112027	0.4468452	0.5257595
16	0.0345941	0.2182002	0.2854987	0.3362757
17	0.0530344	0.2134167	0.2843925	0.4306362
18	0.0461451	0.2685084	0.3142235	0.3980815
19	0.0391031	0.2089457	0.3758995	0.4835842
20	0.0425441	0.2441554	0.3549719	0.3983104
21	0.0351131	0.1909939	0.4028922	0.5114467
22	0.0423839	0.2593151	0.3863974	0.4248987
23	0.0626016	0.3279339	0.4452430	0.4818674
24	0.0636393	0.2293849	0.2781364	0.4227855
25	0.0747476	0.2779766	0.3991995	0.4698285
26	0.0484416	0.2532496	0.3755790	0.4621302
27	0.0591378	0.3141017	0.4077369	0.4486949
28	0.0559489	0.2691417	0.3462670	0.3970972
29	0.0398203	0.2040859	0.4660864	0.5405987
30	0.0471751	0.2433162	0.3915397	0.4352519
31	0.0770899	0.3918834	0.4917212	0.5260187
32	0.0464504	0.2532726	0.2958289	0.3404184
33	0.0579173	0.2190929	0.3126060	0.5030237
34	0.0499826	0.2096324	0.2519827	0.3952205
35	0.0664698	0.2868649	0.4063942	0.5049235
36	0.0447946	0.2356334	0.3301002	0.3779705
37	0.0396831	0.1986231	0.3911352	0.5603664
38	0.0491359	0.2292324	0.4118872	0.4864374
39	0.0527063	0.3086086	0.4716864	0.5054577
40	0.0390194	0.2457346	0.3241265	0.3612619
41	0.0389811	0.2374187	0.3975439	0.5002849
42	0.0492351	0.2049784	0.3195717	0.5261179

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (8/12)

43	0.0849634	0.3204722	0.4228429	0.4695002
44	0.0403696	0.2933039	0.3766547	0.4102809
45	0.0465571	0.2845837	0.4638969	0.4987437
46	0.0507456	0.2312007	0.2955772	0.3680600
47	0.1015421	0.3555369	0.5024024	0.5736187
48	0.0510356	0.2222134	0.2548437	0.3352457
49	0.0422311	0.2224574	0.3524695	0.4335735
50	0.0614496	0.2587199	0.3019325	0.4435679
51	0.0497003	0.2329707	0.4381172	0.5556972
52	0.0898768	0.2475582	0.3600379	0.4114024
53	0.0368679	0.2349771	0.4523230	0.5643565
54	0.0410791	0.2883219	0.4045782	0.4371060
55	0.0860238	0.3719402	0.4466467	0.5002087
56	0.0713221	0.2843776	0.3384545	0.3855922
57	0.0804546	0.2985836	0.4351187	0.5208995
58	0.1059444	0.3113474	0.3819037	0.4978740
59	0.1017711	0.3324274	0.4101324	0.4617947
60	0.0806376	0.3161694	0.3770742	0.4261729
61	0.0403926	0.2475429	0.4897072	0.5424982
62	0.0856348	0.3081356	0.3650580	0.4196422
63	0.1269404	0.3869472	0.4893409	0.5686977
64	0.0434824	0.2360837	0.2780600	0.3133189
65	0.0418651	0.2319484	0.3452827	0.4628092
66	0.0489603	0.2467494	0.2899314	0.3739422
67	0.0426736	0.2517467	0.4503852	0.4987819
68	0.0520273	0.2222057	0.3269034	0.4355877
69	0.0503489	0.2114939	0.3289862	0.4984692
70	0.0378901	0.2339549	0.4012442	0.4437970
71	0.0586421	0.3239054	0.4513007	0.4847742
72	0.0535381	0.2897412	0.3360587	0.4293162
73	0.0435206	0.2215954	0.4267264	0.4775644
74	0.0574901	0.2256541	0.3072959	0.4940670
75	0.1118951	0.3167566	0.4153814	0.4908855
76	0.0484491	0.2971111	0.3449775	0.4052607
77	0.0459774	0.2509532	0.4741964	0.5308712
78	0.0630746	0.2734447	0.3246527	0.4087702
79	0.1140009	0.3457866	0.4602272	0.5491207
80	0.0419336	0.2679897	0.3196480	0.3575847
81	0.0644023	0.2253489	0.3058539	0.4635417
82	0.0552699	0.2786556	0.3315422	0.4527385
83	0.0436731	0.2381434	0.4381552	0.5449397
84	0.0411784	0.2574992	0.3607475	0.3974025
85	0.0383096	0.1968761	0.4218894	0.5767389
86	0.0407663	0.2810131	0.4086752	0.4457804

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (9/12)

87	0.1169688	0.3609462	0.4541922	0.5148569
88	0.0883661	0.2732617	0.3436729	0.4681880
89	0.0476634	0.2308344	0.4719155	0.5617015
90	0.0747171	0.2409359	0.3835822	0.5001094
91	0.0757548	0.3430782	0.4035559	0.4640300
92	0.0566051	0.3101954	0.3611747	0.4506405
93	0.0423839	0.2398449	0.4595174	0.5952937
94	0.0576349	0.2596659	0.4065772	0.4615122
95	0.0971398	0.4130169	0.4855184	0.5255610
96	0.0913036	0.2502972	0.3030310	0.3646799
97	0.0503641	0.2260889	0.4144202	0.5527827
98	0.0605646	0.2422404	0.2842094	0.4255015
99	0.1093623	0.2779919	0.4322502	0.5217844
100	0.0493494	0.2642664	0.3617775	0.4039562
101	0.0521036	0.2444681	0.4186622	0.5673550
102	0.0825449	0.2476344	0.4138785	0.4686992
103	0.0548656	0.3074259	0.5146934	0.5541179
104	0.0539958	0.2613139	0.3199530	0.3615367
105	0.0450084	0.2611614	0.4286489	0.5203197
106	0.0706201	0.2418742	0.3456642	0.5219982
107	0.0877101	0.3536219	0.4546805	0.5136135
108	0.0515926	0.3193812	0.3878394	0.4272944
109	0.0483348	0.2932964	0.4955664	0.5358227
110	0.0527294	0.2580256	0.3515844	0.4058102
111	0.1148021	0.3744349	0.5189962	0.5876337
112	0.0686519	0.2667537	0.3068762	0.4015452
113	0.0533931	0.2166666	0.3641502	0.5085247
114	0.0641199	0.2813259	0.3285132	0.4843852
115	0.0724589	0.3162684	0.4735862	0.5224177
116	0.0869701	0.2875972	0.3906165	0.4303767
117	0.0715816	0.2502589	0.4691917	0.5805079
118	0.0446116	0.2926326	0.4397040	0.4765804
119	0.1499201	0.4114221	0.4777289	0.5370204
120	0.1164194	0.3080439	0.3554907	0.4199474
121	0.0842311	0.3249811	0.4947425	0.5590770
122	0.1098506	0.2977444	0.3564214	0.4961422
123	0.1211038	0.3491437	0.4092247	0.4806315
124	0.1234156	0.3441461	0.4048072	0.4651362
125	0.0524011	0.2811504	0.4945595	0.5828119
126	0.0926081	0.3071819	0.3898002	0.4932505
127	0.1621348	0.4338147	0.5333090	0.6058069
128	0.0648981	0.2275309	0.2610312	0.3271739
129	0.0365091	0.1930004	0.3418265	0.4554392
130	0.0602213	0.2309336	0.2656622	0.3588739

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (10/12)

131	0.0438486	0.2618861	0.4307014	0.4733302
132	0.0524316	0.2228312	0.3358985	0.3837230
133	0.0439936	0.2154919	0.3721762	0.5186182
134	0.0654016	0.2333446	0.3893652	0.4367855
135	0.0609309	0.3061214	0.4297552	0.4773509
136	0.0663706	0.2746349	0.3497535	0.3877285
137	0.0507913	0.2200236	0.3752204	0.4800822
138	0.0658898	0.2191309	0.2723762	0.5218302
139	0.0857568	0.3249584	0.3919060	0.4678905
140	0.0510888	0.2820812	0.3615027	0.3964487
141	0.0435053	0.2400889	0.4488822	0.5082347
142	0.0686898	0.2501292	0.2982244	0.3835094
143	0.1125818	0.3445734	0.4668647	0.5354792
144	0.0447794	0.2313304	0.3013984	0.3391367
145	0.0654934	0.2522654	0.3274832	0.4468487
146	0.0668209	0.2867656	0.3293907	0.4116160
147	0.0484033	0.2200467	0.3980857	0.5116985
148	0.0492503	0.2509532	0.3862982	0.4301250
149	0.0354944	0.2259669	0.4230950	0.5317562
150	0.0581461	0.2644649	0.4033347	0.4441325
151	0.0715891	0.3669964	0.4672004	0.5034739
152	0.0789133	0.2589107	0.3288184	0.4470927
153	0.1139246	0.3210216	0.4397879	0.5167719
154	0.0716501	0.2692714	0.3795997	0.4818905
155	0.0663631	0.3518444	0.4351264	0.4751384
156	0.0617166	0.2877574	0.3555442	0.4069012
157	0.0364558	0.2094494	0.5000524	0.5844677
158	0.0612666	0.2804789	0.3948127	0.4361750
159	0.0999246	0.3671412	0.4950020	0.5368219
160	0.0653026	0.2563624	0.3025580	0.3415477
161	0.0686824	0.2385402	0.3851082	0.4776027
162	0.0567881	0.2113871	0.2563697	0.4290797
163	0.0955606	0.3075862	0.4407264	0.4892527
164	0.0461529	0.2370144	0.3573982	0.4120357
165	0.0585506	0.2267069	0.3928367	0.5940502
166	0.0685373	0.2757259	0.4484320	0.5090052
167	0.0668283	0.3443369	0.4964592	0.5262630
168	0.0670648	0.2546764	0.3434287	0.3818537
169	0.0492046	0.2423779	0.3988029	0.5414377
170	0.0680566	0.2285229	0.3227150	0.5690104
171	0.1107584	0.3421549	0.4525977	0.5085475
172	0.0517603	0.2882914	0.3848182	0.4207407
173	0.0578868	0.2963634	0.4895317	0.5324505
174	0.0640131	0.2550274	0.3114462	0.3835705

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (11/12)

175	0.1544748	0.4096904	0.5156470	0.5879924
176	0.0575511	0.2400356	0.2856360	0.3411662
177	0.0528436	0.2469401	0.4053032	0.4678294
178	0.0967738	0.2679591	0.3181524	0.4619700
179	0.0550409	0.2628246	0.4726935	0.5653637
180	0.0937068	0.2573162	0.3677207	0.4455975
181	0.0487238	0.2746271	0.4612342	0.5563075
182	0.0523401	0.3121944	0.4161367	0.4500912
183	0.1201121	0.3759762	0.4456549	0.5005595
184	0.1080043	0.2845759	0.3477240	0.3965937
185	0.1235148	0.3588862	0.4889595	0.5584437
186	0.0944391	0.3439022	0.4117194	0.4939449
187	0.1016109	0.3391567	0.4401389	0.4938380
188	0.1007411	0.3370434	0.3974752	0.4450022
189	0.0540339	0.2596431	0.5279227	0.5673397
190	0.0870463	0.3195644	0.3878547	0.4382580
191	0.1134364	0.4054332	0.5322714	0.5982234
192	0.0670801	0.2410044	0.2865744	0.3242747
193	0.0788294	0.2325054	0.3401940	0.4652582
194	0.0756021	0.2374721	0.2778159	0.3902080
195	0.0576808	0.2881237	0.4632637	0.5176722
196	0.0581538	0.2466807	0.3444587	0.4303157
197	0.0555521	0.2198024	0.3694219	0.5342662
198	0.0549951	0.2529291	0.3879844	0.4433162
199	0.0780131	0.3515927	0.4754172	0.5141400
200	0.0835139	0.3042826	0.3689719	0.4253337
201	0.0582376	0.2380672	0.4342337	0.5161769
202	0.0831096	0.2510524	0.3028479	0.5050762
203	0.1189371	0.3488001	0.4452354	0.5164209
204	0.0681636	0.3013607	0.3657904	0.4105554
205	0.0768001	0.2594142	0.4852745	0.5414302
206	0.0756403	0.2966304	0.3498297	0.4246852
207	0.1062571	0.3415751	0.4995947	0.5670420
208	0.0590236	0.2657009	0.3157875	0.3621012
209	0.0814006	0.2574917	0.3396444	0.4893977
210	0.0837428	0.2945704	0.3413992	0.4503505
211	0.0560861	0.2737347	0.4816732	0.5740382
212	0.0623729	0.2666012	0.3735725	0.4149042
213	0.0541406	0.2369304	0.4771950	0.6092327
214	0.0666681	0.2811884	0.4226600	0.4634577
215	0.1085993	0.3964077	0.4760200	0.5267207
216	0.0933559	0.2772136	0.3535910	0.5098292
217	0.0744653	0.2784649	0.4890892	0.5669199
218	0.0953166	0.2633509	0.3878164	0.5333202

Table 5.2.1.5-3 Middle Frequency LSP at First Stage (12/12)

219	0.0970788	0.3720621	0.4376517	0.4845759
220	0.0762278	0.3086314	0.3610374	0.4765957
221	0.0549951	0.2615504	0.5204687	0.6131312
222	0.0588481	0.2725291	0.4426794	0.4959210
223	0.0963008	0.4139172	0.5189887	0.5592372
224	0.1057383	0.2738261	0.3212652	0.3748272
225	0.0808971	0.2580104	0.4239187	0.5431697
226	0.0760828	0.2362436	0.2799827	0.4610392
227	0.0933633	0.3015741	0.4497290	0.5428875
228	0.0630136	0.2847056	0.3749610	0.4250362
229	0.0892131	0.2597577	0.4050589	0.5793329
230	0.0814769	0.2578274	0.4304342	0.4917247
231	0.0739009	0.3584514	0.5256187	0.5576502
232	0.0607248	0.2690424	0.3453512	0.3876139
233	0.0756173	0.2485272	0.4342872	0.5419185
234	0.1050441	0.2742839	0.3408957	0.5340679
235	0.1418559	0.3897471	0.4828635	0.5529430
236	0.0725503	0.3377986	0.4003135	0.4466044
237	0.0893886	0.3077769	0.4849997	0.5370814
238	0.0586648	0.2500834	0.3697197	0.4426905
239	0.1119716	0.3776621	0.5348120	0.5973002
240	0.0904183	0.2617259	0.3123160	0.4294995
241	0.0667443	0.2683712	0.4281684	0.5196787
242	0.0967434	0.2734372	0.3337164	0.4817837
243	0.0879009	0.3465877	0.4801627	0.5377604
244	0.1066386	0.2891002	0.4071342	0.4592922
245	0.1012064	0.2887874	0.4747460	0.5907237
246	0.0515543	0.2769999	0.4443045	0.4840419
247	0.1399866	0.4278941	0.4938957	0.5469540
248	0.1317241	0.3102259	0.3708030	0.4463985
249	0.1260018	0.3804317	0.5014104	0.5602977
250	0.1295573	0.3175959	0.3954459	0.5376154
251	0.1561991	0.3768992	0.4334937	0.5081812
252	0.1430536	0.3639599	0.4236517	0.4889552
253	0.0717189	0.3106227	0.5282202	0.5966822
254	0.0755259	0.3028331	0.4163504	0.4733989
255	0.1529261	0.4498669	0.5406562	0.6142527

Table 5.2.1.5-4 Middle Frequency LSP at Second Stage

i	$C_{LSPM2}(i, 1)$	$C_{LSPM2}(i, 2)$	$C_{LSPM2}(i, 3)$	$C_{LSPM2}(i, 4)$
0	-0.0098477	-0.0046091	0.0019502	-0.0051794
1	-0.0030355	0.0025473	-0.0005846	0.0039396
2	0.0001440	-0.0028429	-0.0004988	0.0009575
3	0.0006561	0.0041561	0.0091591	-0.0063229
4	0.0049162	0.0030556	-0.0079308	-0.0080652
5	-0.0080624	0.0003319	-0.0031242	-0.0040951
6	0.0020237	0.0120459	0.0004663	-0.0025511
7	0.0018826	-0.0050707	-0.0063610	-0.0076246

i	$C_{LSPM2}(i, 5)$	$C_{LSPM2}(i, 6)$	$C_{LSPM2}(i, 7)$	$C_{LSPM2}(i, 8)$
0	0.0010262	0.0054140	-0.0013666	-0.0043888
1	-0.0056715	0.0093231	0.0042448	-0.0053721
2	-0.0012703	-0.0060825	-0.0085020	-0.0101642
3	0.0003252	-0.0018520	0.0067015	-0.0038624
4	0.0063419	0.0053177	-0.0005446	0.0003061
5	-0.0063019	-0.0043325	-0.0003643	0.0084753
6	-0.0056457	-0.0009842	-0.0049276	-0.0019226
7	-0.0041666	-0.0031118	0.0076609	0.0000935

5.2.1.6 Frame energy quantization

Mean amplitude in each subframe $amp_n (n=0,1,2,3)$ are computed as follows:

$$amp_n = \sqrt{\frac{\sum_{i=N_{SUB} \cdot n}^{N_{SUB} \cdot n + N_{SUB} - 1} s^2(i)}{N_{SUB}}} \quad (5.2.1.6-1)$$

Then, the μ -law conversion is performed to obtain μamp_n :

$$\mu amp_n = \frac{\log_{10}(10 + \mu \cdot \frac{amp_n}{S_{max}})}{\log_{10}(10 + \mu)} \quad (5.2.1.6-2)$$

where $\mu=100$, S_{max} is the maximum amplitude of the input PCM signal.

Next, the index "i" which minimizes "err" in Eq. 5.2.1.6-3 is denoted as energy parameter "POW".

$$err = \sum_{n=0}^3 (\mu amp_n - C_{pow}(i,n))^2. \quad (5.2.1.6-3)$$

$C_{pow}(i,n)$ is shown in Table 5.2.1.6-1.

Then the quantized frame energy P_n is obtained as follows:

$$P_n = \left(\frac{S_{max} \cdot \{10^{C_{POW}(POW,n) \cdot \log_{10}(10+\mu)} - 1.0\}}{\mu} \right)^2 \quad (5.2.1.6-4)$$

Table 5.2.1.6-1 Power Quantization (1/3)

index[i]	Cpow(i,0)	Cpow(i,1)	Cpow(i,2)	Cpow(i,3)
0	0.000848	0.000805	0.000794	0.000814
1	0.156911	0.222012	0.260220	0.278730
2	0.147144	0.129974	0.113154	0.094369
3	0.030839	0.335109	0.372834	0.638975
4	0.221803	0.087208	0.036648	0.023135
5	0.423130	0.388019	0.326053	0.245727
6	0.023319	0.042255	0.156844	0.232672
7	0.548638	0.583868	0.605232	0.619448
8	0.084952	0.043884	0.021076	0.012710
9	0.431073	0.329244	0.135975	0.064468
10	0.075432	0.151230	0.223427	0.252618
11	0.680453	0.642389	0.555117	0.376477
12	0.017563	0.020096	0.044194	0.170220
13	0.431037	0.434459	0.434589	0.433602
14	0.031805	0.073109	0.333978	0.580125
15	0.642352	0.696218	0.731968	0.748557
16	0.046613	0.020497	0.010464	0.007344
17	0.279426	0.314689	0.338287	0.359635
18	0.162696	0.155725	0.149085	0.141227
19	0.209598	0.545381	0.686975	0.730332
20	0.347846	0.134758	0.062313	0.035237
21	0.407280	0.404410	0.399113	0.391499
22	0.062131	0.196836	0.298008	0.331483
23	0.683102	0.680807	0.674887	0.666479
24	0.029295	0.049877	0.071772	0.086405
25	0.522236	0.469427	0.351821	0.154646
26	0.263649	0.227629	0.173610	0.110074
27	0.576854	0.551181	0.516769	0.469462
28	0.016831	0.016916	0.041089	0.365062
29	0.497780	0.497267	0.496288	0.493353
30	0.056276	0.227367	0.398196	0.464107
31	0.790981	0.803074	0.808633	0.807811
32	0.013281	0.011542	0.010819	0.010725
33	0.279891	0.275494	0.269721	0.263730
34	0.061804	0.118346	0.157254	0.173974
35	0.209126	0.483589	0.581486	0.597777
36	0.206669	0.177856	0.133277	0.074562
37	0.478926	0.441755	0.380755	0.297357
38	0.021051	0.043226	0.277961	0.267497
39	0.628403	0.630166	0.630672	0.630070
40	0.107810	0.090325	0.057064	0.033077
41	0.553776	0.389734	0.174052	0.093454
42	0.241455	0.225388	0.208709	0.188385
43	0.707294	0.675923	0.611071	0.506072

Table 5.2.1.6-1 Power Quantization (2/3)

index[i]	Cpow(i,0)	Cpow(i,1)	Cpow(i,2)	Cpow(i,3)
44	0.009690	0.020450	0.129735	0.114158
45	0.535298	0.512231	0.474617	0.409744
46	0.173917	0.202407	0.412368	0.520873
47	0.732384	0.739357	0.742884	0.742436
48	0.006427	0.007707	0.019317	0.052147
49	0.346510	0.363593	0.373757	0.381238
50	0.190816	0.184881	0.176721	0.166036
51	0.505948	0.635993	0.696703	0.719830
52	0.491541	0.213125	0.102091	0.055625
53	0.418234	0.458650	0.489098	0.512193
54	0.175365	0.258395	0.337708	0.378497
55	0.767604	0.743250	0.689100	0.588069
56	0.088377	0.086740	0.084492	0.082842
57	0.686354	0.590229	0.353261	0.195298
58	0.300207	0.273977	0.236869	0.193545
59	0.383747	0.573724	0.645769	0.658558
60	0.034047	0.067426	0.212503	0.399735
61	0.479796	0.524854	0.551599	0.570380
62	0.233932	0.346887	0.420008	0.444185
63	0.834535	0.843035	0.842104	0.834501
64	0.005254	0.004851	0.004811	0.005268
65	0.235214	0.267897	0.289701	0.306184
66	0.126893	0.130986	0.134028	0.137184
67	0.090294	0.335011	0.608283	0.723095
68	0.245272	0.165390	0.067859	0.034736
69	0.547460	0.381300	0.258866	0.296624
70	0.095232	0.100332	0.121366	0.305212
71	0.611495	0.604564	0.594285	0.578361
72	0.145061	0.058015	0.024365	0.016033
73	0.424244	0.377279	0.276968	0.125230
74	0.212729	0.157818	0.211678	0.359120
75	0.786919	0.754238	0.641560	0.380000
76	0.019758	0.020584	0.039678	0.252726
77	0.472092	0.468946	0.463683	0.458318
78	0.052499	0.112938	0.520970	0.613347
79	0.688345	0.767030	0.805870	0.817938
80	0.041947	0.043873	0.040810	0.040290
81	0.343726	0.339537	0.332583	0.323387
82	0.142348	0.158405	0.172924	0.186167
83	0.348237	0.629376	0.761710	0.782720
84	0.328128	0.242538	0.105041	0.050381
85	0.354380	0.390522	0.431700	0.464621
86	0.063355	0.366779	0.367870	0.344750
87	0.716826	0.713197	0.707384	0.698864

Table 5.2.1.6-1 Power Quantization (3/3)

index[i]	Cpow(i,0)	Cpow(i,1)	Cpow(i,2)	Cpow(i,3)
88	0.067529	0.063596	0.060018	0.057053
89	0.565159	0.532204	0.448578	0.253427
90	0.334375	0.299847	0.217428	0.099648
91	0.647856	0.604163	0.544960	0.505082
92	0.026370	0.024627	0.065026	0.521556
93	0.528781	0.527121	0.524529	0.520847
94	0.124605	0.373675	0.493156	0.514645
95	0.833633	0.819521	0.791542	0.751115
96	0.021050	0.024988	0.027435	0.023307
97	0.305226	0.303853	0.302391	0.299699
98	0.032659	0.248052	0.201449	0.154229
99	0.319986	0.380189	0.647871	0.733753
100	0.289992	0.151095	0.120684	0.194040
101	0.492723	0.459810	0.419067	0.381288
102	0.035533	0.056852	0.380844	0.402556
103	0.613940	0.644949	0.671283	0.688894
104	0.153667	0.127924	0.077605	0.041248
105	0.644562	0.507314	0.233792	0.122570
106	0.238045	0.240930	0.241251	0.241452
107	0.688905	0.666619	0.636415	0.601586
108	0.044616	0.079259	0.110631	0.134941
109	0.636401	0.560107	0.441585	0.365170
110	0.265989	0.260326	0.490677	0.651185
111	0.770345	0.772854	0.771176	0.765307
112	0.009786	0.010961	0.026757	0.104261
113	0.396373	0.378387	0.360281	0.340402
114	0.190952	0.200466	0.207264	0.213288
115	0.505174	0.727144	0.793853	0.805674
116	0.434607	0.268691	0.197921	0.193552
117	0.381544	0.461936	0.548592	0.602094
118	0.362392	0.262446	0.328649	0.524239
119	0.783697	0.765652	0.734640	0.687603
120	0.105809	0.106958	0.107846	0.109943
121	0.713473	0.667399	0.513354	0.222163
122	0.349754	0.323281	0.286033	0.238194
123	0.517666	0.476131	0.617428	0.710688
124	0.146361	0.168891	0.222688	0.504976
125	0.564158	0.562604	0.558768	0.552404
126	0.283663	0.436396	0.494884	0.514327
127	0.879671	0.890789	0.888855	0.875443

5.2.1.7 Long-term predictor lag pre-selection

Long term predictor lag is pre-selected so that the computational complexity for the adaptive codebook search can be reduced. Method and the candidate number of the pre-selection are optional. In the recommended pre-selection method, autocorrelations of LPC residues are used as shown below. The analysis is done twice a frame. The center of each analysis is the middle of the 0th subframe and the 2nd subframe, and the analysis length N_{LTP} is 256. The pre-selection results of the 1st and the 3rd subframe are commonly used for the 0th and the 2nd respectively.

5.2.1.7.1 LPC residue

LPC residues $\text{res}(n,i)$ ($n=0,2$, $0 \leq i \leq N_{LTP} - 1$) are obtained by filtering the input signal "s" with the linear predictive inverse filter $F(n,z)$ ($n=0,2$).

$$F(n,z) = 1 + \sum_{i=1}^{N_0} a(n,i) \cdot z^{-i} \quad (n=0,2) \quad (5.2.1.7.1-1)$$

5.2.1.7.2 Hamming window

$$\text{resw}(n,i) = \text{res}(n,i) \cdot W_{H256}(i) \quad (n=0,2, 0 \leq i \leq N_{LTP} - 1). \quad (5.2.1.7.2-1)$$

$$W_{H256}(i) = 0.54 - 0.46 \cdot \cos\left(\frac{2\pi(i+0.5)}{N_{LTP}}\right) \quad (0 \leq i \leq N_{LTP} - 1). \quad (5.2.1.7.2-2)$$

5.2.1.7.3 Autocorrelation

$$\phi_{\text{res}_{\text{int}}}(n,i) = \sum_{k=0}^{N_{LTP}-1-i} \text{resw}(n,k) \cdot \text{resw}(n,k+i) \quad (n=0,2, L_{\min}-2 \leq i \leq L_{\max}+2), \quad (5.2.1.7.3-1)$$

Where $L_{\min}=16$, $L_{\max}=96$.

5.2.1.7.4 Autocorrelation interpolation

$$\phi_{\text{res}_{\text{zip}}}(n,i) = \begin{cases} \phi_{\text{res}_{\text{int}}}(n,m) & (i = 4m, L_{\min} - 2 \leq m \leq L_{\max} + 2). \\ 0 & (\text{else}) \end{cases} \quad (5.2.1.7.4-1)$$

$$\phi_{\text{res}_{\text{ups}}}(n,i) = \sum_{j=-10}^9 \phi_{\text{res}_{\text{zip}}}(n,i+j) \cdot W_{LIP}(j) \quad (4 \cdot L_{\min} - 1 \leq i \leq 4 \cdot L_{\max} + 2). \quad (5.2.1.7.4-2)$$

$$W_{LIP}(-10) = 0.12732395 \quad (5.2.1.7.4-3)$$

$$W_{LIP}(-9) = 0.10003515$$

$$W_{LIP}(-8) = 0.00000000$$

$$W_{LIP}(-7) = -0.12861662$$

$$W_{LIP}(-6) = -0.21220659$$

$$W_{LIP}(-5) = -0.18006326$$

$$W_{LIP}(-4) = 0.00000000$$

$$W_{LIP}(-3) = 0.30010544$$

$$W_{LIP}(-2) = 0.63661977$$

$$W_{LIP}(-1) = 0.90031632$$

$$\begin{aligned}
W_{LIP}(0) &= 1.00000000 \\
W_{LIP}(1) &= 0.90031632 \\
W_{LIP}(2) &= 0.63661977 \\
W_{LIP}(3) &= 0.30010544 \\
W_{LIP}(4) &= 0.00000000 \\
W_{LIP}(5) &= -0.18006326 \\
W_{LIP}(6) &= -0.21220659 \\
W_{LIP}(7) &= -0.12861662 \\
W_{LIP}(8) &= 0.00000000 \\
W_{LIP}(9) &= 0.10003515
\end{aligned}$$

5.2.1.7.5 Maximum autocorrelation detection

$$\phi res_{\max}(n,i) = \max(\phi res_{\text{ups}}(n,4i-1), \phi res_{\text{ups}}(n,4i), \phi res_{\text{ups}}(n,4i+1), \phi res_{\text{ups}}(n,4i+2)) \\
(L_{\min} \leq i \leq L_{\max}), \quad (5.2.1.7.5-1)$$

where $\max(a, b, c, d)$ is the maximum value of a, b, c , and d .

5.2.1.7.6 Pre-selection of the long-term predictor lag

With each $n=0,2$, the largest 6 $\phi res_{\max}(n,i)$ are selected, and indices "i"s of the selected $\phi res_{\max}()$ are denoted as PSEL (n,j) ($n=0,2, 0 \leq j \leq 5$), where $\phi res_{\max}(n, \text{PSEL}(n,0))$ is the largest. PSEL(1,j) and PSEL(3,j) are set to PSEL(0,j) and PSEL(2,j) respectively.

5.2.1.7.7 Pitch predictor coefficients

Three-tap pitch predictor coefficients $cov(n,j)$ ($n=0,2, j=0,1,2$) shown below are derived from $res(n,i)$ and PSEL($n,0$) by the autocorrelation method using 256-sample length hamming window.

$$Pn(Z) = \frac{1}{1 + \sum_{i=0}^2 cov(n,i) \cdot z^{-(\text{PSEL}(n,0)+i-1)}}. \quad (5.2.1.7.7-1)$$

The values of $cov(1,j)$ and $cov(3,j)$ are set to $cov(0,j)$ and $cov(2,j)$ respectively.

5.2.1.8 Subframe processing

5.2.1.8.1 Perceptual weighting filter coefficients

Design of the perceptual weighting filter described in this section is optional. In the following example, the perceptual weighting filter, $W(z)$, is a cascade of a spectral weighting filter $I(z)$ and a pitch weighting filter $Q(z)$.

$$W(z) = I(z) \cdot Q(z). \quad (5.2.1.8.1-1)$$

5.2.1.8.1.1 Spectral weighting filter

The spectral weighting filter $I(z)$ is given by

$$I(z) = \sum_{i=0}^{N_{\text{fir}}} \alpha_{\text{fir}}(i) \cdot z^{-i}, \quad (5.2.1.8.1.1-1)$$

where $N_{fir}=11$. The $\alpha_{fir}(i)$ is the impulse response of $G(z)$ in Eq. 5.2.1.8.1.1-2, truncated at the $N_{fir}+1$ -th sample, and $\alpha_{fir}(0)=1$.

$$G(z) = \frac{1 + \sum_{i=1}^{N_p} \alpha(n,i) \cdot \lambda_{ma}^i \cdot z^{-i}}{1 + \sum_{i=1}^{N_p} \alpha(n,i) \cdot \lambda_{ar}^i \cdot z^{-i}}, \quad (5.2.1.8.1.1-2)$$

where $\lambda_{ma}=0.9$ and $\lambda_{ar}=0.4$.

5.2.1.8.1.2 Pitch weighting filter

$$Q(z) = 1 + \sum_{i=0}^2 \text{cov}(n,i) \cdot \lambda_{pi} \cdot z^{-(\text{PSEL}(n,0)+i-1)} \quad (5.2.1.8.1.2-1)$$

where $\lambda_{pi}=0.4$, the $\text{cov}(n,i)$ and the $\text{PSEL}(n,0)$ are those obtained in Section 5.2.1.7.

5.2.1.8.2 Perceptually weighted input

Perceptually weighted input $\text{spw}(i)$ ($0 \leq i \leq N_{SUB}-1$) is obtained by filtering the input signal "s" with the perceptual weighting filter $W(z)$.

5.2.1.8.3 Zero-input response subtraction

The zero input response of the perceptually weighted synthesis filter $H(z)$ is subtracted from the $\text{spw}(i)$, then reference speech $r(i)$ ($0 \leq i \leq N_{SUB}-1$) is obtained.

$$H(z) = \frac{1}{1 + \sum_{i=1}^{N_p} \alpha_q(n,i) \cdot z^{-i}} \cdot W(z) \quad (5.2.1.8.3-1)$$

5.2.1.8.4 ACB/FCB search

Using candidates $\text{PSEL}(n, i)$ obtained in the long term predictor lag pre-selection, some adaptive codebook (ACB) outputs are generated considering fractional lag period. In the ACB/FCB final-selection, plural candidates are selected from these codebook outputs for the delayed decision. Recommend candidate number for the final-selection is two. Prior to the final-selection, pre-selection is performed to reduce the computational complexity. Method and the candidate number of the pre-selection are optional. In the recommended pre-selection method, four and two candidates are selected referring to the reference input from ACB and FCB respectively.

5.2.1.8.4.1 ACB selection

5.2.1.8.4.1.1 ACB output computation

By using candidates $\text{PSEL}(n, i)$ obtained in long term predictor lag pre-selection, lag candidates $\text{lagi}(j)$, $\text{lagf}(j)$ ($0 \leq j \leq N_{CAN_LTP}-1$) are generated considering fractional lag period as follows:

$$\begin{array}{lll}
\text{PSEL}(n,i) \leq 45 & : & \begin{array}{ll} \text{lagi}(j) = \text{PSEL}(n,i), & \text{lagf}(j) = 0, \\ \text{lagi}(j+1) = \text{PSEL}(n,i), & \text{lagf}(j+1) = 1, \\ \text{lagi}(j+2) = \text{PSEL}(n,i), & \text{lagf}(j+2) = 2, \\ \text{lagi}(j+3) = \text{PSEL}(n,i), & \text{lagf}(j+3) = 3, \end{array} & (5.2.1.8.4.1.1-1) \\
46 \leq \text{PSEL}(n,i) \leq 65 & : & \begin{array}{ll} \text{lagi}(j) = \text{PSEL}(n,i), & \text{lagf}(j) = 1, \\ \text{lagi}(j+1) = \text{PSEL}(n,i), & \text{lagf}(j+1) = 3, \end{array} \\
66 \leq \text{PSEL}(n,i) & : & \begin{array}{ll} \text{lagi}(j) = \text{PSEL}(n,i), & \text{lagf}(j) = 1, \end{array}
\end{array}$$

where $\text{lagf}(i)=0$ corresponds to the fractional lag period of $+\frac{1}{4}$, $\text{lagf}(i)=1$ to 0, $\text{lagf}(i)=2$ to $-\frac{1}{4}$, $\text{lagf}(i)=3$ to $-\frac{1}{2}$.

With the above values, the ACB output $P_{ACB}(i,k)$ ($0 \leq i \leq \text{NCAN}_{LTP}-1$, $0 \leq k \leq \text{N}_{SUB}-1$) are computed as follows:

$$\begin{aligned}
& P_{ACB} \left(i, n \cdot \text{lagi}(i) + j - \left\lfloor \frac{\text{lagf}(i) + n \cdot (\text{lagf}(i) - 1)}{4} \right\rfloor \right) \\
& = \text{dqint}(j, \text{mod}(\text{lagf}(i) + n \cdot (\text{lagf}(i) - 1), 4)) , \\
& (-1 \leq j \leq \text{lagi}(i) - 1, 0 \leq n \leq \left\lfloor \frac{\text{N}_{SUB}}{\text{lagi}(i)} \right\rfloor)
\end{aligned} \tag{5.2.1.8.4.1.1-2}$$

$$\text{dqint}(j, \text{frac}) = \begin{cases} \sum_{k=-2}^2 \text{dq}(\text{N}_{dq} - \text{lagi}(i) + j + k) \cdot W_{\text{frac}}(k) & (j \leq \text{lagi}(i) - 3) \\ \sum_{k=-2}^2 \text{dq}(\text{N}_{dq} - \text{lagi}(i) + j + k - 1) \cdot W_{\text{frac}}(k - 1) & (j = \text{lagi}(i) - 2) \\ \sum_{k=-2}^2 \text{dq}(\text{N}_{dq} - \text{lagi}(i) + j + k - 2) \cdot W_{\text{frac}}(k - 2) & (j = \text{lagi}(i) - 1) \end{cases}$$

($\text{frac}=0,1,2,3$) , (5.2.1.8.4.1.1-3)

$$W_0(2) = 0.10003515 \tag{5.2.1.8.4.1.1-4}$$

$$W_0(1) = -0.18006326$$

$$W_0(0) = 0.90031632$$

$$W_0(-1) = 0.30010544$$

$$W_0(-2) = -0.12861662$$

$$W_0(-3) = 0.08184694$$

$$W_0(-4) = -0.06002109$$

$$W_1(2) = 0.00000000 \tag{5.2.1.8.4.1.1-5}$$

$$W_1(1) = 0.00000000$$

$$W_1(0) = 1.00000000$$

$$W_1(-1) = 0.00000000$$

$$W_1(-2) = 0.00000000$$

$$W_1(-3) = 0.00000000$$

$$W_1(-4) = 0.00000000$$

$$\begin{aligned}
W_2(2) &= -0.12861662 \\
W_2(1) &= 0.30010544 \\
W_2(0) &= 0.90031632 \\
W_2(-1) &= -0.18006326 \\
W_2(-2) &= 0.10003515 \\
W_2(-3) &= -0.06925510 \\
W_2(-4) &= 0.05295978
\end{aligned}
\tag{5.2.1.8.4.1.1-6}$$

$$\begin{aligned}
W_3(2) &= -0.21220659 \\
W_3(1) &= 0.63661977 \\
W_3(0) &= 0.63661977 \\
W_3(-1) &= -0.21220659 \\
W_3(-2) &= 0.12732395 \\
W_3(-3) &= -0.09094568 \\
W_3(-4) &= 0.07073553
\end{aligned}
\tag{5.2.1.8.4.1.1-7}$$

where $dq(i)$ ($0 \leq i \leq N_{dq}-1$) is the current ACB state, $\lfloor x \rfloor$ gives the maximum integer which doesn't exceed x , and $\text{mod}(x,y)$ indicates the positive remainder of x divided by y .

If there are some combinations of n and j which give same $P_{ACB}(i,k)$ values, the priority listed below is used.

- (1) $j=0$
- (2) $j=\text{lagi}(i)-1$
- (3) $1 \leq j \leq \text{lagi}(i)-2$
- (4) $j=-1$

5.2.1.8.4.1.2 Time-reverse synthesis of reference speech

The time reverse synthesis $rh(k)$ ($0 \leq k \leq N_{SUB}-1$) of the reference speech $r(k)$ ($0 \leq k \leq N_{SUB}-1$) is computed. This is made by rearranging $r(k)$ in time reverse order, synthesizing by zero state $H(z)$, then again rearranging in time reverse order.

5.2.1.8.4.1.3 ACB pre-selection

$\text{prh}(i)$ are calculated for each $P_{ACB}(i,k)$, and $\text{NCAN}_{ACB}=4$ largest $\text{prh}(i)$ are selected. Then the selected $\text{prh}()$ indices i 's are denoted as $\text{APSEL}(j)$ ($0 \leq j \leq \text{NCAN}_{ACB}-1$).

$$\text{prh}(i) = \sum_{k=0}^{N_{SUB}-1} P_{ACB}(i,k) \cdot rh(k) \quad (0 \leq i \leq \text{NCAN}_{LTP}-1). \tag{5.2.1.8.4.1.3-1}$$

5.2.1.8.4.1.4 ACB final selection

The perceptually weighted synthesis outputs $\text{SYN}_{ACB}(j,k)$ are computed from $P_{ACB}(\text{APSEL}(j),k)$ ($0 \leq j \leq \text{NCAN}_{ACB}-1$) using zero state $H(z)$. The two largest $\text{sacbr}(j)$ are selected, and the selected $\text{sacbr}()$ indices j 's are denoted as ACB final-selection results $\text{ASEL}(i)$ ($i=0,1$).

$$\text{sacbr}(j) = \frac{\text{prh}^2(\text{APSEL}(j))}{\sum_{k=0}^{N_{SUB}-1} \text{SYN}_{ACB}^2(j,k)} \quad (0 \leq j \leq \text{NCAN}_{ACB}-1). \tag{5.2.1.8.4.1.4-1}$$

5.2.1.8.4.2 FCB selection

5.2.8.4.2.1 FCB output computation

FCB output candidates $P_{FCB}(j,k)$ ($0 \leq j \leq 31$, $0 \leq k \leq N_{SUB}-1$) are computed from Table 5.2.1.8-1.

$$P_{FCB}(i+8n,k) = C_{FCB}(n, \text{mod}(k+10i, 80)) \quad (0 \leq i \leq 7, n=0,1,2,3, 0 \leq k \leq N_{SUB}-1) \quad (5.2.1.8.4.2.1-1)$$

5.2.1.8.4.2.2 FCB final selection

With each $P_{FCB}(j,k)$, the $\text{prh}(j)$'s are computed by the same way as described in Section 5.2.1.8.4.1.3. The two largest $|\text{prh}(j)|$ are selected, and the selected $\text{prh}()$ indices j 's are denoted as FCB final-selection results $\text{FSEL}(i)$ ($i=0,1$).

Next, the perceptually weighted synthesis output $\text{SYN}_{FCB}(j,k)$ are computed from $P_{FCB}(\text{FSEL}(i),k)$ using zero state $H(z)$. Then $\text{sfcbr}(i)$'s are computed as follows:

$$\text{sfcbr}(i) = \frac{\text{prh}^2(\text{FSEL}(i))}{\sum_{k=0}^{N_{SUB}-1} \text{SYN}_{FCB}^2(i,k)} \quad (i=0,1). \quad (5.2.1.8.4.2.2-1)$$

5.2.1.8.4.3 ACB/FCB final selection

Two largest values in $\text{sacbr}(i)$ and $\text{sfcbr}(i)$ ($i=0,1$) are selected as ACB/FCB final-selection results, including the candidates for the delayed decision. The corresponding ACB/FCB outputs are denoted as $P(0,1)$ and $P(1,i)$, the synthesized ACB/FCB outputs through zero state $H(z)$ are as $\text{SYNp}(0,i)$ and $\text{SYNp}(1,i)$.

Here, if the FCB is selected, the ACB/FCB output is either $P_{FCB}(j,k)$ or $-P_{FCB}(j,k)$ depending on the polarity of the sign of corresponding $\text{prh}(j)$.

With regard to the subframe in which the $r(k)$ is completely zero, all $P(0,i)$, $P(1,i)$, $\text{SYNp}(0,i)$ and $\text{SYNp}(1,i)$ are set to 0, and the corresponding "lagi" is also set to 0.

5.2.1.8.5 SCB search

The stochastic codebook (SCB) has a two channel structure. Each codebook contains $N_{SC}=16$ vectors. The output of each codebook (SCB0,SCB1) is processed by Pitch Synchronous Innovation (PSI) procedure, then multiplied by a sign and added together to produce a stochastic codebook output SC.

In this section, $P(0,i)$ and $\text{SYNp}(0,i)$ which are selected in Section 5.2.1.8.4.3 are denoted as $P(i)$ and $\text{SYNp}(i)$ respectively, and the corresponding long term prediction lag as lagi and lagf . If FCB is selected, $\text{lagi}=N_{SUB}$ and $\text{lagf}=1$ are used.

5.2.1.8.5.1 PSI (Pitch Synchronous Innovation) Procedure

The outputs from the two channel codebook SCB0 and SCB1 are processed by PSI procedure to produce $PSI_0()$ and $PSI_1()$. If the lag_i is 0, however, the PSI procedure is not performed.

$$PSI_m \left(i, n \cdot lag_i + j - \left\lfloor \frac{1 + n \cdot (lagf - 1)}{4} \right\rfloor \right) \\ = C_{SCBm}(i, \text{mod}(1 + n \cdot (lagf - 1), 4), j) \quad (5.2.1.8.5.1-1) \\ (0 \leq i \leq 15, 0 \leq j \leq lag_i, 0 \leq n \leq \left\lfloor \frac{N_{SUB}}{lag_i} \right\rfloor, m = 0, 1)$$

$C_{SCBm}()$ is shown in Table 5.2.1.8-2. If there are plural combinations of n and j , which give the same $PSI_m(i, k)$ values, the one which has smaller " j " has precedence.

5.2.1.8.5.2 Time-reverse synthesis of ACB/FCB output

The time reverse synthesis $ph(i)$ ($0 \leq i \leq N_{SUB}-1$) is computed from $SYN_p(i)$ ($0 \leq i \leq N_{SUB}-1$) in the same way as described in Section 5.2.1.8.4.1.2.

5.2.1.8.5.3 SCB pre-selection

Prior to the SCB final-selection, pre-selection is performed to reduce the computational complexity. Method and the candidate number of the pre-selection are optional.

The pre-selection procedure is applied to $PSI_0(i, j)$ and $PSI_1(i, j)$ independently to select the respective $N_{SCP}=6$ pre-selection output candidates $PSI_0(\text{SSEL}(0, k), j)$ and $PSI_1(\text{SSEL}(1, k), j)$. The $\text{SSEL}()$ are selected to maximize the $|cr(i)|^2$ using Eq. 5.2.1.8.5.3-1.

$$cr(i) = \sum_{j=0}^{N_{SUB}-1} PSI_m(i, j) rh(j) - \frac{pr}{powp} \cdot \sum_{j=0}^{N_{SUB}-1} PSI_m(i, j) ph(j), \quad (0 \leq i \leq N_{SC}-1, m=0, 1) \quad (5.2.1.8.5.3-1)$$

$$powp = \sum_{j=0}^{N_{SUB}-1} SYN_p^2(j), \quad (5.2.1.8.5.3-2)$$

$$pr = \sum_{j=0}^{N_{SUB}-1} SYN_p(j) r(j), \quad (5.2.1.8.5.3-3)$$

where $rh(i)$ is the time reverse synthesized value of the reference input, which is computed in Section 5.2.1.8.4.1.2, and $r(i)$ is the reference input value. The pre-selected $PSI_0(i, j)$ and $PSI_1(i, j)$ are filtered by zero state $H(z)$ to compute synthesized pre-selection outputs $SYN_{PSI0}(i, j)$ and $SYN_{PSI1}(i, j)$, and orthogonalized vectors are also computed. Synthesized and orthogonalized pre-selection outputs $SYNO_{PSI0}(i, j)$ and $SYNO_{PSI1}(i, j)$ may be computed as follows:

$$SYNO_{PSIm}(i, j) = SYN_{PSIm}(i, j) - \frac{SYN_p(j)}{powp} \cdot \sum_{k=0}^{N_{SUB}-1} PSI_m(i, k) ph(k) \\ (m=0, 1, 0 \leq i \leq N_{SCP}-1, 0 \leq j \leq N_{SUB}-1). \quad (5.2.1.8.5.3-4)$$

In the above computation, computational complexity can be reduced by utilizing that the term

$$\sum_{k=0}^{N_{SUB}-1} PSI_m(i, k) ph(k) \text{ appeared in Eq. 5.2.1.8.5.3-1 as well.}$$

5.2.1.8.5.4 SCB final selection

Among the pre-selected SCB candidates, final indices and signs for each channel are determined. The $scr0$ and $scr1$ are computed for all i and j ($0 \leq i, j \leq N_{SCP}-1$) as shown below:

$$scr0 = \frac{cscr0^2}{\sum_{k=0}^{N_{SUB}-1} (SYNO_{PSI0}(i, k) + SYNO_{PSI1}(j, k))^2}, \quad (5.2.1.8.5.4-1)$$

$$cscr0 = \sum_{k=0}^{N_{SUB}-1} SYNO_{PSI0}(i, k) r(k) + \sum_{k=0}^{N_{SUB}-1} SYNO_{PSI1}(j, k) r(k), \quad (5.2.1.8.5.4-2)$$

$$scr1 = \frac{cscr1^2}{\sum_{k=0}^{N_{SUB}-1} (SYNO_{PSI0}(i, k) - SYNO_{PSI1}(j, k))^2}. \quad (5.2.1.8.5.4-3)$$

$$cscr1 = \sum_{k=0}^{N_{SUB}-1} SYNO_{PSI0}(i, k) r(k) - \sum_{k=0}^{N_{SUB}-1} SYNO_{PSI1}(j, k) r(k). \quad (5.2.1.8.5.4-4)$$

The SCB final-selection results $I0$ and $I1$ are the values of $SSEL(0, i)$ and $SSEL(1, j)$ of which the indices " i " and " j " are selected to give the maximum value of either $scr0$ or $scr1$. If $scr0$ gives the maximum value, the signs ($S0, S1$) to be multiplied with each codebook output are set as follows depending on polarity of $cscr0$;

$$\begin{aligned} S0=+1, S1=+1 & \quad (\text{if } cscr0 \geq 0), \\ S0=-1, S1=-1 & \quad (\text{if } cscr0 < 0). \end{aligned} \quad (5.2.1.8.5.4-5)$$

If $scr1$ gives the maximum value, signs are set as follows depending on polarity of $cscr1$;

$$\begin{aligned} S0=+1, S1=-1 & \quad (\text{if } cscr1 \geq 0), \\ S0=-1, S1=+1 & \quad (\text{if } cscr1 < 0). \end{aligned} \quad (5.2.1.8.5.4-6)$$

Then SCB output $SC(k)$ and synthesized SCB output $SYN_{sc}(k)$ are computed as follows:

$$SC(k) = S0 \cdot PSI_0(I0, k) + S1 \cdot PSI_1(I1, k) \quad (0 \leq k \leq N_{SUB}-1), \quad (5.2.1.8.5.4-7)$$

$$SYN_{sc}(k) = S0 \cdot SYN_{PSI0}(I0, k) + S1 \cdot SYN_{PSI1}(I1, k) \quad (0 \leq k \leq N_{SUB}-1). \quad (5.2.1.8.5.4-8)$$

5.2.1.8.6 Gain quantization

Estimated residual power (RS) for each subframe is computed as follows:

$$RS = N_{SUB} \cdot P_n \cdot \prod_{i=1}^{N_p} (1 - kq^2(n, i)). \quad (5.2.1.8.5.6-1)$$

The index " i " which minimizes err_g is searched using the gain quantization codebooks $Cg0(i)$ and $Cg1(i)$ shown in Table 5.2.1.8-3.

$$err_g = \sum_{k=0}^{N_{SUB}-1} \left(\sqrt{\frac{RS}{pow0}} \cdot C_{g0}(i) \cdot SYN_p(k) + \sqrt{\frac{RS}{pow1}} \cdot C_{g1}(i) \cdot SYN_{sc}(k) - r(k) \right)^2, \quad (5.2.1.8.6-2)$$

$$\text{pow0} = \sum_{k=0}^{N_{\text{SUB}}-1} P^2(k) \quad , \quad (5.2.1.8.6-3)$$

$$\text{pow1} = \sum_{k=0}^{N_{\text{SUB}}-1} SC^2(k) \quad , \quad (5.2.1.8.6-4)$$

where the gains $g0$ and $g1$ which are applied to $P(k)$ and $SC(k)$ are computed as follows:

$$g0 = \sqrt{\frac{RS}{\text{pow0}}} \cdot C_{g0}(i) \quad , \quad (5.2.1.8.6-5)$$

$$g1 = \sqrt{\frac{RS}{\text{pow1}}} \cdot C_{g1}(i) \quad . \quad (5.2.1.8.6-6)$$

5.2.1.8.7 Delayed decision

Delayed decision is applied in order to improve the coding quality furthermore.
Method of the delayed decision is optional.

The procedures described in Section 5.2.1.8.5 through 5.2.1.8.6 are performed again with the combination of $P(1,i)$ and $\text{SYNp}(1,i)$, which has been selected as the second candidate in the ACB/FCB search. The two resulted err_g are compared, then the combination which gives smaller err_g is used as the result of delayed decision.

If the ACB/FCB output corresponding to the delayed decision result comes from FCB, the index "j" which corresponds to the selected FCB output candidate ($P_{\text{FCB}}(j,k)$) is used as the FCB index for this subframe ($I_{\text{FCB_n}}$). And the sign for FCB ($S_{\text{FCB_n}}$) is set depending on the sign of $\text{prh}(j)$ in Section 5.2.1.8.4.2.2 as follows:

$$\begin{aligned} S_{\text{FCB_n}} &= 0 & (\text{if } \text{prh}(j) \geq 0), \\ S_{\text{FCB_n}} &= 1 & (\text{if } \text{prh}(j) < 0). \end{aligned} \quad (5.2.1.8.7-1)$$

$I_{\text{SCB0_n}}$ and $I_{\text{SCB1_n}}$ are defined as the selected indices for SCB0 and SCB1 respectively, and the SCB signs $S_{\text{SCB0_n}}$ and $S_{\text{SCB1_n}}$ are set depending on $S0$ and $S1$ as follows:

$$\begin{aligned} S_{\text{SCBk_n}} &= 0 & (\text{if } S_k \geq 0), \\ S_{\text{SCBk_n}} &= 1 & (\text{if } S_k < 0). \end{aligned} \quad (5.2.1.8.7-2)$$

Then the index for the gain quantization codebook $Cg0(i)$ and $Cg1(i)$, which correspond to the delayed decision result is denoted as excitation gain index GAIN_i .

5.2.1.8.8 Updating of filter states

Excitation source $\text{ex}(i)$ is computed with the delayed decision result as follows:

$$\text{ex}(i) = g0 \cdot P(i) + g1 \cdot SC(i) \quad . \quad (5.2.1.8.8-1)$$

The adaptive codebook state $\text{dq}(j)$ is updated using the $\text{ex}(i)$.

$$\text{dq}(i) = \begin{cases} \text{dq}(i + N_{\text{SUB}}) & (0 \leq i \leq N_{\text{dq}} - N_{\text{SUB}} - 1) \\ \text{ex}(i + N_{\text{SUB}} - N_{\text{dq}}) & (N_{\text{dq}} - N_{\text{SUB}} \leq i \leq N_{\text{dq}} - 1) \end{cases} \quad . \quad (5.2.1.8.8-2)$$

$H(z)$ is also updated using this $\text{ex}(i)$.

Table 5.2.1.8-1 C_{FCB} Parameter (1/2)

index	$C_{FCB}(0, i)$	$C_{FCB}(1, i)$	$C_{FCB}(2, i)$	$C_{FCB}(3, i)$
0	-0.031891	0.042175	0.060303	0.050140
1	-0.044983	0.095764	0.007935	0.061615
2	-0.082764	0.019867	0.108093	0.093567
3	-0.014252	0.054413	0.013275	0.047333
4	-0.071533	0.068756	-0.090668	-0.012085
5	-0.035950	0.017548	-0.008331	0.008423
6	-0.103027	0.064880	-0.019287	0.020264
7	-0.026764	-0.068237	0.038574	-0.077301
8	-0.031586	0.005829	0.011261	-0.067505
9	-0.038086	-0.076233	-0.037262	-0.058960
10	0.052155	-0.080688	0.007690	-0.039520
11	0.009277	-0.089203	0.033600	-0.019501
12	0.130432	-0.362213	-0.029236	-0.074341
13	0.107635	-0.196930	-0.062683	-0.030457
14	0.228058	-0.567932	0.030945	-0.065857
15	0.261047	-0.170197	-0.038910	-0.022461
16	0.244568	0.749573	-0.008026	0.022797
17	0.535858	0.221497	-0.043823	-0.021454
18	-0.327606	-0.081665	-0.096039	0.013580
19	-0.624329	0.099030	-0.000549	0.033386
20	0.048859	0.105072	0.030487	-0.038605
21	-0.044067	0.117889	0.070190	0.033752
22	-0.083801	0.057953	-0.001862	0.022736
23	-0.087555	0.029205	-0.010559	0.066437
24	-0.056274	-0.024841	0.063385	0.048462
25	-0.034241	0.062164	0.012146	0.069855
26	0.018616	-0.019867	0.058472	0.003998
27	-0.022766	-0.017120	0.007874	0.033722
28	-0.038483	0.023407	-0.009705	0.067688
29	0.042999	-0.076538	-0.001556	0.036896
30	-0.090942	-0.011993	0.068817	0.031738
31	0.027740	-0.028625	0.047729	0.033478
32	0.029510	0.007172	0.005768	-0.039062
33	-0.017029	-0.012146	0.024902	-0.044861
34	0.054260	-0.047516	-0.010651	0.000793
35	0.040283	-0.008484	0.006012	-0.006287
36	0.060028	-0.034698	-0.048462	0.010925
37	-0.003479	-0.018066	-0.047058	-0.035919
38	0.067535	0.036255	-0.062164	-0.068726
39	-0.009857	-0.033752	-0.039581	-0.049042
40	0.000336	0.037323	-0.006989	0.004333
41	0.083801	-0.079193	-0.123077	-0.090424
42	-0.011017	0.048401	-0.001923	-0.010468

Table 5.2.1.8-1 C_{FCB} Parameter (2/2)

index	$C_{FCB}(0, i)$	$C_{FCB}(1, i)$	$C_{FCB}(2, i)$	$C_{FCB}(3, i)$
43	0.019684	0.016205	-0.121033	-0.007568
44	-0.039001	0.038849	-0.000275	-0.037292
45	-0.041534	-0.050995	-0.051971	0.014526
46	-0.017242	-0.019470	-0.026123	-0.057648
47	-0.000183	-0.014709	-0.006989	0.054779
48	-0.007233	0.002930	-0.059113	0.006775
49	-0.044769	-0.002502	0.101868	0.028381
50	-0.002777	0.000366	-0.013702	0.055542
51	-0.028809	0.072357	0.168243	-0.005096
52	0.004456	0.003113	0.087128	0.004730
53	-0.001068	0.035767	0.131073	-0.028015
54	-0.005615	-0.002197	0.348846	0.008209
55	-0.003296	-0.004242	0.225372	-0.063446
56	-0.033661	0.013702	0.554901	0.057007
57	0.029846	0.021393	-0.221954	0.194702
58	-0.003387	-0.007416	-0.734283	0.052063
59	-0.002075	-0.050262	0.032806	0.068848
60	0.002411	-0.025269	-0.037811	0.040558
61	0.020844	-0.016449	-0.112030	-0.016235
62	0.014618	-0.041260	-0.084198	-0.005707
63	-0.018738	-0.040314	-0.116608	-0.049896
64	0.022980	-0.005127	-0.077698	-0.056152
65	0.014648	-0.031616	-0.009613	-0.103668
66	0.045563	-0.032379	0.010040	-0.049866
67	0.032898	0.015228	-0.068420	-0.156494
68	0.046356	-0.041931	0.074463	-0.272430
69	0.041473	-0.010590	0.026550	-0.218445
70	0.012268	-0.032074	-0.143860	-0.588104
71	0.070801	-0.012085	0.067383	0.094849
72	0.013458	0.017944	-0.084045	0.766602
73	0.022430	-0.035980	0.127197	0.223450
74	0.023804	0.023254	-0.080933	-0.084717
75	-0.047455	0.001892	-0.016052	0.098663
76	0.002045	0.041138	0.057343	0.091614
77	-0.005371	0.019440	0.055237	0.054993
78	0.005432	0.041046	0.043091	0.017181
79	-0.033752	0.064178	0.002380	0.047852

Table 5.2.1.8-2 C_{SCBm} Parameter(1/40)

k	$C_{SCB0}(0, 0, k)$	$C_{SCB0}(1, 0, k)$	$C_{SCB0}(2, 0, k)$	$C_{SCB0}(3, 0, k)$
0	0.000000	0.031250	0.015625	-0.046875
1	-0.031250	0.000000	-0.046875	-0.015625
2	-0.007812	-0.046875	-0.015625	-0.054688
3	-0.039062	0.015625	0.023438	0.031250
4	-0.007812	0.000000	-0.070312	-0.070312
5	-0.039062	0.000000	-0.007812	-0.015625
6	-0.031250	-0.070312	-0.039062	0.015625
7	0.007812	0.000000	0.000000	-0.046875
8	-0.093750	-0.015625	0.015625	0.015625
9	-0.015625	-0.007812	-0.031250	-0.039062
10	0.000000	0.023438	0.039062	0.023438
11	-0.039062	-0.039062	-0.023438	0.007812
12	-0.015625	0.031250	0.023438	0.039062
13	-0.023438	0.000000	0.015625	0.062500
14	0.007812	0.054688	0.015625	-0.007812
15	-0.007812	0.007812	0.085938	0.070312
16	0.039062	-0.007812	0.007812	0.023438
17	0.000000	0.070312	0.078125	0.093750
18	0.023438	-0.007812	0.007812	0.046875
19	0.023438	0.062500	0.078125	0.000000
20	-0.031250	0.039062	0.031250	0.070312
21	0.093750	0.054688	-0.023438	-0.039062
22	0.031250	0.031250	0.070312	0.062500
23	0.148438	0.007812	-0.039062	-0.054688
24	0.023438	0.039062	0.046875	-0.039062
25	-0.015625	-0.031250	-0.093750	-0.023438
26	0.109375	0.070312	-0.046875	-0.195312
27	-0.007812	-0.117188	-0.046875	-0.062500
28	0.015625	-0.007812	-0.187500	-0.296875
29	-0.101562	-0.039062	-0.039062	-0.117188
30	-0.031250	-0.203125	-0.265625	0.359375
31	-0.039062	-0.023438	-0.046875	0.125000
32	-0.179688	-0.382812	0.335938	-0.023438
33	-0.062500	-0.015625	0.140625	-0.015625
34	-0.328125	0.468750	-0.007812	0.054688
35	0.015625	0.046875	-0.007812	0.085938
36	0.359375	0.000000	0.085938	0.062500
37	0.132812	0.000000	0.015625	0.015625
38	0.031250	0.085938	0.054688	-0.023438
39	0.015625	0.046875	0.062500	0.031250
40	0.109375	-0.007812	-0.039062	-0.062500

Table 5.2.1.8-2 C_{SCBm} Parameter(2/40)

k	$C_{SCB0}(4, 0, k)$	$C_{SCB0}(5, 0, k)$	$C_{SCB0}(6, 0, k)$	$C_{SCB0}(7, 0, k)$
0	-0.015625	-0.093750	-0.007812	0.015625
1	-0.046875	0.000000	-0.039062	-0.023438
2	-0.023438	-0.023438	0.015625	0.015625
3	-0.054688	-0.015625	-0.031250	0.015625
4	0.000000	-0.015625	0.046875	-0.031250
5	0.000000	-0.023438	0.015625	0.085938
6	-0.031250	0.078125	0.015625	0.023438
7	-0.015625	-0.015625	0.054688	0.070312
8	0.039062	0.023438	0.000000	0.023438
9	0.000000	0.023438	0.070312	0.062500
10	-0.031250	-0.007812	0.023438	0.078125
11	0.054688	0.070312	0.078125	-0.023438
12	0.023438	0.000000	0.023438	0.062500
13	0.046875	0.070312	0.015625	-0.015625
14	0.046875	0.070312	0.093750	0.007812
15	0.031250	0.015625	-0.054688	-0.039062
16	0.046875	0.062500	0.046875	-0.039062
17	0.000000	-0.031250	-0.031250	-0.039062
18	0.046875	0.023438	-0.039062	-0.171875
19	0.023438	-0.031250	-0.031250	-0.062500
20	0.023438	-0.039062	-0.203125	-0.203125
21	0.000000	-0.031250	-0.054688	-0.140625
22	-0.070312	-0.148438	-0.281250	0.273438
23	0.000000	-0.085938	-0.093750	0.210938
24	-0.156250	-0.289062	0.343750	-0.015625
25	-0.054688	-0.109375	0.125000	-0.054688
26	-0.257812	0.296875	-0.023438	0.062500
27	-0.164062	0.117188	0.000000	0.015625
28	0.382812	0.054688	0.039062	0.031250
29	0.125000	0.023438	0.031250	0.039062
30	-0.031250	0.023438	0.039062	-0.039062
31	0.054688	0.000000	0.000000	0.046875
32	0.062500	0.023438	-0.015625	-0.023438
33	0.015625	0.070312	0.054688	0.000000
34	0.000000	0.000000	-0.023438	0.039062
35	0.000000	0.007812	0.000000	-0.062500
36	-0.078125	-0.015625	0.023438	0.031250
37	0.109375	0.039062	0.000000	-0.039062
38	-0.039062	-0.015625	0.031250	-0.023438
39	-0.015625	0.031250	0.000000	-0.031250
40	0.015625	0.023438	-0.007812	-0.039062

Table 5.2.1.8-2 C_{SCBm} Parameter(3/40)

k	$C_{SCB0}(8, 0, k)$	$C_{SCB0}(9, 0, k)$	$C_{SCB0}(10, 0, k)$	$C_{SCB0}(11, 0, k)$
0	-0.031250	-0.039062	0.000000	-0.046875
1	-0.039062	-0.039062	-0.046875	0.000000
2	-0.015625	-0.039062	-0.054688	-0.046875
3	-0.039062	-0.015625	0.007812	0.062500
4	-0.046875	-0.015625	-0.031250	0.093750
5	0.007812	0.000000	0.062500	0.039062
6	0.000000	-0.015625	0.054688	0.101562
7	0.023438	0.007812	0.023438	0.023438
8	-0.046875	0.062500	0.078125	0.132812
9	0.031250	0.046875	0.062500	0.046875
10	0.070312	0.101562	0.132812	-0.054688
11	0.023438	0.031250	0.000000	-0.015625
12	0.101562	0.109375	-0.085938	-0.039062
13	0.031250	0.078125	0.007812	0.023438
14	0.140625	-0.093750	-0.031250	-0.062500
15	0.054688	-0.023438	0.000000	-0.109375
16	-0.054688	-0.031250	-0.078125	-0.101562
17	0.007812	0.023438	-0.085938	-0.062500
18	-0.031250	-0.062500	-0.070312	0.070312
19	0.000000	-0.109375	-0.070312	0.031250
20	-0.117188	-0.023438	0.039062	0.046875
21	-0.062500	-0.062500	0.054688	0.000000
22	-0.007812	0.031250	0.039062	0.007812
23	-0.085938	0.039062	-0.007812	0.000000
24	0.000000	0.046875	0.000000	-0.031250
25	0.054688	0.039062	0.023438	0.007812
26	0.023438	0.015625	-0.039062	-0.023438
27	0.023438	-0.015625	0.031250	0.039062
28	0.046875	0.000000	-0.031250	0.000000
29	0.000000	0.015625	0.007812	0.007812
30	-0.062500	-0.046875	0.007812	0.023438
31	0.039062	-0.023438	0.007812	-0.007812
32	-0.023438	-0.031250	0.039062	0.023438
33	0.039062	0.023438	-0.015625	0.015625
34	-0.023438	-0.007812	0.078125	0.039062
35	-0.007812	-0.007812	0.007812	0.000000
36	0.070312	0.078125	0.031250	-0.007812
37	-0.054688	-0.015625	-0.023438	0.000000
38	0.054688	0.070312	0.000000	-0.054688
39	0.054688	0.023438	0.085938	0.039062
40	0.000000	0.015625	-0.101562	-0.023438

Table 5.2.1.8-2 C_{SCBm} Parameter(4/40)

k	$C_{SCB0}(12, 0, k)$	$C_{SCB0}(13, 0, k)$	$C_{SCB0}(14, 0, k)$	$C_{SCB0}(15, 0, k)$
0	-0.007812	0.062500	0.093750	0.101562
1	0.039062	0.078125	0.109375	0.046875
2	0.070312	0.039062	0.117188	-0.046875
3	0.039062	0.078125	0.039062	0.023438
4	0.070312	0.164062	-0.031250	-0.023438
5	0.054688	0.039062	-0.007812	0.015625
6	0.125000	-0.046875	-0.031250	-0.078125
7	0.046875	0.023438	0.015625	-0.062500
8	-0.101562	-0.078125	-0.085938	-0.054688
9	-0.031250	0.023438	-0.085938	-0.117188
10	-0.031250	-0.054688	-0.015625	0.054688
11	0.015625	-0.132812	-0.070312	0.023438
12	-0.046875	-0.023438	0.015625	0.039062
13	-0.109375	-0.078125	0.000000	0.046875
14	-0.023438	0.000000	0.054688	-0.007812
15	-0.109375	0.046875	0.062500	-0.015625
16	0.007812	0.085938	-0.015625	-0.046875
17	0.085938	0.023438	-0.039062	0.031250
18	0.015625	-0.023438	-0.054688	-0.015625
19	0.031250	-0.007812	0.023438	-0.015625
20	-0.046875	-0.039062	-0.015625	0.023438
21	-0.039062	0.000000	0.000000	0.000000
22	-0.007812	-0.023438	-0.015625	0.023438
23	-0.007812	-0.015625	-0.007812	0.015625
24	0.000000	0.015625	0.046875	0.015625
25	-0.015625	0.015625	-0.023438	0.000000
26	0.023438	0.023438	0.015625	0.023438
27	0.007812	-0.007812	-0.007812	0.023438
28	0.023438	0.046875	0.054688	0.015625
29	0.023438	0.000000	0.015625	0.046875
30	0.007812	0.015625	-0.007812	-0.007812
31	0.000000	-0.023438	0.031250	0.078125
32	0.023438	-0.015625	-0.039062	-0.023438
33	0.007812	0.046875	0.078125	0.023438
34	0.007812	-0.007812	-0.039062	0.062500
35	0.039062	0.031250	0.023438	-0.046875
36	0.000000	-0.031250	0.000000	0.023438
37	0.093750	0.023438	-0.046875	-0.078125
38	-0.046875	0.039062	0.031250	-0.046875
39	-0.023438	-0.007812	-0.148438	-0.148438
40	0.078125	-0.023438	0.000000	-0.046875

Table 5.2.1.8-2 C_{SCBm} Parameter(5/40)

k	$C_{SCB1}(0, 0, k)$	$C_{SCB1}(1, 0, k)$	$C_{SCB1}(2, 0, k)$	$C_{SCB1}(3, 0, k)$
0	-0.078125	-0.070312	0.000000	0.101562
1	-0.007812	-0.031250	-0.039062	0.000000
2	-0.015625	-0.007812	0.070312	0.085938
3	0.007812	-0.023438	-0.046875	0.109375
4	-0.007812	0.070312	0.140625	0.101562
5	-0.031250	-0.007812	0.054688	0.281250
6	0.062500	0.117188	0.101562	-0.093750
7	-0.007812	0.054688	0.281250	-0.203125
8	0.093750	0.117188	-0.109375	-0.031250
9	0.085938	0.296875	-0.187500	-0.015625
10	0.101562	-0.140625	-0.093750	-0.007812
11	0.250000	-0.218750	0.007812	-0.023438
12	-0.062500	-0.039062	0.046875	0.039062
13	-0.203125	-0.007812	-0.078125	-0.023438
14	-0.085938	0.031250	0.039062	0.007812
15	0.023438	-0.039062	0.000000	0.007812
16	0.031250	0.007812	0.007812	-0.062500
17	-0.070312	0.000000	-0.031250	0.000000
18	0.015625	0.015625	-0.054688	-0.046875
19	0.007812	-0.007812	0.007812	0.015625
20	-0.007812	-0.039062	-0.062500	0.000000
21	-0.007812	0.000000	0.062500	0.000000
22	-0.062500	-0.031250	-0.007812	0.007812
23	-0.007812	0.023438	0.015625	-0.015625
24	-0.015625	0.000000	0.023438	0.000000
25	-0.007812	0.015625	0.000000	0.015625
26	0.015625	0.039062	0.031250	0.046875
27	-0.007812	-0.007812	0.000000	-0.031250
28	0.015625	0.007812	0.031250	0.015625
29	-0.039062	0.046875	-0.007812	0.031250
30	0.062500	0.015625	0.000000	-0.078125
31	0.015625	-0.054688	0.039062	0.062500
32	0.039062	0.031250	-0.015625	-0.039062
33	0.000000	0.007812	0.054688	-0.023438
34	0.000000	-0.023438	-0.015625	-0.015625
35	0.031250	0.023438	0.015625	-0.062500
36	-0.039062	-0.023438	0.015625	-0.054688
37	0.062500	-0.007812	-0.031250	-0.007812
38	-0.054688	0.000000	-0.007812	0.046875
39	0.023438	-0.023438	-0.039062	-0.054688
40	0.023438	-0.015625	0.062500	0.015625

Table 5.2.1.8-2 C_{SCBm} Parameter(6/40)

k	$C_{SCB1}(4, 0, k)$	$C_{SCB1}(5, 0, k)$	$C_{SCB1}(6, 0, k)$	$C_{SCB1}(7, 0, k)$
0	0.156250	0.125000	-0.328125	0.078125
1	0.187500	0.359375	-0.281250	0.023438
2	0.109375	-0.093750	-0.062500	0.007812
3	0.296875	-0.226562	0.015625	-0.007812
4	-0.125000	-0.039062	0.039062	-0.007812
5	-0.226562	-0.007812	-0.015625	-0.015625
6	-0.007812	0.000000	0.062500	0.023438
7	0.031250	-0.023438	-0.023438	-0.015625
8	0.023438	0.046875	0.085938	-0.054688
9	-0.054688	-0.023438	-0.007812	-0.007812
10	0.007812	-0.007812	-0.054688	-0.031250
11	-0.039062	0.023438	0.031250	-0.007812
12	0.015625	-0.085938	-0.054688	-0.007812
13	0.007812	-0.007812	0.031250	-0.031250
14	-0.062500	0.000000	-0.046875	0.000000
15	0.000000	-0.015625	0.015625	0.015625
16	-0.046875	0.007812	0.000000	0.054688
17	-0.007812	0.023438	0.000000	0.007812
18	-0.023438	0.031250	0.070312	0.039062
19	0.015625	0.000000	0.039062	0.023438
20	0.000000	0.015625	0.132812	-0.039062
21	-0.007812	0.015625	-0.078125	0.000000
22	0.031250	0.007812	-0.015625	0.000000
23	0.000000	0.015625	0.039062	0.000000
24	0.023438	0.000000	0.000000	-0.007812
25	-0.039062	0.023438	0.039062	0.007812
26	-0.015625	0.000000	-0.046875	0.039062
27	0.031250	0.031250	0.000000	0.000000
28	0.000000	-0.039062	0.000000	0.023438
29	0.015625	0.023438	0.015625	-0.039062
30	-0.031250	-0.007812	0.023438	0.023438
31	0.015625	-0.015625	-0.023438	-0.039062
32	0.039062	-0.023438	0.039062	-0.085938
33	0.000000	-0.046875	-0.031250	0.062500
34	-0.015625	0.023438	-0.015625	0.015625
35	-0.054688	-0.015625	0.000000	0.007812
36	0.070312	0.031250	0.000000	0.062500
37	0.015625	0.000000	0.015625	0.078125
38	0.015625	-0.039062	-0.015625	0.046875
39	0.015625	0.039062	0.007812	0.007812
40	-0.078125	0.023438	0.000000	0.015625

Table 5.2.1.8-2 C_{SCBm} Parameter(7/40)

k	$C_{SCB1}(8, 0, k)$	$C_{SCB1}(9, 0, k)$	$C_{SCB1}(10, 0, k)$	$C_{SCB1}(11, 0, k)$
0	0.007812	-0.015625	0.000000	0.007812
1	0.046875	0.015625	-0.015625	-0.046875
2	-0.007812	0.015625	0.039062	0.015625
3	0.007812	-0.007812	-0.054688	-0.039062
4	0.000000	0.046875	0.031250	-0.078125
5	-0.023438	-0.062500	-0.117188	-0.031250
6	0.031250	0.007812	-0.015625	-0.210938
7	-0.046875	-0.070312	-0.039062	0.070312
8	0.015625	-0.078125	-0.218750	0.195312
9	-0.046875	-0.015625	0.117188	0.031250
10	-0.070312	-0.179688	0.140625	0.007812
11	-0.031250	0.085938	0.078125	-0.046875
12	-0.187500	0.179688	-0.007812	0.031250
13	0.062500	0.031250	-0.039062	-0.015625
14	0.156250	0.031250	0.062500	0.015625
15	0.054688	-0.039062	-0.023438	0.015625
16	0.031250	0.000000	0.031250	-0.046875
17	-0.039062	-0.023438	-0.031250	0.046875
18	0.007812	0.031250	0.000000	0.046875
19	-0.031250	0.007812	0.023438	0.070312
20	0.015625	-0.031250	0.023438	-0.031250
21	0.000000	0.054688	0.054688	0.000000
22	-0.023438	0.031250	0.000000	0.039062
23	0.046875	0.039062	0.000000	-0.078125
24	0.015625	-0.015625	0.015625	0.000000
25	0.070312	-0.007812	-0.054688	-0.062500
26	0.000000	0.000000	-0.039062	-0.007812
27	0.007812	-0.023438	-0.046875	-0.023438
28	0.007812	-0.031250	0.007812	-0.023438
29	-0.062500	-0.062500	-0.007812	0.023438
30	0.023438	-0.007812	-0.015625	0.007812
31	-0.078125	-0.054688	0.015625	0.070312
32	0.031250	0.023438	-0.007812	-0.046875
33	0.000000	-0.007812	0.023438	0.046875
34	-0.007812	-0.023438	-0.007812	0.007812
35	0.000000	0.062500	-0.007812	-0.023438
36	-0.031250	-0.023438	-0.031250	0.015625
37	0.054688	0.046875	0.031250	-0.046875
38	0.007812	-0.023438	0.007812	0.054688
39	0.054688	0.046875	0.000000	0.000000
40	-0.007812	0.085938	0.085938	0.031250

Table 5.2.1.8-2 C_{SCBm} Parameter(8/40)

k	$C_{SCB1}(12, 0, k)$	$C_{SCB1}(13, 0, k)$	$C_{SCB1}(14, 0, k)$	$C_{SCB1}(15, 0, k)$
0	0.015625	-0.031250	-0.093750	0.015625
1	-0.039062	-0.015625	0.109375	-0.070312
2	-0.054688	-0.195312	0.171875	-0.007812
3	-0.031250	0.085938	0.046875	-0.015625
4	-0.203125	0.187500	0.007812	-0.007812
5	0.085938	0.000000	-0.070312	0.000000
6	0.187500	0.023438	0.031250	-0.007812
7	0.015625	-0.031250	-0.023438	0.000000
8	-0.007812	0.046875	0.000000	-0.023438
9	-0.039062	0.000000	-0.023438	0.062500
10	0.031250	-0.007812	-0.007812	0.015625
11	-0.023438	0.000000	0.031250	0.007812
12	0.000000	-0.015625	0.015625	-0.007812
13	-0.023438	0.070312	0.070312	0.054688
14	0.000000	-0.023438	-0.031250	0.070312
15	0.078125	0.046875	0.000000	-0.070312
16	0.000000	0.015625	-0.015625	0.039062
17	0.062500	0.000000	-0.031250	-0.007812
18	0.023438	0.023438	0.007812	-0.054688
19	0.000000	-0.031250	-0.054688	-0.039062
20	-0.015625	0.007812	0.023438	-0.007812
21	0.007812	-0.039062	-0.062500	0.039062
22	0.000000	0.000000	-0.031250	-0.031250
23	-0.062500	-0.031250	0.007812	0.046875
24	0.000000	-0.023438	-0.031250	-0.054688
25	-0.054688	0.015625	0.007812	0.023438
26	0.031250	0.000000	-0.015625	0.046875
27	0.000000	0.062500	0.023438	-0.054688
28	-0.023438	-0.039062	-0.031250	0.054688
29	0.085938	0.023438	-0.007812	0.000000
30	-0.039062	0.039062	0.000000	0.039062
31	0.023438	-0.062500	-0.007812	-0.015625
32	0.000000	0.007812	0.015625	0.000000
33	0.015625	-0.007812	0.000000	0.093750
34	0.015625	0.062500	0.085938	-0.070312
35	0.023438	-0.039062	0.000000	0.109375
36	0.078125	0.054688	0.000000	0.046875
37	-0.070312	0.046875	0.015625	-0.046875
38	0.015625	-0.054688	-0.015625	0.007812
39	0.078125	0.085938	0.109375	0.007812
40	-0.054688	0.000000	-0.039062	0.039062

Table 5.2.1.8-2 C_{SCBm} Parameter(9/40)

k	$C_{SCB0}(0, 1, k)$	$C_{SCB0}(1, 1, k)$	$C_{SCB0}(2, 1, k)$	$C_{SCB0}(3, 1, k)$
0	-0.007812	0.039062	0.007812	-0.046875
1	-0.031250	-0.015625	-0.062500	-0.023438
2	-0.007812	-0.031250	0.007812	-0.039062
3	-0.039062	0.023438	0.000000	0.031250
4	0.000000	-0.015625	-0.070312	-0.093750
5	-0.054688	-0.007812	0.000000	0.015625
6	-0.007812	-0.070312	-0.054688	-0.007812
7	-0.015625	0.015625	0.015625	-0.039062
8	-0.093750	-0.031250	0.000000	0.015625
9	0.007812	0.000000	-0.023438	-0.046875
10	-0.023438	0.007812	0.039062	0.039062
11	-0.039062	-0.039062	-0.031250	0.000000
12	-0.015625	0.039062	0.031250	0.062500
13	-0.023438	-0.007812	0.000000	0.039062
14	0.007812	0.062500	0.039062	0.000000
15	-0.007812	-0.015625	0.070312	0.078125
16	0.046875	0.015625	0.007812	0.023438
17	0.000000	0.054688	0.085938	0.109375
18	0.031250	-0.015625	0.000000	0.015625
19	0.000000	0.078125	0.093750	0.023438
20	-0.015625	0.023438	0.000000	0.054688
21	0.101562	0.062500	0.000000	-0.031250
22	0.031250	0.015625	0.062500	0.070312
23	0.164062	0.015625	-0.039062	-0.085938
24	-0.023438	0.031250	0.054688	0.000000
25	0.039062	-0.015625	-0.117188	-0.062500
26	0.101562	0.062500	-0.007812	-0.179688
27	-0.023438	-0.132812	-0.093750	-0.101562
28	0.015625	0.039062	-0.164062	-0.351562
29	-0.109375	-0.093750	-0.070312	0.039062
30	0.007812	-0.156250	-0.312500	0.343750
31	-0.078125	-0.093750	0.085938	0.031250
32	-0.156250	-0.421875	0.312500	0.000000
33	-0.117188	0.187500	0.062500	0.000000
34	-0.351562	0.382812	0.000000	0.070312
35	0.171875	-0.031250	0.007812	0.078125
36	0.304688	0.046875	0.085938	0.054688
37	0.078125	-0.007812	0.007812	0.007812
38	0.046875	0.109375	0.085938	-0.007812
39	0.039062	0.007812	0.031250	0.015625
40	0.109375	0.023438	-0.031250	-0.062500
41	-0.023438	0.125000	0.007812	0.046875
42	0.046875	-0.007812	-0.085938	0.000000

Table 5.2.1.8-2 C_{SCBm} Parameter(10/40)

43	0.000000	0.093750	0.046875	-0.007812
44	-0.031250	-0.078125	0.007812	0.015625
45	0.062500	0.007812	0.000000	-0.023438
46	-0.078125	0.015625	0.000000	0.054688
47	0.070312	-0.023438	-0.039062	-0.109375
48	0.031250	0.046875	0.039062	-0.007812
49	-0.062500	-0.132812	-0.039062	0.015625
50	-0.023438	0.039062	0.015625	-0.015625
51	-0.046875	-0.039062	-0.046875	0.015625
52	0.031250	0.000000	-0.015625	-0.054688
53	-0.070312	-0.039062	0.000000	0.039062
54	-0.046875	-0.046875	-0.046875	-0.023438
55	0.007812	-0.007812	0.054688	0.023438
56	0.023438	-0.023438	0.000000	-0.007812
57	-0.007812	0.078125	0.000000	-0.031250
58	-0.062500	0.015625	0.000000	0.046875
59	0.101562	0.031250	-0.070312	-0.007812
60	0.031250	0.015625	0.078125	0.015625
61	0.007812	0.031250	-0.023438	0.070312
62	-0.078125	0.078125	0.070312	0.007812
63	-0.046875	-0.085938	-0.023438	0.015625
64	0.070312	0.054688	0.078125	0.070312
65	0.039062	0.007812	0.085938	0.046875
66	0.156250	-0.039062	-0.023438	0.000000
67	-0.046875	0.054688	0.062500	0.062500
68	0.046875	0.000000	-0.015625	-0.125000
69	0.039062	0.031250	0.046875	0.000000
70	0.031250	-0.054688	0.070312	0.101562
71	0.085938	0.109375	-0.062500	-0.062500
72	-0.078125	-0.015625	0.023438	-0.039062
73	0.070312	-0.062500	0.078125	-0.078125
74	-0.023438	0.132812	0.054688	0.015625
75	-0.085938	-0.062500	-0.023438	-0.023438
76	0.031250	0.000000	-0.054688	0.000000
77	-0.054688	-0.054688	-0.101562	0.085938
78	-0.007812	-0.132812	0.023438	-0.062500
79	0.054688	-0.070312	-0.007812	0.039062

Table 5.2.1.8-2 C_{SCBm} Parameter(11/40)

k	$C_{SCB0}(4, 1, k)$	$C_{SCB0}(5, 1, k)$	$C_{SCB0}(6, 1, k)$	$C_{SCB0}(7, 1, k)$
0	-0.023438	-0.085938	-0.015625	0.007812
1	-0.046875	0.015625	-0.023438	-0.023438
2	-0.031250	-0.046875	0.007812	0.031250
3	-0.054688	0.000000	-0.031250	-0.007812
4	0.007812	-0.031250	0.054688	-0.007812
5	-0.023438	0.000000	-0.007812	0.085938
6	-0.031250	0.078125	0.039062	0.007812
7	0.000000	-0.031250	0.039062	0.078125
8	0.031250	0.046875	0.000000	0.015625
9	-0.023438	0.000000	0.070312	0.085938
10	-0.015625	0.000000	0.015625	0.046875
11	0.062500	0.062500	0.085938	-0.015625
12	0.007812	-0.007812	0.000000	0.070312
13	0.054688	0.093750	0.046875	-0.031250
14	0.039062	0.046875	0.070312	0.015625
15	0.039062	0.031250	-0.054688	-0.046875
16	0.039062	0.046875	0.070312	-0.023438
17	0.000000	-0.031250	-0.054688	-0.070312
18	0.062500	0.031250	-0.007812	-0.156250
19	0.007812	-0.054688	-0.070312	-0.070312
20	0.031250	-0.015625	-0.179688	-0.265625
21	-0.015625	-0.046875	-0.085938	-0.031250
22	-0.054688	-0.132812	-0.335938	0.304688
23	-0.007812	-0.125000	0.054688	0.125000
24	-0.156250	-0.328125	0.328125	-0.023438
25	-0.070312	0.015625	0.039062	-0.023438
26	-0.328125	0.281250	0.000000	0.070312
27	-0.007812	0.054688	0.015625	0.000000
28	0.382812	0.078125	0.046875	0.054688
29	0.007812	0.007812	0.031250	0.015625
30	0.015625	0.023438	0.039062	-0.031250
31	0.062500	-0.007812	-0.007812	0.054688
32	0.046875	0.039062	0.000000	-0.046875
33	0.007812	0.054688	0.054688	0.023438
34	0.007812	-0.015625	-0.039062	0.007812
35	-0.023438	0.015625	0.023438	-0.054688
36	-0.046875	-0.015625	0.007812	0.046875
37	0.117188	0.039062	0.007812	-0.062500
38	-0.085938	-0.023438	0.039062	-0.007812
39	0.023438	0.054688	-0.007812	-0.046875
40	-0.007812	-0.007812	-0.007812	-0.023438
41	-0.015625	-0.070312	-0.031250	-0.007812
42	0.054688	-0.007812	-0.039062	-0.039062

Table 5.2.1.8-2 C_{SCBm} Parameter(12/40)

43	-0.078125	-0.062500	-0.015625	0.000000
44	0.062500	0.054688	-0.039062	-0.023438
45	-0.054688	-0.031250	0.000000	-0.007812
46	0.000000	-0.031250	0.000000	-0.023438
47	0.007812	0.015625	-0.007812	0.007812
48	-0.039062	-0.054688	-0.015625	0.015625
49	-0.015625	0.039062	0.007812	0.007812
50	0.000000	-0.023438	-0.007812	0.023438
51	-0.015625	0.031250	-0.007812	0.007812
52	-0.015625	-0.023438	-0.007812	0.046875
53	0.031250	-0.039062	-0.039062	0.000000
54	0.000000	0.070312	-0.007812	0.007812
55	0.062500	0.000000	-0.007812	0.054688
56	0.007812	0.132812	-0.015625	0.015625
57	-0.046875	-0.046875	-0.007812	0.015625
58	0.140625	-0.007812	0.000000	0.000000
59	-0.062500	0.054688	-0.007812	-0.007812
60	-0.023438	-0.015625	-0.015625	0.000000
61	0.039062	0.023438	0.015625	-0.015625
62	0.000000	-0.031250	0.039062	0.046875
63	0.109375	0.046875	0.039062	-0.039062
64	0.000000	0.007812	0.062500	-0.023438
65	0.015625	-0.031250	0.101562	0.046875
66	-0.054688	0.039062	-0.062500	-0.046875
67	0.007812	-0.046875	-0.031250	0.000000
68	0.031250	0.046875	0.031250	0.007812
69	-0.070312	-0.023438	0.000000	-0.054688
70	-0.054688	-0.078125	0.023438	-0.023438
71	-0.109375	0.046875	0.007812	0.007812
72	0.085938	-0.031250	0.000000	-0.039062
73	-0.070312	-0.046875	-0.023438	-0.015625
74	-0.117188	0.023438	0.000000	0.007812
75	0.218750	-0.015625	0.007812	-0.062500
76	-0.023438	-0.046875	-0.093750	0.007812
77	0.031250	0.000000	0.000000	0.039062
78	0.023438	-0.039062	-0.093750	0.015625
79	0.070312	-0.015625	-0.015625	0.062500

Table 5.2.1.8-2 C_{SCBm} Parameter(13/40)

k	$C_{SCB0}(8, 1, k)$	$C_{SCB0}(9, 1, k)$	$C_{SCB0}(10, 1, k)$	$C_{SCB0}(11, 1, k)$
0	-0.046875	-0.039062	-0.007812	-0.046875
1	-0.031250	-0.039062	-0.062500	0.000000
2	-0.023438	-0.039062	-0.031250	-0.046875
3	-0.054688	-0.015625	0.000000	0.093750
4	-0.039062	-0.015625	-0.023438	0.062500
5	0.007812	0.000000	0.085938	0.046875
6	0.000000	-0.023438	0.023438	0.093750
7	0.007812	0.023438	0.039062	0.031250
8	-0.054688	0.054688	0.078125	0.156250
9	0.062500	0.054688	0.078125	0.000000
10	0.046875	0.093750	0.125000	-0.039062
11	0.031250	0.039062	-0.039062	-0.015625
12	0.093750	0.140625	-0.054688	-0.031250
13	0.046875	0.031250	0.015625	0.031250
14	0.164062	-0.085938	-0.039062	-0.093750
15	0.000000	0.000000	0.000000	-0.101562
16	-0.031250	-0.031250	-0.093750	-0.109375
17	0.007812	0.023438	-0.070312	-0.039062
18	-0.031250	-0.101562	-0.085938	0.070312
19	-0.007812	-0.085938	-0.054688	0.015625
20	-0.132812	-0.031250	0.054688	0.046875
21	-0.023438	-0.062500	0.039062	0.000000
22	-0.039062	0.046875	0.031250	0.023438
23	-0.085938	0.023438	-0.007812	-0.007812
24	0.023438	0.054688	0.015625	-0.023438
25	0.039062	0.031250	0.007812	0.007812
26	0.023438	0.007812	-0.031250	-0.015625
27	0.031250	-0.015625	0.039062	0.046875
28	0.039062	0.015625	-0.046875	-0.015625
29	-0.015625	0.000000	0.031250	0.023438
30	-0.046875	-0.054688	0.000000	0.007812
31	0.039062	-0.015625	0.023438	-0.007812
32	-0.031250	-0.023438	0.023438	0.023438
33	0.046875	0.031250	-0.007812	0.015625
34	-0.046875	-0.023438	0.085938	0.039062
35	0.031250	0.015625	0.000000	-0.007812
36	0.039062	0.062500	0.031250	0.000000
37	-0.054688	-0.015625	-0.031250	-0.007812
38	0.085938	0.093750	0.039062	-0.046875
39	0.023438	0.000000	0.046875	0.046875
40	0.007812	0.023438	-0.101562	-0.062500
41	0.023438	-0.007812	0.078125	-0.046875
42	0.015625	0.007812	0.000000	0.031250

Table 5.2.1.8-2 C_{SCBm} Parameter(14/40)

43	0.039062	0.015625	0.031250	-0.031250
44	0.000000	-0.046875	-0.023438	0.000000
45	0.031250	0.007812	0.023438	-0.062500
46	-0.078125	0.000000	-0.078125	-0.007812
47	0.039062	0.039062	-0.117188	-0.148438
48	0.000000	0.000000	0.046875	-0.085938
49	-0.031250	0.000000	-0.218750	-0.257812
50	0.039062	0.000000	-0.085938	-0.070312
51	-0.078125	-0.125000	-0.226562	0.484375
52	0.085938	-0.093750	-0.343750	0.093750
53	-0.148438	-0.140625	0.617188	-0.140625
54	-0.031250	-0.273438	-0.070312	0.070312
55	-0.265625	0.445312	0.148438	0.015625
56	-0.304688	0.140625	0.039062	0.101562
57	0.351562	-0.046875	0.062500	0.054688
58	-0.023438	0.078125	0.140625	0.117188
59	0.210938	0.039062	0.023438	-0.031250
60	0.250000	0.218750	0.101562	-0.007812
61	-0.054688	0.031250	-0.062500	0.054688
62	0.132812	0.117188	0.062500	0.031250
63	0.023438	-0.062500	0.000000	0.007812
64	0.195312	0.000000	-0.007812	-0.140625
65	-0.101562	0.093750	0.039062	0.062500
66	-0.039062	-0.117188	-0.101562	-0.085938
67	-0.015625	-0.023438	0.093750	-0.148438
68	-0.148438	-0.148438	-0.070312	0.039062
69	0.125000	0.109375	-0.039062	-0.226562
70	-0.039062	-0.093750	-0.062500	-0.054688
71	-0.031250	-0.070312	0.000000	-0.085938
72	-0.085938	-0.023438	0.023438	-0.054688
73	-0.164062	-0.226562	-0.117188	-0.031250
74	0.062500	0.062500	0.031250	-0.015625
75	0.078125	-0.109375	0.007812	0.046875
76	0.015625	-0.031250	-0.023438	0.070312
77	-0.164062	0.156250	-0.007812	0.085938
78	0.000000	-0.117188	-0.039062	0.054688
79	0.015625	0.062500	0.101562	0.054688

Table 5.2.1.8-2 C_{SCBm} Parameter(15/40)

k	$C_{SCB0}(12, 1, k)$	$C_{SCB0}(13, 1, k)$	$C_{SCB0}(14, 1, k)$	$C_{SCB0}(15, 1, k)$
0	0.000000	0.078125	0.117188	0.117188
1	0.062500	0.070312	0.101562	0.007812
2	0.054688	0.039062	0.117188	-0.031250
3	0.046875	0.109375	0.015625	0.031250
4	0.070312	0.148438	-0.015625	-0.023438
5	0.070312	0.000000	0.000000	0.015625
6	0.140625	-0.023438	-0.023438	-0.101562
7	0.000000	0.015625	0.015625	-0.039062
8	-0.093750	-0.070312	-0.109375	-0.085938
9	-0.007812	0.039062	-0.054688	-0.085938
10	-0.031250	-0.101562	-0.031250	0.070312
11	0.015625	-0.101562	-0.070312	0.000000
12	-0.078125	-0.031250	0.031250	0.062500
13	-0.085938	-0.085938	-0.007812	0.031250
14	-0.031250	0.023438	0.078125	-0.007812
15	-0.117188	0.046875	0.039062	-0.023438
16	0.054688	0.085938	-0.023438	-0.031250
17	0.062500	0.007812	-0.039062	0.039062
18	0.015625	-0.015625	-0.039062	-0.039062
19	0.023438	-0.007812	0.031250	0.007812
20	-0.062500	-0.031250	-0.023438	0.015625
21	-0.015625	0.007812	0.000000	0.000000
22	-0.015625	-0.031250	-0.023438	0.031250
23	0.000000	0.000000	0.007812	0.007812
24	0.000000	0.015625	0.031250	0.015625
25	-0.007812	0.015625	-0.031250	0.000000
26	0.031250	0.015625	0.023438	0.031250
27	0.000000	0.000000	-0.007812	0.015625
28	0.031250	0.054688	0.062500	0.023438
29	0.015625	-0.015625	-0.007812	0.039062
30	0.007812	0.023438	0.007812	0.000000
31	0.007812	-0.039062	0.015625	0.070312
32	0.023438	0.000000	-0.023438	-0.046875
33	0.000000	0.039062	0.085938	0.062500
34	0.015625	-0.007812	-0.062500	0.031250
35	0.031250	0.031250	0.054688	-0.031250
36	0.015625	-0.039062	-0.031250	0.023438
37	0.085938	0.046875	-0.015625	-0.093750
38	-0.078125	0.023438	0.007812	-0.039062
39	0.031250	0.000000	-0.148438	-0.140625
40	0.054688	-0.054688	0.015625	-0.039062
41	-0.078125	-0.156250	-0.179688	-0.304688
42	0.000000	0.007812	-0.007812	-0.250000

Table 5.2.1.8-2 C_{SCBm} Parameter(16/40)

43	-0.062500	-0.117188	-0.195312	0.437500
44	-0.046875	-0.007812	-0.312500	0.125000
45	-0.085938	-0.242188	0.523438	-0.070312
46	-0.171875	-0.203125	0.062500	0.046875
47	-0.218750	0.414062	-0.054688	0.046875
48	-0.320312	-0.046875	0.140625	0.156250
49	0.359375	-0.007812	-0.015625	0.023438
50	0.226562	0.039062	0.140625	0.039062
51	-0.007812	-0.015625	0.031250	-0.023438
52	0.085938	0.179688	0.062500	0.054688
53	0.031250	0.000000	0.015625	0.046875
54	0.046875	0.125000	0.000000	-0.093750
55	-0.007812	-0.093750	0.007812	0.039062
56	0.000000	0.085938	-0.039062	0.023438
57	0.046875	0.046875	0.062500	0.000000
58	0.125000	-0.054688	-0.054688	-0.070312
59	-0.046875	0.046875	-0.015625	0.015625
60	-0.046875	-0.117188	0.000000	0.015625
61	0.031250	0.031250	-0.046875	-0.093750
62	-0.062500	0.015625	0.070312	0.023438
63	0.023438	-0.085938	-0.078125	-0.070312
64	-0.062500	-0.031250	0.046875	-0.070312
65	-0.054688	0.000000	-0.054688	0.031250
66	0.000000	0.023438	-0.039062	-0.046875
67	-0.085938	-0.109375	0.046875	0.046875
68	-0.039062	-0.031250	-0.125000	-0.132812
69	-0.031250	0.046875	-0.023438	0.007812
70	0.007812	-0.054688	-0.101562	-0.015625
71	-0.070312	-0.015625	0.023438	-0.023438
72	-0.070312	0.007812	-0.046875	-0.007812
73	0.015625	0.000000	-0.054688	-0.093750
74	0.007812	-0.046875	0.148438	0.062500
75	0.031250	0.062500	-0.031250	-0.015625
76	0.015625	-0.039062	0.085938	0.062500
77	0.039062	-0.062500	0.054688	0.093750
78	0.000000	0.093750	0.078125	0.179688
79	0.031250	0.101562	0.085938	0.226562

Table 5.2.1.8-2 C_{SCBm} Parameter(17/40)

k	$C_{SCB1}(0, 1, k)$	$C_{SCB1}(1, 1, k)$	$C_{SCB1}(2, 1, k)$	$C_{SCB1}(3, 1, k)$
0	-0.078125	-0.078125	-0.023438	0.093750
1	0.000000	-0.007812	-0.007812	-0.007812
2	-0.023438	-0.023438	0.046875	0.109375
3	0.007812	-0.007812	-0.039062	0.093750
4	-0.023438	0.062500	0.156250	0.171875
5	-0.007812	-0.015625	0.015625	0.242188
6	0.046875	0.125000	0.195312	-0.187500
7	-0.015625	0.031250	0.226562	-0.125000
8	0.109375	0.210938	-0.187500	-0.023438
9	0.070312	0.234375	-0.132812	-0.023438
10	0.171875	-0.234375	-0.085938	-0.007812
11	0.218750	-0.140625	0.031250	-0.015625
12	-0.140625	-0.039062	0.007812	0.031250
13	-0.164062	-0.007812	-0.070312	-0.031250
14	-0.054688	0.023438	0.062500	0.031250
15	0.023438	-0.046875	-0.023438	-0.007812
16	0.000000	0.023438	0.015625	-0.054688
17	-0.070312	0.000000	-0.046875	0.000000
18	0.039062	0.023438	-0.031250	-0.054688
19	-0.007812	-0.023438	0.000000	0.031250
20	0.000000	-0.031250	-0.054688	-0.007812
21	-0.023438	0.000000	0.070312	0.015625
22	-0.062500	-0.031250	-0.031250	0.000000
23	0.000000	0.023438	0.039062	-0.015625
24	-0.023438	-0.015625	0.007812	0.007812
25	0.000000	0.031250	0.000000	0.023438
26	0.007812	0.023438	0.023438	0.031250
27	-0.007812	-0.007812	0.000000	-0.031250
28	0.007812	0.031250	0.039062	0.039062
29	-0.031250	0.046875	-0.023438	0.000000
30	0.078125	0.000000	0.015625	-0.054688
31	0.000000	-0.039062	0.023438	0.078125
32	0.046875	0.046875	-0.007812	-0.070312
33	-0.007812	-0.007812	0.054688	0.007812
34	0.015625	0.000000	-0.023438	-0.039062
35	0.015625	0.023438	0.039062	-0.070312
36	-0.023438	-0.039062	0.000000	-0.039062
37	0.054688	0.000000	-0.031250	0.007812
38	-0.070312	-0.007812	-0.015625	0.031250
39	0.062500	-0.031250	-0.023438	-0.054688
40	-0.015625	-0.007812	0.070312	0.046875
41	-0.054688	0.015625	-0.062500	0.007812
42	0.046875	0.000000	-0.007812	-0.007812

Table 5.2.1.8-2 C_{SCBm} Parameter(18/40)

43	-0.015625	-0.023438	0.023438	-0.007812
44	0.031250	-0.023438	-0.023438	0.054688
45	-0.023438	0.085938	0.031250	-0.015625
46	0.039062	0.000000	-0.023438	-0.046875
47	-0.007812	0.015625	-0.031250	0.062500
48	-0.015625	0.039062	0.000000	-0.023438
49	0.078125	-0.007812	-0.031250	-0.062500
50	-0.046875	0.000000	-0.015625	-0.054688
51	0.062500	-0.085938	-0.039062	-0.031250
52	-0.054688	0.039062	0.031250	-0.007812
53	0.000000	-0.054688	-0.070312	-0.117188
54	0.039062	0.000000	-0.023438	0.054688
55	-0.039062	-0.085938	-0.039062	-0.109375
56	-0.031250	-0.031250	-0.031250	0.039062
57	-0.054688	0.000000	-0.015625	-0.007812
58	0.062500	0.000000	-0.015625	-0.054688
59	-0.062500	-0.039062	0.007812	0.062500
60	-0.007812	0.007812	0.023438	0.070312
61	0.000000	0.015625	0.031250	0.031250
62	-0.039062	0.007812	0.015625	0.046875
63	0.039062	0.054688	0.054688	0.046875
64	0.046875	0.039062	0.015625	0.062500
65	0.000000	0.000000	0.046875	0.062500
66	0.015625	0.062500	0.093750	0.054688
67	0.007812	0.054688	0.078125	-0.062500
68	-0.007812	0.023438	0.031250	0.000000
69	0.101562	0.023438	-0.046875	-0.031250
70	0.062500	0.000000	-0.023438	-0.070312
71	0.093750	0.039062	0.000000	-0.070312
72	0.078125	-0.046875	-0.117188	-0.046875
73	0.015625	-0.078125	-0.023438	-0.109375
74	0.007812	-0.046875	-0.179688	-0.062500
75	0.046875	-0.085938	-0.179688	-0.015625
76	-0.164062	-0.054688	0.039062	-0.078125
77	-0.257812	-0.148438	-0.007812	-0.070312
78	-0.156250	-0.031250	-0.031250	-0.007812
79	-0.015625	-0.007812	-0.007812	-0.023438

Table 5.2.1.8-2 C_{SCBm} Parameter(19/40)

k	$C_{SCB1}(4, 1, k)$	$C_{SCB1}(5, 1, k)$	$C_{SCB1}(6, 1, k)$	$C_{SCB1}(7, 1, k)$
0	0.210938	0.218750	-0.390625	0.085938
1	0.125000	0.296875	-0.195312	0.000000
2	0.187500	-0.203125	-0.054688	0.015625
3	0.257812	-0.140625	0.015625	-0.015625
4	-0.226562	-0.031250	0.015625	-0.007812
5	-0.132812	-0.023438	-0.007812	-0.007812
6	0.000000	-0.007812	0.054688	0.023438
7	0.023438	-0.015625	-0.023438	-0.031250
8	0.007812	0.046875	0.101562	-0.039062
9	-0.054688	-0.039062	-0.054688	-0.007812
10	0.015625	0.023438	-0.023438	-0.031250
11	-0.046875	-0.007812	0.015625	0.000000
12	0.039062	-0.085938	-0.054688	-0.015625
13	-0.023438	0.007812	0.039062	-0.031250
14	-0.046875	-0.023438	-0.062500	0.007812
15	0.000000	-0.007812	0.039062	0.023438
16	-0.062500	0.015625	-0.023438	0.054688
17	0.000000	0.023438	0.015625	0.007812
18	-0.023438	0.023438	0.062500	0.054688
19	0.023438	0.000000	0.054688	0.000000
20	-0.007812	0.023438	0.109375	-0.031250
21	0.000000	0.007812	-0.109375	0.000000
22	0.031250	0.015625	0.031250	-0.007812
23	0.000000	0.015625	0.015625	0.000000
24	0.015625	0.000000	0.007812	-0.007812
25	-0.054688	0.023438	0.023438	0.015625
26	0.000000	0.000000	-0.062500	0.031250
27	0.023438	0.023438	0.015625	0.000000
28	0.000000	-0.046875	0.000000	0.023438
29	0.007812	0.039062	0.031250	-0.031250
30	-0.039062	-0.031250	0.007812	0.031250
31	0.039062	-0.007812	-0.015625	-0.078125
32	0.031250	-0.031250	0.039062	-0.046875
33	-0.007812	-0.031250	-0.046875	0.070312
34	-0.023438	0.023438	0.000000	-0.015625
35	-0.039062	-0.023438	0.000000	0.031250
36	0.078125	0.046875	0.007812	0.070312
37	0.000000	-0.031250	0.007812	0.078125
38	0.039062	-0.015625	-0.015625	0.039062
39	-0.007812	0.039062	0.015625	0.015625
40	-0.054688	0.015625	-0.007812	0.007812
41	0.085938	0.031250	0.000000	-0.054688
42	-0.046875	0.007812	0.015625	-0.015625

Table 5.2.1.8-2 C_{SCBm} Parameter(20/40)

43	0.000000	-0.039062	0.015625	-0.070312
44	-0.007812	0.000000	-0.062500	-0.023438
45	-0.078125	-0.070312	0.039062	-0.046875
46	0.039062	-0.023438	-0.007812	-0.078125
47	-0.015625	0.039062	-0.054688	-0.007812
48	-0.007812	-0.046875	0.046875	-0.015625
49	-0.039062	-0.062500	0.000000	0.031250
50	-0.093750	-0.015625	0.054688	0.023438
51	-0.023438	0.000000	-0.015625	-0.023438
52	-0.054688	-0.140625	0.054688	0.031250
53	-0.031250	0.070312	-0.062500	0.046875
54	0.023438	0.031250	0.078125	0.062500
55	0.023438	-0.031250	0.078125	0.085938
56	0.023438	0.109375	0.078125	0.070312
57	-0.023438	0.101562	0.062500	0.039062
58	0.062500	0.015625	0.101562	0.078125
59	0.031250	0.132812	0.054688	0.000000
60	0.062500	0.085938	0.046875	0.054688
61	0.015625	0.015625	0.039062	-0.015625
62	0.070312	0.085938	0.015625	-0.085938
63	0.078125	-0.031250	-0.078125	-0.078125
64	0.039062	0.023438	-0.015625	-0.148438
65	-0.031250	-0.062500	-0.164062	-0.046875
66	-0.039062	-0.070312	-0.156250	-0.070312
67	0.007812	-0.023438	-0.101562	0.062500
68	-0.031250	-0.093750	-0.187500	-0.093750
69	-0.078125	-0.101562	0.000000	0.015625
70	-0.085938	-0.046875	-0.101562	-0.046875
71	-0.125000	-0.046875	0.023438	0.039062
72	-0.015625	-0.117188	0.007812	0.093750
73	0.007812	-0.007812	-0.070312	-0.031250
74	-0.039062	-0.085938	0.093750	0.296875
75	0.000000	-0.054688	0.054688	-0.015625
76	-0.007812	0.085938	0.164062	0.351562
77	0.023438	0.007812	0.156250	0.335938
78	0.039062	0.070312	0.265625	-0.617188
79	0.007812	0.117188	0.351562	0.164062

Table 5.2.1.8-2 C_{SCBm} Parameter(21/40)

k	$C_{SCB1}(8, 1, k)$	$C_{SCB1}(9, 1, k)$	$C_{SCB1}(10, 1, k)$	$C_{SCB1}(11, 1, k)$
0	0.015625	-0.007812	-0.007812	-0.007812
1	0.039062	0.031250	0.000000	-0.046875
2	-0.015625	0.000000	0.031250	0.023438
3	0.023438	0.007812	-0.046875	-0.062500
4	-0.007812	0.031250	0.023438	-0.054688
5	-0.007812	-0.062500	-0.132812	-0.078125
6	0.023438	0.023438	0.023438	-0.195312
7	-0.046875	-0.093750	-0.125000	0.164062
8	0.031250	-0.046875	-0.171875	0.140625
9	-0.070312	-0.062500	0.179688	0.015625
10	-0.046875	-0.164062	0.093750	0.007812
11	-0.078125	0.164062	0.070312	-0.046875
12	-0.179688	0.125000	-0.031250	0.046875
13	0.132812	0.023438	-0.007812	-0.031250
14	0.117188	0.031250	0.054688	0.039062
15	0.054688	-0.046875	-0.031250	-0.015625
16	0.023438	0.007812	0.046875	-0.031250
17	-0.031250	-0.031250	-0.046875	0.070312
18	0.015625	0.039062	0.015625	0.039062
19	-0.031250	-0.015625	0.015625	0.054688
20	0.023438	-0.015625	0.039062	-0.046875
21	-0.023438	0.062500	0.046875	0.031250
22	-0.007812	0.023438	-0.007812	0.007812
23	0.046875	0.039062	0.015625	-0.078125
24	0.015625	-0.023438	0.000000	0.015625
25	0.078125	0.007812	-0.054688	-0.078125
26	-0.015625	0.000000	-0.039062	0.007812
27	0.031250	-0.023438	-0.039062	-0.046875
28	-0.015625	-0.039062	0.015625	-0.007812
29	-0.046875	-0.054688	-0.023438	0.023438
30	0.015625	-0.007812	0.000000	0.015625
31	-0.085938	-0.054688	0.007812	0.054688
32	0.062500	0.039062	-0.007812	-0.054688
33	-0.031250	-0.031250	0.023438	0.070312
34	0.007812	0.000000	-0.015625	-0.023438
35	-0.023438	0.054688	-0.007812	0.000000
36	-0.015625	-0.023438	-0.031250	0.000000
37	0.054688	0.046875	0.039062	-0.031250
38	0.000000	-0.031250	-0.007812	0.062500
39	0.062500	0.085938	0.031250	-0.015625
40	-0.031250	0.070312	0.062500	0.062500
41	0.007812	0.007812	-0.046875	0.023438
42	0.046875	-0.023438	0.054688	0.000000

Table 5.2.1.8-2 C_{SCBm} Parameter(22/40)

43	-0.015625	-0.031250	0.015625	0.054688
44	0.039062	0.101562	0.007812	-0.039062
45	0.007812	0.023438	0.039062	0.085938
46	0.062500	0.117188	0.000000	-0.054688
47	-0.015625	-0.007812	0.093750	0.039062
48	-0.031250	-0.039062	-0.031250	-0.078125
49	0.054688	0.000000	-0.015625	0.054688
50	-0.054688	-0.046875	-0.015625	0.000000
51	-0.007812	0.031250	0.039062	-0.023438
52	0.039062	-0.031250	0.015625	0.000000
53	-0.023438	0.023438	-0.070312	-0.046875
54	-0.023438	0.031250	0.031250	0.031250
55	-0.007812	-0.117188	-0.062500	-0.015625
56	0.000000	0.070312	-0.046875	-0.101562
57	-0.023438	-0.093750	-0.007812	-0.023438
58	-0.007812	-0.062500	-0.031250	-0.125000
59	-0.023438	-0.046875	-0.046875	0.000000
60	-0.085938	-0.062500	-0.062500	0.031250
61	0.015625	0.117188	-0.015625	-0.007812
62	-0.046875	-0.039062	0.039062	0.031250
63	-0.015625	-0.023438	0.007812	0.000000
64	-0.062500	-0.007812	0.148438	0.218750
65	-0.015625	0.023438	0.023438	0.031250
66	0.054688	0.054688	0.101562	0.218750
67	0.007812	0.000000	0.343750	0.359375
68	0.101562	0.187500	0.359375	-0.250000
69	0.101562	0.101562	0.351562	-0.210938
70	0.171875	0.117188	-0.492188	-0.015625
71	0.250000	0.687500	-0.531250	-0.062500
72	0.242188	-0.195312	0.109375	-0.054688
73	0.523438	-0.453125	-0.148438	-0.187500
74	-0.304688	0.078125	-0.125000	-0.007812
75	-0.726562	-0.250000	-0.007812	-0.054688
76	0.000000	-0.101562	0.015625	0.015625
77	-0.039062	0.007812	-0.054688	0.078125
78	-0.148438	-0.062500	-0.054688	-0.046875
79	-0.132812	-0.015625	-0.078125	-0.031250

Table 5.2.1.8-2 C_{SCBm} Parameter(23/40)

k	$C_{SCB1}(12, 1, k)$	$C_{SCB1}(13, 1, k)$	$C_{SCB1}(14, 1, k)$	$C_{SCB1}(15, 1, k)$
0	0.015625	-0.023438	-0.078125	0.000000
1	-0.062500	-0.054688	0.171875	-0.070312
2	-0.039062	-0.179688	0.132812	0.007812
3	-0.085938	0.171875	0.039062	-0.031250
4	-0.179688	0.125000	0.000000	0.000000
5	0.171875	0.000000	-0.054688	0.000000
6	0.125000	0.031250	0.046875	-0.007812
7	0.000000	-0.031250	-0.046875	0.000000
8	0.000000	0.062500	0.007812	-0.007812
9	-0.023438	-0.031250	-0.039062	0.070312
10	0.031250	0.000000	0.007812	0.000000
11	-0.039062	-0.007812	0.031250	0.007812
12	0.000000	0.000000	0.031250	0.000000
13	-0.039062	0.054688	0.062500	0.070312
14	0.023438	-0.031250	-0.039062	0.031250
15	0.062500	0.070312	0.015625	-0.054688
16	0.000000	0.000000	-0.031250	0.062500
17	0.078125	0.015625	-0.015625	-0.046875
18	0.007812	0.015625	0.000000	-0.046875
19	0.007812	-0.031250	-0.046875	-0.039062
20	-0.015625	0.007812	0.023438	0.000000
21	0.015625	-0.046875	-0.085938	0.023438
22	-0.015625	0.000000	0.000000	-0.023438
23	-0.054688	-0.046875	-0.007812	0.046875
24	0.000000	-0.007812	-0.031250	-0.070312
25	-0.054688	0.007812	0.015625	0.062500
26	0.046875	0.007812	-0.015625	0.007812
27	-0.023438	0.054688	0.023438	-0.039062
28	0.007812	-0.054688	-0.039062	0.070312
29	0.070312	0.062500	0.007812	-0.007812
30	-0.054688	0.000000	-0.015625	0.046875
31	0.046875	-0.054688	0.000000	-0.031250
32	-0.015625	0.015625	0.015625	0.039062
33	0.023438	-0.007812	0.015625	0.054688
34	0.007812	0.054688	0.085938	-0.054688
35	0.046875	-0.046875	-0.023438	0.148438
36	0.046875	0.093750	0.015625	-0.007812
37	-0.078125	0.000000	0.000000	-0.023438
38	0.070312	-0.023438	0.015625	0.015625
39	0.031250	0.101562	0.101562	0.007812
40	-0.046875	-0.039062	-0.078125	0.031250
41	0.085938	0.015625	-0.007812	-0.046875
42	-0.031250	0.007812	0.023438	0.062500

Table 5.2.1.8-2 C_{SCBm} Parameter(24/40)

43	0.023438	-0.015625	-0.023438	-0.093750
44	0.031250	0.023438	0.023438	0.085938
45	-0.039062	0.000000	-0.039062	-0.085938
46	0.070312	0.085938	-0.015625	-0.117188
47	-0.031250	-0.046875	-0.109375	0.078125
48	0.015625	-0.031250	-0.031250	-0.125000
49	-0.054688	-0.078125	-0.015625	-0.039062
50	-0.007812	0.007812	-0.109375	-0.148438
51	0.023438	-0.023438	-0.054688	-0.101562
52	-0.093750	-0.085938	-0.164062	-0.085938
53	0.046875	0.015625	-0.046875	0.007812
54	-0.070312	-0.101562	0.023438	0.242188
55	-0.062500	-0.015625	0.046875	-0.023438
56	-0.015625	0.046875	0.109375	0.164062
57	-0.023438	-0.031250	0.000000	0.210938
58	-0.070312	0.109375	0.187500	0.179688
59	0.046875	0.078125	0.164062	0.367188
60	0.117188	0.046875	0.171875	-0.273438
61	-0.023438	0.242188	0.421875	-0.414062
62	0.234375	0.304688	-0.265625	0.039062
63	0.179688	0.328125	-0.421875	0.039062
64	0.242188	-0.335938	0.062500	-0.015625
65	0.429688	-0.562500	0.007812	0.000000
66	-0.359375	0.093750	0.039062	0.000000
67	-0.398438	-0.015625	-0.062500	-0.054688
68	0.062500	-0.023438	-0.085938	0.000000
69	0.015625	-0.078125	0.054688	-0.007812
70	-0.031250	0.039062	0.023438	0.039062
71	-0.039062	-0.023438	-0.015625	-0.007812
72	0.007812	0.000000	-0.039062	-0.046875
73	-0.085938	0.031250	0.015625	-0.054688
74	0.062500	-0.070312	-0.007812	-0.039062
75	0.078125	0.000000	0.031250	-0.039062
76	-0.148438	-0.093750	-0.046875	-0.031250
77	-0.078125	0.031250	-0.093750	-0.085938
78	-0.039062	0.085938	0.046875	-0.054688
79	-0.101562	-0.031250	-0.109375	0.078125

Table 5.2.1.8-2 C_{SCBm} Parameter(25/40)

k	$C_{SCB0}(0, 2, k)$	$C_{SCB0}(1, 2, k)$	$C_{SCB0}(2, 2, k)$	$C_{SCB0}(3, 2, k)$
0	-0.023438	0.031250	0.000000	-0.039062
1	-0.023438	-0.039062	-0.062500	-0.031250
2	-0.015625	-0.015625	0.031250	-0.015625
3	-0.031250	0.023438	-0.031250	0.000000
4	-0.007812	-0.015625	-0.054688	-0.085938
5	-0.054688	-0.023438	0.000000	0.039062
6	0.000000	-0.046875	-0.046875	-0.031250
7	-0.046875	0.007812	0.031250	-0.023438
8	-0.078125	-0.039062	-0.023438	0.000000
9	0.023438	0.015625	0.000000	-0.031250
10	-0.039062	-0.015625	0.023438	0.039062
11	-0.031250	-0.015625	-0.031250	0.000000
12	-0.015625	0.031250	0.039062	0.070312
13	-0.015625	0.000000	-0.007812	0.015625
14	0.000000	0.054688	0.062500	0.015625
15	0.000000	-0.023438	0.046875	0.070312
16	0.039062	0.039062	0.023438	0.031250
17	0.000000	0.031250	0.070312	0.101562
18	0.039062	0.000000	0.015625	0.000000
19	-0.023438	0.078125	0.093750	0.046875
20	0.007812	0.023438	-0.023438	0.023438
21	0.078125	0.062500	0.031250	0.000000
22	0.062500	0.007812	0.039062	0.046875
23	0.140625	0.031250	-0.015625	-0.085938
24	-0.046875	0.015625	0.031250	0.031250
25	0.085938	0.015625	-0.109375	-0.109375
26	0.070312	0.015625	0.015625	-0.132812
27	-0.015625	-0.109375	-0.132812	-0.171875
28	-0.007812	0.054688	-0.109375	-0.351562
29	-0.093750	-0.140625	-0.148438	0.187500
30	0.023438	-0.093750	-0.296875	0.273438
31	-0.117188	-0.218750	0.210938	-0.023438
32	-0.117188	-0.367188	0.250000	0.015625
33	-0.210938	0.351562	0.007812	0.007812
34	-0.296875	0.250000	0.015625	0.085938
35	0.281250	-0.062500	0.039062	0.078125
36	0.218750	0.070312	0.062500	0.046875
37	0.046875	0.007812	0.015625	0.000000
38	0.054688	0.101562	0.093750	0.007812
39	0.070312	-0.015625	0.007812	-0.007812

Table 5.2.1.8-2 C_{SCBm} Parameter(26/40)

k	$C_{SCB0}(4, 2, k)$	$C_{SCB0}(5, 2, k)$	$C_{SCB0}(6, 2, k)$	$C_{SCB0}(7, 2, k)$
0	-0.039062	-0.070312	-0.031250	0.000000
1	-0.039062	0.015625	-0.007812	-0.015625
2	-0.039062	-0.054688	0.000000	0.031250
3	-0.039062	0.000000	-0.015625	-0.023438
4	0.007812	-0.046875	0.054688	0.023438
5	-0.039062	0.031250	-0.015625	0.078125
6	-0.031250	0.046875	0.054688	0.007812
7	0.015625	-0.031250	0.023438	0.070312
8	0.023438	0.054688	0.015625	0.015625
9	-0.039062	-0.015625	0.062500	0.101562
10	0.007812	0.023438	0.031250	0.015625
11	0.054688	0.039062	0.078125	0.007812
12	0.007812	0.007812	-0.007812	0.054688
13	0.062500	0.101562	0.078125	-0.031250
14	0.031250	0.023438	0.023438	0.015625
15	0.046875	0.046875	-0.023438	-0.046875
16	0.023438	0.031250	0.062500	-0.007812
17	0.015625	-0.015625	-0.062500	-0.101562
18	0.054688	0.031250	0.007812	-0.117188
19	0.015625	-0.054688	-0.109375	-0.117188
20	0.031250	0.000000	-0.132812	-0.289062
21	-0.031250	-0.070312	-0.156250	0.078125
22	-0.023438	-0.109375	-0.328125	0.296875
23	-0.039062	-0.187500	0.187500	0.046875
24	-0.125000	-0.312500	0.257812	-0.031250
25	-0.132812	0.132812	-0.015625	0.015625
26	-0.351562	0.218750	0.023438	0.054688
27	0.156250	0.023438	0.023438	0.000000
28	0.312500	0.085938	0.039062	0.070312
29	-0.062500	0.007812	0.031250	-0.007812
30	0.062500	0.023438	0.031250	0.000000
31	0.070312	0.000000	-0.015625	0.031250
32	0.039062	0.062500	0.023438	-0.046875
33	0.015625	0.039062	0.031250	0.046875
34	0.007812	-0.007812	-0.039062	-0.023438
35	-0.046875	0.007812	0.039062	-0.023438
36	0.007812	0.000000	-0.007812	0.039062
37	0.085938	0.023438	0.015625	-0.062500
38	-0.093750	-0.015625	0.031250	0.000000
39	0.046875	0.062500	-0.007812	-0.054688

Table 5.2.1.8-2 C_{SCBm} Parameter(27/40)

k	$C_{SCB0}(8, 2, k)$	$C_{SCB0}(9, 2, k)$	$C_{SCB0}(10, 2, k)$	$C_{SCB0}(11, 2, k)$
0	-0.046875	-0.046875	-0.015625	-0.031250
1	-0.023438	-0.039062	-0.070312	-0.007812
2	-0.031250	-0.039062	-0.015625	-0.031250
3	-0.062500	-0.015625	-0.023438	0.109375
4	-0.023438	-0.015625	-0.007812	0.039062
5	0.007812	-0.015625	0.085938	0.062500
6	0.000000	-0.023438	0.007812	0.070312
7	-0.007812	0.031250	0.054688	0.062500
8	-0.031250	0.046875	0.078125	0.148438
9	0.078125	0.062500	0.101562	-0.031250
10	0.023438	0.078125	0.101562	-0.015625
11	0.046875	0.054688	-0.070312	-0.015625
12	0.078125	0.148438	-0.023438	-0.007812
13	0.070312	-0.015625	0.007812	0.015625
14	0.156250	-0.070312	-0.031250	-0.117188
15	-0.031250	0.000000	-0.007812	-0.101562
16	-0.007812	-0.015625	-0.101562	-0.109375
17	0.007812	0.007812	-0.062500	0.000000
18	-0.023438	-0.117188	-0.093750	0.062500
19	-0.039062	-0.054688	-0.031250	0.015625
20	-0.125000	-0.046875	0.062500	0.046875
21	0.000000	-0.046875	0.031250	0.000000
22	-0.070312	0.046875	0.015625	0.023438
23	-0.070312	0.023438	-0.007812	-0.015625
24	0.046875	0.062500	0.031250	-0.007812
25	0.023438	0.023438	-0.007812	0.000000
26	0.015625	0.000000	-0.015625	-0.007812
27	0.046875	-0.007812	0.023438	0.039062
28	0.031250	0.023438	-0.039062	-0.015625
29	-0.039062	-0.015625	0.031250	0.031250
30	-0.015625	-0.046875	-0.007812	0.000000
31	0.023438	-0.015625	0.031250	0.000000
32	-0.023438	-0.015625	0.000000	0.023438
33	0.031250	0.015625	0.015625	0.023438
34	-0.046875	-0.031250	0.070312	0.031250
35	0.062500	0.046875	-0.007812	-0.015625
36	0.000000	0.031250	0.023438	0.007812
37	-0.031250	0.000000	-0.031250	-0.031250
38	0.093750	0.085938	0.070312	-0.015625
39	0.000000	0.000000	-0.007812	0.039062

Table 5.2.1.8-2 C_{SCBm} Parameter(28/40)

k	$C_{SCB0}(12, 2, k)$	$C_{SCB0}(13, 2, k)$	$C_{SCB0}(14, 2, k)$	$C_{SCB0}(15, 2, k)$
0	0.007812	0.085938	0.125000	0.117188
1	0.070312	0.054688	0.101562	-0.031250
2	0.039062	0.046875	0.109375	0.000000
3	0.046875	0.140625	0.000000	0.023438
4	0.070312	0.125000	-0.007812	-0.015625
5	0.093750	-0.023438	-0.007812	0.000000
6	0.132812	0.007812	0.000000	-0.093750
7	-0.039062	-0.007812	-0.007812	-0.031250
8	-0.070312	-0.039062	-0.117188	-0.117188
9	0.000000	0.031250	-0.031250	-0.046875
10	-0.015625	-0.132812	-0.054688	0.062500
11	0.007812	-0.070312	-0.054688	-0.007812
12	-0.101562	-0.046875	0.023438	0.078125
13	-0.054688	-0.085938	0.000000	0.015625
14	-0.062500	0.039062	0.093750	-0.007812
15	-0.093750	0.054688	0.015625	-0.031250
16	0.085938	0.070312	-0.031250	-0.007812
17	0.039062	-0.007812	-0.046875	0.023438
18	0.031250	0.000000	-0.015625	-0.046875
19	0.007812	-0.015625	0.015625	0.023438
20	-0.062500	-0.023438	-0.031250	0.007812
21	-0.007812	0.000000	0.000000	0.000000
22	-0.015625	-0.039062	-0.023438	0.031250
23	0.000000	0.007812	0.023438	0.007812
24	-0.015625	0.015625	0.015625	0.015625
25	0.000000	0.015625	-0.023438	0.007812
26	0.023438	0.007812	0.015625	0.031250
27	0.000000	0.007812	0.000000	0.007812
28	0.031250	0.046875	0.062500	0.031250
29	0.007812	-0.007812	-0.023438	0.015625
30	0.000000	0.015625	0.023438	0.023438
31	0.015625	-0.046875	0.000000	0.046875
32	0.023438	0.023438	0.007812	-0.039062
33	0.000000	0.023438	0.054688	0.085938
34	0.023438	0.000000	-0.054688	0.000000
35	0.015625	0.015625	0.062500	-0.007812
36	0.046875	-0.031250	-0.046875	0.007812
37	0.046875	0.062500	0.015625	-0.085938
38	-0.078125	0.015625	-0.031250	-0.054688
39	0.078125	0.000000	-0.101562	-0.101562

Table 5.2.1.8-2 C_{SCBm} Parameter(29/40)

k	$C_{SCB1}(0, 2, k)$	$C_{SCB1}(1, 2, k)$	$C_{SCB1}(2, 2, k)$	$C_{SCB1}(3, 2, k)$
0	-0.062500	-0.078125	-0.039062	0.062500
1	0.007812	0.000000	0.031250	0.007812
2	-0.023438	-0.039062	0.007812	0.117188
3	0.007812	0.015625	0.000000	0.078125
4	-0.039062	0.031250	0.132812	0.250000
5	0.015625	0.007812	0.007812	0.156250
6	0.023438	0.101562	0.281250	-0.234375
7	0.000000	0.039062	0.132812	-0.062500
8	0.109375	0.296875	-0.218750	-0.023438
9	0.062500	0.125000	-0.093750	-0.031250
10	0.234375	-0.265625	-0.062500	-0.015625
11	0.148438	-0.070312	0.046875	0.000000
12	-0.195312	-0.039062	-0.031250	0.015625
13	-0.117188	0.000000	-0.039062	-0.023438
14	-0.031250	0.000000	0.062500	0.039062
15	0.023438	-0.039062	-0.023438	-0.031250
16	-0.039062	0.031250	0.015625	-0.039062
17	-0.046875	0.000000	-0.062500	-0.015625
18	0.046875	0.015625	-0.007812	-0.039062
19	-0.015625	-0.039062	-0.023438	0.031250
20	0.007812	-0.015625	-0.023438	-0.015625
21	-0.039062	-0.015625	0.062500	0.023438
22	-0.046875	-0.023438	-0.031250	-0.007812
23	0.000000	0.023438	0.054688	-0.015625
24	-0.023438	-0.015625	-0.007812	0.015625
25	0.007812	0.046875	0.015625	0.031250
26	0.000000	0.000000	0.023438	0.007812
27	0.000000	-0.007812	0.007812	-0.023438
28	-0.007812	0.046875	0.023438	0.054688
29	-0.007812	0.039062	-0.023438	-0.039062
30	0.070312	-0.023438	0.031250	-0.015625
31	0.000000	-0.015625	0.000000	0.054688
32	0.046875	0.046875	0.007812	-0.078125
33	-0.015625	-0.023438	0.039062	0.023438
34	0.031250	0.007812	-0.023438	-0.054688
35	0.000000	0.007812	0.046875	-0.070312
36	0.000000	-0.031250	-0.015625	-0.031250
37	0.031250	0.015625	-0.023438	0.023438
38	-0.054688	-0.015625	-0.023438	0.000000
39	0.078125	-0.031250	0.007812	-0.046875

Table 5.2.1.8-2 C_{SCBm} Parameter(30/40)

k	$C_{SCB1}(4, 2, k)$	$C_{SCB1}(5, 2, k)$	$C_{SCB1}(6, 2, k)$	$C_{SCB1}(7, 2, k)$
0	0.234375	0.296875	-0.406250	0.070312
1	0.078125	0.179688	-0.125000	-0.007812
2	0.273438	-0.250000	-0.054688	0.015625
3	0.156250	-0.070312	0.015625	-0.023438
4	-0.273438	-0.031250	0.000000	-0.007812
5	-0.054688	-0.023438	0.007812	0.000000
6	0.007812	-0.015625	0.031250	0.015625
7	0.015625	0.000000	0.000000	-0.046875
8	-0.015625	0.031250	0.085938	-0.023438
9	-0.031250	-0.039062	-0.078125	-0.015625
10	0.007812	0.039062	0.007812	-0.031250
11	-0.039062	-0.039062	0.000000	0.000000
12	0.046875	-0.070312	-0.031250	-0.031250
13	-0.046875	0.015625	0.015625	-0.023438
14	-0.023438	-0.031250	-0.054688	0.007812
15	-0.007812	0.000000	0.039062	0.031250
16	-0.054688	0.015625	-0.039062	0.039062
17	0.000000	0.031250	0.039062	0.015625
18	-0.015625	0.015625	0.054688	0.054688
19	0.023438	0.000000	0.085938	-0.015625
20	-0.015625	0.023438	0.054688	-0.015625
21	0.015625	0.007812	-0.109375	0.007812
22	0.023438	0.015625	0.062500	-0.007812
23	0.007812	0.007812	0.000000	-0.007812
24	0.007812	0.007812	0.023438	-0.007812
25	-0.054688	0.023438	0.000000	0.031250
26	0.023438	0.007812	-0.046875	0.023438
27	0.015625	0.007812	0.023438	0.007812
28	0.000000	-0.031250	-0.007812	0.007812
29	0.000000	0.039062	0.031250	-0.015625
30	-0.031250	-0.031250	0.000000	0.023438
31	0.054688	0.000000	0.007812	-0.101562
32	0.015625	-0.046875	0.023438	0.000000
33	-0.007812	-0.015625	-0.046875	0.054688
34	-0.039062	0.015625	0.015625	-0.031250
35	-0.007812	-0.007812	-0.007812	0.046875
36	0.070312	0.039062	0.015625	0.070312
37	-0.007812	-0.046875	0.000000	0.070312
38	0.046875	0.000000	-0.007812	0.031250
39	-0.046875	0.039062	0.015625	0.023438

Table 5.2.1.8-2 C_{SCBm} Parameter(31/40)

k	$C_{SCB1}(8, 2, k)$	$C_{SCB1}(9, 2, k)$	$C_{SCB1}(10, 2, k)$	$C_{SCB1}(11, 2, k)$
0	0.031250	0.000000	-0.015625	-0.023438
1	0.023438	0.031250	0.015625	-0.023438
2	-0.007812	0.000000	0.007812	0.015625
3	0.023438	0.023438	-0.023438	-0.070312
4	-0.015625	0.007812	-0.007812	-0.039062
5	0.007812	-0.039062	-0.101562	-0.148438
6	0.007812	0.015625	0.031250	-0.140625
7	-0.031250	-0.101562	-0.203125	0.210938
8	0.023438	-0.023438	-0.085938	0.078125
9	-0.078125	-0.132812	0.203125	0.015625
10	-0.023438	-0.109375	0.062500	0.000000
11	-0.140625	0.203125	0.054688	-0.023438
12	-0.125000	0.070312	-0.039062	0.031250
13	0.171875	0.031250	0.023438	-0.023438
14	0.070312	0.023438	0.031250	0.046875
15	0.054688	-0.039062	-0.015625	-0.039062
16	0.015625	0.007812	0.031250	-0.007812
17	-0.023438	-0.015625	-0.054688	0.070312
18	0.007812	0.039062	0.031250	0.039062
19	-0.023438	-0.039062	0.007812	0.031250
20	0.023438	0.007812	0.046875	-0.039062
21	-0.039062	0.054688	0.031250	0.054688
22	0.007812	0.023438	0.000000	-0.023438
23	0.039062	0.031250	0.031250	-0.054688
24	0.031250	-0.023438	-0.015625	0.000000
25	0.054688	0.015625	-0.054688	-0.070312
26	-0.015625	-0.007812	-0.039062	0.007812
27	0.039062	-0.023438	-0.031250	-0.062500
28	-0.039062	-0.046875	0.007812	0.007812
29	-0.023438	-0.039062	-0.031250	0.015625
30	-0.007812	-0.023438	0.007812	0.031250
31	-0.062500	-0.039062	0.000000	0.023438
32	0.062500	0.031250	0.000000	-0.031250
33	-0.046875	-0.046875	0.023438	0.070312
34	0.015625	0.031250	-0.015625	-0.039062
35	-0.039062	0.023438	-0.015625	0.015625
36	0.007812	-0.015625	-0.015625	-0.015625
37	0.046875	0.031250	0.031250	-0.007812
38	0.015625	-0.023438	-0.015625	0.046875
39	0.054688	0.109375	0.062500	-0.015625

Table 5.2.1.8-2 C_{SCBm} Parameter(32/40)

k	$C_{SCB1}(12, 2, k)$	$C_{SCB1}(13, 2, k)$	$C_{SCB1}(14, 2, k)$	$C_{SCB1}(15, 2, k)$
0	0.000000	-0.007812	-0.031250	-0.023438
1	-0.070312	-0.109375	0.203125	-0.054688
2	-0.023438	-0.117188	0.093750	0.007812
3	-0.156250	0.218750	0.039062	-0.039062
4	-0.117188	0.054688	-0.015625	0.007812
5	0.218750	0.000000	-0.023438	-0.007812
6	0.062500	0.023438	0.039062	-0.007812
7	0.000000	-0.007812	-0.046875	-0.015625
8	0.000000	0.054688	0.007812	0.007812
9	-0.007812	-0.039062	-0.046875	0.062500
10	0.023438	0.007812	0.023438	-0.015625
11	-0.039062	-0.015625	0.023438	0.007812
12	0.000000	0.023438	0.046875	0.015625
13	-0.039062	0.031250	0.039062	0.085938
14	0.046875	-0.015625	-0.031250	-0.007812
15	0.039062	0.070312	0.023438	-0.023438
16	0.007812	-0.015625	-0.039062	0.062500
17	0.078125	0.031250	0.000000	-0.070312
18	0.000000	0.000000	-0.015625	-0.039062
19	0.007812	-0.023438	-0.023438	-0.031250
20	-0.007812	0.000000	0.007812	0.015625
21	0.015625	-0.039062	-0.085938	0.000000
22	-0.031250	0.000000	0.015625	0.000000
23	-0.039062	-0.054688	-0.031250	0.015625
24	-0.015625	0.000000	-0.015625	-0.054688
25	-0.039062	0.000000	0.007812	0.078125
26	0.039062	0.023438	-0.007812	-0.031250
27	-0.046875	0.023438	0.015625	-0.007812
28	0.039062	-0.039062	-0.039062	0.062500
29	0.039062	0.078125	0.015625	-0.007812
30	-0.046875	-0.031250	-0.023438	0.039062
31	0.054688	-0.039062	0.000000	-0.031250
32	-0.023438	0.015625	0.007812	0.070312
33	0.031250	0.000000	0.039062	0.000000
34	0.007812	0.023438	0.062500	-0.007812
35	0.070312	-0.023438	-0.031250	0.156250
36	0.000000	0.101562	0.023438	-0.054688
37	-0.054688	-0.039062	-0.007812	0.000000
38	0.093750	0.015625	0.054688	0.015625
39	-0.015625	0.078125	0.062500	0.015625

Table 5.2.1.8-2 C_{SCBm} Parameter(33/40)

k	$C_{SCB0}(0, 3, k)$	$C_{SCB0}(1, 3, k)$	$C_{SCB0}(2, 3, k)$	$C_{SCB0}(3, 3, k)$
0	-0.031250	0.023438	-0.023438	-0.023438
1	-0.007812	-0.046875	-0.039062	-0.046875
2	-0.023438	0.007812	0.039062	0.007812
3	-0.015625	0.007812	-0.054688	-0.031250
4	-0.023438	-0.007812	-0.031250	-0.054688
5	-0.039062	-0.046875	-0.023438	0.039062
6	0.007812	-0.023438	-0.031250	-0.046875
7	-0.078125	0.000000	0.023438	0.000000
8	-0.046875	-0.031250	-0.031250	-0.015625
9	0.015625	0.023438	0.023438	-0.007812
10	-0.046875	-0.031250	0.000000	0.023438
11	-0.023438	0.000000	-0.007812	0.015625
12	-0.023438	0.015625	0.023438	0.070312
13	-0.007812	0.023438	0.000000	-0.007812
14	0.000000	0.031250	0.078125	0.039062
15	0.015625	-0.023438	0.023438	0.046875
16	0.023438	0.062500	0.046875	0.062500
17	0.007812	0.000000	0.039062	0.078125
18	0.031250	0.023438	0.046875	-0.007812
19	-0.039062	0.062500	0.070312	0.062500
20	0.046875	0.039062	-0.031250	0.000000
21	0.054688	0.054688	0.062500	0.031250
22	0.101562	0.007812	0.000000	0.000000
23	0.085938	0.039062	0.023438	-0.054688
24	-0.039062	0.000000	-0.015625	0.023438
25	0.117188	0.054688	-0.078125	-0.148438
26	0.031250	-0.046875	0.000000	-0.085938
27	0.000000	-0.054688	-0.164062	-0.265625
28	-0.046875	0.031250	-0.062500	-0.273438
29	-0.054688	-0.164062	-0.242188	0.296875
30	0.007812	-0.046875	-0.210938	0.187500
31	-0.140625	-0.343750	0.296875	-0.039062
32	-0.078125	-0.226562	0.179688	0.015625
33	-0.312500	0.445312	-0.007812	0.031250
34	-0.171875	0.117188	0.015625	0.085938
35	0.343750	-0.046875	0.070312	0.070312
36	0.148438	0.062500	0.039062	0.031250
37	0.031250	0.046875	0.031250	-0.007812
38	0.054688	0.078125	0.085938	0.023438

Table 5.2.1.8-2 C_{SCBm} Parameter(34/40)

k	$C_{SCB0}(4, 3, k)$	$C_{SCB0}(5, 3, k)$	$C_{SCB0}(6, 3, k)$	$C_{SCB0}(7, 3, k)$
0	-0.046875	-0.031250	-0.039062	-0.015625
1	-0.031250	0.000000	0.007812	0.000000
2	-0.046875	-0.039062	-0.023438	0.023438
3	-0.023438	-0.007812	0.007812	-0.039062
4	0.000000	-0.039062	0.031250	0.054688
5	-0.039062	0.062500	-0.007812	0.046875
6	-0.023438	0.007812	0.054688	0.031250
7	0.031250	-0.007812	0.007812	0.046875
8	0.007812	0.046875	0.039062	0.039062
9	-0.039062	-0.023438	0.039062	0.093750
10	0.039062	0.046875	0.054688	-0.007812
11	0.031250	0.015625	0.054688	0.039062
12	0.023438	0.031250	0.000000	0.031250
13	0.054688	0.093750	0.093750	-0.015625
14	0.031250	0.015625	-0.015625	0.000000
15	0.054688	0.062500	0.015625	-0.039062
16	0.007812	0.000000	0.031250	-0.007812
17	0.031250	0.000000	-0.046875	-0.132812
18	0.046875	0.007812	0.007812	-0.078125
19	0.015625	-0.046875	-0.148438	-0.179688
20	0.023438	0.000000	-0.085938	-0.257812
21	-0.046875	-0.093750	-0.250000	0.187500
22	0.000000	-0.085938	-0.250000	0.250000
23	-0.078125	-0.265625	0.289062	0.000000
24	-0.078125	-0.242188	0.179688	-0.031250
25	-0.218750	0.226562	-0.031250	0.054688
26	-0.296875	0.148438	0.031250	0.031250
27	0.296875	0.023438	0.031250	0.015625
28	0.203125	0.078125	0.039062	0.062500
29	-0.070312	0.015625	0.039062	-0.031250
30	0.093750	0.007812	0.015625	0.023438
31	0.070312	0.007812	-0.023438	0.000000
32	0.023438	0.070312	0.046875	-0.031250
33	0.015625	0.015625	0.000000	0.054688
34	0.000000	0.000000	-0.023438	-0.046875
35	-0.070312	0.000000	0.031250	0.007812
36	0.070312	0.007812	-0.007812	0.007812
37	0.023438	0.000000	0.031250	-0.054688
38	-0.070312	0.007812	0.015625	-0.007812

Table 5.2.1.8-2 C_{SCBm} Parameter(35/40)

k	$C_{SCB0}(8, 3, k)$	$C_{SCB0}(9, 3, k)$	$C_{SCB0}(10, 3, k)$	$C_{SCB0}(11, 3, k)$
0	-0.046875	-0.039062	-0.031250	-0.007812
1	-0.015625	-0.039062	-0.062500	-0.031250
2	-0.039062	-0.031250	-0.007812	0.007812
3	-0.062500	-0.015625	-0.039062	0.101562
4	-0.007812	-0.007812	0.023438	0.023438
5	0.007812	-0.023438	0.070312	0.078125
6	0.007812	-0.007812	0.000000	0.046875
7	-0.039062	0.046875	0.062500	0.109375
8	0.000000	0.039062	0.070312	0.109375
9	0.078125	0.078125	0.125000	-0.046875
10	0.007812	0.054688	0.062500	-0.007812
11	0.070312	0.093750	-0.085938	-0.023438
12	0.054688	0.132812	0.007812	0.015625
13	0.117188	-0.054688	-0.007812	-0.015625
14	0.117188	-0.039062	-0.015625	-0.117188
15	-0.046875	-0.007812	-0.031250	-0.101562
16	0.015625	0.007812	-0.093750	-0.101562
17	0.000000	-0.023438	-0.062500	0.031250
18	-0.007812	-0.117188	-0.101562	0.039062
19	-0.078125	-0.031250	0.000000	0.023438
20	-0.093750	-0.062500	0.054688	0.031250
21	0.000000	-0.015625	0.023438	0.000000
22	-0.093750	0.046875	0.007812	0.023438
23	-0.039062	0.031250	0.000000	-0.023438
24	0.054688	0.054688	0.031250	0.000000
25	0.015625	0.015625	-0.031250	-0.007812
26	0.015625	-0.007812	0.007812	0.015625
27	0.054688	0.000000	0.000000	0.015625
28	0.023438	0.031250	-0.023438	-0.007812
29	-0.054688	-0.031250	0.023438	0.031250
30	0.015625	-0.039062	0.000000	-0.007812
31	0.000000	-0.023438	0.031250	0.007812
32	0.000000	0.000000	-0.015625	0.023438
33	0.000000	0.000000	0.046875	0.031250
34	-0.031250	-0.023438	0.046875	0.015625
35	0.070312	0.062500	0.007812	-0.007812
36	-0.039062	0.000000	0.000000	0.007812
37	0.007812	0.039062	-0.015625	-0.046875
38	0.085938	0.062500	0.085938	0.015625

Table 5.2.1.8-2 C_{SCBm} Parameter(36/40)

k	$C_{SCB0}(12, 3, k)$	$C_{SCB0}(13, 3, k)$	$C_{SCB0}(14, 3, k)$	$C_{SCB0}(15, 3, k)$
0	0.023438	0.085938	0.117188	0.093750
1	0.070312	0.046875	0.109375	-0.046875
2	0.031250	0.054688	0.085938	0.015625
3	0.054688	0.164062	-0.007812	0.000000
4	0.062500	0.085938	0.000000	0.000000
5	0.125000	-0.039062	-0.015625	-0.031250
6	0.101562	0.023438	0.015625	-0.078125
7	-0.078125	-0.039062	-0.039062	-0.039062
8	-0.039062	0.000000	-0.101562	-0.132812
9	-0.007812	0.000000	-0.015625	0.000000
10	0.000000	-0.148438	-0.070312	0.039062
11	-0.007812	-0.039062	-0.023438	0.007812
12	-0.109375	-0.070312	0.007812	0.070312
13	-0.031250	-0.054688	0.023438	0.007812
14	-0.101562	0.039062	0.093750	-0.007812
15	-0.054688	0.062500	0.000000	-0.031250
16	0.093750	0.054688	-0.031250	0.015625
17	0.015625	-0.015625	-0.046875	0.000000
18	0.039062	0.000000	0.007812	-0.039062
19	-0.015625	-0.023438	0.000000	0.031250
20	-0.054688	-0.007812	-0.023438	0.000000
21	-0.007812	-0.007812	0.000000	0.015625
22	-0.015625	-0.031250	-0.015625	0.023438
23	0.000000	0.007812	0.039062	0.007812
24	-0.015625	0.007812	-0.007812	0.007812
25	0.007812	0.015625	-0.007812	0.015625
26	0.015625	0.000000	0.000000	0.023438
27	0.007812	0.031250	0.023438	0.007812
28	0.031250	0.023438	0.039062	0.039062
29	0.007812	0.000000	-0.023438	0.000000
30	0.000000	0.000000	0.031250	0.054688
31	0.023438	-0.039062	-0.023438	0.007812
32	0.015625	0.039062	0.046875	-0.007812
33	0.000000	0.007812	0.007812	0.085938
34	0.031250	0.015625	-0.015625	-0.023438
35	0.007812	-0.007812	0.039062	0.015625
36	0.078125	-0.007812	-0.054688	-0.023438
37	0.000000	0.062500	0.039062	-0.070312
38	-0.054688	0.007812	-0.085938	-0.078125

Table 5.2.1.8-2 C_{SCBm} Parameter(37/40)

k	$C_{SCB1}(0, 3, k)$	$C_{SCB1}(1, 3, k)$	$C_{SCB1}(2, 3, k)$	$C_{SCB1}(3, 3, k)$
0	-0.039062	-0.062500	-0.046875	0.023438
1	0.000000	0.000000	0.062500	0.046875
2	-0.007812	-0.046875	-0.031250	0.109375
3	0.000000	0.039062	0.054688	0.093750
4	-0.039062	0.000000	0.085938	0.296875
5	0.039062	0.046875	0.046875	0.039062
6	0.000000	0.070312	0.328125	-0.226562
7	0.031250	0.078125	0.015625	-0.015625
8	0.085938	0.343750	-0.210938	-0.023438
9	0.078125	-0.007812	-0.078125	-0.031250
10	0.281250	-0.250000	-0.031250	-0.023438
11	0.054688	-0.031250	0.046875	0.015625
12	-0.210938	-0.031250	-0.062500	0.000000
13	-0.085938	0.007812	0.000000	0.000000
14	0.000000	-0.023438	0.039062	0.039062
15	0.023438	-0.015625	-0.015625	-0.046875
16	-0.062500	0.023438	0.000000	-0.015625
17	-0.007812	0.007812	-0.062500	-0.031250
18	0.031250	0.007812	0.000000	-0.015625
19	-0.015625	-0.039062	-0.046875	0.015625
20	0.000000	-0.007812	0.015625	-0.007812
21	-0.054688	-0.023438	0.023438	0.015625
22	-0.031250	0.000000	-0.015625	-0.015625
23	-0.007812	0.007812	0.046875	-0.007812
24	-0.023438	0.000000	-0.007812	0.015625
25	0.015625	0.046875	0.023438	0.039062
26	-0.007812	-0.007812	0.007812	-0.007812
27	0.007812	0.000000	0.023438	0.000000
28	-0.023438	0.054688	0.007812	0.046875
29	0.023438	0.023438	-0.015625	-0.062500
30	0.046875	-0.046875	0.039062	0.031250
31	0.007812	0.007812	-0.015625	0.015625
32	0.031250	0.031250	0.031250	-0.054688
33	-0.007812	-0.031250	0.015625	0.015625
34	0.039062	0.023438	0.000000	-0.070312
35	-0.023438	-0.007812	0.046875	-0.062500
36	0.031250	-0.023438	-0.031250	-0.015625
37	-0.007812	0.015625	-0.015625	0.039062
38	-0.015625	-0.023438	-0.031250	-0.031250

Table 5.2.1.8-2 C_{SCBm} Parameter(38/40)

k	$C_{SCB1}(4, 3, k)$	$C_{SCB1}(5, 3, k)$	$C_{SCB1}(6, 3, k)$	$C_{SCB1}(7, 3, k)$
0	0.226562	0.351562	-0.367188	0.046875
1	0.070312	0.039062	-0.078125	0.000000
2	0.335938	-0.250000	-0.039062	0.007812
3	0.023438	-0.023438	0.007812	-0.015625
4	-0.265625	-0.031250	-0.015625	-0.015625
5	-0.007812	-0.023438	0.031250	0.015625
6	0.015625	-0.023438	0.000000	0.000000
7	0.007812	0.023438	0.046875	-0.054688
8	-0.039062	0.000000	0.046875	-0.015625
9	-0.007812	-0.023438	-0.078125	-0.023438
10	-0.015625	0.039062	0.031250	-0.015625
11	-0.007812	-0.070312	-0.031250	0.000000
12	0.031250	-0.039062	0.000000	-0.039062
13	-0.062500	0.015625	-0.007812	-0.007812
14	0.000000	-0.039062	-0.023438	0.007812
15	-0.023438	0.000000	0.023438	0.039062
16	-0.039062	0.015625	-0.031250	0.023438
17	-0.007812	0.031250	0.054688	0.031250
18	-0.007812	0.007812	0.046875	0.046875
19	0.015625	0.007812	0.117188	-0.031250
20	-0.015625	0.023438	-0.015625	-0.007812
21	0.023438	0.007812	-0.070312	0.007812
22	0.015625	0.015625	0.070312	-0.007812
23	0.015625	0.007812	-0.007812	-0.015625
24	-0.015625	0.015625	0.039062	0.000000
25	-0.039062	0.015625	-0.023438	0.031250
26	0.031250	0.023438	-0.031250	0.007812
27	0.000000	-0.015625	0.015625	0.015625
28	0.007812	0.000000	0.000000	-0.007812
29	-0.023438	0.015625	0.031250	0.007812
30	-0.007812	-0.031250	-0.015625	-0.007812
31	0.054688	-0.007812	0.031250	-0.109375
32	0.007812	-0.054688	0.000000	0.031250
33	-0.007812	0.007812	-0.039062	0.023438
34	-0.054688	0.000000	0.015625	-0.023438
35	0.031250	0.007812	0.000000	0.054688
36	0.046875	0.023438	0.015625	0.078125
37	0.000000	-0.054688	-0.007812	0.054688
38	0.039062	0.023438	0.000000	0.023438

Table 5.2.1.8-2 C_{SCBm} Parameter(39/40)

k	$C_{SCB1}(8, 3, k)$	$C_{SCB1}(9, 3, k)$	$C_{SCB1}(10, 3, k)$	$C_{SCB1}(11, 3, k)$
0	0.046875	0.007812	-0.015625	-0.039062
1	0.007812	0.023438	0.031250	0.000000
2	0.000000	-0.007812	-0.015625	0.000000
3	0.015625	0.039062	0.000000	-0.070312
4	-0.023438	-0.023438	-0.054688	-0.031250
5	0.031250	-0.007812	-0.054688	-0.210938
6	-0.015625	-0.015625	0.000000	-0.039062
7	0.000000	-0.093750	-0.250000	0.218750
8	0.000000	-0.015625	0.015625	0.039062
9	-0.070312	-0.179688	0.179688	0.015625
10	-0.023438	-0.015625	0.054688	-0.015625
11	-0.187500	0.203125	0.031250	0.000000
12	-0.039062	0.031250	-0.039062	0.007812
13	0.179688	0.039062	0.054688	-0.007812
14	0.046875	0.000000	0.000000	0.031250
15	0.046875	-0.015625	0.007812	-0.054688
16	0.000000	0.000000	0.007812	0.015625
17	0.000000	0.000000	-0.031250	0.062500
18	-0.007812	0.023438	0.031250	0.054688
19	-0.007812	-0.046875	0.007812	0.000000
20	0.015625	0.031250	0.054688	-0.015625
21	-0.039062	0.046875	0.015625	0.062500
22	0.023438	0.031250	0.000000	-0.054688
23	0.023438	0.007812	0.031250	-0.023438
24	0.054688	-0.015625	-0.031250	-0.023438
25	0.031250	0.023438	-0.046875	-0.039062
26	0.000000	-0.015625	-0.039062	0.000000
27	0.031250	-0.023438	-0.007812	-0.054688
28	-0.054688	-0.046875	0.000000	0.015625
29	0.000000	-0.023438	-0.023438	0.007812
30	-0.046875	-0.046875	0.007812	0.054688
31	-0.015625	-0.007812	-0.007812	-0.007812
32	0.031250	0.007812	0.015625	0.007812
33	-0.031250	-0.046875	0.007812	0.046875
34	0.007812	0.054688	-0.015625	-0.039062
35	-0.039062	0.000000	-0.023438	0.015625
36	0.031250	0.007812	0.007812	-0.039062
37	0.023438	0.000000	0.015625	0.023438
38	0.031250	0.000000	-0.007812	0.023438

Table 5.2.1.8-2 C_{SCBm} Parameter(40/40)

k	$C_{SCB1}(12, 3, k)$	$C_{SCB1}(13, 3, k)$	$C_{SCB1}(14, 3, k)$	$C_{SCB1}(15, 3, k)$
0	-0.015625	0.000000	0.031250	-0.046875
1	-0.070312	-0.164062	0.195312	-0.031250
2	-0.023438	-0.023438	0.062500	0.000000
3	-0.210938	0.218750	0.031250	-0.031250
4	-0.015625	0.007812	-0.039062	0.007812
5	0.218750	0.015625	0.007812	-0.007812
6	0.015625	0.000000	0.007812	0.000000
7	0.000000	0.015625	-0.031250	-0.023438
8	-0.007812	0.031250	-0.007812	0.039062
9	0.007812	-0.031250	-0.031250	0.039062
10	0.000000	0.007812	0.023438	-0.007812
11	-0.023438	-0.015625	0.015625	0.000000
12	-0.015625	0.046875	0.062500	0.031250
13	-0.031250	0.000000	0.000000	0.085938
14	0.062500	0.015625	-0.015625	-0.039062
15	0.007812	0.054688	0.007812	0.015625
16	0.039062	-0.015625	-0.039062	0.031250
17	0.062500	0.039062	0.007812	-0.070312
18	-0.007812	-0.015625	-0.039062	-0.039062
19	0.000000	-0.007812	0.000000	-0.023438
20	0.000000	-0.007812	-0.023438	0.023438
21	0.015625	-0.023438	-0.062500	-0.015625
22	-0.046875	-0.015625	0.015625	0.023438
23	-0.015625	-0.046875	-0.039062	-0.023438
24	-0.039062	0.007812	0.000000	-0.015625
25	-0.007812	0.000000	0.000000	0.070312
26	0.023438	0.046875	0.007812	-0.054688
27	-0.046875	-0.007812	-0.007812	0.031250
28	0.070312	0.000000	-0.023438	0.039062
29	0.000000	0.070312	0.007812	0.007812
30	-0.015625	-0.062500	-0.023438	0.007812
31	0.039062	-0.007812	0.007812	-0.015625
32	-0.015625	0.000000	0.000000	0.085938
33	0.023438	0.023438	0.062500	-0.046875
34	0.015625	-0.007812	0.031250	0.054688
35	0.078125	0.015625	-0.023438	0.117188
36	-0.039062	0.078125	0.023438	-0.070312
37	-0.015625	-0.062500	-0.015625	0.015625
38	0.093750	0.062500	0.093750	0.015625

Table 5.2.1.8-3 Gain Quantization (1/3)

i	Cg0(i)	Cg1(i)
0	0.017543	0.026087
1	0.085654	0.068821
2	0.095995	0.150149
3	0.154412	0.121565
4	0.127649	0.175831
5	0.238296	0.081831
6	0.150637	0.215765
7	0.215511	0.161783
8	0.050897	0.273677
9	0.196722	0.199322
10	0.265998	0.148706
11	0.281973	0.085053
12	0.137203	0.288742
13	0.221368	0.230407
14	0.275592	0.192390
15	0.193801	0.281749
16	0.371721	0.108869
17	0.338110	0.107127
18	0.360271	0.199871
19	0.241511	0.289529
20	0.337757	0.171818
21	0.338397	0.067207
22	0.296963	0.245940
23	0.130083	0.385568
24	0.383142	0.137241
25	0.300593	0.203611
26	0.310365	0.276914
27	0.253464	0.341949
28	0.358497	0.260623
29	0.269769	0.458188
30	0.283259	0.369620
31	0.168044	0.445071
32	0.421402	0.081405
33	0.417901	0.145521
34	0.470621	0.142309
35	0.439375	0.200494
36	0.543250	0.090296
37	0.466778	0.224750
38	0.505884	0.234818
39	0.423965	0.287262
40	0.501689	0.080627
41	0.451481	0.103279
42	0.499078	0.134240

Table 5.2.1.8-3 Gain Quantization (2/3)

43	0.417644	0.174584
44	0.538538	0.142586
45	0.460763	0.187079
46	0.508530	0.182376
47	0.425809	0.242557
48	0.392405	0.338004
49	0.356344	0.333664
50	0.530734	0.290227
51	0.485792	0.285102
52	0.417206	0.389860
53	0.340806	0.453130
54	0.508559	0.349483
55	0.433535	0.337604
56	0.508769	0.420735
57	0.432703	0.434634
58	0.635760	0.418963
59	0.558004	0.397743
60	0.396098	0.541203
61	0.252004	0.554263
62	0.559125	0.530358
63	0.551968	0.458726
64	0.535332	0.251661
65	0.572131	0.168647
66	0.533520	0.188602
67	0.559472	0.121742
68	0.580223	0.284595
69	0.628966	0.347271
70	0.568145	0.208912
71	0.579762	0.240326
72	0.602925	0.170771
73	0.621263	0.110367
74	0.595867	0.135977
75	0.590353	0.095981
76	0.636095	0.240630
77	0.628236	0.300098
78	0.631095	0.172969
79	0.614504	0.211719
80	0.676685	0.200664
81	0.647875	0.128804
82	0.678772	0.269400
83	0.653373	0.187065
84	0.708337	0.130592
85	0.665592	0.144330
86	0.701134	0.225600

Table 5.2.1.8-3 Gain Quantization (3/3)

87	0.716557	0.172097
88	0.736304	0.152841
89	0.702024	0.093203
90	0.753644	0.090339
91	0.657941	0.086929
92	0.785677	0.176899
93	0.726645	0.202016
94	0.761235	0.142183
95	0.777839	0.116267
96	0.832351	0.155096
97	0.908505	0.181151
98	0.761467	0.212470
99	0.814692	0.107026
100	0.762032	0.301502
101	0.801739	0.358566
102	0.736796	0.437635
103	0.653118	0.556082
104	0.832255	0.218314
105	0.876937	0.233578
106	0.776541	0.238473
107	0.867560	0.179085
108	0.851111	0.270048
109	0.893896	0.306505
110	0.707646	0.302038
111	0.768460	0.490364
112	0.858988	0.115695
113	0.914099	0.115113
114	0.955131	0.226397
115	0.952124	0.130551
116	0.447304	0.791394
117	1.004539	0.683824
118	1.116511	0.201502
119	1.461294	0.358059
120	1.141643	0.433559
121	1.137856	0.292875
122	1.019032	0.268594
123	1.031678	0.141413
124	0.729077	0.732916
125	1.182631	0.817592
126	1.189718	0.654382
127	1.827527	0.480079

5.2.2 Channel coding

The encoded parameters are code-converted furthermore for packing.

From among converted parameters, bits to be error-protected are represented as "P(x)" and bits not to be error-protected, as "NP(x)". The "P(x)" is encoded by convolutional coder after CRC coding. The code is then combined with the "NP(x)". And after bit interleaving, frame interleaving is performed.

5.2.2.0 Definitions

ACB	adaptive codebook	
FCB	fixed codebook	
SCB	stochastic codebook	
LSP0	zero-th LSP parameter	(7 bit)
LSP1	first LSP parameter	(8 bit)
LSP2	second LSP parameter	(8 bit)
LSP3	third LSP parameter	(8 bit)
POW	power parameter index	(7 bit)
LAG _i	ACB/FCB index code for the i-th subframe	(8 bit)
CODE _i	SCB index code for the i-th subframe	(10 bit)
GAIN _i	gain index for the i-th subframe	(7 bit)
lagi _i	integral part of ACB period for the i-th subframe	(16≤lagi _i ≤96)
lagf _i	fractional part of ACB period for the i-th subframe	(lagf _i =0,1,2,3)
l _{FCB_i}	FCB index for the i-th subframe	(0≤l _{FCB_i} ≤31)
S _{FCB_i}	FCB sign for the i-th subframe	(S _{FCB_i} =0,1)
l _{SCB0_i}	SCB0 index for the i-th subframe	(0≤l _{SCB0_i} ≤15)
S _{SCB0_i}	SCB0 sign for the i-th subframe	(S _{SCB0_i} =0,1)
l _{SCB1_i}	SCB1 index for the i-th subframe	(0≤l _{SCB1_i} ≤15)
S _{SCB1_i}	SCB1 sign for the i-th subframe	(S _{SCB1_i} =0,1)
l _{0_i}	SCB1 temporary index for the i-th subframe	
S _{0_i}	SCB0 temporary sign for the i-th subframe	
l _{1_i}	SCB1 temporary index for the i-th subframe	
S _{1_i}	SCB1 temporary sign for the i-th subframe	
P[x]	protected bits	(0≤x≤65)
NP[x]	unprotected bits	(0≤x≤71)
CRC[x]	cyclic redundancy check bits	(0≤x≤8)
Q(x)	quotient polynomial of CRC codes	
R(x)	remainder polynomial of CRC codes	
G _{CRC} (x)	generation polynomial of CRC codes	
CVin[x]	convolutional encoder input bits	(0≤x≤81)
G ₁ (D), G ₂ (D)	generator polynomials of the convolutional encoder	
V ₁ [x], V ₂ [x]	output bits from the convolutional encoder	(0≤x≤81)
NP _p [x]	unprotected bits of the previous frame	(0≤x≤71)
V _{1p} [x], V _{2p} [x]	output bits from the convolutional encoder of the previous frame	(0≤x≤81)
V _{1R} [x], V _{2R} [x]	de-interleaved output at the decoder	(0≤x≤81)
NP _R [x]	received unprotected bits	(0≤x≤71)
P _R [x]	protected bits after convolutional decoding	(0≤x≤65)
CRC _R [x]	CRC codes after convolutional decoding	(0≤x≤8)
ST	frame masking state variable	(ST=-1,0,1,2,3,4)
LSP _p i	LSP parameter of the previous frame	(i=0,1,2,3)
POW _p	power parameter of the previous frame	
LAG _p i	ACB/FCB index for i-th subframe of the previous frame	

$GAIN_{p_i}$ gain index for i-th subframe of the previous frame

5.2.2.1 Code conversion

From among coded parameters, ACB (Adaptive code book) period/FCB (Fixed codebook) indices and SCB (Stochastic codebook) indices are code-converted.

5.2.2.1.1 ACB/FCB index code conversion

ACB periods ($lagi_i$, $lagf_i$)/FCB indices (I_{FCB_i} , S_{FCB_i}) for each encoded subframe ($i=0,1,2,3$) are converted into an 8-bit ACB period code (LAG_i).

(1) For the subframe selecting ACB

$$LAG_i = \begin{matrix} lagi_i \times 4 + lagf_i & -64 & (16 \leq lagi_i \leq 45) \\ lagi_i \times 2 + ((lagf_i - 1) / 2) & +28 & (45 < lagi_i \leq 65) \\ lagi_i & +94 & (65 < lagi_i \leq 96) \\ 255 & & (lagi_i = 0). \end{matrix} \quad (5.2.2.1.1-1)$$

(2) For the subframe selecting FCB

$$LAG_i = S_{FCB_i} + I_{FCB_i} \times 2 + 191 \quad (5.2.2.1.1-2)$$

5.2.2.1.2 SCB index code conversion

The SCB index (I_{SCB0_i} , S_{SCB0_i} , I_{SCB1_i} , S_{SCB1_i}) are converted to the temporary index (I_0_i , S_0_i , I_1_i , S_1_i) by Table 5.2.2.1-1 and Table 5.2.2.1-2. These indices are further converted into SCB index code ($CODE_i$) in accordance with Eq. 5.2.2.1.2-3.

Table 5.2.2.1-1 : SCB Index Conversion Rule #1 ($[S_{SCB0_i}, I_{SCB0_i}] \Rightarrow [S_0_i, I_0_i]$)

S_{SCB0_i}	I_{SCB0_i}	S_0_i	I_0_i	S_{SCB0_i}	I_{SCB0_i}	S_0_i	I_0_i
0	0	1	1	1	0	0	13
0	1	0	6	1	1	0	12
0	2	1	14	1	2	0	14
0	3	1	12	1	3	1	9
0	4	0	4	1	4	1	11
0	5	0	5	1	5	1	3
0	6	0	9	1	6	1	6
0	7	0	8	1	7	1	4
0	8	0	1	1	8	1	7
0	9	0	11	1	9	1	2
0	10	0	10	1	10	1	0
0	11	1	13	1	11	0	7
0	12	1	5	1	12	0	3
0	13	1	10	1	13	0	0
0	14	1	8	1	14	0	2
0	15	0	15	1	15	1	15

Table 5.2.2.1-2 : SCB Index Conversion Rule #2 ($[S_{SCB1_i}, I_{SCB1_i}] \Rightarrow [S_1_i, I_1_i]$)

S_{SCB1_i}	I_{SCB1_i}	S_1_i	I_1_i	S_{SCB1_i}	I_{SCB1_i}	S_1_i	I_1_i
0	0	0	0	1	0	1	12
0	1	0	1	1	1	1	4
0	2	1	8	1	2	0	10
0	3	1	9	1	3	0	14
0	4	1	13	1	4	0	4
0	5	1	7	1	5	0	5
0	6	0	6	1	6	1	5
0	7	1	3	1	7	0	13
0	8	1	14	1	8	0	2
0	9	1	0	1	9	0	3
0	10	0	8	1	10	1	10
0	11	0	11	1	11	1	11
0	12	0	15	1	12	1	1
0	13	0	7	1	13	1	6
0	14	1	15	1	14	0	9
0	15	0	12	1	15	1	2

$$CODE_i = S_0_i \times 512 + S_1_i \times 256 + I_0_i \times 16 + I_1_i \quad (5.2.2.1.2-3)$$

5.2.2.2 Protected bit selection

The protected bits ($P[x] : 0 \leq x \leq 65$) and the unprotected bits ($NP[x] : 0 \leq x \leq 71$) are made from the speech coder bits. The number of protected bits for each parameter is shown in the Table 5.2.2.2-1. Details of $P(X)$ is shown in the Table 5.2.2.2-2 and that of $NP(x)$ is shown in the Table 5.2.2.2-3.

Table 5.2.2.2-1 : Number of Protected / Unprotected Bits

Parameter	Number of Protected Bits	Number of Unprotected Bits	Total
LSP0	7	0	7
LSP1	8	0	8
LSP2	0	8	8
LSP3	0	8	8
POW	7	0	7
LAG_0	8	0	8
CPDE_0	0	10	10
GAIN_0	3	4	7
LAG_1	8	0	8
CODE_1	0	10	10
GAIN_1	3	4	7
LAG_2	8	0	8
CODE_2	0	10	10
GAIN_2	3	4	7
LAG_3	8	0	8
CODE_3	0	10	10
GAIN_3	3	4	7
Total	66	72	138

Table 5.2.2.2-2 : P[x] Contents

	Parameter Name	Bit Index		Parameter Name	Bit Index		Parameter Name	Bit Index
P [65]	LSP0	0	P [43]	LAG_0	0	P [21]	LAG_2	6
P [64]	LSP0	1	P [42]	LAG_0	1	P [20]	LAG_2	7
P [63]	LSP0	2	P [41]	LAG_0	2	P [19]	LAG_3	0
P [62]	LSP0	3	P [40]	LAG_0	3	P [18]	LAG_3	1
P [61]	LSP0	4	P [39]	LAG_0	4	P [17]	LAG_3	2
P [60]	LSP0	5	P [38]	LAG_0	5	P [16]	LAG_3	3
P [59]	LSP0	6	P [37]	LAG_0	6	P [15]	LAG_3	4
P [58]	LSP1	0	P [36]	LAG_0	7	P [14]	LAG_3	5
P [57]	LSP1	1	P [35]	LAG_1	0	P [13]	LAG_3	6
P [56]	LSP1	2	P [34]	LAG_1	1	P [12]	LAG_3	7
P [55]	LSP1	3	P [33]	LAG_1	2	P [11]	GAIN_0	4
P [54]	LSP1	4	P [32]	LAG_1	3	P [10]	GAIN_0	5
P [53]	LSP1	5	P [31]	LAG_1	4	P [9]	GAIN_0	6
P [52]	LSP1	6	P [30]	LAG_1	5	P [8]	GAIN_1	4
P [51]	LSP1	7	P [29]	LAG_1	6	P [7]	GAIN_1	5
P [50]	POW	0	P [28]	LAG_1	7	P [6]	GAIN_1	6
P [49]	POW	1	P [27]	LAG_2	0	P [5]	GAIN_2	4
P [48]	POW	2	P [26]	LAG_2	1	P [4]	GAIN_2	5
P [47]	POW	3	P [25]	LAG_2	2	P [3]	GAIN_2	6
P [46]	POW	4	P [24]	LAG_2	3	P [2]	GAIN_3	4
P [45]	POW	5	P [23]	LAG_2	4	P [1]	GAIN_3	5
P [44]	POW	6	P [22]	LAG_2	5	P [0]	GAIN_3	6

where bit index 0 indicates the LSB of the parameter.

Table 5.2.2.2-3 : NP[x] Contents

	Parameter Name	Bit Index		Parameter Name	Bit Index		Parameter Name	Bit Index
NP [0]	LSP2	0	NP [24]	CODE_0	4	NP [48]	CODE_2	0
NP [1]	LSP2	1	NP [25]	CODE_0	5	NP [49]	CODE_2	1
NP [2]	LSP2	2	NP [26]	CODE_0	6	NP [50]	CODE_2	2
NP [3]	LSP2	3	NP [27]	CODE_0	7	NP [51]	CODE_2	3
NP [4]	LSP2	4	NP [28]	CODE_0	8	NP [52]	CODE_2	4
NP [5]	LSP2	5	NP [29]	CODE_0	9	NP [53]	CODE_2	5
NP [6]	LSP2	6	NP [30]	GAIN_1	0	NP [54]	CODE_2	6
NP [7]	LSP2	7	NP [31]	GAIN_1	1	NP [55]	CODE_2	7
NP [8]	LSP3	0	NP [32]	GAIN_1	2	NP [56]	CODE_2	8
NP [9]	LSP3	1	NP [33]	GAIN_1	3	NP [57]	CODE_2	9
NP [10]	LSP3	2	NP [34]	CODE_1	0	NP [58]	GAIN_3	0
NP [11]	LSP3	3	NP [35]	CODE_1	1	NP [59]	GAIN_3	1
NP [12]	LSP3	4	NP [36]	CODE_1	2	NP [60]	GAIN_3	2
NP [13]	LSP3	5	NP [37]	CODE_1	3	NP [61]	GAIN_3	3
NP [14]	LSP3	6	NP [38]	CODE_1	4	NP [62]	CODE_3	0
NP [15]	LSP3	7	NP [39]	CODE_1	5	NP [63]	CODE_3	1
NP [16]	GAIN_0	0	NP [40]	CODE_1	6	NP [64]	CODE_3	2
NP [17]	GAIN_0	1	NP [41]	CODE_1	7	NP [65]	CODE_3	3
NP [18]	GAIN_0	2	NP [42]	CODE_1	8	NP [66]	CODE_3	4
NP [19]	GAIN_0	3	NP [43]	CODE_1	9	NP [67]	CODE_3	5
NP [20]	CODE_0	0	NP [44]	GAIN_2	0	NP [68]	CODE_3	6
NP [21]	CODE_0	1	NP [45]	GAIN_2	1	NP [69]	CODE_3	7
NP [22]	CODE_0	2	NP [46]	GAIN_2	2	NP [70]	CODE_3	8
NP [23]	CODE_0	3	NP [47]	GAIN_2	3	NP [71]	CODE_3	9

where bit index 0 indicates the LSB of the parameter.

5.2.2.3 CRC coding

A 9-bit CRC code (CRC [x]) is computed from the protected bit sequence (P [x]) following the definition below:

$$R(x) = \sum_{i=0}^8 \text{CRC}[i] \cdot x^i, \quad (5.2.2.3-1)$$

where

$$x^9 \cdot \sum_{i=0}^{65} P[i] \cdot x^i = Q(x) \cdot G_{\text{CRC}}(x) + R(x), \quad (5.2.2.3-2)$$

$$G_{\text{CRC}}(x) = 1 + x + x^2 + x^5 + x^8 + x^9 \quad (5.2.2.3-3)$$

5.2.2.4 Convolutional coding

The input sequence of the convolutional coder (CVin[x]) is constructed by Eq. 5.2.2.4-1. It is coded by R = 1/2, m = 7 convolutional coder. Initially all coder memory elements are cleared to 0.

$$CV_{in}[x] = \begin{cases} CRC[2x] & (0 \leq x \leq 4) \\ P[74 - 2x] & (5 \leq x \leq 37) \\ P[2x - 75] & (38 \leq x \leq 70) \\ CRC[149 - 2x] & (71 \leq x \leq 74) \\ 0 & (75 \leq x \leq 81) \end{cases} \quad (5.2.2.4-1)$$

Generator polynomials are defined as:

$$G_1(D) = 1 + D + D^2 + D^5 + D^7, \quad (5.2.2.4-2)$$

$$G_2(D) = 1 + D^3 + D^4 + D^5 + D^6 + D^7. \quad (5.2.2.4-3)$$

These definitions follow the notation in Shu Lin and Daniel J. Costello, Jr., "Error Control Coding: Fundamentals and Applications", Prentice-Hall, 1983, pp330.

The input order of the convolutional coder starts from CVin [0] to CVin [81]. The output sequence from G₁(D) is represented as V₁ [x] (0 ≤ x ≤ 81), and that from G₂(D) as V₂ [x] (0 ≤ x ≤ 81).

5.2.2.5 Puncturing and interleaving

The current frame V₁ [x], V₂ [x], NP [x] and the previous frame V_{1p} [x], V_{2p} [x], NP_p [x], are punctured and interleaved according to Table 5.2.2.5-1, then 224-bit frame are made. V₂ [2], V₂ [9], V₂ [16], V₂ [23], V₂ [30], V₂ [37], V₂ [44], V₂ [51], V₂ [58], V₂ [65], V₂ [72], and V₂ [79] are not transmitted because of the bit puncturing. Detailed transmission order are shown in Table 5.2.2.5-1, where the order 0 represents the first transmitted bit in the frame.

Table 5.2.2.5-1 : Bit Transmission Order

Order	Trans. bit	Order	Trans. bit	Order	Trans. bit	Order	Trans. bit	Order	Trans. bit
0	V ₁ [0]	45	NP _p [25]	90	V ₂ [74]	135	V _{1p} [50]	180	V ₁ [29]
1	V _{2p} [0]	46	NP [32]	91	V _{1p} [75]	136	V ₁ [54]	181	V _{2p} [29]
2	V ₂ [4]	47	NP _p [33]	92	V ₂ [78]	137	V _{2p} [54]	182	V ₂ [33]
3	V _{1p} [5]	48	NP [40]	93	V _{1p} [79]	138	V ₁ [58]	183	V _{1p} [34]
4	V ₂ [8]	49	NP _p [41]	94	NP [2]	139	V _{2p} [59]	184	V ₁ [38]
5	V _{1p} [9]	50	NP [48]	95	NP _p [3]	140	V ₂ [62]	185	V _{2p} [38]
6	V ₁ [13]	51	NP _p [49]	96	NP [10]	141	V _{1p} [63]	186	V ₁ [42]
7	V _{2p} [13]	52	NP [56]	97	NP _p [11]	142	V ₁ [67]	187	V _{2p} [42]
8	V ₂ [17]	53	NP _p [57]	98	NP [18]	143	V _{2p} [67]	188	V ₂ [46]
9	V _{1p} [18]	54	NP [64]	99	NP _p [19]	144	V ₁ [71]	189	V _{1p} [47]
10	V ₂ [21]	55	NP _p [65]	100	NP [26]	145	V _{2p} [71]	190	V ₂ [50]
11	V _{1p} [22]	56	V ₁ [1]	101	NP _p [27]	146	V ₂ [75]	191	V _{1p} [51]
12	V ₁ [26]	57	V _{2p} [1]	102	NP [34]	147	V _{1p} [76]	192	V ₁ [55]
13	V _{2p} [26]	58	V ₂ [5]	103	NP _p [35]	148	V ₁ [80]	193	V _{2p} [55]
14	V ₁ [30]	59	V _{1p} [6]	104	NP [42]	149	V _{2p} [80]	194	V ₂ [59]
15	V _{1p} [31]	60	V ₁ [10]	105	NP _p [43]	150	NP [4]	195	V _{1p} [60]
16	V ₂ [34]	61	V _{2p} [10]	106	NP [50]	151	NP _p [5]	196	V ₂ [63]
17	V _{1p} [35]	62	V ₁ [14]	107	NP _p [51]	152	NP [12]	197	V _{1p} [64]
18	V ₁ [39]	63	V _{2p} [14]	108	NP [58]	153	NP _p [13]	198	V ₁ [68]
19	V _{2p} [39]	64	V ₂ [18]	109	NP _p [59]	154	NP [20]	199	V _{2p} [68]
20	V ₁ [43]	65	V _{1p} [19]	110	NP [66]	155	NP _p [21]	200	V ₁ [72]
21	V _{2p} [43]	66	V ₂ [22]	111	NP _p [67]	156	NP [28]	201	V _{1p} [73]
22	V ₂ [47]	67	V _{1p} [23]	112	V ₁ [2]	157	NP _p [29]	202	V ₂ [76]
23	V _{1p} [48]	68	V ₁ [27]	113	V _{1p} [3]	158	NP [36]	203	V _{1p} [77]
24	V ₁ [52]	69	V _{2p} [27]	114	V ₂ [6]	159	NP _p [37]	204	V ₁ [81]
25	V _{2p} [52]	70	V ₂ [31]	115	V _{1p} [7]	160	NP [44]	205	V _{2p} [81]
26	V ₁ [56]	71	V _{1p} [32]	116	V ₁ [11]	161	NP _p [45]	206	NP [6]
27	V _{2p} [56]	72	V ₂ [35]	117	V _{2p} [11]	162	NP [52]	207	NP _p [7]
28	V ₂ [60]	73	V _{1p} [36]	118	V ₁ [15]	163	NP _p [53]	208	NP [14]
29	V _{1p} [61]	74	V ₁ [40]	119	V _{2p} [15]	164	NP [60]	209	NP _p [15]
30	V ₂ [64]	75	V _{2p} [40]	120	V ₂ [19]	165	NP _p [61]	210	NP [22]
31	V _{1p} [65]	76	V ₁ [44]	121	V _{1p} [20]	166	NP [68]	211	NP _p [23]
32	V ₁ [69]	77	V _{1p} [45]	122	V ₁ [24]	167	NP _p [69]	212	NP [30]
33	V _{2p} [69]	78	V ₂ [48]	123	V _{2p} [24]	168	V ₂ [3]	213	NP _p [31]
34	V ₂ [73]	79	V _{1p} [49]	124	V ₁ [28]	169	V _{1p} [4]	214	NP [38]
35	V _{1p} [74]	80	V ₁ [53]	125	V _{2p} [28]	170	V ₂ [7]	215	NP _p [39]
36	V ₂ [77]	81	V _{2p} [53]	126	V ₂ [32]	171	V _{1p} [8]	216	NP [46]
37	V _{1p} [78]	82	V ₁ [57]	127	V _{1p} [33]	172	V ₁ [12]	217	NP _p [47]
38	NP [0]	83	V _{2p} [57]	128	V ₂ [36]	173	V _{2p} [12]	218	NP [54]
39	NP _p [1]	84	V ₂ [61]	129	V _{1p} [37]	174	V ₁ [16]	219	NP _p [55]
40	NP [8]	85	V _{1p} [62]	130	V ₁ [41]	175	V _{1p} [17]	220	NP [62]
41	NP _p [9]	86	V ₁ [66]	131	V _{2p} [41]	176	V ₂ [20]	221	NP _p [63]

42	NP [16]	87	V_{2p} [66]	132	V_2 [45]	177	V_{1p} [21]	222	NP [70]
43	NP_p [17]	88	V_1 [70]	133	V_{1p} [46]	178	V_1 [25]	223	NP_p [71]
44	NP [24]	89	V_{2p} [70]	134	V_2 [49]	179	V_{2p} [25]		

5.2.3 Channel decoding

Received sequence is de-interleaved and decoded for error correction. After error correction decoding is done, 9-bit CRC is computed in the same way as in the encoder. This computed CRC is compared with transmitted CRC. If these CRCs are not identical, then parameter replacement is performed before speech decoding.

5.2.3.1 De-interleaving

Received bit sequences ($V_{1R}[x]$, $V_{2R}[x]$, $NP_R[x]$) are recovered by de-interleaving according to Table 5.2.2.5-1.

5.2.3.2 Error correction decoding

After entire data of one frame is received at the reception side, decoded sequences $P_R[x]$, $CRC_R[x]$ are obtained through error correction decoding from $V_{1R}[x]$ and $V_{2R}[x]$. In order to execute effective error correction decoding, Viterbi algorithm or another decoding method with at least the same performance is used. If soft decision information is available, it is recommended to be used for the path metric calculation. These error correction decoding methods are optional.

5.2.3.3 CRC decoding

Nine-bit CRC sequence ($CRC[x]$) is calculated from the decoded sequence $P_R[x]$ in the same way as Eq. 5.2.2.3-1 through Eq. 5.2.2.3-3. If the calculated $CRC[x]$ is different from $CRC_R[x]$, the frame is considered as a bad frame.

5.2.3.4 Bad frame masking

In order to improve the decoded speech quality in the presence of coded bit errors, bad frame masking shall be performed by making use of the previous reception sequence. Detailed techniques for the bad frame masking described in this section and 5.2.3.5 are optional.

Frame masking state (ST) of the current frame is updated depending on the CRC decoding result. Fig. 5.2.3.4-1 shows the ST state transitions. The initial state is $ST = 0$. The arrow with a letter "1" represents the transition with a bad frame, and that with a letter "0" an error-free frame.

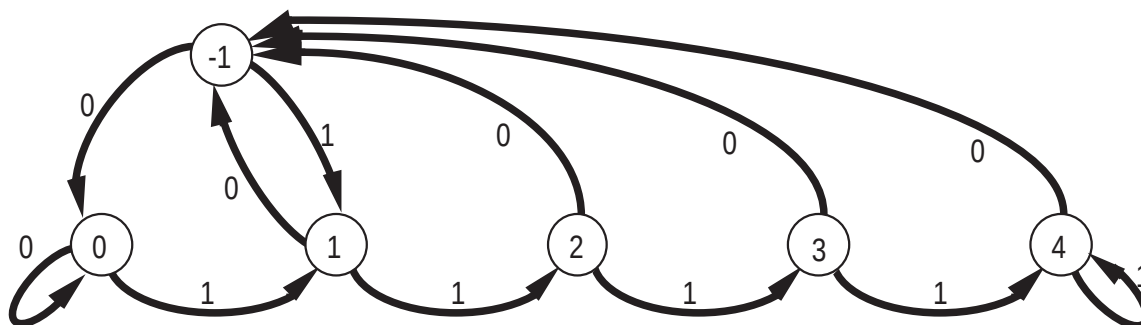


Fig 5.2.3.4-1 Frame Masking State Transition

5.2.3.5 Parameter replacement

The following parameter replacements are performed in the case that $ST = 1, 2, 3$ or 4 .

$LSP_i = LSP_{p_i}$	$(i=0,1)$	(5.2.3.5 - 1)
$POW = POW_p$		(5.2.3.5 - 2)
$LAG_i = LAP_{p_3}$	$(i=0,1,2,3),$	(5.2.3.5 - 3)
$GAIN_i = GAIN_{p_3}$	$(i=0,1,2,3)$	(5.2.3.5 - 4)

where LSP_{p_i} , POW_p , LAG_{p_i} , and $GAIN_{p_i}$ indicate LSP_i , POW , LAG_i , and $GAIN_i$ of the previous frame respectively.

5.2.3.6 Parameter reverse conversion

The speech decoding parameters are recomposed by a reverse conversion of Section 5.2.2.1.

5.2.4 Speech decoding

5.2.4.0 Definitions

$lagi_i$	integral part of ACB period for the i -th subframe	$(16 \leq lagi_i \leq 96)$
$lagf_i$	fractional part of ACB period for the i -th subframe	$(lagf_i = 0,1,2,3)$
I_{FCB_i}	FCB index for the i -th subframe	$(0 \leq I_{FCB_i} \leq 31)$
S_{FCB_i}	FCB sign for the i -th subframe	$(S_{FCB_i} = 0,1)$
I_{SCB0_i}	SCB# 0 index for the i -th subframe	$(0 \leq I_{SCB0_i} \leq 15)$
S_{SCB0_i}	SCB# 0 sign for the i -th subframe	$(S_{SCB0_i} = 0,1)$
I_{SCB1_i}	SCB# 1 index for the i -th subframe	$(0 \leq I_{SCB1_i} \leq 15)$
S_{SCB1_i}	SCB# 1 sign for the i -th subframe	$(S_{SCB1_i} = 0,1)$
LSP_i	LSP parameters	$(i = 0,1,2,3)$
ST	bad frame masking state variable for the current frame	$(ST = -1,0,1,2,3,4)$
ST_p	bad frame masking state variable for the previous frame	$(ST_p = -1,0,1,2,3,4)$
$qlsp1[x], qlsp2[x]$	LSP parameters before interpolation	$(1 \leq x \leq 10)$
$sqlsp_i[x]$	LSP parameters for the i -th subframe	$(1 \leq x \leq 10)$
$sqlsp_p_i[x]$	LSP parameters for the previous frame's i -th subframe	$(1 \leq x \leq 10)$
$a_i[x]$	short term predictor coefficients for the i -th subframe	$(1 \leq x \leq 10)$
$k_i[x]$	PARCOR coefficients for the i -th subframe	$(1 \leq x \leq 10)$
$rspow_i$	decoded energy for the i -th subframe	$(i = 0,1,2,3)$
$spow_i$	adjusted power for the i -th subframe	$(i = 0,1,2,3)$
$spow_{p_i}$	adjusted power for the i -th subframe of the previous frame	$(i = 0,1,2,3)$
$g0_i, g1_i$	excitation gain for the i -th subframe	$(i = 0,1,2,3)$
$C_0[x]$	ACB/FCB excitation vector	$(0 \leq x \leq 79)$
$C_1[x]$	SCB excitation vector	$(0 \leq x \leq 79)$
$ex[x]$	combined excitation vector	$(0 \leq x \leq 79)$
$1/A(z)$	transfer function of synthesis filter	
$dq[x]$	ACB state	$(0 \leq x \leq 79)$
$sq[x]$	output of synthesis filter	$(0 \leq x \leq 79)$
$P(z)$	transfer function of pitch emphasis filter	

$P_P(z)$	transfer function of pitch emphasis filter of the previous subframe	
g_c	weighting factor of the pitch emphasis filter	
$B(z)$	transfer function of high frequency emphasis filter	
$B_P(z)$	transfer function of high frequency emphasis filter for the previous subframe	
g_b	weighting factor of high frequency emphasis filter	
$H(z)$	transfer function of the spectral shaping filter	
$H_P(z)$	transfer function of the spectral shaping filter of the previous subframe	
g_n, g_d	weighting factor of the spectral shaping filter	$(0 \leq x \leq 79)$
$sp1[x]$	filter output by the current subframe parameters	$(0 \leq x \leq 79)$
$sp_c1[x]$	$sp1[x]$ after power scaling	
SC	power scale factor	
$sp2[x]$	filter output by the previous subframe parameters	$(0 \leq x \leq 19)$
$sp_c2[x]$	$sp2[x]$ after power scaling	$(0 \leq x \leq 19)$
SC_P	power scale factor for the previous subframe	
$W1[n], W2[n]$	window function for subframe overlap	$(0 \leq n \leq 79)$

5.2.4.1 Decoder block diagram

Figure 5.2.4-1 shows the block diagram of the decoder.

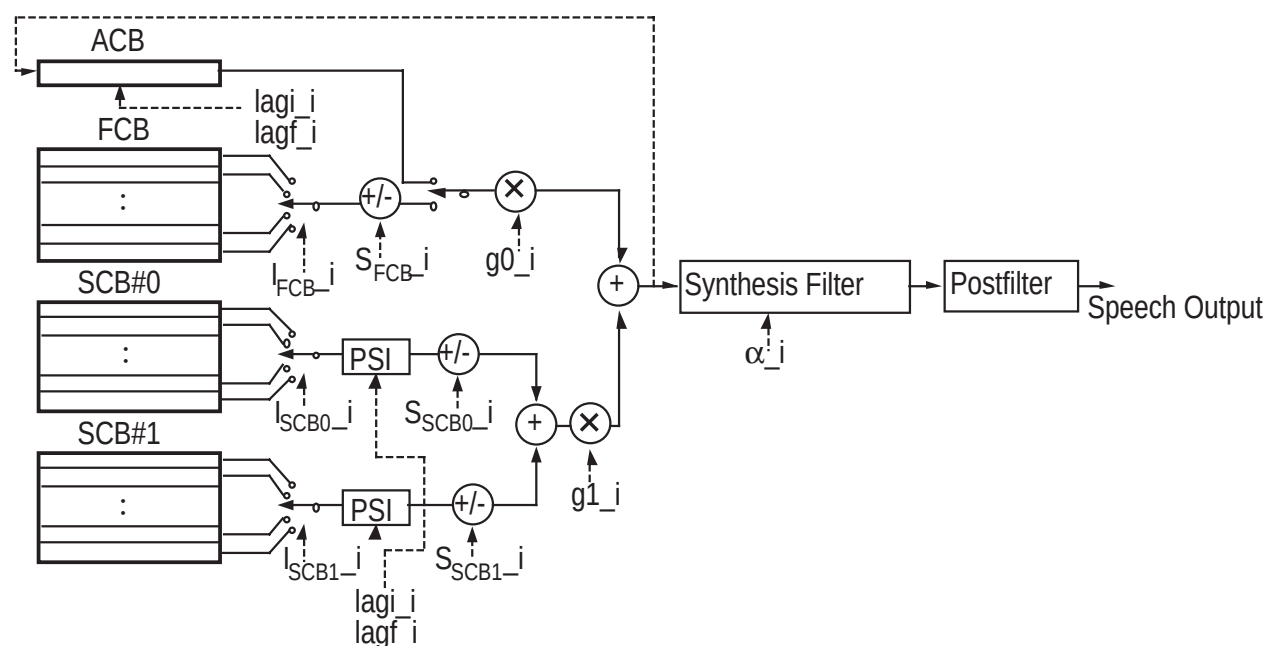


Fig.5.2.4.1-1 Block Diagram of Decoder

5.2.4.2 Decoding of synthesis filter coefficients

The $qlsp1[x]$, $qlsp2[x]$ ($1 \leq x \leq 10$) are decoded from LSP_i ($i = 0, 1, 2, 3$) in the same way as Section 5.2.1.5.1. From these parameters and the last subframe LSP parameters of the previous frame " $sqlsp_p_3[x]$ ", the LSP parameters " $sqlsp_i[x]$ " of each subframe are interpolated. The interpolation is achieved in accordance with equation (5.2.4.2-1) through (5.2.4.2-4). Here, additional interpolation shall be executed depending on the frame masking state "ST", so that the quality in the coded bit errors can be improved. The interpolation method in the coded bit error state is optional.

(1) $ST = 0$

$$sqlsp_0[x] = \frac{sqlsp_p_3[x] + qlsp1[x]}{2} \quad (1 \leq x \leq 10), \quad (5.2.4.2-1)$$

$$sqlsp_1[x] = qlsp1[x] \quad (1 \leq x \leq 10), \quad (5.2.4.2-2)$$

$$sqlsp_2[x] = \frac{qlsp1[x] + qlsp2[x]}{2} \quad (1 \leq x \leq 10), \quad (5.2.4.2-3)$$

$$sqlsp_3[x] = qlsp2[x] \quad (1 \leq x \leq 10), \quad (5.2.4.2-4)$$

(2) $ST = -1$

$$sqlsp_0[x] = sqlsp_1[x] = qlsp1[x] \quad (1 \leq x \leq 10), \quad (5.2.4.2-5)$$

$$sqlsp_2[x] = \frac{qlsp1[x] + qlsp2[x]}{2} \quad (1 \leq x \leq 10), \quad (5.2.4.2-6)$$

$$sqlsp_3[x] = qlsp2[x] \quad (1 \leq x \leq 10), \quad (5.2.4.2-7)$$

(3) $ST \geq 1$

$$sqlsp_0[x] = sqlsp_1[x] = sqlsp_2[x] = sqlsp_3[x] = sqlsp_p_3[x] \quad (1 \leq x \leq 10). \quad (5.2.4.2-8)$$

Next, the LSP parameters $sqlsp_i[x]$ of the i -th subframe are converted into the linear predictor coefficients $\alpha_i[x]$ and the PARCOR coefficients $k_i[x]$. If the PARCOR coefficients do not satisfy the stable condition shown in Eq(5.2.4.2-9), protection shall be executed by making use of the previous reception sequence. The protection method is optional. In the recommended protection method, the linear predictor coefficients are replaced with those of the previous subframe. Here, if the linear predictor coefficients of the 0-th subframe are unstable, they are replaced with the last subframe parameters of the previous frame.

$$|k_i[x]| < 1 \quad (1 \leq x \leq 10). \quad (5.2.4.2.9)$$

5.2.4.3 Power recovery and frame masking

The subframe power "rspow_i" ($i=0, 1, 2, 3$) is recovered from POW in the same manner as 5.2.1.6. Then the adjusted power "spow_i" shall be calculated depending on both the frame masking state and the earlier internal state, in order to get the recovered power suited for the channel state. The calculation method of the adjusted power is optional, while a method shown in Table 5.2.4.3-1 is recommended.

Table 5.2.4.3-1 Frame Masking Process

ST _p ST	
0 0	spow _i = rspow _i • 1.44 (i = 0,1,2,3)
-1 0	spow _i = rspow _i (i = 0,1,2,3)
X -1	spow _i = $\frac{(4-i) \cdot spow_{p_3} + (1+i) \cdot rspow_1}{5}$ (i = 0,1,2,3)
X 1	if spow _{p_0} < spow _{p_3} $spow_i = \frac{spow_{p_0} + spow_{p_3}}{2} \cdot 0.64^{i+1}$ else $spow_i = spow_{p_3} \cdot 0.64^{i+1}$ (i = 0,1,2,3)
X 2	spow _i = spow _{p_3} • 0.25 ⁱ⁺¹ (i = 0,1,2,3)
X 3	spow _i = spow _{p_3} • 0.09 ⁱ⁺¹ (i = 0,1,2,3)
X 4	spow _i = 0.0 (i = 0,1,2,3)

In Table 5.2.4.3.1, X represents any states.

5.2.4.4 Subframe process

The processes described in this section is performed by the subframe with changing $\alpha_i[x]$ to $\alpha[x]$.

5.2.4.4.1 ACB/FCB excitation vector decoding

See Section 5.2.1.8.4 and 5.2.1.8.7.

5.2.4.4.2 SCB excitation vector decoding

See Section 5.2.1.8.5 and 5.2.1.8.7.

5.2.4.4.3 Excitation gain decoding

See Section 5.2.1.8.6.

5.2.4.4.4 Decoding speech synthesis

The combined excitation $ex[x]$ is computed from the ACB/FCB excitation source $C_0[x]$, the SCB excitation vector $C_1[x]$, and the excitation gains $g0_i$, $g1_i$ as shown in Eq. 5.2.4.4.4-1:

$$ex[n] = g0_i \cdot C_0[n] + g1_i \cdot C_1[n] \quad (0 \leq n \leq 79). \quad (5.2.4.4.4-1)$$

This is used for the synthesis filter to generate its output $sq[x]$. The transfer function of this synthesis filter, $1/A(z)$, is shown in Eq. 5.2.4.4.4-2:

$$\frac{1}{A(z)} = \frac{1}{1 + \sum_{i=1}^{10} \alpha[i] \cdot z^{-i}}. \quad (5.2.4.4.4-2)$$

5.2.4.4.5 ACB state updating

ABC state ($dq[x]$) is updated every subframe. The ACB updating is achieved in accordance with equation (5.2.4.4.5-2). Here, the updating scheme shall be altered depending on the frame protection state. The updating method in the error occurrence is optional, while a scheme shown in equation (5.2.4.4.5-1) is recommended.

(1) In the 0-th subframe, $ST_P \neq 0$ and ACB is selected in the ACB/FCB selection,

$$dq[n] = \begin{cases} dq[N_{SUB} + n] & (0 \leq n \leq 17) \\ 0.1 \cdot ex[n-18] & (18 \leq n \leq 97) \end{cases} \quad (5.2.4.4.5-1)$$

(2) In other cases,

$$dq[n] = \begin{cases} dq[N_{SUB} + n] & (0 \leq n \leq 17) \\ ex[n-18] & (18 \leq n \leq 97) \end{cases} \quad (5.2.4.4.5-2)$$

5.2.4.4.6 Postfilter

Postfilter shall be applied to the synthesized subframe output $sq[x]$.

Construction of the postfilter is optional. For example, a postfilter can be formed by cascading a pitch emphasis filter, a high frequency emphasis filter and a spectral shaping filter.

5.2.4.4.6.1 Pitch emphasis filter

If ACB is selected in the ACB/FCB selection of the current subframe, the pitch emphasis filter $P(z)$ described in Eq. 5.2.4.4.6.1-1 is applied. If FCB is selected or $lag=0$, the pitch emphasis is not performed but only the $P(z)$ internal state is updated.

$$P(z) = \frac{1}{1 + g_c \cdot \sum_{i=lag-1}^{lag+1} \xi_i \cdot z^{-i}} \quad (5.2.4.4.6.1-1)$$

where g_c is

$$g_c = \begin{cases} 0.7 & (\text{if } ST = 0 \text{ and } ST_p = 0) \\ 0.4 & (\text{else}) \end{cases} \quad (5.2.4.4.6.1-2)$$

ξ_i are the 3-tap pitch predictor coefficients which are derived from the combined excitation of the current subframe $ex[x]$, the previous subframe $ex_p[x]$ and the second previous subframe $ex_{pp}[x]$ by an autocorrelation method using 240-sample rectangular window. The integral part of the lag period of the current subframe "lag_i" is used as the "lag" above.

5.2.4.4.6.2 High-frequency emphasis filter

The transfer function of the high frequency emphasis filter $B(z)$ is shown in Eq. 5.2.4.4.6.2-1:

$$B(z) = 1 - g_b z^{-1}, \quad (5.2.4.4.6.2-1)$$

where g_b is

$$g_b = \begin{cases} 0.4 & (\text{if } ST = 0 \text{ and } ST_p = 0) \\ 0.0 & (\text{else}) \end{cases} \quad (5.2.4.4.6.2-2)$$

5.2.4.4.6.3 Spectral shaping filter

The transfer function of the spectral shaping filter $H(z)$ is shown in Eq. 5.2.4.4.6.3-1:

$$H(z) = \frac{1 + \sum_{i=1}^{10} g_n^i \cdot \alpha[i] \cdot z^{-i}}{1 + \sum_{i=1}^{10} g_d^i \cdot \alpha[i] \cdot z^{-i}}, \quad (5.2.4.4.6.3-1)$$

$$\begin{cases} g_n = 0.5 \\ g_d = 0.8 \end{cases} \quad (5.2.4.4.6.3-2)$$

where $\alpha[x]$'s are the linear predictor coefficients for the current subframe.

5.2.4.4.6.4 Power scaling

The filters described from 5.2.4.4.6.1 through 5.2.4.4.6.3 are applied to $sq[x]$, and the output is denoted as $sq1[x]$. The postfilter output is scaled as follows:

$$sp_c1[x] = SC \cdot sp1[x] \quad (0 \leq x \leq 79), \quad (5.2.4.4.6.4-1)$$

$$SC = \sqrt{\frac{\sum_{i=0}^{79} sq^2[i]}{\sum_{i=0}^{79} sp1^2[i]}}, \quad (5.2.4.4.6.4-2)$$

5.2.4.4.6.5 Subframe overlap

In order to smooth the boundary between subframes, $sp_c2[x]$, which is the filtered output of the first 20 samples of the current subframe by utilizing the filter coefficients for the previous subframe, and $sp_c1[x]$, which is the result in Section 5.2.4.4.6.4, are combined to make a postfilter output as described in Fig. 5.2.4.4.6.5-1, Eq. 5.2.4.4.6.5-1, and Eq. 5.2.4.4.6.5-2.

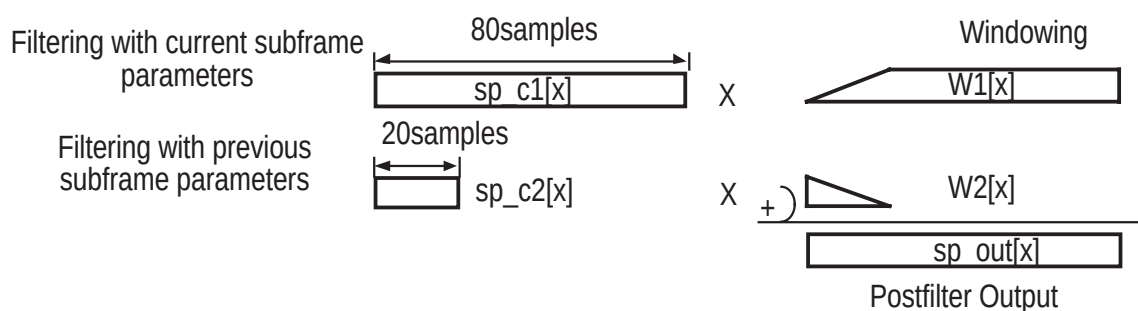


Fig.5.2.4.4.6.5-1 Subframe Overlap

$$W1[n] = \begin{cases} \frac{n+1}{21} & (0 \leq n \leq 19) \\ 1 & (20 \leq n \leq 79) \end{cases}, \quad (5.2.4.4.6.5-1)$$

$$W2[n] = \begin{cases} \frac{20-n}{21} & (0 \leq n \leq 19) \\ 0 & (20 \leq n \leq 79) \end{cases}, \quad (5.2.4.4.6.5-2)$$

$sp_c2[x]$ is obtained as follows:

$$sp_c2[x] = SC_P \cdot sp2[x] \quad (0 \leq x \leq 19), \quad (5.2.4.4.6.5-3)$$

where SC_P is the power scale factor used for the previous subframe. The transfer function of the filter to compute the $sp2[x]$ from $sq[x]$ is

$$P_P(z) \cdot B_P(z) \cdot H_P(z), \quad (5.2.4.4.6.5-4)$$

where $P_P(z)$, $B_P(z)$ and $H_P(z)$ are $P(z)$, $B(z)$ and $H(z)$ used for filtering in the previous subframe, respectively.

5.3 CS-ACELP speech CODEC

The speech coding algorithm used for this speech Codec is CS-ACELP “Conjugate-Structure Algebraic-Code-Excited Linear-Prediction”. This speech Codec is an optional function.

The CS-ACELP is based on the code-excited linear-prediction (CELP) coding model and operates on a 10ms speech frame (corresponding to 80 samples at the sampling frequency 8000 Hz.). The speech signal is analyzed for every 10ms speech frame and the parameters including linear-prediction filter coefficients, adaptive and fixed-codebook indices and gains used for the CELP model are extracted for coding of the information source. Following this, they are subject to channel coding and transmitted as 11.2kbit/s full-rate signal.

5.3.1 Speech coding processing

Follow ITU-T Recommendation G.729, “General description of the coder”, 2.1 “Encoder” for the summary of the speech encoder. Follow ITU-T Recommendation G.729, “Functional description of the encoder” for the functional description of the encoder.

5.3.1.1 Pre-processing

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.1 “Pre-processing”.

5.3.1.2 Linear prediction analysis and quantization

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.2 “Linear prediction analysis and quantization”.

5.3.1.3 Windowing and autocorrelation computation

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.2.1 “Windowing and autocorrelation computation”.

5.3.1.3.1 Levinson-Durbin algorithm

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.2.2 “Levinson-Durbin algorithm”.

5.3.1.3.2 LP to LSP conversion

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.2.3 “LP to LSP conversion”.

5.3.1.3.3 Quantization of the LSP coefficients

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.2.4 “Quantization of the LSP coefficients”.

5.3.1.3.4 Interpolation of the LSP coefficients

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.2.5 “Interpolation of the LSP coefficients”.

5.3.1.3.5 LSP to LP conversion

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.2.6 "LSP to LP conversion".

5.3.1.4 Perceptual weighting

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.3 "Perceptual weighting".

5.3.1.5 Open-loop pitch analysis

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.4 "Open-loop pitch analysis".

5.3.1.6 Computation of the impulse response

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.5 "Computation of the impulse response".

5.3.1.7 Computation of the target signal

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.6 "Computation of the target signal".

5.3.1.8 Adaptive-codebook search

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.7 "Adaptive-codebook search".

5.3.1.8.1 Generation of the adaptive-codebook vector

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.7.1 "Generation of the adaptive-codebook vector".

5.3.1.8.2 Codeword computation for adaptive-codebook delays

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.7.2 "Codeword computation for adaptive-codebook delays".

5.3.1.8.3 Computation of the adaptive-codebook gain

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.7.3 "Computation of the adaptive-codebook gain".

5.3.1.9 Fixed codebook – Structure and search

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 3.8 "Fixed codebook – Structure and search".

5.3.1.9.1 Fixed-codebook search procedure

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.8.1 “Fixed-codebook search procedure”.

5.3.1.9.2 Codeword computation of the fixed-codebook

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.8.2 “Codeword computation of the fixed-codebook”.

5.3.1.10 Quantization of the gains

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.9 “Quantization of the gains”.

5.3.1.10.1 Gain prediction

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.9.1 “Gain prediction”.

5.3.1.10.2 Codebook search for gain quantization

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.9.2 “Codebook search for gain quantization”.

5.3.1.10.3 Codeword computation for gain quantizer

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.9.3 “Codeword computation for gain quantizer”.

5.3.1.11 Memory update

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 3.10 “Memory update”

5.3.2 Channel coding

Fig.5.3.2-1 Shows the block diagram of the channel coding used for the CS-ACELP speech Codec

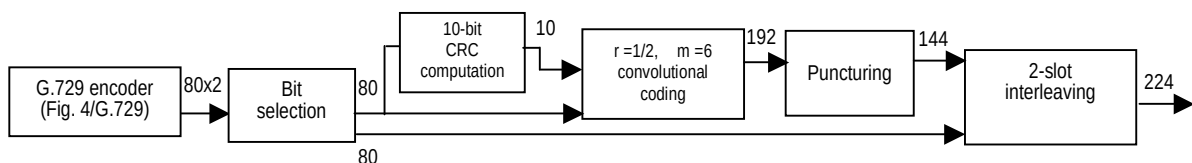


Fig.5.3.2-1 Channel Coding

5.3.2.1 Protected bit selection

The protected bits $p_n[x]$ ($0 \leq x \leq 39$) and unprotected bits $np_n[x]$ ($0 \leq x \leq 39$) are selected from the coded speech bitstream $sp_n[x]$ ($0 \leq x \leq 79$) encoded using G.729 (8kbit/s) in the n th frame. The details on the protected bits and

unprotected bits are shown in Table 5.3.2.1-1. The codewords in the table correspond to the ITU-T Recommendation G.729 and the smallest value of x is equivalent to the MSB to be transmitted first.

By using 2 frames of the selected speech bitstreams, a 20ms (160 bit) channel coding frame is constructed. The protected bits $P_m[x](0 \leq x \leq 79)$ and unprotected bits $NP_m[x](0 \leq x \leq 79)$ are defined as Eq. 5.3.2.1-1 and 5.3.2.1-2.

$$P_m[x] = p_n[x], NP_m[x] = np_n[x] \quad (n=2m) \quad (5.3.2.1-1)$$

$$P_m[x+40] = p_n[x], NP_m[x+40] = np_n[x] \quad (n=2m+1) \quad (5.3.2.1-2)$$

Table 5.3.2.1-1 Protected/unprotected bit allocation of CS-ACELP codec

x	Codeword	$sp_n[x]$	x	Codeword	$sp_n[x]$
0	<i>LO</i>	$p_n[0]$	40	<i>S1</i>	$np_n[18]$
1	<i>L1</i>	$p_n[1]$	41		$np_n[19]$
2		$p_n[2]$	42		$np_n[20]$
3		$p_n[3]$	43		$np_n[21]$
4		$p_n[4]$	44	<i>GA1</i>	$p_n[22]$
5		$p_n[5]$	45		$p_n[23]$
6		$p_n[6]$	46		$p_n[24]$
7		$p_n[7]$	47	<i>GB1</i>	$p_n[25]$
8	<i>L2</i>	$p_n[8]$	48		$p_n[26]$
9		$p_n[9]$	49		$p_n[27]$
10		$p_n[10]$	50	<i>P2</i>	$p_n[28]$
11		$p_n[11]$	51		$p_n[29]$
12	<i>L3</i>	$p_n[12]$	52		$p_n[30]$
13		$np_n[0]$	53		$p_n[31]$
14		$np_n[1]$	54	<i>C2</i>	$p_n[32]$
15		$np_n[2]$	55		$np_n[22]$
16	<i>P1</i>	$np_n[3]$	56		$np_n[23]$
17		$np_n[4]$	57		$np_n[24]$
18		$p_n[13]$	58		$np_n[25]$
19		$p_n[14]$	59		$np_n[26]$
20		$p_n[15]$	60		$np_n[27]$
21		$p_n[16]$	61		$np_n[28]$
22		$p_n[17]$	62		$np_n[29]$
23		$p_n[18]$	63		$np_n[30]$
24		$p_n[19]$	64		$np_n[31]$
25		$p_n[20]$	65		$np_n[32]$
26	<i>PO</i>	$p_n[21]$	66		$np_n[33]$
27	<i>C1</i>	$np_n[5]$	67		$np_n[34]$
28		$np_n[6]$	68		$np_n[35]$
29		$np_n[7]$	69	<i>S2</i>	$np_n[36]$
30		$np_n[8]$	70		$np_n[37]$
31		$np_n[9]$	71		$np_n[38]$
32		$np_n[10]$	72		$np_n[39]$
33		$np_n[11]$	73	<i>GA2</i>	$p_n[33]$
34		$np_n[12]$	74		$p_n[34]$
35		$np_n[13]$	75		$p_n[35]$
36		$np_n[14]$	76	<i>GB2</i>	$p_n[36]$
37		$np_n[15]$	77		$p_n[37]$
38		$np_n[16]$	78		$p_n[38]$
39		$np_n[17]$	79		$p_n[39]$

5.3.2.2 CRC coding

A 10-bit CRC code (CRC[x]) is computed from the protected bitstream P[x] by Eq.5.3.2.2-1, Eq.5.3.2.2-2, and Eq.5.3.2.2-3.

$$R(x) = \sum_{i=0}^9 \text{CRC}[i] \cdot x^i \quad (5.3.2.2-1)$$

where

$$x^{10} \cdot \sum_{i=0}^{79} P[i] \cdot x^i = Q(x) \cdot g(x) + R(x) \quad (5.3.2.2-2)$$

$$g(x) = 1 + x + x^4 + x^5 + x^9 + x^{10} \quad (5.3.2.2-3)$$

5.3.2.3 Convolutional coding

The input bitstream of the convolutional coder (Cin[x]) is constructed by Eq.5.3.2.3-1. Then it is coded by r=1/2 and m=6 convolutional coder.

$$\text{Cin}[x] = \begin{cases} \text{CRC}[2x] & (0 \leq x \leq 4) \\ P[88 - 2x] & (5 \leq x \leq 44) \\ P[2x - 89] & (45 \leq x \leq 84) \\ \text{CRC}[179 - 2x] & (85 \leq x \leq 89) \\ 0 & (90 \leq x \leq 95) \end{cases} \quad (5.3.2.3-1)$$

Generator polynomials of the convolutional coder are defined as :

$$G_1(D) = 1 + D^2 + D^3 + D^5 + D^6 \quad (5.3.2.3-2)$$

$$G_2(D) = 1 + D + D^2 + D^3 + D^6 \quad (5.3.2.3-3)$$

The output bitstream V[x] ($0 \leq x \leq 191$) of the convolutional coder is obtained from the output bitstream V₁[x] ($0 \leq x \leq 95$) from G₁(D), and the output bitstream V₂[x] ($0 \leq x \leq 95$) from G₂(D) using Eq.5.3.2.3-4.

$$V[2x] = V_1[x], V[2x+1] = V_2[x] \quad (5.3.2.3-4)$$

5.3.2.4 Puncturing and interleaving

The bitstream VP[x] ($0 \leq x \leq 143$) is created by puncturing the output bitstream V[x] ($0 \leq x \leq 191$) from the convolutional coder. The removed bits are V[x] ($x=4m+3, m=0,1,\dots,47$).

From the VP[x] and NP[x] of the current frame and the corresponding bitstream VPp[x] and NPp[x] of the previous frame, interleaving is performed as shown in Table 5.3.2.4-1 to create a 224-bit frame.

The order of transmission bits is shown in Table 5.3.2.4-1.

Table 5.3.2.4-1 : Transmission bitstream ordering

Number in order	Transmis- sion bit	Number in order	Transmis- sion bit	Number in order	Transmis- sion bit	Number in order	Transmis- sion bit	Number in order	Transmis- sion bit
0	VP _P [1]	45	VP[6]	90	NP[14]	135	VP[128]	180	VP _P [79]
1	VP[0]	46	VP[34]	91`	VP _P [54]	136	VP _P [101]	181	VP[78]
2	VP _P [29]]	47	NP _P [45]	92	NP[15]	137	NP[24]	182	VP _P [107]
3	VP[28]	48	VP[62]	93	NP _P [55]	138	VP _P [129]	183	VP[106]
4	VP[56]	49	NP _P [46]	94	NP[16]	139	NP[25]	184	VP _P [135]
5	VP _P [57]	50	VP[90]	95	VP _P [56]	140	NP _P [64]	185	VP[134]
6	VP[84]	51	NP _P [47]	96	VP _P [13]	141	NP[26]	186	NP _P [73]
7	VP _P [85]	52	VP[118]	97	VP[12]	142	NP _P [65]	187	NP _P [74]
8	VP[112]	53	VP _P [7]	98	VP _P [41]	143	VP[18]	188	NP[33]
9	VP[140]	54	VP _P [35]	99	VP[40]	144	NP _P [66]	189	NP _P [75]
10	VP _P [113]	55	NP[8]	100	VP _P [69]	145	VP _P [19]	190	NP[34]
11	NP[18]	56	VP _P [63]	101	VP[68]	146	VP[46]	191	NP[35]
12	VP _P [141]	57	NP[9]	102	VP[96]	147	VP _P [47]	192	VP[24]
13	NP[19]	58	VP _P [91]	103	VP _P [97]	148	VP[74]	193	VP[52]
14	NP _P [58]	59	NP[10]	104	VP[124]	149	VP _P [75]	194	VP _P [25]
15	VP[2]	60	VP[8]	105	VP _P [125]	150	VP[102]	195	VP[80]
16	NP _P [59]	61	VP _P [119]	106	NP[17]	151	VP _P [103]	196	VP _P [53]
17	VP[30]	62	VP[36]	107	NP[36]	152	VP _P [181]	197	VP[108]
18	VP[58]	63	NP _P [48]	108	NP _P [57]	153	VP[180]	198	NP _P [81]
19	VP _P [3]	64	VP[64]	109	NP[37]	154	NP _P [67]	199	VP[136]
20	VP[86]	65	VP[92]	110	NP _P [76]	155	NP[27]	200	NP[0]
21	VP _P [81]	66	NP _P [49]	111	VP[14]	156	NP _P [68]	201	VP _P [109]
22	VP[114]	67	NP _P [50]	112	NP _P [77]	157	NP[28]	202	NP[1]
23	VP[142]	68	VP _P [9]	113	VP[42]	158	NP[29]	203	VP _P [137]
24	VP _P [59]	69	VP[120]	114	VP _P [15]	159	VP _P [69]	204	NP[2]
25	NP[20]	70	VP _P [37]	115	VP[70]	160	VP[20]	205	NP _P [40]
26	NP _P [87]	71	NP[11]	116	VP _P [43]	161	VP _P [21]	206	VP[26]
27	NP[21]	72	VP _P [65]	117	VP _P [71]	162	VP[48]	207	NP _P [41]
28	VP _P [115]	73	NP[12]	118	VP[98]	163	VP _P [49]	208	NP _P [42]
29	VP[4]	74	VP _P [93]	119	VP _P [99]	164	VP[76]	209	VP[54]
30	VP _P [143]	75	VP _P [121]	120	VP[126]	165	VP _P [77]	210	VP _P [27]
31	VP[32]	76	VP[13]	121	NP[38]	166	VP _P [105]	211	VP[82]
32	VP[60]	77	NP _P [51]	122	NP[39]	167	VP[104]	212	VP _P [55]
33	NP _P [60]	78	VP[10]	123	VP _P [127]	168	VP _P [133]	213	VP[110]
34	VP[88]	79	VP[38]	124	NP _P [78]	169	VP[132]	214	VP _P [83]
35	NP _P [61]	80	NP _P [52]	125	NP[28]	170	NP _P [70]	215	VP _P [111]
36	VP[116]	81	NP _P [53]	126	NP _P [79]	171	NP[30]	216	VP[138]
37	VP[5]	82	VP _P [11]	127	VP[16]	172	NP _P [71]	217	VP _P [139]
38	VP _P [33]	83	VP[66]	128	NP _P [63]	173	NP _P [72]	218	NP[3]
39	NP[5]	84	VP _P [39]	129	VP[44]	174	NP[31]	219	NP _P [48]

40	VP _P [61]	85	VP[94]	130	VP _P [17]	175	VP _P [23]	220	NP _P [44]
41	NP[6]	86	VP _P [67]	131	VP _P [45]	176	NP[32]	221	NP[4]
42	VP _P [89]	87	VP[122]	132	VP[72]	177	VP _P [51]	222	NP _P [62]
43	NP[7]	88	VP _P [95]	133	VP _P [73]	178	VP[22]	223	NP[22]
44	VP _P [117]	89	VP _P [123]	134	VP[100]	179	VP[50]		

5.3.3 Channel decoding

By reverse tracing the channel coding procedure described in the preceding section, channel decoding is performed. The soft decision Viterbi algorithm or a decoding method with an equivalent or higher performance is used for decoding convolutional code.

After de-interleaving the received bitstream, it is decoded by soft decision error correction decoding. Then, 10-bit CRC is computed from the decoded bitstream in the same way as the transmission side for comparison with the received CRC. If those CRC sequences are different from each other, then it is assumed that the frame has been lost and concealment of frame erasures described in 5.3.4.3 is applied.

5.3.3.1 De-interleaving

De-interleaving is performed using Table 5.3.2.4-1 and the received bitstreams (VP_r[x] and NP_r[x]) are re-constructed.

5.3.3.2 Error correction decoding

After all data in one frame is received, error correction decoding is performed for VP_r[x] to compute the decoded bitstream P_r[x] and CRC_r[x]. In order to perform efficient error correction, the soft decision Viterbi algorithm which computes path metric using the reliability information per bit and the envelop information per symbol, or a decoding method with an equivalent or higher performance, is used. Punctured bits are not used for path metric computation.

5.3.3.3 CRC decoding

A 10-bit CRC sequence (CRC[x]) is computed from the decoded bitstream P_r[x] in the same way as that described in 5.3.2.2 and is compared to the received CRC_r[x]. If the CRC sequences are different from each other, it is assumed that the frame has been lost.

5.3.4 Speech decoding

For the summary of the speech decoder, follow ITU-T Recommendation G.729, "General description of the coder", 2.2 "Decoder". Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4 "Functional description of the decoder".

5.3.4.1 Parameter decoding procedure

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.1 "Parameter decoding procedure".

5.3.4.1.1 Decoding of LP filter parameters

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.1.1 "Decoding of LP filter parameters".

5.3.4.1.2 Computation of the parity bit

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.1.2 "Computation of the parity bit".

5.3.4.1.3 Decoding of the adaptive-codebook vector

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.1.3 "Decoding of the adaptive-codebook vector".

5.3.4.1.4 Decoding fixed-codebook vector

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.1.4 "Decoding fixed-codebook vector".

5.3.4.1.5 Decoding of the adaptive and fixed-codebook gains

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.1.5 "Decoding of the adaptive and fixed-codebook gains".

5.3.4.1.6 Computing the reconstructed speech

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.1.6 "Computing the reconstructed speech".

5.3.4.2 Post-processing

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.2 "Post-processing".

5.3.4.2.1 Long-term postfilter

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.2.1 "Long-term postfilter".

5.3.4.2.2 Short-term postfilter

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.2.2 "Short-term postfilter".

5.3.4.2.3 Tilt compensation

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.2.3 "Tilt compensation".

5.3.4.2.4 Adaptive gain control

Follow ITU-T Recommendation G.729, "Functional description of the encoder", 4.2.4 "Adaptive gain control".

5.3.4.2.5 High-pass filtering and upscaling

Follow ITU-T Recommendation G.729 “Functional description of the encoder”, 4.2.5 “High-pass filtering and upscaling”.

5.3.4.2.6 Encoder and decoder initialization

Follow ITU-T Recommendation G.729 “Functional description of the encoder”, 4.3 “Encoder and decoder initialization”.

5.3.4.3 Concealment of frame erasures

The choice of the method of the concealment of frame erasures is left free. An example is given below.

5.3.4.3.1 Lost frame counter

A variable counter which counts the number of lost voice frames is defined as Eq.5.3.4.3.1-1.

$$\begin{cases} \text{counter} + + & (\text{if a frame is lost}) \\ \text{counter} = 0 & (\text{else}) \end{cases} \quad (5.3.4.3.1-1)$$

5.3.4.3.2 Adaptive prefilter control

The pitch gain β described in ITU-T Recommendation G.729 “Functional description of the encoder”, 3.8 “Fixed codebook - structure and search”, Eq (47) is defined as Eq.5.3.4.3.2-1.

$$\begin{cases} \beta = \hat{g}_p^{(m-1)}, & \text{bounded by } 0.2 \leq \beta \leq 0.8. (\text{if } \text{counter} \leq 2) \\ \beta = 0 & (\text{else}) \end{cases} \quad (5.3.4.3.2-1)$$

5.3.4.3.3 Repetition of synthesis filter parameters

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 4.4.1 “Repetition of synthesis filter parameters”.

5.3.4.3.4 Attenuation of adaptive and fixed-codebook gains

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 4.4.2 “Attenuation of adaptive and fixed-codebook gains”.

5.3.4.3.5 Attenuation of the memory of the gain predictor

Follow ITU-T Recommendation G.729, “Functional description of the encoder”, 4.4.3 “Attenuation of the memory of the gain predictor”.

5.3.4.3.6 Generation of the replacement excitation signal

Follow ITU-T Recommendation G.729 “Functional description of the encoder”, 4.4.4 “Generation of the replacement excitation”.

5.3.4.3.7 Reconstruction of the past excitation

When the counter value changes to “0” from a positive value, the past excitation signal $u(n-40)$ ($n=0,\dots,39$) is re-constructed using pitch delay T_1 of the decoded current subframe by Eq.5.3.4.3.7-1.

$$u(n-40) = g_p^{(m-1)} v'(n) + g_c^{(m-1)} c(n) \quad (5.3.4.3.7-1)$$

The $g_p^{(m-1)}$ and $g_c^{(m-1)}$ are respectively the adaptive and fixed-codebook gains which are used in the previous subframe decoding, and $c(n)$ is the replacement excitation generated in the previous sub-frame decoding by the method described in Section 5.3.4.3.4. The $v'(n)$ is computed by Eq.5.3.4.3.7-2 which is equivalent to Eq (40) described in ITU-T Recommendation G.729, “Functional description of the encoder”, 3.7.1 “Generation of the adaptive-codebook vector”.

$$v'(n) = \sum_{i=0}^9 u(n-40-k-i) b_{30}(t+i \cdot 3) + \sum_{i=0}^9 u(n-40-k+1+i) b_{30}(3-t+i \cdot 3), \quad (5.3.4.3.7-2)$$

$$n = 0,\dots,39, \quad t = 0,1,2.$$

where k and t are respectively the integer and fractional parts of the pitch computed from T_1 .

5.3.4.3.8 Gain control of the excitation

When the counter value changes to “0” from a positive value, the adaptive and fixed-codebook gains are converted according to Eq.5.3.4.3.8-1 using adequate m and n . However, if the previous counter value is equal to or less than two, this cannot be applied.

$$\begin{cases} g'_p = m \cdot g_p \\ g'_c = n \cdot g_c \end{cases} \quad (5.3.4.3.8-1)$$

5.3.5 VOX control

The CS-ACELP speech Codec uses voice activity detection (VAD) algorithm described in ITU-T Recommendation G.729, Annex B (A silence compression scheme for G.729 optimized for terminals conforming to Recommendation V.70) and uses silence insertion description (SID) for Confort Noise Generation (follows the descriptions in ITU-T Recommendation G. 729, Annex B for details of the VAD and SID computation).

5.3.5.1 SID bitstream description

The SID bitstream is shown in Table 5.3.5.1-1. The MSB of each parameter is transmitted first.

Table 5.3.5.1-1 SID bit sequence

Parameter Description	Bits
Switched predictor index of LSF quantizer	1
First stage vector of LSF quantizer	5
Second stage vector quantizer	4
Gain (Energy)	5

5.3.5.2 Bit pattern during SID transmission

The bit pattern during SID transmission (when PST1 is transmitted) is specified as Table 5.3.5.2-1. The P[x] and NP[x] correspond respectively to the protected and unprotected described in Section 5.3.2.1 and are coded by channel coding and transmitted as 224-bit PST1.

Table 5.3.5.2-1 Bit pattern during SID Transmission

P[0] - P[14]	:	SID information bit (15 bit)
P[15]	:	1 (1)
P[16] - P[23]	:	0 (8)
P[24] - P[31]	:	1 (8)
P[32] - P[39]	:	0 (8)
P[40] - P[47]	:	1 (8)
P[48] - P[55]	:	0 (8)
P[56] - P[63]	:	1 (8)
P[64] - P[71]	:	0 (8)
P[72] - P[79]	:	1 (8)
NP[0] - NP[7]	:	0 (8)
NP[8] - NP[15]	:	1 (8)
NP[16] - NP[23]	:	0 (8)
NP[24] - NP[31]	:	1 (8)
NP[32] - NP[39]	:	0 (8)
NP[40] - NP[47]	:	1 (8)
NP[48] - NP[55]	:	0 (8)
NP[56] - NP[63]	:	1 (8)
NP[64] - NP[73]	:	0 (10)
NP[74] - NP[79]	:	1 (6)

5.3.5.3 Transmission timing of pre-amble and post-amble

The pre-amble and post-amble are transmitted at the timing shown in Fig.5.3.5-1. During the voice absence state, post-amble PST0 and PST1 are transmitted when SID(silence insertion description) becomes ready for transmission. Even if the SID is not ready for transmission, the PST0 and PST1 shall be transmitted when 1 second has passed after the transmission of the last PST0 and PST1.

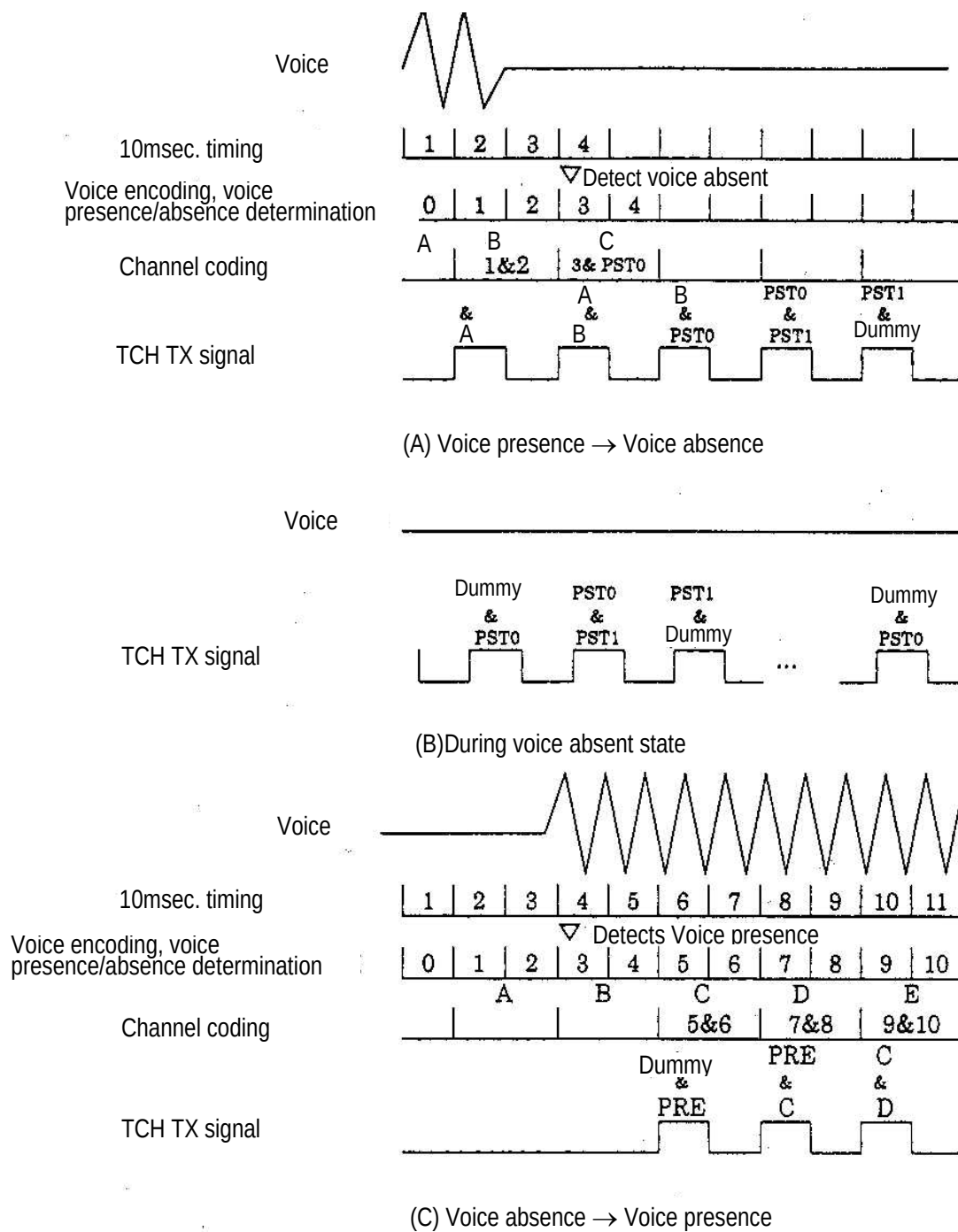


Fig.5.3.5-1 Transmission timings of pre-amble and post-amble

5.4 ACELP Speech CODEC

5.4.1 Preface

This document gives a description of the ACELP 6.7 kbit/s speech codec.

Section 5.4.2 gives a general description of the speech coder and the sections 5.4.3 and 5.4.4 give more detailed descriptions of the speech encoder and the speech decoder, respectively. Section 5.4.5 gives a detailed description of how the speech codec bits are mapped into the VSELP channel coding. The bad frame masking described in the section 5.4.6 is provided as an example solution; other solutions are allowed.

This specification consists of this document and a soft copy distribution which contain a bit-exact description in the form of a fixed point ANSI C code and a number of test vectors. The ANSI C code defines the speech codec with a set of fixed point basic operators. All implementations shall be bit-exact. The ANSI C code is described in more detail in the section 5.4.7.

In the case of discrepancy between the high level and the bit-exact descriptions, the description in the bit-exact ANSI C program will prevail. This version of the specification corresponds to version 2.0 of the bit-exact ANSI C program.

Test vectors to test bitexactness are included in the soft copy distribution of the specification. These testvectors are described in the section 5.4.8. The contents of the soft copy distribution is described in the section 5.4.9

5.4.2 General description of the speech codec

5.4.2.1 Principles of the ACELP encoder

The codec is based on the code-excited linear predictive (CELP) coding model [1]. A 10th order linear prediction (LP), or short-term, synthesis filter is used which is given by

$$H(z) = \frac{1}{\hat{A}(z)} = \frac{1}{1 + \sum_{i=1}^m \hat{a}_i z^{-i}}, \quad (5.4.2.1-1)$$

where $\hat{a}_i, i = 1, \dots, m$, are the (quantized) LP parameters, and $m = 10$ is the predictor order. The long-term, or pitch, synthesis filter is given by

$$\frac{1}{B(z)} = \frac{1}{1 - g_p z^{-T}}, \quad (5.4.2.1-2)$$

where T is the pitch delay and g_p is the pitch gain. The pitch synthesis filter is implemented using the so-called adaptive codebook approach.

The CELP speech synthesis model is shown in Figure 5.4.2.1-1. In this model, the excitation signal at the input of the short-term LP synthesis filter is constructed by adding two excitation vectors from adaptive and fixed codebooks. The speech is synthesized by feeding the two properly chosen vectors from these codebooks through the short-term synthesis filter. The optimum excitation sequence in a codebook is chosen using an analysis-by-synthesis search procedure in which the error between the original and synthesis speech is minimized according to a perceptually weighted distortion measure.

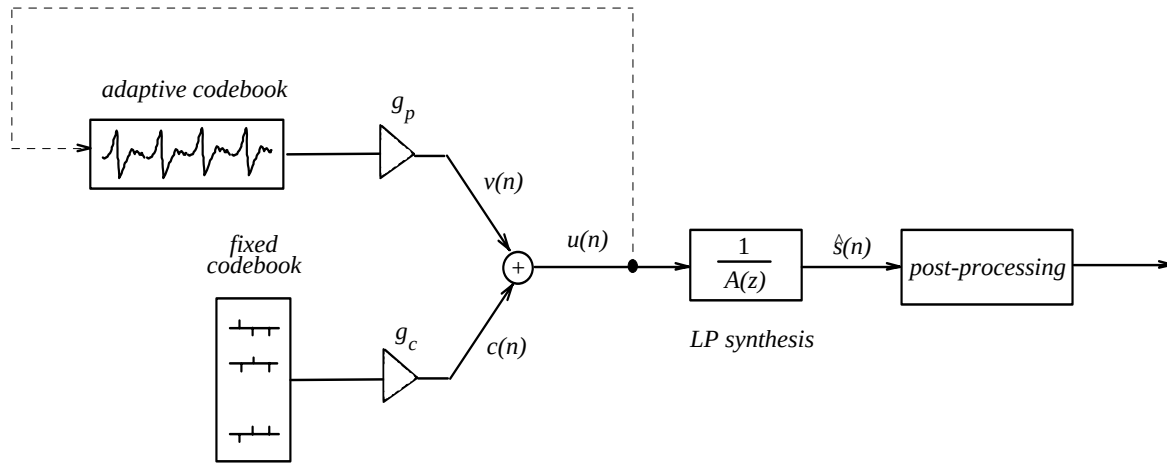


Fig. 5.4.2.1-1 Simplified block diagram of the CELP synthesis model

The perceptual weighting filter used in the analysis-by-synthesis search technique is given by

$$W(z) = \frac{A(z / \gamma_1)}{A(z / \gamma_2)}, \quad (5.4.2.1-3)$$

where $A(z)$ is the unquantized LP filter and $0 < \gamma_2 < \gamma_1 \leq 1$ are the perceptual weighting factors. The values $\gamma_1 = 0.94$ and $\gamma_2 = 0.6$ are used. The weighting filter uses the unquantized LP parameters while the formant synthesis filter uses the quantized ones.

The coder operates on speech frames of 20 ms corresponding to 160 samples at the sampling frequency of 8000 samples/sec. At each 160 speech samples, the speech signal is analyzed to extract the parameters of the CELP model (LP filter, adaptive and innovative codebooks' indices and gains). These parameters are encoded and transmitted. At the decoder, these parameters are decoded and speech is synthesized by filtering the reconstructed excitation signal through the LP synthesis filter.

The signal flow at the encoder is shown in Figure 5.4.2.1-2. The LP analysis is performed once per frame. The set of LP parameters is converted to line spectrum pairs (LSP) and vector quantized using split vector quantization (SVQ) with 26 bits. The speech frame is divided into 4 subframes of 5 ms each (40 samples). The adaptive and fixed codebook parameters are transmitted every subframe. The quantized and unquantized LP parameters are used for the fourth subframe while in the first, second, and third subframes interpolated LP parameters are used (both quantized and unquantized). An open-loop pitch lag is estimated twice per frame (every 10 ms) based on the perceptually weighted speech signal.

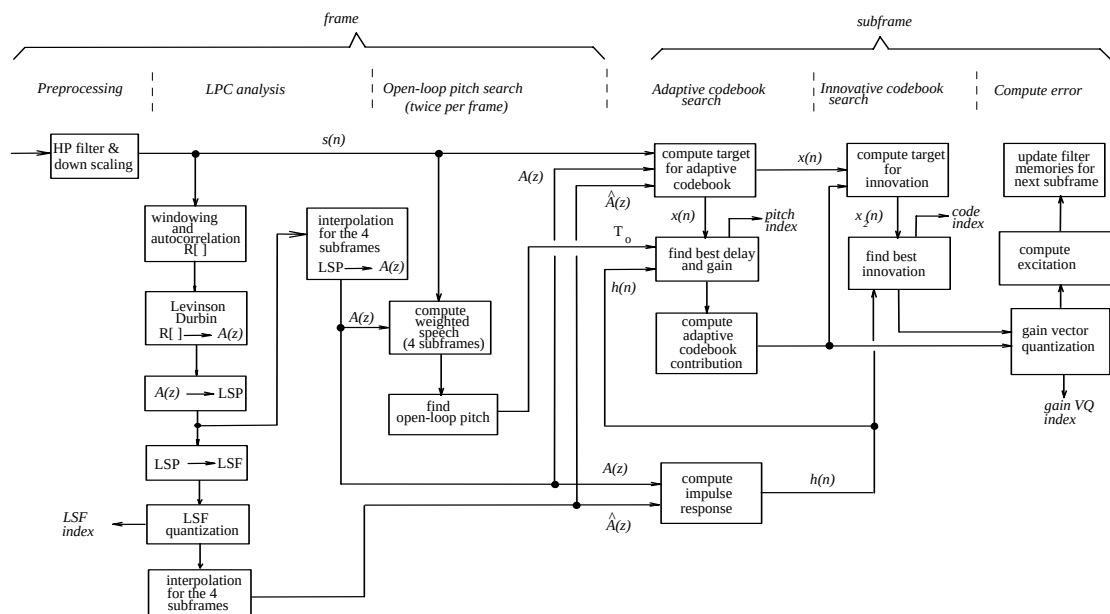


Fig. 5.4.2.1-2 Simplified block diagram of the ACELP encoder

Then the following operations are repeated for each subframe:

- The target signal $x(n)$ is computed by filtering the LP residual through the weighted synthesis filter $W(z)H(z)$ with the initial states of the filters having been updated by filtering the error between LP residual and excitation (this is equivalent to the common approach of subtracting the zero-input response of the weighted synthesis filter from the weighted speech signal).
- The impulse response, $h(n)$ of the weighted synthesis filter is computed.
- Closed-loop pitch analysis is then performed (to find the pitch lag and gain), using the target $x(n)$ and impulse response $h(n)$, by searching around the open-loop pitch lag. Fractional pitch with 1/3 resolution is used. The pitch lag is encoded with 8 bits in the first and third subframes and relatively encoded with 4 bits in the second and fourth subframes.
- The target signal $x(n)$ is updated by removing the adaptive codebook contribution (filtered adaptive codevector), and this new target, $x_2(n)$, is used in the fixed algebraic codebook search (to find the optimum innovation). An algebraic codebook with 14 bits is used for the innovative excitation.
- The gains of the adaptive and fixed codebook are vector quantized with 7 bits (with MA prediction applied to the fixed codebook gain).
- Finally, the filters memories are updated (using the determined excitation signal) for finding the target signal in the next subframe.

The bit allocation of the codec is shown in Table 5.4.2.1-1. In each 20 ms speech frame, 134 bits are produced, corresponding to a bit rate of 6.7 kbit/s.

Table 5.4.2.1-1 Bit allocation of the 6.7 kbit/s coding algorithm.

Parameter	1st & 3rd subframes	2nd & 4th subframes	total per frame
LSP set			26
Pitch delay	8	4	24
Algebraic code	14	14	56
Gain VQ	7	7	28
Total			134

5.4.2.2 Principles of the ACELP decoder

The signal flow at the decoder is shown in Figure 5.4.2.2-1. At the decoder, the transmitted indices are extracted from the received bitstream. The indices are decoded to obtain the coder parameters at each transmission frame. These parameters are the LSP vector, the 4 fractional pitch lags, the 4 innovative codevectors, and the 4 sets of vector quantized pitch and innovative gains. The LSP vector is converted to the LP filter coefficients and interpolated to obtain LP filters at each subframe. Then, at each 40-sample subframe:

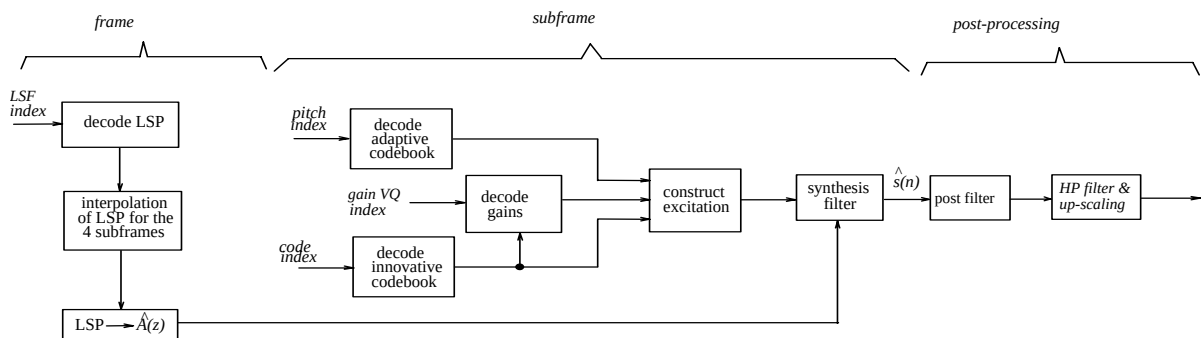


Fig. 5.4.2.2-1 Simplified block diagram of the ACELP decoder

- The excitation is constructed by adding the adaptive and innovative codevectors scaled by their respective gains.
- The speech is reconstructed by filtering the excitation through the LP synthesis filter.

Finally, the reconstructed speech signal is passed through an adaptive postfilter.

5.4.2.3 Audio Interface

Whether the input is analog or digital, the signal presented to the input of the speech codec shall be sampled at a rate of 8000 samples per second and shall be quantized to a uniform Pulse Code Modulation, PCM, format with at least 13 bits of dynamic range.

The speech encoder input and the speech decoder output assume a 16-bit integer input normalization with a range from -32,768 through +32,767. If an input audio interface uses a different normalization scheme, then appropriate scaling shall be used.

At the network side, the normalization shall be done such that a sinusoidal signal with a level of -18 dBm0 on the digital trunk network (0 dBr) has an amplitude of 21.17 dB below the maximum undistorted amplitude in the speech codec.

The following examples show how the speech samples may be processed to achieve minimum distortion of the signal.

Example 1: Linear Normalization Scheme

A linear A/D or D/A converter must provide a resolution of at least 13 bits.

The output from the A/D converter must be left-justified in 16-bit words before being input to the speech encoder.

Pseudo-Code (assuming 13-bit A/D converter):

```
AccA = <input sample> : 13-bit right-justified
AccA = AccA << 3      : 13-bit left-justified
```

At the speech decoder output, the data is rounded and right-justified to fit the resolution of the D/A converter.

Pseudo-Code (assuming 13-bit D/A converter):

```
AccA = <output sample> : 16-bit linear sample
AccA = AccA + 0x0004    : Add rounding bit
AccA = AccA & 0xFFF8    : 13-bit left-justified
AccA = AccA >> 3        : 13-bit right-justified
```

Example 2: Compressed PCM Normalization Scheme

The conversion from 8-bit μ -Law-compressed PCM into uniform PCM at the speech encoder input is performed in two steps. First, the received data shall be expanded to 14-bit linear data. Second, this data shall be placed left-justified into a 16-bit word.

Pseudo-Code:

```
AccA = <input sample> : 8-bit u-Law sample
AccA = ulawExpand(AccA) : 14-bit linear sample right-justified
AccA = AccA << 2      : 14-bit left-justified
```

At the speech decoder output, the data is rounded to 14-bit resolution and left-justified prior to feeding the data into the μ -Law compressor.

Pseudo-Code:

```
AccA = <output sample> : 16-bit linear sample
AccA = AccA + 0x0002    : Add rounding bit
AccA = AccA & 0xFFFC    : 14-bit left-justified
AccA = AccA >> 2        : 14-bit right-justified
AccA = ulawCompr(AccA) : 8-bit u-Law sample
```

5.4.3 Speech encoding

In this section we describe the different functions of the speech encoder represented in the blocks of Figure 3.

5.4.3.1 Pre-processing

Two pre-processing functions are applied prior to the encoding process: high-pass filtering and signal down-scaling.

Down-scaling consists of dividing the input by a factor of 2 to reduce the possibility of overflows in the fixed-point implementation.

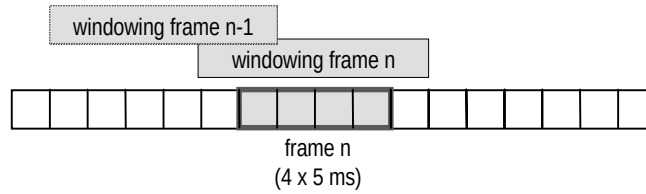
The high-pass filter serves as a precaution against undesired low frequency components. A filter at a cut off frequency of 80 Hz is used, and it is given by

$$H_{h1}(z) = \frac{0.927246093 - 1.8544941z^{-1} + 0.927246093z^{-2}}{1 - 1.906005859z^{-1} + 0.911376953z^{-2}}. \quad (5.4.3.1-1)$$

Both down-scaling and high-pass filtering are combined by dividing the coefficients at the numerator of $H_{h1}(z)$ by 2.

5.4.3.2 Linear prediction analysis and quantization

Short-term prediction, or LP, analysis is performed once per speech frame using the autocorrelation approach with 30 ms asymmetric windows. An overhead of 40 samples (5 ms) is used in the autocorrelation computation. The frame structure is depicted below.



The autocorrelations of windowed speech are converted to the LP coefficients using the Levinson algorithm. Then the LP coefficients are transformed to the LSP domain for quantization and interpolation purposes. The interpolated quantized and unquantized filters are converted back to the LP filter coefficients (to construct the synthesis and weighting filters at each subframe).

5.4.3.2.1 Windowing and autocorrelation computation

LP analysis is performed once per frame using an asymmetric window. The window has its weight concentrated at the fourth subframe and it consists of two parts: the first part is a half of a Hamming window and the second part is a quarter of a cosine function cycle. The window is given by:

$$\begin{aligned} w(n) &= 0.54 - 0.46 \cos\left(\frac{2\pi n}{2L_1 - 1}\right), & n = 0, \dots, L_1 - 1, \\ &= \cos\left(\frac{2\pi(n - L_1)}{4L_2 - 1}\right), & n = L_1, \dots, L_1 + L_2 - 1 \end{aligned} \quad (5.4.3.2.1-1)$$

where the values $L_1 = 200$ and $L_2 = 40$ are used.

The autocorrelations of the windowed speech $s'(n), n = 0, \dots, 239$, are computed by

$$r(k) = \sum_{n=k}^{239} s'(n)s'(n-k), \quad k = 0, \dots, 10, \quad (5.4.3.2.1-2)$$

and a 60 Hz bandwidth expansion is used by lag windowing the autocorrelations using the window [2]

$$w_{lag}(i) = \exp \left[-\frac{1}{2} \left(\frac{2\pi f_0 i}{f_s} \right)^2 \right], \quad i = 1, \dots, 10, \quad (5.4.3.2.1-3)$$

where $f_0 = 60$ Hz is the bandwidth expansion and $f_s = 8000$ Hz is the sampling frequency. Further, $r(0)$ is multiplied by the white noise correction factor 1.0001 which is equivalent to adding a noise floor at -40 dB.

5.4.3.2.2 Levinson-Durbin algorithm

The modified autocorrelations $r'(0) = 1.0001 r(0)$ and $r'(k) = r(k)w_{lag}(k)$, $k = 1, \dots, 10$, are used to obtain the LP filter coefficients a_k , $k = 1, \dots, 10$, by solving the set of equations.

$$\sum_{k=1}^{10} a_k r'(|i-k|) = -r'(i), \quad i = 1, \dots, 10. \quad (5.4.3.2.2-1)$$

The set of equations in (5.4.3.2.2-1) is solved using the Levinson-Durbin algorithm [3]. This algorithm uses the following recursion:

$$\begin{aligned} E(0) &= r'(0) \\ \text{For } i &= 1 \text{ to } 10 \text{ do} \\ k_i &= -\left[r'(i) + \sum_{j=1}^{i-1} a_j^{(i-1)} r'(i-j) \right] / E(i-1) \\ a_i^{(i)} &= k_i \\ \text{For } j &= 1 \text{ to } i-1 \text{ do} \\ a_j^{(i)} &= a_j^{(i-1)} + k_i a_{i-j}^{(i-1)} \\ E(i) &= (1 - k_i^2) E(i-1) \end{aligned}$$

The final solution is given as $a_j = a_j^{(10)}$, $j = 1, \dots, 10$.

The LP filter coefficients are converted to the LSP representation [4] for quantization and interpolation purposes. The conversions to the LSP domain and back to the LP filter domain are described in the next section.

5.4.3.2.3 LP to LSP conversion

The LP filter coefficients a_k , $k = 1, \dots, 10$, are converted to the LSP representation for quantization and interpolation purposes. For a 10th order LP filter, the LSPs are defined as the roots of the sum and difference polynomials

$$f_1'(z) = A(z) + z^{-11} A(z^{-1}) \quad (5.4.3.2.3-1)$$

and

$$f_2'(z) = A(z) - z^{-11} A(z^{-1}) \quad (5.4.3.2.3-2)$$

respectively. The polynomials $f_1'(z)$ and $f_2'(z)$ are symmetric and antisymmetric, respectively. It can be proven that all roots of these polynomials are on the unit circle and they alternate each other [5]. $f_1'(z)$ has a root $z = -1$ ($\omega = \pi$) and $f_2'(z)$ has a root $z = 1$ ($\omega = 0$). To eliminate these two roots, we define the new polynomials

$$f_1(z) = f_1'(z) / (1 + z^{-1}) \quad (5.4.3.2.3-3)$$

and

$$f_2(z) = f_2'(z) / (1 - z^{-1}). \quad (5.4.3.2.3-4)$$

Each polynomial has 5 conjugate roots on the unit circle $(e^{\pm j\omega_i})$, therefore, the polynomials can be written as

$$F_1(z) = \prod_{i=1,3,\dots,9} (1 - 2q_i z^{-1} + z^{-2}) \quad (5.4.3.2.3-5)$$

and

$$F_2(z) = \prod_{i=2,4,\dots,10} (1 - 2q_i z^{-1} + z^{-2}) \quad (5.4.3.2.3-6)$$

where $q_i = \cos(\omega_i)$ with ω_i being the line spectral frequencies (LSF) and they satisfy the ordering property $0 < \omega_1 < \omega_2 < \dots < \omega_{10} < \pi$. We refer to q_i as the LSPs in the cosine domain.

Since both polynomials $f_1(z)$ and $f_2(z)$ are symmetric only the first 5 coefficients of each polynomial need to be computed. The coefficients of these polynomials are found by the recursive relations (for $i = 0$ to 4)

$$\begin{aligned} f_1(i+1) &= a_{i+1} + a_{m-i} - f_1(i), \\ f_2(i+1) &= a_{i+1} - a_{m-i} + f_2(i). \end{aligned} \quad (5.4.3.2.3-7)$$

where $m = 10$ is the predictor order.

The LSPs are found by evaluating the polynomials $F_1(z)$ and $F_2(z)$ at 60 points equally spaced between 0 and π and checking for sign changes. A sign change signifies the existence of a root and the sign change interval is then divided 4 times to better track the root. The Chebyshev polynomials are used to evaluate $F_1(z)$ and $F_2(z)$ [6]. In this method the roots are found directly in the cosine domain $\{q_i\}$. The polynomials $F_1(z)$ or $F_2(z)$ evaluated at $z = e^{j\omega}$ can be written as

$$F(\omega) = 2e^{-j5\omega} C(x),$$

with

$$C(x) = T_5(x) + f(1)T_4(x) + f(2)T_3(x) + f(3)T_2(x) + f(4)T_1(x) + f(5)/2, \quad (5.4.3.2.3-8)$$

where $T_m(x) = \cos(m\omega)$ is the m th order Chebyshev polynomial, and $f(i)$, $i = 1, \dots, 5$, are the coefficients of either $F_1(z)$ or $F_2(z)$, computed using the equations in (5.4.3.2.3-7). The polynomial $C(x)$ is evaluated at a certain value of $x = \cos(\omega)$ using the recursive relation:

$$\begin{aligned} &\text{for } k = 4 \text{ down to } 1 \\ &\quad b_k = 2xb_{k+1} - b_{k+2} + f(5-k) \\ &\text{end} \\ &C(x) = xb_1 - b_2 + f(5)/2, \end{aligned}$$

with initial values $b_5 = 1$ and $b_6 = 0$. The details of the Chebyshev polynomial evaluation method are found in [6].

5.4.3.2.4 LSP to LP conversion

Once the LSPs are quantized and interpolated, they are converted back to the LP coefficient domain $\{a_k\}$. The conversion to the LP domain is done as follows. The coefficients of $F_1(z)$ or $F_2(z)$ are found by expanding equations (5.4.3.2.3-5) and (5.4.3.2.3-6) knowing the quantized and interpolated LSPs q_i , $i = 1, \dots, 10$. The following recursive relation is used to compute $f_1(i)$

```

for  $i = 1$  to  $5$ 
     $f_1(i) = -2q_{2i-1}f_1(i-1) + 2f_1(i-2)$ 
    for  $j = i-1$  down to  $1$ 
         $f_1(j) = f_1(j) - 2q_{2i-1}f_1(j-1) + f_1(j-2)$ 
    end
end

```

with initial values $f_1(0) = 1$ and $f_1(-1) = 0$. The coefficients $f_2(i)$ are computed similarly by replacing q_{2i-1} by q_{2i} .

Once the coefficients $f_1(i)$ and $f_2(i)$ are found, $F_1(z)$ and $F_2(z)$ are multiplied by $1+z^{-1}$ and $1-z^{-1}$, respectively, to obtain $F_1'(z)$ and $F_2'(z)$; that is

$$\begin{aligned} f_1'(i) &= f_1(i) + f_1(i-1), & i = 1, \dots, 5, \\ f_2'(i) &= f_2(i) - f_2(i-1), & i = 1, \dots, 5, \end{aligned} \quad (5.4.3.2.4-1)$$

Finally the LP coefficients are found by

$$\begin{aligned} a_i &= 0.5f_1'(i) + 0.5f_2'(i), & i = 1, \dots, 5, \\ a_i &= 0.5f_1'(11-i) - 0.5f_2'(11-i), & i = 6, \dots, 10. \end{aligned} \quad (5.4.3.2.4-2)$$

This is directly derived from the relation $A(z) = (F_1'(z) + F_2'(z)) / 2$, and considering the fact that $F_1'(z)$ and $F_2'(z)$ are symmetric and antisymmetric polynomials, respectively.

5.4.3.2.5 Quantization of the LSPs

The LP filter coefficients are quantized using the LSP representation in the frequency domain; that is

$$f_i = \frac{f_s}{2\pi} \arccos(q_i), \quad i = 1, \dots, 10,$$

where f_i are the LSFs in Hz $[0, 4000]$ and $f_s = 8000$ is the sampling frequency. The LSF vector is given by $\mathbf{f}^t = [f_1 \ f_2 \ \dots \ f_{10}]$, with t denoting transpose.

A 1st order MA prediction is applied, and the residual LSF vector is quantized using SVQ. The prediction and quantization are performed as follows. Let $\mathbf{z}(n)$ denote the mean-removed LSF vector at frame n . The prediction residual vector $\mathbf{r}(n)$ is given by

$$\mathbf{r}(n) = \mathbf{z}(n) - \mathbf{p}(n),$$

where $\mathbf{p}(n)$ is the predicted mean-removed LSF vector at frame n . First order moving-average (MA) prediction is used where

$$p_j(n) = \alpha_j \hat{r}_j(n-1), \quad j = 1, \dots, 10 \quad (5.4.3.2.5-1)$$

where $\hat{r}_j(n-1)$ are the quantized residual vector at the past frame and α_j are the MA prediction coefficients corresponding to the j th element of the LSF vector.

The LSF residual vector $\mathbf{r}$ is quantized using SVQ. The vector is split into 3 subvectors of dimensions 3, 3, and 4. The 3 subvectors are quantized with 8, 9, and 9 bits, respectively.

A weighted LSP distortion measure is used in the quantization process. In general, for an input LSP vector $\mathbf{f}$ and a quantized vector at index k , $\hat{\mathbf{f}}^k$, the quantization is performed by finding the index k which minimizes

$$E_k = \sum_{i=1}^{10} \left[f_i w_i - \hat{f}_i^k w_i \right]^2. \quad (5.4.3.2.5-2)$$

The weighting factors $w_i, i = 1, \dots, 10$, are given by

$$w_i = \begin{cases} 3.347 - \frac{1.547}{450} d_i, & \text{for } d_i < 450 \\ 18 - \frac{0.8}{1050} (d_i - 450), & \text{otherwise} \end{cases}$$

where $d_i = f_{i+1} - f_{i-1}$ with $f_0 = 0$ and $f_{11} = 4000$ Hz. In the quantization of each subvector, the weighting coefficients are used with their corresponding LSFs.

5.4.3.2.6 Interpolation of the LSPs

The set of quantized (and unquantized) LP parameters is used for the fourth subframe whereas the first, second, and third subframes use a linear interpolation of the parameters in the adjacent frames. The interpolation is performed on the LSPs in the $\mathbf{q}$ domain. Let $\hat{\mathbf{q}}_4^{(n)}$ be the LSP vector at the 4th subframe of the frame, and $\hat{\mathbf{q}}_4^{(n-1)}$ the LSP vector at the 4th subframe of the past frame $n-1$. The interpolated LSP vectors at the 1st, 2nd, and 3rd subframes are given by

$$\begin{aligned} \hat{\mathbf{q}}_1^{(n)} &= 0.75\hat{\mathbf{q}}_4^{(n-1)} + 0.25\hat{\mathbf{q}}_4^{(n)}, \\ \hat{\mathbf{q}}_2^{(n)} &= 0.5\hat{\mathbf{q}}_4^{(n-1)} + 0.5\hat{\mathbf{q}}_4^{(n)}, \\ \hat{\mathbf{q}}_3^{(n)} &= 0.25\hat{\mathbf{q}}_4^{(n-1)} + 0.75\hat{\mathbf{q}}_4^{(n)}. \end{aligned}$$

The same formula is used for interpolation of the unquantized LSPs. The interpolated LSP vectors are used to compute a different LP filter at each subframe (both quantized and unquantized) using the LSP to LP conversion method described in Section 5.4.3.2.4.

5.4.3.3 Open-loop pitch analysis

Open-loop pitch analysis is performed twice per frame (each 10 ms) to find two estimates of the pitch lag in each frame. This is done in order to simplify the pitch analysis and confine the closed loop pitch search to a small number of lags around the open-loop estimated lags.

Open-loop pitch estimation is based on the weighted speech signal $s_w(n)$ which is obtained by filtering the input speech signal through the weighting filter $W(z) = \frac{A(z/\gamma_1)}{A(z/\gamma_2)}$. That is, in a subframe of size L , the weighted speech is given by

$$s_w(n) = s(n) + \sum_{i=1}^{10} a_i \gamma_1^i s(n-i) - \sum_{i=1}^{10} a_i \gamma_2^i s_w(n-i), \quad n = 0, \dots, L-1. \quad (5.4.3.3-1)$$

Open-loop pitch analysis is performed as follows. In the first step, 3 maxima of the correlation

$$C_k = \sum_{n=0}^{79} s_w(n) s_w(n-k) \quad (5.4.3.3-2)$$

are found in the three delay ranges, $k = 20 \dots 39$, $40 \dots 79$, and $80 \dots 143$, respectively. Negative indices of s and s_w refer to the previous frame. The retained maxima for each range i , C_{k_i} , $i = 1$ to 3 , are normalized by dividing by $\sqrt{\sum_n s_w^2(n-k_i)}$, $i = 1$ to 3 , respectively. The normalized maxima and corresponding delays are denoted by (R_i, k_i) $i = 1$ to 3 . The winner, T_{op} , among the three normalized correlations is selected by favoring the delays in the lower ranges. That is, k_i is selected if $R_i > 0.85 R_{i+1}$. This procedure of dividing the delay range into 3 sections and favoring the lower sections is used to avoid choosing pitch multiples.

5.4.3.4 Impulse response computation

The impulse response, $h(n)$, of the weighted synthesis filter $H(z)W(z) = A(z/\gamma_1) / [\hat{A}(z)A(z/\gamma_2)]$ is computed each subframe. This impulse response is needed for the search of adaptive and fixed codebooks. The impulse response $h(n)$ is computed by filtering the vector of coefficients of the filter $A(z/\gamma_1)$ extended by zeros through the two filters $1/\hat{A}(z)$ and $1/A(z/\gamma_2)$.

5.4.3.5 Target signal computation

The target signal for adaptive codebook search is usually computed by subtracting the zero-input response of the weighted synthesis filter $H(z)W(z) = A(z/\gamma_1) / [\hat{A}(z)A(z/\gamma_2)]$ from the weighted speech signal $s_w(n)$. This is performed on a subframe basis.

An equivalent procedure for computing the target signal, which is used in this codec, is the filtering of the LP residual signal $r(n)$ through the combination of synthesis filter $1/\hat{A}(z)$ and the weighting filter $A(z/\gamma_1) / A(z/\gamma_2)$. After determining the excitation for the subframe, the initial states of these filters are updated by filtering the difference between the LP residual and excitation. The memory update of these filters is explained in Section 5.4.3.9.

The residual signal $r(n)$ which is needed for finding the target vector is also used in the adaptive codebook search to extend the past excitation buffer. This simplifies the adaptive codebook search procedure for delays less than the subframe size of 40 as will be explained in the next section. The LP residual is given by

$$r(n) = s(n) + \sum_{i=1}^{10} \hat{a}_i s(n-i), \quad n = 0, \dots, 39. \quad (5.4.3.5-1)$$

5.4.3.6 Adaptive codebook search

Adaptive codebook search is performed on a subframe basis. It consists of performing closed loop pitch search, and then computing the adaptive codevector by interpolating the past excitation at the selected fractional pitch lag.

The adaptive codebook parameters (or pitch parameters) are the delay and gain of the pitch filter. In the adaptive codebook approach for implementing the pitch filter, the excitation is repeated for delays less than the subframe length. In the search stage, the excitation is extended by the LP residual to simplify the closed-loop search.

In the first and third subframes, a fractional pitch delay is used with resolutions $1/3$ in the range $\left[19\frac{1}{3}, 84\frac{2}{3}\right]$ and integers only in the range $[85, 143]$. For the second and fourth subframes, integer pitch resolution is used in the range $[T_1 - 5, T_1 + 4]$, where T_1 is the nearest integer to the fractional pitch lag of the previous (1st or 3rd) subframe, bounded by 20...143. Additionally, a fractional resolution of $\frac{1}{3}$ is used in the range $\left[T_1 - 1\frac{2}{3}, T_1 + \frac{2}{3}\right]$.

Closed-loop pitch analysis is performed around the open-loop pitch estimates on a subframe basis. In the first (and third) subframe the range $T_{op} \pm 3$, bounded by 20...143, is searched. For the other subframes, closed-loop pitch analysis is performed around the integer pitch selected in the previous subframe, as described above. The pitch delay is encoded with 8 bits in the first and third subframes and the relative delay of the other subframes is encoded with 4 bits.

The closed loop pitch search is performed by minimizing the mean-square weighted error between the original and synthesized speech. This is achieved by maximizing the term

$$T_k = \frac{\sum_{n=0}^{39} x(n)y_k(n)}{\sqrt{\sum_{n=0}^{39} y_k(n)y_k(n)}}, \quad (5.4.3.6-1)$$

where $x(n)$ is the target signal and $y_k(n)$ is the past filtered excitation at delay k (past excitation convolved with $h(n)$). Note that the search range is limited around the open-loop pitch as explained earlier.

The convolution $y_k(n)$ is computed for the first delay in the searched range, and for the other delays, it is updated using the recursive relation

$$y_k(n) = y_{k-1}(n-1) + u(-k)h(n) \quad (5.4.3.6-2)$$

where $u(n)$, $n = -(143+11), \dots, 39$, is the excitation buffer. Note that in search stage, the samples $u(n)$, $n = 0, \dots, 39$, are not known, and they are needed for pitch delays less than 40. To simplify the search, the LP residual is copied to $u(n)$ in order to make the relation in Equation (5.4.3.6-2) valid for all delays.

Once the optimum integer pitch delay is determined, the fractions from $\frac{-2}{3}$ to $\frac{2}{3}$ with a step of $\frac{1}{3}$ around that integer are tested. The fractional pitch search is performed by interpolating the normalized correlation in Equation (5.4.3.6-1) and searching for its maximum. Once the fractional pitch lag is determined, the adaptive codebook vector $v(n)$ is computed by interpolating the past excitation signal $u(n)$ at the given phase (fraction). The interpolation is performed using two FIR filters (Hamming windowed sinc functions); one for interpolating the term in Equation (5.4.3.6-1) with the sinc truncated at ± 11 and the other for interpolating the past excitation with the sinc truncated at ± 29 . The filters have their cut-off frequency (-3 dB) at 3600 Hz in the oversampled domain.

The adaptive codebook gain is then found by

$$g_p = \frac{\sum_{n=0}^{39} x(n)y(n)}{\sum_{n=0}^{39} y(n)y(n)}, \quad \text{bounded by } 0 \leq g_p \leq 1.2, \quad (5.4.3.6-3)$$

where $y(n) = v(n) * h(n)$ is the filtered adaptive codebook vector (zero-state response of $H(z)W(z)$ to $v(n)$).

5.4.3.7 Algebraic codebook structure and search

The codebook structure is based on interleaved single-pulse permutation (ISPP) design. In this codebook, the innovation vector contains 3 non-zero pulses. All pulses can have the amplitudes $+1$ or -1 . The 40 positions in a subframe are divided into 3 tracks, where each track contains one pulse, as shown in Table 5.4.3.7-1.

Table 5.4.3.7-1 Potential positions of individual pulses in the algebraic codebook.

Pulse	Positions
i_0	0, 5, 10, 15, 20, 25, 30, 35
i_1	1, 6, 11, 16, 21, 26, 31, 36 3, 8, 13, 18, 23, 28, 33, 38
i_2	2, 7, 12, 17, 22, 27, 32, 37 4, 9, 14, 19, 24, 29, 34, 39

The first pulse position is coded with 3 bits and the second and third pulse positions with 4 bits, and the sign of the each pulse is encoded with 1 bit. This makes total of 14 bits per subframe.

The algebraic codebook is searched by minimizing the mean square error between the weighted input speech and the weighted synthesis speech. The target signal used in the closed-loop pitch search is updated by subtracting the adaptive codebook contribution. That is

$$x_2(n) = x(n) - g_p y(n), \quad n = 0, \dots, 39, \quad (5.4.3.7-1)$$

where $y(n) = v(n) * h(n)$ is the filtered adaptive codebook vector and g_p is the unquantized adaptive codebook gain.

The matrix $\mathbf{H}$ is defined as the lower triangular Toeplitz convolution matrix with diagonal $h(0)$ and lower diagonals $h(1), \dots, h(39)$, and $\mathbf{d} = \mathbf{H}^t \mathbf{x}_2$ is the correlation between the target signal $x_2(n)$ and the impulse response $h(n)$, and $\Phi = \mathbf{H}^t \mathbf{H}$ is the matrix of correlations of $h(n)$.

The elements of the vector $\mathbf{d}$ are computed by

$$d(n) = \sum_{i=n}^{39} x_2(i)h(i-n), \quad n = 0, \dots, 39, \quad (5.4.3.7-2)$$

and the elements of the symmetric matrix Φ are computed by

$$\phi(i, j) = \sum_{n=j}^{39} h(n-i)h(n-j), \quad i = 0, \dots, 39, \quad j = i, \dots, 39. \quad (5.4.3.7-3)$$

If c_k is the algebraic codevector at index k , then the algebraic codebook is searched by maximizing the term

$$T_k = \frac{(C_k)^2}{E_k} = \frac{(\mathbf{d}^t \mathbf{c}_k)^2}{\mathbf{c}_k^t \Phi \mathbf{c}_k}. \quad (5.4.3.7-4)$$

The vector $\mathbf{d}$ and the matrix Φ are computed prior to the codebook search

The algebraic structure of the codebooks allows for very fast search procedures since the innovation vector $\mathbf{c}_k$ contains only a few nonzero pulses. The correlation in the numerator of the equation (5.4.3.7-4) is given by

$$C = \sum_{i=0}^{N_p-1} a_i d(m_i) \quad (5.4.3.7-5)$$

where m_i is the position of the i th pulse, a_i is its amplitude, and $N_p = 3$ is the number of pulses.

The energy in the denominator of the equation (5.4.3.7-4) is given by

$$E = \sum_{i=0}^{N_p-1} \phi(m_i, m_i) + 2 \sum_{i=0}^{N_p-2} \sum_{j=i+1}^{N_p-1} a_i a_j \phi(m_i, m_j) \quad (5.4.3.7-6)$$

To simplify the search procedure, the pulse amplitudes are predetermined by quantizing the signal $d(n)$. This is done by setting the amplitude of a pulse at a certain position equal to the sign of $d(n)$ at that position. Before the codebook search, the following steps are done. First, the signal $d(n)$ is decomposed into two parts: its absolute value $|d(n)|$ and its sign $\text{sign}[d(n)]$. Second, the matrix Φ is modified by including the sign information; that is,

$$\phi'(i, j) = \text{sign}[d(i)] \text{sign}[d(j)] \phi(i, j), \quad i = 0, \dots, 39, j = i + 1, \dots, 39. \quad (5.4.3.7-7)$$

The correlation in the equation (5.4.3.7-5) is now given by

$$C = \sum_{i=0}^{N_p-1} |d(m_i)| \quad (5.4.3.7-8)$$

and the energy in the equation (5.4.3.7-6) is given by

$$E = \sum_{i=0}^{N_p-1} \phi'(m_i, m_i) + 2 \sum_{i=0}^{N_p-2} \sum_{j=i+1}^{N_p-1} \phi'(m_i, m_j) \quad (5.4.3.7-9)$$

Having preset the pulse amplitudes, as explained above, the optimal pulse positions are determined using an efficient non-exhaustive analysis-by-synthesis search technique. In this technique, the term in the equation (5.4.3.7-4) is tested for a small percentage of position combinations.

A special feature incorporated in the codebook is that the selected codevector is filtered through an adaptive prefilter $F(z)$ which enhances special spectral components in order to improve the synthesis speech quality. Here the filter $F(z) = 1 / (1 - \beta z^{-T})$ is used, where T is the integer part of the pitch lag and β is a pitch gain. β is given by the quantized pitch gain, $\hat{g}_p$, from the previous subframe bounded by $[0.0, 0.8]$. Note that prior to the codebook search, the impulse response $h(n)$ must include the prefilter $F(z)$. That is, $h(n) \leftarrow h(n) + \beta h(n - T)$.

5.4.3.8 Quantization of the gains

The adaptive codebook gain (pitch gain) and the fixed (algebraic) codebook gain are vector quantized using a 7-bit codebook.

The fixed codebook gain quantization is performed using MA prediction with fixed coefficients. The 4th order MA prediction is performed on the innovation energy as follows. Let $E(n)$ be the mean-removed innovation energy (in dB) at subframe n , and given by

$$E(n) = 10 \log \left(\frac{1}{N} g_c^2 \sum_{i=0}^{N-1} c^2(i) \right) - \bar{E} \quad (5.4.3.8-1)$$

where $N = 40$ is the subframe size, $c(i)$ is the fixed codebook excitation, and $\bar{E} = 28.75$ is the mean of the innovation energy. The predicted energy is given by

$$\tilde{E}(n) = \sum_{i=1}^4 b_i \hat{R}(n-i) \quad (5.4.3.8-2)$$

where $[b_1 \ b_2 \ b_3 \ b_4] = [0.68 \ 0.58 \ 0.34 \ 0.19]$ are the MA prediction coefficients, and $\hat{R}(k)$ is the quantized energy prediction error at subframe k . The predicted energy is used to compute a predicted fixed-codebook gain g_c' as in the equation (5.4.3.8-1) (by substituting $E(n)$ by $\tilde{E}(n)$ and g_c by g_c'). This is done as follows. First, the mean innovation energy is found by

$$E_i = 10 \log \left(\frac{1}{N} \sum_{i=0}^{N-1} c^2(i) \right) \quad (5.4.3.8-3)$$

and then the predicted gain g_c' is found by

$$g_c' = 10^{0.05(\tilde{E}(n) + \bar{E} - E_i)}. \quad (5.4.3.8-4)$$

A correction factor between the gain g_c and the estimated one g_c' is given by

$$\gamma = g_c / g_c'. \quad (5.4.3.8-5)$$

Note that the prediction error is given by

$$R(n) = E(n) - \tilde{E}(n) = 20 \log(\gamma). \quad (5.4.3.8-6)$$

The pitch gain, g_p , and correction factor γ are jointly vector quantized using a 7-bit codebook. The gain codebook search is performed by minimizing the mean-square of the weighted error between original and reconstructed speech which is given

$$E = x^t x + g_p^2 y^t y + g_c^2 z^t z - 2g_p x^t y - 2g_c x^t z + 2g_p g_c y^t z, \quad (5.4.3.8-7)$$

where the x is the target vector, y is the filtered adaptive codebook vector, and z is the filtered fixed codebook vector. Each gain vector in the codebook also has an element representing the quantized energy prediction error. The one associated with the chosen gains is used to update $\hat{R}(n)$. $\hat{R}(n)$ is related to the variable `past_qua_en` in the C-code.

5.4.3.9 Memory update

An update of the states of the synthesis and weighting filters is needed in order to compute the target signal in the next subframe.

After the two gains have been quantized, the excitation signal, $u(n)$, in the present subframe is found by

$$u(n) = \hat{g}_p v(n) + \hat{g}_c c(n), \quad n = 0, \dots, 39, \quad (5.4.3.9-1)$$

where $\hat{g}_p$ and $\hat{g}_c$ are the quantized adaptive and fixed codebook gains, respectively, $v(n)$ the adaptive codebook vector (interpolated past excitation), and $c(n)$ is the fixed codebook vector (algebraic code including pitch sharpening). The states of the filters can be updated by filtering the signal $r(n) - u(n)$ (difference between residual and excitation) through the filters $1/\hat{A}(z)$ and $A(z/\gamma_1)/A(z/\gamma_2)$ for the 40 sample subframe and saving the states of the filters. This would require 3 filterings. A simpler approach which requires only one filtering is as follows. The local synthesis speech, $\hat{s}(n)$, is computed by filtering the excitation signal through $1/\hat{A}(z)$. The output of the filter due to the input $r(n) - u(n)$ is equivalent to $e(n) = s(n) - \hat{s}(n)$. So the states of the synthesis filter $1/\hat{A}(z)$ are given by $e(n)$, $n = 30, \dots, 39$. Updating the states of the filter $A(z/\gamma_1)/A(z/\gamma_2)$ can be done by filtering the error signal $e(n)$ through this filter to find the perceptually weighted error $e_w(n)$. However, the signal $e_w(n)$ can be equivalently found by

$$e_w(n) = x(n) - \hat{g}_p y(n) - \hat{g}_c z(n). \quad (5.4.3.9-2)$$

Since the signals $x(n)$, $y(n)$, and $z(n)$ are available, the states of the weighting filter are updated by computing $e_w(n)$ as in the equation (5.4.3.9-2) for $n = 30, \dots, 39$. This saves two filterings.

5.4.3.10 Vox processing

According to the section 4.1.20 a mobile station shall have a VOX function. The VOX function includes a Voice Activity Detector (VAD) that detects voice presence/absence. The voice activity detector is not specified in the section 4.1.20 and it is not specified in this specification.

In the case of voice absence, the speech encoder in the mobile station shall encode the information that is needed, denoted PST1, for generating comfort noise in the base station. This information consists of three parameters, the same 26-bit LSP set as is used when voice is present, a 3-bit index for the LSP prediction initialization and a 6-bit energy parameter. The PST1 frame contains the same number of bits as a normal speech frame and it is protected with the normal full rate channel coding. The 26-bit LSP set is transmitted in the same bit positions as in the case of voice present. The 6 bit frame energy is placed in the 6 MSB:s of the 1:st subframe adaptive codebook index parameter. The 3-bits representing the initial value for the LSP prediction are placed in the 3 MSB:s of the 3:rd subframe adaptive codebook index parameter. All other bits are set to zero.

5.4.3.10.1 VOX handler

The VOX handler controls the generation of the VOX parameters in the PST1 frame. It also controls the transmission of the POST and PRE frames defined in the section 4.1.20 as well as controlling the TCH transmission on and off. Input to the VOX handler is the voice presence/absence decision from the voice activity detector.

The purpose of the VOX handler is to control the timing of the calculation of the PST1 parameters and the synchronization of this timing with the POST (PST0 and PST1) and PRE frames.

A signal `sp_flag` is generated from the voice presence/absence signal. The `sp_flag` signal controls the VOX handler state machine. The transmitted frame type (Normal Speech, PST0, PST1, PRE or No Transmission) is controlled by the state machine.

When the voice activity detection signals a voice absence segment the `sp_flag` stays in the voice mode (`sp_flag` = 1) for an additional 8 frames to ensure that the PST1 parameters are derived from a voice absence segment. After this a PST0 and PST1 frame is generated once every second as long as the voice activity detector signals voice absence. Figure 5.4.3.10.1-1 shows the timing diagram of the `sp_flag` signal and the associated frame types.

In order to improve performance in conditions where the voice activity detector toggles between voice presence and voice absence the following additions have been made:

- If a speech present segment is shorter than 20 frames the `sp_flag` will not stay in the voice mode for the additional 8 frames but it will directly switch to the no voice mode (`sp_flag` = 0). In this case no new parameter set for the PST1 frame will be calculated, instead the parameters from the previous PST1 frame will be transmitted.
- A second set of PST0 and PST1 frames are transmitted 8 frames after the first PST0 and PST1 frames when a new voice absence segment starts.

The update rate of the PST0 and PST1 frames in the fixed point C-code is once every 50:th frame, i.e. once every second. The update rate may be changed according to the section 4.1.20.

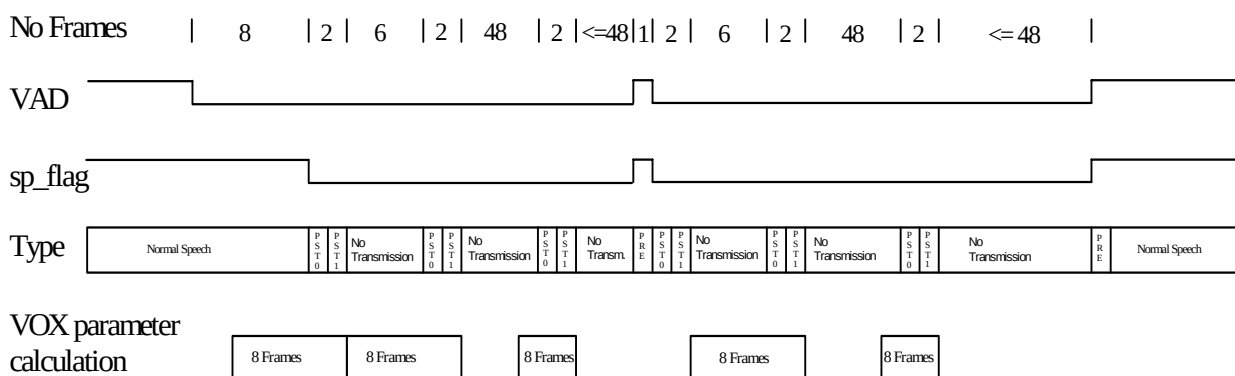


Fig 5.4.3.10.1-1 VOX handler timing

5.4.3.10.2 LSP

The comfort noise parameters to be encoded into a PST1 frame are calculated over 8 consecutive frames marked with voice absence, as follows:

The averaged LSF parameter vector of the frame i , $\mathbf{f}^{mean}(i)$, shall be computed according to the equation:

$$\mathbf{f}^{mean}(i) = \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{f}(i-n) \quad (5.4.3.10.2-1)$$

where:

$\mathbf{f}(i)$ is the unquantized LSF parameter vector of the current frame i ($n = 0$);

$\mathbf{f}(i-n)$ is the unquantized LSF parameter vector of the previous frames ($n = 1, \dots, 7$);

i is the frame index;

N is the number of averaging frames = 8;

The averaged LSF parameter vector $\mathbf{f}^{mean}(i)$ of the frame i is quantized using the same quantization tables as for the non-averaged LSF parameter vectors in ordinary speech encoding mode but the quantization algorithm is slightly modified. The parameter $\hat{r}_j(n-1)$ in the equation (5.4.3.2.5-1) is initialized from a lookup table containing 8 vectors with indices corresponding to subvectors of the SVQ. The predictor coefficients, α_j in the equation (5.4.3.2.5-1), are all set to one. The quantization process includes a search for the best of these 8 vectors. The vector which yields the lowest prediction error is selected.

After the above step the LSF parameter encoding procedure is performed as normal. A 3-bit index is used for the selected $\hat{r}_j(n-1)$ initialization vector.

5.4.3.10.3 Frame energy

The frame energy is computed for each frame marked with voice absence according to the equation :

$$en_{\log}(i) = \frac{1}{2} \log_2 \left(\frac{1}{N_1} \sum_{n=0}^{N_1-1} s_n^2 \right)$$

where $N_1 = 160$ is the number of samples in a frame. The parameter $en_{\log}(i)$ represent the logarithmic RMS level of the input signal of frame i .

The averaged logarithmic energy is computed from

$$en_{mean} = \frac{1}{N} \sum_{n=0}^N en_{\log}(i-n)$$

where $N = 8$ is the number of frames. The averaged logarithmic energy is quantized by a 6-bit algorithmic quantizer according to:

$$en_{index} = \max(0, \min(63, \lfloor (en_{mean} + 2.5) * 4 + 0.5 \rfloor))$$

The index 0 is assumed to represent a level of zero.

The quantized value of en_{mean} is used for initialization of all four old values of the quantized energy prediction error in subframe k , $\hat{R}(k)$, see the equation (5.4.3.8-2). The initialization is done every PST1 frame.

5.4.3.11 Instability protection

The purpose of the instability protection is to prevent such divergence to occur in the contents of the adaptive codebook memories in the encoder and decoder that would make the decoder unstable. Typically, this divergence is due to channel errors. The speech codec allows pitch gains greater than unity which may result the pitch filter to become temporarily unstable. In the encoder, the stability is controlled by the analysis-by-synthesis loop. However, incorrect contents of adaptive codebook memory in the decoder may result the decoder to become unstable for

highly correlated continuous (stationary) signals which keep the pitch gain continuously in a high value. A typical signal having these characteristics is a sinusoidal tone or a combination of tones.

The instability protection detects possible problem conditions by monitoring the occurrence of resonance in LPC-spectrum in the encoder. If a resonance is found in a several consecutive frames, periodicity detection is activated. The periodicity detection calculates the average pitch gain of the present and seven previous subframes, and limits the present pitch gain below unity if a threshold is exceeded. This guarantees the stability of the pitch synthesis filter.

In the speech decoder, an additional safety feature is used. Possible overflows in the LPC synthesis filter are monitored and, if necessary, the adaptive codebook memory is scaled down by a factor of 4.

5.4.3.11.1 Monitoring a Resonance in LPC Spectrum

The monitoring of a resonance signal is made in LSP domain using the unquantized LSPs $q_i, i = 1, \dots, 10$. The LSPs are available after LP to LSP conversion presented in the section 5.4.3.2.3. The values of LSPs are in range -1.0 to 1.0 corresponding to frequency range 4000 to 0 Hz. In the following, all the values referring to LSPs or distances between adjacent LSPs are presented in Q15 format as they are presented in the fixed point C-source code.

The algorithm utilizes the fact that LSPs are closely located around a peak in the spectrum. First, two minimum distances between adjacent LSPs are searched in two sections:

$$\begin{aligned} dist_1 &= \min(q_i - q_{i+1}), \quad i = 4, \dots, 8 \\ dist_2 &= \min(q_i - q_{i+1}), \quad i = 2, 3 \end{aligned}$$

Either of the minimum distances conditions must be fulfilled to classify the frame to resonance frame. The counter calculating consecutive resonance frame is updated according to the decision.

$$\begin{aligned} & \text{if } (dist_1 < TH_1) \quad \text{OR} \quad \text{if } (dist_2 < TH_2) \\ & \quad counter = counter + 1 \\ & \text{else} \\ & \quad counter = 0 \end{aligned}$$

A fixed threshold distance, $TH_1 = 1500$, is used for the first section. For the second section, the threshold distance depends on the location of q_2 .

$$TH_2 = \begin{cases} 600, & q_2 > 32000 \\ 800, & 30500 < q_2 \leq 32000 \\ 1100, & \text{otherwise} \end{cases}$$

12 consecutive resonance frames are needed to set the *LSP_flag* which indicates possible problem condition. Otherwise, *LSP_flag* is cleared and the instability protection algorithm is completed

$$\begin{aligned} & \text{if } (counter \geq 12) \\ & \quad counter = 12 \\ & \quad LSP_flag = 1 \\ & \text{else} \\ & \quad LSP_flag = 0 \end{aligned}$$

5.4.3.11.2 Pitch Gain Control

In case that 12 consecutive resonance frames are found ($LSP_flag = 1$) and the present pitch gain exceeds the gain threshold $GP_{th} = 0.95$ (15565 in Q14), the average pitch gain is calculated over the present unquantized pitch gain and quantized pitch gains of the seven previous subframes

$$GP_{ave} = \text{mean}\{g_p(n), \hat{g}_p(n-1), \hat{g}_p(n-2), \dots, \hat{g}_p(n-7)\},$$

where the n refers to the present subframe. The average pitch gain calculation can be done after calculating the pitch gain of the present subframe as presented in the section 5.4.3.6. If the average pitch gain exceeds the GP_{th} the gain is limited to the threshold value and the GpC_flag is set to indicate the limitation.

```

if ( $GP_{ave} > GP_{th}$ )
     $g_p = GP_{th}$ 
     $GpC\_flag = 1$ 
else
     $GpC\_flag = 0$ 

```

If the GpC_flag is set, the limitation of the pitch gain is taken into consideration in the gain Vector Quantization (section 5.4.3.8) where the codebook search range is limited to include only pitch gain values less than GP_{th} .

```

if ( $GpC\_flag = 1$ )
     $VQsearch\_range = 96$ 
else
     $VQsearch\_range = 128$ 

```

After the gain Vector Quantization, the buffer of past quantized pitch gains is updated always regardless the value of GpC_flag .

$$\hat{g}_p(n-i) = \hat{g}_p(n-i+1), \quad i = 7, \dots, 1$$

5.4.4 Speech decoding

The function of the decoder consists of decoding the transmitted parameters (LP parameters, adaptive codebook vectors, adaptive codebook gains, fixed codebook vectors, fixed codebook gains) and performing synthesis to obtain the reconstructed speech. The reconstructed speech is then postfiltered and upsampled. The signal flow at the decoder is shown in Figure 5.4.2.2-1.

5.4.4.1 Decoding and speech synthesis

The decoding process is performed in the following order:

Decoding of LP filter parameters: The received indices of LSP quantization are used to reconstruct the quantized LSP vector. The interpolation described in the section 5.4.3.2.6 is performed to obtain 4 interpolated LSP vectors (corresponding to 4 subframes). For each subframe, the interpolated LSP vector is converted to the LP filter coefficient domain a_k , which is used for synthesizing the reconstructed speech in the subframe.

The following steps are repeated for each subframe:

(1) Decoding of the adaptive codebook vector: The received pitch index (adaptive codebook index) is used to find the integer and fractional parts of the pitch lag. The adaptive codebook vector $v(n)$ is found by interpolating the past excitation $u(n)$ (at the pitch delay) using the FIR filter described in the section 5.4.3.6.

(2) Decoding of the innovative vector: The received algebraic codebook index is used to extract the positions and amplitudes (signs) of the excitation pulses and to find the algebraic codevector $c(n)$. If the integer part of the pitch lag is less than the subframe size 40, the pitch sharpening procedure is applied which translates into modifying $c(n)$ by $c(n) \leftarrow c(n) + \beta c(n-T)$, where β is the decoded pitch gain from the previous subframe, $\hat{g}_p$, bounded by [0.0,0.8].

(3) Decoding of the adaptive and innovative codebook gains: The received index gives the fixed codebook gain correction factor $\hat{\gamma}$. The estimated fixed codebook gain g'_c is found as described in the section 5.4.3.8. First, the predicted energy for every subframe n is found by

$$\tilde{E}(n) = \sum_{i=1}^4 b_i \hat{R}(n-i) \quad (5.4.4.1-1)$$

and then the mean innovation energy is found by

$$E_i = 10 \log \left(\frac{1}{N} \sum_{i=0}^{N-1} c^2(i) \right) \quad (5.4.4.1-2)$$

The predicted gain g'_c is found by

$$g'_c = 10^{0.05(\tilde{E}(n) + \bar{E} - E_i)} \quad (5.4.4.1-3)$$

The quantized fixed codebook gain is given by

$$\hat{g}_c = \hat{\gamma} g'_c. \quad (5.4.4.1-4)$$

The received index is used to readily find the quantified adaptive codebook gain, $\hat{g}_p$ from the quantization table.

(4) Smoothing of the fixed codebook gain: An adaptive smoothing of the fixed codebook gain is performed to avoid unnatural fluctuations in the energy contour. The smoothing is based on a measure of the stationarity of the short-term spectrum in the q domain. The smoothing strength is computed from this measure. An averaged q -value is computed for each frame n by:

$$\bar{q}(n) = 0.84 \cdot \bar{q}(n-1) + 0.16 \cdot \hat{q}_4(n) \quad (5.4.4.1-5)$$

For each subframe m , a difference measure between the averaged vector and the quantized and interpolated vector is computed by:

$$diff_m = \sum_j \sum_m \frac{|\bar{q}^{(j)}(n) - \hat{q}_m^{(j)}(n)|}{\bar{q}^{(j)}(n)}, \quad (5.4.4.1-6)$$

where j runs over the 10 LSPs. Furthermore, a smoothing factor, k_m , is computed by:

$$k_m = \min(K_2, \max(0, diff_m - K_1)) / K_2, \quad (5.4.4.1-7)$$

where the constants are set to $K_1 = 0.4$ and $K_2 = 0.25$. A hangover period of 40 subframes is used where the k_m -value is set 1.0 if the $diff_m$ has been above 0.65 for 10 consecutive frames. A value of 1.0 corresponds to no smoothing. An averaged fixed codebook gain value is computed for each subframe by:

$$\bar{g}(m) = \frac{1}{5} \sum_{i=0}^4 \hat{g}_c(m-i). \quad (5.4.4.1-8)$$

The fixed codebook gain used for synthesis is now replaced by a smoothed value given by:

$$\hat{g}_c := \hat{g}_c \cdot k_m + \bar{g} \cdot (1 - k_m). \quad (5.4.4.1-9)$$

(5) Anti-sparseness processing: An adaptive anti-sparseness post-processing procedure is applied to the fixed codebook vector in order to reduce perceptual artifacts arising from the sparseness of the algebraic fixed codebook vectors with only a few non-zero samples per subframe. The anti-sparseness processing consists of circular convolution of the fixed codebook vector with an impulse response. Three pre-stored impulse responses are used and a number $impNr = 0,1,2$ is set to select one of them. A value of 2 corresponds to no modification, a value of 1 corresponds to medium modification, while a value of 0 corresponds to strong modification. The selection of the impulse response is performed adaptively from the adaptive and fixed codebook gains. The following procedure is employed:

```

IF  $\hat{g}_p < 0.6$  THEN  $impNr = 0$ ;
ELSE IF  $\hat{g}_p < 0.9$  THEN  $impNr = 1$ ;
ELSE  $impNr = 2$ ;

```

Detection of onset is done by comparing the fixed codebook gain to the previous codebook gain. If the current value is more than twice the previous value an onset is detected.

If an onset is not detected and $impNr = 0$, the median filtered value of the current and the previous four adaptive codebook gains are computed. If this value is less than 0.6, $impNr$ is set to 0.

If an onset is not detected, the $impNr$ -value is restricted to increase by one step from the previous subframe.

If an onset is detected, the $impNr$ -value is increased by one if it is less than 2.

(6) Computing the reconstructed speech: The following steps are for $n = 0, \dots, 39$. The total excitation is constructed by:

$$u(n) = \hat{g}_p v(n) + \hat{g}_c c(n), \quad . \quad (5.4.4.1-10)$$

Before the speech synthesis, a post-processing of excitation elements is performed. This means that the total excitation is modified by emphasizing the contribution of the adaptive codebook vector:

$$\hat{u}(n) = \begin{cases} u(n) + 0.5\beta_c \hat{g}_p v(n), & \hat{g}_p > 0.5 \\ u(n), & \hat{g}_p \leq 0.5 \end{cases} \quad (5.4.4.1-11)$$

where β_c is the decoded pitch gain of the current subframe, $\hat{g}_p$, bounded by [0.0,0.8]. Adaptive gain control (AGC) is used to compensate for the gain difference between the nonemphasized excitation $u(n)$ and emphasized excitation $\hat{u}(n)$. The gain scaling factor γ for the emphasized excitation is computed by:

$$\gamma = \begin{cases} \sqrt{\frac{\sum_{n=0}^{39} u^2(n)}{\sum_{n=0}^{39} \hat{u}^2(n)}}, & \hat{g}_p > 0.5 \\ 1.0 & \hat{g}_p \leq 0.5 \end{cases} \quad (5.4.4.1-12)$$

The gain-scaled emphasized excitation signal $\hat{u}'(n)$ is given by

$$\hat{u}'(n) = \hat{u}(n)\gamma \quad (5.4.4.1-13)$$

The reconstructed speech for the subframe of size 40 is given by

$$\hat{s}(n) = \hat{u}'(n) - \sum_{i=1}^{10} \hat{a}_i \hat{s}(n-i), \quad (5.4.4.1-14)$$

where $\hat{a}_i$ are the interpolated LP filter coefficients.

The synthesis speech $\hat{s}(n)$ is then passed through an adaptive postfilter which is described in the following section.

5.4.4.2 Instability protection

In the speech decoder, an additional protection is used by monitoring overflows in the synthesis filter (equation (5.4.4.1-14)). In case of overflow(s), the whole adaptive codebook memory, $v(n)$, $n = -(143+11), \dots, 39$ in Equation (5.4.4.1-10), is scaled down by a factor of 4, and the the synthesis filtering is repeated using this down-scaled memory. If the down-scaling has occurred, the post-processing of the excitation signal in the equations (5.4.4.1-11) – (5.4.4.1-13) is by-passed, and the synthesis filtering in the equation (5.4.4.1-14) is performed for the down-scaled excitation $u(n)$ instead of the emphasized excitation $\hat{u}'(n)$.

5.4.4.3 Post-processing

Post-processing consists of three functions: adaptive postfiltering, high-pass filtering and signal up-scaling.

5.4.4.3.1 Adaptive postfiltering

The adaptive postfilter is the cascade of two filters: a formant postfilter, and a tilt compensation filter. The postfilter is updated every subframe of 5 ms.

The formant postfilter is given by

$$H_f(z) = \frac{\hat{A}(z / \gamma_n)}{\hat{A}(z / \gamma_d)}, \quad (5.4.4.3.1-1)$$

where $\hat{A}(z)$ is the received quantized (and interpolated) LP inverse filter (LP analysis is not performed at the decoder), and the factors γ_n and γ_d control the amount of the formant postfiltering.

Finally, the filter $H_t(z)$ compensates for the tilt in the formant postfilter $H_f(z)$ and is given by

$$H_t(z) = (1 - \mu z^{-1}), \quad (5.4.4.3.1-2)$$

where $\mu = \gamma_t k_1$ is a tilt factor, with k_1 being the first reflection coefficient calculated on the truncated impulse response, $h_f(n)$, of the filter $\hat{A}(z / \gamma_n) / \hat{A}(z / \gamma_d)$. k_1 is given by

$$k_1 = \frac{r_h(1)}{r_h(0)}; \quad r_h(i) = \sum_{j=0}^{L_h-i-1} h_f(j) h_f(j+i), \quad (L_h = 22, \quad i = 0, 1). \quad (5.4.4.3.1-3)$$

The postfiltering process is performed as follows. First, the synthesis speech $\hat{s}(n)$ is inverse filtered through $\hat{A}(z / \gamma_n)$ to produce the residual signal $\hat{r}(n)$. The signal $\hat{r}(n)$ is filtered by the synthesis filter $1 / \hat{A}(z / \gamma_d)$. Finally, the signal at the output of the synthesis filter $1 / \hat{A}(z / \gamma_d)$ is passed to the tilt compensation filter $h_t(z)$ resulting in the postfiltered synthesis speech signal $s_f(n)$.

AGC is used to compensate for the gain difference between the synthesis speech signal $\hat{s}(n)$ and the postfiltered signal $s_f(n)$. The gain scaling factor γ for the present subframe is computed by

$$\gamma = \sqrt{\frac{\sum_{n=0}^{39} \hat{s}^2(n)}{\sum_{n=0}^{39} s_f^2(n)}}. \quad (5.4.4.3.1-4)$$

The gain-scaled postfiltered signal $s'(n)$ is given by

$$s'(n) = \beta(n) s_f(n), \quad (5.4.4.3.1-5)$$

where $\beta(n)$ is updated in sample-by-sample basis and given by

$$\beta(n) = \alpha \beta(n-1) + (1 - \alpha) \gamma. \quad (5.4.4.3.1-6)$$

where α is a AGC factor given by 0.9. The adaptive postfiltering factors are given by: $\gamma_n = 0.55$, $\gamma_d = 0.7$ and $\gamma_t = 0.8$.

5.4.4.3.2 High-pass filtering and up-scaling

Two post-processing functions are applied after the decoding process: high-pass filtering and signal up-scaling.

The high-pass filter serves as a precaution against undesired low frequency components. A filter at a cut off frequency of 60 Hz is used, and it is given by

$$H_{h2}(z) = \frac{0.939819335 - 1.879638672z^{-1} + 0.939819335z^{-2}}{1 - 1.933105469z^{-1} + 0.935913085z^{-2}}. \quad (5.4.4.3.2-1)$$

Up-scaling consists of multiplying the output from the high-pass filtering by a factor of 2 in order to compensate the down-scaling at the pre-processing stage.

5.4.5 Ordering of the speech encoder bitstream

The speech codec is using the same channel coding scheme as the VSELP codec specified in the section 5.1 ("VSELP codec" hereinafter). Table 5.4.5-1 describes how the speech codec bits are mapped into the VSELP codec bits (b0 is LSB).

Table 5.4.5-1 Ordering of speech parameters for the channel encoder

PDC-FR Parameter	PDC Bit	PDC-EFR Parameter	EFR Bit	CHC Class
R0	b0	Position of pulse 3 of 1st subframe	b0	2
R0	b1	Index of 3rd LSF subvector	b6	1
R0	b2	Index of 1st LSF subvector	b3	1 CRC
R0	b3	Index of 1st LSF subvector	b1	1 CRC
R0	b4	Index of 1st LSF subvector	b0	1 CRC
LPC1	b0	VQ-gain index 2nd subframe	b0	2
LPC1	b1	Adaptive codebook index 3rd subframe	b0	1
LPC1	b2	Index of 2nd LSF subvector	b1	1 CRC
LPC1	b3	Index of 1st LSF subvector	b6	1 CRC
LPC1	b4	Index of 1st LSF subvector	b2	1 CRC
LPC2	b0	VQ-gain index 4th subframe	b0	2
LPC2	b1	VQ-gain index 3rd subframe	b0	2
LPC2	b2	Adaptive codebook index 4th subframe	b0	1
LPC2	b3	Index of 2nd LSF subvector	b0	1 CRC
LPC2	b4	Index of 1st LSF subvector	b4	1 CRC
LPC3	b0	Sign of pulse 1 of 1st subframe	b0	2
LPC3	b1	VQ-gain index 1st subframe	b5	1
LPC3	b2	Index of 2nd LSF subvector	b2	1 CRC
LPC3	b3	Index of 1st LSF subvector	b5	1 CRC
LPC4	b0	Sign of pulse 3 of 1st subframe	b0	2
LPC4	b1	Sign of pulse 2 of 1st subframe	b0	2
LPC4	b2	Index of 3rd LSF subvector	b7	1
LPC4	b3	Index of 1st LSF subvector	b7	1 CRC
LPC5	b0	Sign of pulse 3 of 2nd subframe	b0	2
LPC5	b1	Sign of pulse 2 of 2nd subframe	b0	2
LPC5	b2	Sign of pulse 1 of 2nd subframe	b0	2
LPC5	b3	Index of 2nd LSF subvector	b3	1 CRC
LPC6	b0	Sign of pulse 2 of 3rd subframe	b0	2

LPC6	b1	Sign of pulse 1 of 3rd subframe	b0	2
LPC6	b2	VQ-gain index 3rd subframe	b1	1
LPC7	b0	Sign of pulse 2 of 4th subframe	b0	2
LPC7	b1	Sign of pulse 1 of 4th subframe	b0	2
LPC7	b2	Sign of pulse 3 of 3rd subframe	b0	2
LPC8	b0	Position of pulse 1 of 1st subframe	b1	2
LPC8	b1	Position of pulse 1 of 1st subframe	b0	2
LPC8	b2	Sign of pulse 3 of 4th subframe	b0	2
LPC9	b0	Position of pulse 2 of 1st subframe	b1	2
LPC9	b1	Position of pulse 2 of 1st subframe	b0	2
LPC9	b2	Position of pulse 1 of 1st subframe	b2	2
LPC10	b0	Position of pulse 2 of 1st subframe	b3	2
LPC10	b1	Position of pulse 2 of 1st subframe	b2	2
SOFINT	b0	Index of 3rd LSF subvector	b2	1
LAG_1	b0	Adaptive codebook index 2nd subframe	b2	1 CRC
LAG_1	b1	VQ-gain index 3rd subframe	b3	1 CRC
LAG_1	b2	VQ-gain index 1st subframe	b3	1 CRC
LAG_1	b3	Adaptive codebook index 3rd subframe	b7	1 CRC
LAG_1	b4	Adaptive codebook index 1st subframe	b4	1 CRC
LAG_1	b5	Index of 2nd LSF subvector	b8	1 CRC
LAG_1	b6	Index of 2nd LSF subvector	b4	1 CRC
CODE_1	b0	Position of pulse 2 of 2nd subframe	b1	2
CODE_1	b1	Position of pulse 2 of 2nd subframe	b0	2
CODE_1	b2	Position of pulse 1 of 2nd subframe	b2	2
CODE_1	b3	Position of pulse 1 of 2nd subframe	b1	2
CODE_1	b4	Position of pulse 1 of 2nd subframe	b0	2
CODE_1	b5	Position of pulse 3 of 1st subframe	b3	2
CODE_1	b6	Position of pulse 3 of 1st subframe	b2	2
CODE_1	b7	Position of pulse 3 of 1st subframe	b1	2
CODE_1	b8	Position of pulse 2 of 2nd subframe	b2	2
GSP0_1	b0	VQ-gain index 3rd subframe	b4	1
GSP0_1	b1	VQ-gain index 4th subframe	b1	1
GSP0_1	b2	VQ-gain index 4th subframe	b4	1
GSP0_1	b3	Adaptive codebook index 2nd subframe	b0	1
GSP0_1	b4	Index of 3rd LSF subvector	b3	1
GSP0_1	b5	Index of 3rd LSF subvector	b4	1
GSP0_1	b6	Adaptive codebook index 3rd subframe	b3	1 CRC
LAG_2	b0	Adaptive codebook index 2nd subframe	b1	1 CRC
LAG_2	b1	VQ-gain index 4th subframe	b6	1 CRC
LAG_2	b2	VQ-gain index 2nd subframe	b6	1 CRC
LAG_2	b3	Adaptive codebook index 3rd subframe	b6	1 CRC
LAG_2	b4	Adaptive codebook index 1st subframe	b3	1 CRC
LAG_2	b5	Adaptive codebook index 1st subframe	b7	1 CRC
LAG_2	b6	Index of 2nd LSF subvector	b5	1 CRC
CODE_2	b0	Position of pulse 1 of 3rd subframe	b2	2
CODE_2	b1	Position of pulse 1 of 3rd subframe	b1	2
CODE_2	b2	Position of pulse 1 of 3rd subframe	b0	2
CODE_2	b3	Position of pulse 3 of 2nd subframe	b3	2
CODE_2	b4	Position of pulse 3 of 2nd subframe	b2	2
CODE_2	b5	Position of pulse 3 of 2nd subframe	b1	2
CODE_2	b6	Position of pulse 3 of 2nd subframe	b0	2
CODE_2	b7	Position of pulse 2 of 2nd subframe	b3	2
CODE_2	b8	Position of pulse 2 of 3rd subframe	b0	2
GSP0_2	b0	VQ-gain index 2nd subframe	b2	1

GSP0_2	b1	VQ-gain index 2nd subframe	b5	1
GSP0_2	b2	VQ-gain index 1st subframe	b1	1
GSP0_2	b3	VQ-gain index 1st subframe	b4	1
GSP0_2	b4	Index of 3rd LSF subvector	b8	1
GSP0_2	b5	Index of 3rd LSF subvector	b0	1
GSP0_2	b6	Adaptive codebook index 3rd subframe	b2	1 CRC
LAG_3	b0	Adaptive codebook index 4th subframe	b3	1 CRC
LAG_3	b1	VQ-gain index 4th subframe	b3	1 CRC
LAG_3	b2	VQ-gain index 2nd subframe	b3	1 CRC
LAG_3	b3	Adaptive codebook index 3rd subframe	b5	1 CRC
LAG_3	b4	Adaptive codebook index 1st subframe	b2	1 CRC
LAG_3	b5	Adaptive codebook index 1st subframe	b6	1 CRC
LAG_3	b6	Index of 2nd LSF subvector	b6	1 CRC
CODE_3	b0	Position of pulse 1 of 4th subframe	b0	2
CODE_3	b1	Position of pulse 3 of 3rd subframe	b3	2
CODE_3	b2	Position of pulse 3 of 3rd subframe	b2	2
CODE_3	b3	Position of pulse 3 of 3rd subframe	b1	2
CODE_3	b4	Position of pulse 3 of 3rd subframe	b0	2
CODE_3	b5	Position of pulse 2 of 3rd subframe	b3	2
CODE_3	b6	Position of pulse 2 of 3rd subframe	b2	2
CODE_3	b7	Position of pulse 2 of 3rd subframe	b1	2
CODE_3	b8	Position of pulse 1 of 4th subframe	b1	2
GSP0_3	b0	VQ-gain index 3rd subframe	b2	1
GSP0_3	b1	VQ-gain index 3rd subframe	b5	1
GSP0_3	b2	VQ-gain index 4th subframe	b2	1
GSP0_3	b3	VQ-gain index 4th subframe	b5	1
GSP0_3	b4	Adaptive codebook index 1st subframe	b0	1
GSP0_3	b5	Index of 3rd LSF subvector	b5	1
GSP0_3	b6	Adaptive codebook index 3rd subframe	b1	1 CRC
LAG_4	b0	Adaptive codebook index 4th subframe	b2	1 CRC
LAG_4	b1	Adaptive codebook index 2nd subframe	b3	1 CRC
LAG_4	b2	VQ-gain index 3rd subframe	b6	1 CRC
LAG_4	b3	Adaptive codebook index 3rd subframe	b4	1 CRC
LAG_4	b4	Adaptive codebook index 1st subframe	b1	1 CRC
LAG_4	b5	Adaptive codebook index 1st subframe	b5	1 CRC
LAG_4	b6	Index of 2nd LSF subvector	b7	1 CRC
CODE_4	b0	Position of pulse 3 of 4th subframe	b2	2
CODE_4	b1	Position of pulse 3 of 4th subframe	b1	2
CODE_4	b2	Position of pulse 3 of 4th subframe	b0	2
CODE_4	b3	Position of pulse 2 of 4th subframe	b3	2
CODE_4	b4	Position of pulse 2 of 4th subframe	b2	2
CODE_4	b5	Position of pulse 2 of 4th subframe	b1	2
CODE_4	b6	Position of pulse 2 of 4th subframe	b0	2
CODE_4	b7	Position of pulse 1 of 4th subframe	b2	2
CODE_4	b8	Position of pulse 3 of 4th subframe	b3	2
GSP0_4	b0	VQ-gain index 2nd subframe	b1	1
GSP0_4	b1	VQ-gain index 2nd subframe	b4	1
GSP0_4	b2	VQ-gain index 1st subframe	b0	1
GSP0_4	b3	VQ-gain index 1st subframe	b2	1
GSP0_4	b4	Adaptive codebook index 4th subframe	b1	1
GSP0_4	b5	Index of 3rd LSF subvector	b1	1
GSP0_4	b6	VQ-gain index 1st subframe	b6	1 CRC

5.4.6 Example solution for substitution and muting of lost speech frames (Bad Frame Masking)

In the following sections BFI = 1 (BFI = Bad Frame Indicator) is equivalent to a CRC error as it is specified in the section 5.1.3.3.

5.4.6.1 State Machine

This example solution for substitution and muting is based on a state machine with seven states. The system starts in state 0. Each time a bad frame is detected, the state counter is incremented by one and is saturated when it reaches 6. Each time a good speech frame is detected, the state counter is reset to zero, except when in state 6, where the state counter is set to 5. The state indicates the quality of the channel: the higher the value of the state counter, the worse the channel quality is. The control flow of the state machine can be described with the following C-code (BFI = bad frame indicator, State = state variable):

```
if(BFI != 0 )
State = State + 1;
else if(State == 6)
State = 5;
else
State = 0;
if(State > 6 )
State = 6;
```

In addition to this state machine, the BFI from the previous frame is checked (prevBFI). The processing depends on the value of the State-variable. In states 0 and 5, the processing depends also on the two flags BFI and prevBFI.

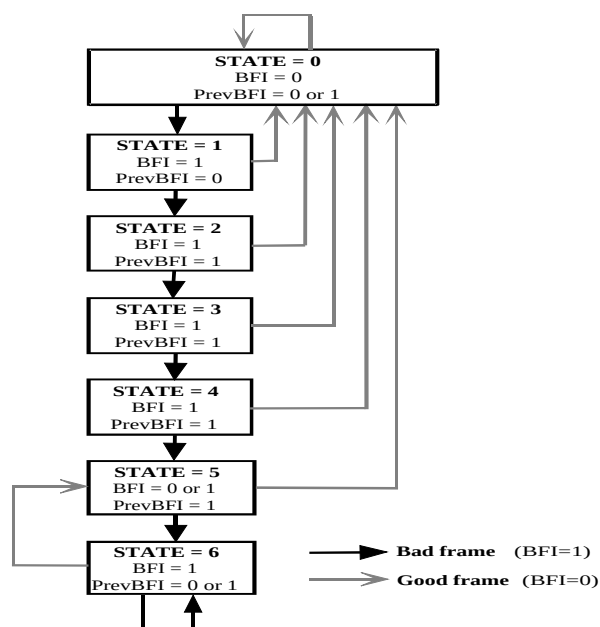


Fig. 5.4.6.1-1 State machine for controlling the bad frame substitution

5.4.6.2 Bad Frame Masking actions

5.4.6.2.1 BFI = 0, prevBFI = 0, State = 0

No error is detected in the received or in the previous received speech frame. The received speech parameters are used normally in the speech synthesis. The current frame of speech parameters is saved.

5.4.6.2.2 BFI = 0, prevBFI = 1, State = 0 or 5

No error is detected in the received speech frame but the previous received speech frame was bad. The LTP-gain and fixed codebook gain are limited below the values used for the last received good subframe:

$$g^p = \begin{cases} g^p, & g^p \leq g^p(-1) \\ g^p(-1), & g^p > g^p(-1) \end{cases}$$

where g^p = current decoded LTP-gain, $g^p(-1)$ = LTP-gain used for the last good subframe (BFI = 0), and

$$g^c = \begin{cases} g^c, & g^c \leq g^c(-1) \\ g^c(-1), & g^c > g^c(-1) \end{cases}$$

where g^c = current decoded fixed codebook-gain and $g^c(-1)$ = fixed codebook gain used for the last good subframe (BFI = 0).

The rest of the received speech parameters are used normally in the speech synthesis. The current frame of speech parameters is saved.

5.4.6.2.3 BFI = 1, prevBFI = 0 or 1, State = 1...6

An error is detected in the received speech frame and the substitution and muting procedure is started. The LTP-gain and fixed codebook gain are replaced by attenuated values from the previous subframes:

$$g^p = \begin{cases} P(state) g^p(-1), & g^p(-1) \leq \text{median5}(g^p(-1), \dots, g^p(-5)) \\ P(state) \text{median5}(g^p(-1), \dots, g^p(-5)), & g^p(-1) > \text{median5}(g^p(-1), \dots, g^p(-5)) \end{cases}$$

where g^p = current decoded LTP-gain, $g^p(-1), \dots, g^p(-n)$ = LTP-gains used for the last n subframes, $\text{median5}()$ = 5-point median operation, $P(state)$ = attenuation factor ($P(1) = 0.98$, $P(2) = 0.98$, $P(3) = 0.8$, $P(4) = 0.3$, $P(5) = 0.2$, $P(6) = 0.2$), $state$ = state number, and

$$g^c = \begin{cases} C(state) g^c(-1), & g^c(-1) \leq \text{median5}(g^c(-1), \dots, g^c(-5)) \\ C(state) \text{median5}(g^c(-1), \dots, g^c(-5)), & g^c(-1) > \text{median5}(g^c(-1), \dots, g^c(-5)) \end{cases}$$

where g^c = current decoded fixed codebook gain, $g^c(-1), \dots, g^c(-n)$ = fixed codebook gains used for the last n subframes, $\text{median5}()$ = 5-point median operation, $C(state)$ = attenuation factor ($C(1) = 0.98$, $C(2) = 0.98$, $C(3) = 0.98$, $C(4) = 0.98$, $C(5) = 0.98$, $C(6) = 0.7$), and $state$ = state number.

The higher the state value is, the more the gains are attenuated. Also the memory of the predictive fixed codebook gain is updated by using the average value of the past four values in the memory:

$$ener(0) = \frac{1}{4} \sum_{i=1}^4 ener(-i)$$

The past LSFs are used by shifting their values towards their mean:

$$lsf_q(i) = \alpha \cdot past_lsf_q(i) + (1 - \alpha) \cdot mean_lsf(i), i = 0 \dots 9$$

where $\alpha = 0.95$, lsf_q is the set of LSF-values for the current frame, $past_lsf_q$ is lsf_q from the previous frame, and $mean_lsf$ is the average LSF-vector.

5.4.6.2.4 LTP-lag update

The LTP-lag values are replaced by the lag value from the previous subframe incremented by one. The value in the first subframe of the first frame in a sequence of bad frames is the last correctly received value incremented by one.

5.4.6.2.5 Innovation sequence

The received fixed codebook innovation pulses from the erroneous frame are always used in the state they were received.

5.4.7 Bit-exact description of the PDC enhanced full rate codec

The various components of the 6.7 kbit/s ACELP codec are described in the form of a fixed-point bit-exact ANSI C-code. This software consists of the speech codec and bad frame handler functions. It also includes the VOX processing in the speech encoder.

The fixed point C-code uses a binary file interface between encoder and decoder. Each encoded frame consists of 134 speech encoder parameter bits and 3 bits of additional information. Each bit is stored in the file as a 16-bit integer. An integer value of 1 indicates a logical one and the value 0 indicates a logical zero. The bits are ordered according to table 5.4.5-1. Table 5.4.7.1-4 and 5.4.7.1-5 describes the file interface in more detail.

Note that the decoder part of the C code does not use the two bits associated with VOX and that the BFI output from the encoder is set to one if the VOX handler is in the no transmission state. Thus, the decoder can not be used to test the function of the VOX mode. Instead test vector 6 could be used to check the correctness of an implementation of the VOX functionality in the encoder.

In the fixed-point ANSI C-code, all the computations are performed using a predefined set of basic operators.

Two types of variables are used in the fixed-point implementation. These two types are signed integers in 2's complement representation, defined by:

Word16	16-bit variables
Word32	32-bit variables

5.4.7.1 Description of the constants and variables used in the C code

The ANSI C code simulation of the codec is, to a large extent, self-documented. However, a description of the variables and constants used in the code is given to facilitate the understanding of the code. The fixed-point precision (in terms of Q format, double precision (DP), or normalized precision) of the vectors and variables is given, along with the vectors dimensions and constant values.

Table 5.4.7.1-1 gives the coder global constants and table 5.4.7.1-2 describes the variables and vectors used in the encoder routine with their precision. Table 5.4.7.1-3 describes the fixed tables in the codec.

Table 5.4.7.1-1 Codec global constants.

Parameter	Value	Description
L_TOTAL	240	Size of speech buffer
L_WINDOW	240	Size of LP analysis window
L_FRAME	160	Size of speech frame
L_FRAME_BY2	80	Half the speech frame size
L_SUBFR	40	Size of subframe
M	10	Order of LP analysis
MP1	11	M+1
AZ_SIZE	44	4*M+4
GAMMA 1	30802	Weighting factor in numerator of W(z) (0.94 in Q15)
GAMMA 2	19661	Weighting factor in denominator of W(z) (0.6 in Q15)
PIT_MAX	143	Maximum pitch lag
PIT_MIN	20	Minimum pitch lag
L_INTER10	10	Order of sinc filter for interpolating The excitations is $2*L_INTER10*3+1$
PRM_SIZE	21	Size of vector of analysis parameters
SERIAL_SIZE	137	Number of speech bits + bfi + vox
GAMMA 3	18022	Formant postfilter factor (numerator) (0.55 in Q15)
GAMMA 4	22938	Formant postfilter factor (denominator) (0.70 in Q15)
MU	26214	Tilt compensation filter factor (0.8 in Q15)
GP_CLIP	15565	Pitch gain clipping (0.95 in Q14)
N_FRAME	7	Old pitch gains in average calculation
AGC_FAC	29491	Automatic gain control factor (0.9 in Q15)

Table 5.4.7.1-2 Description of the coder vectors and variables

Parameter	Precision	Initial value	Description
speech [-40..199]	Q0	0	Speech buffer
wsp [-143..-1]	Q0	0	Weighted speech buffer
wsp [0..79]	Q0	---	Weighted speech buffer
exc [-(143+11)..-1]	Q0	0	LP excitation
exc [0..39]	Q0	---	LP excitation
lsp_old [0..9]	Q15	30000, 26000, 21000, 15000, 8000, 0, -8000, -15000, -21000, -26000	LSP vector in past frame
lsp_old_q [0..9]	Q15	Same as for lsp_old	Quantized LSP vector in past frame
mem_syn [0..9]	Q0	0	Memory of synthesis filter
mem_w [0..9]	Q0	0	Memory of weighting filter (applied to input)
mem_w0 [0..9]	Q0	0	Memory of weighting filter (applied to error)
error [-10..-1]	Q0	0	Error signal (input minus synthesis)
error [0..39]	Q0	---	Error signal (input minus synthesis)
r_l [0..10] & r_h [0..10]	Normalized DP	---	Correlations of windowed speech (low and hi)
A_t [11x4]	Q12	---	LP filter coefficients in 4 subframes
Aq_t [11x4]	Q12	---	Quantized LP filter coefficients in 4 subframes
Ap1 [0..10]	Q12	---	LP coefficients with spectral expansion
Ap2 [0..10]	Q12	---	LP coefficients with spectral expansion
lsp_new [0..9]	Q15	---	LSP vector in 4th subframe
lsp_new_q [0..9]	Q15	---	Quantized LSP vector in 4th subframe
code [0..39]	Q13	---	Fixed codebook excitation vector
h1 [-40..-1]	Q12	0	Impulse response of weighted synthesis filter
h1 [0..39]	Q12	---	Impulse response of weighted synthesis filter
xn [0..39]	Q0	---	Target vector in pitch search
xn2 [0..39]	Q0	---	Target vector in algebraic codebook search
dn [0..39]	scaled max < 8192	---	Backward filtered target vector
y1 [0..39]	Q0	---	Filtered adaptive codebook vector
y2 [0..39]	Q12	---	Filtered fixed codebook vector
zero [-11..39]		0	Zero vector
gain_pit	Q14	---	Adaptive codebook gain
gain_code	Q1	---	Algebraic codebook gain
past_qua_en [0..3]	Q10	-14336	Past quantized energy prediction errors

Table 5.4.7.1-3 Codec fixed tables

Parameter	Vector size	Precision	Description
grid []	61	Q15	Grid points at which Chebyshev polynomials are evaluated
lag_h [] and lag_l []	10	DP ¹	Higher and lower parts of the lag window table
F_gamma1	10	Q15	Spectral expansion factors
F_gamma2	10	Q15	Spectral expansion factors
F_gamma3	10	Q15	Spectral expansion factors (post filter)
F_gamma4	10	Q15	Spectral expansion factors (post filter)
window_200_40 []	240	Q15	LP analysis window
table [] in Lsf_lsp ()	65	Q15	Table to compute cos(x) in Lsf_lsp ()
slope [] in Lsp_lsf ()	64	Q12	Table to compute acos(x) in LSP_lsf ()
table [] in Inv_sqrt ()	49		Table used in inverse square root computation
table [] in Log2 ()	33		Table used in base 2 logarithm computation
table [] in Pow2 ()	33		Table used in 2 to the power computation
mean_lsf []	10	Q15	LSF means in normalized frequency [0.0, 0.5]
pred_fac []	10	Q15	MA coefficients for LSF prediction
dico1_lsf []	256 x 3	Q15	1st LSF quantizer in normalized frequency [0.0, 0.5]
dico2_lsf []	512 x 3	Q15	2nd LSF quantizer in normalized frequency [0.0, 0.5]
dico3_lsf []	512 x 4	Q15	3rd LSF quantizer in normalized frequency [0.0, 0.5]
table_gain []	128*3	Q14 Q12 Q10	Quantization table of Adaptive codebook gain Fixed codebook gain and Energy prediction error
inter_3[] in Interpol_3()	13	Q15	Interpolation filter coefficients in Interpol_3 ()
inter_3[] in Pred_lt_3()	31	Q15	Interpolation filter coefficients in Pred_lt_3 ()
ph_imp_low	40	Q15	Anti-sparseness filter coefficients
ph_imp_mid	40	Q15	Anti-sparseness filter coefficients
REORD_efr2fr	134	Q0	Reordering table
b [] in Pre_Process()	3	Q12	HP filter coefficients (numerator)
a [] in Pre_Process()	3	Q12	HP filter coefficients (denominator)
b [] in Post_Process()	3	Q13	HP filter coefficients (numerator)
past_rq_init	80	Q15	LSP prediction initialization
a [] in Post_Process()	3	Q13	HP filter coefficients (denominator)
bitno []	21	Q0	Number of bits corresponding to transmitted parameters

Table 5.4.7.1-4 Encoder – Decoder file interface

Word number	Name	Description
0	BFI	Bad Frame Indicator, indicates CRC error
1-2	VOX	VOX frame type indicator
3-136	Parameters	Speech coder parameters ordered according to table 4.1

¹ Special double precision format, see oper_32b.c listing.

Table 5.4.7.1-5 VOX frame types

Bit 1	Bit 2	Description
0	0	Normal speech frame or PST1 frame
0	1	PRE frame
1	0	PST0 frame
1	1	No transmission

5.4.8 Description of the test vectors

The purpose of the test vectors is to check the bit-exactness of the speech codec implementation. All test vectors are in MS-DOS binary format. The low and high bytes must be swapped in order to use them in other platforms (for example HP-UNIX or SUN). The input files have 13-bit resolution: The 3 LSBs of the 16-bit samples are masked to zero.

5.4.8.1 Speech encoder input test sequences:

tstseq1.inp - test vector for LPC quantization codebooks

- A test vector which selects every index in each of the three SVQ codebooks (LSF residual) at least once.
- The file is composed of various kinds of input material including speech, noise, and synthetic signals
- File length: 1060 frames

tstseq2.inp - test vector for LTP (adaptive) codebook

- A test vector which selects every possible long term prediction filter delay (lag). The file contains pulse like harmonic signal and noise.
- File length: 966 frames

tstseq3.inp - test vector containing speech signal

- A test vector containing female speech samples with flat characteristics.
- File length: 360 frames

tstseq4.inp - test vector containing low level speech signal

- A test vector containing low level (-42 dB) male speech samples with flat characteristics.
- File length: 296 frames

tstseq5.inp - test vector for instability protection

- A test vector containing a sequence of DTMF tones as well as single-frequency and swept sinusoidal signals.
- File length: 350 frames.

tstseq6.inp - test vector for VOX parameters

- The file is composed of different types of input material, mainly noise.

- File length: 450 frames.

The implementation of the VOX parameter encoding is tested using this input sequence while forcing the speech encoder to VOX mode by setting VAD_flag according to table 5.4.8.1-1. VAD_flag = 1 indicates voice present, VAD_flag = 0 indicates voice absent.

Table 5.4.8.1-1 VAD_flag settings for test sequence 6

Frame number	Setting
0 - 98	1
99 - 138	0
139	1
140 - 168	0
169 - 170	1
171 - 198	0
199 - 201	1
202 - 248	0
249 - 298	1
299 - 348	0
349	1
350 - 378	0
379 - 380	1
381 - 418	0
419 - 421	1
422 - 449	0

5.4.8.2 Speech encoder reference output sequences:

tstseq1.cod
tstseq2.cod
tstseq3.cod
tstseq4.cod
tstseq5.cod
tstseq6.cod

The implementation of the speech encoder can be verified by encoding the speech encoder input sequences and comparing the output with the corresponding speech encoder reference output sequences.

5.4.8.3 Speech decoder input sequences:

The speech encoder reference output sequences (tstseq1.cod, tstseq2.cod, tstseq3.cod, tstseq4.cod, tstseq5.cod) are used as the speech decoder input test sequences. Note that test sequence 6 should not be used as input to the speech decoder. The only purpose of test sequence 6 is to verify the VOX part of the speech encoder.

5.4.8.4 Speech decoder reference output sequences:

tstseq1.out
tstseq2.out
tstseq3.out
tstseq4.out
tstseq5.out

The implementation of the speech decoder can be verified by decoding the speech decoder input sequences and comparing the output with the corresponding speech decoder reference output sequences.

5.4.9 Contents of the soft copy distribution

The soft copy distribution consists of a CD-ROM in MS-DOS format. The detailed contents of the CD-ROM is listed below.

Directory src:

a_refl.c	lsp_lsf.tab
agc.c	makefile.pc
autocorr.c	makefile.sun
az_lsp.c	n_stack.h
basic_op.h	oper_32b.c
basicop2.c	oper_32b.h
bits2prm.c	ph_disp.c
c3_14pf.c	ph_disp.tab
c_g_aver.c	pitch.c
cnst.h	post_pro.c
cod_6k7.c	postfilt.c
codec.h	pow2.c
coder.c	pow2.tab
coder.rsp	pre_proc.c
convolve.c	pred_lt3.c
copy.c	preemph.c
d3_14pf.c	prm2bits.c
d_plsf_3.c	proc_hd.c
dec_6k7.c	q_plsf_3.c
dec_gain.c	q_plsf_3.tab
dec_lag3.c	qua_gain.c
decoder.c	qua_gain.tab
decoder.rsp	readme.txt
ec_gains.c	refl_a.c
gmed5.c	reord_ch.c
grid.tab	reord_ch.tab
int_lpc.c	reorder.c
int_lsf.c	residu.c
inv_sqrt.c	set_zero.c
inv_sqrt.tab	sig_proc.h
lag_wind.c	sqrt_s.c
lag_wind.tab	syn_filt.c
levinson.c	typedef.h
log2.c	vox_hand.c
log2.tab	weight_a.c
lsp_avg.c	weight_f.c
lsp_az.c	window.tab
lsp_lsf.c	

Directory testinp:

tstseq1.inp
tstseq2.inp
tstseq3.inp

tstseq4.inp
 tstseq5.inp
 tstseq6.inp

Directory testcod:

tstseq1.cod
 tstseq2.cod
 tstseq3.cod
 tstseq4.cod
 tstseq5.cod
 tstseq6.cod

Directory testout:

tstseq1.out
 tstseq2.out
 tstseq3.out
 tstseq4.out
 tstseq5.out

5.4.10 References

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- [6] P. Kabal and R.P. Ramachandran, "The computation of line spectral frequencies using Chebyshev polynomials", *IEEE Trans. on ASSP*, vol. 34, no. 6, pp. 1419-1426, Dec. 1986.
- [7] C. Laflamme, J-P. Adoul, R. Salami, S. Morissette, and P. Mabillean, "16 kpbs wideband speech coding technique based on algebraic CELP" *Proc. ICASSP'91*, pp. 13-16.

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Chapter 6 Measurement Methods

As for the items stipulated in Ordinance concerning Technical Regulations Conformity Certification, etc. of Specified Radio Equipment Appendix table 1 item 1(3), measurement methods are specified by Ministry of Internal Affairs and Communications Notification (Note 1).

(Note 1) Public Notice of Ministry of Internal Affairs and Communications No. 88 (January 26, 2004)

6.1 Transmission system

Detailed examples on measurement methods are given in the following Items 6.1.1 through 6.1.12.

The following items are common for all the measurement methods.

- i) The mobile stations must be set for repetitious transmission and base stations must be set for continuous transmission.
- ii) The standard encoding test signal used for modulation must be pseudo noise random test signal with a 511-bit code length and must be set for the user information transfer channel or all of the slot areas.
- iii) Definition for the inner burst section used for mobile stations must be at least 120 symbols ranging from the beginning of the symbol immediately after ramp up to the end of the symbol immediately before ramp down.

Definition for the outer burst section is to be at least 240 symbols from the end of the symbol immediately before ramp down minus 3 symbols to the beginning of the symbol immediately after ramp up of the next slot minus 3 symbols.

6.1.1 Frequency deviation

Measurement methods shall comply with the test methods specified by Ministry of Internal Affairs and Communications Notification (Note 1) stipulated in Ordinance concerning Technical Regulations Conformity Certification, etc. of Specified Radio Equipment Appendix table 1 item 1(3).

6.1.2 Strength of spurious emissions or unwanted emissions

Measurement methods shall comply with the test methods specified by Ministry of Internal Affairs and Communications Notification (Note 1) stipulated in Ordinance concerning Technical Regulations Conformity Certification, etc. of Specified Radio Equipment Appendix table 1 item 1(3).

6.1.3 Occupied bandwidth

Measurement methods shall comply with the test methods specified by Ministry of Internal Affairs and Communications Notification (Note 1) stipulated in Ordinance concerning Technical Regulations Conformity Certification, etc. of Specified Radio Equipment Appendix table 1 item 1(3).

6.1.4 Antenna power deviation

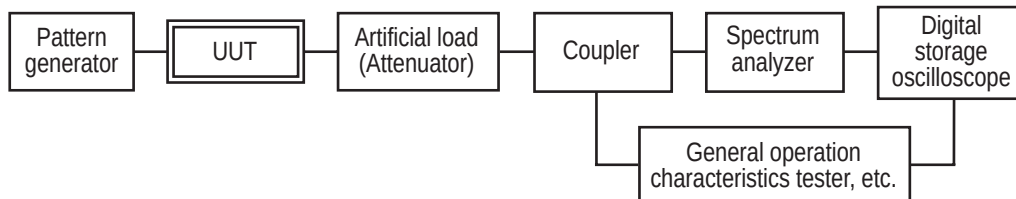
Measurement methods shall comply with the test methods specified by Ministry of Internal Affairs and Communications Notification (Note 1) stipulated in Ordinance concerning Technical Regulations Conformity Certification, etc. of Specified Radio Equipment Appendix table 1 item 1(3).

6.1.5 Leakage power during carrier off

Measurement methods shall comply with the test methods specified by Ministry of Internal Affairs and Communications Notification (Note 1) stipulated in Ordinance concerning Technical Regulations Conformity Certification, etc. of Specified Radio Equipment Appendix table 1 item 1(3).

6.1.6 Transmission ramp-up and ramp-down power

(1) Measuring system diagram



(2) Conditions for measuring instruments

- a. The general operation characteristics tester or equivalent which can output the trigger signal corresponding to the timing of the transmission burst must be used. The gating signal may be generated by a broadband detector instead of the general operation characteristics tester.
- b. A spectrum analyzer attaching a video output terminal must be used. Also, the vertical axis of the oscilloscope for the entire measuring system comprised of spectrum analyzer and oscilloscope must be pre-calibrated using the power meter.
- c. The spectrum analyzer is to be set as shown below:

Center frequency : Carrier frequency

Frequency sweeping range : 0 Hz

Resolution bandwidth : Approx. 100 KHz

Y-axis scale : 10 dB/div

Input level : No greater than the input power tolerance. The average noise level of the spectrum analyzer shall be at least 10dB below the standard leakage power during carrier off.

- d. The video output signal must be obtained using a digital storage oscilloscope.

Sweep: : External trigger

Sweep time : Approx. 200 μ s

(3) Conditions for unit under test

The test frequency must be set for transmission.

(4) Measuring procedure

Measure the video output signal from the spectrum analyzer with the oscilloscope.

6.1.7. Modulation accuracy

(1) Definition

a. Principle

When the ideal transmitter output is made to pass through the ideal root roll-off reception filter and subjected to sampling at a 1-symbol interval, no interference between codes occurs. Thus, the phase and amplitude are represented by the formula below.

$$S(k) = S(k-1) e^{j\{\pi/4 + B(k) \cdot \pi/2\}}$$

$$K = \frac{n}{21\text{KHz}}, |S(k)| = 1 \quad \text{.....(6.1.7-1)}$$

In this instance, B(k) = 0, 1, 2, 3 (must accord with the table below.)

Xk	Yk	B(k)
0	0	0
0	1	1
1	1	2
1	0	3

Xk, Yk are two data that are obtained from binary data which have undergone serial-parallel conversion.

When an actual transmission signal is used, interference occurs between codes. By measuring the amount of interference error, modulation accuracy can be defined.

b. Modulation accuracy definition formula

The signal transmitted by an actual transmitter is passed through the ideal reception filter and the signal obtained at moment "k" (which is a 1-symbol interval) is Z(k). Then, the Modulation accuracy can be indicated by using S(k) as shown below.

$$Z(k) = [C_0 + C_1 \cdot \{S(k) + E(k)\}] \cdot W^k \quad \text{..... (6.1.7-2)}$$

In this instance:

$W = e^{dr+jda}$: Frequency offset which is equivalent to the phase rotation of da [rad/symbol] and the amplitude variation of dr [neper/symbol].

C_0 : Specific origin offset which represents an unbalance for the orthogonal modulator.

C_1 : Complex constant which represents the specified phase and output power of the transmitter.

$E(k)$: Residual vector error with sample S(k)

The sum of the squares of vector errors is represented by the following formula.

$$\sum_{k=\text{MIN}}^{k=\text{MAX}} |E(k)|^2 = \sum_{k=\text{MIN}}^{k=\text{MAX}} | \{ [Z(k) \cdot W^{-k} - C0] / C1 \} - S(k) |^2 \quad \dots\dots (6.1.7-3)$$

CO, C1 and W are selected to minimize the sum which is obtained using this formula and then used to calculate the vector error for each symbol. The symbol timing position for reception output is selected to minimize vector error.

The maximum and minimum sums for the uplink channel (individual assignment slot) are given below:

MIN = 2 (vector right after ramp up)

MAX = 136 (vector right before ramp down)

The maximum and minimum sums for the downlink channel (individual assignment slot) are given below:

MIN = 2 (vector right after ramp up)

MAX = 139 (vector right after TCH or FACCH symbol)

The effective value of the vector error is calculated as the square root of the sum of the square of the vector errors divided by the number of symbols in the slot (uplink channel: 135, downlink channel: 138).

The effective value for this vector error is defined as the modulation accuracy.

(2) Measuring system diagram



(3) Conditions for measuring instrument

The modulation accuracy meter must have a reception root roll-off filter function and be capable of measuring the difference of the effective value between the ideal signal and the signal which is actually transmitted.

(4) Conditions for unit under test

The test frequency must be set for transmission.

(5) Measuring procedure

- a. Measure the difference between the ideal vector convergence point in the signal space and the actual transmission wave.
- b. Sum up the square of the vector error at each point obtained in a. above and divide the sum by the

number of symbols in the slot. Modulation accuracy is obtained as a square root of this value.

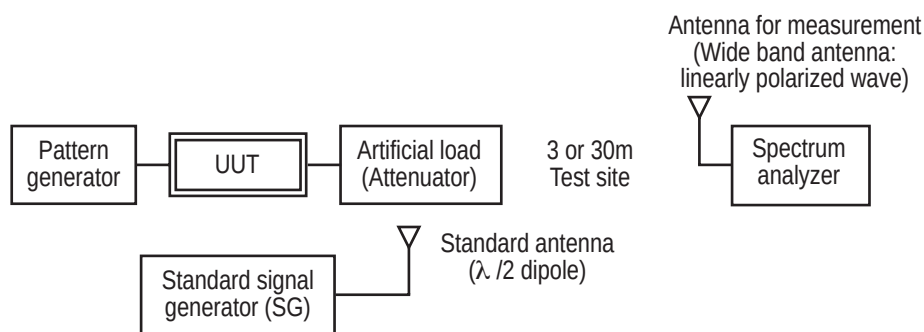
- The mobile station : 135 symbols (individual assignment slots)
- The base station : 138 symbols (individual assignment slots)

6.1.8. Leakage power of adjacent channel

Measurement methods shall comply with the test methods specified by Ministry of Internal Affairs and Communications Notification (Note 1) stipulated in Ordinance concerning Technical Regulations Conformity Certification, etc. of Specified Radio Equipment Appendix table 1 item 1(3).

6.1.9. Cabinet radiation

(1) Measuring system diagram



(2) Conditions for measuring instrument

- The antenna terminal of the unit under test is terminated with an artificial load.
- Measurement is done in the open test site according to IEC Pub. 489-6 Appendix K if one side of the unit under test exceeds 60 cm or the measuring frequency is less than 100MHz. In other cases, measurement is done in the open test site according to Appendix K or L.
- In case of a frequency of 300 MHz or over, it is better to suppress the ground reflected wave to avoid trouble due to complication of the height pattern. Therefore, it is better to make the antenna for measurement highly directional and increase the height of the unit under test.
- The standard antenna is $\lambda/2$ dipole type and the measurement range shall be 25 MHz up to 3 GHz.

(3) Conditions for unit under test

The test frequency is set for transmission.

(4) Measuring procedure

- Place the unit under test on a turntable and check if the spectrum is present within the specified frequency range.
- Synchronize the spectrum analyzer with one of the frequency components checked in a. above.

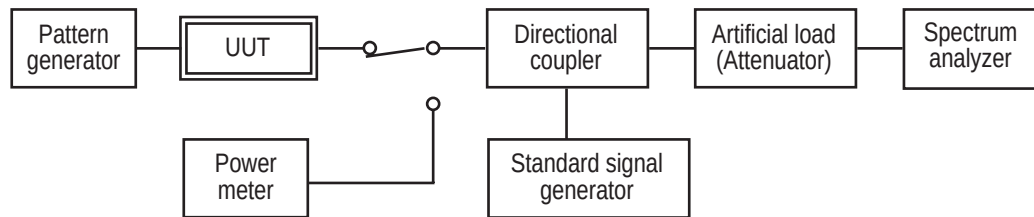
- c. The antenna for measurement is set for vertical polarized wave.
- d. Rotate the turntable to obtain the maximum display value of spurious power (average power in the burst section). Measurement is the same as "6.1.2 strength of the spurious emission".
- e. Move the antenna for measurement up and down between 1m and 4m above the ground to obtain the maximum display value.
- f. Repeat d. and e. above until the maximum value does not increase then record the frequency, maximum display value and the antenna height at that time.
- g. The antenna for measurement is set for horizontal polarized wave and repeat d. to f. above.
- h. Do the above steps b. to g. on all spectrum frequencies obtained in a. above.
- i. Replace the unit under test by the standard antenna to which RF power is supplied by the standard signal generator.
- j. Synchronize the standard antenna with the frequency of the spectrum measured in f. above.
- k. Both of the standard antenna and antenna for measurement are set for vertical detection.
- l. Move the antenna for measurement up and down between 1m and 4m above the ground and adjust SG output level so that the maximum display value of the spectrum analyzer at that time coincides with the maximum value obtained in f. above. Record the SG output level and the measurement antenna height at that time.
- m. Both standard antenna and antenna for measurement are set for horizontal polarized wave and l. is repeated.
- n. Repeat j. to m. on all frequency components measured up to h. above.

(5) Calculation of results

Cabinet radiation is obtained by adding the SG output level obtained in the above (4) measurement to the calibration of standard antenna gain and cable loss between the SG and standard antenna.

6.1.10 Transmission intermodulation

(1) Measuring system diagram



(2) Conditions for measuring instrument

- a. The transmitting wave must have enough directivity so that no leak nor measurement error occurs in the standard signal generator.
- b. Setting of standard signal generator

Center frequency: Transmission average frequency ± 500 kHz

- c. Setting of digital storage type spectrum analyzer

Same as "6.1.2 strength of spurious emission" measurement.

(3) Conditions for unit under test

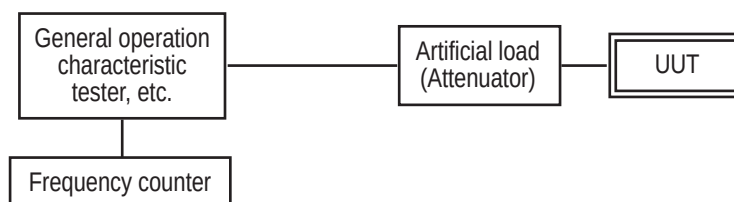
The test frequency is set for transmission.

(4) Measuring procedure

- a. Set the center frequency of the standard signal generator at the carrier + 500 kHz (or -500 kHz) and set the output level of the standard signal generator so that a maximum transmission output of -30 dB is obtained in terms of the unit under test antenna output end calculation.
- b. Set the switching selector on the unit under test side and obtain power of transmission intermodulated wave by the spectrum analyzer.
Measurement is the same as "6.1.2 Strength of spurious emission" measurement.
- c. The ratio of the maximum transmission output and the maximum value of power obtained in b. corresponds to the transmission intermodulation.

6.1.11 Time alignment

(1) Measuring system diagram



(2) Conditions for measuring instruments

- a. The general operation characteristics tester must feature the base station functions to establish connection to the unit under test and be capable of changing the specific parameters.
- b. The general operation characteristics tester must be capable of yielding the difference between the tester's transmission timing and unit under test's transmission timing.

The accuracy of the difference in the output timing must be over 10 times the standard set accuracy.

- c. The frequency counter must be capable of time interval measurement. The accuracy of time interval measurement must be over 10 times of the standard set accuracy.
- d. The total measurement accuracy in b. and c. above must be over 10 times the standard set accuracy.
- e. The unit under test must be set for the maximum output and no fading for the above measurement.

(3) Conditions for unit under test

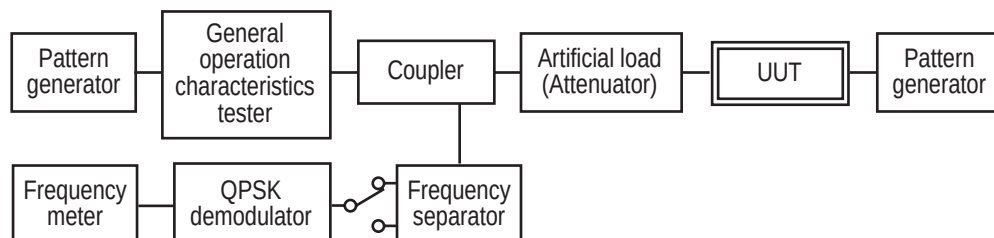
The unit under test must be set for normal operation.

(4) Measuring procedure

- a. The general operation characteristics tester must activate the TCH and transmit the command to transmit at the reference timing.
- b. The general operation characteristics tester must yield the difference between the tester's communication channel transmission timing and unit under test's communication channel transmission timing.
- c. Measure the timing gap with the time interval mode of the frequency counter.
- d. Change the amount of time alignment and repeat procedures a. and b. above when specifying the time alignment 6 symbols faster or 1 symbol faster.

6.1.12 Signal transmission rate

(1) Measuring system diagram



(2) Conditions for measuring instruments

- a. The general operation characteristics tester must have the functions of the base station against the UUT and possess the characteristic parameters can be varied.
- b. The QPSK demodulator must be capable of recovering and outputting the clock frequency which determines the signal transmission rate.
- c. The frequency precision for the frequency meter must be at least 1 digit higher than the standard resolution, required by the operator company.

(3) Conditions for UUT

Set the unit under test to the test frequency for transmission.

(4) Measuring procedure

- a. Set the output frequency from the general operation characteristics tester to the test frequency. Then adjust the output level so that -103 dBm can be obtained for the input of the UUT.
- b. Measure the clock frequency from the UUT. (F1)
- c. Connect the output from the general operation characteristics tester to the frequency counter and measure the clock frequency. (F2)
- d. Calculation of base station tracking accuracy. The base station tracking accuracy (ΔF) is shown below.

$$\Delta F = | F1 - F2 | / F2$$

(5) Other method

In case the clock frequency of the tester and UUT recovered clock are measurable, these may be used for the base station tracking accuracy measurement.

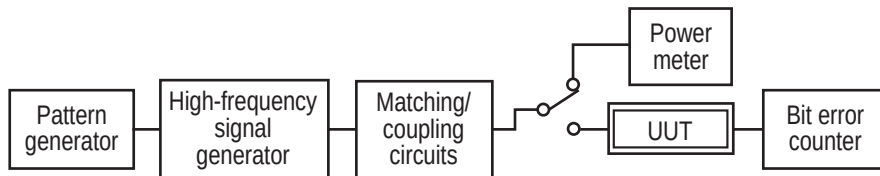
6.2 Reception system

Detailed examples on measurement methods are given in the following Items 6.2.1 through 6.2.10.

6.2.1 Reception sensitivity

6.2.1.1. Reception sensitivity (Static)

(1) Measuring system diagram



(2) Conditions for measuring instruments

a. High-frequency signal generator

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r.m.s. vector error must be 3% or less (recommended value)

Adjacent channel leakage power : ± 50 kHz detuning, 65 dB or more (recommended value)

± 100 kHz detuning, 80 dB or more (recommended value)

b. Pattern generator

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Must continuously generate the standard encoded test signal (a pseudo-noise sequence binary code with the code length of 511 bits) which accords with ITU-T V.52 for certain portions of the user data transfer channel. It must also generate the specified patterns necessary for communications for portions other than the user data transfer channel. Refer to Fig. 6.2-1 for an example of transmission and reception timing.

(3) Conditions for the unit under test

a. The unit under test must be set for the test frequency for reception.

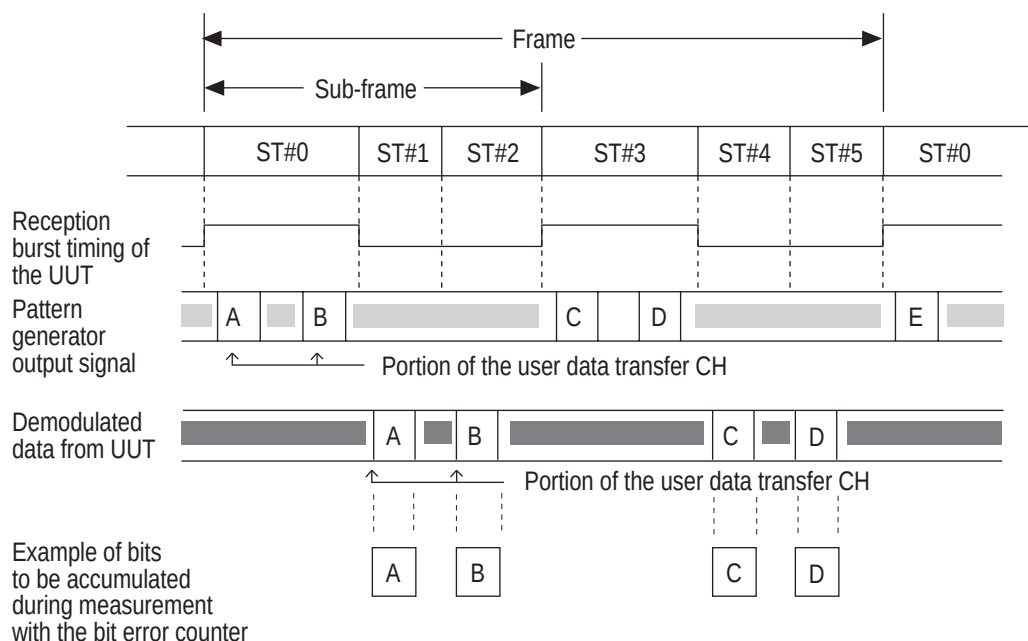
b. Demodulation data must be a portion of user data transfer channel (the standard encoded test signal portion).

Note: Refer to Fig. 6.2-1 for the output timing for the demodulated data.

(4) Measuring procedure

- Set the center frequency for the high frequency signal generator to the test frequency.
- The high-frequency signal generator must be set for continuous or burst transmission. Adjust the signal level so that the specified sensitivity level (static) is obtained with the power meter. Also, set the switch to apply signals to the unit under test.
- Accumulate the bit series for the data for the user data transfer channel specified slot using the bit error counter, then obtain the bit error rate for the bits more than above 2556 bits.

* (Criteria) When the measured bit error rate is below standard, it is rated as successful.



(Note 1) By connecting A B C D E signals, the standard encoded test signal can be accumulated.

(Note 2) Specific patterns required for communication are inserted in the  sections.

(Note 3) Specific demodulated patterns required for communication are included in the  sections.

(Note 4) ST#0 Time slot 0 ST#3 Time slot 3
 ST#1 Time slot 1 ST#4 Time slot 4
 ST#2 Time slot 2 ST#5 Time slot 5

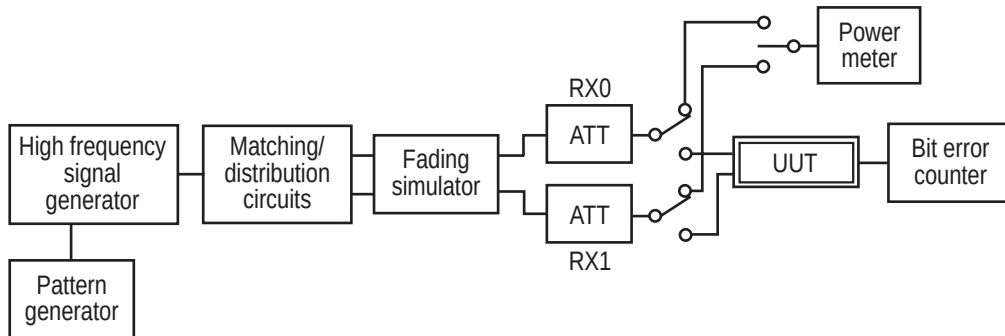
(Note 5) The figure illustrates the case of the full rate. In the half rate case, the standard encoded test signal is transmitted and received in each TDMA frame (6 slots).

Fig. 6.2-1: Standard encoded test signal transmission and reception methods using the pattern generator and bit error counter. (An example)

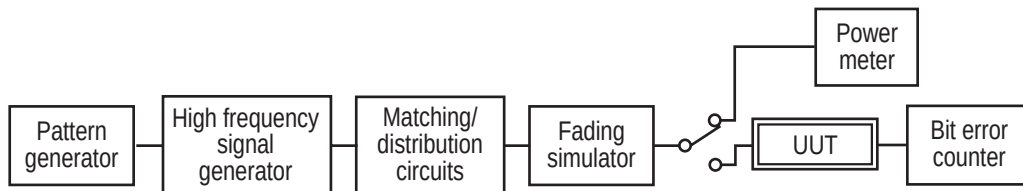
6.2.1.2. Reception sensitivity (Fading)

(1) Measuring system diagram

a. With diversity



b. Without diversity



(2) Conditions for measuring instruments

a. High-frequency signal generator

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r.m.s. vector error must be 3% or less (recommended value)

Adjacent channel leakage power : ± 50 kHz detuning, 65 dB or more (recommended value)
 ± 100 kHz detuning, 80 dB or more (recommended value)

b. Pattern generator

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Generates a reference coding test signal continuously (pseudo binary noise signal of 511-bit code length which accords with ITU-T V.52) to the user data transfer channel. Also, generates specific patterns that are necessary for communications to other portions. Fig. 6.2-1 is an example of transmission/reception timings.

c. Fading simulator : Rayleigh fading of maximum Doppler frequency 40 Hz.

Must accord with IEC Pub. 489-6 APPENDIX C.

(3) Conditions for the unit under test

- a. The unit under test must be set to the test frequency for reception.
- b. Demodulation data must be part of the data for the user data transfer channel (the standard encoded test signal portion).

(4) Measuring procedure

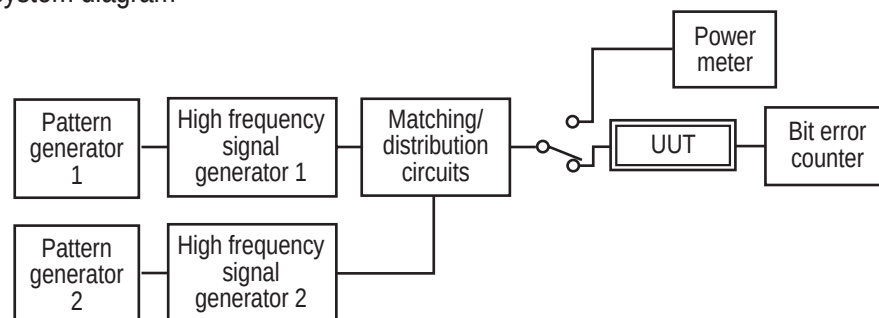
- a. Set the center frequency of the high frequency generator to the test frequency.
- b. The high-frequency generator must be set for continuous or burst transmission. Adjust the signal level to a value which yields the specified sensitivity level (median during fading) at the power meter. Then, set the switch to apply signals to the unit under test.
- c. Accumulate the bit series of the user data transfer channel using the bit error counter and obtain the bit error rate with more than 2556 bits or the bits obtained by the following formula, whichever is larger:

$$\frac{43,200 \times \text{bit rates (bps)}}{\text{Running speed (Km / Hr)} \times \text{Test frequency (MHz)}} \quad \text{Bit rate (bps): 42,000}$$

*(Criteria) When the measured bit error rate is below standard, it is rated as successful.

6.2.2 Adjacent channel selectivity

(1) Measuring system diagram



(2) Conditions for measuring instruments

- a. High-frequency signal generator 1 and 2

Frequency : Within the specified frequency range

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r.m.s. vector error must be less than 3% (recommended value)

Adjacent channel leakage power : ± 50 kHz detuning, 65 dB or more (recommended value)
 ± 100 kHz detuning, 80 dB or more (recommended value)

- b. Pattern generator 1

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Generates a reference coding test signal continuously (pseudo binary noise signal of 511 bit code length which accords with ITU-T V.52) to the user data transfer channel. Also, generates specific patterns that are necessary for communications to other portions. Fig. 6.2-1 is an example of transmission/reception timings.

c. Pattern generator 2

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Must continuously generates digital signals (a pseudo-noise sequence binary code with code length of 32,767 bits, ITU-T 0.151).

(3) Conditions for the unit under test

- a. The unit under test must be set to the test frequency for reception.
- b. Demodulated data must be part of the data for the user data transfer channel (the standard encoded test signal portion).

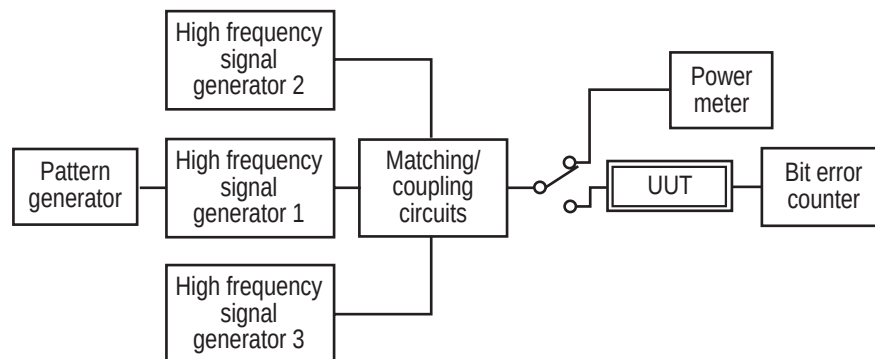
(4) Measuring procedure

- a. Set the center frequency of high frequency signal generator 1 to the test frequency.
- b. Set the center frequency of high frequency signal generator 2 to the adjacent channel frequency.
- c. The high-frequency signal generator 1 must be set for continuous or burst transmission. Also adjust the signal level so that the specified sensitivity level +3dB is obtained at the power meter.
- d. The high-frequency signal generator 2 must be set for continuous or burst transmission. Adjust the signal level so that the following value is obtained at the power meter. [(specified sensitivity level + 3 dB) + (specified adjacent channel selectivity)dB]dBμV. Then set the switch to apply signals to the unit under test.
- e. Accumulate the bit series of the user data transfer channel using the bit error counter and obtain the bit error rate with more than 2556 bits.

(Criteria) When the measured bit error rate is below 1×10^{-2} , it is rated as successful.

6.2.3 Intermodulation characteristics

(1) Measuring system diagram



(2) Conditions for measuring instruments

a. High-frequency signal generators 1,2 and 3

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r.m.s. vector error must be 3% or less (recommended value)

Adjacent channel leakage power : ± 50 kHz detuning, 65 dB or more (recommended value)
 ± 100 kHz detuning, 80 dB or more (recommended value)

b. Pattern generator

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Generates a reference coding test signal continuously (pseudo binary noise signal of 511-bit code length which accords with ITU-T V.52) to the user data transfer channel. Also, generates specific patterns that are necessary for communications to other portions. Fig. 6.2-1 is an example of transmission/reception timings.

(3) Conditions for the unit under test

- a. The unit under test must be set to the test frequency for reception.
- b. Demodulated data must be part of the data for the user data transfer channel (the standard encoded test signal portion).

(4) Measuring procedure

- a. Set the center frequency of high frequency signal generator 1 to the test frequency.

- b. Set the frequency of high frequency signal generator 2 to the test frequency +100 KHz (or -100 KHz) and high frequency signal generator 3 to the test frequency +200 KHz (or -200 KHz).
- c. The high-frequency signal generator 1 must be set for continuous or burst transmission. The signal level is set so that the reference sensitivity + 3dB can be obtained at the power meter.
- d. The high-frequency signal generators 2 and 3 must be set for continuous or burst transmission and non-modulation. The signal levels for the high frequency signal generators 2 and 3 must be set so that the following value is obtained at the power meter:

$$[(\text{Specified usable sensitivity level} + 3\text{dB}) + (\text{Specified IM characteristics value})\text{dB}]\text{dB}\mu\text{V}$$

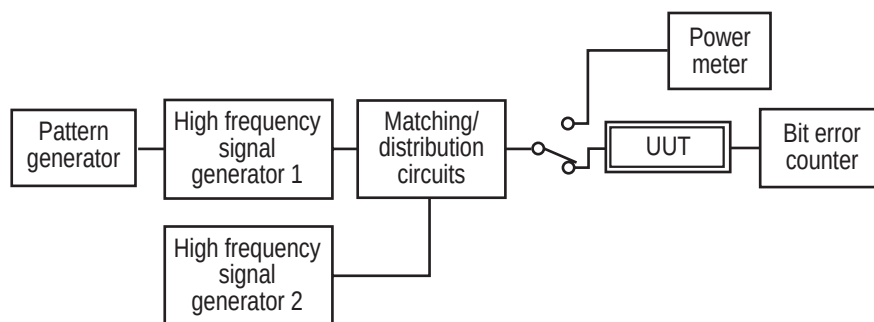
Then, set the switch to apply signals to the unit under test.

- e. Accumulate the bit series of the user data transfer channel using the bit error counter and obtain the bit error rate with more than 2556 bits.

*(Criteria) When the measured bit error rate is below 1×10^{-2} , it is rated as successful.

6.2.4 Spurious sensitivity

(1) Measuring system diagram



(2) Conditions for measuring instruments

- a. High-frequency signal generators 1 and 2

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r.m.s. vector error must be 3% or less (recommended value)

Adjacent channel leakage power : ± 50 kHz detuning, 65 dB or more (recommended value)
 ± 100 kHz detuning, 80 dB or more (recommended value)

b. Pattern generator

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Generates a reference coding test signal continuously (pseudo binary noise signal of 511-bit code length which accords with ITU-T V.52) to the user data transfer channel. Also, generates specific patterns that are necessary for communications to other portions. Fig. 6.2-1 is an example of transmission/reception timings.

(3) Conditions for the unit under test

- a. The unit under test must be set to the test frequency for reception.
- b. Demodulated data must be part of the user data transfer channel (the standard encoded test signal portion).

(4) Measuring procedure

- a. Set the center frequency of high frequency signal generator 1 to the test frequency.
- b. Set the frequency for high frequency signal generator 2 to the spurious frequency(*).
- c. The high-frequency signal generator 1 must be set for continuous or burst transmission. Set the signal level so that the specified sensitivity level +3dB is obtained at the power meter.
- d. The high-frequency signal generator 2 must be set for continuous or burst transmission and non-modulation.

Set the signal level so that the following value is obtained at the power meter:
 [(specified sensitivity level + 3 dB) + (specified spurious sensitivity value) dB] dBμV

Then, set the switch to apply signals to the unit under test.

- e. Accumulate the bit series of the user data transfer channel using the bit error counter and obtain the bit error rate with more than 2556 bits.

(Note*) The spurious frequency must be the frequency of the unnecessary signals which deteriorates the bit error rate. One of the methods to set the spurious frequency is to derive the spurious frequency from the IF level of the unit under test using a receiver capable of receiving IF frequency.

* (Criteria) When the measured bit error rate is below 1×10^{-2} , it is rated as successful.

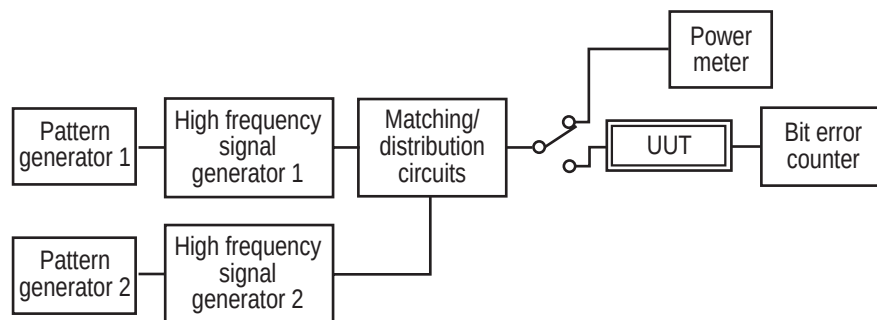
6.2.5 Strength of secondary radiowave emission

Measurement methods shall comply with the test methods specified by Ministry of Internal Affairs and Communications Notification (Note 1) stipulated in Ordinance concerning Technical Regulations Conformity Certification, etc. of Specified Radio Equipment Appendix table 1 item 1(3).

6.2.6 Interference level

6.2.6.1 Interference level (Static)

(1) Measuring system diagram



(2) Conditions for measuring instruments

a. High-frequency signal generators 1 and 2

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r.m.s. vector error must be 3% or less (recommended value)

Adjacent channel leakage power : ± 50 kHz detuning, 65 dB or more (recommended value)
 ± 100 kHz detuning, 80 dB or more (recommended value)

b. Pattern generator 1

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Generates a reference coding test signal continuously (pseudo binary noise signal of 511 bit code length which accords with ITU-T V.52) to the user data transfer channel. Also, generates specific patterns that are necessary for communications to other portions. Fig. 6.2-1 is an example of transmission/reception timings.

c. Pattern generator 2

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Must generate digital signals (a pseudo-noise sequence binary code with code length of 32,767 bits, ITU-T 0.151) continuously.

(3) Conditions for the unit under test

a. The unit under test must be set to the test frequency for reception.

- b. Demodulated data must be part of the data for the user data transfer channel (the standard encoded test signal portion).

(4) Measuring procedure

- a. Set the center frequency of high frequency signal generator 1 to the test frequency.
- b. Set the center frequency of high frequency signal generator 2 to the test frequency.
- c. The high-frequency signal generator 1 must be set for continuous or burst transmission. The signal level must be set so that the [specified sensitivity level +30 dB] dBμV is obtained at the power meter.
- d. The high-frequency signal generator 2 must be set for continuous or burst transmission. Also adjust the signal level so that the following value is obtained at the power meter:

$$[(\text{Specified sensitivity level} + 30\text{dB}) - (\text{specified interference level}) \text{ dB}] \text{ dB}\mu\text{V}$$

Then, set the switch to apply signal to the unit under test.

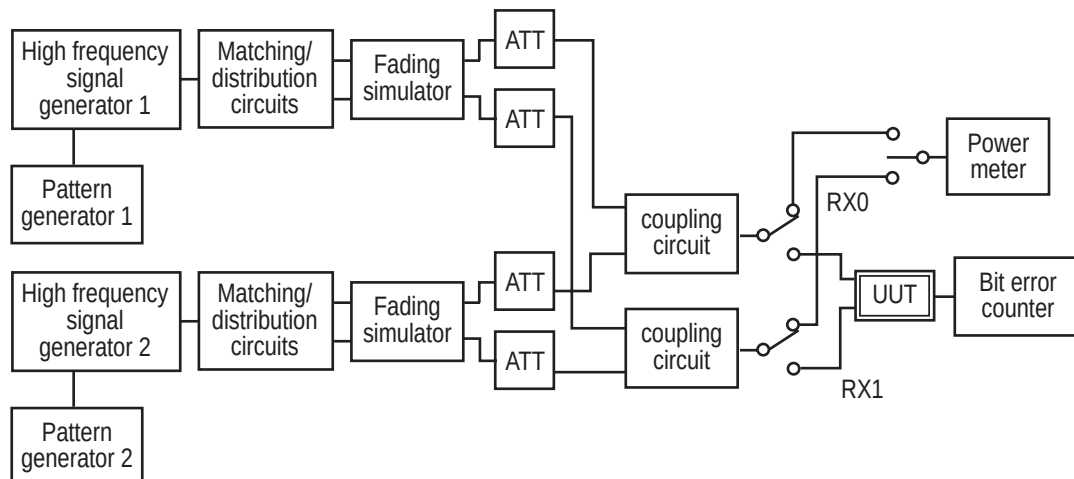
- e. Accumulate the bit series of the user data transfer channel using the bit error counter and obtain the bit error rate with more than 2556 bits.

*(Criteria) When the measured bit error rate is below standard, it is rated as successful.

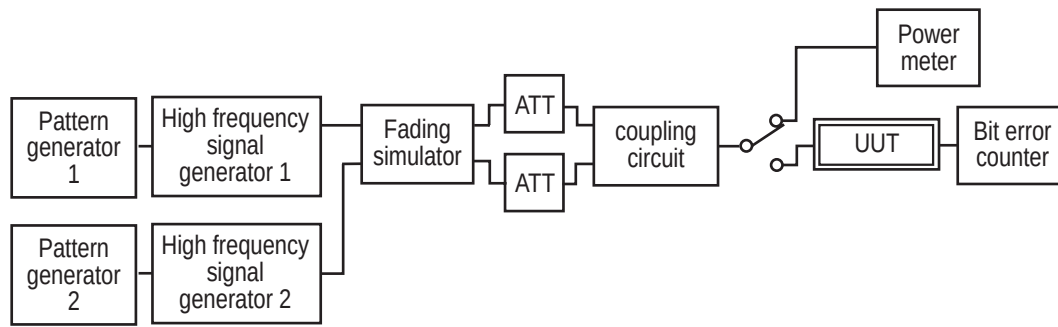
6.2.6.2 Interference level (Fading)

(1) Measuring system diagram

- a. With diversity



- b. Without diversity



(2) Conditions for measuring instruments

a. High-frequency signal generators 1 and 2

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r.m.s. vector error must be 3% or less (recommended value)

Adjacent channel leakage power : ± 50 kHz detuning, 65 dB or more (recommended value)
 ± 100 kHz detuning, 80 dB or more (recommended value)

b. Pattern generator 1

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Generates a reference coding test signal continuously (pseudo binary noise signal of 511-bit code length which accords with ITU-T V.52) to the user data transfer channel. Also, generates specific patterns that are necessary for communications to other portions. Fig. 6.2-1 is an example of transmission/reception timings.

c. Pattern generator 2

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Must generate digital signals (a pseudo-noise sequence binary code with code length of 32,767 bits, ITU-T 0.151) continuously.

d. Fading simulator : Must be Rayleigh fading with max. Doppler frequency of 40 Hz.
Must accord with IEC Pub.489-6 APPENDIX C.

(3) Conditions for the unit under test

- a. The unit under test must be set to the test frequency for reception.
- b. Demodulated data must be the user data transfer channel (the standard encoded test signal portion).

(4) Measuring procedure

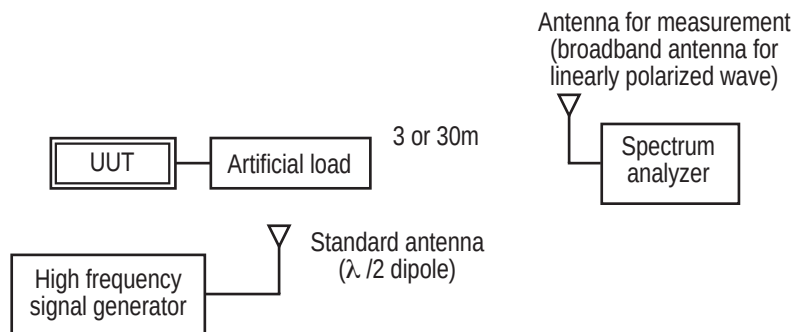
- a. Set the center frequency of high frequency signal generator 1 to the test frequency.
- b. Set the center frequency of high frequency signal generator 2 to the test frequency.
- c. The high-frequency signal generator 1 must be set for continuous or burst transmission. Set the signal level so that [specified sensitivity level +30dB] dB μ V (median) is obtained at the power meter.
- d. The high-frequency signal generator 2 must be set for continuous or burst transmission. Set the signal level so that the following value is obtained at the power meter:
[(Specified sensitivity level + 30 dB) - (specified interference level)dB] dB μ V
Then, set the switch to apply signals to the unit under test.
- e. Accumulate the bit series of the user data transfer channel using the bit error counter and obtain the bit error rate with more than 2556 bits or the bits obtained by the following formula, whichever is larger:

$$\frac{43,200 \times \text{bit rates (bps)}}{\text{Running speed (Km / Hr)} \times \text{Test frequency (MHz)}} \quad \text{Bit rate (bps): 42,000}$$

*(Criteria) When the measured bit error rate is below standard, it is rated as successful.

6.2.7 Cabinet radiation

(1) Measuring system diagram



(2) Conditions for measuring instruments

- a. The antenna terminal of the unit under test must be terminated with an artificial load.
- b. If the length of one of the sides of the unit under test is in excess of 60 cm, or else the measurement frequency is below 100 MHz, measurement must be performed in an open test site as prescribed in IEC Pub. 489-6 APPENDIX K. In other cases, measurement must be performed in an open test site as prescribed in the APPENDIX K or L.
- c. At frequencies above 300 MHz, the ground reflection wave should be suppressed in order to prevent troubles caused by a complex height pattern. For this reason, a high directivity antenna must be used as the measurement antenna and the unit under test should be positioned higher.
- d. Use the same measurement method described in Item 6.1.2 "Strength of spurious emission." For setting the spectra analyzer of digital storage type.
- e. The standard antenna is $\lambda/2$ dipole type and the measurement range shall be 25 MHz up to 3 GHz.

(3) Conditions for the unit under test

The unit under test must be set to the test frequency for reception.

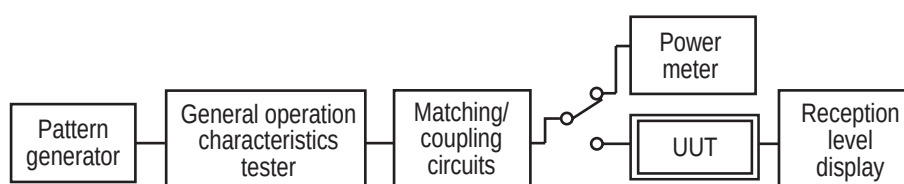
(4) Measuring procedure

- a. Place the unit under test on the turntable and measure the spectrum for cabinet radiation.
- b. Tune the spectrum analyzer to one of the frequency components spectrums measured in a. above.
- c. The measurement antenna must be set for vertically polarized plane waves.
- d. Rotate the turntable to obtain the maximum spurious power (average power in the bursts) readout. Use the same measurement method described in Item 6.1.2 "Strength of spurious emission"
- e. Vary the height of the measurement antenna between 1m to 4m from the ground to obtain the maximum spurious power readout.

- f. Repeat steps d. and e. above until the maximum spurious power stops increasing and record the frequency, maximum read out and the height of the measurement antenna.
- g. Set the measurement antenna for horizontally polarized plane waves and repeat steps d. thru f. above.
- h. Carry out the procedures outlined in b. thru g. above for the frequencies for all the spectrums obtained in a.
- i. Replace the unit under test with a standard antenna that is powered by the high frequency signal generator.
- j. Tune the standard antenna to the frequency of the spectrum measured in step f. above.
- k. Set both the standard and measurement antennas for vertically polarized plane waves.
- l. Vary the height of the measurement antenna between 1m and 4 m from the ground and adjust the output level of the high frequency signal generator so that the maximum display value for the spectrum analyzer is equivalent to the maximum value obtained in step f.. Record the output level of the high frequency signal generator and the measurement antenna height.
- m. Set both the standard and measurement antennas for horizontally polarized plane waves and repeat the procedure in step l. above.
- n. Repeat steps j. through m. for all the frequencies measured thru step h. above.
- o. Cabinet radiation can be obtained by adding the gain of the standard antenna and the calibration of the cable loss between the high frequency generator and standard antenna to the high frequency signal generator output level obtained in l. above.

6.2.8. Reception level detection

(1) Measuring system diagram



(2) Conditions for measuring instruments

- a. General operation characteristics tester : must have the function of the base station communicating with the UUT and the specific parameter can be varied.
- b. Pattern generator
 - Clock frequency : 42 kHz
 - Clock accuracy : within $\pm 1 \times 10^{-6}$

Generated pattern : Generates a reference coding test signal continuously (pseudo binary noise signal of 511-bit code length which accords with ITU-T V.52) to the user data transfer channel. Also, generates specific patterns that are necessary for communications to other portions. Fig. 6.2-1 is an example of transmission/reception timings.

(3) Conditions for the unit under test

- a. The unit under test must be set to the test frequency for reception.
- b. The reception level output must be the mean value of the reception levels in 36 slots (Full rate: 1 superframe, Half rate: 2 superframes).

(4) Measuring procedure

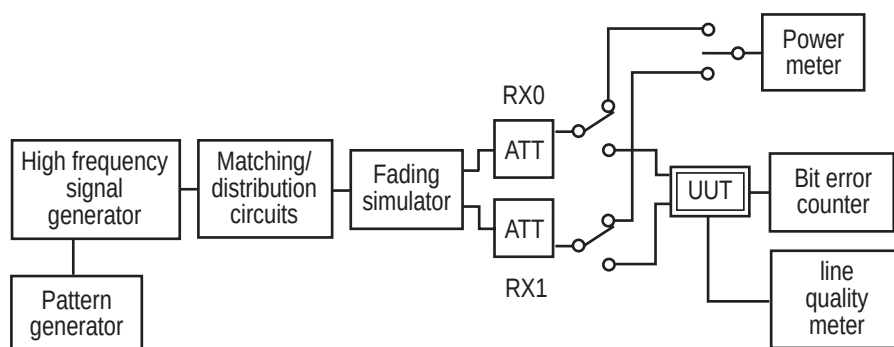
- a. Set the center frequency of the general operation characteristics tester to the test frequency.
- b. Set the general operation characteristics tester for continuous transmission.
- c. Set the output level of the general operation tester to the minimum value (or the maximum value) in the detection range specified by the technical requirements for absolute accuracy using the power meter. Then, select the switch to apply signals to the UUT.
- d. The general operation characteristics tester generates a command for setting the UUT for the standby state.
- e. The general operation characteristics tester activates the TCH and transmits a command for reporting the detected reception level by the home reception slot for the entire detection range. Also, it transmits a command for reporting the detected reception level by empty slots within the detection range specified by the relative accuracy.
- f. The UUT detects the reception level for the home reception and empty slots and indicates it on the reception level display or reports to the general operation characteristics tester. (*see note.)
- g. Measure the reception level detected by the home reception and empty slots appearing on the reception level display or the general operation characteristics tester.
- h. Vary the output level for the general operation characteristics tester between the maximum value and the minimum value for the specified detection range using the power meter and repeat the procedures in e) through g) above.

*Note : The reception level of the same frequency is detected for the home reception slot and empty slots. However, even for the empty slots, frequency for the synthesizer must be set in the same manner as for normal operation.

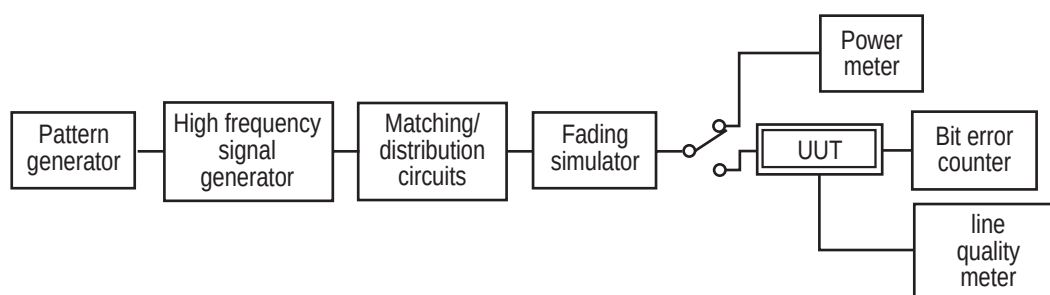
6.2.9 Line quality detection accuracy

(1) Measuring system diagram

a. With diversity



b. Without diversity



(2) Conditions for measuring instruments

a. High-frequency signal generator

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r.m.s. vector error must be 3% or less (recommended value)

Adjacent channel leakage power : ± 50 kHz detuning, 65 dB or more (recommended value)
 ± 100 kHz detuning, 80 dB or more (recommended value)

b. Pattern generator

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Generates a reference coding test signal continuously (pseudo binary noise signal of 511-bit code length which accords with ITU-T V.52) to the user data transfer channel. Also, generates specific patterns that are necessary for communications to other portions. Fig. 6.2-1 is an example of transmission/reception timings.

- c. Fading simulator : Rayleigh fading of max. Doppler frequency of 40 Hz. (Must accord with the IEC Pub. 489-6 APPENDIX C.)
- d. Line quality meter : Yields the estimated bit error rate for each superframe.

(3) Conditions for unit under test

- a. Set the UUT to the test frequency for reception.
- b. Demodulation data shall be the channel (portion corresponding to the reference coding test signal) for user information transfer.
- c. Line quality shall be detected from 36 slots (Full rate: 1 superframe, Half rate: 2 superframes).

(4) Measuring procedure

- a. Set the center frequency of the high-frequency signal generator to the test frequency.
- b. The high-frequency generator must be set for continuous transmission or burst transmission and apply it to the UUT.
- c. Accumulate bit series of the user information transfer channel with the bit error meter and measure the bit error rate. Use the values listed in Table 6.2.9-1 for number of measured bit samples required for each reference error rate.
- d. Repeat the above measurement several times to obtain the average rate. The number of measurement for each reference error rate is listed in Table 6.2.9-1.
- e. Adjust the signal level in less than 1dB steps so that the average error rate becomes closest to the reference error rate.
- f. Measure the estimated error rate with a line quality meter.
- g. Repeat steps c. to f. above at reference error rates listed in the table.

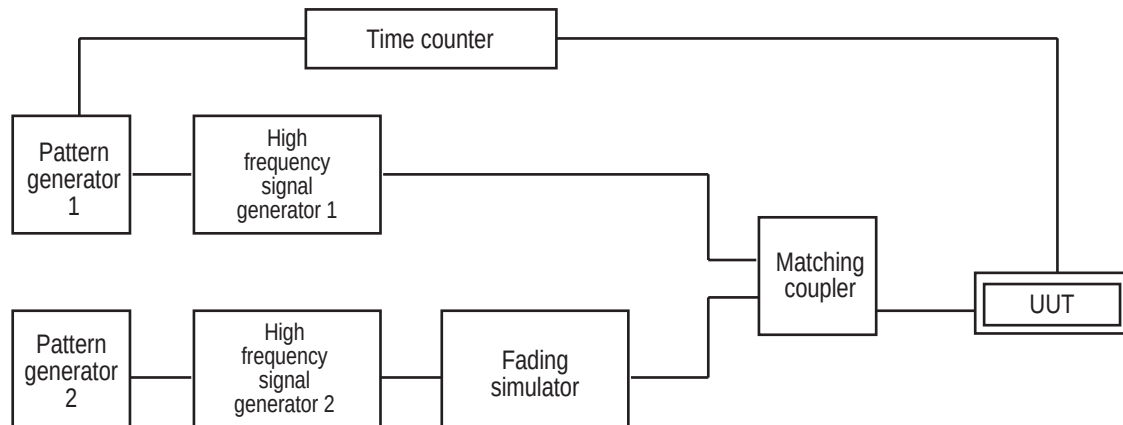
Table 6.2.9-1

Ref. error rate (%)	The number of measured bits required for each measurement	Measurement count (number of measurements)
5.5	$\geq$ The number of bits given by equation (1).	≥ 5
1.6	$\geq$ The number of bits given by equation (1).	≥ 5
0.55	$\geq$ Twice the number of bits given by equation (1)	≥ 5
0.1	≥ 10 times the number of bits given by equation (1)	≥ 10
$\frac{43,200 \times \text{bit rates (bps)}}{\text{Running speed (Km / Hr)} \times \text{Test frequency (MHz)}} \dots\dots \text{Equation (1)}$		

- * If the estimated error rate is within the standard range no less than 8 times out of 10 measurements in comparison to the reference error rate, it is rated as successful.

6.2.10 Handover time

(1) Measuring system diagram



(2) Conditions for measuring instruments

a. High-frequency signal generators 1 and 2

Stability : within $\pm 2 \times 10^{-9}$

Modulation accuracy : r. m. s. vector error must be 3% or less (recommended value)

Adjacent channel leakage power : $\pm 50\text{kHz}$ detuning, 65 dB or more

(recommended value)

$\pm 100\text{kHz}$ detuning, 80 dB or more

(recommended value)

b. Pattern generators 1 and 2

Clock frequency : 42 kHz

Clock accuracy : within $\pm 1 \times 10^{-6}$

Generating pattern : Pattern generator 1 must continuously generate the standard encoded test signal (a pseudo-noise sequence binary code, in compliance with ITU-T V.52, with the code length of 511 bits). It must also generate the specific pattern required for communications to the portions other than the user information transfer channel.

It also can generate a command to handover the information channel to a channel which corresponds with high-frequency signal generator 2. Refer to Fig. 6.2-1 for an example of transmission and reception timing.

Pattern generator 2 continuously generates downlink initial bursts (SB1).

c. Fading simulator

A fading simulator with Doppler frequency equivalent to pseudo mobile speed 8km/h and 100km/h (in accordance with IEC Pub. 489-6 APPENDIX C) shall be used.

(3) Conditions for unit under test

- a. Set the UUT to the test frequency before channel handover for reception.
- b. Set up the UUT so that the transmission timing for SB2 can be monitored.

(4) Measuring procedure

- a. Set the center frequency for high-frequency signal generator 1 to the test frequency.
- b. Set the high-frequency signal generator 1 for continuous transmission. Set the signal level to -60 dBm using the power meter.
- c. Set high-frequency signal generator 2 to the test frequency which is detuned at least 50 kHz from the test frequency set for high-frequency signal generator 1. Apply fading specified in Item 4.1.8 and the frame timing shall be asynchronous with high-frequency signal generator 1. Set the signal level to -90 dBm and perform continuous transmission.
- d. Connect the radio circuit with the TCH for the test frequency set for high-frequency signal generator 1.
- e. Allow pattern generator 1 to generate a handover command to switch the UUT to the channel for the test frequency set for high-frequency signal generator 2.
- f. Measure the duration of time after the handover command is generated to the UUT until the UUT transmits the first SB2 at the correct position, within 1 symbol.
- g. If the measured handover time satisfies the range listed in Table 4.1.18-1 with 95% or higher probability, it is rated as successful.
- h. Change the fading speed for high-frequency signal generator 2 and repeat the procedures outlined in above a. thru g..

Chapter 7 Glossary

Associated control channel (ACCH):

A point-to-point bidirectional control channel appended to the TCH for transmitting signaling information and user packet data.

ACELP:

Abbreviation of Algebraic Code Excited Linear Prediction. This is an algorithm compressing redundant information in speech by sending codebook parameters to generate speech in the same manner as VSELP and PSI-CELP. However, it realizes further memory and processing volume reductions and quantization efficiency improvements by utilizing an algebraic codebook.

Acknowledged information transfer service:

An information transferring operation which transfers layer 3 information by frame acknowledgment on the data link layers. This service also performs error recovery by retransmitting unacknowledged frames. When errors cannot be recovered within the data link layer, this service sends a report to the management entity. This service also specifies the flow procedure.

Assignment source point (ASP):

Layer management entities on the network for SMSI assignment.

Authentication:

A function to check if the mobile station is authorized for communications.

Authentication result:

The result obtained from the random authentication pattern by the mobile's predetermined operation.

Authentication random pattern:

The random pattern sent from the network to the mobile station for authentication.

Basic retransfer control:

Retransfer control which is carried out for all layer 2 under HDLC procedure using N(S) and N(R).

Broadcast channel (BCCH):

A unidirectional channel used by the base station to broadcast control information to land mobile units, such as location registration, channel structure and system information.

Bearer capability:

Transmission service which the mobile station requests to the network.

Bit interleave:

The method by which the data written horizontally in matrix is read vertically (column direction) to create radio signals which have strong burst error resistance.

Bit transparency:

Transferring the bit series as it is sent from the originating party to the called party.

Burst ID number:

The number used to distinguish between synchronization burst and end burst.

Cause indication:

An information field used for indicating the cause for call disconnection.

Common access channel (CAC):

A common access channel (BCCH, PCH, SCCH, UPCH). One physical channel is commonly used by several users.

Call control (CC):

The layer 3 module which carries out service control for calls.

Call number:

A number used to identify each call over the user-network interface.

Called party number:

An information field which is used for specifying the called party.

Calling party number:

An information field which is used for identifying the calling party.

Code Excited Linear Prediction (CELP):

An abbreviation for Code Excited Linear Prediction, which is a type of speech coding system wherein voice waveforms are analyzed into parameters before they are transmitted. The reception end recovers the speech signal using the parameter information and a filter and reproduces the voice with white noise. The characteristics of this system is to vector quantize the excitation signal using the codebook and the system has excellent performance for less than 8 kbps, but numerous steps are required for signal processing.

Channel identifier:

An information field which is used for identifying the channel for the interface controlled by the signaling procedure.

Ciphering:

Encryption of information.

To prevent possible illegal use of mobile telephone systems.

Ciphering mode:

Field for bit assignment of ciphering key for encryption of radio signal.

Class 1:

The segment in the 134 bit voice coded frame to which convolutional encoding applies. This section corresponds to a total of 75 bits.

Class 2:

The segment of the 134 bit voice coded frame to which convolutional encoding does not apply. Class 2 bits correspond to the remaining 59 bits which are not included in Class 1.

Collision control bits:

Bits which contain transmission enable/disable, signal reception/non-reception and partial echo. These are used for random access control for the SCCH uplink channel.

Common control channel (CCCH):

A point-to-multipoint bidirectional channel which can also be used as the UPCH. It includes the PCH and SCCH.

Common Platform:

A function that simultaneously transmits CC, MM and RT information on one signal in order to increase signal transfer efficiency on layer 3.

Condition report information:

An information field which is used to specify the conditions under which the communications quality which is detected by the mobile station is reported to the base station.

Connection endpoint:

A connection endpoint for (N) SAPs.

Connection endpoint suffixes (CES):

Element which, together with the SAPI, constructs the connection end point identifiers.

The connection endpoint identifier recognizes the message units that are transmitted between the data link layer and layer 3.

Connection endpoint identifiers:

A concept which is used in place of DLCI by the layer 3 entities or management entities.

CON Primitive:

A primitive which is used by the layer which provides the requested service to confirm that the operation of the primitive is ended.

Control channel structure information:

An information element which indicates the physical structure (frequency, slot, etc.) of the CCH for each radio channel. The frequency and slots for the common control channel can be specified by both mobile and base stations by using a specially designated number, such as the MS identification number.

CS-ACELP:

Abbreviation of Conjugate Structure Algebraic Code Excited Linear Prediction. This is an algorithm compressing redundant information in speech by sending codebook parameters to generate speech in the same manner as VSELP and PSI-CELP. However, it is characterized both by further memory and processing volume reductions and by improvements in quantization efficiency and error resiliency, resulting from the use of conjugate structure codebooks and algebraic codebooks.

Cyclic redundancy check (CRC):

A method for detecting errors by dividing the reception polynomial, which expresses the reception data by the generator polynomial and then detecting the errors by residuals.

Datalink connection:

In order to realize an exchange of information between two or more layer 3 entities, a correspondence must be established between the layer 3 entities inside the data link layer using the appropriate data link protocol.

Datalink connection identifier (DLC I):

The datalink connection identifier consists of SAPI and MSI (or SMSI) and is used for identifying the message unit transferred between the data link layer and layer 3.

EC Control field extension bit (EC):

The top bit of each control field octet, which is added to extend the control field length.

Entity:

A conceptual model used to illustrate a functional module which a layer employs to proceed communication.

Expanded information element length:

The information length of the information element. The length of the information element can be set at the operator's option.

Fast associate control channel (FACCH):

An ACCH which steals the TCH temporarily and performs high speed data transfer.

Forward error correction (FEC):

An error detection and correction method by which the transmission side attaches the error detection correction code and the reception side detects and corrects errors by means of the pertinent code.

Flow control process:

A signal transfer function which transfers signals while acknowledging whether or not the signal is received by the other party and if the other party is ready to receive signals.

Frame:

A signal interval which consists of 6 TDMA slots.

Frame synchronous word:

Frame synchronization bits. Several patterns are used within one frame to identify the slot to be used in the carrier.

Guard time:

The time period when no signal provided between burst signals so to prevent collisions with adjacent channels while burst signals are being transmitted.

Handover Type:

An information field which indicates the handover type.

Housekeeping bit (RCH):

The housekeeping bit (RCH) is composed of the bits on the slot including TCH. It contains the layer 1 information which is used for real-time transmission of radio channel control bits, such as interference level report, transmission power control, that are used for keeping the radio circuit operation.

ID control field (AC):

A control area used by the network to assign or cancel the SMSI to the mobile station.

ID display field (AI):

A display field used for identifying the MSI and SMSI contained in the address field.

IND Primitive:

A primitive used by the layer which provides service to report some kind of operation related to the service to the adjacent higher layer.

Interference countermeasure codes (color codes):

A code which is assigned to each cluster (frequency repetition unit) to distinguish signals sent from a station which is causing interference. 256 8-bit code types can be used as interference countermeasure codes.

LAPDM:

A link access procedure (layer 2) on the CCH for the digital mobile communications system. The purpose of the LAPDM is to transfer the information between layer 3 entities over the digital mobile telephone system radio zone interface.

Location identity:

A number that indicates the location registration area that the mobile station is situated in.

Maximum number of reporting channels:

The maximum number of channels to report the results of zone/sector supervision.

Measurement mode:

An information field which is used to specify the number of measurement frames for the specified frequency to the mobile station.

Message type:

An information field which is used to identify the function of the transferred messages.

Mobile management (MM):

A functional module which performs location registration and authentication functions for the MS.

Mobile station identifier (MSI):

An ID that distinguishes given mobile stations from others. It is used as part of the address field in layer 2. 127 (1111111) indicates a global address.

Multiframe:

A function which permits transmitting lframes up to modular -1 consecutively, without the acknowledgement.

Multiframe operation mode:

Acknowledged information transfer mode using the modulo 64 sequence number.

(N) Connection:

Linkage between (N + 1) entities is referred to as (N) connection. The (N) connection is provided using the (N) layer as the communications route to interconnects (N) SAPs which each (N + 1) entity accesses.

Number of location registration area:

The number of location registration areas multiplexed in a particular radio zone.

Number of selected zones:

The number of zones selected as prospects for the circuit connection zone.

Outstanding number:

The number of information frames which have not yet been acknowledged in the acknowledged operation.

Paging channel (PCH):

A one-way channel that transfers the same information from the network to mobile stations over a broad "paging" area.

Partial echo (PE) generation processing:

By means of slot type random access radio channel control the base station returns part of the uplink signal that was received in the previous slot in the downlink signal to acknowledgment whether the signal which the mobile station transmitted was received successfully or not, and determine mobiles further transmission of succeeding slots.

Partial retransmission control:

Retransmission control where by error correction is performed by retransmission of only unreceived units in a layer 2 frame during an acknowledgment functional process.

Peripheral level measurement:

Measurement of the reception level of the frequency from the peripheral zone, by which the mobile station detects the zone where the mobile station is located during the standby state or in communications.

Perch channel:

A candidate channel for the BCCH for which the mobile station starts searching when the power is turned on.

Perch channel number for supervision:

An information source which indicates the channel number supervising the level of peripheral zone/sector when the mobile station is in the standby state or when active communications are taking place.

Postamble:

Layer 1 bit series which indicates that the TCH transmission output will be OFF in the VOX control mode.

Preamble:

A layer 1 signal pattern used to establish bit synchronization.

Primitive:

Information which is exchanged between adjacent upper and lower layers through service access points.

Priority order identifier:

An identifier used to determine which message unit has higher priority when message units sent from layer 2 to layer 1 become competitive.

Progress indicator:

An information field which is used to report an event which occurs while a call exists.

Protocol identifier:

An identifier used for distinguishing the message for the user-network call control from the messages regulated by the RCR standards. It is also used to distinguish the messages regulated by the RCR standards from OSI network layer protocol units which are coded in accordance with other standards.

PSI-CELP

Abbreviation for the Pitch Synchronous Innovation-Code Excited Linear Prediction Coding. This is a speech coding algorithm compressing redundant information in speech by using codebook parameters to generate the speech in the same manner as in the VSELP. However, the PSI-CELP further improves the quantization efficiency by processing the codebook synchronizing with the fundamental period of the speech as the feature.

Radio link random access control with partial echo:

A method to prohibit a mobile station from sending further bursts to avoid collision, according to the result of reception of the first burst. This control method also acknowledges whether the transmission of a mobile station was successful or not by echoing back a portion of the up-link bit sequence in the collision control bits. This is a suitable random access method to apply to balanced layer 2 procedures.

Radio transmission control (RT):

A functional module which sets, up maintains and switches the radio channel.

Regulatory information restriction indicator:

An information field which is used for reporting the regulatory conditions on the network to mobile stations.

Reserved:

The information element area is specified, however, other coded values than those defined are set as "reserved" for future standardization purposes, and must not be used until they are defined.

REQ Primitive:

A primitive used when a high layer request service from a lower layer.

Received signal strength Indicator (RSSI):

Reception level

RSP Primitive:

A primitive used by the higher layer to acknowledge the reception of the IND primitive sent from the lower layer.

Slow associate control channel (SACCH):

Another name for ACCH in the normal mode.

Scramble:

Randomizing operation of transmitting bit sequence by taking an EOR (exclusive OR) between the transmitting data sequence and a Maximum length bit sequence.

Scramble codes:

Codes which indicate the pattern used for bit scrambling for the RF section.

The scramble code is specified by the initial value of 9-level PN and the 9th bit is fixed to "1" while the remaining 8 bits indicate 256 types of patterns.

Security:

To prevent illegal use of mobile telephone systems, assignment of an exclusive mobile unit number, application of the authentication procedure, ciphering functions for communications information and so forth can be specified.

Service access point (SAP):

Access points where (N) entities receive (N - 1) services from the layer (N - 1).

Service access point identifiers:

SAPIs are used to identify service access points on the network or user side of the user/network interface.

Shortened mobile station identifier (SMSI):

A 1-octet mobile station identification which is tentatively assigned by the network to the mobile station to simplify the MSI search by the network. The SMSI is controlled by each channel in individual cells and it is released when the layer 1 channel is switched.

Signal disassembly/assembly bits:

Bit series which indicates the first and last flags in the message, the number of units remaining in the message and the number of valid octets in the last unit, and used for disassembly and assembly of messages.

SID:

Abbreviation of Silence Insertion Description. This is information, described with 15 bits, used to generate background noise during silences resulting from VOX control. Refer to ITU-T G.729 Annex B for details.

Single cell signaling channel (SCCH):

A point-to-multipoint bidirectional channel used for transmitting information between the network and mobile stations. This channel is used when the cell location of the mobile station is known by the network. Information is transmitted independently to each cell using different frequency sources. Up-link is performed by random access control.

Slots:

A signal segment which has six 40 ms frames. The scramble segment is 6.6 ms in length and consists of two slot types: single assignment slots and common slots. One user uses 2 slots in 1 frame on a full-rate channel, while one user uses 1 slot on a half-rate channel.

Spare:

Spare bits are ignored at the reception end in the standard. (Refer to appendix K.)

Superframe synchronization counter (SSC):

A synchronization bit for 7 superframe cycle.

Superframe synchronization word (SSW):

Bits which are used to establish superframe synchronization for determining the positions for BCCH, PCH and SCCH in the control channel and the positions for SACCH and RCH in the traffic channel.

Superframe:

A signal area consisting of 18 frames.

Steal flag :

FACCH steals TCH and performs signal transfer. The steal flag is one of the steal identification methods. When the steal flag indicates "steal" and no error is detected from the CRC of FACCH, it is recognized as FACH.

Symbol:

2 bits which correspond to 47.6 μ s.

Synchronization burst:

A signal which is sent to establish synchronization either when setting up the traffic channel or switching channels. A sync burst include a 32-bit sync word, burst ID bit, time alignment bit, superframe synchronization word.

System Identifier:

An information field which reports system identification.

Information transfer channel (TCH):

A point-to-point bidirectional channel which transfers the user information and user information control signal. The TCH transfers voice and facsimile information.

Time alignment control:

Time alignment control is performed in order to prevent burst signals from colliding due to transmission delays in the radio channel. The transmission timing of the mobile station is adjusted by the symbol unit when a new traffic channel is activated, or while communications are in progress on the traffic channel.

Transmission output control:

The transmission level of the mobile station is controlled by the reception level of the signals from the mobile station at the base station in order to reduce the interference to other stations and to save power consumption of portable units.

Transmission output control for the base station is left up to the operator.

Transparent:

Transfer of bit sequence from the sender to the receiver without any modification.

Unacknowledged operating mode:

An information transmission of layer 3 information by the unnumbered information (UI) frames. Transmission is not acknowledged by the data link layer. i.e., the error correction function or flow control function of layer 2 do not operate.

Unit:

The minimum unit of layer 1 bits which forms layer 2 frame. One unit consists of several to somewhere over ten octets length. The number of remaining units is indicated at the top of each unit and used for disassembly and assembly of the layer 2 frame.

User packet channel (UPCH):

A point-to-multipoint bidirectional channel which transfers the control signal information and user packet data. The uplink channel is a random access.

Vector Sum Excited Linear Prediction (VSELP):

A variation of CELP (Code Excited Linear Predictive Coding). VSELP uses a codebook which has a predefined structure such that the computations required for the codebook search process can be significantly reduced, subsequently reduces the memory required for the codebook search processing.

VOX control:

A function that the mobile station during call in progress turns transmission output ON and OFF according to presence/absence of voice input. This is performed to reduce power consumption of the mobile station.

PERSONAL DIGITAL CELLULAR TELECOMMUNICATION SYSTEM

ARIB STANDARD

RCR STD-27L

Fascicle 2

Published History of Japanese Versions

I S S U E D	APRIL	1991
REVISION A	JANUARY	1992
REVISION B	DECEMBER	1992
REVISION C	NOVEMBER	1994
REVISION D	JUNE	1995
REVISION E	SEPTEMBER	1996
REVISION F	FEBRUARY	1997
REVISION G	MAY	1998
REVISION H	FEBRUARY	1999
REVISION I	JULY	2000
REVISION I-1	MARCH	2001
REVISION J	MAY	2002
REVISION K	JULY	2003
REVISION L	NOVEMBER	2005

Published by

Association of Radio Industries and Businesses

Nittochi Bldg. 1-4-1 Kasumigaseki, Chiyoda-ku, Tokyo 100-0013 Japan

TEL 81-3-5510-8590

FAX 81-3-3592-1103

Printed in Japan

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Bit Exact ANSI-C Program for 5.4 ACELP Speech Code

**ACELP EFR
Speech Codec
Version 27 L
Fixed point code and specification**

RCR STD-27
ACELP EFR
Speech Codec
Fix point code and specification
Ver.1